

SKEMA BUSINESS SCHOOL

Master Financial Markets & Investment – Second Year

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## **Fundamentals of Finance**

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# Mathematical and statistical elements

The whole set of points addressed in this review covers more or less the mathematics syllabus of the first and second years of a degree in economics and management. While the mastery of these elements makes it possible to understand in depth the construction of the economic models addressed in this course, they might nevertheless frighten the least mathematically inclined candidates among you. Do not hesitate, therefore, to skip the reading of this section at first, since the economics paper in administrative competitive examinations is above all a literary and compositional exercise: the absence of mathematical considerations in a dissertation, an operational note, a short question or a text analysis will never be held against you and is not penalised. Priority therefore to mastering the spirit of economic models, failing an apprehension of all their mathematical arcana! Only part 3.1, pages 38 and following, devoted to descriptive statistics, must be perfectly mastered for the day itself.

## 1 A review of matrices and linear algebra

### 1.1 Matrix calculus: definitions and elementary operations

#### 1.1.1 Matrices, transpose and trace

A **matrix** is an array of real numbers comprising  $n$  rows and  $p$  columns, where  $n$  and  $p$  are non-zero natural numbers. For example, we set  $\mathbf{A}$  to be the following matrix:

$$\mathbf{A} := \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,p} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,p} \\ \vdots & \vdots & & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,p} \end{pmatrix}$$

Usually, if  $\mathbf{A}$  is a matrix with  $n$  rows and  $p$  columns, we indeed denote, for every integer  $i$  between 1 and  $n$  and every integer  $j$  between 1 and  $p$ , by  $a_{i,j}$  the term of the  $i$ -th row and the  $j$ -th column of  $\mathbf{A}$ . Consequently, the pair  $(n, p)$  is then called the **dimension** of the matrix  $\mathbf{A}$ ; the set of matrices of dimension  $(n, p)$  with real coefficients is denoted  $\mathcal{M}_{n,p}(\mathbb{R})$ . In particular, if we have  $n = p$ , the matrix  $\mathbf{A}$  is said to be a **square matrix** (of order  $n$ ); the set of square matrices of order  $n$  with real coefficients is usually denoted  $\mathcal{M}_n(\mathbb{R}) := \mathcal{M}_{n,n}(\mathbb{R})$ . If we have  $p = 1$ , the matrix  $\mathbf{A}$  is said to be a **column vector** (or quite simply a **vector**) with  $n$  rows (or with  $n$  dimensions, or even with  $n$  coordinates): by an abuse of language we denote by  $\mathbb{R}^n = \mathcal{M}_{n,1}(\mathbb{R})$  the set of real vectors with  $n$  coordinates. By analogy, if we have  $n = 1$ , the matrix  $\mathbf{A}$  is said to be a **row vector** with  $p$  columns (or with  $p$  dimensions, or even with  $p$  coordinates). Consequently, if  $\mathbf{A} := (a_{i,j})_{1 \leq i, j \leq n}$  is a square matrix of order  $n$ , then we call the **trace** of  $\mathbf{A}$  the sum of the **diagonal elements** of  $\mathbf{A}$ ; otherwise put, we have  $\text{Tr}(\mathbf{A}) = a_{1,1} + \cdots + a_{n,n}$ . For example, let us set:

$$\mathbf{A} := \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Then we have  $\text{Tr}(\mathbf{A}) = 1 + 5 + 9 = 15$ . Moreover, if  $\mathbf{A} := (a_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p}$  is a matrix of dimension  $(n, p)$ , then we call the **transpose** of  $\mathbf{A}$  the matrix with  $p$  rows and  $n$  columns which ‘swaps the rows and the columns’: otherwise put, we have  $\mathbf{A}^T := (a_{j,i})_{1 \leq j \leq p, 1 \leq i \leq n}$  (the transpose matrix may likewise be denoted  $\mathbf{A}'$  or even  ${}^t\mathbf{A}$ ). For example, we have:

$$\mathbf{A} := \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \quad \text{and} \quad \mathbf{A}^T = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

Let us note moreover that, for every matrix  $\mathbf{A}$  of dimension  $(n, p)$ , we have  $(\mathbf{A}^T)^T = \mathbf{A}$ : the transposition operator is then said to be **involutive**. In addition, for every square matrix  $\mathbf{A}$  of order  $n$ , we have  $\text{Tr}(\mathbf{A}^T) = \text{Tr}(\mathbf{A})$ . Finally, a square matrix  $\mathbf{A}$  of order  $n$  is said to be a **symmetric matrix** if, and only if, we have  $\mathbf{A}^T = \mathbf{A}$ .

### 1.1.2 Diagonal and triangular matrices

Let  $n$  be a non-zero natural number. Let  $\mathbf{A} := (a_{i,j})_{1 \leq i, j \leq n}$  be a square matrix of order  $n$ . Then  $\mathbf{A}$  is said to be a **diagonal matrix** (of order  $n$ ) if, and only if, for all distinct integers  $i$  and  $j$  between 1 and  $n$ , we have  $a_{i,j} = 0$ . Otherwise put, the matrix  $\mathbf{A}$  is of the form:

$$\mathbf{A} = \begin{pmatrix} a_{1,1} & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & a_{n,n} \end{pmatrix} =: \text{diag}(a_{1,1}, \dots, a_{n,n})$$

We then note that every diagonal matrix is symmetric. Consequently,  $\mathbf{A}$  is said to be an **upper triangular matrix** (of order  $n$ ) if, and only if, for all integers  $i$  and  $j$  between 1 and  $n$ , if we have  $i > j$ , then we have  $a_{i,j} = 0$ . Otherwise put, the matrix  $\mathbf{A}$  is of the form:

$$\mathbf{A} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1,n} \\ 0 & \cdots & 0 & a_{n,n} \end{pmatrix}$$

By analogy,  $\mathbf{A}$  is said to be a **lower triangular matrix** (of order  $n$ ) if, and only if, for all integers  $i$  and  $j$  between 1 and  $n$ , if we have  $i < j$ , then we have  $a_{i,j} = 0$ . Otherwise put, the matrix  $\mathbf{A}$  is of the form:

$$\mathbf{A} = \begin{pmatrix} a_{1,1} & 0 & \cdots & 0 \\ a_{2,1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ a_{n,1} & \cdots & a_{n,n-1} & a_{n,n} \end{pmatrix}$$

In general,  $\mathbf{A}$  is said to be a **triangular matrix** if, and only if,  $\mathbf{A}$  is a lower triangular matrix or an upper triangular matrix. We note that a diagonal matrix is in particular a triangular matrix.

### 1.1.3 Addition of matrices and multiplication by a scalar

Let  $n$  and  $p$  be two non-zero natural numbers. Let  $\mathbf{A} := (a_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p}$  and  $\mathbf{B} := (b_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p}$  be two matrices of dimension  $(n, p)$ . Then the **sum** of the matrices  $\mathbf{A}$  and  $\mathbf{B}$  is defined by the matrix  $\mathbf{A} + \mathbf{B} := (a_{i,j} + b_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p}$ . For example, we have:

$$\mathbf{A} := \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \quad \text{and} \quad \mathbf{B} := \begin{pmatrix} 0 & 1 \\ 3 & 4 \\ 2 & 5 \end{pmatrix} \quad \text{and} \quad \mathbf{A} + \mathbf{B} = \begin{pmatrix} 1 & 5 \\ 5 & 9 \\ 5 & 11 \end{pmatrix}$$

Let us note moreover that, for all matrices  $\mathbf{A}$ ,  $\mathbf{B}$  and  $\mathbf{C}$  of dimension  $(n, p)$ , we have  $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$ : matrix addition is then said to be **associative**; we may therefore write  $\mathbf{A} + \mathbf{B} + \mathbf{C}$  without brackets and without ambiguity. In addition, for all matrices  $\mathbf{A}$  and  $\mathbf{B}$  of dimension  $(n, p)$ , we have  $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ : matrix addition is then said to be **commutative**. Moreover, for all square matrices  $\mathbf{A}$  and  $\mathbf{B}$  of order  $n$ , we have  $\text{Tr}(\mathbf{A} + \mathbf{B}) = \text{Tr}(\mathbf{A}) + \text{Tr}(\mathbf{B})$ . Finally, for all matrices  $\mathbf{A}$  and  $\mathbf{B}$  of dimension  $(n, p)$ , we have  $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$ . Consequently, let  $\lambda$  be a real number. Then we define the matrix  $\lambda \mathbf{A} := (\lambda a_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p}$ . For example, for  $\lambda := 2$ , we have:

$$\mathbf{A} := \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \quad \text{and} \quad 2\mathbf{A} := \begin{pmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{pmatrix}$$

In particular, for every matrix  $\mathbf{A}$  of dimension  $(n, p)$ , we set  $-\mathbf{A} := (-1)\mathbf{A}$ . This makes it possible to define a **matrix subtraction**: let  $\mathbf{A} := (a_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p}$  and  $\mathbf{B} := (b_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p}$  be two matrices of dimension  $(n, p)$ ; then we define the matrix  $\mathbf{A} - \mathbf{B} := \mathbf{A} + (-1)\mathbf{B}$ . In addition, we have  $\lambda(\mathbf{A} + \mathbf{B}) = \lambda\mathbf{A} + \lambda\mathbf{B}$ : scalar multiplication is said to be **distributive** over addition. Moreover, for every square matrix  $\mathbf{A}$  of order  $n$ , we have  $\text{Tr}(\lambda\mathbf{A}) = \lambda\text{Tr}(\mathbf{A})$ . Finally, for every matrix  $\mathbf{A}$  of dimension  $(n, p)$ , we have  $(\lambda\mathbf{A})^T = \lambda(\mathbf{A}^T)$ , and we may therefore write  $\lambda\mathbf{A}^T$  without any ambiguity.

### 1.1.4 Matrix multiplication

Let  $n$ ,  $p$  and  $m$  be three non-zero natural numbers,  $\mathbf{A} := (a_{i,k})_{1 \leq i \leq n, 1 \leq k \leq m}$  a matrix of dimension  $(n, m)$  and  $\mathbf{B} := (b_{k,j})_{1 \leq k \leq m, 1 \leq j \leq p}$  two matrices of dimension  $(m, p)$ . Then the **product** of the matrices  $\mathbf{A}$  and  $\mathbf{B}$  is the matrix of dimension  $(n, p)$  defined by:

$$\mathbf{AB} = \begin{pmatrix} \sum_{k=1}^m a_{1,k} b_{k,1} & \sum_{k=1}^m a_{1,k} b_{k,2} & \cdots & \sum_{k=1}^m a_{1,k} b_{k,p} \\ \sum_{k=1}^m a_{2,k} b_{k,1} & \sum_{k=1}^m a_{2,k} b_{k,2} & \cdots & \sum_{k=1}^m a_{2,k} b_{k,p} \\ \vdots & \vdots & & \vdots \\ \sum_{k=1}^m a_{n,k} b_{k,1} & \sum_{k=1}^m a_{n,k} b_{k,2} & \cdots & \sum_{k=1}^m a_{n,k} b_{k,p} \end{pmatrix}$$

Otherwise put, for every integer  $i$  between 1 and  $n$  and every integer  $j$  between 1 and  $p$ , the term of the  $i$ -th row and the  $j$ -th column of the matrix  $\mathbf{AB}$  is given by  $a_{i,1}b_{1,j} + \cdots + a_{i,m}b_{m,j}$ . We then say that we have multiplied  $\mathbf{B}$  **on the left** by  $\mathbf{A}$  or, equivalently, that we have multiplied  $\mathbf{A}$  **on the right** by  $\mathbf{B}$ . More generally, in order to know whether a matrix product can be carried out, we verify whether the dimensions are compatible; with colours:

$$\underbrace{\mathbf{A}}_{(n,m)} \times \underbrace{\mathbf{B}}_{(m,p)} = \underbrace{\mathbf{AB}}_{(n,p)} = \begin{pmatrix} \sum_{k=1}^m a_{1,k} b_{k,1} & \sum_{k=1}^m a_{1,k} b_{k,2} & \cdots & \sum_{k=1}^m a_{1,k} b_{k,p} \\ \sum_{k=1}^m a_{2,k} b_{k,1} & \sum_{k=1}^m a_{2,k} b_{k,2} & \cdots & \sum_{k=1}^m a_{2,k} b_{k,p} \\ \vdots & \vdots & & \vdots \\ \sum_{k=1}^m a_{n,k} b_{k,1} & \sum_{k=1}^m a_{n,k} b_{k,2} & \cdots & \sum_{k=1}^m a_{n,k} b_{k,p} \end{pmatrix}$$

The matrix product  $\mathbf{BA}$ , by contrast, is not defined: the dimensions are said not to be compatible. Before carrying out a matrix product, one must therefore make quite sure that it is indeed well defined. We see here a property that we were not accustomed to with real numbers: whereas, for all real numbers  $a$  and  $b$ , we have  $ab = ba$ , this is therefore not the case in general for matrices: the matrix product is **not commutative** (unlike multiplication on the real numbers which, for its part, is). In general, the **multiplication on the left** of a matrix  $\mathbf{A}$  of dimension  $(n, p)$  is therefore the operation  $\mathbf{XA}$ , which can be carried out for every matrix  $\mathbf{X}$  with  $n$  columns (whatever the number of rows of  $\mathbf{X}$ ), while the **multiplication on the right** of a matrix  $\mathbf{B}$  of dimension  $(n, p)$  is the operation  $\mathbf{BX}$  which can be carried out for every matrix  $\mathbf{X}$  with  $p$  rows (whatever the number of columns of  $\mathbf{X}$ ). For example, let us set the following matrices  $\mathbf{A}$  and  $\mathbf{B}$ :

$$\mathbf{A} := \begin{pmatrix} 2 & 3 \\ 1 & 5 \\ 4 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{B} := \begin{pmatrix} 6 & 7 & 8 \\ 9 & 10 & 11 \end{pmatrix}$$

We note that the matrix  $\mathbf{A}$  is of dimension  $(3, 2)$  and the matrix  $\mathbf{B}$  of dimension  $(2, 3)$ . So the product  $\mathbf{AB}$  can be carried out. The matrix  $\mathbf{AB}$  is of dimension  $(3, 3)$  and is then worth:

$$\mathbf{AB} = \begin{pmatrix} 2 \times 6 + 3 \times 9 & 2 \times 7 + 3 \times 10 & 2 \times 8 + 3 \times 11 \\ 1 \times 6 + 5 \times 9 & 1 \times 7 + 5 \times 10 & 1 \times 8 + 5 \times 11 \\ 4 \times 6 + 0 \times 9 & 4 \times 7 + 0 \times 10 & 4 \times 8 + 0 \times 11 \end{pmatrix} = \begin{pmatrix} 39 & 44 & 49 \\ 51 & 57 & 63 \\ 24 & 28 & 32 \end{pmatrix}$$

Consequently, the matrix  $\mathbf{B}$  is of dimension  $(2, 3)$  and the matrix  $\mathbf{A}$  of dimension  $(3, 2)$ . So the product  $\mathbf{BA}$  can be carried out. The matrix  $\mathbf{BA}$  is of dimension  $(2, 2)$  and is then worth:

$$\mathbf{BA} = \begin{pmatrix} 6 \times 2 + 7 \times 1 + 8 \times 4 & 6 \times 3 + 7 \times 5 + 8 \times 0 \\ 9 \times 2 + 10 \times 1 + 11 \times 4 & 9 \times 3 + 10 \times 5 + 11 \times 0 \end{pmatrix} = \begin{pmatrix} 51 & 53 \\ 72 & 77 \end{pmatrix}$$

Moreover, for every matrix  $\mathbf{A}$  of dimension  $(n, p)$  and every matrix  $\mathbf{B}$  of dimension  $(p, n)$ , we have  $\text{Tr}(\mathbf{AB}) = \text{Tr}(\mathbf{BA})$ : the trace is said to be **invariant under cyclic permutation**. Consequently, for every matrix  $\mathbf{A}$  of dimension  $(n, m)$  and every matrix  $\mathbf{B}$  of dimension  $(m, p)$ , we have  $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ . Finally, if  $\mathbf{A}$  is a square matrix of order  $n$ , we define the **matrix power**  $\mathbf{A}^k$  by induction by setting  $\mathbf{A}^1 := \mathbf{A}$  and, for every non-zero natural number  $k$ , by  $\mathbf{A}^{k+1} = \mathbf{A} \mathbf{A}^k$  (let us note that here, the matrix product is commutative: we have  $\mathbf{A} \mathbf{A}^k = \mathbf{A}^k \mathbf{A}$ ). If  $\mathbf{A}$  is a diagonal matrix of order  $n$ , the matrix

power is computed trivially: if  $\mathbf{A} := \text{diag}(\lambda_1, \dots, \lambda_n)$  then, for every natural number  $k$ , we have  $\mathbf{A}^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$ . Moreover, a square matrix  $\mathbf{A}$  of order  $n$  satisfying  $\mathbf{A}^2 = \mathbf{A}$  is said to be **idempotent** (one sometimes also says that  $\mathbf{A}$  constitutes a **projection matrix**): for every integer  $k$  at least equal to 2, we therefore trivially have  $\mathbf{A}^k = \mathbf{A}$ . Moreover, matrix multiplication is associative: for every matrix  $\mathbf{A}$  with  $n$  rows and  $m$  columns, every matrix  $\mathbf{B}$  with  $m$  rows and  $q$  columns, and every matrix  $\mathbf{C}$  with  $q$  rows and  $p$  columns, we have  $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$ .

### 1.1.5 Identity matrix and zero matrix

Let  $n$  be a non-zero natural number. Then we call the **identity matrix** of order  $n$  the diagonal matrix of order  $n$  all of whose diagonal elements are equal to 1; it is usually denoted  $\mathbf{I}_n$ . For example, we have:

$$\mathbf{I}_3 := \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The identity matrix  $\mathbf{I}_n$  and all its scalar multiples (that is to say all the matrices of the form  $\lambda \mathbf{I}_n$  where  $\lambda$  is a real number) commute with all the square matrices of order  $n$ : we therefore have, for every real number  $\lambda$  and every square matrix  $\mathbf{A}$  of order  $n$ ,  $(\lambda \mathbf{I}_n)\mathbf{A} = \mathbf{A}(\lambda \mathbf{I}_n)$ . In addition, for every square matrix  $\mathbf{A}$  of order  $n$ , we have  $\mathbf{I}_n \mathbf{A} = \mathbf{A} \mathbf{I}_n = \mathbf{A}$ :  $\mathbf{I}_n$  is said to be the **identity element** for matrix multiplication. We thus verify that, for every non-zero natural number  $k$ , we have  $\mathbf{I}_n^k = \mathbf{I}_n$  and  $\mathbf{I}_n^T = \mathbf{I}_n$ . Moreover, by convention, for every square matrix  $\mathbf{A}$  of order  $n$ , we set  $\mathbf{A}^0 := \mathbf{I}_n$ . Consequently, we call the **zero matrix** of order  $n$  the diagonal matrix of order  $n$  all of whose diagonal elements are equal to 0; it is usually denoted  $\mathbf{0}_n$ . Likewise, we call the **zero vector** (with  $n$  rows) the column vector with  $n$  rows all of whose coefficients are zero: it is usually denoted  $\mathbf{0}_{n,1}$ . For example, we have:

$$\mathbf{0}_3 := \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{0}_{3,1} := \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The matrix satisfies, for every square matrix  $\mathbf{A}$  of order  $n$ ,  $\mathbf{A} + \mathbf{0}_3 = \mathbf{A}$ :  $\mathbf{0}_n$  is said to be the **identity element** for matrix addition. In addition, for every square matrix  $\mathbf{A}$  of order  $n$ ,  $\mathbf{0}_3 \mathbf{A} = \mathbf{A} \mathbf{0}_3 = \mathbf{0}_3$ :  $\mathbf{0}_n$  is said to be the **absorbing element** for matrix multiplication. We thus verify that, for every non-zero natural number  $k$ , we have  $\mathbf{0}_n^k = \mathbf{0}_n$ .

### 1.1.6 Inverse of a matrix

Let  $n$  be a non-zero natural number. Let  $\mathbf{A} := (a_{i,j})_{1 \leq i,j \leq n}$  be a square matrix of order  $n$ . If there exists a square matrix  $\mathbf{B}$  of order  $n$  satisfying  $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$ , then this matrix  $\mathbf{B}$  is unique and is called the **inverse** of  $\mathbf{A}$ : we then denote  $\mathbf{A}^{-1} := \mathbf{B}$ . For example, let us set:

$$\mathbf{A} := \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \quad \text{and} \quad \mathbf{B} := \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix}$$

Then we have:

$$\begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Thus, the matrix  $\mathbf{A}$  is invertible and we have  $\mathbf{A}^{-1} = \mathbf{B}$ . In particular, if  $\mathbf{A}$  is a triangular matrix of order  $n$  whose diagonal elements are the real numbers  $\lambda_1, \dots, \lambda_n$  (in that order), then  $\mathbf{A}$  is invertible if, and only if, for every integer  $i$  between 1 and  $n$ , we have  $\lambda_i \neq 0$ ; in that case,  $\mathbf{A}^{-1}$  is a triangular matrix whose diagonal elements are  $\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$  (in that order). More precisely, if  $\mathbf{A}$  is a diagonal matrix of order  $n$  whose diagonal elements are the real numbers  $\lambda_1, \dots, \lambda_n$  (in that order), then  $\mathbf{A}$  is invertible if, and only if, for every integer  $i$  between 1 and  $n$ , we have  $\lambda_i \neq 0$ ; in that case,  $\mathbf{A}^{-1}$  is the diagonal matrix whose diagonal elements are  $\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$  (in that order). More generally, a square matrix is invertible if, and only if, its **determinant** is non-zero. We retain in particular the computation of the determinant for square matrices of order 2 and of order 3. We have:

$$\text{— } \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc: \text{ if } ad - bc \neq 0, \text{ then we have } \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\text{— } \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - afh - bdi - ceg.$$

In practice, except in the framework of a square matrix of order 2, the computation of the inverse of a matrix does not answer to a simple formula. We therefore proceed thanks to an algorithmic method known by the name of **Gaussian elimination**: we write the square matrix  $\mathbf{A}$  of order  $n$  to be inverted on the left, and the matrix  $\mathbf{I}_n$ ; we apply in parallel a succession of **elementary operations** on the two matrices (that is to say operations of the type  $L_i$  becomes  $\lambda L_i + \mu L_j$ , where  $L_k$  denotes the  $k$ -th row of the matrices, with  $\lambda$  a non-zero real number and  $\mu$  any real number) until the matrix  $\mathbf{I}_n$  appears on the left: the corresponding matrix on the right is then precisely the matrix  $\mathbf{A}^{-1}$ . For example, let us set:

$$\mathbf{A} := \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$$

We verify that  $\det(\mathbf{A})$  is non-zero, and therefore that  $\mathbf{A}$  is invertible. We may therefore apply Gaussian elimination:

$$\begin{aligned} \left( \begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array} \right) &\equiv \left( \begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right) & L_3 \leftarrow L_3 - L_1 \\ &\equiv \left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & -1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right) & L_1 \leftarrow L_1 - L_2 - L_3 \end{aligned}$$

We thus have:

$$\mathbf{A}^{-1} = \begin{pmatrix} 2 & -1 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

Moreover, if  $\mathbf{A}$  is an invertible square matrix of order  $n$  then, for every non-zero real number  $\lambda$ ,  $\lambda\mathbf{A}$  is likewise invertible and we have  $(\lambda\mathbf{A})^{-1} = \frac{1}{\lambda}(\mathbf{A}^{-1})$ : we may therefore write  $\frac{1}{\lambda}\mathbf{A}^{-1}$  without ambiguity. In addition, if  $\mathbf{A}$  is an invertible square matrix of order  $n$ , then  $\mathbf{A}^T$  is likewise invertible and we have  $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$ . Consequently, if  $\mathbf{A}$  and  $\mathbf{B}$  are two invertible square matrices of order  $n$ , then  $\mathbf{BA}$  and  $\mathbf{AB}$  are invertible and we have  $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$  and  $(\mathbf{BA})^{-1} = \mathbf{A}^{-1}\mathbf{B}^{-1}$ . Finally, a square matrix  $\mathbf{A}$  of order  $n$  is said to be an **orthogonal matrix** if, and only if, we have  $\mathbf{A}^{-1} = \mathbf{A}^T$ .

### 1.1.7 Block matrices

A **block matrix** is obtained by partitioning the set of row indices and the set of column indices. We suppose that these partitions are contiguous (reordering the rows and the columns if need be). For example, let us set  $\mathbf{A}$  to be the following matrix:

$$\mathbf{A} := \left( \begin{array}{ccc|cc} 3 & 2 & 2 & 1 & 2 \\ -5 & -2 & -1 & -5 & 4 \\ \hline 0 & 0 & 7 & -6 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & -3 & 3 & 1 \end{array} \right)$$

We have here four blocks corresponding to the partition  $\{1, 2\} \cup \{3, 4, 5\}$  for the rows and  $\{1, 2, 3\} \cup \{4, 5\}$  for the columns. We then set:

$$\mathbf{A}_{1,1} := \begin{pmatrix} 3 & 2 & 2 \\ -5 & -2 & -1 \end{pmatrix} \quad \text{and} \quad \mathbf{A}_{1,2} := \begin{pmatrix} 1 & 2 \\ -5 & 4 \end{pmatrix} \quad \text{and} \quad \mathbf{A}_{2,1} := \begin{pmatrix} 0 & 0 & 7 \\ 0 & 0 & -1 \\ 0 & 0 & -3 \end{pmatrix} \quad \text{and} \quad \mathbf{A}_{2,2} := \begin{pmatrix} -6 & 2 \\ 1 & 0 \\ 3 & 1 \end{pmatrix}$$

Then we may rewrite the matrix  $\mathbf{A}$  symbolically in the following manner:

$$\mathbf{A} := \left( \begin{array}{c|c} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \\ \hline \mathbf{A}_{2,1} & \mathbf{A}_{2,2} \end{array} \right)$$

The use of block matrices often makes it possible to simplify matrix computations. Thus, let  $\lambda$  be a real number,  $\mathbf{A}_{1,1}$  a matrix with  $n_1$  rows and  $p_1$  columns,  $\mathbf{A}_{2,1}$  a matrix with  $n_1$  rows and  $p_2$  columns,  $\mathbf{A}_{2,1}$  a matrix with  $n_2$  rows and  $p_1$  columns and  $\mathbf{A}_{2,2}$  a matrix with  $n_2$  rows and  $p_2$  columns (where  $n_1$ ,  $n_2$ ,  $p_1$  and  $p_2$  are natural numbers). Then we have:

$$\underbrace{\lambda \left( \begin{array}{c|c} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \\ \hline \mathbf{A}_{2,1} & \mathbf{A}_{2,2} \end{array} \right)}_{n_1 + n_2 \text{ rows, } p_1 + p_2 \text{ columns}} = \underbrace{\left( \begin{array}{c|c} \lambda\mathbf{A}_{1,1} & \lambda\mathbf{A}_{1,2} \\ \hline \lambda\mathbf{A}_{2,1} & \lambda\mathbf{A}_{2,2} \end{array} \right)}_{n_1 + n_2 \text{ rows, } p_1 + p_2 \text{ columns}}$$



We likewise have:

$$\underbrace{\begin{pmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} \end{pmatrix}^T}_{p_1 + p_2 \text{ rows, } n_1 + n_2 \text{ columns}} = \underbrace{\begin{pmatrix} \mathbf{A}_{1,1}^T & \mathbf{A}_{2,1}^T \\ \mathbf{A}_{1,2}^T & \mathbf{A}_{2,2}^T \end{pmatrix}}_{p_1 + p_2 \text{ rows, } n_1 + n_2 \text{ columns}}$$

If moreover  $\mathbf{B}_{1,1}$  is a matrix with  $n_1$  rows and  $p_1$  columns,  $\mathbf{B}_{2,1}$  a matrix with  $n_1$  rows and  $p_2$  columns,  $\mathbf{B}_{2,1}$  a matrix with  $n_2$  rows and  $p_1$  columns and  $\mathbf{B}_{2,2}$  a matrix with  $n_2$  rows and  $p_2$  columns, then we also have:

$$\underbrace{\begin{pmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} \end{pmatrix}}_{n_1 + n_2 \text{ rows, } p_1 + p_2 \text{ columns}} + \underbrace{\begin{pmatrix} \mathbf{B}_{1,1} & \mathbf{B}_{1,2} \\ \mathbf{B}_{2,1} & \mathbf{B}_{2,2} \end{pmatrix}}_{n_1 + n_2 \text{ rows, } p_1 + p_2 \text{ columns}} = \underbrace{\begin{pmatrix} \mathbf{A}_{1,1} + \mathbf{B}_{1,1} & \mathbf{A}_{1,2} + \mathbf{B}_{1,2} \\ \mathbf{A}_{2,1} + \mathbf{B}_{2,1} & \mathbf{A}_{2,2} + \mathbf{B}_{2,2} \end{pmatrix}}_{n_1 + n_2 \text{ rows, } p_1 + p_2 \text{ columns}}$$

Finally, let  $\mathbf{C}_{1,1}$  be a matrix with  $p_1$  rows and  $m_1$  columns,  $\mathbf{C}_{1,2}$  a matrix with  $p_2$  rows and  $m_1$  columns,  $\mathbf{C}_{2,1}$  a matrix with  $p_1$  rows and  $m_2$  columns and  $\mathbf{C}_{2,2}$  a matrix with  $p_2$  rows and  $m_2$  columns where  $m_1$  and  $m_2$  are natural numbers. Then we have:

$$\underbrace{\begin{pmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} \end{pmatrix}}_{n_1 + n_2 \text{ rows, } p_1 + p_2 \text{ columns}} \times \underbrace{\begin{pmatrix} \mathbf{C}_{1,1} & \mathbf{C}_{1,2} \\ \mathbf{C}_{2,1} & \mathbf{C}_{2,2} \end{pmatrix}}_{p_1 + p_2 \text{ rows, } m_1 + m_2 \text{ columns}} = \underbrace{\begin{pmatrix} \mathbf{A}_{1,1}\mathbf{C}_{1,1} + \mathbf{A}_{1,2}\mathbf{C}_{2,1} & \mathbf{A}_{1,1}\mathbf{C}_{1,2} + \mathbf{A}_{1,2}\mathbf{C}_{2,2} \\ \mathbf{A}_{2,1}\mathbf{C}_{1,1} + \mathbf{A}_{2,2}\mathbf{C}_{2,1} & \mathbf{A}_{2,1}\mathbf{C}_{1,2} + \mathbf{A}_{2,2}\mathbf{C}_{2,2} \end{pmatrix}}_{n_1 + n_2 \text{ rows, } m_1 + m_2 \text{ columns}}$$

These results (multiplication by a scalar, transposition, addition by blocks and multiplication by blocks) apply in particular for block column vectors (the case  $p_2 = 0$ ) and for block row vectors (the case  $n_2 = 0$ ). We then recover the results of the exercise for block vectors by considering naturally that a matrix with zero rows or zero columns is quite simply a matrix that one does not write, and which therefore does not appear in the aforementioned formulas. In particular, we therefore have on the one hand:

$$\underbrace{\begin{pmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \end{pmatrix}}_{n_1 \text{ rows, } p_1 + p_2 \text{ columns}} \times \underbrace{\begin{pmatrix} \mathbf{C}_{1,1} \\ \mathbf{C}_{2,1} \end{pmatrix}}_{p_1 + p_2 \text{ rows, } m_1 \text{ columns}} = \underbrace{\begin{pmatrix} \mathbf{A}_{1,1}\mathbf{C}_{1,1} + \mathbf{A}_{1,2}\mathbf{C}_{2,1} \end{pmatrix}}_{n_1 \text{ rows, } m_1 \text{ columns}}$$

And we have on the other hand:

$$\underbrace{\begin{pmatrix} \mathbf{A}_{1,1} \\ \mathbf{A}_{2,1} \end{pmatrix}}_{n_1 + n_2 \text{ rows, } p_1 \text{ columns}} \times \underbrace{\begin{pmatrix} \mathbf{C}_{1,1} & \mathbf{C}_{1,2} \end{pmatrix}}_{p_1 \text{ rows, } m_1 + m_2 \text{ columns}} = \underbrace{\begin{pmatrix} \mathbf{A}_{1,1}\mathbf{C}_{1,1} & \mathbf{A}_{1,1}\mathbf{C}_{1,2} \\ \mathbf{A}_{2,1}\mathbf{C}_{1,1} & \mathbf{A}_{2,1}\mathbf{C}_{1,2} \end{pmatrix}}_{n_1 + n_2 \text{ rows, } m_1 + m_2 \text{ columns}}$$

## 1.2 Diagonalisation of square matrices

### 1.2.1 Why diagonalise a matrix?

It is often difficult to compute  $\mathbf{A}^k$ , unless  $\mathbf{A}$  is a diagonal matrix, as we mentioned *supra*. The aim of this part will then be to examine under what conditions a square matrix  $\mathbf{A}$  may be written in the form  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ , where  $\mathbf{P}$  is an invertible matrix and  $\mathbf{D}$  a diagonal matrix. Such a form indeed considerably facilitates computations: if we have  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$  then, for every natural number  $k$ , we find  $\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}$ ; there are therefore only two matrix products to carry out. This result is proved by induction. Let  $\mathbf{A}$  be a square matrix of order  $n$  which may be written  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ , where  $\mathbf{D}$  is a diagonal matrix and where  $\mathbf{P}$  is an invertible matrix. First of all, we have on the one hand  $\mathbf{A}^0 = \mathbf{I}_n$  by convention and, on the other hand, we have  $\mathbf{P}\mathbf{D}^0\mathbf{P}^{-1} = \mathbf{P}\mathbf{I}_n\mathbf{P}^{-1}$  by convention, whence  $\mathbf{P}\mathbf{I}_n\mathbf{P}^{-1} = \mathbf{P}\mathbf{P}^{-1}$  by the neutrality of  $\mathbf{I}_n$  for the matrix product, then  $\mathbf{P}\mathbf{P}^{-1} = \mathbf{I}_n$  by definition of the inverse. We do therefore indeed have  $\mathbf{A}^0 = \mathbf{P}\mathbf{D}^0\mathbf{P}^{-1}$ , and the base case is proved. Let now  $k$  be a natural number. Let us suppose that we have  $\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}$  and let us show that we have  $\mathbf{A}^{k+1} = \mathbf{P}\mathbf{D}^{k+1}\mathbf{P}^{-1}$ . Indeed, we do have:

$$\begin{aligned} \mathbf{A}^{k+1} &= \mathbf{A}\mathbf{A}^k \quad \text{by definition of the matrix power} \\ &= (\mathbf{P}\mathbf{D}\mathbf{P}^{-1})(\mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}) \quad \text{by definition of } \mathbf{A} \text{ and by the induction hypothesis} \\ &= \mathbf{P}(\mathbf{D}\mathbf{P}^{-1}\mathbf{P})\mathbf{D}^k\mathbf{P}^{-1} \quad \text{by associativity of the matrix product} \\ &= \mathbf{P}\mathbf{D}\mathbf{D}^k\mathbf{P}^{-1} \quad \text{since } \mathbf{P}^{-1}\mathbf{P} = \mathbf{I}_n, \text{ the identity element for matrix multiplication} \\ &= \mathbf{P}\mathbf{D}^{k+1}\mathbf{P}^{-1} \quad \text{by definition of the matrix power} \end{aligned}$$

The inductive step is therefore proved, and the induction is complete. Notably, every symmetric matrix  $\mathbf{A}$  with real coefficients may be written in the form  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ , where  $\mathbf{P}$  is invertible and  $\mathbf{D}$  is diagonal. More specifically, if  $\mathbf{A}$

is real symmetric, one can even find an orthogonal matrix  $\mathbf{P}$  (therefore invertible and satisfying  $\mathbf{P}^{-1} = \mathbf{P}^T$ ) and a diagonal matrix  $\mathbf{D}$  satisfying  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ : this result is known by the name of the **spectral theorem** (the matrix  $\mathbf{A}$  is then said to be diagonalisable in the orthogonal group or in an **orthogonal basis**). It remains in all cases to know how to determine the matrices  $\mathbf{P}$  and  $\mathbf{D}$ ...

### 1.2.2 How is a matrix to be diagonalised?

A square matrix  $\mathbf{A}$  of order  $n$  admits a real number  $\lambda$  as an **eigenvalue** if, and only if, there exists a vector  $\mathbf{X}$  with  $n$  rows (different from the zero vector) satisfying  $\mathbf{A}\mathbf{X} = \lambda\mathbf{X}$ :  $\mathbf{X}$  is then said to be an eigenvector associated with the eigenvalue  $\lambda$ . The first step is therefore to determine the eigenvalues of  $\mathbf{A}$ . In order to do this, we compute the determinant of the matrix  $\mathbf{A} - \lambda\mathbf{I}_n$ . The computation of  $\chi(\lambda) := \det(\mathbf{A} - \lambda\mathbf{I}_n)$  then gives a polynomial of degree  $n$  in  $\lambda$ , called the **characteristic polynomial** of  $\mathbf{A}$ : the roots of  $\chi$  (that is to say the solutions of the equation  $\chi(\lambda) = 0$ ) are then precisely the eigenvalues of  $\mathbf{A}$ . Once the eigenvalues  $\lambda_1, \dots, \lambda_k$  (with  $k \leq n$ ) have been determined (the set of eigenvalues of  $\mathbf{A}$  constitutes the **spectrum** of  $\mathbf{A}$ ), we compute the **eigenspace** associated with each of them; otherwise put, we solve the equation  $\mathbf{A}\mathbf{X} = \lambda_i\mathbf{X}$  for every integer  $i$  between 1 and  $k$ . The solutions of this system form precisely the eigenspace associated with the eigenvalue  $\lambda_i$ , each element of this solution set constituting an **eigenvector** associated with the eigenvalue  $\lambda_i$ ; we then denote by  $n_i$  the dimension of this set, then called the **multiplicity of the eigenvalue**  $\lambda_i$ , that is to say the number of 'degrees of freedom' of the vectors  $\mathbf{X}$  solving the system  $\mathbf{A}\mathbf{X} = \lambda_i\mathbf{X}$ . From then on, the matrix  $\mathbf{A}$  is **diagonalisable**, that is to say that there exists a diagonal matrix  $\mathbf{D}$  and an invertible matrix  $\mathbf{P}$  satisfying  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$  if, and only if, we have  $n_1 + \dots + n_k = n$  (*a contrario*, if we have  $n_1 + \dots + n_k < n$ , then the matrix  $\mathbf{A}$  is not diagonalisable). In particular, a square matrix of order  $n$  admitting  $n$  distinct eigenvalues is necessarily diagonalisable. If  $\mathbf{A}$  is diagonalisable, we have more precisely:

$$\mathbf{D} := \text{diag}(\underbrace{\lambda_1, \dots, \lambda_1}_{n_1 \text{ times}}, \dots, \underbrace{\lambda_k, \dots, \lambda_k}_{n_k \text{ times}})$$

In addition, the columns of  $\mathbf{P}$  form a sequence of linearly independent eigenvectors (see *infra*) associated with the corresponding eigenvalues (for example, if the  $j$ -th diagonal term of  $\mathbf{D}$  is  $\lambda_i$ , then the  $j$ -th column of  $\mathbf{P}$  must be an eigenvector associated with  $\lambda_i$ ). But  $\mathbf{P}$  is not only a matrix all of whose columns  $\mathbf{X}_1, \dots, \mathbf{X}_n$  are independent; we shall moreover be able to write any vector  $\mathbf{X}$  of  $\mathbb{R}^n$  as a **linear combination** of the vectors  $\mathbf{X}_1, \dots, \mathbf{X}_n$ , that is to say that there exist real numbers  $\lambda_1, \dots, \lambda_n$  satisfying  $\mathbf{X} = \lambda_1\mathbf{X}_1 + \dots + \lambda_n\mathbf{X}_n$ : we shall thus say that the columns of  $\mathbf{P}$  form a **basis** of  $\mathbb{R}^n$  (otherwise put, the eigenvectors of  $\mathbf{A}$  form a basis of  $\mathbb{R}^n$ ). It therefore only remains to invert the matrix  $\mathbf{P}$ , which can be done thanks to Gaussian elimination. Moreover, let  $\mathbf{A}$  be a square matrix of order  $n$  (where  $n$  is a non-zero natural number) assumed to be diagonalisable, whose eigenvalues we then denote  $\lambda_1, \dots, \lambda_n$  (not necessarily distinct, so as to take account of the fact that certain eigenvalues are possibly of multiplicity at least 2). Then we verify  $\lambda_1 + \dots + \lambda_n = \text{Tr}(\mathbf{A})$  and  $\lambda_1 \times \dots \times \lambda_n = \det(\mathbf{A})$ : the sum (resp. the product) of the eigenvalues of a diagonal matrix  $\mathbf{A}$ , taking account of the multiplicity of each of them, is equal to the trace (resp. to the determinant) of  $\mathbf{A}$ .

### 1.2.3 An example of the diagonalisation of a matrix

Let  $a$  be a real number. We set  $\mathbf{A}$  to be the following matrix:

$$\mathbf{A} = \begin{pmatrix} a & 0 & 0 \\ -5 & 4 & 0 \\ -8 & 0 & 1 \end{pmatrix}$$

— We look for the eigenvalues of  $\mathbf{A}$ . We must therefore determine the roots of the characteristic polynomial of  $\mathbf{A}$ . Now we have:

$$\chi(\lambda) := \det(\mathbf{A} - \lambda\mathbf{I}_3) = \det \begin{pmatrix} a - \lambda & 0 & 0 \\ -5 & 4 - \lambda & 0 \\ -8 & 0 & 1 - \lambda \end{pmatrix}$$

Now we have  $\chi(\lambda) = (a - \lambda)(4 - \lambda)(1 - \lambda)$ . We then note trivially that  $\chi(\lambda)$  vanishes for  $\lambda = a$ ,  $\lambda = 4$  and  $\lambda = 1$ . Consequently,  $\mathbf{A}$  admits as eigenvalues  $a$ , 1 and 4. From then on, let us note that, if  $a$  is different from 1 and from 4, then the matrix  $\mathbf{A}$ , square of order 3, admits three distinct values: it will therefore necessarily be diagonalisable.

- The aim is now to determine the eigenvectors of  $\mathbf{A}$ . Let us compute the eigenspace associated with the eigenvalue  $a$ . Let  $\mathbf{X}$  be a column vector formed of the real numbers  $x_1, x_2, x_3$  (in that order). Then we have:

$$\begin{aligned} \mathbf{AX} = a\mathbf{X} &\iff \begin{pmatrix} a & 0 & 0 \\ -5 & 4 & 0 \\ -8 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} ax_1 \\ ax_2 \\ ax_3 \end{pmatrix} \\ &\iff \begin{pmatrix} ax_1 \\ -5x_1 + 4x_2 \\ -8x_1 + x_3 \end{pmatrix} = \begin{pmatrix} ax_1 \\ ax_2 \\ ax_3 \end{pmatrix} \\ &\iff \begin{cases} ax_1 = ax_1 \\ -5x_1 + 4x_2 = ax_2 \\ -8x_1 + x_3 = ax_3 \end{cases} \end{aligned}$$

- Let us suppose first of all  $a = 1$ . Then we have:

$$\begin{aligned} \mathbf{AX} = a\mathbf{X} &\iff \begin{cases} x_1 = x_1 \\ 3x_2 = 5x_1 \\ 8x_1 = 0 \end{cases} \quad \text{since } a = 1 \text{ here} \\ &\iff \begin{cases} x_1 = 0 \\ x_2 = \frac{5}{3}x_1 = 0 \end{cases} \end{aligned}$$

Otherwise put, the value of  $x_3$  is fixed freely while those of  $x_1$  and  $x_2$  are imposed by the system: there is therefore one degree of freedom, and the eigenspace associated with the eigenvalue 1 is of dimension 1. More precisely, the eigenspace associated with the eigenvalue  $a$  then satisfies:

$$S_1 := \left\{ \begin{pmatrix} 0 \\ 0 \\ \lambda \end{pmatrix}; \lambda \in \mathbb{R} \right\} = \left\{ \lambda \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}; \lambda \in \mathbb{R} \right\}$$

- Let us suppose first of all  $a = 4$ . Then we have:

$$\begin{aligned} \mathbf{AX} = a\mathbf{X} &\iff \begin{cases} ax_1 = ax_1 \\ (4 - a)x_2 = 5x_1 \\ (1 - a)x_3 = 8x_1 \end{cases} \\ &\iff \begin{cases} 4x_1 = 4x_1 \\ 5x_1 = 0 \\ -3x_3 = 8x_1 \end{cases} \quad \text{since } a = 4 \text{ here} \\ &\iff \begin{cases} x_1 = 0 \\ x_3 = -\frac{8}{3}x_1 = 0 \end{cases} \end{aligned}$$

Otherwise put, the value of  $x_2$  is fixed freely while those of  $x_1$  and  $x_3$  are imposed by the system: there is therefore one degree of freedom, and the eigenspace associated with the eigenvalue 4 is of dimension 1. More precisely, the eigenspace associated with the eigenvalue  $a$  then satisfies:

$$S_4 := \left\{ \begin{pmatrix} 0 \\ \lambda \\ 0 \end{pmatrix}; \lambda \in \mathbb{R} \right\} = \left\{ \lambda \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; \lambda \in \mathbb{R} \right\}$$

- Let us suppose now  $a \neq 1$  and  $a \neq 4$ . Then we have:

$$\mathbf{AX} = a\mathbf{X} \iff \begin{cases} ax_1 = ax_1 \\ x_2 = \frac{5}{4-a}x_1 \\ x_3 = \frac{8}{1-a}x_1 \end{cases}$$

Otherwise put, the value of  $x_1$  is fixed freely (since the first equation ‘serves no purpose’), while those of  $x_2$  and  $x_3$  are imposed by the system: there is therefore one degree of freedom, and the eigenspace associated with the eigenvalue  $a$  is of dimension 1. More precisely, the eigenspace associated with the eigenvalue  $a$  then satisfies:

$$S_a := \left\{ \begin{pmatrix} \lambda \\ \frac{5\lambda}{4-a} \\ \frac{8\lambda}{1-a} \end{pmatrix}; \lambda \in \mathbb{R} \right\} = \left\{ \lambda \begin{pmatrix} 1 \\ \frac{5}{4-a} \\ \frac{8}{1-a} \end{pmatrix}; \lambda \in \mathbb{R} \right\}$$

Thus, if  $a = 1$  or if  $a = 4$ , then the eigenvalues are 1 and 4. In that case, the dimension of the eigenspace associated with the eigenvalue 1 is 1, and that of the eigenspace associated with the eigenvalue 4 is 1. The sum of the dimensions of the eigenspaces of  $\mathbf{A}$  is therefore equal to 2. Now  $\mathbf{A}$  is a square matrix of order 3. Since  $2 < 3$ , then  $\mathbf{A}$  is not diagonalisable. By contrast, if  $a \neq 1$  and  $a \neq 4$ , then  $\mathbf{A}$  is diagonalisable (see *supra*). In fine,  $\mathbf{A}$  is diagonalisable if, and only if, we have  $a \neq 1$  and  $a \neq 4$ . In this last case, we have  $\mathbf{A} = \mathbf{PDP}^{-1}$ , with:

$$\mathbf{D} := \begin{pmatrix} a & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \quad \text{and} \quad \mathbf{P} := \begin{pmatrix} 1 & 0 & 0 \\ \frac{5}{4-a} & 0 & 1 \\ \frac{8}{1-a} & 1 & 0 \end{pmatrix}$$

### 1.3 Some reminders of linear and bilinear algebra

#### 1.3.1 Kernel of a matrix

Let  $n$  and  $p$  be two non-zero natural numbers. Let  $\mathbf{A}$  be a matrix of dimension  $(n, p)$ . Then we call the **kernel** of  $\mathbf{A}$  the set  $\text{Ker}(\mathbf{A}) := \{\mathbf{X} \in \mathcal{M}_{n,1}(\mathbb{R}); \mathbf{AX} = \mathbf{0}_{n,1}\}$ . Otherwise put, the kernel is the eigenspace associated with the eigenvalue  $\mathbf{A}$ . A square matrix whose kernel is not reduced to the zero vector  $\mathbf{0}_{n,1}$  is non-invertible. From then on, for every square matrix  $\mathbf{A}$  of order  $n$ ,  $\mathbf{A}$  is non-invertible if, and only if, the dimension of its kernel is at least equal to 1 (we then write  $\dim(\text{Ker}(\mathbf{A})) \geq 1$ ) if, and only if, 0 is an eigenvalue of  $\mathbf{A}$ . Conversely, for every square matrix  $\mathbf{A}$  of order  $n$ ,  $\mathbf{A}$  is invertible if, and only if, the dimension of its kernel is zero (we then write  $\dim(\text{Ker}(\mathbf{A})) = 0$ ) if, and only if, 0 is not an eigenvalue of  $\mathbf{A}$ .

#### 1.3.2 Vector space spanned

Let  $n$  and  $p$  be two non-zero natural numbers. Let  $\mathbf{X}_1, \dots, \mathbf{X}_p$  be vectors with  $n$  rows. We suppose that the vectors  $\mathbf{X}_1, \dots, \mathbf{X}_p$  are linearly independent. Then we denote  $\text{Vect}(\mathbf{X}_1, \dots, \mathbf{X}_p) := \{\lambda_1 \mathbf{X}_1 + \dots + \lambda_p \mathbf{X}_p; (\lambda_1, \dots, \lambda_p) \in \mathbb{R}^p\}$ , the set of vectors with  $n$  coordinates spanned by the family  $(\mathbf{X}_1, \dots, \mathbf{X}_p)$ , that is to say the set of **linear combinations** which may be formed with the help of the vectors  $\mathbf{X}_1, \dots, \mathbf{X}_p$ : such a set is therefore of **dimension**  $p$  (each vector indeed constituting a ‘degree of freedom’). For example, we have:

$$\left\{ \lambda \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; \lambda \in \mathbb{R} \right\} = \text{Vect} \left( \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right) \quad \text{and} \quad \left\{ \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}; (\lambda, \mu) \in \mathbb{R}^2 \right\} = \text{Vect} \left( \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right)$$

#### 1.3.3 Rank of a matrix

The **rank** of a matrix  $\mathbf{A}$ , denoted  $\text{rg}(\mathbf{A})$ , is the maximum number of linearly independent columns; vectors  $\mathbf{X}_1, \dots, \mathbf{X}_k$  with  $n$  coordinates ( $n$  and  $k$  being two non-zero natural numbers) are said to be **linearly independent** if, and only if, for all real numbers  $\lambda_1, \dots, \lambda_k$  satisfying  $\lambda_1 \mathbf{X}_1 + \dots + \lambda_k \mathbf{X}_k = \mathbf{0}$ , we necessarily find  $\lambda_1 = \dots = \lambda_k = 0$ . The **rank-nullity theorem** states that, for every matrix  $\mathbf{A}$  of dimension  $(n, p)$  ( $n$  and  $p$  being two non-zero natural numbers), we have  $\dim(\text{Ker}(\mathbf{A})) + \text{rg}(\mathbf{A}) = n$ . The matrix  $\mathbf{A}$  is then said to be of **full rank** if, and only if, all its columns are linearly independent, that is to say if, and only if, we have  $\text{rg}(\mathbf{A}) = n$  if, and only if, by virtue of the rank-nullity theorem, we have  $\dim(\text{Ker}(\mathbf{A})) = 0$ . Moreover, if  $\mathbf{A}$  is a matrix of dimension  $(n, p)$  ( $n$  and  $p$  being two non-zero natural numbers satisfying  $n \geq p$ ) assumed to be of dimension  $p$ , then the matrix  $\mathbf{A}^T \mathbf{A}$  (square of order  $p$ ) is of rank  $p$ : being of full rank,  $\mathbf{A}^T \mathbf{A}$  is therefore invertible. Finally, for every matrix  $\mathbf{A}$  of dimension  $(n, p)$  ( $n$  and  $p$  being two non-zero natural numbers satisfying  $n \geq p$ ), we have  $\text{rg}(\mathbf{A}) = \text{rg}(\mathbf{A}^T)$ .

### 1.3.4 Scalar product and norm

Let  $n$  be a non-zero natural number. Let  $\mathbf{X}_1$  and  $\mathbf{X}_2$  be two vectors with  $n$  coordinates, whose coordinates we denote respectively  $x_{1,1}, \dots, x_{1,n}$  and  $x_{2,1}, \dots, x_{2,n}$  (in that order). Then we call the (Euclidean) **scalar product** the following real number:

$$\langle \mathbf{X}_1, \mathbf{X}_2 \rangle := \mathbf{X}_1^T \mathbf{X}_2 = \sum_{i=1}^n x_{1,i} x_{2,i} = x_{1,1} x_{2,1} + \dots + x_{1,n} x_{2,n}$$

The scalar product is **symmetric**: we have  $\langle \mathbf{X}_1, \mathbf{X}_2 \rangle = \langle \mathbf{X}_2, \mathbf{X}_1 \rangle$ . The scalar product is likewise **positive definite**: for every vector  $\mathbf{X}$  with  $n$  coordinates, we have  $\langle \mathbf{X}, \mathbf{X} \rangle \geq 0$ , and the equality  $\langle \mathbf{X}, \mathbf{X} \rangle = 0$  is satisfied if, and only if, we have  $\mathbf{X} = \mathbf{0}_{n,1}$ . The scalar product is finally a **bilinear form**: for every vector  $\mathbf{X}$  with  $n$  coordinates and every real number  $\lambda$ , we have (i)  $\langle \mathbf{X} + \mathbf{X}_1, \mathbf{X}_2 \rangle = \langle \mathbf{X}, \mathbf{X}_2 \rangle + \langle \mathbf{X}_1, \mathbf{X}_2 \rangle$ , (ii)  $\langle \mathbf{X}_1, \mathbf{X} + \mathbf{X}_2 \rangle = \langle \mathbf{X}_1, \mathbf{X} \rangle + \langle \mathbf{X}_1, \mathbf{X}_2 \rangle$  and (iii)  $\langle \lambda \mathbf{X}_1, \mathbf{X}_2 \rangle = \langle \mathbf{X}_1, \lambda \mathbf{X}_2 \rangle = \lambda \langle \mathbf{X}_1, \mathbf{X}_2 \rangle$ . Consequently, the vectors  $\mathbf{X}_1$  and  $\mathbf{X}_2$  are then said to be **orthogonal** if, and only if, we have  $\langle \mathbf{X}_1, \mathbf{X}_2 \rangle = 0$ . Moreover, the (Euclidean) **norm** of the vector  $\mathbf{X}_1$  is defined by  $\|\mathbf{X}_1\| := \sqrt{\langle \mathbf{X}_1, \mathbf{X}_1 \rangle} = \sqrt{x_{1,1}^2 + \dots + x_{1,n}^2}$ . A vector of norm 1 is then called a **unit vector**. We finally show that we have  $\|\mathbf{X}_1 + \mathbf{X}_2\|^2 = \|\mathbf{X}_1\|^2 + \|\mathbf{X}_2\|^2 + 2\langle \mathbf{X}_1, \mathbf{X}_2 \rangle$ . From then on, if the vectors  $\mathbf{X}_1$  and  $\mathbf{X}_2$  are orthogonal, this result may be rewritten  $\|\mathbf{X}_1 + \mathbf{X}_2\|^2 = \|\mathbf{X}_1\|^2 + \|\mathbf{X}_2\|^2$ : this result is known by the name of **Pythagoras's theorem**.

### 1.3.5 Orthogonal projection

Let  $n$  and  $p$  be two non-zero natural numbers. Let  $\mathbf{X}_1, \dots, \mathbf{X}_p$  be vectors with  $n$  rows, assumed to be linearly independent. Let  $\mathbf{X}$  be a vector with  $n$  rows. Then there exists a unique vector  $\tilde{\mathbf{X}}$  with  $n$  rows belonging to  $\text{Vect}(\mathbf{X}_1, \dots, \mathbf{X}_p)$  such that, for every integer  $j$  between 1 and  $p$ , we have  $\langle \tilde{\mathbf{X}} - \mathbf{X}, \mathbf{X}_j \rangle = 0$ : the vector  $\tilde{\mathbf{X}}$  is then called the **orthogonal projection** of the vector  $\mathbf{X}$  onto the vector space  $E := \text{Vect}(\mathbf{X}_1, \dots, \mathbf{X}_p)$ , and we write  $\tilde{\mathbf{X}} = p_E^\perp(\mathbf{X})$ . It may be proved that we have  $\tilde{\mathbf{X}} = \frac{\langle \mathbf{X}_1, \mathbf{X} \rangle}{\|\mathbf{X}_1\|^2} \mathbf{X}_1 + \dots + \frac{\langle \mathbf{X}_p, \mathbf{X} \rangle}{\|\mathbf{X}_p\|^2} \mathbf{X}_p$ : the real numbers  $\frac{\langle \mathbf{X}_1, \mathbf{X} \rangle}{\|\mathbf{X}_1\|^2}, \dots, \frac{\langle \mathbf{X}_p, \mathbf{X} \rangle}{\|\mathbf{X}_p\|^2}$  then form the **coordinates of the orthogonal projection** of the vector  $\mathbf{X}$ . Moreover, orthogonal projection has a geometrical interpretation: we thus say that the orthogonal projection of  $\mathbf{X}$  onto  $E$  is the vector  $\tilde{\mathbf{X}}$  whose (Euclidean) **distance** to  $E$  is the smallest possible, that is to say that we have  $\tilde{\mathbf{X}} = \arg \min_{\mathbf{Y} \in E} \|\mathbf{X} - \mathbf{Y}\|$ .

### 1.3.6 Symmetric matrices and orthogonal projection

If  $\mathbf{A}$  is a real symmetric matrix, square of order  $n$ , one can even find an orthogonal matrix  $\mathbf{P}$  whose columns  $\mathbf{X}_1, \dots, \mathbf{X}_p$ , besides being linearly independent, each constitute a unit vector, as well as a diagonal matrix  $\mathbf{D}$  satisfying  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ ; in addition, the columns of  $\mathbf{P}$  are necessarily pairwise orthogonal, that is to say that, for all distinct integers  $i$  and  $j$  between 1 and  $n$ , we have  $\langle \mathbf{X}_i, \mathbf{X}_j \rangle = 0$ . This is why we shall say that every symmetric matrix is diagonalisable **in an orthonormal basis**, the family  $(\mathbf{X}_1, \dots, \mathbf{X}_p)$  constituting the said orthonormal basis (if we had not additionally imposed that these vectors be of norm 1, we would then have said more generally that the family  $(\mathbf{X}_1, \dots, \mathbf{X}_p)$  forms an **orthogonal basis**). In that case, the orthogonal projection of any vector  $\mathbf{X}$  with  $n$  rows onto the vector space spanned by the columns  $\mathbf{X}_1, \dots, \mathbf{X}_p$  will satisfy  $\tilde{\mathbf{X}} := \langle \mathbf{X}_1, \mathbf{X} \rangle \mathbf{X}_1 + \dots + \langle \mathbf{X}_p, \mathbf{X} \rangle \mathbf{X}_p$  (since the columns of  $\mathbf{P}$  have been chosen so as to be unit vectors). The real numbers  $\langle \mathbf{X}_1, \mathbf{X} \rangle, \dots, \langle \mathbf{X}_p, \mathbf{X} \rangle$  are then known by the name of the coordinates of the orthogonal projection of the vector  $\mathbf{X}$  in the orthonormal basis  $(\mathbf{X}_1, \dots, \mathbf{X}_p)$ .

## 2 A review of analysis

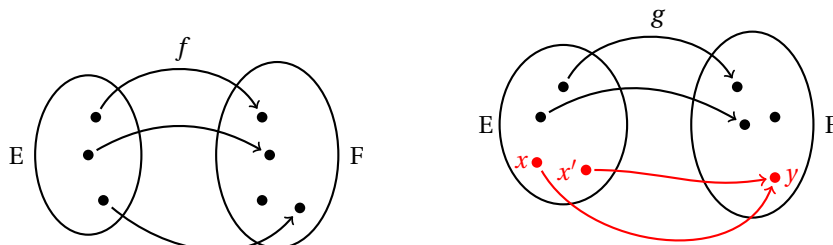
### 2.1 General points on functions

#### 2.1.1 Function

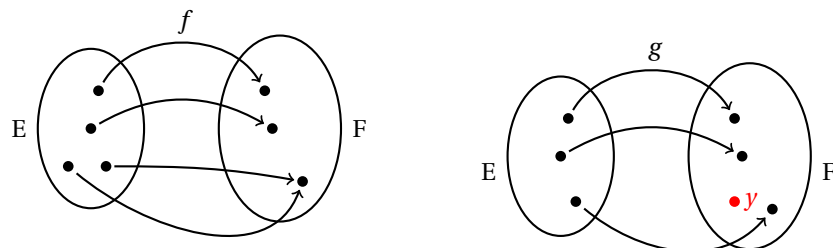
Let  $E$  and  $F$  be two sets. Then a **function**  $f$  is a relation (that is to say a kind of ‘mapping’) which links the elements of the set  $E$ , called the **domain**, to the elements of the set  $F$ , called the **codomain**. Let  $x$  be an element of  $E$  and  $y$  an element of  $F$ . If the elements  $x$  and  $y$  are related by the function  $f$ , we shall write  $f(x) = y$ , and we shall say that  $y$  is the **image** of  $x$  under the function  $f$  or, equivalently, that  $x$  is a **preimage** of  $y$  under the function  $f$ . Careful! A function returns, for each element  $x$  of the domain, at most one image, that is to say that one cannot have  $f(x) = y_1$  and  $f(x) = y_2$  (one would then speak no longer of a function but of a **correspondence**). By contrast, it is entirely possible for two distinct elements of the domain to have the same image. For example, the square function  $h : x \mapsto x^2$  satisfies  $h(-2) = h(2) = 4$ : an element of the codomain may therefore admit more than a single preimage.

### 2.1.2 Injection, surjection, bijection

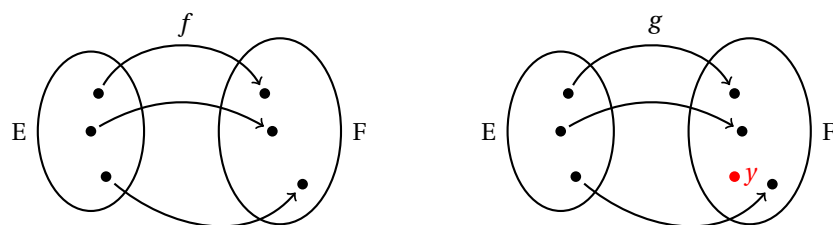
Let  $E$  and  $F$  be two sets and  $f$  a function from  $E$  into  $F$ . When every element of the codomain  $F$  admits at most one preimage under the function  $f$ , the function  $f$  is said to be **injective**. For example, the square root function  $h : x \mapsto \sqrt{x}$  realises an injection from  $E = \mathbb{R}_+$  (the set of non-negative real numbers) into  $F = \mathbb{R}$  (the set of real numbers): indeed, the strictly negative real numbers of  $F$  admit no image under the function  $h$  (since the square root is not defined for strictly negative real numbers), while the non-negative real numbers of  $F$  admit a unique preimage under  $f$  (if  $y$  is a non-negative real number, this preimage is indeed the unique non-negative real number  $x$  satisfying  $x^2 = y$ ). Below, the map  $f$  is injective, but not  $g$  (since  $y$  admits two distinct preimages):



Consequently, if each element of the codomain  $F$  is reached at least once by the function  $f$ , the function  $f$  is said to be **surjective**. For example, the square function  $h : x \mapsto x^2$  realises a surjection from  $E = \mathbb{R}$  onto  $F = \mathbb{R}_+$ : indeed, for every element  $y$  of  $\mathbb{R}_+$ , if we set  $x := \sqrt{y}$  (or  $x := -\sqrt{y}$ ), we do indeed find  $h(x) = y$ ; each element of the codomain therefore does indeed admit at least one preimage under the function  $h$ . Below, the map  $f$  is surjective, but not  $g$  (since  $y$  admits no preimage under  $g$ ):



Finally, when the function  $f$  is both injective and surjective, this means that there exists a one-to-one correspondence between the elements of the domain  $E$  and those of the codomain  $F$ : for every element  $y$  of  $F$ , there exists one and only one element  $x$  of  $E$  satisfying  $f(x) = y$ . The function  $f$  is then said to be **bijective**. And since the correspondence between the sets  $E$  and  $F$  is one-to-one, it may be ‘reversed’ by defining  $f^{-1}$  as the function which, to every element  $y$  of  $F$ , associates the unique element  $x$  of  $E$  satisfying  $f(x) = y$  (we therefore have  $f^{-1}(y) := x$ ): this function is then called the **inverse bijection** of the function  $f$ . Below, the map  $f$  is bijective, but not  $g$  (since the element  $y$  admits no preimage under  $g$ ):

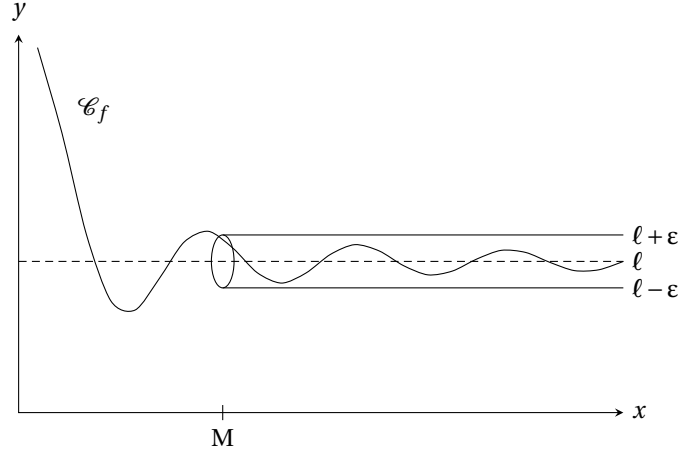


Careful, however: a function which is not injective is not necessarily surjective, and *vice versa*. Thus, the function  $g$  presented in the first point of the preceding examples is neither injective nor surjective.

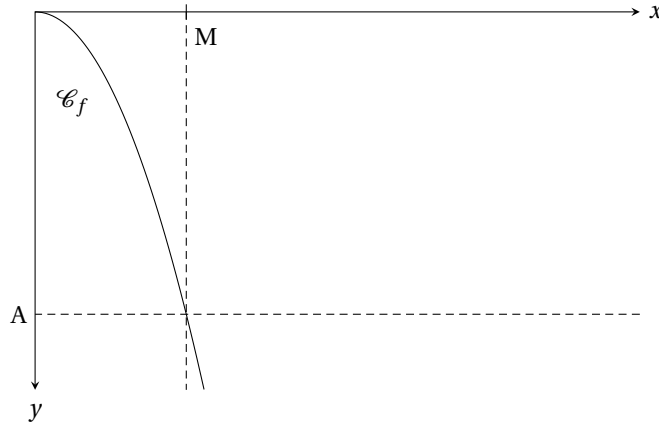
### 2.1.3 Limits of a function

Let  $f$  be a function defined on an interval  $I$  of  $\mathbb{R}$  and let  $a$  be a point (possibly  $\pm\infty$ ) adherent to  $I$  (that is to say situated in  $I$  or equal to one of the endpoints of the interval  $I$ ). The function  $f$  is said to **admit a limit**  $\ell$  at  $a$  when, intuitively, the values  $f(x)$  come arbitrarily close to  $\ell$  as  $x$  comes close to  $a$ . We shall then write  $\lim_{x \rightarrow a} f(x) = \ell$ , or equivalently,  $f(x) \rightarrow \ell$  as  $x \rightarrow a$ . Rigorously, three cases are distinguished according to the nature of  $a$ :

- **Finite limit at a finite point:** let  $a$  be a real number. We say that  $\lim_{x \rightarrow a} f(x) = \ell$  (where  $\ell$  is a real number) if and only if, for every real number  $\varepsilon > 0$ , there exists a real number  $\delta > 0$  such that, for every  $x \in I$  satisfying  $0 < |x - a| < \delta$ , we have  $|f(x) - \ell| < \varepsilon$ . In other words,  $f(x)$  may be made as close as one wishes to  $\ell$  by choosing  $x$  sufficiently close to  $a$  (but distinct from  $a$ ).
- **Finite limit at infinity:** let  $\ell$  be a real number. We say that  $\lim_{x \rightarrow +\infty} f(x) = \ell$  if and only if, for every real number  $\varepsilon > 0$ , there exists a real number  $M$  such that, for every real number  $x$  of  $I$  with  $x > M$ , we have  $|f(x) - \ell| < \varepsilon$ . Likewise, we say that  $\lim_{x \rightarrow -\infty} f(x) = \ell$  if and only if, for every real number  $\varepsilon > 0$ , there exists a real number  $M$  such that, for every real number  $x$  of  $I$  with  $x < M$ , we have  $|f(x) - \ell| < \varepsilon$ .



- **Infinite limit:** we say that we have  $\lim_{x \rightarrow a} f(x) = +\infty$  (where  $a$  is a real number or  $a = \pm\infty$ ) if and only if, for every real number  $x$  belonging to a neighbourhood of  $a$  (that is to say an interval of the form  $]M, +\infty[$  if  $a = +\infty$ ,  $]-\infty, M[$  if  $a = -\infty$ , or  $]m_1, m_2[$  with  $m_1 < a < m_2$  if  $a$  is real), we have  $f(x) > A$ . Analogously, we say that we have  $\lim_{x \rightarrow a} f(x) = -\infty$  if  $f(x) < A$  for all the real numbers  $x$  belonging to a certain neighbourhood of  $a$ .



**Careful:** a function may fail to admit a limit at a given point. For example, the function  $h : x \mapsto \sin(1/x)$  admits no limit at 0, since it oscillates indefinitely between  $-1$  and  $1$  in the neighbourhood of 0. However, a function can admit at most one single limit at a given point: if  $\lim_{x \rightarrow a} f(x) = \ell_1$  and  $\lim_{x \rightarrow a} f(x) = \ell_2$ , then necessarily  $\ell_1 = \ell_2$  (the limit is said to be unique).

#### 2.1.4 Operations on limits

Let  $f$  and  $g$  be two functions admitting respective limits at a point  $a$  (finite or infinite), and let  $\lambda$  be a real number. The following properties make it possible to compute the limits of combinations of functions:

- **Linearity:** if  $\lim_{x \rightarrow a} f(x) = \ell_f$  and  $\lim_{x \rightarrow a} g(x) = \ell_g$ , then we have  $\lim_{x \rightarrow a} (f(x) + g(x)) = \ell_f + \ell_g$  and  $\lim_{x \rightarrow a} \lambda f(x) = \lambda \cdot \ell_f$ . In particular, we have  $\lim_{x \rightarrow a} (f(x) - g(x)) = \ell_f - \ell_g$ .

- **Multiplication:** if  $\lim_{x \rightarrow a} f(x) = \ell_f$  and  $\lim_{x \rightarrow a} g(x) = \ell_g$ , then we have  $\lim_{x \rightarrow a} f(x)g(x) = \ell_f \ell_g$ .
- **Division:** if  $\lim_{x \rightarrow a} f(x) = \ell_f$  and  $\lim_{x \rightarrow a} g(x) = \ell_g$  with  $\ell_g \neq 0$ , then we have  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\ell_f}{\ell_g}$ .
- **Composition:** if  $\lim_{x \rightarrow a} f(x) = b$  and  $\lim_{y \rightarrow b} g(y) = \ell$ , and if  $f$  does not locally take the value  $b$  (except possibly at  $a$ ), then we have  $\lim_{x \rightarrow a} g(f(x)) = \ell$ .
- **Power:** if  $\lim_{x \rightarrow a} f(x) = \ell$  and  $n$  is a non-zero natural number, then we have  $\lim_{x \rightarrow a} (f(x))^n = \ell^n$ .

These properties remain valid when  $\ell_f$ ,  $\ell_g$ , or  $a$  take infinite values, provided that the operations make sense. Indeed, certain combinations lead to **indeterminate forms**, that is to say that the result cannot be determined directly by the preceding rules:  $\frac{\infty}{\infty}$ ,  $\frac{0}{0}$ ,  $\infty - \infty$ ,  $0 \times \infty$ ,  $1^\infty$ ... In these cases, one must have recourse to more refined computational techniques, such as factorisation, multiplication by the conjugate quantity, or the use of reference limits.

### 2.1.5 Monotonicity

Let  $f$  be a function defined on an interval  $I$  of  $\mathbb{R}$ . We say that  $f$  is **increasing** on  $I$  if, for all real numbers  $x_1$  and  $x_2$  of  $I$  satisfying  $x_1 < x_2$ , we have  $f(x_1) \leq f(x_2)$ . More restrictively, we say that  $f$  is **strictly increasing** on  $I$  if, for all real numbers  $x_1$  and  $x_2$  of  $I$  satisfying  $x_1 < x_2$ , we have  $f(x_1) < f(x_2)$ . Conversely, we say that  $f$  is **decreasing** on  $I$  if, for all real numbers  $x_1$  and  $x_2$  of  $I$  satisfying  $x_1 < x_2$ , we have  $f(x_1) \geq f(x_2)$ . Finally, we say that  $f$  is **strictly decreasing** on  $I$  if, for all real numbers  $x_1$  and  $x_2$  of  $I$  satisfying  $x_1 < x_2$ , we have  $f(x_1) > f(x_2)$ . A function which is increasing or decreasing on an interval is said to be **monotone** on that interval; a function which is strictly increasing or strictly decreasing is said to be **strictly monotone**.

### 2.1.6 Linear functions

Let  $a$  and  $b$  be two real numbers. Let  $f$  be the function which, to every real number  $x$ , associates the image  $f(x) := a + bx$ . Then  $f$  is said to be an **affine function** (by an abuse of language,  $f$  will often be said to be **linear** whereas, *stricto sensu*, this is the case only when we have  $a = 0$ ). We then call  $a$  the **intercept** of the function  $f$  and  $b$  its slope or its **gradient**. In the usual plane, affine functions may be represented by a straight line:

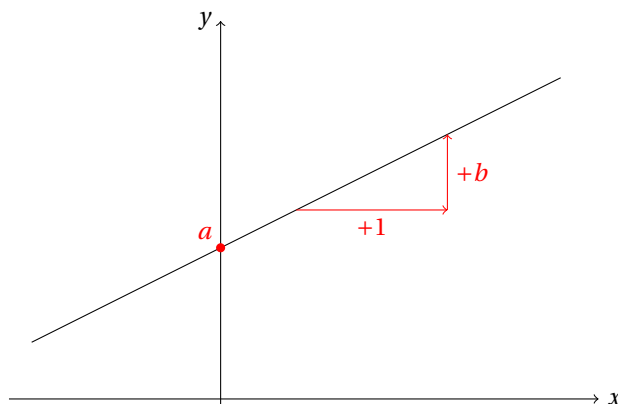


Figure 1.1: Graphical representation of an affine function

#### Exercise

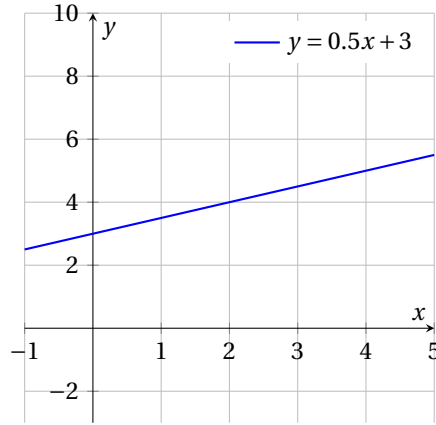


#### Graphs, slopes and the solution of a system of two linear equations.

- (i) Consider the equation of the line  $y = 0.5x + 3$ . Draw the line corresponding to the above equation.
- (ii) Compute the slope of the above equation.
- (iii) Consider now the line with equation  $y = 8 - 2x$ . Draw the line corresponding to this new equation.
- (iv) Compute the values of  $x$  and  $y$  which simultaneously satisfy the two equations  $y = 0.5x + 3$  and  $y = 8 - 2x$ .
- (v) Same question but adopting a graphical method of solution.

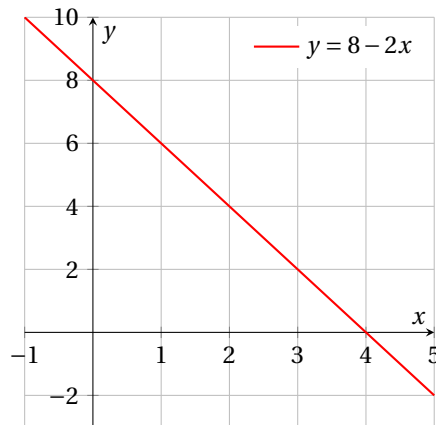


**Solution.** (i) Consider the equation of the line  $y = 0.5x + 3$ . Draw the line corresponding to the above equation. An equation of a line is written  $y = a + bx$ : the parameter  $a$  is called the intercept and the parameter  $b$  is called the gradient of the line. For  $x = 0$ , we therefore find  $y = a$ ; in particular, for the equation  $y = 0.5x + 3$ , we find  $y = 3$  for  $x = 0$ . Moreover, we compute the point where the line crosses the horizontal axis, that is to say in order to determine the number  $x^*$  satisfying  $0 = a + bx^*$ , finding  $x^* = -\frac{a}{b}$ . Thus, for the equation  $y = 0.5x + 3$ , if we set  $x = -\frac{3}{0.5} = -6$ , then we find  $y = 0$ . Graphically, the line with equation  $y = 0.5x + 3$  may be represented as follows:



(ii) Compute the slope of the above equation. Let us recall that the slope of a line indicates by how much the ordinate increases when the abscissa increases by one unit. It corresponds to the ratio of the variation of  $y$  to the variation of  $x$ . In order to determine the slope of the preceding line, we therefore write  $\frac{0-3}{-6-0} = 0.5$ . Let us note that, for an equation of the form  $y = a + bx$ ,  $b$  corresponds exactly to the slope.

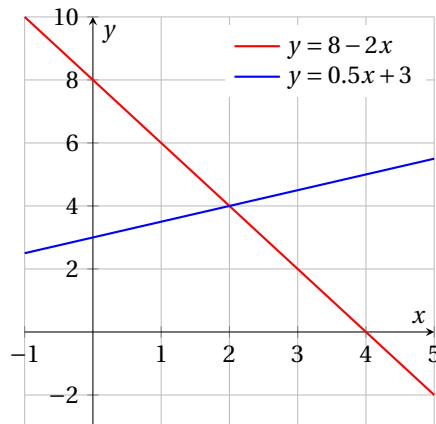
(iii) Consider now the line with equation  $y = 8 - 2x$ . Draw the line corresponding to this new equation. For the equation  $y = 8 - 2x$ , we find  $y = 8$  for  $x = 0$ , and if we set  $x = -\frac{8}{-2} = 4$ , then we find  $y = 0$ . Graphically, the line with equation  $y = 8 - 2x$  may be represented as follows:



(iv) Compute the values of  $x$  and  $y$  which simultaneously satisfy the two equations  $y = 0.5x + 3$  and  $y = 8 - 2x$ . We have:

$$\begin{aligned} \begin{cases} y = 0.5x + 3 \\ y = 8 - 2x \end{cases} &\Leftrightarrow \begin{cases} 8 - 2x = 0.5x + 3 \\ y = 8 - 2x \end{cases} \\ &\Leftrightarrow \begin{cases} 8 - 3 = 0.5x + 2x \\ y = 8 - 2x \end{cases} \\ &\Leftrightarrow \begin{cases} 5 = 2.5x \\ y = 8 - 2x \end{cases} \\ &\Leftrightarrow \begin{cases} x = \frac{5}{2.5} = 2 \\ y = 8 - 2 \times 2 = 4 \end{cases} \end{aligned}$$

- (v) *Same question but adopting a graphical method of solution.* In order to solve graphically the system of the previous question, we determine the point of intersection between the two lines. We recover the point of intersection with coordinates (2, 4) determined in the previous question:



□

### Exercise



**Non-linear relation between two variables.** Below, we transcribe the evolution of the enrolment rate in France in 2009 as a function of age:

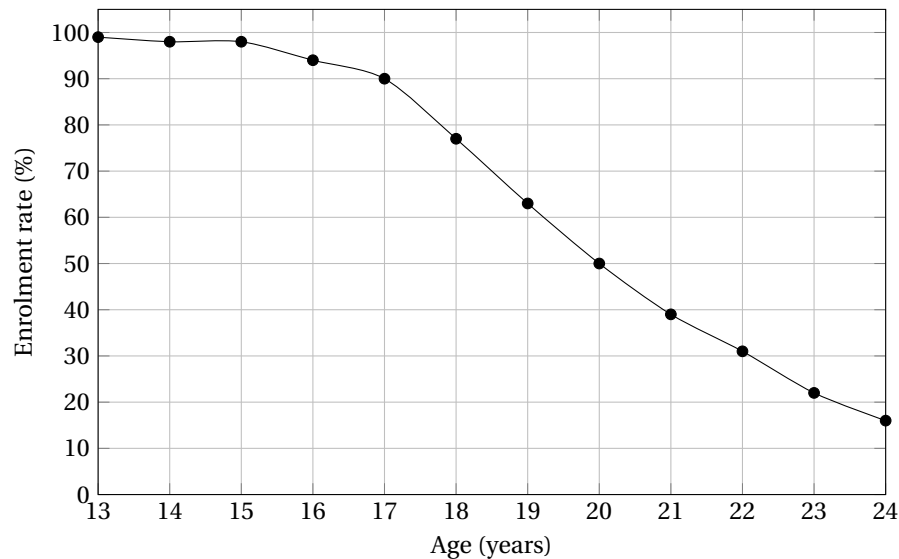
| Age (years) | Enrolment rate (%) | Age (years) | Enrolment rate (%) |
|-------------|--------------------|-------------|--------------------|
| 13          | 98.6               | 19          | 62.6               |
| 14          | 98.3               | 20          | 49.9               |
| 15          | 97.9               | 21          | 39.1               |
| 16          | 94.2               | 22          | 30.6               |
| 17          | 90.4               | 23          | 22.4               |
| 18          | 76.5               | 24          | 15.6               |

- Round the enrolment rates to the nearest unit in a corresponding column  $S^*$ . Represent the link between the enrolment rate  $S^*$  and age  $A$ . Considering that one graduation on the vertical axis (enrolment rate) corresponds to 10 points and one graduation on the horizontal axis (age) corresponds to 1 year, place the origin of the years at 13 years. Interpret.
- Is the relation between age and the enrolment rate linear or non-linear, increasing or decreasing? Is the link between  $A$  and  $S$  positive, negative or nil?
- Consider the tangent to this curve when  $A = 15$ , and compute its slope graphically.
- Consider the tangent when  $A = 22$  and compute its slope graphically. Interpret. What would happen if the relation between age and the enrolment rate were linear?

**Solution.** (i) *Round the enrolment rates to the nearest unit in a corresponding column  $S^*$ . Represent the link between the enrolment rate  $S^*$  and age  $A$ . Considering that one graduation on the vertical axis (enrolment rate) corresponds to 10 points and one graduation on the horizontal axis (age) corresponds to 1 year, place the origin of the years at 13 years. Interpret.* We begin by rounding to the nearest unit (not forgetting that 0.5 rounds to 1) and we find:

| Age | Enrolment rate (%) | Rounded enrolment rate (%) |
|-----|--------------------|----------------------------|
| 13  | 98.6               | 99                         |
| 14  | 98.3               | 98                         |
| 15  | 97.9               | 98                         |
| 16  | 94.2               | 94                         |
| 17  | 90.4               | 90                         |
| 18  | 76.5               | 77                         |
| 19  | 62.6               | 63                         |
| 20  | 49.9               | 50                         |
| 21  | 39.1               | 39                         |
| 22  | 30.6               | 31                         |
| 23  | 22.4               | 22                         |
| 24  | 15.6               | 16                         |

Graphically, we may represent the evolution of the enrolment rate as a function of age in France in 2009 as follows:



- (ii) *Is the relation between age and the enrolment rate linear or non-linear, increasing or decreasing? Is the link between A and S positive, negative or nil?* It appears that the relation between age and the enrolment rate does not follow a linear trend (that is to say does not have the form of a straight line): the relation between these two parameters is therefore non-linear. Moreover, the more age increases, the more the enrolment rate tends to decrease: the relation between age and the enrolment rate is therefore decreasing; otherwise put, the link (or, to use the consecrated statistical term, the correlation) between A and S is negative.
- (iii) *Consider the tangent to this curve when A = 15, and compute its slope graphically.* When one has a function  $f$  differentiable on an interval I, the equation of the tangent to the curve representing  $f$  at a point  $a$  of I is written  $y = f'(a)(x - a) + f(a) = f'(a)x + (f(a) - af'(a))$ : this is the equation of a line with slope  $f'(a)$  and intercept  $f(a) - af'(a)$ . Here, we do not have a defined function  $f$  and, since we define the points discretely, *a fortiori*, we cannot speak of a derivative. But noting that  $f'(a)$  is the slope of the tangent, we may define a 'discrete tangent' at a point whose slope would correspond to the slope at that point. Now the variation of the enrolment rate between 15 and 16 years is written  $\frac{94.2-97.9}{16-15} = -3.7$ : this means that between 15 and 16 years, the enrolment rate loses 3.7 percentage points.
- (iv) *Consider the tangent when A = 22 and compute its slope graphically. Interpret. What would happen if the relation between age and the enrolment rate were linear?* As in the previous question, the variation of the enrolment rate between 22 and 23 years is written  $\frac{22.4-30.6}{23-22} = -8.2$ : this means that between 22 and 23 years, the enrolment rate loses 8.2 percentage points. If the trend had been linear, then this decrease should have been appreciably the same as in the previous question, which is clearly not the case: the trend is therefore not linear. □

### 2.1.7 Quadratic functions

Let  $a, b$  and  $c$  be three real numbers. We suppose that  $a$  is non-zero. We then call the **discriminant** of the polynomial  $P(X) := aX^2 + bX + c$  the value  $\Delta := b^2 - 4ac$ . If  $\Delta$  is strictly positive, then the polynomial  $P$  admits two roots  $x_1 := \frac{-b + \sqrt{\Delta}}{2a}$  and  $x_2 := \frac{-b - \sqrt{\Delta}}{2a}$ . We then have  $P(X) = a(X - x_1)(X - x_2)$ . The real polynomial function  $P : x \mapsto ax^2 + bx + c$  then has the sign of  $a$  on  $]-\infty, x_1] \cup [x_2, +\infty[$  and the sign opposite to that of  $a$  on  $]x_1, x_2[$  and it vanishes at  $x_1$  and  $x_2$ . If now  $\Delta$  is zero, then the polynomial  $P$  admits a double root  $x_0 := \frac{-b}{2a}$ . We then have  $P(X) = a(X - x_0)^2$ . The real polynomial function  $P : x \mapsto ax^2 + bx + c$  then has the sign of  $a$  on  $\mathbb{R}$  and vanishes at  $x_0$ . Finally, if  $\Delta$  is strictly negative, then the polynomial  $P$  admits no real root. The real polynomial function  $P : x \mapsto ax^2 + bx + c$  then has the sign of  $a$  on  $\mathbb{R}$  and never vanishes. Moreover, every polynomial function  $P : x \mapsto ax^2 + bx + c$  is differentiable on  $\mathbb{R}$  and, for every real number  $x$ , we have  $P'(x) = 2ax + b$ . Finally,  $P$  tends to  $+\infty$  at  $+\infty$  as at  $-\infty$  if we have  $a > 0$ ; conversely,  $P$  tends to  $-\infty$  at  $+\infty$  as at  $-\infty$  if we have  $a < 0$ .

### 2.1.8 Exponential and logarithm functions

The exponential function ( $\exp$ ) is the function defined from  $\mathbb{R}$  onto  $\mathbb{R}_+^*$  which, to every real number  $x$ , associates the number  $\exp(x) := e^x$ , where  $e$  is a real number worth roughly 2.718. The exponential function is strictly increasing on  $\mathbb{R}$ , strictly positive on  $\mathbb{R}$ , tends to 0 at  $-\infty$  and tends to  $+\infty$  at  $+\infty$ . Moreover, we have  $e^0 = 1$  (therefore, for every strictly negative real number  $x$ , we have  $0 < e^x < 1$  and, for every strictly positive real number, we have  $e^x > 1$ ). Consequently, the natural logarithm function ( $\ln$ ) is the function defined from  $\mathbb{R}_+^*$  onto  $\mathbb{R}$  which, to every strictly positive real number  $x$ , associates the (unique) real number  $y$  satisfying  $e^y = x$ : the logarithm is therefore the inverse bijection of the exponential function. For every strictly positive real number  $x$ , we therefore have  $\ln(e^x) = e^{\ln(x)} = x$ . The logarithm function is strictly increasing on  $\mathbb{R}_+^*$ , strictly negative on  $]0, 1[$ , strictly positive on  $]1, +\infty[$ , tends to  $-\infty$  at  $0^+$  and tends to  $+\infty$  at  $+\infty$ . Moreover, we have  $\ln(1) = 0$  and  $\ln(e) = 1$ .

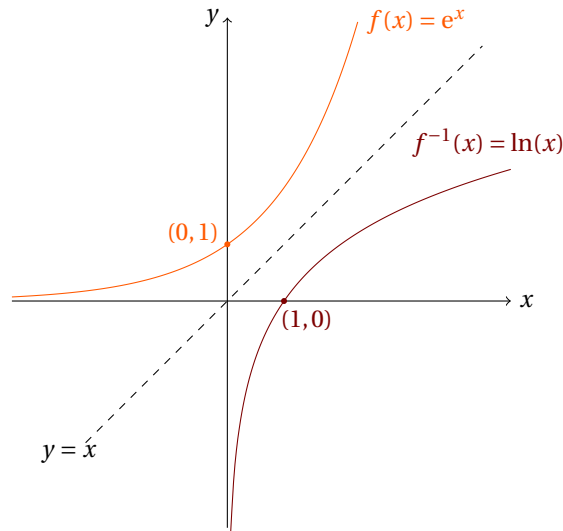


Figure 1.2: Graphical representation of the exponential and logarithm functions

Moreover, the use of the exponential and logarithm functions makes it possible to generalise **power functions**: for every real number  $\alpha$ , we set  $x^\alpha := \exp(\alpha \ln(x))$ . Finally, for all real numbers  $x$  and  $y$ , we have  $e^{x+y} = e^x e^y$ ,  $e^{-x} = \frac{1}{e^x}$ ,  $e^{x-y} = \frac{e^x}{e^y}$  and  $e^{xy} = (e^x)^y$ . By analogy, for all strictly positive real numbers  $x$  and  $y$ , we have  $\ln(xy) = \ln(x) + \ln(y)$ ,  $\ln(\frac{1}{x}) = -\ln(x)$ ,  $\ln(\frac{x}{y}) = \ln(x) - \ln(y)$  and  $\ln(x^y) = y \ln(x)$ . The real power notation therefore does indeed extend the usual integer power: indeed, if  $n$  is a natural number, we do have  $\exp(n \ln(x)) = \exp(\ln(x^n)) = x^n$ .

## 2.2 Numerical sequences and series

### 2.2.1 Real sequences

A **real sequence**  $(a_n)_{n \in \mathbb{N}}$  is a function from  $\mathbb{N}$  into  $\mathbb{R}$  which associates to each natural number  $n$  a real number  $a_n := a(n)$ , the **general term** ( $n$ -th term) of the sequence. A sequence may be defined explicitly (by a closed formula,

for example  $a_n = n^2 + 1$  for every natural number  $n$ ) or recursively (by a recurrence relation, for example  $a_n = 2a_{n-1}$  for every non-zero natural number  $n$ , with an initial condition  $a_0 = 1$ ). A sequence  $(a_n)$  is said to **converge** to a real number  $\ell$  when the terms of the sequence come arbitrarily close to  $\ell$  as  $n$  increases. We then write  $\lim_{n \rightarrow +\infty} a_n = \ell$ . Rigorously, the sequence  $(a_n)_{n \in \mathbb{N}}$  converges to the real number  $\ell$  if, and only if, for every real number  $\varepsilon > 0$ , there exists a natural number  $N$  such that, for every integer  $n > N$ , we have  $|a_n - \ell| < \varepsilon$ . A sequence which does not converge is said to be **divergent**. For example, the sequence  $a_n = \frac{1}{n}$  converges to 0; the sequence  $a_n = n^2$  diverges (it tends to  $+\infty$ ); and the sequence  $a_n = (-1)^n$  diverges (it oscillates between  $-1$  and  $1$ ).

### 2.2.2 The summation symbol

The **summation symbol**  $\sum$  is a compact notation which makes it possible to represent the addition of several terms of a sequence. More precisely, if  $(a_n)_{n \in \mathbb{N}}$  is a sequence and if  $p$  and  $q$  are two natural numbers with  $p \leq q$ , we write:

$$\sum_{n=p}^q a_n := a_p + a_{p+1} + \cdots + a_{q-1} + a_q$$

The index  $n$  is called the summation index, the letter  $p$  is the lower bound and the letter  $q$  is the upper bound of the sum. For example, we have  $\sum_{n=1}^5 n^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$ . The following properties facilitate the computation of sums:

- **Linearity:** if  $\lambda$  is a real number and  $(a_n)$ ,  $(b_n)$  are two sequences, then we have  $\sum_{n=p}^q (\lambda a_n + b_n) = \lambda \sum_{n=p}^q a_n + \sum_{n=p}^q b_n$ .
- **Change of index:** the index  $n$  may be replaced by a bijective function of  $n$  without modifying the sum. For example, setting  $m = n - p$ , we have  $\sum_{n=p}^q a_n = \sum_{m=0}^{q-p} a_{m+p}$ .
- **Chasles's relation:** if  $p \leq r \leq q$ , then we have  $\sum_{n=p}^q a_n = \sum_{n=p}^r a_n + \sum_{n=r+1}^q a_n$ .

### 2.2.3 The product symbol

The **product symbol**  $\prod$  is a compact notation which makes it possible to represent the multiplication of several terms of a sequence. More precisely, if  $(a_n)_{n \in \mathbb{N}}$  is a sequence and if  $p$  and  $q$  are two natural numbers with  $p \leq q$ , we write:

$$\prod_{n=p}^q a_n := a_p \times a_{p+1} \times \cdots \times a_{q-1} \times a_q$$

The index  $n$  is called the multiplication index, the letter  $p$  is the lower bound and the letter  $q$  is the upper bound of the product. For example, we have  $\prod_{n=1}^3 n^3 = 1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36$ . The following properties facilitate the computation of products:

- **Linearity:** if  $\lambda$  is a real number and  $(a_n)$ ,  $(b_n)$  are two sequences, then we have  $\prod_{n=p}^q (\lambda a_n b_n) = \lambda^{q-p+1} \left( \prod_{n=p}^q a_n \right) \left( \prod_{n=p}^q b_n \right)$ .
- **Change of index:** the index  $n$  may be replaced by a bijective function of  $n$  without modifying the sum. For example, setting  $m = n - p$ , we have  $\prod_{n=p}^q a_n = \prod_{m=0}^{q-p} a_{m+p}$ .
- **Chasles's relation:** if  $p \leq r \leq q$ , then we have  $\prod_{n=p}^q a_n = \left( \prod_{n=p}^r a_n \right) \left( \prod_{n=r+1}^q a_n \right)$ .

### 2.2.4 Proof by induction

**Proof by induction** is a proof technique making it possible to demonstrate that a property  $P(n)$  depending on a natural number  $n$  is true for all integers  $n \geq n_0$  (where  $n_0$  is generally 0 or 1). This reasoning rests on two fundamental steps:

- **Base case:** we verify that the property  $P(n_0)$  is true for the smallest integer  $n_0$  considered.

- **Inductive step:** we suppose that the property  $P(n)$  is true for a certain integer  $n \geq n_0$  (the induction hypothesis), and we demonstrate that, under this hypothesis, the property  $P(n+1)$  is true for the following integer  $n+1$ .

If these two steps are established, we conclude that the property  $P(n)$  is true for all integers  $n \geq n_0$ . The intuition is the following: the property is true for  $n_0$  (step 1); since if it is true for  $n_0$  then it is true for  $n_0+1$  (step 2 applied to  $n = n_0$ ), it is therefore true for  $n_0+1$ ; by the same argument, it is true for  $n_0+2$ ; and so on. For example, let us show by induction that, for every non-zero natural number  $n$ , we have  $1 + 2 + \dots + n = \frac{n(n+1)}{2}$ .

- **BASE CASE.** We do indeed have  $\sum_{k=1}^1 k = 1 = \frac{1 \times (1+1)}{2}$ .

- **INDUCTIVE STEP.** Let  $n$  be a non-zero natural number. Let us suppose that we have  $\sum_{k=1}^n k = \frac{n(n+1)}{2}$  and let us show that we have  $\sum_{k=1}^{n+1} k = \frac{(n+1)(n+2)}{2}$ . Indeed, we have:

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \left( \sum_{k=1}^n k \right) + (n+1) \quad \text{by definition of the summation operator} \\ &= \frac{n(n+1)}{2} + (n+1) \quad \text{by the induction hypothesis} \\ &= \frac{n(n+1) + 2(n+1)}{2} = \frac{(n+2)(n+1)}{2} \end{aligned}$$

- **CONCLUSION.** By the principle of induction, we have shown that, for every natural number  $n \geq 1$ , we have  $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ .

### 2.2.5 Arithmetic sequences

A sequence  $(a_n)$  is **arithmetic** if there exists a constant real number  $d$  (called the **common difference**) such that, for every natural number  $n$ , we have  $a_{n+1} - a_n = d$ . In other words, each term is obtained by adding a constant to the preceding term. The general term of an arithmetic sequence with first term  $a_p$  and common difference  $d$  is then written, for every integer  $p$  and every natural number  $p \leq n$ ,  $a_n = a_p + (n-p)d$ . Moreover, the sum of the first terms of an arithmetic sequence is given by  $a_p + a_{p+1} + \dots + a_{n-1} + a_n = \frac{(n-p+1)(a_p + a_n)}{2}$ , for every integer  $p$  and every natural number  $p \leq n$ . Intuitively, the sum of an arithmetic sequence is therefore the number of terms multiplied by the average of the first and the last term. Finally, an arithmetic sequence converges if and only if its common difference  $d$  is zero (in which case it is constant and converges to that constant); if  $d \neq 0$ , the sequence diverges (it tends to  $+\infty$  if  $d > 0$ , or to  $-\infty$  if  $d < 0$ ).

### 2.2.6 Geometric sequences

A sequence  $(a_n)$  is **geometric** if there exists a constant real number  $r$  (called the **common ratio**) such that, for every natural number  $n$ , we have  $a_{n+1} = r a_n$ . In other words, each term is obtained by multiplying the preceding term by a constant. The general term of an arithmetic sequence with first term  $a_p$  and common difference  $d$  is then written, for every integer  $p$  and every natural number  $p \leq n$ ,  $a_n = a_p r^{n-p}$ . Moreover, if  $r \neq 1$ , then the sum of the first terms of an arithmetic sequence is given by  $a_p + a_{p+1} + \dots + a_{n-1} + a_n = \frac{a_p - a_{n+1}}{1-r}$ , for every integer  $p$  and every natural number  $p \leq n$ . Finally, a geometric sequence converges when its common ratio  $r$  is different from 0 (in which case it converges to 0); otherwise, the sequence diverges (except when the common ratio is equal to 1: in which case, the sequence remains constant, equal to its first term).

### 2.2.7 Arithmetico-geometric sequences

A sequence  $(a_n)$  is **arithmetico-geometric** if it satisfies a recurrence relation of the form  $a_{n+1} = r \cdot a_n + p$ , where  $r$  and  $p$  are two constant real numbers with  $r \neq 1$  and  $p \neq 0$ . This sequence generalises both arithmetic sequences (the case  $r = 1$ ) and geometric sequences (the case  $p = 0$ ). In order to solve such a recurrence, we first look for the **fixed point**  $\ell$  satisfying  $\ell = r\ell + p$ , whence  $\ell = \frac{p}{1-r}$ . We then set  $b_n := a_n - \ell$ , and we verify that, for every integer  $n$ , we have  $b_{n+1} = a_{n+1} - \ell = (r a_n + p) - \ell = r a_n + p - r\ell - p = r(a_n - \ell) = r b_n$ . Thus, the sequence  $(b_n)$  is geometric with common ratio  $r$ . The general term of  $(b_n)$  is therefore  $b_n = b_0 \cdot r^n = (a_0 - \ell) r^n$ , whence  $a_n = \ell + (a_0 - \ell) r^n = \frac{p}{1-r} + \left(a_0 - \frac{p}{1-r}\right) r^n$ , for every natural number  $n$ . From then on, an arithmetico-geometric sequence converges if and only if  $|r| < 1$ : in which case it converges to the fixed point  $\ell = \frac{p}{1-r}$ .

### 2.2.8 Numerical series and convergence

A **numerical series** is an infinite sum of the form  $a_0 + a_1 + a_2 + \dots$ , where  $(a_n)$  is a sequence of real numbers. In order to give a meaning to this infinite sum, we introduce the notion of **partial sums**  $S_N := a_0 + a_1 + \dots + a_N$ . The series with general term  $a_n$  is said to **converge** if the sequence of partial sums  $(S_N)_{N \in \mathbb{N}}$  converges to a finite limit  $S$ , called the **sum of the series**. Otherwise put, we have  $\sum_{n=0}^{\infty} a_n := \lim_{N \rightarrow +\infty} S_N = S$ . In the contrary case (if the limit does not exist or is infinite), the series is said to **diverge**. Let us note that, if the series with general term  $a_n$  converges, then necessarily the sequence  $(a_n)_{n \in \mathbb{N}}$  tends to 0. Otherwise put, the terms of the series must tend to zero. However, this condition is not sufficient: there exist series whose general term tends to zero but which diverge (for example, the harmonic series, see *infra*).

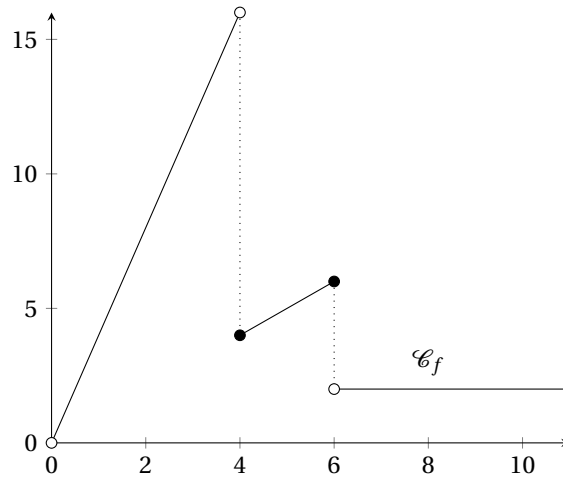
### 2.2.9 Convergence criteria for series

- **Geometric series:** the series with general term  $r^n$  (where  $r$  is a real constant) converges if and only if  $|r| < 1$ , in which case its sum is worth  $\frac{1}{1-r}$ . Conversely, if  $|r| \geq 1$ , the series diverges. For example, in a simple Keynesian model, an autonomous increase in public spending  $\Delta G$  provokes an increase in income which depends on the Keynesian multiplier: if firms invest a constant fraction  $c$  (the marginal propensity to consume) of each increase in income, the cumulative effect is a geometric series:  $\Delta Y = \Delta G + c\Delta G + c^2\Delta G + \dots = \frac{\Delta G}{1-c}$ . The multiplier is therefore  $\frac{1}{1-c}$ . In order for the series to converge, one must verify  $c < 1$  (the marginal propensity to consume must be less than 1), which is economically reasonable.
- **Riemann series:** the series with general term  $\frac{1}{n^\alpha}$  converges if and only if  $\alpha > 1$ . In particular, the series with general term  $\frac{1}{n}$  diverges (the harmonic series), whereas the series with general term  $\frac{1}{n^2}$  converges (and its sum is worth  $\frac{\pi^2}{6}$ ).
- **Comparison criterion (series with positive terms):** if  $0 \leq a_n \leq b_n$  for all the terms, and if the series with general term  $b_n$  converges, then the series with general term  $a_n$  likewise converges. Conversely, if  $a_n \leq b_n$  and the series with general term  $a_n$  diverges, then the series with general term  $b_n$  likewise diverges.
- **D'Alembert's criterion (or the ratio test):** if  $a_n > 0$  for every natural number  $n$ , let us set  $\ell := \lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n}$ . Then if  $\ell < 1$ , the series with general term  $a_n$  converges; if  $\ell > 1$ , the series with general term  $a_n$  diverges; if  $\ell = 1$ , the criterion does not permit a conclusion (other criteria must be used).
- **Cauchy's criterion (or the root test):** if  $a_n > 0$  for every natural number  $n$ . Let us set  $\ell := \lim_{n \rightarrow +\infty} \sqrt[n]{a_n}$ . Then if  $\ell < 1$ , the series with general term  $a_n$  converges; if  $\ell > 1$ , the series with general term  $a_n$  diverges; if  $\ell = 1$ , the criterion does not permit a conclusion.
- **Leibniz's criterion (alternating series):** let  $(a_n)$  be a decreasing sequence of positive terms converging to 0. Then the alternating series with general term  $(-1)^n a_n$  converges, that is to say the series  $a_0 - a_1 + a_2 - a_3 + \dots$  converges.
- **Absolute convergence:** the series with general term  $a_n$  is said to converge absolutely if the series with general term  $|a_n|$  converges. If a series converges absolutely, it converges in the ordinary sense (the converse is not necessarily true: a series may converge without converging absolutely).

## 2.3 The study of a real function

### 2.3.1 Differentiation and continuity

Let  $f$  be a function defined on an interval  $I$  of  $\mathbb{R}$  and let  $a$  be a point interior to  $I$  (that is to say a real number  $a$  such that there exists  $\delta > 0$  such that  $]a - \delta, a + \delta[$  is contained in  $I$ ). The function  $f$  is said to be **differentiable** at  $a$  when the limit  $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$  exists and is finite. This limit is denoted  $f'(a)$  and is called the **derivative** of  $f$  at  $a$ . Equivalently, we may write  $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a)$ . Intuitively, the derivative  $f'(a)$  represents the **slope of the tangent** to the curve representing  $f$  at the point of abscissa  $a$ , or again the **instantaneous rate of change** of  $f$  at  $a$ . Geometrically, when  $f$  is differentiable at  $a$ , we may therefore write the equation of the tangent line to the curve at that point  $y = f'(a)(x - a) + f(a)$ . Consequently,  $f$  is said to be **differentiable on the interval  $I$**  if it is differentiable at each point of  $I$ : we may then define the **derivative function**  $f'$ , which associates to each real number  $x$  of  $I$  its derivative  $f'(x)$ .



**Careful!** A function may fail to be differentiable at a point, even if it is continuous at that point. For example, the absolute value function  $h : x \mapsto |x| := \max(x, -x)$  is continuous at 0, but is not differentiable at 0 since the left and right derivatives do not coincide (the ‘left’ derivative is worth  $-1$  and the ‘right’ one  $+1$ ). Every function differentiable at a point is necessarily continuous at that point. Let us recall in this respect that a function is **continuous** at a point  $a$  of its interval of definition  $I$  if, and only if, we have  $\lim_{x \rightarrow a} f(x) = f(a)$ . Otherwise put, we shall consider that a real function  $f : x \mapsto f(x)$  is continuous if, to infinitesimal variations of the real variable  $x$ , there correspond infinitesimal variations of the image  $f(x)$ . For functions of a single variable, this amounts mathematically to saying that, if  $f$  is a function defined on an interval  $I$ , then  $f$  is continuous at a point  $a$  of  $I$  if, and only if, for every strictly positive real number  $\epsilon$ , there exists a strictly positive real number  $\eta$  such that, for every real number  $x$  of  $I$  satisfying  $|x - a| < \eta$ , we have  $|f(x) - f(a)| < \epsilon$ . For example, for the function  $f$  above is continuous at 5 (among other points), but not at 4 nor at 6.

### 2.3.2 Operations on derivatives

| Function $f$                               | Derivative function $f'$      | Differentiable on...                                                                     |
|--------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------|
| $f(x) = k, k \in \mathbb{R}$               | $f'(x) = 0$                   | $\mathbb{R}$                                                                             |
| $f(x) = x^n, n \in \mathbb{N}^*$           | $f'(x) = nx^{n-1}$            | $\mathbb{R}$                                                                             |
| $f(x) = \frac{1}{x}$                       | $f'(x) = -\frac{1}{x^2}$      | $\mathbb{R}^* := ]-\infty, 0[ \cup ]0, +\infty[$                                         |
| $f(x) = \frac{1}{x^n}, n \in \mathbb{N}$   | $f'(x) = -\frac{n}{x^{n+1}}$  | $\mathbb{R}^* := ]-\infty, 0[ \cup ]0, +\infty[$                                         |
| $f(x) = \sqrt{x}$                          | $f'(x) = \frac{1}{2\sqrt{x}}$ | $]0, +\infty[$                                                                           |
| $f(x) = x^\alpha, \alpha \in \mathbb{R}^*$ | $f'(x) = \alpha x^{\alpha-1}$ | $\mathbb{R}^* (\alpha < 0), \mathbb{R}_+^* (0 < \alpha < 1), \mathbb{R} (\alpha \geq 1)$ |
| $f(x) = e^x$                               | $f'(x) = e^x$                 | $\mathbb{R}$                                                                             |
| $f(x) = \ln(x)$                            | $f'(x) = \frac{1}{x}$         | $]0, +\infty[$                                                                           |

Table 1.1: Derivatives of the standard functions

Let  $f$  and  $g$  be two functions differentiable on an interval  $I$ , and let  $\lambda$  be a real number. The following properties make it possible to compute the derivatives of combinations of functions:

- **Linearity:** if  $f$  and  $g$  are differentiable at a point  $a$ , then  $f + g$  and  $\lambda f$  are likewise differentiable at  $a$ , and we have  $(f + g)'(a) = f'(a) + g'(a)$  and  $(\lambda f)'(a) = \lambda f'(a)$ . In particular, we have  $(f - g)'(a) = f'(a) - g'(a)$ .
- **Product (Leibniz's rule):** if  $f$  and  $g$  are differentiable at a point  $a$ , then the product  $fg$  is likewise differentiable at  $a$ , and we have  $(f \cdot g)'(a) = f'(a)g(a) + f(a)g'(a)$ .
- **Quotient:** if  $f$  and  $g$  are differentiable at a point  $a$  with  $g(a) \neq 0$ , then the quotient  $\frac{f}{g}$  is likewise differentiable at  $a$ , and we have  $\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{[g(a)]^2}$ .
- **Composition (the chain rule):** let  $f$  be a function differentiable at a point  $a$  and  $g$  a function differentiable at  $f(a)$ . Then the composite  $g \circ f$  is differentiable at  $a$ , and we have  $(g \circ f)'(a) = g'(f(a)) \cdot f'(a)$ .



- **Power:** let  $n$  be a non-zero natural number. If  $f$  is differentiable at a point  $a$ , then  $f^n$  is likewise differentiable at  $a$ , and we have  $(f^n)'(a) = n(f(a))^{n-1} \cdot f'(a)$ .

Moreover, let  $f$  be a function differentiable on a subset  $D$  of  $\mathbb{R}$ . Then the function  $e^f : x \mapsto e^{f(x)}$  is also differentiable on  $D$  and, for every real number  $x$  of  $D$ , we have  $(e^f)'(x) = f'(x)e^{f(x)}$ ; moreover, the function  $\ln \circ f : x \mapsto \ln(f(x))$  is likewise differentiable on  $D_+^* := D \setminus \{x \in \mathbb{R}; f(x) \leq 0\}$  (that is to say on the set of points of  $D$  on which the function  $f$  is strictly positive) and we have, for every real number  $x$  of  $D_+^*$ ,  $(\ln \circ f)'(x) = \frac{f'(x)}{f(x)}$ ; finally, for every non-zero real number  $\alpha$ ,  $f^\alpha : x \mapsto (f(x))^\alpha$  is likewise differentiable on  $D$  and we have, for every real number  $x$  of  $D$ ,  $(f^\alpha)'(x) = \alpha f'(x)(f(x))^{\alpha-1}$ .

### 2.3.3 The link between the derivative and monotonicity

Let  $f$  be a function defined and differentiable on an open interval  $I$  (that is to say an interval without endpoints or with excluded endpoints). The following theorems establish a fundamental link between the sign of the derivative and the monotonicity of the function:

- If  $f'(x) > 0$  for every real number  $x$  of  $I$ , then  $f$  is strictly increasing on  $I$ . If  $f'(x) < 0$  for every real number  $x$  of  $I$ , then  $f$  is strictly decreasing on  $I$ .
- If  $f'(x) \geq 0$  for every real number  $x$  of  $I$ , then  $f$  is real number  $x$  of  $I$  increasing on  $I$ . If  $f'(x) \leq 0$  for every real number  $x$  of  $I$ , then  $f$  is decreasing on  $I$ .
- If  $f'(x) \geq 0$  for every real number  $x$  of  $I$  and if  $f'(x) = 0$  only on a finite set of points (or more generally, on a set of measure zero), then  $f$  is strictly increasing on  $I$ . Analogously, if  $f'(x) \leq 0$  for every real number  $x$  of  $I$  with isolated zeros, then  $f$  is strictly decreasing on  $I$ .

In other words, the sign of the derivative informs us about the 'local' direction of variation of the function: a positive derivative means that the function increases locally, while a negative derivative means that it decreases. This makes it possible to establish the **table of variation** (or table of the sign of the derivative) of a function, which synthesises the whole monotone behaviour of  $f$  on its domain of definition. In order to construct such a table, we generally proceed according to the following steps:

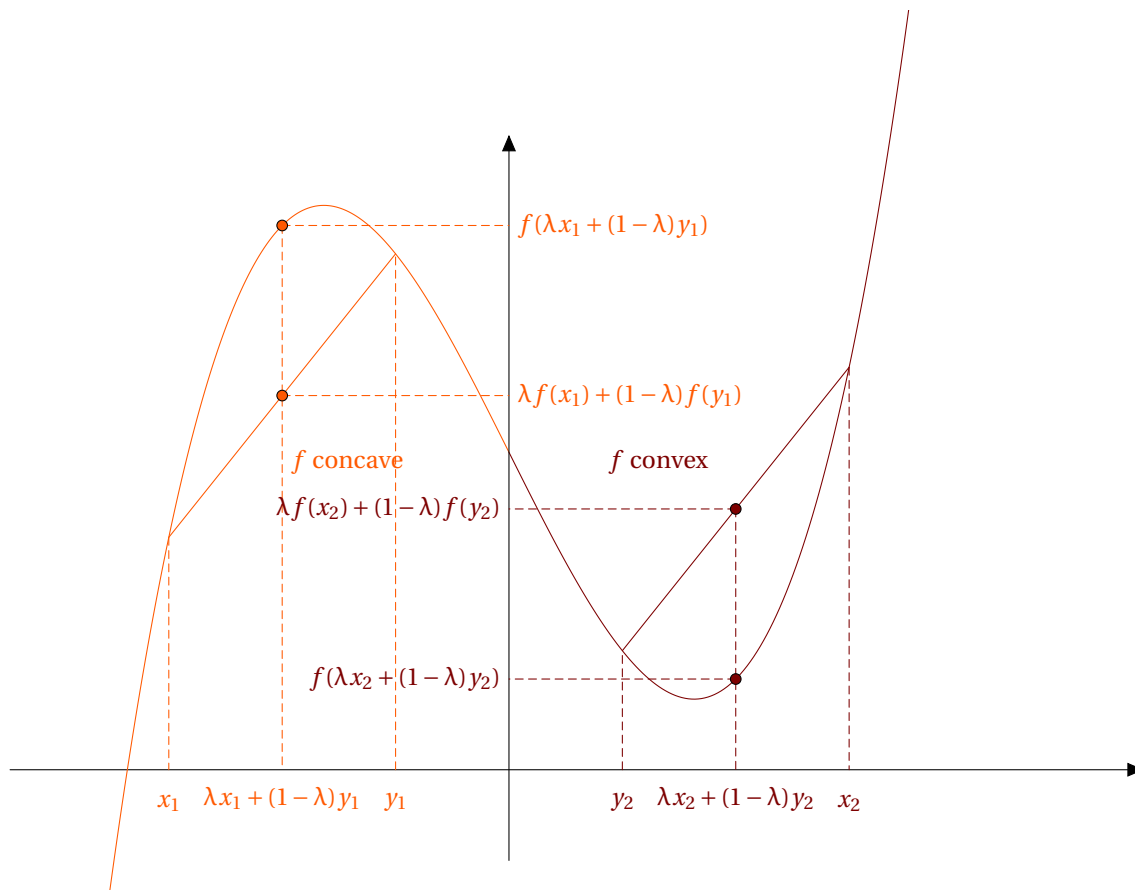
- Determine the domain of definition and of differentiability of  $f$ .
- Compute the derivative  $f'(x)$  and determine the points where  $f'(x) = 0$  and where  $f'(x)$  does not exist.
- Study the sign of  $f'(x)$  on each interval determined by the critical points and the endpoints of the domain.
- Deduce from this the intervals of increase (where  $f' > 0$ ) and of decrease (where  $f' < 0$ ).
- Identify the **local extrema**: a point  $a$  of the domain is a **local maximum** if  $f(a) \geq f(x)$  for all the  $x$  in a neighbourhood of  $a$ ; a point  $a$  is a **local minimum** if  $f(a) \leq f(x)$  for all the  $x$  in a neighbourhood of  $a$ . A local extremum generally occurs at critical points where the derivative changes sign.

**Careful!** A function may possess a local extremum at a point where it is not differentiable, or where its derivative vanishes without changing sign. For example, the absolute value function  $h(x) = |x|$  possesses a global minimum at  $x = 0$ , but is not differentiable there. In addition, a critical point where  $f'(x) = 0$  is not necessarily an extremum: if the derivative does not change sign (for example if  $f'(x) = 0$  but  $f'$  remains positive on either side), the point is a **point of inflection** with a horizontal tangent. Finally, it is essential to distinguish local extrema from global (or *absolute*) extrema: a function may admit several distinct local maxima or minima, but only one global maximum or minimum on a closed and bounded interval.

### 2.3.4 Convex functions and concave functions

Let  $f$  be a function defined on an interval  $I$  of  $\mathbb{R}$ . We say that  $f$  is **convex** on  $I$  if, for all real numbers  $x_1$  and  $x_2$  of  $I$  and for every real number  $\lambda \in [0, 1]$ , we have  $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$ . Intuitively, this condition means that the graph of  $f$  'remains below' any chord joining two points of the curve. Geometrically, this amounts to saying that, for every line segment joining two points of the curve, the points of the segment are situated above or on the curve. Equivalently, a function is convex if the **line segment** joining any two points of its representative curve never passes below the curve itself. Conversely, we say that  $f$  is **concave** on  $I$  if, for all real numbers  $x_1$  and  $x_2$  of

I and for every real number  $\lambda \in [0, 1]$ , we have  $f(\lambda x_1 + (1 - \lambda)x_2) \geq \lambda f(x_1) + (1 - \lambda)f(x_2)$ . Thus, a concave function is characterised by the fact that the graph ‘remains above’ any chord; the line segment joining two points of its curve never passes above the curve. Finally, we say that  $f$  is **strictly convex** (or **strictly concave**) if the preceding inequalities are strict for all distinct real numbers  $x_1 \neq x_2$  and for every  $\lambda \in ]0, 1[$ .



**Careful!** A function may be neither convex nor concave on an interval. For example, the function  $f(x) = \sin(x)$  is convex on the interval  $[\pi, 2\pi]$  and concave on the interval  $[0, \pi]$ . A point where the function changes convexity (or concavity) is called a **point of inflection**: it is a point where the curve ‘changes curvature’.

### 2.3.5 The link between the second derivative and convexity

Let  $f$  be a function defined and twice differentiable on an open interval  $I$  (that is to say whose derivative  $f'$  is itself differentiable). We denote by  $f''(x)$  the **second derivative** of  $f$  at  $x$ , that is to say the derivative of  $f'(x)$ . The following theorems establish a fundamental link between the sign of the second derivative and the convexity of the function:

- If  $f''(x) > 0$  for every real number  $x$  of  $I$ , then  $f$  is strictly convex on  $I$ . If  $f''(x) < 0$  for every real number  $x$  of  $I$ , then  $f$  is strictly concave on  $I$ .
- If  $f''(x) \geq 0$  for every real number  $x$  of  $I$ , then  $f$  is convex on  $I$ . If  $f''(x) \leq 0$  for every real number  $x$  of  $I$ , then  $f$  is concave on  $I$ .
- If  $f''(x_0) = 0$  and if  $f''$  changes sign at  $x_0$  (that is to say that  $f''$  is positive on one side and negative on the other), then  $x_0$  is a point of inflection of  $f$ . At a point of inflection, the tangent to the curve ‘crosses’ the curve itself.

In other words, the second derivative measures the curvature of the function: a positive second derivative means that the slope increases (the curve ‘steepens’), while a negative second derivative means that the slope decreases (the curve ‘flattens’). This makes it possible to establish the **table of convexity** (or table of the sign of the second derivative) of a function, which synthesises the behaviour of the curvature of  $f$  on its domain of definition. In order to construct such a table, we generally proceed according to the following steps:

- Determine the domain of definition and of differentiability (up to order 2) of  $f$ .
- Compute the second derivative  $f''(x)$  and determine the points where  $f''(x) = 0$  or where  $f''(x)$  does not exist.
- Study the sign of  $f''(x)$  on each interval determined by the potential points of inflection and the endpoints of the domain.
- Deduce from this the intervals of convexity (where  $f'' > 0$ ) and of concavity (where  $f'' < 0$ ).
- Identify the genuine points of inflection: these are the points where  $f''(x) = 0$  and where  $f''$  changes sign.

### 2.3.6 The link between convexity and extrema

The convexity of a function makes it possible to characterise the nature of its extrema. In particular, if a function  $f$  is convex on an interval  $I$  and if  $f'(a) = 0$  for a point  $a \in I$ , then  $a$  is a **global minimum** of  $f$  on  $I$ . Analogously, if  $f$  is concave and  $f'(a) = 0$ , then  $a$  is a **global maximum**. This result turns out to be extremely useful in economics: a convex cost function with a zero derivative at a point guarantees that this point is the minimal cost; a concave utility function indicates that the economic agent has a preference for diversity (risk aversion). In addition, convex and concave functions play a fundamental role in optimisation: every local minimum of a strictly convex function is a global minimum, and every local maximum of a strictly concave function is a global maximum. **Careful!** A potential point of inflection (where  $f''(x) = 0$ ) is a genuine point of inflection only if the second derivative effectively changes sign at that point. For example, the function  $f(x) = x^4$  satisfies  $f''(0) = 0$ , but  $f''(x) = 12x^2 \geq 0$  for all real numbers  $x$ : the function remains convex in the neighbourhood of  $x = 0$ , and  $x = 0$  is therefore not a point of inflection.

### 2.3.7 The intermediate value theorem and the solution of equations

Let  $f$  be a real function defined on a set  $\mathcal{D}_f$  specified in the statement or to be determined. In order to justify the existence of a solution to the equation  $f(x) = 0$ :

- We begin by justifying the continuity of  $f$  on a real interval  $I$  included in  $\mathcal{D}_f$  to be determined: let  $I$  be a real interval included in  $\mathcal{D}_f$ ; then  $f$  may be continuous on  $I$  as a composite, sum, product or quotient of functions continuous on  $I$ ; in the case of a quotient, we specify that the denominator does not vanish on the interval  $I$ .
- One must then determine two values  $\alpha$  and  $\beta$  of  $I \cup \{-\infty, +\infty\}$  satisfying  $f(\alpha) < 0$  (if  $\alpha$  is real) or  $\lim_{x \rightarrow \alpha} f(x) < 0$  (if  $\alpha$  is an endpoint of  $I$  at which  $f$  is not defined) on the one hand, and  $f(\beta) > 0$  (if  $\beta$  is real) or  $\lim_{x \rightarrow \beta} f(x) > 0$  (if  $\beta$  is an endpoint of  $I$  at which  $f$  is not defined) on the other hand.
- Two cases then present themselves:
  - If it is a matter of justifying that there exists **at least one** solution to the equation  $f(x) = 0$ , we then write as follows: ‘We have  $f(\alpha) < 0$  (or  $\lim_{x \rightarrow \alpha} f(x) < 0$  according to the situation) on the one hand, and  $f(\beta) > 0$  (or  $\lim_{x \rightarrow \beta} f(x) > 0$  according to the situation) on the other hand. Now the function  $f$  is continuous on  $]\alpha, \beta[$  (or  $]\beta, \alpha[$  according to the situation). Therefore, by virtue of the **intermediate value theorem**, there exists a real number  $x_0$  belonging to  $]\alpha, \beta[$  (or  $]\beta, \alpha[$  according to the situation) satisfying  $f(x_0) = 0$ .’
  - If it is a matter of justifying that there exists **a unique** solution to the equation  $f(x) = 0$ , one must in addition show that  $f$  is strictly monotone on  $]\alpha, \beta[$  (or  $]\beta, \alpha[$ ). We then denote by  $J$  the image of  $]\alpha, \beta[$  (or  $]\beta, \alpha[$ ) under  $f$ . We then write as follows: ‘The function  $f$  is continuous on  $]\alpha, \beta[$  (or  $]\beta, \alpha[$ ) and strictly monotone from  $]\alpha, \beta[$  (or  $]\beta, \alpha[$ ) onto  $J$ . Now 0 belongs to  $J$ . Therefore, by virtue of the **bijection theorem**, there exists a unique real number  $x_0$  belonging to  $]\alpha, \beta[$  (or  $]\beta, \alpha[$  according to the case) satisfying the equality  $f(x_0) = 0$ .’

## 2.4 Differential and integral calculus

### 2.4.1 Differential equations

An (ordinary) **differential equation** is a relation which links an unknown function  $y$  of a real variable  $t$  (often time) and its successive derivatives  $y'(t)$ ,  $y''(t)$ , and so on. For example, the equation  $y'(t) = ay(t)$ , where  $a$  is a given real number, is a first-order differential equation, since it involves the first derivative of  $y$  (and no derivative of higher order). More generally, a differential equation of order  $n$  is written in a form involving  $y, y', \dots, y^{(n)}$ . In this course, we limit ourselves to the most standard first-order linear equations:

- **Homogeneous case:** a homogeneous first-order linear differential equation is written  $y'(t) + ay(t) = 0$ , where  $a$  is a real constant. All its solutions are given by  $y(t) = Ce^{-at}$ , where  $C$  is a real constant determined by an initial condition (for example the value of  $y(t_0)$  at a given date  $t_0$ ).
- **Non-homogeneous case with a constant right-hand side:** an equation of the form  $y'(t) + ay(t) = b$ , where  $a$  and  $b$  are real constants, admits as solutions  $y(t) = y_{\text{part}}(t) + y_{\text{hom}}(t) = \frac{b}{a} + Ce^{-at}$ , when  $a \neq 0$ . The term  $\frac{b}{a}$  is a constant particular solution (or steady state), and  $Ce^{-at}$  is the general solution of the associated homogeneous equation  $y'(t) + ay(t) = 0$ .

When  $a > 0$ , the term  $Ce^{-at}$  converges to 0 as  $t \rightarrow +\infty$ , which means that every solution  $y(t)$  converges to the equilibrium solution  $y^* = \frac{b}{a}$ : the equilibrium is stable. By contrast, if  $a < 0$ , the term  $e^{-at} = e^{|a|t}$  diverges exponentially, and the solution moves away from the equilibrium: the equilibrium is said to be unstable. This type of differential equation serves to model dynamics towards an equilibrium level. For example, a capital accumulation function  $K(t)$  satisfying  $K'(t) = sY - \delta K(t)$  (the contribution of an investment flow  $sY$  and depreciation at rate  $\delta$ ) leads to an equation of the preceding form, whose solution converges to an equilibrium level of capital  $K^* = \frac{sY}{\delta}$ .

## 2.4.2 Taylor expansions

**Taylor expansions** make it possible to approximate a sufficiently regular function by a polynomial in the neighbourhood of a given point. More precisely, if a function  $f$  is  $n$  times differentiable at a point  $a$ , we may write, for  $x$  close to  $a$ , we have  $f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$ , where  $R_n(x)$  is a remainder which becomes negligible as  $x$  comes closer to  $a$ . When the remainder is negligible relative to  $(x-a)^n$ , we often write  $f(x) = f(a) + f'(a)(x-a) + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + o((x-a)^n)$  as  $x$  tends to  $a$ , where the notation  $o((x-a)^n)$  means that the remainder decreases faster than  $(x-a)^n$ : that is to say that  $\lim_{x \rightarrow a} \frac{R_n(x)}{(x-a)^n} = 0$ .

| Function $f$                  | Taylor expansion in the neighbourhood of 0                                                                                 |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| $f(x) = e^x$                  | $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + o(x^n)$                                          |
| $f(x) = \ln(1+x)$             | $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n+1} \frac{x^n}{n} + o(x^n)$                 |
| $f(x) = (1+x)^\alpha$         | $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \dots + o(x^n)$ |
| $f(x) = \frac{1}{1-x}$        | $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots + x^n + o(x^n)$                                                                 |
| $f(x) = \frac{1}{1+x}$        | $\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + o(x^n)$                                                          |
| $f(x) = \sqrt{1+x}$           | $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + \dots + o(x^n)$                                           |
| $f(x) = \frac{1}{\sqrt{1+x}}$ | $\frac{1}{\sqrt{1+x}} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x^3}{16} + \dots + o(x^n)$                               |

Table 1.2: Standard Taylor expansions in the neighbourhood of 0

Taylor expansions are useful for:

- **Approximating complex functions** by linear or quadratic forms that are easy to interpret, notably around an equilibrium or a reference point. For example, a non-linear demand function may be approximated in the neighbourhood of the current price by its tangent, which facilitates the local study of sensitivity to price variations.
- **Deriving linearised versions** of non-linear models, extremely useful for local analysis (the stability of an equilibrium, low-dimensional dynamics, macroeconomic models linearised around a steady state). For example, a neoclassical growth model may be linearised around its balanced growth path with the help of first-order Taylor expansions.
- **Obtaining quantitative approximations** useful for numerical models. For example, the logarithmic approximation  $\ln(1+x) \approx x$  for small  $x$  is very much used in economics in order to linearise growth rates and rates of return.

**Careful!** Taylor expansions are valid only for values of  $x$  sufficiently close to the reference point (here 0). For large variations, the polynomial approximation departs from the real behaviour of the function, and one must either use an expansion of higher order, or abandon the linear or quadratic approximation. This is why Taylor expansions are above all tools of local analysis, suited to studying the behaviour of a function in the neighbourhood of a given point, rather than over a wide interval.

### 2.4.3 Antiderivatives of a function

Let  $f$  be a function defined on an interval  $I$  of  $\mathbb{R}$ . We call an **antiderivative** of  $f$  on  $I$  any function  $F$  defined and differentiable on  $I$  such that we have  $F'(x) = f(x)$  for every real number  $x$  of  $I$ . In other words, an antiderivative of  $f$  is a function whose derivative is precisely  $f$ . For example, an antiderivative of the function  $f(x) = 2x$  is the function  $F(x) = x^2$  (since  $(x^2)' = 2x$ ), but the function  $G(x) = x^2 + 5$  is likewise an antiderivative of it (since  $(x^2 + 5)' = 2x$ ). **Careful!** A function may fail to admit an antiderivative. However, if a function  $f$  possesses an antiderivative  $F$  on an interval  $I$ , then it possesses infinitely many: indeed, if  $F$  is an antiderivative of  $f$ , then for every real number  $C$ , the function  $G(x) = F(x) + C$  is likewise an antiderivative of  $f$ , since  $(F(x) + C)' = F'(x) + 0 = f(x)$ . Conversely, if  $F$  and  $G$  are two distinct antiderivatives of  $f$  on an interval  $I$ , then their difference  $F - G$  is a constant function on  $I$ . Thus, two antiderivatives of the same function differ by a constant, and the set of all the antiderivatives of  $f$  on  $I$  may be described by the expression  $F(x) + C$ , where  $F$  is a particular antiderivative of  $f$  and  $C$  runs over the set of real numbers.

| Function $f$                                              | Antiderivative $F$                              | Valid on...                                      |
|-----------------------------------------------------------|-------------------------------------------------|--------------------------------------------------|
| $f(x) = k, k \in \mathbb{R}$                              | $F(x) = kx + C$                                 | $\mathbb{R}$                                     |
| $f(x) = x^n, n \in \mathbb{N}^*$                          | $F(x) = \frac{x^{n+1}}{n+1} + C$                | $\mathbb{R}$                                     |
| $f(x) = \frac{1}{x}$                                      | $F(x) = \ln x  + C$                             | $\mathbb{R}^* := ]-\infty, 0[ \cup ]0, +\infty[$ |
| $f(x) = \frac{1}{x^n}, n \in \mathbb{N}^*$                | $F(x) = -\frac{1}{(n-1)x^{n-1}} + C$            | $\mathbb{R}^* := ]-\infty, 0[ \cup ]0, +\infty[$ |
| $f(x) = \frac{1}{\sqrt{x}}$                               | $F(x) = 2\sqrt{x} + C$                          | $]0, +\infty[$                                   |
| $f(x) = x^\alpha, \alpha \in \mathbb{R} \setminus \{-1\}$ | $F(x) = \frac{x^{\alpha+1}}{\alpha+1} + C$      | $\mathbb{R}_+^*$                                 |
| $f(x) = e^x$                                              | $F(x) = e^x + C$                                | $\mathbb{R}$                                     |
| $f(x) = \frac{u'(x)}{u(x)}$                               | $F(x) = \ln u(x)  + C$                          | According to the domain of $u$                   |
| $f(x) = u'(x)e^{u(x)}$                                    | $F(x) = e^{u(x)} + C$                           | According to the domain of $u$                   |
| $f(x) = u'(x)(u(x))^\alpha, \alpha \neq -1$               | $F(x) = \frac{[u(x)]^{\alpha+1}}{\alpha+1} + C$ | According to the domain of $u$                   |

Table 1.3: Antiderivatives of the standard functions ( $C$  denotes an arbitrary constant)

### 2.4.4 The Riemann integral

Let  $f$  be a function continuous on a closed interval  $[a, b]$  with  $a < b$ . We call the **Riemann integral** of  $f$  on the interval  $[a, b]$  the real number denoted  $\int_a^b f(x) dx$ . Intuitively, this integral represents the **signed area** between the curve of  $f$  and the horizontal axis, on the interval  $[a, b]$ : the portions where  $f(x) > 0$  contribute positively to the integral, while the portions where  $f(x) < 0$  contribute negatively. Rigorously, this integral may be defined as the common limit of the lower and upper Riemann sums, as the mesh of the subdivision tends to zero. In particular, if  $f$  is a function continuous on an interval  $[a, b]$  and if  $F$  is any antiderivative of  $f$  on that interval, then we have  $\int_a^b f(x) dx = F(b) - F(a) =: [F(x)]_a^b$ . This property establishes the fundamental link between antiderivatives and integrals: computing an integral amounts to finding an antiderivative and evaluating the difference of its values at the endpoints of the interval. It is therefore thanks to this theorem that most of the integrals encountered in practice may be computed explicitly.

### 2.4.5 Properties and operations on integrals

The following properties make it possible to compute integrals and to combine several of them:

— **Linearity:** if  $f$  and  $g$  are two functions continuous on  $[a, b]$  and  $\lambda$  a real number, then:

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx \quad \text{and} \quad \int_a^b \lambda f(x) dx = \lambda \int_a^b f(x) dx$$

— **Chasles's relation:** if  $f$  is continuous on an interval containing the real numbers  $a, b$  and  $c$ , then:

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

- **Integration by parts:** if  $u$  and  $v$  are two differentiable functions whose derivatives  $u'$  and  $v'$  are continuous on  $[a, b]$ , then:

$$\int_a^b u(x) v'(x) dx = [u(x) v(x)]_a^b - \int_a^b u'(x) v(x) dx$$

- **Change of variable (substitution):** if  $\varphi$  is a differentiable and bijective function from  $[\alpha, \beta]$  onto  $[a, b]$ , and if  $f$  is continuous on  $[a, b]$ , then:

$$\int_a^b f(x) dx = \int_\alpha^\beta f(\varphi(t)) \varphi'(t) dt$$

- **Comparison:** if  $f$  and  $g$  are two functions continuous on  $[a, b]$  with  $f(x) \leq g(x)$  for every  $x \in [a, b]$ , then:

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$

#### 2.4.6 Improper integrals

The integrals we have studied so far bore on closed and bounded intervals  $[a, b]$  with  $a < b$ , and on functions continuous on those intervals. However, it is often useful to consider integrals over unbounded intervals (for example  $[a, +\infty[$ ) or over intervals containing points of discontinuity or of singularity. We call these integrals **improper integrals** (or generalised integrals).

- **Case 1 (interval unbounded on the right).** Let  $f$  be a function continuous on the interval  $[a, +\infty[$  where  $a$  is a fixed real number. We say that the integral  $\int_a^{+\infty} f(x) dx$  **converges** if the following limit exists and is finite and we set  $\int_a^{+\infty} f(x) dx := \lim_{t \rightarrow +\infty} \int_a^t f(x) dx$ . If this limit is infinite or does not exist, we say that the integral **diverges**. Analogously, we say that the integral  $\int_{-\infty}^b f(x) dx$  converges if  $\int_{-\infty}^b f(x) dx := \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$  exists and is finite. Finally, for an integral over the whole interval  $\mathbb{R}$ , we set  $\int_{-\infty}^{+\infty} f(x) dx := \int_{-\infty}^c f(x) dx + \int_c^{+\infty} f(x) dx$  for any real number  $c$  (the result does not depend on the choice of  $c$ , provided that the two integrals on the right-hand side converge).
- **Case 2 (singularity at an endpoint).** Let  $f$  be a function continuous on the half-open interval  $[a, b[$  (where the function may have a singularity at  $b$ ), but not necessarily bounded in the neighbourhood of  $b$ . We say that the integral  $\int_a^b f(x) dx$  **converges** if the following limit exists and is finite and we set  $\int_a^b f(x) dx := \lim_{t \rightarrow b^-} \int_a^t f(x) dx$ . Analogously, if the singularity is at  $a$ , we set  $\int_a^b f(x) dx := \lim_{t \rightarrow a^+} \int_t^b f(x) dx$ . If the singularity is at a point  $c$  of the interval  $]a, b[$ , we decompose the integral into  $\int_a^b f(x) dx := \int_a^c f(x) dx + \int_c^b f(x) dx$ , where each of the two integrals on the right is defined as a limit relative to the singularity.

#### 2.4.7 Convergence criteria for improper integrals

- **Comparison (positive integrals):** if  $0 \leq f(x) \leq g(x)$  for every real number  $x$  of  $[a, +\infty[$  and if  $\int_a^{+\infty} g(x) dx$  converges, then  $\int_a^{+\infty} f(x) dx$  likewise converges. Conversely, if  $f(x) \leq g(x)$  for every  $x$  and if  $\int_a^{+\infty} f(x) dx$  diverges, then  $\int_a^{+\infty} g(x) dx$  likewise diverges.
- **Absolute convergence:** an integral  $\int_a^{+\infty} f(x) dx$  is said to **converge absolutely** if  $\int_a^{+\infty} |f(x)| dx$  converges. If an integral converges absolutely, it converges in the ordinary sense. However, the converse is false: an integral may converge without converging absolutely (one then speaks of **conditional convergence**).
- **Riemann's criterion (comparison with powers):** let  $f$  be a function continuous and positive on  $[a, +\infty[$ . If  $\lim_{x \rightarrow +\infty} x^\alpha f(x)$  exists and is finite for a real number  $\alpha > 1$ , then  $\int_a^{+\infty} f(x) dx$  converges. If this limit is positive and infinite for a real number  $\alpha \leq 1$ , then the integral diverges. Analogously, in the neighbourhood of a singularity at  $x = a$ , if  $\lim_{x \rightarrow a^+} (x - a)^\beta f(x)$  exists and is finite for a real number  $\beta < 1$ , then  $\int_a^b f(x) dx$  converges; if this limit is positive and infinite for a real number  $\beta \geq 1$ , then the integral diverges.

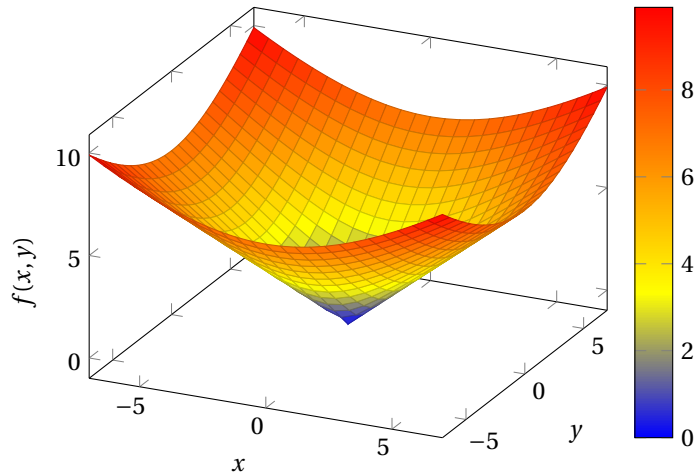
— **Reference example:** the integral  $\int_{-\infty}^{+\infty} \frac{1}{x^\alpha} dx$  converges if and only if  $\alpha > 1$ . Analogously, the integral  $\int_0^1 \frac{1}{x^\beta} dx$  converges if and only if  $\beta < 1$ . These reference integrals are particularly useful for applying the criteria of comparison and of equivalence.

**Careful!** The criteria of comparison and of equivalence apply to functions of constant sign (positive or negative). For functions of variable sign, it is preferable to have recourse to the notion of absolute convergence.

## 2.5 Functions of several variables

### 2.5.1 Functions of several variables

A real function of several variables is a function whose domain is  $\mathbb{R}^n$  (where  $n$  is a natural number at least equal to 2) and whose codomain is  $\mathbb{R}$ . For example, a Cobb-Douglas production function  $f : (K, L) \mapsto K^\alpha L^{1-\alpha}$  (with  $\alpha$  a real number between 0 and 1) is a function which indicates the quantity  $f(K, L)$  produced by a firm when it employs a certain quantity  $L$  of labour and  $K$  of capital: it is therefore a function of two variables. Consequently, let  $n$  be a natural number at least equal to 2 and  $f$  a real function from  $\mathbb{R}^n$  into  $\mathbb{R}$ . Let  $\mathbf{a} := (a_1, \dots, a_n)$  be an element of  $\mathbb{R}^n$  for which  $f$  admits an image (otherwise put, there exists a real number  $y$  satisfying  $f(a_1, \dots, a_n) = y$ ). Let  $i$  be an integer between 1 and  $n$ . Then we call the **partial function** associated with  $f$  and linked to the  $i$ -th variable the real function  $f_{\mathbf{a},i} : x \mapsto f(a_1, \dots, a_{i-1}, x, a_{i+1}, \dots, a_n)$ : it is therefore a real function of **a single variable** and not of several; all the variables with the exception of the  $i$ -th are thus fixed, which is at the source of the ‘all other things being equal’ (*ceteris paribus*) reasoning in economics. For example, let  $f$  be the map from  $\mathbb{R}^2$  into  $\mathbb{R}$  which, to every pair  $(x, y)$  of real numbers, associates the real number  $f(x, y) := \sqrt{x^2 + y^2}$ . A representation in space of such a function is given by:



Let us set  $\mathbf{a} := (0, 2)$ . Then we have  $f_{\mathbf{a},1} : x \mapsto \sqrt{x^2 + 4}$  and  $f_{\mathbf{a},2} : y \mapsto \sqrt{y^2} = |y|$ , functions defined on  $\mathbb{R}$ . Finally, we call an **affine function** (or linear, by an abuse of language) of  $n$  variables ( $n$  being a non-zero natural number) any function for which there exist real numbers  $b_0, \dots, b_n$  satisfying, for every family  $(x_1, \dots, x_n)$  of real numbers,  $f(x_1, \dots, x_n) = b_0 + b_1 x_1 + \dots + b_n x_n$ . From then on, for every integer  $i$  between 1 and  $n$ , the function  $f$  admits a partial derivative at every point of  $\mathbb{R}^n$  with respect to the  $i$ -th variable and, for every family  $(x_1, \dots, x_n)$  of real numbers, we have  $\frac{\partial f}{\partial x_i} \Big|_{(a_1, \dots, a_n)} = a_i$  (see *infra*).

### 2.5.2 Partial derivatives

Let  $n$  be a natural number at least equal to 2 and  $f$  a real function from  $\mathbb{R}^n$  into  $\mathbb{R}$ . Let  $\mathbf{a} := (a_1, \dots, a_n)$  be an element of  $\mathbb{R}^n$  for which  $f$  admits an image (otherwise put, there exists a real number  $y$  satisfying  $f(a_1, \dots, a_n) = y$ ). Let  $i$  be an integer between 1 and  $n$ . Then  $f$  admits a **partial derivative** at  $\mathbf{a} := (a_1, \dots, a_n)$  with respect to the  $i$ -th variable if, and only if, the partial function associated with  $f$  and linked to the  $i$ -th variable  $f_{\mathbf{a},i}$  is differentiable at  $a_i$ , otherwise put if, and only if, the **difference quotient**  $\tau : t \mapsto \frac{f_{\mathbf{a},i}(x_i) - f_{\mathbf{a},i}(a_i)}{x_i - a_i}$  admits a finite limit  $\ell$ , then denoted  $f'_{\mathbf{a},i}(a_i)$  as the non-zero real number  $x_i$  tends to  $a_i$ . We then write  $\frac{\partial f}{\partial x_i} \Big|_{(a_1, \dots, a_n)} = f'_{\mathbf{a},i}(a_i)$ . We then implicitly define a real function of  $n$  variables, called the **partial derivative** with respect to the  $i$ -th variable, by setting  $\frac{\partial f}{\partial x_i} : (a_1, \dots, a_n) \mapsto f'_{\mathbf{a},i}(a_i)$ . Moreover, the function  $\nabla f$  (in the absence of ambiguity) is known by the name of the **gradient** of  $f$ . It is a vector-valued function simply denoted  $\nabla f$  from  $\mathbb{R}^n$  into  $\mathbb{R}^n$ . It is a

of  $f$ . For example, let the function from  $\mathbb{R}^3$  into  $\mathbb{R}$  which, to every triple  $(x, y, z)$  of real numbers, associates  $f(x, y, z) := x^2 + z + xyz + y^2 + x + 1$ . Then, for every triple  $(x, y, z)$  of real numbers, we have  $\frac{\partial f}{\partial x} \Big|_{(x, y, z)} = 2x + yz + 1$ ,  $\frac{\partial f}{\partial y} \Big|_{(x, y, z)} = xz + 2y$  and  $\frac{\partial f}{\partial z} \Big|_{(x, y, z)} = 1 + xy$ .



### 2.5.3 Second partial derivatives and the Hessian matrix

We denote by  $\frac{\partial^2 f}{\partial x_j \partial x_i} : (x_1, \dots, x_n) \mapsto \frac{\partial}{\partial x_j} \left( \frac{\partial f}{\partial x_i}(x_1, \dots, x_n) \right) \Big|_{(x_1, \dots, x_n)}$  the **second partial derivative** with respect to the  $i$ -th then to the  $j$ -th variable, that is to say the partial derivative with respect to the  $j$ -th variable of the partial derivative of  $f$  with respect to the  $i$ -th variable, defined naturally when the partial derivatives themselves admit partial derivatives. In particular, if we have  $i = j$ , we may write more simply  $\frac{\partial^2 f}{\partial x_i^2} := \frac{\partial^2 f}{\partial x_i \partial x_i}$  the partial derivative with respect to the  $i$ -th then again with respect to the  $i$ -th variable. We often summarise the second partial derivatives in the matrix  $\mathbf{H}(x_1, \dots, x_n) := \left( \frac{\partial^2 f}{\partial x_j \partial x_i}(x_1, \dots, x_n) \right)_{1 \leq i, j \leq n}$ , called the **Hessian matrix** of the function  $f$  evaluated at the point  $(x_1, \dots, x_n)$ . For example, let the function from  $\mathbb{R}^3$  into  $\mathbb{R}$  which, to every triple  $(x, y, z)$  of real numbers, associates  $f(x, y, z) := x^2 \exp(yz) + x^2 + \ln(x)y^4 + 1$ . Then, for every triple  $(x, y, z)$  of real numbers, we have  $\frac{\partial f}{\partial x} \Big|_{(x, y, z)} = 2x \exp(yz) + yz + \frac{1}{x}$ ,  $\frac{\partial f}{\partial y} \Big|_{(x, y, z)} = x^2 z \exp(yz) + xz + 2y + 4 \ln(x)y^3$  and  $\frac{\partial f}{\partial z} \Big|_{(x, y, z)} = x^2 y \exp(yz) + 1 + xy$ . We therefore have, for every triple  $(x, y, z)$  of real numbers:

$$\begin{cases} \frac{\partial^2 f}{\partial x^2} = 2 \exp(yz) - \frac{y^4}{x^2} & \text{and} & \frac{\partial^2 f}{\partial x \partial y} = 2xz \exp(yz) + z + \frac{4y^3}{x} & \text{and} & \frac{\partial^2 f}{\partial x \partial z} = 2xy \exp(yz) + y \\ \frac{\partial^2 f}{\partial y \partial x} = 2xz \exp(yz) + z + \frac{4y^3}{x} & \text{and} & \frac{\partial^2 f}{\partial y^2} = x^2 z^2 \exp(yz) + 2 + 12 \ln(x)y^2 & \text{and} & \frac{\partial^2 f}{\partial y \partial z} = x^2(1 + yz) \exp(yz) + x \\ \frac{\partial^2 f}{\partial z \partial x} = 2xy \exp(yz) + y & \text{and} & \frac{\partial^2 f}{\partial z \partial y} = x^2(1 + yz) \exp(yz) + x & \text{and} & \frac{\partial^2 f}{\partial z^2} = x^2 y^2 \exp(yz) \end{cases}$$

For every triple  $(x, y, z)$  of real numbers, the Hessian matrix of  $f$  evaluated at the point  $(x, y, z)$  therefore satisfies:

$$\mathbf{H}(x, y, z) = \begin{pmatrix} 2 \exp(yz) - \frac{y^4}{x^2} & 2xz \exp(yz) + z + \frac{4y^3}{x} & 2xy \exp(yz) + y \\ 2xz \exp(yz) + z + \frac{4y^3}{x} & x^2 z^2 \exp(yz) + 2 + 12 \ln(x)y^2 & x^2(1 + yz) \exp(yz) + x \\ 2xy \exp(yz) + y & x^2(1 + yz) \exp(yz) + x & x^2 y^2 \exp(yz) \end{pmatrix}$$

### 2.5.4 Convex functions and concave functions

Let  $n$  be a non-zero natural number. Then a real function  $f : (x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$  of  $n$  variables is **concave** if, and only if, we have  $f(\lambda x_1 + (1 - \lambda)x'_1, \dots, \lambda x_n + (1 - \lambda)x'_n) \geq \lambda f(x_1, \dots, x_n) + (1 - \lambda)f(x'_1, \dots, x'_n)$  whatever the families  $(x_1, \dots, x_n)$  and  $(x'_1, \dots, x'_n)$  of real numbers and the real number  $\lambda$  of  $[0, 1]$  chosen. Dually, a real function  $f : (x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$  of  $n$  variables is **convex** if, and only if, we have  $f(\lambda x_1 + (1 - \lambda)x'_1, \dots, \lambda x_n + (1 - \lambda)x'_n) \leq \lambda f(x_1, \dots, x_n) + (1 - \lambda)f(x'_1, \dots, x'_n)$  whatever the families  $(x_1, \dots, x_n)$  and  $(x'_1, \dots, x'_n)$  of real numbers and the real number  $\lambda$  of  $[0, 1]$  chosen. In particular, in the case where  $f$  is a function of two variables (and only in that case), it is easy to verify whether a function is concave or convex: a function  $f : (x_1, x_2) \mapsto f(x_1, x_2)$  admitting first and second partial derivatives is concave (resp. convex) when, on the one hand the function  $\left( \frac{\partial^2 f}{\partial x_1^2} \right) \left( \frac{\partial^2 f}{\partial x_2^2} \right) - \left( \frac{\partial^2 f}{\partial x_1 \partial x_2} \right) \left( \frac{\partial^2 f}{\partial x_2 \partial x_1} \right)$  (that is to say the determinant of the Hessian matrix, or **Hessian**) is strictly positive, and on the other hand the function  $\frac{\partial^2 f}{\partial x_1^2}$  is negative (resp. positive).

### 2.5.5 Total differentiation and *ceteris paribus* reasoning

Let  $f$  be a differentiable function of several variables, for example  $f(x, y)$ . The **total differential** (or complete differential) of  $f$  is written  $df := \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$ . This expression represents the total infinitesimal variation of  $f$  when the variables  $x$  and  $y$  undergo respectively variations  $dx$  and  $dy$ . More generally, for a function  $f(x_1, x_2, \dots, x_n)$  depending on  $n$  variables, we set  $df := \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n$ . The major interest of this formulation resides in what is called *ceteris paribus* reasoning ('all other things being equal' in Latin). This reasoning consists in studying how a function reacts to the variation of one variable **on the assumption that all the other variables remain constant**. Mathematically, this amounts to fixing all the differentials except one in the preceding expression. For example, from the total differential  $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$ , if we impose the assumption that  $y$  remains constant (that is to say  $dy = 0$ ), we obtain  $df = \frac{\partial f}{\partial x} dx$ : this result expresses the fact that the total variation of  $f$  arises exclusively from the variation of  $x$ , the effect of  $y$  being 'frozen' by the assumption of constancy. The partial derivative  $\frac{\partial f}{\partial x}$  measures precisely this isolated effect of the variable  $x$  on the function  $f$ . This reasoning then makes it possible to 'dissect' complex phenomena by progressively isolating the effect of each variable. Without this simplification, it would be extremely difficult to understand how the different variables interact and how each contributes individually to the changes observed. In particular, *ceteris paribus* reasoning makes it possible to:

- **Identify causal relations:** by maintaining other variables constant, we may attribute the changes observed to the single variable which we make vary, thus isolating its direct 'causal contribution'. In addition, studying how a function reacts to the variation of a single variable (while keeping the others fixed) is much simpler than

analysing the simultaneous variations of several variables. This is why partial derivatives are easier to compute and to interpret than 'complete total differentials'.

- **Facilitate empirical tests:** in econometrics, empirical studies often seek to estimate the effect of one variable on another 'all other things being equal'. For example, estimating the effect of the minimum wage on employment (while maintaining other factors such as technology or global economic conditions constant) or the effect of schooling on income (while taking account of differences in individual abilities). Without this assumption, it is impossible to separate the effects of variables which vary simultaneously.

**Careful!** *Ceteris paribus* reasoning is valid only for small variations of the variables: the total differential gives a local linear approximation of the function. For large changes, this approximation departs from the true behaviour of the function. In addition, the assumption that 'all the other variables remain constant' may be unrealistic in practice: economic variables are often strongly interdependent, so that when one variable changes, other variables change as well. For example, an increase in the key interest rate by the central bank simultaneously affects borrowing rates, investment, employment, prices, and many other variables; one cannot suppose that everything else remains constant. This is why, when one seeks to understand the total effect of a policy or of an economic shock, one must often go beyond *ceteris paribus* analysis and study how all the variables evolve in interaction.

## 2.5.6 The implicit function theorem

The **implicit function theorem** is a fundamental tool of analysis which makes it possible to solve equations of the form  $F(x, y) = 0$  by expressing  $y$  as an implicit function of  $x$ , without needing to solve it explicitly. More precisely, the theorem asserts that, under certain conditions, such an equation locally defines  $y$  as a function of  $x$ . Let  $F(x, y)$  be a continuously differentiable function defined in the neighbourhood of a point  $(x_0, y_0)$ . Let us suppose that we have  $F(x_0, y_0) = 0$  (the point  $(x_0, y_0)$  satisfies the equation) and  $\frac{\partial F}{\partial y}(x_0, y_0) \neq 0$  (the partial derivative with respect to  $y$  is non-zero). Then there exists a neighbourhood of  $x_0$  on which we may define an implicit function  $y = f(x)$  such that we have  $f(x_0) = y_0$ ,  $F(x, f(x)) = 0$  for every  $x$  in that neighbourhood, and for which  $f$  is continuously differentiable, with  $\frac{dy}{dx} = f'(x) = -\frac{\frac{\partial F}{\partial x}(x, y)}{\frac{\partial F}{\partial y}(x, y)}$ . More generally, let  $F(x_1, \dots, x_n, y)$  be a continuously differentiable function defined in the neighbourhood of a point  $(x_1^0, \dots, x_n^0, y_0)$ . If we have  $F(x_1^0, \dots, x_n^0, y_0) = 0$  and  $\frac{\partial F}{\partial y} \neq 0$  at that point, then we may locally define  $y = f(x_1, \dots, x_n)$  with  $\frac{\partial y}{\partial x_i} = -\frac{\frac{\partial F}{\partial x_i}}{\frac{\partial F}{\partial y}}$ . Intuitively, the theorem essentially asserts that if we have an equation  $F(x, y) = 0$  and the curve is not 'vertical' (that is to say that  $\frac{\partial F}{\partial y} \neq 0$ ), then we may locally solve this equation in order to express  $y$  as a function of  $x$ . The condition  $\frac{\partial F}{\partial y} \neq 0$  means that  $y$  really varies when one changes the value of  $F$  by acting on  $y$ , which makes it possible to invert this relation locally.

## 2.5.7 Comparative statics and equilibrium analysis

The major importance of the implicit function theorem in economics resides in **comparative statics**, which studies how an endogenous variable changes in reaction to a variation of an exogenous parameter, on the assumption that the system remains in equilibrium. Let us suppose for example that an economic model is described by an equilibrium equation  $F(x, p) = 0$ , where  $x$  is the endogenous variable (for example the equilibrium quantity) and  $p$  is an exogenous parameter (for example a price or an income). The implicit function theorem asserts that, locally, we may express  $x = f(p)$  implicitly, with  $\frac{dx}{dp} = -\frac{\frac{\partial F}{\partial p}}{\frac{\partial F}{\partial x}}$ . This derivative tells us how the endogenous variable reacts to a variation of the exogenous parameter, on the assumption that the system remains in equilibrium. For example, let us consider a simple supply-demand model, where the demand function is given by  $Q_d = a - bp + cY$  (where  $Y$  is income) and the supply function  $Q_s = d + ep$  (where  $d, e > 0$ ). Then the equilibrium output  $F(Q, p, Y) := Q_d - Q_s = a - bp + cY - (d + ep) = 0$ . We may then rewrite this relation in the form  $F(p, Y) := (a - d + cY) - (b + e)p = 0$ . In order to study the effect of a variation of income on the equilibrium price, we therefore use the relation  $\frac{dp}{dY} = -\frac{\frac{\partial F}{\partial Y}}{\frac{\partial F}{\partial p}} = -\frac{c}{-(b+e)} = \frac{c}{b+e} > 0$ : this formula indicates that if income increases (which shifts the demand curve to the right), the equilibrium price increases as well. **Careful!** The implicit function theorem is a local result: it guarantees the existence and differentiability of the implicit function in the neighbourhood of a point, but says nothing about the global existence of such a function over the whole domain. In addition, the condition  $\frac{\partial F}{\partial y} \neq 0$  is crucial: if this partial derivative vanishes, the theorem does not apply, and it may be that one cannot solve the equation for  $y$  as a function of  $x$  (for example if the curve is vertical at that point). Finally, in economic applications, one must verify that the conditions of the theorem are satisfied for comparative statics to be valid.

### 2.5.8 Homogeneity and Euler's theorem

A function  $f(x_1, x_2, \dots, x_n)$  of several variables is said to be **homogeneous of degree  $k$**  (where  $k$  is a real number) if, for every positive real number  $\lambda > 0$  and for all arguments  $x_1, x_2, \dots, x_n$ , we have  $f(\lambda x_1, \lambda x_2, \dots, \lambda x_n) = \lambda^k f(x_1, x_2, \dots, x_n)$ . Intuitively, this means that if we multiply all the arguments by the same factor  $\lambda$ , the function is multiplied by  $\lambda^k$ . For example, the function  $f : (x, y) \mapsto 3x^2 + 2xy + y^2$  is homogeneous of degree 2, since  $f(\lambda x, \lambda y) = 3(\lambda x)^2 + 2(\lambda x)(\lambda y) + (\lambda y)^2 = \lambda^2(3x^2 + 2xy + y^2) = \lambda^2 f(x, y)$ , for all real numbers  $x$  and  $y$ . **Euler's theorem** then establishes a fundamental relation between a homogeneous function, its partial derivatives and its degree of homogeneity. Let  $f(x_1, x_2, \dots, x_n)$  be a function homogeneous of degree  $k$ . Then, for all arguments  $x_1, x_2, \dots, x_n$ , we have  $x_1 \frac{\partial f}{\partial x_1} + x_2 \frac{\partial f}{\partial x_2} + \dots + x_n \frac{\partial f}{\partial x_n} = k f(x_1, x_2, \dots, x_n)$ .

## 2.6 Optimisation of functions of several variables

### 2.6.1 The notion of a neighbourhood of a point

Let  $n$  be a natural number. Let  $\mathbf{a} := (a_1, \dots, a_n)$  be an element of  $\mathbb{R}^n$ . Let  $r$  be a non-negative real number. Then we call the **closed ball** with centre  $\mathbf{a}$  and radius  $r$  the set  $B_F(\mathbf{a}, r) := \{\mathbf{x} \in \mathbb{R}^n; \|\mathbf{x} - \mathbf{a}\| \leq r\}$ , that is to say the set of points of the space  $\mathbb{R}^n$  situated at a distance less than or equal to  $r$  (in the sense of the norm  $\|\cdot\| : \mathbf{a} := (a_1, \dots, a_n) \mapsto \sqrt{a_1^2 + \dots + a_n^2}$ ) from the point  $\mathbf{a}$  (in particular, if  $r$  is zero,  $B_F(\mathbf{a}, r)$  is reduced to the point  $\mathbf{a}$ ). By analogy, we call the **open ball** with centre  $\mathbf{a}$  and radius  $r$  the set  $B_O(\mathbf{a}, r) := \{\mathbf{x} \in \mathbb{R}^n; \|\mathbf{x} - \mathbf{a}\| < r\}$ , that is to say the set of points of the space  $\mathbb{R}^n$  situated at a distance strictly less than  $r$  (in the sense of the usual Euclidean norm) from the point  $\mathbf{a}$  (in particular, if  $r$  is zero,  $B_O(\mathbf{a}, r)$  is the empty set). Any open ball with centre  $\mathbf{a}$  and non-zero radius  $r$  is then called a **neighbourhood** of the point  $\mathbf{a}$ . Consequently, let  $f$  be a real function of  $\mathbb{R}^n$ . We suppose that  $f$  is defined at the point  $\mathbf{a}$ . We then say that  $\mathbf{a}$  is an **interior point** of  $D$  if, and only if, there exists a strictly positive real number  $r$  such that, for every element  $\mathbf{x}$  of the open ball  $B_O(\mathbf{a}, r)$ , the function  $f$  is well defined at the point  $\mathbf{x}$  (this means roughly that  $\mathbf{a}$  is not situated on the boundaries of the domain of definition of the function  $f$ ).

### 2.6.2 Extrema of a real function

Let  $n$  be a natural number. Let  $f$  be a function from  $\mathbb{R}^n$  into  $\mathbb{R}$ , defined on a subset  $D$  of  $\mathbb{R}^n$ . Let  $\mathbf{a} := (x, y)$  be an element of  $D$  (that is to say that  $(x, y)$  is not situated on the boundaries of the domain of definition of the function  $f$ ). Then the function  $f$  presents:

- a **global maximum** if, for every  $\mathbf{x} \in D$ , we have  $f(\mathbf{x}) \leq f(\mathbf{a})$ .
- a **global minimum** if, for every  $\mathbf{x} \in D$ , we have  $f(\mathbf{x}) \geq f(\mathbf{a})$ .
- a **strict global maximum** if, for every  $\mathbf{x} \in D \setminus \{\mathbf{a}\}$ , we have  $f(\mathbf{x}) < f(\mathbf{a})$ .
- a **strict global minimum** if, for every  $\mathbf{x} \in D \setminus \{\mathbf{a}\}$ , we have  $f(\mathbf{x}) > f(\mathbf{a})$ .
- a **local maximum** if there exists  $r > 0$  such that, for every  $\mathbf{x} \in B_O(\mathbf{a}, r) \cap D$ , we have  $f(\mathbf{x}) \leq f(\mathbf{a})$ .
- a **local minimum** if there exists  $r > 0$  such that, for every  $\mathbf{x} \in B_O(\mathbf{a}, r) \cap D$ , we have  $f(\mathbf{x}) \geq f(\mathbf{a})$ .
- a **strict local maximum** if there exists  $r > 0$  such that, for every  $\mathbf{x} \in B_O(\mathbf{a}, r) \cap D \setminus \{\mathbf{a}\}$ ,  $f(\mathbf{x}) < f(\mathbf{a})$ .
- a **strict local minimum** if there exists  $r > 0$  such that, for every  $\mathbf{x} \in B_O(\mathbf{a}, r) \cap D \setminus \{\mathbf{a}\}$ ,  $f(\mathbf{x}) > f(\mathbf{a})$ .

Finally, we say that the function  $f$  presents no extremum at  $\mathbf{a}$  if, for every  $r > 0$ , we have two elements  $\mathbf{x}_1$  and  $\mathbf{x}_2$  belonging to  $B_O(\mathbf{a}, r) \cap D$  such that we have  $f(\mathbf{x}_1) > f(\mathbf{a})$  and  $f(\mathbf{x}_2) < f(\mathbf{a})$ .

### 2.6.3 Quadratic form associated with a matrix

Let  $n$  be a non-zero natural number and  $\mathbf{A}$  a square matrix of order  $n$ . Then we call the **quadratic form** associated with the matrix  $\mathbf{A}$  the function  $q_{\mathbf{A}}$  of  $n$  variables which, to every vector  $\mathbf{x} := (x_1, \dots, x_n)$  of real numbers, associates the square matrix of order 1 (identified with a real number)  $q_{\mathbf{A}}(\mathbf{x}) := \mathbf{x} \mathbf{A} \mathbf{x}^T$ . We then say that the matrix  $\mathbf{A}$  is **positive definite** (resp. **negative definite**) if, and only if, for every vector  $\mathbf{x} := (x_1, \dots, x_n)$  of real numbers, we have  $q_{\mathbf{A}}(\mathbf{x}) > 0$  (resp.  $q_{\mathbf{A}}(\mathbf{x}) < 0$ ). Likewise, we say that the matrix  $\mathbf{A}$  is **positive semi-definite** (resp. **negative semi-definite**) if, and only if, for every vector  $\mathbf{x} := (x_1, \dots, x_n)$  of real numbers, we have  $q_{\mathbf{A}}(\mathbf{x}) \geq 0$  (resp.  $q_{\mathbf{A}}(\mathbf{x}) \leq 0$ ). Moreover, the matrix  $\mathbf{A}$  is positive definite (resp. negative definite, positive semi-definite, negative semi-definite) if, and only if, the eigenvalues of  $\mathbf{A}$  are all strictly positive (resp. strictly negative, non-negative, non-positive). If there exists for the matrix  $\mathbf{A}$  at least one strictly negative eigenvalue and one strictly positive eigenvalue, we shall say that the matrix  $\mathbf{A}$  is **indefinite**, like the quadratic form associated with it (this means that there exist two elements  $\mathbf{x}_1$  and  $\mathbf{x}_2$  of  $\mathbb{R}^n$  satisfying  $q_{\mathbf{A}}(\mathbf{x}_1) < 0$  and  $q_{\mathbf{A}}(\mathbf{x}_2) > 0$ ). For example, we set  $\mathbf{A}$  to be the following matrix:

$$\mathbf{A} := \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 3 & 1 \end{pmatrix}$$

Then the quadratic form associated with the matrix  $\mathbf{A}$  satisfies, for every triple  $\mathbf{x} := (x_1, x_2, x_3)$  of real numbers:

$$q_{\mathbf{A}}(\mathbf{x}) = \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} x_1 + 2x_2 + 3x_3 \\ 3x_1 + 2x_2 + x_3 \\ 2x_1 + 3x_2 + x_3 \end{pmatrix} = x_1^2 + 2x_2^2 + x_3^2 + 5x_1x_2 + 5x_1x_3 + 4x_2x_3$$

#### 2.6.4 Unconstrained optimisation of functions of several variables

Let  $n$  be a natural number at least equal to 2. Let  $f$  be a function from  $\mathbb{R}^n$  into  $\mathbb{R}$  and defined on a subset  $D$  of  $\mathbb{R}^n$ . Let  $(x_1^*, \dots, x_n^*)$  be an interior point of  $D$ . If  $f$  admits a local extremum at  $(x_1^*, \dots, x_n^*)$ , then we necessarily have  $\nabla(f)(x_1^*, \dots, x_n^*) := \left( \frac{\partial f}{\partial x_1}(x_1^*, \dots, x_n^*), \dots, \frac{\partial f}{\partial x_n}(x_1^*, \dots, x_n^*) \right) = (0, \dots, 0)$ : this is the **first-order condition** (for the existence of a local extremum at  $(x_0, y_0)$ ); any point satisfying the first-order condition will then be called a **critical point** of the function  $f$ . Let us suppose that  $(x_1^*, \dots, x_n^*)$  is a critical point of  $f$ . We then compute the Hessian matrix  $\mathbf{H}(x_1^*, \dots, x_n^*)$  of  $f$  evaluated at the point  $(x_1^*, \dots, x_n^*)$ . From then on, (i) if  $\mathbf{H}(x_1^*, \dots, x_n^*)$  is positive definite, then  $(x_1^*, \dots, x_n^*)$  constitutes a local minimum for  $f$ ; (ii) if  $\mathbf{H}(x_1^*, \dots, x_n^*)$  is negative definite, then  $(x_1^*, \dots, x_n^*)$  constitutes a local maximum for  $f$ ; (iii) if  $\mathbf{H}(x_1^*, \dots, x_n^*)$  is indefinite, then  $(x_1^*, \dots, x_n^*)$  constitutes a **saddle point** for  $f$ , that is to say neither a minimum nor a maximum; (iv) if  $\mathbf{H}(x_1^*, \dots, x_n^*)$  is positive semi-definite or negative semi-definite, we cannot conclude as to the nature of the critical point  $(x_1^*, \dots, x_n^*)$  and more thorough analyses will have to be carried out. Moreover, if the function  $f$  is convex or concave, we can go further: when  $f$  is concave, every critical point is a **global** extremum for  $f$ ; conversely, when  $f$  is convex, every critical point is a **global** extremum for  $f$ .

#### 2.6.5 The particular case of two variables

Let  $f$  be a function from  $\mathbb{R}^2$  into  $\mathbb{R}$  and defined on a subset  $D$  of  $\mathbb{R}^2$ . Let  $(x_0, y_0)$  be an interior point of  $D$ . If  $f$  admits a local extremum at  $(x_0, y_0)$ , then we necessarily have  $\nabla(f)(x_0, y_0) := \left( \frac{\partial f}{\partial x}(x_0, y_0), \frac{\partial f}{\partial y}(x_0, y_0) \right) = (0, 0)$ : this is the **first-order condition** (for the existence of a local extremum at  $(x_0, y_0)$ ); any point satisfying the first-order condition will then be called a **critical point** of the function  $f$ . Let us suppose that  $(x_0, y_0)$  is a critical point of  $f$ . We then compute the Hessian matrix of  $f$  evaluated at the point  $(x_0, y_0)$ : this is the **second-order condition**. We therefore have:

$$\mathbf{H}(x_0, y_0) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} \Big|_{(x_0, y_0)} & \frac{\partial^2 f}{\partial x \partial y} \Big|_{(x_0, y_0)} \\ \frac{\partial^2 f}{\partial y \partial x} \Big|_{(x_0, y_0)} & \frac{\partial^2 f}{\partial y^2} \Big|_{(x_0, y_0)} \end{pmatrix} =: \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Then the Hessian of  $f$  at  $(x_0, y_0)$  is worth  $\Delta := ad - bc$ . From then on, (i) if we have  $\Delta > 0$  and  $a > 0$ , then  $f$  admits at  $(x_0, y_0)$  a local minimum; (ii) if we have  $\Delta > 0$  and  $a < 0$ , then  $f$  admits at  $(x_0, y_0)$  a local maximum; (iii) if we have  $\Delta < 0$ , then  $f$  admits at  $(x_0, y_0)$  a saddle point; (iv) otherwise, we cannot conclude and more thorough analyses will have to be carried out. Moreover, if the function  $f$  is convex or concave (that is to say if the Hessian keeps a constant sign on the whole of  $D$ ), then we already know that every critical point will be a local extremum; but we can go further: when  $f$  is concave, every critical point is a **global** extremum for  $f$ ; conversely, when  $f$  is convex, every critical point is a **global** extremum for  $f$ .

#### 2.6.6 Constrained optimisation: the case of functions of two variables

Let  $f: (x_1, x_2) \mapsto f(x_1, x_2)$  and  $g: (x_1, x_2) \mapsto (x_1, x_2)$  be two functions from  $\mathbb{R}^2$  into  $\mathbb{R}$  admitting admitting first and second partial derivatives on an open subset  $\mathcal{D}$  of  $\mathbb{R}^2$ . We seek to determine a local optimum on  $\mathcal{D}$  of the function  $f$  under the constraint  $\mathcal{C} := \{(x_1, x_2) \in \mathcal{D}; g(x_1, x_2) = 0\}$ . We verify that, for every pair  $(x_1, x_2)$  of  $\mathcal{D}$ , the partial derivatives evaluated at the point  $(x_1, x_2)$ , namely  $\frac{\partial g}{\partial x_1} \Big|_{(x_1, x_2)}$  and  $\frac{\partial g}{\partial x_2} \Big|_{(x_1, x_2)}$ , are not simultaneously zero: the constraint  $\mathcal{C}$  is then said to be **non-critical**. We may then apply the **method of Lagrange** in order to search for the local optima of  $f$  on the constraint  $\mathcal{C}$ . We seek to solve the following programme:

$$(P) : \begin{cases} \arg \min_{(x_1, x_2) \in \mathcal{D}} f(x_1, x_2) \\ \text{subject to: } g(x_1, x_2) = 0 \end{cases} \quad \text{or} \quad (P) : \begin{cases} \arg \max_{(x_1, x_2) \in \mathcal{D}} f(x_1, x_2) \\ \text{subject to: } g(x_1, x_2) = 0 \end{cases}$$

We then set  $\mathcal{L}$  to be the function from  $\mathcal{D} \times \mathbb{R}^*$  into  $\mathbb{R}$  defined, for every element  $(x_1, x_2, \lambda)$  of  $\mathcal{D} \times \mathbb{R}^*$ , by the value  $\mathcal{L}(x_1, x_2, \lambda) := f(x_1, x_2) - \lambda g(x_1, x_2)$ : this function is called the **Lagrangian** associated with the programme

(P). If (P) admits a solution  $(x_1^*, x_2^*)$ , then we necessarily satisfy the following **first-order conditions**, namely the three conditions  $\frac{\partial \mathcal{L}}{\partial x_1} \Big|_{(x_1^*, x_2^*, \lambda)} = 0$ ,  $\frac{\partial \mathcal{L}}{\partial x_2} \Big|_{(x_1^*, x_2^*, \lambda)} = 0$  and  $\frac{\partial \mathcal{L}}{\partial \lambda} \Big|_{(x_1^*, x_2^*, \lambda)} = 0$  (this last condition corresponds simply to the constraint  $g(x_1^*, x_2^*) = 0$ ). Solving the system formed by these three equations therefore makes it possible to determine the **critical points** of the Lagrangian, which are potential solutions of the programme (P). It remains to determine whether the point (or points) found do effectively correspond to a minimum or to a maximum for  $f$  under the constraint  $\mathcal{C}$ . For this we shall need to study the **second-order condition**. We then write the **Hessian matrix** of the Lagrangian which is worth, for every pair  $(x_1, x_2)$  of real numbers and every strictly positive real number  $\lambda$ :

$$H(\mathcal{L}(x_1, x_2, \lambda)) := \begin{pmatrix} \frac{\partial^2 \mathcal{L}}{\partial x_1^2} \Big|_{(x_1, x_2, \lambda)} & \frac{\partial^2 \mathcal{L}}{\partial x_1 \partial x_2} \Big|_{(x_1, x_2, \lambda)} & \frac{\partial^2 \mathcal{L}}{\partial x_1 \partial \lambda} \Big|_{(x_1, x_2, \lambda)} \\ \frac{\partial^2 \mathcal{L}}{\partial x_1 \partial x_2} \Big|_{(x_1, x_2, \lambda)} & \frac{\partial^2 \mathcal{L}}{\partial x_2^2} \Big|_{(x_1, x_2, \lambda)} & \frac{\partial^2 \mathcal{L}}{\partial x_2 \partial \lambda} \Big|_{(x_1, x_2, \lambda)} \\ \frac{\partial^2 \mathcal{L}}{\partial x_1 \partial \lambda} \Big|_{(x_1, x_2, \lambda)} & \frac{\partial^2 \mathcal{L}}{\partial x_2 \partial \lambda} \Big|_{(x_1, x_2, \lambda)} & \frac{\partial^2 \mathcal{L}}{\partial \lambda^2} \Big|_{(x_1, x_2, \lambda)} \end{pmatrix}$$

We then compute the determinant of the Hessian at the point  $(x_1^*, x_2^*, \lambda)$ . Three cases are then possible: if we verify  $\det(H(x_1^*, x_2^*, \lambda)) > 0$ , then  $(x_1^*, x_2^*)$  constitutes a local minimum for  $f$  under the constraint  $\mathcal{C}$ ; if we satisfy the condition  $\det(H(x_1^*, x_2^*, \lambda)) < 0$ , then  $(x_1^*, x_2^*)$  constitutes a local maximum for  $f$  under the constraint  $\mathcal{C}$ ; if we verify the equality  $\det(H(x_1^*, x_2^*, \lambda)) = 0$ , then we cannot conclude. Moreover, if  $\mathcal{L}$  is concave (which is in particular the case if  $f$  is concave and if  $g$  is affine), then  $(x_1^*, x_2^*)$  is even a global maximum under the constraint  $\mathcal{C}$ ; likewise, if  $\mathcal{L}$  is convex (which is in particular the case if  $f$  is concave and if  $g$  is affine), then  $(x_1^*, x_2^*)$  is even a global minimum under the constraint  $\mathcal{C}$ . We recall that a function  $f : (x_1, x_2) \mapsto f(x_1, x_2)$  admitting first and second partial derivatives is concave (resp. convex) when, on the one hand the function  $\left(\frac{\partial^2 f}{\partial x_1^2}\right)\left(\frac{\partial^2 f}{\partial x_2^2}\right) - \left(\frac{\partial^2 f}{\partial x_1 \partial x_2}\right)^2$  is strictly positive, and on the other hand the function  $\frac{\partial^2 f}{\partial x_1^2}$  is negative (resp. positive). **Careful!** This property of concave and convex functions does not generalise to functions of three or more variables (in particular, it does not apply to the Lagrangian  $\mathcal{L}$  which constitutes a function of three variables).

### 2.6.7 Optimisation under an equality constraint

Let  $n$  be a non-zero natural number. Let  $f : (x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$  and  $g : (x_1, \dots, x_n) \mapsto g(x_1, \dots, x_n)$  be two functions from  $\mathbb{R}^n$  into  $\mathbb{R}$  admitting first and second partial derivatives on an open subset  $\mathcal{D}$  of  $\mathbb{R}^n$ . We seek to determine a local maximum or minimum on  $\mathcal{D}$  of the function  $f$  under the constraint  $\mathcal{C} := \{(x_1, \dots, x_n) \in \mathcal{D}; g(x_1, \dots, x_n) = 0\}$ . We verify that, for every tuple  $(x_1, \dots, x_n)$  of  $\mathcal{D}$ , the partial derivatives evaluated at the point  $(x_1, \dots, x_n)$ , namely the  $\frac{\partial g}{\partial x_i} \Big|_{(x_1, \dots, x_n)}$  for every integer  $i$  between 1 and  $n$ , are not simultaneously zero: the constraint  $\mathcal{C}$  is then said to be **non-critical**. We may then apply the **method of Lagrange** in order to search for the local optima of  $f$  on the constraint  $\mathcal{C}$ . We seek to solve the following programme:

$$(P) : \begin{cases} \arg \min_{(x_1, \dots, x_n) \in \mathcal{D}} f(x_1, \dots, x_n) \\ \text{subject to: } g(x_1, \dots, x_n) = 0 \end{cases} \quad \text{or} \quad (P) : \begin{cases} \arg \max_{(x_1, \dots, x_n) \in \mathcal{D}} f(x_1, \dots, x_n) \\ \text{subject to: } g(x_1, \dots, x_n) = 0 \end{cases}$$

We then set  $\mathcal{L}$  to be the function from  $\mathcal{D} \times \mathbb{R}^*$  into  $\mathbb{R}$  defined, for every element  $(x_1, \dots, x_n, \lambda)$  of  $\mathcal{D} \times \mathbb{R}^*$ , by the value  $\mathcal{L}(x_1, \dots, x_n, \lambda) := f(x_1, \dots, x_n) - \lambda g(x_1, \dots, x_n)$ : this function is called the **Lagrangian** associated with the programme (P). If (P) admits a solution  $(x_1^*, \dots, x_n^*, \lambda^*)$ , then we necessarily satisfy the following **first-order conditions**, namely the  $n+1$  conditions  $\frac{\partial \mathcal{L}}{\partial x_i} \Big|_{(x_1^*, \dots, x_n^*, \lambda^*)} = 0$ , for every integer  $i$  between 1 and  $n$ , and  $\frac{\partial \mathcal{L}}{\partial \lambda} \Big|_{(x_1^*, \dots, x_n^*, \lambda^*)} = 0$  (this last condition corresponds quite simply to the constraint  $g(x_1^*, \dots, x_n^*) = 0$ ). Solving the system formed by these  $n+1$  equations therefore makes it possible to determine the **critical points** of the Lagrangian, which are potential solutions of the programme (P). It remains to determine whether the point (or points) found do effectively correspond to a minimum or to a maximum for  $f$  under the constraint  $\mathcal{C}$ . In particular, if  $\mathcal{L}$  is concave (which is in particular the case if  $f$  is concave and if  $g$  is affine or, more generally, if  $f$  is concave and if  $\lambda^* g$  is convex), then  $(x_1^*, \dots, x_n^*)$  is a global maximum under the constraint  $\mathcal{C}$ ; likewise, if  $\mathcal{L}$  is convex (which is in particular the case if  $f$  is concave and  $g$  affine or, more generally, if  $f$  is concave and if  $\lambda^* g$  is convex), then  $(x_1^*, \dots, x_n^*)$  is a global minimum under the constraint  $\mathcal{C}$ .

### 2.6.8 Optimisation under equality constraints

Let  $n$  and  $k$  be two non-zero natural numbers. Let  $f : (x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$  and  $g_1 : (x_1, \dots, x_n) \mapsto g_1(x_1, \dots, x_n), \dots, g_k : (x_1, \dots, x_n) \mapsto g_k(x_1, \dots, x_n)$  be functions from  $\mathbb{R}^n$  into  $\mathbb{R}$  admitting first and second partial derivatives on an open subset  $\mathcal{D}$  of  $\mathbb{R}^n$ . We seek to determine a local maximum or minimum on  $\mathcal{D}$  of the function  $f$  under the

constraints  $\mathcal{C} := \{(x_1, \dots, x_n) \in \mathcal{D}; \forall \ell \in [1, k] \ g_\ell(x_1, \dots, x_n) = 0\}$ . For the constraint  $\mathcal{C}$  to be non-critical, it suffices that the functions  $g_1, \dots, g_k$  be affine and non-zero (other criteria of constraint qualification exist, but we shall not dwell any further on this point). We may then apply the **method of Lagrange** in order to search for the local optima of  $f$  on the constraint  $\mathcal{C}$ . We seek to solve the following programme:

$$(P) : \begin{cases} \arg \min_{(x_1, \dots, x_n) \in \mathcal{D}} f(x_1, \dots, x_n) & \text{or} & \arg \max_{(x_1, x_2) \in \mathcal{D}} f(x_1, \dots, x_n) \\ \text{subject to: } \forall \ell \in [1, k] & g_\ell(x_1, \dots, x_n) = 0 \end{cases}$$

We set  $\mathcal{L}$  to be the function from  $\mathcal{E} := \mathcal{D} \times (\mathbb{R}^k)^*$  into  $\mathbb{R}$  defined, for every element  $(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k)$  of  $\mathcal{E}$ , by the value  $\mathcal{L}(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k) := f(x_1, \dots, x_n) - \lambda_1 g_1(x_1, \dots, x_n) - \dots - \lambda_k g_k(x_1, \dots, x_n)$ : this function is called the **Lagrangian** associated with the programme (P). Let us set:

$$\mathbf{g} : (x_1, \dots, x_n) \mapsto \begin{pmatrix} g_1(x_1, \dots, x_n) \\ \vdots \\ g_k(x_1, \dots, x_n) \end{pmatrix} \quad \text{and} \quad \mathbf{x} := \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad \boldsymbol{\lambda} := \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_k \end{pmatrix}$$

Then the Lagrangian satisfies, rewriting its arguments in matrix form, for every vector  $\mathbf{x}$  of  $\mathcal{D}$  and every vector  $\boldsymbol{\lambda}$  of  $(\mathbb{R}^k)^*$ ,  $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) := f(\mathbf{x}) - \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x})$ . If (P) admits a solution  $(x_1^*, \dots, x_n^*, \lambda_1^*, \dots, \lambda_k^*)$ , then we satisfy the following **first-order conditions**, namely the  $n + k$  conditions  $\frac{\partial \mathcal{L}}{\partial x_i} \Big|_{(x_1^*, \dots, x_n^*, \lambda_1^*, \dots, \lambda_k^*)} = 0$ , for every integer  $i$  of  $[1, n]$ , and  $\frac{\partial \mathcal{L}}{\partial \lambda_\ell} \Big|_{(x_1^*, \dots, x_n^*, \lambda_1^*, \dots, \lambda_k^*)} = 0$ , for every integer  $\ell$  of  $[1, k]$  (this last condition corresponds quite simply to the constraint  $g_\ell(x_1^*, \dots, x_n^*) = 0$ ). The whole set of these first-order conditions may be summarised in matrix form, denoting  $\mathbf{x}^* := (x_1^*, \dots, x_n^*)^T$  and  $\boldsymbol{\lambda}^* := (\lambda_1^*, \dots, \lambda_k^*)^T$ , in the form  $\nabla_{x_1, \dots, x_n, \lambda_1, \dots, \lambda_k} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = (0, \dots, 0)$ , that is to say  $\nabla_{x_1, \dots, x_n} f(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \lambda_1 \nabla_{x_1, \dots, x_n} g_1(\mathbf{x}^*) + \dots + \lambda_k \nabla_{x_1, \dots, x_n} g_k(\mathbf{x}^*)$  on the one hand and, for every integer  $\ell$  of  $[1, k]$ ,  $g_\ell(x_1^*, \dots, x_n^*) = 0$  on the other hand. Solving the system formed by these  $n + k$  equations therefore makes it possible to determine the **critical points** of the Lagrangian, which are potential solutions of the programme (P). It remains to determine whether the point (or points) found do effectively correspond to a minimum or to a maximum for  $f$  under the constraint  $\mathcal{C}$ . In particular, if  $\mathcal{L}$  is concave (which is in particular the case if  $f$  is concave and if  $g_1, \dots, g_k$  are affine or, more generally, if  $f$  is concave and if  $\lambda_1 g_1, \dots, \lambda_k g_k$  are convex), then  $(x_1^*, \dots, x_n^*)$  is a global maximum under the constraint  $\mathcal{C}$ ; likewise, if  $\mathcal{L}$  is convex (which is in particular the case if  $f$  is concave and if  $g_1, \dots, g_k$  are affine or, more generally, if  $f$  is concave and if  $\lambda_1 g_1, \dots, \lambda_k g_k$  are convex), then  $(x_1^*, \dots, x_n^*)$  is a global minimum under the constraint  $\mathcal{C}$ .

### 2.6.9 Constrained optimisation: the general case

Let  $n$  and  $k$  be two non-zero natural numbers. Let  $f : (x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$  and  $g_1 : (x_1, \dots, x_n) \mapsto g_1(x_1, \dots, x_n), \dots, g_k : (x_1, \dots, x_n) \mapsto g_k(x_1, \dots, x_n)$  be functions from  $\mathbb{R}^n$  into  $\mathbb{R}$  admitting admitting first and second partial derivatives on an open subset  $\mathcal{D}$  of  $\mathbb{R}^n$ . We seek to determine a local maximum or minimum on  $\mathcal{D}$  of the function  $f$  under the constraints  $\mathcal{C} := \{(x_1, \dots, x_n) \in \mathcal{D}; \forall \ell \in [1, k] \ g_\ell(x_1, \dots, x_n) \leq 0\}$ . For the constraint  $\mathcal{C}$  to be non-critical, it suffices that the functions  $g_1, \dots, g_k$  be affine and non-zero (other criteria of constraint qualification exist, but we shall not dwell any further on this point). We may then apply the **Kuhn-Karush-Tucker method** in order to search for the local optima of  $f$  on the constraint  $\mathcal{C}$ . We seek to solve the following programme:

$$(P) : \begin{cases} \arg \min_{(x_1, \dots, x_n) \in \mathcal{D}} f(x_1, \dots, x_n) & \text{or} & \arg \max_{(x_1, x_2) \in \mathcal{D}} f(x_1, \dots, x_n) \\ \text{subject to: } \forall \ell \in [1, k] & g_\ell(x_1, \dots, x_n) \leq 0 \end{cases}$$

We set  $\mathcal{L}$  to be the function from  $\mathcal{E} := \mathcal{D} \times (\mathbb{R}^k)^*$  into  $\mathbb{R}$  defined, for every element  $(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k)$  of  $\mathcal{E}$ , by the value  $\mathcal{L}(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k) := f(x_1, \dots, x_n) - \lambda_1 g_1(x_1, \dots, x_n) - \dots - \lambda_k g_k(x_1, \dots, x_n)$ : this function is called the **Lagrangian** associated with the programme (P). If (P) admits a solution  $(x_1^*, \dots, x_n^*, \lambda_1^*, \dots, \lambda_k^*)$ , then we satisfy the following **first-order conditions**, namely the following  $n + 3k$  conditions:

- $\forall i \in [1, n] \quad \frac{\partial \mathcal{L}}{\partial x_i} \Big|_{(x_1^*, \dots, x_n^*, \lambda_1^*, \dots, \lambda_k^*)} = 0$
- $\forall \ell \in [1, k] \quad \frac{\partial \mathcal{L}}{\partial \lambda_\ell} \Big|_{(x_1^*, \dots, x_n^*, \lambda_1^*, \dots, \lambda_k^*)} \geq 0$
- $\forall \ell \in [1, k] \quad (\lambda_\ell^* \geq 0 \quad \text{and} \quad (\lambda_\ell^* = 0 \quad \text{or} \quad g_\ell(x_1^*, \dots, x_n^*) = 0))$

Solving the system formed by these  $n + 3k$  equations therefore makes it possible to determine the **critical points** of the Lagrangian, which are potential solutions of the programme (P). It remains to determine whether the point (or points) found do effectively correspond to a minimum or to a maximum for  $f$  under the constraint  $\mathcal{C}$ . In particular, if  $\mathcal{L}$  is concave (which is in particular the case if  $f$  is concave and if  $g_1, \dots, g_k$  are affine or, more generally, if  $f$  is concave and if  $g_1, \dots, g_k$  are convex), then  $(x_1^*, \dots, x_n^*)$  is a global maximum under the constraint  $\mathcal{C}$ ; likewise, if  $\mathcal{L}$  is convex (which is in particular the case if  $f$  is concave and if  $g_1, \dots, g_k$  are affine or, more generally, if  $f$  is concave and if  $g_1, \dots, g_k$  are convex), then  $(x_1^*, \dots, x_n^*)$  is a global minimum under the constraint  $\mathcal{C}$ .

### 3 A review of probability and statistics

#### 3.1 Elements of descriptive statistics

##### 3.1.1 Quantitative variables and qualitative variables

There exist two types of variable that one may be confronted with in statistics: quantitative variables and qualitative variables. Let us recall that a variable is a characteristic common to the whole set of individuals of a study (the value of this characteristic varies between individuals, whence its name of ‘variable’). Among them:

- **Quantitative variables** (or numerical variables) are quantifiable characteristics whose values are numbers, to the exclusion of numbers which in fact correspond to codes. Numerical variables may be continuous or discrete: a variable is said to be *continuous* if it takes an infinite number of possible real values within a given interval (for example, height); a variable is said to be *discrete* when it can take only a finite or countable number of possible values within a given interval (for example, the mark obtained by a gymnast at the Olympic Games).
- **Qualitative variables** (or categorical variables) refer to a characteristic which is not quantifiable. A categorical variable may be nominal or ordinal. A *nominal* variable describes a name, a label or a category with no natural order (for example, sex). An *ordinal* variable, by contrast, is a variable whose values are defined by an order relation between the possible categories (excellent, very good, good, fairly good, satisfactory, poor).

**Careful!** It is important to note that although categorical variables are not quantifiable, they may appear in the form of numbers in a data set (1 for excellent, 2 for very good, and so on). The correspondence between these numbers and the corresponding categories is established in the course of the coding of the data. In order to identify the types of variables correctly, one must therefore make sure to have the metadata (the data about the data) which must include the sets of codes used for each categorical variable.

##### 3.1.2 Parameters of position

Let  $X$  be a quantitative variable,  $n$  a non-zero natural number, and  $\mathbf{X} := (X_1, \dots, X_n)$  measurements of the quantitative variable  $X$  taken on  $n$  different individuals (or  $n$  different periods), thus forming a *statistical series*. We first of all define several parameters of position:

- **The mode.** A measure of central tendency, the mode is the value most frequently found in the sample  $(X_1, \dots, X_n)$ . It is therefore the value associated with the greatest frequency. The mode is therefore not necessarily unique.
- **The empirical mean.** A measure of central tendency, the mean is computed by dividing the sum of the values by the number of observations. For our sample, the empirical mean is therefore  $\bar{X} := \frac{X_1 + \dots + X_n}{n}$ . The empirical mean presents the advantage of being sensitive to sampling fluctuations. By contrast, it is particularly sensitive to extreme values. Finally, the mean is a poor summary when the distribution is strongly asymmetric.
- **The median.** A measure of central tendency, the median is the value for which 50 % of the values of  $(X_1, \dots, X_n)$  are below it and 50 % of the values of  $(X_1, \dots, X_n)$  are above it. If there is an odd number of data points, then we shall find an integer  $j$  between 1 and  $n$  such that half the values are less than or equal to  $X_j$  and the other half greater than or equal to  $X_j$ . By contrast, if  $n$  is even, we shall rank the values in increasing order: the median will then be defined as the mean of the  $(n/2)$ -th value and the  $(1 + n/2)$ -th value of the  $(X_1, \dots, X_n)$  ranked in increasing order. The median presents the advantage of being little sensitive to extreme values. However, it does not lend itself to mathematical computations.
- **Quantiles.** Quantiles are likewise measures of position. For every real number  $\alpha$  between 0 and 1, the quantile of level  $100\alpha$  % is the value such that  $100\alpha$  % of the values of  $(X_1, \dots, X_n)$  are below it and  $100(1 - \alpha)$  % of the values

are above it. In particular, the median is the quantile of level 50 %. The quantile of level 25 % is known by the name of the first quartile, while the quantile of level 75 % is known by the name of the third quartile. We likewise define the deciles: for every integer  $i$  between 1 and 10, the  $i$ -th decile is the quantile of level  $10i$  %.

### 3.1.3 Parameters of dispersion

The parameters of position of a statistical series are generally complemented by the mention of several parameters of dispersion, among which:

- **The range.** The first parameter of dispersion, the range of a statistical series is the gap between the greatest value (maximum) and the smallest value (minimum) taken within the sample  $(X_1, \dots, X_n)$ . Very easy to compute and independent of the value of position, the range nevertheless suffers from a great sensitivity to extreme values, while it takes no account of the values lying between the minimum and the maximum of a statistical series.
- **The interquartile range.** A measure of dispersion of a statistical series, the interquartile range (IQR) is the difference between the third and the first quartile. Little sensitive to extreme values, it moreover presents the advantage of being independent of the parameters of position such as the mean.
- **The interdecile ratio.** A measure of dispersion of a statistical series, the interdecile ratio is the ratio of the ninth decile to the first. The interdecile ratio is therefore little sensitive to extreme values and is notably used in the social sciences in order to measure inequalities within a population (for example: by how many times are the richest 10 % richer than the poorest 10 %?).
- **The variance.** The main measure of dispersion along with the standard deviation, the variance is the mean of the squared deviations from the mean. We indeed define the empirical variance as the value  $\widehat{\sigma^2} = \frac{(X_1 - \bar{X})^2 + \dots + (X_n - \bar{X})^2}{n-1}$ : the division by  $n - 1$  rather than  $n$  is explained by the wish to obtain an unbiased estimator of the variance of the general population (which for its part is divided by  $n$ ). Always positive, the variance thus gives an account of the distance of each value from the mean. The variance therefore depends on the value of position and presents the disadvantage of not being in the same unit as the value studied (if the  $X_i$  are in euros, then the variance is in euros squared, which is hard to interpret).
- **The standard deviation.** The main measure of dispersion, the empirical standard deviation is the square root of the empirical variance. Always positive, the standard deviation therefore measures the dispersion around the mean. Unlike the variance, the unit of the empirical standard deviation is the same as that of the data. The standard deviation therefore makes it possible to have an idea of the distribution even if, like the variance, it remains in a certain manner sensitive to extreme values (since the mean itself is).
- **The coefficient of variation.** A combination of a parameter of position and a parameter of dispersion, the coefficient of variation is the ratio of the standard deviation to the mean and is generally expressed as a percentage. We then speak of a measure of relative dispersion. The coefficient of variation is generally employed in order to compare two samples whose scales of measurement are not comparable.

#### Exercise



**Measuring a phenomenon and rates of change.** Below, we transcribe the evolution of the percentage of unemployed people in the labour force (15–24 years), for three countries, according to OECD data:

| Year    | Germany | France | United Kingdom |
|---------|---------|--------|----------------|
| 2017 T1 | 6.7     | 22.5   | 12.3           |
| 2017 T2 | 6.6     | 23.0   | 12.1           |
| 2017 T3 | 6.5     | 22.0   | $a$            |
| 2017 T4 | 6.6     | 21.3   | 12.1           |
| 2018 T1 | 6.5     | 21.3   | $b$            |
| 2018 T2 | 6.2     | 20.1   | 11.9           |
| 2018 T3 | 6.1     | 21.4   | 11.8           |
| 2018 T4 | 6.0     | 19.5   | $c$            |

- (i) What is the population studied and the characteristic? Are the measurements expressed in level or in variation?
- (ii) Compute the levels, quarterly variations in the form of multiplier coefficients, and quarterly variations in the form of rates of change for Germany.



- (iii) Transform the quarterly variations into annual variations.
- (iv) Considering an index of 100 in the first quarter of 2017, compute the index values for France. Interpret the index obtained for the row 2018 Q1.
- (v) Complete the United Kingdom column knowing that the 2018 Q4 index is worth 89.7 and that the unemployment rate in the United Kingdom fell by 0.88 % between the first quarter of 2017 and the first quarter of 2018 and by 8 % between the second and the third quarter of 2017.

**Solution.** (i) *What is the population studied and the characteristic? Are the measurements expressed in level or in variation?* The population designates the whole set of individuals on which a statistical study bears. Here, three populations are studied: the German, French and English labour force. The characteristic studied, that is to say the property studied for each population, is the unemployment rate, expressed as a percentage. Inasmuch as this percentage is the corresponding value of the unemployment rate for the quarter of study and not the evolution of the unemployment rate between the quarter of study and the preceding one, the measurements in the table are expressed in level.

- (ii) *Compute the levels, quarterly variations in the form of multiplier coefficients, and quarterly variations in the form of rates of change for Germany.* We take up again the unemployment rates for the German population:

| Year    | Unemployment rate | Multiplier coefficient   | Rate of change                        |
|---------|-------------------|--------------------------|---------------------------------------|
| 2017 T1 | 6.7               | —                        | —                                     |
| 2017 T2 | 6.6               | $6.6/6.7 \approx 0.9851$ | $\frac{6.6-6.7}{6.7} \approx -1.49\%$ |
| 2017 T3 | 6.5               | $6.5/6.6 \approx 0.9848$ | $\frac{6.5-6.6}{6.6} \approx -1.52\%$ |
| 2017 T4 | 6.6               | $6.6/6.5 \approx 1.0153$ | $\frac{6.6-6.5}{6.5} \approx +1.53\%$ |
| 2018 T1 | 6.5               | $6.5/6.6 \approx 0.9848$ | $\frac{6.5-6.6}{6.6} \approx -1.52\%$ |
| 2018 T2 | 6.2               | $6.2/6.5 \approx 0.9538$ | $\frac{6.2-6.5}{6.5} \approx -4.62\%$ |
| 2018 T3 | 6.1               | $6.1/6.2 \approx 0.9839$ | $\frac{6.1-6.2}{6.2} \approx -1.51\%$ |
| 2018 T4 | 6.0               | $6.0/6.1 \approx 0.9836$ | $\frac{6.0-6.1}{6.1} \approx -1.64\%$ |

This means that the unemployment rate in Germany in the fourth quarter of 2017 is 1.0153 times higher than that of the preceding quarter; otherwise put, the unemployment rate increased by 1.53 % over this period (this represents a rise of 0.1 percentage point or of 10 basis points). Likewise, the unemployment rate in Germany in the third quarter of 2018 is worth 0.9839 times that of the preceding quarter; otherwise put, the unemployment rate decreased by 1.51 % over this period (this represents a rise of 0.1 percentage point or of 10 basis points).

- (iii) *Transform the quarterly variations into annual variations.* We take up the preceding table again for annual variations:

| Year    | Unemployment rate | Multiplier coefficient   | Rate of change                        |
|---------|-------------------|--------------------------|---------------------------------------|
| 2017 T1 | 6.7               | —                        | —                                     |
| 2017 T2 | 6.6               | —                        | —                                     |
| 2017 T3 | 6.5               | —                        | —                                     |
| 2017 T4 | 6.6               | —                        | —                                     |
| 2018 T1 | 6.5               | $6.5/6.7 \approx 0.9701$ | $\frac{6.5-6.7}{6.7} \approx -2.99\%$ |
| 2018 T2 | 6.2               | $6.2/6.5 \approx 0.9538$ | $\frac{6.2-6.5}{6.5} \approx -4.62\%$ |
| 2018 T3 | 6.1               | $6.1/6.6 \approx 0.9242$ | $\frac{6.1-6.6}{6.6} \approx -7.58\%$ |
| 2018 T4 | 6.0               | $6.0/6.6 \approx 0.9091$ | $\frac{6.0-6.6}{6.6} \approx -8.09\%$ |

This means that the unemployment rate in Germany in the third quarter of 2018 is worth 0.9242 times the rate one year earlier; otherwise put, the unemployment rate decreased by 7.58 % over one year (this represents a fall of 0.5 percentage point or of 50 basis points). More generally, transformations into indices make it possible to compare the evolution of data over time more easily by normalising a reference period to a value of 100: they facilitate comparison between different series, even if their absolute values are very different, and make it possible to visualise relative variations and to follow trends of evolution with greater clarity.

- (iv) *Considering an index of 100 in the first quarter of 2017, compute the index values for France. Interpret the index obtained for the row 2018 Q1.* We consider an index of 100 in the first quarter of 2017. The evolution of this index is then the following:

| Year    | Unemployment rate | Multiplier coefficient   | Index                         |
|---------|-------------------|--------------------------|-------------------------------|
| 2017 T1 | 6.7               | —                        | 100                           |
| 2017 T2 | 6.6               | $6.6/6.7 \approx 0.9851$ | $100.0 \times 0.9851 = 98.51$ |
| 2017 T3 | 6.5               | $6.5/6.6 \approx 0.9848$ | $98.51 \times 0.9848 = 97.01$ |
| 2017 T4 | 6.6               | $6.6/6.5 \approx 1.0153$ | $97.01 \times 1.0153 = 98.50$ |
| 2018 T1 | 6.5               | $6.5/6.6 \approx 0.9848$ | $98.50 \times 0.9848 = 97.00$ |
| 2018 T2 | 6.2               | $6.2/6.5 \approx 0.9538$ | $97.00 \times 0.9538 = 92.52$ |
| 2018 T3 | 6.1               | $6.1/6.2 \approx 0.9839$ | $92.52 \times 0.9839 = 91.03$ |
| 2018 T4 | 6.0               | $6.0/6.1 \approx 0.9836$ | $91.03 \times 0.9836 = 89.54$ |

In the first quarter of 2018, the unemployment rate in the German labour force presents an index of 97 (base 100 in the first quarter of 2017): this means that the unemployment rate in the first quarter of 2018 is 3 % lower than in the first quarter of 2017.

- (v) *Complete the United Kingdom column knowing that the 2018 Q4 index is worth 89.7 and that the unemployment rate in the United Kingdom fell by 0.88 % between the first quarter of 2017 and the first quarter of 2018 and by 8 % between the second and the third quarter of 2017.* In the second quarter of 2017, the unemployment rate in the United Kingdom was 12.1 %: the fall of 8 % in this rate over one quarter brings it to  $(1 - 8\%) \times 12.1\% = 0.92 \times 12.1\% = 11.1\%$  in the third quarter of 2017 (the value of *a*). Likewise, in the first quarter of 2017, the unemployment rate in the United Kingdom was 12.3 %: the fall of 0.88 % in this rate over one year brings it to  $(1 - 0.88\%) \times 12.1\% = 0.9912 \times 12.3\% = 12.2\%$  in the first quarter of 2018 (the value of *b*). Finally, considering an index of 100 for the unemployment rate in the first quarter of 2017 (which is worth 12.3 %), and knowing that this index is worth 89.7 in the fourth quarter of 2014, then the value of the English unemployment rate at that latter date is  $\frac{89.7}{100} \times 12.3\% \approx 11.0\%$  (the value of *c*). □

### 3.1.4 Standard representations in descriptive statistics

The objective of graphs is to represent a statistical series in a way that is *legible* (the data represented must be able to be read), *faithful* (the reality of the data must not be distorted by the production of the graph) and *self-sufficient* (all the information must be placed in the legend, including the reference set).

- **The case of a qualitative variable.** First of all, the **bar chart** makes it possible to compare the different modalities of the qualitative variable considered: the length of the bar is proportional to the frequencies or to the relative frequency of each modality. When moreover the modalities are ordered by decreasing frequency, we obtain a **Pareto chart**. Consequently, the **sector diagram**, circular or semi-circular (or 'pie chart', 'half-pie chart') makes it possible to compare the part to the whole: the area of the sector is proportional to the frequency or to the relative frequency. These types of representation require one to have a good mental perception of a percentage, either in the domain of the lengths of the bars, or in that of the areas of the sectors.
- **The case of a continuous quantitative variable.** The **histogram** is constituted by contiguous rectangles having as their base each of the classes and an area proportional to the frequency or to the relative frequency of the corresponding class. If the widths are all equal, the height of the rectangle is proportional to the frequency or to the relative frequency; if the widths are unequal, the height is proportional to the density of the class. We may likewise represent the **frequency polygon** by joining the midpoints of the upper bases of the bands of the histogram. We may finally represent the **curve of increasing cumulative frequencies** which is akin to an empirical cumulative distribution function for the quantitative variable considered. Likewise, the **curve of decreasing cumulative frequencies** which is akin to an empirical survival function for a quantitative variable.
- **The case of a discrete quantitative variable.** The **stick diagram** makes it possible easily to compare frequencies and relative frequencies with one another. **Careful!** Joining the tops of the stick diagram in order to constitute what is called the **frequency polygon** makes no sense, since the points of the segments other than the endpoints have no significance. We may likewise a **sector diagram** if we wish to compare the part to the whole: the semi-circular sector diagram must then be preferred in order to respect an order structure to be brought out in the modalities. It is likewise possible to represent a curve of cumulative frequencies for a discrete quantitative variable; in which case, the curve takes the form of a staircase, like the cumulative distribution function or the survival function of a discrete distribution.
- A **box plot** for a quantitative variable *X* of which we have *n* observations  $X_1, \dots, X_n$  (where *n* is a non-zero natural number) may be represented horizontally or vertically. The central value of the graph is the median. In addition, the edges of the rectangle are the quartiles (for the lower edge, the first quartile Q1; for the upper edge, the third quartile Q3). Finally, the extremities of the whiskers are generally determined using the interquartile

range IQR. Otherwise put, the lower bound is  $\max(\min(X_1, \dots, X_n), Q1 - 1.5(Q3 - Q1))$  and the upper bound is  $\min(\max(X_1, \dots, X_n), Q3 + 1.5(Q3 - Q1))$ . The values outside the whiskers are represented by points and may indicate outliers (at least values to be examined in detail). Sometimes, a cross likewise figures on the box plot: in which case, it denotes the mean of the series  $(X_1, \dots, X_n)$ . *Box plot* graphs are generally used to visualise the symmetry, the dispersion and the centrality of distributions; they are likewise used to compare variables based on similar scales and to compare the values of the observations of groups of individuals on the same variable.

## 3.2 Probability measures and random variables

### 3.2.1 Random experiment, event, sample space

A **random experiment** is an action or a procedure which produces one of the possible results among a set of possible **outcomes**, this result not being known in advance. The set of possible outcomes constitutes the **sample space**, usually denoted  $\Omega$ , and any subset of  $\Omega$  constitutes an **event**, that is to say a set of one or more results. The set of possible events answers to a structure which is called a  $\sigma$ -**algebra** and which is usually denoted  $\mathcal{A}$ ; the details of the construction of  $\sigma$ -algebras will not, however, concern us in this course. Each of the events of the sample space is then associated with a probability (chance or risk) of occurrence, between 0 and 1: we thus define a **probability measure**  $\mathbb{P}$  for each of the possible events. The set  $(\Omega, \mathcal{A}, \mathbb{P})$  then constitutes a structure known by the name of a **probability space**. Let us note moreover that the empty set is likewise considered an event (no result), called the **impossible event**, while  $\Omega$  is itself an event (obtaining heads or obtaining tails), called the **certain event**; we then have  $\mathbb{P}(\emptyset) = 0$  and  $\mathbb{P}(\Omega) = 1$ . For example, tossing a fair coin and noting its result is a random experiment whose sample space  $\Omega$  is constituted by the two events 'Obtain heads' and 'Obtain tails'; a probability distribution is given by  $\mathbb{P}(\text{Obtain heads}) = \mathbb{P}(\text{Obtain tails}) = \frac{1}{2}$  (since the coin is fair).

### 3.2.2 Random variables

There exists a technical distinction in probability theory between a probability distribution and a **random variable**: in the case of a random variable, each distinct possible event is associated with a real number. For example, if we toss a fair coin and we associate for example 0 with the result 'Obtain heads' and 1 with the event 'Obtain tails', then the corresponding random result  $X$  constitutes a real random variable. Otherwise put, a real random variable is a function  $X$  from  $\Omega$  into  $\mathbb{R}$ ; the set of values taken by it from then on constitutes the **support** of  $X$ , which we denote  $X(\Omega)$ . For example, if we take up the preceding example of the tossing of a fair coin and we link to this random experiment the random variable  $X$  taking the value 0 if we obtain heads and the value 1 if we obtain tails, we find  $X(\Omega) = \{0, 1\}$ . Among real random variables, we distinguish essentially two types:

- **Discrete random variables**, whose set of possible results is finite or countably infinite. For example, if we denote by  $X$  the random variable associated with the number of attempts necessary before obtaining one's driving licence, then we have  $X(\Omega) = \mathbb{N}^*$ , a countably infinite set.
- **Continuous random variables** (or variables with a density), whose set of possible results is uncountably infinite, such as an interval of real numbers. For example, if we denote by  $X$  the random variable associated with the distance separating any given star from the Sun, then we have  $X(\Omega) = \mathbb{R}_+$ , an uncountably infinite set.

Let us note that in the framework of continuous random variables, it is very difficult if not impossible to attribute a non-numerical value to each distinct result; this is why the probability measure is generally conflated with the random variable in this particular case. Careful, however: there exist random variables which are neither discrete nor continuous, but we shall hardly need them in this course.

### 3.2.3 The distribution of a random variable

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space and  $X$  a real random variable defined on that space. Then the **distribution** of  $X$  relates the elements of the support of  $X$  to the probability measure  $\mathbb{P}$ . In particular, if  $X$  is a discrete random variable, the distribution of  $X$  is the datum of the set  $\{\mathbb{P}([X = x]); x \in X(\Omega)\}$ : to each distinct result is associated a probability, which moreover satisfies the condition  $\sum_{x \in X(\Omega)} \mathbb{P}([X = x]) = 1$ . If now  $X$  is a random variable with a density, we can no longer speak of the probability of obtaining a given result (in fact, the probability of obtaining a precise value is even equal to 0: continuous distributions are said to be **atomless**): each result is from then on associated with a **density**  $f$ , that is to say with a real function, positive and continuous (except possibly at a finite number of points) which satisfies the equalities  $\int_{X(\Omega)} f(x) dx = 1$  and, for all real numbers  $a$  and  $b$  satisfying  $a \leq b$ ,  $\mathbb{P}([a \leq X \leq b]) = \int_a^b f(x) dx$ .

Thus, for every real number  $x$ , the derivative of the **cumulative distribution function**  $F : x \mapsto \mathbb{P}([X \leq x])$  satisfies  $F'(x) = f(x)$ , the density of  $X$ .

### 3.2.4 Independence

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $A$  and  $B$  be two events defined on that set ( $A$  and  $B$  are therefore elements of  $\mathcal{A}$ ). Then  $A$  and  $B$  are **independent** if, and only if, we have  $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ . Let now  $X$  and  $Y$  be two real random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ . Then  $X$  and  $Y$  are two **independent** random variables if, and only if, for all real numbers  $x$  and  $y$ , we have  $\mathbb{P}([X \leq x] \cap [Y \leq y]) = \mathbb{P}([X \leq x])\mathbb{P}([Y \leq y])$ . In particular, if  $X$  and  $Y$  are discrete random variables, it suffices to verify that, for every real number  $x$  of  $X(\Omega)$  and every real number  $y$  of  $Y(\Omega)$ , we have  $\mathbb{P}([X = x] \cap [Y = y]) = \mathbb{P}([X = x])\mathbb{P}([Y = y])$  in order to conclude that the random variables  $X$  and  $Y$  are independent. The notion of independence generalises naturally to any finite number of events or of random variables. Indeed, let  $n$  be an integer at least equal to 2. Let  $A_1, \dots, A_n$  be events defined on that set. Then  $A_1, \dots, A_n$  are **mutually independent** if, and only if, we have  $\mathbb{P}(A_1 \cap \dots \cap A_n) = \mathbb{P}(A_1) \times \dots \times \mathbb{P}(A_n)$ . Let now  $X_1, \dots, X_n$  be real random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ . Then  $X_1, \dots, X_n$  are **mutually independent** random variables if, and only if, for all real numbers  $x_1, \dots, x_n$ , we have  $\mathbb{P}([X_1 \leq x_1] \cap \dots \cap [X_n \leq x_n]) = \mathbb{P}([X_1 \leq x_1]) \times \dots \times \mathbb{P}([X_n \leq x_n])$ . In particular, if  $X_1, \dots, X_n$  are discrete random variables, it suffices to verify that, for every integer  $i$  between 1 and  $n$  and every real number  $x_i$  of  $X_i(\Omega)$ , we have  $\mathbb{P}([X_1 = x_1] \cap \dots \cap [X_n = x_n]) = \mathbb{P}([X_1 = x_1]) \times \dots \times \mathbb{P}([X_n = x_n])$  in order to conclude that the random variables  $X_1, \dots, X_n$  are mutually independent. Moreover, if the random variables  $X_1, \dots, X_n$  are mutually independent and follow the same distribution, then we shall say that  $(X_1, \dots, X_n)$  forms an **independent and identically distributed** (i.i.d.) sample. Finally, let  $X_1, \dots, X_n$  (where  $n$  is an integer at least equal to 2) be mutually independent variables and  $k$  a natural number between 1 and  $n-1$ . Then, for every function  $f$  from  $\mathbb{R}^k$  into  $\mathbb{R}$  and every function  $g$  from  $\mathbb{R}^{n-k}$  into  $\mathbb{R}$ , the random variable  $f(X_1, \dots, X_k)$  is independent of the random variable  $g(X_{k+1}, \dots, X_n)$ . In addition, for all functions  $f_1, \dots, f_n$  from  $\mathbb{R}$  into  $\mathbb{R}$ , the random variables  $f_1(X_1), \dots, f_n(X_n)$  are likewise mutually independent. These two results are known by the name of the **grouping lemma**.

### 3.2.5 The convolution theorem

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $X$  and  $Y$  be two random variables with a density defined on that space. We denote by  $f$  a density of  $X$  and by  $g$  a density of  $Y$ . We suppose that  $X$  and  $Y$  are independent. Then, for every real number  $x$ , the two integrals  $\int_{\mathbb{R}} f(t)g(x-t)dt$  and  $\int_{\mathbb{R}} f(x-u)g(u)du$  are of the same nature (convergent or divergent) and equal in the case of convergence. Subject to convergence, we set, for every real number  $x$ ,  $h(x) := \int_{\mathbb{R}} f(t)g(x-t)dt = \int_{\mathbb{R}} f(x-u)g(u)du$ . If  $h$  is defined and continuous on  $\mathbb{R}$ , except possibly at a finite number of points (in which case, we choose for  $h(x)$  an arbitrary positive value in order to define the function  $h$  on  $\mathbb{R}$ ), then  $h$  constitutes a density of the random variable  $X+Y$ . Moreover, if  $f$  or  $g$  is bounded on  $\mathbb{R}$ , then the function  $h$  is necessarily defined and continuous on  $\mathbb{R}$ , and therefore immediately constitutes a density of  $X+Y$ . Finally, the function  $h$  is called the **convolution product** of the functions  $f$  and  $g$  and is usually denoted  $f * g$ .

### 3.2.6 The law of total probability

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $(B_i)_{i \in I}$  (where  $I$  is a finite or countably infinite set) be a **complete system of events** (that is to say a family of events of  $\mathcal{A}$  satisfying  $\sum_{i \in I} \mathbb{P}(B_i) = 1$  and, for all distinct elements  $i$  and  $j$  of  $I$ ,  $\mathbb{P}(B_i \cap B_j) = 0$ :  $(B_i)_{i \in I}$  thus constitutes a **partition** of the sample space  $\Omega$ ). Then, for every event  $A$  of  $\mathcal{A}$ , we have  $\sum_{i \in I} \mathbb{P}(A \cap B_i) = \mathbb{P}(A)$ .

## 3.3 Moments of a random variable and of a random vector

### 3.3.1 Expectation

The **expectation** of a random variable, if it exists, designates the average value taken by it in theory. Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $X$  be a real random variable defined on that set. If  $X$  is a discrete random variable, then we define the expectation of  $X$  by the value  $\sum_{x \in X(\Omega)} x\mathbb{P}([X = x])$ : this is the average value of the random variable  $X$  weighted by the probability of occurrence of each event. If now  $X$  is a continuous random variable with density  $f$ , then we define the expectation of  $X$  by the value  $\int_{X(\Omega)} xf(x)dx$ ; the integral being a kind of ‘continuous sum’, the average value of the random variable  $X$  weighted by the density of each event. Consequently, let  $Y$  be another real random variable defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ . Let  $\lambda$  and  $\mu$  be two real numbers. We suppose that  $X$  and  $Y$  admit an

expectation. Then  $\lambda$  (a **certain** random variable since it is constant),  $\mu X$  and  $X + Y$  admit an expectation, and we have  $\mathbb{E}[\lambda] = \lambda$ ,  $\mathbb{E}[\mu X] = \mu \mathbb{E}[X]$  and  $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ ; the last two equalities are expressed by saying that expectation is **linear**. We thus have  $\mathbb{E}[\lambda + \mu X + Y] = \lambda + \mu \mathbb{E}[X] + \mathbb{E}[Y]$ . In addition, if the random variables  $X$  and  $Y$  are independent, then  $XY$  likewise admits an expectation and we have  $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$  (careful, however: if the random variables  $X$  and  $Y$  are not independent, this equality is *a priori* false). Finally, let  $f$  be a function from  $\mathbb{R}$  into  $\mathbb{R}$  defined at least at  $\mathbb{E}[X]$ . If the function  $g(X)$  admits an expectation, then we verify:

$$\begin{cases} \mathbb{E}[g(X)] = \sum_{x \in X(\Omega)} g(x) \mathbb{P}([X = x]) & \text{if } X \text{ is a discrete random variable} \\ \mathbb{E}[g(X)] = \int_{X(\Omega)} g(x) f(x) dx & \text{if } X \text{ is a continuous random variable with density } f \end{cases}$$

This last result is known by the name of the **law of the unconscious statistician**. In particular, we have:

$$\begin{cases} \mathbb{E}[X^2] = \sum_{x \in X(\Omega)} x^2 \mathbb{P}([X = x]) & \text{if } X \text{ is a discrete random variable} \\ \mathbb{E}[X^2] = \int_{X(\Omega)} x^2 f(x) dx & \text{if } X \text{ is a continuous random variable with density } f \end{cases}$$

### 3.3.2 Variance

The **variance** of a random variable  $X$  defined on a probability space  $(\Omega, \mathcal{A}, \mathbb{P})$ , if it exists, is by definition equal to  $\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2]$ . The variance informs us about the dispersion of  $X$ : the lower it is, the more concentrated the values taken by  $X$  (around its expectation as it happens); *a contrario*, the higher the variance of  $X$ , the more dispersed the values taken by  $X$  (it is therefore riskier to make forecasts about the value taken by  $X$ ). This aspect is fundamental on the stock exchange: the higher the variance presented by the returns of an asset, the less stable these are over time and the riskier the asset; conversely, the less risky an asset, the lower its variance. On the stock exchange moreover, one more usually employs the **volatility**  $\sigma(X) := \sqrt{\text{Var}(X)}$ , likewise called the **standard deviation**. Indeed, since the variance corresponds to a mean squared deviation from the mean, the unit of  $X$  is raised to the square. For example, if  $X$  designates the random variable associated with the floor area of the dwelling in  $\text{m}^2$  of any given household, then  $\text{Var}(X)$  is expressed in  $\text{m}^4$ , which is hardly practical; taking the square root of the variance therefore makes it possible to recover  $\text{m}^2$ , which facilitates interpretation. Consequently, let  $Y$  be another real random variable defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ . Let  $\lambda$  be a real number. We suppose that  $X$  and  $Y$  admit a variance. Then  $\lambda$  (a **certain** random variable since it is constant) and  $\mu X$  admit a variance, and we have  $\text{Var}(\lambda) = 0$ ,  $\text{Var}(\mu X) = \mu^2 \text{Var}(X)$  (every certain random variable is indeed independent of any other random variable). In addition, if the random variables  $X$  and  $Y$  are independent, then  $X + Y$  likewise admits a variance and we have  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ . We thus have  $\text{Var}(\lambda + \mu X) = \mu^2 \text{Var}(X)$ . Finally, the computation of the variance may be obtained by the **Koenig-Huygens formula**: we have  $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ .

### 3.3.3 Conditioning

How is one to update one's beliefs when a result has just been realised? For example, if we have thrown a fair six-sided die and someone tells us that the number obtained is at least equal to 3, there remain to us only four equiprobable possibilities for the value of the die: 3, 4, 5 or 6. Let  $X$  be the random variable corresponding to the number of the die thrown. Then we shall have  $\mathbb{P}([X = 3] | [X \geq 3]) = \mathbb{P}([X = 4] | [X \geq 3]) = \mathbb{P}([X = 5] | [X \geq 3]) = \mathbb{P}([X = 6] | [X \geq 3]) = \frac{1}{4}$ , the notation  $\mathbb{P}(A|B)$  meaning the **conditional probability** of  $A$  given  $B$ , that is to say the probability that an event  $A$  is realised given that we know that the event  $B$  has been realised. For all events  $A$  and  $B$  such that we have  $\mathbb{P}(B) \neq 0$ , we have  $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ . In particular, if the events  $A$  and  $B$  are independent, we have  $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ , whence  $\mathbb{P}(A|B) = \mathbb{P}(A)$ . Once conditional probability measures have been defined, we may likewise define a **conditional expectation** of a random variable, that is to say the expectation which takes account of all the information we know concerning its realisation. Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space and  $X$  a real random variable defined on that set. Let  $B$  be an event of  $\mathcal{A}$  of non-zero probability. We then define the **conditional expectation** of  $X$  given  $B$  by:

$$\begin{cases} \mathbb{E}[X|B] := \frac{1}{\mathbb{P}(B)} \sum_{x \in X(\Omega)} x \mathbb{P}([X = x] \cap B) = \sum_{x \in X(\Omega)} x \mathbb{P}([X = x] | B) & \text{if } X \text{ is discrete} \\ \mathbb{E}[X|B] := \frac{1}{\mathbb{P}(B)} \int_{X(B)} x f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

where  $X(B)$  is the set of values that the random variable  $X$  can take given that the event  $B$  has been realised (it is therefore a kind of 'updated support').

### 3.3.4 Covariance and correlation

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $X$  and  $Y$  be two random variables defined on that set. We have already seen what independence between  $X$  and  $Y$  meant. But when the random variables  $X$  and  $Y$  depend on one another (for example, if  $X$  designates the professional experience of an individual in a given sector and  $Y$  the salary offered by a firm in the sector to the same individual), how is one to quantify the ‘strength of the link’ between  $X$  and  $Y$ ? A first measure is the **covariance** between  $X$  and  $Y$ , which sets the deviation from the mean of the variable  $X$  against that of the variable  $Y$ :  $\mathbf{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$  (we therefore have in particular  $\mathbf{Cov}(X, Y) = \mathbf{Cov}(Y, X)$ ,  $\mathbf{Cov}(X, \lambda) = 0$  whatever the real number  $\lambda$  chosen, and  $\mathbf{Cov}(X, X) = \mathbf{Var}(X)$ ). Thus, the covariance is a kind of measure of relative dispersion which indicates the simultaneous variation of two random variables: when we have  $\mathbf{Cov}(X, Y) > 0$ , the random variables  $X$  and  $Y$  tend to vary in the same direction (that is to say that, if  $X$  increases, the variable  $Y$  increases as well on average, and *vice versa*), while, when we have  $\mathbf{Cov}(X, Y) < 0$ , the random variables  $X$  and  $Y$  tend to vary in opposite directions (that is to say that, if  $X$  increases, the variable  $Y$  decreases on average, and *vice versa*). In particular, when the random variables  $X$  and  $Y$  are independent, we find  $\mathbf{Cov}(X, Y) = 0$ . Careful, however: the converse is false: it is not because we have the equality  $\mathbf{Cov}(X, Y) = 0$  that the random variables  $X$  and  $Y$  are necessarily independent. In order to facilitate the interpretation of the joint variation between  $X$  and  $Y$ , we often ‘normalise’ the covariance by considering the **linear correlation coefficient** between the variables  $X$  and  $Y$ , namely  $\rho(X, Y) := \frac{\mathbf{Cov}(X, Y)}{\sigma(X)\sigma(Y)}$ ,  $\sigma(X)$  and  $\sigma(Y)$  designating the standard deviations of the random variables  $X$  and  $Y$ , respectively:  $\rho(X, Y)$  is indeed always between  $-1$  and  $1$ . Moreover, the computation of the covariance may be facilitated by the **Koenig-Huygens formula**: we have  $\mathbf{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$  with:

$$\begin{cases} \mathbb{E}[XY] := \sum_{x \in X(\Omega)} \sum_{y \in Y(\Omega)} xy \mathbb{P}([X = x] \cap [Y = y]) & \text{if } X \text{ and } Y \text{ are discrete} \\ \mathbb{E}[XY] := \int_{X(\Omega)} \int_{Y(\Omega)} xy f_{X,Y}(x, y) dy dx & \text{if } X \text{ and } Y \text{ are continuous} \end{cases}$$

In the notation above,  $f_{X,Y} := \frac{\partial^2 F_{X,Y}}{\partial x \partial y}$  designates the **joint density** of the random variables  $X$  and  $Y$  and where, for all real numbers  $x$  and  $y$ ,  $F_{X,Y}(x, y) := \mathbb{P}([X \leq x] \cap [Y \leq y])$  the **joint cumulative distribution function** of those same random variables. Finally, the covariance is bilinear: if  $a, b, \lambda, \mu, \theta$  and  $\tau$  are real numbers and  $Z$  and  $W$  are two random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ , we have the equality  $\mathbf{Cov}(a + \lambda X + \theta W, b + \mu Y + \tau Z) = \lambda \mu \mathbf{Cov}(X, Y) + \lambda \tau \mathbf{Cov}(X, Z) + \theta \mu \mathbf{Cov}(W, Y) + \theta \tau \mathbf{Cov}(W, Z)$  (we therefore have  $\mathbf{Cov}(X + W, Y) = \mathbf{Cov}(X, Y) + \mathbf{Cov}(W, Y)$ ,  $\mathbf{Cov}(\lambda X, \mu Y) = \lambda \mu \mathbf{Cov}(X, Y)$  and  $\mathbf{Cov}(a + X, b + Y) = \mathbf{Cov}(X, Y)$ ). Consequently,  $\mathbf{Var}(\lambda X + \mu Y) = \lambda^2 \mathbf{Var}(X) + \mu^2 \mathbf{Var}(Y) + 2\lambda \mu \mathbf{Cov}(X, Y)$ . In particular, when the random variables  $X$  and  $Y$  are independent, we do indeed recover the equality  $\mathbf{Var}(\lambda X + \mu Y) = \lambda^2 \mathbf{Var}(X) + \mu^2 \mathbf{Var}(Y)$ . More generally, if  $n$  is a natural number at least equal to 2 and  $(X_1, \dots, X_n)$  is a family of mutually independent random variables admitting a variance, then we have  $\mathbf{Var}(\lambda_1 X_1 + \dots + \lambda_n X_n) = \lambda_1^2 \mathbf{Var}(X_1) + \dots + \lambda_n^2 \mathbf{Var}(X_n)$ , whatever the real numbers  $\lambda_1, \dots, \lambda_n$  chosen.

### 3.3.5 Random vectors

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $n$  be a non-zero natural number and  $X_1, \dots, X_n$  random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ . Then the vector  $\mathbf{X}$  with  $n$  rows whose coordinates are given by  $X_1, \dots, X_n$  (in that order) constitutes a **random vector**. If the moments of order 1 (expectation) and 2 (variance) of each random variable  $X_1, \dots, X_n$  are well defined, we may define an **expectation** and a **variance-covariance matrix** by:

$$\mathbb{E}[\mathbf{X}] := \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{pmatrix} \quad \text{and} \quad \mathbf{Var}(\mathbf{X}) := \begin{pmatrix} \mathbf{Var}(X_1) & \mathbf{Cov}(X_1, X_2) & \cdots & \mathbf{Cov}(X_1, X_{n-1}) & \mathbf{Cov}(X_1, X_n) \\ \mathbf{Cov}(X_2, X_1) & \mathbf{Var}(X_2) & \cdots & \mathbf{Cov}(X_2, X_{n-1}) & \mathbf{Cov}(X_2, X_n) \\ \vdots & & \ddots & & \vdots \\ \mathbf{Cov}(X_{n-1}, X_1) & \mathbf{Cov}(X_{n-1}, X_2) & \cdots & \mathbf{Var}(X_{n-1}) & \mathbf{Cov}(X_{n-1}, X_n) \\ \mathbf{Cov}(X_n, X_1) & \mathbf{Cov}(X_n, X_2) & \cdots & \mathbf{Cov}(X_n, X_{n-1}) & \mathbf{Var}(X_n) \end{pmatrix}$$

The symmetry of the covariance, that is to say the fact that we have  $\mathbf{Cov}(X_i, X_j) = \mathbf{Cov}(X_j, X_i)$ , then implies that the variance-covariance matrix is real symmetric, and therefore diagonalisable: its eigenvalues are all non-negative; the matrix  $\mathbf{Var}(\mathbf{X})$  is therefore positive semi-definite. Moreover, if  $p$  is a non-zero natural number and  $(X_{i,j})_{1 \leq i \leq n, 1 \leq j \leq p}$  constitutes a family of random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$  each admitting an expectation, we may even define an expectation for a **random matrix**:

$$\mathbb{E} \left[ \begin{pmatrix} X_{1,1} & \cdots & X_{1,p} \\ \vdots & & \vdots \\ X_{n,1} & \cdots & X_{n,p} \end{pmatrix} \right] := \begin{pmatrix} \mathbb{E}[X_{1,1}] & \cdots & \mathbb{E}[X_{1,p}] \\ \vdots & & \vdots \\ \mathbb{E}[X_{n,1}] & \cdots & \mathbb{E}[X_{n,p}] \end{pmatrix}$$

In particular, if we have  $n = p$ , then the random matrix above, which we shall denote  $\tilde{X}$ , is a square matrix; we then verify in this case that we have  $\text{Tr}(\mathbb{E}[\tilde{X}]) = \mathbb{E}[\text{Tr}(\tilde{X})]$ .

### 3.4 Standard distributions

#### 3.4.1 Standard discrete distributions

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $X$  be a discrete real random variable defined on that set.

- **Uniform distribution.** We suppose that there exist a non-zero natural number and a finite family  $(x_1, \dots, x_n)$  of real numbers ranked in increasing order such that  $X$  admits as support  $X(\Omega) = \{x_1, \dots, x_n\}$ . Then  $X$  follows a uniform distribution on  $\{x_1, \dots, x_n\}$  if, and only if, for every natural number  $k$  between 1 and  $n$ , we have  $\mathbb{P}(X = x_k) = \frac{1}{n}$ . The random variable  $X$  then admits an expectation and a variance. In particular, if we have  $\{x_1, \dots, x_n\} = \{p, \dots, n\}$  (with  $n \geq p$ ), then we have  $\mathbb{E}[X] = \frac{p+n}{2}$  and  $\text{Var}(X) = \frac{(n-p+1)^2-1}{12}$ . The discrete uniform distribution is employed in order to model equiprobable outcomes; for example for a throw of a fair die (each face has the same probability of appearing).
- **Bernoulli distribution.** We suppose that  $X$  admits as support  $X(\Omega) = \{0, 1\}$ . Then  $X$  follows a Bernoulli distribution with parameter  $p$  (with  $p$  a real number between 0 and 1) if, and only if, we have  $\mathbb{P}(X = 1) = p$  and  $\mathbb{P}(X = 0) = 1 - p$ . The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = p$  and  $\text{Var}(X) = p(1 - p)$ . The Bernoulli distribution serves to model a random experiment with two outcomes, a ‘success’ and a ‘failure’, the probability of ‘success’ constituting the parameter of the Bernoulli distribution; for example in order to model success in an examination.
- **Binomial distribution.** We suppose that there exists a natural number  $n$  at least equal to 2 such that  $X$  admits as support  $X(\Omega) = \{0, 1, \dots, n\}$ . Then  $X$  follows a binomial distribution with parameters  $n$  and  $p$  (with  $p$  a real number between 0 and 1) if, and only if, for every natural number  $k$  between 0 and  $n$ , we have  $\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ . The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = np$  and  $\text{Var}(X) = np(1-p)$ . The binomial distribution is a repetition of  $n$  independent and identically distributed Bernoulli trials with distribution  $\mathcal{B}(p)$ . The binomial distribution counts the number of ‘successes’ within the  $n$  repetitions of experiments, the probability of obtaining a ‘success’ at each stage being  $p$ ; for example the number of times one hits the target in a game where one throws  $n$  darts.
- **Hypergeometric distribution.** Let  $N$  be a non-zero natural number and  $n$  a natural number at most equal to  $N$ . We suppose that  $X$  admits as support  $X(\Omega) = \{0, 1, \dots, n\}$ . Let  $p$  be a real number between 0 and 1 such that  $pN$  is an integer. Then  $X$  follows a hypergeometric distribution with parameters  $n$ ,  $p$  and  $N$  if, and only if, for every natural number  $k$  between 0 and  $n$ , we have:

$$\mathbb{P}(X = k) = \frac{\binom{pN}{k} \binom{(1-p)N}{n-k}}{\binom{N}{n}}$$

The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = np$  and  $\text{Var}(X) = \frac{(N-n)np(1-p)}{N-1}$ . The hypergeometric distribution models the number of ‘successes’ in a sample of size  $n$  selected at random from a population of given size  $N$  (unlike the binomial distribution, where the size of the population is unknown), given that the probability of a ‘success’ is given by  $p$ ; for example the number of Covid-19 patients in critical care in a given region, given the probability of developing a severe form and given the size of the French population.

- **Poisson distribution.** We suppose that  $X$  admits as support  $X(\Omega) = \mathbb{N}$ . Then  $X$  follows a Poisson distribution with parameter  $\lambda$  (with  $\lambda$  a strictly positive real number) if, and only if, for every natural number  $k$ , we have  $\mathbb{P}(X = k) = \frac{\lambda^k \exp(-\lambda)}{k!}$ . The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = \lambda$  and  $\text{Var}(X) = \lambda$ . The Poisson distribution models a discrete waiting time, the mean waiting time being given by the parameter  $\lambda$ ; for example, if on average a website receives 30 visits a day, then the random variable  $X$  which counts the number of visits *per day* to the site follows a Poisson distribution with parameter 30.
- **Geometric distribution.** We suppose that  $X$  admits as support  $X(\Omega) = \mathbb{N}^*$ . Then  $X$  follows a geometric distribution with parameter  $p$  (with  $p$  a real number between 0 and 1) if, and only if, for every non-zero natural number  $k$ ,

we have  $\mathbb{P}([X = k]) = (1 - p)^{k-1}p$ . The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = \frac{1}{p}$  and  $\mathbf{Var}(X) = \frac{1-p}{p^2}$ . The geometric distribution models the number of repetitions necessary of independent and identically distributed Bernoulli trials with distribution  $\mathcal{B}(p)$ ; it is the waiting time for the first success; for example, the number of attempts to obtain one's driving licence.

- **Multinomial distribution.** Let  $m$  be a natural number at least equal to 2 and  $n$  a natural number. Let  $(X_1, \dots, X_m)$  be a family of random variables such that, for every natural number  $i$  between 1 and  $m$ , there exists a real number  $p_i$  belonging to  $]0, 1[$  such that  $X_i$  follows a binomial distribution with parameters  $n$  and  $p_i$ . We suppose that we have  $X_1 + \dots + X_m = n$  and  $p_1 + \dots + p_m = 1$ . Then, for every family  $(n_1, \dots, n_m)$  of natural numbers satisfying  $n_1 + \dots + n_m = n$ , we verify the equality  $\mathbb{P}([N_1 = n_1] \cap \dots \cap [N_m = n_m]) = \frac{n!}{n_1! \dots n_m!} p_1^{n_1} \dots p_m^{n_m}$ . We then say that the real random vector  $(X_1, \dots, X_m)$  follows a multinomial distribution with parameters  $(p_1, \dots, p_m, n)$ .

### 3.4.2 Standard continuous distributions

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $X$  be a continuous real random variable defined on that set.

- **Uniform distribution.** We suppose that there exist two real numbers  $a$  and  $b$  satisfying  $a < b$  such that  $X$  admits as support  $X(\Omega) = [a, b]$ . Then  $X$  follows a uniform distribution on  $[a, b]$  if, and only if, its density  $f$  satisfies, for every real number  $x$ :

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

The cumulative distribution function of  $X$  is then given, for every real number  $x$ , by:

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } x \in [a, b] \\ 1 & \text{if } x \geq b \end{cases}$$

The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = \frac{a+b}{2}$  and  $\mathbf{Var}(X) = \frac{(b-a)^2}{12}$ .

- **Exponential distribution.** We suppose that  $X$  admits as support  $X(\Omega) = \mathbb{R}_+$ . Let  $\lambda$  be a strictly positive real number. Then  $X$  follows an exponential distribution with parameter  $\lambda$  if, and only if, its density  $f$  satisfies, for every real number  $x$ :

$$f(x) = \begin{cases} \lambda \exp(-\lambda x) & \text{if } x \in \mathbb{R}_+ \\ 0 & \text{otherwise} \end{cases}$$

The cumulative distribution function of  $X$  is then given, for every real number  $x$ , by:

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - \exp(-\lambda x) & \text{otherwise} \end{cases}$$

The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = \frac{1}{\lambda}$  and  $\mathbf{Var}(X) = \frac{1}{\lambda^2}$ .

- **Gamma distribution.** We suppose that  $X$  admits as support  $X(\Omega) = \mathbb{R}_+$ . Let  $\theta$  and  $k$  be two strictly positive real numbers. Then  $X$  follows a Gamma distribution with parameters  $k$  and  $\theta$  if, and only if, its density  $f$  satisfies, for every real number  $x$ :

$$f(x) = \begin{cases} \frac{x^{k-1} \exp(-x/\theta)}{\Gamma(k) \theta^k} & \text{if } x \in \mathbb{R}_+ \\ 0 & \text{otherwise} \end{cases}$$

where we have  $\Gamma(k) := \int_0^{+\infty} \exp(-x) x^{k-1} dx$ . Moreover, for every non-zero natural number  $k$ , we have  $\Gamma(k) = (k-1)!$ . The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = k\theta$  and  $\mathbf{Var}(X) = k\theta^2$ .

- **Normal distribution.** We suppose that  $X$  admits as support  $X(\Omega) = \mathbb{R}$ . Let  $\mu$  be a real number and  $\sigma$  a strictly positive real number. Then  $X$  follows a normal distribution with parameters  $\mu$  and  $\sigma^2$  if, and only if, its density  $f$  satisfies, for every real number  $x$ ,  $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ . The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = \mu$  and  $\mathbf{Var}(X) = \sigma^2$ .



### 3.4.3 Distributions deriving from the normal distribution

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space.

- **The  $\chi^2$  distribution.** Let  $k$  be a non-zero natural number. Let  $(X_1, \dots, X_k)$  be independent and identically distributed random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ . We suppose that, for every integer  $j$  between 1 and  $k$ ,  $X_j$  follows a standard normal distribution. Then the random variable  $X := X_1^2 + \dots + X_k^2$  follows by definition a  $\chi^2$  distribution with  $k$  degrees of freedom. The density  $f$  of  $X$  satisfies, for every real number  $x$ :

$$f(x) = \begin{cases} \frac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{\sqrt{2}^k \Gamma(k/2)} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

The random variable  $X$  then admits an expectation and a variance and we have  $\mathbb{E}[X] = k$  and  $\mathbf{Var}(X) = 2k$ .

- **Student's distribution.** Let  $k$  be a non-zero natural number. Let  $Z$  be a random variable defined on  $(\Omega, \mathcal{A}, \mathbb{P})$  assumed to follow a standard normal distribution. Let  $U$  be a random variable defined on  $(\Omega, \mathcal{A}, \mathbb{P})$  assumed to follow a  $\chi^2$  distribution with  $k$  degrees of freedom. We suppose that  $Z$  and  $U$  are independent. Then the variable  $X := \frac{Z}{\sqrt{U/k}}$  follows by definition a Student's distribution with  $k$  degrees of freedom. The density  $f$  of  $X$  then satisfies, for every real number  $x$ ,  $f(x) = \left(1 + \frac{x^2}{k}\right)^{-\frac{k+1}{2}} \Gamma\left(\frac{k+1}{2}\right) / \sqrt{k\pi} \Gamma\left(\frac{k}{2}\right)$ . The random variable  $X$  then admits an expectation when we have  $k > 1$  and a variance when we have  $k > 2$  and we have  $\mathbb{E}[X] = 0$  and  $\mathbf{Var}(X) = \frac{k}{k-2}$ .
- **Fisher's distribution.** Let  $k_1$  and  $k_2$  be two non-zero natural numbers. Let  $U_1$  and  $U_2$  be two random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$  assumed to follow a  $\chi^2$  distribution with  $k_1$  and  $k_2$  degrees of freedom respectively. We suppose that  $U_1$  and  $U_2$  are independent. Then the variable  $X := \frac{U_1/k_1}{U_2/k_2}$  follows by definition a Fisher's distribution with  $k_1$  and  $k_2$  degrees of freedom. The density  $f$  of  $X$  then satisfies, for every real number  $x$ :

$$f(x) = \begin{cases} \frac{\left(\frac{k_1 x}{k_1 x + k_2}\right)^{\frac{k_1}{2}} \left(1 - \frac{k_1 x}{k_1 x + k_2}\right)^{\frac{k_2}{2}}}{x \int_0^1 t^{\frac{k_1}{2}-1} (1-t)^{\frac{k_2}{2}-1} dt} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

The random variable  $X$  then admits an expectation when we have  $k_2 > 2$  and a variance when we have  $k_2 > 4$  and we have  $\mathbb{E}[X] = \frac{k_2}{k_2-2}$  and  $\mathbf{Var}(X) = \frac{2k_2^2(k_1+k_2-2)}{k_1(k_2-2)^2(k_2-4)}$ .

### 3.4.4 Stability of the standard distributions

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $X$  and  $Y$  be two independent real random variables defined on that set.

- **Stability of the binomial distribution.** We suppose that there exist a real number  $p$  belonging to  $[0, 1]$  and two natural numbers  $n_1$  and  $n_2$  such that  $X$  and  $Y$  follow respectively a binomial distribution  $\mathcal{B}(n_1, p)$  and a binomial distribution  $\mathcal{B}(n_2, p)$ . Then the random variable  $X + Y$  follows a binomial distribution  $\mathcal{B}(n_1 + n_2, p)$ .
- **Stability of the Poisson distribution.** We suppose that there exist two strictly positive real numbers  $\lambda_1$  and  $\lambda_2$  such that  $X$  and  $Y$  follow respectively a Poisson distribution  $\mathcal{P}(\lambda_1)$  and a Poisson distribution  $\mathcal{P}(\lambda_2)$ . Then the random variable  $X + Y$  follows a Poisson distribution  $\mathcal{P}(\lambda_1 + \lambda_2)$ .
- **Stability of the discrete uniform distribution.** We suppose that there exists a finite set  $A$  such that  $X$  follows a uniform distribution on  $A$ . Then, for all real numbers  $\lambda$  and  $\mu$ ,  $(\lambda X + \mu)$  follows a uniform distribution on the set  $\{\lambda x + \mu; x \in A\}$ . Careful, however: the sum of two uniform distributions does not *a priori* follow a uniform distribution.
- **Stability of the continuous uniform distribution.** We suppose that there exist two real numbers  $a$  and  $b$  satisfying  $a < b$  for which  $X$  follows a (continuous) uniform distribution on the interval  $[a, b]$ . Then, for all real numbers  $\lambda$  and  $\mu$ ,  $\lambda X + \mu$  likewise follows a uniform distribution on the interval bounded by the real numbers  $\lambda a + \mu$  and  $\lambda b + \mu$ .
- **Stability of the normal distribution.** We suppose that there exist two real numbers  $\mu_1$  and  $\mu_2$  and two strictly positive real numbers  $\sigma_1$  and  $\sigma_2$  such that  $X$  and  $Y$  follow respectively a normal distribution  $\mathcal{N}(\mu_1, \sigma_1^2)$  and a normal distribution  $\mathcal{N}(\mu_2, \sigma_2^2)$ . Then the random variable  $X + Y$  follows a normal distribution  $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ .

### 3.4.5 Standard approximations

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $X$  be a random variable defined on that space.

- **Normal approximation to the binomial distribution.** We suppose that  $X$  follows a binomial distribution with parameters  $n$  and  $p$ , where  $n$  is a non-zero natural number and where  $p$  is a real number between 0 and 1. If we have  $n \geq 30$ ,  $np \geq 5$  and  $n(1-p) \geq 5$ , otherwise put if  $n$  is ‘sufficiently large’ and  $p$  is ‘not too close’ to 0, then the distribution of  $X$  may be approximated by a normal distribution with expectation  $np$  and variance  $np(1-p)$ .
- **Approximation of the binomial distribution by a Poisson distribution.** We suppose that  $X$  follows a binomial distribution with parameters  $n$  and  $p$ , where  $n$  is a non-zero natural number and where  $p$  is a real number between 0 and 1. If we have  $n \geq 30$ ,  $np \leq 15$  and  $p \leq 0.1$ , otherwise put if  $n$  is ‘sufficiently large’ and  $p$  is ‘small’, then the distribution of  $X$  may be approximated by a Poisson distribution with parameter  $np$ .
- **Normal approximation to the Poisson distribution.** We suppose that  $X$  follows a Poisson distribution with parameter  $\lambda$ , where  $\lambda$  is a strictly positive real number. If we have  $\lambda \geq 20$ , otherwise put if  $\lambda$  is ‘sufficiently large’, then the distribution of  $X$  may be approximated by a normal distribution with expectation  $\lambda$  and variance  $\lambda$ .

## 3.5 Inferential statistics

### 3.5.1 Convergence in probability

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $(X_n)_{n \in \mathbb{N}^*}$  be a family of real random variables defined on that space. Let  $X$  be a real random variable defined on that same space. We say that  $(X_n)_{n \in \mathbb{N}^*}$  **converges in probability** to  $X$ , and we write  $X_n \xrightarrow[n \rightarrow +\infty]{\mathbb{P}} X$  if, and only if, for every strictly positive real number  $\epsilon$ ,  $\mathbb{P}(|X_n - X| \geq \epsilon)$  tends to 0 as the non-zero natural number  $n$  tends to  $+\infty$ . Moreover, let  $f$  be a real function continuous on  $X(\Omega)$ . If  $(X_n)_{n \in \mathbb{N}^*}$  converges in probability to  $X$ , then  $(f(X_n))_{n \in \mathbb{N}^*}$  converges in probability to  $f(X)$ . In particular, if the sequence of random variables  $(X_n^2)_{n \in \mathbb{N}^*}$  converges in probability to  $\sigma^2$ , where  $\sigma$  is a non-negative real number and the  $X_i$  are non-negative random variables then, since the function  $x \mapsto \sqrt{x}$  is continuous on  $\mathbb{R}_+$ , the sequence  $(X_n)_{n \in \mathbb{N}^*}$  converges in probability to  $\sigma$ . Consequently, let  $(Y_n)_{n \in \mathbb{N}^*}$  be a family of real random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$  and  $Y$  a real random variable defined on that same space. Let us suppose that the sequences  $(X_n)_{n \in \mathbb{N}^*}$  and  $(Y_n)_{n \in \mathbb{N}^*}$  converge in probability to real numbers  $\lambda$  and  $\mu$  respectively. Then the sequence  $(X_n + Y_n)_{n \in \mathbb{N}^*}$  converges in probability to the certain variable  $\lambda + \mu$ . Careful, however: if the said sequences converge respectively to random variables  $X$  and  $Y$ , we do not necessarily have convergence in probability of the sequence  $(X_n + Y_n)_{n \in \mathbb{N}^*}$  to the random variable  $X + Y$ . Finally, if the family  $(X_n)_{n \in \mathbb{N}^*}$  forms an i.i.d. sample and the  $X_i$  admit an expectation (since these are identically distributed, the expectation is in that case the same for all the  $X_i$ , and we may therefore denote it  $\mu$ ) and a variance  $\sigma^2$  (where  $\sigma$  is a non-negative real number), then the **empirical mean**, that is to say the sequence of random variables  $(\bar{X}_n)_{n \in \mathbb{N}^*}$  defined, for every non-zero natural number  $n$ , by  $\bar{X}_n := \frac{X_1 + \dots + X_n}{n}$ , converges in probability to  $\mu$ : this is the **weak law of large numbers**. Moreover, we also have  $\mathbb{E}[\bar{X}_n] = \mu$  and, by independence of the  $X_i$ ,  $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$ : the weak law of large numbers therefore indicates that, under the assumptions of the theorem, the dispersion of the empirical mean tends to 0; otherwise put, the empirical mean tends to stabilise around its theoretical value as the size of the sample  $n$  increases.

### 3.5.2 Convergence in distribution

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $(X_n)_{n \in \mathbb{N}^*}$  be a family of real random variables defined on that space. Let  $X$  be a real random variable defined on that same space. We say that  $(X_n)_{n \in \mathbb{N}^*}$  **converges in distribution** to  $X$ , and we write  $X_n \xrightarrow[n \rightarrow +\infty]{\mathcal{L}} X$  if, and only if, for every real number  $x$  at which the cumulative distribution function  $F: x \mapsto \mathbb{P}([X \leq x])$  is continuous,  $\mathbb{P}([X_n \leq x])$  converges to  $\mathbb{P}([X \leq x])$  as the non-zero natural number  $n$  tends to  $+\infty$ . Moreover, let  $f$  be a real function continuous on  $X(\Omega)$ . If  $(X_n)_{n \in \mathbb{N}^*}$  converges in distribution to  $X$ , then  $(f(X_n))_{n \in \mathbb{N}^*}$  converges in distribution to  $f(X)$ . For example, if the sequence of random variables  $(X_n)_{n \in \mathbb{N}^*}$  converges in distribution to a random variable  $X$  then, since the function  $x \mapsto e^x$  is continuous on  $\mathbb{R}$ , the sequence  $(e^{X_n})_{n \in \mathbb{N}^*}$  converges in distribution to the random variable  $e^X$ . Consequently, let  $\lambda$  and  $\mu$  be two real numbers. Let us suppose that the sequence  $(X_n)_{n \in \mathbb{N}^*}$  and  $(Y_n)_{n \in \mathbb{N}^*}$  converges in distribution to a random variable  $X$ . Then the sequence  $(\lambda X_n + \mu)_{n \in \mathbb{N}^*}$  converges in distribution to the random variable  $\lambda X + \mu$ . Finally, let  $(Y_n)_{n \in \mathbb{N}^*}$  be a family of real random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$  and  $Y$  a real random variable defined on that same space. Let us suppose that the sequence  $(X_n)_{n \in \mathbb{N}^*}$  converges **in distribution** to the random variable  $X$  and that the sequence  $(Y_n)_{n \in \mathbb{N}^*}$  converges **in probability** to the random variable  $Y$ . Then, on the one hand, the sequence  $(X_n + Y_n)_{n \in \mathbb{N}^*}$  converges **in distribution** to the random variable  $X + Y$  and, on the other hand, the sequence  $(X_n Y_n)_{n \in \mathbb{N}^*}$  converges **in distribution** to the random variable  $XY$ :

this result is known by the name of **Slutsky's theorem**. Moreover, convergence in probability implies convergence in distribution, but the converse is false as a general rule (it is nevertheless true if the convergence in distribution is to a certain variable).

### 3.5.3 Estimators

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $n$  be a non-zero natural number. Let  $(X_k)_{k \in [1, n]}$  be a sample of independent and identically distributed real random variables following a distribution  $\mathcal{L}_\theta$ , where  $\theta$  (belonging to a certain set  $\Theta$ ) is the parameter to be estimated. Then we call a **statistic** any function of  $n$  variables such that  $\phi(X_1, \dots, X_n)$  is a real random variable. From then on, an **estimator** of  $\theta$  is a statistic  $\phi$  such that the real random variable  $\phi(X_1, \dots, X_n)$  can be evaluated from the data of  $X_1, \dots, X_n$  alone (we say that  $\phi$  is independent of  $\theta$ ). Let  $\hat{\theta}_n$  be an estimator of a given parameter  $\theta$ . Then the **mean squared error** of  $\hat{\theta}_n$ , also known by the name of the quadratic risk of  $\hat{\theta}_n$ , satisfies the equality (denoting by  $b(\hat{\theta}_n) := \hat{\theta}_n - \theta$  its bias):  $\mathbb{E}[(\hat{\theta}_n - \theta)^2] = \text{Var}(\hat{\theta}_n) + (b(\hat{\theta}_n))^2$ . Moreover,  $\hat{\theta}_n$  is said to be **(weakly) convergent** (or consistent) if, and only if,  $\hat{\theta}_n$  converges in probability to  $\theta$ . For  $\hat{\theta}_n$  to be a convergent estimator, it suffices that it be unbiased and that its variance tend to 0 as the number of realisations  $n$  tends to  $+\infty$  (careful, this is not a necessary condition).

- **Estimator of the expectation.** For every non-zero natural number  $n$ , we set  $\bar{X}_n$  to be the empirical mean of  $\mathcal{X}_n$ . Otherwise put, for every non-zero natural number  $n$ , we have  $\bar{X}_n := \frac{X_1 + \dots + X_n}{n}$ . Then  $\bar{X}_n$  constitutes an unbiased estimator of  $\mu$ , and its variance tends to 0 as the integer  $n$  tends to  $+\infty$ :  $\bar{X}_n$  is therefore a convergent estimator of  $\mu$ .
- **Estimator of the variance.** For every non-zero natural number  $n$ , we set  $\hat{\sigma}_n^2$  to be the empirical variance of  $\mathcal{X}_n$ . Otherwise put, for every non-zero natural number  $n$ , we have  $\hat{\sigma}_n^2 := \frac{(X_1 - \bar{X}_n)^2 + \dots + (X_n - \bar{X}_n)^2}{n-1}$ . Then  $\hat{\sigma}_n^2$  constitutes an unbiased and convergent estimator of  $\sigma^2$ .

### 3.5.4 Likelihood and the maximum likelihood estimator

The likelihood function is a function of the parameters of a statistical model computed from the observed data. Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $X$  be a random variable defined on that space.

- Let us suppose that  $X$  is a discrete random variable whose distribution depends on an unknown parameter  $\theta$ . Let  $n$  be a non-zero natural number. Let  $(X_1, \dots, X_n)$  be an independent and identically distributed sample with the same distribution as  $X$ . Then the likelihood  $L_n$  is a function of  $\theta$ , given a realisation  $(x_1, \dots, x_n)$  of the random vector  $(X_1, \dots, X_n)$ , which is then written  $L_n(\theta|x_1, \dots, x_n) = \mathbb{P}_\theta([X_1 = x_1] \cap \dots \cap [X_n = x_n]) = \prod_{i=1}^n \mathbb{P}_\theta([X_i = x_i])$ . We then define the log-likelihood of the sample  $(x_1, \dots, x_n)$  by  $\ell_n(\theta|x_1, \dots, x_n) = \sum_{i=1}^n \ln(\mathbb{P}_\theta([X_i = x_i]))$ .
- Let us suppose that  $X$  is a continuous random variable whose density  $f$  depends on an unknown parameter  $\theta$ . Let  $n$  be a non-zero natural number. Let  $(X_1, \dots, X_n)$  be an independent and identically distributed sample with the same distribution as  $X$ . Then the likelihood  $L_n$  is a function of  $\theta$ , given a realisation  $(x_1, \dots, x_n)$  of the random vector  $(X_1, \dots, X_n)$ , which is then written  $L_n(\theta|x_1, \dots, x_n) = \prod_{i=1}^n f_\theta(x_i)$ . We then define the log-likelihood of the sample  $(x_1, \dots, x_n)$  by  $\ell_n(\theta|x_1, \dots, x_n) = \sum_{i=1}^n \ln(f_\theta(x_i))$ .

We then look for  $\hat{\theta}_n$  such that  $L_n(\theta|x_1, \dots, x_n)$  is maximal: such a  $\hat{\theta}_n$ , if it exists, is called the **maximum likelihood estimator** of the parameter  $\theta$ . In practice, since  $\ln$  is strictly increasing, maximising  $L_n(\theta|x_1, \dots, x_n)$  is equivalent to maximising  $\ell_n(\theta|x_1, \dots, x_n)$ . In order to determine  $\hat{\theta}_n$ , one must compute the first-order condition  $\frac{\partial \ell_n(\theta|x_1, \dots, x_n)}{\partial \theta} = 0$  then, if a candidate point is found, confirm that it satisfies the second-order condition for a local maximum, namely  $\frac{\partial^2 \ell_n(\theta|x_1, \dots, x_n)}{\partial \theta^2} < 0$ .

### 3.5.5 Confidence interval, fluctuation interval

A **fluctuation interval** of order  $100(1 - \alpha) \%$ , where  $\alpha$  is a real number between 0 and 1, for a random phenomenon  $Y$  is an interval  $I(Y)$  in which the realisation of  $Y$  is supposed to be found with a probability at least equal to  $1 - \alpha$ : otherwise put, we must have  $\mathbb{P}(Y \in I(Y)) \geq 1 - \alpha$ . A fluctuation interval is expressed as a function of theoretical quantities linked to the distribution of the random phenomenon, within which the realisation of  $Y$  is supposed to be found with a high probability at least equal to  $1 - \alpha$ . Conversely, a **confidence interval** with confidence level  $100(1 - \alpha) \%$  of an unobserved parameter  $\gamma$  linked to  $Y$  (for example, its expectation or its variance) is an estimated window within which the said quantity is supposed to be found with a probability at least equal to  $1 - \alpha$ . A confidence

interval is therefore a bounding of  $\gamma$  between two random variables; the latter must in particular not depend on unknown quantities, thus making the confidence interval always computable.

- **Asymptotic fluctuation interval.** Let  $(X_n)_{n \in \mathbb{N}^*}$  be a sequence of independent and identically distributed random variables admitting an expectation  $\mathbb{E}[X]$  and a finite variance  $\text{Var}(X)$ . For every real number  $\alpha$  between 0 and 1, we denote by  $z_\alpha$  the quantile at  $100(1 - \alpha/2)$  % of a standard normal distribution. Let therefore  $\alpha$  be a real number between 0 and 1. Then an asymptotic fluctuation interval for the empirical mean of the  $X_i$  at the confidence level  $100(1 - \alpha)$  % is given by  $\text{IF}_{100(1-\alpha)\%} := \left[ \mathbb{E}[X] - \frac{z_\alpha \sqrt{\text{Var}(X)}}{\sqrt{n}}, \mathbb{E}[X] + \frac{z_\alpha \sqrt{\text{Var}(X)}}{\sqrt{n}} \right]$ . If  $\text{Var}(X)$  is unknown, then we estimate the variance of  $X$  by  $\hat{\sigma}_n^2 := \frac{(X_1 - \bar{X})^2 + \dots + (X_n - \bar{X})^2}{n-1}$ , where  $\bar{X}$  designates the empirical mean of the i.i.d. sample  $(X_1, \dots, X_n)$ . From then on, an asymptotic fluctuation interval for the empirical mean of the  $X_i$  at the confidence level  $100(1 - \alpha)$  % is given by  $\text{IF}_{100(1-\alpha)\%} := \left[ \mathbb{E}[X] - \frac{z_\alpha \sqrt{\hat{\sigma}_n^2}}{\sqrt{n}}, \mathbb{E}[X] + \frac{z_\alpha \sqrt{\hat{\sigma}_n^2}}{\sqrt{n}} \right]$ .
- **Confidence interval for the mean of a normal i.i.d. sample.** Let  $n$  be a non-zero natural number. Let  $\mu$  be an unknown real number and  $\sigma$  a strictly positive real number, known or unknown. Let  $(X_1, \dots, X_n)$  be a family of real random variables all following the same normal distribution  $\mathcal{N}(\mu, \sigma^2)$ . We denote by  $\bar{X}_n$  the empirical mean of the random variables  $X_1, \dots, X_n$ . For every real number  $\alpha$  of  $]0, 1[$ , we denote by  $z_\alpha$  and  $t_\alpha$  the quantiles of order  $1 - \alpha$  of the standard normal distribution and of Student's distribution of order  $n - 1$ , respectively. Let  $\mu_0$  be the empirical mean of the sample  $(X_1, \dots, X_n)$ . If we suppose that the variance  $\sigma^2$  is known, then a confidence interval for  $\mu$  at the confidence level  $100(1 - \alpha)$  % is given by  $\text{IC}_{100(1-\alpha)\%} := \left[ \mu_0 - \frac{z_{\alpha/2} \sigma}{\sqrt{n}}, \mu_0 + \frac{z_{\alpha/2} \sigma}{\sqrt{n}} \right]$ . By contrast, if we suppose that the variance  $\sigma^2$  is unknown, then a confidence interval for  $\mu$  at the confidence level  $100(1 - \alpha)$  % is given by  $\text{IC}_{100(1-\alpha)\%} := \left[ \mu_0 - \frac{t_{\alpha/2} \sqrt{\hat{\sigma}_n^2}}{\sqrt{n}}, \mu_0 + \frac{t_{\alpha/2} \sqrt{\hat{\sigma}_n^2}}{\sqrt{n}} \right]$ .
- **Asymptotic confidence interval for a proportion.** Let  $n$  be a non-zero natural number. Let  $p$  be a real number strictly between 0 and 1. Let  $(X_1, \dots, X_n)$  be a family of real random variables all following the same Bernoulli distribution  $\mathcal{B}(p)$ . We denote by  $\bar{X}_n$  the empirical mean of the random variables  $X_1, \dots, X_n$ . For every real number  $\alpha$  of  $]0, 1[$ , we denote by  $z_\alpha$  the quantile of order  $1 - \alpha$  of the standard normal distribution. Then an asymptotic confidence interval for  $p$  at the confidence level  $100(1 - \alpha)$  % (valid concretely as soon as we have  $n \geq 30$ ) is given by  $\text{IC}_{100(1-\alpha)\%} := \left[ \bar{X}_n - z_{\alpha/2} \sqrt{\frac{\bar{X}_n(1-\bar{X}_n)}{n}}, \bar{X}_n + z_{\alpha/2} \sqrt{\frac{\bar{X}_n(1-\bar{X}_n)}{n}} \right]$ .

### 3.5.6 Statistical tests

Let  $(\Omega, \mathcal{A}, \mathbb{P})$  be a probability space. Let  $n$  be a non-zero natural number. Let  $(X_k)_{k \in [1, n]}$  be a sequence of real random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ . Let us consider the following situation: we seek to test a hypothesis relating to all or part of the random variables  $(X_k)_{k \in [1, n]}$ . Then we call the **null hypothesis**, which we usually denote  $(H_0)$ , the hypothesis we wish to test, while we call the **alternative hypothesis**, which we usually denote  $(H_1)$ , the proposition  $\neg(H_0)$ , that is to say the proposition which is true if, and only if, the null hypothesis is false. Consequently, we call a **statistical test** any Boolean function  $\phi$ , that is to say any function such that, for every sequence  $(X_k)_{k \in [1, n]}$  of real random variables defined on  $(\Omega, \mathcal{A}, \mathbb{P})$ ,  $\phi(X_1, \dots, X_n)$  takes 0 or 1 as its value:  $\phi(X_1, \dots, X_n)$  is therefore a random variable. Let us then denote  $\alpha := \mathbb{P}_{H_0}(\phi(X_1, \dots, X_n) = 1)$  and  $\beta := \mathbb{P}_{H_1}(\phi(X_1, \dots, X_n) = 0)$ . From then on, as a function of the value returned by  $\phi(X_1, \dots, X_n)$ , we shall accept or reject the null hypothesis, wrongly or rightly:

|                            | Case A. We observe $[\phi(X_1, \dots, X_n) = 0]$                                                                      | Case B. We observe $[\phi(X_1, \dots, X_n) = 1]$                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| $(H_0)$ is true in reality | We rightly accept the hypothesis $(H_0)$ with a probability $(1 - \alpha)$ ( <b>confidence level</b> ): true positive | We wrongly reject the hypothesis $(H_0)$ with a probability $\alpha$ ( <b>type I error</b> ): false negative |
| $(H_1)$ is true in reality | We wrongly accept the hypothesis $(H_0)$ with an error $\beta$ ( <b>type II error</b> ): false positive               | We rightly reject the hypothesis $(H_0)$ with a probability $(1 - \beta)$ ( <b>power</b> ): true negative    |

Table 1.4: Carrying out a statistical test: the trade-off between the null hypothesis and the alternative hypothesis

A subtlety of language: in a statistical test of a null hypothesis  $(H_0)$  against an alternative hypothesis  $(H_1)$ , one never accepts the null hypothesis: either one rejects the null hypothesis, and therefore one accepts the alternative hypothesis, or one cannot reject the null hypothesis (which is what is improperly translated by 'accepting the null hypothesis', which is rigorously false). Thus, if the objective of the modeller is to prove a given fact, the said fact must be the alternative hypothesis, while its contrary (possibly simplified) will be the null hypothesis. Finally, in any statistical test whatsoever, the **p-value** represents the minimum level of risk at which the null hypothesis may be rejected. It is generally computed directly by statistical software. In practice, when we test a hypothesis at the

confidence level 95 % (or, equivalently, at the 5 % risk level), we reject the null hypothesis with a confidence level of 95 % if, and only if, the  $p$ -value computed is less than 0.05.

### Key Takeaways



The notions of median, quartiles and interdecile ratio are often employed in the economic outlook notes of INSEE. Thus, if we order a distribution, the median divides that distribution into two parts of equal frequency; for example, for a distribution of wages, 50 % of wages are situated below the median and 50 % above. Moreover, interdecile ratios make it possible to measure the inequalities of a distribution: the ratio of the deciles  $D9/D1$  thus brings out the gap between the top (ninth decile  $D9$ : if we are speaking of wages, 90 % of wages are situated below this value and 10 % above) and the bottom of the distribution (first decile  $D1$ : if we are speaking of wages, 10 % of wages are situated below this value and 90 % above), while the ratio  $D9/D5$  compares the top of the distribution to the median value ( $D5$  therefore) and the ratio  $D5/D1$  compares the median to the bottom of the distribution.

# Money and the financing of the economy

Would you rather receive one hundred euros today or one hundred euros in a year's time? In answering with a preference for immediacy, you bring to light the fact that time has a monetary value: it is precisely the **interest rate** which will capture this value. But what is money? How and why is it created? What roles does it play in relation to economic policy and to the financing of the economy? These are all questions to which elements of an answer will be given in this part.

## 1 Money: characteristics, roles and evolution

Guillaumin (2018) defines **money** as the whole set of means of payment immediately usable and accepted by the community for the settlement of exchanges. Money is an asset which presents two particularities: an **absence of yield** (holding a hundred-euro note earns no interest) and maximum **liquidity** (the same note indeed makes it possible to settle all transactions of an amount less than or equal to one hundred euros).

### 1.1 Typology of money

Money therefore plays a central role in the emergence of the gains from trade, by making indirect exchange possible. In a barter system, exchange requires a **double coincidence of wants**, that is to say each party must want what the other one holds: as Krugman & Wells (2019) illustrates, a heart surgeon could only trade his services if the owner of a household appliance shop happened to need heart surgery and if the surgeon happened to want a refrigerator. Money solves this problem by allowing everyone to exchange their goods or services for money, which can then be used to obtain whatever they desire. Money thus makes it easier to exploit the gains from trade and increases welfare.

#### 1.1.1 The three functions of money

As Clower (1964) points out, in a barter economy, where each individual produces according to his capacities and exchanges his surpluses for goods that he does not produce himself, the double coincidence of wants is essential for exchange to take place. However, this form of exchange rapidly becomes inefficient as wants become more diverse and as the distances between individuals increase. Indeed, in a society with  $n$  types of goods,  $n(n-1)/2$  relative prices have to be established (finding out how many units of good  $i$  are worth one unit of good 1, for every  $i$  between 2 and  $n$ , then finding out how many units of good  $i$  are worth one unit of good 2, for every  $i$  between 3 and  $n$ , and so on), which generates **computation and information costs**, on top of the costs of exchange proper. These costs comprise, on the one hand, **transaction costs**, associated with travelling and with the time lost searching for the double coincidence of wants, and, on the other hand, **waiting costs**. Waiting manifests itself in the time needed to find a compatible trading partner, and this may generate subjective frustration, especially where the goods are perishable, such as meat or eggs. Storing goods for a prolonged period before exchange also entails objective costs, including losses due to spoilage. Consequently, Clower (1964) summarises that, in a barter economy, the cost of exchange is not limited to travelling and to the search for partners, but also includes the management of information and of storage, which makes the introduction of money indispensable in order to smooth exchange and reduce these multiple sources of inefficiency. While, as Delaplace (2021a) specifies, **fairs**, those places where large numbers of individuals gather, make the search for the double coincidence of wants easier, the desire to limit the costs of exchange rapidly took concrete form in the arrival of a **monetary economy**: by using a standard of value in the shape of money, there are no longer  $n(n-1)/2$  relative prices to be established (that is, 4950 relative prices for an economy with a hundred goods), but only  $n-1$  money prices (that is, only 99 money prices for an economy with a hundred goods).

The first function of money is therefore to be a **unit of account**, which plays a fundamental role by providing a common benchmark for expressing the economic value of goods and services. It thereby makes it possible to quantify economic relations, thus facilitating the comparison and the confrontation of economic values. As Simmel (1987) emphasised, this common 'economic language' abstracts from the non-economic aspects of goods in order to reduce them to a homogeneous monetary representation. Guillaumin (2018) thus indicates that money constitutes a **numéraire**, that is to say a reference value in which the whole set of prices of the goods of an economy can be expressed. But this unit-of-account function is not the only function of money. As early as the *Nicomachean Ethics* (fourth century BCE), Aristotle identified three essential functions of money. As we have already suggested in the previous paragraph, money also constitutes a **medium of exchange**. However, unlike the unit of account, the medium-of-exchange function is not exclusive to money. Fisher (1930) nevertheless specifies that the acceptability of money in exchange stems from its representation as a property right and as wealth, although this view is often confined to cash and demand deposits: money is regarded as an external good which facilitates exchange by reducing uncertainty and transaction costs, as Davidson (1983) explains. Money thus allows for greater market efficiency, by providing a stable benchmark for transactions, similar to the use of a unit of measurement. More precisely, money fulfils its function as an intermediary of exchange because it satisfies five qualities listed by Mishkin (2007): being standardised (so that its value can easily be assessed), having legal tender status, being divisible (so that change can be given), being easily transportable and being durable. Finally, money is also a **store of value**. This last function is generally linked to the role of money as a medium of exchange. As Keynes (1936) emphasised, money makes it possible to carry purchasing power forward into the future, thereby creating a link between the present and what is to come: this store-of-value role is particularly relevant when receipts and payments are not synchronised. Thus, an individual paid at the beginning of the month can spread his spending over several weeks by holding money. Moreover, holding money provides protection against the uncertainty attached to future spending and income. While other assets, such as precious metals, buildings or securities, can also serve as a store of value, money has decisive advantages which Delaplace (2021b) lists: in the first place, it is perfectly **liquid** within the area where it has currency, that is to say immediately available for transactions without conversion cost, whereas assets such as buildings or shares require brokerage or transaction fees; in addition, money is perceived as a **risk-free** asset, since it does not lose nominal value, unlike shares, whose price may fluctuate (thus, a hundred-euro note will always be worth a hundred euros, whereas a share may see its value fall, which generates a risk of capital loss). However, in order fully to fulfil this store-of-value function, money must retain its stability over time, which raises the question of the preservation of its value in the face of **inflation** (purchasing power) or of other economic factors.

### Focus...



**The general price level and inflation.** As Delaplace (2021b) indicates, in order to determine the value of money in terms of all goods, it is necessary to construct a **composite price index** which represents the average of the money prices of the various commodities, weighted by their respective shares in transactions. Let  $P_i$ , with  $i$  running from 1 to  $n$ , be the money prices of the  $n$  commodities and  $k_i$  the weighting coefficients assigned to each commodity  $i$ , representing the share of each commodity in exchange (so that  $k_1 + \dots + k_n = 1$ ). We then set  $P = k_1 P_1 + \dots + k_n P_n$ , where  $P$  represents the general price level. Thus, while the value of money relative to a single good is given by the inverse of its money price, the value of money relative to a set of commodities traded within a country is equal to the inverse of this composite index, that is the inverse of the general price level, or  $1/P$ .

Since the value of money is inversely proportional to the general price level, it retains its value over time only if the latter remains stable, that is to say in the absence of inflation, where **inflation** is defined as a sustained rise in the general price level. Conversely, a rise in the general price level entails a fall in the value of money, thereby reducing its purchasing power and its capacity to fulfil the store-of-value function. As Delaplace (2021b) specifies, the variation in the **purchasing power** of money therefore depends on which goods are retained in the computation of the general price index and on the respective variations in their prices. In France, the computation of the **general price index** and of inflation is carried out by INSEE (the French national statistical institute) on the basis of twelve consumption functions (grouping together 305 weighted items) taken to represent the typical consumer basket.

More precisely, INSEE (2021) indicates that the **classification of household consumption functions** (*classification of individual consumption by purpose*, COICOP) is an international nomenclature making it possible to break household consumption down by units of need. It is one of the 'functional' nomenclatures of the System of National Accounts (SNA) and is used for the dissemination of consumer price indices (CPI). Its first three levels of grouping (coded on four characters) are standardised at international level and used for the dissemination of household consumption figures drawn from the national accounts (volumes and prices). In the version given by INSEE (2021), the next two levels (coded on five and six characters) are specific to the dissemination of consumer price indices on

the 1998 base; since 2016, it is the European ECOICOP classification which has been used, the first four levels of which are standardised at European level.

### Key Takeaways



Money emerges as a solution to the inefficiency of barter by reducing the computation, transaction and waiting costs associated with the double coincidence of wants [Clower (1964)]. It fulfils three functions highlighted by Aristotle in the *Nicomachean Ethics*: unit of account providing a common numéraire [Simmel (1987)], medium of exchange facilitating transactions by reducing uncertainty [Davidson (1983)], and store of value making it possible to carry purchasing power forward through time [Keynes (1936)]. This last function requires money to preserve its nominal stability in the face of inflation, that is to say in the face of the rise in the general price level measured by a weighted composite index [Delaplace (2021b)].

#### 1.1.2 A brief history of money

Delaplace (2021a) emphasises that the earliest forms of money consisted of objects or commodities whose value initially rested on their usefulness as goods, that is to say that they were recognised and valued by the whole of the community within which they circulated: this is the era of **commodity money**. Thus, certain commodities met essential needs, such as cattle, which played a monetary role among stock breeders, and from which the term 'pecuniary' incidentally derives (the Latin term *pecus* meaning precisely 'cattle'). As Delaplace (2021a) indicates, the ox in particular was used as commodity money in Egypt under the Ramessides, as well as among the Greeks and the Romans until the sixth and fifth centuries BCE, and likewise among certain Germanic peoples. Other examples include cod in Newfoundland, the ear of barley in Mesopotamia six thousand years ago, the tea leaf in southern Asia, tobacco in Virginia, or again cocoa beans among the Aztecs. Delaplace (2021a) adds moreover that certain commodities had an ornamental value, such as shells (like the cowrie in China, in India, in Brazil and even in Africa until the beginning of the twentieth century) or precious stones such as turquoises among certain Amerindian peoples; others, finally, were working tools, such as iron axes and spades in Central Africa, or again fish hooks on certain Pacific islands. It therefore appears that commodity money combined a twofold function: on the one hand, it satisfied a particular need as a specific good, and on the other hand, it played the role of a universal instrument of exchange. Nevertheless, despite this diversity of commodities used as money, one particular form rapidly came to prevail: metallic money, which became predominant in economic exchange.

As Davies (2002) points out, the invention of **metallic money** goes back to Antiquity, where the first forms of money appear in the shape of precious metals used to facilitate commercial exchange: the first coins are thought to have been struck around the seventh century BCE in the kingdom of Lydia and in several independent cities of Asia Minor. These precious metals were prized for their scarcity and their durability, which made them an efficient means of exchange and of storing value. As Malthus (1846) emphasised, precious metals are **divisible**, that is to say that it is possible, thanks to their **malleability**, to divide them into small, easily transportable elements whose value is proportional to their weight, and **unalterable**, that is to say that they do not deteriorate on contact with air, which allows easy storage without risk of deterioration, unlike the case of commodity monies of a foodstuff nature: such a characteristic therefore makes it possible to defer purchases into the future. The evolution of the forms in which precious metals were used began with gold and silver circulating in the shape of pieces of metal of no defined weight or shape, which slowed exchange down because of the need to weigh them and to check their purity. This stage, known as **weighed money**, was followed by the introduction of **counted money**, where the metal was divided into pieces of standardised weight, a practice observed as early as the third millennium BCE in China and Mesopotamia according to Delaplace (2021a). Later, **struck money**, where distinctive marks and the weight were engraved on the coins, appeared in the seventh century BCE around the Aegean Sea. This system evolved so as to become a monopoly of the political and religious authorities (sovereign right), as in 755 under Pepin the Short in France, where money became a symbol of power. **Bimetallism**, using gold and silver, was instituted in France in 1803 under Napoleon Bonaparte with the *franc germinal*, setting a legal ratio of 1 gram of gold to 15.5 grams of silver. However, this system showed its limits in the nineteenth century because of the fluctuations in the commercial ratio between these metals, leading to the disappearance of gold, the 'good money', from the market, in accordance with Gresham's law: as Delaplace (2021a) indicates, in 1873 the commercial ratio of gold to silver reached 16, 18 in 1876, 20 in 1886, 33 in 1900 and 38 in 1910 in France. Faced with these problems, **monometallism** came to prevail, gold becoming the standard in most countries, notably in Great Britain in 1816, in France in 1876, and in the United States in 1853. On the eve of the First World War, gold dominated as a monetary instrument, although a new means of exchange, the banknote, had begun to spread.



## Focus...



**Gresham's law and its application to the German hyperinflation crisis.** The Englishman Gresham, Chancellor of the Exchequer in the sixteenth century, analyses that, in a bimetallic monetary system, if one of the two metals (the 'good money') appreciates relative to the other on the commercial market, then the former ought to disappear from circulation: 'bad money drives out good'. Three reasons explain this apparently counter-intuitive phenomenon according to Delaplace (2021a): first of all, individuals wishing to save will tend to keep the good money, since its value has appreciated, and to use the bad money in transactions; in addition, individuals making payments abroad will have every interest in using the good money abroad, where only the commercial ratio between the two metals is recognised; finally, speculators can carry out an **arbitrage** by exchanging the good money abroad at the commercial parity, then coming back with the bad money to exchange it at the Hôtel des Monnaies (the French mint) for good money, and repeating the process *ad vitam æternam*.

Gresham's law was moreover implemented in an original way in order to resolve the German hyperinflation crisis. That crisis was triggered by a combination of several factors. First of all, Germany, heavily indebted in the aftermath of the war, was compelled to pay massive reparations imposed by the Treaty of Versailles in 1919. In order to finance these debts and reparations, the German government chose to print money on a massive scale, hoping thereby to meet its financial needs without raising taxes, which contributed to runaway inflation. As Clavert (2006) explains, between 1921 and 1923, the value of the German mark collapsed spectacularly; in January 1921, one American dollar was worth about 60 marks; in November 1923, one dollar exchanged for 4 200 billion marks. This phenomenon of **hyperinflation** completely destabilised the German economy: prices rose exponentially, sometimes several times a day, and the population lost confidence in the currency. Household savings were wiped out, wages lost their value almost immediately after being paid, and barter became a commonplace method of transacting. In parallel, this hyperinflation was also fuelled by the reparations crisis when, in 1923, France and Belgium occupied the industrial region of the Ruhr, exacerbating economic tensions and reducing German industrial production, which further accentuated the depreciation of the currency.

The return to stability was finally obtained in November 1923 with the introduction of a new currency, the **rentenmark**, and thanks to monetary and fiscal reforms supported by the Dawes Plan in 1924, which rescheduled war reparations. More precisely, as Clavert (2006) explains, Hjalmar Schacht, governor of the German central bank during the 1920s and 1930s, relied on Gresham's law to create the rentenmark, a new currency intended to be used in parallel with the failing mark: the parity between these two currencies was fixed at one rentenmark for one million billion marks, thereby making it possible to give the rentenmark a stable value. Although the rentenmark was not widely used in everyday exchange, it served as a store of value and made it possible to fix the exchange rate of the mark by pegging it to the rentenmark; this stabilisation mechanism helped to restore confidence in the German currency and to put an end to the hyperinflation.

## Short-Answer Question



**Does bad money always drive out good?** The maxim according to which 'bad money drives out good' refers to Gresham's law, formulated on the basis of the monetary disorders of Tudor England and long held to be a general law of monetary circulation. The work of Rolnick & Weber (1986) has nevertheless shown that this formula is valid only within a set of precise institutional conditions, notably the existence of legal tender status which imposes acceptance at par of monies of unequal quality. In a contemporary context of competing fiat currencies and of dollarisation, the hypothesis of an automatic dominance of the 'bad' money becomes more debatable, as the historical analyses of Selgin (1996) suggest. At international level, the COFER statistics of FMI (2025) thus indicate that the dollar still accounts for 56.3 % of allocated official foreign exchange reserves in the second quarter of 2025, against nearly 65 % in 2017 according to the backward-looking series, which reflects both strong inertia and increased competition from other currencies. The question is therefore not whether the mechanism described by Gresham is false, but whether it is universal. In other words, does bad money always drive out good, or only in particular institutional configurations where private arbitrage is constrained? We shall first show that, within the framework for which it was designed, Gresham's law describes a robust mechanism of adverse selection among monetary instruments; we shall then see that, as soon as these assumptions are relaxed (flexible exchange rates, absence of forced currency status, competition from foreign currencies), the law ceases to be general and may even be reversed in favour of a 'Thiers' law' whereby good money replaces bad.

In metallic and fiduciary systems accompanied by strong legal tender arrangements, Gresham's law correctly describes the way in which a 'debased' money tends to push a 'sound' money out of circulation. The theoretical analysis of Oppers (2000), in a model of gold and silver bimetalism, shows that when the state fixes a legal conversion rate between two metallic monies which diverges from their market price ratio, agents have an interest in settling their debts with the overvalued money and in hoarding or exporting the undervalued money. This mechanism is

confirmed by the historical reconstructions of Meissner (2015), who emphasises how nineteenth-century France avoided 'losing all its gold' by limiting the minting of silver when the market ratio shifted, precisely in order to contain Gresham effects. From this perspective, the 'bad money' is that whose intrinsic (metallic) value is lower than its face value, but which the law requires to be accepted at par: it then becomes the preferred means of payment, while the 'good' money is set aside as a store of value. The interpretation put forward by Selgin (1996) stresses the central role of legal constraints (legal tender status, taxation, rules for the payment of taxes) in the dominance of bad money: in the absence of these constraints, agents spontaneously select the better monetary quality. Even though modern data bear more on fiat currencies, the work of Levy Yeyati (2006) on financial dollarisation shows that the coexistence of monetary assets of differing quality remains associated with systematic arbitrage: in heavily dollarised countries, the share of foreign currency deposits frequently exceeds 50 % of bank savings, which reflects a preference for the 'good' foreign money as a store of value. Gresham's law thus retains strong analytical relevance, but within a framework where relative prices are made rigid by regulation and where agents' scope for choice is restricted.

As soon as one moves away from this narrow framework to consider open monetary systems, characterised by currency competition and freedom of choice of the unit of account, Gresham's law, by contrast, ceases to be universal and may even be reversed. This is the thrust of the criticisms of Rolnick & Weber (1986), who show, on the basis of examples of bimetallism and of the joint circulation of metallic money and paper, that good and bad money often coexist at different market rates, one of them even being able to trade at a premium rather than disappear. For his part, Selgin (1996) proposes to 'save' Gresham's law by restricting it to situations where the public authorities impose a legal exchange rate between monies of differing quality; as soon as agents are able to quote and accept payments at a free price, the better-quality money tends on the contrary to establish itself as means of payment and store of value, which the French tradition designates as 'Thiers' law'. Contemporary dollarisation phenomena support this reversal: Reinhart et al. (2003) show that, at the beginning of the 2000s, 22 economies displayed a high composite dollarisation index, with ratios of foreign currency assets far exceeding 50 % of deposits or of public debt, a sign that the bad domestic money had been partially crowded out in favour of a currency judged safer. The study by Alvarez-Plata & Garcia-Herrero (2007) recalls that inflation exceeded 300 % in several heavily dollarised Latin American countries at the end of the 1980s, illustrating the link between macroeconomic instability and the preference for foreign currency. Likewise, the empirical analysis of Levy Yeyati (2006) shows that financially dollarised economies experience on average more volatile growth and more banking crises, which reinforces the idea that the good foreign money tends to supplant, rather than to be driven out by, the national currency when the credibility of the latter has been durably impaired. Finally, in a more global register, FMI (2025) emphasises that, despite a slight erosion, the dollar still accounts for nearly 57 % of allocated foreign exchange reserves in 2025, which attests that a currency judged better by virtue of its stability and the depth of its financial markets continues to dominate, without being driven out by currencies judged less safe.

In conclusion, the adage according to which 'bad money drives out good' cannot be held to be a universal truth: it aptly describes a mechanism of adverse selection specific to systems in which monies of differing quality are required to circulate at the same legal rate, but it loses its generality as soon as these conditions cease to be met. Gresham's law presupposes an institutional locking of relative prices, failing which the better-quality money may circulate at a premium or establish itself as the standard of value: in the presence of competition between national and foreign currencies, it is often the good money, the more stable one, which tends to replace the bad rather than the reverse. The challenge for contemporary monetary authorities is then to maintain a sufficient quality of the national currency so as to prevent the onset, in a context of capital mobility and currency competition, of a lasting dynamic of substitution which would this time consecrate the victory of good money over bad.

As Davies (2002) points out, money evolved over the centuries, from the coins struck by the city states of ancient Greece to the paper banknotes introduced in China in the seventh century; this last innovation moreover marked a crucial stage in the development of modern economies, by simplifying transactions and fostering the expansion of markets. While individuals could deposit their gold or silver with bankers in exchange for **certificates of deposit**, facilitating transactions by avoiding the physical handling of precious metals, these certificates were no more than substitutes for metallic money: as Delaplace (2021a) indicates, around 1668 in Great Britain goldsmiths issued receipts (*goldsmith's notes*) to the merchants of London who deposited their gold with them. True paper money proper appeared in 1656 with Palmstruch, who received authorisation to found in Sweden a 'bank of exchange and lending' which was to become the Bank of Stockholm, and to issue notes against the surrender of metallic money. Later, in 1844, the *Bank Charter Act* in England granted the Bank of England the monopoly of note issue. Confidence in the banknote was strengthened when the state conferred **legal tender** status upon it, entailing the obligation imposed by law to accept notes and coins in payment of a debt, as was the case in France in 1870. We then enter the era of **fiat money**, which, devoid of intrinsic value, rests on the confidence that economic agents have in the fact

that it will be accepted by others in the future: in other words, its value depends on the collective confidence in the issuing authority, generally a government or a central bank. The difference between the 'real' value of a banknote (a piece of cotton) and its face value (and, to a lesser extent, the difference between the production cost of coins and their face value) therefore creates an income for the central bank, the issuer of physical money. More generally, Banque de France (2021b) indicates that the difference between the interest received by the central bank on its assets (claims on commercial banks as the counterpart of the notes in circulation) and the central bank's expenditure on manufacturing, distributing and maintaining the notes (and the coins) generates **seigniorage** income for the central bank.

### Focus...



**Banking school versus Currency school.** As Delaplace (2021a) emphasises, a major debate in the nineteenth century set the *Currency school* (Ricardo, Say), which advocated a strict correspondence between notes and metallic reserves, against the *Banking school* (Tooke, Fullarton), which held that issue should respond to the needs of the economy. The *Currency school* triumphed in Great Britain with the adoption of the **gold specie standard** in 1816 through the *Coinage Act*, which remained in force until 1914: gold coins effectively circulated and any holder of a note could exchange his notes for gold, whatever the amount; it is this regime which enabled the pound sterling to become an international currency at that time according to Delaplace (2021a).

Other regimes inspired by the *Banking school* followed, such as the **gold bullion standard** adopted in France in 1928, which limited the convertibility of notes to gold bars of at least twelve kilograms, in order to remedy the insufficiency of gold reserves in many countries. Delaplace (2021a) recalls that this system was also adopted in England from May 1925: notes were then convertible not into gold coins but into gold bars of a minimum weight of four hundred ounces (roughly 11.34 kilograms). However, the convertibility of notes was progressively suspended, notably in France in 1936, until **inconvertible legal tender**, dispensing banks from redeeming notes in gold, became permanent after the 1929 crisis, definitively detaching money from its metallic reserves. The banknote then becomes detached from the precious metal and derives its value solely from being recognised and accepted by all economic agents as a means of payment.

The most recent development in the history of money lies in the dichotomy between physical money and deposit money, according to Banque de France (2023c). **Physical money** comprises coins and banknotes, which are tangible forms of money issued by central banks. These forms of money are increasingly being taken over by **deposit money**, which corresponds to accounting entries and comprises bank deposits, electronic transfers, and card payments. Deposit money today accounts for the majority of transactions in modern economies, facilitating rapid and secure exchange, often through digital systems. In reality, as Delaplace (2021a) recalls, traces of deposit and loan accounting, the forerunners of this deposit money, have been discovered as far back as 1800 BCE on clay tablets, while the Greeks and Romans already practised a form of transfer between current accounts held with a money changer (*trapezite*), as did the Arabs in the ninth and tenth centuries. Delaplace (2021a) adds that it is truly in the twelfth century that deposit money appears in Genoa, before developing in the thirteenth century with Italian and Flemish merchants; from the fourteenth century onwards, the **bill of exchange**, a claim between merchants involving a change of currency, thus became commonplace. In France, however, this type of money was not widely adopted by households until the 1960s, as Delaplace (2021a) qualifies.

### Focus...



**The development of deposit money in France.** As the bulletin of the Banque de France (2023c) indicates, in 2022, 29.5 billion transactions representing 42 500 billion euros exchanged were carried out in France, across all non-cash means of payment, against 24.9 billion transactions representing 28 500 billion euros exchanged in 2019. In 2022, the **payment card** consolidated its place as the dominant means of payment in France: it accounted for 62 % of non-cash payments (representing, however, less than 2 % of the value of transactions carried out in deposit money), a marked increase on the 46 % recorded in 2012. This growth is due in part to the increase in **contactless payments**, which have grown rapidly, with 60 % of proximity card payments made contactless. Card transactions by mobile phone have also soared, rising by 137 % on 2021, to reach nearly 6 % of card payments.

In parallel, **credit transfers** remained the main means of transferring funds in terms of the amounts exchanged, representing more than 92 % of the volumes exchanged in 2022 (and about 18 % of the total number of transactions in deposit money). Although cheques continue to decline, with a 9 % fall in the number of transactions in 2022, France remains the country of the European Union which uses cheques the most, with 88 % of EU cheques issued in France: cheques account for about 3 % of the number of transactions and less than 2 % of the value of transactions

in deposit money. The **instant credit transfer**, launched in 2018, is beginning to carve out a place for itself, with an increase of 85 % in volume and of 138 % in value in 2022, according to the Banque de France (2023c).

Finally, despite the strong growth of electronic payments, **cash** continues to play an important role in France, the Banque de France (2023c) qualifies. In 2022, it accounted for a little more than 20 % of transactions at the point of sale, although this figure is down on the 34 % recorded in 2012. Cash withdrawals have, however, risen by 7 % in amount, reaching 133 billion euros, indicating a resilience of cash, motivated in part by hoarding needs or needs linked to tourism, according to the Banque de France (2023c).

### Key Takeaways



Historically, money first took the form of a commodity endowed with intrinsic usefulness (cattle, precious metals, shells) [Delaplace (2021a)], before precious metals, prized for their divisibility, malleability and unalterability, imposed a lasting dominance [Malthus (1846)]. The evolution of metallic forms — from weighed money to struck money, from bimetallism to monometallism — progressively concentrated monetary issue in the hands of the political authorities, consolidating a sovereign monopoly. The introduction of fiat money, devoid of intrinsic value but resting on collective confidence in the issuing authority, marked a decisive break [Delaplace (2021a)]. This confidence was institutionalised through the granting of legal tender status to banknotes and through seigniorage — the income derived from the difference between the face value and the production cost of physical money.

#### 1.1.3 Monetary aggregates

As the Banque de France (2023b) defines it, the **money supply** of an economy corresponds to the total quantity of money in circulation within that economy. Besides money directly in circulation, the money supply also includes financial investments which can rapidly be converted into cash, that is to say assets which can easily be used by agents without suffering any depreciation. The money supply, whether it consists of money in circulation or of liquid financial investments, represents the overall stock of money held by economic agents at a given moment. This stock is assessed by means of a series of specific indicators, known as **monetary aggregates**. Monetary aggregates are key indicators used to measure the quantity of money in circulation in an economy. They are classified according to the liquidity of the assets they include, from the most liquid to the least liquid.

- The monetary aggregate  $M_1$ , which represents the most liquid forms of money, consists of currency in circulation (banknotes and coins) and overnight deposits. In December 2022, according to the Banque de France (2023b),  $M_1$  stood at 11 404 billion euros in the euro area, with 9859 billion euros of overnight deposits and 1545 billion euros of banknotes and coins in circulation.
- The aggregate  $M_2$ , for its part, encompasses  $M_1$  together with interest-bearing deposits easily transformable into money, such as savings accounts and time deposits with a maturity of up to two years. In December 2022, according to the Banque de France (2023b),  $M_2$  reached 15 336 billion euros, reflecting the importance of three-month deposits (2555 billion euros) and two-year deposits (1377 billion euros) in the economy of the euro area Banque de France (2023b).
- Finally, the aggregate  $M_3$ , the broadest one, includes  $M_2$  and adds financial instruments such as two-year debt securities, shares in money market funds (*organismes de placement collectif*, OPC), and securities sold under repurchase agreements. In 2022, the Banque de France (2023b) estimated that  $M_3$  stood at 16 151 billion euros, including 50 billion euros of two-year debt securities and 655 billion euros of money market fund shares.

In parallel, the **monetary base**, also known as the aggregate  $M_0$  or central bank money, constitutes the foundation of the money supply in circulation. In December 2022, the monetary base of the euro area was 6280 billion euros, made up of currency in circulation, the required and excess reserves deposited by commercial banks with the central bank and the deposit facilities, according to the figures of the Banque de France (2023b). Care is therefore needed: while the notes and deposits in circulation also belong to  $M_1$  (and therefore to  $M_2$  and to  $M_3$ ), by contrast the required reserves (214 billion euros in December 2022) and the deposit facilities (4521 billion euros in December 2022) belong to none of the three aggregates mentioned above.

### Exercise



**Computing monetary aggregates.** In December 2022, the amount of banknotes and coins in circulation held by the private sector (excluding monetary financial institutions) and general government (excluding central government)

in the euro area stood at 1545 billion euros (GEUR), that of overnight deposits at 9859 GEUR, that of three-month deposits at 2555 GEUR, that of two-year deposits at 1377 GEUR, that of two-year debt securities at 50 GEUR, that of securities sold under repurchase agreements at 110 GEUR, and money market fund (OPC) shares at 655 GEUR. Moreover, required and excess reserves stood at 214 GEUR while deposit facilities amounted to 4521 GEUR. Define and then compute the various monetary aggregates defined by the European Central Bank (ECB) as well as the monetary base of the euro area.

**Solution.** The money supply of the euro area represents the quantity of money in circulation within the economy of the euro zone, used by economic actors for their transactions and their financing. It also encompasses financial investments easily convertible into cash, ready to be used by economic actors without any depreciation of value. This money supply, whether in circulation or in the form of liquid investments, constitutes a stock held by economic actors at a given moment. It is measured through a series of three main monetary aggregates:

- The aggregate M1 encompasses the most liquid forms of money, notably currency in circulation (banknotes and coins in circulation) and deposit money (money recorded in the form of figures in an overnight deposit account) immediately usable through various means of payment. In December 2022, the aggregate M1 therefore corresponds to the sum of banknotes and coins in circulation (1545 GEUR) and overnight deposits (9859 GEUR), that is 11 404 GEUR.
- The aggregate M2 encompasses M1 and adds to it interest-bearing deposits (such as *livrets A*, *comptes d'épargne-logement*, *livrets de développement durable et solidaire*, and time deposits with a maturity of up to two years, for instance) regarded as liquid, that is to say easily transferable to an overnight deposit account for use in the economy. In December 2022, the aggregate M2 therefore totals the sum of M1 (11 404 GEUR), three-month deposits (2555 GEUR) and two-year deposits (1377 GEUR), that is 15 336 GEUR.
- As for the aggregate M3, it represents the broadest aggregate, encompassing M2 and financial instruments issued by financial institutions and subscribed by savers and investors for terms of less than two years. These assets are regarded as relatively available and transferable to an overnight deposit account so as to be used by economic actors. In December 2022, the aggregate M3 therefore aggregates the sum of M2 (15 336 GEUR), money market fund shares (655 GEUR), securities sold under repurchase agreements (110 GEUR) and two-year debt securities (50 GEUR), that is 16 151 GEUR.
- In parallel with this money held by economic actors, there also exists the monetary base, also known as the aggregate M0 or central bank money. This monetary base is composed of currency in circulation (1545 GEUR) to which are added the required and excess reserves that commercial banks deposit with the central bank (214 GEUR), as well as the deposit facilities (4521 GEUR). The monetary base, which therefore represents 6280 GEUR in December 2022, feeds the money supply through currency in circulation and influences the creation of deposit money by commercial banks. □

### Short-Answer Question



**Variations in the money supply and their effects.** The money supply designates the whole set of means of payment held by non-financial economic agents, measured by hierarchically ordered aggregates (M1, M2, M3) which reflect increasing degrees of liquidity. According to Trading Economics (2025b), the broad monetary aggregate M3 of the euro area reaches 17.2 trillion euros in November 2025, growing by 3 % year on year, after having experienced a deceleration in 2023 when the growth rate had fallen to 0.6 %, and even to negative values for the aggregate M1 (−8 % in June 2023 according to Trading Economics (2025a)). These large-scale fluctuations raise questions about the mechanisms through which variations in the quantity of money in circulation affect economic activity, prices and the allocation of resources. Now according to Friedman (1970), ‘inflation is always and everywhere a monetary phenomenon’, which suggests a direct causal relationship between excessive monetary growth and a general rise in prices. However, the recent empirical work of Ireland (2024) shows that this relationship has weakened considerably since the 1980s in the advanced economies, notably because of the instability of the velocity of circulation of money and of the long and variable lags in monetary transmission. The problem therefore lies at the intersection between the theoretical validity of the quantity equation and the empirical constraints which limit its operational scope. To what extent do variations in the money supply produce the effects expected by the quantity theory, and what institutional and cyclical conditions moderate their magnitude? This question invites us to examine first the effects predicted by the quantity theory on inflation, output and the allocation of resources, before showing that these



effects depend on the velocity of circulation, on the structure of transmission lags and on the macroeconomic context.

Variations in the money supply exert, according to the quantity theory inherited from Friedman (1970), long-run effects on the general price level and, in the short run, on real activity through the channel of nominal rigidities and expectations. The quantity equation  $MV = PY$ , where  $M$  denotes the money supply,  $V$  the velocity of circulation,  $P$  the price level and  $Y$  real output, postulates that an increase in  $M$ , at constant velocity, translates proportionally into a rise in  $PY$ . The study by Jung (2024) covering 150 years of monetary data in the industrialised countries confirms that, over long horizons (beyond two years), a robust cointegration relationship exists between excess monetary growth and inflation, with a median adjustment lag of about two years; the results of panel cointegration tests show that this relationship remains valid when sufficiently long series are used and when cross-sectional dependencies between countries are controlled for. On the empirical level, the euro area episode between 2008 and 2012 illustrates this mechanism: the growth of M3 fell from more than 12 % in 2007 to values close to zero in 2010 according to Trading Economics (2025b), contributing to keeping inflation under control despite very accommodative interest rate policies. Conversely, the sharp acceleration of M3 observed in 2020 and 2021 (growth rates above 10 %) preceded by a few quarters the rise of inflation to levels unprecedented since the 1980s, reaching more than 10 % in the euro area at the end of 2022. Moreover, variations in the money supply also affect real activity in the short run via the credit and interest rate channel: a monetary expansion increases bank liquidity, reduces real interest rates and stimulates investment and consumption, which translates into a temporary rise in output as long as prices have not fully adjusted. The work of Ireland (2024) recalls that this transmission operates with long and variable lags, in line with the Friedmanite prediction, which complicates the operational use of monetary aggregates in the conduct of contemporary monetary policy.

While the quantity theory establishes a long-run link between money and inflation, the effects of monetary variations depend crucially on the evolution of the velocity of circulation, itself influenced by saving behaviour, financial innovations and macroeconomic shocks. As the analysis of the Banque de France (2023a) shows, the velocity of circulation of money (the ratio of nominal GDP to M3) has fallen sharply in the euro area since 2008, going from about 2 to less than 1.5 in 2020, and has stabilised at this low level since then. This structural decline means that a given increase in the money supply no longer translates into a proportional rise in nominal demand, because a growing share of money is hoarded or held for precautionary rather than transaction motives. The data of Banque de France (2024f) confirm that the growth of note issuance is accompanied by an increased importance of domestic hoarding and of foreign demand, which mechanically induces a lower velocity of circulation. This instability of  $V$  invalidates the constancy assumption which underpins the quantity equation and explains why the quantitative easing policies conducted after 2008 did not trigger inflation until 2021: the massive increase in M3 was absorbed by a parallel fall in the velocity of circulation. Moreover, Jung (2024) emphasise that the heterogeneity between countries in the speed of adjustment of inflation to monetary growth reflects institutional differences in the structure of the banking system, in the degree of wage indexation and in the credibility of monetary policy. Finally, the real effects of monetary variations depend on the macroeconomic context: in a situation of underemployment of productive capacity, a monetary expansion stimulates output more than prices (Keynesian effect), whereas in a situation of full employment, it translates mainly into inflation (long-run monetary neutrality). Recent euro area data illustrate this duality: the strong growth of M3 in 2020 first supported the post-Covid recovery in activity, before translating into an inflationary surge once capacity was saturated in 2021–2022.

In conclusion, variations in the money supply do therefore produce effects on inflation and on real activity, but these effects depend decisively on the evolution of the velocity of circulation, on the length and variability of transmission lags, as well as on the macroeconomic context within which monetary policy is set. The work of Friedman (1970) and the recent empirical analyses of Jung (2024) confirm the existence of a robust long-run relationship between excess monetary growth and inflation, validating the core of the quantity theory over multi-year horizons. However, the analysis of the Banque de France (2023a) shows that the velocity of circulation is not constant but has been declining structurally since 2008, which weakens the transmission of monetary impulses to nominal demand and explains the temporary decoupling observed between monetary expansion and inflation during the 2010s. The challenge for central banks is then to maintain a rate of monetary growth compatible with the objective of price stability (close to 2 %) while taking account of unpredictable variations in the velocity of circulation and in transmission lags.

### Key Takeaways



Contemporary money is divided between physical money (banknotes and coins) and deposit money (bank deposits, credit transfers, electronic payments), the latter dominating in quantitative terms. The quantity of money in circulation is apprehended through the monetary aggregates —  $M_1$  (the most liquid: currency in circulation –

banknotes, coins – and overnight deposits),  $M_2$  (interest-bearing deposits easily transformable into money),  $M_3$  (the least liquid: debt securities of at least two years, money market fund shares...) — which include assets of decreasing liquidity. The monetary base ( $M_0$ ), made up of central bank money (currency in circulation, required and excess reserves of commercial banks and deposit facilities), provides the foundation for the whole system, although only currency in circulation enters into the money supply aggregates *supra* [Banque de France (2023b)].

## 1.2 The demand for money

Money does not circulate in the economy of its own accord: its presence in circulation results from the decisions of agents who choose to hold it rather than other assets. Now this question — how much money do agents wish to keep, and on what conditions? — has divided economists for two centuries and structures two antagonistic visions of the monetary economy. On the one hand, the classical quantity theory sets out a simple answer: the demand for money is neutral, determined by transaction needs and stable over time; it therefore affects prices only proportionally, without any influence on output or employment. On the other hand, Keynesian analysis reveals a more nuanced reality: the demand for money responds to three distinct motives — transaction, precaution, speculation — the last of which introduces uncertainty and instability into the system. This distinction explains why an increase in the money supply may, depending on the circumstances, translate into a mere rise in prices or into a real stimulation of activity.

### 1.2.1 The classical analysis of the neutrality of money: the quantity theory

The quantity theory of money rests on the principle that the quantity of money in circulation is a key determinant of the general price level in an economy. According to this theory, popularised by Friedman (1956), **money is neutral in the long run** and does not affect real variables, such as output or employment: it serves only to facilitate exchange, without modifying economic fundamentals; ‘money is a veil’, as Jean-Baptiste Say (1767-1832) put it. From this perspective, any variation in the money supply entails a proportional variation in prices, which means that the quantity of money in circulation determines only the absolute level of prices, without any impact on the relative prices of goods and services. This classical conception assumes that the volume of transactions, regarded as a real datum, remains independent of the money supply, thereby confirming the neutrality of money. The most rigorous formulation of this theory is to be found in the equation of exchange developed by Fisher (1911): he proposes that the relationship between the money supply ( $M$ ), the velocity of circulation of money ( $V$ ), the general price level ( $P$ ), and the volume of transactions ( $T$ ) be expressed by the equation  $MV = PT$ , where the **velocity of circulation of money** ( $V$ ) designates the number of times a monetary unit is used in transactions during a given period. This relationship expresses the fact that there is at all times equality between the flow of money in circulation during the period ( $MV$ ) and the value at current prices of the transactions carried out over that period ( $PT$ ): in the total of all the exchanges of a year, the total value of the money paid ( $MV$ ) is equal to the total value of the goods purchased ( $PT$ ). Fisher (1911) assumes that  $V$  is relatively constant in the short run, so that any variation in  $M$  directly affects  $P$ , the general price level. It is also necessary to assume that the volume of goods and services available, that is to say the level of output, is relatively stable. From then on, an increase in the money supply mechanically entails a proportional increase in the price level, since  $T$  is likewise assumed to be constant in the short run. However, as Couppey-Soubeyran (2012a) qualifies, neither of these two assumptions goes without saying: the constancy of  $V$  requires monetary behaviour to be stable and predictable and the demand for money to grow just at the pace of the expenditure to be made, while the relative stability of the volume of goods and services implies full employment of the available factors of production: it is only under these conditions that monitoring an aggregate of money representative of the money supply constitutes a good indicator of inflationary pressures.

#### Focus...



**How is the velocity of circulation of money to be measured?** The velocity of circulation  $V$  of money is, in the tradition of Fisher (1911), defined by the equation of exchange  $MV = PT$ , where  $M$  denotes the average outstanding stock of money over the period,  $P$  the general price level and  $T$  the volume of all the transactions carried out. From this one derives the theoretical definition  $V = \frac{PT}{M}$ . This magnitude is, however, purely conceptual, since in practice one observes neither the whole set of transactions  $T$ , nor the instantaneous velocity of money. This is why, as Bordes et al. (2007) explains, recourse is had to an empirical indicator by replacing  $PT$  with nominal GDP and  $M$  with a broad monetary aggregate, which leads to the ratio  $V_{PIB, M_3} = \frac{PIB}{M_3}$ , which is no more than a proxy for the velocity of circulation in the theoretical sense.

This ratio is not an exact measure, for several reasons. First, nominal GDP does not coincide with the value of the whole set of transactions PT: it retains only the production of final goods and services on the territory and excludes numerous operations for which money does effectively circulate (transactions in financial assets, resales of existing real estate, sales of second-hand goods, intermediate exchanges internal to firms, and so on). Next, the aggregate  $M_3$  is not reducible to 'transaction money' alone: besides  $M_1$  (banknotes, coins, overnight deposits), it comprises time deposits, highly liquid savings instruments, money market fund shares and other short-term claims. A significant part of  $M_3$  is therefore held for investment or store-of-value motives, and takes only a marginal part in the settlement of the expenditure recorded in GDP, so that relating the flow of GDP to this composite stock could not provide the exact velocity of a homogeneous category of money. Moreover, a non-negligible fraction of the transactions entering into GDP is carried out with the help of credit instruments and clearing mechanisms which do not translate in any simple way into variations in  $M_3$ , so that the monetary flow effectively mobilised exceeds or diverges from what the aggregate suggests.

Finally, measurement considerations reinforce the approximate character of the ratio  $\frac{PIB}{M_3}$ . The velocity of circulation refers conceptually to a number of rotations per unit of time, whereas what is being divided here is a flow over a period (annual or quarterly GDP) by a stock of money observed at a given date or averaged over the period, neglecting intra-period variations in  $M_3$ . Moreover, velocity is a concept dependent on the model retained (the formulation of Fisher (1911) in terms of transactions, the Cambridge approach in terms of average money holdings, and so on) and the choice of the monetary aggregate as well as of the reference variable (GDP rather than PT) rests on institutional and financial assumptions which are never perfectly verified. Under these conditions, the ratio  $\frac{PIB}{M_3}$  must be interpreted as a practical indicator of trend, useful for tracking the overall evolution of monetary circulation, but not as the exact measure of the velocity of circulation of  $M_3$  in the strictly theoretical sense.

The **Cambridge school** enriched this approach by formulating the demand for money as a function of agents' income. Marshall (1922) thus postulates that individuals hold a fraction  $k$  of their income in the form of money, and expresses this relationship by the equation  $M_d = kPY$ , where  $M_d$  represents the demand for money,  $P$  the price level,  $Y$  nominal income and  $k$  the proportion of income that agents wish to keep in the form of liquid balances. By coupling this relationship with the equation  $MV = PT$  and considering that  $T = \alpha Y$  (transactions represent a share  $\alpha$  of GDP), it follows that  $k = \alpha/V$ :  $k$  is therefore inversely proportional to the velocity of circulation of money, and may accordingly be assumed to be stable. This approach makes it possible better to understand agents' behaviour with regard to the demand for money and brings out the link between the money supply and purchasing power: according to this view, agents adjust their demand for money so as to maintain constant real balances, so that any increase in the price level entails a corresponding rise in the demand for money. More precisely, Pigou (1917) shows that if the price level rises, agents will seek to maintain their **real money balances**, that is to say their balances deflated by the general price level, so that their purchasing power does not vary (the demand for money will therefore increase): agents will thus increase the amount of their income which they hold in money balances, without for all that falling victim to **money illusion**, which Delaplace (2021c) defines as behaviour consisting in confusing an increase in money balances with an increase in purchasing power.

## Debate



**Is the velocity of circulation of money stable?** An analysis by Bordes et al. (2007) revealed that, from 2001 onwards, the growth of the monetary aggregate  $M_3$  had systematically exceeded the reference value of 4.5 % established by the ECB in order to ensure price stability. During this same period, inflation in the euro area nevertheless remained close to 2 % on average. This phenomenon raises questions as to the stability of the velocity of circulation of  $M_3$  and the reliability of the relationship between money and the price level: now the study by Bordes et al. (2007) shows precisely that the velocity of circulation of  $M_3$ , approximated by the ratio of GDP to  $M_3$ , and the monetary aggregate itself are subject to shocks of a permanent and not a transitory nature. More precisely, since the 1970s, the velocity of circulation of  $M_3$  has followed a downward trend marked by several breaks, notably during periods associated with major economic upheavals: **financial deregulation** in the 1980s, the crises of the European exchange rate mechanism at the beginning of the 1990s, and the changeover to the euro at the turn of the twenty-first century. These periods, particularly between 1992-1993 and 2000-2002, saw significant changes in the velocity of circulation, affecting both its level and its slope.

As Couppey-Soubeyran (2012a) emphasises, some researchers have proposed examining the influence of financial asset prices on the demand for money in order better to understand its evolution. This relationship is, however, complex and difficult to isolate. The **development of financial markets** has offered agents the possibility of substituting liquid and more remunerative financial assets for traditional monetary assets: this ought, in theory, to have reduced the demand for money and increased the velocity of circulation, at least until these new assets were



taken into account in the definition of money. However, contrary to these expectations, the demand for money has often increased, while the velocity of circulation has continued to fall, especially during periods of rupture: in periods of falling asset prices, as at the time of the 1987 stock market crash, the EMS crises of 1992-1993 or the collapse of Internet stocks in 2000, it is understandable that agents should have accumulated precautionary balances, thereby increasing the demand for money and slowing down the velocity of circulation. Another possible explanation lies in the **wealth effects** associated with the holding of financial assets. According to this theory, the holding of financial assets would strengthen the demand for money, thus contributing to the fall in the velocity of circulation. As Couppey-Soubeyran (2012a) analyses, this would presuppose the existence of a stable long-run relationship between interest rates, which influence these wealth effects, and the velocity of circulation. Yet the empirical data do not clearly confirm this hypothesis, making it difficult to identify any obvious correlation between these two variables.

Friedman (1956) goes further than the Cambridge school in the implications of the quantity theory for monetary policy: according to him, inflation is always and everywhere a monetary phenomenon, that is to say it results mainly from the excessive growth of the money supply relative to the supply of goods and services. More precisely, the quantity theory of money of the Cambridge school lays stress on the demand for money for the transaction motive: Marshall (1922) and Pigou (1917) developed an approach in which the demand for money is a function of nominal income and of the proportion of income that individuals wish to hold in the form of money (the proportion  $k$ ); from this perspective, money is above all regarded as a means of transaction and of short-term saving. By contrast, the **monetarist theory** of Friedman (1956) proposes a money demand function in which money is regarded as a good in its own right, possessing a return in terms of future purchasing power: the demand for money is then determined by several factors, notably **permanent income** (the expected measure of individuals' future incomes) and the interest rate. This theory puts forward the neutrality of money in the long run, asserting that changes in the money supply affect only nominal variables (such as prices), but not real variables (such as output or employment). *In fine*, while both theories link the money supply to the price level, the main difference between them lies in the role that each assigns to money and to its relationship with the economy. In monetarist theory, money is regarded as a key lever of macroeconomic regulation, with a direct influence on inflation in the long run: Friedman (1956) postulates that the stability of the money supply is crucial in order to control inflation, which results directly from an excessive growth of the money supply. By contrast, the Cambridge approach lays greater stress on the demand for money, influenced by income and by agents' preferences, without insisting as much on the stability of demand or on its direct impact on inflation. Thus, where Friedman (1956) sees money as an active and central factor in the regulation of the economy, the Cambridge school regards it mainly as a means of transaction, with a more passive role in inflationary dynamics.

## Debate



**Does raising wages create inflation?** The indexation of wages to inflation, which consists in adjusting remuneration in line with the evolution of consumer prices, gives rise to debate as to its potential impact on inflation. Historically, this practice was used in contexts of high inflation in order to protect households' purchasing power, as in 1952 in France, at the time of the introduction of the SMIG (the forerunner of the SMIC, the national minimum wage), when inflation exceeded 20 %, as Le Monde (2023) indicates. However, this method was abandoned in the 1980s, notably because of its role in amplifying the **wage-price spiral**: when wages rise in response to a rise in prices, this increases production costs, which in turn pushes firms to raise their prices, thereby fuelling further inflation. This phenomenon was observed after the oil shocks of 1973 and 1979, when inflation reached 13 % in 1980.

The economists of the Groupe d'experts sur le SMIC (2023) warn against a generalised reintroduction of wage indexation: in their view, this could harm the competitiveness of firms, notably because of the increase in labour costs. Although the SMIC in France remains indexed to inflation (rising automatically when inflation exceeds 2 %), these experts even recommend dissociating the SMIC from this indexation. The argument put forward is that this would risk weighing still more heavily on the wage bill, notably in the private sector, and bringing about negative consequences for employment. Other European Union countries, such as Belgium and Luxembourg, nevertheless maintain an indexation mechanism for all wages. In Belgium, for example, indexation led to a 9 % increase in wages in 2022, but this rise also raised fears of a loss of competitiveness for private sector firms.

There are, however, nuances in the analysis: the members of the France insoumise (2023), in their private member's bill before the National Assembly (sixteenth legislature), consider that the inflation of 2022-2023, largely fuelled by the rise in the prices of energy and of food commodities, affects above all low-income households, which justifies wage adjustments in order to protect their purchasing power; moreover, in their view, the effects of the wage-price spiral are not mechanical and may be attenuated by other economic policies, such as the control of firms' margins or the regulation of the prices of certain essential goods. Nevertheless, as Le Monde (2023) qualifies, past experience

shows that while generalised indexation may temporarily protect purchasing power, it also risks perpetuating an inflationary dynamic in the long run, notably in a context of slowing inflation such as that observed in France, where the rate fell from 6.2 % in October 2022 to 4 % a year later.

### Key Takeaways



The demand for money is not reducible to a neutral transaction phenomenon: it reflects optimising behaviour in the face of uncertainty and of the trade-offs between liquidity and return. The classical quantity theory postulates that money is neutral in the long run and that any increase in the money supply  $M$  entails a proportional rise in prices  $P$ , according to the equation  $MV = PT$  where the velocity of circulation  $V$ , that is to say the average number of transactions for which a single monetary unit is used ( $T$  denoting the total number of transactions), is assumed to be stable [Fisher (1911)]. The Cambridge school enriches this approach by formalising the demand for money  $M_d$  as a fraction  $k$  of nominal income  $Y$  ( $M_d = kPY$ ), which implies that agents adjust their real balances so as to preserve their purchasing power [Marshall (1922), Pigou (1917)]. The monetarism of Friedman (1956) generalises this logic by asserting that inflation is always a monetary phenomenon resulting from an excessive growth of the money supply.

#### 1.2.2 The overlapping generations model as a way of understanding the demand for money

The **overlapping generations model** (OLG), introduced by Samuelson (1958), constitutes a break with the canonical models of neoclassical growth by virtue of the temporal structure that it imposes on economic analysis. Unlike the **Ramsey model**, where all economic agents are present simultaneously over an infinite horizon, the analytical framework proposed by Samuelson (1958) rests on the fundamental assumption that new generations arrive perpetually in the economy, while older generations disappear after having lived a finite lifespan. This generational configuration makes it possible to understand why fiat money, which *a priori* possesses no intrinsic value, may nevertheless be accepted as a store of value and a medium of exchange in a decentralised economy. As Samuelson (1958) establishes, each young generation accepts the money of the old generation in exchange for consumption goods because it anticipates being able to use it in its turn to acquire goods when it reaches retirement age. This system of **reciprocal expectations** constitutes an equilibrium in which money retains a positive purchasing power in each period, although it produces no real return. The extension of the model proposed by Diamond (1965) enriches this structure by introducing the accumulation of physical capital and by making it possible to analyse the **intertemporal trade-offs between consumption and saving** within a framework where agents live for two distinct periods: a period of youth during which they work, receive a wage and decide on their level of saving, and a period of old age during which they consume the fruits of their accumulated saving. According to Diamond (1965), the saving decisions of the young generations determine the stock of capital available for production in the following period, which directly influences the future wages and interest rates that later generations will receive.

The demand for money within the framework of the overlapping generations model is expressed as a function of the anticipated purchasing power of money in the following period, as Weil (2008) and Diamond (1965) formalise it. Individuals do not value money for itself, but because it **facilitates intertemporal exchange** by serving as a store of value between youth and old age. The real demand for money thus depends on **anticipated inflation**: if the intertemporal elasticity of substitution is greater than unity, a rise in inflation reduces the demand for money because the substitution effect dominates, whereas if this elasticity is less than unity, the income effect may dominate and lead to an increase in the demand for money. The statistics of the European Central Bank published by Statista (2024a) reveal that the monetary aggregate M1 of the euro area, which comprises currency in circulation and overnight deposits, reached 10.57 trillion euros in December 2024, that is a modest increase on the previous year, while the aggregate M2, which further includes time deposits with a maximum maturity of two years and deposits redeemable at three months' notice, stood at 15.6 trillion euros at the end of 2024 according to Statista (2024b). These figures testify to the quantitative importance of the holding of liquid balances in developed economies, a phenomenon which the overlapping generations model makes it possible to justify theoretically without any *ad hoc* assumption about the direct utility of money. Moreover, as Weil (2008) emphasises, the velocity of circulation of money in the overlapping generations model may fluctuate as a function of inflation expectations and of the real interest rate, which contrasts with the simplified vision of the quantity theory of money where velocity is assumed to be constant. The empirical data compiled by the IMF and analysed by Park (1988) show that the variability of the velocity of circulation of money differs considerably across countries and periods, with significant correlations between the variability of monetary growth and that of the inflation rate in thirty countries studied over the period 1974-1985. This variability is explained notably by the adjustment lags between variations in the money supply and

changes in the price level, as well as by changes in the expectations of economic agents, all of them mechanisms that the overlapping generations model makes it possible to analyse rigorously.

### Exercise



**The OLG model and the demand for money.** At any moment, the total population is assumed to be equal to  $N$  (in other words, there are as many old people dying as there are young people being born). We consider that there is only one consumption good in the economy, which is non-storable: it is impossible here to prepare for one's retirement by storing the good for the future. It is assumed here that all agents receive an endowment of  $y$  units of the good when they are young, but none when they are old: there is therefore no national solidarity mechanism here. The only way for elderly people to prepare for their retirement will be to trade with the young of the following generation. The intertemporal preferences of the representative agent are given by  $W(c_1, c_2) = u(c_1) + \beta u(c_2)$ , where  $c_1$  is consumption in youth,  $c_2$  consumption in the old-age period and  $\beta$  the discount factor: this factor, assumed to lie between 0 and 1, measures the preference for present consumption: the closer  $\beta$  is to 0, the more utility is formed solely by consumption in youth; conversely, the closer  $\beta$  is to 1, the more present consumption and future consumption are placed on the same footing (that is to say, generate the same level of utility). Marginal utility is assumed to be positive ( $u' > 0$ ) and decreasing ( $u'' < 0$ ): the more one consumes, the more well-being one derives, but at an ever slower rate.

- (i) **CENTRALISED EQUILIBRIUM.** We shall first assume that the allocation of the economy (that is to say the parameter  $y$  mentioned above) is decided in a centralised manner by a benevolent planner who seeks to maximise social utility or welfare. Since agents are assumed to be identical, maximising the social utility of all agents amounts to maximising the utility of a single agent. The planner is likened to a 'benevolent dictator', since he has the power to dictate the quantities exchanged and consumed by all agents. We then denote by  $c_1^P$  the consumption, decided by the planner, of agents during their youth, and by  $c_2^P$  that, likewise decided by the planner, of agents in their old-age period.
  - (a) What is the feasibility constraint faced by the planner in allocating the endowments of goods in a representative period  $t$ ?
  - (b) Explain why it is optimal, for the planner, for the feasibility constraint to bind.
  - (c) Write the planner's programme, given that the decision variables are  $c_1$  and  $c_2$ .
  - (d) Use the binding feasibility constraint to express  $c_2$  as a function of  $c_1$  and rewrite the planner's programme with  $c_1$  as the only decision variable.
  - (e) Solve the social planner's programme, *i.e.* determine  $c_1^P$  and  $c_2^P$ . We shall consider here that  $u = \ln$ .
  - (f) Is one of the two consumption quantities chosen by the planner larger than the other? If so, why? What condition must be satisfied for  $c_1^P = c_2^P$  to hold?
  - (g) Explain why  $(c_1^P, c_2^P)$  may be called 'the optimal allocation'.
- (ii) **DECENTRALISED EQUILIBRIUM WITHOUT MONEY.** We now assume that the allocation of the economy is determined in a decentralised manner, that is to say without the intervention of any planner whatsoever: the market is self-regulating here. It is assumed that there is no money: we are necessarily in a barter economy. It is assumed in this context that agents may exchange the good in a perfectly competitive market (no agent has market power).
  - (a) What are the exogenous variables and the endogenous variables? Among the endogenous variables, which are the decision variables and which are the control variables?
  - (b) Compute the allocation of resources that prevails in equilibrium.
- (iii) **DECENTRALISED EQUILIBRIUM WITH MONEY.** As in the previous part, it is assumed that the allocation of resources is determined in a decentralised manner, but that money exists in the economy. We then consider a consolidated public authority (government and central bank). It is assumed that the money supply issued by the public authorities is constant and equal to  $M$  in each period (after the initial period of the economy, there are no further money issues). Agents may exchange the good and money in a perfectly competitive market. At date  $t$ , we denote by  $v_t \geq 0$  the price of one monetary unit in terms of goods; this means that  $p_t := 1/v_t$  is the unit price of the good in terms of money.

- (a) What are the exogenous variables and the endogenous variables? Among the endogenous variables, which are the decision variables and which are the control variables?
- (b) Compute the allocation of resources that prevails in a non-monetary autarkic equilibrium, that is to say when  $v_t = 0$  at every date  $t$ .
- (c) We therefore consider a monetary equilibrium, that is to say a situation in which the price of money in terms of goods is positive for every period  $t$ : we thus have  $v_t > 0$ .
  - i. What is the budget constraint of the representative agent in his first period of life? Does his constraint bind in equilibrium?
  - ii. What is the budget constraint of the representative agent in his second period of life? Does his constraint bind in equilibrium?
  - iii. What is the intertemporal budget constraint of the representative agent?
  - iv. What is the maximisation programme of the representative agent?
  - v. Solve this constrained maximisation programme.
- (d) We consider the monetary equilibrium just determined, namely  $c_1^* = \frac{y}{1+\beta}$  and  $c_2^* = \frac{v_{t+1}}{v_t} \frac{\beta y}{1+\beta}$ .
  - i. How does the optimal choice of  $c_1$  depend on  $y$ ?
  - ii. For given  $v_t$  and  $v_{t+1}$ , how does the optimal choice of  $c_2$  depend on  $y$ ?
  - iii. For given  $v_t$ , how does the optimal choice of  $c_2$  depend on  $v_{t+1}$ ?
  - iv. For given  $v_{t+1}$ , how does the optimal choice of  $c_2$  depend on  $v_t$ ?
  - v. Compute the optimal choice of  $m_t$  for the representative agent. Comment.
- (e) We complete the solution of the model by determining the missing values of the various variables.
  - i. Give the expression of aggregate supply and aggregate demand for money in a period  $t$ .
  - ii. Deduce from this the equilibrium condition of the money market which makes it possible to determine the price of money in terms of goods at date  $t$ . Comment.
  - iii. In an analogous way, determine the equilibrium condition of the money market which makes it possible to determine the price of money in terms of goods at date  $t + 1$ . Deduce from this the return on money between  $t$  and  $t + 1$ .
  - iv. Deduce from this the definitive solution for the optimal choice of second-period consumption  $c_2$ . Comment on the efficiency of the market allocation.

(iv) CONCLUSION

- (a) Conclude by comparing the autarkic equilibrium (part (ii)) and the monetary equilibrium (part (iii)).
- (b) Discuss the limitations of this exercise.

**Solution.** (i) CENTRALISED EQUILIBRIUM. We shall first assume that the allocation of the economy (that is to say the parameter  $y$  mentioned above) is decided in a centralised manner by a benevolent planner who seeks to maximise social utility or welfare. Since agents are assumed to be identical, maximising the social utility of all agents amounts to maximising the utility of a single agent. The planner is likened to a 'benevolent dictator', since he has the power to dictate the quantities exchanged and consumed by all agents. We then denote by  $c_1^P$  the consumption, decided by the planner, of agents during their youth, and by  $c_2^P$  that, likewise decided by the planner, of agents in their old-age period.

- (a) *What is the feasibility constraint faced by the planner in allocating the endowments of goods in a representative period  $t$ ?* In order to allocate the goods in the economy, the planner is constrained by resources: the feasibility constraint means that the total quantity of resources devoted to consumption in the two periods ( $Nc_1 + Nc_2$ : since agents are identical, each of the  $N$  agents indeed consumes  $c_1$  during his youth and  $c_2$  during his old-age period) cannot exceed the whole of the resources in the economy ( $Ny$ : indeed, each of the  $N$  individuals present receives an endowment  $y$  when he is young). *In fine*, the feasibility constraint is written  $Nc_1 + Nc_2 \leq Ny$ , that is to say  $c_1 + c_2 \leq y$ .
- (b) *Explain why it is optimal, for the planner, for the feasibility constraint to bind.* Suppose, for the sake of contradiction, that  $c_1 + c_2 < y$  at the optimum. This means that there remains a residual  $r := y - c_1 - c_2 > 0$

not consumed by the agent. But this remainder is of no use whatsoever: indeed, the form of the utility function  $W$  indicates that the agent derives well-being only from his consumption, and not from any saving he might hold. It would have been better to consume  $r$  by consuming more during youth (for instance), and he would then have generated a higher utility, since the function  $u$  is increasing: we have  $u(c_1 + r) + \beta u(c_2) > u(c_1) + \beta u(c_2)$ . This means that an allocation satisfying  $c_1 + c_2 < y$  cannot be optimal (since we have found a better one). By contradiction, we have therefore proved that at the optimum the feasibility constraint binds:  $c_1 + c_2 = y$ .

- (c) Write the planner's programme, given that the decision variables are  $c_1$  and  $c_2$ . The planner's problem consists in maximising the utility  $W(c_1, c_2) = u(c_1) + \beta u(c_2)$  of the representative agent subject to the binding feasibility constraint  $c_1 + c_2 = y$ . Recall that we denote by  $(c_1^P, c_2^P)$  the respective levels of consumption during youth and old age which make it possible to maximise the utility  $W(c_1, c_2)$  subject to the binding feasibility constraint. From then on, the planner's programme ( $\mathcal{P}$ ) is written:

$$(\mathcal{P}) : (c_1^P, c_2^P) := \begin{cases} \arg \max_{(c_1, c_2) \in \mathbb{R}^2} u(c_1) + \beta u(c_2) \\ \text{subject to: } c_1 + c_2 = y \end{cases}$$

- (d) Use the binding feasibility constraint to express  $c_2$  as a function of  $c_1$  and rewrite the planner's programme with  $c_1$  as the only decision variable. Consequently, we may write  $c_2 = y - c_1$  for any levels of consumption  $c_1$  and  $c_2$ . From then on, the constrained maximisation programme ( $\mathcal{P}$ ) may be rewritten, by reintegrating the constraint into the objective function, as follows:

$$(\mathcal{P}) \iff \begin{cases} c_1^P := \arg \max_{c_1 \in \mathbb{R}} u(c_1) + \beta u(y - c_1) \\ c_2^P := y - c_1^P \end{cases}$$

- (e) Solve the social planner's programme, i.e. determine  $c_1^P$  and  $c_2^P$ . We shall consider here that  $u = \ln$ . The programme ( $\mathcal{P}$ ) thus rewritten is a simple programme for maximising a function of a single variable. In order to determine  $c_1^P$ , we solve the first-order condition. Now  $\frac{d(u(c_1) + \beta u(y - c_1))}{dc_1} = 0$  if, and only if,  $u'(c_1) - \beta u'(y - c_1) = 0$ , if, and only if,  $\frac{u'(c_1)}{u'(y - c_1)} = \beta$ , if, and only if, given the constraint  $c_2 = y - c_1$ ,  $\frac{u'(c_1)}{u'(c_2)} = \beta$ . This condition, sometimes known as the 'Euler equation', indicates that at the optimum  $u'(c_1^P) = \beta u'(c_2^P)$ , and therefore that the consumption of an additional unit during youth is 'beta times more important' for the agent than the consumption of an additional unit during the old-age period; here we recover the parameter of preference for present consumption. In particular, if  $u = \ln$ , we know that the derivative of the logarithm function is the reciprocal function: the Euler equation in this case is then written  $c_2^P / c_1^P = \beta$ , that is to say  $c_2^P = \beta c_1^P$ . It remains to determine the values of  $c_1^P$  and  $c_2^P$ ; we then recall the feasibility constraint. We have:

$$\begin{aligned} \begin{cases} c_2^P = \beta c_1^P & (\text{Euler}) \\ c_2^P = y - c_1^P & (\text{feasibility}) \end{cases} & \iff \begin{cases} y - c_1^P = \beta c_1^P \\ c_2^P = y - c_1^P \end{cases} \\ & \iff \begin{cases} (1 + \beta)c_1^P = y \\ c_2^P = y - c_1^P \end{cases} \\ & \iff \begin{cases} c_1^P = \frac{y}{1 + \beta} \\ c_2^P = y - \frac{y}{1 + \beta} = \frac{\beta y}{1 + \beta} \end{cases} \end{aligned}$$

- (f) Is one of the two consumption quantities chosen by the planner larger than the other? If so, why? What condition must be satisfied for  $c_1^P = c_2^P$  to hold? Recall that  $\beta$  lies between 0 and 1 and that  $c_2^P = \beta c_1^P$ . Consequently, one consumes more during youth than during the old-age period: here again we recover the idea, already mentioned in the previous question, that present consumption is preferred to future consumption. The equality  $c_2^P = c_1^P$  will hold if, and only if,  $\beta = 1$ : in this last case, future consumption would be placed on the same footing as present consumption.
- (g) Explain why  $(c_1^P, c_2^P)$  may be called 'the optimal allocation'. What is the problem that the social planner has solved? He has determined the allocation which made it possible to obtain the maximum level of utility of the representative agent, and therefore the maximum level of utility for society as a whole: no other possible allocation of resources can increase social welfare. It is in this sense that the allocation  $(c_1^P, c_2^P)$  may be described as the optimal allocation.

(ii) **DECENTRALISED EQUILIBRIUM WITHOUT MONEY.** We now assume that the allocation of the economy is determined in a decentralised manner, that is to say without the intervention of any planner whatsoever: the market is self-regulating here. It is assumed that there is no money: we are necessarily in a barter economy. It is assumed in this context that agents may exchange the good in a perfectly competitive market (no agent has market power).

(a) *What are the exogenous variables and the endogenous variables? Among the endogenous variables, which are the decision variables and which are the control variables?* The exogenous variables are the data of the statement, over which agents have no power of modification or decision. Among these variables are the discount factor  $\beta$ , that is to say the preference for present consumption, which is assumed to be given, but also the size of the population  $N$  and the utility function  $u$ , characteristic of agents' preferences (like the parameter  $\beta$ ), which is assumed not to be controlled by them. As regards the endogenous variables, that is to say those which will be determined by agents in the light of the overlapping generations model, there are the levels of present consumption  $c_1$  and of future consumption  $c_2$ : these two parameters are determined by agents; they are decision variables.

(b) *Compute the allocation of resources that prevails in equilibrium.* Here, there is no money. The only possibility for exchange is therefore barter. However, barter is possible only if there is an exchange of one good against another. Now, when they are old, individuals would be tempted to exchange goods with the young of the following generation. But elderly people receive no endowment, and they have been unable to store the endowment acquired in the present, since the good is non-storable. They therefore have nothing to offer the young: no exchange can take place. Necessarily, the young consume their whole endowment  $y$  during their youth, and find themselves with nothing in old age. It may be said that we are faced with a market failure. This is the reason why we shall now consider a monetary equilibrium: the good may be exchanged against a financial counterpart.

(iii) **DECENTRALISED EQUILIBRIUM WITH MONEY.** As in the previous part, it is assumed that the allocation of resources is determined in a decentralised manner, but that money exists in the economy. We then consider a consolidated public authority (government and central bank). It is assumed that the money supply issued by the public authorities is constant and equal to  $M$  in each period (after the initial period of the economy, there are no further money issues). Agents may exchange the good and money in a perfectly competitive market. At date  $t$ , we denote by  $v_t \geq 0$  the price of one monetary unit in terms of goods ('one euro is worth two bananas'); this means that  $p_t := 1/v_t$  is the unit price of the good in terms of money ('one banana costs 50 cents': it is this definition that we are more usually accustomed to using).

(a) *What are the exogenous variables and the endogenous variables? Among the endogenous variables, which are the decision variables and which are the control variables?* The exogenous variables are the data of the statement, over which agents have no power of modification or decision. Among these variables are the discount factor  $\beta$ , that is to say the preference for present consumption, which is assumed to be given, but also the size of the population  $N$  and the utility function  $u$ , characteristic of agents' preferences (like the parameter  $\beta$ ), which is assumed not to be controlled by them. As regards the endogenous variables, that is to say those which will be determined by agents in the light of the overlapping generations model, there are the levels of present consumption  $c_1$  and of future consumption  $c_2$ : these two parameters are determined by agents; they are decision variables. But, unlike in the second part, new endogenous variables appear: the nominal demand for money  $m_t$  that the young of generation  $t$  hold in order to buy goods from the young of generation  $t + 1$  when they themselves are old; the price of money in terms of goods  $v_t$  (or the price of the good in monetary terms  $p_t = 1/v_t$ ). While the nominal demand for money is likewise a decision variable, the price  $v_t$ , for its part, is an equilibrium variable: its value results from interactions on the market and is not decided by any agent individually.

(b) *Compute the allocation of resources that prevails in a non-monetary autarkic equilibrium, that is to say when  $v_t = 0$  at every date  $t$ .* Unlike in the second question of the second part, money exists here. On the other hand, the price of money is zero: money can be acquired at will without any counterpart; why pay for a good if money can be acquired without doing anything? Therefore, if monetary exchange there be, that exchange has no value whatsoever: nobody, then, wishes to use money. We therefore return *de facto* to the non-monetary equilibrium of the second part: agents consume the whole of their endowment during their youth and find themselves with nothing during their old-age phase. This example is found in situations of hyperinflation: money is worth nothing (people paper their walls with banknotes); why use

it when it has no value? In Venezuela, since the hyperinflation crisis, the bolívar has been shunned while a dollarisation of the economy is observed. . .

- (c) We therefore consider a monetary equilibrium, that is to say a situation in which the price of money in terms of goods is positive for every period  $t$ : we thus have  $v_t > 0$ ; money is worth something.
- i. *What is the budget constraint of the representative agent in his first period of life? Does his constraint bind in equilibrium?* When the agent is young, he has a total resource  $y$  (initial endowment, in terms of goods). He then has two possibilities: to consume a part  $c_1$  of it (in terms of goods) and to sell the other part to the elderly people of the following generation against a monetary counterpart  $m_t$ , which corresponds to  $v_t m_t$  goods (since  $v_t$  is the price of money in terms of goods). The budget constraint (expressed in terms of goods) is then written  $c_1 + v_t m_t \leq y$ . Suppose, for the sake of contradiction, that  $c_1 + v_t m_t < y$ . This means that there remains to the young agent a residual  $r := y - c_1 - v_t m_t$ . This could be used to consume more (indeed, it cannot be stored for future consumption  $c_2$ , since the good is perishable): if it is not consumed, this residual is therefore lost. As previously, he may as well make his budget constraint bind so as to maximise his utility: we must therefore have at the optimum  $c_1 + v_t m_t = y$ .
  - ii. *What is the budget constraint of the representative agent in his second period of life? Does his constraint bind in equilibrium?* When the agent is old, his only resource is the money  $m_t$  that he has kept thanks to the sale of what he did not consume during his youth. But the price of money in terms of goods has varied: we are now in period  $t + 1$ , so  $m_t$  monetary units make it possible to buy  $v_{t+1} m_t$  goods. From then on, the second-period budget constraint is written  $c_2 \leq v_{t+1} m_t$ . Here again, in equilibrium, the budget constraint binds: there is no valuation of the bequest that one might leave to children; one may as well consume everything since one will be dead in the following period. We therefore have at the optimum  $c_2 = v_{t+1} m_t$ .
  - iii. *What is the intertemporal budget constraint of the representative agent?* When we turn to the intertemporal budget constraint, we must take account of the two constraints mentioned above,  $c_1 + v_t m_t = y$  and  $c_2 = v_{t+1} m_t$ . In order to combine these two constraints, we use the fact that  $m_t$  appears in both. We have on the one hand  $m_t = (y - c_1)/v_t$  and on the other hand  $m_t = c_2/v_{t+1}$ . We therefore have  $(y - c_1)/v_t = c_2/v_{t+1}$ , that is to say  $(y - c_1)/c_2 = v_t/v_{t+1}$ , or again  $v_{t+1}y - v_{t+1}c_1 = v_t c_2$  or  $v_{t+1}c_1 + v_t c_2 = v_{t+1}y$ , that is to say  $c_1 + (v_t/v_{t+1})c_2 = y$ .
  - iv. *What is the maximisation programme of the representative agent?* Fundamentally, the representative agent seeks to maximise his intertemporal utility subject to the budget constraints of the two periods  $c_1 + v_t m_t \leq y$  and  $c_2 \leq v_{t+1} m_t$ . We may therefore write the maximisation programme ( $\mathcal{P}'$ ) as follows:

$$(\mathcal{P}') : (c_1^*, c_2^*) := \begin{cases} \arg \max_{(c_1, c_2) \in \mathbb{R}^2} u(c_1) + \beta u(c_2) \\ \text{subject to: } c_1 + v_t m_t \leq y \text{ and } c_2 \leq v_{t+1} m_t \end{cases}$$

But, according to the three previous questions, we have proved that at the optimum these two budget constraints bind. This has allowed us to summarise the two constraints in a single intertemporal constraint  $c_1 + (v_t/v_{t+1})c_2 = y$ . From then on, the constrained maximisation programme ( $\mathcal{P}'$ ) may therefore be rewritten as follows:

$$(\mathcal{P}') \iff (c_1^*, c_2^*) = \begin{cases} \arg \max_{(c_1, c_2) \in \mathbb{R}^2} u(c_1) + \beta u(c_2) \\ \text{subject to: } c_1 + (v_t/v_{t+1})c_2 = y \end{cases}$$

- v. *Solve this constrained maximisation programme.* We may first express  $c_2$  as a function of  $c_1$  according to the intertemporal budget constraint so as to reduce the problem to an unconstrained maximisation problem in a single variable  $c_1$ . Indeed,  $c_1 + (v_t/v_{t+1})c_2 = y$  if, and only if,  $c_2 = (y - c_1)(v_{t+1}/v_t)$ . It then follows that:

$$(\mathcal{P}') \iff \begin{cases} c_1^* := \arg \max_{c_1 \in \mathbb{R}} u(c_1) + \beta u((y - c_1)(v_{t+1}/v_t)) \\ c_2^* := (y - c_1^*)(v_{t+1}/v_t) \end{cases}$$

In order to determine  $c_1^*$ , we solve the first-order condition. Now  $\frac{d(u(c_1) + \beta u((y - c_1)(v_{t+1}/v_t)))}{dc_1} = 0$  if, and only if,  $u'(c_1) - \beta(v_{t+1}/v_t)u'((y - c_1)(v_{t+1}/v_t)) = 0$ , if, and only if, given the constraint  $c_2 =$

$(y - c_1)(v_{t+1}/v_t)$ ,  $u'(c_1) - \beta(v_{t+1}/v_t)u'(c_2) = 0$ , if, and only if,  $\frac{u'(c_1)}{u'(c_2)} = \beta(v_{t+1}/v_t)$  (the 'Euler equation'). Under the assumption that  $u = \ln$ , we therefore find as Euler equation the relation  $\frac{c_2}{c_1} = \beta \frac{v_{t+1}}{v_t}$ . It remains to determine the values of  $c_1^*$  and  $c_2^*$ . We have:

$$\begin{aligned} \begin{cases} c_2^* = \beta c_1^* v_{t+1}/v_t & \text{(Euler)} \\ c_2^* = (y - c_1^*)(v_{t+1}/v_t) & \text{(feasibility)} \end{cases} &\iff \begin{cases} (y - c_1^*)(v_{t+1}/v_t) = \beta c_1^* v_{t+1}/v_t \\ c_2^* = \beta c_1^* v_{t+1}/v_t \end{cases} \\ &\iff \begin{cases} y - c_1^* = \beta c_1^* \\ c_2^* = \beta c_1^* v_{t+1}/v_t \end{cases} \\ &\iff \begin{cases} c_1^* = \frac{y}{1+\beta} \\ c_2^* = \frac{v_{t+1}}{v_t} \frac{\beta y}{1+\beta} \end{cases} \end{aligned}$$

(d) We consider the monetary equilibrium just determined, namely  $c_1^* = \frac{y}{1+\beta}$  and  $c_2^* = \frac{v_{t+1}}{v_t} \frac{\beta y}{1+\beta}$ .

- i. *How does the optimal choice of  $c_1$  depend on  $y$ ?* We observe that  $c_1^*$  increases with  $y$ , and therefore with the initial endowment. This is logical: the higher the endowment initially received by the agent, the more goods the agent has in the first period: he can consume more in both periods (since he is 'richer'), and therefore, in particular, he can consume more in the first period.
- ii. *For given  $v_t$  and  $v_{t+1}$ , how does the optimal choice of  $c_2$  depend on  $y$ ?* Here again,  $c_2^*$  increases with  $y$ . The individual who receives more in the first period sees his wealth increase. In particular, he can sell his surplus in the first period so as to enable himself to acquire more goods in the second period, and therefore to consume more in old age. More precisely, the function  $u$  is concave (since  $u'' < 0$ ): this means in our case that total utility will be higher if consumption is increased in both periods at the same time rather than consumption in a single period. Indeed, since marginal utility  $u'$  is decreasing, increasing only  $c_1$  by  $x$  or only  $c_2$  by  $x$  (where  $x$  is any positive real number) yields less utility than increasing  $c_1$  by  $x/2$  and  $c_2$  by  $x/2$  (for instance).
- iii. *For given  $v_t$ , how does the optimal choice of  $c_2$  depend on  $v_{t+1}$ ?* If  $v_{t+1}$  increases, all other things being equal,  $c_2^*$  increases as well. Indeed, the higher  $v_{t+1}$  is, that is to say the higher the price of money in terms of goods in period  $t + 1$ , the more goods the money held by agents allows them to buy: hoarding permits a greater consumption of goods in old age.
- iv. *For given  $v_{t+1}$ , how does the optimal choice of  $c_2$  depend on  $v_t$ ?* If  $v_t$  increases, all other things being equal,  $c_2^*$  decreases. Indeed, the higher  $v_t$  is, that is to say the higher the price of money in terms of goods in period  $t$ , the more agents are compelled to sell goods in order to acquire more money. For a given initial endowment, young agents are therefore less able to acquire monetary units if  $v_t$  increases. If they have acquired fewer monetary units in their youth, they will have fewer possibilities of purchasing goods in old age: their future consumption therefore contracts.
- v. *Compute the optimal choice of  $m_t$  for the representative agent. Comment.* We showed previously that  $m_t = c_2/v_{t+1}$ . At the optimum, we therefore find  $m_t^* := \frac{\beta y}{v_t(1+\beta)}$ . We then observe that the choice of money in period  $t$  increases with the discount factor  $\beta$ . Indeed, the more the preference for the present decreases (that is to say the more patient agents are, which an increase in  $\beta$  models well), the more importance agents attach to their future consumption. Therefore the more they wish to 'set aside' so as to be able to buy when they are old. In order to do so, they must buy more money in their youth. By contrast,  $m_t^*$  decreases with  $v_t$ : the higher the price of money in the first period, the less of it agents can acquire, all other things being equal.

(e) We complete the solution of the model by determining the missing values of the various variables.

- i. *Give the expression of aggregate supply and aggregate demand for money in a period  $t$ .* According to the statement, the stock of money (aggregate supply) is constant over time, equal to the exogenous value  $M$ . As for aggregate demand, it is written  $Nm_t^*$ , since the representative agent demands  $m_t^* = \frac{\beta y}{v_t(1+\beta)}$  and the population amounts to  $N$  individuals.
- ii. *Deduce from this the equilibrium condition of the money market which makes it possible to determine the price of money in terms of goods at date  $t$ . Comment.* The equilibrium condition of the money market consists in equalising supply and demand. We therefore have in equilibrium  $M = \frac{N\beta y}{v_t(1+\beta)}$ . We then observe that  $v_t$  depends negatively on  $M$ . Indeed, the greater the money supply, the lower the price of money in terms of goods (this is the principle of inflation). Otherwise put, the money supply  $M$  increases when the price of money in terms of goods  $v_t$  decreases.



- iii. *In an analogous way, determine the equilibrium condition of the money market which makes it possible to determine the price of money in terms of goods at date  $t + 1$ . Deduce from this the return on money between  $t$  and  $t + 1$ .* The equality  $M = \frac{N\beta y}{v_t(1+\beta)}$  (\*) of the previous question must hold in every period  $t$ . In particular, we may likewise write the equality  $M = \frac{N\beta y}{v_{t+1}(1+\beta)}$  (\*\*). The conjunction of the two equalities (\*) and (\*\*) then leads to the equality  $v_t = v_{t+1}$ : there is no inflation between the two periods. Since  $v_t = v_{t+1}$  in equilibrium, the return on money is necessarily zero.
- iv. *Deduce from this the definitive solution for the optimal choice of second-period consumption  $c_2$ . Comment on the efficiency of the market allocation.* We proved previously that  $c_2^* = \frac{v_{t+1}}{v_t} \frac{\beta y}{1+\beta}$ . Under the condition of an absence of inflation between the two periods, proved in the previous question, we therefore find  $c_2^* = \beta y / (1 + \beta)$ : we thus arrive at the same solution as the centralised equilibrium which maximised social welfare. This is the reason why the public authorities must fight inflation so that the market may arrive of its own accord at the 'best equilibrium'. By fighting inflation, the decentralised equilibrium is then superimposed on the centralised equilibrium of the benevolent social planner: here we have an efficient market allocation. By contrast, if the equality  $v_t = v_{t+1}$  did not hold, we would no longer have  $c_2^* = c_2^P$ : in that case the market would not arrive at an efficient allocation.

(iv) CONCLUSION

- (a) *Conclude by comparing the autarkic equilibrium (part (ii)) and the monetary equilibrium (part (iii)).* The monetary equilibrium of part (iii) makes it possible to remedy the market failures of the non-monetary (or autarkic) equilibrium of part (ii) and even makes it possible to attain the maximum level of social utility when the public authorities fight inflation effectively. This is a logical result: in the autarkic equilibrium, the young consume their whole initial endowment in the first period and find themselves with nothing in old age. By contrast, with the monetary equilibrium, agents smooth their consumption between the two periods. Given that the utility function is strictly concave, spreading one's consumption systematically yields more than consuming everything in one period or the other. The monetary equilibrium with the fight against inflation is therefore indeed the most desirable situation, and the only one which permits an efficient self-regulation of the market.
- (b) *Discuss the limitations of this exercise.* In this exercise, we first assumed that the good was non-storable, which is not always the case in reality. One might imagine a model (which would then be far more complex to solve computationally) in which the good could be stored from one period to the next, taking account of the fact that a part  $p$  (where  $p$  is a real number between 0 and 1) of the stock would rot or disappear between the two periods. More interesting still, in a model of *overlapping* generations, would be to take account of the possibility of a bequest. Indeed, in our model, the intertemporal utility function takes absolutely no account of any residual that generation  $t$  might transmit to generation  $t + 1$  on its death (generation  $t + 1$  would therefore enjoy a new income when its members were old: their parents' bequest). One might thus imagine a utility function which took account of the bequest left to one's descendants. Here again, however, the computations of the model would be considerably heavier. *In fine*, the overlapping generations model (OLG) as described in this exercise already provides a framework, simplified to be sure, but one which already makes it possible to formulate some macroeconomic prescriptions, since it brings out mathematically the necessity for the public authorities to fight inflation, at least in a non-centralised system.  $\square$

### 1.2.3 The Keynesian analysis of the motives for holding money

Money plays a central role in the economy, not only as a means of payment, but also as a tool for risk management and wealth accumulation. Keynes (1936) thus identifies three main uses of money: the transaction motive, the precautionary motive and the speculative motive; money is therefore no longer neutral and may be held for its own sake. The **transaction motive** is the most obvious one and concerns the use of money to make purchases of goods and services. This transactional function is at the heart of the economy, since it enables economic agents to carry out exchanges in a smooth and efficient manner. The holding of active balances falls precisely within this transactional logic, corresponding to a demand for money intended for payments and transactions. Unlike other assets, such as real estate or securities, money is liquid and involves no transformation cost, which makes it particularly well suited to everyday exchange. This demand for money depends strongly on the level of overall income: the more an individual's or a firm's income rises, the more the demand for money grows. This demand is also influenced by the synchronisation of receipts and expenditure, as Delaplace (2021c) specifies: if an individual receives a daily

income, his average balance held for the transaction motive will be lower, since on average he keeps only what he earns each day. Moreover, Keynes (1936) enriched his analysis by introducing, alongside the transaction motive, a **motive of investment finance**: when firms seek to invest, they must have liquid resources available to finance that investment, thus constituting working capital; this fund varies with anticipated investment (the larger the latter, the more finance balances increase). However, for a given investment, the demand for liquid balances remains constant.

### Focus...



**The Baumol-Tobin model.** Baumol (1952) and Tobin (1956) put forward a theoretical framework which explains the demand for money for the **transaction motive**. This model aims to resolve the dilemma between keeping non-remunerated liquid balances and investing in remunerative assets, while minimising the costs associated with the management of liquid balances. It rests on the idea that agents must manage their liquid balances optimally in order to finance their everyday transactions, while taking account of the transaction costs incurred when they convert remunerative assets into liquid balances. Thus, the model establishes a balance between the opportunity costs associated with holding money (loss of return on other assets) and the transaction costs linked to the conversion of financial assets into money.

The equilibrium of this model is represented by the optimal quantity of money that agents should hold in order to minimise their total costs. According to this model, the demand for money is proportional to the total amount of transactions ( $T$ ) and to transaction costs ( $C$ ), while being inversely proportional to the interest rate ( $i$ ). This means that the larger the transactions and the higher the costs of converting the asset into liquid balances, the more money economic agents will keep. Conversely, if interest rates rise, agents will tend to reduce the quantity of money held, so as to maximise the return on their financial assets. We then have  $M^* = \sqrt{2TC/i}$ , where  $M^*$  is the optimal quantity of money for the transaction motive,  $T$  is the amount of transactions,  $C$  is the transaction cost, and  $i$  is the interest rate on the remunerative asset. This model has influenced the understanding of liquidity management behaviour, by showing that the demand for money is a function of economic variables such as the volume of transactions and the interest rate.

Beyond the transaction motive, economic agents may also hold money in order to guard against the uncertainty of the future, which brings us to the **precautionary motive**. The latter refers to the holding of money in order to face unforeseen events. In other words, economic agents keep a certain quantity of money in order to cover unexpected expenditure or emergency situations. This approach is particularly important in periods of economic uncertainty, where liquidity becomes a priority for households and firms (liquidity preference). As Tobin (1958) explains, the precautionary demand for money increases when economic uncertainty is high, since agents seek to protect themselves against potential risks. However, as Delaplace (2021c) recalls, incomes being generally stable in the short run, the demand for money for the transaction and precautionary motives may be regarded as stable in the short run.

### Focus...



**Regulated savings in France.** As the Direction de l'information légale et administrative (2024a) explains, there exist in France several privileged savings accounts, that is to say ones benefiting from a tax exemption on interest. The most emblematic is the **Livret A**, whose rate is set at 3 % until January 2025, with a ceiling of 22 950 euros per taxpayer. This account has its rate theoretically adjusted so as to remain a quarter of a point above the rate of inflation, even though the Minister of the Economy may derogate from this rule. More precisely, as the Ministère de l'Économie (2024) explains, the interest rate on the Livret A is theoretically the higher index, rounded to the nearest quarter of a point, between, on the one hand, inflation over the last twelve months as measured by the variation over the last twelve known months of the INSEE consumer price index for all households, increased by a quarter of a point and, on the other hand, the arithmetic mean of inflation over the last twelve months and half of the sum of the monthly average of three-month Euribor and the monthly average of Eonia (for the last known month). Recall that the **Eonia** rate, replaced by the *euro short term rate* at the end of 2019, is the rate of remuneration of overnight interbank deposits in the euro area, while **Euribor** (*Euro interbank offered rate*) is the interbank rate offered between banks of the highest standing for the remuneration of deposits in the euro area.

In addition, the **Livret de développement durable et solidaire** (LDDS), formerly the Codevi, shares the Livret A rate of 3 %, although its deposits are limited to 12 000 euros: intended to encourage responsible saving, it has since its overhaul in 2007 been the successor to the Codevi. The **Livret d'épargne populaire** (LEP), with a rate of 4 % since August 2024, aims for its part to protect the savings of low-income taxpayers who are not liable to income tax; each tax household may open up to two LEPs, with a ceiling per account recently raised to 10 000 euros. Moreover, for young people aged twelve to twenty-five, the **Livret jeune** offers a rate at least equivalent to that of the Livret A,

although freely set by the banks, with a ceiling of 1 600 euros. Finally, regulated savings also comprise the CEL and the PEL: the **Plan épargne logement** (PEL) is a savings product intended to finance a property project, offering a fixed rate of remuneration of 2.25 % for new plans opened since January 2024, with a payment ceiling of 61 200 euros; the PEL also makes it possible to benefit from an entitlement to a loan at a preferential rate for the purchase of a property (3.45 % for PELs opened since January 2024). As for the **Compte épargne logement** (CEL), it offers a remuneration of 2 % in August 2024, with a deposit ceiling of 15 300 euros, and likewise gives access to loans for works or for the purchase of a dwelling, on advantageous terms (interest rate of 3.5 % for CELs opened since January 2023). The maximum amount of the housing savings loan is 23 000 euros if one holds only a CEL and 92 000 euros if one combines a PEL and a CEL.

As the Banque de France (2024g) specifies, 2023 was an exceptional year for regulated savings in France, with net inflows reaching 61.4 billion euros, a level unmatched for more than ten years. The total outstanding amount of regulated savings thus rose by 7 %, reaching 935.5 billion euros, which represents 15.1 % of the financial wealth of French households. The number of Livrets A stood at 55.7 million, with a total outstanding amount of 403.9 billion euros, while the number of LDDS reached 24.9 million, for an outstanding amount of 143.9 billion euros. The LEP, for its part, stood out with record net inflows of 16.6 billion euros and a 50 % increase in its outstanding amount, reaching 71.8 billion euros at the end of the year, with 11.4 million accounts open (out of the 19.5 million French people potentially concerned). In parallel, the household saving ratio remained at a high level, namely 16.6 % in 2023, while flows into overnight deposits and ordinary savings accounts of banks and savings banks decreased.

Finally, the **speculative motive** designates the use of money as a means of investing in anticipation of future price movements. Individuals and institutions hold money in order to take advantage of investment opportunities or to protect themselves against fluctuations in the prices of other financial assets. This motive for holding money is linked to the volatility of financial markets, since money is perceived as a safe asset compared with other, riskier investments. According to Baumol (1952), the speculative demand for money varies notably as a function of expectations about interest rates and the prices of financial assets: when the current interest rate is lower than the anticipated rate, the value of securities falls, since the price of bonds and shares varies inversely with interest rates. In such a situation, agents holding securities seek to sell them, anticipating a fall in their value, while those who hold none prefer to keep their money while waiting for a better buying opportunity. Consequently, when the interest rate is low, the demand for money for the speculative motive increases, since agents prefer to keep liquid balances rather than to run the risk of a capital loss: when the interest rate is sufficiently low, one speaks of a '**liquidity trap**', in the sense that the preference for liquidity is such that agents avoid buying securities, maintaining a high demand for money. The aggregate demand for money therefore results both from income and from interest rates. Although the demand for money for the transaction and precautionary motives is relatively stable in the short run, speculative demand is, for its part, more volatile, since it depends on agents' expectations. This analysis is enriched by Tobin (1958), who applies a portfolio approach to analysing the holding of speculative balances: in an uncertain environment, agents prefer to hold money in order to limit risks, even if this reduces the overall financial performance of their portfolio.

### Key Takeaways



Contrary to classical theory, Keynesian analysis reveals that the demand for money rests on three distinct motives: the transaction motive (linked to income and to the synchronisation of flows), the precautionary motive (protection against uncertainty) and the speculative motive (trade-off between liquidity and financial return) [Keynes (1936)]. The Baumol-Tobin model shows that transaction demand depends on the volume of exchange, on conversion costs and on the interest rate, formalised by  $M^* = \sqrt{2TC/i}$  [Baumol (1952), Tobin (1956)], where  $M^*$  is the optimal quantity of money for the transaction motive,  $T$  is the amount of transactions,  $C$  is the transaction cost and  $i$  is the interest rate on the remunerative asset. Speculative demand increases when interest rates fall or when uncertainty rises, generating a preference for liquidity which may degenerate into a liquidity trap [Tobin (1958)]. From then on, money is neutral only in the long run and under restrictive conditions (stability of the velocity of circulation, full employment): in the short run, it affects real variables through its effects on aggregate demand and on agents' expectations.

## 1.3 Money creation

As Delaplace (2021d) defines it, **money creation** corresponds to the ways in which new money comes to increase the quantity of money held by resident non-financial agents and other financial intermediaries (**ANFR-AIF**), and therefore comes to increase the monetary aggregates.

### 1.3.1 The role of commercial banks

Banks play a central role in the financial system by performing several key functions, notably the management of the means of payment, the transformation of deposits into loans and the reduction of information asymmetries. Although other financial institutions and the capital markets may likewise fulfil some of these functions, banks are distinguished by their capacity to allow financial transactions which could not be carried out through the markets alone, as Diamond & Dybvig (1983) explain. In particular, as Leland & Pyle (1977) specify, banks are indispensable where there exists a divergence of horizon between lenders, who prefer short-term investments, and borrowers, who need long-term financing. One of the main functions of banks is thus the **management of the means of payment**. As Diamond (1984) explains, banks facilitate the circulation of deposits through instruments such as cash, cheques, payment cards or credit transfers. This service is essential in order to guarantee the fluidity of economic exchange, all the more so as the efficiency of interbank settlement is crucial to the proper functioning of the economy. However, beyond the mere management of payments, banks are also major providers of **liquidity insurance**. They allow depositors to access their funds at any time, while lending part of these resources over a longer term. This maturity transformation, although essential in order to finance investment, makes banks vulnerable to liquidity crises, hence the importance of public deposit guarantee schemes, as Diamond & Dybvig (1983) mention. Finally, banks play a fundamental role in the **reduction of information asymmetries** between lenders and borrowers. When a bank grants a loan, it has a better capacity to assess the quality of the investment project thanks to the centralisation of information and to its monitoring tools, as Leland & Pyle (1977) explain. Moreover, as financial intermediaries, banks put in place incentive contracts so as to ensure that borrowers honour their commitments. In the event of non-repayment, they can guard against losses by requiring guarantees or collateral, as Diamond (1984) indicates. The diversification of bank investments likewise contributes to the reduction of risks, although it is necessary to regulate banks in order to avoid opportunistic behaviour according to Diamond & Dybvig (1983).

As Couppey-Soubeyran (2012a) mentions, the history of banks has its source in Antiquity, where the first forms of financial institution appear in the temples of Sumer, in Mesopotamia, in the fourth millennium BCE: at that time, priests received on deposit surpluses from harvests which they lent to those who needed them, thereby introducing the first forms of lending and of financial transformation. This system rested on the confidence that depositors placed in the priests, who thus embodied the security necessary to the management of deposits. The use of coins, from the second millennium BCE onwards, strengthened the development of banking operations by facilitating commercial exchange: historians report a great prosperity of banking activities during Antiquity, particularly in regions such as Babylon and the Mediterranean basin. However, the fall of the Roman Empire in 476 marks a break in banking development, which was to know a genuine revival only at the time of the crusades and of the fairs of Champagne in the ninth century CE. The evolution of banks continued in parallel with that of commercial exchange according to Couppey-Soubeyran (2012a): in the twelfth and thirteenth centuries, the great trade routes and the multiplication of exchanges of precious goods, metals or spices favoured the rebirth of banking activities in the West. The banking sector then underwent a considerable expansion with the industrial revolution, first in England in the eighteenth century, then in France in the nineteenth century, where institutions such as the *Crédit Lyonnais* came into being; at that time, French banks specialised mainly in the management of overnight deposits and the discounting of short-term commercial paper, as Couppey-Soubeyran (2012a) specifies. This model of the **'narrow bank'** was distinguished by a strict separation between deposit management activities and long-term lending. This specialisation nevertheless began to soften with the second industrial revolution at the beginning of the twentieth century, and several banks adopted mixed models, combining deposit and investment activities. The process of despecialisation accelerated in the twentieth century, notably with the **Debré reforms** of 1966 and 1967, which led to the universalisation of banking activity in France: this model of the **universal bank** came to prevail in most European countries with the banking law of 1984, allowing credit institutions to diversify their activities. In the United States, this evolution came later because of the *Banking Act* of 1933, which separated deposit banks and investment banks. It was only with the adoption of the *Gramm-Leach-Bliley Act* in 1999 that American banks were able to engage in the diversification of their activities. However, as Couppey-Soubeyran (2012a) qualifies, this diversification was not without risk, as the *subprime* crisis of 2008 bears witness, which particularly affected banks specialising in investment, such as Lehman Brothers.

#### Focus...



**Retail banking versus investment banking.** Two main models of banking activity are generally distinguished: retail banking and corporate and investment banking. These two models address distinct client bases and answer to different strategic objectives. As Couppey-Soubeyran (2012a) explains, **retail banking** targets mainly private individuals, small and medium-sized enterprises, as well as local authorities and associations: it offers a wide range

of services including account management, savings products, financing solutions and, increasingly, insurance. This model is therefore not limited to a single service, and the retail bank constantly seeks to diversify its offerings so as to retain its customers. Moreover, although traditionally focused on a national market, retail banking today tends to internationalise, in part thanks to cross-border mergers and acquisitions, as European groups such as Santander and BNP Paribas bear witness.

By contrast, **corporate and investment banking** is oriented towards a client base composed of large firms, which makes it a 'wholesale bank'. As Couppey-Soubeyran (2012a) specifies, activities are far larger in volume, including services such as advice on mergers and acquisitions, initial public offerings, bond issues, and other complex financial operations. In particular, the investment bank, which likewise encompasses market operations in securities and derivatives, plays a crucial role in strategic advice to large firms. Over time, the rise of these activities has led to a notable increase in fees within net banking income (NBI), these now representing about one third of the NBI of French banks, against only 10 % in the 1980s according to Couppey-Soubeyran (2012a).

On the financial side, market activities have taken an increasingly important place in bank balance sheets over the decades, notably through the issuance and acquisition of securities. Thus, the share of securities in bank balance sheets, which hardly exceeded 5 % in the 1980s, reached nearly 50 % before the financial crisis of 2008 according to Couppey-Soubeyran (2012a). This phenomenon is explained in part by the new accounting standards and by the growth of stock markets. However, the crisis reminded banks of the importance of aligning market activities with those of traditional banking, bringing about a reduction in the share of securities portfolios: the rise of operations on forward markets has also led to a significant expansion of 'off-balance-sheet' commitments, reflecting commitments which do not translate immediately into monetary flows, as Couppey-Soubeyran (2012a) specifies.

### 1.3.2 The agents of money creation: banks and the central bank

Money creation by banks has evolved over time. Originally, when an individual deposited a hundred euros of metallic money in a bank, the latter recorded this amount as an overnight deposit on its liabilities side, while placing the equivalent in metallic cash on its assets side; this guaranteed that the money deposited was always available to the depositor. However, in the seventeenth century, banks realised that they could lend part of the deposits, without risking an immediate shortage of liquidity (bankruptcy): as Brana & Cazals (2014a) explain, they thus began to lend up to 90 % of the funds deposited, while keeping a fraction of 10 % in liquid form so as to meet withdrawal demands. This mechanism, known as **fractional reserve**, allowed banks to increase their lending activities while ensuring a certain stability of their commitments towards depositors. Over time, this process became more sophisticated, and at the end of the nineteenth century, modern money creation through credit emerged. Today, when a bank grants a loan, it no longer lends existing deposits alone, as Brana & Cazals (2014a) detail: instead, it creates a deposit *ex nihilo* by crediting the borrower's account with the amount lent, which then appears on the liabilities side of its balance sheet; simultaneously, the bank records a claim of an equivalent amount on its assets side. This mechanism then allows the bank to create deposit money every time it grants a loan: **loans make deposits**; as long as the loan is not repaid, the money created continues to exist in the system, and it is when the loan is repaid that this money is destroyed.

However, in an economy where several banks coexist, the newly created money does not always remain within the initial bank. For example, when the borrowing firm pays its employees, the latter may have accounts in other banks. This entails a transfer of funds (a **leakage**) towards other institutions: each bank thus sees part of the deposits it has created being redeposited elsewhere. Nevertheless, banks can offset these leakages by borrowing funds on the interbank market. Ultimately, as Brana & Cazals (2014a) explain, while each individual bank sees its power of money creation limited by these leakages, the banking system as a whole can potentially create an unlimited quantity of money. This collective mechanism therefore underlines the necessity of putting in place regulations so as to avoid an uncontrolled monetary expansion: this is the reason why commercial banks must constitute **required reserves**, which represent a percentage  $r$  of deposits that banks are obliged to place with the central bank.

#### Focus...

**The structure of banks' balance sheets.** For a bank, the balance sheet represents the whole of its holdings (what it owns and what it lends, on the assets side) and of its debts and its customers' deposits (on the liabilities side). A bank's balance sheet therefore takes the following form:

| Assets                                                        | Liabilities                                                               |
|---------------------------------------------------------------|---------------------------------------------------------------------------|
| C: the loans granted to customers                             | D: customers' deposits                                                    |
| RE: excess reserves, placed voluntarily with the central bank | Ref: the refinancing granted by the central bank to cover liquidity needs |
| RO: required reserves                                         |                                                                           |

The structure of a commercial bank's balance sheet then makes it possible to understand why loans make deposits. As Brana & Cazals (2014a) illustrate, suppose that there exists only one bank in the economy and that it grants a loan of 100 to a firm: C increases by 100 on the assets side, while D increases by 100 on the liabilities side (the accounting entry is therefore balanced): this constitutes the act by which it creates money. As Brana & Cazals (2014a) add, this creation likewise takes place when the bank buys financial assets (a bond, a negotiable debt security) or foreign currencies. To be sure, money creation on the occasion of a loan (or of the purchase of a debt security) is only provisional since, when the loan is repaid, there is monetary destruction: C and D will both decrease by 100 relative to the moment when the loan was contracted once the firm has settled its debt. But there is indeed net money creation if the speed at which the bank grants loans is greater than that at which it is repaid by its customers. For a central bank, the balance sheet differs slightly since its customers are not households or firms but essentially commercial banks. The structure of a central bank's balance sheet is therefore the following, as Delaplace (2021d) indicates:

| Assets                                                                           | Liabilities                                                           |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Ref:</b> the refinancing granted by the central bank to cover liquidity needs | <b>RE:</b> excess reserves, placed voluntarily with the central bank  |
| <b>T:</b> the Treasury securities held by the central bank                       | <b>B:</b> the banknotes issued by the central bank and held by agents |
| <b>De:</b> the foreign currencies kept in stock                                  | <b>RO:</b> required reserves                                          |
| <b>Etr:</b> net claims on the rest of the world (gold and foreign currencies)    | <b>APU:</b> current account of general government                     |

As the BCE (2015) indicates, a **central bank** is a public institution which manages the currency of a country or of a group of countries and controls the money supply, that is to say the quantity of money in circulation. As Brana & Cazals (2014a) recall, the central bank likewise creates money in currency form (since it has the monopoly of the issuance of coins and banknotes) and in deposit form. Indeed, in the same way as ANFR-AIF have an account with one or more commercial banks, commercial banks hold an account with the central bank (see the focus *supra*). This account allows commercial banks to record their reciprocal commitments in **central bank money** (recall that central bank money is the monetary base  $M_0$  studied *supra*): part of the reserves held by banks with the central bank is imposed by the monetary authorities in the form of **required reserves**, as is the case in the euro area, while the rest of the reserves, called **excess reserves**, is determined by the banks themselves according to their liquidity needs. In addition, the central bank also plays the role of the state's bank, which explains the presence of the credit accounts of general government (APU) on its liabilities side. Money creation by the central bank may then take place through several mechanisms. Traditionally, the central bank could create money for the benefit of the Treasury by granting it advances or by buying public securities, although this practice is henceforth prohibited in the European Union, as Brana & Cazals (2014a) recall. It may likewise create money in exchange for foreign currencies brought in by commercial banks or in responding to their liquidity needs: this may be done either by granting loans to banks, or by buying financial securities held by them, via specific procedures which will be detailed *infra*. In every case, the central bank credits the account of the commercial banks, thereby creating central bank money and thus increasing  $M_0$ : this process of **refinancing** the banks is similar to the way in which the latter create deposit money by financing non-financial agents. However, money creation by the central bank and by the state turns out to be rather small: Brana & Cazals (2014a) recall that more than 80 % of money is created by commercial banks.

### 1.3.3 Money multiplier *versus* money divisor

The theory of the **money multiplier** rests on the idea that the creation of money in an economy is directly linked to the monetary base issued by the central bank. Monetarists, notably under the influence of Friedman (1956), hold that the money supply is a multiple of the monetary base, computed according to the formula  $M = mM_0$ , where  $M$  represents the money supply,  $M_0$  the monetary base, and  $m$  the said multiplier. This multiplier depends on factors such as the ratio of banks' required reserves and the proportion of liquid balances held by the public in the form of banknotes. Within this framework, **money creation is exogenous**, determined by the actions of the central bank which directly controls the supply of money by adjusting the monetary base. Monetarist theory therefore recommends that the central bank adjust its issuance so as to maintain price stability and to avoid an excessive growth of the money supply, a source of inflation according to Friedman (1968).

#### Focus...



**Computing the money multiplier.** In order to demonstrate the value of the money multiplier, let us begin by establishing the basic relations concerning the money supply ( $M$ ) and the monetary base ( $M_0$ ). The money supply

M, in the sense of the aggregate  $M_1$ , is composed of the banknotes issued by the central bank (B) and the overnight deposits created by banks (D). We therefore have  $M = B + D$ . The monetary base, denoted  $M_0$ , comprises for its part bank reserves (R) and the banknotes issued (B). Thus, we may write  $M_0 = B + R$ . Suppose that non-financial agents hold a fraction  $b$  of the money supply in the form of banknotes (B), and a fraction  $1 - b$  in the form of overnight deposits (D). Thus, we may express B as a function of M:  $B = bM$ . Likewise, we assume that banks keep a fraction  $r$  of their deposits in reserves with the central bank, which allows us to write  $R = rD$ . Using the relation  $M = B + D$ , it then follows that  $D = (1 - b)M$ . Let us substitute these expressions for B and R into the equation of the monetary base  $M_0$ : we then obtain  $M_0 = B + R = bM + rD$ . Then, replacing D by  $(1 - b)M$ , we obtain  $M_0 = bM + r(1 - b)M$ . Factorising M, it follows that  $M_0 = [b + r(1 - b)]M$ . Thus, we may isolate M so as to obtain the relation between the money supply and the monetary base  $M = M_0 / [b + r(1 - b)]$ . The term  $\frac{1}{b + r(1 - b)}$  is the money multiplier, denoted  $m$ , which gives us  $M = mM_0$ . Since  $b$  and  $r$  lie between 0 and 1 (according to Delaplace (2021e),  $b$  is worth about 14 % and  $r$  does not exceed 10 %), it is clear that  $m$  is greater than 1. This means that the money supply in circulation in an economy is a multiple of the monetary base issued by the central bank: following an increase in bank liquidity, money creation has been  $m$  times greater than that increase.

By contrast, the theory of the money divisor, developed by Le Bourva (1962), adopts a different perspective. Unlike the multiplier, this approach insists on the **endogenous character of money creation**: according to this theory, it is not the central bank which determines the quantity of money in circulation, but the demand for credit emanating from non-financial agents. As Delaplace (2021e) specifies, the demand for credit may be lower than what banks could grant given their excess reserves in central bank money (in that case, money creation will be lower than it could have been given those excess reserves), banks may use their surpluses in central bank money differently (they may for instance take back securities which they had previously refinanced with the central bank) and, in France and more generally in the euro area, banks are on the whole indebted towards the central bank and therefore hold no surpluses in central bank money. The central bank is then obliged to refinance the commercial banks: it is, with respect to them, in the position of a constrained lender of last resort. Thus, commercial banks create money not according to their surpluses in central bank money but according to the demand for credit: the issuance of central bank money becomes a fraction of the money already created by the banking system, hence the use of the term 'divisor' rather than multiplier: the mathematics is identical, but conceptually one will write  $M_0 = M/m$ . The fundamental distinction between these two approaches therefore lies in the direction of causality between the money supply and the monetary base: in the multiplier model, the central bank has a leading role and directly controls the supply of money by adjusting the monetary base; by contrast, in the divisor approach, the central bank plays a more passive role, responding to the refinancing needs of the banking system after the event. This dichotomy reflects a divergence over the exogenous or endogenous nature of money, monetarists considering that money is an exogenous instrument controlled by the central bank, while Keynesians and the proponents of the divisor see money as an endogenous phenomenon, determined by internal economic dynamics, as Hicks (1974) for example emphasises.

### Key Takeaways



Money creation designates the increase in the monetary aggregates held by non-financial agents and financial intermediaries [Delaplace (2021d)]. Commercial banks create more than 80 % of it by granting loans according to Brana & Cazals (2014a): when a bank lends 100 euros, it credits the customer's account by creating a deposit *ex nihilo* (assets increase by the claim, liabilities by the money created). Banks therefore do not lend pre-existing deposits, but create the money needed for exchange. However, in a multi-bank system, inter-system leakages (payments of wages into other banks) compel each bank to borrow on the interbank market, hence the necessity of required reserves (a fraction  $r$  of deposits) in order to regulate collective expansion [Brana & Cazals (2014a)]. The central bank completes this creation: it issues currency (banknotes) and creates central bank money via the refinancing of banks (loans, purchases of securities, purchases of foreign currencies) [BCE (2015)]. Two theories are opposed as to causality. The money multiplier postulates an exogenous relation  $M = mM_0$  where  $m = \frac{1}{b + r(1 - b)}$ ,  $b$  being the fraction of money held in banknotes [Friedman (1956)]: the central bank controls the money supply M by adjusting the monetary base  $M_0$ . The money divisor, conversely, asserts that money creation is endogenous: banks create money according to the demand for credit, compelling the central bank to refinance passively via  $M_0 = M/m$  [Le Bourva (1962), Delaplace (2021e)]. This distinction reflects a fundamental disagreement over the active (monetarist) or passive (Keynesian) role of the central bank in monetary regulation.



## 2 Monetary policy

As Delaplace (2021f) defines it, **monetary policy** may be understood as the whole set of actions undertaken by the monetary authorities with the aim of supplying the economy with the adequate quantity of money to support economic growth and attain full employment, while guaranteeing the stability of the currency both internally, through the control of the general price level, and externally, via the management of the exchange rate. This definition brings to light three essential aspects. First of all, it rests on the assumption that the quantity of money in circulation directly influences economic activity. Next, the objectives of monetary policy belong to the broader objectives of economic policies, but it may prove necessary to give priority to one or more of these objectives, which raises the question of the priority choice of the monetary authorities. Finally, in order to attain the objectives set, the authorities have at their disposal a set of levers of action which they may mobilise according to the economic situation and economic priorities.

### 2.1 Monetary policy in Europe

Monetary policy is one of the instruments of economic policy, alongside fiscal policy and exchange rate policy. As Brana & Cazals (2014b) emphasises, in the euro area since 1999 it is to the **Eurosystem**, composed of the European Central Bank (ECB) and the national central banks, that full responsibility for the definition and implementation of this policy falls. More generally, the objectives of a central bank, the architect of monetary policy, are articulated around three strands: (i) issuing fiat money so as to regulate the money supply, in order to foster economic growth while maintaining the stability of exchange rates; (ii) guaranteeing price stability through the **key interest rates**, which correspond to the rates set on a day-to-day basis for lending money to commercial banks and for remunerating the deposits that the latter make with the central bank; (iii) lending money to commercial banks, the central bank then playing the role of '**lender of last resort**'. The central bank thus helps to maintain financial stability by ensuring the continuity of the financial system, by preventing cascading failures and by preserving confidence in the banking system.

#### 2.1.1 The evolution of the role of central banks

Central banks play a fundamental role in the management of monetary policy, with the main objective of maintaining price stability, while fostering sustainable economic growth. Through their mission of **controlling the money supply** and **regulating interest rates**, they see to it that inflation is kept at an acceptable level, generally around 2 %, according to the objectives set by the central banks of the advanced economies such as the ECB. This regulatory function makes it possible to avoid the dangers of hyperinflation, but also those of deflation, which can paralyse an economy by making indebtedness more costly and by delaying investment and consumption decisions, as Couppey-Soubeyran (2012a) explains. Indeed, central banks draw on the theories developed by economists such as Kaldor (1961) and his '**magic square**', which brings out four major objectives of demand-management policy: growth, employment, price stability and external balance. These objectives cannot be attained simultaneously with a single instrument of economic policy. Tinbergen (1959) emphasised that for each objective a specific instrument is necessary if it is to be effective, thereby justifying the **separation of monetary and fiscal tools**. For example, according to Mundell (1963), monetary policy must target external stability, in particular exchange rates and price stability, while fiscal policy must concentrate on employment and domestic growth. Historically, from the 1980s onwards, with the growing influence of **monetarists** such as Friedman (1956), the fight against inflation became the main objective of central banks, which led to the progressive adoption of inflation targeting. This approach rests on the control of the money supply in order to avoid an excessive increase in prices, in line with the quantity theory of money studied *supra*. Thus, as Couppey-Soubeyran (2012a) recalls, central banks, particularly after the two oil shocks of the 1970s, progressively acquired a greater independence so as to concentrate on price stability, notably by limiting the growth of the monetary aggregate.

#### Debate



**Should central banks be independent?** When one speaks of independence for a central bank, what is considered is its independence with respect to the state, to the political authorities. Two typologies of the notion of independence may then be sketched out. According to the first, there are two levels of independence for a central bank: **organisational**, that is to say independence with respect to the state as regards the whole set of provisions governing the institutional organisation of the central bank (composition of decision-making bodies, method of appointing and dismissing its leaders, and so on) and **economic**, that is to say independence in matters of monetary policy (with respect to



objectives and instruments). According to the second typology, set out by Bordes (2008), there are three levels of independence of the monetary authorities: at the level of **instruments**, with the freedom to implement monetary policy; at the level of **leadership**, in relation to the power available to a country's government to appoint or dismiss the leaders of the central bank and to its participation in the elaboration of monetary policy decisions; and at the **financial** level, with the possibility of financing state expenditure through direct or indirect recourse to central bank credit.

The independence of central banks appears at first sight as a guarantee of the effectiveness of monetary policy. First of all, a monetary policy conducted by the public authorities suffers from a **credibility deficit**, whereas the credibility of the central bank rests on its capacity to abide by its own announcements of economic policy. Now, if monetary policy is conducted by the public authorities, the central bank suffers from a lack of credibility owing to the state's benevolence. To this hazard is added the risk of **time inconsistency** which Kydland & Prescott (1977) set out: an announcement that is optimal at date  $t$  may cease to be optimal at date  $t + 1$ , which gives the central bank an incentive to depart from its choice at  $t$ . In other words, the optimal short-run policy (which must take account of surprise effects) may differ from the optimal long-run policy. Such a situation generates an **inflationary bias**, as Barro & Gordon (1983) explain: indeed, agents are aware of this inconsistency (they know the central bank's reaction function) and anticipate this incentive towards more inflation. They therefore incorporate it into their decisions in such a way that *in fine* one always ends up with a level of inflation higher than the one announced, without the level of employment being effectively higher than its long-run level. This bias, linked to the temptation not to abide by initial commitments in matters of inflation, is due to a lack of credibility of the monetary authorities. Moreover, the lack of credibility entails a fall in investors' confidence in the country, which increases the country's risk premium (and therefore the interest rates required by investors) and may provoke capital withdrawals. The independence of the central bank may then be a solution to this lack of credibility: in order to fight this bias and to attain the announced inflation objective, the authorities must indeed invest in credibility and banish discretionary policies. This goes by way of putting in place a **conservative central banker** (with a strong aversion to inflation) according to Rogoff (1985). At the same time, the central bank must also seek to tie its own hands – this is the model of Ulysses and the Sirens of Elster (1986) – by announcing target rules (an objective for the growth of the money supply), thus banishing the use of discretionary policies; for example, a price stability rule. The independence of the central bank with respect to political power makes it possible to resolve the credibility deficit. Such a situation then allows a better anchoring of expectations, hence a greater effectiveness of monetary policy.

However, the independence between the central bank and the public authorities rapidly reaches its limits. The first problem that arises when the central bank is independent is that of the **coordination between fiscal and monetary policies**. In the case of a system with central bank independence, monetary policy is entrusted to the central bank, while fiscal policy remains the exclusive competence of the state: there may then exist coordination problems between the two economic policies. Thus, it may be that two policies conducted jointly and independently do not lead to an optimum: for example, in the case of a lax fiscal policy (translating into an increase in public deficits), a restrictive monetary policy translating into a rise in interest rates could weigh down the burden of the debt and set it on an explosive trajectory (snowball effect, unsustainability). Moreover, even if there is theoretically central bank independence, monetary policy remains linked to fiscal policy, which may hamper its credibility, as the 'unpleasant monetarist arithmetic' (*chicken game*) of Sargent & Wallace (1981) bears witness. Indeed, if a government conducts an expansionary fiscal policy, it risks ending up in a situation where the widening of deficits is such that its debt becomes unsustainable. Two possibilities are open to it: in a regime of **monetary dominance**, it is the government that adapts, by reducing its spending and raising its taxes, whereas in a regime of **fiscal dominance**, it is the central bank that adapts, by accepting a monetisation of the debt and, *in fine*, more inflation. From then on, while independence reduces the risk of monetisation, that risk always exists and independence may not be sufficient to guarantee the credibility of the central bank if, in parallel, the government conducts a lax fiscal policy. Finally, a credible central bank may destabilise financial markets: this is the **credibility paradox** of Borio & Shim (2003). Indeed, banks enjoy a strong credibility in matters of fighting inflation, which favours keeping agents' inflation expectations at very low levels: the result is interest rates lower than in the past, for similar levels of inflation, which favours risky investments; the degree of risk-taking in the economy is therefore stronger, which favours the formation of bubbles and, in due course, the crises linked to their bursting.

In the 1990s, a crucial change took place in the conduct of monetary policy, marked by the move from monetary targeting to **inflation targeting**, and by the emergence of the rule of Taylor (1993). This rule offers a simple but effective framework for adjusting the short-term nominal interest rate ( $i$ ) set by the central bank as a function of two essential gaps: the inflation gap and the output gap. Mathematically, the **Taylor rule** is expressed by the relation  $i = r^* + \pi + \alpha(\pi - \pi^*) + \beta(y - y^*)$ , where  $i$  represents the short-term nominal interest rate set by the central bank,

$r^*$  is the equilibrium real interest rate (generally estimated according to Couppey-Soubeyran (2012a) at around 2 %),  $\pi$  is the observed rate of inflation,  $\pi^*$  is the central bank's target rate of inflation (often set at around 2 %),  $y$  is the effective GDP of the economy,  $y^*$  is potential or full-employment GDP (that is to say the maximum level of output that the economy can attain without generating inflationary pressures),  $\alpha$  and  $\beta$  are coefficients which weight respectively the importance of the inflation and output gaps (in the original version of Taylor (1993),  $\alpha = \beta = 0,5$ ). The rule of Taylor (1993) thus makes it possible to determine an interest rate which reacts symmetrically to the gaps relative to the inflation target and to the level of potential output. Concretely, when inflation ( $\pi$ ) exceeds its objective ( $\pi^*$ ), the central bank must raise the interest rate in order to cool the economy and reduce inflationary pressures. Conversely, if inflation is below the target or if the economy is operating below its potential ( $y < y^*$ ), the central bank must lower rates in order to stimulate demand. This approach rests on the idea that a proportionate and systematic response to output and inflation gaps makes it possible to stabilise both economic activity and the price level. As Couppey-Soubeyran (2012a) indicates, the rule of Taylor (1993) has been widely adopted, notably because it offers a transparent and credible guideline for monetary policy decisions, while leaving the monetary authorities a certain flexibility. However, Couppey-Soubeyran (2012a) recalls that the 'Nobel prize' in economics Joseph Stiglitz, vice-president and chief economist of the World Bank from 1997 to 2000, emphasised that the low world inflation of the 1990s and 2000s is also explained by external structural factors, such as low production costs in emerging countries like China, and not solely by the strict application of monetary principles such as the Taylor rule.

### Short-Answer Question



**The Fed and the ECB: similarities and differences.** The financial crisis of 2007-2008, and then the Covid-19 pandemic, propelled onto the international stage the two most powerful central banks in the world: the United States Federal Reserve System (the Fed) and the European Central Bank (ECB). These institutions, created respectively in 1913 and in 1998, exercise considerable monetary power over economic areas together representing nearly 40 % of world GDP and directly influence global financial conditions through their interest rate decisions and their asset purchase programmes. At the height of the pandemic crisis, the balance sheet of the BCE (2023) reached a historic peak of 8.8 trillion euros in June 2022, while that of the Fed peaked at 8.93 trillion dollars in the same year according to The Daily Economy (2025), attesting to the unprecedented scale of their expansionary interventions. While these two institutions share an institutional architecture founded on independence from political power — a principle recalled by Bernanke (2010) and Draghi (2018) as a necessary condition of anti-inflationary credibility — they differ substantially in their statutory mandates, their governance structures and their operational strategies. The Fed has a *dual mandate* aiming simultaneously at price stability and full employment, whereas the ECB has a hierarchical mandate granting absolute priority to price stability, without prejudice to support for the general economic policies of the European Union, as Fontana (2006) explains. These differences, far from being purely formal, reflect distinct political choices and institutional legacies which shape the reactions of the two central banks in the face of macroeconomic shocks. To what extent do the Fed and the ECB share a common institutional foundation of independence and accountability, while diverging in their statutory objectives and their modes of action? In order to answer this question, it is appropriate to examine first the structural convergences which place these two institutions within a common model of an independent and accountable central bank, before analysing the substantial divergences which flow from their differentiated mandates and their specific politico-economic contexts.

The Fed and the ECB share a common institutional foundation resting on three pillars: operational independence from political power, the anchoring of a quantified price stability objective around 2 %, and a framework of accountability towards democratic authorities and the public. As Bernanke (2010) establishes in his analysis, central bank independence rests on a widely shared consensus according to which the objectives of monetary policy must be set by the political authorities, but the operational conduct of that policy must be removed from direct political control so as to avoid the inflationary bias linked to the time inconsistency of discretionary decisions. This institutional architecture, which Walsh (1995) formalises in his model of an optimal contract between the government (the principal) and the central bank (the agent), makes it possible to combine anti-inflationary credibility and stabilising flexibility: the central bank is penalised when inflation departs from the target, but retains an instrumental freedom to react to demand or supply shocks, as Dall'Orto Mas et al. (2020) explain. Empirically, the two institutions have adopted symmetric inflation targets close to 2 %: the Fed explicitly aims at PCE (*personal consumption expenditures*) inflation of 2 % on average over the medium term since the strategic review of 2020, while the ECB redefined its objective in 2021 as '2 % over the medium term', abandoning the earlier asymmetric formulation ('below, but close to 2 %') so as to affirm the symmetry of its target, as CaixaBank Research (2020) recalls. Moreover, the two central banks have developed collegial governance structures aiming to balance technical expertise and geographical representativeness: the *Federal Open Market Committee* (FOMC) of the Fed comprises twelve members — the seven

governors appointed by the American president and five of the twelve presidents of the regional banks, with annual rotation except for New York — while the Governing Council of the ECB brings together the six members of the Executive Board and the governors of the twenty national central banks of the euro area, as La Finance pour tous (2024) explains. Finally, the two institutions are subject to strict obligations of transparency and accountability: the Fed publishes its meeting minutes, testifies regularly before Congress and reports on its actions in semi-annual reports, while the ECB appears quarterly before the European Parliament, publishes its monetary policy accounts and justifies its decisions through statements by their presidents, Jerome Powell (Fed) and Christine Lagarde (ECB) in January 2026.

While these structural convergences attest to the adoption of a widely shared institutional model, the Fed and the ECB diverge substantially in their statutory mandates, their operational strategies and the politico-economic constraints which bear upon their decisions. The main difference lies in the nature of the mandate: whereas the *Federal Reserve Act* of 1977 explicitly entrusts the Fed with a *dual mandate* aiming at ‘maximum employment, stable prices and moderate long-term interest rates’, the Treaty on the Functioning of the European Union (article 127) establishes a strict hierarchy of objectives by assigning to the ECB ‘the maintenance of price stability’ as its primary objective, ‘without prejudice’ to which it supports the general economic policies of the Union, as Fontana (2006) explains. This formal asymmetry reflects distinct historical legacies: the American dual mandate flows from the Humphrey-Hawkins Act of 1978, adopted in the post-Vietnam and post-oil-shock context in which Congress wished to compel the Fed to take account of inflation and unemployment simultaneously, whereas the European hierarchical mandate belongs to the lineage of the German *Bundesbank*, an institution whose success in the fight against the inflation of the 1970s and 1980s shaped the European consensus on the primacy of price stability, as Nagel (2022) analyses. However, as the analysis of Mody et al. (2020) emphasises, this statutory difference does not necessarily translate into divergent behaviour in practice: during the great recession of 2008-2009, the ECB acted ‘as if it had a dual mandate’, aggressively lowering its key rates and deploying unconventional measures to support economic activity despite its hierarchical mandate. Moreover, the two institutions differ in the scale and the nature of their unconventional interventions: while both had massive recourse to asset purchases (*quantitative easing*), the Fed concentrated its purchases on American Treasury securities and mortgage-backed securities (MBS), whereas the ECB had to contend with the heterogeneity of the euro area by distributing its purchases across each member country, which raises recurrent questions of ‘fiscal dominance’ and of implicit mutualisation of national sovereign debts, as Mody et al. (2020) explain. Finally, the two central banks face differentiated challenges of democratic legitimacy: the Fed, backed by a federal state endowed with unified political sovereignty, enjoys a robust institutional legitimacy despite the recurrent criticisms of the Congress which influences it, whereas the ECB, operating in full independence within a framework of monetary union without fiscal union or federal government, must justify its actions before nineteen national parliaments and face recurrent legal challenges, notably from the German Constitutional Court, over the compatibility of its asset purchase programmes with the European treaties, as Calliess & Tuncel (2023) mention.

The Fed and the ECB therefore embody a largely convergent institutional model of an independent, credible and accountable central bank, founded on the anchoring of a symmetric inflation target around 2 % and on a collegial governance architecture subject to strict obligations of transparency and accountability. This operational independence, combined with a clear mandate and effective democratic accountability, constitutes the optimal framework for maximising anti-inflationary credibility while preserving the stabilising flexibility needed in the face of macroeconomic shocks. However, these structural similarities must not obscure the substantial differences which flow from their differentiated statutory mandates — the American *dual mandate* versus the European hierarchical mandate — and from their specific politico-economic contexts. These formal differences influence the cyclical reactions of the two institutions, both of which acted accommodatively during the recent crises, less than they influence their long-run structural constraints: the Fed enjoys a democratic legitimacy backed by a sovereign federal state, whereas the ECB must contend with the heterogeneity of the euro area and the absence of a federal government, which limits its room for manoeuvre and exposes it to recurrent legal and political challenges. The future of these two institutions will depend on their capacity to adapt their strategies to new challenges — the ecological transition, digital finance, political polarisation — while preserving the foundation of credibility and independence on which their legitimacy rests.

### 2.1.2 The architecture of the European System of Central Banks

The **Economic and Monetary Union** (EMU) of the European Union (EU) rests on three key elements: a single currency, a central bank and a common monetary policy. As Moschetto & Deguen (2022) indicates, the **euro**, officially introduced in 1999, became the single currency of eleven European countries, replacing their national

currencies. Its adoption took place in three major stages: the liberalisation of capital (1990-1993), the creation of the European Monetary Institute (EMI) in 1994, and finally the introduction of the euro as physical currency in 2002; the conversion rates of the national currencies were irrevocably fixed in 1999. Among the advantages according to Moschetto & Deguen (2022), the **elimination of exchange rate risk** within EMU greatly secured European firms, particularly those operating internationally; the reduction of transaction costs and the growing importance of the euro in international exchange likewise strengthened the economic power of the EU. However, these benefits come with constraints linked to the **volatility of the currencies** of external trading partners, notably the American dollar: for example, after its introduction at 1.20 dollars in 1999, the euro fell to 0.80 dollars in 2002 before rising again to 1.60 dollars in 2008 and then reaching parity once more in July 2022.

The **European System of Central Banks** (ESCB) brings together the European Central Bank (ECB) and the national central banks (NCBs) of the EU countries, whether or not they are members of the euro area. Within this system, the Eurosystem represents the ECB and the NCBs which use the euro. The federal structure of the ESCB is reminiscent of that of the *Federal Reserve* (the Fed) in the United States, although the NCBs pre-existed the Eurosystem: as at 1 January 2024, as the BCE (2024b) indicates, the capital of the ECB is held 81.77 % by the NCBs of the nineteen euro area countries, and 18.23 % by the NCBs of the seven EU countries not using the euro (the ECB is therefore a public limited company to which all EU countries must contribute their share, proportional to their GDP). The ECB, established in 1999 in Frankfurt, has three bodies which Moschetto & Deguen (2022) enumerate: the Governing Council, the Executive Board and the General Council. The **Governing Council**, which brings together the governors of the euro area NCBs and the members of the Executive Board, takes the decisions relating to monetary policy; chaired by the president of the ECB, it meets twice a month. The **Executive Board of the ECB**, composed of six members, implements these decisions with the help of the NCBs; this managing body carries out the monetary policy defined by the Governing Council. Finally, the **General Council of the ECB** includes the governors of all the NCBs of the EU and the six members of the Executive Board: it serves as a forum for coordination between the euro area countries and those not taking part in it; it is therefore a consultative body.

The ECB pursues a single monetary policy whose main objective is **price stability**, defined as annual inflation below but close to 2 %. This target rate is based on the Harmonised Index of Consumer Prices (HICP). As Moschetto & Deguen (2022) explain, in order to attain this objective, the ECB uses two main pillars: economic analysis and monetary analysis. The first rests on a regular assessment of the economic factors influencing prices in the short and medium term, while the second concentrates on the evolution of the monetary aggregate  $M_3$ , whose growth must remain below 4.5 % per year in order to keep inflation under control. Moreover, the ECB has intervention instruments such as refinancing operations and standing facilities, making it possible to manage the liquidity of commercial banks and to ensure the stability of the banking system. Moschetto & Deguen (2022) recall that, since 1999, required reserves for commercial banks have been set at 2 %, before being reduced to 1 % in 2012 in response to the interbank refinancing difficulties encountered by certain European banks.

### Short-Answer Question



**Are there different monetary cultures within the euro area?** European monetary union, which came into force in 1999, rests on the hypothesis of a progressive convergence of economies and monetary practices around a common institutional framework embodied by the European Central Bank. Yet, more than twenty-five years after the introduction of the euro, substantial divergences persist in the representations, preferences and monetary behaviour of European citizens and decision-makers. According to the SPACE survey of the BCE (2024c), 62 % of euro area citizens consider it very or fairly important to have the option of paying in cash in 2024, but this proportion varies considerably from country to country: it reaches 73 % in Austria, 69 % in Germany, 60 % in France, against only 46 % in the Netherlands. These gaps reflect deep cultural differences in the relationship to money, to trust, and to the role of the state in the economic sphere. As Dyson & Marcussen (2009) establish in their political analysis of central banks in the age of the euro, national historical trajectories have produced distinct monetary cultures which continue to influence political preferences and economic behaviour within the Union. German ordoliberalism, founded on the primacy of price stability and fiscal discipline, thus stands in head-on opposition to the French neo-Keynesian tradition which favours growth and employment, as Tokarski (2019) emphasises. From then on, the question is no longer whether monetary union has abolished national differences, but to what extent these heterogeneous monetary cultures coexist, or even come into conflict, within a common institutional framework. Do there exist different monetary cultures within the European Union, and how do these differences affect the functioning of monetary union? This question invites us to examine first the structural divergences in representations of money and in preferences as regards monetary policy, before showing how these differences translate into political tensions and asymmetries in the transmission of the common monetary policy.

National monetary cultures within the European Union are distinguished by deep divergences in representations of ‘good’ money, in inflation aversion and in payment practices, legacies of specific historical and institutional trajectories. As Berlemann et al. (2011) establishes, inflation aversion varies considerably between member states: on average over the period 2007–2010, 41 % of Austrian citizens regarded inflation as one of the most important problems of their country, against only 18 % of Spaniards and 20 % of Irish and Finnish citizens. Germany, often presented as the archetype of the stability culture, displays a rate of 76 % of citizens judging the fight against inflation to be an essential economic task, that is 20 points above the European average. This difference is explained by the collective memory of the hyperinflationary episodes of the 1920s and by the influence of the *Bundesbank*, which durably anchored in German society the idea that monetary stability constitutes the foundation of economic prosperity. Conversely, the countries of southern Europe, historically confronted with higher rates of inflation and frequent currency devaluations, developed a relative tolerance of inflation and a preference for more accommodative monetary policies favouring growth and employment. These ideological divergences are likewise embodied in the dominant schools of economic thought: German ordoliberalism, heir to the Freiburg school, advocates a rigid institutional framework favouring fiscal discipline and central bank independence, whereas the French neo-Keynesian tradition regards monetary policy as an instrument of macroeconomic regulation in the service of employment, as Tokarski (2019) analyses. Finally, payment practices attest to persistent cultural differences: the SPACE survey of the BCE (2024c) shows that in 2024, 30 % of Germans declared a preference for paying in cash in shops, against only 12 % of Finns and 14 % of Dutch citizens, which reflects differentiated levels of confidence in banking institutions and digital payment systems.

These cultural differences translate into recurrent political tensions over the conduct of the common monetary policy and into asymmetries in the transmission of monetary impulses, revealing the limits of monetary integration in the absence of cultural convergence. As Dyson & Marcussen (2009) analyse, the creation of the ECB certainly permitted the formal adoption of a single mandate of price stability, but the interpretation of that mandate and preferences as regards monetary action remain deeply divided along national lines. The sovereign debt crisis from 2010 to 2012 revealed these divergences: while Germany and the countries of northern Europe opposed the ECB’s programmes of purchases of government bonds, regarded as a violation of the principle prohibiting monetary financing, the countries of the south called for a massive intervention in order to avoid the break-up of the euro area. The analysis of Högenauer & Howarth (2016) shows that the ECB’s decisions from 2010 onwards aroused strong frustration within the German political class, where the ‘political sensitivity of low inflation’ and attachment to the monetary orthodoxy inherited from the *Bundesbank* remain deeply rooted. Likewise, the ECB’s interest rate rises in April and July 2011, motivated by inflationary fears widely judged premature, illustrate the persistent influence of German monetary culture within the governance of the institution. Moreover, the work of Schmidt (2015) emphasises that the uniform application of austerity fiscal rules (*one size fits all*), combining the ordoliberal philosophy of fiscal discipline and the neoliberal philosophy of structural reforms, has produced asymmetric effects on structurally different economies, aggravating the divergences between surplus countries and deficit countries. Finally, the empirical analysis of Berend & Prüser (2024) shows that the transmission of the common monetary policy remains heterogeneous: the economies of southern Europe experienced a marked divergence relative to the common output and inflation cycles after the financial crisis, which implies that monetary policy shocks propagate in a differentiated manner according to national economic and financial structures.

In conclusion, the euro area does indeed harbour different monetary cultures, legacies of distinct historical and institutional trajectories which continue to influence political preferences, economic behaviour and the transmission of the common monetary policy. These cultural differences are not residual, but structuring: they shape the debates on the governance of the ECB, generate recurrent political tensions in periods of crisis, and explain in part the asymmetries in the transmission of monetary impulses to national economies. German and Austrian ordoliberalism, founded on the primacy of stability and fiscal discipline, stands in head-on opposition to the neo-Keynesian traditions of southern Europe which favour growth and employment, creating a cultural fault line running through monetary union. The challenge for the future of monetary union is to know whether a progressive cultural convergence can take place as the generations renew themselves and as the memory of the euro takes the place of that of the national currencies, or whether these structural differences will continue to fuel recurrent political conflicts threatening the cohesion of the euro area. In the absence of fiscal federalism and of effective mechanisms of transfer between regions, the diversity of monetary cultures remains a major challenge for the long-run sustainability of European monetary union.

### 2.1.3 Some debates around monetary policy in Europe

The debate on monetary policy in Europe has been largely marked by the criticisms addressed to the **Stability and Growth Pact** (SGP), as well as to its successive reforms according to Delaplace (2021g): introduced in 1997 at the Amsterdam European Council, this pact imposed strict fiscal constraints on member states, notably the limit of a **public deficit not exceeding 3 % of GDP**. Although reformed on several occasions, notably in 2005 with the introduction of measures taking account of national specificities and in 2011 with the **six pack**, the pact did not make it possible to avoid the sovereign debt crisis of 2010. The golden rule imposing a structural deficit below 0.5 % of GDP (see the focus *infra*) gave rise to lively debate, notably as to its capacity to respond to asymmetric economic crises affecting the countries of the euro area differently, on account of their economic and social heterogeneity. Thus, an opinion piece by Bénassy-Quéré et al. (2018) (fourteen Franco-German economists in all) suggests abandoning the SGP rule requiring debt below 60 % of GDP, in favour of a rule according to which public spending must not grow faster than GDP in volume. In 2020, the Covid-19 pandemic further exacerbated these tensions, forcing the temporary suspension of the SGP so as to allow exceptional public spending, in the face of deficits reaching on average 5.8 % of GDP in the European Union, with a peak of 9.2 % in France in 2021, as Delaplace (2021g) recalls.

#### Focus...



**The ECB's *six pack*.** As Delaplace (2021g) sets out, the *six pack* is a set of reforms introduced by the European Union in 2011 in order to strengthen the economic and fiscal surveillance of the member states of the euro area, following the financial crisis of 2008 and the sovereign debt crisis of 2010. It consists of five regulations and one directive, which aim mainly to provide a better framework for national fiscal policies and to prevent the accumulation of macroeconomic imbalances within the euro area. Here are the main objectives and mechanisms of the *six pack*:

- **Strengthening of fiscal surveillance.** The *six pack* strengthens the rules of fiscal surveillance on the basis of the Stability and Growth Pact (SGP). It requires member states to keep their public deficits below the threshold of 3 % of GDP and their public debt below 60 % of GDP. If a country exceeds these limits, corrective measures must be put in place. The *six pack* likewise introduces the notion of the **structural deficit**, that is to say the budget balance adjusted for cyclical economic fluctuations, in order to assess the health of the public finances in the long run.
- **The fiscal golden rule.** One of the main innovations of the *six pack* is the introduction of a strict rule to limit the structural deficit. Member states must aim at a structural deficit below 0.5 % of GDP, which means that even in periods of economic growth, they must maintain a rigorous management of their public finances. This rule is written into the national law of the member states and subject to the surveillance of the European authorities.
- **Prevention of macroeconomic imbalances.** The *six pack* is not limited to fiscal criteria: it likewise introduces a mechanism to detect and correct the macroeconomic imbalances which could threaten the stability of the euro area. These imbalances include factors such as excessive trade deficits, property bubbles or excessively high levels of private debt. In order to monitor these imbalances, scoreboards are put in place for each member state. If imbalances are detected, an alert mechanism is activated, followed by specific recommendations and, potentially, by sanctions in the event of non-compliance.
- **Strengthening of sanctions.** The *six pack* introduces a stricter and more rapid regime of sanctions against member states which do not comply with the fiscal and macroeconomic rules. For example, fines of up to 0.2 % of GDP may be imposed on countries which do not take the measures needed to reduce their deficit or to correct their macroeconomic imbalances. Sanctions may be activated in a more automatic manner thanks to the principle of 'reverse majority': sanctions are decided automatically, unless a qualified majority of member states opposes them.
- **Medium-term fiscal planning.** The *six pack* requires member states to establish a multi-annual fiscal plan over three years. Each year, national budgets must be submitted for examination by the European Commission and the Council of the EU before their adoption, within the framework of the European Semester, in order to ensure that they comply with long-run fiscal objectives and with the rules of the Union.
- **Strengthened economic coordination.** The *six pack* aims to improve the coordination of economic policies between the member states of the euro area. It encourages structural reforms and a more harmonised management of national economic policies, in order to reduce the divergences within the Union and to guarantee long-run stability.

As the Commission européenne (2013) indicates, the prerogatives of the *six pack* were completed by the *two pack*, a set of two regulations adopted by the European Union in 2013 in order to strengthen the fiscal and economic surveillance of the countries of the euro area. On the one hand, the member states of the euro area must submit their draft national budgets to the European Commission and to the Eurogroup before their adoption: this allows the Commission to check that the drafts comply with European fiscal objectives and to propose adjustments if necessary; this stage, which falls within the framework of the 'European Semester', aims to strengthen transparency and fiscal discipline. On the other hand, member states faced with major financial difficulties (excessive deficits or debts) are subject to enhanced surveillance, including the obligation to prepare economic adjustment programmes: this surveillance likewise includes monitoring missions by the European Commission in order to guarantee that the fiscal consolidation measures are implemented. Thus, the *two pack* aims to prevent financial crises and to ensure a stricter fiscal management within the euro area, by increasing transparency and by coordinating the fiscal policies of the member states.

Beyond the debates on the SGP, it is the ECB's mandate itself that is under discussion: the main objective of the ECB's monetary policy of maintaining a rise in the HICP below but close to 2 % over the medium term is indeed the subject of several debates. First, the possibility of imposing a single inflation limit for all the countries of the euro area raises questions, given the **economic heterogeneity of the euro area**, according to Delaplace (2021g): a higher rate of inflation in certain countries leads to a rise in the overall HICP, which obliges the ECB to raise its key rates. This rise in rates may then harm consumption and investment, including in countries where inflation is low, thus weighing on their economic growth. Moreover, inflation disparities within the euro area translate into different real interest rates across countries, paradoxically favouring those whose inflation is higher, to the detriment of those which control their prices better. Second, as Delaplace (2021g) explains, the ECB is often criticised for having only one objective, price stability, unlike the American Federal Reserve (the Fed), which must also **promote full employment**. Although the ECB has progressively softened its position in response to crises, as in 2008 with the lowering of rates under the presidency of Mario Draghi, its mandate remains focused on the control of inflation. This single framework has sometimes been regarded as ill-suited, notably in 2001 or in 2007, when the Fed reacted more rapidly to the risks of a slowdown in growth. Finally, another debate concerns the taking into account of financial asset prices in price indices, a dimension that the ECB monitors without, however, integrating it directly into its monetary policy decisions, on account of the volatility of these assets and of the absence of a systematic correlation with the prices of goods and services, as Delaplace (2021g) recalls.

### Focus...



**The Phillips curve.** The curve of Phillips (1958), enriched by Samuelson & Solow (1960), established an inverse relationship between unemployment and inflation: according to this curve, a rise in inflation reduced unemployment, since firms, faced with increased demand, hired more. This relationship served as the foundation of the economic policies of the *Trente Glorieuses* (the three post-war decades of high growth in France), notably through *fine tuning* and *stop and go* policies. **Fine tuning** designates an approach to macroeconomic regulation in which the public authorities, notably via fiscal and monetary policy, finely adjust economic variables so as to maintain growth while controlling inflation; the underlying idea is that the state can stabilise the economy by intervening in a subtle and continuous manner, in order to correct imbalances in demand, without provoking strong fluctuations in economic activity. In parallel, **stop and go** policies rest on an alternation of expansions and restrictions of aggregate demand according to the economic situation: in periods of overheating, governments adopt restrictive measures (the *stop*) so as to contain inflation, while in periods of slowdown, they conduct expansionary policies (the *go*) so as to stimulate demand and reduce unemployment.

However, the oil shock of 1973 upset this dynamic. While inflation was exploding, unemployment was also rising, a phenomenon that Blinder (1979) names '**stagflation**': Sterdyniak & Debonneuil (1984) specify that in France, inflation reached 11 % in 1974 and unemployment rose from 3 % in 1970 to more than 10 % at the beginning of the 1980s: this observation led to a questioning of the validity of the curve of Phillips (1958) in the long run. The analysis of the **wage-price spiral**, developed by Sterdyniak & Debonneuil (1984), showed how inflation could become self-sustaining through the indexation of wages, reinforcing an inflationary spiral. Moreover, Friedman (1968) holds that the trade-off between unemployment and inflation was only a short-run illusion: he introduced the concept of the **natural rate of unemployment**, asserting that inflationary policies could not durably influence that rate. Natural unemployment refers to the level of unemployment which exists even when the economy is in a situation of full employment, that is to say when all the forces of the labour market are operating optimally. It includes mainly two types of unemployment: **frictional unemployment**, which results from the normal movements in the labour market, such as periods of transition between two jobs or the search for a first job after training, and **structural**



**unemployment**, which arises when workers' qualifications do not correspond to the needs of the labour market or when certain regions or industries decline.

The introduction of the euro in 1999 and the creation of the ECB further strengthened this more orthodox monetary orientation. With a mandate strictly oriented towards price stability, the ECB established itself as the guardian of an inflation kept under control around 2 %: the member states of the euro area, constrained by the convergence criteria of the SGP, lost control of their monetary policy, no longer being able to use tools such as devaluation or money creation in order to adjust inflation and influence employment. This centralisation of monetary policy means that the very idea of a trade-off between inflation and unemployment has been abandoned, in favour of a management oriented exclusively towards price stability. Consequently, the capacity of member states to adjust inflation in order to revive employment has been institutionally reduced, depriving national governments of this tool of macroeconomic regulation. Today, this responsibility belongs entirely to an independent ECB, whose priorities do not explicitly take account of growth or unemployment, unlike the Fed, as De Grauwe (2016) recalls.

### Key Takeaways



Monetary policy in the euro area falls to the Eurosystem (the ECB and the national central banks), whose mission is articulated around three strands: the issuance of fiat money regulating the money supply, the guarantee of price stability via the key interest rates, and the function of lender of last resort ensuring financial stability [Brana & Cazals (2014b)]. Historically, the magic square of Kaldor (1961) reveals the simultaneous incompatibility of four major objectives (growth, employment, price stability, external balance), justifying by the principle of Tinbergen (1959) a clear separation of monetary and fiscal instruments. From the 1980s onwards, under the influence of monetarists such as Friedman (1956), the fight against inflation becomes the priority objective of central banks, accompanied by a growing independence in order to limit the inflationary bias linked to the time inconsistency of discretionary decisions [Kydland & Prescott (1977), Barro & Gordon (1983)]. The independence of the central bank, by establishing a conservative central banker and by imposing rules rather than discretionary policies [Rogoff (1985), Elster (1986)], makes it possible to resolve the credibility deficit and to improve the anchoring of expectations. However, this independence reaches its limits: it does not remove the risks of inconsistency between fiscal and monetary policies (the snowball effect of indebtedness), it remains vulnerable to fiscal dominance if the government pursues an unsustainable expansionary fiscal policy, and it may paradoxically give rise to financial bubbles by keeping rates abnormally low [Sargent & Wallace (1981), Borio & Shim (2003)]. In the 1990s, inflation targeting replaces monetary targeting, crystallised by the rule of Taylor (1993) which systematically adjusts the key rate as a function of inflation and output gaps, although external structural factors (offshoring to low-cost countries) also explain the low inflation observed [Coupey-Soubeyran (2012a)].

## 2.2 The management of recent crises

Economic crises do not resemble one another: where 2008-2009 threatened a prolonged deflation, 2022 saw inflation reach levels unprecedented for forty years. The ECB managed these two opposite shocks with radically different instruments, to the point of overturning established monetary doctrine. Faced with the financial crisis of 2009 and the Covid crisis, it implemented **quantitative easing**; by contrast, faced with the return of inflation in 2022, it drastically raised its key rates, abandoning at the same stroke **unconventional policies**.

### 2.2.1 Unconventional monetary policy in the face of the 2008 crisis

The financial crisis of 2008-2009 compelled the ECB to adopt unconventional monetary policy measures in order to support the European economy. After lowering its key rates to historically low levels, the ECB launched several innovative programmes to stabilise financial markets and avoid a prolonged deflation. Among these measures, quantitative easing (QE), qualitative easing, and *forward guidance* played a key role. **Quantitative easing**, launched in January 2015 by the ECB (and as early as November 2008 by the Fed), consisted in the massive purchase of financial assets, mainly government bonds, for an initial monthly amount of 60 to 80 billion euros from March 2015 to December 2017 (an envelope progressively reduced to zero by January 2019 before returning to 20 billion of monthly purchases from November 2019 onwards): this programme made it possible to inject liquidity massively into the economy, reaching a total of 4968 billion euros of assets held by the Eurosystem at the end of 2022 according to the Banque de France (2024d). By increasing the demand for financial securities, the ECB thus contributed to a significant fall in long-term interest rates, favouring credit and investment. Indeed, the ECB's massive purchases create a strong demand for bonds and other financial assets, which drives their price up; the increase in the prices of



securities entails a fall in their interest rates, since bonds and securities are generally traded on the basis of a yield inversely proportional to their price. For example, the interest rate on ten-year French bonds fell from 1.64 % in 2014 to 0.84 % in 2015 (against 4.23 % in 2008 and  $-0.15$  % in 2020), just after the announcement of QE, as INSEE (2024d) indicates. This fall in interest rates then makes government bonds (Treasury bills) less attractive to private investors who are seeking higher yields; consequently, these investors are given an incentive to turn to other, riskier types of asset offering higher yields, such as corporate bonds or shares, thereby favouring investment. The Banque de France (2024d) thus estimates at 1.9 points of GDP the additional growth generated by the asset purchase programme in the euro area between 2016 and 2020.

### Short-Answer Question



**How is the existence of negative interest rates to be explained?** The appearance of negative interest rates constitutes an economic anomaly defying the intuition according to which lending money should always procure a positive return. Whereas interest rates had reached their historic peak in the 1980s, with nominal yields above 15 % in the United States, the trajectory progressively reversed until it crossed the zero threshold from 2014 onwards in several advanced economies. The European Central Bank thus maintained a negative deposit rate from June 2014 to July 2022, reaching a floor of  $-0.5$  %, while Switzerland was still maintaining in 2023 a key rate of  $-0.75$  % according to La Finance pour tous (2023). The interest rate represents the price of time and of the renunciation of immediate liquidity; when it becomes negative, the borrower is paradoxically remunerated for borrowing, and the lender agrees to pay in order to place his funds. This inversion of roles contradicts the logic of preference for the present and calls into question the very foundations of the theory of interest. As Aglietta & Valla (2016) emphasise, negative rates can be understood only by articulating long-run structural factors and short-run monetary policies. What are the economic mechanisms which allow the emergence and the persistence of negative interest rates in contemporary economies? This question requires us to examine first the structural factors which have oriented the equilibrium interest rate downwards, before showing how accommodative monetary policies crossed the zero threshold and durably maintained negative rates.

The existence of negative interest rates is explained first of all by a structural evolution of the determinants of the equilibrium real interest rate, linked to the conjunction of demographic ageing, weak potential growth and the rise of a preference for safety in a context of high macroeconomic uncertainty. As Buchner (2020) analyse in their work on secular stagnation, the differential between the interest rate paid on public debt and the growth rate of GDP has narrowed continuously since the 1980s, becoming negative in the majority of advanced economies during the 2010s. This phenomenon reflects a fall in the natural rate of interest, that is to say the level of rate which ensures equilibrium between saving and investment in a situation of full employment. Demographic ageing mechanically increases aggregate saving, since cohorts approaching retirement accumulate assets while the younger generations, being fewer in number, borrow less to finance their consumption or their housing. In parallel, potential growth has slowed in the advanced economies, falling from about 3 % in the 1990s to less than 1.5 % on average in the euro area during the 2010s according to Buchner (2020), which reduces the anticipated profitability of productive investments and therefore the demand for financing. Finally, the perception of a high macroeconomic risk, notably after the financial crisis of 2008 and the European sovereign debt crisis, led agents to favour holding liquid and low-risk assets, even at the price of a negative return. Khoury & Pal (2020) show that this ‘flight to safety’ translates into a liquidity and safety premium which may largely offset the negative nominal yield, notably for German or Swiss sovereign bonds regarded as safe havens. From then on, the existence of negative rates does not reflect an irrationality on the part of agents, but a trade-off in which the value of insurance and of capital preservation prevails over the search for current yield.

While these structural factors prepared the ground, it is the conduct of exceptionally accommodative monetary policies which permitted the effective move into negative territory and the durable maintenance of negative nominal rates across a wide range of maturities and financial instruments. As the Banque de France (2025d) recalls, the European Central Bank lowered its deposit rate to  $-0.1$  % in June 2014, then progressively to  $-0.5$  % in September 2019, in order to fight the risk of deflation and to stimulate demand in a context of persistent under-utilisation of productive capacity. This policy of negative key rates falls within a logic of classical monetary transmission: by remunerating negatively the excess reserves of commercial banks, the central bank gives them an incentive to lend more to the real economy rather than to accumulate sterile liquid balances. However, Aglietta & Valla (2016) emphasise that the effectiveness of this transmission remains contested, notably because of the downward rigidity of bank deposit rates, which protects savers but compresses intermediation margins. In parallel, the programmes of *quantitative easing* conducted by the ECB, the Bank of Japan and other central banks amplified the movement by creating a structural imbalance between the supply of and demand for sovereign securities on bond markets. The massive purchases of government bonds by central banks (more than 2 500 billion euros of cumulative net purchases

by the ECB between 2015 and 2018) drove bond yields down across all maturities, pushing even ten-year rates into negative territory for Germany, France, the Netherlands and several other euro area countries, as La Finance pour tous (2023) recalls. Finally, the analysis of Khoury & Pal (2020) shows that negative rates produce differentiated effects across economic agents: while borrowing states and well-rated firms benefit from exceptional financing conditions, savers see their real return eroded, which may paradoxically increase their precautionary saving rate, as the empirical study of Stall (2023) on twenty countries between 2000 and 2020 confirms.

The existence of negative interest rates results from the combination of profound structural transformations of the saving-investment equilibrium and of deliberately expansionary monetary policies aimed at countering deflationary risks in the advanced economies. Demographic factors, weak potential growth and the increased preference for safety have lowered the natural rate of interest to historically low levels, and even to negative levels in real terms. Central banks then crossed the psychological threshold of nominal zero by instituting negative key rates and by conducting massive asset purchase programmes which tipped bond yields into negative territory. These negative rates, although counter-intuitive, constitute a coherent response to an economic environment characterised by an excess of global saving and a chronic insufficiency of aggregate demand. However, their macroeconomic effectiveness remains debated, notably because of their redistributive effects unfavourable to savers and of their potentially perverse impact on financial stability by encouraging excessive risk-taking. Since the end of 2021, the return of inflation has led central banks to raise their key rates, putting a provisional end to the era of negative rates. The future will tell whether this episode was no more than a historical parenthesis or whether, in the face of new crises, the monetary authorities will once again be compelled to cross the zero threshold in order to preserve macroeconomic stability.

In parallel, the ECB also eased the conditions of access to liquidity for commercial banks, a policy termed **qualitative easing**: as Valla (2014) explains, this approach involved broadening the range of assets accepted as collateral for the loans granted to banks, including assets of lower quality or issued by private firms. This arrangement thus aimed to strengthen the capacity of banks to lend to the real economy, while supporting the stability of the financial system. At the end of 2022, the outstanding amount of Eurosystem loans to banks within the framework of these long-term refinancing operations stood at 1300 billion euros according to the Banque de France (2024e). This massive support made it possible to maintain favourable credit conditions for firms and households, thereby avoiding a severe credit contraction and supporting economic activity.

### Focus...



**How did asset purchase programmes restore market confidence?** The asset purchase programmes launched by the ECB played a key role in restoring the confidence of financial markets during the successive crises which struck the euro area. A striking example which the Banque de France (2024d) mentions is the **Securities markets programme** (SMP), put in place in 2010 in response to the sovereign debt crisis, notably the Greek crisis: this programme allowed the Eurosystem to buy government bonds on the secondary market, thereby contributing to the stabilisation of debt markets by providing liquidity and by lowering interest rates on sovereign bonds. In 2011 and 2012, at a time when tensions on the markets were acute, the ECB launched two *very long term refinancing operations* (**TLTRO**) with a maturity of three years, injecting nearly 1000 billion euros into the banking system so as to ensure the solvency of European banks and, indirectly, to strengthen investors' confidence in the capacity of states to honour their debts, as the CVCE (2014) explains.

Another crucial moment in the restoration of confidence was the summer of 2012, when the president of the ECB, Mario Draghi, declared that the Eurosystem would do 'whatever it takes' to save the euro: this announcement, followed by the establishment of a programme of unlimited purchases of public debt securities (**outright monetary transactions**, or OMT), was enough to restore market confidence without the programme even being activated: as the Banque de France (2024d) explains, this strong signal made it possible to contain the risk premia on the sovereign debts of the countries in difficulty, notably Spain and Italy. In parallel, the *targeted long term refinancing operations* (**TLTRO**) launched in 2014, as well as the quantitative easing begun in 2015, likewise strengthened market confidence by injecting massive liquidity into the economy and by keeping interest rates at historically low levels. Indeed, between 2015 and 2016, the ECB broadened its purchase programme to corporate bonds, thereby stimulating private investment and supporting growth, as the Banque de France (2024d) sets out. These measures made it possible to prevent a deflationary spiral and to support the real economy, while consolidating the stability of financial markets.

Finally, the ECB used **forward guidance**, a communication tool aimed at influencing market expectations. From 2012 onwards, it began to announce explicitly that key rates would remain low for as long as necessary, thereby providing economic actors with better visibility over the future evolution of monetary policy: as the Banque de France (2024d) explains, this strategy made it possible to reduce uncertainty on financial markets, giving firms an incentive to invest

and households an incentive to consume, while forestalling expectations of a rate rise which would have held back the recovery. This approach thus played a crucial role in the effectiveness of the other unconventional measures by ensuring them a performative impact even before their effective implementation.

### 2.2.2 The Covid-19 crisis and the risk of deflation

The economic crisis engendered by the Covid-19 pandemic required the ECB to adopt unprecedented measures in order to stabilise financial markets and support the economy of the euro area, and to pursue its unconventional monetary policy: as early as March 2020, faced with the risks of deflation and with the fragmentation of sovereign debt markets, the ECB launched the *Pandemic emergency purchase programme (PEPP)*, initially endowed with 750 billion euros, before being extended to 1850 billion in December 2020, as Blot (2021) indicates. The PEPP made it possible to absorb a substantial part of the debts issued by member states to finance their recovery plans, reaching 86 % of the public debts issued by euro area countries in 2020 according to Martin et al. (2021). These massive purchases lowered long-term interest rates, stabilised bond markets, and reduced the rate spreads between countries, notably between Germany and more vulnerable economies such as Italy and Spain. Thus, the ECB effectively contained the risk of fragmentation of the euro area, by modulating its purchases of securities so as to respond to specific tensions on national markets, as Blot (2021) explains. One of the major challenges of this period was therefore to preserve financial stability while guaranteeing low-cost financing of the massive public debts issued in response to the crisis. The PEPP was distinguished from the preceding asset purchase programmes by its flexibility, allowing the ECB to adjust its weekly purchases according to market conditions: this flexibility was essential in order to avoid an explosion of sovereign rate spreads between member countries, in particular for Italy, where rates had risen sharply at the beginning of 2020, as Martin et al. (2021) recall. The announcement of the PEPP by Christine Lagarde had an immediate effect on the markets, lowering interest rates by several basis points in the most fragile countries: as Blot (2021) indicates, a 1 % increase in weekly purchases under the PEPP reduced the Italian spread by 1.9 basis points, and that of Spain by 1.2 points. This massive support for public debt therefore allowed states to maintain their economic support plans while avoiding a liquidity or solvency crisis.

#### Focus...



**What are the risks of deflation?** As the Ministère de l'Économie (2014) indicates, between October 2012 and July 2014, the annual rate of inflation in the euro area fell from 2.5 % to only 0.4 %, bringing to light an increased risk of deflation, notably in countries such as Greece, which have already experienced a prolonged fall in prices. **Deflation**, defined as a continuous fall in the general price level, is feared because it can bring about a deflationary spiral in which the fall in prices gives consumers and firms an incentive to delay their purchases and investments, thus aggravating the economic contraction. For example, as the Ministère de l'Économie (2014) adds, energy prices in the euro area fell by 2 % year on year in August 2014, accentuating this deflationary trend. The main risk associated with deflation is a continuous fall of inflation towards negative levels, which can make public and private debts more costly in real terms, thereby harming economic growth. Indeed, when inflation is positive, it progressively erodes the real value of debt, making its repayment easier; by contrast, in a period of deflation, the reverse occurs: the real value of debts increases. For example, if a country or a firm has contracted a debt of a hundred million euros, a fall in prices means that the income generated by these economic agents falls, but the nominal amount of the debt remains unchanged: repaying this debt therefore becomes more costly in real terms. For governments, deflation can likewise weigh down the burden of public debt, making the financing of public projects more difficult and increasing the risk of budgetary crises.

The Covid-19 crisis also offered the ECB opportunities to reassess its role and its instruments in the management of future crises. The **extension of forward guidance measures** made it possible to stabilise market expectations and to strengthen the effectiveness of the other unconventional tools. However, as Villeroy de Galhau (2022) recalls, the ECB put an end to this policy in July 2022 in order to adopt a more flexible approach of 'meeting-by-meeting' decision-making. This decision was motivated by the situation of extreme uncertainty which has prevailed since the Covid-19 pandemic: according to Villeroy de Galhau (2022), *forward guidance*, although relevant in a period of relative economic stability, could limit the capacity to react rapidly in the face of unforeseen economic shocks. Here we find, between the lines, the reference to the risk of **time inconsistency** which Kydland & Prescott (1977) set out: an announcement that is optimal at date  $t$  may cease to be optimal at date  $t + 1$ , which gives the central bank an incentive to depart from its choice at  $t$ . Thus, by abandoning this strategy, the ECB chose to give itself more flexibility in the management of monetary policy, avoiding locking itself into overly rigid commitments concerning future interest rates, even though, as Villeroy de Galhau (2022) indicates, it remains essential that the yield curve should coherently reflect the intentions of monetary policy decision-makers, notably because medium- and long-term

interest rates play a crucial role in the transmission of monetary policy: central bankers must continue to provide a certain predictability, without, however, tying their hands with explicit commitments. The abandonment of *forward guidance* therefore does not mean the adoption of an overly 'empiricist' approach, which would consist in mechanically adjusting key rates according to immediate economic data. Indeed, as Villeroy de Galhau (2022) warns, such an approach would risk leading to an overreaction of monetary policy in the face of volatile cyclical data, especially in a context of significant time lags between monetary policy decisions and their effects on the real economy.

### 2.2.3 The return of inflation and of conventional policies since 2022

The president of the ECB, Lagarde (2022), emphasised that 'headline inflation, still negative in December 2020, has risen by 9.4 percentage points relative to the trough reached during the pandemic, standing at its highest level' in August 2022. The inflationary pressures of recent years, far stronger and more durable than anticipated, were exacerbated by the Covid-19 crisis and the Russian-Ukrainian conflict, which disrupted supply chains and drove up the prices of energy and commodities. To be sure, the annual rate of inflation in the euro area fell back to 2.4 % in November 2023, as Eurostat (2023) indicates, but this disinflation results from an unprecedented rise in the ECB's key rates, which jumped from 0.5 % in July 2022 to 4.5 % in September 2023, as the figures of the BCE (2024a) qualify. Indeed, the first lever available to central banks in order to curb inflation is to raise their key rate, which makes investment by firms more difficult: the demand for capital and, consequently, long-term investment are closely linked to the cost of capital; this relationship flows from standard microeconomic theory, according to which firms opt for the optimal combination of capital and labour so as to maximise their profits. The tightening of interest rates thus sounds the death knell of unconventional monetary policies and marks the return to the use of standard levers of monetary policy. Indeed, the policy of QE is reversed: from December 2022 onwards, as Schnabel (2023), a member of the Executive Board of the ECB, indicates, the latter adopts a policy of *quantitative tightening* (a choice which the Fed made as early as 2017). **Quantitative tightening** (QT) is a monetary policy used by central banks to reduce the size of their balance sheet, by reversing the process of quantitative easing (QE) which had been put in place to stimulate the economy. Unlike QE, where central banks massively buy financial assets such as government bonds so as to inject liquidity into the economy and drive long-term interest rates down, QT consists in withdrawing that liquidity. This is generally done by not renewing securities reaching maturity or, in some cases, by directly selling assets on the financial markets. The main objective of QT is to normalise monetary policy after a period of prolonged easing, by progressively bringing interest rates back up to more neutral levels and by controlling inflationary pressures. By withdrawing liquidity from the financial system, QT raises long-term interest rates and tightens financial conditions, making access to credit more costly for households and firms.

#### Focus...



**The steering of short-term rates by the ECB.** As the Banque de France (2025c) explains, in managing their treasury, commercial banks have two main options: borrowing or depositing liquidity with the central bank, or borrowing and lending on the interbank market. As regards the first option, every Tuesday, banks may **borrow liquidity** from the Eurosystem for a period of one week. The interest rate is determined by tender, based on the main key rate, called the **main refinancing operations rate**. Banks also have the possibility of borrowing for a period of twenty-four hours from the central bank, but at a higher rate, known as the **marginal lending facility rate**. Conversely, if a bank has a surplus of liquidity, it may deposit it with the central bank for twenty-four hours, which will remunerate it at the **deposit facility rate**, generally lower than the borrowing rate. On the **interbank market**, banks lend one another liquidity at very short term. The rate applicable to these loans, such as the *euro short term rate* (formerly Eonia), settles between the central bank's deposit facility rate, which constitutes a floor, and the marginal lending facility rate, which plays the role of a ceiling; this mechanism forms what is called the **interest rate corridor**. Complementing this system, conventional monetary policy also relies on a mechanism of **remunerated required reserves**, requiring commercial banks to deposit a fraction of their customers' deposits with the central bank. The amount of required reserves is computed as an average over a period of about six weeks. This arrangement makes it possible to stabilise the interbank market rate, since banks can adjust the amounts deposited on a daily basis according to the evolution of interbank rates, such as the *euro short term rate*. As the Banque de France (2025c) explains, a bank may take advantage of a high *euro short term rate* to lend its reserves on the interbank market, which will exert downward pressure on the rate. In addition, this mechanism increases the effectiveness of the transmission of key rates to the real economy, since the obligation to deposit reserves strengthens the constraint on the liquidity of commercial banks, increasing their financing needs and reinforcing the influence of key rates.

The rise in interest rates may therefore imperil economic growth: the **Mésange model**, the quarterly macroeconomic model of INSEE and of the Directorate General of the Treasury (DGT) taken from Allard-Prigent et al. (2002), indeed recalls that a monetary shock of one hundred basis points (that is, one percentage point) on the central bank's key rates translates into a fall in gross domestic product (GDP) of 0.2 % after one year, 0.8 % after three years and 3 % in the long run; this fall is explained notably by the sharp drop in investment of 2.7 % after three years according to Bardaji et al. (2017). Moreover, an increase in the cost of capital then gives firms an incentive to substitute labour for capital, entailing a decrease in investment: the long-run elasticity of investment with respect to this user cost is estimated at 0.44 in the Mésange model; assuming a capital depreciation rate of 10 %, initial nominal rates at 0 %, and neglecting risk premia, a one percentage point increase in the interest rate entails in the long run a reduction in investment of 5 %, as Aurissergues (2022) indicates. In addition, the increase in the cost of capital translates into a rise in production costs, affecting selling prices, inducing *in fine* inflation and a loss of competitiveness. Although this positive effect of interest rate rises on prices via the cost-of-capital channel has recently been explored, notably by Beaudry et al. (2022), it is important to note that its detection is complex with more agnostic empirical methods, such as unrestricted VAR models or local projections. And while positive short-run effects on prices are sometimes observed, the results often become non-significant or clearly negative at more extended time horizons: far from curbing inflation, the rise in key rates therefore appears *in fine* to reinforce it in the long run for Miranda-Agrippino & Ricco (2021).

### Key Takeaways



Confronted with the financial crisis of 2008-2009, the ECB adopted massive unconventional measures: *quantitative easing* (QE) launched in January 2015 – imitating the Fed's decision of November 2008 – consisted in massive purchases of government bonds (60 to 80 billion euros per month initially, reaching 4968 billion euros at the end of 2022), which by increasing the demand for securities drove long rates down and generated 1.9 points of GDP of additional growth between 2016 and 2020 [Banque de France (2024d)]. *Qualitative easing* in parallel broadened the range of assets accepted as collateral, strengthening credit to the real economy (1300 billion euros outstanding at the end of 2022) [Valla (2014)], while *forward guidance* announcing that rates would be kept low reduced uncertainty and exerted a performative effect on the markets [Banque de France (2024d)]. Faced with the Covid-19 crisis, the *Pandemic emergency purchase programme* (PEPP), endowed with 1850 billion euros and flexible on a weekly basis, absorbed 86 % of the public debts issued in 2020, containing sovereign rate spreads [Blot (2021), Martin et al. (2021)]. However, post-pandemic uncertainty led in July 2022 to the abandonment of *forward guidance*, favouring reactive flexibility over rigid commitments liable to generate time inconsistency [Kydland & Prescott (1977), Villeroy de Galhau (2022)]. The return of inflation from 2022 onwards subsequently required an unprecedented rise in the ECB's key rates (from 0.5 % in July 2022 to 4.5 % in September 2023), reversing QE into *quantitative tightening* (December 2022), which tightens financial conditions by reducing the balance sheet through the non-renewal of securities reaching maturity [Schnabel (2023)]. This restrictive monetary transmission generates significant economic costs: according to the Mésange model, a shock of 100 basis points on key rates reduces GDP by 3 % in the long run, with investment falling by 2.7 % after three years [Allard-Prigent et al. (2002), Bardaji et al. (2017), Aurissergues (2022)]. The cost-of-capital channel on prices moreover remains ambiguous, the positive short-run effects often becoming non-significant or negative at extended horizons, suggesting that the rise in rates potentially reinforces long-run inflation rather than curbing it [Beaudry et al. (2022), Miranda-Agrippino & Ricco (2021)].

## 2.3 The regulatory challenges of the financial system

Recent crises have brought to light a paradox of contemporary finance: never have banking and insurance systems been so tightly framed by **prudential standards**, and yet the question of their stability is constantly raised. The challenges of regulation are no longer reducible to checking that each institution is solvent on paper, but consist in ensuring that the system as a whole can absorb extreme shocks without interrupting the financing of the real economy.

### 2.3.1 Banks and the Basel Accords

The **Basel Accords** are a set of international banking regulations drawn up by the Basel Committee on Banking Supervision, which aim to guarantee global financial stability by imposing risk management rules on banks. Since their creation in 1988 with Basel I, these accords have evolved so as to incorporate the lessons drawn from the various financial crises. The first accord, **Basel I**, introduced in 1988, aimed to guarantee that banks maintain a minimum level of own funds in order to cover credit risks. As the management consultancy Sia Partners (2017) recalls, the

**Cooke ratio**, which is one of its main contributions, requires banks to hold own funds equal to at least 8 % of their assets weighted according to the risks of counterparty default (according to ACPR (1991): 0 % for OECD states, 20 % for OECD banks and local authorities and 100 % for other parties, with the exception of certain particular operations such as leasing and mortgage lending, for which the risk of default is estimated at 50 %). More precisely, this ratio is composed of three levels of capital: *tier 1* own funds (essentially share capital, reserves and minority interests), with a minimum requirement of 4 %, *tier 2* own funds (essentially subordinated securities of infinite maturity and of fixed term) and *tier 3* own funds (subordinated securities covering market risks), the latter being reserved for market risks. Recall, with Pernaud-Orliac (2012), that **subordinated securities** are financial instruments representing a debt issued by a firm, a bank or another type of institution, but which are repaid last in the event of the liquidation or bankruptcy of the issuer. As ACPR (1991) indicates, the introduction of the prudential rules of Basel I led to an increase in the own funds of French banks of 14 % between 1988 and 1989, and banks concentrated more heavily on market operations which do not consume own funds, with a rise in the share of off-balance-sheet instruments linked to interest rates and exchange rates, going from 34.2 % to 44.5 % of the total balance sheet between 1988 and 1990. This strategic reorientation raised concerns as to a possible decrease in the supply of credit: as Sia Partners (2017) emphasises, the growth of over-the-counter derivatives from 1996 onwards brought about considerable off-balance-sheet risks, not captured by the regulatory framework then in force, the Basel I accords concentrating essentially on credit risk and taking no account of market risks and operational risks.

The Basel I framework therefore rapidly proved insufficient in the face of the growing complexity of financial products in the 1990s, notably with the rise of over-the-counter derivatives. In order to respond to these limitations, the Basel Committee introduced in 2004 **Basel II**, a more comprehensive reform aiming to apprehend the whole range of banking risks more effectively. This new regulation rests on three pillars: (i) minimum own funds requirements now covering not only credit risk, but also market and operational risks, (ii) more thorough prudential supervision, allowing regulators to assess banks' risk management strategies in a finer-grained manner, and (iii) market discipline, which requires banks to disclose information on their risk exposures so as to increase transparency. One of the striking aspects of Basel II is the introduction of finer methods of risk assessment, such as the approaches based on banks' **internal ratings-based** (IRB) systems for assessing credit risk. As Sia Partners (2017) explains, Basel II introduces the **McDonough ratio**, which requires banks to maintain a level of own funds at least equal to 8 % of their risk-weighted assets, like the Cooke ratio, but with several significant improvements in terms of computation. Indeed, as Navarro (2008) emphasises, under Basel I a loan to an OECD state was weighted at zero risk whatever the country considered, whereas Mexico's credit risk differs from that of France; likewise, all private firms, whatever their size, their sector and their status, saw their risk weighted by the same factor. With Basel II, the level of the risks taken by banks is better apprehended: banks may, if they so wish and by making their method of computation explicit, set the risk weightings themselves internally. As Navarro (2008) explains, the idea is that banks are best placed to assess their risks, on condition that the method used to compute the level of the risks taken be made public; banks moreover retain the possibility of choosing the weighting grid proposed by the Basel Committee, which has been enriched. In France, the McDonough ratio has replaced the Cooke ratio since 1 January 2007 and applies to all banks, whereas in the United States the McDonough ratio applies only to large international banks (about twenty of them, whereas there are more than seven thousand banks in the country according to Navarro (2008)). However, despite these advances, the financial crisis of 2008 revealed important gaps in Basel II, notably as regards systemic risk and liquidity risk, which were not sufficiently covered. As Azzouzi Idrissi & Madiès (2012) recall, the failure of banks such as Northern Rock, which nevertheless complied with the requirements of Basel II, showed that solvency was not sufficient to guarantee banking stability if liquidity risks were neglected: between 14 and 15 September 2007, Northern Rock experienced a **bank run** in which depositors withdrew two billion pounds sterling, that is 10 % of deposits, which led the British government to nationalise Northern Rock in February 2008, for want of a buyer.

Faced with the shortcomings revealed by the crisis, the Basel Committee introduced in 2010 the **Basel III** accords, a set of reforms aiming to strengthen the resilience of the global banking system. Basel III marks a break by broadening the regulatory framework to include measures intended to improve the quality of own funds and to manage **liquidity risks** more effectively: the solvency ratio rises from 8 % to 10.5 %, with a stricter requirement as to the quality of own funds. Thus, *tier 1* own funds must henceforth represent at least 6 % of risk-weighted assets, and the **core equity tier 1** ratio (ordinary own funds: share capital net of holdings, reserves and minority interests) must be 4.5 %, as Sia Partners (2017) indicates. In addition, a supplementary 'safety buffer' equal to 2.5 % is instituted, on which banks will be able to draw in the event of difficulties so that they may preserve a minimum level of capital: according to Azzouzi Idrissi & Madiès (2012), this conservation buffer is intended to absorb losses without banks having to reduce their lending operations during crisis periods. Basel III likewise introduces two new liquidity ratios. On the one hand, the **liquidity coverage ratio** or **LCR** requires banks to hold high-quality liquid assets sufficient to cover their financing needs over a period of thirty days in the event of a liquidity crisis. This LCR ratio must be above 100 %, and

as Azzouzi Idrissi & Madiès (2012) recalls, which means that the bank must possess liquid reserves sufficient to cover net cash outflows during that period (for example, eligible liquid assets include high-quality sovereign debt securities or reserves held with central banks). On the other hand, the *net stable funding ratio* or **NSFR** requires banks' available stable resources to exceed their stable financing needs over a one-year horizon: as Azzouzi Idrissi & Madiès (2012) explains, this ratio aims to limit banks' excessive dependence on short-term funding and to encourage a prudent management of liquidity over the longer term. By introducing these ratios, Basel III thus seeks to limit the accumulation of liquidity imbalances which contributed to the 2008 crisis, in particular in the large international banks which operate on a global scale.

### Focus...



**'Basel IV'.** Regarded as an extension of the Basel III regulation, the Basel IV reform (or Basel III *Endgame*), published on 7 December 2017, aims to improve and harmonise risk assessment and to strengthen the financial stability of the banking sector; its application is expected on 1 January 2025 with a transition phase extending until 2030. The introduction of Basel IV brings substantial changes which mainly affect the computation of *risk-weighted assets* (RWA) and the methods of assessing exposures. One of the main innovations of Basel IV is the establishment of a capital floor or **output floor**, which requires that the RWA computed by banks' internal models may not be lower than 72.5 % of those computed by the standardised approaches, as Nexialog Consulting (2022) indicates; this floor will be implemented progressively, beginning at 50 % in 2025 so as to reach 72.5 % in 2030. This measure aims to limit the variability of RWA across banks and to strengthen the comparability of solvency ratios, thereby reducing the risk of an underestimation of risks by internal models. Moreover, Basel IV introduces an entry floor or **input floor**, which sets minimum values for the parameters used in internal models, such as the probability of default, the loss given default and the exposure at default: these thresholds guarantee that capital requirements cannot fall below levels judged prudent, as Nexialog Consulting (2022) explains.

Another major change brought by Basel IV concerns the revision of the methods of computing risks. The **advanced IRB approach**, which allowed banks to use their own models to assess certain exposures, is henceforth prohibited for exposures to financial institutions and to large firms, as Nexialog Consulting (2022) recalls. Moreover, the standardised approach is revised so as to include more risk weightings and to take account of new methods of computing market risk, in accordance with the requirements of the **FRTB** (*fundamental review of the trading book*) framework; Basel IV likewise introduces a revision of the method of computing the **CVA** (*credit valuation adjustment*) in order to protect banks more effectively against losses linked to variations in the market prices of derivatives in the event of a deterioration in the solvency of counterparties. The stakes of Basel IV are therefore high for banks, since this reform will bring about an increase in capital needs, in particular for banks making extensive use of internal models: according to Sia Partners (2017), the capital needs of European banks could increase by 50 %, that is about 850 billion euros of additional capital to be raised by 2030.

### 2.3.2 Stress tests and the stability of the banking system

Banking **stress tests** constitute a fundamental prudential instrument allowing supervisory authorities to assess the capacity of financial institutions to withstand severe economic and financial shocks, and thus to ensure that the banking system has sufficient capital to absorb losses in a period of crisis while continuing to finance the real economy. As the Comité de Bâle sur le contrôle bancaire (2009) explains in its guiding principles, *stress tests* are forward-looking exercises which make it possible to simulate the impact of adverse macroeconomic scenarios on the solvency and liquidity of banks by projecting their potential losses, their revenues, their expenses and their capital ratios over a determined time horizon. These exercises took on considerable importance after the global financial crisis of 2008, when it appeared that traditional own funds requirements and static solvency ratios did not make it possible fully to apprehend the vulnerabilities of the banking system in the face of systemic shocks. *Stress tests* differ from the Basel capital requirements by their dynamic and forward-looking nature: whereas solvency ratios such as the McDonough ratio or the tier 1 capital ratio measure the balance sheet position at a given instant, *stress tests* project the evolution of bank balance sheets over several quarters by incorporating coherent macroeconomic assumptions about the growth of gross domestic product, the unemployment rate, residential and commercial property prices, interest rates and credit spreads. According to Blaschke et al. (2001), who analyse the first *stress testing* experiences conducted within the framework of the **IMF Financial Sector Assessment Program**, the methodology rests on three pillars: the definition of macroeconomic scenarios that are coherent and severe enough to test banks' resilience, the application of econometric models relating macroeconomic variables to bank losses and revenues, and the aggregation of the results at systemic level so as to identify contagion risks and common vulnerabilities.



In the European Union, the **European Banking Authority** has conducted since 2011 euro-area-wide *stress test* exercises in collaboration with the ECB and the European Systemic Risk Board: the 2023 exercise, whose results were published by the Autorité bancaire européenne (2023), covered seventy banks representing 75 % of the assets of the European banking sector and assessed their resilience over a three-year horizon in an adverse scenario characterised by a severe recession combining a contraction of gross domestic product, a rise in unemployment, an increase in interest rates and a widening of credit spreads. The results of this exercise show that the aggregate tier 1 capital ratio of European banks stood at 15.0 % at the beginning of the exercise and would fall to 10.4 % in the adverse scenario, that is a capital depletion of 459 basis points, which testifies to an overall resilience of the European banking system despite cumulative losses estimated at 496 billion euros over the projection period. As the Autorité bancaire européenne (2023) emphasises, this loss absorption capacity is explained in large part by the substantial strengthening of banks' capital positions since the financial crisis of 2008, as well as by the improvement in asset quality observed at the beginning of 2023.

In the United States, the Fed conducts annually the *Comprehensive Capital Analysis and Review* or **CCAR**, a supervised *stress testing* exercise which assesses the capacity of large American banks to maintain adequate capital ratios in a severely adverse economic scenario: the results of the 2024 exercise, published by the Federal Reserve (2024), indicate that the 31 banks subjected to the test that year have sufficient capital to absorb nearly 683 billion dollars of losses while continuing to lend to households and firms under conditions of strong economic stress. The adverse scenario used in 2024 envisaged a rise in the unemployment rate of 6.3 percentage points to reach 10 %, a contraction of real gross domestic product of 8.5 % between peak and trough, a fall in residential property prices of 36 %, a drop in commercial property prices of 40 % and a decline in share prices of 55 %. Under this scenario, the aggregate tier 1 capital ratio of the 31 banks tested would decrease from 12.7 % in the fourth quarter of 2023 to a minimum of 9.9 % before rising again to 10.4 % at the end of the nine-quarter projection horizon, thus remaining above the minimum regulatory level of 4.5 % throughout the period.

The results of *stress tests*, whether conducted by the European Banking Authority or by the American Federal Reserve, play an essential role in the process of prudential supervision by allowing supervisors to adjust banks' capital requirements according to their risk profile and their vulnerability to macroeconomic shocks: as Ahmed & Calice (2024) analyse in their study on the impact of the European Banking Authority's *stress tests* on the supply of bank credit, these exercises have led European banks to reduce their lending to firms and consumers, which suggests that *stress tests* have a disciplinary effect by giving institutions an incentive to adopt more prudent lending strategies and to strengthen their capital ratios. Moreover, the **increased transparency** resulting from the publication of individual results bank by bank contributes to market discipline by allowing investors, creditors and depositors to assess the financial soundness of each institution and to adjust their capital allocation decisions accordingly. The publication of the capital requirements arising from *stress tests* and of the requirements of the second pillar of Basel thus exerts a disciplinary effect by giving banks an incentive to maintain higher capital ratios, which fosters financial stability and strengthens market discipline in the European banking sector.

### 2.3.3 Insurance and the Solvency I and II rules

The Solvency I and Solvency II rules represent European regulatory frameworks specifically dedicated to the **insurance** sector, aiming to frame risk management and to protect policyholders. **Solvency I**, introduced in the 1970s, rested on a traditional and prudent approach to risks, centred on the accounting balance sheet and on an industrial view of the insurance business. The main objective was to guarantee that insurers have sufficient own funds to honour their commitments towards policyholders. This regulatory framework imposed a minimum level of capital based on flat-rate percentages of certain accounting indicators such as premiums and claims provisions. As Frézal (2021) indicates, in the non-life insurance sector, own funds represented about 20 % of insurers' balance sheets, while in the life insurance sector this ratio stood at around 4 %. Solvency I was distinguished by an explicit prudence in provisioning, with provisions often overestimated so as to cover future risks, without necessarily discounting commitments in the non-life sector, unlike life insurance where the mortality tables were calibrated so as to strengthen reserves, as Frézal (2021) explains.

However, this approach showed its limits in the face of the growing complexity of financial markets and of the evolution of insurance products, in particular after the financial crisis of 2008. This led to the introduction of **Solvency II**, which marked a major turning point in the prudential regulation of the sector by introducing a much more sophisticated and risk-based approach. Coming into force in 2016, Solvency II rests on three pillars similar to those of the Basel Accords for banks: (i) quantitative capital requirements via the SCR (*solvency capital requirement*) and the MCR (*minimum capital requirement*), (ii) governance and risk management through the own risk and solvency assessment (ORSA) and (iii) transparency requirements towards the public and the regulatory authorities. As Frézal (2021) explains, the **SCR** represents the level of capital required to absorb extreme shocks, computed over a



period of twelve months with a probability of 99.5 %, that is a catastrophic event occurring once every two hundred years; the **MCR**, for its part, corresponds to the minimum capital threshold below which the regulatory authority intervenes automatically to withdraw the insurance undertaking's authorisation.

One of the most important changes brought by Solvency II is the move to an **economic valuation of balance sheets**, based on the fair value of assets and liabilities. This means that provisions, which constituted the bulk of insurance companies' liabilities under Solvency I, are henceforth valued according to their market value, with adjustments to take account of the risks linked to their future management. This approach rests on advanced actuarial techniques and on the use of stochastic projection models in order to estimate future cash flows and the associated risks. Moreover, Solvency II introduces a new way of computing available own funds, taking account not only of traditional own funds (*adjusted net asset value*, ANAV), but also of the value of the contracts in the portfolio (*value of in-force*, VIF), which reflects the prospects of future profits, as Frézal (2021) details. Thus, the prudential balance sheet under Solvency II is more closely aligned with the practices of financial markets, with provisions and assets valued at their market value, within a logic of **shareholder value management** as much as of policyholder protection.

### Key Takeaways



The Basel Accords aim to strengthen the stability of the banking system by imposing minimum own funds requirements and, since Basel III, liquidity constraints, in order to cover credit, market, operational and liquidity risks more effectively [Sia Partners (2017), Azzouzi Idrissi & Madiès (2012)]. Basel I introduces the Cooke ratio (8 % of risk-weighted assets) but captures neither market risks nor the growing off-balance-sheet risks linked to derivatives, which leads to Basel II and the McDonough ratio, founded on three pillars (capital requirements, prudential supervision, market discipline) and on internal ratings-based approaches, without, however, preventing the systemic and liquidity risk revealed by the 2008 crisis [Sia Partners (2017), Navarro (2008), Azzouzi Idrissi & Madiès (2012)]. Basel III then raises the quality and the level of own funds (*core equity tier 1* at 4.5 %, *tier 1* at 6 %, overall ratio at 10.5 % with a conservation buffer) and introduces two liquidity ratios — the LCR (high-quality liquid assets covering thirty days of stress) and the NSFR (stable funding over one year) — so as to limit dependence on short-term funding and liquidity imbalances [Sia Partners (2017), Azzouzi Idrissi & Madiès (2012)]. In parallel, the *stress tests* conducted by the European Banking Authority and the Fed complement these static requirements with a forward-looking assessment of banks' resilience to severe macroeconomic scenarios, showing a substantial but bearable depletion of capital ratios and exerting a disciplinary effect on the supply of credit and on risk management [Autorité bancaire européenne (2023), Federal Reserve (2024), Ahmed & Calice (2024)]. In insurance, the move from Solvency I — a simple framework founded on flat-rate percentages of premiums and provisions — to Solvency II aligns prudential regulation with a logic of risk management comparable to that of banks, by introducing quantitative requirements, strengthened governance and an economic valuation of balance sheets at fair value [Frézal (2021)]. The whole of these arrangements builds an integrated prudential architecture in which capital requirements, liquidity ratios, stress tests and economic valuation converge so as to limit the risk of banking or insurance failure while framing internal risk assessment models more strictly.

## 3 Corporate finance

**Corporate finance** designates the whole set of financial decisions taken within an organisation in order to optimise the allocation of financial resources, to maximise its long-run value and to ensure its continuity, through the management of investments, of financing, of cash and of risks.

### 3.1 Interest rates and investment

Interest rates do not merely measure the cost of time: they constitute the pivot which links investment decisions to the **creation of economic value**, by making it possible to compare present and future flows rigorously. Now this comparison reveals a fundamental paradox: while the mathematical formulae appear simple, their practical application raises crucial choices between accounting conventions, methods of computation and assessment criteria, which determine whether or not a project is profitable.

#### 3.1.1 The different interest rates

Simple interest and compound interest are two distinct methods of computing interest. As Couppey-Soubeyran (2012b) indicates, **simple interest** is computed solely on the initial capital throughout the investment period, whereas

compound interest is computed not only on the initial capital, but also on the interest accumulated over time: the total interest  $I_n$  accumulated after  $n$  periods is given by the formula  $I_n = V_0 \times n \times i$ , where  $V_0$  is the initial capital,  $i$  is the interest rate per period, and  $n$  is the number of periods. The total sum to be repaid, or accrued value, is therefore  $V_n = V_0 + I_n = V_0 \times (1 + n \times i)$ . For example, if one borrows 1000 euros for three years at a rate of 6 % per year, the total interest is  $I_3 = 1000 \times 3 \times 0.06 = 180$  euros; the total sum to be repaid will therefore be  $V_3 = 1000 + 180 = 1180$  euros. With **compound interest**, by contrast, interest is capitalised at the end of each period, as Couppey-Soubeyran (2012b) explains: this means that in each period, the interest is added to the capital, and interest is computed on the new capital; the formula for computing the accrued value  $V_n$  at the end of  $n$  periods is then given by  $V_n = V_0 \times (1 + i)^n$ . To return to the previous example, if one invests 1000 euros for three years at a rate of 6 % with compound interest, the accrued value will be  $V_3 = 1000 \times (1 + 0.06)^3 \approx 1191.02$  euros: we see here that compound interest makes it possible to generate more interest than simple interest over the same period, since the interest is computed on an ever larger capital. Usually, financial investments are computed using compound interest.

### Focus...



**Interest in advance and interest in arrears.** **Interest in arrears** is computed at maturity on the whole of the capital: if  $V_0$  is the initial capital borrowed or invested at rate  $i$  over one period, the final value is  $V_n = V_0(1 + i)$ . **Interest in advance** is deducted from the outset: the borrower receives  $V_0 = (1 - i_p)V_n$ , where  $V_n$  is the sum to be repaid at maturity. This convention *a priori* favours the lender since, for the same posted rate, the real cost is higher on an in-advance basis: a loan of 350 euros at  $i_p = 6.8\%$  in advance gives a net disbursement of  $350 \times (1 - 0.068) = 326.2$  euros (interest of 23.8 euros received immediately), whereas at 6.8 % in arrears the interest of 23.8 euros is received only at the end. This is why the in-advance rate must necessarily be lower than the in-arrears rate. More precisely, as Morlais (2010) explains, if  $i_p$  is the in-advance rate and  $i$  the in-arrears rate that are consistent with one another (same economic cost), the relation  $(1 - i_p)(1 + i) = 1$  must hold, that is  $i_p = \frac{i}{1+i} < i$ . Otherwise, an arbitrage opportunity would be observed, that is to say a possibility of making money without taking any risk. Let us prove this result:

- Indeed, let us borrow  $N$  at the in-arrears rate  $i$  (repayment  $N(1 + i)$ ) and simultaneously lend  $x = \frac{N}{1 - i_p}$  at the in-advance rate  $i_p$  (so as to disburse only  $x(1 - i_p) = N$ ): the operation is nil at the initial date. At maturity, we are repaid  $x$ , and we must repay the loan for  $N(1 + i)$ . We therefore receive at maturity  $x - N(1 + i) = \frac{N}{1 - i_p} - N(1 + i)$ . For there to be an absence of arbitrage opportunity, we must have  $\frac{N}{1 - i_p} - N(1 + i) \leq 0$ , that is to say  $\frac{N}{1 - i_p} \leq N(1 + i)$ , or again  $\frac{1}{1 - i_p} \leq (1 + i)$  or  $(1 + i)(1 - i_p) \geq 1$ .
- Indeed, let us borrow  $x$  at the in-advance rate  $i_p$  (we then receive  $N = x(1 - i_p)$ ) and simultaneously lend  $N$  at the in-arrears rate  $i$ : the operation is nil at the initial date. At maturity, we repay  $x$ , and we are repaid the loan for  $N(1 + i)$ . We therefore receive at maturity  $N(1 + i) - x = N(1 + i) - \frac{N}{1 - i_p}$ . For there to be an absence of arbitrage opportunity, we must have  $N(1 + i) - \frac{N}{1 - i_p} \leq 0$ , that is to say  $\frac{N}{1 - i_p} \geq N(1 + i)$ , or again  $\frac{1}{1 - i_p} \geq (1 + i)$  or  $(1 + i)(1 - i_p) \leq 1$ .

Combining the two results, we then observe that there is indeed an absence of arbitrage opportunity if, and only if, the relation  $(1 + i)(1 - i_p) = 1$  holds.

It is therefore through the interest rate that the link can be made between a present monetary flow and a future monetary flow. More precisely, compounding and discounting are two fundamental notions in the field of interest rates and financial mathematics: compounding consists in computing the future value of a capital invested today, whereas discounting makes it possible to determine the present value of a future capital; these two operations are based on the interest rate and on the number of periods over which the interest is computed. As Bossu & Henrotte (2009a) explains, within the framework of **compounding**, if an initial capital  $V_0$  is invested for  $n$  periods at the rate  $i$  per period, the future value  $V_n$  (or accrued value) may be obtained by applying the following compounding formula  $V_n = V_0 \times (1 + i)^n$ . This formula shows that the accrued value at the end of the  $n$  periods depends on the initial capital, on the interest rate and on the number of investment periods. Let us take an example: if one invests 1000 euros for five years at a rate of 5 % per year, the accrued value will be  $V_5 = 1000 \times (1 + 0.05)^5 \approx 1276.28$  euros. Compounding thus makes it possible to know the final sum available after a certain time. **Discounting**, by contrast, is the reverse process of compounding: it consists in computing the present value of a future capital. If one knows that a sum  $V_n$  will be available at the end of  $n$  periods, the present value  $V_0$  is given by  $V_0 = \frac{V_n}{(1 + i)^n}$ , as Bossu & Henrotte (2009a) details. Thus, if one wishes to know the present value of 1276.28 euros to be received in five years' time, assuming an interest rate of 5 % per year on the market, the discounting formula gives  $V_0 = \frac{1276.28}{(1 + 0.05)^5} \approx 1000$  euros. Compounding and discounting are thus two inverse operations. Compounding projects a present value into the future, whereas

discounting brings a future value back to its equivalent today. These two concepts are essential in order to assess financial investments and investment projects.

### Focus...



**The discount rate and the *in fine* rate.** If I repay my creditor before maturity, it is as though I were making him a gift since, as we saw at the beginning of the chapter, one euro today is not equivalent to one euro in a month's time. But then, how is this difference in the appreciation of value over time to be quantified? This is precisely the role of the **discount rate**: this rate is, in the creditor's eyes, the one which renders a present flow and a future flow equivalent, and more precisely the one which 'brings a future flow back to the present'. It is according to this rate (which therefore depends on the creditor's subjectivity) that he grants us a financial reduction for cash payment or early settlement. Conversely, if I was supposed to settle an invoice today and, for want of cash, I cannot honour it today, I shall ask my creditor to let me pay in a few days' time (a month later, for example). But then, the financial flow that he was expecting will no longer be equivalent, from an economic point of view, to a financial flow of the same face value in a month's time, since money will have 'lost some of its value' in the creditor's eyes in the meantime. He will then generally grant me a payment deadline, but will charge me fees (sometimes statutory fees, such as a forty-euro surcharge for late payment) and notably fees which will make the future flow equivalent to the flow that ought to have been received: the surcharge will then follow a certain rate, called the ***in fine* rate**, which will render a present flow and a future flow equivalent. The *in fine* rate is more precisely the rate which 'carries a present flow into the future'. The discount rate and the *in fine* rate are therefore two sides of the same coin. And for good reason: if we consider an original financial flow  $D_0$  and a final financial flow  $D_1$ , the time separating these two flows being  $\Delta$  days, then the **discount rate**  $t$  between the final financial flow and the original financial flow satisfies by definition the following equation  $D_0 = -D_1 \left(1 - \frac{t\Delta}{360}\right)$ . Note that we have used here a proportional daily rate and not an equivalent rate, inasmuch as settlement discounts bear on operations of less than one year (the length of the accounting period).

Symmetrically, the ***in fine* rate**  $T$  between the final financial flow and the original financial flow satisfies by definition the following equation  $D_1 = -D_0 \left(1 + \frac{T\Delta}{360}\right)$ . Note here again that we have used a proportional daily rate and not an equivalent rate, inasmuch as payment deadlines are generally granted for a period shorter than that of the accounting period (a few days or a few weeks). From then on, the discount rate  $t$  and the *in fine* rate are said to be **equivalent** over a period of  $\Delta$  days if, and only if,  $T = \frac{t}{1 - t\Delta/360}$ . Note then that, for a period of one year, the discount rate  $t$  and the *in fine* rate  $T$  are equivalent if, and only if, they satisfy the simple relation  $(1 - t)(1 + T) = 1$ . The case of the discount and of *in fine* repayment therefore arises when a repayment has not taken place at the scheduled maturity (early in the case of the discount, late in the case of *in fine* repayment).

In what precedes, we have each time mentioned annual interest rates. But in practice, payments are often made at periodicities shorter than the year (for example, property loans are repaid through monthly instalments and not annuities, as a general rule). How, then, is a rate to be proposed which corresponds to a period shorter than the year? As Bossu & Henrotte (2009a) recalls, the **30/360 convention** is often used in finance: a year lasts 360 days and each month lasts 30 days; a half-year therefore lasts 0.5 year, a quarter 0.25 year, a month 1/12 year, a week 7/360 year and a day 1/360 year. Thus, in order to compute the number of days separating 12 January 2002 and 10 March 2025, one computes the number of whole years separating these two dates (23 years: which brings us to 12 January 2025), then the number of whole months separating 12 January 2025 from 10 March 2025 (only one: which brings us to 12 February 2025) and finally the number of remaining days (18 days between 12 and '30' February, then 10 days between '30' February and 10 March). There are therefore  $23 + 1/12 + 28/360 \approx 23.161$  years separating the two dates according to the 360/30 convention. Knowing this convention, there are two methods for applying an annual rate to infra-annual periods, which Couppey-Soubeyran (2012b) analyses: the **proportional rate**, used within the framework of simple interest, and the equivalent rate, relevant for compound interest. If an interest rate  $i$  is given for one period, but one wishes to know the rate for a sub-period, the proportional rate  $i_k$  is computed by dividing the interest rate by the number of sub-periods  $k$ , that is  $i_k = i/k$ . For example, if the annual interest rate is 6 %, the monthly proportional rate will be  $i_{12} = 0.06/12 = 0.005 = 0.5$  %: this means that with a proportional interest rate, the monthly interest is 0.5 %. The **equivalent rate** is used for compound interest and makes it possible to determine an interest rate applicable to a sub-period in such a way that the accrued value is the same as the one obtained with the overall rate: the equivalent rate  $i_k$  is therefore given by  $1 + i_k = (1 + i)^{1/k}$  (and  $ki_k$  is then the annual rate for  $k$  compoundings per year<sup>1</sup>). Thus, returning to the example of an annual rate of 6 %, the monthly equivalent

<sup>1</sup>Care is therefore needed in examining the terms of loans and investments: if one speaks of a monthly rate of 1 %, then the nominal annual rate will be 12% (the periodic rate multiplied by the number of periods in a year), but the effective annual rate will be  $(1 + 0.01)^{12} - 1 \approx 12.68$  % (which is quite simply the corresponding equivalent rate): it is the latter that must be taken into account in practice in order to determine the repayment instalment.

rate satisfies  $1 + i_{12} = (1 + 0.06)^{1/12} \approx 1.00487$ , that is  $i_{12} \approx 0.00487 = 0.487\%$ . We then note, as Couppey-Soubeyran (2012b) emphasises, that for compound interest, the equivalent rate is always lower than the proportional rate, since it takes account of the capitalisation of interest in each sub-period. Regulated savings accounts, in particular, are capitalised by fortnight: as the BRED (2024) explains, the sums paid in bear interest from the first day of the fortnight following the payment or transfer (the 16th of the same month or the 1st of the following month), while the sums withdrawn cease to bear interest from the first day of the fortnight in which the withdrawal occurs (either the 1st or the 16th): in order to avoid unremunerated periods, it is therefore advisable to make payments before the 16th or the 1st of the month and withdrawals after the 15th or the 30th.

#### Focus...



**The continuous rate.** The continuous interest rate is a more advanced notion resting on the idea that interest is capitalised permanently, at each instant. Unlike compound interest, where capitalisation takes place at discrete intervals (for example, monthly or annually), interest under continuous compounding is computed continuously through time. As Baudot (2024) explains, denoting by  $i$  the annual interest rate, if a capital  $C_0$  is compounded  $k$  times a year over  $n$  years, then as the number of periods  $k$  tends to  $+\infty$ , one proves that  $C_n$  tends to  $C_0 e^{in}$ , this value constituting the continuous compounding of  $C_0$  over  $n$  years.

### 3.1.2 The investment decision: net present value and internal rate of return

When one seeks to convince an investor or a banker to finance a project to the tune of an investment  $I$ , one presents to him a sequence of estimated future **net cash flows**  $(F_n)_{n \in \mathbb{N}^*}$ . If the natural number  $T$  denotes the investment horizon of the project (in years), then the investor or the banker determines whether the sum of the flows  $F_1, \dots, F_T$  at least allows him to offset his initial investment  $I$ . But the criterion is not to check whether  $F_1 + \dots + F_T > I$ . Indeed, as we have just seen, one euro received in  $n$  years' time does not repay one euro financed initially. Account must therefore be taken of a certain **discount rate**  $r$  for future flows: a flow  $F$  received after  $n$  years will from then on be equivalent to a flow  $\frac{F}{(1+r)^n}$  received today, where  $r$  represents an implicit interest rate required by the project's financiers. Indeed, suppose that this implicit annual interest rate is worth  $r$ . Then investing one euro today will yield  $(1+r)^n$  euros in  $n$  years' time, on account of compound interest. Thus, by applying a rule of three, a flow of one euro in  $n$  years' time is equivalent to  $(1+r)^{-n}$  today. From then on, as Charreaux (2014a) explains, the investor will agree to finance the project if, and only if,  $\frac{F_1}{1+r} + \frac{F_2}{(1+r)^2} + \dots + \frac{F_T}{(1+r)^T} > I$  or  $\text{NPV}(r, T) := \frac{F_1}{1+r} + \frac{F_2}{(1+r)^2} + \dots + \frac{F_T}{(1+r)^T} - I > 0$  or  $\text{PI}(r, T) := \left( \frac{F_1}{1+r} + \frac{F_2}{(1+r)^2} + \dots + \frac{F_T}{(1+r)^T} \right) / I > 1$ , where  $\text{NPV}(r, T)$  denotes the **net present value** and  $\text{PI}(r, T)$  the **profitability index** of the investment with a horizon of  $T$  years for a discount rate  $r$ . The NPV criterion then offers a decision (comparison) rule which Charreaux (2014a) mentions: at a fixed discount rate  $r$ , a project A will be more profitable than a project B (that is to say it will yield more in discounted absolute value *in fine*) if, and only if, the NPV associated with project A is greater than that of project B; in which case, it is project A that should be financed as a priority.

#### Focus...



**The overall effective rate.** Couppey-Soubeyran (2012b) takes the example of a loan with fixed instalments: suppose that one has borrowed 100 euros from a bank over two years, repaying 55 in the first year and 55 in the second year. Then the nominal rate is equal to 10 % over two years, that is a nominal proportional annual rate of 5 %. If one computes the rate which equalises the discounted flows of payments associated with a financial instrument and its present value, one finds  $100 = 55/(1+i) + 55/(1+i)^2$ , that is roughly  $i \approx 7\%$ . This rate, known as the overall effective rate (*taux effectif global*) according to *Journal officiel* no. 134 of 11 June 2002, computed from the repayments (including in particular insurance and arrangement fees, which bankers are obliged to communicate to their borrowing customers), then measures the true cost of the loan from the borrower's point of view and the true return from the lender's point of view.

We then note that the value  $r^*$  such that  $\text{NPV}(r^*, T)$  is zero denotes the threshold rate from which the investment becomes attractive to the investor: this is the **internal rate of return** (IRR). Investors therefore accept only those investment projects for which the cost of capital is lower than the IRR, thus making it possible to generate a positive NPV or a profitability index greater than 1. As Charreaux (2014a) indicates, the discount rate represents the rate of return required by shareholders and lenders, who take account notably of inflation expectations, which induce all other things being equal a fall in the profitability of the investment, but also of the intrinsic level of risk of the project and of the financial structure (indebtedness) of the firm. For example, as Fisher (1930) brings out, if the nominal interest rate is  $i = 5\%$  but the anticipated rate of inflation is  $\pi_a = 3\%$ , then the **real interest rate** will

be  $i_r := i - \pi_a = 2\%$ : in the event of high anticipated inflation, agents expect a lower purchasing power and will therefore require a higher nominal remuneration in order to offset the rise in prices. Moreover, the NPV and IRR criteria are generally complemented by that of the **discounted payback period** (DPP), that is to say the duration  $T^*$  from which the initial investment has been recovered (in present value). As Charreaux (2014a) explains,  $T^*$  is therefore the smallest natural number such that  $NPV(r, T^*)$  is positive: investors will from then on agree to finance only those projects for which the investment horizon exceeds the discounted payback period.

### Exercise



**Investment decision.** The Café-Théâtre business unit, which specialises in the manufacture of machines for bars and restaurants, wishes to invest in new machine tools; the manager informs you of the following parameters: acquisition of 55 000 euros excluding VAT at the beginning of year N, straight-line depreciation over five years, additional turnover of 100 000 euros per year, annual charges of 60 000 euros, zero residual value, tax rate of 33.33 %.

- (i) Draw up the table of the firm's net cash flows from year N to year N + 4.
- (ii) Complete the previous table by adding the net flows discounted at a rate of 11 %.
- (iii) Compute the net present value (NPV) of the project. Should the firm invest in the new machines?
- (iv) Recover the result of the previous question using the profitability index (PI) criterion.
- (v) Compute the discounted payback period (DPP). Comment.
- (vi) Suppose that, for a discount rate of 18 %, the NPV would stand at -45 896.93 euros. Propose an approximation of the internal rate of return (IRR) of the project.

**Solution.** (i) Taking up the data of the statement, we observe that the increase in turnover is 100 000 euros per year over the whole period; straight-line depreciation is computed by dividing 55 000 euros by five, that is 11 000 euros per year; profit before tax is equal to receipts minus disbursements, that is  $100\,000 - 60\,000 - 11\,000 = 29\,000$  euros; corporation tax corresponds to  $29\,000 \times 33.33\% = 9\,666.66$  euros; profit after tax is  $29\,000 - 9\,666.66 = 19\,333.34$  euros; depreciation is then added back, being a non-cash charge, of 11 000 euros; the annual net cash flow is then  $19\,333.34 + 11\,000 = 30\,333.34$  euros; the completed table of net cash flows (NCF) takes up these constant magnitudes for each financial year from N to N + 4:

| Items                         | N         | N + 1     | N + 2     | N + 3     | N + 4     |
|-------------------------------|-----------|-----------|-----------|-----------|-----------|
| Turnover                      | 100 000   | 100 000   | 100 000   | 100 000   | 100 000   |
| Annual charges                | 60 000    | 60 000    | 60 000    | 60 000    | 60 000    |
| Depreciation                  | 11 000    | 11 000    | 11 000    | 11 000    | 11 000    |
| Profit before corporation tax | 29 000    | 29 000    | 29 000    | 29 000    | 29 000    |
| Corporation tax               | 9 666.66  | 9 666.66  | 9 666.66  | 9 666.66  | 9 666.66  |
| Net profit                    | 19 333.34 | 19 333.34 | 19 333.34 | 19 333.34 | 19 333.34 |
| Depreciation                  | 11 000    | 11 000    | 11 000    | 11 000    | 11 000    |
| Net cash flows                | 30 333.34 | 30 333.34 | 30 333.34 | 30 333.34 | 30 333.34 |

- (ii) For each period  $N + i$ , the discounted net flow is computed from the annual net cash flow by means of the formula  $\text{Discounted NCF} = \text{Annual NCF} \times (1 + \text{rate})^{-(i+1)}$ ; applying a discount rate of 11 % and taking for example an annual NCF of 30 333.34 euros, one obtains for period 2 (year N + 1) the following computation:  $30\,333.34 \times (1 + 0.11)^{-2} = 24\,619.22$  euros; one proceeds in the same way for each year in order to complete the column of discounted NCF, and one obtains:

| Items                         | N         | N + 1     | N + 2     | N + 3     | N + 4     |
|-------------------------------|-----------|-----------|-----------|-----------|-----------|
| Turnover                      | 100 000   | 100 000   | 100 000   | 100 000   | 100 000   |
| Annual charges                | 60 000    | 60 000    | 60 000    | 60 000    | 60 000    |
| Depreciation                  | 11 000    | 11 000    | 11 000    | 11 000    | 11 000    |
| Profit before corporation tax | 29 000    | 29 000    | 29 000    | 29 000    | 29 000    |
| Corporation tax               | 9 666.66  | 9 666.66  | 9 666.66  | 9 666.66  | 9 666.66  |
| Net profit                    | 19 333.34 | 19 333.34 | 19 333.34 | 19 333.34 | 19 333.34 |
| Depreciation                  | 11 000    | 11 000    | 11 000    | 11 000    | 11 000    |
| Net cash flows                | 30 333.34 | 30 333.34 | 30 333.34 | 30 333.34 | 30 333.34 |
| Discounted cash flow          | 27 327.33 | 24 619.22 | 22 179.47 | 19 981.51 | 18 001.36 |

- (iii) The net present value is determined by subtracting the amount of the investment from the sum of the discounted NCF, that is  $NPV = \sum \text{discounted NCF} - \text{Investment}$ ; using the values 27 327.33 euros, 24 619.22 euros, 22 179.47 euros, 19 981.51 euros and 18 001.36 euros, the sum reaches 112 108.89 euros and the NPV is therefore  $112 108.89 - 55 000 = 57 108.89$  euros; the NPV being positive, the business unit may contemplate the investment in the new self-cleaning machine.
- (iv) The profitability index is equal to the ratio between the sum of the discounted NCF and the amount of the investment, that is  $PI = \frac{\sum \text{discounted NCF}}{\text{Investment}}$ ; taking the sum 112 108.89 euros and an investment of 55 000 euros, one obtains  $PI = 112 108.89 \div 55 000 = 2.03$ ; since the index is greater than 1, the business unit may contemplate investing in the software solution.
- (v) The DPP is obtained by cumulating the discounted NCF year after year until the amount invested is reached or exceeded; one carries over for  $N$  the discounted NCF of the year, then one adds successively for the following years, for example  $27 327.33 + 24 619.22 = 51 946.55$  euros for  $N + 1$ , and one continues until one observes by simple inspection that the cumulative total exceeds 55 000 euros (it reaches  $51 946.55 + 22 179.47 = 74 126.02$  euros) as early as the end of year  $N + 2$ . In order to know the DPP precisely, one proceeds by linear interpolation. On the last day of year  $N + 1$ , the cumulative total is 51 946.55 euros; on the last day of year  $N + 2$ , the cumulative total is 74 126.02 euros. From when has exactly 55 000 euros been recovered? In one year, the sum of the discounted NCF has grown by  $74 126.02 - 51 946.55 = 22 179.47$  euros, that is roughly  $22 179.47 \div 360 \approx 61.61$  euros per day on average (recall that in financial analysis, every month has 30 days). In order to go from 51 946.55 to 55 000 euros, 3053.45 euros are needed, that is  $3053.45 \div 61.61 \approx 49.56$  days. It is therefore after 50 days, that is one month and 20 days, following the end of year  $N + 1$  that the DPP is attained. Thus, the DPP is 2 years, 1 month and 20 days. The payback period being shorter than the duration of the project (5 years), the investment may be retained.
- (vi) The IRR is the rate which cancels the NPV and is deduced by linear interpolation between two rates bracketing zero. We have two points: for a rate of 11 %, the NPV is 57 108.89 euros; for a rate of 18 %, it is -45 896.93 euros. We assume that  $NPV = a + b \times \text{Rate}$ . We are therefore led to solve the following system, in two real unknowns  $a$  and  $b$ :

$$(S): \begin{cases} 57 108.89 &= a + b \times 11\% \\ -45 896.93 &= a + b \times 18\% \end{cases}$$

Now we have:

$$\begin{aligned} (S) &\Leftrightarrow \begin{cases} 57 108.89 = a + b \times 11\% \\ -45 896.93 - 57 108.89 = (a - a) + b \times (18\% - 11\%) \end{cases} \\ &\Leftrightarrow \begin{cases} 57 108.89 = a + b \times 11\% \\ -103 005.82 = b \times 7\% \end{cases} \\ &\Leftrightarrow \begin{cases} b = \frac{-103 005.82}{7\%} \approx -14 715 11.71 \\ a = 57 108.89 - (-14 715 11.71) \times 11\% \approx 218 975.18 \end{cases} \end{aligned}$$

We now seek the value IRR such that  $a + b \times \text{IRR} = 0$ , that is to say  $\text{IRR} = -\frac{a}{b}$ . We therefore find  $\text{IRR} \approx \frac{218 975.18}{14 715 11.71} \approx 14.88\%$ . The cost of capital (11 %) being lower than the IRR, the investment is indeed profitable.  $\square$

### Key Takeaways



The assessment of an investment project rests on the discounting of future net cash flows ( $F_1, \dots, F_T$ ) at the rate  $r$  required by the financiers, in order to determine whether their discounted sum exceeds the initial investment  $I$ : the decision criterion is  $NPV(r, T) := \frac{F_1}{1+r} + \frac{F_2}{(1+r)^2} + \dots + \frac{F_T}{(1+r)^T} - I > 0$  or the profitability index  $PI(r, T) := \frac{1}{I} \left( \frac{F_1}{1+r} + \frac{F_2}{(1+r)^2} + \dots + \frac{F_T}{(1+r)^T} \right) > 1$ , where the NPV compares the absolute profitability of projects (a project A is preferable to a project B if  $NPV_A > NPV_B$  at a fixed rate) [Charreaux (2014a)]. The internal rate of return (IRR)  $r^*$  is the threshold rate such that  $NPV(r^*, T) = 0$ , the investor accepting the project if his cost of capital is lower than the IRR [Charreaux (2014a)]. The discounted payback period (DPP)  $T^*$ , finally, complements these criteria by identifying the smallest horizon such that  $NPV(r, T^*) > 0$ , investors requiring that the horizon of the project  $T$  exceed this threshold [Charreaux (2014a)].

### 3.1.3 The choice of how to invest: the Modigliani-Miller theorem

Let us now show that the way in which a firm's investment, or a firm itself, is financed (self-financing thanks to shareholders or indebtedness via bank borrowing) will not alter the value of its economic assets. This result was proved by Modigliani & Miller (1958) under very restrictive assumptions: absence of frictions on markets (no transaction costs), pure and perfect competition (firms and individuals are *price-takers*), absence of information asymmetry (no privileged information held by *insiders*), absence of taxation... A utopian hypothetical framework in the light of the reality of the market. The idea of the proof rests on an argument by contradiction: consider two firms differing only in their capital structure (with the same cash flows in particular); if they have different market values, then an **arbitrage opportunity** exists (that is to say one can make a riskless profit by exchanging the assets between them), which is incompatible with the idea of an equilibrium. More precisely, let  $a$  and  $d$  be two firms generating in every period  $t$  of  $\mathbb{N}^*$  the same income  $X$  (where  $X$  is a random variable). At  $t = 0$ , we assume that these two firms adopt different financial structures: firm  $d$  will be composed of shares and riskless debt ( $d$  for 'debt'), whereas firm  $a$  will have no debt ( $a$  for 'all-equity'). Let  $r_f$  be the riskless rate (denoted  $r$  hereafter). At date  $t = 0$ , we denote by  $D_d$  the market value of firm  $d$ 's debt and we denote respectively by  $E_a$  and  $E_d$  ( $E$  for 'equity') that of the shares of firms  $a$  and  $d$ . The market value of firm  $a$  is therefore  $V_a := E_a$ , while that of firm  $d$  is  $V_d := D_d + E_d$ . Consequently, for every period  $t \geq 1$ , the holders of firm  $d$ 's debt receive  $rD_d$  and the shareholders of firm  $d$  therefore receive  $X - rD_d$ , while the shareholders of firm  $a$  receive  $X$ .

- Suppose, for the sake of contradiction, that  $V_a > V_d$  and consider an investor holding a share  $\alpha$  (where  $\alpha$  is a real number between 0 and 1) of firm  $a$ . For every period  $t \geq 1$ , he would receive  $\alpha X$ . But instead, he could also sell his holding for a value  $\alpha V_a$  and subsequently buy back a fraction  $\frac{\alpha V_a}{V_d}$  of firm  $d$  (debt and shares), so that  $\alpha V_a = \frac{\alpha V_a}{V_d} D_d + \frac{\alpha V_a}{V_d} E_d$ . The investor would then receive  $\frac{\alpha V_a}{V_d} rD_d + \frac{\alpha V_a}{V_d} (X - rD_d)$ , that is  $\frac{\alpha V_a}{V_d} X$ , a value systematically greater than  $\alpha X$ , whatever the income  $X$ . There does therefore exist an arbitrage opportunity (AO).
- Suppose, for the sake of contradiction, that  $V_a < V_d$  and consider an investor holding a share  $\alpha$  (where  $\alpha$  is a real number between 0 and 1) of firm  $d$ . For every period  $t \geq 1$ , he would receive  $\alpha(X - rD_d)$ . But instead, he could also sell his holding for a value  $\alpha E_d$ , borrow  $\alpha D_d$  and then invest the total  $\alpha(E_d + D_d) = \alpha V_d$  to buy a share  $\frac{\alpha V_d}{V_a}$  of firm  $a$ , so that the equality  $\alpha E_d + \alpha D_d = \frac{\alpha V_d}{V_a} E_a$  holds. The investor would then receive  $\frac{\alpha V_d}{V_a} X$  and would pay interest  $r\alpha D_d$ , that is  $\alpha \left( \frac{V_d}{V_a} X - rD_d \right)$ , a value systematically greater than  $\alpha(X - rD_d)$ , whatever the income  $X$ . There does therefore exist an AO.

The only possible case for an equilibrium is therefore that of the equality of the values  $V_a = V_d$ . Indeed, when an AO arises, investors will sell their holdings in firm  $a$  (if  $V_a > V_d$ ) or sell their holdings in firm  $d$  (if  $V_d > V_a$ ). The law of supply and demand then tells us that the value of firm  $a$  (if  $V_a > V_d$ ) or of firm  $d$  (if  $V_d > V_a$ ) will decrease until the situation is reached where the two values  $V_a$  and  $V_d$  are equal: there is then no further AO possible, and the equilibrium is *de facto* attained.

However, this theorem comes up against the resistance of the facts, for it seems that debt is *a priori* cheaper than equity: whereas the interest rate on corporate debt hovers around 5 %, the rate of return on own funds is rather of the order of 10 %. How, then, could financing be neutral? Modigliani & Miller (1958) answer this problem by showing that **the cost of a firm's shares increases with its debt-to-equity ratio**. The intuition is the following: increasing debt makes shares riskier (indeed, in the event of bankruptcy, the repayment of debts takes priority over that of shareholders) and therefore more costly; the lower cost of debt is thus only an illusion. Let us prove this result. Let  $R_a$  be the return on the shares of firm  $a$ . Then  $R_a = \frac{E[X]}{V_a}$ , that is to say  $R_a V_a = E[X]$ . Likewise, the return  $R_{d_E}$  on the shares of firm  $d$  satisfies  $R_{d_E} = \frac{E[\max(X - rD_d, 0)]}{E_d}$  (indeed, the indebted firm must repay its creditors the amount  $rD_d$ ; the residual  $X - rD_d$  is what can be paid out to shareholders; if  $rD_d$  exceeds  $X$ , the shareholders receive nothing and the firm is in a state of suspension of payments). We therefore have  $R_{d_E} E_d = E[\max(X - rD_d, 0)]$ . Consequently, the expected return  $R_{d_D}$  on firm  $d$ 's debt satisfies  $R_{d_D} = \frac{E[\min(X, rD_d)]}{D_d}$  (indeed, if the firm cannot repay what it owes, that is to say  $rD_d$ , the creditors will be paid with what is left in the firm, that is  $X$ , and the firm will then go bankrupt for want of having been able to honour its repayment obligation). We therefore have  $R_{d_D} D_d = E[\min(X, rD_d)]$ . We then observe, by linearity of expectation, that  $E[\max(X - rD_d, 0)] + E[\min(X, rD_d)] = E[\max(X - rD_d, 0) + \min(X, rD_d)] = E[X]$  (indeed, if  $X > rD_d$ , then  $\max(X - rD_d, 0) = X - rD_d$  and  $\min(X, rD_d) = rD_d$ , and if  $X \leq rD_d$ , then  $\max(X - rD_d, 0) = 0$  and  $\min(X, rD_d) = X$ ). We therefore have  $R_{d_E} E_d + R_{d_D} D_d = E[X]$ . Recalling the equality  $R_a V_a = E[X]$ , it then follows that  $R_{d_E} E_d + R_{d_D} D_d = R_a V_a$ . Now we have:

$$\begin{aligned}
 R_{d_E} E_d + R_{d_D} D_d = R_a V_a &\iff R_a = R_{d_E} \frac{E_d}{V_a} + R_{d_D} \frac{D_d}{V_a} \\
 &\iff R_a = R_{d_E} \frac{E_d}{V_d} + R_{d_D} \frac{D_d}{V_d} \quad \text{by the previous theorem} \\
 &\iff R_a = R_{d_E} \frac{E_d}{E_d + D_d} + R_{d_D} \frac{D_d}{E_d + D_d}
 \end{aligned}$$



This relation defines the **weighted average cost of capital (WACC)**, which serves as the reference rate for computing the net cash flows (and therefore the NPV and the profitability index) of each investment project<sup>2</sup>. Given that  $R_{dE}$  is the return required by shareholders ( $R_{dE} = r$ ) and  $R_{dD}$  the cost of riskless debt, we have:

$$\begin{aligned} R_a &= R_{dE} \frac{E_d}{E_d + D_d} + R_{dD} \frac{D_d}{E_d + D_d} \iff R_{dE} \frac{E_d}{E_d + D_d} = R_a - R_{dD} \frac{D_d}{E_d + D_d} \\ &\iff R_{dE} E_d = R_a (E_d + D_d) - R_{dD} D_d \\ &\iff R_{dE} E_d = (R_a - R_{dD}) D_d + R_a E_d \\ &\iff R_{dE} = (R_a - R_{dD}) \frac{D_d}{E_d} + R_a \end{aligned}$$

The illusion of debt is therefore indeed brought out: firms take on debt because the return required by bankers is lower than that required by shareholders and, as indebtedness increases, shareholders require ever higher returns in order to offset the risk, and the cost of own funds becomes higher and higher. It is the part  $(R_a - R_{dD}) \frac{D_d}{E_d}$  which measures the **risk premium** required by shareholders in order to accept the indebtedness  $D_d$ . There is therefore a game of ‘communicating vessels’ between risk and required returns: if markets are in pure and perfect competition, the increase in the value of debt is entirely offset by a rise in the return required on shares which, for given anticipated future flows, *ipso facto* decreases the value of the shares. Here we recover the idea that the value of the firm must not change according to the financing policy.

### Key Takeaways



The theorem of Modigliani & Miller (1958) demonstrates that the financing structure (self-financing or indebtedness) does not affect the economic value of a firm, under restrictive assumptions of an absence of market frictions, pure and perfect competition, information asymmetries and taxation. The weighted average cost of capital ( $WACC = R_{dE} E_d / V_d + R_{dD} D_d / V_d$ , where  $R_{dD}$  is the return on the debt  $D_d$  and  $R_{dE}$  the return on the shares  $E_d$  of an indebted firm of value  $V_d$ ), which serves as the reference rate for computing discounted cash flows (and therefore the NPV of an investment project), remains constant, the increase in debt making shares riskier and their required return proportionally higher, which is captured by the relation  $R_{dE} = (R_a - R_{dD}) D_d / E_d + R_a$ , where  $R_a$  is the return of a self-financed firm offering the same turnover as the indebted firm *supra*.

## 3.2 Estimating the value of the firm

The **valuation of a firm** never produces a single figure: it reveals the gap between what the firm owns in accounting terms, what it will generate as future flows, and what buyers are prepared to pay. Now these three logics — asset-based, actuarial, transactional — lead to radically different estimates. This divergence structures private equity operations.

### 3.2.1 A few words on private equity

**Private equity** operations consist in taking majority or minority stakes in the capital of small and medium-sized enterprises which are generally unlisted, for amounts which often exceed a hundred million euros: the activity therefore consists, starting from unlisted funds, in investing by way of equity contributions in unlisted companies (but also sometimes in listed companies). Three parties then take part in this type of operation: the **capital providers**, namely institutional investors, insurance companies, pension funds, banks, large industrial groups and *family offices*; the **private equity** firms, which invest the capital entrusted to them by the providers; and the **entrepreneurs** in search of equity. Naturally, to each phase of the life cycle its favoured type of private equity, as Glachant et al. (2008) explain:

— During the pre-start-up phase of the activity, the project is developed and its feasibility studied. This is a transitional period between research and industrialisation. The necessary capital is then generally provided by

<sup>2</sup>**Careful!** It is therefore not the rate of inflation that must be used to discount a firm's cash flows. Indeed, if future flows are expressed in nominal terms (with inflation included), a nominal discount rate must be used which incorporates both anticipated inflation and a risk premium, since the NPV must compare a cash flow with the opportunity cost of money. Inflation taken in isolation is therefore not sufficient: it is a possible component of the rate, but the discount rate must above all reflect the cost of capital and the risk of the project. This is why, in corporate finance, the discount rate is in general the weighted average cost of capital (WACC), that is to say a rate which combines the after-tax cost of debt and the cost of equity, weighted by their respective weights in the firm's financing. This rate reflects what the financing of the project ‘costs’ the firm, and therefore the return that must at least be obtained in order to create value. By contrast, if one reasons from the point of view of an individual investor (and no longer of the firm), one could take as discount rate the return that this investor could obtain on an alternative investment of comparable risk (for example, a secure investment at 4 % if the project is as low-risk as a guaranteed bank deposit).



family and close relations ('*love money*'): this is the **seed financing** phase. Venture capital companies, by contrast, generally keep their distance since getting the firm off the ground demands an investment of time and research (not to mention the risk that the seed may not grow).

- At the start-up of the activity, one moves from setting up production to bringing the product to market. This is a stage which presents a technical and a commercial risk (in relation to positioning against the competition), but also a management risk (what of the founders' capacity to coordinate actions and decisions?). The necessary capital is generally provided in the first instance by *business angels*: this is the stage of start-up capital or **start-up financing**.
- When the firm meets its market, it gradually approaches its break-even point, but it is not always in a position to self-finance its growth: the search for capital at this moment is then called **expansion financing** or expansion capital. And when the firm is mature enough to conquer new markets and to make capacity and productivity investments, it will be able to have recourse to the leverage of debt for this stage of **bridge financing** or deployment capital.
- Once the firm is fully launched and at the summit of its success, its developers generally want to sell it on. They therefore seek to carry out a **buyout** operation. The point here is to have the firm taken over by outside managers (*buy-in*) or by executives of the firm (*buy-out*).

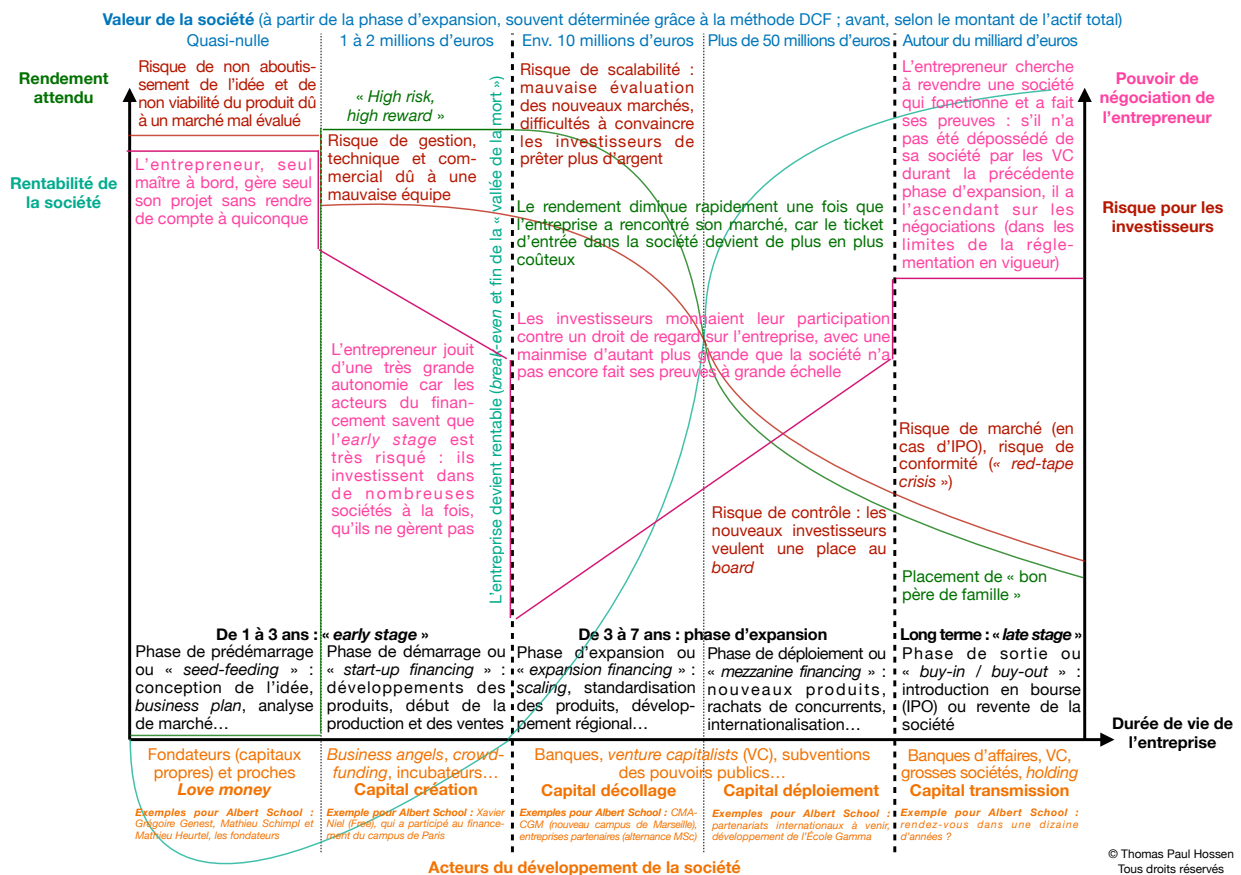


Figure 2.1: An example of the development of a firm, Albert School, in the light of *private equity*

In particular, among buyout operations are **leveraged buy-outs** (LBO). As the Banque de France (2024c) explains, this is a financial arrangement enabling the takeover of a firm by means of a **holding company**. The latter may be created by an investment fund or by employees wishing to acquire their firm without having sufficient funds. They then create this company, which has recourse to borrowing in order to finance the investment. Having become the majority shareholder of the company taken over, the *holding* repays the loan contracted thanks to the profits generated by the target. It is therefore the latter which bears the repayment of the acquirers' debt. The setting up of an LBO therefore begins with the creation of an **acquisition holding**: it frequently contributes roughly 25 % of the

value of the target company, then borrows the balance. In a second stage, the target company's available funds are passed up to the *holding* thanks to the payment of regular (or exceptional) dividends: it is therefore the company bought out which finances the operation, if it has sufficient profitability. The feasibility of an LBO then depends on the capacity to repay the acquisition loans out of the target's future cash. In addition, the LBO may give rise to **tax advantages** (consolidation, tax credit for the *holding*, tax relief on loan interest, and so on), notably when it is a takeover organised by the employees. More precisely, the acquirers benefit from a **tax leverage effect**, since the *holding* will be able, for example, to deduct the loan interest from the tax of the controlling *holding*. With the tax consolidation mechanism, the target's profit is thus offset by the *holding's* loss, and the taxable share decreases.

### Key Takeaways



Private equity consists, for unlisted funds, in taking majority or minority stakes in unlisted (or listed) firms, involving three categories of actor: institutional capital providers (pension funds, insurers, banks, *family offices*), private equity companies, and entrepreneurs in search of financing. The private equity life cycle distinguishes four phases: pre-start-up *seed financing* (financed by *love money*), start-up capital or *start-up financing* with high technical and commercial risks (provided by *business angels*), expansion capital (*expansion financing*) when the firm approaches profitability but cannot self-finance its growth, deployment capital (*bridge financing*) for capacity investments, and finally buyout capital (*buy-in* or *buy-out*) at the time of resale [Glachant et al. (2008)]. *Leveraged buy-outs* (LBO) constitute a specific transfer arrangement: an acquisition *holding* contributes a minority share of the target's value and borrows the balance, the company bought out then financing the repayment via its dividends and its cash, feasibility therefore depending on its future profitability [Banque de France (2024c)]. The LBO offers significant tax advantages: the *holding* deducts the loan interest from its tax, tax consolidation offsets the target's profit by the *holding's* loss, thereby reducing the taxable share and generating a tax leverage effect [Banque de France (2024c)].

### 3.2.2 Methods for valuing a firm

Determining the value of a firm constitutes a central question in corporate finance, inseparable from that of value creation, which guides the major investment and financing decisions as well as the strategic diagnosis of the organisation. As Charreaux (2014b) sets out, depending on whether the firm is listed or not, the question of valuation arises differently: for listed companies, the **stock market value** provides a permanent estimate of the value of the firm and makes it possible to appreciate the creation or destruction of value from the evolution of the share price, whereas for unlisted firms, recourse must be had to alternative valuation methods founded on the discounting of cash flows or on asset-based approaches. However, the implicit assumption that financial markets are **informationally efficient** raises important questions: a financial market is efficient relative to a set of information if it is impossible to derive an abnormal profit from holding that information, and three levels of efficiency are traditionally distinguished: weak, semi-strong and strong. But, as Charreaux (2014b) indicates, this assumption is increasingly contested on account of the existence of speculative bubbles, excessive volatility and systematic anomalies observed on markets. For listed companies, several indicators make it possible to measure value creation based on stock market values. **Created shareholder value** (CSV) measures the value created for shareholders over a given period by comparing added shareholder value, computed as the variation in market capitalisation adjusted for movements of funds (capital increases, dividends, share buybacks), with the shareholder value required given the rate of return demanded on equity as assessed by the capital asset pricing model (CAPM, see *infra*). Likewise, **Jensen's alpha** measures the realised return on equity relative to the required return, which allows a comparison of performance between firms of different sizes. **Market value added** (MVA) constitutes an alternative indicator which measures the difference between the market value of the firm (market capitalisation plus the market value of debt) and the book value of the capital invested, thereby revealing the economic rent that the market anticipates the firm will create in the future. As Charreaux (2014b) explains, when MVA is positive, this means that the firm generates an economic profitability higher than its weighted average cost of capital (WACC), which corresponds to a situation in which there exists a **goodwill** relative to the initial investment. **Tobin's q** and the **market-to-book ratio** express this same idea in the form of ratios, allowing a relative assessment of MVA and facilitating comparisons between firms: a Tobin's *q* greater than 1 or a *market-to-book* ratio greater than 1 indicate a value creation anticipated by the markets, as Charreaux (2014b) analyses. According to Bain & Company (2025), 2025 is on its way to becoming the second-best year in terms of merger and acquisition activity, with an increase of 40 % in value to reach 4.9 trillion dollars, which underlines the practical importance of firm valuation for decision-makers.

The **valuation method by discounting available cash flows** (FCE, for *free cash flows*) constitutes the most fundamental and the most theoretically justified approach for valuing a firm in its entirety, taking into account all the providers of capital, shareholders and financial creditors. As Charreaux (2014b) establishes, this method takes up the logic of

investment appraisal by discounting the free cash flows secreted by the economic assets at the WACC: the total value of the firm results from discounting, over a determined investment horizon (of  $T$  years), the annual FCF augmented by a terminal value received at the end of the horizon. The assessment of the value created is then obtained by subtracting from this discounted value the amount of the initial economic assets. In order to estimate the terminal value, several approaches coexist: the first assumption considers that the terminal value is zero after a very long horizon; the second supposes that the terminal value equals the book value of the economic assets, implying that there is no further value creation after the final year; the third retains a constant growth of FCF to infinity at the rate  $g^*$  (with  $g^*$  lower than the WACC), giving a terminal value equal to  $FCF_{T+1}/(WACC - g^*)$ ; the fourth and last method uses periods with differentiated growth rates, better reflecting the reality of a firm whose competitive advantage erodes progressively over time. Moreover, the **CFROI** (*cash flow return on investment*) represents the internal rate of return of the overall investment that the firm constitutes, that is to say the discount rate for which the discounted value of the FCF and of the terminal value equals the initial economic assets: according to Charreaux (2014b), if this CFROI exceeds the WACC, there is value creation. A complementary approach consists in valuing equity directly by discounting the **flows available to shareholders** (FDA), which comprise dividends, share buybacks and capital increases, at the required rate of return on equity; the **total shareholder return** (TSR) represents the internal rate of return of these flows and must exceed the cost of equity for there to be value creation. **Economic value added** (EVA), finally, constitutes an equivalent alternative measure founded on the discounting of the economic super-profits secreted each year, defined as the difference between after-tax operating profit and the cost of the capital invested, making it possible to measure the economic rent created annually.

**Asset-based and hybrid approaches** offer complementary perspectives by considering the firm no longer as an entity generating value, but as a portfolio of distinct assets, or by incorporating non-tradable intangible assets. As Charreaux (2014b) explains, asset-based methods value the firm either from the seller's perspective (envisaging a voluntary or forced liquidation), or from the buyer's perspective (considering the costs of reconstituting the economic apparatus). **Net book assets** constitute the book value of the firm, equal to the difference between accounting assets and liabilities, while **adjusted net assets**, or revalued net assets, bring this book value closer to its realisation value by correcting for worthless items and by adjusting fixed assets to their probable negotiated price, as Charreaux (2014b) explains. **Substantial value** measures, from a reconstitution perspective, the replacement cost of the assets necessary to operations, including leased durable goods, while **realisable value** and **liquidation value** reflect respectively a progressive or a forced realisation of assets. Hybrid methods founded on *goodwill*, finally, seek to reconcile asset-based and actuarial approaches by valuing equity as the sum of adjusted net assets and goodwill, the latter being obtained by discounting a super-profit representing the additional economic rent that the firm can generate thanks to its intangible assets (know-how, competences, competitive advantages). As Charreaux (2014b) analyses, goodwill exists only if there is value creation, that is to say if the expected return on equity exceeds the required return. In practice, valuers frequently have recourse to relative methods founded on the valuation multiples observed on comparable transactions. According to CLFI (2025), the global median **enterprise value to EBITDA ratio** (EV/EBITDA) for merger and acquisition transactions stood at 9.3 in June 2025, but this multiple varies appreciably according to the type of acquirer: private equity funds pay a median multiple of 12.8 in the United States compared with 9.9 for corporate acquirers, whereas in Europe these multiples stand respectively at 11.2 and 8.5, reflecting the superior access of *private equity* sponsors to leverage and their capacity to identify and realise synergies. By sector, technology firms and health services display the highest multiples, at 12.5 and 12.8 respectively, on account of their scalable business models and their predictable cash flows, whereas the energy sector presents the lowest multiples at 7.4 on account of capital intensity and exposure to commodity cycles.

### Key Takeaways



The valuation of a firm mobilises three complementary categories of method. The actuarial method of discounting free cash flows (FCF) at the weighted average cost of capital (WACC) constitutes the theoretically grounded approach, complemented by the discounting of flows available to shareholders (FDA) and by *economic value added* (EVA) [Charreaux (2014b)]. Asset-based approaches (net book assets, substantial value) value the firm as a portfolio of real assets, while hybrid methods based on goodwill reconcile asset-based and actuarial perspectives by valuing intangible assets [Charreaux (2014b)].

### 3.2.3 Case study: Picard Surgelés

On exiting an LBO, shareholders may merge the *holding* and the company before floating it on the stock market or selling it on to other investors or to another LBO fund. This disposal often generates comfortable capital gains for the acquirers. An unavoidable stage is then to assess correctly the **resale value** of the company. Let us study a concrete

case, recalled by Le Monde (2001) and Décideurs (2010): on 20 February 2001, Carrefour sold Picard Surgelés (an unlisted firm) within the framework of an LBO. The acquirers were two financial investors, the British firm Candover and the French firm Chevrillon & Associés, together with the founding family Decelle and some forty executives. The transaction amounted to 920 million euros, of which 75 million euros of exceptional dividends paid to Carrefour. How were these 920 million euros found in order to carry out the transaction? The family shareholders contributed 67 million euros and the financial shareholders 253 million euros: the total capital contribution therefore stood at 320 million euros, that is 35 % of the amount of the transaction; the remainder (600 million euros) was financed by bank debt. The following data are provided, which will be useful in order to propose a valuation of the firm:

- At the end of 1999, total assets stood (in millions of euros) at 259.963 (of which 26.929 of available funds), equity at 133.727, provisions at 5.727, financial debts at 8.076 (of which 0 in the medium or long term), trade payables at 112.43, and depreciation and provision charges stood at 17.62.
- In 2000, Picard's net profit amounted to 36 million euros. In general, Picard's dividend distribution policy was to pay out 30 % of annual net profit on average to its shareholders.
- Extending economic growth over five years and considering the absence of any notable new competition, it is estimated that net profit will be able to grow by 20 % per year from 2001 to 2005 and that turnover will grow by 16 % per year.
- Toupargel, a firm listed on the Paris stock exchange and Picard's main competitor, presents a *price earnings ratio* of 8.7. The rate on French Treasury bills at the beginning of 2001 was 5 % per year and the expected annual return of the CAC 40 was 8 % at that time.
- The acquirers consider that investments will be limited to the construction of stores, to their maintenance and to marketing; there are no research and development investments. The sum of investments and of the change in working capital requirement is estimated at 21.3 million euros in 2001.

#### (i) The asset-based method

The first method of estimating the value of a firm is called asset-based: it rests on the valuation of the firm's **net assets**. At the end of year  $t$ , we know that net assets are worth the net assets at the end of year  $t - 1$ , to which is added the net profit of year  $t$  and from which are subtracted the dividends paid for year  $t$ . Net assets, moreover, are equal to the amount of total assets from which are subtracted financial debts (short term as well as medium and long term), trade payables and any accruals. Indeed, equity belongs to the firm (it is therefore not repayable to shareholders), while provisions are still in the firm's accounts and do not (yet) correspond to a certain debt.

Thus, for Picard, at the end of 1999, net assets are determined by  $259.963 - 8.076 - 112.43 = 139.46$  million euros. Consequently, the net profit of the year 2000 is equal to 36 million euros, as indicated in the introductory remarks. Finally, if we make the assumption (resulting from a historical analysis of the dividend distribution policy) that the amount of dividends paid amounts to 30 % of net profit, we find as the amount of dividends  $0.30 \times 36 = 10.8$  million euros. Net assets at the end of the year 2000 therefore amount to  $139.46 + 36 - 10.8 = 164.66$  million euros. Moreover, we know that Carrefour is in addition to receive **exceptional dividends** of 75 million euros, linked to the buyout operation. Now, according to the asset-based method, these exceptional dividends must also be subtracted *in fine* from the valuation of the firm. Thus, we arrive at a final assessment of net assets of  $139.455 + 36 - 10.8 - 75 = 89.655$  million euros at the beginning of 2001, at the time of the buyout operation.

The asset-based valuation method is, however, criticised inasmuch as it does not take account of the **revaluation of property**, an element all the more important in that, in the Île-de-France region, real estate appreciates regularly: it is highly improbable that the property has depreciated even if, in accounting terms, it has been written down. In our case study, Picard is a firm which rests on stores located all over France. In accounting terms, the tangible fixed assets corresponding to these stores were acquired long ago (Picard was indeed created in 1973, that is 27 years before the buyout operation), and have therefore been almost entirely written down. Yet we know full well that property prices have soared over 30 years, and the market value of the buildings far exceeds their net book value: account will therefore have to be taken of this. In addition, the asset-based method does not take account of **brand image** and of all the intangible elements which are not valued in accounting terms, which is what is called *goodwill*.

#### (ii) Taking goodwill into account

This is why the asset-based method is generally complemented by the estimation of goodwill, that is to say the sum of the discounted annual **super-profits** generated recurrently by the firm thanks to its organisation, its reputation, its brands: it is therefore an economic rent capitalised over several years. This super-profit is assessed each year by subtracting from net profit the value  $rA$ , where  $A$  is the net assets committed to operations and  $r$  the usual rate of

return on assets in the firm's sector. How is  $r$  to be determined in the first place? One may consult the **price earnings ratio** (PER): the ratio of the value of a share divided by the dividend per share. From then on, the inverse of the PER denotes the 'rate of return' of a share, which may be useful as a proxy variable for assessing the rate of return on assets (it must nevertheless be noted that this approach is somewhat questionable inasmuch as the rate of return on the net assets committed to operations is not exactly the rate of return on equity — which, this time, corresponds exactly to the PER). By contrast, in practice the PER is used because the data on dividends paid and on the valuation of a listed firm are very easily accessible; conversely, determining precisely the net assets committed to operations presupposes an extremely fine-grained knowledge of all competing firms, which in practice is not possible. For example, Toupargel, a firm listed on the Paris stock exchange, presents a PER of 8.7, whence an estimate of  $r$  equal to  $1/8.7$ , that is roughly 11.49 %. What of the net assets committed to operations? We take up the net assets computed previously, that is roughly 164.7 million euros at the end of 2000.

In order to determine goodwill, it is then necessary to establish how net profit on the one hand and net assets excluding exceptional dividends on the other hand will evolve over a given time horizon, generally five years (the average lifespan of the project before exit). Assumptions must then be made about the evolution of these two variables on the basis of a market study. Now, for the Picard case, extending economic growth over five years and considering the absence of any notable new competition, it is estimated that net profit will be able to grow by 20 % per year from 2001 to 2005. From this we deduce the **forecast net profits**, knowing that the net profit for 2000 is 36 million euros:  $36 \times 1.2 = 43.2$  million euros in 2001,  $43.2 \times 1.2 = 51.8$  million euros in 2002,  $51.8 \times 1.2 = 62.2$  million euros in 2003,  $62.2 \times 1.2 = 74.6$  million euros in 2004 and  $74.6 \times 1.2 = 89.5$  million euros in 2005. What of the evolution of net assets? For want of further detail, we may suppose that net assets will grow at the same pace as turnover, that is 16 % per year. Thus, for 2001, we find as **forecast net assets**  $164.5 \times 1.16 = 190.9$  million euros, then  $190.9 \times 1.16 = 221.4$  million euros in 2002, then  $221.4 \times 1.16 = 256.9$  million euros in 2003, then  $256.9 \times 1.16 = 298.0$  million euros in 2004, then  $298.0 \times 1.16 = 345.7$  million euros in 2005. The super-profit (given the estimate of roughly 11.5 % for the usual rate of return on assets in the sector) is therefore written  $43.2 - 0.115 \times 190.9 = 21.2$  million euros in 2001, then  $51.8 - 0.115 \times 221.4 = 26.3$  million euros in 2002, then  $62.2 - 0.115 \times 256.9 = 32.7$  million euros in 2003, then  $74.6 - 0.115 \times 298.0 = 40.3$  million euros in 2004, then  $89.5 - 0.115 \times 345.7 = 49.7$  million euros in 2005. We suppose moreover that beyond 2005, that is to say after five years, super-profits will remain stable at 46.4 million euros (this is generally a precautionary measure: the saturation of available space may weigh on growth, competitors may emerge, an economic recession may occur, and so on). Finally, these forecast super-profits must be discounted. What of the discount rate?

The **discount rate** corresponds to the rate required by investors on the market. It is therefore *a priori* equal to the riskless rate augmented by a **risk premium**. This risk premium corresponds to the beta coefficient of the CAPM. However, for unlisted firms, liquidity is not necessarily forthcoming. A **liquidity premium** must therefore be added, which may for example be set at 1 % for unlisted companies lacking liquidity (obviously, this liquidity premium must flow from market estimates which are not detailed here). We must therefore estimate the CAPM, augmented by the liquidity premium, for our firm Picard. Identifying the expected return on the Picard share, denoted  $\bar{r}_P$ , with the discount rate, we must have  $\bar{r}_P = r_f + \beta(\bar{r}_M - r_f) + \ell$ , where  $r_f$  is the rate of return on the riskless asset (for a French firm, it may notably be approximated by the annual rate on French Treasury bills, that is 5 % at the beginning of 2001),  $\bar{r}_M$  is the expected return of the market (for the French market, the average annual return of the CAC 40 is generally taken, that is roughly 8 % at the beginning of 2001) and where  $\ell$  denotes the liquidity premium. The coefficient  $\beta$  may be estimated econometrically. For want of an econometric estimate, a macroeconomic analysis may be proposed: Picard is not a growth company like those of the tech sector (characterised by  $\beta$  coefficients greater than 1), but neither is it (at least it was not at the beginning of the millennium) in a mature market like most large retail companies (which, for their part, are characterised by  $\beta$  coefficients well below 1). As a first approach, we may therefore consider that Picard's  $\beta$  is roughly equal to 1. From then on, the discount rate is  $\bar{r}_P = 0.05 + (0.08 - 0.05) \times 1 + 0.01 = 9\%$ . This will be our estimate of the discount rate. We may then compute the discounted sum of the super-profits of the first five years:  $\frac{21.2}{1.09} + \frac{26.3}{1.09^2} + \frac{32.7}{1.09^3} + \frac{40.3}{1.09^4} + \frac{46.4}{1.09^5}$ , that is roughly 125.5 million euros over a five-year horizon: this is the goodwill over a five-year horizon. It remains to take account of the super-profits of the years to come, that is  $\frac{46.4}{1.09^6 \times 0.09} = 365.2$  million euros (the sum of the terms of a geometric sequence). The goodwill over an infinite horizon therefore stands at  $125.5 + 365.2 = 490.7$  million euros. Thus, the value of equity at the beginning of 2001 is estimated by the sum of the net assets at the time of the operation (89.7 million euros) and of the goodwill, that is, for an infinite horizon,  $89.7 + 490.7 = 580.4$  million euros.

### (iii) The **discounted cash flows (DCF) method**

The objective of valuation by the actuarial method of discounted cash flows (DCF) consists in estimating the **forecast free cash flows**, that is to say the forecast self-financing capacity (CAF), from which are subtracted the forecast dividends and investments, as well as the change in working capital requirement, denoted  $\Delta WCR$ . Let us begin by

computing the **forecast CAF**: we have already estimated above the future net profits: 43.2 million euros in 2001, 51.8 million euros in 2002, 62.2 million euros in 2003, 74.6 million euros in 2004 and 89.5 million euros in 2005. We are tempted to estimate the CAF by means of the subtractive method, that is to say starting from profit. After the buyout, we consider that there will be no disposals of fixed assets: indeed, the objective of the buyout operation is not to save the firm Picard but to continue to develop it; it is therefore best to capitalise on the existing infrastructure and not to sell off fixed assets. We also suppose that the acquirers will receive no investment grant: there is therefore no reason to show any share of investment grants in the computation of the CAF. It remains to determine the forecast evolution of depreciation, impairment and provision charges (DADP); we must therefore compute the difference between charges and reversals. It is generally supposed that the evolution of DADP follows the same pace as that of turnover, that is 16 % per year according to the estimates. The only difficulty is therefore to determine the forecast DADP for the first year (2001 in our case). We know that Picard's DADP at the close of the 1999 financial year amounted to 17.62 million euros. We may therefore suppose that this was also the case at the close of the 2000 financial year. Then, on account of the investments which will have to be made from 2001 onwards (we then suppose that these investments will likewise follow the growth of turnover, that is 16 % per year), the forecast DADP for 2001 will stand at  $17.6 \times 1.16 = 20.4$  million euros, then  $20.4 \times 1.16 = 23.7$  million euros for 2002, then  $23.7 \times 1.16 = 27.5$  million euros for 2003, then  $27.5 \times 1.16 = 31.9$  million euros for 2004, then  $31.9 \times 1.16 = 37.0$  million euros for 2005. The forecast CAF therefore stands at  $43.2 + 20.4 = 63.6$  million euros in 2001, then  $51.8 + 23.7 = 75.5$  million euros in 2002, then  $62.2 + 27.5 = 89.7$  million euros in 2003, then  $74.6 + 31.9 = 106.5$  million euros in 2004, then  $89.5 + 37.0 = 126.5$  million euros in 2005.

Consequently, we must estimate the **evolution of investments and of the change in working capital requirement** (this corresponds roughly to the first two lines of a financing plan, *modulo* the fact that we identify the working capital requirement with the operating working capital requirement, which is not problematic in the case of Picard, which is a food retail firm). We have already supposed that the evolution of investments would follow that of turnover. It is generally supposed that the same will hold for the evolution of the change in working capital requirement. Now the preliminary data indicate that the sum of investments and of the change in working capital requirement is estimated at 21.3 million euros in 2001. This sum should therefore be worth  $21.3 \times 1.16 = 24.7$  million euros in 2002, then  $24.7 \times 1.16 = 28.7$  million euros in 2003, then  $28.7 \times 1.16 = 33.2$  million euros in 2004, then  $33.2 \times 1.16 = 38.6$  million euros in 2005. Consequently, inasmuch as the new shareholders give priority to the success of the LBO, we consider that they will give priority to the repayment of the debt and to the payment of financial charges: we may therefore suppose *a priori* that no dividend will come to reduce the free cash flows. We thus deduce the forecast free cash flows from 2001 to 2005, by subtracting from the CAF the forecast investments and the forecast changes in working capital requirement:  $63.6 - 21.3 = 42.3$  million euros in 2001, then  $75.5 - 24.7 = 50.8$  million euros in 2002, then  $89.7 - 28.7 = 61.0$  million euros in 2003, then  $106.5 - 33.2 = 73.3$  million euros in 2004, then  $126.5 - 38.6 = 87.9$  million euros in 2005. We suppose moreover that beyond 2005, that is to say after five years, the forecast free cash flows will remain stable at 87.9 million euros (this is generally a precautionary measure). It is now appropriate to discount the free cash flows. We shall take up the same methodology for determining the discount rate as previously (the goodwill method) and consider that it stands at 9 %. We may then compute the discounted sum of the free cash flows of the first five years:  $\frac{42.3}{1.09} + \frac{50.8}{1.09^2} + \frac{61.0}{1.09^3} + \frac{73.3}{1.09^4} + \frac{87.9}{1.09^5}$ , that is roughly 237.7 million euros over a five-year horizon. It remains to take account of the discounted sum of the free cash flows of the years to come, that is  $\frac{87.9}{1.09^6 \times 0.09} = 582.4$  million euros (the sum of the terms of a geometric sequence). The sum of the free cash flows over an infinite horizon therefore stands at  $237.7 + 582.4 = 820.1$  million euros.

We have therefore estimated the sum of the discounted free cash flows over an infinite horizon. It is then possible to estimate the **value of the firm's equity** by subtracting from it the sum of medium- and long-term financial debts. Now Picard Surgelés presents no medium- or long-term debt. The estimate of the firm's equity therefore corresponds exactly to the discounted free cash flows over an infinite horizon, that is roughly 820.1 million euros. Analogously, the **value of the firm** may be estimated by subtracting net financial debt from the sum of the discounted free cash flows over an infinite horizon. Recall in this respect that **net financial debt** corresponds to the difference between every form of financial debt and every form of cash. Now Picard Surgelés has no long- or medium-term debt, and its short-term financial debt (8.076 million euros) is lower than its available funds (26.929 million euros). We therefore find a net financial debt of  $8.076 - 26.929 = -18.853$  million euros, negligible relative to the sum of the discounted free cash flows. Thus, we may consider that the value of the firm stands at 820.1 million euros: this is a value very close to the **acquisition price** of the firm paid by the acquirers, that is 845 million euros if the exceptional dividends are subtracted (an error of less than 3 %). We then see that it is the DCF method which makes it possible to obtain the value of the firm closest to the acquisition price; it is moreover often this method which is used in practice, rather than the asset-based method (very little used because too heavily contested) and the goodwill method (which presupposes a very good knowledge of the sector's competitors).



### 3.3 Single-lender loans

When a firm chooses to have recourse to indebtedness for its financing, it contracts a loan from a commercial bank. What are the different modes of repaying a bank loan? Two possibilities may be envisaged by the customer: repaying the same amount every month — one then speaks of a **loan with constant repayments** (which is generally what French banks offer as standard) — or writing down the same amount of the loan every month, and therefore repaying more at the beginning than at the end — one then speaks of a **loan with constant principal repayments** (which is generally what German banks offer as standard).

#### 3.3.1 Amortisation schedule for a loan with constant principal repayments

Let  $n$  be a non-zero natural number,  $i$  a real number belonging to  $[0, 1]$  and  $E$  a strictly positive real number. Then the amortisation schedule for a loan with constant principal repayments over  $n$  years of  $E$  monetary units at the interest rate  $i$  is determined as follows, the values being expressed in monetary units:

| Year       | Outstanding balance    | Interest              | Principal repayment | Annuity                  | Net value            |
|------------|------------------------|-----------------------|---------------------|--------------------------|----------------------|
| Year 1     | $V_1 := E$             | $I_1 := iV_1$         | $A := E/n$          | $a_1 := I_1 + A$         | $V_2 := E - A$       |
| Year 2     | $V_2 = E - A$          | $I_2 := iV_2$         | $A$                 | $a_2 := I_2 + A$         | $V_3 := V_2 - A$     |
| $\vdots$   | $\vdots$               | $\vdots$              | $\vdots$            | $\vdots$                 | $\vdots$             |
| Year $n-1$ | $V_{n-1} = E - (n-2)A$ | $I_{n-1} := iV_{n-1}$ | $A$                 | $a_{n-1} := I_{n-1} + A$ | $V_n := V_{n-1} - A$ |
| Year $n$   | $V_n = E - (n-1)A = A$ | $I_n := iV_n$         | $A$                 | $a_n := I_n + A$         | $0$                  |

Table 2.1: Structure of an amortisation schedule for a loan with constant principal repayments

We therefore observe that, for every integer  $k$  between 1 and  $n$ , we have  $V_k = E - (k-1)A = E - \frac{(k-1)E}{n} = \left(1 - \frac{k-1}{n}\right)E$ . From then on, the repayment annuity in the  $k$ -th year (where  $k$  is an integer between 1 and  $n$ ) satisfies  $a_k = I_k + A = iV_k + A = i\left(1 - \frac{k-1}{n}\right)E + \frac{E}{n}$ . We then indeed observe that the further one advances in the repayment of the loan (that is to say the more  $k$  increases), the lower the repayment annuity  $a_k$ : one repays more at the beginning of the loan, as Amzallag (2022) analyses.

#### Exercise



**Loan with constant principal repayments.** Draw up the amortisation schedule for a loan with constant principal repayments of 20 000 euros to be repaid over five years at the constant rate of 2.9 %. What is the total amount of interest paid?

**Solution.** The amortisation schedule for a loan with constant principal repayments of 20 000 euros to be repaid over five years at the constant rate of 2.9 %, with values expressed in euros, is the following:

| Year   | Outstanding balance | Interest | Principal repayment | Annuity | Net value |
|--------|---------------------|----------|---------------------|---------|-----------|
| Year 1 | 20 000              | 580      | 4 000               | 4 580   | 16 000    |
| Year 2 | 16 000              | 464      | 4 000               | 4 464   | 12 000    |
| Year 3 | 12 000              | 348      | 4 000               | 4 348   | 8 000     |
| Year 4 | 8 000               | 232      | 4 000               | 4 232   | 4 000     |
| Year 5 | 4 000               | 116      | 4 000               | 4 116   | 0         |

Indeed, for a loan with constant principal repayments, the loan is written down uniformly over the period: the outstanding capital is divided by the number of years ( $20\,000/5 = 4\,000$  euros per year). The interest for each year is computed on the capital outstanding at the beginning of the year at the effective rate of 2.9%. The annuity (equivalent annual instalment) is the sum of the interest and the fixed principal repayment. The net value (outstanding capital) is updated at the end of the year. More precisely:

— **Year 1.** The interest for the first year is obtained by multiplying the capital outstanding at the beginning of the period by the annual interest rate, that is  $20\,000 \times 0.029 = 580$  euros. The fixed annual principal repayment is equal to the initial capital divided by the duration of the loan, that is  $20\,000/5 = 4\,000$  euros. The total annuity paid at the end of year 1 is therefore the sum of the interest and the principal repayment, namely  $580 + 4\,000 = 4\,580$  euros. Finally, the net book value at the end of year 1, that is to say the outstanding capital, amounts to  $20\,000 - 4\,000 = 16\,000$  euros.

- **Year 2.** The interest for the second year is computed on the new capital outstanding at the beginning of the period, that is  $16\,000 \times 0.029 = 464$  euros (where  $16\,000 \times 0.02 = 320$  and  $16\,000 \times 0.009 = 144$ , totalling 464 euros). The principal repayment remains fixed at 4000 euros. The corresponding annuity is therefore  $464 + 4\,000 = 4\,464$  euros and the capital outstanding at the end of the year stands at  $16\,000 - 4\,000 = 12\,000$  euros.
- **Year 3.** The interest for the third year is obtained by  $12\,000 \times 0.029 = 348$  euros. The constant annual principal repayment of 4000 euros is maintained, which gives an annuity of  $348 + 4\,000 = 4\,348$  euros. The capital outstanding at the end of the period is then  $12\,000 - 4\,000 = 8\,000$  euros.
- **Year 4.** The interest for the fourth year amounts to  $8\,000 \times 0.029 = 232$  euros. With the fixed principal repayment of 4000 euros, the total annuity paid is  $232 + 4\,000 = 4\,232$  euros and the outstanding capital becomes  $8\,000 - 4\,000 = 4\,000$  euros.
- **Year 5.** The final interest for the last year is computed on the residual balance, that is  $4\,000 \times 0.029 = 116$  euros. The last principal repayment of 4000 euros closes the loan, for a final annuity of  $116 + 4\,000 = 4\,116$  euros. The outstanding capital is thus reduced to  $4\,000 - 4\,000 = 0$  euros, confirming the full repayment of the loan.

The total interest paid over five years is therefore  $580 + 464 + 348 + 232 + 116 = 1\,740$  euros, and the total of the annuities  $20\,000 + 1\,740 = 21\,740$  euros.  $\square$

### 3.3.2 Amortisation schedule for a loan with constant annuities

Let  $n$  be a non-zero natural number,  $i$  a real number belonging to  $[0, 1]$  and  $E$  a strictly positive real number. Then the amortisation schedule for a ‘classic’ loan with constant annuities (determined according to a formula recalled earlier) over  $n$  years of  $E$  monetary units at the interest rate  $i$  is determined as follows, the values being expressed in monetary units:

| Year       | Outstanding balance | Interest              | Principal repayment      | Annuity  | Net value                  |
|------------|---------------------|-----------------------|--------------------------|----------|----------------------------|
| Year 1     | $V_1 := E$          | $I_1 := iV_1$         | $A_1 := a - I_1$         | $a$      | $V_2 := E - A_1$           |
| Year 2     | $V_2$               | $I_2 := iV_2$         | $A_2 := a - I_2$         | $a$      | $V_3 := V_2 - A_2$         |
| $\vdots$   | $\vdots$            | $\vdots$              | $\vdots$                 | $\vdots$ | $\vdots$                   |
| Year $n-1$ | $V_{n-1}$           | $I_{n-1} := iV_{n-1}$ | $A_{n-1} := a - I_{n-1}$ | $a$      | $V_n := V_{n-1} - A_{n-1}$ |
| Year $n$   | $V_n$               | $I_n := iV_n$         | $A_n := a - I_n$         | $a$      | 0                          |

Table 2.2: Structure of an amortisation schedule for a loan with constant annuities

In the amortisation schedule for a loan with constant annuities, we note that the annuity  $a$  must be determined in such a way that  $V_{n+1} := V_n - A_n = 0$ . Amzallag (2022) then notes that, for every integer  $k$  between 2 and  $n+1$ , we have  $V_k = (1+i)^{k-1}E + \frac{a(1-(1+i)^{k-1})}{i}$  ( $\star$ ). Let us prove it. Recall that  $V_1 = E$ . From then on,  $V_2 = E - A_1 = E - a + iE = (1+i)E - a = (1+i)^{2-1}E - a(1+i)^0$ . Consequently, let  $k$  be a natural number between 2 and  $n$ . Suppose (induction hypothesis) that the relation ( $\star$ ) holds. Then we indeed have:

$$\begin{aligned}
 V_{k+1} &= V_k - A_k = V_k - a + iV_k = (1+i)V_k - a \\
 &= (1+i) \left( (1+i)^{k-1}E - a \sum_{j=0}^{k-2} (1+i)^j \right) - a \\
 &= (1+i)^k E - a \sum_{j=0}^{k-2} (1+i)^{j+1} - a \\
 &= (1+i)^k E - a \sum_{j=1}^{k-1} (1+i)^j - a(1+i)^0 \\
 &= (1+i)^k E - a \sum_{j=0}^{k-1} (1+i)^j \\
 &= (1+i)^{k-1} E - a \sum_{j=0}^{k-2} (1+i)^j \\
 &= (1+i)^{k-1} E - \frac{a - a(1+i)^{k-1}}{1-(1+i)}
 \end{aligned}$$



Now  $V_{n+1} = 0$  must hold in order to complete the repayment of the loan. From then on, the (constant) repayment annuity satisfies:

$$\begin{aligned} V_{n+1} = 0 &\iff (1+i)^n E + \frac{a(1-(1+i)^n)}{i} = 0 \\ &\iff a = -\frac{i(1+i)^n E}{1-(1+i)^n} \\ &\iff a = \frac{i(1+i)^n E}{(1+i)^n - 1} \\ &\iff a = \frac{iE}{1-(1+i)^{-n}} \end{aligned}$$

### Exercise



**Loan with constant annuities.** Draw up the amortisation schedule for a loan with constant annuities of 15 000 euros to be repaid over four years at the constant rate of 3.5 %. What is the total amount of interest paid?

**Solution.** The amortisation schedule for a loan with constant annuities of 15 000 euros to be repaid over four years at the constant rate of 3.5 %, with values expressed in euros, is the following:

| Year   | Outstanding balance | Interest | Principal repayment | Annuity  | Net value |
|--------|---------------------|----------|---------------------|----------|-----------|
| Year 1 | 15 000.00           | 525.00   | 3 558.77            | 4 083.77 | 11 441.23 |
| Year 2 | 11 441.23           | 400.44   | 3 683.33            | 4 083.77 | 7 757.90  |
| Year 3 | 7 757.90            | 271.53   | 3 812.24            | 4 083.77 | 3 945.66  |
| Year 4 | 3 945.66            | 138.11   | 3 945.66            | 4 083.77 | 0.00      |

Indeed, for a loan with constant annuities, each year the borrower repays the same fixed sum. This annuity  $A$  is determined by the formula for discounting an annuity  $a = \frac{iE}{1-(1+i)^{-n}} = \frac{0.035 \times 15\,000}{1-(1+0.035)^{-4}} \approx 4\,083.77$  euros. The interest for each year is computed on the capital outstanding at the beginning of the year at the effective rate of 3.5 %. The principal repayment is obtained by difference ( $A - \text{interest}$ ). The net value is updated at the end of the year. More precisely:

- **Year 1.** The interest for the first year is obtained by multiplying the capital outstanding at the beginning of the period by the annual interest rate, that is  $15\,000.00 \times 0.035 = 525.00$  euros. The fixed annuity being 4083.77 euros, the principal repayment amounts to  $4083.77 - 525.00 = 3\,558.77$  euros. The net book value at the end of year 1, that is to say the outstanding capital, thus stands at  $15\,000.00 - 3\,558.77 = 11\,441.23$  euros.
- **Year 2.** The interest for the second year is computed on the new capital outstanding at the beginning of the period, that is  $11\,441.23 \times 0.035 = 400.44$  euros (exact computation:  $11\,441.23 \times 0.035 = 400.44305 \approx 400.44$ ). The principal repayment is  $4083.77 - 400.44 = 3\,683.33$  euros and the capital outstanding at the end of the year stands at  $11\,441.23 - 3\,683.33 = 7\,757.90$  euros.
- **Year 3.** The interest for the third year is obtained by  $7\,757.90 \times 0.035 = 271.53$  euros. With the constant annuity of 4083.77 euros, the principal repayment amounts to  $4083.77 - 271.53 = 3\,812.24$  euros. The capital outstanding at the end of the period is then  $7\,757.90 - 3\,812.24 = 3\,945.66$  euros.
- **Year 4.** The final interest for the last year is computed on the residual balance, that is  $3\,945.66 \times 0.035 = 138.11$  euros. The last principal repayment of  $4083.77 - 138.11 = 3\,945.66$  euros closes the loan (this principal repayment exceeds the outstanding capital by one cent on account of the successive roundings to the nearest cent). The outstanding capital is thus reduced to  $3\,945.66 - 3\,945.66 = 0.00$  euros, confirming the full repayment of the loan.

The total interest paid over four years is therefore  $525.00 + 400.44 + 271.53 + 138.11 = 1\,335.08$  euros, and the total of the annuities  $4 \times 4\,083.77 = 16\,335.08$  euros (that is  $15\,000.00 + 1\,335.07$  with a one-cent adjustment for roundings).  $\square$

### Debate



**Loan with constant principal repayments or loan with constant annuities?** On a strictly accounting view, the total nominal cost of a loan with constant principal repayments (said to be ‘German-style’) is necessarily lower than that of a loan with constant annuities (said to be ‘French-style’). Indeed, the accelerated repayment structure of the capital in the first case mechanically reduces the base on which future interest is computed. For a loan of amount  $E$ ,

of duration  $n$  and of rate  $i$ , the sum of the interest paid in the case of constant principal repayments is given by:

$$\begin{aligned}
 S_{DE} &= \sum_{k=1}^n i \left( E - (k-1) \frac{E}{n} \right) \\
 &= niE - \frac{iE}{n} \sum_{k=1}^n (k-1) \\
 &= niE - \frac{iE}{n} \times \frac{n(n-1)}{2} \\
 &= niE - \frac{(n-1)iE}{2} = \frac{(n+1)iE}{2}
 \end{aligned}$$

By contrast, the total cost of the loan with constant annuities is given by  $S_{FR} = n \times \frac{iE}{1-(1+i)^{-n}} - E$ . However, this nominal comparison must be qualified by the time value of money: if the cash flows are discounted at the rate of the loan, the two formulae are financially equivalent. In a context where the average rate on new housing loans stands at 3.08 % in December 2025 according to the data of the Banque de France (2026), the comparative advantage of the German-style loan lies above all in minimising interest rate risk in the event of a future renegotiation, but it imposes a far greater initial cash effort. This additional initial effort inherent in the loan with constant principal repayments then considerably penalises the borrowing capacity of households subject to strict solvency constraints. As Scanlon et al. (2008) emphasise in their international comparison of mortgage products, markets where fixed rates and constant annuities predominate, such as France, favour access to home ownership by smoothing the repayment burden. In France, the Haut conseil de stabilité financière (HCSF, the French high council for financial stability) imposes a maximum debt-service ratio of 35 %, which often makes the German-style loan unworkable for first-time buyers: its first instalment  $a_1 = \frac{E}{n} + iE$  is greater than the constant annuity  $a = \frac{iE}{1-(1+i)^{-n}}$ , which saturates the debt ratio faster for a given income. Thus, the optimisation choice depends on the wealth objective: if the investor wishes to minimise the total cost and has abundant savings, the German scheme is optimal; conversely, if the objective is to maximise leverage for the acquisition of a principal residence, the French scheme prevails, since it makes it possible to borrow a larger capital  $E$  at constant income.

### Exercise



**Borrowing capacity.** You have 100 000 euros of liquid funds in a current account following a gift. You have just begun your professional career and your annual net salary before tax is 30 000 euros (a managerial post on a permanent contract in a CAC 40 firm). You estimate that your salary progression will be 4 % per year over the next twenty-five years. You are hesitating between investing in your own name or with your partner in 2026. Two types of project present themselves to you:

- **Project 1: purchase of a principal residence.** You have spotted a recently built two-room flat at Mairie de Saint-Ouen (line 14) of 45 square metres for 300 000 euros. You plan to make it your principal residence and to live there for the next twenty-five years with your other half (no children planned).
  - **Project 2: buy-to-let investment.** You prefer to rent your 38 square-metre principal residence at Alésia (14th arrondissement) for 950 euros a month and to invest 240 000 euros to buy a large studio flat in Montrouge, convertible into a two-room flat for 30 000 euros of works and furnishing. Letting the property (non-professional furnished letting, or LMNP) would bring you 800 euros a month excluding service charges.
- (i) Are these projects financeable by you alone? Compute your maximum borrowing capacity (loan with constant principal repayments and loan with constant annuities), on the assumption of an interest rate negotiated at 2.5 % per year over 25 years (insurance included) and notary fees of 10 %. It is assumed that the bank will take account of 70 % of the estimated rental income. It may be supposed that rents rise by 4 % per year on average. Are the projects financeable by adding your partner, who will be assumed to earn the same salary with the same prospects of progression?
  - (ii) Compare the two projects from a tax and financial point of view for an investment as a couple in the first year. Recall that the allowance on salaries is 10 % (for the sake of simplicity, net salaries before tax will be identified with taxable salaries), that the allowance on rents under LMNP is 50 %, and that the two persons have no income other than their salaries and their possible buy-to-let investment. The couple will be supposed to be in a civil partnership or married.

- (iii) In the long run, which project would you lean towards for an investment as a couple? It may be assumed that the brackets of the income tax scale will evolve by 5 % per year and that the depreciation of capital will likewise stand at 5 % per year. Give here a few intuitions without getting lost in delicate computations: there is no right or wrong answer *a priori*.

| Annual income brackets             | Tax rate on the income bracket |
|------------------------------------|--------------------------------|
| Up to 11 600 euros                 | 0 %                            |
| From 11 601 euros to 29 579 euros  | 11 %                           |
| From 29 580 euros to 84 577 euros  | 30 %                           |
| From 84 578 euros to 181 917 euros | 41 %                           |
| Over 181 917 euros                 | 45 %                           |

Table 2.3: Scale of income tax on 2025 income per tax part, after the Direction de l'information légale et administrative (2026b)

**Solution.** (i) The borrowing capacity of a household is determined in accordance with the rules issued by the Haut conseil de stabilité financière, which limits the debt-service ratio to 35 % of net monthly income, incorporating 70 % of estimated gross rents. In the present case, the initial net monthly salary amounts to 2500 euros for the single borrower, that is a maximum authorised effort of  $2500 \times 0.35 = 875$  euros per month for the principal residence project; with the partner on identical income, this ceiling doubles to  $2 \times 2500 \times 0.35 = 1750$  euros per month. For the buy-to-let investment, the rental income taken into account adds  $800 \times 0.70 = 560$  euros per month, bringing the effective income to  $2500 + 560 = 3060$  euros and the maximum effort to  $3060 \times 0.35 = 1071$  euros alone, or  $5560 \times 0.35 = 1946$  euros for two with doubled salaries. Consequently, the formula for the constant annuity  $a$ , incorporating the equivalent monthly rate  $i = (1 + 0.025)^{1/12} - 1 \approx 0.205984$  % over  $n = 300$  months, is written  $a_{\max} = \frac{i E_{\max}^{\text{FR}}}{1 - (1+i)^{-n}}$ , whence the maximum capacity  $E_{\max}^{\text{FR}} = a_{\max} \times \frac{1 - (1+i)^{-n}}{i}$ . For the loan with constant principal repayments, the highest initial instalment  $a_{\max} = \frac{E_{\max}^{\text{DE}}}{300} + i E_{\max}^{\text{DE}}$  constrains the capacity via  $E_{\max}^{\text{DE}} = \frac{a_{\max}}{i + 1/300}$ . Thus, the maximum borrowing capacities stand at the following values, expressed in euros and rounded to the nearest euro:

|    | Individual financing                                                   |                                                    | Joint financing                                                        |                                                    |
|----|------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------|
|    | Constant annuities                                                     | Constant principal                                 | Constant annuities                                                     | Constant principal                                 |
| P1 | $875 \frac{1 - (1 + 0.00205984)^{-300}}{0.00205984} \approx 195\,063$  | $\frac{875}{0.00205984 + 1/300} \approx 162\,242$  | $1750 \frac{1 - (1 + 0.00205984)^{-300}}{0.00205984} \approx 391\,325$ | $\frac{1750}{0.00205984 + 1/300} \approx 324\,485$ |
| P2 | $1071 \frac{1 - (1 + 0.00205984)^{-300}}{0.00205984} \approx 239\,491$ | $\frac{1071}{0.00205984 + 1/300} \approx 198\,585$ | $1946 \frac{1 - (1 + 0.00205984)^{-300}}{0.00205984} \approx 435\,154$ | $\frac{1946}{0.00205984 + 1/300} \approx 360\,827$ |

The notary fees of 10 % bring the total cost of project 1 (P1) to 330 000 euros (300 000 euros of price excluding fees), requiring a loan of 230 000 euros after a contribution of 100 000 euros; for project 2 (P2), the acquisition of 240 000 euros (plus 30 000 euros of works and furnishing) totals  $240\,000 \times 1.10 + 30\,000 = 294\,000$  euros with fees, that is a required loan of 194 000 euros (it only just works under 'German-style' financing). Alone, project 1 is financeable neither with constant annuities nor with constant principal repayments, whereas project 2 is financeable in both cases. Indeed, for two, both projects become financeable whatever the type of loan, the capacities largely exceeding the needs.

- (ii) Consequently, let us compare the two types of project for joint financing. The tax comparison between the two projects rests on the progressive income tax scale applicable to 2025 income, which distinguishes five marginal brackets per part of the family quotient for couples in a civil partnership subject to joint taxation. For the couple in a civil partnership, the annual net salaries before tax total 60 000 euros, that is  $60\,000 \times (1 - 0.10) = 54\,000$  euros taxable after the flat-rate allowance of 10 %; the family quotient of two parts sets the income per part at  $54\,000/2 = 27\,000$  euros for project 1, generating a tax of  $(27\,000 - 11\,600) \times 0.11 = 1694$  euros per part on the 11 % bracket, that is  $1694 \times 2 = 3398$  euros in total. By contrast, project 2 incorporates  $800 \times 12 = 9600$  euros of gross annual rents, reduced to  $9600 \times 0.5 = 4800$  euros taxable after the 50 % allowance for non-professional furnished letting, bringing taxable income to  $54\,000 + 4800 = 58\,800$  euros, that is  $58\,800/2 = 29\,400$  euros per part, which gives an income tax of  $(29\,400 - 11\,600) \times 0.11 = 1958$  euros per part, that is  $1958 \times 2 = 3916$  euros in total: the buy-to-let investment therefore increases the tax burden by  $3916 - 3398 = 518$  euros per year in the first year. Financially, the net cash flows incorporate the rental costs of the principal dwelling for project 2, where the  $9600 - 518 = 9082$  euros of net after-tax receipts are offset by  $950 \times 12 = 11\,400$  euros of rental expenditure

at Alésia, generating a net deficit of  $9082 - 11\,400 = -2318$  euros per year. Moreover, the annual annuity of project 1 is  $\frac{0.025 \times 230\,000}{1 - (1 + 0.025)^{-25}} \approx 12\,483$  euros, while that of project 2 amounts to  $\frac{0.025 \times 194\,000}{1 - (1 + 0.025)^{-25}} \approx 10\,530$  euros. There therefore remains an advantage in terms of net cash flow in favour of project 1 of  $44\,119 - 43\,754 = 365$  euros, as the following summary shows:

|                         | Project 1     | Project 2     |
|-------------------------|---------------|---------------|
| Earned income           | 60 000        | 60 000        |
| Property loan           | -12 483       | -10 530       |
| Property income         | 0             | 9 600         |
| Rental outgoings        | 0             | -11 400       |
| Taxes                   | -3 398        | -3 916        |
| <b>Total net income</b> | <b>44 119</b> | <b>43 754</b> |

- (iii) In the long run, project 1 likewise prevails for an investment as a couple, since the repayment annuities remain fixed unlike rents, which are bound to rise every year: with project 2, the initial net rental burden of 950 euros per month thus climbs to roughly  $950 \times (1 + 0.04)^9 \approx 1352$  euros after nine years, aggravating the cash flow needed to finance project 2 despite the relative tax relief afforded by tax brackets indexed at 5 % per year. In addition, the location of the asset in Montrouge — a tight market but less dynamic than Saint-Ouen — and the operational letting risks (vacancy, unpaid rent) make the conservative choice of project 1 preponderant over a twenty-five-year horizon. That said, project 2 offers better wealth stability in the event of a break-up of the couple (avoiding the constraint of a forced resale of jointly owned property).  $\square$

### 3.3.3 Repayment deferral and loan smoothing

It also happens that a **repayment deferral** following a loan is provided for from the moment of signature: for private individuals, this is often the case with student loans, which one begins to repay only when one lands a job, while for firms, this was the case with the State-guaranteed loans made within the framework of the lockdowns following the Covid crisis, loans for which the State granted a repayment deferral of eighteen to twenty-four months before firms were required to begin repaying their loan, as Bpifrance (2023) recalls. The treatment of such a loan is then very simple: consider a loan of  $V_0$  euros of a duration of  $n$  years at the rate  $i$ , with a repayment deferral of  $k$  years (with  $k < n$ : this means that nothing is repaid in the first  $k$  years). Now the value  $V_k$  at the end of  $k$  years of borrowing satisfies  $V_k = V_0(1 + i)^k$  euros: it is therefore ‘as if’ the borrower were contracting a loan of  $E := V_k$  euros over  $(n - k)$  years, without any repayment deferral.

#### Focus...



**Payment facilities following the Covid loans. State-guaranteed loans** (PGE) were put in place within the framework of the support measures for firms affected by the economic crisis linked to the Covid-19 pandemic. These loans, intended to preserve firms' cash, are granted by banking establishments or by crowdfunding platforms, as the ministerial order of 6 May 2020 stipulates. They offer a temporary financing solution to help firms overcome the financial difficulties engendered by the collapse of economic activity during the health crisis. As Bpifrance (2023) recalls, the amount of the PGE is capped according to the firm's profile and its specific needs: for the majority of firms, the loan may reach a maximum amount equivalent to 3 months of 2019 turnover; for innovative firms or those created after 1 January 2019, this ceiling is nevertheless adapted, allowing them to borrow up to the equivalent of two years of payroll. This flexibility makes it possible better to answer the needs of firms whose business model is not yet fully mature or which require substantial human capital in order to operate. The PGE offers considerable flexibility in its repayment arrangements. In the first instance, firms benefit from a repayment deferral of one year, which means that they begin to repay neither the capital nor the interest before that period, as Bpifrance (2023) explains. In addition, the amortisation of the PGE may be spread over a maximum duration of six years, thus permitting a flexible distribution of payments over a long period. The government strengthened this arrangement within the framework of the second lockdown, by offering the possibility of spreading the amortisation over one to five additional years. This measure is accompanied by attractive interest rates for SMEs, negotiated between 1 % and 2.5 %, including the State guarantee, which makes it possible to reduce interest charges for firms, as Bpifrance (2023) indicates. In order to answer the repayment difficulties encountered by certain firms, additional measures were taken. On 14 January 2021, the government announced that all firms, irrespective of their size or their sector of activity, could benefit from an additional one-year deferral before beginning the repayment of their PGE, as Bpifrance (2023) mentions. Moreover, since 15 February 2022, micro-enterprises and SMEs having taken out a PGE of an amount lower than 50 000 euros may request a restructuring of their loan if they encounter proven difficulties in repaying their instalments in 2022. This restructuring may include an extension of the repayment period of the loan. Depending on

the firm's situation, this extension may reach two additional years, thus bringing the total repayment period to eight years. Exceptionally, for the most critical cases, the repayment period may be extended by up to four additional years, that is a total repayment period of ten years, without additional costs for the firm. Bpifrance (2023) specifies that this rescheduling procedure was extended until the end of 2023. The eligibility criteria remain unchanged: only micro-enterprises and SMEs having taken out a PGE lower than or equal to 50 000 euros may access this measure, as may associations, which are likewise eligible for the arrangement. In addition to the extension of the repayment period, it is also possible to benefit from an additional repayment deferral of six months at most, allowing firms to postpone the start of their repayments while easing their cash position.

### Exercise



**Loan with a repayment deferral.** Determine the amount  $V_0$  of a loan of a duration of five years, two of which are of repayment deferral, knowing that the rate is 2 % and that the annuity amounts to 9019 euros. Deduce from this the amortisation schedule of this loan.

**Solution.** The statement implies that the borrower repays nothing in the first two years. Now the value  $V_2$  at the end of one year of a loan  $V_0$  is  $V_2 = V_0 \times (1 + 0.02)^2 = 1.0404 V_0$  euros. It is therefore 'as if' the borrower were contracting a loan with constant annuities of  $E := V_2 = 1.0404 V_0$  euros over  $n = 3$  years, without any repayment deferral. The annuity of the loan is then determined as follows, to the nearest cent, by  $a = \frac{iE}{1 - (1+i)^{-n}} = \frac{1.0404 i V_0}{1 - (1+i)^{-n}}$ . Now we have:

$$a = \frac{1.0404 i V_0}{1 - (1+i)^{-n}} \iff V_0 = \frac{a(1 - (1+i)^{-n})}{1.0404 i}$$

$$\iff V_0 = \frac{9019 \times (1 - (1+0.02)^{-3})}{1.0404 \times 0.02} \approx 25\,000$$

From then on, the cost of the loan is  $9019 \times 3 - 25\,000$ , that is roughly 2057 euros, while the amortisation schedule of such a loan is determined as follows, to the nearest euro:

| Year   | Outstanding balance | Interest | Principal repayment | Annuity | Net value |
|--------|---------------------|----------|---------------------|---------|-----------|
| Year 1 | 25 000              | 500      | (500)               | 0       | 25 500    |
| Year 2 | 25 500              | 510      | (510)               | 0       | 26 010    |
| Year 3 | 26 010              | 520.20   | 8 498.80            | 9 019   | 17 511.20 |
| Year 4 | 17 511.20           | 350.22   | 8 668.78            | 9 019   | 8 842.42  |
| Year 5 | 8 842.42            | 176.85   | 8 842.15            | 9 019   | 0         |

More precisely:

- **Year 1.** During the first year of deferral, the interest due is computed on the initial capital, that is  $25\,000 \times 0.02 = 500$  euros. Since no repayment is made (zero annuity), this interest is capitalised and is added to the capital: the principal repayment is therefore negative, equal to  $-500$  euros (shown in brackets). The capital outstanding at the end of year 1 thus amounts to  $25\,000 + 500 = 25\,500$  euros.
- **Year 2.** During the second year of deferral, the interest due is computed on the new, increased capital, that is  $25\,500 \times 0.02 = 510$  euros. Once again, no annuity is paid and the interest is capitalised: the negative principal repayment is  $-510$  euros. The capital outstanding at the end of year 2 therefore reaches  $25\,500 + 510 = 26\,010$  euros.
- **Year 3.** From the third year the phase of effective repayment begins. The interest due on the capital at the beginning of the year is  $26\,010 \times 0.02 = 520.20$  euros. The constant annuity set at 9019 euros permits a principal repayment of  $9019 - 520.20 = 8\,498.80$  euros. The capital outstanding at the end of year 3 is then  $26\,010 - 8\,498.80 = 17\,511.20$  euros.
- **Year 4.** The interest for the fourth year is computed on the residual capital, that is  $17\,511.20 \times 0.02 = 350.22$  euros. With the constant annuity of 9019 euros, the principal repayment amounts to  $9019 - 350.22 = 8\,668.78$  euros. The capital outstanding at the end of year 4 becomes  $17\,511.20 - 8\,668.78 = 8\,842.42$  euros.
- **Year 5.** The final interest for the fifth year is computed on the remaining balance, that is  $8\,842.42 \times 0.02 = 176.85$  euros (rounded). The last principal repayment of  $9019 - 176.85 = 8\,842.15$  euros settles the loan (a slight difference of 0.27 euros due to the successive roundings). The capital outstanding is thus brought back to  $8\,842.42 - 8\,842.15 \approx 0$  euros, confirming the complete repayment of the loan.

The total interest paid over five years (two of which capitalised) is  $500 + 510 + 520.20 + 350.22 + 176.85 = 2\,057.27$  euros, and the total of the annuities effectively paid (years 3 to 5) is  $3 \times 9\,019 = 27\,057$  euros.  $\square$

Consequently, let us imagine that you have contracted several bank loans, each with its own maturity; for the sake of greater readability, you will wish to group all your loans into a single one: one then speaks of **loan smoothing**. How may this operation be envisaged? Consider a principal loan: the amount borrowed is  $E_0$ , the interest rate is  $i_0$  and its duration is  $n_0$ . Consider next  $N$  secondary loans (where  $N$  is a non-zero natural number). For every integer  $k$  between 1 and  $n$ , we denote by  $a_k$  the annuity of the  $k$ -th secondary loan,  $n_k$  its duration and  $d_k$  its payment deferral relative to the beginning of the principal loan. Then the constant (or smoothed) annuity  $a^*$  for the principal loan together with the secondary loans over the whole duration of the principal loan satisfies, as Papin (2025) indicates:

$$a^* := \frac{1}{1 - (1 + i_0)^{-n_0}} \left( i_0 E + \sum_{k=1}^N a_k (1 + i_0)^{-d_k} (1 - (1 + i_0)^{-n_k}) \right)$$

### Exercise



**Loan smoothing.** Consider a combination of a loan of 250 000 euros at a rate of 5 % of a duration of 25 years, and an interest-free loan of 50 000 euros of a duration of 10 years without deferral. Compute the cost of the smoothed loan and comment.

**Solution.** Consider a combination of a classic loan of  $D_0 := 250\,000$  euros at a rate  $r_0 := 5\%$  of a duration of  $n_0 := 25$  years and an interest-free loan, repayable over  $n_1 := 10$  years, with a repayment deferral of  $d_1 := 0$  year for an amount of 50 000 euros. The annuity of the interest-free loan (PTZ) is then  $a_1 := \frac{50\,000}{10} = 5\,000$  euros, while the annuity  $a_0$  of the principal loan is worth, to the nearest cent,  $a_0 = \frac{D_0 i}{1 - (1 + i)^{-n}} = \frac{250\,000 \times 0.05}{1 - (1 + 0.05)^{-25}} \approx 17\,738.11$ . The cost  $R$  of the unsmoothed loan therefore satisfies, to the nearest cent,  $R = n_0 a_0 + n_1 a_1 - D_0 - D_1 = 25 \times 17\,738.11 + 10 \times 5\,000 - 250\,000 - 50\,000 = 193\,452.86$ . Consequently, the constant annuity  $a^*$  over  $n_0 := 25$  years corresponding to the smoothing of the two loans stands, to the nearest cent, at:

$$a^* := \frac{1}{1 - (1 + 0.05)^{-25}} \times (250\,000 \times 0.05 + 5\,000 \times (1 + 0.05)^{-0} \times (1 - (1 + 0.05)^{-10})) \approx 20\,477.49$$

The cost  $R^*$  of the smoothed loan therefore satisfies, to the nearest cent,  $R^* = n_0 a^* - D_0 - D_1 = 25 \times 20\,477.49 - 250\,000 - 50\,000 = 211\,937.25$ . The cost of the smoothed loan is therefore here greater than that of the classic loan. Why? Without smoothing, the PTZ is repaid rapidly (5000 euros per year for 10 years), which then makes it possible to concentrate payments on the loan at 5 %. With smoothing, everything is spread over 25 years: during years 11 to 25, one therefore continues to pay interest at 5 % on an outstanding capital larger than it would be without smoothing, since the principal loan has been repaid more slowly. Despite this additional cost, smoothing has the advantage of regularising the monthly instalments and of preserving the borrower's borrowing capacity by avoiding repayment peaks.  $\square$

### Exercise



**A 'tailor-made' loan.** Consider a loan of 15 000 euros of a duration of 3 years at a rate of 2 %. The first two annuities are 5 000 euros. Draw up the amortisation schedule of this loan.

**Solution.** Consider a loan of  $V_1 := 15\,000$  euros of a duration of 3 years at a rate of  $i := 2\%$ . The amortisation schedule is then the following:

| Year   | Outstanding balance | Interest | Principal repayment | Annuity  | Net value |
|--------|---------------------|----------|---------------------|----------|-----------|
| Year 1 | 15 000              | 300      | 4 700               | 5 000    | 10 300    |
| Year 2 | 10 300              | 206      | 4 794               | 5 000    | 5 506     |
| Year 3 | 5 506               | 110.12   | 5 506               | 5 616.12 | 0         |

Indeed:

— The first annuity is  $a_1 := 5\,000$  euros. The interest paid during the year is therefore  $I_1 := 5\,000 \times 0.02 = 300$  euros, which gives a principal repayment  $A_1 := 5\,000 - 300 = 4\,700$  euros. The net value at the end of the first year is therefore  $V_2 := 15\,000 - 4\,700 = 10\,300$  euros.

- The second annuity is  $a_2 := 5000$  euros. The interest paid during the year is therefore  $I_2 := 10\,300 \times 0.02 = 206$  euros, which gives a principal repayment  $A_2 := 5000 - 206 = 4794$  euros. The net value at the end of the second year is therefore  $V_3 := 10\,300 - 4794 = 5506$  euros.
- Let  $a_3$  be the third and last annuity. The interest paid during the year is therefore  $I_3 := 5506 \times 0.02 \approx 110.12$  euros. Now the principal repayment  $A_3 = a_3 - I_3$  must satisfy  $A_3 = 5506$  euros, so that the net value of the loan at the end of the third year  $V_3$  is zero. We therefore have  $a_3 = I_3 + A_3$ , that is 5616.12 euros, to the nearest cent.  $\square$

### 3.4 Cash investments

A banking establishment is not only a partner from which one can borrow money; it is also an institution which can allow economic agents to invest money in order to make a cash investment grow. Investing one's money, that is to say presently renouncing the effective enjoyment of one's capital, is then remunerated by the regular and periodic payment of interest or of other forms of payment (rents, dividends, and so on). The interest income that a lender receives at the end of a period may then in turn be reinvested, thereby increasing the investor's loan capital and hence the overall amount of interest that a lender or investor receives: this is the phenomenon of **compound interest** which we have already studied *supra*.

#### 3.4.1 The value of a regular investment

The compounding of interest therefore makes it possible to increase an initial capital by the interest income of each period, which makes it possible in the following periods to earn interest on the interest of the preceding periods in addition to that derived from the initial capital. Thus, as we have already studied *supra*, if one invests an initial capital  $V_0$  at the interest rate  $i$ , the **accrued capital** at the end of  $n$  years (where  $n$  is a non-zero natural number) satisfies  $V_n = (1 + i)^n V_0$ . In general, however, one rarely contents oneself with a single initial investment: one regularly invests the same amount  $E$ , every year for instance. Thus, at the end of year  $n$ , the accrued capital is that of the preceding year, augmented by the interest at the rate  $i$ , to which is added the regular investment  $E$ . Otherwise put, if we denote by  $V_n$  the capital accrued at the end of year  $n$ , for every non-zero natural number  $n$ , then  $V_n = (1 + i)V_{n-1} + E$ . Thus, the sequence  $(V_n)_{n \in \mathbb{N}}$  is an **arithmetico-geometric sequence** of the form  $V_n = \lambda V_{n-1} + \mu$ , with  $\lambda := 1 + i$  and  $\mu := E$ . In order to determine the capital accrued at the end of the  $n$ -th year, as Major Prépa (2026) explains, one begins by solving the **characteristic equation**  $x = \lambda x + \mu$ , in the real unknown  $x$ . Since  $i$  is assumed to be strictly positive,  $\lambda$  is different from 1 and one finds as solution  $x^* := \frac{\mu}{1-\lambda}$ . Let us then show that the sequence  $(V_n - x^*)_{n \in \mathbb{N}}$  is geometric. Indeed, for every natural number  $n$ , we have  $V_{n+1} - x^* = \lambda V_n + \mu - x^*$  by definition of the sequence  $(V_n)_{n \in \mathbb{N}}$ , whence  $V_{n+1} = \lambda V_n + \mu - \lambda x^* - \mu$ , since  $x^*$  is a solution of the equation  $x = \lambda x + \mu$ . Thus, it follows that  $V_{n+1} = \lambda(V_n - x^*)$ . From then on, for every non-zero natural number  $n$ , the equality  $V_n - x^* = \lambda^n(V_0 - x^*)$  holds, whence  $V_n = (1 + i)^n \left( V_0 + \frac{E}{i} \right) - \frac{E}{i}$ .

#### Exercise



##### Multiple-choice questions on interest rates and investments

- (i) On 1 January 2010, 500 euros were paid into a savings account remunerated at 4 %. Two further payments were made into the same account on 1 January 2011 and on 1 January 2012. Over these three years, the sum paid in progressed geometrically (with a common ratio of 1.05). What will be the value accrued by the savings account at the very beginning of 2013?
  - (a) 1 623.23 euros.
  - (b) 1 704.66 euros.
  - (c) 1 638.05 euros.
  - (d) 1 703.57 euros.
  - (e) 1 788.55 euros.
- (ii) Consider a compound-interest investment at 3 %. The initial amount of the investment is 2 000 euros. Interest is paid in arrears. Today, rounding up to the nearest cent, the value of the investment is 2 185.46 euros. How much time has elapsed since the beginning of the investment?
  - (a) 2 years.
  - (b) 3 years.

(c) 4 years.

In two years' time, the value of the investment will be, rounding down to the nearest euro:

(c) 2 316 euros.

(d) 2 318 euros.

(iii) A private individual invests, in an account yielding 4 % per year, six annuities of 2 000 euros. The capital available in this account 2 years after the last payment is:

(a) 13 265.95 euros.

(b) 14 348.45 euros.

(c) 13 165.98 euros.

(d) 12 987.45 euros.

(e) 14 875.90 euros.

(iv) A private individual invests in an account at 2.1 % a sum of 10 000 euros. At the end of two years, he withdraws a sum  $x$  from this account. Two years later, the value of his investment is 2 187.36 euros. What is  $x$ ?

(a) 8 000 euros.

(b) 8 022.64 euros.

(c) 8 679.47 euros.

(d) 8 326.10 euros.

(e) 8 150 euros.

**Solution.** (i) On 1 January 2010, 500 euros were paid into a savings account remunerated at 4 %. Two further payments were made into the same account on 1 January 2011 and on 1 January 2012. Over these three years, the sum paid in progressed geometrically (this progression having a common ratio of 1.05). This therefore means that the second payment amounts to  $500 \times 1.05 = 525$  euros and that the third amounts to  $525 \times 1.05 = 551.25$  euros. From then on, the sum accrued by the savings account (with compound interest) once the year 2012 has elapsed stands, to the nearest cent, at  $V_3 := (((500 \times 1.04) + 525) \times 1.04 + 551.25) \times 1.04 \approx 1\,703.57 \rightarrow$  **Answer (d)**.

(ii) Consider a compound-interest investment at 3 %. The initial amount of the investment is  $D_0 := 2\,000$  euros. Interest is paid in arrears. At the end of  $n$  years, rounding up to the nearest cent, the value of the investment is  $D_n := 2\,185.46$  euros. Now we have:

$$\begin{aligned}
 D_n &= 1.03^n D_0 \iff 1.03^n = \frac{D_n}{D_0} \\
 &\iff \exp(n \ln(1.03)) = \frac{D_n}{D_0} \\
 &\iff n \ln(1.03) = \ln\left(\frac{D_n}{D_0}\right) \\
 &\iff n \ln(1.03) = \ln(D_n) - \ln(D_0) \\
 &\iff n = \frac{\ln(D_n) - \ln(D_0)}{\ln(1.03)} \\
 &\iff n = \frac{\ln(2\,185.46) - \ln(2\,000)}{\ln(1.03)} \approx 3
 \end{aligned}$$

So the money was invested for three years. Two years later (that is to say at the end of 5 years), the investment will then be worth  $D_5 := 1.03^2 D_3 = 1.03^2 \times 2\,185.46 \approx 2\,318$  (rounded down to the nearest euro)  $\rightarrow$  **Answers (a) and (e)**.

(iii) A private individual invests, in an account yielding  $\tau := 4$  % per year, six annuities each of  $A := 2\,000$  euros (the investments are therefore made at the end of the period). At the end of the  $n := 6$  years, the value of



the portfolio is therefore, to the nearest cent,  $V_6 := \frac{A((1+\tau)^n-1)}{\tau} = \frac{2\,000 \times ((1+0.04)^6-1)}{0.04} \approx 13\,265.95$ . Two years later, that is two years after the last payment, the accrued capital is therefore, to the nearest cent,  $V_8 := (1+\tau)^2 V_6 = (1+0.04)^2 \times \frac{2\,000 \times ((1+0.04)^6-1)}{0.04} \approx 14\,348.45 \rightarrow$  **Answer (b)**.

- (iv) A private individual invests in an account at 2.1 % a sum of 10 000 euros. At the end of two years, he withdraws a sum  $x$  from this account. At the end of the two years, the value therefore stands at  $V_2 := 10\,000 \times (1+0.021)^2 - x = 10\,424.41 - x$ . Two years later, the value of this investment is  $V_4 := 2\,187.36$ . We know that we also have  $V_4 = (1+0.021)^2 V_2$ . Now we have:

$$\begin{aligned} V_4 &= (1+0.021)^2 V_2 \iff V_4 = (1+0.021)^2 (10\,424.41 - x) \\ &\iff 10\,424.41 - x = \frac{V_4}{(1+0.021)^2} \\ &\iff x = 10\,424.41 - \frac{V_4}{(1+0.021)^2} \\ &\iff x = 10\,424.41 - \frac{2\,187.36}{(1+0.021)^2} \approx 8\,326.10 \end{aligned}$$

The amount withdrawn at the end of two years was therefore roughly 8 326.10 euros  $\rightarrow$  **Answer (d)**.  $\square$

### 3.4.2 Financial fixed assets and marketable securities

Banking partners are not, however, the only establishments with which one can invest money. The firm itself can lend money (or goods) and derive benefits from it: as Collain et al. (2021) indicate, **financial fixed assets**, recorded in accounts 26 and 27 of the French general chart of accounts (PGC), comprise long-term financial investments, equity stakes in other companies and long-term claims such as guarantee deposits paid and loans. More precisely, financial fixed assets group together investments in securities whose maturity is greater than one year, such as **shares**, which may be kept for as long as the company which issued them exists (their maturity is not determined), and **bonds** issued by a firm, whose maturity is more than one year, as Collain et al. (2021) specify. More generally, firms have a vast range of products for investing their surplus funds: as Charreaux (2014c) indicates, classic financial products, in addition to the remuneration of overnight deposits when the balance is positive, mainly include term deposits and cash bonds. The interest rate associated with these products is freely negotiated, generally computed as a function of the money market rate from which a margin is subtracted. A **term deposit** imposes a commitment for a predefined duration, with penalties in the event of early withdrawal. **Cash bonds**, which may be at a fixed rate or at a progressive rate — where the remuneration, freely set, increases with the duration of the investment — are mostly issued by banking establishments. Their duration generally varies between one month and five years, and the repayment conditions are fixed at the time of issue, as Charreaux (2014c) specifies.

#### Focus...



**UCITS.** As Charreaux (2014c) indicates, UCITS (*organismes de placement collectif en valeurs mobilières*) are classified by the AMF into six main categories, according to risk exposure:

- **Equity UCITS:** these funds permanently invest at least 60 % of their assets in shares. They are subdivided into several categories: French equities, equities of euro area countries, equities of European Union countries and international equities.
- **Bond and other debt security UCITS:** these funds invest in interest rate instruments. They must indicate their level of risk in terms of sensitivity to variations in interest rates. These funds are exposed to credit risk and are classified into euro-denominated UCITS and international UCITS.
- **Money market UCITS:** these funds place their portfolio in money market products. Their sensitivity to interest rates is low, between 0 and 0.5 (that is to say that, if the interest rate varies by one percentage point, the value of the portfolio must vary by at most half a percentage point). They come in two types: those denominated in euros and those with an international vocation.
- **Alternative fund UCITS:** these funds invest more than 10 % of their assets in other UCITS adopting alternative strategies, that is to say strategies independent of the evolution of market indices. These funds generally offer reduced liquidity.
- **Formula fund UCITS:** the performance of these funds depends on stock market indicators or on other values (such as prices), and it is computed according to a predefined mathematical formula. They may offer partial or total protection of capital, depending on the conditions.

- **Diversified UCITS:** this category groups together the funds which do not correspond to the other categories mentioned. They present varied and diversified investment strategies.

Another way for the firm to make its cash grow in the short term is to acquire **marketable securities** (VMP), namely securities acquired by the firm with a view to realising a gain at short maturity (less than one year); marketable securities therefore form part of current assets. Marketable securities may notably be acquired by a firm through undertakings for collective investment in transferable securities (**UCITS**). Among UCITS are notably open-ended investment companies (SICAV) and mutual funds (FCP). As the AMF (2018a) specifies, the **SICAV** is a public limited company with variable capital which issues shares as and when subscription requests come in (any investor who buys shares therefore becomes a shareholder and may express a view on the management of the company at general meetings and/or stand for election to the board of directors), whereas the **FCP** is a co-ownership of transferable securities which issues units (the unit holder therefore has none of the rights conferred on a shareholder, but the management company which runs the fund acts in the name of the holders and in their exclusive interest).

### Key Takeaways



The firm has two categories of financial investment for making its cash grow: financial fixed assets (long-term investments, maturity greater than one year) group together shares, bonds, loans and long-term claims, while marketable securities (VMP, less than one year) constitute current assets and may be acquired via undertakings for collective investment in transferable securities (UCITS) [Collain et al. (2021)]. Ordinary cash products, for their part, include term deposits and cash bonds at fixed or progressive rates. UCITS take two forms: SICAVs (public limited companies with variable capital, issuing shares conferring shareholder rights) and FCPs (co-ownerships of transferable securities, issuing units without voting rights). The Autorité des marchés financiers classifies UCITS into six categories: equities (at least 60 % of the portfolio), bonds and claims, money market (interest rate sensitivity between 0 and 0.5), alternative funds (at least 10 % in alternative strategies), formula funds (performance computed mathematically) and diversified funds [Charreaux (2014c)].

#### 3.4.3 Short-term financial planning

Short-term financial planning, also designated by the generic term **working capital management**, constitutes a central field of corporate finance which aims to optimise the management of current assets and current liabilities so as to guarantee sufficient liquidity to meet short-term obligations while maximising profitability. As Pike & Neale (2012) explain, the often contradictory objectives of working capital management consist on the one hand in increasing the firm's profitability by minimising the resources tied up in the operating cycle, and on the other hand in ensuring that the firm has enough liquidity to continue its operations and honour its short-dated debts. The heart of short-term financial planning rests on an understanding of the **operating cycle**, which measures the number of days elapsing between the cash outflow made for the purchase of raw materials or stocks and the collection of sales revenue. According to Deloof (2003), the **cash conversion cycle** (CCC) is computed according to the formula  $CCC = DIO + DSO - DPO$ , where **DIO** denotes the days of stock to be sold off (*days inventory outstanding*, or the inventory turnover period of table ?? page ??), **DSO** represents the days taken to recover trade receivables (*days sales outstanding*, or the trade receivables turnover period of table ?? page ??), and **DPO** measures the days taken to pay suppliers (*days payables outstanding*, or the trade payables turnover period of table ?? page ??). Indeed, inasmuch as purchases create debts which are eventually extinguished through payments to suppliers, products or services in the course of preparation appear in stock — until the day the sale takes place and the stocks disappear — and the sales invoices issued to customers are settled at the time of collection, the economic cycle described *supra* makes it possible to measure with the CCC the number of days needed to transform a firm's flows into bank liquidity.

The data compiled by Efficio Consulting (2024) reveal that in 2023, the cash conversion cycle in Europe stood at 64 days on average, which represents an increase of four days on 2022, mainly on account of an increase in inventory days of 4.5 days, reflecting strategic decisions to build up precautionary stocks so as to protect firms against persistent geopolitical tensions. The data published by Atradius (2025) further indicate that average payment periods in Western Europe remain stable between 31 and 60 days from invoicing, while the average period for paying suppliers stands at 70 days according to Hackett Group (2025), reflecting a strategy on the part of many firms aimed at preserving their cash by delaying their payments to suppliers. This structure of the operating cycle varies considerably according to sector of activity and to the strategic profile of firms: a firm presenting a short operating cycle has more flexibility in its cash management since inflows of funds occur rapidly after outflows, whereas a firm with a long cycle must finance the gap between these flows by having recourse to external sources of financing.

The composition of the cash conversion cycle then offers distinct levers for optimising the cash position: **reducing inventory days** by improving stock turnover, accelerating the **recovery of trade receivables** by implementing a rigorous invoicing and reminder policy, or again **negotiating with suppliers** to extend payment periods.

Indeed, the building up and **management of stocks** constitute a central element of short-term financial planning, since, as Charreaux (2014c) explains, firms hold stocks for several distinct economic motives: minimising the cost of orders by reducing their number and benefiting from discounts granted by suppliers for large orders; avoiding a breakdown in the manufacturing process which would entail a loss of productivity or a halt in activity; protecting themselves against fluctuations in the prices of raw materials by building up speculative stocks when a price rise is anticipated; and finally rationing production by dissociating the production process from the level of sales so as to minimise the costs linked to periods of under-activity or over-activity. The profitability of investments in stocks must be assessed differentially by confronting the gains procured — losses avoided thanks to the absence of stockouts, cost reductions resulting from the regularity of production — with the costs entailed, notably the financial costs of storage, insurance, deterioration and obsolescence, after excluding the financing costs which must be taken into account at the level of the discount rate.

Likewise, the **management of trade credit** constitutes an investment decision in the financial sense of the term, since extending the credit periods granted to customers ties up cash resources in exchange for an increase in turnover. As Charreaux (2014c) indicates, the desirability of lengthening trade credit periods must be assessed by comparing the additional flows of sales and margins expected with the additional costs entailed, including notably a provision for the increased risk of non-recovery generated by a longer period. The data compiled by Upflow (2024) indicate that 55 % of commercial invoices issued in the United States remain unpaid beyond their due date, a situation which is even more serious in the United Kingdom where this percentage reaches 58 %, even though late payment by customers is responsible for 25 % of business failures in Europe. An often neglected but critical element of trade credit management remains the *float*, that is to say the delay between the payment instruction given by the customer and the effective crediting of the payment to the seller's account, a delay which depends notably on the value dates applied by banks. As Charreaux (2014c) indicates, value dates constitute the gap between the date on which an operation is recorded in the accounts and the date used by the bank to perform its interest computations, thereby creating an additional financial cost which may be substantial over a year. The effective management of the float therefore implies a rigorous administrative organisation of invoicing and collection, as well as active negotiation with banks so as to minimise these time lags.

### Focus...



**Cash management and the cost of overdrafts.** A firm's cash represents the whole of the liquid funds available at a given date and constitutes the minimum level of cash necessary to cope with the gaps between inflows and outflows. As ACT (2021) explain, day-to-day cash management rests on the production of detailed cash flow forecasts which make it possible to anticipate short-term financing needs and to decide on the instruments to be used in order to fill temporary cash deficits. When a firm faces a temporary cash deficit and does not have sufficient own resources, it may have recourse to a **bank overdraft**, which constitutes a very short-term credit granted by a bank allowing the current account to become temporarily overdrawn. But the cost of a bank overdraft generally turns out to be very high compared with other sources of financing, which makes its use expensive in the event of a prolonged need. Thus, for the customers of the bank Santander (2026), the cost of an arranged overdraft stands at around 28 % in overall effective annual rate for standard customer accounts in the United Kingdom. In France, according to the data compiled by CCSF (2025), the interest rate for a bank overdraft ranges between 7 % (Boursobank) and 20.04 % (BRED) in annual rate. In the United States, according to the report of the Consumer Federation of America (2025), banks collected roughly five billion dollars of unauthorised overdraft fees in 2024, with an average cost of 35 dollars per overrun, which generates a substantial burden for households and small firms that do not have sophisticated cash management. The **real economic cost** of an overdraft nevertheless far exceeds the nominal interest rate, since to it must be added the set-up costs — arrangement fees from 8 euros (Crédit mutuel de Bretagne) to a minimum of 20 euros (Crédit agricole Nord-Est) according to CCSF (2025) — the overrun commissions and the opportunity costs resulting from the obligation to maintain higher minimum balances in order to avoid going overdrawn.

The annual **treasury plan**, produced on a monthly basis, then constitutes the fundamental tool of short-term financial planning, making it possible to visualise month by month the balances or imbalances between stable cash resources and uses and to determine the bank facilities needed. According to Charreaux (2014c), the construction of a treasury plan proceeds in two stages: first the assessment of the annual profile of cash needs independently of any consideration of short-term financing, then the putting in place of the appropriate short-term financing while taking account of their differential costs. Indeed, effective cash management founded on reliable forecasts makes it possible to reduce substantially the recourse to overdrafts by optimising the payment periods of customers and

suppliers, as well as by managing the levels of stocks and receivables in such a way as to minimise the working capital financing requirement. In addition, **day-to-day cash management** requires a shorter forecasting horizon, generally of four to five months organised by ten-day period and revised on a rolling basis, permitting fine adjustments of the trade-offs between different modes of financing and forms of investment so as to minimise the overall cost of financing or to maximise the return on investments while respecting the liquidity and flexibility constraints imposed by the variability of flows. According to Karnik (2025), modern cash management software now offers real-time visibility over the consolidated cash position across all of the firm's entities, which makes it possible to detect cash deficits well in advance and to put in place optimal financing solutions before unauthorised overdrafts become necessary.

### Exercise



**Comparison of forms of cash financing.** An industrial firm, the company Alpha, faces a one-off cash requirement of 500 000 euros for a period of 60 days. It has a portfolio of trade receivables of 500 000 euros (bills at 60 days, well spread across several solvent customers) and a good bank *rating*. Its bank offers it several modes of short-term financing:

- **Solution A: discounting of bills.** The company has all its bills discounted for a face value of 500 000 euros over 60 days, at the nominal annual rate of 5 % (360-day basis). The bank charges in addition a discounting commission of 0.10 % of the face value. Otherwise put, the bank immediately advances the funds to Alpha before maturity, but it withholds discount interest and in addition levies a discounting commission.
- **Solution B: Daily credit.** The company assigns its receivables to the bank, by means of a Dailly schedule (law of 2 January 1981, amended in 1984), for a face value of 500 000 euros: this is a document by which it legally transfers these receivables to the bank. The bank finances 90 % of the assigned receivables (the remainder constituting a safety margin, in case some customers should default), at the nominal annual rate of 4.5 % on the amount paid out (360-day basis). A service commission of 0.20 % is levied on the face value assigned.
- **Solution C: cash credit by mobilisation of a note.** The company issues a treasury note (a written promise to repay a sum on a given date) with a face value of 500 000 euros, maturing in 60 days. The bank mobilises it (that is to say it accepts the note and, in exchange, immediately advances the funds to the firm) at the nominal annual rate of 4.8 % (360-day basis), with interest deducted in advance, and charges flat-rate arrangement fees of 400 euros.
- **Solution D: account overdraft (account advance).** The company obtains an overdraft authorisation with a ceiling of 600 000 euros. It plans to use on average 500 000 euros of overdraft over 60 days (with a maximum overdraft of 600 000 euros during a period of 10 days). The nominal annual rate of the overdraft is 6 % (360-day basis). The bank charges an annual commitment commission of 0.25 % on the authorised ceiling, *prorata temporis* for the duration of use, and a largest-overdraft commission (CPFD) of 0.05 % over the period.
  - (i) For each of the four solutions, compute the total cost of the financing over the 60 days and the equivalent annual cost (effective annual rate). Rank the four solutions from the cheapest to the most expensive.
  - (ii) Beyond the criterion of cost alone, discuss qualitatively the advantages and disadvantages of the different forms of cash financing for Alpha (flexibility, security, suitability to the need, risks, administrative burden).

**Solution.** (i) For each solution we compute the total cost over 60 days, then the effective rate over 60 days, and finally a corresponding annual rate assuming the proportionality of rates (common practice for the short term), that is  $i_{\text{annual}} \approx i_{60 \text{ days}} \times \frac{360}{60} = 6 i_{60 \text{ days}}$ . We denote by  $N = 500\,000$  euros the face value and by  $T = 60$  days the duration.

- **Solution A: discounting of bills.** The bank discounts bills with a face value of  $N = 500\,000$  euros at 60 days at the annual rate of 5 % (360-day basis) and levies in addition a commission of 0.10 % of the face value. The discount interest  $I_A$  is deducted in advance and is computed on the face value and on the time remaining until maturity:  $I_A = N \times 0.05 \times \frac{T}{360} = 500\,000 \times 0.05 \times \frac{60}{360} = 500\,000 \times 0.05 \times \frac{1}{6} = 500\,000 \times 0.008\bar{3} \approx 4\,166.67$  euros. The discounting commission  $C_A$  is proportional to the face value:  $C_A = 0.001 \times N = 0.001 \times 500\,000 = 500$  euros. The company does not receive the face value  $N$ , but a net resource  $R_A$  equal to the face value less the interest and the commission:  $R_A = N - I_A - C_A = 500\,000 - 4\,166.67 -$

500 = 495333.33 euros. The total cost of the operation over 60 days is the sum of the interest and the commission:  $\text{Cost}_A = I_A + C_A = 4166.67 + 500 = 4666.67$  euros. The effective rate over 60 days is then defined as the ratio between the cost and the resource effectively made available:  $i_{A,60} = \frac{\text{Cost}_A}{R_A} = \frac{4666.67}{495333.33} \approx 0.0094$ , that is 0.94 % over 60 days. The equivalent annual rate (by proportionality) is therefore:  $i_{A,\text{annual}} \approx 6 i_{A,60} \approx 6 \times 0.0094 \approx 0.056$ , that is roughly 5.65 % per year.

- **Solution B: Daily credit.** The company assigns receivables with a face value of 500 000 euros, but the bank finances only 90 % of this face value, the remainder constituting a safety margin. The amount effectively advanced by the bank is therefore  $M_B^{\text{fin}} = 0.90 \times 500\,000 = 450\,000$  euros. Interest is computed on this financed amount at the annual rate of 4.5 % for 60 days:  $I_B = M_B^{\text{fin}} \times 0.045 \times \frac{T}{360} = 450\,000 \times 0.045 \times \frac{60}{360} = 450\,000 \times 0.045 \times \frac{1}{6} = 450\,000 \times 0.0075 = 3\,375$  euros. The bank levies in addition a service commission of 0.20 % on the face value assigned (and not on the amount effectively financed):  $C_B = 0.002 \times 500\,000 = 1\,000$  euros. The net resource effectively received by Alpha is therefore the amount financed less the interest and the commission:  $R_B = M_B^{\text{fin}} - I_B - C_B = 450\,000 - 3\,375 - 1\,000 = 445\,625$  euros. The total cost over 60 days is:  $\text{Cost}_B = I_B + C_B = 3\,375 + 1\,000 = 4\,375$  euros. The effective rate over 60 days is:  $i_{B,60} = \frac{\text{Cost}_B}{R_B} = \frac{4\,375}{445\,625} \approx 0.0098$ , that is 0.98 % over 60 days. The equivalent annual rate is:  $i_{B,\text{annual}} \approx 6 i_{B,60} \approx 6 \times 0.0098 \approx 0.0588$ , that is roughly 5.88 % per year.
- **Solution C: cash credit by mobilisation of a note.** The company issues a treasury note with a face value of  $N = 500\,000$  euros, at 60 days, which the bank mobilises at the rate of 4.8 % per year, with interest deducted in advance. The interest deducted in advance  $I_C$  is therefore computed on the face value for 60 days:  $I_C = N \times 0.048 \times \frac{T}{360} = 500\,000 \times 0.048 \times \frac{60}{360} = 500\,000 \times 0.048 \times \frac{1}{6} = 500\,000 \times 0.008 = 4\,000$  euros. The bank charges in addition flat-rate arrangement fees of 400 euros, independent of the amount of the interest. The net resource effectively made available is the face value less the interest deducted in advance and the fees:  $R_C = N - I_C - 400 = 500\,000 - 4\,000 - 400 = 495\,600$  euros. The total cost of the operation over 60 days is the sum of the interest and the fees:  $\text{Cost}_C = I_C + 400 = 4\,000 + 400 = 4\,400$  euros. The effective rate over 60 days is therefore:  $i_{C,60} = \frac{\text{Cost}_C}{R_C} = \frac{4\,400}{495\,600} \approx 0.0089$ , that is 0.89 % over 60 days. The equivalent annual rate is:  $i_{C,\text{annual}} \approx 6 i_{C,60} \approx 6 \times 0.0089 \approx 0.053$ , that is roughly 5.3 % per year.
- **Solution D: account overdraft (account advance).** The company has an overdraft authorisation with a ceiling of 600 000 euros and plans to use on average 500 000 euros over 60 days, with a largest overdraft of 600 000 euros over 10 days. The interest on the overdraft (agios) is computed on the average debit balance. As a first approximation, we use the announced average of 500 000 euros as the basis of computation, which means that the variations around this average (including the peak at 600 000 euros) offset one another over the period. The interest  $I_D$  due over 60 days at the annual rate of 6 % is therefore:  $I_D = 500\,000 \times 0.06 \times \frac{T}{360} = 500\,000 \times 0.06 \times \frac{60}{360} = 500\,000 \times 0.06 \times \frac{1}{6} = 500\,000 \times 0.01 = 5\,000$  euros. The commitment commission is computed on the authorised ceiling of 600 000 euros, at the annual rate of 0.25 %, *prorata temporis* 60 days out of 360 days. It is therefore worth:  $C_{\text{comm}} = 600\,000 \times 0.0025 \times \frac{60}{360} = 600\,000 \times 0.0025 \times \frac{1}{6} = 600\,000 \times 0.000416\bar{6} \approx 250$  euros. The largest-overdraft commission (CPFD) is expressed at 0.05 % over the period, and bears, by definition, on the maximum amount effectively reached, that is 600 000 euros during the 10 days of largest overdraft. It is therefore equal to:  $C_{\text{PFD}} = 0.0005 \times 600\,000 = 300$  euros. The total cost of the commissions is the sum of the commitment commission and the CPFD:  $C_D = C_{\text{comm}} + C_{\text{PFD}} = 250 + 300 = 550$  euros. The total cost of the operation over 60 days, interest and commissions combined, is then:  $\text{Cost}_D = I_D + C_D = 5\,000 + 550 = 5\,550$  euros. Since this is an overdraft, the gross resource of 500 000 euros is made available continuously, and the interest and commissions are paid only at the end of the period (by debit of the account). The average net resource may therefore be identified with 500 000 euros over the period, which leads to an effective rate over 60 days:  $i_{D,60} = \frac{\text{Cost}_D}{500\,000} = \frac{5\,550}{500\,000} = 0.0111$ , that is 1.11 % over 60 days. The equivalent annual rate by proportionality is then:  $i_{D,\text{annual}} \approx 6 i_{D,60} \approx 6 \times 0.0111 \approx 0.0666$ , that is roughly 6.7 % per year.

We may now compare the four solutions using the equivalent annual rates obtained: solution C (mobilisation of a note) leads to a rate of roughly 5.3 % per year, solution A (discounting) to roughly 5.6 % per year, solution B (Daily) to roughly 5.9 % per year, and solution D (overdraft) to roughly 6.7 % per year. From a strictly financial point of view, for a one-off need well calibrated over 60 days, the mobilisation of a note (solution C) is the most advantageous, whereas the overdraft (solution D) is the most expensive form of financing.

- (ii) Beyond the criterion of cost alone, the qualitative characteristics of each form of financing must be discussed.

- Solution A (discounting of bills) is directly backed by trade receivables materialised in the form of bills: it makes it possible to transform these receivables into cash while offering a certain visibility over flows (the

maturities of the bills). It nevertheless remains relatively burdensome administratively (management of the bills, remittances to the bank) and its base is limited to the bills of exchange available; where the portfolio is thin or poorly spread, the firm may find itself under-financed or, on the contrary, over-mobilised.

- Solution B (Daily credit) considerably broadens the base for mobilisation: it makes it possible to assign, by means of a single schedule, a set of receivables of diverse nature (commercial or otherwise, existing or future) to the bank, which confers great flexibility in the management of the working capital requirement. In return, the bank finances only a fraction of the face value (90 % in the statement), which reduces the net resource for the firm, and the procedure has a more marked legal dimension (assignment of receivables) with specific commissions.
- Solution C (cash credit by mobilisation of a note) rests not on trade receivables but on the firm's own signature: the firm issues a treasury note of known face value and determined maturity, which the bank mobilises. This technique is well suited to a one-off need of clearly identified amount and duration: it is simple to monitor (one face value, one maturity, one rate), the headline rates are generally competitive, and the arrangement may operate on a *revolving* basis if the note is renewed regularly. By contrast, since the interest is deducted in advance and fixed fees are added to the cost, the net resource is slightly lower than the face value, and this form of financing presupposes a good *rating* for the firm.
- Finally, solution D (account overdraft) is the most flexible: the financing adjusts in real time to cash fluctuations, without its being necessary to set up a specific operation for each need; it is therefore particularly suited to the day-to-day management of an uncertain or irregular cash position. This flexibility nevertheless has a price: the overall cost (interest rate, commitment commission, CPFD, and so on) is generally higher than in the other forms, and the overdraft, especially when it is not formally confirmed, exposes the firm to a risk of unilateral withdrawal on the part of the bank, which may weaken its liquidity in the event of tension.

*In fine*, for the company Alpha, faced with a one-off need of clearly defined amount and duration (500 000 euros over 60 days), an arrangement of the mobilisation-of-a-note type (solution C) appears optimal from the point of view of cost while offering good readability. Solutions A and B are of interest for realising the value of a portfolio of trade receivables and for financing the working capital requirement more structurally. The overdraft (solution D), despite its high cost, retains a role as a tool for fine adjustment and as a safety net for absorbing peaks in need or cash contingencies, but it should not be the main instrument for financing a well anticipated one-off need. □

### Key Takeaways



Working capital management rests on the optimisation of the cash conversion cycle (CCC), measuring the number of days between the cash outflow for the purchase of stocks and the collection of sales, computed as the sum of the days of stock to be sold off and the days taken to recover trade receivables, from which are subtracted the days taken to pay suppliers [Deloof (2003)]. Three levers make it possible to optimise the cycle: reducing inventory days by improving stock turnover, accelerating the recovery of receivables by putting in place a rigorous invoicing policy, and negotiating the extension of supplier payment periods [Pike & Neale (2012)]. The management of stocks constitutes an investment decision confronting the gains (absence of stockouts, regularity of production) with the costs (financing, insurance, deterioration) [Charreaux (2014c)]. Likewise, the granting of trade credit ties up resources but must be assessed by comparing the additional flows with the costs of non-recovery. Moreover, the float, the delay between the payment instruction and the effective crediting of the payment to bank accounts on account of value dates, generates a financial cost which is often neglected. The annual treasury plan, drawn up on a monthly basis, then constitutes the fundamental tool for visualising balances and determining the bank facilities needed, complemented by day-to-day management over a rolling horizon of four to five months [Charreaux (2014c)].



# Finance and financial markets

Finance constitutes the nervous system of modern economies, ensuring the allocation of capital between economic agents and determining the financing conditions of investment, consumption and public spending. We shall examine the structural transformations of the world financial system since the 1980s, marked by the passage from a bank-based overdraft economy to a decompartmentalised **financial market economy**, then analyse the major crises revealing the systemic vulnerability of this financialised architecture, before focusing on the main types of financial assets traded on markets.

## 1 Towards financial globalisation

**Financial globalisation** designates the progressive integration of national capital markets into a borderless globalised financial system, characterised by the instantaneous mobility of capital flows between jurisdictions, the liberalisation of exchanges of currencies and securities, and the emergence of banking and investment networks of planetary scope. What are the structural transformations which permitted the emergence of this system?

### 1.1 A paradigm shift since the 1980s

Since the 1980s, the global financial system has swung from an overdraft economy centred on bank credit to an open and decompartmentalised financial market economy, a transformation summarised by the 3 Ds of Bourguinat (1986): **deregulation, decompartmentalisation and disintermediation**. Now this transition, far from being neutral, has deepened asymmetries: while financial globalisation claims to allocate capital optimally worldwide, capital flows are massively concentrated within the **Triad**: the United States, Western Europe, South-East Asia.

#### 1.1.1 From an overdraft economy to a financial market economy

The **overdraft economy**, as it dominated the financial landscape of the preceding decades, rests essentially on recourse to bank credit to finance both firms and households. According to the typology proposed by Hicks (1974), this system is characterised by a strong dependence of firms on banks, the latter playing a central role in mobilising savings to finance economic activity; in this configuration, commercial banks borrow from the central bank to refinance the loans they grant, in a mechanism which creates a kind of cycle of **institutional dependence between the central bank and commercial banks**. However, this system rapidly showed its limits: the growing inflation of the 1970s, exacerbated by the oil shocks, weighed heavily on the viability of this model, since inflation erodes the real value of debts and makes credit financing ever more costly for banks. Indeed, as Castel (2020) recalls, in the United States inflation did not exceed 2 % to 3 % in the 1960s and up to 1972, but reached 11.1 % in 1974 and 13.6 % in 1980; in France, inflation remained rather high at the beginning of the 1960s (between 4 % and 5 %), falling back to 2.5 % in 1964, then rising again from 1968 onwards to reach 6.1 % in 1972, 13.6 % in 1974 and in 1980 and 13.4 % in 1981. More generally, Castel (2020) emphasises that from 1974 onwards, the first full year for the effects of the first oil shock (beginning in October 1973), inflation reached a peak of 14 % for the OECD countries as a whole; with the second oil shock of 1979, a new inflationary peak was reached in 1980 with more than 12 % for the OECD countries as a whole. Beyond inflation, the overdraft economy suffers from a structural problem which Castel (2020) brings to light: Keynesian-inspired policies, advocating an active role for the state in financing the economy, are often associated with recourse to low or even subsidised interest rates; the main instrument of monetary policy then translates above all into a direct control of the volume of credit via **credit controls**; now these controls limit the capacity of the monetary authorities to influence economic activity through interest rates and make the economy less adaptable to external shocks. As a consequence of this state interventionism, the overdraft economy was distinguished by a



strong **fragmentation of financial markets**, with specialised circuits for different types of financing such as housing or exports, benefiting from specific support policies, as Renversez (2008) explains. Now this fragmentation limits the fluidity of capital and prevents an optimal allocation of financial resources: administered interest rates do not then necessarily reflect market conditions, which can lead to economic distortions.

In this context, the move to a **financial market economy** was the result of an organised and deliberate transformation, initiated at the beginning of the 1980s in order to adapt the financial system to the constraints of an open economy. The objective was to put in place a decompartmentalised capital market of sufficient size to face international capital movements. This transition formed part of an approach aimed at facilitating the access of firms and states to a broadened capital market, where the interest rate would become a genuine market price. Bourguinat (1986) then synthesised this evolution in three dimensions, the **3 Ds**: deregulation, decompartmentalisation and disintermediation. **Deregulation** implied a lightening of the rules and standards framing the functioning of financial markets, thereby favouring greater freedom in transactions and facilitating access to capital. **Decompartmentalisation**, for its part, consisted in eliminating the barriers between the different segments of the financial market, allowing banks to carry on both deposit and investment activities, and thus to diversify their business. **Disintermediation**, finally, opened up the possibility for firms, households and states to finance themselves directly on the markets, without systematically going through traditional financial intermediaries; this entails a reallocation of risks and resources on an international scale. Henrat-Lemée (2021) even adds a fourth dimension to the typology of Bourguinat (1986): the **dematerialisation of transactions** thanks to new technologies. This transformation finally allowed firms to strengthen their equity and to reduce their dependence on bank loans, while offering investors increased opportunities to diversify their portfolios. More generally, Henrat-Lemée (2021), drawing on the work of Sharpe (1964), considers that the liberalisation of financial markets leads to an **efficient allocation of resources** within the world economy; on the one hand, world savings are directed towards the investment projects and towards the regions of the world which guarantee the highest return for a given level of risk; on the other hand, financial globalisation makes it possible to benefit from a lower interest rate than in a situation of autarky, which increases investment and offers savers a better diversification of their asset portfolio thanks to the breadth of investment possibilities. With financial globalisation, even states therefore benefit from new financing opportunities: indeed, according to Bénassy-Quéré (2015), financial globalisation offers countries the possibility of consuming beyond their production by making the financing of trade deficits easier, which relaxes their **external financing constraint**; a country may therefore temporarily disconnect its consumption and its investment from its macroeconomic production, by increasing its imports without the necessity of selling as many goods and services in return, thanks to incoming foreign financial flows, that is to say by borrowing from other countries.

#### Focus...



**France's conversion to financial liberalism.** France's conversion to financial liberalism from the end of 1983 onwards profoundly modified the financing circuits of the economy and the nature of the relations between the state, the banking sector and the capital markets. As Castel (2020) analyses, this transformation, occurring only two years after the election of President Mitterrand on 10 May 1981, represents a reversal relative to the initial political programme, which was articulated around 110 proposals advocating the development of the public sector, the nationalisation of the banking sector, the control of credit and a substantial increase in purchasing power. The economic difficulties encountered during the first two years — rising unemployment, a growing balance of payments deficit and the **recurrent weakness of the franc** marked by three devaluations between October 1981 and March 1983 — imposed on the French authorities a reassessment of their strategy. However, as Castel (2020) emphasises, these cyclical data alone do not explain the radicalism of the change of course: other countries confronted with the same difficulties maintained their initially planned policies. From the end of March 1983, when the French government decided that the country would remain anchored to the **European Monetary System** (EMS), the public authorities embarked on a series of structural financial reforms orienting the French economy towards financial liberalism, progressively transforming an economy of bank debt into an economy financed more heavily by capital markets. This reorientation was not the object of a conscious and deliberate strategy mapped out in advance, but rather the product of successive decisions taken in response to **liquidity and interest rate crises** engendered by the non-monetary financing of growing budget deficits.

The international macroeconomic context largely explains this bifurcation towards financial liberalism. As Castel (2020) documents, two categories of reason underlie this conversion: the international reasons common to all OECD member countries, and the reasons specific to France. On the international level, the deterioration of Western economies from 1973 onwards, accentuated by the oil shock of October of the same year which multiplied the price of oil by four in less than six months, generated a persistent **stagflation** combining inflation and unemployment at levels unprecedented since the post-war period. Thus, according to Castel (2020), unemployment in the OECD

countries recorded a brutal increase, going from 3 % on average during the 1960s to more than 4 % in 1974-1979, with a rising trend during the 1980s. In parallel with this cyclical deterioration, the **growing ineffectiveness of Keynesian-inspired economic policies** became manifest and gave ground to monetarist approaches emphasising the central role of financial markets, the reduction of state intervention and the strict control of the money supply. This intellectual mutation was emblematically translated by the arrival in power of Margaret Thatcher in the United Kingdom on 4 May 1979 and by the rise of monetarist thinking in the United States, where Paul Volcker at the head of the Federal Reserve raised interest rates to nearly 20 % in 1981 in order to fight galloping inflation. As Castel (2020) recalls, this American policy engendered an **appreciation of the dollar**, going from 4.3 francs in 1980 to 10 francs in February 1985, thereby creating considerable pressure on the French franc and making it unsustainable to maintain an expansionary economic policy in France.

The reasons specific to France reinforced this orientation towards financial liberalism, notably through the unintended consequences of the decision to finance public deficits by non-monetary resources. As Castel (2020) explains, the French government had promised to bring inflation down sharply (it was close to 13 % at the time) and decided to that end that the public deficit would be financed exclusively by **bond resources** so as to avoid any drift in money creation. This decision produced two major consequences: first, the first government bonds had to be issued at fixed rates above inflation, the first public bond issue of eight billion francs in September 1981 having been offered at a rate of 16.75 %, generating a considerable additional cost for the budget once inflation progressively decelerated; second, the state had to have massive recourse to SICAVs (open-ended investment companies) and mutual funds in order to channel savings towards the purchase of public debt, relegating banks to a reduced intermediation role. The success of these collective investment vehicles brought about an accumulation of capital gains in savers' portfolios for as long as interest rates were falling, but this situation became untenable from 1985 onwards when the need to defend the franc against the *deutsche Mark* demanded on the contrary a rise in rates. In order to manage this systemic risk, the state had progressively to create the instruments and markets making it possible to reduce exposure to interest rate risks: the **MATIF** (*marché à terme des instruments financiers*, the financial futures market), which became operational in 1986, and the **MONEP** (*marché des options négociables d'Europe*, the traded options market), created in 1987, thus reproduced the innovations of American finance. In parallel, by allowing large firms direct access to capital markets through the issuance of commercial paper from the end of 1985 onwards — the French equivalent of American *commercial paper* — the French government aimed to **reduce the cost of bank intermediation** and to compete directly with banks in financing their most important customers. The share of bank credit in the total financing of the economy thus declined from roughly 80 % during the 1970s to 63 % in 1985 and 55 % a decade later according to Castel (2020), progressively bringing France closer to the Anglo-Saxon model where capital markets play a preponderant role. Finally, the law of 23 December 1988 introducing **securitisation** in France, allowing banks to sell off their loans rather than keeping them to maturity, marked the culmination of this shift towards a growing financialisation of French financing circuits, reducing the mutualisation of credit risk in bank balance sheets and transferring more of that risk onto capital markets and institutional investors.

### Key Takeaways



The overdraft economy, dominant until the 1970s, rested on bank credit financing firms and households via an institutional dependence between the central bank and commercial banks, but came up against growing inflation, which erodes the viability of the model, against credit controls hampering control through interest rates, and against the fragmentation of markets limiting the optimal allocation of resources [Hicks (1974), Castel (2020)]. The move to a financial market economy, initiated at the beginning of the 1980s, took place along the three dimensions of the 3 Ds: deregulation (lightening of standards), decompartmentalisation (removal of barriers between segments), and disintermediation (direct financing on markets), to which was added technological dematerialisation, permitting an efficient allocation of world resources and a reduction of external financing constraints via international borrowing [Bourguinat (1986), Henrat-Lemée (2021), Bénassy-Quéré (2015)].

#### 1.1.2 Paradoxes and asymmetries of financial globalisation

Despite a certain consensus among economists on the merits of financial globalisation, debates persist concerning its redistributive effects, particularly as regards the asymmetries of financial power between the three poles of the **Triad** (the United States, Western Europe and Japan) and the rest of the world. As Kose et al. (2009) demonstrate, these divergences are not recent and already find their roots in the work of Keynes (1936), which identified financial instability as an inevitable consequence of herd behaviour and of the collective psychology of the economic agents operating on markets. The global financial crisis of 2008 revived these concerns, calling into question the supposed

efficiency of financial markets. According to Albert (1991), the Triad represents the heart of the global capitalist system, characterised by distinct models of **financial organisation**: the neo-American model founded on capital markets, the Rhenish model of the social market economy centred on bank intermediation, and the Japanese model of interlocking financial capitalism.

### Focus...



**The Japanese model of interlocking financial capitalism.** The Japanese model of financial capitalism, characterised by an architecture of close interlocking between banking institutions, large industrial firms and the public authorities, stands in fundamental opposition to the Anglo-Saxon model founded on the primacy of capital markets and the maximisation of shareholder value. Like the Rhenish capitalism analysed by Albert (1991), the Japanese model rests on a philosophy in which *stakeholders* — employees, suppliers, main banks, stable shareholders — enjoy an institutional recognition well anchored in corporate governance practices. But the Japanese model finds its specificity in the centrality of the **main bank system**, a system in which a main bank acts simultaneously as principal creditor, significant shareholder and strategic adviser for its affiliated firms, as Kang & Stulz (1997) explain. This structure of financial interlocking is itself set within broader economic groups called **keiretsu**, built around systematic cross-shareholdings between firms of the same group, including the central bank of the network. According to Scher (2001) in his analysis of the ownership structure of large listed companies in Japan at the end of the 1990s, stable cross-shareholdings — that is to say held over the long term under tacit arrangements of mutual non-interference — represented between 65 % and 70 % of the whole of the shares issued by listed companies, while the remaining 25 % to 35 % constituted shares freely traded on capital markets. These **cross-shareholding** arrangements aimed above all to insulate the management of firms against threats of hostile takeover, by constituting a majority of stable shareholders to whom company management could be accountable without fear of sudden dismissal. The preponderant bank financing system meant that Japanese companies depended largely on bank credit rather than on bond or share issues to finance themselves, the share of bank credit financing in total financing reaching roughly 80 % during the 1970s and 1980s, according to the data compiled by Ozkan (2010).

The historical origins of the *keiretsu* system go back to the pre-war structures called **zaibatsu**, family conglomerates grouping together dozens of industrial and commercial firms placed under the centralised control of a single family holding all the shareholdings. As Ozkan (2010) documents, after the Second World War and the American occupation of Japan (1945-1952), the American state, with a view to economic democratisation, imposed the formal dissolution of these family *zaibatsu* by forbidding families to retain control of their industrial empires. However, rather than disappearing, these structures reorganised themselves by transforming the main banks — formerly mere financing institutions — into genuine orchestrators of the economic groups renamed *keiretsu*. This institutional transformation reveals a remarkable continuity with the pre-war structures: whereas the *zaibatsu* was controlled by a single patriarchal family, the *keiretsu* now operates according to a collective model in which the main bank plays the role formerly devolved to the *zaibatsu* patriarch. The system consolidated progressively from the 1950s onwards and reached its classic form in the 1960s to 1980s, a period during which Japan experienced vigorous economic growth — the average annual growth rate of GDP reaching roughly 9 % between 1960 and 1973 according to the Banque mondiale (1994). The Japanese economic success of this period consolidated the conviction, shared by political decision-makers, bankers and academics, that the *keiretsu* system constituted an economic model superior to Anglo-Saxon capitalism, marked by instability, stock market speculation and mass redundancies in the event of a cyclical contraction.

The economic rationality of the *keiretsu* system rested on several mechanisms: first, financing by bank credit rather than by capital markets allowed a **fine-grained and continuous assessment of credit risk** by specialised institutions, thereby reducing the adverse selection problem which characterises decentralised financial markets; second, the stable cross-shareholding structure eliminated the **uncertainty linked to changes of control**, allowing management to adopt a long-term perspective in its investment decisions; third, the system of long-term financing by the main bank implied that the latter would continue to support its clients in periods of difficulty, reducing the **risk of a bank run** or of a *credit crunch* — this arrangement was formalised, according to Kang & Stulz (1997), by implicit clauses of mutual obligation to assist in the event of a crisis; fourth, **vertical integration** within the *keiretsu*, associated with cross-shareholding arrangements between suppliers and customers, permitted a reduction of transaction costs and an efficient coordination of investment across value chains. According to Kang & Stulz (1997), large Japanese firms affiliated to *keiretsu* did indeed present significantly lower failure rates than their non-affiliated counterparts, suggesting that the security procured by the support of the main bank constituted a real competitive advantage. Japanese corporate governance, focused on the interests of employees and internal stakeholders rather than on the maximisation of short-term shareholder return, likewise encouraged investment in human capital and vocational

training, which strengthened labour productivity in the long run and facilitated the adoption of new technologies by employees.

A first way of putting the virtues of financial globalisation into perspective is to qualify the real extent of this integration of financial markets on a world scale, by observing that the benefits of integration are disproportionately concentrated within the Triad. As Feldstein & Horioka (1980) establish, in a world of perfect capital mobility (neoclassical theory), domestic saving ought not to be correlated with national investment, since capital ought to move freely towards its most productive uses whatever the place where it is formed; however, the results of their study, based on a sample of sixteen OECD member countries over the period 1960-1974, show that nearly 90 % of domestic investment is financed by national saving, which indicates a **very limited mobility of capital**. While capital mobility has evolved since the oil crisis of 1974, a certain persistence of the **Feldstein-Horioka paradox** in fact reveals the dominance of the Triad: capital now circulates freely within this closed circle, but remains largely compartmentalised within the national economies of developing countries, thereby creating a world financial hierarchy, as Balaresque & Oster (2021) show. More precisely, according to Giannone & Lenza (2008), the saving-retention coefficient (which measures the fraction of a country's additional saving that remains invested domestically rather than exported to other countries) remains high in the 1970s but decreases from the 1980s onwards, which testifies to the progressive increase in capital mobility within the OECD countries; in the years 1990-2004, the saving-retention coefficient, once controlled for the global shocks affecting the world economy, becomes almost nil between OECD countries, thus indicating a **substantial financial integration within the Triad**.

Financial globalisation therefore constitutes above all an intra-Triad phenomenon rather than a genuine phenomenon of planetary redistribution of capital, even though, according to Lucas (1990), capital ought to be invested where it is most productive, that is to say in developing countries where the marginal productivity of capital is *a priori* higher on account of the relative scarcity of capital compared with labour. Paradoxically, according to Sassen (2018), the three financial megacities of the Triad — the '**world cities**' New York, London and Tokyo — appropriate the majority of international capital flows. Conversely, as Prasad et al. (2004) document, over the period 1970-2006, the few portfolio equity investments and the foreign direct investment destined for developing countries were disproportionately concentrated in a small group of well institutionalised emerging countries (47 nations), while the poorest countries, and those least well integrated into world financial markets, received negligible flows relative to their capital accumulation needs. The persistence of this **Lucas paradox** calls into question one of the great postulates of financial globalisation, namely that it would automatically favour the optimal allocation of resources across the whole globe and would allow developing countries to accelerate their economic growth and their technological catch-up by attracting substantial masses of foreign capital.

### Key Takeaways



The benefits of globalisation are disproportionately concentrated within the Triad (the United States, Europe, Japan), revealing asymmetries of financial power and calling into question the efficiency of markets, already questioned by Keynes (1936) on herd behaviour and revived by the 2008 crisis [Kose et al. (2009)]. The paradox of Feldstein & Horioka (1980) demonstrates that if domestic saving were freely mobile, it ought not to be correlated with national investment, yet 90 % of domestic investment remains financed by national saving (1960-1974), a sign of very limited mobility; although this mobility increased in the years 1980-2004 within the OECD (saving-retention coefficient close to zero), capital remains largely compartmentalised within the national economies of developing countries, creating a world financial hierarchy [Giannone & Lenza (2008), Balaresque & Oster (2021)]. Globalisation thus remains an intra-Triad phenomenon: three financial megacities (New York, London, Tokyo) concentrate the majority of international flows, while the poorest countries receive negligible flows, contradicting the theory of Lucas (1990) according to which capital ought to flow towards developing economies where its marginal productivity is highest.

#### 1.1.3 The link between finance and growth

In the age of globalisation, finance and growth are therefore concomitant. Goldsmith (1959) is one of the first to propose an empirical study bringing to light a positive and significant correlation between finance and growth, taking growth as the explained variable and financial development as the explanatory variable: the proposed test then consists in regressing a variable representative of the average growth of thirty-six countries over a long period (1860-1963) on a variable of financial development. Does there, however, exist a causal relation between these two phenomena? For Schumpeter (1934), finance favours growth: financial institutions are necessary to the technological innovation which underpins growth. By contrast, for Robinson (1952), the causality runs the other way: where growth leads, finance follows. As Couppey-Soubeyran (2012c) emphasises, estimating the direction of causality is complex:

first of all, there are several ways of apprehending and measuring financial development, which complicates the analysis of its link with economic growth; some indicators concentrate on the development of bank intermediation, by assessing for example the relative weight of deposits or of loans in relation to gross domestic product (GDP), whereas other approaches favour the examination of stock markets, taking account of market capitalisation or of share issues, always in relation to GDP. However, in order to establish a link between finance and growth, it is necessary to introduce into the estimates a set of control variables which make it possible to neutralise the influence of other explanatory factors, since economic growth is not conditioned solely by financial development: variables such as the country's initial level of development, the quality of the education system, or again the economic policies in place, must likewise be taken into account.

### Short-Answer Question



**Does finance produce externalities?** The notion of externality designates, in economic theory, a situation in which the action of one agent affects the welfare of other agents without there existing any monetary compensation via the price system. Applied to the financial sphere, this question interrogates the capacity of the financial sector to produce effects — positive or negative — which overflow the simple contractual relation between borrowers and lenders. According to the data of Eurostat (2024), the financial assets of financial corporations in the European Union represented in 2023 roughly 502 % of GDP, that is more than five times the annual wealth produced, which testifies to the structural weight of the sector. This expansion raises the question of whether finance merely reflects real economic activity, or whether it exerts an autonomous influence on growth, stability and the distribution of resources. As King & Levine (1993) established, financial development is positively correlated with long-run economic growth, suggesting the existence of positive externalities via the efficient allocation of capital and the financing of innovation. However, Jackson & Pernoud (2020) recall that contemporary financial crises likewise reveal massive negative externalities, notably through the mechanisms of contagion and *spillover* which propagate shocks beyond the institutions initially affected. The problem is therefore situated at the intersection of these two dimensions: does finance generate more positive externalities through its role of intermediation and allocation of capital, or do the negative externalities linked to systemic risk and to crises prevail in the social balance sheet? In order to answer this question, it is appropriate first to examine the positive externalities that finance produces through its functions of intermediation, risk mutualisation and financing of growth, before showing that these benefits are counterbalanced by structural negative externalities linked to systemic risk and to the social costs of financial crises.

Finance produces substantial positive externalities by performing intermediation functions which go beyond the framework of individual transactions and benefit the economy as a whole. King & Levine (1993) demonstrate, on the basis of a sample of 80 countries over the period 1960–1989, that the initial level of financial development (measured by private credit in relation to GDP, the share of bank assets or the liquidity of the financial system) significantly predicts future economic growth, the accumulation of physical capital and the improvement of allocative efficiency. King & Levine (1993) show notably that an increase of one standard deviation in the private credit indicator is associated with an increase of 0.8 percentage point in the annual growth rate of GDP per capita, which constitutes a considerable externality in the long run. This result is explained by the fact that financial intermediaries, by mobilising dispersed savings, by assessing investment projects and by monitoring borrowers, reduce information asymmetries and permit the financing of innovative projects which, without this intermediation, would have remained unexploited. As Levine (2004) emphasises, these financial services generate positive externalities since they increase the overall productivity of the economy beyond the private benefits alone captured by financial institutions: a developed banking system facilitates the entry of new firms, improves the selection of projects and stimulates competition, which benefits economic agents as a whole. Moreover, finance plays an essential role in the mutualisation of idiosyncratic risks, by allowing households and firms to smooth their consumption and their investment in the face of sectoral or individual shocks. The data of the Banque mondiale (2013) indicate that in countries where private credit represents more than 100 % of GDP, the volatility of household consumption is on average 30 % lower than in less financialised countries, which attests to a better absorption of shocks thanks to credit and insurance markets.

While these positive externalities are well documented, they are largely counterbalanced by the negative externalities that finance generates, notably via systemic risk and the considerable social costs of financial crises. As Jackson & Pernoud (2020) summarise, the interconnections between financial institutions create channels of contagion which permit the rapid propagation of liquidity or solvency shocks from one institution to the whole system. This phenomenon of *spillover* constitutes a classic negative externality which Chan-Lau (2014) analyses: when a bank adopts an excessive level of risk or suffers losses, it increases the probability of failure of other institutions with which it is connected, without fully internalising this social cost. Subsequently, the idiosyncratic shocks affecting systemic

institutions propagate asymmetrically: a negative shock to Citigroup or Morgan Stanley significantly increases the risk of distress of European banks, whereas the reverse effect is far more limited, which testifies to a one-way network externality. Moreover, financial crises engender massive budgetary and macroeconomic costs which are largely borne by taxpayers and by the economy as a whole, rather than by financial actors themselves. According to the database of Laeven & Valencia (2020) on 151 systemic banking crises observed between 1970 and 2017, the median net budgetary cost of a crisis reaches 10 % of GDP, while the cumulative *output loss* exceeds on average 25 % of trend GDP. By way of illustration, according to the FMI (2009), the financial crisis of 2008–2009 entailed a direct budgetary cost estimated at more than 5 % of GDP in the euro area, to which must be added the losses of growth and the rise in unemployment which affected millions of people. Finally, Jackson & Pernoud (2020) emphasise that the very structure of financial markets favours excessive risk-taking and the accumulation of leverage in periods of expansion, creating a procyclical dynamic which amplifies macroeconomic fluctuations: these behaviours produce a negative externality since each institution, in seeking to maximise its individual profit, contributes to the accumulation of aggregate risks which materialise in synchronised fashion at the time of downturns.

Finance therefore undeniably produces externalities, both positive and negative, whose respective magnitude conditions the assessment of the overall social balance sheet of the financial sector. The positive externalities attest to the essential role of financial intermediation in the efficient allocation of capital, the financing of innovation and the mutualisation of idiosyncratic risks, all functions which support long-run growth. However, these benefits cannot obscure the structural negative externalities that the financial sector generates, notably through the mechanisms of contagion and systemic risk, and the considerable macroeconomic and budgetary costs of crises. The fact that European financial assets represent more than 500 % of GDP in 2023 signals both the functional importance of the sector and the potential magnitude of the negative externalities in the event of a systemic shock. The challenge for the regulatory authorities is therefore to devise a prudential framework which preserves the positive externalities of finance while internalising the negative externalities, notably via countercyclical capital requirements, contributions to resolution mechanisms and strengthened surveillance of systemic interconnections.

From the point of view of econometric method, even estimating the magnitude of the link between growth and finance is complex. Couppey-Soubeyran (2012c) thus emphasises the existence of an **endogeneity problem** in this type of analysis: growth, which one attempts to explain by financial development, may in return influence financial development; even the control variables, introduced in order better to isolate the effect of financial development on growth, may themselves also be influenced, directly or indirectly, by finance. In order to correct this endogeneity bias, the most common method consists in ‘instrumenting’ financial development, that is to say in using a variable which determines financial development without itself being affected by growth; an example often cited in the literature is the use of variables representative of the legal framework (see the focus *infra*). Another methodological issue to consider, brought out by Couppey-Soubeyran (2012c), is that of empirical studies carried out on a cross-country basis, that is to say based on several countries: these studies often rest on average data by country over a long period; otherwise put, they seek to explain the average growth of a given period by the average level of financial development over that same period. Now, by aggregating the data in this way, one loses sight of the temporal dynamics of growth, which may distort the analysis of the links between finance and growth, which are often long-period phenomena, as Couppey-Soubeyran (2012c) emphasises: this is why some studies favour **time series** and **panel data**, which make it possible to examine the fluctuations and interactions between finance and growth over longer periods.

#### Focus...



**The impact of law and culture on financial systems.** The research initiated by Porta et al. (1998) brought to light a link between financial systems and the legal traditions of different countries, notably by distinguishing **common law** countries, founded on case law, from **civil law** countries, based on codified law. These differences in legal systems are said to influence the financial development of countries: in common law countries, shareholders and creditors are said to benefit from more extensive protection, which would favour direct financial markets; by contrast, in civil law countries, notably those of the French legal tradition, investor protection is said to be lower, thereby encouraging a centralisation of transactions within financial intermediaries. Beck et al. (2003) add that the **flexibility** inherent in Anglo-Saxon case law plays an essential role in the ease of financial transactions, making the legal framework more favourable to economic innovation.

**Culture** is another determining factor in the structuring of financial systems. According to the work of Hofstede (1984), the culture of a group or of a country influences its relation to concepts such as power, uncertainty and risk. Thus, the study conducted by Kwok & Tadesse (2006) shows that in countries where uncertainty avoidance is high, financial systems tend to orient themselves towards banks, which smooth risks over time, whereas in countries where uncertainty is less feared, financial markets are more developed, offering instruments for transferring risks,



such as derivatives: cultural preferences with regard to risk may therefore orient the evolution of financial systems, by favouring either bank intermediation or market financing.

However, the idea that culture and law influence financial systems in an immutable manner may be qualified: while culture and the legal framework do indeed play an important role, Grjebine & Tripier (2016) thus emphasise that financial systems evolve at a faster pace than these two factors: the recent evolution of financial and banking markets shows that banks have progressively adopted practices closer to those of markets, notably as regards risk management, by transferring risks rather than retaining them. Moreover, as Arcand et al. (2012) recall, legal reforms, although not decisive, may accelerate the **transformation of financial systems**, as was the case in Europe in the 1980s and 1990s. It thus appears that neither law nor culture suffices, on its own, to explain the dynamics of financial systems, the latter being shaped by a complex combination of institutional, economic and social factors.

*In fine*, the relation between financial globalisation and growth should be thought of as not necessarily linear. Arcand et al. (2012) thus show that beyond a certain threshold, namely a volume of credit greater than 110 % of GDP, finance may have harmful effects on long-run economic growth: a hypertrophied financial development may therefore become a source of instability. Indeed, according to Grjebine & Tripier (2016), financial globalisation, when it becomes excessive and detaches itself from the real economy, engenders financial and property crises whose repercussions on society are considerable: each crisis has the effect of reducing the potential growth trajectory of economies, on account of **hysteresis** effects, that is to say that, even after the end of a financial crisis, its negative effects, such as the reduction of productivity, of productive capacity or of human capital, persist in the long run. Now, in 2020, the total volume of credit in developed countries represented 432 % of their GDP according to Henrat-Lemée (2021), prompting some economists such as Artus (2019) to advocate a stricter regulation of world finance. It finally seems that the link between finance and growth, which is non-linear, has durably changed with the crisis of 2009: Rousseau & Wachtel (2011) show indeed that the positive link that most empirical studies brought out when covering a period running from the end of the 1960s to the end of the 1980s is not found when these studies are replicated using more recent data covering the decades of the 1990s and 2000s.

### Key Takeaways



The relation between financial development and economic growth, although generally positive and significant, remains complex as to its direction of causality: Schumpeter (1911) maintains that finance favours technological innovation and therefore growth, while Robinson (1952) reverses the causality by asserting that growth attracts finance. Beyond a critical threshold (credit at least equal to 110 % of GDP), financial development nevertheless seems to have negative effects on long-run growth via financial instability and property crises, whose hysteresis effects (persistent losses of productivity, productive capacity, human capital) persist long after the crisis [Arcand et al. (2012)]. The positive link between finance and growth well established between the 1960s and the 1980s is therefore no longer found in the data covering the 1990s and 2000s, suggesting a durable change in the relation after 2009 [Rousseau & Wachtel (2011)].

## 1.2 The world of finance in crisis

Financial history since the beginning of the twentieth century has been marked by a succession of structural crises presenting recurrent mechanisms: accumulation of speculative debt, disconnection between valuations and economic fundamentals, then systemic contagion.

### Focus...



**The tulip crisis, the first speculative bubble of modern economic history.** The tulip crisis, which occurred in Holland in February 1637, is generally regarded as the **first recorded speculative bubble** in modern economic history. As Goldgar (2007) establishes, enthusiasm for the tulip — an exotic flower introduced into the Netherlands from the Ottoman Empire in the middle of the sixteenth century — intensified particularly from 1634 onwards when a shortage of certain rare varieties provoked a rapid acceleration of prices among the Dutch *bloemisten* (florists and collectors), who began to buy bulbs with a view to speculative profit rather than to simple enjoyment. At the peak of '*tulipmania*', during the winter of 1636-1637, certain particularly prized varieties such as the *Semper Augustus* reached unprecedented valuations: according to Goldgar (2007), a single *Semper Augustus* bulb in 1635 was worth more than a sumptuous house on the main canal of Amsterdam, while the average price of the bulbs sold at the Alkmaar auctions would have been enough to employ a master carpenter for two whole years. The institutional infrastructure which permitted this unprecedented speculation was the creation of a rudimentary

**derivatives market** allowing the trading of contracts for the future delivery of tulips during the winter months when the bulbs were dormant and could not be planted: these standardised contracts, called '*tavern colleges*' — since the transactions took place in taverns in Amsterdam, Haarlem and elsewhere — facilitated a multiplication of the volume of speculation, allowing investors to own the same physical bulb several times over in contracts.

The revisionist analysis proposed by Goldgar (2007) nevertheless brings an important correction to the traditional narrative: far from constituting a generalised economic catastrophe affecting the whole of the Netherlands, the tulip crisis remained largely confined to a small group of merchant collectors and well-off artisans, and the macroeconomic impact on the Dutch economy — then the most prosperous in Europe with the highest income per head in the world — turned out to be minimal. In February 1637, at a routine auction in Haarlem, the first lot of bulbs put up for sale received no bid despite the repeated calls of the auctioneer. This absence of bidders triggered a generalised panic: the few thousand speculators involved suddenly realised that prices would not rise indefinitely and attempted to liquidate their positions, causing a collapse of prices of 95 % to 98 % within the space of a few weeks. Contrary to the popular myth perpetuated since Mackay (1856), very few contracts were actually settled at the agreed prices during the crash, which explains why the subsequent social upheavals finally remained rather limited.

### 1.2.1 The 1929 crisis, harbinger of the Great Depression of the 1930s

The **stock market crash of October 1929** constitutes far more than a cyclical correction of financial markets: it is the symptom of a structural instability which progressively accumulated throughout the decade of the 1920s and which triggered the greatest economic contraction of the twentieth century. As Shiller (2000) recalls, the Standard & Poor's composite index presented an average price-earnings ratio of 32.6 in September 1929, far above historical norms, which testifies to a manifest disconnection between the prices of securities and the real profitability of the underlying firms. The pioneering analysis proposed by De Long & Shleifer (1991) using closed-end mutual fund data as an indicator of overvaluation reveals that the shares composing the Standard & Poor's index were overvalued by at least 30 % at the end of the summer of 1929, that is to say that the market granted a negative risk premium to real assets, implying a classic **speculative bubble** before its bursting. This situation is itself the logical product of a period of unbridled financial expansion characterised by **margin buying**, a mechanism by which investors could buy shares while paying only a fraction of the total price, the bulk being financed by loans from brokers. According to Schlesinger (1957), this practice of margin buying multiplied during the 1920s, creating a situation in which millions of ordinary investors holding limited capital exposed the whole of their savings to highly leveraged positions in shares. This structure of massive indebtedness for speculative ends meant that even a moderate fall in asset prices would have had catastrophic consequences for indebted investors, who not only saw the value of their securities eroded, but also had to repay the loans granted by brokers at high rates. As Temin (1975) analyses in his study of the monetary causality of the depression, this progressive accumulation of leveraged positions during a continuous credit expansion constitutes precisely the mechanism of financial instability described later by Minsky (1975), according to which a capitalist market economy naturally tends to evolve from a **robust** financial structure (where operating cash flows largely cover repayment obligations) towards a **fragile** structure (where economic agents depend on continuous refinancing in order to honour their commitments).

The bursting of the speculative bubble was determined by the convergence of several macroeconomic shocks which transformed investors' self-fulfilling expectations into generalised panic. As Cecchetti & Karras (1992) documents in analysing the sources of economic fluctuations during the interwar period, the monetary tightening carried out by the Federal Reserve Board in August 1929, which raised the official discount rate from 5 % to 6 %, played a determining role by signalling to financial markets that the monetary authority did not intend to maintain a policy of abundant credit. Moreover, the early signs of economic slowdown observed from the summer of 1929 onwards, after the sustained growth of the 1920s, were enough to undermine investors' confidence: the agricultural sector had already been in deep depression since the beginning of the 1920s on account of chronic overproduction and the collapse of commodity prices, while the manufacturing sector presented clear signs of slowdown with persistent overcapacity despite the overall expansion. According to Bernanke (1995), unemployment in the United States, which had stood at nearly 3 % in 1929, reached more than 25 % at the height of the depression in 1932-1933, reflecting an economic contraction without equivalent since the great depression of 1893. It was in March 1929 that the first signs of reversal manifested themselves, when the Federal Reserve warned of excessive speculation, provoking a first massive sale of shares which exposed the fragility of the market's foundations. Subsequently, in September 1929, the more astute investors began to unwind their positions, which provoked a stagnation of prices and encouraged others to sell, initiating the vicious circle characteristic of **crises of confidence**. On Monday 28 October 1929 (*Black Monday*), the Dow Jones index fell by nearly 13 % in a single session, while on Tuesday 29 October (*Black Tuesday*),



the volume of trading reached a record level of 16.4 million securities, never observed before, with a contraction of prices of the order of 12 %, as the Banque de France (2025b) recalls.

The transmission mechanisms linking the stock market collapse to the Great Depression which ensued operated through several distinct economic channels whose interaction produced an unprecedented contraction of real economic activity. First, the **destruction of financial wealth** provoked by the 89 % fall of the Dow Jones index between 1929 and 1932 engendered a drastic reduction of households' net worth, which reduced the propensity to consume and anticipated permanent incomes according to the theses developed by Friedman (1957) and formalised subsequently in microeconomics. Second, the collapse of the stock market initiated an unprecedented wave of **bank failures**, since American banks had invested massively in transferable securities and held substantial positions on the share market: between 1930 and 1933, nearly nine thousand banks disappeared in the United States through bankruptcy, destroying the deposits of millions of small savers who lost all confidence in the viability of the banking system, as the Banque de France (2025b) recalls. According to Wheelock (1992), the American money supply (the aggregate M1) underwent a contraction of one third between 1929 and 1933, which constitutes the sharpest reduction of the money supply ever observed in peacetime. This monetary contraction was itself endogenous to the banking collapse: banks, confronted with massive withdrawals of deposits, had to reduce their credit portfolios, thereby contracting the supply of credit money to the real economy and amplifying the cycle of contraction initially provoked by the stock market collapse. Third, as Keynes (1933) establishes in his contemporary correspondence and his press articles relating the events of 1929-1933, the failure of economic policy considerably aggravated the depression: not only did the authorities refuse to increase compensatory public spending, but they even adopted **fiscal austerity** at exactly the wrong point in the cycle, thereby reducing public revenues at the very moment when the contraction of the private sector was accelerating. This combination of destruction of financial wealth, of monetary contraction induced by bank failures, of reduction of investment spending due to the collapse of the expected profitability of capital, and of absence of countercyclical macroeconomic intervention transformed a classic stock market crash into a great depression which lasted more than a decade and required the massive spending of the Second World War to recover full employment.

### Key Takeaways



The stock market crash of October 1929 revealed a structural accumulation of instability during the 1920s: the price-earnings ratio of 32.6 in September 1929 and an overvaluation of shares of 30 % reflect a classic speculative bubble, while margin buying allowed investors to finance share positions through massive borrowing from brokers, creating a fragile financial structure in the sense of Minsky (1975) in which continuous refinancing was necessary [Shiller (2000), Schlesinger (1957)]. The bursting of the bubble resulted from the convergence of three shocks: the Fed's monetary tightening in August 1929, the signs of economic slowdown (agricultural overcapacity since the beginning of the 1920s, manufacturing slowdown), and the reversal of investors' expectations from March 1929 onwards, culminating in the crashes of Thursday 24 October, Monday 28 October and Tuesday 29 October 1929 [Bernanke (1995)]. Transmission to the Great Depression operated through four channels: the destruction of financial wealth reducing consumption, bank failures, endogenous monetary contraction, and the countercyclical fiscal austerity adopted by the authorities, transforming the crash into a depression [Wheelock (1992)].

### 1.2.2 The other major crises of the twentieth century

The **oil crisis of the 1970s**, triggered by the embargo imposed by the Organisation of Arab Petroleum Exporting Countries (OAPEC) in October 1973 in reaction to American support for Israel during the Yom Kippur War, constitutes a major macroeconomic shock. According to the Federal Reserve (2025b), the price of crude went from 2.90 dollars a barrel before the embargo to 11.65 dollars in January 1974, representing a quadrupling of the cost of oil in less than three months. This rise came at a moment of macroeconomic vulnerability for the industrialised economies: as Arthur Burns, the then chairman of the Federal Reserve (2025b), explains, 'the American economy was operating at full production capacity, the prices of industrial commodities were already rising at more than 10 % a year, and many critical industrial materials were in extreme short supply'. This context of pre-existing economic overheating transformed the oil shock into a crisis of **stagflation** — an economic phenomenon combining high inflation and economic stagnation simultaneously, contradicting the curve of Phillips (1958). The analysis proposed by Kilian (2009) on the role of monetary policy compared with oil shocks establishes that the 1970s were characterised not solely by exogenous oil supply shocks, but also by fundamental modifications of the monetary policy regime: the collapse of the Bretton Woods system in 1971 released the constraints previously imposed on national monetary policy, allowing central banks to adopt a far more aggressive monetary expansion than they could have done before. This combination of unprecedented monetary expansion and restrictions on oil supply thus provoked the highest

inflation recorded since the immediate post-war period in most developed countries: in the United States, the rate of inflation reached more than 12 % in 1974, while in Great Britain and Italy it exceeded 18 %, as Castel (2020) recalls. In parallel, the unemployment rate increased substantially, breaking with the inverse relation historically observed between inflation and unemployment posited by the curve of Phillips (1958), thereby demonstrating that the standard Keynesian understanding of macroeconomic trade-offs turned out to be inadequate in the face of this type of combined supply and demand shock. The incapacity of the monetary authorities to control inflation and unemployment simultaneously at the end of the 1970s obliged the Federal Reserve, under the direction of Paul Volcker from 1979 onwards, to impose a substantial monetary tightening — the key rate reaching nearly 20 % in 1981 according to Castel (2020) — in order to break the inflationary expectations anchored in the behaviour of economic agents. This ultra-restrictive monetary discipline provoked a deep recession (1981-1982) with an unemployment rate exceeding 10 %, but it finally made it possible to bring inflation back to normalised levels by the beginning of the 1980s.

The **Asian financial crisis of 1997-1998**, for its part, constitutes one of the major crises of financial contagion in modern economic history, demonstrating the extent of the vulnerabilities created by the rapid and partially unregulated integration of capital markets in a region where macroeconomic fundamentals had seriously deteriorated. According to the Federal Reserve (2025a), the crisis process began in July 1997 when the Thai government, having exhausted its foreign exchange reserves in a vain attempt to defend the fixed parity of the baht against the American dollar, was forced to let its currency float. This decision, which could have constituted an ordinary correction of exchange rate imbalances, on the contrary triggered an international panic whose dynamic was amplified according to the classic mechanisms of financial contagion: as the baht depreciated by more than 50 % against the dollar — as Mister Prépa (2024) recalls, before the crisis the exchange rate was roughly 25 baht to one American dollar; at its lowest point in January 1998, the baht had fallen to nearly 56 baht to the dollar — international investors began to reassess risk across the whole Asian region, supposing that the fragilities which had permitted the Thai crisis might exist elsewhere. The analysis conducted by Corsetti et al. (1998a) and Corsetti et al. (1998b) on the causes of the Asian crisis establishes that the countries affected — Thailand, Indonesia, Malaysia, South Korea — had all accumulated substantial macroeconomic vulnerabilities without external observers having sufficiently appreciated them: current account deficits had worsened in the 1990s, banking systems had granted substantial quantities of doubtful loans, external indebtedness had accumulated at dangerous short-term levels, and the composition of capital inflows had progressively shifted towards speculative flows rather than towards stable foreign direct investment. The analysis of Mishkin (2000) suggests that the real problem was not first of all an overvaluation of currencies, but rather the existence of an extreme fragility of the financial sector: banks, having borrowed massively in hard currencies on international markets, had in turn lent in local currency to firms with no real capacity to generate the foreign currency necessary to repay the bank's debts in dollars. This currency *mismatch* between the assets and liabilities of Asian financial institutions meant that any devaluation of the local currency translated immediately into a deterioration of banks' balance sheets, creating a vicious cycle of bank failures, credit tightening and compression of economic activity.

The speculative bubble in internet stocks of the 1990s, now known as the **dot-com bubble** or TMT crisis (technology-media-telecoms), finally, constitutes an example of the irrational exuberance of financial markets in the face of a revolutionary technology whose real implications remained profoundly uncertain. As Google Finance (2026) recalls, the NASDAQ index, composed in the main of technology and internet stocks, rose by 486 % in only five years, going from 1541 points in December 1994 to 5048 points in March 2000, in the absence of any concomitant growth in corporate profits. More precisely, as DeLong (2006) explains in his econometric analysis of the size of the bubble, several firms in the internet sector had infinitely high price-earnings ratios simply because they recorded no profit at all, having rather deeply negative operating cash flows, which implied an implicit assumption that they would find a profitable business model in a future sufficiently distant to justify contemporary investment at these high prices. The particular case of Pets.com, analysed by Coulter & Vogel (2004), illustrates the disconnection between expectations and reality: this firm, founded in 1998 to sell pet food over the internet, succeeded in raising three hundred million dollars in venture capital and in achieving a stock market flotation with a high valuation before proceeding to a massive restructuring and finally disappearing in November 2000, having burned through these three hundred million dollars in less than two years without ever generating a single dollar of profit. More generally, as Cassidy (2002) recalls, the twenty best internet companies of the moment — including Priceline, whose valuation reached at one moment nearly ten billion dollars, higher than each of the three largest American airlines combined (United, Continental and Northwest) even though the firm was making losses — fell by 90 % or more in the crash. The mechanism amplifying the bubble lies in a combination of factors: the particularly low interest rates of 1998-1999 facilitated a proliferation of venture capital seeking placements; the growing deregulation of telecommunications and the less strict regulation of the securities market created incentives to launch multiple competing firms with

no clear model of profitability; Wall Street analysts such as Mary Meeker (Morgan Stanley) aggressively promoted internet firms to institutional and individual investors, creating a dynamic of informational cascade in which the mere fact that a firm obtained the ‘internet’ label was enough to justify a massive revaluation; the mass media amplified the narratives of the ‘new economy’ in which the old rules of profitability and return on capital no longer applied. As Google Finance (2026) shows, the crash came abruptly in March 2000 when, after the NASDAQ reached its peak of 5048 points on 10 March, the more astute investors began seriously to doubt that any of the hundreds of competing internet firms could ever justify the valuations granted. Now once this realisation began to spread — a process that Eyrolle (2019) describes as a reversal of **informational cascades** — the reverse mechanics of the bubble were inexorably set in motion: venture capitalists ceased to finance new internet firms, institutional investors once again chose to favour return on capital, and the flotations of new dot-coms were frozen. Between March 2000 and October 2002, the NASDAQ thus lost 78 % of its value, wiping out more than five thousand billion dollars of market capitalisation, a destruction of wealth without equivalent since 1929.

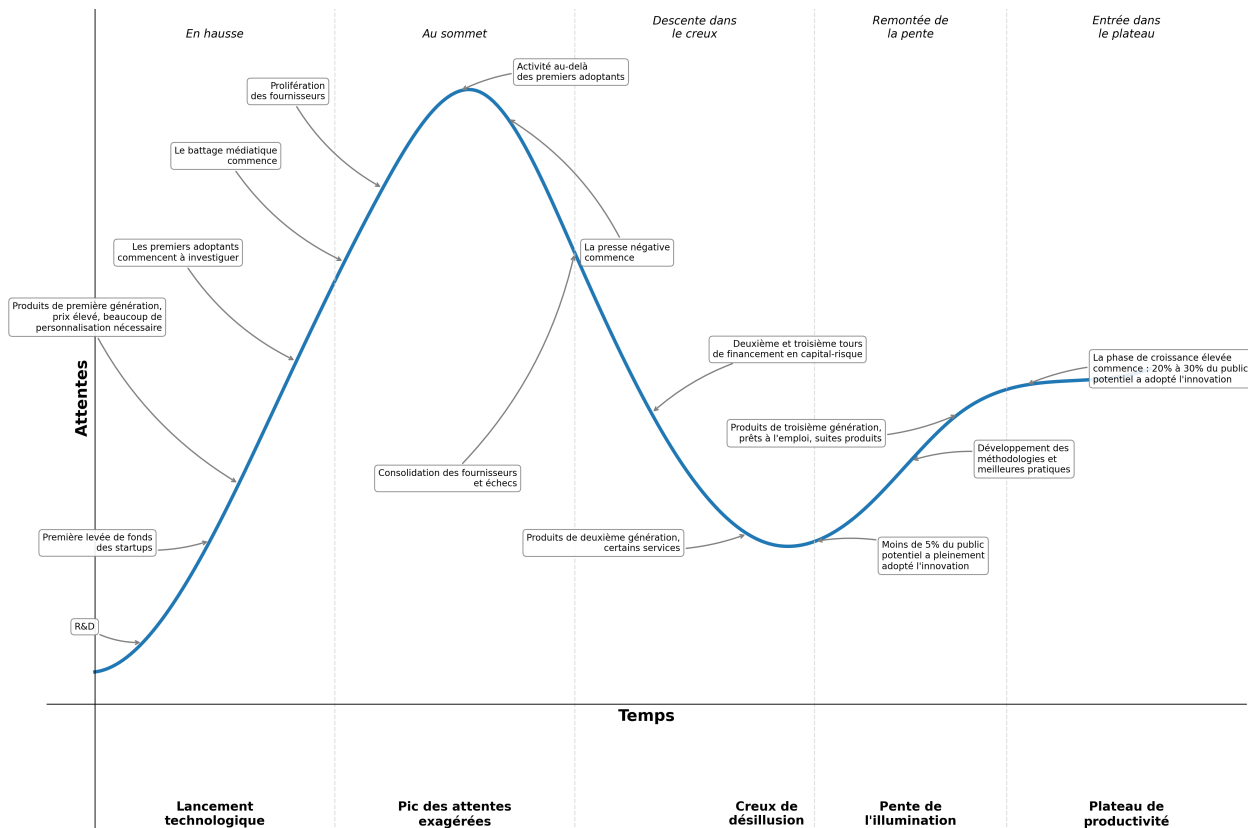


Figure 3.1: The Gartner (2025) *hype cycle*

## Debate



**Does artificial intelligence constitute a new speculative bubble?** According to Goldman Sachs (2026), the valuations of firms exposed to artificial intelligence (AI) have experienced a substantial increase, which gives rise to concerns regarding the real extent of the economic fundamentals justifying these prices. To be sure, according to Tool Advisor (2025), the market for generative AI should grow at an average annual rate of roughly 37 % between 2025 and 2030, which suggests a real economic potential justifying a substantial part of the investments made. McKinsey & Company (2023) thus estimates that generative AI will create between 2600 billion and 4400 billion dollars of additional economic value annually in steady state, which would theoretically justify substantial initial investments in infrastructure and development. However, the valuation criteria present distinct warning signals: Goldman Sachs (2026) observes that estimates of return on investment (ROI) remain difficult to justify given the 3000 to 4000 billion dollars of cumulative AI spending that Nvidia estimates will be made by the end of the decade, while Acemoglu (2024) estimates that AI will make it possible to generate an increase in total factor productivity only of the order of 0.71 % over ten years, which contrasts with the exuberant expectations of the market. This moderate perspective suggests that the Gartner (2025) cycle for generative AI currently positions this technology at the *peak of*

*inflated expectations*, meaning that substantial valuation corrections remain possible as firms confront the concrete challenges of operational integration. The situation of the AI market thus presents ambivalent characteristics: on the one hand, long-run macroeconomic fundamentals justify substantial investment in AI as a transformative technology; on the other hand, current valuations incorporate particularly optimistic assumptions as to the temporal trajectory of monetisation and to massive commercial adoption, thereby creating a significant risk of price correction in the short and medium term.

### Key Takeaways



The oil crisis of 1973 quadrupled the price of oil in three months, producing a stagflation (inflation and stagnation simultaneously) inexplicable by Keynesian models, arising from monetary expansion and the oil shock, resolved by the Fed chairman Paul Volcker, who raised key rates to nearly 20 % in 1981 [Castel (2020)]. The Asian crisis of 1997, for its part, revealed financial contagion via a currency *mismatch* between bank liabilities in dollars and assets in local currency, provoking a depreciation of the baht of 50 % [Mishkin (2000)]. The internet bubble of the 1990s, finally, saw the NASDAQ triple despite the absence of profits [DeLong (2006)].

### 1.2.3 The 2009 crisis, a consequence of financial globalisation

As Bricogne et al. (2009) analyse, the ‘**Great Moderation**’ of the 2000s, that is to say the economic conjuncture characterised by a low volatility of growth and inflation which could give the impression that macroeconomic risk had disappeared, brought about a fall in risk premia across a wide spectrum of assets: this situation led to rises in asset prices which in turn brought about an expectation of further rises, feeding **speculative bubbles**. In order to continue to provide investors with high returns, fund managers then turned towards ever riskier assets, benefiting moreover from a **leverage effect** offered by low interest rates and abundant liquidity. Indeed, as Dai (2009) emphasises, following the collapse of the speculative bubble in internet stocks in 2000, the Fed, under the direction of Alan Greenspan, rapidly reduced its key interest rate in order to attenuate the negative repercussions on the economy, amplified by the events of 11 September 2001; this rate was then maintained at an exceptionally low level, around 0.75 % for a prolonged period. From 2000 onwards, a considerable flow of liquidity was redirected towards the property sector. Now, as Dai (2009) explains, the rise in prices in this sector generated a **wealth effect**, thereby stimulating consumption and favouring new construction projects, while the maintenance of interest rates at low levels significantly encouraged corporate investment. The American mortgage market provides an illustration of this enthusiasm: as Bricogne et al. (2009) recall, even the least creditworthy households began to have access to property loans, subject to an ‘extra’ risk premium<sup>1</sup>, so much so that the total amount of *subprime* loans went from 2.4 % to 13 % of the total outstanding property loans in the United States between 1998 and 2007, even representing 40 % of new loans taken out in 2007. As Bricogne et al. (2009) detail, lenders even took account of the unrealised capital gains on the properties mortgaged by these *subprime* loans in order to allow households to raise more debt, thereby increasing risk-taking both for lending banks and for borrowing households. It should then be recalled that, unlike France, which favours fixed-rate loans, American households generally contract property loans at variable rates: any rise in rates therefore puts borrowers at risk. Now, Bricogne et al. (2009) recall that the Fed began to raise its key rates as early as 2004, so much so that roughly 18 % (resp. roughly 21 %) of borrowers having contracted a fixed-rate (resp. variable-rate) *subprime* loan found themselves in a situation of payment default. As Bricogne et al. (2009) specify, when the debt of an American household exceeds the value of its property, it may cancel the repayment of its debt subject to seizure of the property, which is then put up for sale by the creditor: roughly 6.5 % of households having contracted a variable-rate *subprime* loan thus found themselves engaged in foreclosure proceedings on their property. This sudden excess of supply contributed to a fall in property prices from the middle of 2006, until it reached an annual fall of nearly 20 % in 2008: stocks of new individual dwellings represented four months of sales in 2005, against thirteen months of sales at the end of 2018, marking the bursting of the property bubble.

### Focus...



***Too big to fail and its limits.*** The doctrine of *too big to fail* designates the principle according to which certain financial institutions are so large, interconnected or systemically important that their failure would provoke catastrophic effects on the economy as a whole, thereby justifying their rescue by the public authorities. This doctrine

<sup>1</sup>In the context of a property loan, a *prime* borrower is regarded as low-risk by lenders on account of his good creditworthiness, which allows him to benefit from more favourable interest rates. By contrast, a *subprime* borrower presents a higher risk of default on account of weaker creditworthiness, which entails higher interest rates to compensate for the risk: loan insurance for *subprime* borrowers is therefore more costly, since it aims to protect lenders against potential payment defaults.

materialised massively during the crisis of 2008-2009, when governments mobilised unprecedented amounts in order to avoid the collapse of the world banking system. In the United States, according to the computations of Harbert (2019), the total direct cost of the **bailouts** linked to the crisis, measured at *fair value*, amounted to roughly 498 billion dollars, that is 3.5 % of 2009 GDP, mainly including the TARP programme (*troubled asset relief program*), the bailouts of Fannie Mae and Freddie Mac, and the expanded federal guarantee programmes. In Europe, as Le Point (2017) reports, state aid to the banking sector between October 2008 and the end of December 2011 reached roughly 1600 billion euros, that is 13 % of European GDP in 2011, 67 % of this aid consisting of public guarantees on bank funding.

However, these massive rescues revealed the limits and dangers of *too big to fail*: by implicitly guaranteeing the creditors of large banks, the authorities create a **moral hazard** giving these institutions an incentive to take excessive risks, knowing that they will be rescued in the event of difficulties. In order to remedy this problem, several major reforms were adopted. In the United States, the *Dodd-Frank Wall Street Reform and Consumer Protection Act* of 2010 explicitly aimed to put an end to *too big to fail* by imposing strengthened **prudential standards** on financial institutions of systemic importance (those exceeding 50 billion dollars of assets), by creating mechanisms of orderly resolution (*orderly liquidation authority*) and by requiring *living wills* making it possible to dismantle these institutions without recourse to public funds, as Wilmarth (2011) analyses. At international level, the Basel III framework adopted in 2010 strengthened capital requirements for globally systemically important banks, imposing a *capital surcharge* which may reach 3.5 % of risk-weighted assets over and above the basic requirements. In Europe, the BRRD (*Bank Recovery and Resolution Directive*) and the single resolution mechanism introduced from 2014 onwards the principle of **bail-in**, obliging the shareholders and creditors of failing banks to bear losses before any eventual recourse to public funds, thereby reversing the logic of the *bailout*, as Hadjiemmanuil (2022) explain. Nevertheless, despite these reforms, the problem of *too big to fail* persists: the six largest American banks held 6.8 trillion dollars of assets in 2006 against 9.5 trillion in 2011, demonstrating that banking concentration has intensified rather than diminished after the crisis, according to the data reported by The Week (2011).

The element which triggered the bursting of this bubble was the failure on 15 September 2008 of **Lehman Brothers**, the forty-second largest bank in the world at the time, with 639 billion dollars of assets according to Couppey-Soubeyran & Renault (2018). The clients of this large bank were not, however, the most modest households. In order to explain the contagion of the property crisis due to *subprime* loans to the whole banking system, it is appropriate to analyse the mechanism of **securitisation**, that is to say the transformation by banks of claims into securities tradable on financial markets. Now, as Dai (2009) analyses, by selling off their claims transformed into financial securities, banks' commitments are no longer recorded on their balance sheet: the risk therefore no longer appears so clearly in accounting terms. Otherwise put, one moves from an **originate to hold model**, in which banks originated the loan and then held it until repayment, to an **originate to distribute model**, in which they still originated the loan, but resold it immediately and therefore no longer bore the risk, according to the terminology of Aglietta & Valla (2017). More precisely, taking advantage of deregulation, commercial banks created subsidiaries called **special purpose vehicles** (SPV) endowed with little capital and specialising in the sale of risky financial securities to global investors: with the bursting of the property bubble, the SPVs were imperilled and their assets were losing value. Bricogne et al. (2009) thus indicate that, up to January 2009, roughly 792 billion dollars of write-downs were recorded by banking establishments worldwide, in application of the Basel II regulations (see *infra*) under which banks had to revise their method of accounting for assets at their market value (**marking-to-market**) from the summer of 2007 onwards. More precisely, while recourse to SPVs excluded certain assets from banks' balance sheets, the loss of value of these assets threatened the survival of the SPVs, which compelled banks to reintegrate them into their balance sheets. Now, in order to comply with the Basel prudential ratios, banks then had to sell part of their assets while proceeding to capital increases: the sudden increase in the supply of securities on the market then engendered a downward spiral on financial markets. In addition, the urgent sale of short-term securities in order to finance long-term property claims limits the mitigation of risk and does not make it possible to reassure financial markets sufficiently, and they lose confidence: the American stock market index S&P 500 thus fell by roughly 40 % in 2008, the worst loss in its history after the collapses of 1932 and 1938, while financial establishments worldwide recorded losses assessed at 2200 to 2800 billion dollars as a result of the financial crisis according to Bricogne et al. (2009). The magnitude of these losses is explained, according to Henrat-Lemée (2021), by the opacity of these risk transfer mechanisms: **systemic risk** was neglected by market actors, since the complexity of the banks' arrangements was not perceived by the rating agencies, in which investors place blind confidence according to Dai (2009). Consequently, financial securities of poor quality (**junk bonds**) found themselves mixed with securities of good quality: this is the principle of **tranching**, whereby 'packages' of assets are created. Now the problem, according to Henrat-Lemée (2021), is that the shock to these *subprime* loans was so strong that it brought about the collapse of the whole package of assets in

which the *junk bonds* figured: one then understands why banks apparently sound and reputed to be ‘*too big to fail*’, such as Lehman Brothers, finally went bankrupt. . .

### Key Takeaways



The 2009 crisis resulted from the accumulation of speculative bubbles during the ‘Great Moderation’ of the 2000s: the low interest rates after 2000 combined with abundant liquidity engendered massive leverage and a search for yield on risky assets, notably American *subprime* mortgage credit [Bricogne et al. (2009), Dai (2009)]. The rise in property prices generated a wealth effect stimulating consumption, while variable rates and the rise in the Fed’s key rates from 2004 onwards put one *subprime* borrower in five into default and triggered a fall in property prices and the bursting of the bubble from 2006 onwards [Bricogne et al. (2009)]. The failure of Lehman Brothers on 15 September 2008 triggered systemic contagion via the mechanism of securitisation [Aglietta & Valla (2017)], while the bursting of the bubble forced banks to reintegrate the failing SPVs into their balance sheets and to sell assets massively in order to comply with the Basel II ratios, generating a downward stock market spiral [Henrat-Lemée (2021), Dai (2009)].

## 1.3 Issues and debates surrounding contemporary finance

Contemporary world finance faces several major structural transformations which are redrawing geo-economic hierarchies. First of all, **tax havens** allow multinationals to reduce their tax obligations considerably via profit shifting, while institutional reforms remain largely ineffective. In parallel, the **BRICS+**, representing 30 % of world GDP, are launching initiatives to reduce their dependence on the dominant American dollar, notably via the creation of alternative institutions and decentralised payment systems.

### 1.3.1 The question of tax havens

**Tax havens** constitute a set of jurisdictions offering abnormally low rates of taxation and exceptional regimes of financial secrecy, allowing multinational firms and wealthy taxpayers to reduce their tax obligations considerably relative to their domicile or to their source states. As Hines & Rice (1994) establish, tax havens are characterised not only by nominally low rates of taxation, but also by an institutional architecture which ‘artificially separates taxable income from the activities which generate it’, thereby creating opportunities for **profit shifting** at low apparent cost. The data of Network (2024) show that at world level, multinational firms shift 1130 billion dollars of profits annually to tax havens; this profit shifting activity deprives governments of 348 billion dollars in direct annual tax revenue. Network (2024) further specifies that the **race to the bottom in corporate tax rates**, far from reducing tax abuses as the Laffer curve suggests, on the contrary aggravates the losses of public revenue: between 2016 and 2021, 38 countries lowered their corporate tax rates by six percentage points on average (including 17 OECD members), which generated an additional loss of 68 billion dollars in 2021 for these countries alone, on top of the 138 billion dollars already lost through the tax abuses of multinationals. Moreover, as Davies et al. (2018) set out, profit shifting operates mainly through three distinct channels: first, through the **manipulation of intra-group transfer prices**, the prices at which the subsidiaries of the same multinational sell each other goods and services being set well below comparable market prices in order to transfer income towards low-tax jurisdictions; second, through the **transfer of intangible assets** (patents, algorithms, financial portfolios) from the country of the head office towards tax havens, thereby generating flows of *royalties* which reduce profits in high-tax countries; and third, through **intra-group debt shifting**, whereby subsidiaries in high-tax countries borrow at high interest rates from sister subsidiaries in tax havens, thereby reducing the local taxable base through interest payments. This architecture of tax avoidance, although technically in conformity with existing regulations, rests on the **arm’s length principle**, a principle whose implementation comes up against the intrinsic indeterminacy of comparable market prices, particularly for intangible goods and services, which leaves multinationals considerable latitude in setting their internal transfer prices.

### Focus...



**The European Union’s list of tax havens.** As Toute l’Europe (2025) explains, since 2016 the European Union has maintained a grey list and a blacklist recording jurisdictions that are non-cooperative in tax matters, thereby distinguishing jurisdictions presenting substantial failings in tax transparency and in the exchange of financial information. On 10 October 2025, the **blacklist** groups together the jurisdictions judged to be persistently and seriously non-cooperative: it historically includes states or territories such as Anguilla, Trinidad and Tobago, the United States Virgin Islands, Panama, Fiji, Vanuatu, Guam, Palau, Samoa, American Samoa and Russia. The **grey**



**list**, more substantial, groups together the jurisdictions which have embarked on reforms with a view to meeting the transparency criteria but have not yet fully brought their regulatory framework into conformity: it comprises countries as diverse as Belize, the British Virgin Islands, Antigua and Barbuda, Greenland, Morocco, the Seychelles, Montenegro, Turkey, Jordan and Brunei. The European Union has progressively strengthened the criteria since the financial crisis of 2008: jurisdictions must henceforth demonstrate substantial tax transparency, put in place the automatic exchange of financial information, and comply with the OECD's transfer pricing standards. However, the effectiveness of this listing mechanism remains contested, since the absence of a jurisdiction from the blacklist in no way guarantees the absence of aggressive tax practices, as the notorious cases of Ireland, the Netherlands and Luxembourg — all three members of the European Union — illustrate, these being among the jurisdictions most active in the profit shifting of multinationals, as *Toute l'Europe* (2025) recalls.

The magnitude of the tax haven phenomenon varies considerably according to the jurisdictions responsible and according to the shifting mechanisms employed. According to Network (2024), the United Kingdom and its dependent territories — constituting the British 'second empire' — are responsible for 23 % of world tax losses linked to the tax abuses of multinational firms, which corresponds to roughly 80 billion dollars a year. The '**axis of tax avoidance**', composed of the United Kingdom and its dependencies, the Netherlands, Luxembourg and Switzerland, is jointly responsible for 33 % of the whole of world corporate tax losses, that is more than 115 billion dollars every year. By contrast, developing countries, although representing the majority of the world population, are responsible for only a negligible fraction of these abuses. This imbalance reveals an asymmetry: rich high-tax countries and the tax havens in their orbit — notably the Caribbean jurisdictions, the Channel Islands (Jersey, Guernsey) and the British overseas territories (the Cayman Islands, Bermuda, the British Virgin Islands) — organise a system of income extraction to the detriment of developing countries and of developed countries with normal taxation. The jurisdictions termed 'tax havens' according to the original definition of Hines & Rice (1994), which include Ireland with an effective tax rate on foreign multinationals of less than 7 % according to *The Curious Economist* (2024), the Netherlands with elaborate *holding* structures, Luxembourg with its preferential regimes, Switzerland with its historic banking confidentiality agreements, as well as jurisdictions such as Singapore, Hong Kong, Bermuda and Puerto Rico, use a combination of nominally low (or even zero) tax rates, financial secrecy regimes, weak reporting requirements and weak audit capacity in order to attract and retain world income flows. The data of the Tax Justice Network likewise show that the eight countries which opposed the negotiation of a global tax convention at the United Nations in 2024 — namely the United States, the United Kingdom, Australia, Canada, Israel, Japan, New Zealand and South Korea — although representing only 8 % of the world population, are collectively responsible for 43 % of world tax losses according to Network (2024), which underlines the disproportionate involvement of certain major economic powers in maintaining the system of tax havens.

Institutional and regulatory responses to the problem of tax havens have intensified over the last decade, although their effectiveness remains largely contested and numerous gaps persist. The **BEPS** (*base erosion and profit shifting*) action plan, launched by the OECD in 2015 after the idea was floated by the G20 in 2013, thus comprised fifteen actions intended to eliminate the tax avoidance practices of multinationals, including the development of stricter transfer pricing documentation, the improvement of the automatic exchange of financial information between tax authorities, and the establishment of revised transfer pricing guidelines, as Thiria (2017) explains. However, according to Alstadsæter et al. (2024) and their *Global Tax Evasion Report*, the data covering the period of implementation of the BEPS plan (2016-2021) reveal that the total volume of profit shifting to tax havens has hardly decreased, thus suggesting that the BEPS plan has had little deterrent effect on the behaviour of multinationals. In 2021, more than 140 countries committed themselves at the G20 to implementing a global minimum tax of 15 % on the profits of multinationals, a mechanism theoretically aimed at eliminating the incentive to shift profits by establishing a global tax floor. However, since this initial political agreement, the framework of this minimum tax has been considerably weakened by a succession of exemptions and derogations according to Alstadsæter et al. (2024): the Pillar Two standard of the OECD's international tax framework, as progressively implemented from 2024 onwards, includes a derogation for 'economic substance' allowing multinationals to maintain effective tax rates below 15 % for as long as they maintain a level of 'real substance' (employee, factory and research activities) in very low-tax countries; this derogation, aimed at preserving the economic development policies of developing nations, in reality provides tax havens with a theoretical justification for maintaining regimes at or close to a 0 % rate. Moreover, the United States has not ratified the Pillar Two safeguards, which means that the domestic profits of American multinationals benefit from an exemption until at least 2026, thereby considerably reducing the revenue potential of this minimum regime according to the findings of Network (2024). More precisely, Alstadsæter et al. (2024) estimates that, on account of the multiple loopholes built into the global minimum framework, this system will yield only 4.8 % of world tax revenue, that is only a fifth of what a minimum tax of 15 % without exemptions would have generated. In

reaction to the perceived ineffectiveness of multilateral mechanisms, certain countries such as France, Italy and Spain have unilaterally introduced taxes on digital services and taxes on financial transactions explicitly targeting transnational technology giants such as Google, Amazon, Facebook and Apple through digital services taxes, and large financial institutions through financial transaction taxes, according to Borders et al. (2023). In parallel, the **automatic exchange of financial information** (*common reporting standard*, CRS), introduced in 2015 after decades of advocacy by tax justice organisations, has permitted a reduction of *offshore* tax evasion by wealthy individuals of roughly two thirds in certain OECD jurisdictions according to Alstadsæter et al. (2024), although this benefit is partially counteracted by the displacement of assets towards categories not covered by the CRS, notably real estate, cryptocurrencies and non-transparent holding companies.

### Key Takeaways



Tax havens allow multinationals to shift 1130 billion dollars of profits annually, generating 348 billion dollars of world tax losses, amplified by the race to the bottom in tax rates [Hines & Rice (1994), Network (2024)]. Shifting operates through the manipulation of intra-group transfer prices, the transfer of intangible assets to tax havens, and intra-group debt shifting [Davies et al. (2018)]. The United Kingdom and its dependencies (responsible for 23 % of tax losses, that is 80 billion dollars), and the ‘axis of tax avoidance’ (the United Kingdom, the Netherlands, Luxembourg, Switzerland: 33 % of tax losses) dominate, while eight countries opposing a global tax convention (8 % of the population) cause 43 % of world losses [Network (2024)]. The OECD’s BEPS plan (2015) and the global minimum tax at 15 % (2021) have done little to reduce shifting; Pillar Two (2024) includes derogations permitting a 0 % rate, and the United States has not ratified the safeguards, reducing the potential to 4.8 % of the expected revenue [Alstadsæter et al. (2024)].

### 1.3.2 The emergence of the BRICS in financial globalisation

The **BRICS**, an acronym designating the alliance of the five largest emerging market economies, namely Brazil, Russia, India, China and South Africa — a grouping recently joined by Egypt, Ethiopia, Iran and the United Arab Emirates in 2024, and Indonesia in 2025 (BRICS+) — henceforth constitute a major geo-economic bloc whose emergence and assertion as a leading actor of financial globalisation represent a shift in the international monetary system under construction since 1944. According to Nugrahaeni (2025), the American dollar remains dominant, representing 59 % of world official foreign exchange reserves in 2024, a level of dominance practically unshaken since the Bretton Woods agreements (1944) establishing the post-war monetary system. Now this anchoring of the world financial system in a single currency — the dollar — has conferred on the government of the United States an extraordinary power of economic coercion over other nations: in 2022, the **exclusion of Russia from the SWIFT system** (the Society for Worldwide Interbank Financial Telecommunication), which interconnects more than ten thousand financial institutions in 230 countries, demonstrated the vulnerability of any economy dependent on the world monetary payment system dominated by Washington. According to Shaarawy (2024), the dollar indeed represents 90 % of world foreign exchange transactions, which means that even bilateral transactions between two countries *a priori* outside American dominance pass through American intermediaries and are therefore subject to controls and potentially to sanctions imposed by the federal authorities. The BRICS, conscious of this structural vulnerability, have launched since the Kazan summit in October 2024 a set of initiatives intended to **reduce their dependence on the dollar** and to build progressively an alternative monetary and financial system escaping unipolar American dominance. These initiatives represent not an attempt to abolish the dollar as world reserve currency — a chimerical undertaking in the eyes of any serious analyst — but rather a pragmatic strategy of risk reduction aimed at improving the **financial autonomy** of emerging economies and at limiting their exposure to the unilateral monetary policies of the United States Federal Reserve according to Nugrahaeni (2025). The BRICS+ bloc henceforth collectively represents an extraordinary share of the world economy: according to the IRIS (2024), the BRICS+ aggregated 30 % of world GDP and 46 % of the population, against respectively 45 % and 10 % for the G7. Moreover, the historic BRICS have overtaken the G7 in terms of GDP at purchasing power parity: 31 % for the former against 29 % for the latter, a signal of the dynamism of their economies and of the relative decline of the G7 economies.

The institutional architecture put in place by the BRICS to realise their vision of a multipolar financial system rests on several distinct but complementary pillars. First, the **New Development Bank** (NDB), created in 2014 and based in Shanghai, represents a financial institution deliberately conceived as an alternative to the Bretton Woods institutions — the World Bank and the IMF — which have long embodied Western control over the finance of developing countries according to Nugrahaeni (2025). Unlike the IMF, which imposes strict conditions of neoliberal economic policy (reduction of public spending, privatisation, deregulation) in exchange for its financing, the NDB grants loans in local currencies without imposing these conditions, thereby allowing borrowing states to preserve an



**autonomy of economic policy** and to implement strategies suited to their national realities rather than conforming to Washington's doctrines. Second, the **contingent reserve arrangement** (BRICS CRA), initially endowed with one hundred billion dollars according to Mahanand & Naik (2025), functions as a collective financial safety net allowing the BRICS to mobilise resources rapidly in the event of a balance of payments shock or an exchange rate crisis, without having to submit to IMF requirements. Third, the **BRICS Pay** project, announced at the Kazan summit, constitutes an attempt to create a decentralised cross-border payment system using distributed ledger technologies (*blockchain*) to facilitate transactions in local currencies by bypassing the payment rails controlled by the United States, as Shaarawy (2024) explains. This platform, although still in the pilot phase, should technically allow BRICS firms and governments to settle their transactions without going through the SWIFT system and therefore without exposure to American sanctions or controls. In addition, according to Nugrahaeni (2025), the Chinese cross-border interbank payment system (CIPS), already operational and built explicitly as an alternative to the SWIFT system, has extended its network to 1467 indirect participants in 119 countries and links 4800 banks in 185 countries as at 1 January 2025, demonstrating the technical feasibility of a payment infrastructure that is not dollar-centred. Finally, the **BRICS Clear** project, described at the Kazan summit, envisages an independent settlement and depository infrastructure for cross-border transactions operating in parallel with the existing international financial architecture, potentially offering a 'non-discriminatory' alternative to the current structures dominated by the United States, as BRICS Grain (2024) explains.

Despite these institutional advances, the prospects of success of the BRICS de-dollarisation project remain ambiguous, confronted with structural obstacles both internal and external. On the internal level, according to Mahanand & Naik (2025), the BRICS suffer from an **absence of fundamental coherence**: the bloc brings together on the one hand liberal market democracies such as India and Brazil, strongly integrated into Western financial institutions and whose geo-economic interests remain partially aligned with the Western economic order, and on the other hand more centralised and state-directed economies such as China and Russia, confronted with Western sanctions and therefore more inclined to build alternatives to the system of American dominance. This economic and ideological diversity creates permanent tensions: while China actively promotes the internationalisation of the yuan as a world reserve currency, India and Brazil remain cautious, fearful of alienating their Western trading partners and conscious of the risks of monetary concentration that excessive dependence on the Chinese yuan might introduce, according to Nugrahaeni (2025). On the currency front, the BRICS currencies remain significantly less liquid and more volatile than the dollar: the Chinese renminbi represents only roughly 5 % of world foreign exchange reserves compared with the dollar's 59 %, as Nugrahaeni (2025) recalls, while the Russian rouble has experienced very high volatility on account of sanctions and geopolitical instability, limiting its acceptability as a credible reserve currency. On the external front moreover, American geopolitical resistance has intensified: according to BRICS Grain (2025), President Trump explicitly threatened to impose tariffs of 100 % on all exports from BRICS countries to the United States if the latter attempted to circumvent the dollar in their international payments, a threat intended to dissuade nations from engaging in de-dollarisation. Furthermore, the SWIFT system retains a definite technological and organisational advantage: created in 1973 — with the network opening in 1977 and opening to private firms from 2001 — by the largest financial institutions in the world, SWIFT has developed a depth of liquidity, a density of interconnection network, and an architecture of trust that emerging alternative systems cannot match within a few years, as Nugrahaeni (2025) qualifies. Nevertheless, the de-dollarisation movement initiated by the BRICS, although incapable of replacing the dollar in the short term, is likely gradually to reduce its predominance and to create 'spaces of flexibility' allowing emerging economies greater monetary autonomy — not through a radical swing towards a new currency, but through a progressive pluralisation of the world financial system.

### Key Takeaways



The BRICS+ (Brazil, Russia, India, China, South Africa + Egypt, Ethiopia, Iran, the United Arab Emirates, Indonesia since 2024-2025), representing 30 % of world GDP and 46 % of the population (against 45 % and 10 % for the G7), constitute a major geo-economic bloc contesting the dominance of the American dollar, which represents 59 % of official foreign exchange reserves and 90 % of world foreign exchange transactions [Nugrahaeni (2025), IRIS (2024)]. Since October 2024 the BRICS have launched a set of initiatives to reduce their dependence on the dollar: the New Development Bank (NDB) grants loans in local currencies without neoliberal conditions, unlike the IMF and the World Bank; the BRICS CRA contingent reserve arrangement (100 billion dollars) offers a safety net without IMF requirements; BRICS Pay proposes a decentralised blockchain payment system bypassing SWIFT; and BRICS Clear would envisage an independent settlement infrastructure [Mahanand & Naik (2025), BRICS Grain (2024)]. The structural obstacles nevertheless remain major: absence of internal coherence (the liberal democracies India and Brazil *versus* the state-directed economies China and Russia), liquidity and volatility of BRICS currencies far inferior

to the dollar (the yuan representing only 5 % of world reserves), the durable technological and trust advantage of SWIFT created in 1973, and external threats (Donald Trump's trade threats) [Nugrahaeni (2025)].

### 1.3.3 A few words on Islamic finance

**Islamic finance** constitutes an alternative financial system founded on the principles of Islamic law (Sharia) and governed by a coherent set of prescriptions derived from the Quran, the Sunnah and the interpretation of Muslim jurists over the centuries. As Uddin (2015) establishes, the fundamental principles of Islamic finance — notably the prohibition of **Riba** (interest or usury), the prohibition of **Gharar** (excessive uncertainty) and the prohibition of **Maysir** (speculation or gambling) — constitute the pillars on which every financial transaction conforming to Islamic precepts is built. These principles do not represent mere regulatory constraints, but rather a **global economic architecture** which claims to rest on the creation of equitable value, the sharing of risks, and the anchoring of transactions in the real economy rather than in purely financial speculation. According to Iqbal & Llewellyn (2002), Islamic finance offers a distinct philosophical perspective on the way in which economic and financial operations must be conducted, going beyond the sole objective of profit maximisation in order, according to its defenders, to encompass collective welfare, distributive justice and the morality of operations. The Islamic finance industry has experienced strong growth: according to LSEG (2025), the sector's world assets reached 5980 billion dollars in 2024, that is growth of 21 % in a single year. Despite this expansion, Islamic finance still represents barely 1 % of world financial assets, which underlines its potential for future development. The market remains strongly concentrated geographically: according to the market analysis report of Mordor Intelligence (2026), Saudi Arabia controls 49.05 % of the regional Middle Eastern market, while LSEG (2025) emphasises that, on the basis of world assets, Malaysia remains the largest Islamic market in the world, followed by Saudi Arabia and the United Arab Emirates. Contrary to received ideas, Islamic finance is moreover no longer limited to Muslim countries: LSEG (2025) recalls that the United Kingdom, Luxembourg and Ireland have issued **sovereign Sukuk bonds** conforming to Islamic standards, while France and Germany now host Islamic investment funds.

The concrete architecture of Islamic finance rests on a set of distinct contracts and financial instruments, each realising a specific form of intermediation while respecting the Quranic prohibitions and the **absolute prohibition of interest**. According to Wise (2025), the main instruments include **Murabahah** (cost-plus sale contract), where an Islamic bank buys a good and resells it to the client at a price fixed in advance including a commercial margin, thus ensuring that both parties know precisely the amount to be repaid and the due date without any interest being charged; **Mudarabah** (limited partnership), where the bank provides the capital while the borrower or entrepreneur contributes his expertise and his labour, profits being shared according to a previously agreed ratio, while any losses are borne by the capital investor rather than through interest-bearing debt; and **Musharakah** (joint venture partnership), where all the partners (bank and clients) contribute capital and share both profits and losses in proportion to their contributions, thereby aligning everyone's interests and distributing risks equitably. Other instruments analysed by the ICMA et al. (2024) include **Ijarah** (leasing contract), where the bank buys an asset and leases it to the client against the payment of regular rents, thereby generating income for the bank without charging interest, and above all **Sukuk** (Islamic bonds), genuine investment certificates representing undivided shares in the ownership of tangible assets, services, or particular project activities. More precisely, Sukuk constitute the fundamental instrument of Islamic capital markets and are distinguished from conventional bonds: whereas a bond confers on its holder a fixed claim to repayment together with periodic interest, a Sukuk confers on the holder a share of the ownership of a real asset or a stream of income generated by that asset. This structure of *ownership* means that Sukuk returns fluctuate as a function of the real economic performance of the underlying asset — whether a property project, an infrastructure, transport equipment, or a set of *royalties* arising from commercial activities — rather than representing a fixed repayment commitment independent of performance. According to S&P Global (2025a), the global Sukuk market exceeded one thousand billion dollars in 2024 in outstanding amount for the first time. The transparency built into this asset structure requires that Sukuk must imperatively be backed by 'Sharia-compliant' assets, which implies the systematic exclusion of the alcohol, gambling, tobacco, armaments and adult entertainment sectors, and of all sectors whose activities would violate Islamic principles. This requirement of 'negative *screening*' of proscribed sectors introduces a filter which claims to be ethical in the allocation of capital: Sukuk investors know with certainty that their funds do not finance industries judged contrary to Islamic 'morality'. According to the data of the London Stock Exchange, Sukuk represented roughly 1031 billion dollars of total Islamic finance assets in 2024 according to LSEG (2025). In periods of financial stress, this anchoring in the real asset and this concentration on the productive economy appear to confer on Islamic finance a relative resilience compared with purely speculative markets: the global financial crisis of 2007-2009 revealed that Islamic banks, on account of

their strict adherence to the prohibition of complex and highly leveraged derivative instruments, withstood better the collapses of liquidity and the solvency crises which paralysed conventional financial institutions.

The contemporary dynamic of Islamic finance and its future challenges are shaped by the tension between strict Sharia compliance and financial innovation, on the one hand, and by the growing integration of its principles with world objectives of sustainability and environmental and social responsibility, on the other. According to Norchaeva (2024), the basic principles of Islamic finance — the prohibition of Riba, the prohibition of Gharar, the prohibition of Maysir, and the requirement of real assets — align with the United Nations sustainable development goals and with the **ESG standards** increasingly adopted by world institutional investors. This convergence — between the objectives of distributive justice, equity and risk sharing inherent in Islamic finance, and the objectives of environmental responsibility, social justice and ethical governance carried by the ESG movements — opens up new market opportunities. ‘Green Sukuk’ and ‘sustainable Sukuk’, issued to finance renewable energy, sustainable transport, social housing and environmental protection projects, constitute the current frontier of innovation: according to S&P Global (2025a), issues of sustainable Sukuk reached 11.9 billion dollars in 2024. However, Islamic finance also faces substantial challenges. The first is that of **regulatory fragmentation**: interpretations of Sharia principles vary considerably across jurisdictions; identical transactions may thus be judged compliant in Malaysia and non-compliant in Saudi Arabia or Egypt. This lack of international harmonisation constitutes a major obstacle to the integration of Islamic finance into world capital markets and limits the liquidity of Islamic instruments. The second challenge is the **limitation of the investment universe**: the universe of ‘Sharia-compliant’ investments remains significantly narrower than the conventional universe, which creates substantial challenges of portfolio diversification for Islamic asset managers and potentially limits comparative returns. The third challenge lies in the **opacity and complexity** of certain instruments: although Islamic finance is founded on the principle of transparency and clarity, certain sophisticated Sukuk structures — notably hybrid Sukuk, convertible Sukuk, and synthetic Sukuk using complex Sharia-compliant contracts — have introduced a degree of complexity which threatens to undermine the founding principle of clarity and creates risks of ‘Sharia arbitrage’ in which institutions exploit the loopholes in order to circumvent the spirit if not the letter of Sharia. Finally, according to Waemustafa & Sukri (2017), Islamic finance remains confronted with the challenge of the operational definition of Gharar (uncertainty): while the Quran and the Hadith collections clearly prohibit excessive Gharar, the precise quantification of what constitutes ‘excessive uncertainty’ remains contested among Islamic jurists, thereby creating a regulatory grey area. Despite these challenges, the potential of Islamic finance to transform world financial markets is real, and the projections of LSEG (2025) indicate growth to 9700 billion dollars by 2029 if current trends continue.

### Key Takeaways



Islamic finance, founded on the principles of Sharia prohibiting Riba (interest), Gharar (excessive uncertainty) and Maysir (speculation), constitutes an alternative economic architecture anchored in the real economy and in risk sharing rather than in speculation [Uddin (2015), Iqbal & Llewellyn (2002)]. The main instruments include Murabahah (cost-plus sale without interest), Mudarabah (profit sharing), Musharakah (equitable risk-profit partnership), Ijarah (leasing), and Sukuk (bonds backed by real assets representing ownership rather than a fixed claim), the latter having exceeded 1031 billion dollars in 2024 [Wise (2025), LSEG (2025)]. Islamic finance concentrates on ‘Sharia-compliant’ sectors (excluding alcohol, gambling, tobacco, armaments), offering a filter which claims to be ethical; during the 2008-2009 crisis, Islamic banks withstood the shock better thanks to the prohibition of complex derivative instruments. Future challenges include regulatory fragmentation between jurisdictions, the limitation of the ‘Sharia-compliant’ investment universe, the growing complexity of certain Sukuk threatening transparency, and the opacity of the operational definition of Gharar [Norchaeva (2024), Waemustafa & Sukri (2017)].

## 2 Shares, indices and bonds

Financial markets divide into two functionally distinct segments: the primary market and the secondary market. The **primary market** is the place where financial securities (shares, bonds...) are issued for the first time by their issuers, whether firms raising capital, states financing their debt or financial institutions refinancing themselves. It is on this market that securities are born and that funds circulate directly from the investor to the issuer, thus permitting the creation of new financial wealth and the financing of the real economy. The **secondary market**, by contrast, is the ‘second-hand market’ where already issued securities circulate between investors: it comprises stock exchanges (Euronext for shares in France), over-the-counter bond markets, and all the places where existing securities are exchanged without the initial issuer being a party to the transaction. Although the secondary market does not directly generate new capital for the real economy, it plays a fundamental economic role by ensuring the

**liquidity of investments**, that is to say the capacity of the investor to resell his securities rapidly, which is precisely what gives investors an incentive to buy massively on the primary market (they know that they will be able to divest themselves of them if necessary).

## 2.1 Bonds

As Lucotte (2018) defines it, a **bond** is a debt security issued by a borrower, whether a state, a firm or a public authority, with a view to raising funds to finance projects or to fill a deficit: the investor who buys a bond becomes a creditor of the issuer and, in exchange for his capital, receives periodic interest payments, called **coupons**, as well as the repayment of the principal at maturity. Unlike shares, bonds confer no ownership right, but they offer a fixed remuneration (except in the case of floating-rate bonds). Jokung Nguéna (2004) specifies that, unlike so-called single-lender loans, a bond issue is contracted between a single borrower and multiple lenders: it is the non-uniqueness of the lender which implies precisely that the overall loan is subdivided into several bonds, freely tradable on an organised market or otherwise. As Jokung Nguéna (2004) explains, bonds are issued with a **face value**, which represents the amount of the initial loan, and **coupons** which are paid regularly to the investor. For example, a bond with a face value of 1000 euros and a coupon rate of 5 % per year will pay 50 euros each year to the investor, and at the end of the period (say, ten years), the investor will recover the initial 1000 euros. The date on which the issuer repays the investor is called the maturity date or **maturity**.

### 2.1.1 A typology of bonds

As Jokung Nguéna (2004) explains, bonds come in various forms depending on their characteristics and on the needs of issuers or investors. Among the most common are classic bonds, also known as **fixed-rate bonds**: the cash flows associated with these bonds are known from the moment of issue; they include regular coupon payments, generally annual, as well as the repayment of the capital at maturity. The repayment of the principal may take place *in fine* (the whole of the capital is repaid at maturity), by constant annuities (capital and interest are repaid on a staggered basis) or by staggered amortisation, where parts of the principal are repaid at regular intervals before the maturity date. As Jokung Nguéna (2004) specifies, this type of bond offers security to investors since the flows are guaranteed and fixed in advance, but they do not make it possible to benefit from any rise in interest rates after issue. In particular, **zero-coupon bonds** pay no coupons during their lifetime: the investor simply receives a sum of money at the end of the term, which corresponds to the repayment of the principal but, as Lucotte (2018) notes, the bond is generally issued at a price below its face value: for example, a zero-coupon bond issued at a price of 950 euros will be redeemed at 1000 euros (face value) at maturity; the investor's interest therefore lies in this difference, which constitutes the yield (the interest rate of the bond). Lucotte (2018) specifies that the prices of these bonds are less sensitive to fluctuations in interest rates than classic bonds, since they involve no intermediate coupon payments. Jokung Nguéna (2004) adds that this type of bond is often used to finance projects whose operating receipts are expected in the long run, since the absence of intermediate payments allows the issuer to retain liquidity over the life of the bond: zero-coupon bonds are then particularly prized by investors wishing to obtain a fixed yield without the intermediate fluctuations of coupon payments.

#### Focus...



**The bond yield curve.** The yield curve graphically represents the relation between the yields of bonds of different maturities and those maturities themselves. In normal circumstances, the yield curve is upward-sloping — that is to say that yields increase with maturity — which reflects the fact that investors require additional compensation (the **term premium**) for accepting prolonged exposure to interest rate risk and to long-run economic uncertainty according to Kekre et al. (2024). However, since October 2022, bond markets have experienced an exceptional phenomenon of **yield curve inversion**. More precisely, Confluence Investment Management (2024) emphasises that American two-year bond yields remained higher than ten-year yields for twenty-four months, from October 2022 to August 2024 — the longest period of inversion in 45 years; at the maximum inversion in July 2023, the *spread* between the ten-year and the two-year yield had reached  $-0.93$  percentage points (or  $-93$  basis points), with two-year yields reaching  $4.83\%$ , while ten-year yields fell to  $3.90\%$ . This prolonged inversion created an environment of **demand for long bonds** and **repulsion from short instruments**, reflecting an existential fear on the part of the market concerning an imminent recession. Indeed, as D'Haese (2024) recalls, in the United States the last ten recessions were preceded by a curve inversion: from the precise point of inversion, eighteen months elapsed on average before the recession itself. However, unlike most previous inversions, the long inversion of 2022 to 2024 was not followed by a recession

in the United States. The curve finally returned to a normal shape in September 2024, with a positive *spread* of ten basis points according to Confluence Investment Management (2024).

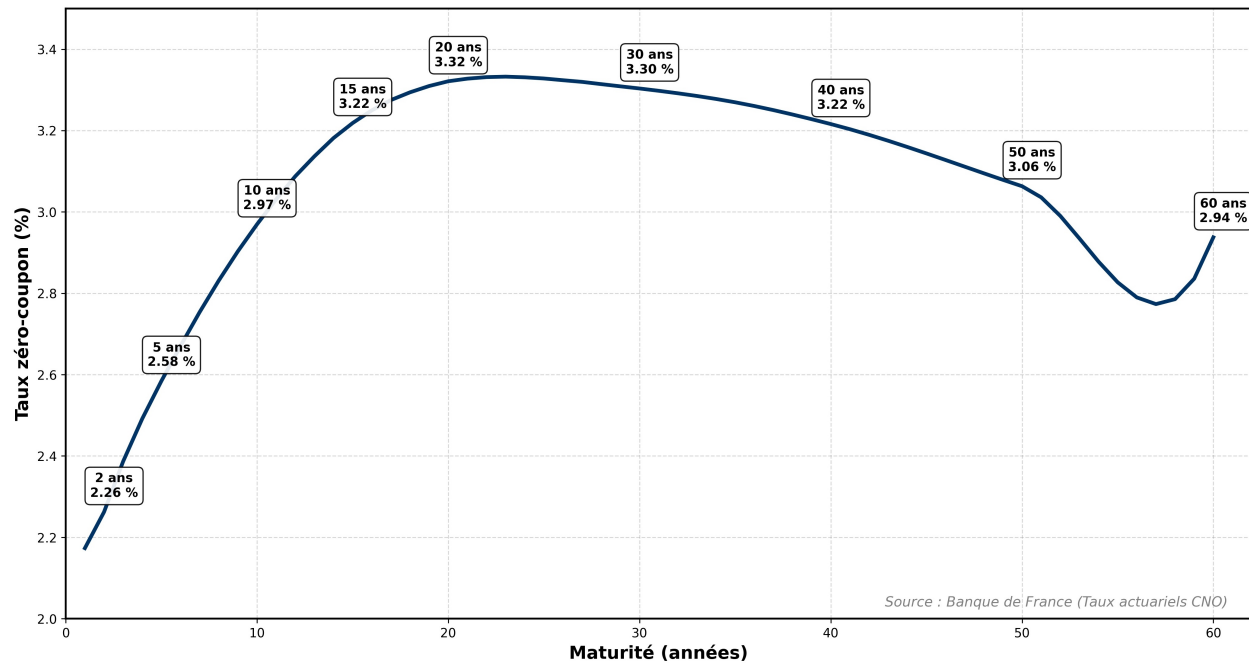


Figure 3.2: The French bond yield curve (zero-coupon bond) at 31 December 2025 according to Banque de France (2025a)

While this course focuses on classic bonds, it is appropriate to master the typology of more exotic bonds. As Jokung Nguéna (2004) recalls, **variable-maturity bonds**, such as the renewable Treasury bonds (*obligations renouvelables du Trésor*, ORT) whose maturity varies between six and eight years in France, offer flexibility as to their maturity: they may indeed be exchanged halfway through for new identical bonds; the interest is capitalised and paid either at final maturity or at the moment of renewal, depending on the conditions provided for at issue. These bonds make it possible to adapt to investors' liquidity needs while offering a cumulative yield at maturity. **Extendable bonds**, such as the fungible Treasury bonds (*obligations assimilables du Trésor*, OAT) whose maturities vary from two to fifty years, are for their part characterised by successive issues at different dates, but possess the same basic characteristics. More precisely, the issuer launches a first loan, called a 'reservoir', which defines the interest rate, the conditions for paying interest and the maturity date of the successive issues intended to feed it; the reservoir is then fed by new tranches with identical characteristics which increase its amount: this allows greater flexibility for issuers while maintaining a coherence in the conditions of issue. Jokung Nguéna (2004) specifies that private individuals have also had access to these securities in France since October 1994.

#### Focus...



**Rating agencies.** Financial rating agencies constitute key actors of modern bond markets by assessing the credit-worthiness of the sovereign debts of states, thus distinguishing the capacity and the political will of governments to honour their financial commitments. The market is dominated by three private firms — **Standard & Poor's**, **Moody's** and **Fitch Ratings** — which represent more than 90 % of the market share in Europe according to Vie publique (2025a), and uses a methodology combining quantitative indicators (GDP per head, ratio of debt to budgetary revenue, rate of inflation) and qualitative ones (political stability, history of default), as Antonin (2009) explains. The economic impact is not negligible: a **rating downgrade** may entail a rise in the interest rates required by investors and may restrict access to the market, as the rating of France, downgraded from AAA in 2012 to A+ in 2025, has illustrated, as Club Patrimoine (2025) recalls. However, the agencies face three major recurrent criticisms: their incapacity to anticipate crises (the Asian crisis in 1997, the *subprime* crisis in 2008, the European sovereign debt crisis from the end of 2009...), their procyclical effect aggravating difficulties during turbulence, and their inherent conflicts of interest since they are remunerated by the issuers they assess, as Vie publique (2025a) explains. In order

to counter the Anglo-Saxon dominance of the rating agencies, the European Union reacted by adopting a specific regulatory framework supervised by the European Securities and Markets Authority (ESMA), and authorised in 2023 the European agency **Scope Ratings** to assess the sovereign debts of the Eurosystem, thereby strengthening competition and the taking into account of regional specificities, as Vie publique (2025a) recalls.

Another interesting type is that of **bonds with warrants**, which offer a purchase right on other securities, such as shares or other bonds of the issuer. Jokung Nguéna (2004) specifies that these bonds may be divided into bonds with share warrants (*obligations à bons de souscription d'actions*, OBSA), giving the right to acquire shares, or bonds with bond warrants (*obligations à bons de souscription d'obligations*, OBSO), giving the right to acquire other bonds; these warrants may be exercised at predefined dates or at any time before the redemption date of the bond, the second option being the more common. This type of bond attracts investors who seek to benefit from greater exposure to shares while enjoying the security of bonds. **Bonds redeemable in shares** (*obligations remboursables en actions*, ORA), for their part, have no optional character. As Jokung Nguéna (2004) explains, they are redeemed directly in shares at maturity, according to a parity fixed in advance; this type of bond is riskier for investors, since redemption is linked to the performance of the shares, but it may offer a higher return in the event of a rise in the share price. **Convertible bonds**, for their part, give investors the possibility of converting their bonds into shares of the issuing company at a predetermined conversion rate, the conditions of conversion, such as the conversion dates and the conversion parity, being fixed in the bond contract. As Jokung Nguéna (2004) mentions, this type of bond is attractive to investors who wish to retain the security of bonds while having the possibility of profiting from the appreciation of the shares; convertible bonds represent for the firm an opportunity to carry out deferred capital increases, since the conversion of the bonds entails their disappearance and the issuance of new shares.

### Key Takeaways



Classic fixed-rate bonds pay regular coupons and repay the capital *in fine*, by annuities or by staggered amortisation, offering the security of known flows but without any benefit from a rise in rates after issue [Jokung Nguéna (2004)]. Zero-coupon bonds, by contrast, issue no coupons, being issued below par (for example, 950 euros redeemed at 1000 euros at maturity) with the yield arising from the difference, and are less sensitive to rate fluctuations than classic bonds [Lucotte (2018)]. Variable-maturity and extendable bonds for their part offer flexibility in maturity or via successive issues feeding a reservoir [Jokung Nguéna (2004)]. Bonds with warrants (OBSA, OBSO) for their part offer the right to acquire future shares or bonds; bonds redeemable in shares (ORA) convert automatically into shares at maturity; convertible bonds give the holder the option of conversion into shares at a predetermined parity [Jokung Nguéna (2004)]. Moreover, the bond yield curve (the relation between yield and maturity) is normally upward-sloping (term premium), but experienced an exceptional inversion of 24 months (between October 2022 and August 2024) reflecting fears of recession, without any recession actually following, however [Confluence Investment Management (2024), D'Haese (2024)].

#### 2.1.2 Price, duration and sensitivity of a bond

The **present value** or price of a bond depends directly on interest rates on the market. As Lucotte (2018) specifies, the higher the market interest rate, the more the price of a bond already issued (at a fixed rate) will tend to fall, and conversely: this is explained by the fact that new investors will prefer to buy bonds which offer higher interest rates, which makes older bonds less attractive, unless their price is reduced to compensate for the difference in rates. The general formula for computing the price of a fixed-rate bond is given by the discounted sum of future cash flows, that is to say the coupons and the principal, and is written thus, as Lucotte (2018) recalls:

$$P = \sum_{t=1}^n \frac{C_t}{(1+i)^t} + \frac{M}{(1+i)^n}$$

In the relation above,  $P$  is the price of the bond,  $n$  corresponds to the maturity,  $C_t$  represents the annual coupon for year  $t$  (for every integer  $t$  between 1 and  $n$ ),  $M$  denotes the amount of the principal to be repaid at maturity and  $i$  denotes the market interest rate.

The **duration** of a bond, for its part, measures the sensitivity of its price to variations in interest rates. As Lucotte (2018) explains, duration corresponds to the time it takes for the holding of a given bond, with reinvestment of the coupons, to procure the equivalent of the amount promised by holding that bond to maturity without reinvestment of the coupons: the higher the duration, the more sensitive the bond is to variations in interest rates. Duration



thus makes it possible to assess the interest rate risk to which the investor is exposed. For a fixed-rate bond paying coupons, the formula for duration is the following:

$$D = \frac{\sum_{t=1}^n \frac{tC_t}{(1+i)^t} + \frac{nM}{(1+i)^n}}{P}$$

where  $D$  denotes the duration,  $C_t$  the annual coupons for year  $t$  (for every integer  $t$  between 1 and the maturity  $n$ ),  $M$  the principal (face value of the bond),  $i$  the market interest rate, and  $P$  the price of the bond determined *supra*. We then note that, in the particular case of zero-coupon bonds, the duration is equal to the maturity of the bond, since there is only one cash flow (the repayment of the principal at the end). The formula for the duration of a zero-coupon bond is written simply  $D = n$ . Indeed, as Lucotte (2018) explains, the whole value of the bond is concentrated at maturity; consequently, a zero-coupon bond is particularly sensitive to variations in interest rates, since any change in rates directly affects the present value of this single future payment.

### Focus...



**Floating-rate bonds.** As Jokung Nguéna (2004) explains, the interest on certain bonds is sometimes paid as a function of a reference rate that evolves over time: this rate may be **floating**, that is to say known at the end of the period to which it applies, or **revisable** if it is a rate determined before the period to which it applies. This rate may be a **money market reference rate** — such as the T4M (*taux moyen mensuel du marché monétaire*, the monthly average money market rate, computed by the French Banking Association), the TAM (*taux annuel monétaire*: the rate of return obtained by a twelve-month investment at the T4M with monthly compounding) or Euribor — which is a short-term rate, or a **bond market reference rate** — such as the TME (the arithmetic mean of the last twelve monthly rates on government loans of more than seven years, computed by the Caisse des dépôts et consignations in France) or the TMO (the arithmetic mean of the last twelve monthly rates of return at settlement on fixed-rate loans guaranteed by the state and equivalents, likewise computed by the Caisse des dépôts et consignations in France) — which is a long-term rate. These bonds allow investors to protect themselves against inflation or fluctuations in interest rates, since their yield adjusts as a function of these variations.

The formulae for the price and the duration of floating-rate bonds are then adjusted. Consider indeed a bond of face value  $M$  and maturity  $n$ , where  $n$  is a non-zero natural number. For every integer  $t$  between 1 and  $n$ , we denote by  $C_t$  the coupon paid for year  $t$  and by  $i_t$  the interest rate applicable for year  $t$ . Then the price  $P$  of the floating-rate bond satisfies [Université numérique (2009)]:

$$P = \sum_{t=1}^n \frac{C_t}{(1+i_1) \times \cdots \times (1+i_t)} + \frac{M}{(1+i_1) \times \cdots \times (1+i_n)}$$

Consequently, the duration  $D$  of the said bond is computed as follows:

$$D = \frac{\sum_{t=1}^n \frac{tC_t}{(1+i_1) \times \cdots \times (1+i_t)} + \frac{nM}{(1+i_1) \times \cdots \times (1+i_n)}}{P}$$

Finally, the **sensitivity** of a fixed-rate bond is a measure of the variation in the price of a bond as a function of the variation in interest rates. It is directly linked to duration and makes it possible to quantify the impact of a rise or a fall in rates on the price of the bond. As Lucotte (2018) specifies, the sensitivity of a bond on the stock market corresponds to the variation in the price of a bond engendered by a rise or a fall of one basis point in interest rates: if  $D$  denotes the duration of the bond and  $r$  its yield, then its sensitivity  $S$  is defined by  $S = \frac{D}{1+i}$ , where  $i$  is the market interest rate. Concretely, a sensitivity  $S$  means that a fall of 1 % (resp. a rise of 1 %) in the interest rate causes the price of the bond to increase (resp. decrease) by  $S$  %:  $S$  therefore measures the price elasticity of the bond with respect to the interest rate. The sensitivity of a bond is therefore a tool for measuring risk for the investor; it allows him to assess the impact of a variation in rates on the profitability of his investment.

### Exercise



#### Fixed-rate bonds

- (i) What is the rate of a zero-coupon bond with a maturity of 2 years and a face value of 100 quoted at 101.51? Give its duration.

- (ii) What is the present value of a bond with a maturity of 3 years and a face value of 100 paying a coupon of 3 with a three-year market rate of 0.5 %? Compute and interpret its duration and its sensitivity.

**Solution.** (i) Consider a zero-coupon bond of maturity  $n = 2$  years, at the price  $P = 101.51$  and of face value  $V_2 = 100$ . Then its yield  $r$  satisfies the equality  $P = \frac{V_2}{(1+r)^2}$ . Now we have:

$$\begin{aligned} P &= \frac{V_2}{(1+r)^2} \iff (1+r)^2 = \frac{V_2}{P} \\ &\iff 1+r = \sqrt{\frac{V_2}{P}} \quad \text{or} \quad 1+r = -\sqrt{\frac{V_2}{P}} \\ &\iff r = \sqrt{\frac{V_2}{P}} - 1 \quad \text{or} \quad r = -\sqrt{\frac{V_2}{P}} - 1 \\ &\iff r \approx -0.0075 \quad \text{or} \quad r = -1.9925 \\ &\iff r \approx -0.75 \% \quad \text{since } r \text{ is greater than } -1 \end{aligned}$$

Moreover, the duration of a zero-coupon bond is equal to its maturity, that is 2 years.

- (ii) The present value  $P$  of a bond of maturity three years and face value 100 paying a coupon of 3 monetary units with a three-year market rate of 0.5 % amounts, in monetary units, to:

$$P = \frac{3}{(1+0.005)^1} + \frac{3}{(1+0.005)^2} + \frac{100+3}{(1+0.005)^3} \approx 107.43$$

Consequently, the duration  $D$  of this bond stands at:

$$D = \frac{\frac{1 \times 3}{(1+0.005)^1} + \frac{2 \times 3}{(1+0.005)^2} + \frac{3 \times (100+3)}{(1+0.005)^3}}{\frac{3}{(1+0.005)^1} + \frac{3}{(1+0.005)^2} + \frac{100+3}{(1+0.005)^3}} \approx 2.92$$

This means that from 2.92 years onwards, that is 2 years, 11 months and 2 days (on the convention of a year lasting 360 days), the price of the bond is no longer affected by a variation in the market interest rate. Consequently, the sensitivity  $S$  of the bond stands at  $S = \frac{D}{1+r} = \frac{2.92}{1.005} \approx 2.91$ : on average, when interest rates rise by 1 %, the price of the bond falls by 2.91 %.  $\square$

### Key Takeaways



The price of a fixed-rate bond is obtained by discounting future cash flows (coupons and principal):  $P = \frac{C_1}{(1+i)^1} + \dots + \frac{C_n}{(1+i)^n} + \frac{M}{(1+i)^n}$ , inversely related to market rates, since new bonds at higher rates make older ones less attractive [Lucotte (2018)]. Consequently, the duration  $D = \left( \frac{1C_1}{(1+i)^1} + \dots + \frac{nC_n}{(1+i)^n} + \frac{nM}{(1+i)^n} \right) / P$  measures the sensitivity of the price to variations in interest rates; for a zero-coupon bond,  $D = n$  (the maturity), since the whole value is concentrated at maturity, making these bonds particularly sensitive to variations in rates [Lucotte (2018)]. Finally, the sensitivity  $S = \frac{D}{1+i}$  measures the price elasticity with respect to the interest rate: a variation of 1 % in the rate entails a variation of  $S$  % in the price (in the opposite direction), allowing investors to assess the impact of a variation in rates on profitability [Lucotte (2018)].

### 2.1.3 Bond financing of the French state

Government bonds, also called **sovereign bonds**, constitute one of the main instruments used by governments to finance their public debt. In France, the outstanding public debt exceeded 3000 billion euros in 2023, a historic level which has aroused concerns as to the sustainability of the debt in a context of rising interest rates, as the iFRAP (2023) emphasises: the **debt service**, which represents the cost of the interest paid by the state on its borrowings, has increased considerably, going from 35.5 billion euros in 2021 to 51.8 billion euros in 2022. This jump is largely explained by the rise in world interest rates, but also by the impact of inflation on indexed government bonds, namely fungible Treasury bonds (OAT). While rates remain high and the state continues to borrow to finance its current spending and its investments, debt service should continue to increase in the years to come: according to



the iFRAP (2023), this phenomenon could entail substantial budgetary tensions, especially if the yield gap (*spread*) between French and German bonds continues to widen.

Indeed, the yield gap on ten-year bonds between France and Germany is often used as a key indicator of the risk perceived by investors concerning French debt; Germany, regarded as one of the financially safest countries with a AAA credit rating awarded by the main rating agencies, offers a benchmark rate. Now, during the sovereign debt crisis in the euro area, between 2010 and 2015, this gap increased sharply for the most indebted countries, notably Greece, Italy and Spain, while German bonds, regarded as the safest, retained lower yields. As the iFRAP (2023) recalls, in 2012, at the height of this crisis, the *spread* between French and German bonds reached a peak of 1.3 percentage points (or 130 basis points), thus revealing an increased distrust on the part of investors towards French debt. Although this gap subsequently decreased to stabilise between 0.3 and 0.4 points from 2015 to 2021, it stood at 0.69 percentage points (or 69 basis points) at the beginning of July 2024, as Reuters (2024) mentions. This evolution testifies to a certain anxiety on the part of markets in the face of the accumulation of French public debt, particularly in a context in which rating agencies might reassess the sovereign rating of France. Now, according to the iFRAP (2023), the crossing of the *spread* above the threshold of 0.7 percentage points (or 70 basis points) would be a worrying sign, indicating growing tensions on France's capacity to honour its financial commitments without further weighing down debt service.

### Focus...



**The sovereign debt crisis in Europe.** The sovereign debt crisis of the euro area, which unfolded between 2009 and 2018 with particular intensity between 2010 and 2015, revealed the fundamental architectural flaws of the European single currency project: the incapacity of member states to use devaluation as an instrument of macroeconomic adjustment, the absence of a federal mechanism of fiscal transfers, and the structural asymmetry between the economically less productive South and the prosperous North. In November 2009, the Greek government revealed that its public deficit had reached 12.7 % of GDP, that is double the previous estimates, triggering a spiral of financial contagion which rapidly spread to Ireland, Portugal and Spain, as De Santis (2012) explains. The financing cost of the Greek government had soared as private investors, conscious of the risk of imminent default, demanded exorbitant yields: the yield *spreads* of Greek (resp. Irish, Portuguese) bonds against the German *Bund* reached levels of 1600 basis points (resp. 1200, 1100 basis points) in July 2011, as De Santis (2012) recalls. The European authorities and the IMF then reacted with a succession of massive rescue plans: an emergency loan of 110 billion for Greece in May 2010, followed by the setting up of the European Financial Stability Facility (EFSF), endowed with 440 billion euros, then by the creation in January 2011 of the European Financial Stabilisation Mechanism (EFSM), funded to the tune of 60 billion euros; the two would finally be merged in October 2012 under the name of the European Stability Mechanism (ESM), a permanent institution capable of raising 700 billion euros on financial markets, as Toute l'Europe (2018) explains. Despite these interventions, existential doubt about the viability of the euro area itself persisted: in 2012, at the height of the crisis, only four countries among the seventeen members of the euro area possessed a Standard & Poor's bond rating in the AAA category, while five countries — Cyprus, Ireland, Italy, Portugal and Spain — had bonds classified in the speculative category (BBB+ or even CCC for Greece), as Kehoe et al. (2012) recall.

The macroeconomic impact of this crisis was particularly substantial for the peripheral economies. **Unemployment** exploded in the countries subject to intervention: according to INSEE (2025a), Greece and Spain still display in 2024 unemployment rates of 10.1 % and 11.4 % respectively against 6.4 % for the euro area on average and 7.4 % for France. The **austerity policies** imposed as conditions of the rescue plans — reductions in public spending, privatisation of state enterprises, labour market reforms — created a perverse procyclical dynamic in which austerity aggravated the recession and reduced tax bases, which paradoxically increased the debt-to-GDP ratio according to Ikonen & Lehmus (2024). The implicit net transfers from the European Union to the stricken countries varied enormously according to negotiating position: while Ireland received the equivalent of barely 0.5 % of its 2010 GDP in net assistance, Greece benefited from transfers equivalent to 43 % of its 2010 GDP according to Gourinchas et al. (2023). This implicit hierarchy of rescues reflected the geopolitical calculations of the creditors: Greece, as a pivotal country between NATO and Russia, and Cyprus, as a strategic financial platform, justified massive interventions in order to prevent *exit* from the euro area and systemic contagion.

Although the acute phase of the crisis abated from 2015 onwards thanks to the announcement by the ECB president Mario Draghi that the institution would do 'whatever it takes' to preserve the euro, the structural after-effects remain visible fifteen years after the outbreak of the crisis. In the fourth quarter of 2024, the euro area displayed a debt-to-GDP ratio of 87.4 % according to Eurostat (2025a), a level which remains significantly above the limit of 60 % set by the Maastricht Treaty. Worse still, the profile of this debt has deteriorated: while the yields on sovereign bonds fell artificially on account of the ECB's unprecedented quantitative easing policies, the average duration of

sovereign debts has shortened, opening up refinancing risks. The most indebted peripheral countries according to Eurostat (2025a) — Greece (153.6 %), Italy (135.3 %), France (113 %), Belgium (104.7 %), and Spain (101.8 %) — remain vulnerable to any interest rate or growth shock. The new regulatory architecture put in place after 2015 — the Treaty on Stability, Coordination and Governance (**TSCG**) imposing a structural deficit ceiling of 0.5 % — is regarded by many economists as excessively procyclical, forbidding governments to use countercyclical fiscal expansion to cushion recessions according to Ikonen & Lehmus (2024).

Another major factor which contributes to the rise in French debt service is the issuance of **inflation-indexed OAT** (OATi). As Ecalte (2023) explains, their redemption value is equal to their face value multiplied by a coefficient representing the inflation observed from their date of issue to their date of redemption; each year, the increase in their redemption value which results from the inflation observed during the year is the object of a provision equal to that increase in the state's budgetary accounts, and this provision increases the interest burden of the debt by the same amount. In 2022, the outstanding amount of these bonds represented 11.5 % of the state's negotiable debt, that is roughly 262 billion euros. The indexation mechanism of these securities means that the redemption value increases in proportion to the inflation observed between the date of issue and the date of redemption: thus, in 2022, the rise in inflation led to an increase in the indexation provision, which went from 3.0 billion euros in 2021 to 15.6 billion euros in 2022 according to Ecalte (2023). This rise in the interest burden linked to inflation explains most of the increase of 16.3 billion euros in debt service that year. Ecalte (2023) emphasises that recourse to this type of bond exposes the state to increased volatility of debt service, notably in periods of unforeseen inflation, which raises questions about the appropriateness of continuing to issue indexed bonds in an uncertain economic context; despite these criticisms, the issuance of indexed OAT has nevertheless allowed the state to limit long-run borrowing costs, by offering investors protection against inflation in exchange for a real interest rate lower than that of classic non-indexed bonds.

### Short-Answer Question



**Are inflation-indexed bonds time bombs?** Inflation-indexed bonds constitute a category of public debt instruments whose principal and coupons are adjusted as a function of the evolution of a price index, generally the consumer price index excluding tobacco for French OATi or the CPI (*consumer price index*) for American TIPS (*Treasury inflation-protected securities*). Their development accelerated from the 1990s onwards, to the point that these securities henceforth represent roughly 8 % of American federal debt in circulation at the end of 2022 according to Meketa Investment Group (2023). This massive expansion raises questions about the budgetary sustainability of these instruments when inflation rises sharply again, as the shock of 2021–2022 illustrated: the indexation charges on French debt thus went from 3.0 billion euros in 2021 to 15.6 billion euros in 2022, according to Ecalte (2023). As Garcia & Van Rixtel (2008) recall, the issuance of indexed bonds transfers inflation risk from the investor to the sovereign issuer, which may generate *ex ante* savings via the disappearance of the inflation risk premium, but mechanically increases budgetary exposure to price shocks. The metaphor of the 'time bomb' then suggests that these securities could, in the event of a durable return of inflation, cause debt service to explode and threaten budgetary sustainability. To what extent do inflation-indexed bonds constitute a structural budgetary risk rather than an efficient debt management tool? It will be a matter first of assessing the *ex ante* advantages of these instruments for sovereign issuers, before showing that their asymmetric risk profile transforms them into a potentially destabilising contingent liability in periods of high inflation.

Inflation-indexed bonds present, in normal conditions of price stability, budgetary and technical advantages which justify their issuance by sovereign authorities. The central argument, developed by Garcia & Van Rixtel (2008), rests on the saving of the *inflation risk premium* that investors usually require on nominal bonds in order to protect themselves against uncertainty about the future evolution of prices. The empirical estimates of Grischenko & Huang (2012) for the United States over the period 2000–2012 situate this premium between 35 and 100 basis points depending on maturity, which represents a significant borrowing cost that the state can avoid by transferring inflation risk via indexation. In other words, in a period of moderate and predictable inflation, the state borrows at a real rate lower than the nominal rate it would have had to pay to compensate for inflationary risk, which improves *ex ante* the sustainability of public debt. Moreover, the analysis of the US Treasury Borrowing Advisory Committee (2023) shows that TIPS contribute to reducing the volatility of the American budget deficit, since their real cost remains stable independently of fluctuations in inflation, thus permitting a better predictability of debt charges: the data of the American Treasury indicate that past studies estimate a reduction in deficit volatility of up to 13 % thanks to the presence of TIPS in the federal debt portfolio. Finally, as BNP Paribas (2022) emphasises, the issuance of indexed bonds constitutes a signal of the credibility of monetary policy: a state which agrees to index part of its debt

manifests its confidence in the capacity of the central bank to keep inflation under control, which may contribute to anchoring markets' inflation expectations and reduce risk premia across the whole yield curve.

However, this *ex ante* attractiveness turns into an *ex post* vulnerability when inflation experiences an unforeseen and durable acceleration. As Ecalte (2023) recalls, the inflationary shock of 2022 increased the indexation charges on French debt by more than 12 billion euros in a single year, illustrating the transfer of risk from the investor to the sovereign issuer. This additional charge, which represents an immediate accounting provision even though the repayment of the increased principal will not occur until maturity, weighs directly on the budget deficit and mechanically increases the public debt-to-GDP ratio in the event of persistent inflation. Indeed, let us imagine that a state has borrowed 100 euros in nominal bonds at 2 %: if inflation suddenly reaches 8 % (unanticipated inflation), the repayment of 100 euros will become much less burdensome in terms of real purchasing power; moreover, the state will collect more tax revenue (VAT, corporation tax. . .) since prices and wages rise. This double effect — lightening of the debt and increase in revenue — constitutes the traditional channel through which inflation helps to reduce the debt burden. Conversely, if this state has borrowed massively in inflation-indexed bonds, this benefit disappears: the amount to be repaid increases exactly with inflation, cancelling out the effect of debt erosion. More precisely, the debt ratio ( $r - g$ , where  $r$  denotes the average cost of the debt and  $g$  the nominal growth rate of GDP) measures whether a state's debt will increase or decrease, as BNP Paribas (2022) explains: if  $r > g$ , the debt increases, while if  $g > r$ , the debt decreases; indeed, inflation helps a state since it increases  $g$  (nominal growth rises) without increasing  $r$  for nominal debt; by contrast, if the debt is indexed,  $r$  also increases with inflation, which cancels out the advantage. Hilscher et al. (2021) confirm this mechanism empirically for the United States: since the majority of American debt is nominal (non-indexed), inflation has historically helped governments to reduce the fiscal burden; conversely, a state which had massively issued indexed bonds loses this lever of adjustment and enters into a spiral in which every point of inflation directly increases its future repayments without generating the usual benefits. Finally, the problem has a definitive and irreversible character: a thirty-year indexed bond issued today will remain in circulation for three decades; the state cannot simply cancel it: if it wishes to be rid of it, it must buy it back on the secondary market at a potentially very high price.

Inflation-indexed bonds do not therefore constitute time bombs strictly speaking, in that they embody a transfer of budgetary risk which, if managed prudently, improves the efficiency of debt management; however, in the event of excess or of a major inflationary shock, these bonds may indeed weaken the sustainability of public finances. In an environment of price stability and monetary credibility, indexation makes it possible to reduce the borrowing cost by saving the inflation risk premium. However, this saving is paid for at the price of increased vulnerability to inflationary shocks. The 'explosive' character of these instruments therefore depends essentially on two parameters: the share of indexed debt in total liabilities and the capacity of the central bank to keep inflation under control so as to avoid a durable divergence between realised and anticipated inflation. The challenge for managers of public debt is to maintain a balanced proportion of indexed bonds which makes it possible to benefit from the advantages of diversification and credibility, without creating an excessive exposure which would effectively transform these instruments into a destabilising liability.

### Key Takeaways



French public debt exceeded 3000 billion euros in 2023; its interest burden moreover exploded from 35.5 billion euros in 2021 to 51.8 billion euros in 2022 on account of the rise in world money market rates and of inflation on indexed bonds (OATi) [iFRAP (2023)]. Moreover, the yield *spread* between French and German ten-year bonds (an indicator of perceived risk) reached 130 basis points in 2012 (the sovereign crisis), stabilised at 30–40 points between 2015 and 2021, before rising again to 69 points at the beginning of July 2024, a sign of increased anxiety on the markets [iFRAP (2023), Reuters (2024)]. The European sovereign debt crisis (2009–2018) indeed revealed the flaws of the monetary union project; fifteen years later, the euro area still displays a debt-to-GDP ratio of 87.4 % (beyond the Maastricht threshold of 60 %) [Eurostat (2025a)].

## 2.2 Shares

As the Banque de France (2021a) explains, a **share** is an ownership title issued by a capital company (for example a *société anonyme* or SA, a *société par actions simplifiée* or SAS, or again a *société en commandite par actions* or SCA) which confers on its holder the ownership of a part of the share capital of the said company: a share is therefore a right over the assets and the profits of the firm, proportional to the share of the capital that it represents. As negotiable securities, shares are issued in return for the contributions made by the members at the time of the creation of the company or on the occasion of capital increases.

### 2.2.1 Shares: characteristics and issuance

Shares therefore offer both pecuniary and political rights to their holders: from a financial point of view, they make it possible to receive **dividends**, determined by the firm's profits and voted at the general meeting. As the Banque de France (2021a) recalls, dividends are payable within nine months of the close of an accounting period; the payment of dividends is not an obligation, even where the firm has made substantial profits (the company may for instance increase its reserves in order to finance a future project). From a political point of view, shares confer a **voting right** at general meetings, allowing shareholders to take part in the governance of the company, notably on the approval of the accounts or the appointment of directors, as the Banque de France (2021a) explains. Holding shares also gives a **pre-emptive subscription right** (DPS) in the event of a capital increase: if new shares are issued, existing shareholders have priority in acquiring them for an amount called the **issue value**; this right allows former shareholders to maintain their share of the capital in the firm if they so wish. The **par value** of a share corresponds to the fraction of the share capital divided by the number of shares issued; this value is generally recorded in the company's articles of association, although law no. 98-546 of 2 July 1998 removed this obligation for joint-stock companies, allowing the issuance of shares said to be 'without par value' or of 'quotity'. The Banque de France (2021a) explains that the acquisition value may differ from the issue value, since subscribers are sometimes required to pay a share premium seen as a counterpart to the right to benefit from the reserves already built up by the company. The Banque de France (2021a) adds that the par value may likewise differ from the **open market value** of the company, which is computed by relating the revalued net assets of the firm to the total number of shares, which presupposes a delicate valuation of all the elements of the company's assets (if the company has suffered losses, the open market value is lower than the par value; conversely, if the company generates and puts into reserve a profit, the open market value is higher than the par value). Moreover, for companies listed on the stock exchange, the **stock market value**, determined as a function of the supply of and demand for securities, may likewise diverge from the open market value.

#### Focus...



**The pre-emptive subscription right (DPS).** The DPS is a legal mechanism intended to protect the existing shareholders of a company at the time of a capital increase. This right allows them to subscribe as a priority to the new shares issued in proportion to the number of shares they already hold, and thus to maintain their percentage holding in the share capital. The DPS is governed by several legal provisions, notably articles L225-132 to L225-134 of the French Commercial Code. According to article L225-132, at the time of any capital increase by contribution in cash (that is to say by payment of money), existing shareholders benefit from this priority subscription right. This means that, before the shares are offered to new investors, the shareholders already present are offered the possibility of acquiring the new shares to be issued. The main interest of the DPS lies in the **protection against dilution of the holding** of existing shareholders. Indeed, without this mechanism, a capital increase could result in a decrease in each shareholder's share in the company, which would reduce both their influence in governance and their right to future dividends. Thanks to the DPS, a shareholder may subscribe to the new shares in proportion to his current holding. For example, if a shareholder owns 10 % of a company's shares and that company issues one thousand new shares, the shareholder will have the right to acquire one hundred of these new shares in order to maintain his 10 % share.

This right is not, however, obligatory for the shareholder. The latter may choose not to exercise his pre-emptive subscription right. In that case, the DPS may be sold on the market, since it has an intrinsic value, notably for companies listed on the stock exchange. Indeed, the DPS represents a purchase option at a price often lower than the market price, which confers an economic advantage. Article L225-133 of the Commercial Code specifies that the DPS is negotiable as soon as the capital increase is decided, and this for the duration of the subscription period, which may not be shorter than five trading days after the opening of the subscription. This negotiation period offers flexibility to shareholders, allowing them either to strengthen their position by subscribing, or to sell their right and realise an immediate gain. Moreover, it is possible for the company to exclude the DPS at the time of a capital increase. In accordance with article L225-135 of the Commercial Code, the extraordinary general meeting (EGM) may, on a proposal from the board of directors or the management board, decide to remove the pre-emptive subscription right; this removal must be motivated by a corporate interest, for example in order to attract new strategic investors or to facilitate a rapid fundraising operation. The EGM must decide on this point by a two-thirds majority, and the decision to remove the right must be supported by a report from the statutory auditor or the conversion auditor, certifying that the conditions of the issue are equitable for existing shareholders. In addition, article L225-136 requires that the issue price of the new shares be fixed in such a way as not to prejudice the former shareholders, and the issuance of equity securities must be limited to 20 % of the share capital per year.

The **capital increase by new contributions** is a common procedure making it possible to strengthen the financial structure of a firm. It consists in issuing new shares in return for contributions in cash or in kind. This operation requires a decision of the extraordinary general meeting, in accordance with article L225-129 of the Commercial Code, since it modifies the company's articles of association. A capital increase may take place at par (par value) or with a share premium (see *supra*). Besides contributions in cash, other means exist for carrying out a capital increase, such as the capitalisation of reserves or the conversion of debts. The **capital increase by capitalisation of reserves** is an accounting operation which consists in converting part of the undistributed profits, of the reserves or of the share premiums into share capital. Carried out under the conditions of article L225-130 of the Commercial Code, this operation does not modify the company's equity, but makes it possible to increase the par value of the shares or to distribute free shares to existing shareholders, thereby strengthening the apparent solidity of the firm. Another option is the **capital increase by conversion of debts**, which allows a firm to transform its debts into capital in exchange for shares: this mechanism is often used to strengthen the financial structure of a company in difficulty, since it allows creditors to become shareholders rather than to receive a partial repayment in the event of judicial liquidation, as the Banque de France (2021a) explains. This type of increase is governed by strict rules, notably the necessity of a closing of the accounts and of a valuation by a statutory auditor.

#### Focus...



**The *plan d'épargne en actions* (PEA).** Created in 1992, the *plan d'épargne en actions* (equity savings plan) is a savings arrangement in France making it possible to invest in European shares while benefiting from tax advantages. It was created to encourage long-term investment in shares, while limiting the tax levied on capital gains and dividends. As the Direction de l'information légale et administrative (2024b) indicates, a PEA may be composed of shares, of company units or of eligible UCITS (undertakings for collective investment in transferable securities), on condition that these are mainly invested in European firms. The payment ceiling is 150 000 euros for a classic PEA and 225 000 euros for a PEA-PME (intended for investment in small and medium-sized enterprises). Gains are exempt from income tax after five years, but social levies remain due. This attractive tax framework thus favours saving in shares over the long run, while supporting European firms.

#### Key Takeaways



A share is an ownership title in a capital company (SA, SAS, SCA) conferring a proportional right over the assets and profits: dividends (not obligatory, payable nine months after the close of the accounting period), the right to vote at the general meeting, and the pre-emptive subscription right (DPS) at the time of a capital increase in order to maintain one's share [Banque de France (2021a)]. The capital increase by new contributions (whether in cash or in kind) requires the convening of an extraordinary general meeting; it may also be carried out by capitalisation of reserves (increasing the par value of the shares without modifying equity) or by conversion of debts (the creditors then becoming shareholders) [Banque de France (2021a)].

### 2.2.2 Determining the fair value of a share

#### Focus...



**The Gordon-Shapiro model and the *price earnings ratio*.** The **Gordon-Shapiro model**, also called the perpetual dividend growth model, constitutes a tool for valuing shares based on the discounting of future dividends on the assumption that these grow at a constant rate to infinity. As Gordon (1959) establishes, the intrinsic value of a share corresponds to the discounted sum of expected dividends, and under the assumption of perpetual growth at the constant rate  $g$ , this sum converges towards a simple expression relating the anticipated dividend, the required rate of return and the growth rate. Let  $P_0$  be the value of the share at the current date,  $D_0$  the dividend paid in the initial period,  $D_t$  the dividend paid in period  $t$ ,  $r_c$  the required rate of return on equity and  $g$  the constant growth rate of dividends with  $g < r_c$ . Under the assumption of perpetual growth, the dividend of period  $t$  is expressed as  $D_t = D_0(1+g)^t$  for every  $t \geq 1$ . The value of the share is obtained by discounting all future dividends at the rate  $r_c$ , which is written  $P_0 = \frac{D_1}{(1+r_c)} + \frac{D_2}{(1+r_c)^2} + \frac{D_3}{(1+r_c)^3} + \dots$ . Replacing each dividend by its expression as a function of the growth rate, we obtain  $P_0 = \frac{D_0(1+g)}{(1+r_c)} + \frac{D_0(1+g)^2}{(1+r_c)^2} + \frac{D_0(1+g)^3}{(1+r_c)^3} + \dots$ . Factorising  $D_0$ , this expression becomes  $P_0 = D_0 \left( \frac{(1+g)}{(1+r_c)} + \frac{(1+g)^2}{(1+r_c)^2} + \frac{(1+g)^3}{(1+r_c)^3} + \dots \right)$ . The value of the share is therefore expressed as  $P_0 = D_0 \times \frac{(1+g)}{r_c - g} = \frac{D_1}{r_c - g}$ , where  $D_1 = D_0(1+g)$  represents the anticipated dividend of the following period. This Gordon-Shapiro formula makes it possible to obtain directly the intrinsic value of a share from the future dividend, the required rate of return and the anticipated perpetual growth rate.



Consequently, the **price earnings ratio** (PER), also called the price-earnings ratio or earnings capitalisation coefficient, constitutes a complementary valuation indicator expressing the number of years of net profit that investors agree to pay in order to acquire a share. As Freihaat (2019) explains, the PER is computed by dividing the share price by the net earnings per share, and may be derived from the Gordon-Shapiro model by introducing the **payout ratio**  $d$ , defined as the proportion of earnings distributed in the form of dividends. If we denote by  $B_1$  the earnings per share of the following period, we have  $D_1 = dB_1$ , and assuming that the growth rate of earnings equals that of the dividend, the Gordon-Shapiro model becomes  $P_0 = \frac{dB_1}{r_c - g}$ , whence we deduce that the theoretical PER is expressed as  $PER = \frac{P_0}{B_1} = \frac{d}{r_c - g}$ . This relation shows that the PER increases with the payout ratio  $d$  and the growth rate  $g$ , but decreases with the required rate of return  $r_c$ , thus establishing a direct link between relative valuation and the economic fundamentals of the firm. According to the data compiled by Alvarez & Marsal (2024), the average forward PER for European firms in the Spanish IBEX index stood at 11.4 in September 2024, reflecting expectations of moderate growth in a context of high interest rates and persistent macroeconomic uncertainty.

### Exercise



**Gordon-Shapiro model.** Estimate the value of a share whose dividend is 5 euros, the expected rate of return on this security being 5 %, and the growth rate of earnings being estimated at 1 %. Deduce from this the *price earnings ratio* of this company.

**Solution.** According to the Gordon-Shapiro model, the current price  $P_0$  of a share is given by the formula  $P_0 = \frac{D_1}{r_c - g}$ , where  $D_1$  is the anticipated dividend of the following period,  $r_c$  the rate of return expected by the shareholder and  $g$  the growth rate of gross earnings per share. Otherwise put, with the data of the statement, we find  $P_0 = \frac{5}{0.05 - 0.01} = 125$ . From then on, the *price-earnings ratio* (PER) is  $PER = \frac{P_0}{D_1} = \frac{125}{5} = 25$ : all other things being equal, if the dividend remained constant at 5 euros a year, it would take 25 years to ‘repay’ the price of the share (this is obviously an abusive interpretation inasmuch as we assume that the dividend will grow, and neglect the fact that the share itself is included in our wealth; nevertheless, this interpretation gives a concrete meaning to the PER). □

Introduced notably by Sharpe (1964), the *capital asset pricing model* (CAPM) makes it possible to analyse the behaviour of a financial asset (generally a share) relative to the **market risk** relevant to that asset. For example, if we consider the Total share, the behaviour of its relevant market may be approximated by that of the CAC 40 index, which aggregates forty of the largest French market capitalisations, or, if we wish to take a broader perspective, by that of the EuroStoxx 50, the benchmark index of the euro area markets. But what do we call risk in finance? When one is an investor, one speculates on assets whose return is **uncertain**: shares, commodity derivatives, cryptocurrencies. ... The annual return of an asset then makes it possible to quantify its performance. But let us imagine an asset offering an expected average return (therefore uncertain) of 3 % a year: if there exist riskless (or almost riskless) investments which yield 5 % a year, would we not say that one is taking an inconsiderate risk in placing one’s money in such a product? Quantifying the performance of an asset therefore means taking an interest in its **risk premium**, that is to say its additional return relative to a riskless asset, as Bossu & Henrotte (2009b) explains. For example, if there exists a riskless investment offering a (certain) annual return of 5 %, then a risky investment offering an expected annual return of 12 % will present a risk premium of  $12 - 5 = 7$  %. In particular, every national market (the French market with the CAC 40 index, the German market with the DAX 40 index, the American market with the S&P 500 index...) presents a certain risk premium, measured by the difference between the average annual return of an index and that of a riskless asset (measured for example by the rate on the Treasury bills of the country or economic area in question). By way of information, if we take the S&P 500, we observe empirically that its average annual return is around 8 % (it suffices to retrieve the index returns from Yahoo Finance (2024) to be convinced of this), with of course many jolts due to economic crises or speculative bubbles.

As Bossu & Henrotte (2009b) indicates, the CAPM therefore seeks to relate the risk premium of a given asset  $A$  to that of its relevant market  $M$ , according to the formula  $\bar{r}_A - r_f = \beta(\bar{r}_M - r_f)$ , where  $\bar{r}_A$  is the expected return of the asset  $A$ ,  $\bar{r}_M$  that of the relevant market and  $r_f$  the rate of the riskless investment. The coefficient  $\beta$  then measures the **reactivity of the risk** of the asset relative to a fluctuation in market risk. Let us take a numerical example: let us imagine for a given asset  $A$  a market offering an expected return of  $\bar{r}_M = 6$  %, with a riskless rate of  $r_f = 2$  %. If the risk premium of the market (here  $6 - 2 = 4$  %) increases by one point and moves to  $7 - 2 = 5$  %, then the risk premium of the asset will move from  $4\beta$  % to  $5\beta$  %, that is a rise of  $\beta$  percentage point(s). And this is the whole interest of the model according to Chancellor (2021):

— If  $\beta$  is **lower than 1**, this means that a variation of one point in the market risk premium will be associated with a variation of less than one point in that of the asset; the asset will therefore **smooth the fluctuations** of the market:

if the market plunges, the risk premium of the asset should fall a little less sharply than the market, whereas if the market soars, the premium of the asset should grow but in a less pronounced way. Such an asset therefore plays the role of a shield. In particular, an asset whose  $\beta$  is zero is riskless.

- Conversely, if  $\beta$  is **greater than 1**, a variation of one point in the market risk premium will be correlated with a variation of more than one point in that of the asset; the asset will therefore **amplify the fluctuations** of the market: if the market plunges, the risk premium of the asset should fall even more sharply than the market, but if the market rises, the premium of the asset will soar all the more; a perfect asset for the ‘risk-loving’ profiles who take a ‘*high risk, high reward*’ view.

The CAPM is moreover a precious tool for the financial analyst, who can determine whether or not an asset presents an investment opportunity, as Chancellier (2021) indicates. Consider a market (typically, a stock market index) offering an expected average return  $\bar{r}_M = 6\%$ , the riskless rate being  $r_f = 2\%$ . We then take an interest in an asset on this market (a share in a firm constituting the index, as it happens), which we know presents a coefficient  $\beta$  of 1.2 and pays its shareholders an annual dividend of 4%. Suppose that the price of such a share is 100 euros: **is the share over-valued or under-valued?** Thanks to the CAPM, we can sketch an answer: the expected return of the asset ought to be  $\bar{r}_A = r_f + \beta(\bar{r}_M - r_f) = 2\% + 1.2 \times (6\% - 2\%)$ , that is 6,8%. The asset therefore presents, according to the CAPM at least, a return (dividends paid) lower than what it ought to offer, so that its price would tend to be over-valued and it is then appropriate to avoid buying it.

While the previous numerical example makes it possible to understand the use of the CAPM, it nevertheless tells us nothing about how the values used are concretely obtained. To be sure, it is relatively simple to determine the riskless rate by consulting for example the rate on Treasury bills or the Euribor / Libor rates (that is to say the interbank rates for the euro area and the United States, respectively) according to Hossen (2020). What then of the determination of the expected returns and above all of the coefficient  $\beta$ ? Most often, the expected return of the asset as of the market is determined in the light of the **historical return**: one considers a certain period, generally of several years, for which one determines the average return. This return may be computed from daily, weekly or monthly returns alike. It should, however, be noted that the granularity of the data must be coherent: if one considers the average daily return for the asset and the market, the riskless rate must likewise be expressed on a daily basis. As a reminder, if the riskless rate amounts to  $x\%$  a year, then the equivalent daily riskless rate is equal to  $(100 \times (1 + x\%)^{1/365} - 100)\%$ , while the equivalent weekly riskless rate is equal to  $(100 \times (1 + x\%)^{1/52} - 100)\%$ . Once one has determined the average return of the asset and of the market, the ratio of the risk premium of the asset to that of the market  $\frac{\bar{r}_A - r_f}{\bar{r}_M - r_f}$  gives us a first estimate of the value of the coefficient. In practice, however, the coefficient is obtained with the help of **econometric techniques**. For example, if one wishes to estimate the  $\beta$  coefficient of Total, one may compute the daily returns of Total over the last five years (for example), those of the market and look up the evolution of the daily Euribor rates over that same period. As Hossen (2020) explains, one will then compute, for each day, the difference between the daily return of Total and Euribor (variable Y) on the one hand, and that between the daily return of the CAC 40 and Euribor (variable X) on the other. The quotient of the empirical covariance of X and Y over the empirical variance of X will then provide an estimate of the coefficient of Total (here one is merely applying the method of ordinary least squares to a simple linear model without a constant in which the risk premium of Total constitutes the explained variable and the risk premium of the CAC 40 the explanatory variable).

### Exercise



**CAPM.** We place ourselves under the assumptions of the capital asset pricing model. Suppose that the market portfolio is composed of three risky securities, denoted A, B and C. The values (in billions of euros) of these assets in the market portfolio and the expected returns  $\mu_A$ ,  $\mu_B$  and  $\mu_C$  are given by the following table:

| Asset           | A   | B     | C   |
|-----------------|-----|-------|-----|
| Capitalisation  | 250 | 150   | 100 |
| Expected return | 3 % | 2.5 % | 5 % |

Give the weight vector of the market portfolio M and deduce from it its expected return. Compute for each asset its  $\beta$ , knowing that the riskless return is  $r_f = 3\%$ . Consequently, consider a portfolio in which one places (values in thousands of euros), 10 in the riskless asset, 45 in asset A, 27 in asset B and 18 in asset C. Give then the  $\beta$  of such a portfolio and its expected return. What do you make of it?

**Solution.** The capitalisation of the market is equal to the sum of the capitalisations of each asset constituting the market, namely  $250 + 150 + 100 = 500$  billion euros. Of these 500 billion euros, the share of asset A is  $\frac{250}{500} = 50\%$ , that of asset B is  $\frac{150}{500} = 30\%$  and that of asset C is  $\frac{100}{500} = 20\%$ . The weight vector of the market portfolio is therefore written

$(\pi_A, \pi_B, \pi_C) = (0.5, 0.3, 0.2)$ . From then on, the expected return of the market portfolio  $\mu_M$  is the weighted average of each asset constituting the market, namely  $\mu_M = \pi_A \mu_A + \pi_B \mu_B + \pi_C \mu_C$ , that is  $\mu_M = 0.5 \times 3\% + 0.3 \times 2.5\% + 0.2 \times 5\% = 3.25\%$ . We then deduce the  $\beta$  of each asset  $i$  by the formula  $\beta_i = \frac{\mu_i - r_f}{\mu_M - r_f}$ . We therefore have  $\beta_A = \frac{0.03 - 0.03}{0.0325 - 0.03} = 0$  (this is a risky asset which yields as much as a riskless asset: no wonder its  $\beta$  is zero);  $\beta_B = \frac{0.025 - 0.03}{0.0325 - 0.03} = -2$  (a negative  $\beta$  is a hedging asset: it varies in the opposite direction to the market, and naturally its expected return is lower than the riskless return);  $\beta_C = \frac{0.05 - 0.03}{0.0325 - 0.03} = 8$  (this is an ultra-aggressive asset which very strongly amplifies the fluctuations of the market, for better and for worse). Consequently, we consider a portfolio in which one places (values in thousands of euros), 10 in the riskless asset, 45 in asset A, 27 in asset B and 18 in asset C. We have therefore placed in total  $10 + 45 + 27 + 18 = 100$  thousand euros. We may then compute the expected return of this portfolio  $\mu_P$  as  $\frac{10}{100} \times 3\% + \frac{45}{100} \times 3\% + \frac{27}{100} \times 2.5\% + \frac{18}{100} \times 5\%$ , that is  $\mu_P = 3.225\%$ . We therefore obtain for the  $\beta$  of this portfolio the value  $\beta_P = \frac{\mu_P - r_f}{\mu_M - r_f}$ , that is to say  $\beta_P = \frac{0.03225 - 0.03}{0.0325 - 0.03} = 0.9$ . This value being lower than 1, we are dealing with a rather defensive portfolio which smooths the variations of the market (the risk premium of the portfolio increases when that of the market increases, but less than the latter; conversely, when the market plunges, the portfolio plunges as well but less than the market).  $\square$

While the CAPM provides us with a wealth of information on the risk of an asset, the problem is that this model rests on an ideal axiomatic framework, very far removed from the reality of markets. Indeed, the CAPM belongs to the **modern portfolio theory** developed by Markowitz (1952), a theory in which the assumptions are simplifying ones. First of all, one assumes that one is in **pure and perfect competition**: no transaction costs, no taxes, no barriers to entry (all assets can be traded and all investors have access to the market), no information asymmetry (therefore no privileged information for stock market *insiders*)... This is after all a very utopian framework, which means that the CAPM is, at best, no more than a simplified approximation of the real world. Even in practice, approximating the behaviour of the market by that of the index is all in all a very imperfect approximation of reality, made necessary, however, by the difficulty of finding other, more relevant estimates in open access. But more than this, can one really say that the risk premium of an asset depends exclusively (and linearly) on that of its relevant market? This simplistic assumption is refuted in reality. Indeed, one observes empirically that value shares and small capitalisations regularly outperform the markets: this is the reason why Fama & French (1993) completed the CAPM by seeking to explain the risk premium of an asset not only by that of its relevant market, but also by a **size factor** (called SMB for ‘*small minus big*’, which measures the difference in return between the securities of small market capitalisations and those of large ones) and a **value factor** (called HML for ‘*high minus low*’, which measures the difference in return between value firms, that is to say those whose ratio of book value to market value, or *book-to-market* (BTM) ratio, is low, and growth firms, whose BTM ratio is higher): this is the **three-factor Fama-French model**. Fama & French (1993) also proposed an extension of their model, in order to take account of the fact that the highest returns generally come from firms whose investment policy is more prudent. This is why the two authors added to the three-factor model an **investment factor** (called CMA for ‘*conservative minus aggressive*’, which measures the difference in return between firms which conserve their investment and those which invest massively) and a **profitability factor** (called RMW for ‘*robust minus weak*’, which measures the difference in return between the portfolios of highest profitability and those of lowest profitability): this is the **five-factor Fama-French model**.

### Key Takeaways



The Gordon-Shapiro model values a share by discounting dividends growing perpetually:  $P_0 = \frac{D_1}{r_c - g}$  where  $D_1$  is the anticipated dividend,  $r_c$  the return required by shareholders and  $g < r_c$  the estimated growth rate of earnings [Gordon (1959)]. The PER (*price earnings ratio*) is deduced from it via the payout ratio  $d$ , that is to say the share of anticipated earnings  $B_1$  distributed in the form of dividends ( $D_1 = dB_1$ ):  $PER = \frac{P_0}{B_1} = \frac{d}{r_c - g}$  [Freihat (2019)]. Consequently, the CAPM, introduced by Sharpe (1964), relates the risk premium of an asset to that of the market:  $\bar{r}_A - r_f = \beta(\bar{r}_M - r_f)$  (where  $\bar{r}_A$  is the average return of the asset,  $\bar{r}_M$  that of the market, and  $r_f$  the return of a riskless investment). The coefficient  $\beta$  then measures the systematic reactivity of the asset ( $\beta < 1$  for a defensive asset smoothing the fluctuations of the market;  $\beta > 1$  for an aggressive asset amplifying them) [Bossu & Henrotte (2009b)]. This model is completed by the three-factor Fama-French model (the addition of a size factor to compare firms of small and large capitalisation, and of a value factor to compare high and low *book-to-market* ratios) and the five-factor model (the addition of an investment factor for aggressive investment relative to a conservative policy of dividend redistribution, and of a profitability factor to compare ‘robust’ and ‘weak’ portfolios) [Fama & French (1993)].



### 2.2.3 Investment strategies: short selling, leverage

Setting about stock market prediction therefore means understanding the determinants of the price of an asset. Naturally, one would tend to say that prices depend on the macroeconomic, geopolitical or societal context, or even that they would depend on private information (a new product, a new regulation), as Petrusheva & Jordanoski (2016) indicate. Analysing the market in all its dimensions is precisely the object of **fundamental analysis** (FA): without this global view, it would be impossible to sense correctly ‘the mood of the market’. From then on, is predicting future prices solely in the light of past prices and traded volumes a pious hope? Yet this is the wager of **technical analysis** (TA). Otherwise put, no macroeconomic element enters into technical analysis. Technical analysis is therefore ontologically poorer than fundamental analysis, in that it focuses on two microeconomic variables only. Technical analysis then rests on the postulate that price and volumes form the best synthesis of the market’s appreciation of an asset, as Petrusheva & Jordanoski (2016) explain. Indeed, for a price to be quoted, there must be sellers and buyers, that is to say two forces with fundamentally antagonistic opinions: the price therefore synthesises the confrontation between sellers and buyers, while volumes materialise the enthusiasm for this combat; each transaction is indeed the result of a conflict resolved between a buyer and a seller who have succeeded in agreeing on an exchange price, as Schwager (1995) analyses.

The sinews of war of technical analysis are therefore not some macroeconomic or scientific component: the foundation of TA is purely psychological and is apprehended according to *band wagon* patterns arising from group psychology. The behaviour of agents in a risky universe is indeed generally instilled by the group; this mimetic (that is to say herd-like) behaviour generally prevents the making of free-will decisions, these finding themselves marginalised by the group. Thus, in group psychology, the law of the greatest number generally imposes itself as the natural inclination accepted by the members of the group. Keynes (1936) analyses this phenomenon with his allegory of the **beauty contest**, with which he compares financial markets to a beauty contest in a newspaper, where the participants must choose, among several photographs, the one which will be judged the most beautiful by the majority of the other participants. The point for each participant is therefore not to vote according to his own aesthetic preference (what he considers the most beautiful), but to anticipate which photograph will be the most popular among the whole set of other participants; in other words, instead of choosing the photograph he personally prefers, each player tries to guess which photograph the majority of the other participants will choose. This introduces an additional level of reflection: one does not vote for one’s own preference, but for what one thinks the majority will prefer. And when financial markets are dynamic, they too do not escape this rule: the more sellers (resp. buyers) there are, the more the buyers (resp. the sellers) will be put in a minority. Subject to the pressure of the sellers (resp. the buyers), the buyers (resp. the sellers) will then doubt their analysis. Their bullish (resp. bearish) conviction will be less strong because it will not be sustained by other buyers (resp. sellers); rapidly, demand (resp. supply) will contract, which will exert downward (resp. upward) pressure on prices. The study of the history of prices then makes it possible to identify the extent to which buyers and sellers have agreed over time. More precisely, this observation indicates which of the two groups (seller or buyer) is serene because it is winning and which is losing because prices are not evolving in the direction of its position. Now, confronted with loss, everyone reacts in the same way; the difference is merely a question of degree: stress and fear obscure the faculties of sound reflection, and one gives in to panic or to the hope of making good one’s losses. It is this psychological bias that *traders* seek to exploit: the point is then to imitate the market; as Keynes (1936) explains, everyone tries to anticipate not the true value of an asset, but the reactions of other investors in the short term, which can entail a disconnection between the fundamental value of assets and their market price, leading to speculative bubbles.

Thus, because technical analysis is as close as possible to the psychological reactions of the market, it is more appropriate to *scalping* (positions of five minutes to one hour), to *intraday* trading (positions of one hour to one day) or even to *swing* trading (positions of two days to two months). Otherwise put, technical analysis helps to determine the opportune moment to buy or sell in the short term (next year, the state of portfolios will in all likelihood be different). The underlying trend, for its part, rests above all on the economic context. Thus, while fundamental analysis makes it possible to determine the underlying trend of the market, technical analysis, for its part, makes it possible to anticipate the short-term fluctuations of the market (notably in order to optimise one’s entry point). Technical analysis is therefore sometimes perceived by short-termist *traders* as self-sufficient: it matters little whether the asset traded has good fundamentals or not; the essential thing is to anticipate short-term fluctuations in order to profit from them, as Hossen (2020) analyses. Technical analysis then rests on a twofold approach, whose aim is to study past prices and the volumes traded up to that point: on the one hand, **chart analysis** aims to analyse the figures (notably Japanese candlesticks) representing market prices over a determined period; on the other hand, **technical analysis** strictly speaking uses complex mathematical indicators. As a general rule, chart analysis gives far better results than technical analysis, as Hossen (2020) indicates. But a chart analysis *is seen*: it is difficult to automate for a computer, unlike the analysis of purely mathematical technical indicators. So technical analysis, even

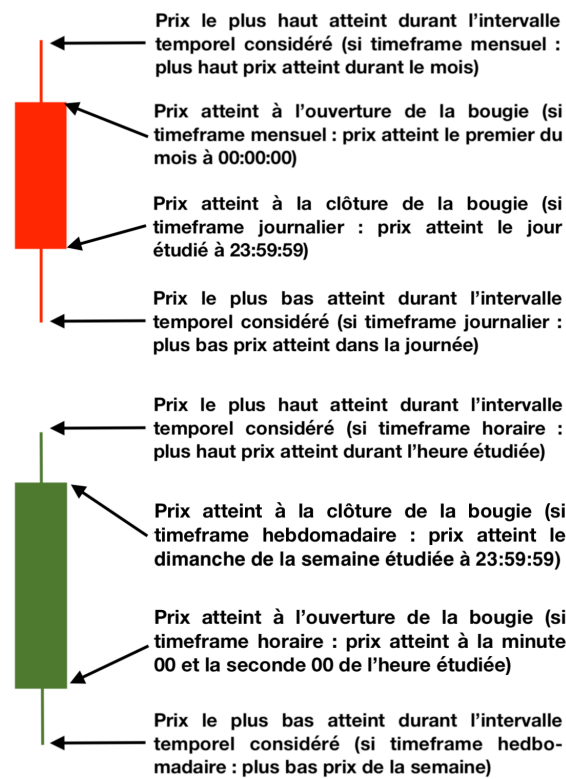


Figure 3.3: Representation of stock market prices known as 'Japanese candles' or 'Japanese candlesticks': colour (in general: red and green, or black and white), *open* (opening price of the candle), *high* (highest price reached on the candle), *low* (lowest price reached on the candle) and *close* (closing price of the candle).

though it presents poorer results than chart analysis, is automatable and makes it possible to *trade* many products at the same time, unlike chart analysis, which forces one to concentrate on one or two products at a time (since the *trader* has to look at them). *In fine*, chart analysis is more linked to **performance**, while technical analysis is more linked to **volume** according to Hossen (2020): with chartism, one takes few positions, but wins big; with indicators, one wins a little but takes very many positions.

But *trading* does not consist solely in buying an asset in order to resell it later. Other approaches, riskier and cumulative, may be employed: leverage on the one hand, and short selling on the other. Such strategies multiply the chances of gain but also the risks of loss. What do these two strategies involve?

- **Short selling:** one would finally like to be able to speculate on a fall in a given asset, even if one does not have the said asset in one's portfolio; this is the principle of short selling which IG (2024b) explains. Let us give an example with bitcoin (BTC): I *short sell* 1 BTC at 40 000 euros because I anticipate that its price will fall to 30 000 euros (it is as if I were borrowing 1 BTC from the broker; I am therefore in debt to the tune of 1 BTC, but I have 40 000 euros in exchange). Two cases are then possible: if BTC falls to 30 000 euros (this is the favourable case), I need only 30 000 euros to liquidate my debt and I gain 10 000 euros; by contrast, if BTC rises again to 50 000 euros (the unfavourable case), I need 50 000 euros to liquidate my debt and I lose 10 000 euros. And if the price rises too far, my position may even be liquidated, which makes me lose all the money I have committed in my portfolio.
- **Leverage (leverage effect):** if one has only 4 000 euros in one's account, it is nevertheless possible to buy a whole bitcoin at 40 000 euros, that is 10 times more; the leverage is then said to be equal to 10. A leverage of 1 therefore designates an absence of leverage. Concretely, with leverage, it is as if the broker (for example Binance) lent money to *traders* so that they could invest more than their capital allows them. This is a risky strategy which multiplies both gains and losses, as IG (2024a) explains. Suppose that on 1 May, one buys 1 BTC worth 40 000 euros with 4000 euros (leverage 10):
  - Suppose that on 2 May, 1 BTC is now worth 42 000 euros (that is a rise of 5 % since the previous day). If one resells at that moment, then one receives  $42\,000 - 40\,000 = 2\,000$  euros (from which the fees of the *broker*

who lent the money must then be subtracted). The gross gain of the portfolio, before fees, is therefore 50 % ( $10 \times 5 \% = 50 \%$ , whence the leverage of 10).

- Suppose that on 3 May, 1 BTC is worth no more than 38 000 euros (that is a fall of 5 % since 1 May). If one resells at that moment, then one owes the *broker*  $40\,000 - 38\,000 = 2\,000$  euros (plus the fees of the *broker* who lent the money). The gross loss of the portfolio, before fees, is therefore  $10 \times 5 \% = 50 \%$ .

With our previous example (purchase of 1 BTC at 40 000 euros with leverage 10), it is therefore enough for the price of Bitcoin to fall to 36 000 euros for you to owe 4000 euros to the broker and for your position to be liquidated. In reality, the liquidation price is even a little higher than 36 000 euros given brokerage fees. More generally, if one borrows with a leverage of  $n$ , it is enough for the price to fall by at most  $(100/n) \%$  for one's position to be liquidated and, in reality, the liquidation threshold is even lower in practice given brokerage fees, as Hossen (2020) indicates.

### Focus...



**Market order, limit order.** A **market order** consists in buying or selling a financial asset at the best price available on the market at the moment of transmission, guaranteeing immediate execution but without precise control over the final price, whereas a **limit order** specifies a maximum purchase or minimum sale price, being executed only if this threshold is reached or favourably crossed. As Hossen (2020) explains, market orders take priority over all other limit orders, the latter then being executed in chronological order: it is therefore possible that a limit order, although triggered, may *in fine* not be executed or be only partially executed (insufficient liquidity at the limit price). The market order is therefore reasonable on liquid and deep markets such as the CAC 40 or S&P 500 indices, where the risk of *slippage* — the gap between the expected and executed price — remains minimal for orders of small size (less than 0.1 % of the average daily volume according to Hossen (2020)), permitting rapid entry or exit at times of urgent technical signals or arbitrages; for example, a *retail* trader executing one hundred Apple shares will benefit from near-instantaneous execution without significant impact. Conversely, as Hossen (2020) analyses, the limit order is called for on illiquid markets (*small caps*, sovereign bonds or low-volume cryptocurrencies), where a large market order (greater than 1 % of the order book) could durably displace the price via market impact. Recall that the **order book** is the real-time electronic register which summarises the purchase orders, at decreasing prices, and sale orders, at increasing prices, awaiting execution for a specific financial asset: it therefore indicates for each price level the number of orders and the associated quantities so as to reflect supply and demand on the market; it is the tool of choice for apprehending the liquidity of a market.

### Key Takeaways



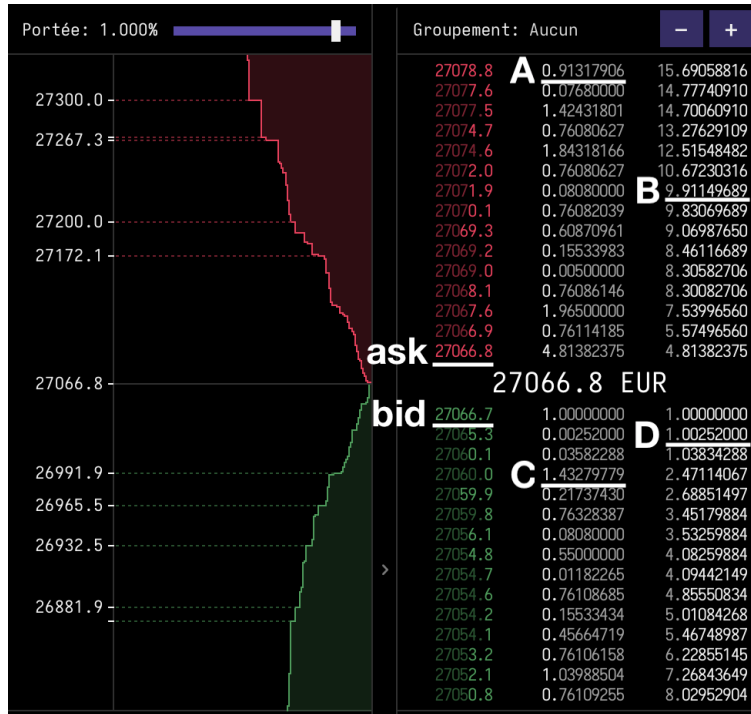
In order to predict stock market movements, fundamental analysis examines the macroeconomic and geopolitical context and private information, while technical analysis rests exclusively on the history of prices and volumes in order to exploit mimetic psychological biases via chart analysis or mathematical indicators so as to anticipate short-term fluctuations [Petrusheva & Jordanoski (2016)]. Thus, investment strategies go beyond simple purchase: short selling consists in borrowing an asset in order to sell it high in anticipation of a fall, then buying it back low in order to return it, with limited gains but unlimited losses and a risk of liquidation [IG (2024b)]. Complementarily, leverage multiplies exposure to an asset, amplifying gains and losses [IG (2024a)]. Finally, execution orders distinguish the market order, which executes immediately at the best price available (with priority but with a risk of *slippage* on illiquid markets), from the limit order, which executes only at a specified price, the order book indicating the levels of supply and demand by price level [Hossen (2020)].

## 2.3 Risk diversification

**Portfolio management** rests on a fundamental principle: combining assets whose returns are not perfectly correlated makes it possible to reduce overall risk without sacrificing expected return. This analytical framework, extended by the capital asset pricing model (CAPM) of Sharpe (1964) distinguishing non-diversifiable systematic risk from eliminable idiosyncratic risk, grounds the theoretical superiority of passive index management and justifies recourse to stock market indices as benchmark instruments for the optimal allocation of capital.

### 2.3.1 An introduction to portfolio theory

Combining financial assets whose returns do not vary in perfectly synchronous fashion makes it possible to reduce the overall risk of the portfolio without necessarily sacrificing expected return: this is what Markowitz (1952)



**Reading:** (A) 0.91317906 bitcoin is offered at the price of 27078.8 euros; (B) 9.91149689 BTC are offered at the price of 27071.9 euros or less; (C) 1.43279779 bitcoin is demanded at the price of 27060.0 euros; (D) 1.00252 bitcoin is demanded at the price of 27065.3 euros or more. Moreover, the *mid* price is equal to the average between the *ask* and the *bid*: it is an artificial price which therefore does not exist strictly speaking.

Figure 3.4: Order book of the BTC/EUR pair on the Kraken platform on 28 January 2021 at 21:22

establishes in his article, which constitutes the birth certificate of **modern portfolio theory** (MPT). Consider indeed a portfolio composed of two risky financial assets A and B, with respective random returns  $R_A$  and  $R_B$ , expected gains  $\mu_A = \mathbb{E}[R_A]$  and  $\mu_B = \mathbb{E}[R_B]$ , and standard deviations  $\sigma_A$  and  $\sigma_B$ . An investor allocates a fraction  $w$  (between 0 and 1) of his wealth to asset A and the remaining fraction  $1 - w$  to asset B. The **expected return of the portfolio**  $\mu_P$  is then a linear combination of the individual returns:  $\mu_P = w\mu_A + (1 - w)\mu_B$ . The **variance of the portfolio**, by contrast, is not a simple linear combination of the variances: it involves a cross term which reflects the way in which the two assets vary jointly, measured by their **covariance**  $\sigma_{AB} = \text{Cov}(R_A, R_B) = \rho_{AB}\sigma_A\sigma_B$ , where  $\rho_{AB} \in [-1, 1]$  denotes the **correlation coefficient** between the two assets. The variance of the portfolio is then written  $\sigma_P^2 = w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 + 2w(1 - w)\rho_{AB}\sigma_A\sigma_B$ . As Markowitz (1952) emphasises, it is precisely this cross term which makes diversification advantageous: as soon as  $\rho_{AB} < 1$ , the variance of the portfolio is strictly lower than the weighted average of the individual variances. In the limiting case where  $\rho_{AB} = -1$ , that is to say when the two assets move in exactly opposite directions, there even exists a weighting  $w^*$  which entirely cancels the variance of the portfolio. Indeed, in this limiting case,  $\sigma_P^2 = w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 - 2w(1 - w)\sigma_A\sigma_B = (w\sigma_A - (1 - w)\sigma_B)^2$ , so that  $\sigma_P^2 = 0$  for  $w\sigma_A = (1 - w)\sigma_B$ , that is to say for  $w^* = \frac{\sigma_B}{\sigma_A + \sigma_B}$ . This phenomenon illustrates the fact that the reduction of risk is all the greater the more the assets are anti-correlated. Conversely, when  $\rho_{AB} = 1$ , the assets move in step and diversification procures no benefit in terms of risk reduction: the variance of the portfolio then reduces to  $\sigma_P^2 = w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 - 2w(1 - w)\sigma_A\sigma_B = (w\sigma_A + (1 - w)\sigma_B)^2$ , whence  $\sigma_P = w\sigma_A + (1 - w)\sigma_B$ , that is to say a strict linear combination of the individual risks. It is therefore the correlation structure between assets which constitutes the true lever of portfolio management, and not the mere fact of holding several securities: as Markowitz (1952) specifies, a portfolio composed of sixty railway securities is no better diversified than a portfolio of twenty securities belonging to distinct sectors, since firms in the same sector tend to suffer the same systemic shocks simultaneously. This approach succeeded the **expected returns rule** of Williams (1938), which prescribed that investors concentrate their wealth on the single asset offering the maximum expected return. Now this latter rule totally ignores variance and the interactions between assets, leading to sub-optimal portfolios because they are needlessly risky.

One of the most enduring contributions of Markowitz (1952) is the notion of the **efficient frontier**, whose rigorous analytical derivation was proposed by Merton (1972). In the mean-variance space  $(\sigma_P^2, \mu_P)$  — or equivalently in the mean-standard-deviation space  $(\sigma_P, \mu_P)$ , which is the most widespread graphical convention since Markowitz (1952)

— the set of feasible portfolios forms a ‘bullet’ delimited by a curve called the **portfolio frontier**. In the two-asset case that we have introduced, this curve is parameterised by  $\omega \in [0, 1]$  and takes the form of a hyperbola in the plane  $(\sigma_P, \mu_P)$ . The upper branch of this hyperbola constitutes the **efficient frontier** in the sense of Markowitz (1952): it groups together the whole set of portfolios which, for a given level of risk  $\sigma_P$ , maximise the expected return  $\mu_P$ , or symmetrically, which for a fixed expected return  $\mu_P$  minimise risk. A portfolio is said to be **dominant** in the sense of second-order stochastic dominance if it offers an expected return greater than or equal to that of another portfolio for identical risk, or a risk lower than or equal for an identical expected return. The efficient frontier is thus the set of non-dominated portfolios. Merton (1972) demonstrates formally, with the help of Lagrange multipliers, that in the general case with  $m$  risky assets, the efficient frontier is a parabola in the plane  $(\sigma_P^2, \mu_P)$ : its parametric equation is written  $\sigma_P^2 = \frac{C\mu_P^2 - 2A\mu_P + B}{D}$ , where  $A, B, C$  and  $D = BC - A^2 > 0$  are scalars determined by the variance-covariance matrix of the risky assets and by the vector of their expected returns.

The analytical framework of Markowitz (1952) was extended by the simultaneous work of Sharpe (1964) and Lintner (1965), who introduced the riskless asset into the modelling and derived, under conditions of general market equilibrium, the capital asset pricing model (CAPM). The central contribution of Sharpe (1964) consists in showing that, when all investors share the same expectations about the returns, variances and covariances of risky assets — what is called the **homogeneous beliefs assumption** — and when they can borrow and lend freely at the riskless rate  $r_f$ , then the efficient frontier is transformed into a **capital market line** (CML): every efficient portfolio is a linear combination of the market portfolio  $M$  — which contains all risky assets in proportions equal to their weights in total market capitalisation — and of the riskless asset. This **two-fund separation theorem**, generalised analytically by Merton (1972), implies that the investment decision breaks down into two strictly independent stages: on the one hand, the determination of the optimal risky portfolio  $M$  (which is the same for all investors, whatever their preferences), and on the other hand, the choice of the desired overall level of risk, which determines the proportion invested in  $M$  versus the riskless asset. On the CML, the expected return of an efficient portfolio  $P$  is written  $\mu_P = r_f + \frac{\mu_M - r_f}{\sigma_M} \sigma_P$ , where the ratio  $(\mu_M - r_f)/\sigma_M$  is called the **market Sharpe ratio** and represents the risk premium per unit of total risk. In equilibrium, Sharpe (1964) and Lintner (1965) show that the expected return of each individual asset  $i$  is determined not by its total risk  $\sigma_i$ , but by its **systematic risk**, measured by the coefficient  $\beta_i = \text{Cov}(R_i, R_M)/\sigma_M^2$ : the part of the risk specific to the asset (**idiosyncratic risk** or diversifiable risk) can indeed be entirely eliminated by diversification, so that the market remunerates only non-diversifiable risk. The CAPM equation is then written  $\mathbb{E}[R_i] = r_f + \beta_i (\mu_M - r_f)$ , where  $\mu_M - r_f$  is the **market risk premium**. It should, however, be recalled, as Fama & French (2004) do, that the empirical validity of the CAPM is seriously called into question by the data: studies bearing on American stock markets show that the linear relation between  $\beta$  and expected return is too flat relative to the theoretical prediction, and that other variables — such as the size of the market capitalisation or the ratio of book value to market value (*book-to-market ratio*) — retain a residual explanatory power over cross-sectional returns, which contradicts the assumption that  $\beta$  is the only relevant risk factor. Notwithstanding these empirical limitations, which Fama & French (2004) attribute sometimes to behavioural inefficiencies of markets, sometimes to rational risk premia not captured by a single-factor model, the CAPM remains a valuable conceptual tool: the portfolio management models of collective funds authorised to market their units in France often rest on an extended or multifactorial version of the CAPM logic, and the assets under management of alternative investment funds (AIF) and of undertakings for collective investment in transferable securities (UCITS), generally managed according to a quantitative optimisation approach, reached 1 984 billion euros in France at the end of 2023, according to the AMF (2024).

### Exercise



**Two-risky-asset portfolio, efficient frontier and capital market line.** Consider two risky assets  $A_1$  and  $A_2$  with expected average annual returns  $\mu_1 = 10\%$  and  $\mu_2 = 6\%$  and estimated annual standard deviations  $\sigma_1 = 20\%$  and  $\sigma_2 = 12\%$  respectively. The correlation coefficient between the returns of the two assets is assumed to be equal to  $\rho_{12} = -0.40$ . We further have a riskless asset with return  $r_f = 2\%$ .

- (i) Recall the general formula for the expected return and the variance of a portfolio  $P$  made up of a fraction  $w \in \mathbb{R}$  invested in  $A_1$  and  $1 - w$  in  $A_2$ . Compute the covariance  $\sigma_{12}$ . Interpret the sign.
- (ii) Determine analytically the minimum variance portfolio (MVP). Compute its expected return and its standard deviation. Compare with investment in a single asset and conclude on the benefit of diversification.
- (iii) Show analytically that the return of the minimum variance portfolio and the variance of that same portfolio satisfy respectively  $\mu_{MVP} = \mu_2 + \frac{(\mu_1 - \mu_2)(\sigma_2^2 - \sigma_{12})}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$  and  $\sigma_{MVP}^2 = \frac{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$ .

- (iv) Determine analytically the portfolio frontier. Specify which part of the frontier constitutes the efficient frontier in the sense of Markowitz.
- (v) Determine the tangency portfolio T. Compute its Sharpe ratio and compare it with those of the assets taken in isolation.
- (vi) Write the equation of the capital market line (CML). Interpret its slope economically. Synthesise the results of this exercise graphically.

**Solution.** (i) Let  $w$  be the real number corresponding to the fraction of wealth invested in  $A_1$ , so that  $1 - w$  is invested in  $A_2$  (short selling corresponds to the cases  $w < 0$  or  $w > 1$ ). The expected return of the portfolio is then  $\mu_P(w) = w\mu_1 + (1 - w)\mu_2$ , while the variance of the portfolio is written  $\sigma_P^2(w) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\sigma_{12}$ , where  $\sigma_{12} = \rho_{12}\sigma_1\sigma_2$  denotes the covariance between  $A_1$  and  $A_2$ . We have directly  $\sigma_{12} = -0.40 \times 0.20 \times 0.12 = -0.0096$ . The negative sign means that the two assets move in opposite directions on average: when  $A_1$  outperforms,  $A_2$  tends to underperform, and conversely. It is precisely this negative correlation which will make diversification effective: the losses on one asset are partially offset by the gains on the other.

- (ii) In order to minimise  $\sigma_P^2(w)$ , we set its derivative with respect to  $w$  to zero. Now we have:

$$\begin{aligned}\frac{d\sigma_P^2(w)}{dw} &= 2w\sigma_1^2 - 2(1 - w)\sigma_2^2 + 2(1 - 2w)\sigma_{12} \\ &= 2w\sigma_1^2 + 2w\sigma_2^2 - 2\sigma_2^2 - 4w\sigma_{12} + 2\sigma_{12} \\ &= 2w(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - 2(\sigma_2^2 - \sigma_{12})\end{aligned}$$

We deduce the optimal weighting  $w_{MVP}$  by setting this value equal to 0. It then follows that:

$$\begin{aligned}w_{MVP} &= \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \\ &= \frac{0.12^2 - (-0.0096)}{0.20^2 + 0.12^2 - 2 \times (-0.0096)} \\ &= \frac{0.0144 + 0.0096}{0.0400 + 0.0144 + 0.0192} \approx 32.61\%\end{aligned}$$

In order to minimise variance, one therefore invests 32.61 % of one's wealth in asset  $A_1$  and 67.39 % of one's wealth in asset  $A_2$ . The standard deviation and the expected return of the minimum variance portfolio (MVP) stand respectively at  $\sigma_{MVP} = \sqrt{0.3261^2 \times 0.20^2 + 0.6739^2 \times 0.12^2 + 2 \times 0.3261 \times 0.6739 \times (-0.0096)} \approx 8.11\%$  and  $\mu_{MVP} = 0.3261 \times 10\% + 0.6739 \times 6\% \approx 7.30\%$ . This result illustrates the benefit of diversification: invested alone,  $A_1$  presents a standard deviation of 20 % and  $A_2$  of 12 %; the optimal combination reduces the risk to 8.11 %, that is a standard deviation lower than that of the less risky of the two assets taken in isolation.

- (iii) We can give a theoretical expression for  $\mu_{MVP}$  and  $\sigma_{MVP}$ . On the one hand, we have:

$$\begin{aligned}\mu_{MVP} &= w_{MVP}\mu_1 + (1 - w_{MVP})\mu_2 \\ &= w_{MVP}\mu_1 + \mu_2 - w_{MVP}\mu_2 \\ &= \mu_2 + w_{MVP}(\mu_1 - \mu_2) \\ &= \mu_2 + \frac{(\mu_1 - \mu_2)(\sigma_2^2 - \sigma_{12})}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}\end{aligned}$$

On the other hand, we have  $\sigma_{MVP}^2 = w_{MVP}^2\sigma_1^2 + (1 - w_{MVP})^2\sigma_2^2 + 2w_{MVP}(1 - w_{MVP})\sigma_{12}$  (relation  $(\star)$ ). Now, by definition of the minimum variance portfolio (setting the derivative to zero), we have  $2w_{MVP}(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - 2(\sigma_2^2 - \sigma_{12}) = 0$ . We then have:

$$\begin{aligned}2w_{MVP}(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - 2(\sigma_2^2 - \sigma_{12}) &= 0 \iff w_{MVP}(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - (\sigma_2^2 - \sigma_{12}) = 0 \\ &\iff \sigma_1^2 w_{MVP} + \sigma_2^2 w_{MVP} - 2\sigma_{12} w_{MVP} - \sigma_2^2 + \sigma_{12} \\ &\iff w_{MVP}\sigma_1^2 - (1 - w_{MVP})\sigma_2^2 + (1 - 2w_{MVP})\sigma_{12} = 0\end{aligned}$$

Taking up this result in relation ( $\star$ ), it then follows that:

$$\begin{aligned}
 \sigma_{\text{MVP}}^2 &= w_{\text{MVP}}^2 \sigma_1^2 + (1 - w_{\text{MVP}})^2 \sigma_2^2 + 2w_{\text{MVP}}(1 - w_{\text{MVP}}) \sigma_{12} \\
 &= w_{\text{MVP}}^2 \sigma_1^2 - w_{\text{MVP}}(1 - w_{\text{MVP}}) \sigma_2^2 + w_{\text{MVP}}(1 - w_{\text{MVP}}) \sigma_{12} + w_{\text{MVP}}(1 - w_{\text{MVP}}) \sigma_2^2 + (1 - w_{\text{MVP}})^2 \sigma_2^2 + w_{\text{MVP}}(1 - w_{\text{MVP}}) \sigma_{12} \\
 &= w_{\text{MVP}}(w_{\text{MVP}} \sigma_1^2 - (1 - w_{\text{MVP}}) \sigma_2^2 + (1 - 2w_{\text{MVP}}) \sigma_{12}) + (1 - w_{\text{MVP}}) \sigma_2^2 + w_{\text{MVP}} \sigma_{12} \\
 &= w_{\text{MVP}} \times 0 + (1 - w_{\text{MVP}}) \sigma_2^2 + w_{\text{MVP}} \sigma_{12} \\
 &= \sigma_2^2 - w_{\text{MVP}} \sigma_2^2 + w_{\text{MVP}} \sigma_{12} = \sigma_2^2 - w_{\text{MVP}} (\sigma_2^2 - \sigma_{12})
 \end{aligned}$$

It then suffices to substitute the expression for  $w_{\text{MVP}}$ :

$$\begin{aligned}
 \sigma_{\text{MVP}}^2 &= \sigma_2^2 - \frac{(\sigma_2^2 - \sigma_{12})^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \\
 &= \frac{\sigma_2^2 (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - (\sigma_2^2 - \sigma_{12})^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \\
 &= \frac{\sigma_1^2 \sigma_2^2 + \sigma_2^4 - 2\sigma_2^2 \sigma_{12} - \sigma_2^4 + 2\sigma_2^2 \sigma_{12} - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \\
 &= \frac{\sigma_1^2 \sigma_2^2 - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} = \frac{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}
 \end{aligned}$$

- (iv) The frontier may be obtained explicitly by eliminating the parameter  $w$  between the expression for  $\mu_P$  and that for  $\sigma_P^2$ . We have two relations: on the one hand,  $\mu_P = w\mu_1 + (1 - w)\mu_2$ , whence  $w = \frac{\mu_P - \mu_2}{\mu_1 - \mu_2}$  (relation (1)); on the other hand,  $\sigma_P^2 = w^2 \sigma_1^2 + (1 - w)^2 \sigma_2^2 + 2w(1 - w)\sigma_{12}$  (relation (2)). Substituting (1) into (2), it then follows that:

$$\begin{aligned}
 \sigma_P^2 &= w^2 \sigma_1^2 + (1 - w)^2 \sigma_2^2 + 2w(1 - w)\sigma_{12} \\
 &= \frac{(\mu_P - \mu_2)^2}{(\mu_1 - \mu_2)^2} \sigma_1^2 + \frac{(\mu_1 - \mu_P)^2}{(\mu_1 - \mu_2)^2} \sigma_2^2 + \frac{2(\mu_P - \mu_2)(\mu_1 - \mu_P)}{(\mu_1 - \mu_2)^2} \sigma_{12} \\
 &= \frac{1}{(\mu_1 - \mu_2)^2} ((\mu_P - \mu_2)^2 \sigma_1^2 + (\mu_1 - \mu_P)^2 \sigma_2^2 + 2(\mu_P - \mu_2)(\mu_1 - \mu_P)\sigma_{12})
 \end{aligned}$$

We therefore have:

$$\begin{aligned}
 (\mu_1 - \mu_2)^2 \sigma_P^2 &= (\mu_P - \mu_2)^2 \sigma_1^2 + ((\mu_1 - \mu_2) - (\mu_P - \mu_2))^2 \sigma_2^2 + 2(\mu_P - \mu_2)((\mu_1 - \mu_2) - (\mu_P - \mu_2))\sigma_{12} \\
 &= (\mu_P - \mu_2)^2 \sigma_1^2 + (\mu_1 - \mu_2)^2 \sigma_2^2 - 2(\mu_P - \mu_2)(\mu_1 - \mu_2) \sigma_{12} + (\mu_P - \mu_2)^2 \sigma_2^2 + 2(\mu_P - \mu_2)(\mu_1 - \mu_2) \sigma_{12} - 2(\mu_P - \mu_2)^2 \sigma_{12} \\
 &= (\mu_P - \mu_2)^2 (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) + 2(\mu_P - \mu_2)(\mu_1 - \mu_2) (\sigma_{12} - \sigma_2^2) + (\mu_1 - \mu_2)^2 \sigma_2^2
 \end{aligned}$$

We recognise a second-degree trinomial in  $(\mu_P - \mu_2)$ . We factor out  $\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$  in order to complete the square:

$$\begin{aligned}
 (\mu_1 - \mu_2)^2 \sigma_P^2 &= (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) \left( (\mu_P - \mu_2)^2 + \frac{2(\mu_P - \mu_2)(\mu_1 - \mu_2)(\sigma_{12} - \sigma_2^2)}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \right) + (\mu_1 - \mu_2)^2 \sigma_2^2 \\
 &= (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) \left( (\mu_P - \mu_2) + \frac{(\mu_1 - \mu_2)(\sigma_{12} - \sigma_2^2)}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \right)^2 - \frac{(\mu_1 - \mu_2)^2 (\sigma_{12} - \sigma_2^2)^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} + (\mu_1 - \mu_2)^2 \sigma_2^2 \\
 &= (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) \left( \mu_P - \mu_2 - \frac{(\mu_1 - \mu_2)(\sigma_2^2 - \sigma_{12})}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \right)^2 + (\mu_1 - \mu_2)^2 \left( \sigma_2^2 - \frac{(\sigma_2^2 - \sigma_{12})^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \right)
 \end{aligned}$$

We simplify the constant term by reducing to a common denominator:

$$\begin{aligned}
 \sigma_2^2 - \frac{(\sigma_2^2 - \sigma_{12})^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} &= \frac{\sigma_2^2 (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - (\sigma_2^2 - \sigma_{12})^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \\
 &= \frac{\sigma_1^2 \sigma_2^2 + \sigma_2^4 - 2\sigma_2^2 \sigma_{12} - \sigma_2^4 + 2\sigma_2^2 \sigma_{12} - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \\
 &= \frac{\sigma_1^2 \sigma_2^2 - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}
 \end{aligned}$$



We note that  $\sigma_1^2 \sigma_2^2 - \sigma_{12}^2 = \sigma_1^2 \sigma_2^2 - \rho_{12}^2 \sigma_1^2 \sigma_2^2 = \sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2) \geq 0$  as soon as  $|\rho_{12}| \leq 1$  (which is always the case). Dividing by  $(\mu_1 - \mu_2)^2$ , we therefore obtain:

$$\begin{aligned}\sigma_P^2 &= \frac{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}{(\mu_1 - \mu_2)^2} \left( \mu_P - \left( \mu_2 + \frac{(\mu_1 - \mu_2)(\sigma_2^2 - \sigma_{12})}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \right) \right)^2 + \frac{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \\ &= \frac{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}{(\mu_1 - \mu_2)^2} (\mu_P - \mu_{MVP})^2 + \sigma_{MVP}^2 \\ &= \frac{0.12^2 - (-0.0096)}{0.20^2 + 0.12^2 - 2 \times (-0.0096)} (0.10 - 0.06)^2 \times (\mu_P - 0.0730)^2 + 0.0811^2 \approx 46(\mu_P - 0.0730)^2 + 0.006575\end{aligned}$$

This establishes the canonical form of the parabola, whose vertex is the minimum variance portfolio, with coordinates  $(\sigma_{MVP}^2, \mu_{MVP})$ . As Merton (1972) demonstrates, this parabolic structure is universal: whatever the number of assets and the structure of the variance-covariance matrix (provided it is positive definite), the minimum variance frontier is always a parabola in the plane  $(\sigma_P^2, \mu_P)$ , or equivalently a sideways parabola in the plane  $(\sigma_P, \mu_P)$ . The efficient frontier in the sense of Markowitz (1952) is the upper branch of this sideways parabola, that is to say the half-curve satisfying  $\mu_P \geq \mu_{MVP}$ : any portfolio below it is dominated, since there always exists a portfolio offering the same risk for a strictly higher return. More specifically, the efficient frontier is here the upper branch of the parabola with equation  $\sigma_P^2 = 46(\mu_P - 0.0730)^2 + 0.006575$ , that is to say the half-parabola with equation  $\sigma_P = \sqrt{46(\mu_P - 0.0730)^2 + 0.006575}$  satisfying  $\mu_P \geq \mu_{MVP} \approx 7.30\%$ .

- (v) The tangency portfolio T maximises the Sharpe ratio  $S(w) = \frac{\mu_P(w) - r_f}{\sigma_P(w)}$  among all the portfolios of the frontier. The Sharpe ratio is a quotient of two functions of  $w$ . We apply the rule for differentiating a quotient: if  $f$  and  $g$  are two differentiable functions with  $g \neq 0$ , then  $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ . Here,  $f(w) = \mu_P(w) - r_f$  and  $g(w) = \sigma_P(w)$ . We therefore have:

$$\frac{dS}{dw} = \frac{\frac{d\mu_P}{dw} \times \sigma_P - (\mu_P - r_f) \times \frac{d\sigma_P}{dw}}{\sigma_P^2}$$

Since  $\sigma_P^2$  is strictly positive, setting this derivative to zero amounts to setting only its numerator to zero: we therefore have  $\frac{dS}{dw} = 0$  if, and only if,  $\frac{d\mu_P}{dw} \cdot \sigma_P - (\mu_P - r_f) \cdot \frac{d\sigma_P}{dw} = 0$  (relation (3)). We now compute each derivative separately. The first is immediate: since  $\mu_P(w) = w\mu_1 + (1-w)\mu_2$ , we have  $\frac{d\mu_P}{dw} = \mu_1 - \mu_2$ . For the second, we note that  $\sigma_P$  is the square root of  $\sigma_P^2$ . We therefore apply the rule for differentiating composite functions (the chain rule): if  $h(w) = \sqrt{u(w)}$ , then  $h'(w) = \frac{u'(w)}{2\sqrt{u(w)}}$ . Setting  $u(w) = \sigma_P^2(w)$ , we obtain :

$$\frac{d\sigma_P}{dw} = \frac{\frac{d\sigma_P^2}{dw}}{2\sqrt{\sigma_P^2}} = \frac{1}{2\sigma_P} \times \frac{d\sigma_P^2}{dw}$$

Substituting these two results into condition (3), it therefore follows that  $(\mu_1 - \mu_2)\sigma_P^2 = (\mu_P - r_f) \times \frac{1}{2} \times \frac{d\sigma_P^2}{dw}$  (relation (4)). We then substitute into relation (4) the three expressions already established, namely  $\sigma_P^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12}$ ,  $\frac{1}{2} \frac{d\sigma_P^2}{dw} = w\sigma_1^2 - (1-w)\sigma_2^2 + (1-2w)\sigma_{12}$  and  $\mu_P - r_f = w(\mu_1 - r_f) + (1-w)(\mu_2 - r_f)$ . Relation (4) then becomes:

$$\begin{aligned}(\mu_1 - \mu_2)(w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12}) &= (w(\mu_1 - r_f) + (1-w)(\mu_2 - r_f))(w\sigma_1^2 - (1-w)\sigma_2^2 + (1-2w)\sigma_{12}) \\ &= w^2(\mu_1 - r_f)\sigma_1^2 - w(1-w)(\mu_1 - r_f)\sigma_2^2 + w(1-2w)(\mu_1 - r_f)\sigma_{12} \\ &\quad + w(1-w)(\mu_2 - r_f)\sigma_1^2 - (1-w)^2(\mu_2 - r_f)\sigma_2^2 + (1-w)(1-2w)(\mu_2 - r_f)\sigma_{12}\end{aligned}$$

We subtract the right-hand side from the left-hand side. The objective is to show that the terms in  $w^2$  cancel out, so as to obtain a linear equation in  $w$  from which we can extract  $w_T$ .

— **Cancellation of the terms in  $w^2$ .** On the left-hand side, the coefficient of  $w^2$  is  $(\mu_1 - \mu_2)(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})$ , since we expand  $w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12}$  and collect the terms in  $w^2$ , which give  $w^2(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})$ . On the right-hand side, we identify the coefficient of  $w^2$  by expanding each of the six terms arising from the product:

- $w^2(\mu_1 - r_f)\sigma_1^2$  contributes  $+(\mu_1 - r_f)\sigma_1^2$  directly;
- $-w(1-w)(\mu_1 - r_f)\sigma_2^2 = -w\sigma_2^2(\mu_1 - r_f) + w^2\sigma_2^2(\mu_1 - r_f)$  contributes  $+(\mu_1 - r_f)\sigma_2^2$ ;



- $w(1-2w)(\mu_1 - r_f)\sigma_{12} = w\sigma_{12}(\mu_1 - r_f) - 2w^2\sigma_{12}(\mu_1 - r_f)$  contributes  $-2(\mu_1 - r_f)\sigma_{12}$ ;
- $w(1-w)(\mu_2 - r_f)\sigma_1^2 = w\sigma_1^2(\mu_2 - r_f) - w^2\sigma_1^2(\mu_2 - r_f)$  contributes  $-(\mu_2 - r_f)\sigma_1^2$ ;
- $-(1-w)^2(\mu_2 - r_f)\sigma_2^2 = -\sigma_2^2(\mu_2 - r_f) + 2w\sigma_2^2(\mu_2 - r_f) - w^2\sigma_2^2(\mu_2 - r_f)$  contributes  $-(\mu_2 - r_f)\sigma_2^2$ ;
- $(1-w)(1-2w)(\mu_2 - r_f)\sigma_{12} = (1-3w+2w^2)(\mu_2 - r_f)\sigma_{12}$  contributes  $+2(\mu_2 - r_f)\sigma_{12}$ .

Adding these six contributions, the total coefficient of  $w^2$  on the right-hand side is :

$$\begin{aligned} & (\mu_1 - r_f)\sigma_1^2 + (\mu_1 - r_f)\sigma_2^2 - 2(\mu_1 - r_f)\sigma_{12} - (\mu_2 - r_f)\sigma_1^2 - (\mu_2 - r_f)\sigma_2^2 + 2(\mu_2 - r_f)\sigma_{12} \\ &= [(\mu_1 - r_f) - (\mu_2 - r_f)]\sigma_1^2 + [(\mu_1 - r_f) - (\mu_2 - r_f)]\sigma_2^2 - 2[(\mu_1 - r_f) - (\mu_2 - r_f)]\sigma_{12} \\ &= (\mu_1 - \mu_2)(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) \end{aligned}$$

This is identical to the coefficient of  $w^2$  on the left-hand side. The terms in  $w^2$  therefore do indeed cancel out when moving to the left-hand side.

- **Identification of the linear equation in  $w$ .** After cancellation of the terms in  $w^2$ , there remain the constant terms (in  $w^0$ ) and the terms in  $w^1$ . The constant term on the left-hand side is  $(\mu_1 - \mu_2)\sigma_2^2$  (setting  $w = 0$ ); that on the right-hand side is  $-(\mu_2 - r_f)\sigma_2^2$ . Their difference is therefore  $(\mu_1 - \mu_2)\sigma_2^2 + (\mu_2 - r_f)\sigma_2^2 = \sigma_2^2[(\mu_1 - \mu_2) + (\mu_2 - r_f)] = (\mu_1 - r_f)\sigma_2^2$ . For the coefficient of  $w^1$ , we identify in the same way the linear contribution of each of the six terms on the right-hand side, by expanding the factors  $(1-w)$ ,  $(1-w)^2$  and  $(1-2w)$  :

- $w^2(\mu_1 - r_f)\sigma_1^2$  : no contribution in  $w^1$  ;
- $-w(1-w)(\mu_1 - r_f)\sigma_2^2 = -w(\mu_1 - r_f)\sigma_2^2 + w^2(\mu_1 - r_f)\sigma_2^2$  : contributes  $-(\mu_1 - r_f)\sigma_2^2$  ;
- $w(1-2w)(\mu_1 - r_f)\sigma_{12} = w(\mu_1 - r_f)\sigma_{12} - 2w^2(\mu_1 - r_f)\sigma_{12}$  : contributes  $+(\mu_1 - r_f)\sigma_{12}$  ;
- $w(1-w)(\mu_2 - r_f)\sigma_1^2 = w(\mu_2 - r_f)\sigma_1^2 - w^2(\mu_2 - r_f)\sigma_1^2$  : contributes  $+(\mu_2 - r_f)\sigma_1^2$  ;
- $-(1-w)^2(\mu_2 - r_f)\sigma_2^2 = -(\mu_2 - r_f)\sigma_2^2 + 2w(\mu_2 - r_f)\sigma_2^2 - w^2(\mu_2 - r_f)\sigma_2^2$  : contributes  $+2(\mu_2 - r_f)\sigma_2^2$  ;
- $(1-w)(1-2w)(\mu_2 - r_f)\sigma_{12} = (1-3w+2w^2)(\mu_2 - r_f)\sigma_{12}$  : contributes  $-3(\mu_2 - r_f)\sigma_{12}$  .

Adding these six contributions, the coefficient of  $w^1$  on the right-hand side is  $-(\mu_1 - r_f)\sigma_2^2 + (\mu_1 - r_f)\sigma_{12} + (\mu_2 - r_f)\sigma_1^2 + 2(\mu_2 - r_f)\sigma_2^2 - 3(\mu_2 - r_f)\sigma_{12}$ . The coefficient of  $w^1$  on the left-hand side is obtained in the same way by expanding  $(\mu_1 - \mu_2)(w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12})$ , which gives  $(\mu_1 - \mu_2)[-2\sigma_2^2 + 2\sigma_{12}] = -2(\mu_1 - \mu_2)\sigma_2^2 + 2(\mu_1 - \mu_2)\sigma_{12}$ . Moving to the left-hand side ('left minus right'), the coefficient of  $w^1$  is:

$$\begin{aligned} & -2(\mu_1 - \mu_2)\sigma_2^2 + 2(\mu_1 - \mu_2)\sigma_{12} + (\mu_1 - r_f)\sigma_2^2 - (\mu_1 - r_f)\sigma_{12} - (\mu_2 - r_f)\sigma_1^2 - 2(\mu_2 - r_f)\sigma_2^2 + 3(\mu_2 - r_f)\sigma_{12} \\ &= [-2(\mu_1 - \mu_2) + (\mu_1 - r_f) - 2(\mu_2 - r_f)]\sigma_2^2 + [-(\mu_2 - r_f)]\sigma_1^2 + [2(\mu_1 - \mu_2) - (\mu_1 - r_f) + 3(\mu_2 - r_f)]\sigma_{12} \\ &= [-2\mu_1 + 2\mu_2 + \mu_1 - r_f - 2\mu_2 + 2r_f]\sigma_2^2 - (\mu_2 - r_f)\sigma_1^2 + [2\mu_1 - 2\mu_2 - \mu_1 + r_f + 3\mu_2 - 3r_f]\sigma_{12} \\ &= -(\mu_1 - r_f)\sigma_2^2 - (\mu_2 - r_f)\sigma_1^2 + (\mu_1 + \mu_2 - 2r_f)\sigma_{12} \end{aligned}$$

The coefficient of  $w^1$  after moving to the left is then  $-(\mu_1 - r_f)\sigma_2^2 - (\mu_2 - r_f)\sigma_1^2 + (\mu_1 + \mu_2 - 2r_f)\sigma_{12}$ . Equation (4) is therefore rewritten  $(\mu_1 - r_f)\sigma_2^2 - (\mu_2 - r_f)\sigma_{12} + w[-(\mu_1 - r_f)\sigma_2^2 - (\mu_2 - r_f)\sigma_1^2 + (\mu_1 + \mu_2 - 2r_f)\sigma_{12}] = 0$ , whose immediate solution provides the weighting  $w_T$  of asset  $A_1$  in the tangency portfolio:

$$w_T = \frac{(\mu_1 - r_f)\sigma_2^2 - (\mu_2 - r_f)\sigma_{12}}{(\mu_1 - r_f)\sigma_2^2 + (\mu_2 - r_f)\sigma_1^2 - (\mu_1 + \mu_2 - 2r_f)\sigma_{12}}$$

We can verify this result intuitively: if  $\sigma_{12} = 0$  (which is the case when the assets are independent), the formula reduces to  $w_T = \frac{(\mu_1 - r_f)\sigma_2^2}{(\mu_1 - r_f)\sigma_2^2 + (\mu_2 - r_f)\sigma_1^2}$ , which allocates more weight to the asset whose risk premium is high relative to its variance, which corresponds well to economic intuition. Substituting the numerical values with  $\mu_1 - r_f = 8\%$  and  $\mu_2 - r_f = 4\%$ , we obtain:

$$\begin{aligned} w_T &= \frac{0.08 \times (0.12)^2 - 0.04 \times (-0.0096)}{0.08 \times (0.12)^2 + 0.04 \times (0.20)^2 - (0.08 + 0.04) \times (-0.0096)} \\ &= \frac{0.08 \times 0.0144 + 0.04 \times 0.0096}{0.08 \times 0.0144 + 0.04 \times 0.0400 + 0.12 \times 0.0096} \\ &= \frac{0.001152 + 0.000384}{0.001152 + 0.001600 + 0.001152} = \frac{24}{61} \approx 39.34\%. \end{aligned}$$

The expected return and the standard deviation of the tangency portfolio then stand respectively at  $\mu_T =$

$\frac{24}{61} \times 10\% + \frac{37}{61} \times 6\% \approx 7.57\%$  and  $\sigma_T = \sqrt{\left(\frac{24}{61}\right)^2 \times (0.20)^2 + \left(\frac{37}{61}\right)^2 \times (0.12)^2 + 2 \times \frac{24}{61} \times \frac{37}{61} \times (-0.0096)} = \sqrt{0.006908} \approx 8.31\%$ . The Sharpe ratio of the tangency portfolio is from then on  $S_T = \frac{\mu_T - r_f}{\sigma_T} = \frac{7.57\% - 2\%}{8.31\%} \approx 0.671$ , to be compared with the individual ratios  $S_1 = \frac{10\% - 2\%}{20\%} = 0.400$  and  $S_2 = \frac{6\% - 2\%}{12\%} = 0.333$ . The tangency portfolio therefore offers a higher remuneration of risk than each asset taken in isolation: for each percentage point of risk borne, the investor receives 0.671 % of premium, against 0.400 % in holding only  $A_1$  and 0.333 % in holding only  $A_2$ . This gain constitutes the most direct measure of the benefit of diversification in return-risk space.

- (vi) By virtue of the two-fund separation theorem of Tobin (1958), every efficient portfolio is written as a combination of the riskless asset and the tangency portfolio T. If the real number  $\alpha$  denotes the fraction of wealth invested in T (and  $1 - \alpha$  in the riskless asset), the expected return and the standard deviation of the portfolio thus constituted are respectively  $\mu_P = (1 - \alpha)r_f + \alpha\mu_T = r_f + \alpha(\mu_T - r_f)$  and  $\sigma_P = |\alpha|\sigma_T$ . The second relation is explained by the fact that the riskless asset has a zero standard deviation and a zero covariance with any other asset: the variance of the mixed portfolio is therefore indeed  $\sigma_P^2 = \alpha^2\sigma_T^2$ . For  $\alpha \geq 0$ , we may therefore write  $\alpha = \sigma_P/\sigma_T$ . Substituting this expression into that for  $\mu_P$ , we eliminate the parameter  $\alpha$  and obtain a direct relation between  $\mu_P$  and  $\sigma_P$ :

$$\begin{aligned}\mu_P &= r_f + \frac{\mu_T - r_f}{\sigma_T} \times \sigma_P \\ &\approx 2\% + \frac{7.57\% - 2\%}{8.31\%} \times \sigma_P \approx 2\% + 0.671\sigma_P\end{aligned}$$

This is the equation of a straight line in the plane  $(\sigma_P, \mu_P)$ , with intercept  $r_f$  and slope  $(\mu_T - r_f)/\sigma_T$ : the capital market line (CML). The slope  $(\mu_T - r_f)/\sigma_T \approx 0.671$  represents the ‘price of risk’: each additional percentage point of standard deviation accepted by the investor is remunerated by 0.671 % of additional expected return. The CML thus expresses the trade-off between ‘patience’ — remunerated by  $r_f = 2\%$  independently of risk, this being the pure price of time — and ‘risk-taking’ — remunerated by the premium  $(\mu_T - r_f)/\sigma_T$  per unit of standard deviation, this being the pure price of risk. Any portfolio situated below the CML is sub-optimal: the rational investor can always improve on it by combining T and the riskless asset so as to attain the same level of risk with a strictly higher expected return. No portfolio can, however, be situated above the CML without generating an arbitrage opportunity, which expresses the efficiency of the market in the absence of frictions, a central assumption of the model of Sharpe (1964). More precisely, as figure 3.5 illustrates, the portfolio frontier describes a hyperbola whose vertex is the MVP ( $\sigma_{MVP} \approx 8.11\%$ ,  $\mu_{MVP} \approx 7.30\%$ ): this is the only point where the variance is strictly lower than that of each asset taken in isolation ( $\sigma_1 = 20\%$  and  $\sigma_2 = 12\%$ ). The CML, with slope  $(\mu_T - r_f)/\sigma_T \approx 0.671$ , is the only straight line passing through  $r_f$  and tangent to the efficient frontier: any portfolio situated to the right of or below the CML is sub-optimal in the sense of Sharpe (1964), and the rational investor will systematically prefer to it a combination of the riskless asset and the tangency portfolio T.  $\square$

### Key Takeaways



The modern portfolio theory of Markowitz (1952) establishes that combining assets whose returns are not perfectly correlated reduces overall risk without sacrificing return: for two assets A and B with weights  $w$  and  $1 - w$ , the expected return is  $\mu_P = w\mu_A + (1 - w)\mu_B$  while the variance  $\sigma_P^2 = w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 + 2w(1 - w)\rho_{AB}\sigma_A\sigma_B$  is strictly lower than the weighted average of the variances as soon as  $\rho_{AB} < 1$ . This result invalidates the maximum expected return rule of Williams (1938), which prescribed concentrating all one's wealth on the single most profitable asset while ignoring the interactions between assets. The efficient frontier, for which Merton (1972) establishes the equation  $\sigma_P^2 = \frac{C\mu_P^2 - 2A\mu_P + B}{D}$  for  $m$  risky assets, groups together the non-dominated portfolios maximising return for a given risk. Sharpe (1964) and Lintner (1965) extend this framework by introducing a riskless asset and the homogeneous beliefs assumption: under these conditions, the efficient frontier is transformed into the capital market line  $\mu_P = r_f + \frac{\mu_M - r_f}{\sigma_M} \sigma_P$ , and the two-fund separation theorem establishes that every investor holds the same combination of the market portfolio M and the riskless asset. The CAPM which flows from this remunerates only non-diversifiable systematic risk:  $\mathbb{E}[R_i] = r_f + \beta_i(\mu_M - r_f)$  with  $\beta_i = \text{Cov}(R_i, R_M)/\sigma_M^2$ , although its empirical validity is contested by the excessively flat relation between  $\beta$  and returns, and by the residual explanatory power of size or of the *book-to-market* ratio [Fama & French (2004)]. Despite these limitations, the CAPM remains central: the assets under management of AIF and UCITS funds managed according to quantitative approaches reached 1984 billion euros in France at the end of 2023 according to AMF (2024).

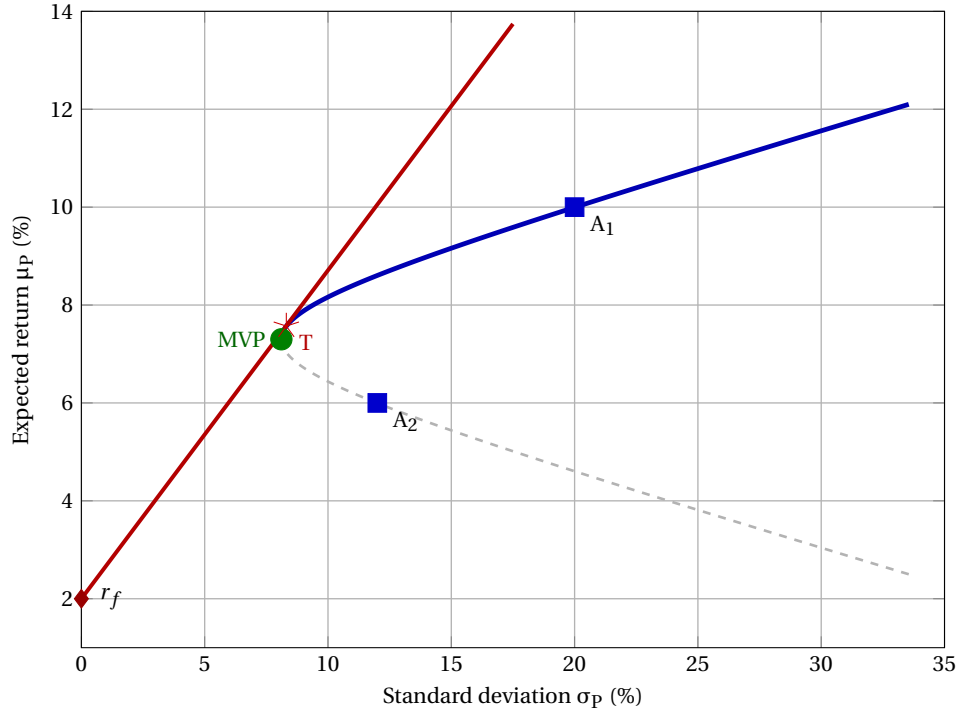


Figure 3.5: Frontier of risky portfolios, tangency portfolio (T), minimum variance portfolio (MVP) and capital market line (CML) in the plane  $(\sigma_P, \mu_P)$ : the dominated frontier is represented by a grey dashed line; the efficient frontier is drawn as a solid blue line; the CML, in red, is tangent to the efficient frontier at the point T.

### 2.3.2 Risk diversification and index markets

The theoretical developments initiated by Markowitz (1952) and formalised by Sharpe (1964) lead to a practical conclusion of considerable scope: the market portfolio, which contains all risky assets in proportion to their weight in total free-float market capitalisation, constitutes the efficient portfolio par excellence. This theoretical abstraction finds its empirical embodiment in stock market indices, whose construction obeys precise and publicly documented methodological rules, and which serve as a universal benchmark both for the measurement of performance and for the passive replication of diversified portfolios. A **stock market index** is, strictly speaking, a dimensionless number — expressed in ‘points’ — which measures the variation in the value of a basket of assets selected according to defined criteria, relative to a base value fixed arbitrarily at a reference date: the CAC 40 was thus initialised at 1000 points on 31 December 1987, the date at which it represented the aggregate capitalisation of the forty stocks composing the index; any subsequent variation in the level of the index measures the relative variation of this aggregate capitalisation. The general formula for computing an index weighted by free-float capitalisation, as defined by the index manager Euronext (2016) in the official documentation of the rules of the CAC family of indices, is:

$$I_t = I_0 \times \frac{\sum_{i=1}^N P_{i,t} S_{i,t} F_{i,t} f_{i,t}}{\sum_{i=1}^N P_{i,0} S_{i,0} F_{i,0} f_{i,0}}$$

where  $P_{i,t}$  denotes the price of security  $i$  at date  $t$ ,  $S_{i,t}$  the total number of shares in issue,  $F_{i,t}$  the **free-float factor**, which measures the fraction of the capital genuinely available for trading — that is to say after deducting strategic holdings, treasury shares and blocks held by reference shareholders regarded as illiquid — and  $f_{i,t}$  a possible **capping factor** intended to prevent a single security from structurally dominating the index. For the CAC 40, this individual cap is set at 15 % of the total free-float capitalisation of the index, so that no security may, even in the event of a spectacular rise in its price, represent more than 15 % of the index portfolio; this *capping* factor is recomputed at each quarterly review by the Euronext Index Scientific Council, an independent body which decides on the entries and exits of stocks according to two cumulative criteria: free-float market capitalisation and the volume of trading observed over the last twelve months, in accordance with the rules published by Euronext (2016). The selection

process incorporates a 'buffer zone': the top 35 stocks in the ranking are automatically retained; for the five remaining places, a tolerance zone covering the stocks ranked between thirty-sixth and forty-fifth makes it possible to avoid too frequent a rotation of the composition and to limit the transaction costs that permanent reconstitution of the portfolios replicating the index would entail. This relative stability of the composition is economically important, since the announcement of an entry into a major index is generally accompanied by an abnormal rise in the price of the share concerned of the order of 3 % to 5 % in the days preceding the effective date of inclusion, a phenomenon documented empirically on the S&P 500 as early as 1986 and attributed to the captive demand of index funds which are obliged to acquire the incoming security. As Harris & Gurel (1986) show, in a study bearing on additions to the S&P 500 over the period 1976–1983, the announcement of an inclusion is accompanied on average by an abnormal rise in the price of more than 3 % within a few days, a rise which is almost entirely reabsorbed within the space of two weeks — which is consistent with the **price pressure hypothesis**: market makers temporarily accommodate the captive demand of index funds and are remunerated by this transitory premium. Shleifer (1986) documents, over the same period, a permanent abnormal rise of roughly 2.79 % at the announcement of inclusion and interprets this result as proof that the demand curves for individual securities are downward-sloping: as soon as index funds constitute a rigid and inelastic source of demand, the equilibrium price increases structurally at the announcement, independently of any fundamental signal about the intrinsic value of the firm concerned.

On a world scale, the main organised markets on which the securities composing the major indices are traded may be classified according to their liquidity, their aggregate capitalisation and their role in the international financial system. Orders on CAC 40 stocks are transmitted and executed on the Euronext Paris platform, which is the stock exchange operating the French regulated market within the meaning of the **European Markets in Financial Instruments Directive** (MiFID II); Euronext is also the operator of the Amsterdam, Brussels, Dublin, Lisbon, Milan and Oslo exchanges. Orders may be placed in several ways : an institutional investor or a private individual may transmit a **market order**, executed immediately at the best price available in the order book, a **limit order**, executed only if the price reaches a threshold set by the investor, or again a **stop order**, which turns into a market order when the price crosses a trigger threshold. Continuous trading extends from 9:00 to 17:35 (Paris time) for CAC 40 stocks, with a pre-opening phase between 7:15 and 9:00 during which orders accumulate in the book without any transaction being concluded, the opening price being fixed algorithmically at the supply-demand equilibrium. On the other side of the Atlantic, the New York Stock Exchange (NYSE) and the Nasdaq constitute the two main execution platforms for the stocks of the S&P 500 and the Nasdaq Composite; unlike the NYSE, which is a market with 'specialists' (*designated market makers*), the **Nasdaq** is an entirely decentralised electronic market, founded on a system of *competing market makers*. The **S&P 500** index, managed by S&P Dow Jones Indices, is weighted by free-float capitalisation (*float-adjusted market capitalization weighted*) and reviewed quarterly in March, June, September and December; to be eligible for the index, a firm must present an unadjusted market capitalisation above 14.5 billion American dollars, an **investable weight factor** (IWF) of at least 0.10, and must have made positive accounting profits over the last four cumulative quarters, according to the methodology published by S&P Global (2023b). As S&P Dow Jones Indices (2023) formally defines it, the IWF is computed for each security by dividing the shares available as free float by the total shares in issue, given that the 'shares available as free float' are obtained by subtracting from the total shares in issue all the shares held by **strategic shareholders**: excluded from the free float are the members of the board and the officers (*officers and directors*, O&D) as a group, as well as any external shareholder holding, directly or indirectly, a stake of at least 5 % of the capital and whose holding is judged strategic — that is to say not intended to be sold in the normal course of business. An IWF of 0.72 therefore means that 72 % of the firm's shares are genuinely available for trading on public markets, the remaining 28 % being held by blocking shareholders whose stake does not contribute to the real liquidity of the security. The IWF is reviewed annually on the basis of the filings made with regulators, and is subject to a quarterly update if a capital operation entails a variation of at least 5 % in the total number of shares in issue, according to the rules of S&P Dow Jones Indices (2023). In Asia, the Tokyo Stock Exchange (TSE), whose main benchmark index is the **TOPIX** (*Tokyo Stock Price Index*), brings together a market capitalisation of the order of 6 100 billion dollars in 2023 according to the data of the Banque mondiale (2023), while the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE) jointly constitute the second largest market in the world by the capitalisation of listed domestic shares, with a total value approaching 8 400 billion dollars at the same date. In total, the aggregate world market capitalisation of all regulated markets amounted to roughly 103 000 billion dollars at the end of 2023, according to the same source, which represents approximately 107 % of world GDP, a proportion which illustrates the systemic importance of financial markets in the international allocation of capital.

The construction of an index weighted by free-float capitalisation constitutes an operational approximation of the 'market portfolio' theorised by Sharpe (1964), and it is precisely this property which grounds the economic justification of **passive management**. Indeed, if the CAPM is valid — that is to say if asset prices correctly incorporate all available information, in accordance with the market efficiency hypothesis of Fama (1970) — then no active

manager can systematically outperform the benchmark index after deduction of management fees: any apparent  $\alpha$  is merely the reflection of additional risk-taking, of passing luck, or of a temporary exploitation of inefficiencies destined to be arbitrated away. This thesis has been corroborated empirically time and again: the annual SPIVA report (*S&P indices versus active*) published by S&P Global (2025b) indicates that over a fifteen-year horizon, more than 88 % of actively managed American equity funds underperform their benchmark index net of fees. The diversification permitted by holding a broad index is not merely theoretical: Fama & French (2004) recall that empirical studies on American stock market data show that a portfolio of twenty to thirty securities selected at random from among the S&P 500 stocks eliminates most of the measurable idiosyncratic risk, the residual non-diversifiable risk corresponding to the residual market  $\beta$ . More precisely, the variance of an equally weighted portfolio of  $n$  assets each of whose individual variance is  $\sigma^2$  and whose average cross-covariance is  $\bar{\sigma}_{ij}$  decomposes into  $\sigma_p^2 = \frac{1}{n} \sigma^2 + (1 - \frac{1}{n}) \bar{\sigma}_{ij}$ . Indeed, we have:

$$\begin{aligned}\sigma_p^2 &= \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma_i^2 + \frac{1}{n^2} \sum_{i \neq j} \sigma_{ij} \\ &= \frac{1}{n^2} \times n \bar{\sigma}^2 + \frac{1}{n^2} \times n(n-1) \bar{\sigma}_{ij} \quad \text{setting } \bar{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \sigma_i^2 \text{ and } \bar{\sigma}_{ij} = \frac{1}{n(n-1)} \sum_{i \neq j} \sigma_{ij} \\ &= \frac{\bar{\sigma}^2}{n} + \frac{n-1}{n} \bar{\sigma}_{ij} = \frac{1}{n} \bar{\sigma}^2 + \left(1 - \frac{1}{n}\right) \bar{\sigma}_{ij}\end{aligned}$$

Thus, when  $n$  tends to  $+\infty$ , the variance of the portfolio converges towards the average covariance  $\bar{\sigma}_{ij}$ , which is the **irreducible systematic risk**: the idiosyncratic risk  $(\sigma^2 - \bar{\sigma}_{ij})/n$  vanishes as the number of securities increases, but the market risk  $\bar{\sigma}_{ij}$  remains, whatever the degree of diversification attained. It is this result, formalised within the framework of the CAPM, which justifies the idea that holding a broad index — which constitutes an approximation of the market portfolio — is not only optimal from the point of view of diversification, but also economically neutral from the point of view of residual risk: the investor who holds the index bears only market risk, unavoidable and remunerated by the premium  $\mu_M - r_f$ , and no longer bears any unremunerated specific risk. This analysis joins the criticism formulated by Markowitz (1952) against portfolios concentrated in a single sector and illustrates why broad indices, such as the MSCI World — which covers more than 1 400 securities spread across twenty-three developed countries and represented a capitalisation of the order of 54 000 billion dollars in 2023 according to the data of the Banque mondiale (2023) — have become benchmark instruments for institutional as well as individual portfolio management.

### Focus...



**Risk-adjusted performance measures.** The mean-variance framework of Markowitz (1952) and the CAPM of Sharpe (1964) gave rise to a family of **risk-adjusted performance ratios**, whose common objective is to relate the excess return of a portfolio or an asset to a measure of its risk, so as to permit comparisons between assets of different risk levels. These ratios constitute the standard tools of portfolio analysis and management control, both within the European regulatory framework (the UCITS V directive, the PRIIPs regulation) and in the practice of fund managers. Four measures deserve a rigorous presentation:

- **The Sharpe ratio.** Introduced implicitly within the framework of the CAPM by Sharpe (1964) and formalised in its contemporary version, the Sharpe ratio measures the risk premium obtained per unit of total risk (measured by the standard deviation of returns):  $S_p = \frac{\mu_p - r_f}{\sigma_p}$ , where  $\mu_p$  is the expected return of the portfolio,  $r_f$  the riskless rate and  $\sigma_p$  the standard deviation of returns. A Sharpe ratio greater than 1 indicates that the excess return exceeds volatility, which is generally regarded as satisfactory in the context of developed equity markets; a ratio lower than 0 means that the portfolio underperforms the riskless asset. The fundamental limitation of the Sharpe ratio is that it penalises upside and downside volatility symmetrically: yet a rational investor is mainly preoccupied by losses, and not by unexpected gains, so that the total variance  $\sigma_p^2$  overestimates the risk actually borne for positively skewed distributions.
- **The Sortino ratio.** Proposed by Sortino & Price (1994) to remedy the symmetric limitation of the Sharpe ratio, the Sortino ratio replaces the total standard deviation  $\sigma_p$  by the downside deviation  $\sigma_d$ , defined as the standard deviation of returns below a target return  $\tau$  (*minimum acceptable return*, MAR) set by the investor:  $\sigma_d = \sqrt{\mathbb{E}[\min(R_p - \tau, 0)^2]}$ , so that the Sortino ratio is written  $\mathcal{S}_p = \frac{\mu_p - \tau}{\sigma_d}$ . When  $\tau = r_f$ , the Sortino ratio is directly

comparable with the Sharpe ratio; it is systematically higher for right-skewed distributions, since only ‘bad’ volatility — that which corresponds to returns below the target — is counted in the denominator. This property makes the Sortino ratio particularly suited to the assessment of strategies presenting asymmetric return profiles, such as alternative funds or option strategies, for which the Sharpe ratio leads to biased comparisons according to the observations of Sortino & Price (1994).

- **The Treynor ratio.** Contemporary with the CAPM, the Treynor ratio is proposed by Treynor (1965) within the framework of a reflection on the assessment of fund managers: rather than relating the risk premium to total volatility, it relates it to the systematic risk  $\beta_P$  of the portfolio, estimated by regressing the returns of the portfolio on those of the market:  $T_P = \frac{\mu_P - r_f}{\beta_P}$ . The Treynor ratio is therefore relevant for the investor who holds the portfolio being assessed as one component among others of a broader, diversified portfolio: in that case, only non-diversifiable systematic risk is relevant to the assessment of the marginal contribution of the portfolio to overall risk, and idiosyncratic risk is already eliminated by the diversification of the aggregate portfolio. Conversely, the Sharpe ratio is relevant for the investor who holds the portfolio exclusively, that is to say for whom this portfolio constitutes the entirety of his invested wealth: in that case, total volatility — systematic and idiosyncratic — is the appropriate risk measure. The distinction between these two ratios therefore illustrates a fundamental tension between the perspective of an isolated investor and that of a market in which specific risks are aggregated and diversified.
- **The Calmar ratio.** Published by Young (1991), the Calmar ratio — an acronym of *California managed accounts reports*, its author’s management newsletter — measures the annualised return relative to the maximum drawdown (MDD) observed over a rolling window of thirty-six months:  $C_P = \frac{\bar{r}_P^{36}}{MDD^{36}}$ , where  $\bar{r}_P^{36}$  is the average annualised return over the last 36 months and  $MDD^{36} = \max_{t \in [0, T]} \left( \frac{\text{peak}_t - \text{trough}_t}{\text{peak}_t} \right)$  is the maximum loss of value recorded between a high point and a subsequent low point over the same window. The maximum drawdown constitutes a measure of tail risk and of the risk of partial ruin: where the standard deviation captures the average dispersion around the trend, the MDD captures the worst historical scenario actually borne by the investor who would have held the portfolio continuously over the period. The Calmar ratio is particularly prized for the assessment of *commodity trading advisors* (CTA) and hedge funds, for which the capacity to limit maximum losses is often as important as the average return. By way of numerical illustration: a fund displaying an annualised return of 10 % over 36 months but having suffered a maximum drawdown of 40 % presents a Calmar ratio of 0.25, well below that of a fund displaying 7 % annualised return with a drawdown of 10 % (a Calmar ratio of 0.70), which in practice is regarded as more attractive by institutional investors averse to the risk of capital loss.

| Ratio   | Numerator        | Denominator                     | Risk captured   | Typical use              |
|---------|------------------|---------------------------------|-----------------|--------------------------|
| Sharpe  | $\mu_P - r_f$    | $\sigma_P$ (total volatility)   | Total symmetric | Exclusive portfolio      |
| Sortino | $\mu_P - \tau$   | $\sigma_d$ (downside deviation) | Downside only   | Asymmetric strategies    |
| Treynor | $\mu_P - r_f$    | $\beta_P$ (systematic risk)     | Systematic only | Component of a portfolio |
| Calmar  | $\bar{r}_P^{36}$ | $MDD^{36}$                      | Tail risk       | CTA, alternative funds   |

### Exercise



**Sharpe ratio and diversification.** Consider five uncorrelated assets each presenting a Sharpe ratio of 1. For the sake of simplicity, we may consider that each of them presents an expected annual return of 1 % and an annualised volatility of 1 % (the riskless rate will be assumed to be zero). Consider a portfolio constituted from the five aforementioned assets, in the same proportions.

- What is the Sharpe ratio of the portfolio?
- What is the probability that the portfolio ends the year in the red?
- Comment on the previous result.

**Solution.** (i) Consider a portfolio  $P := \frac{A_1 + \dots + A_5}{5}$  each of whose assets  $A_i$  has an average return  $R_i := \mathbb{E}[A_i] = 1\%$  and an annualised volatility  $\sigma_i := \sqrt{\text{Var}(A_i)} = 1\%$ . In order to compute the Sharpe ratio of the portfolio, we must compute its return  $R$  and its volatility  $\sigma$ . Now the return of the portfolio is, by linearity of expectation,  $R := \mathbb{E}[P] = \mathbb{E}\left[\frac{A_1 + \dots + A_5}{5}\right] = \frac{\mathbb{E}[A_1] + \dots + \mathbb{E}[A_5]}{5} = \frac{1 + \dots + 1}{5} = 1\%$ . As regards the variance of the portfolio, we have:

$$\begin{aligned}
\text{Var}(P) &= \text{Var}\left(\frac{A_1 + \dots + A_5}{5}\right) \\
&= \frac{\text{Var}(A_1 + \dots + A_5)}{25} \quad \text{by bilinearity of the variance} \\
&= \frac{\text{Var}(A_1) + \dots + \text{Var}(A_5)}{25} \quad \text{since the } A_i \text{ are uncorrelated} \\
&= \frac{1 + \dots + 1}{25} = \frac{1}{5}
\end{aligned}$$

We deduce that the volatility  $\sigma$  of the portfolio is  $\sigma := \sqrt{\text{Var}(P)} = \sqrt{\frac{1}{5}}\%$ . We deduce that the Sharpe ratio of the portfolio P is  $S := \frac{E[P] - r_f}{\sigma} = \sqrt{5} \approx 2.23$ .

- (ii) We shall make the simplifying assumption that the returns of each asset follow a normal distribution. Since the assets are uncorrelated with one another, the aforementioned portfolio follows a normal distribution, with expectation 1 % and volatility  $\sqrt{\frac{1}{5}}\%$  as it happens. Thus, by centring and standardising this normal distribution to facilitate the computations, it follows that:

$$\begin{aligned}
\mathbb{P}([P < 0]) &= \mathbb{P}\left(\left[\frac{P - 1}{\sqrt{1/5}} < \frac{0 - 1}{\sqrt{1/5}}\right]\right) \\
&= \mathbb{P}\left(\left[\frac{P - 1}{\sqrt{1/5}} < -\sqrt{5}\right]\right) \\
&\approx \mathbb{P}([\mathcal{N}(0, 1) \leq -2.23]) \approx 1\%
\end{aligned}$$

- (iii) Thus, with five uncorrelated strategies with a Sharpe ratio of 1, one constructs a portfolio whose Sharpe ratio is greater than 2 and whose probability of ending the year in the red is of the order of 1 %. By contrast, if one traded only a single asset among the five, for example the first, then the probability that the asset closes the year down satisfies:

$$\begin{aligned}
\mathbb{P}([A_1 < 0]) &= \mathbb{P}\left(\left[\frac{A_1 - 1}{\sqrt{1}} < \frac{0 - 1}{\sqrt{1}}\right]\right) \\
&= \mathbb{P}\left(\left[\frac{A_1 - 1}{\sqrt{1}} < -1\right]\right) \\
&\approx \mathbb{P}([\mathcal{N}(0, 1) \leq -1]) \approx 15\%
\end{aligned}$$

This shows the importance of diversification: it suffices to multiply the number of uncorrelated strategies in order to obtain an identical return but for a far lower level of risk.  $\square$

### Key Takeaways



A stock market index is a dimensionless number measuring the variation of a basket of assets weighted by their free-float capitalisation [Euronext (2016)]. The list of assets constituting a stock market index remains relatively stable, given that entry into an index generates a transitory abnormal price rise of 3 % to 5 % according to the price pressure theory [Harris & Gurel (1986), Shleifer (1986)]. Moreover, the variance of an equally weighted portfolio of  $n$  assets converges towards  $\sigma_p^2 = \frac{\bar{\sigma}^2}{n} + \left(1 - \frac{1}{n}\right)\bar{\sigma}_{ij}$ , the idiosyncratic risk vanishing as  $n$  tends to  $+\infty$  in favour of the irreducible systematic risk  $\bar{\sigma}_{ij}$  alone, which theoretically justifies passive management [Markowitz (1952), Sharpe (1964)]: more than 88 % of active American funds underperform their benchmark index over fifteen years, and the MSCI World (more than 1400 securities from 23 developed countries for 54 000 billion dollars in 2023) constitutes the benchmark instrument of institutional management [S&P Global (2025b), Banque mondiale (2023)].

### 2.3.3 Replicability by ETF

**Replicability** by listed index fund (*exchange-traded fund*, ETF) constitutes the most direct operational translation of the portfolio theory of Markowitz (1952): an **ETF** is an investment fund whose units are listed continuously on a regulated market, exactly like an ordinary share, and whose objective is to reproduce as faithfully as possible the performance of a benchmark index. The connection with the theory developed *supra* is immediate: it has been established, in the wake of the work of Sharpe (1964) and Merton (1972), that the market portfolio — approximated by a broad index weighted by free-float capitalisation — constitutes the efficient portfolio par excellence, and that every rational investor should, all other things being equal, seek to hold it. An ETF replicating the S&P 500 or the

MSCI World is precisely the practical implementation of this theoretical result: it allows any investor, whatever the amount invested, to hold a fraction of the market portfolio with management fees considerably lower than those of actively managed funds. Technically, two main methods of replication are distinguished. **Physical replication** consists in buying directly, into the fund's portfolio, all or a representative sample of the securities composing the index, in the proportions dictated by the index weightings; it may be full (*full replication*) or by optimized sampling, this latter variant buying only a subset of securities selected so as to minimise the **tracking error** — that is to say the standard deviation of the difference between the return of the fund and that of the index — while retaining a representative sectoral and factor exposure. The iShares Core S&P 500 UCITS ETF (ISIN IE00B5BMR087), managed by BlackRock (2026) and domiciled in Ireland, constitutes an example of full physical replication: it holds directly the 503 shares of the S&P 500 in proportions close to their weights in the index, displays a total expense ratio (TER) of 0.07 % per year and managed assets of more than 114 billion euros in February 2026, which makes it one of the largest ETFs in Europe. Physical replication has the advantage of transparency — the investor knows exactly which securities constitute his portfolio — but engenders significant transaction and rebalancing costs at the time of quarterly index reviews, as well as a friction linked to dividends withheld at source in the countries of origin of the securities (*dividend withholding tax drag*), as Charupat & Miu (2011) document in their analysis of the sources of ETF *tracking error* for equities.

**Synthetic replication** constitutes the second method, and it is this one which permits an application of portfolio theory within the French tax framework. Its principle rests on the use of a performance swap contract (*equity swap*): the ETF manager constitutes a portfolio of 'substitute' securities (*substitute basket*) — which are not necessarily the securities composing the index being replicated — and concludes with a bank counterparty (generally its parent bank or a large investment bank) a contract by which it undertakes to hand over the performance of the substitute basket in exchange for the performance of the target index. The mechanism comes in two main variants: the **unfunded swap** structure, in which the fund actually holds the substitute basket and exchanges its performance daily, and the **funded swap** structure, in which the fund transfers its assets to the counterparty, which hands it securities as collateral. The European regulation applicable to undertakings for collective investment in transferable securities (UCITS), as formalised by the guidelines published by European Securities and Markets Authority (2012), strictly frames the counterparty risk inherent in these structures: net exposure to a single counterparty may not exceed 10 % of the net asset value of the fund, and any collateral received must be sufficiently diversified and liquid to cover the exposure in full in the event of default. The main advantage of synthetic replication is that, by eliminating the transaction costs linked to the physical purchase and sale of securities, as well as the tax frictions on dividends, it makes it possible to obtain a *tracking error* often lower than that of physical replication, notably for indices with a high turnover rate or comprising illiquid securities. European Securities and Markets Authority (2012) notes that this method makes it possible to avoid the high rebalancing costs and replication errors inherent in the physical holding of securities, while introducing a counterparty risk which must be framed by regulation.

It is synthetic replication which makes possible a fiscally optimal application of portfolio theory in the French legal context. Indeed, the **plan d'épargne en actions** (PEA), instituted by the law of 14 November 1992 and codified at article L. 221-30 of the French Monetary and Financial Code, offers an exemption from tax on capital gains after five years of holding, but is accessible only to instruments invested to the tune of at least 75 % in shares of companies having their registered office in the European Economic Area (EEA). A physically replicating ETF on the S&P 500 or the MSCI World — whose weighting in American shares exceeds 60 % — could therefore not structurally be eligible for the PEA. Synthetic replication resolves this constraint: the substitute basket is made up of liquid European shares eligible for the PEA, which confers on the fund its legal eligibility, while the swap contract guarantees that the performance delivered to the investor is indeed that of the target American or world index. Three products concretely illustrate this mechanism. The Amundi PEA S&P 500 Screened UCITS ETF (PE500, ISIN FR0013412285), domiciled in France, is eligible for the PEA, replicates the S&P 500 index synthetically, displays a TER of 0.25 % and managed assets of more than one billion euros in 2025 according to the data of Amundi Asset Management (2025). Likewise, the Lyxor PEA Monde (MSCI World) UCITS ETF – Capi. (EWLD, ISIN FR0011869353), also domiciled in France and eligible for the PEA, replicates the MSCI World synthetically with a TER of 0.45 % according to the product factsheet published by Amundi Asset Management (2024). These two products illustrate the theoretical chain described *supra*: the substitute basket of European shares plays the role of a 'physically held' portfolio which ensures regulatory eligibility, while the swap guarantees the investor exposure to the true world market portfolio. The notion of **tracking difference** — to be distinguished from *tracking error*: the *tracking difference* measures the cumulative gap between the performance of the fund and that of the index over a given period, whereas the *tracking error* measures the volatility of that gap — is the relevant indicator for comparing these products from the point of view of the long-term investor: a fund may display a low *tracking error* (good stability of the gap) but a negative *tracking difference* (systematic cumulative underperformance), the second indicator being more decisive for the



investor who does not rebalance his portfolio frequently. As the AMF (2025b) records in its dashboard of active investors, roughly 509 000 private individuals carried out transactions on ETFs admitted to trading in Europe in 2024 — against 296 000 in 2023, that is a rise of more than 70 % in one year — and the AMF counted 5.2 million transactions on these instruments in 2024, after 2.8 million in 2023, testifying to the massive and accelerating adoption of these instruments by French retail investors.

### Key Takeaways



An ETF is a continuously listed fund replicating a benchmark index, constituting the practical implementation of the efficient market portfolio with fees considerably lower than those of active funds. Physical replication (full or by sampling) buys the securities of the index directly in their weightings, while synthetic replication uses a performance swap contract (*equity swap*): the fund holds a substitute basket and exchanges its performance for that of the target index with a bank counterparty. This method notably permits eligibility for the PEA (exemption from capital gains tax after five years): the substitute basket being composed of European shares, the constraint of 75 % of assets issued in the European Economic Area is respected, while the *swap* guarantees exposure to the target world index. Moreover, the *tracking difference* (cumulative performance gap) is the relevant indicator for the long-term investor, to be distinguished from the *tracking error* (the volatility of that gap).

## 3 Currencies, commodities and alternative assets

Beyond equity and bond markets, currencies, commodities and alternative assets constitute asset classes whose integration into a diversified portfolio improves the return-risk profile by exploiting their low correlations with traditional financial assets. The **foreign exchange market**, the largest market in the world by daily volume, offers specific strategies — *carry trade*, interest rate parity arbitrage, hedging of exchange rate risk — while **commodities** (hydrocarbons, precious metals, rare earths, water, agricultural products) expose the investor to geopolitical dynamics and to the cycles of the real economy. **Commercial property** and **infrastructure**, finally, complete this universe by offering stable and predictable income flows, at the price of the reduced liquidity characteristic of real assets.

### 3.1 The foreign exchange market

The economic analysis of the foreign exchange market (**Forex**) presupposes distinguishing several complementary and successive levels of apprehension, notably the microstructure and liquidity of the market, which account for the precise mechanisms by which price formation and the meeting of supply and demand operate in a decentralised market.

#### 3.1.1 Microstructure and liquidity of the foreign exchange market

The Forex market constitutes, by the volume of transactions carried out on it daily, the largest and most liquid financial market in the world. The Bank for International Settlements (2025c), within the framework of its fourteenth triennial survey conducted among more than 1100 banks and other reporting institutions spread across 52 jurisdictions, reports an average daily volume of transactions on the over-the-counter (**OTC**) market reaching 9600 billion dollars, that is a rise of 28 % on the previous survey: that of 2022 already recorded a record figure of 7500 billion dollars a day according to the Bank for International Settlements (2022b), itself up 14 % on 2019 where the amount stood at 6600 billion dollars traded each day. These orders of magnitude bear no comparison with those observed on stock markets; they underline the scale of the financial flows which pass through each day in a market organisation radically different from that of organised exchanges: there exists, on the Forex, no single central marketplace, no aggregated order book accessible to all participants, and the prices displayed by a given market maker are visible only to its own counterparties. This structural **informational opacity** is, as Lyons (2001) emphasises, one of the essential characteristics of the Forex which distinguishes it from organised markets and which has profound implications for the way in which the information relevant to the formation of parities is diffused in the economy. The market breaks down into two complementary segments: the **inter-dealer market**, in which the large banks and market makers exchange currencies among themselves — mainly via specialised electronic platforms such as EBS (Electronic Broking Services) or Refinitiv Matching —; and the **dealer-client market**, in which these same institutions serve as counterparties to non-financial firms, investment funds, central banks and other agents of the non-reporting financial sector. From then on, while the Forex does not have a single central exchange, the interbank market nevertheless relies on anonymous central order books — notably those of EBS (CME

Group) and Refinitiv Matching — even though the dealer-client segment remains largely opaque: as the Krohn et al. (2025) emphasises, more than 80 % of client orders there are handled directly within the internal liquidity *pools* of banks by a mechanism of **internalisation**, without ever appearing on the visible market. Moreover, according to the Bank for International Settlements (2025c), *inter-dealer* trading represented in April 2025 roughly 46 % of overall daily volume (that is 4400 billion dollars), while trading with ‘other financial institutions’ — a category grouping together non-reporting banks, hedge funds, institutional investors and proprietary trading firms — accounted for 50 % of volume; conversely, non-financial clients represented no more than 5 % of volumes in 2025, against 6 % in 2022 and 9 % in 2013, testifying to a marked and continuous financialisation of the market. The geographical distribution of trading is highly concentrated: the four leading world financial centres — the United Kingdom, the United States, Singapore and Hong Kong — alone intermediated 75 % of all world currency transactions in April 2025, the United Kingdom retaining its position as leading world centre with a share of roughly 38 % of global volumes, the United States maintaining second position with 19 %, and Singapore having appreciably increased its share to 11.8 % (against 9.5 % in 2022), a reflection of the rise of the Asian centres according to the Bank for International Settlements (2025c).

More globally, the analysis of the internal mechanisms of the Forex falls within the theory of **financial market microstructure**, which is concerned with price formation in markets characterised by the heterogeneity of agents, information asymmetries and the diversity of trading institutions. The central notion of this theory is the *bid-ask spread* (**bid-ask spread**), that is to say the gap between the price at which a market maker is prepared to buy a currency (*bid*) and that at which he is prepared to sell it (*ask*); this gap constitutes the market maker’s remuneration for the liquidity service he provides, but its economic decomposition is far richer than this simple definition suggests. As Hasbrouck (2007) sets out, the *spread* in reality incorporates three distinct components: an **order processing costs** component, linked to operational and technological expenses; an **inventory holding costs** component, which remunerates the exchange rate risk that the market maker assumes when he absorbs his clients’ orders without being able immediately to unwind his position; and, the most fundamental component from an informational point of view, the **adverse selection** component, which results from the information asymmetry between the market maker and some of his clients. Indeed, the market maker knows that some of his counterparties — the informed agents — have private information about the future exchange rate superior to his own, and that in dealing with them, he will systematically find himself on the wrong side of the transaction; he therefore widens his spread in proportion to the risk of being confronted with such agents, which implies, as the sequential models of price formation set out by Hasbrouck (2007) formalise, that *spreads* are positively correlated with the degree of informational asymmetry on the market. On the most liquid currency pairs according to the Bank for International Settlements (2025c) — EUR/USD (21.2 % of trading in 2025), USD/JPY (14.3 % of trading in 2025), GBP/USD (7.6 % of trading in 2025) — the *bid-ask spread* is structurally between 0.5 and 2 pips<sup>2</sup> during hours of high volume, testifying to a certain efficiency of these segments of the market. It is precisely within this framework that the major contribution of Evans & Lyons (2002) to financial economics belongs: starting from daily transaction data on the DM/USD market (German mark against American dollar) covering the period between May and August 1996, these authors demonstrate that **order flow** — defined as the signed sum of transactions, positive for those initiated by net buyers and negative for those initiated by net sellers — constitutes an excellent proxy for the exchange rate. Where traditional macroeconomic models of exchange rate determination produce in-sample coefficients of determination  $R^2$  rarely above 10 % and are regularly outclassed out of sample by a simple random walk, the microstructural model of Evans & Lyons (2002) produces  $R^2$  above 50 % for daily variations in the exchange rate: a net positive order flow of one billion dollars on the DM/USD market entails an appreciation of the dollar of roughly 0.5 %. This result means that the information relevant to the formation of the exchange rate is not entirely and instantaneously incorporated into public macroeconomic fundamentals, but is diffused progressively through the trading process itself, by the mechanism known as **price discovery**.

<sup>2</sup>The acronym **pip** — a contraction of *percentage in point* — designates the smallest standardised variation of an exchange rate quotation in the interbank market convention. For virtually all the major currency pairs — EUR/USD, GBP/USD and USD/CHF among others — one pip corresponds to a variation of one unit in the fourth decimal place of the quotation, that is  $10^{-4}$  of the unit of account; thus, a movement of the EUR/USD rate from 1.0800 to 1.0801 represents exactly one pip. An exception to this convention are the pairs involving the Japanese yen (USD/JPY, EUR/JPY...), whose quotation is expressed only to the second decimal place on account of the absolute level of the Japanese currency: a pip there is then worth  $10^{-2}$ , that is 0.01 yen per unit of foreign currency. It should be noted that one pip is equivalent to one basis point (bps), that is to say one hundredth of a percent (0.01 %) for four-decimal pairs. The concrete monetary value of a pip depends both on the instrument traded, on the lot size and on the quotation currency: on a standard lot (100 000 units of the base currency), one pip on EUR/USD represents approximately 10 dollars. Modern electronic platforms — EBS Market, Refinitiv Matching — now quote the main pairs to the fifth decimal place, which is equivalent to a tenth of a pip and reflects the refinement of price competition between market makers induced by high-frequency algorithmic trading.

## Focus...



**Transaction and refinancing costs on Forex.** The operational architecture of the foreign exchange market engenders specific carry costs, inherent in the time lag between trading and the effective settlement of transactions. On the spot market, the standard convention stipulates that a transaction initiated on the trade date (T) is the object of settlement and delivery of currencies two working days later (T + 2), as both the Chaboud et al. (2023) in its institutional analysis of the foreign exchange market and the ISDA (2021) in its legal guidelines relating to foreign exchange contracts specify. Consequently, any investor or financial institution wishing to maintain a speculative or hedging position open beyond the daily close of the New York session — conventionally fixed at five o'clock in the afternoon — without proceeding to the physical delivery of the capital must imperatively roll over its position: this accounting and financial operation takes the name of *overnight* and is technically executed by means of a foreign exchange *swap* known as *tomorrow-next*. Suppose indeed that an operator has bought euros against dollars on Monday (T): his value date is Wednesday (T + 2), the date at which he ought theoretically to receive the euros and deliver the dollars. If he does not wish to proceed to this delivery but to keep his position open one day longer, his broker will carry out simultaneously two operations in opposite directions: on the one hand, he will unwind the initial position with a value date of the following day (*tomorrow*, that is T + 1 or Tuesday), and on the other hand, he will immediately reopen an identical position with a new value date the day after (*next*, that is T + 2 or Wednesday). The operation is therefore entirely transparent for the end operator: his position remains open, only his settlement date has been shifted by one day. But this rollover is not financially neutral: in order to offset exactly the differential in the time value of money between the two currencies over the period of one day, the broker adjusts the reopening price of the position by applying *swap* points, expressed in pips, whose amount is rigorously determined by the overnight interest rate differential between the two currencies in application of the covered interest rate parity condition. The theoretical formula for the *tomorrow-next swap* rate for a currency pair A/B is thus  $S_{\text{tom-next}} = \frac{(r_B - r_A)}{360} \times \text{Rate}_{A/B}$ , where  $r_B$  denotes the overnight interest rate of the quoted currency,  $r_A$  that of the base currency, and the denominator 360 reflects the interest computation convention in force on the interbank foreign exchange market (360-day basis, with the exception of pairs in pounds sterling and in a few other *Commonwealth* currencies which use a 365-day basis). If  $r_B > r_A$ , the *swap* points are positive and the buyer of the base currency receives a credit at the *rollover*; if  $r_B < r_A$ , the points are negative and the buyer bears a debit. A calendar peculiarity of this continuous mechanism lies in the phenomenon of the *overweek* or triple Wednesday *swap*: since interbank markets are closed during the weekend, a transaction rolled over from Wednesday evening to Thursday morning sees its value date shifted from Friday to the Monday of the following week, which obliges the market maker to charge or credit three consecutive days of interest in a single accounting operation in order to compensate very exactly for the tying up of capital over seventy-two hours.

The structure by currency and by instrument of the foreign exchange market, as the most recent data published by the Bank for International Settlements (2025c) document, illustrates the persistence of a very marked international monetary hierarchy, coupled with a notable evolution in the composition of instruments. The American dollar figures in one of the two legs of 89.2 % of all foreign exchange transactions carried out in April 2025, slightly up on the 88.4 % recorded in 2022, which confirms its role as a **vehicle currency**: in a decentralised market, it is indeed less costly, for any currency pair, to go through an intermediate conversion into dollars than to organise liquid markets for all the possible combinations of currencies, which explains the structural hegemony of the greenback. The euro maintains second position, present in 28.9 % of transactions (against 30.6 % in 2022 and 32.3 % in 2019 according to the Bank for International Settlements (2022b), that is a continuous retreat), followed by the Japanese yen with 16.8 % (almost stable since 2019), the pound sterling with 10.2 % (significantly down from 12.9 % in 2022 and 12.8 % in 2019), the Chinese renminbi with 8.5 % (against 7 % in 2022 and only 4 % in 2019, that is a doubling in six years testifying to the accelerated internationalisation of Beijing's currency) and the Swiss franc with 6.4 % (against 5.2 % in 2022 and 4.9 % in 2019)<sup>3</sup>. As regards instruments, the Bank for International Settlements (2025c) signals a notable structural evolution: **FX swaps** — which combine a spot transaction and a forward transaction in the opposite direction and are mainly used to manage short-term foreign currency liquidity and to hedge exchange rate risks — remain the most traded instrument with a volume of 4000 billion dollars a day in April 2025, but their share in the global total has gone from 51 % in 2022 to 42 %, its lowest level since the 2010 survey, in favour of sustained growth in spot transactions (whose share rises from 28 % to 31 %), outright forwards (from 15 % to 19 %), and above all foreign exchange options, whose volume has more than doubled to represent henceforth 7 % of the total against 4 % in 2022; this evolution in the structure of instruments reflects, according to the Bank for International Settlements (2025c), an increasing sophistication of the hedging strategies of institutional actors, in particular in response to the strong volatility of foreign exchange markets in April 2025 consequent upon the trade policy announcements of the major economic powers, which led many asset managers exposed to the dollar to hedge massively via forward sales.

<sup>3</sup>It should be noted that these shares add up to 200 % and not 100 % inasmuch as each transaction necessarily involves two currencies.

## Key Takeaways



The Forex market is, with 9600 billion dollars traded daily in April 2025, the largest and most liquid financial market in the world; its architecture is radically different from that of organised exchanges — absence of a single central marketplace, of an aggregated order book and of pre-trade transparency — the interbank market resting on anonymous electronic platforms (EBS, Refinitiv Matching) while more than 80 % of client orders are internalised in the private liquidity pools of the large banks without appearing on the visible market [Bank for International Settlements (2025c), Lyons (2001)]. The distribution of volumes reveals a twofold concentration: geographical — the United Kingdom, the United States, Singapore and Hong Kong alone intermediating 75 % of world transactions — and monetary, the dollar figuring in one of the two legs of nearly 90 % of all transactions as a structural vehicle currency, the euro retreating to less than 30 %, the renminbi progressing to nearly 10 % [Bank for International Settlements (2025c)].

### 3.1.2 Trading strategies: *carry trade*, momentum and arbitrage

The *carry trade* constitutes the most documented speculative strategy on the foreign exchange market: the investor borrows a currency with a low interest rate — the funding currency — in order to place the funds obtained in a currency with a higher yield — the target currency — and pockets the rate difference daily at the *rollover overnight*. If the condition of **uncovered interest rate parity** (UIP) were verified empirically, this strategy would be sterile: as Obstfeld & Rogoff (1996) formalise it, UIP is stated as  $i_t - i_t^* = \mathbb{E}_t[\Delta s_{t+1}]$ , where  $i_t$  denotes the domestic interest rate,  $i_t^*$  the foreign interest rate and  $\mathbb{E}_t[\Delta s_{t+1}]$  the expected logarithmic variation of the exchange rate, in units of national currency per foreign unit; if this condition held, the profitability of the *carry trade* would therefore be nil in expectation. Now, as Fama (1984) demonstrated, the regression coefficient of the variation of the exchange rate on the forward premium is not only lower than the unity predicted by UIP, but often negative: currencies with high interest rates therefore tend in reality to appreciate rather than to depreciate, which constitutes the heart of the **forward premium puzzle**. In this context of structurally violated UIP, the *carry trade* turns out to be profitable on average: according to Norges Bank (2014), the average annual return of *carry trade* strategies on G10 currencies — dollar (USD), euro (EUR), yen (JPY), pound sterling (GBP), Swiss franc (CHF), Australian dollar (AUD), Canadian dollar (CAD), Swedish krona (SEK), New Zealand dollar (NZD) and Norwegian krone (NOK) — reached 5.3 % over the period 1999–2013, with an annualised volatility of 8.5 %, resulting in a Sharpe ratio of 0.6, a level of performance comparable to that of equity markets over the long run but with a different risk structure. In order to explain the persistence of these excess returns, Lustig & Verdelhan (2007) propose an interpretation in terms of a **consumption risk premium**. More precisely, an asset is all the riskier the more its return is negatively correlated with the investor's consumption: an asset which depreciates precisely when consumption contracts is particularly painful to hold, since the capital loss occurs at a moment when the marginal utility of wealth is at its maximum — the investor is already constrained — and therefore deserves to offer in expectation an excess return (risk premium) to convince agents to hold it. Now, according to Lustig & Verdelhan (2007), high interest rate currencies — those which the American investor buys within the framework of the *carry trade* — present precisely this procyclical risk profile: on a sample of 53 countries covering the period 1953–2002, they show indeed that portfolios of high-yielding currencies tend to depreciate significantly against the dollar during phases of contraction of American consumption, that is to say during recessions, while funding currencies — the yen, the Swiss franc — appreciate in the same circumstances, playing the role of safe haven. The negative correlation between the return of the *carry trade* and the growth of American consumption is thus the central mechanism of their model: the operator who sells yen to buy Australian dollars (a classic example of *carry trade*) exposes himself to a capital loss in the event of an American recession, precisely when he can least afford it. From then on, Lustig & Verdelhan (2007) conclude that the excess return of the *carry trade* is not a market anomaly but a fair remuneration of this systemic consumption risk: it constitutes a premium for holding an asset whose return is procyclical from the point of view of the American investor. The fundamental fragility of the *carry trade* nevertheless lies in its asymmetric risk profile: according to Brunnermeier et al. (2009), the returns of this strategy present a very pronounced **negative skewness**, with a strongly negative skewness coefficient and excessive kurtosis, which reflects the phenomenon known as *going up by the stairs, coming down by the elevator*: profits accumulate slowly and regularly during calm phases, only to be entirely wiped out during episodes of sudden collapse (*currency crash*). Brunnermeier et al. (2009) show that crash risk is linked to the liquidity conditions of the interbank market, measured by the TED **spread** (the gap between the three-month LIBOR rate and the rate on American Treasury bills of the same maturity): when this *spread* widens — signalling tension in interbank liquidity, as was the case in October 2008, when the TED *spread* exceeded 450 basis points — *carry trade* operators unwind their positions simultaneously, triggering self-fulfilling spirals of depreciation of target currencies and appreciation of funding currencies which retrospectively validate their decision to exit. The Japanese yen and

the Swiss franc, as structural funding currencies of the world *carry trade*, are thus exposed to brutal and massive appreciations during episodes of financial stress.

### Exercise



**EUR/JPY carry trade.** In January 2024, the key rate of the European Central Bank (ECB) stands at 4.50 % per year and the key rate of the Bank of Japan (BoJ) at 0.10 % per year. An institutional investor decides to implement a *carry trade* strategy: he borrows 100 000 euros at 0.10 % per year (yen funding rate, reflected in the cost of the *rollover overnight*) and places the proceeds in euros at 4.50 % per year. The spot EUR/JPY exchange rate is 158.50 yen to one euro on the day the position is initiated.

- (i) Compute the net daily carry income for a position of 100 000 euros, applying the 360-day convention.
- (ii) Compute the cumulative carry income over a *holding* period of 90 consecutive working days, assuming a daily rollover of the position without any variation in key rates.
- (iii) Determine the level of the EUR/JPY exchange rate below which the strategy becomes loss-making after 90 days (*break-even* threshold).
- (iv) Compute the relative variation of the EUR/JPY rate (as a percentage) corresponding to this *break-even* threshold. Comment.

**Solution.** (i) The annual interest rate differential is  $\Delta r = 4.50\% - 0.10\% = 4.40\%$  per year. On a 360-day basis, the daily carry income for a position of 100 000 euros is therefore written  $G_{\text{day}} = 100\,000 \times \frac{4.40\%}{360} \approx 12.22$  euros per day. This amount corresponds to the *overnight* credit posted to the investor's account at each *rollover tomorrow-next*, in net remuneration for holding the high-yielding currency.

- (ii) Assuming key rates unchanged and a daily rollover without any quotation gap, the cumulative gain over 90 working days is then:  $G_{90} = 12.22 \times 90 \approx 1100$  euros. Expressed as a percentage of the amount of the initial position (100 000 euros), this gain represents approximately 1.10 % over 90 days, that is an annualised return of the order of 4.40 % (consistent with the rate differential).
- (iii) The cumulative gain of 1100 euros corresponds, at the initial rate of 158.50 yen per euro, to a counterpart in yen of  $1100 \times 158.50 = 174\,350$  yen. Now the initial position of 100 000 euros is worth 15 850 000 yen at the initiation rate. For the strategy to break even, the capital loss due to an appreciation of the yen must therefore be strictly lower than the cumulative carry income of 174 350 yen. In other words, we must have  $100\,000 \times (158.50 - S) \leq 174\,350$ , that is  $158.50 - S \leq 1.7435$  or again  $S \geq 158.50 - 1.74 = 156.76$  yen to one euro. The strategy therefore becomes loss-making if the EUR/JPY exchange rate falls below the *break-even* threshold of  $S_{\text{BE}} := 156.76$ , that is to say if the euro depreciates by more than 1.74 yen relative to its entry rate.
- (iv) The relative variation corresponding to the *break-even* threshold is therefore  $\frac{156.76 - 158.50}{158.50} \approx -1.10\%$ . An appreciation of the yen of only 1.10 % over 90 days therefore suffices to wipe out the entirety of the accumulated carry income. This computation illustrates precisely the asymmetric risk profile described by Brunnermeier et al. (2009): the gain is slow, gradual and capped at 4.40 % per year, whereas the risk of loss is in principle unbounded and may materialise brutally — during the USD/JPY collapse of October 1998 for example, the yen appreciated by more than 15 % in less than 48 hours. Carry income accumulated over several weeks may thus be wiped out in a few minutes during an episode of forced liquidation, in accordance with the vivid formula '*going up by the stairs, coming down by the elevator*': the strategy offers regular returns which induce a false sense of security, masking the low but non-zero probability of a catastrophic and sudden loss linked to the collective unwinding of *carry* positions.  $\square$

Subsequently, **momentum** strategies — founded not on rate differentials but on the persistence of past exchange rate trends — constitute the second great family of systematic strategies on the Forex: a currency which has appreciated over the last few days or weeks tends, outside episodes of brutal reversal, to continue to appreciate over a short-term horizon, and conversely for a currency in a downward trend; the momentum strategy therefore consists in buying currencies in an upward trend and selling those in a downward trend, on the basis of the past return signal alone. Menkhoff et al. (2012) thus conducted a cross-sectional analysis on a universe of 48 currencies covering the period 1976–2010: they establish that *cross-sectional momentum* strategies — consisting in buying the quintile of the best-performing currencies over the preceding month and selling the quintile of the worst-performing — generate an average annualised return *spread* of roughly 10 % and a Sharpe ratio reaching 0.95 for the monthly strategy

with one-month formation, thus surpassing the performance of the *carry trade* over the same period. In addition, Menkhoff et al. (2012) show that these currency momentum returns are not explained by the classic risk factors of the *carry trade* — in particular the dollar factor (which captures the average return of all foreign currencies against the dollar and reflects world demand for dollar liquidity) and the HML-FX factor (which captures the *carry* premium properly speaking by extracting the cross-sectional variation of return as a function of the rate differential) identified by Lustig et al. (2011). This suggests that the momentum strategy and the *carry trade* constitute two independent sources of risk premia on the Forex, opening the way to combined strategies that are even more effective: the two strategies present a slightly negative correlation, so that a portfolio combining the two approaches in equal proportion substantially improves the Sharpe ratio relative to each strategy taken in isolation, while reducing the negative *skewness* characteristic of the pure *carry trade*. The explanation of the profitability of momentum has been the object of debate: behaviouralist interpretations invoke **cognitive biases** such as initial underreaction to information and subsequent overreaction, leading investors to extrapolate past trends mechanically well beyond their real informational content; interpretations in terms of **risk**, advanced notably by Menkhoff et al. (2012), argue that currencies in an upward momentum tend to depreciate during episodes of abnormally high global volatility — they therefore constitute risky assets in terms of hedging against world financial turbulence — so that their excess return in reality remunerates a systemic risk that is difficult to diversify.

Finally, **arbitrage** on the foreign exchange market covers a spectrum of very heterogeneous strategies, ranging from classic triangular arbitrage — which exploits momentary inconsistencies between three exchange rate quotations — to covered interest rate parity (CIP) arbitrage — which exploits the gaps between the implicit forward exchange rate and the forward exchange rate observed on derivatives markets. In its canonical theoretical conception, arbitrage must be instantaneous, riskless and self-extinguishing: as soon as an opportunity appears, its exploitation by arbitrageurs destroys it. If markets were efficient, such *free lunch* strategies ought not to exist. Yet, by mobilising transaction data on *timeframes* of the order of the microsecond on three major capital and foreign exchange markets over a period of seven months, Akram et al. (2008) establish that CIP and triangular arbitrage opportunities really do exist on the Forex, but that their lifespan is extremely brief — of the order of a few seconds, or even a few tenths of a second for the largest in terms of economic value — and that the exploitable amounts generally remain below one million dollars, negligible relative to the daily volumes of 9600 billion dollars documented by the Bank for International Settlements (2025c). This result empirically validates the **weak efficiency of the foreign exchange market** in its electronic interbank segment: the market is therefore not efficient in the sense that no opportunity would ever exist, but the opportunities are so ephemeral and of such small magnitude that they are unexploitable by a human agent and are captured only by **high-frequency algorithmic trading** (HFT) systems capable of placing orders in a few microseconds. The arbitrage landscape has, however, been reshaped from the financial crisis of 2008 onwards, by a more durable and structurally more significant phenomenon: the persistent violation of CIP, which stipulates that the return on an investment in dollars must be strictly identical to the return on an investment in euros hedged against exchange rate risk by a forward contract, so that the CIP basis — defined as the gap between the cost of synthetic borrowing of dollars via a foreign exchange *swap* and the direct borrowing rate on the American money market — must be identically nil in the absence of an arbitrage opportunity. Formally, the condition is therefore written  $1 + i_{USD} = \frac{F_{t,t+n}}{S_t} (1 + i_{EUR})$ , where  $i_{USD}$  and  $i_{EUR}$  denote the interest rates over the period  $[t, t + n]$  in dollars and in euros respectively,  $S_t$  the spot exchange rate and  $F_{t,t+n}$  the forward exchange rate contractually fixed at  $t$  for a maturity  $t + n$ . If this condition is violated — if the left-hand side exceeds the right-hand side — a bank could borrow in euros, convert them spot into dollars, invest on the American money market and sell the dollars forward: a *riskless profit* arbitrage opportunity in principle instantaneously eliminated. Now, as Du et al. (2018) establish definitively from interbank rate (LIBOR) and repo<sup>4</sup> data on all the G10 currencies covering the period 2000–2016, the CIP basis remained systematically and significantly close to zero before 2007, before collapsing at the time of the bankruptcy of Lehman Brothers in September 2008, where certain bases momentarily reached –200 basis points, before stabilising thereafter, from 2010 onwards, at a persistent and non-zero level. Over the period 2010–2016, the estimates of Du et al. (2018) report negative and statistically significant three-month CIP bases for the whole of the G10, with arbitrage opportunities of between 9 and 20 basis points on annual average — a magnitude which, within the framework of classical theory, ought to have triggered massive and immediate arbitrage flows. The origin of this distortion lies in what Du et al. (2018) describe as a **‘quarter-end effect’**: CIP deviations are systematically

<sup>4</sup>A **repo** (*repurchase agreement*) is a short-term financing operation by which a financial institution temporarily transfers securities — most often government bonds — to a counterparty in exchange for cash, with a firm undertaking to buy them back at an agreed date and at a slightly higher price. The difference between the initial transfer price and the buyback price constitutes the repo rate, which represents the implicit cost of financing for the borrower of cash and the remuneration of the investor. Economically, the repo is therefore the equivalent of a loan secured on securities: the borrower of cash ‘pledges’ his securities, while the lender of cash carries out a ‘reverse repo’. In the context of measuring deviations from CIP, Du et al. (2018) use repo rates on government bonds as a proxy for the credit-risk-free cost of financing, in order to ensure that the CIP bases they measure do indeed reflect a friction of financial intermediation and not a simple gap in counterparty risk between borrowers.



and significantly larger at quarter-end dates (31 March, 30 June, 30 September and 31 December) than on other days. This pattern is explained by the quarterly reporting obligations to which banks subject to the Basel III leverage ratio requirements are bound: institutions artificially inflate and deflate their balance sheets at regular intervals, which directly affects their capacity to supply arbitrage and generates predictable and repeated tensions on CIP bases. Du et al. (2018) thus show that the distortion is regulatory in origin and not attributable to simple credit risk or to transaction costs, since it affects even strategies constructed from bonds issued by top-quality government agencies. This result belongs fully within the theoretical framework of the limits to arbitrage formalised by Shleifer & Vishny (1997): in a market where arbitrage requires capital and is conducted by a small number of specialised intermediaries managing third-party funds, opportunities may persist durably if arbitrageurs face balance sheet constraints or suffer withdrawals of capital at the precise moment when prices diverge most from fundamentals; in the case of the post-2008 foreign exchange market, it is prudential regulation itself which plays the role of the capital constraint, transforming what would have been an ephemeral anomaly into a structural and persistent distortion inherent in the new international regulatory architecture.

### Debate



**Are financial markets efficient?** The Nobel prize in economics was awarded jointly in 2013 to Eugene Fama and Robert Shiller: Fama (1965) is the father of the theory of market efficiency, whereas Shiller (2000) mobilised behavioural finance to explain speculative bubbles. This joint award appears schizophrenic inasmuch as the two theories appear at first sight diametrically opposed: indeed, the keystone of the theory of **efficient markets** is the systematic return of the market price of a good or an asset to its **fundamental value**, that is to say an intrinsic price of the good or the asset, which depends on various real variables called ‘fundamentals’ (production costs, stocks, growth rate of the sector, and so on). The theory of efficient markets of Fama (1965) then stipulates that the market price can never depart for long from fundamental value, since agents, seeking to maximise their utility, ceaselessly carry out cost-benefit and return-risk arbitrage operations, which tend to bring the market price closer to fundamental value.

Let us illustrate this mechanism through a drought affecting maize production. In this case, since world demand for maize is constant or even growing, a slowdown in maize production provokes an increase in maize prices, thereby moving the price of the cereal away from its fundamental value. Indeed, maize buyers cannot, at first, defer their consumption of maize, a good for which there is no immediate substitute, while the supply of maize is rigid in the short run. However, in the longer run, substitutability of goods and flexibility of supply operate: discouraged by the runaway prices, some buyers, notably the poorest, defer their consumption of maize or substitute another cereal for it, thus leaving the maize market and contracting demand. At the same time, encouraged by the rise in prices and cherishing the hope of a larger profit, maize producers increase production, provoking a rise in supply. This twofold movement then leads to a relaxation of upward pressure on the price of maize, which returns *in fine* to its fundamental value. From then on, if a **speculative bubble** designates, as Shiller (2000) defines it, a substantial gap between the market price of a good or an asset and its fundamental value, then the theory of efficient markets is incompatible with the persistence of speculative bubbles in the long run. Indeed, according to the theory of efficient markets, bubbles may be explained by a failure in agents’ reaction in the face of the non-substitutability of assets and the rigidity of supply in the short run, but these are always destined to be reabsorbed in the long run. More precisely, the market price can never move away durably from its fundamental value, but fluctuates around a given level, assessable and therefore predictable. Consequently, financial markets are stable as soon as fundamentals are. However, it is appropriate to go beyond this dichotomy between market efficiency and speculative bubbles in order to articulate these two concepts dialectically. We must then redefine the notion of efficiency so as to bring it closer to the original definition of Fama (1965). Whereas our first paragraphs posited efficiency as **allocative efficiency of markets**, Fama (1965) thought of it rather as **informational efficiency**: a market may be described as efficient as soon as its prices fully and always incorporate the available information. Otherwise put, efficient markets react almost instantaneously to any new information. Fama (1965) then distinguishes three levels of efficiency: at the level of **weak efficiency**, only the information contained in past market prices is incorporated by the prices of goods or assets; at the level of **semi-strong efficiency** this time, in addition to past information, all public information is reflected by market prices; finally, the level of **strong efficiency** likewise incorporates all private information into price formation. Whatever the level of efficiency retained to explain a financial market, nothing therefore presupposes the stability of the said market. Otherwise put, efficiency does not imply effectiveness. Understood in the informational sense, market efficiency is therefore in no way incompatible with the formation of speculative bubbles and more generally with the presence of instabilities on markets.

The consequences of the explosion of the Challenger shuttle in 1986, studied by Maloney & Mulherin (2003), constitute a striking illustration of this: whereas the commission of inquiry set up in the aftermath of the catastrophe

took weeks to formulate its conclusions, barely an hour after the event, while the share prices of all the firms partnering the Challenger project were plunging, one firm saw its price collapse further than the others: Morton-Thiokol, the group responsible precisely for the manufacture of the defective part. This example illustrates the efficiency of the aerospace market at that instant, inasmuch as the new information was incorporated into prices within the space of less than an hour. In such an example, markets turned out to be efficient, in the sense that the new information was very rapidly incorporated into prices. The efficiency of the market is then in no way contradictory with a brutal fall in prices, a speculative bubble, downward in our example. In its informational acceptance, the theory of market efficiency therefore permits the formation of speculative bubbles, which it explains by the **occurrence of unpredictable information** and the reaction of markets to that information, which may be private in the case of markets of strong efficiency, or even public in the case of markets of lesser efficiency.

### Key Takeaways



The *carry trade* — borrowing a currency with a low interest rate in order to place it in a currency with a high yield, with daily collection of the differential at the *rollover overnight* — is profitable only because uncovered interest rate parity (UIP) is structurally violated, which Fama (1984) precisely brings to light: Lustig & Verdelhan (2007) interpret this excess return not as an anomaly but as a systemic consumption risk premium, high-yielding currencies (the Australian dollar) tending to depreciate precisely during American recessions, while funding currencies (the yen, the Swiss franc) play the role of safe haven. Subsequently, momentum strategies ('following the trend', whether upward or downward) constitute an independent source of risk premium: the slightly negative correlation with the *carry trade* allows a combined portfolio substantially to improve the Sharpe ratio while reducing the negative *skewness* of the pure *carry*, which is interpreted either through cognitive biases of initial underreaction followed by overreaction, or through an exposure to global volatility risk [Menkhoff et al. (2012), Lustig et al. (2011)]. Finally, triangular and CIP arbitrage exist empirically but in ephemeral fashion: Akram et al. (2008) detect opportunities lasting a few tenths of a second and of an amount below one million dollars, unexploitable by a human agent and captured only by high-frequency *trading*, which validates the weak efficiency of the electronic interbank market.

#### 3.1.3 Exposure to exchange rate risk and hedging instruments

Exposure to exchange rate risk constitutes one of the main preoccupations of the financial management of multinational firms and of institutional investors operating internationally. Papaioannou (2006) thus distinguishes three exposure risks. The first type is **transaction exposure**: it designates the risk that future cash flows contractually committed in foreign currencies — trade receivables, trade payables, repatriation of dividends — may be altered by a variation in the exchange rate between the date of signature of the contract and the date of effective settlement. This risk is the most immediate and the most easily measurable: a French firm having signed in January a contract for the sale of industrial equipment denominated in dollars for delivery in July receives exactly the EUR/USD exchange rate differential between these two dates. The second type is **translation exposure** or accounting exposure: it concerns the consolidation of the financial statements of foreign subsidiaries, whose assets, liabilities and results must be converted into the group's functional currency; an appreciation of the euro mechanically generates a reduction in the consolidated book value of subsidiaries whose assets are denominated in foreign currencies, without this necessarily representing a real economic loss. The third type, the most diffuse and the most difficult to hedge, is **economic exposure** (or *competitive exposure*): it designates the impact of a durable variation in the exchange rate on the structural competitiveness of the firm, its future flows not yet contracted and its fundamental value. A durable appreciation of the euro, for example, durably weakens the margins of French exporters even on markets not directly invoiced in dollars, as soon as their American or Asian competitors see their price competitiveness improve. The extent of the use of hedging instruments by non-financial firms has been documented by Bodnar et al. (1998), who report the results of the third edition of the biennial survey conducted on a sample of 399 American non-financial firms: 50 % of them used derivatives to manage their financial risks; this rate rose to 83 % among large firms and, within these, 44 % had recourse to currency options, which testifies to an increasing sophistication of hedging strategies as the size of the firm increases. Hedging exchange rate risk represents not only an issue of cash flow management, but also an issue of stock market valuation, as Allayannis & Weston (2001) establish, whose empirical work on a panel of 720 large American firms between 1990 and 1995 reveals that an active policy of hedging exchange rate risk via derivatives is associated on average with a valuation premium (*hedging premium*) of the order of 4.87 % of the value of the firm (*Tobin's Q*), a result which argues strongly for the integration of exchange rate risk management into the value creation strategy of international groups and not for its accounting or tax dimension alone.



The instruments for **hedging exchange rate risk** available on financial markets form a continuum ranging from the simplest and most standardised solutions to the most complex and bespoke structures. The first and most widely used is the **forward exchange contract**: it makes it possible to fix today the exchange rate at which a future transaction will be carried out, thereby eliminating any uncertainty about the flow in national currency. A French importer awaiting payment of 10 million dollars in 90 days may buy these dollars forward today at the EUR/USD *forward* rate, computed in application of covered interest rate parity — that is, for example, EUR/USD 1.0550 if the spot rate is 1.0500 and the rate differential favours the dollar — and thus ensure a predictable conversion rate for his accounts. The forward rate is not a forecast of the future rate; it is strictly determined by the interest rates in force, as Smith & Stulz (1985) explain. The second major instrument is the **currency option**: unlike the forward contract, which *obliges* both parties to execute the transaction at the agreed rate, the option confers on its buyer the *right* — but not the obligation — to buy (*call*) or to sell (*put*) a currency at a given exercise price (*strike*) before a determined expiry date. This asymmetry of rights represents the value of the option, whose price is determined like that of Black & Scholes (1973): in return for this premium, the firm retains the possibility of benefiting from a favourable exchange rate movement if the *spot* rate is more advantageous than the *strike* at expiry. Foreign exchange derivatives markets reach massive volumes: according to the Bank for International Settlements (2025b), the notional outstanding amount of all over-the-counter foreign exchange derivatives stood at 155 000 billion dollars at the end of June 2025 (more than world GDP of 113 800 billion dollars according to Agarwal (2025)), up 20 % on the end of June 2024, of which the vast majority — that is 100 000 billion dollars — corresponds to contracts of maturity below one year, which confirms the predominance of short-term hedging in the practice of corporate treasurers and risk managers. The third instrument is the long-term **cross-currency swap**: it makes it possible to exchange interest flows and a principal in two different currencies over a period ranging from one to thirty years, which makes it the benchmark instrument for hedging bond issues in foreign currencies. Firms as well as sovereign states use it systematically to finance themselves in the most favourable currency and simultaneously to hedge the associated exchange rate risk. To these financial instruments is added **natural hedging**, which consists in aligning revenues and costs geographically in the same currency: a European multinational which finances its American factories with debt denominated in dollars mechanically reduces its net exposure to the dollar, since the depreciation of the euro appreciates its ‘dollar-denominated’ revenues in euros but simultaneously increases its debt service in dollars, the two effects partially offsetting one another.

From the point of view of the individual or institutional investor holding a portfolio of international assets, exchange rate risk constitutes a component of total risk which may represent a substantial fraction of the volatility of a portfolio invested in foreign assets. The question of whether it is worth hedging this exchange rate risk bears on the trade-off between the **cost of hedging** and the benefits in terms of **volatility reduction**. Campbell et al. (2010) thus show that the optimal hedge ratio for an investor in international equities is neither 0 % nor 100 %, but depends on the correlation between asset returns and exchange rate variations: for investors in foreign bonds, full hedging of exchange rate risk is generally optimal since bonds and currencies are weakly correlated with one another; by contrast, for investors in foreign equities, partial hedging is called for since the equities and currencies of developed countries present a sufficiently positive correlation for the currency component to play the role of a partial shock absorber for portfolio volatility in periods of crisis. The effective cost of exchange rate hedging via forward contracts is directly determined by the interest rate differential between the two currencies in application of covered interest rate parity: a French investor who hedges his portfolio of American equities against EUR/USD risk pays an annualised hedging cost equal to the difference between the American riskless rate and the European riskless rate. In the rate environment of 2024–2025, where the *Fed funds* stand between 4.25 and 4.50 % and the ECB rate between 2.50 and 3.50 %, this annualised hedging cost lies between 100 and 200 basis points, a levy which, on an S&P 500 ETF whose historical average annual return is of the order of 10 % in dollars, remains manageable but significant. As Bodnar et al. (1998) show, many institutional investors consequently adopt **selective hedging**, not systematically hedging the entirety of their currency exposure but dynamically adjusting their hedge ratio as a function of their market expectations.

### Focus...



**Reading the name of an ETF.** The commercial name of an ETF constitutes a genuine condensed technical datasheet, each component of which refers to a precise legal, tax or financial characteristic. Let us take as an example the benchmark product *Amundi PEA S&P 500 UCITS ETF EUR Hedged Acc*, whose **ISIN code** is FR0011871136, listed on Euronext Paris. The first element is the **manager** — here Amundi Asset Management, a subsidiary of Crédit Agricole and the leading European asset manager with 2240 billion euros of assets under management at 31 December 2024 according to Amundi (2025). The term **UCITS** (*undertaking for collective investment in transferable securities*), for its part, means that the fund complies with European directive 2009/65/EC of the European Parliament and of

the Council, transposed into French law within the Monetary and Financial Code. This directive imposes a set of strict regulatory constraints: minimum diversification of assets (a single issuer may not represent more than 10 % of net assets), daily liquidity, rules for managing counterparty risk, publication of a key investor information document (KIID), daily valuation and authorisation by the competent authority, namely the Autorité des marchés financiers (AMF) in France. A UCITS may moreover be marketed throughout all the member states of the European Union under the European passport procedure, which confers on it first-rank liquidity and regulatory transparency. Subsequently, the mention **PEA** indicates that the fund is eligible for the French *plan d'épargne en actions*, which implies that it respects the eligibility conditions defined at article L221-31 of the Monetary and Financial Code: its assets must comprise at least 75 % of shares in firms having their registered office in a member state of the European Economic Area. For an ETF replicating the American S&P 500 index, this constraint is satisfied by the technique of the synthetic *swap*. Moreover, the mention **EUR Hedged** indicates that the fund neutralises exchange rate risk between the American dollar (the currency of the underlying index) and the euro (the reference currency of the French investor): each month or each quarter, the manager rolls over forward contracts selling dollars (buying euros) for an amount close to the net assets of the fund, so that the performance delivered to the investor is almost exclusively that of the S&P 500 index expressed in euros, without the amplifying or damping effect of the EUR/USD exchange rate. Finally, the mention **Acc** (*accumulating*) means that the dividends received by the fund on the underlying shares are automatically reinvested in the fund rather than distributed in *cash*; the investor therefore receives no periodic flows but sees the net asset value of his units increase. The **Dist** (*distributing*) variant would be the object of an annual or half-yearly distribution of dividends. Beyond the product *Amundi PEA S&P 500 UCITS ETF EUR Hedged Acc*, only one other PEA-eligible S&P ETF is hedged against EUR/USD exchange rate risk: *BNP Paribas Easy S&P 500 UCITS ETF EUR Hedged (Acc)* (ISIN FR0013041530).

### Key Takeaways



Exposure to exchange rate risk comes in three analytically distinct forms: transaction exposure, which arises from any contractual commitment in foreign currency whose settlement is deferred and which is the most directly measurable and hedgeable; translation exposure, which results mechanically from the consolidation of foreign subsidiaries without necessarily representing a real economic loss; and economic exposure, the most diffuse, which designates the impact of a durable variation in parity on the structural competitiveness and fundamental value of the firm, even on markets not directly invoiced in foreign currencies [Papaioannou (2006)]. Hedging exchange rate risk is a widespread and value-creating practice [Bodnar et al. (1998)]. The instruments available are ordered along a continuum ranging from the simplest to the most bespoke: the *forward* contract, whose rate is strictly determined by covered interest rate parity and not by a forecast of the future rate; the currency option, whose premium remunerates the asymmetry of rights and makes it possible to benefit from a favourable movement of the *spot* rate beyond the *strike*; and the *cross-currency swap*, the benchmark instrument for long hedges (one to thirty years) on bond issues in foreign currencies [Smith & Stulz (1985)].

## 3.2 Commodities

The markets for **commodities** constitute a structuring component of the world financial system, at the intersection of the real economy and speculative finance: their prices determine world macroeconomic balances, inflationary dynamics and the geopolitical balance of power between producing and consuming nations. **Hydrocarbons**, whose prices are subject to geopolitical shocks, condition the whole set of industrial production costs and the monetary policies of the major central banks; **precious metals** and **rare earths**, having become indispensable to the technologies of the energy transition, concentrate growing issues of industrial sovereignty; **water** and **agricultural commodities**, finally, are establishing themselves as the *commodities* of the twenty-first century, whose progressive scarcity under the effect of climate disruption makes them strategic assets as much as vital goods.

### 3.2.1 Hydrocarbons

Understanding the hydrocarbon market presupposes mastering three dimensions simultaneously: the geology of reserves and the logic of their depletion, the transformation chain which leads from extracted crude to consumable refined products, and finally the mechanics of prices, intimately linked to the decisions of the major producing powers and to the expectations of financial actors. These three dimensions are closely interlocked: the location of reserves determines geopolitical issues, which in turn command production levels and therefore prices, which condition the profitability of the whole industrial chain and the allocation of capital on financial markets. More

precisely, crude oil and natural gas are **fossil hydrocarbons**, formed by the thermal and mechanical decomposition of organic matter buried over millions of years. Their extraction is fundamentally an activity of drawing down a resource that is non-renewable on a human timescale, which confers on the notion of **proven reserves** a central economic importance. Proven reserves designate the quantities of hydrocarbons whose extraction is regarded as technically and economically viable with the technologies and prices in force: they are thus distinguished from **probable reserves** (whose exploitation is plausible but uncertain) and from **resources** (geologically present but not yet accessible). According to the annual data compiled by the Oil & Gas Journal (2024), world proven reserves of crude oil stood at 1765 billion barrels at the end of 2024, while proven reserves of natural gas reached 7538 trillion cubic feet. The geographical distribution of these reserves is extremely concentrated and constitutes the foundation of contemporary geopolitical tensions: Venezuela holds the world's leading proven oil reserves, with roughly 303 billion barrels — that is approximately 17 % of the world total — according to the data of the U.S. Energy Information Administration (2025b), followed by Saudi Arabia, Iran, Iraq and Canada. This geographical concentration explains why the oil market does not function like an ordinary competitive market: it is structured by **OPEC+** (the *Organization of the petroleum exporting countries* enlarged to non-member partners, foremost among which figures Russia), whose collective decisions on **production quotas** constitute the main short-term determinant of world prices. Since October 2022, OPEC+ has thus proceeded to a series of coordinated reductions in its extraction quotas, bringing the cumulative contraction to 3.66 million barrels a day; to this collective reduction are added the unilateral voluntary cuts of Saudi Arabia, which maintained an additional reduction of one million barrels a day throughout 2024 in order to support prices, according to the analysis published by the Institute of Energy Economics Japan (2024). The **reserves-to-production ratio** (R/P), defined as the quotient of proven reserves by the annual volume of production, constitutes the standard indicator of the 'longevity' of extractable resources: computed at world level, this ratio stood at roughly 47 years for oil according to the Energy Institute (2025), which means that, at the current rate of production, proven reserves would be exhausted by 2071. This computation must, however, be interpreted with caution, as Brandt (2017) emphasises in his critique of Hubbert's model: proven reserves are an economic magnitude as much as a geological one, which evolves as a function of oil prices — a rise in prices makes the exploitation of deposits previously judged too costly profitable — and of technological progress, so that the R/P ratio has paradoxically increased since the 1980s despite the growth of production, new discoveries and upward revisions largely offsetting consumption. The Agence internationale de l'énergie (2024) further notes that, in its stated policies scenario, world oil demand should reach a peak before the end of the decade under the combined effect of the advance of electric vehicles and the improvement of energy efficiency, even though it notes that the renewed withdrawal of the United States in 2025 from the Paris climate agreement signed in 2015 and the slowdown in the penetration of electric vehicles in that country could prolong the growth of consumption until the mid-2030s.

### Focus...



**The macroeconomic impact of oil shocks: from the cost of energy to the mechanisms of transmission.** Oil is an intermediate *input* in practically every sector of the economy: transport, manufacturing industry, mechanised agriculture, thermal electricity generation. A durable rise in the price of crude is transmitted to the economy through several simultaneous channels, which Hamilton (2009) formalised within the framework of a structural VAR model. The **production cost channel** is the most direct: a rise in the price of oil increases the variable costs of energy-intensive firms (transport, chemicals, aluminium, cement), compresses their margins and leads them to pass part of this inflation on to consumer prices, feeding **cost-push inflation**. The **disposable income channel**, for its part, operates on the household side: the rise in spending on fuel and heating mechanically reduces the share of disposable income allocated to other goods and services, entailing a negative income effect which depresses demand. The **uncertainty channel**, finally, is less immediate but often more powerful: oil shocks generate uncertainty about future energy prices, which leads firms to defer their investments in energy-intensive equipment and households to delay their purchases of vehicles, which Hamilton (2009) analyses as a *shifting composition effect*.

By way of framing data: the oil shock of 2007–2008 took the price of WTI from 50 dollars to 145 dollars a barrel in eighteen months; Hamilton (2009) concludes that, in the absence of the contractions in consumption induced by the rise in energy prices — notably the collapse of the American automobile market — the period running from the fourth quarter of 2007 to the third quarter of 2008 would in all likelihood not have been described as a recession by the NBER, which makes the oil shock a determining cause, and not merely an amplifying one, of the cyclical downturn which preceded the financial crisis of autumn 2008. A rule of practical scope is thus established by the convergence of several institutional studies: the Banque de France (2024a) establishes that a rise of 10 euros in the price of a barrel entails an increase of roughly 0.4 percentage point in the level of consumer prices in France and in the euro area; Raabe & Burgstaller (2004) record, on the basis of a meta-analysis of comparative macroeconomic simulations, that a rise of 10 % in the price of oil translates on average into a deceleration of the growth rate of GDP

of roughly 0.1 percentage point per year over the first three years in the net importing economies of the OECD, and into a rise in inflation of 0.1 to 0.2 percentage point; finally, the Fonds monétaire international (2000) estimates, on the basis of simulations of the MULTIMOD model, that a rise of 50 % in the price of oil maintained for one year reduces the GDP of industrialised countries by 0.25 % on average over two years and increases inflation by less than 0.2 point.

Subsequently, the economic understanding of the oil market requires distinguishing the different stages of the **oil value chain** which transform extracted crude into consumable products, since it is these stages which determine the cost structure and the margins of industrial actors. Everything begins with the *upstream*, which covers exploration, drilling and the extraction of crude: extraction costs vary considerably according to the nature of the deposits: according to the analysis published by Rystad Energy (2024), the average *breakeven price* of new *onshore* projects in the Middle East stands at roughly 27 dollars a barrel — the lowest of all production segments — against 50 to 80 Canadian dollars a barrel (that is roughly 36 to 58 American dollars) for Canadian oil sands, whose full costs (investments included) are structurally the highest in the world, and between 43 and 47 dollars a barrel for *offshore deepwater* projects. Next comes the **refining (downstream)** phase, which constitutes the central industrial transformation. As Penn State College of Earth and Mineral Sciences (2009) describes it, the process rests first on **fractional distillation**: the crude is heated to roughly 350 degrees Celsius in an atmospheric distillation unit (ADU), which vaporises it into a mixture of gases; these rise up the column and condense at different heights as a function of their boiling temperature, thus separating the crude into **petroleum fractions** according to their number of carbon atoms: the light fractions (liquefied petroleum gases, petrols; C<sub>1</sub> to C<sub>10</sub>, boiling temperature below roughly 175 degrees Celsius) condense at the top of the column, while the heavy fractions (heavy fuel oil, bitumen; beyond C<sub>20</sub>, boiling temperature above 350 degrees Celsius) remain at the bottom of the column. The intermediate fractions comprise notably naphtha (a petrochemical feedstock), kerosene (aviation fuel) and gas oil (*diesel*). Since demand is structurally oriented towards the light and middle fractions (petrol, diesel, kerosene), whose commercial value is higher than that of heavy residues, refiners do not stop at distillation: they have recourse to **catalytic cracking** (*fluid catalytic cracking*, FCC), a process which breaks heavy molecules into lighter and more valuable fractions, and to **catalytic reforming**, which rearranges the molecular structure of naphthenes and paraffins in order to improve the octane rating of petrols. The **refining margin** (*crack spread*) — that is to say the gap between the price of the refined products obtained and the cost of the crude processed — constitutes a major financial indicator for investors active on commodity markets: a rise in the price of gas oil without an equivalent rise in the price of crude widens the *crack spread* and improves the profitability of integrated refiners (such as TotalEnergies, Shell, or Valero), while a compression of this margin — for example when the price of crude rises faster than that of finished products — weighs on their operating results. This is precisely what the U.S. Energy Information Administration (2022b) documents during the energy shock of 2022: the *crack spread* between ultra-low sulfur diesel (ULSD) and Brent crude on the New York Harbor benchmark market, which stood at roughly 0.51 dollar per gallon before the conflict, reached a historic peak of 2.14 dollars per gallon in October 2022, that is a quadrupling in twelve months relative to October 2021. In other words, where refining a barrel of crude into diesel historically yielded of the order of 21 dollars of margin, it yielded more than 90 dollars at the peak of the crisis — a level which the Federal Reserve Bank of Dallas (2022) describes as 410 % above the pre-pandemic average. This surge did not reflect a rise in crude: it is precisely the divergence between the price of the refined product and that of crude — that is to say the widening of the *crack spread* — which reflects the conjunction of a collapse of distillate stocks (gas oil stocks on the American East Coast were 32 % below their five-year average in February 2022, according to the U.S. Energy Information Administration (2022b)) and the redirection of Russian distillate flows away from Europe, suddenly depriving the market of one of its most important historic suppliers. This architecture of the value chain likewise illustrates the strategic stakes of hydrocarbons, including natural gas: the latter is indeed often extracted jointly with oil (*associated gas*), and its commercial valorisation depends on the existence of specific transport infrastructures — gas pipelines or liquefaction terminals (*liquefied natural gas*, LNG) — which involve considerable fixed investments and massive economies of scale, making the geography of deposits even more structuring than for crude oil.

#### Focus...



**Production and trading of commodities.** The high capital intensity of mining has favoured a highly concentrated market, historically structured around a small number of groups whose geographical base partly reflects the legacy of the colonial empires. The two Anglo-Australian giants **BHP** (formerly BHP Billiton) and **Rio Tinto**, joined by the Brazilian **Vale**, maintain their leading position on the markets for iron ore, copper and coal. Their dominance must, however, be qualified: since the merger of May 2013 between the Swiss trader **Glencore** and the miner **Xstrata** — after 18 months of negotiation and the belated obtaining of the approval of the Chinese authorities, conditional

on the disposal of the Las Bambas mining project in Peru — there has emerged a hybrid giant, both producer and trader, which constitutes an unprecedented economic model in the sector. According to the data compiled by London Mining Network (2013), at its birth the GlencoreXstrata group employed roughly 190 000 staff spread across 50 countries and operated more than 100 mines; it simultaneously became the fourth largest mining group in the world, the leading world exporter of thermal coal and the third largest world producer of copper. In 2024, the Glencore group — renamed after the disposal of the Xstrata brand — declared a turnover of 231 billion dollars and employed roughly 140 000 people, according to the annual report of Glencore (2025). As with energy, tensions on mineral commodity markets are structurally linked to the **strong rigidity of supply**: it is most often necessary to allow six to ten years between the investment decision and the ramp-up of a deposit, without counting the risks of obstruction of an environmental, regulatory or political nature, so that the adjustment of supply to demand shocks is fundamentally asymmetric. The **fall in the price of iron ore in 2015** illustrates this dynamic in a particularly striking manner: the price of the ore reached a historic floor of 38.80 dollars a tonne in December 2015, against a peak of 191.70 dollars in 2011, that is a loss of 80 % of its value in four years; according to the estimates published by Transtainer (2015), BHP and Rio Tinto maintained at that date *breakeven costs* of roughly 25 dollars to 34 dollars a tonne, allowing them to remain profitable when virtually all second-tier producers were loss-making — whence the suspicion, relayed by Bloomberg (2015), that the three majors were deliberately maintaining their capacity at full throttle in order to accelerate the exit from the market of the most fragile producers.

The international **trading** of commodities, an activity whose roots go back to the merchant companies of the first colonial expansion in the sixteenth century, has for its part experienced a revival since the disturbances of the 1970s and 1980s: the end of the Bretton Woods system, the loss of control by cartels and the calling into question of agricultural policies have made traders indispensable in the **management of commercial risks**, in terms of price as well as of exchange rate and counterparty. A small group of specialised firms has established itself at the centre of the markets: **Vitol** (the leading independent world trader, whose turnover reached 400 billion dollars in 2023 with a net profit of 13 billion, according to Insights Global / Financial Times (2024)), Trafigura, Glencore and Cargill in agricultural products. Note that the investment banks which had established themselves there — Morgan Stanley, Goldman Sachs — progressively withdrew after the 2008 crisis, constrained by the Basel III regulatory requirements to lighten their balance sheets of real assets. **Geneva** remains the world hub of commodity trading. According to the data published by Public Eye (2023) and the professional association STSA (2023), Switzerland concentrates roughly 35 % of world crude oil trading, 60 % of that in base metals, 50 % of cereals and 65 % of cotton; the trading sector represented roughly 3.8 % of Swiss GDP in 2023 and mobilised roughly 950 companies directly employing 10 000 people. **Singapore** positions itself as Geneva's main competitor, in particular in Asian hydrocarbons; it benefits from a corporation tax rate of 5 % for approved trading activities, against roughly 8 to 10 % in Switzerland. The actors may be grouped into four categories: the multi-commodity groups (Glencore, Trafigura); traders specialised by commodity or by energy (Vitol); the agricultural specialists (Cargill, Louis Dreyfus, Bunge); and the integrated physical *trading* of the large *majors* (Shell, BP, TotalEnergies) and of the mining groups. Regularly accused of profiting from the corruption of producing countries and from tax optimisation via *offshore* structures — GlencoreXstrata having notably been targeted for the sheltering of its profits on Zambian copper in subsidiaries in the British Virgin Islands — the trading sector is likewise called into question for its environmental externalities: in 2006, the serious pollution caused in Abidjan by the discharge of 528 cubic metres of toxic petroleum residues transported by the vessel Probo Koala, chartered by Trafigura, provoked the death of 17 people and poisoned tens of thousands of inhabitants; Trafigura paid 100 million pounds sterling to the Ivorian government for the clean-up, and a further 30 million to victims in compensation, without formally acknowledging its responsibility. In 2023, the African Court on Human and Peoples' Rights declared the case admissible and engaged the responsibility of the Ivorian state for failure to protect, opening the way to further reparations, as the *Fédération internationale pour les droits humains* (2023) explains.

Founded in 1993 by Claude Dauphin (1951–2015) and five associates, from a spin-off of the Marc Rich group, **Trafigura** is today run by the Australian Jeremy Weir, appointed CEO in March 2014. Trafigura's primary business consists in conveying commodities — oil and derivatives, non-ferrous metals — between producing and consuming countries, by organising transport, storage logistics and, where appropriate, processing. Its five main operational centres (Geneva, Singapore, Houston, Johannesburg, Montevideo) feed a world network of *pipelines*, ports, silos, tanks and terminals. In parallel, the company practises physical *trading*, consisting in buying and selling cargoes by taking advantage of the geographical or technical anomalies of physical markets. Moreover, Trafigura is squarely engaged in finance: futures contracts on the exchanges of London (the offices of ICE), New York (NYMEX) and Singapore (SGX) constitute its instruments for hedging against fluctuations in price, exchange rate and freight. Trafigura likewise manages financial assets via **Galena Asset Management**, its *hedge fund* specialising in mining private equity. In 2024, the group declared a turnover of 243 billion dollars, an underlying EBITDA of 8.1 billion and

employed on average 13 086 people, according to the report of Trafigura (2024). Unlike its rival **Glencore**, listed on the London Stock Exchange since 2011, Trafigura is a company entirely owned by its employees: 100 % of the capital is in the hands of roughly 1400 executives and associates, according to the data published by Trafigura (2024), which spares it any obligation of public financial disclosure and confers on it greater strategic freedom. The group is topped by a Dutch *holding*, while its main trading company is incorporated in Singapore.

The geopolitical dimension of the oil market has thus reasserted itself since 2022, illustrating the **resource curse**: the holding of hydrocarbons certainly confers a considerable geopolitical rent on its beneficiaries, but likewise exposes those holders to international pressures liable to destabilise their economy. The **invasion of Ukraine by Russia in February 2022** thus constituted a significant exogenous shock on the hydrocarbon market since the oil counter-shock of 1986, in the course of which Saudi Arabia, in order to defend its market share, decided at the end of 1985 unilaterally to increase its production, provoking the collapse of the price of a barrel from 30 dollars to less than 10 dollars within a few months, as the Observatoire français des conjonctures économiques (1986) recalls. As Hamilton (2009) shows in his analysis of oil shocks, supply disturbances induce price rises with simultaneously recessionary and inflationary macroeconomic effects — a structural mechanism of *stagflation*. Russia being, before the conflict, the leading world exporter of natural gas in 2021 — with 45 % of the European Union's total natural gas imports conveyed by pipeline from Russia, representing 155 billion cubic metres according to the U.S. Energy Information Administration (2022a) — the second largest world exporter of crude oil behind Saudi Arabia (4.7 million barrels a day exported in 2021 by Russia, of which nearly half towards OECD Europe according to the U.S. Energy Information Administration (2022a)), and a major exporter of distillate products — notably gas oil bound for Europe, representing 11 % of world exports of refined products in 2023 according to the U.S. Energy Information Administration (2025a) — the Western sanctions decided from March 2022 onwards provoked a considerable supply shock. The partial redirection of Russian flows towards Asia (China and India absorbing the greater part of sanctioned Russian crude exports) and the reconstruction of alternative logistical and financial circuits have, however, attenuated the impact on production: as the analyses published by Bruegel (2025) show, the sanctions did not entail a collapse of Russian production — the latter remaining broadly consistent with OPEC+ quotas — but they did compel Russia to sell its crude at a discount (*Urals discount*) to Brent which reached its maximum between December 2022 and March 2023 (at the time of the implementation of the price cap of 60 dollars a barrel decided by the G7), before stabilising at around 10 % during the summer of 2024, then rising again at the end of 2025 following the new American sanctions against Rosneft and Lukoil, exceeding 15 % at that point. The Russian current account surplus for its part collapsed, going from 124 billion dollars over the first five months of 2022 to 23 billion over the same period of 2023, and the Russian state's oil revenues fell by roughly 50 % year on year over the first five months of 2023 according to Bruegel (2025). The second major theatre of geopolitical tension is Venezuela. This country holds the world's leading proven reserves of crude oil (303 billion barrels according to the U.S. Energy Information Administration (2025b)), but its production has fallen to roughly one million barrels a day in 2025 — against more than three million at the beginning of the 2000s — as a result of the deindustrialisation of PDVSA (the Venezuelan state oil company), systemic corruption and the accumulation of American sanctions since 2019. The **American military operation of 3 January 2026 which led to the capture of Nicolás Maduro** has from then on opened a period of unprecedented uncertainty over the country's production prospects: according to Reuters (2026b), the Trump administration announced that between 30 and 50 million barrels of 'sanctioned oil' would be transferred to the United States, and in parallel authorised the return of American oil companies — foremost among them Chevron, which was exporting between 150 000 and 200 000 barrels a day of Venezuelan oil continuously despite the blockade — with the declared objective of restoring the infrastructure and increasing production. The Agence internationale de l'énergie (2024) nevertheless emphasises that the recovery of Venezuelan production could only be gradual and limited in the short run, given the dilapidation of the infrastructure and the chronic under-investment of several decades. These two crises — the Russo-Ukrainian and the Venezuelan — thus illustrate a fundamental structural feature of the oil market: its sensitivity to exogenous political shocks, which Hamilton (2009) carefully distinguishes from demand shocks in his model of structural identification of oil market disturbances, and which he shows induce asymmetric macroeconomic effects — rises in the oil price weighing more heavily on growth than falls support it, on account of sectoral adjustment costs and of the uncertainty channel affecting investment.

#### Focus...



**The correlation between the oil price and the S&P 500: an unstable and conditional relation.** The question of the correlation between the price of crude and equity markets is one of the most discussed in the financial literature applied to commodities. The answer depends on the nature of the shock at the origin of the variation in the oil price, a distinction of which Kilian & Park (2009) have provided a formalisation by decomposing variations in the oil

price into three types of structural shock: **supply shocks**, which reflect unforeseen variations in world production; **aggregate demand shocks**, which capture the acceleration or slowdown of world industrial activity; and oil-specific demand shocks (*precautionary demand shocks*), which reflect revisions of expectations about the future availability of supply. Their main result, obtained on monthly data from January 1975 to September 2005, is that the correlation between the oil price and equity markets is positive when the rise in crude is driven by aggregate demand (world economic expansion, simultaneous rise in equities and oil), and negative when the rise is driven by a negative supply shock or by precautionary speculative demand (stagflationary shock, fall in equities and rise in oil).

This asymmetry explains apparently contradictory observations: as the U.S. Energy Information Administration (2023) and the Federal Reserve Bank of St. Louis (2026) recall, during the period of strong world growth 2003–2007, WTI rose by more than 190 % — from roughly 31 dollars to more than 90 dollars a barrel — while the S&P 500 simultaneously gained 63 %, illustrating the dominance of demand shocks; conversely, at the time of the Russo-Ukrainian supply shock of 2022, Brent leapt by roughly 65 % between January and March (from roughly 78 to 128 dollars a barrel) while the S&P 500 fell back by 18.5 % over the year as a whole, illustrating the dominance of supply shocks. In a portfolio management framework, this implies that the correlation between an oil ETF (or a *futures* contract on crude) and a diversified equity portfolio is not a stable parametric constant: it is dynamic and conditional on the state of the economy, which makes **dynamic conditional correlation models** — DCC-GARCH developed by Engle (2002) for example — particularly suited to the estimation of diversification benefits in a mixed portfolio made up of equities and commodities. An institutional investor who wishes to integrate crude into a diversified portfolio must therefore distinguish two regimes: a regime of world growth, in which the equity-oil correlation is positive and diversification less effective; and a regime of geopolitical supply shock, in which this correlation becomes negative and the long position on crude acts as a natural *hedge* against the fall in equities.

#### Focus...



**The oil futures market and the term structure of prices: contango and backwardation.** Futures contracts on oil are traded on two main exchanges: the **NYMEX** (*New York Mercantile Exchange*), a subsidiary of the CME Group (*Chicago Mercantile Exchange*), for **WTI** (*West Texas Intermediate*), the American benchmark crude; and the **ICE** (*Intercontinental Exchange*) of Atlanta, for **Brent**, the European and international benchmark crude. Brent and WTI are the two underlying price indices of the most liquid oil derivative instruments in the world — they represent roughly 40 % of the total weighting of the S&P GSCI (*Goldman Sachs Commodity Index*), making energy the structurally dominant sector of this benchmark index for *commodity investors*, according to the weighting data published annually by S&P Global (2023a). Subsequently, the **term structure** designates the relation between the prices of *futures* contracts and their maturity: when the prices of distant contracts are higher than spot prices, the market is said to be in **contango**; in the opposite case, it is said to be in **backwardation**. Contango reflects the storage and carry costs (*cost of carry*) which accumulate until delivery, as well as a liquidity premium on short contracts; backwardation, more frequent in tight oil markets, reflects a *convenience yield* which the immediate physical holding of crude confers (the capacity to meet urgent demand or to feed a refinery continuously). For an investor who replicates a commodity index via *futures* — for example via an ETF on the S&P GSCI or the *Bloomberg Commodity Index* (BCI) — the term structure generates a **roll yield**: in contango, rolling the positions (selling the near contract reaching maturity, buying the following contract at a higher price) is costly and generates a negative *roll yield*; in backwardation, rolling is profitable (positive *roll yield*). This is why the returns of an oil ETF may diverge significantly from the return of spot crude over a long period.

#### Key Takeaways



World proven oil reserves make it possible to envisage the end of the hydrocarbon era within 47 years, a reminder that proven reserves are an economic magnitude as much as a geological one [Oil & Gas Journal (2024), Brandt (2017)]. OPEC+ steers prices in the short run via its quotas [Institute of Energy Economics Japan (2024)]. Now a rise of 10 % in the price of oil reduces the growth of the importing economies of the OECD by roughly 0.1 point per year over three years and increases inflation by 0.1 to 0.2 point [Raabe & Burgstaller (2004)]. The Banque de France (2024a) for its part establishes that a rise of 10 euros a barrel raises consumer prices by roughly 0.4 point in the euro area. Beyond the economic aspect, Glencore and Trafigura dominate world trading alongside Vitol; Geneva concentrates 35 % of world crude oil trading, 60 % of base metals and 50 % of cereals [Glencore (2025), Trafigura (2024), Insights Global / Financial Times (2024), Public Eye (2023)].



### 3.2.2 Precious metals and rare earths

First of all, **gold** occupies a place apart in the universe of commodities: it is simultaneously an **industrial commodity**, with roughly 7 % of world gold demand in 2024, that is 326 tonnes, of which 271 tonnes for electronics — driven by the rise of AI servers — and 9 tonnes for dentistry, in structural decline, according to the World Gold Council (2025b), a **symbolic consumer good**, jewellery aggregating 38 % of demand in 2024 (1877 tonnes out of 4974 tonnes in total, according to the World Gold Council (2025b)), and above all a **sovereign reserve asset** and an instrument of financial hedging. This threefold nature makes it the most closely followed asset on *commodity* markets, and one of the most revealing of world macroeconomic and geopolitical tensions. Thus, central bank demand reached 1045 tonnes in 2024 — the third consecutive year above 1000 tonnes — that is roughly 21 % of total demand, against an average of 473 tonnes over the period 2010–2021 (roughly 10 to 11 % of annual demand), according to the World Gold Council (2025a). This movement illustrates an underlying phenomenon: the **partial de-dollarisation of foreign exchange reserves**, accelerated since the freezing of Russian dollar assets in 2022 (roughly 300 billion dollars immobilised by the United States and its allies). The lesson drawn by the central banks of emerging countries is that even dollar-denominated assets — reputedly free of counterparty risk — may be mobilised as an economic weapon. Consequently, gold, which constitutes the debt of no state, has become the only truly neutral reserve. In September 2025, gold had thus for the first time since 1996 overtaken the dollar to constitute the leading world reserve asset by value (27 % of total world reserves against 23 % for American Treasury bonds), according to Reuters (2025b). The spot price of gold had reached a historic peak of 3728 dollars an ounce in September 2025 and 4381 dollars an ounce in October 2025, that is a doubling relative to the end of 2022, according to BestBrokers (2025). Gold is likewise a **trading instrument** which presents several structural correlations:

- **Inverse correlation with the American dollar:** the price of gold is quoted in dollars; all other things being equal, an appreciation of the dollar makes the metal more expensive for buyers in foreign currencies, compressing demand; the historical correlation is negative and robust over the long run, as Interactive Brokers (2025) explains.
- **Inverse correlation with real rates:** since gold pays neither coupon nor dividend, its opportunity cost is the real interest rate (the nominal rate less anticipated inflation). Thus, when real rates fall — which is the case in periods of uncontrolled inflation or of accommodative monetary policy — gold becomes relatively more attractive; it is this mechanism which fuelled the rise of 2020–2025 according to J.P. Morgan (2026).
- **Decorrelation with equities in periods of supply shock:** as Yun et al. (2023) analyse, gold and the S&P 500 are positively correlated in periods of demand shock (both rise with world growth), but gold behaves as a safe-haven asset in periods of systemic stress (negative or nil correlation with equities).
- **Positive correlation with implied volatility (VIX):** Yun et al. (2023) add that a rise in the VIX — the index of implied volatility of options on the S&P 500, known as the ‘fear index’ — is generally accompanied by an inflow into gold, which makes it an effective portfolio hedging instrument in the short run.

The main instruments of exposure to gold on organised markets are *futures* on the COMEX (New York), ETFs backed by physical gold (of which the SPDR Gold Trust is the most liquid, with roughly 900 tonnes of physical gold in deposit at 31 December 2025 in the vaults of the banks HSBC and JPMorgan Chase Bank in London, with an annual expense ratio of 0.40 %, according to State Street Investment Management (2025)), the shares of gold mining companies (*gold miners*, which offer leverage on the price of gold thanks to their high fixed cost structure), and the *royalty companies* (Franco-Nevada, Royal Gold) which receive royalties on production without bearing mining operational risk.

#### Focus...



**Fort Knox gold, Trump and the destabilisation of American reserves.** In 2025, the Trump administration revived the idea of a revaluation of American gold reserves. The United States officially declares 8133 tonnes of gold in its reserves, valued in the Fed’s books at 42.22 dollars an ounce — the official price frozen since 1973 — that is a book value of roughly 11 billion dollars, whereas the market value now exceeds 1000 billion dollars, as BestBrokers (2025) explains. Indeed, the American **Gold Reserve Act** of 30 January 1934, still in force, authorises the President of the United States to fix the statutory price of gold by decree. Now this price is currently 42.22 dollars an ounce, fixed by the **Par Value Modification Act** of 1973. People close to the president, including Elon Musk (before the latter distanced himself from Trump), publicly demanded an audit of the reserves stored at Fort Knox, going so far as to contest their existence. If an accounting revaluation to market value were carried out, it would theoretically make it possible to release a book capital gain of more than several hundred billion dollars — without money creation *stricto sensu*, but with potentially destabilising effects on the American bond market, since such an operation would



mechanically reduce the need to issue Treasury bonds. More precisely, the Treasury is authorised to **issue gold certificates** to the reserve banks in order to monetise the gold it holds: in exchange for these certificates, the Federal Reserve Bank of New York (FRBNY) credits the Treasury General Account (TGA) with an equivalent amount. No physical sale of gold is required. If Congress raised the statutory price to market value (roughly 3000 dollars an ounce at the beginning of 2026), the Treasury could issue new gold certificates for the difference in value: this would mechanically create roughly 990 billion dollars of liquidity in the TGA, without any issuance of Treasury bonds, as BusinessAM (2025) and Bloomberg (2025) explain. Since these gold certificates are **not counted as federal debt**, the operation does not run up against the debt ceiling — which explains its attraction in a period of budgetary constraint. But the mechanism functions in the opposite direction to the issuance of Treasury bonds: instead of the Treasury borrowing on the markets (by issuing Treasury bonds bought by private or foreign investors), the revaluation of gold certificates supplies it directly with liquidity via the TGA. As the Treasury uses these funds to repay maturing debt without renewing it, the outstanding amount of Treasury bonds in circulation is reduced: mechanically, the fall in the supply of bonds pushes their price up — which amounts to a **compression of yields**, similar in its effects to a *quantitative easing* programme, but without expansion of the Fed's balance sheet in the strict sense. Bloomberg (2025) describes this effect as a '*messy outcome*' for monetary policy, since it would increase bank reserves and complicate the *quantitative tightening* under way. This debate is in reality profoundly geopolitical: it signals a calling into question of the international monetary architecture founded on the dollar, and explains in part why foreign central banks massively bought gold in 2024–2025 even before this revaluation was formally contemplated, out of pure sovereign precaution, as Reuters (2025b) explains.

Moreover, **rare earths** (*rare earth elements*, REE) designate a group of 17 chemical elements — the 15 lanthanides, to which are added yttrium and scandium — with magnetic, luminescent and catalytic properties that are irreplaceable in the technologies of the future (permanent magnets for wind turbine and electric vehicle motors, lasers, missile guidance, medical MRI, OLED screens). Their name is misleading: geologically abundant — cerium is more common than copper — they are, however, rarely so in concentrations sufficient to justify profitable exploitation. Thus, China concentrates 69.2 % of world production in 2024 (270 000 tonnes of REO equivalent, *rare earth oxide*), and processes roughly 90 % of all the world's rare earths, including those extracted elsewhere, according to the U.S. Geological Survey (2025) and China Briefing (2025). This dominance results from a strategy initiated by Deng Xiaoping as early as 1986 (**Programme 863**), which led China to develop simultaneously extraction, metallurgy and user industries, at costs that no competitor has been able to match to this day. The export of rare earths has been subject to quotas since 2010 — a practice condemned by the WTO in 2014 following complaints by the United States, the EU and Japan — and to export restrictions on processing technologies since 2023, which places Western supply chains in a situation of **structural dependence on China**. The United States, the second largest producer with 11.5 % of the world total (the Mountain Pass mine, California), nevertheless exports virtually all of its rare earths to China for processing, for want of domestic refining infrastructure, as the U.S. Geological Survey (2025) shows. This is notably what explains American ambitions regarding Greenland. Indeed, the subsoil of **Greenland** contains some of the largest rare earth deposits in the world, concentrated for the most part in the Ilímaussaq complex, at the southern extremity of the island. The Tanbreez deposit, held by the company Critical Metals, is for its part one of the largest known deposits of *heavy REE* in the world: its host rock (the kakortokites) represents 4.7 billion tonnes of mineralisation, of which an initial JORC (*joint ore reserve committee*) resource estimated at 44.97 million tonnes of rare earth materials (indicated and inferred grade), particularly rich in heavy rare earths (dysprosium, terbium, yttrium) and light ones (neodymium, praseodymium) indispensable to the permanent magnets of electric motors and wind turbines; its low uranium (20 ppm) and thorium (53 ppm) content distinguishes it favourably from its neighbour as regards metallurgical processing, according to the technical report of Critical Metals (2025) filed with the SEC. The Kvanefjeld deposit (*Kuannersuit* in Greenlandic), in the same Ilímaussaq complex, is for its part classified as the third largest known rare earth deposit in the world, with a total JORC resource estimated at 1.01 billion tonnes at a grade of 1.10 % in rare earth oxides, and proven and probable reserves of 108 million tonnes; it moreover holds the second largest world uranium deposit (266 ppm of uranium oxide), according to the Center for Strategic and International Studies (2026). It is precisely this co-presence of uranium and rare earths which led the Greenlandic government to reject the exploitation project in 2021, at the end of elections won by the Inuit Ataqatigiit party, hostile to nuclear mining — at the risk of exposing itself to international arbitration on the part of the Australian promoter Greenland Minerals, which is claiming billions of euros in damages. Since his first term, Donald Trump has expressed the intention of acquiring Greenland, an autonomous Danish territory; in December 2025 and January 2026, the Trump administration crossed a rhetorical threshold by publicly evoking the possibility of an American 'protectorate', explicitly invoking the necessity of securing access to rare earths in order to reduce dependence on China, as CNN (2026) recalls. Reuters (2025a) thus revealed that the Trump administration was discussing taking a stake in the

company Critical Metals (formerly Tanbreez), under the policy of securing critical minerals. This ambition falls within the American **Critical Minerals Strategy**, which aims to reduce the share of rare earths imported from China from 70 % to less than 25 % by 2030, as Z2Data (2026) explains.

#### Focus...



**Rare earths and nickel: the assets of the French EEZ.** France has the second largest **exclusive economic zone** in the world by area: 11 million square kilometres, that is 9.5 million square kilometres of seabed at more than 1000 metres depth, which makes it the leading world power for deep seabeds, according to the Adventon consultancy for the Fondation de la Mer (2023). These seabeds hold three types of strategic mineralisation: **polymetallic nodules** (manganese, nickel, copper, cobalt), particularly present in the Clarion-Clipperton zone (North Pacific), where France holds exploration permits; **cobalt-rich crusts** rich in rare earths; and **polymetallic sulphides** on mid-ocean ridges. At 30 June 2023, France thus held a significant share of the 30 exploration permits issued by the International Seabed Authority (ISA), of which 19 for nodules and four for cobalt-rich crusts, as the Ministère de la Transition écologique (2024) recalls. Commercial exploitation nevertheless comes up against considerable technical obstacles (depths of 4000 to 6000 metres, unknown environmental impact) and against the *de facto* moratorium defended by France since 2021 within the framework of the negotiations at the ISA.

**New Caledonia** moreover concentrates 25 % of the known world reserves of **nickel**, a key metal of the energy transition (lithium-nickel-manganese-cobalt batteries, stainless steel). The institutional crisis which broke out in May 2024 — thirteen dead, several billion euros of damage, a state of emergency declared by the Attal government following the riots against the unfreezing of the electoral roll — paralysed the archipelago's three pyrometallurgical plants. Koniambo Nickel SAS (KNS), held 51 % by the Xstrata/Glencore group and 49 % by the SMSP, definitively halted its activities in August 2024, Glencore having announced that it would finance its closure; the Société Le Nickel (SLN), a subsidiary of Eramet, mothballed its Thio mining centre in October 2024; only Prony Resources restarted in November 2024, with reduced capacity, as Le Figaro (2024) and Mining Beyond Hot Air (2024) relate. These closures come in an already difficult context: competition from low-cost Indonesian nickel — HPAL (*high pressure acid leaching*) processing developed by Chinese firms in Indonesia — halved nickel prices between 2022 and 2024, eliminating the profitability of New Caledonian extractors, whose production costs are among the highest in the world. The question of the future of New Caledonian nickel — between French sovereignty, Kanak interests and European industrial strategy for the battery value chain — constitutes one of the important dossiers of French overseas policy.

Finally, the dependence of developed economies on critical minerals goes beyond the question of rare earths alone. The European Union published in 2023 its **Critical Raw Materials Act** (regulation 2024/1252 of 11 April 2024), distinguishing 34 critical raw materials — economically important and exposed to supply risks — among which a subset of 17 strategic raw materials (bismuth, boron, cobalt, copper, gallium, germanium, natural graphite, lithium, magnesium, manganese, nickel, platinum group metals, scandium, silicon metal, heavy rare earths, light rare earths and tungsten), essential to the green, digital, aerospace and defence sectors, and fixing binding objectives for 2030: extraction of at least 10 % of the EU's needs on its own soil, refining of at least 40 %, recycling of at least 25 %; for each of the 17 strategic raw materials, no third country must supply more than 65 % of EU consumption — a threshold currently exceeded for four strategic raw materials: lithium, magnesium, natural graphite and heavy rare earths, according to special report 04/2026 of the Cour des comptes européenne (2026). In particular, **lithium**, whose reserves are concentrated in the 'lithium triangle' (Chile, Argentina, Bolivia, which together hold 53 % of known world resources, that is 56 million tonnes out of 105 million in total), nevertheless sees the bulk of its production provided by Australia (48 %) and Chile (24 %), Bolivia remaining virtually unexploited; by contrast, the bulk of refining is located in China (more than 60 % of world capacity), as Fastmarkets (2024) and Our World in Data (2024) explain. **Cobalt**, indispensable to lithium-ion batteries, is extracted 70 to 80 % in the Democratic Republic of Congo — the upper bound incorporating direct and indirect production, as Gregory & Milas (2024) explain — whose mines are owned or operated roughly 72 % by Chinese firms, which likewise control between 60 and 90 % of world cobalt refining capacity, according to Tearline (2025). **Uranium** presents a different configuration: the EU's supply in 2023 is distributed between Canada (33 %), Russia (23 %), Kazakhstan (21 %) and Niger (14 %), the latter having experienced a coup d'état in July 2023 which led France to seek to diversify the supplies of Orano (formerly Areva), according to Connaissance des énergies (2025). For **platinum** and **palladium**, the dependence is doubly concentrated: South Africa produces 74 % of world platinum (120 000 kg in 2024 according to Investing News Network (2025)), and Russia — mainly via Norilsk Nickel — is the leading world producer of palladium with 75 000 kg in 2024, that is roughly 40 % of the world total, the two countries together concentrating more than 80 % of world production of platinum group metals. Faced with these dependencies, three complementary strategies are taking shape at the level of states and

regional blocs. The first strategy is **minerals diplomacy**: the United States signed a *Critical Minerals Agreement* (CMA) with Japan on 28 March 2023 within the framework of the *Inflation Reduction Act*; similar agreements have been signed with Australia (July 2023) and the United Kingdom (March 2023 as well), integrating these countries into the list of qualifying free trade partners for electric vehicle tax credits, as the U.S. Trade Representative (2023) and the firm White & Case (2023) explain. For its part, the EU has concluded 14 strategic partnerships on raw materials since 2021, notably with Canada and Ukraine (2021), Namibia and Kazakhstan (2022), then Argentina, Chile, Zambia and the Democratic Republic of Congo (2023), as Transport & Environment (2023) and SWP (2025) recall. However, none of these partnerships is legally binding: they are memoranda of understanding which will have to be converted into effective industrial cooperation in order to produce their effects on supply chains. The second strategy for combating strategic dependencies is **recycling**: according to Eurostat (2025b), the circularity rate of metals reaches 24.7 % on average in the EU in 2023 — the highest of all material categories — with particularly high performance for lead (a secondary recycling rate exceeding 85 %, according to the batteries regulation adopted in July 2023). **Batteries Regulation 2023/1542** indeed sets binding objectives: collection of 73 % of portable batteries by 2030, recovery of 80 % of lithium and 16 % of cobalt by 2031, and minimum recycled content rates in electric vehicle batteries (6 % for lithium and nickel, 16 % for cobalt), as Hache & Normand (2024) emphasise. The sector nevertheless remains in its infancy for the most strategic minerals: less than 1 % of the lithium and cobalt consumed in Europe currently comes from secondary sources. Finally, **technological substitution** constitutes a complementary medium-term lever: the Chinese giant CATL (2025) launched its sodium-ion battery Naxtra, entirely without lithium or cobalt, with an energy density of 175 Wh/kg and a certified range of 500 kilometres in a passenger vehicle, whose mass production began in December 2025, marking the passage of this type of chemistry from the research stage to that of mass commercialisation. In parallel, the European programme Passenger Project (2025) is developing permanent magnets entirely without rare earths, resting on ferrites and iron-cobalt alloys, whose performance (magnetic energy density) remains inferior to neodymium-iron-boron (NdFeB) magnets but sufficient for traction applications at moderate speed.

### Key Takeaways



Gold combines three functions: industrial commodity (electronics, AI), symbolic consumer good (jewellery) and sovereign reserve asset; accelerated by the freezing of Russian assets in 2022, partial de-dollarisation made gold the leading world reserve asset (27 %) ahead of American Treasury bonds (23 %) in September 2025 [World Gold Council (2025b), World Gold Council (2025a), Reuters (2025b)]. Moreover, China concentrates 69.2 % of world rare earth production and 90 % of refining, the legacy of Deng Xiaoping's Programme 863 (1986); the United States, the second largest producer (11.5 %), exports virtually all its rare earths to China for want of refining infrastructure [U.S. Geological Survey (2025), China Briefing (2025)]. Thus, the European *Critical Raw Materials Act* (2024) sets for 17 strategic raw materials binding objectives for 2030 (10 % extraction, 40 % refining, 25 % recycling, no third country above 65 % of consumption), a threshold already exceeded for lithium, magnesium, graphite and heavy rare earths [Cour des comptes européenne (2026)]. Critical dependencies include cobalt, platinum, palladium and uranium [Gregory & Milas (2024), Investing News Network (2025), Connaissance des énergies (2025)]. Three response strategies are emerging: minerals diplomacy, recycling and technological substitution [SWP (2025), Hache & Normand (2024), CATL (2025)].

### 3.2.3 Water and agriculture

**Fresh water** constitutes one of the few natural resources to have no organised world market: its volumes are too considerable to be transported on a large scale, and its dimension as a fundamental common good — which the General Assembly of the United Nations recognised as a fundamental human right on 28 July 2010, by resolution A/RES/64/292, adopted with 122 votes in favour and 41 abstentions (including the United States, the United Kingdom, Canada and Australia) — makes any attempt at international quotation politically delicate. Yet the structural scarcity linked to climate change, agricultural expansion and demographic growth — the world water deficit is estimated at 40 % by 2030 in the benchmark scenarios of the OCDE (2012) — creates the objective conditions for growing financialisation. Agriculture alone absorbs 70 % of world fresh water withdrawals according to the FAO (2017), which places the resource at the heart of agricultural commodity markets. The first attempt at a regulated quotation of water took place in the United States: the *Nasdaq Veles California Water Index* (NQH20), launched on 31 October 2018, tracks the volume-weighted average price of water rights transactions on the five most active markets in California (the surface market and four groundwater basins: Central Basin, Chino Basin, Main San Gabriel Basin, Mojave Basin

Alto), expressed in dollars per *acre-foot*<sup>5</sup> as the CME Group (2020) explains. In December 2020, the CME Group backed regulated *futures* on it, allowing Californian farmers, municipalities and industrial firms to hedge against the risk of a rise in the price of water. The index was trading at roughly 261 dollars per *acre-foot* in February 2026 according to the quotation Nasdaq Global Index Watch (2026); by way of comparison, in the summer of 2021, during a historic drought, it had reached 964 dollars per *acre-foot*, revealing the extreme sensitivity of the price to climatic variability. The liquidity of the market nevertheless remains very low relative to that of traditional agricultural commodities: the absence of a homogeneous and transportable resource mechanically limits the depth of the order book. Failing the ability to invest directly in the resource, markets have developed **sectoral ETFs** exposing investors to firms in the water value chain. The four main instruments listed on American and European markets are the *Invesco Water Resources ETF* (PHO), which replicates the *Nasdaq OMX US Water Index* and selects water utility, equipment and filtration firms based in the United States, with quarterly rebalancing, as Invesco (2025) explains; the *First Trust Water ETF* (FIW), which replicates the *ISE Clean Edge Water Index* on an equally weighted basis, thus offering less concentration in large capitalisations as AquaVest (2025) sets out; the *Invesco S&P Global Water Index ETF* (CGW), the international version of PHO; and the *iShares Global Water UCITS ETF* (IH20 in Europe, backed by the S&P Global Water 50 index), more accessible to European investors. These ETFs present a high correlation with the large infrastructure and *utilities* stocks, which confers on them a defensive profile, relatively insensitive to economic cycles but highly sensitive to regulatory decisions on water pricing.

### Focus...



**The determinants of the NQH2O price in California.** The NQH2O, published each Wednesday by WestWater Research (the exclusive data provider), reflects the volume-weighted average of water rights transactions on five Californian markets. Its price dynamics respond to a hierarchy of factors:

- **Hydrology: the dominant factor.** Herman (2024) shows that the NQH2O can be predicted, with statistically significant precision, from two hydrological indicators: reservoir filling anomalies (relative to the seasonal norm) and inflow forecasts for the main catchments. The relation is non-linear: in a year of severe drought, the reduction in reservoir filling generates a convex rise in price, amplified by the retention behaviour of senior rights holders. The Sierra Nevada *snowpack* constitutes the most closely followed leading indicator: its measurement in April (the annual survey of the *California Department of Water Resources*) conditions the forecasts of meltwater availability for the irrigation season (from May to September). The optimal hedging strategy thus consists in buying NQH2O *futures* in winter (between November and March) when six-month price forecasts are above the current level, which generates an average saving of 17 % on the average seasonal price and exceeds 35 % when entering a dry year.
- **Regulation.** The *Sustainable Groundwater Management Act* (SGMA), adopted in 2014, requires over-exploited basins to achieve hydrological balance by 2042. Its entry into the operational phase (first probationary hearings in 2024) caused the value of farmland to fall by up to 40 % in certain areas of the San Joaquin Valley according to the Sierra Daily News (2025), and structurally increases the demand for surface water rights: farmers progressively substitute surface water for groundwater that has become particularly expensive or rationed, which supports the NQH2O over the long run.
- **The structure of actors: farmers and cities.** Farmers hold roughly four times more water rights than urban authorities and constitute the main supply side of the market; the cities of Southern California (Greater Los Angeles, San Diego) are the main net buyers: Southern Californian conurbations now draw 15 % of their supply from market transactions according to the PPIC (2025). This dichotomy creates a structural seasonality: farmers sell during the winter *off-peak* when their needs are low, and prices rise again as the irrigation season approaches.
- **The inelasticity of demand.** The demand for water is strongly price-inelastic: Ferguson (2024) estimates the price elasticity of demand at  $-0.18$  at average price levels, which means that a rise of 10 % in the price reduces demand by only 1.8 %. This inelasticity amplifies price movements during supply shocks: the tensions of the summer of 2021 (a historic drought) were precisely the result of this conjunction between a low reservoir and incompressible demand.

<sup>5</sup>An *acre-foot* corresponds to the volume of water needed to cover one acre (4047 square metres) with one foot of water (30.48 cm), that is 1233 cubic metres or 325 851 gallons.



## Debate

**Is water a market like any other?** Faced with the growing scarcity of the resource, the pricing of water is a necessary signal in order to internalise its production and infrastructure costs, reduce waste and finance investment. Indeed, network losses (*non-revenue water* or **NRW**: water produced but not billed, through physical leaks or fraud) reach on average 40 to 50 % in the cities of developing countries; in capitals such as Nairobi, Lagos or Addis Ababa, losses of 40 to 50 % are commonplace, and Mexico City locally exceeds 55 %, according to the We Are Water (2024) foundation. The UN resolution of July 2010 does not proclaim that water should be free, but that it should be accessible at an ‘affordable cost’; progressive pricing (on the model of South Africa, where the first six cubic metres of water have been free since 2000) can reconcile human rights with a price signal. The multinationals Veolia and Suez, pioneers of public service delegation since the 1980s, have demonstrated that the private operator can invest more efficiently than the municipal authority in infrastructure.

However, several arguments distinguish water from ordinary financial commodities: its **captive use character** (one cannot substitute another good for water for drinking), its immobility (it cannot be transported on a world scale), and its dimension as a collective good. Thus, the failure of the water distribution concession in Cochabamba (Bolivia) in 1999 demonstrated the risks of commodification without a social safety net. More precisely, in September 1999, the Bolivian government — under pressure from the World Bank, which made its loans conditional on the privatisation of urban water services — granted a 40 year concession to Aguas del Tunari, a consortium controlled by the American multinational Bechtel (via its subsidiary International Water) and the Spanish group Abengoa. Within a few weeks, the company raised tariffs by 50 to 200 %, or even by 300 % in some cases according to Public Citizen (2002): in a country where the monthly minimum wage was below 100 dollars, some families had to devote up to half their income to paying for water. Massive demonstrations broke out between February and April 2000; the repression left at least one dead and more than a hundred injured. The government declared a state of siege and then terminated the concession in April 2000. Bechtel then sued Bolivia before the International Centre for Settlement of Investment Disputes (ICSID) for 25 million dollars of lost future profits; the case was settled out of court in January 2006, Bechtel abandoning its claims after an international civil society campaign, as The Democracy Center (2006) recalls.

*In fine*, the institutional economist Elinor Ostrom (Nobel prize 2009) offers a way out through the **governance of the commons**: neither pure market nor nationalisation, but local collective arrangements, including rules of access and use co-constructed by users. On the international scale, this logic nevertheless comes up against the absence of a binding legal framework on transboundary waters: the **New York Convention** of 1997 on international watercourses only came into force in August 2014 and has been ratified by only 38 states.

**Agricultural futures markets**, for their part, constitute one of the oldest forms of organised finance: the *Chicago Board of Trade* (CBOT, founded in 1848, today integrated into CME Group) was created precisely to allow farmers and millers to hedge against the volatility of cereal prices. **Wheat**, the leading source of vegetable protein in human food, is the most traded agricultural commodity on futures markets: its benchmark quotation is the *CBOT Wheat* contract (which is quoted in dollars per bushel, one bushel representing roughly 27.2 kilograms of wheat), while the **MATIF** (*marché à terme international de France*, now integrated into Euronext Paris) quotes the *Milling Wheat* contract in euros per tonne — which serves as the benchmark for Europe. Despite its symbolic importance, wheat remains a largely domestic commodity: only 25 to 27 % of world wheat production is the object of international trade in 2024 and 2025: the FAO (2024) estimates world wheat trade at 198 million tonnes for a production of 787 million tonnes; the U.S. Department of Agriculture (2024) for its part projects trade of 216.5 million tonnes for a record production of 796.9 million tonnes. The fundamental indicator for valuing a cereal market is the **stocks-to-use ratio** (S/U), defined as the ratio of end-of-season stocks to total use in a given year. This ratio measures the capacity of the market to absorb a negative supply shock: an S/U of 30 % means that stocks cover roughly 110 days of world consumption. The empirical relation between S/U and price is robust and non-linear: Flanders (2016) shows a strong inverse correlation such that, as S/U decreases, prices increase in an increasingly convex manner, the last percentage points of stocks being valued exponentially (scarcity effect). For wheat, the world S/U for the 2025–2026 season is estimated at 32.9 %, its lowest level since 2012–2013, against a historical average of 34 % over the period 2009–2024, according to the USDA projections relayed by Mecardo (2025). More precisely, the **WASDE** (*World agricultural supply and demand estimates*) is the monthly report published in the second week of each month by the USDA (*United States Department of Agriculture*), regarded as the most influential publication on world agricultural markets. It presents projections of production, consumption, stocks and trade for around ten agricultural commodities (wheat, maize, soya, cotton, sugar), including revisions of the current and projected S/U. The publication systematically generates a detectable **price reaction**: Massa et al. (2009) have shown that post-harvest revisions of the wheat S/U make it possible to construct precise empirical 90 % confidence intervals for end-of-season prices. For the agricultural commodity trader, the basic strategy therefore consists in (i) anticipating the direction of the S/U revision relative to the Bloomberg consensus; (ii) taking a position on the *CBOT Wheat* or *MATIF Milling Wheat* contracts before

publication; (iii) unwinding the position within the 48 hours following the announcement. This *WASDE play* is asymmetric: downside surprises on stocks (a downward revision of the S/U) generate more violent reactions than upside surprises, on account of the asymmetry of the price curve relative to the S/U ratio. Finally, agricultural markets present a marked **seasonality**: prices generally fall as harvests approach (between July and August for the northern hemisphere) and rise again on entering the lean season<sup>6</sup>. Moreover, the S/U ratio constitutes an indicator of geopolitical and geo-economic tensions. Thus, the crisis triggered by the Russian invasion of Ukraine in February 2022 highlights the link between the S/U ratio and geopolitics. According to Paulson et al. (2022), Russia and Ukraine together represented a little less than 30 % of world wheat exports between 2017 and 2021 (roughly 18 % for Russia and 10 % for Ukraine), roughly 17 % of those of maize (of which 15 % for Ukraine alone) and nearly a third of world barley exports (of which 17 % for Russia alone). According to Glauben et al. (2022), Ukraine likewise provided 66 % of world sunflower oil exports between 2015 and 2022. Glauben et al. (2022) adds that between 2018 and 2020, 56 million tonnes of wheat and 31 million tonnes of maize were exported annually from Russia and Ukraine to 123 and 95 countries respectively. The main export markets for wheat are Egypt (19 %) and Turkey (13 %), while the main markets for maize are China (16 %), the Netherlands and Spain (11 % each) and Egypt (10 %). Ten million tonnes of sunflower oil were shipped annually to 166 countries, the main outlets being India (27 %) and China (15 %). Russian and Ukrainian mineral fertilisers for their part were exported to 143 countries, Brazil (21 %), the United States (9 %) and China (8 %) constituting the main destinations. The shock of the Russian invasion of Ukraine consequently sent the price of the *CBOT Wheat* contract leaping from 7 dollars per bushel in January 2022 to 13.6 dollars per bushel on 7 March 2022, that is a near doubling in six weeks. The **Black Sea Grain Initiative**, signed on 22 July 2022 in Istanbul by Russia, Ukraine, Turkey and the UN, certainly permitted the resumption of Ukrainian exports via a secured maritime corridor from three key ports — Odesa, Chornomorsk and Pivdennyi — to the rest of the world; in one year of application, roughly 33 million tonnes of cereals and agricultural products were thus able to be exported, as Glauber et al. (2023) explain. The agreement, extended on three occasions (November 2022, March 2023 and May 2023, this last time for only 60 days at Moscow's request), was, however, unilaterally terminated by Russia on 17 July 2023: Moscow considered that its own exports of cereals and fertilisers remained hampered by Western sanctions and that the guarantees given in this regard had not been honoured. The very next day, Russia launched missile strikes on the port of Odesa and its cereal infrastructure, sending wheat futures prices to their sharpest one-day rise since February 2022, according to Glauber et al. (2023). This episode shows that the world wheat S/U has become a **leading indicator of geopolitical tensions**: any unforeseen reduction in the exports of a major producer translates immediately into a deterioration of the ratio and upward pressure on food prices, notably in the importing countries of North Africa and the Middle East.

#### Focus...



**The financialisation of agricultural markets and the food crisis of 2007–2008.** The surge in agricultural prices of 2007–2008 — wheat reached 13 dollars a bushel in March 2008, its highest level since 1974, against three dollars a bushel in 2005 — revived the debate on the role of financial investors in commodity markets. The thesis of **speculation as an amplifying factor** rests on a structural fact: between 2003 and March 2008, the positions of non-commercial investors on American commodity futures markets — mainly via **commodity index funds** (CTI) — were multiplied by twenty, going from roughly 13 billion to 260 billion dollars, according to the IATP (2008) and the WEED (2008). The proponents of the thesis of **financial harmlessness** — according to which fund speculation would not have caused the price surge — argue that these investors provide market liquidity and that the price surge reflected real fundamentals: low stocks (world wheat S/U at roughly 20–22 % in 2007–2008, against an average of 29 % over the five preceding years according to the U.S. Department of Agriculture (2008)); concomitant droughts in Australia, Ukraine and Canada; a rise in Chinese demand; and the massive conversion of American maize into ethanol within the framework of the **Renewable Fuel Standard** (RFS), instituted by the *Energy Independence and Security Act* of 2007: the share of American maize used to produce ethanol went from 24 % of total production in 2006–2007 to 37 % in 2010–2011 according to McPhail & Babcock (2012). The debate remains open; the institutional response has nevertheless been clear: under the French presidency of the G20 in 2011, the *Agricultural Market Information System* (AMIS) was launched in order to improve the transparency of stocks and to limit information retention behaviour, as the official site of the AMIS (2011) explains.

<sup>6</sup>The **lean season** designates the interval which separates the exhaustion of the reserves from the previous harvest and the arrival of the new harvest: the granaries are empty, demand for cereals on the markets increases, and prices soar, sometimes to the point of food crisis if a climatic hazard delays the harvest. It generally corresponds to the period between June and September in the Sahelian countries, and to equivalent periods in other agricultural zones. On futures markets, it manifests itself in a seasonal rise in cereal prices in the months preceding the harvests, as Géoconfluences (2024) analyses.



Agricultural tensions are moreover inseparable from **international food aid**, which represents a specific component of official development assistance (ODA), treated separately from financial transfers on account of its humanitarian dimension and its direct impact on commodity markets. The Programme alimentaire mondial (2025), a United Nations agency created in 1961 and winner of the Nobel peace prize in 2020, is the leading world operator: it served 124 million people in 2024, the voluntary contributions of member states amounting to 9.8 billion dollars. This aid takes two main forms: **in-kind transfers** (deliveries of cereals bought in donor countries — a historic American practice, criticised because it disturbs local markets) and **cash transfers** (vouchers or transfers allowing beneficiaries to obtain supplies locally, now the dominant practice). The **Democratic People's Republic of Korea** (DPRK) constitutes an example of the use of food aid as a bargaining chip in an international security negotiation. The famine of 1995–1998 — which caused between 600 000 and three million deaths according to estimates, a range this wide reflecting the statistical opacity of the regime<sup>7</sup> — led the regime to accept massive external aid for the first time since the foundation of the state in 1948. As early as 1994, the United States had negotiated the *Agreed Framework* with Pyongyang: in exchange for halting the plutonium nuclear programme, Washington undertook to supply 500 000 tonnes of fuel oil a year and to finance the construction of two light water reactors (KEDO). Between 1995 and 2008, the United States supplied North Korea with more than 1.3 billion dollars of assistance, of which more than 50 % in food form — roughly 2 million tonnes of cereals and food substitutes — according to the Congressional Research Service (2014). This system of conditional aid reached its climax with the *Leap-day Agreement*, signed on 29 February 2012 by the Obama administration and the new regime of Kim Jong-un, installed after the death of his father Kim Jong-il in December 2011: North Korea undertook to observe a **moratorium on its nuclear and ballistic tests** and accepted a monitoring arrangement for its key nuclear sites; in exchange, the United States promised the delivery of 240 000 tonnes of nutritional aid (maize and soya mixes, fortified infant foods), with the prospect of additional aid conditional on progress in denuclearisation, as *The Diplomat* (2016) recalls. The agreement collapsed six weeks after its signature: in April 2012, North Korea proceeded to launch a ballistic rocket (*Kwangmyongsong-3*) under cover of a civilian satellite launch, explicitly violating the moratorium. The Obama administration suspended the delivery of food aid. This episode reveals the logic of **brinkmanship** in which food aid is instrumentalised: Pyongyang alternates signals of openness (obtained in exchange for aid) and provocations (demonstrations of ballistic capabilities to strengthen its negotiating position), in a cycle which the CISSM (2025) likens to a 'hunger diplomacy': the international aid supplied since 1995 is said to have allowed the regime to finance its nuclear and ballistic programme in full, by freeing domestic budgetary resources from the food constraint.

### Key Takeaways



Fresh water is the only major natural resource without an organised world market, on account of its immobile character and its recognition as a fundamental human right by the UN in July 2010 (resolution A/RES/64/292); yet the world water deficit estimated at 40 % by 2030 creates the conditions for growing financialisation, illustrated by the *Nasdaq Veles California Water Index* (NQH2O) [OCDE (2012), CME Group (2020)]. Agricultural futures markets for their part allow producers to hedge against the volatility of cereal prices. The fundamental indicator is the stocks-to-use ratio (S/U): its relation with prices is robust and convex, the last percentage points of stocks being valued exponentially [Flanders (2016), Mecardo (2025)]. The WASDE, published monthly by the USDA, is the most influential publication on these markets: revisions of the S/U make it possible to construct precise empirical 90 % confidence intervals for end-of-season prices, and downside surprises on stocks generate more violent price reactions than upside surprises on account of the asymmetry of the price curve [Massa et al. (2009)]. The food crisis of 2007–2008 for its part revealed the risks of excessive financialisation: between 2003 and March 2008, the positions of commodity index funds were multiplied by twenty, while the world wheat S/U fell to 20–22 % under the combined effect of droughts, the rise in Chinese demand and the massive conversion of American maize into ethanol; the debate on the amplifying role of speculation remains open, but led to the launch of AMIS in 2011 under the French presidency of the G20 [AMIS (2011)].

### 3.3 Property and infrastructure

Property and infrastructure are characterised by low liquidity offset by stable income flows and a structurally low correlation with listed financial assets, thus offering institutional investors effective diversification in the sense of Markowitz (1952). On the one hand, **professional property investment** — offices, retail, logistics, managed residential — rests on specialised vehicles (SCPI, OPCI, REIT) permitting indirect exposure to property markets. On the other hand, **infrastructure** — energy, transport, digital, water — represents for its part a rapidly expanding

<sup>7</sup>The famine mainly affected rural regions and the social classes least favoured by the distribution system, relatively sparing the army and the nomenclature of the Workers' Party.

investment frontier, driven by the financing needs of the energy transition and by the insufficiency of public spending.

### 3.3.1 Professional property investment

Commercial real estate (CRE) constitutes an asset class apart in the universe of institutional management: heterogeneous, illiquid, indivisible and highly leveraged, it offers in return a structural **illiquidity premium**, partial protection against inflation — commercial leases are frequently indexed to price indices such as the commercial rent index (ILC) or the tertiary activities rent index (ILAT) in France — and a low correlation with equity markets over the long run. More precisely, the ILC applies to classic commercial leases, that is to say to tenants carrying on a commercial or artisanal activity registered in the trade and companies register. Its formula since 2022 is weighted 75 % by the consumer price index (IPC) excluding tobacco and rents and 25 % by the construction cost index (ICC), as the INSEE (2025b) explains. By contrast, the ILAT concerns offices, warehouses and premises of the liberal professions — that is precisely the assets of interest to the professional property investor. Its composition is different: 50 % of the IPC, 25 % of the ICC and 25 % of the economic growth of GDP in value terms in the market sector, as Arthur Loyd (2023) analyses. Institutional investors — insurance companies, pension funds, sovereign funds — generally devote between 5 and 10 % of their asset allocations to it: according to Option finance (2025), the share of property in the portfolios of the large European institutional investors was 10.4 % in 2020 and fell back to 8.5 % at the end of 2024 under the effect of asset depreciation; on a world scale, the proportion stabilises at around 10 %. The dispersion remains substantial: the most active Anglo-Saxon pension funds allocate up to 15 to 20 % to property, while French insurers remain below 4 % of their balance sheet, partly on account of the capital requirements of Solvency II, as the EDHEC (2019) explains. The central valuation metric is the **capitalisation rate** (*cap rate*), defined as the ratio of annual net operating income to the open market value of the asset. This rate maintains an inverse relation with valuations: all other things being equal, a compression of the *cap rate* from 5 % to 4 % mechanically raises the value of the asset by 25 %. Between 2015 and 2021, the zero interest rate policy of the major central banks provoked a historic compression of *cap rates* on *prime* Parisian offices: the prime yield of the central business district (CBD), which still stood at roughly 5 % in 2010, compressed to 2.75 % at the end of 2021 — that is a fall of 225 basis points in a decade, mechanically fuelling a rise in valuations according to Knight Frank (2021) and BNP Paribas Real Estate (2022). The rise in the ECB's key rates since 2022 subsequently provoked a symmetrical decompression: the *prime* CBD yield again exceeded 4.0 % at the beginning of 2025 according to Knight Frank (2025).

#### Focus...



**The décret tertiaire and the risk of stranded assets.** The tertiary sector decree no. 2019-771 of 23 July 2019, arising from article 175 of the **loi Elan**, which came into force progressively from 2022, imposes on all buildings for tertiary use of more than 1000 square metres obligations to reduce final energy consumption relative to a reference year: –40 % by 2030, –50 % by 2040 and –60 % by 2050, according to Légifrance (2019). Consumption data must be declared annually on ADEME's OPERAT platform; in the event of failure to comply, the sanctions include a *name and shame* formal notice (publication on an official site) and fines of up to 7500 euros per year. The financial stake is considerable: buildings incapable of attaining these objectives without heavy investment risk becoming **stranded assets** — assets whose value collapses as a result of a regulatory or technological constraint, by analogy with the concept initially formulated for hydrocarbons that cannot be exploited within decarbonisation trajectories. For tertiary property, a building rated DPE F or G presenting prohibitive renovation costs will progressively be shunned by solvent tenants, then by institutional investors: its value will converge towards its bare land value, erasing the premium of a built location. The **flight to quality** — the phenomenon by which tenant firms concentrate their demand on *grade A* space meeting the HQE, BREEAM or LEED standards — is already reflected in the statistics of the Parisian market: the vacancy rate for *prime* offices in Paris remains below 5 % in July 2024, while that of the inner suburbs exceeds 18 % and that of La Défense reaches 15.2 % according to BNP Paribas Real Estate (2024).

Access to commercial real estate takes several forms according to the investor's profile: (i) **SCPI** (*sociétés civiles de placement immobilier*), known as 'paper property', allow individual savers to acquire units in a diversified portfolio of offices, retail premises or warehouses, in exchange for a flow of quarterly dividends; with a total capitalisation of 88.5 billion euros at the end of 2024, their market remains dominated by office and retail SCPI, whose average yield stood at 4.72 % in 2024, as Weelim (2025) explains; (ii) **OPCI** (*organismes de placement collectif immobilier*), more liquid since they are required to hold at least 10 % of financial assets; (iii) **SIIC** (*sociétés d'investissement immobilier cotées*), the French equivalent of American *real estate investment trusts* (**REIT**): these are listed property companies, exempt from corporation tax in return for high distribution obligations, thus offering liquid access (via the stock



exchange) to large property portfolios. The main French SIIC are Gecina (Parisian offices), Unibail-Rodamco-Westfield (pan-European shopping centres) and Covivio (offices and hotels in Europe). While conventional offices were faltering, three segments demonstrated their resilience or their dynamism: logistics, data centres and managed residential. **Logistics** (*last mile* warehouses, parcel handling, large platforms) has benefited from the acceleration of online commerce: BNP Paribas Real Estate (2026) records 3.2 million square metres of logistics take-up in 2025 and 3.1 billion euros invested in the asset class. **Data centres** for their part are experiencing growth of 33 % in installed capacity in megawatts in 2024, driven by demand from artificial intelligence: Knight Frank (2026) projects 500 *data centres* in France by 2030. Finally, **managed residential** (student residences, care homes for dependent elderly people, *coliving*, senior residences) recorded 1.7 billion euros of investment in 2025, that is a tripling relative to 2024 (598 million euros), according to Cushman & Wakefield (2026).

### Focus...



**SIIC and REIT: two models of listed property company.** *Real estate investment trusts* (REIT) constitute the world benchmark vehicle for listed property, created in the United States by the **Real Estate Investment Trust Act** of 1960. At the end of 2024, American REITs listed on the NYSE represented a market capitalisation of 1424 billion dollars and had paid out more than 60 billion dollars in dividends during the year, according to the NAREIT (2025). Their long-run performance is remarkable: the *FTSE Nareit All Equity REITs Index* displays a compound annual return of 12.6 % since 1972, surpassing the S&P 500 (8.0 % over the same period), according to The Motley Fool (2025). The outperformance is explained by the combination of a structurally high *dividend yield* on property (3.96 % at the end of 2024 according to the NAREIT (2025)) and a revaluation of capital driven by the growth of rents; it is, however, weaker over short horizons: over the last ten years, the S&P 500 has regained the advantage under the effect of the rise of technology stocks (an average annual return of 11.1 % between 2014 and 2024 on the S&P 500 against 7.2 % for the aforementioned REIT index, as The Motley Fool (2025) relates). *Sociétés d'investissement immobilier cotées* (SIIC) constitute the French equivalent of REITs, introduced by the 2003 Finance Act. Their tax regime rests on **full tax transparency**: the SIIC is exempt from corporation tax in return for three strict distribution obligations which Actu juridique (2024) sets out: 95 % of profits arising from rental income must be distributed before the end of the following financial year; 60 % of net capital gains on disposals of property assets must be distributed; 100 % of dividends received from subsidiaries having themselves opted for the SIIC regime. One of the distinctive characteristics of SIIC and listed REITs is the possibility of trading at prices discounted relative to the book value of the assets held, a phenomenon measured by the **discount to net asset value** EPRA NTA (*European public real estate association net tangible assets*). At 30 June 2025, Unibail-Rodamco-Westfield thus displayed a discount of 23 %, against 35 % for Gecina and 33 % for Covivio, according to BMF Bourse (2025). For the contrarian investor, a discount to net asset value above 30 % historically constitutes a buy signal: it means that it is possible to acquire a professionally managed property portfolio at 70 cents for one euro of assets — a structural advantage that SCPI, whose units are valued at the cost price of the underlying assets, can never offer.

### Key Takeaways



Commercial real estate constitutes a heterogeneous and illiquid asset class whose valuation rests on the capitalisation rate (*cap rate*), the inverse ratio between net operating income and open market value. The risk of *stranded assets* now constitutes the structural constraint of the asset class: the *décret tertiaire* of 23 July 2019 imposes reductions in energy consumption of 40 % by 2030, 50 % by 2040 and 60 % by 2050, entailing a flight to quality which widens the gap between *grade A* assets and secondary or peripheral assets, whose vacancy rate exceeds 18 % in the inner suburbs [Légifrance (2019), BNP Paribas Real Estate (2024)]. Moreover, institutional access to this asset class is organised along a continuum combining liquidity and ticket size: SCPI offer fractional access for individual savers; SIIC and their American equivalents, REITs, offer stock market liquidity in return for strict distribution obligations, with a structural discount permitting the acquisition of a managed portfolio at lower cost [Weelim (2025), NAREIT (2025), BMF Bourse (2025)]. Finally, while conventional offices were faltering, three segments demonstrated structural resilience or dynamism: logistics, *data centres* and managed residential [BNP Paribas Real Estate (2026), Knight Frank (2026), Cushman & Wakefield (2026)].

### 3.3.2 The state of the property market in France

The French property market went through, between 2022 and 2024, the most brutal correction observed since the crisis of 2008, after a decade of almost uninterrupted rises. The period 2013–2023 had seen prices progress by nearly 30 % on a national average according to Nexity (2024), driven by **historically accommodative credit conditions** —

the average rate on mortgage loans to private individuals stood at 1.06 % in 2021, its historic low, against an average of roughly 4 % over the period 2000–2020 according to Observatoire Crédit logement (2026) — and by a powerful post-Covid attraction to individual housing, which had propelled the volume of transactions to a historic record of 1 198 000 transactions over the rolling twelve months to the end of October 2021, according to Capitaine (2021). The reversal of ECB monetary policy from the summer of 2022 onwards broke this cycle: mortgage rates leapt from 1.06 % to more than 4.2 % in twenty-three months, mechanically reducing household solvency and contracting demand, as Immobilier danger (2026) relates. Consequently, the volume of transactions in existing homes fell to 780 000 over the rolling twelve months to the end of September 2024 — its lowest level since 2015, that is a contraction of roughly 35 % relative to the peak of 2021 — while prices recorded a fall of 3.9 % between September 2023 and September 2024 across the whole territory, according to the annual review of the Notaires de France (2024). This correction, while it may appear severe in terms of volumes, has in reality been insufficient to bring prices back to the level of economic fundamentals: the ratio of housing prices to household disposable income, initially formalised by the economist Jacques Friggitt of the IGEDD (2026), remains significantly above the ‘tunnel’ which characterised the market between 1965 and 2000. More precisely, expressed as multiples of income, prices fluctuated within a narrow tunnel of roughly plus or minus 10 % around a stable central value. Since the bubble of 2000–2007, prices have left this tunnel to settle at a durably higher level: according to the long series updated in February 2026, the ratio remains between 70 and 100 % above the central value depending on the vintage, which Friggitt attributes to the conjunction of the trend decline in real interest rates over thirty years, the lengthening of loan durations, and self-fulfilling expectation effects. This disconnection has two major operational implications. First of all, the price correction observed in 2023–2024 is quite insufficient to bring the ratio back inside the tunnel, which means that the market remains structurally expensive in terms of **property purchasing power for first-time buyers**. To be sure, the dynamics of the market began an appreciable reversal during 2025, mainly under the effect of the monetary easing embarked upon by the ECB, whose deposit rate was brought down to 2.0 % during the first half of 2025 at the end of a series of successive cuts. This easing was transmitted to the mortgage market with the usual lag of two to three quarters: the average rate granted to private borrowers stood at 3.17 % on average in the last quarter of 2025, down on the average of nearly 4 % observed a year earlier according to Observatoire Crédit logement (2026). This improvement in financing conditions restored a measurable purchasing capacity to households, and rapidly translated into transaction volumes: at the end of September 2025, the number of transactions in existing homes over the preceding twelve months reached 921 000 deeds, up 11 % year on year, that is a progressive return towards the long-run equilibrium of the market, according to the data of the Conseil supérieur du notariat (2025). However, this recovery risks sustaining the bubble without resolving the affordability constraint: household solvency improves by way of the cost of credit, and not by way of a fall in prices, deferring the affordability problem onto future generations. The **Friggitt tunnel model** thus suggests that, barring a deflationary shock or a regulatory rupture, prices will not spontaneously return within the tunnel: public housing policy must therefore act on the supply side (construction) rather than on the side of demand made solvent by credit, on pain of feeding a new upward cycle disconnected from fundamentals.

#### Focus...



**The rent reference index.** While inflation is likewise felt in rents, it is nevertheless regulated through the rent reference index (IRL). Created by the ENL law of 13 July 2006, the IRL falls within the framework of residential leases regulated by the law of 6 July 1989: it is computed quarterly by the INSEE (2024b) from the annual average of the evolution of consumer prices, excluding tobacco and rents, and serves as the basis for revising the rents of unfurnished or furnished dwellings. Landlords who wish to revise the rent from one year to the next must therefore do so within the limit of the annual variation of the IRL, which thus fixes the maximum increase. The principle is to adapt the rent to the increase in the cost of living without inflicting a disproportionate rise on tenants. In an inflationary phase, the IRL displayed growing variations: after a rise of only +0.83 % in the third quarter of 2021, it progressed by +2.48 % in the first quarter of 2022, then reached its peak at +3.60 % in the second quarter of 2022, as the INSEE (2024c) mentions.

In order to limit the impact of inflation on household budgets, the state then instituted a ‘rent shield’ within the framework of the law of 16 August 2022 on emergency measures for the protection of purchasing power. This arrangement capped the rise in the IRL at 3.5 % in metropolitan France, 2 % in Corsica and 2.5 % in the overseas territories, as Légifrance (2022) indicates. Without this cap, the annual variation of the IRL would have stood at +6.26 % in the first quarter of 2023 according to the INSEE (2023). Faced with still high inflation, law no. 2023-568 of 7 July 2023 extended this arrangement unchanged until 31 March 2024, maintaining the cap at 3.5 % for residential leases and for the commercial rents of SMEs as Légifrance (2023) indicates. Moreover, in the cities having implemented a **rent control** arrangement, the Purchasing Power law of 16 August 2022 available at Légifrance (2022) likewise removed the possibility for the landlord to apply a ‘rent supplement’ when the dwelling presents degraded

characteristics — notably infiltration or flooding from outside — thereby reducing housing expenditure for the most modest households.

Furthermore, tensions in the supply of affordable housing are reaching historic levels, revealing the structural limits of the French model of housing production. Social housing organisations (public housing offices and social housing enterprises in particular) manage, within the **HLM** (*habitations à loyer modéré*, moderate-rent housing) stock, 4.8 million ordinary dwellings (of which 4.6 million occupied, housing 10.4 million people, that is 15 % of households), to which are added 380 000 supported housing units, as the Union sociale pour l'habitat (2025b) explains. The total social rental stock, all landlords combined, amounts to 5.9 million dwellings; HLM organisations constitute in this respect the leading residential investor in France with 17.6 billion euros of investment in 2024 according to the Union sociale pour l'habitat (2025b), financed over long cycles backed by loans from the Caisse des dépôts et consignations (CDC). Now at the end of June 2025, demand for **social housing** reached an absolute record of 2.87 million households waiting, up 103 000 new applications (+3.7 %) in the first half of 2025 alone according to the Union sociale pour l'habitat (2025a). The ratio between allocations and the stock of applications illustrates the scale of the imbalance: in 2024, the 363 000 allocations made satisfied only roughly one application in seven, the stock of applicants reaching 2 767 000 households at the end of the year according to the Union sociale pour l'habitat (2025a). Moreover, the 98 682 social dwellings financed (approved) by the state in 2024 (+8.2 % relative to 2023) remain quite insufficient to absorb this structural imbalance, according to Létard (2025). The roadmap signed in February 2025 by the minister Valérie Létard and the HLM Movement indeed aims at a production of 100 000 to 110 000 new dwellings per year and the renovation of 120 000 to 130 000 existing dwellings, relying on the **reduction of the Livret A rate** enacted by the government — since the financing of social landlords is backed by the resources of regulated savings, the reduction of the Livret A rate mechanically lightens their debt burden. This financial equation nevertheless remains constrained by the twofold **scissor effect** that landlords have suffered since 2022: on the one hand the rise in construction costs and interest rates, on the other the capping of the IRL which has restrained the growth of their rental receipts. Indeed, the financial model of HLM organisations rests on a fragile balance between **rental receipts** (framed by the IRL and the solidarity rent reduction) and **financial charges** (variable-rate debt indexed to the Livret A). Now the shock of 2022 broke this balance by activating three adverse levers simultaneously: (i) the rise in the Livret A rate from 0.5 % (January 2022) to 1.0 % (February 2022), then to 2.0 % (August 2022) and finally to 3.0 % (February 2023), which increased interest charges by roughly 3.75 billion euros on a full-year basis across the whole sector, for an outstanding debt of 150 billion euros indexed to the Livret A, according to the Union sociale pour l'habitat (2023); (ii) the inflation of construction costs (materials, energy) which increased the cost of new-build and rehabilitation operations; (iii) the capping of the IRL at 3.5 % (the rent shield of 2022–2024) which restrained the revaluation of rents below the real inflation borne. The government's decision to bring the Livret A rate back to 2.4 % (February 2025), then to 1.7 % (August 2025) and finally to 1.5 % (February 2026) constitutes a significant easing, of the order of several hundred euros of financial capacity gained per dwelling, as the Ministère de l'Économie et des Finances (2025b) explains. This easing largely conditions the sector's capacity to honour the commitment of 100 000 new dwellings per year, of which the 85 000 dwellings started in 2024 — against 98 682 planned by Létard (2025) — constituted a first signal of stabilisation.

### Focus...

#### Personal housing benefits

- **Programme 109** 'Assistance with access to housing' constitutes the state's main instrument of intervention in favour of making the most modest households solvent in the face of their housing expenditure. It carries the personal housing benefits (APL, ALF, ALS), paid by the family allowance funds (CAF) and the agricultural social mutual fund (MSA) via the national housing assistance fund (FNAL). In the course of 2024, the total amount of benefits paid reached 13.1 billion euros, according to the analysis of budgetary execution by the Cour des comptes (2025a). Since the 2025 finance bill, the financing circuit has been simplified: employer contributions, which until then co-financed the FNAL, have been abolished; programme 109 is henceforth financed entirely by the state, to the tune of 17 billion euros in the initial finance act for 2025 as the Sénat (2024) recalls.
- **The solidarity rent reduction (RLS)**, created in the initial finance act for 2018, is an automatic rebate on rent applied by social landlords to HLM tenants on low incomes. Its mechanism is inseparable from the APL: when a household benefits from an RLS, its APL is reduced by roughly 98 % of the amount of the rebate, so that the tenant's overall out-of-pocket cost remains broadly stable, while public expenditure is lightened by as much: the bulk of the financial effort is thus transferred onto social landlords, whose investment capacity has consequently been durably constrained — a point raised by the Cour des comptes (2025a) in its work on the sector.

- **The reform of ‘real-time APL’**, which came into force on 1 January 2021, constituted a break in the method of computing benefits. Entitlement to the APL is henceforth computed on the basis of the last twelve months of income, updated every quarter thanks to data from the tax administration (deduction at source), in place of the income of year  $N - 2$  previously, as CDC Habitat (2024) explains. This bringing up to date makes it possible to adapt the amount paid to the real situation of beneficiaries: a rise in income entails a reduction of the benefit, a fall increases it, avoiding phenomena of overpayment. The first official review estimated the budgetary saving at 1.1 billion euros in the very first year, that is well beyond the 750 million euros anticipated in the 2021 finance act, 38 % of recipients having seen their benefit fall (rising income) and 25 % having seen it rise (falling income) according to the data of Banque des Territoires (2021).

#### Focus...



**Tax incentives for affordable letting.** In order to reduce the consequences of inflation on the most modest households, the state has put in place tax incentives intended to mobilise the private stock towards affordable housing. Coming into force on 1 January 2022, the **Loc’Avantages** arrangement — instituted by the 2022 finance act (article 67) to replace the ‘Louer abordable’ arrangement (the Cosse law of 27 January 2017) — aims to be simpler and more advantageous for landlords, as the ANIL (2025) and Direction de l’information légale et administrative (2025) explain. Concretely, it offers landlords a **tax reduction** computed on gross rental income, subject to letting their dwelling (new or existing) as a principal residence, at a rent below local market prices and to a tenant whose resources do not exceed ceilings fixed by the national housing improvement agency (**ANAH**). The tax reduction is structured at three levels, defined by the discount granted on the market rent and by whether or not rental intermediation is used: **Loc1** (–15 % on the market rent): a reduction of 15 % for direct letting, 20 % with intermediation; **Loc2** (–30 %): a reduction of 35 % for direct letting, 40 % with intermediation; **Loc3** (–45 %, only with rental intermediation): a reduction of **65 %**. Additional bonuses are likewise provided for the Loc2 and Loc3 levels with rental intermediation: 1 000 euros for sub-letting formulas or 2 000 euros for a management mandate, increased by a further 1 000 euros when the surface area of the dwelling is less than or equal to 40 square metres, that is up to 3 000 euros per dwelling, as France Rénov’ (2023) details. The Loc’Avantages arrangement thus combines the wealth interests of the private landlord and the general interest, likewise constituting a lever in the fight against **housing vacancy**: according to the INSEE (2025c), France still counted 3.0 million vacant dwellings in 2025, representing 7.7 % of the total residential stock — a level nevertheless falling for the fifth consecutive year, attesting to the progressive effectiveness of public policies of incentive to bring dwellings back onto the market.

#### Key Takeaways



The French property market underwent between 2022 and 2024 its most severe correction since 2008: after a decade of rises of nearly 30 % driven by historically low rates and by the post-Covid shock to housing demand, the reversal of ECB monetary policy contracted transaction volumes by 35 %, while prices fell back by 3.9 % year on year [Notaires de France (2024), Observatoire Crédit logement (2026)]. This correction has, however, remained insufficient to bring the ratio of prices to household disposable income back within the Friggitt tunnel, above whose historic central value it remains by 70 to 100 % [IGEDD (2026), Conseil supérieur du notariat (2025)]. Tensions in the supply of social housing are simultaneously reaching a record level — 2.87 million households waiting at the end of June 2025, of which only one application in seven was satisfied in 2024 — revealing the limits of the financial equation of HLM organisations, caught since 2022 between the rise in the cost of their debt indexed to the Livret A and the capping of their rental receipts by the rent shield [Union sociale pour l’habitat (2025a), Létard (2025), Ministère de l’Économie et des Finances (2025b)].

#### 3.3.3 Infrastructure

**Infrastructure assets** designate the physical equipment essential to the functioning of the economy and of society: energy networks (transmission, distribution, renewable generation), transport (toll roads, airports, ports, rail), utilities (water, waste, telecommunications), and social infrastructure (hospitals, schools). Their common feature is to generate stable income flows over the very long run — often 20 to 40 years — backed by concession contracts, regulated tariffs or *take-or-pay* type contracts, thus offering a visibility on *cash flows* that no other private asset class can match, as BlackRock (2023) explains. For institutional investors, the asset class thus presents three major differentiating advantages relative to property. First, the position of **natural monopoly** or regulated quasi-monopoly of most infrastructure assets confers a resistance to economic cyclicity markedly superior to that of other asset

classes. As Medda & Panayiotou (2014) explain, an infrastructure asset indeed presents the classic characteristics of a natural monopoly: massive *sunk costs* making the entry of competitors economically irrational, increasing economies of scale, and above all **inelasticity of demand** — the consumption of electricity, drinking water or public transport varies little with the cycle, which guarantees the stability of income flows whatever the economic situation. Second, revenues are often explicitly indexed to inflation — via contractual clauses or regulated tariff formulas — making infrastructure a **natural hedge against price rises**: between the last quarter of 2019 and the third quarter of 2024, *network utilities* and transport infrastructure assets thus markedly outperformed under an inflationary regime relative to a low-inflation environment — transport assets displayed quarterly returns increasing by more than one percentage point in periods of high inflation relative to periods of low inflation — and dominated equities (MSCI ACWI) and bonds (*Bloomberg Global Aggregate*) according to Chen & Shen (2025): more precisely, between 2007 and 2022, the annualised total return of unlisted infrastructure stands at roughly 14.2 % per year (an *income return* of 7.0 % per year and a *price return* of 7.2 % per year). Third, the **low correlation with listed markets** improves the efficient frontier of portfolios from a diversification standpoint, as Chen & Shen (2025) explains. Professionals thus classically distinguish four segments, ordered by increasing level of risk, whose boundaries are defined by the maturity of the asset, the degree of regulation of revenues and exposure to construction risk, as Meketa Investment (2025) analyses:

- **The core segment** is made up of mature, fully operational assets with quasi-contractual flows (regulated energy transmission networks, historic toll motorways, infrastructure with public-private partnerships). The risk is minimal: revenues are known in advance and indexed to inflation. The remuneration comes essentially from the annual *cash yield*, little or not at all from the capital gain on exit. The target IRR stands at 7 or 8 % per year according to Private Corner (2023).
- **The core+ segment** is composed of real assets, not always regulated, exposed to limited market risk (airports, telecom towers, *data centres* with long leases). It is the dominant segment of the market: with 141.7 billion dollars of target capital in funds being raised in 2024, it clearly leads *core* (88.0 billion dollars) and *value-add* (86.2 billion dollars) according to Infrastructure Investor (2025). The target IRR is 8 to 10 % per year, combining dividends and capital gain on exit over a horizon of 10 to 15 years, as Private Corner (2023) analyses.
- **The value-add segment** for its part brings together assets requiring significant transformation (network extension, change of regulation, operational optimisation). The distribution is more restricted; the IRR comes more from resale after 5 to 7 years. InfraRed Capital Partners (2025) estimates the entry IRR on this segment at 14.7 % per year on average over ten years, while Private Corner (2023) gives a range between 10 % and 15 %.
- **The greenfield segment**, finally, aggregates projects in development, carrying construction risk, permitting risk and ramp-up risk. Very consuming of capital and time, these assets offer the highest risk premium. Cambridge Associates (2025) puts them at target IRRs of 12 to 16 % net or more: infrastructure funds specialising in digital (*data centres*, fibre) thus delivered a median IRR of 14.0 % on the *vintages*<sup>8</sup> 2009–2020, without a single fund in negative territory; by way of comparison, the median IRR across all strategies combined on the *vintages* 2009–2020 stands at 9.8 %.

### Focus...



#### The main contractual mechanisms linked to infrastructure investment

- **Concession contracts.** A concession is a long-term contract by which a public authority entrusts a private operator (the concessionaire) with the financing, construction, operation and maintenance of an infrastructure, in return for the right to collect fees from users (tolls in particular) throughout the duration of the contract, the works reverting thereafter to the state, as the ART (2023) explains. France offers the most accomplished example in Europe with its **motorway concessions**, orchestrated by the law of 18 April 1955: the concession companies (ASF, COFIROUTE, Sanef, APRR...) operate the conceded network (roughly 9200 kilometres) for durations running to 2036, collect tolls indexed to inflation (the IPC augmented by a works component), and in return pay a fee to the state, as Vie publique (2025c) details. Melchior (2021) thus specifies that the acquirers paid 22 billion euros at the time of the privatisation of 2002–2005 and took over 17 billion euros of public debt, bringing the value of the transaction to 39 billion euros for the concessionaires. But this model thereby generates highly predictable cash flows. More precisely, the historic concession companies (SCA) display margin ratios (the ratio

<sup>8</sup>The *vintage* of a fund designates the year of its first investment. It makes it possible to compare funds launched in the same year and to isolate the effect of the economic cycle on performance.

of EBITDA to turnover) structurally between 65 and 70 % — that is three to six times the levels observed in most industrial sectors (10 to 20 %) — and net results of the order of 20 % of turnover, sharply rising since privatisation, a reflection of the optimal operating phase attained as the debt is amortised, as Melchior (2021) details. These high margins do not, however, constitute an ‘exceptional’ profitability: assessed over the whole concession cycle, the return on capital invested is not atypical; it is its temporal profile which is: a long loss-making period (break-even after 25 to 30 years), followed by a phase of increasing profitability, the works being lost without compensation at the expiry of the contract. From the institutional investor’s point of view, the conceded asset is therefore valued as a *discounted cash flow* of very long duration, whose sensitivity to the discount rate requires active management of interest rate risk.

- **Regulated tariffs (*regulated asset base*, RAB).** When a network is in a position of natural monopoly (distribution of electricity, gas, water), the state delegates to an independent regulator the fixing of an access tariff which covers operating costs and remunerates the capital invested, according to the ***regulated asset base*** (RAB) model: the regulator authorises a return on the regulated asset base, and adjusts the tariff every four to five years at tariff *resets*. Thus, in France, the Commission de régulation de l’énergie (**CRE**, the French energy regulator) fixes the TURPE (the tariff for the use of public electricity networks), which represents roughly 30 % of consumers’ electricity bill including tax according to EDF (2025), and finances the investments of Enedis (distribution) and RTE (transmission): **TURPE 7** (August 2025–2028) thus accompanies an Enedis investment programme of 25.6 billion euros over the period (that is 6.4 billion euros per year), up 34 % on the average of 4.8 billion euros per year observed between 2021 and 2023, while Enedis’s regulated asset base will progress from 61 billion euros to 71 billion euros at 1 January 2028 according to Légifrance (2025). For the investor, the RAB mechanism is the most transparent: the authorised return, called the **regulatory WACC**, is computed explicitly by the regulator (the CRE in France, OFWAT in the United Kingdom, FERC in the United States) and adjusted for inflation; the value of the asset is therefore directly proportional to the RAB, which facilitates valuation. Concretely, if the regulator authorises a WACC of 5 % on a RAB of 10 billion euros, the operator is entitled to receive 500 million euros a year of regulated revenue: the valuation of the asset is thus a simple discounting of these contractually guaranteed future revenues, without exposure to market or demand risk, unlike a property asset or a *private equity* holding.
- ***Take-or-pay contracts.*** The *take-or-pay* mechanism (or **firm offtake clause**) is used for infrastructure whose construction cost is massively fixed and whose marginal cost is low: LNG terminals, gas pipelines, electricity interconnection lines, energy generation plants. The buyer (often a network operator or a state) undertakes contractually to pay for a minimum quantity, whether or not it physically takes delivery; this guarantee of revenues allows the concessionaire to raise project debt (*project finance*) on the strength of the contract alone, without recourse to the operator’s balance sheet, as Village de la justice (2021) for example explains. Thus, in a **liquefied natural gas** (LNG) *take-or-pay* contract, a European importer such as Engie or Total undertakes for a duration of 15 to 20 years to pay for a minimum regasification capacity — typically expressed in millions of tonnes per year — independently of the volumes actually taken, as Engie (2006) and Hwang & Rogers (2017) explain. The contractual structure breaks down into a fixed component (*capacity charge*), corresponding to the construction and financing costs of the terminal, due in all circumstances, and a variable component (*commodity charge*), proportional to the volumes delivered. This sharing is precisely what makes the financing of the asset possible: a study by Ruester (2014) bearing on 26 LNG terminal projects thus shows that debt leverage increases significantly with the share of capacity covered by long-term *take-or-pay* contracts, the guaranteed flow serving as collateral for the borrowing. The case of the Dunkerque LNG terminal — whose investment amounted to 1.3 billion euros, EDF being the majority shareholder (65 %) according to Noble (2014) — illustrates this arrangement: the regasification capacities (13 billion cubic metres per year, that is roughly 20 % of Franco-Belgian gas consumption according to Connaissance des énergies (2017)) were entirely pre-sold to gas operators ahead of commissioning via long-term capacity reservation contracts. On the Qatari side, the signature in February 2020 by Qatar Petroleum of a reservation contract on the Montoir-de-Bretagne terminal until 2035, covering 3 million tonnes per year, illustrates the same *take-or-pay* mechanism making it possible to secure long-term financing, as the Direction générale du Trésor (2020) mentions.

### Short-Answer Question



**The governance of water in France and in Europe.** In October 2022, the European Commission launched infringement proceedings against seventeen member states for non-compliance with the Water Framework Directive (WFD), emphasising that fewer than 40 % of European water bodies had attained the ‘good ecological status’ required for



2015 and postponed to 2027, as Voulvoulis et al. (2016) recall. This situation reflects a structural tension between the logic of economic efficiency which has historically guided water management and the environmental imperatives now codified in European law. In France, the governance of water is understood in the sense of institutional coordination between public, private and territorial actors to ensure the allocation, protection and financing of the water resource. In Europe, it refers to the normative framework and the economic instruments deployed by the European Union in order to harmonise profoundly heterogeneous national practices. As Kaika (2003) emphasises, the directive constitutes a response to a paradigm shift — from government to governance — which displaces responsibilities from the central state towards multi-level configurations integrating hitherto marginal actors. Yet the persistent failures in attaining environmental objectives, combined with chronic under-investment in infrastructure and growing pressures linked to climate change, show that this governance remains incomplete. To what extent does the framework of water governance, as it has been structured in France and in Europe, permit an effective and sustainable management of this resource subject to growing tensions? The analysis of the strengths and limitations of the French institutional model invites us first to examine its architecture, before measuring the contribution and the shortcomings of the European framework.

The governance of water in France rests on a model of management by river basin instituted by the water law of 1964, which created six water agencies and their corresponding basin committees in metropolitan France, to which are added the water offices in each overseas department, as the Ministère de la Transition écologique (2023a) explains. This model, founded on the principle that ‘water pays for water’, according to which the charges levied on users directly finance the protection and improvement of the resource, constituted a remarkable institutional innovation whose decentralised architecture prefigured the recommendations formulated later by Ostrom (1990) in her analysis of institutions for managing common goods. The revenues of the water agencies amount to 2355 million euros in 2023, of which 2217 million coming from the charges levied on users (that is 94 % of the total), to which are added 21 million from the France Relance plan and 56 million from the Green Fund, as Économie de l’Eau (2024) relates. In parallel, in 2022, expenditure devoted to waste water management reached 14.1 billion euros, up 0.6 % in nominal terms on 2021 but down 2.5 % in real terms according to the Service de la donnée et des études statistiques (2025a). This dualist system — which entrusts roughly 60 % of the distribution service to three private operators (Veolia, Suez, Saur) — is nevertheless subject to growing pressure for remunicipalisation since the 2000s, a phenomenon that Kishimoto (2016) interprets as a collective response to the failures of the delegated model, notably under-investment in infrastructure and tariff increases. At 1 January 2024, the overall average price of water indeed stands at 4.69 euros including tax per cubic metre for a reference consumption of 120 cubic metres per year (2.32 euros per cubic metre for drinking water, 2.37 euros per cubic metre for collective sanitation) according to the Observatoire des services publics d’eau et d’assainissement (2025), with substantial territorial disparities ranging from 2.98 euros per cubic metre in Réunion to more than 5.41 euros per cubic metre in Hauts-de-France and 5.34 euros per cubic metre in Brittany according to the Banque des Territoires (2025). The main limitation of this model lies in the chronic imbalance between the needs for renewing infrastructure and actual investment, which translates into network losses estimated at 18.7 % of the volume put into distribution in 2022 — that is roughly 900 million cubic metres, the equivalent of the annual consumption of 16.5 million inhabitants — on a network of 875 000 kilometres of pipes the majority of which was built in the 1960s and 1970s, as the Office international de l’eau (2024) emphasises.

The European framework of water governance, whose central instrument is the Water Framework Directive adopted in October 2000, constitutes an ambitious normative architecture aiming to attain ‘good ecological and chemical status’ of all surface and groundwater bodies, initially fixed for 2015 then postponed to 2027 on account of accumulated delays. As Kaika (2003) and Voulvoulis et al. (2016) show, the WFD departs from the classic administrative paradigm by imposing an integrated approach by hydrographic basin, by instituting the obligation of public participation in management plans, and by introducing the principle of cost recovery (article 9) according to which member states must ensure that water pricing gives users an incentive to efficient use. To this central architecture are added complementary sectoral directives: the nitrates directive (91/676/EEC, adopted on 12 December 1991) against diffuse agricultural pollution, the urban waste water directive (91/271/EEC, adopted on 21 May 1991) requiring compliant sanitation systems for agglomerations, and the drinking water directive revised on 23 October 2020, whose new standards were to be transposed before January 2023. The shortcomings nevertheless remain substantial. First, according to the data of the Office international de l’eau (2023), France reuses only 0.6 % of its treated waste water, against 14 % in Italy, 8 % in Spain and 91 % in Israel; this structural lag is all the more worrying in that European regulation (EU) 2020/741 on the reuse of waste water, which came into force on 26 June 2023, sets as its objective the move from 1.7 billion cubic metres of annual reuse to 6.6 billion cubic metres by 2025, which would reduce water stress in the European Union by 5 % according to Mendret (2022), while the Water Plan of March 2023 sets itself for its part a national objective of 10 % reuse by 2030, as the Ministère de la Transition écologique (2023b) recalls. For the time being, in 2022, 29.3 billion cubic metres of fresh water were withdrawn in France (excluding hydroelectric

generation); 80 % of these total withdrawals come from surface water and 20 % from groundwater, according to the Service de la donnée et des études statistiques (2025c). Second, according to the review published by the Service de la donnée et des études statistiques (2025b), agriculture represents 58 % of the water consumed in France in 2020 — that is to say the share not returned to aquatic environments after use — ahead of drinking water (26 %) and the cooling of power stations (12 %), and irrigation constitutes almost its exclusive use, with 92 % of agricultural volumes withdrawn; this sector, the leading structural consumer of the resource, nevertheless remains insufficiently regulated despite the diffuse pollution charge instituted at the end of the 1980s, as the increase in irrigated areas of 14.8 % between 2010 and 2020 shows according to the INSEE (2024a). Third, global warming in Europe exceeds twice the world average, with an increase of 0.5 degree per decade over the period 1991–2021, entailing notably the loss of thirty metres of thickness of Alpine glaciers between 1997 and 2021, which compromises the basic assumptions on which basin management plans were constructed, as the Organisation météorologique mondiale (2022) emphasises. The governance of water in France and in Europe therefore presents a mixed record. The French model of management by river basin constitutes a coherent institutional architecture, theoretically validated by the analysis of the commons of Ostrom (1990), which has permitted the financing of an integrated management of the resource for sixty years. The work of Kaika (2003) and Voulvoulis et al. (2016) confirms that the WFD represented a major advance by unifying a fragmented European law and by imposing a transnational ecosystem approach. However, the persistent failures — structural under-investment in networks, massive lag in the reuse of waste water, insufficient regulation of agricultural and industrial pressures — attest that the institutional framework, however sophisticated it may be, remains subordinate to political trade-offs and budgetary constraints. Faced with the acceleration of climate change documented by the Organisation météorologique mondiale (2022), the underlying question is whether governance by river basin, conceived in a context of relatively abundant resource, can adapt to a durably deficient water regime. The answer to this question will depend less on the normative sophistication of the existing frameworks than on the capacity of member states and European institutions to mobilise structural investment — the Water Plan announced by Macron (2023) providing for only 180 million euros a year of network renovation, an amount insufficient in comparison with the 600 million euros a year in the Netherlands (whose distribution network is roughly half as extensive as that of France), as the Sénat (2016) analyses — and on their capacity to correct the asymmetries of regulation between the sectors using the water resource.

International institutional investors (pension funds, insurers, sovereign funds...) have thus raised their target allocation to infrastructure to 5.5 % of assets in 2024 (up 42 basis points on 2023), a record level, according to the annual survey Hodes Weill / Cornell University (2024). The realised return over three years reaches 10.1 % on average, exceeding target return objectives by 78 basis points; most institutions nevertheless remain under-invested by 123 basis points relative to their target, on account of liquidity constraints linked to the slowdown in *private equity* and property distributions in 2023–2024, as Hodes Weill / Cornell University (2024) qualify. Moreover, the French market constitutes the leading European market for infrastructure: in 2024, French unlisted infrastructure funds raised 13.7 billion euros, up 16 % on 2023, prolonging the dynamic begun after the retreat of 2022–2023; since 2018, never had so many infrastructure projects been supported simultaneously (189 projects, for a total amount invested of 10.9 billion euros, up 7.4 % on the previous year), according to the Bird & Bird (2025) barometer. France thus confirms its international attractiveness: 80 % of the amounts raised come from foreign investors, a stable share over the whole period 2018–2024 (74 % on annual average) according to Bird & Bird (2025). The structure of fundraising remains dominated by *brownfield* assets (+21 % year on year in amount, even though the number of *brownfield* projects has remained stable), a reflection of the maturity of the French concession market and of the preference of large institutional investors for assets with low operational risk in a phase of normalisation of the rate cycle. Indeed, a *brownfield* asset designates an asset already built and in operation (a motorway concession in service, an electricity distribution network, an operational LNG terminal...): the infrastructure exists, generates revenues, and its operating history makes it possible to estimate future flows precisely. The risk is therefore low: no construction risk, no uncertainty about demand. In parallel, the democratisation of the asset class is accelerating: in application of **law no. 2023-973 of 23 October 2023 on green industry**, PER (retirement savings plans) must henceforth include an unlisted asset pocket (3 to 8 % depending on the investment horizon); at the end of 2024, seven infrastructure funds thus represented 16 % of the total outstanding amount of roughly 11 billion euros of unlisted funds accessible to individual savers in France according to Funds Magazine (2025).

#### Focus...



**Listed versus unlisted infrastructure: liquid proxies and the illiquidity premium.** Two vectors of exposure to infrastructure *equity* coexist for institutional investors seeking a stake in the capital of a company which owns and operates an infrastructure asset. **Unlisted infrastructure** (*unlisted infrastructure equity / debt*) is direct access; it



requires substantial entry tickets (generally above 10 million euros for institutional funds), a lock-up of capital of 10 to 15 years and a quarterly valuation founded on *discounted cash flow* (DCF) models or on comparable multiples; its benchmark indices are the EDHEC 300 and MSCI Infrastructure series. **Listed infrastructure** for its part offers the liquidity of equity markets via operating companies (Vinci, Engie, Atlantia, Transurban) or index funds (*liquid proxies*) such as the *S&P Global Infrastructure Index*; it trades in real time but correlates more with equity markets, especially in periods of stress, reducing its diversification advantage, as BlackRock (2023) emphasises. This is why the **illiquidity premium** of unlisted infrastructure *equity* relative to listed infrastructure is estimated at roughly 150 basis points in the normalised rate environment of 2024–2025, that is a forecast entry IRR of 10.6 % gross per year for a diversified portfolio of unlisted infrastructure, according to CBRE Investment Management (2024).

**Infrastructure debt** constitutes a third route: the investor becomes a creditor and not a shareholder of the company operating the infrastructure, which spares it first-rank exposure to losses if operating flows deteriorate. Infrastructure debt rests on debt financing — in the form of syndicated loans, private placements or project bonds — of infrastructure assets generating stable and predictable cash flows (energy networks, transport, digital infrastructure, public-private partnerships), as Schroders (2025) explains. Unlike classic *corporate* bonds, whose downgrade risk is correlated with the credit quality of the issuer, infrastructure debt is backed by the cash flows of the underlying asset itself — often contractual or regulated — and not by the borrower's balance sheet, as Allianz Global Investors (2024) analyses. It comes in two main families: **senior debt** (*investment grade*), of long maturity (10 to 30 years) and **junior debt** (mezzanine or *sub-investment grade*), whose yield may rival that of infrastructure *equity* for a higher rank of repayment in the capital structure according to All News (2023). This class moreover benefits from favourable prudential treatment under the Solvency II directive: whereas standard *equity* bears a solvency capital requirement (SCR) of 49 %, *qualifying infrastructure equity investments*, benefiting from a derogatory treatment instituted in 2016 by the EIOPA (2016), see their SCR reduced to 36 %; qualifying infrastructure debt (*qualifying infrastructure corporate bonds*, QIICB) is indeed subject to the *spread* risk module, whose calibration is more favourable than that of classic *corporate* bonds of the same rating.

Moreover, the comparative analysis of realised returns between infrastructure and *private equity* reveals a structural inversion since the turn of the financial cycle. In 2024, unlisted infrastructure generated an IRR of 10.2 %, bringing the cumulative return over the last three years to +34.5 %; it thus establishes itself as the leading alternative asset class in terms of performance over this period according to the Preqin (2025) report. By contrast, *private equity buyout* is going through a phase of structurally compressed returns: according to McKinsey & Company (2026), the returns of the first quartile of world *buyout* funds reached only 8 % in 2025, that is less than half the return generated by the S&P 500 (18 %) and the MSCI World (22 %) over the same period. Over the period 2015–2025, the average of *buyout* funds stands at roughly 6 %, dragged down by the 2015–2017 *vintages*, whose IRRs no longer exceed 2 %, a reflection of the compression of valuation multiples, of the rise in interest rates and of the slowdown in disposals, as McKinsey & Company (2026) details. By contrast, over the long run (*vintages* 2009–2020), the median IRR of infrastructure stands at 9.8 %, relatively stable over time, whereas the long-run median of *private equity buyout* is roughly 14.1 % according to Cambridge Associates (2025); infrastructure is therefore structurally less remunerative than *buyout* in normal conditions, but offers two decisive advantages in a stress phase: **near-zero dispersion** and inflationary resilience. This comparison must, however, be conducted with caution: unlisted infrastructure indices rest on quarterly *fair value* valuations computed by the managers, and not on observed market prices; the research of the EDHEC (2025) has brought to light a tendency towards the **smoothing** of these valuations, which underestimates the real volatility of the asset class and artificially inflates its apparent Sharpe ratio.

#### Focus...



**The Grand Paris Express.** The Grand Paris Express (GPE) is a project for an automatic metro of 200 kilometres of new lines and 68 new stations, that is a near doubling of the existing Île-de-France network according to the Société des Grands Projets (2023). It is composed of four entirely automatic lines, of which the longest — **line 15** — will form a great orbital ring around Paris, linking the suburban employment zones to one another without transiting through the centre of the capital. The project is placed under the exclusive contracting authority of the **Société des Grands Projets** (SGP, formerly Société du Grand Paris), a public industrial and commercial establishment (EPIC) created by the law of 3 June 2010, held 100 % by the French state. According to Vie publique (2020), the parliamentary working group of the Senate relied on the work carried out by the Court of Auditors, which had revalued at 34.6 billion euros at the end of 2017 the financing cost of the GPE project against 22.6 billion euros in March 2013. The commissioning timetable has moreover been significantly pushed back, line 15 South (Pont de Sèvres to Noisy-Champs, 33 kilometres, 16 stations) no longer being expected before the beginning of 2027, after having missed the symbolic date of the 2024 Olympic Games; by a domino effect, lines 16 (25 kilometres, Paris

Roissy-CDG airport) and 17 will not be delivered before the second quarter of 2027; the complete line 15 (orbital, 75 kilometres in total) is now expected for 2031, according to the Société des Grands Projets (2025).

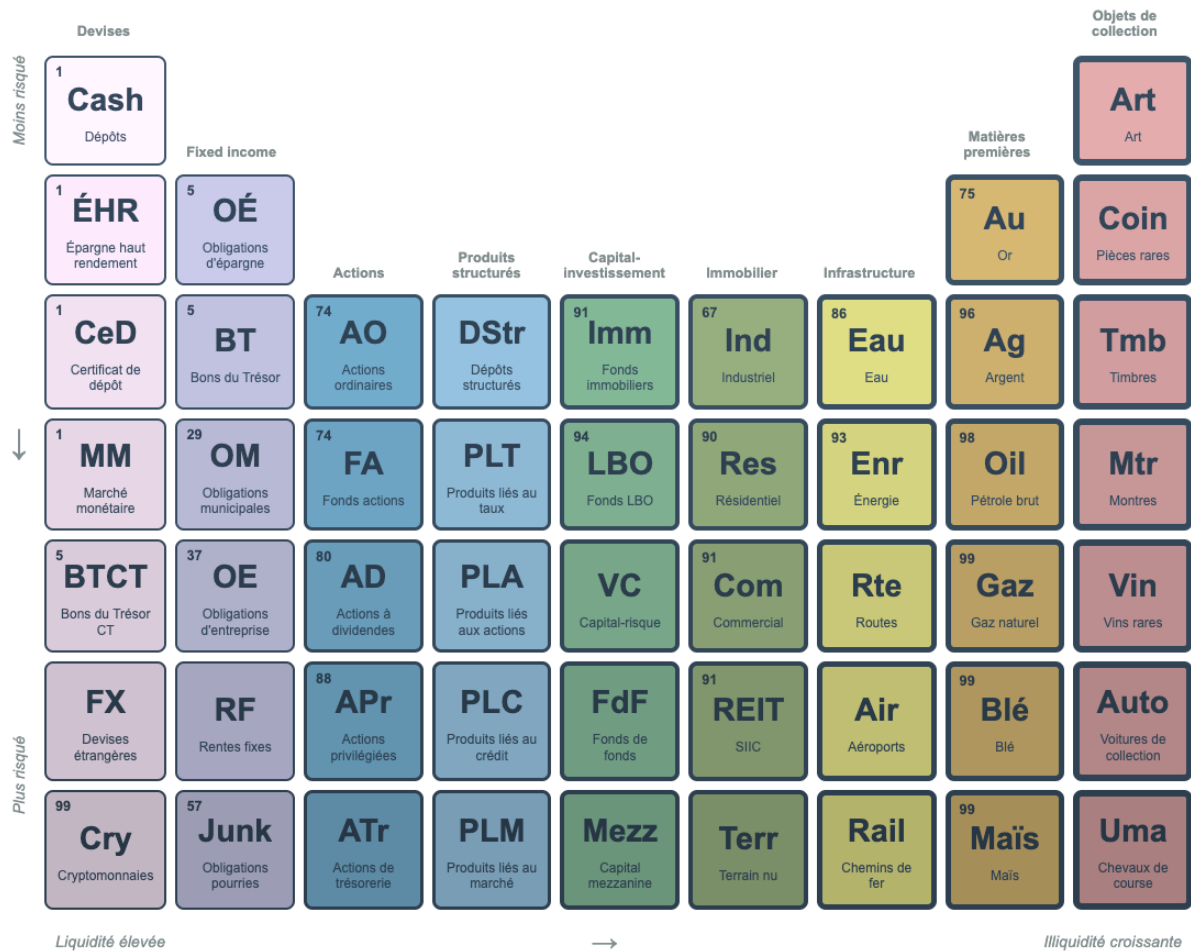
Unlike the motorways, the GPE does not rest on a private concession model: the Société des Grands Projets (SGP) is an EPIC held 100 % by the French state, rated AA/Aa3/A+ (Morningstar DBRS / Moody's / Fitch), with a credit quality aligned with that of the French Republic, as the Société des Grands Projets (2025) sets out. The project is financed by **long-term borrowing** raised on the bond markets (*debt capital markets*, a 100 % green EMTN programme) and from the European Investment Bank (contracts signed for 3.5 billion euros between 2015 and 2023); at 31 December 2024, 30.6 billion euros had already been raised on the markets, with an average maturity of 21.3 years and an average yield of 1.26 % per year, as the Société des Grands Projets (2025) details. The legal debt ceiling, subject to an annual report to Parliament, is fixed at 39 billion euros (to be reached at the end of the construction phase), for a total construction cost of the GPE of 36.1 billion euros (in constant 2012 euros) and 4.1 billion euros for the modernisation of the existing network, as the Société des Grands Projets (2025) explains. The debt is repaid over several generations (horizon 2070) thanks to five dedicated tax resources levied in Île-de-France, totalling 0.9 billion euros in 2024 (up 8.3 % on 2023) and estimated at 1.0 billion euros in 2025: the office tax (TSB, 718 million euros), the flat-rate tax on network enterprises (IFER, 84 million euros), the special equipment tax (TSE, 67 million euros), the additional regional tourist tax (TATS, 21 million euros) and the additional tax on parking areas (TASS, 14 million euros). These resources are earmarked by law: they can benefit only the GPE and cannot be allocated to any other use, as the Société des Grands Projets (2025) recalls. The SGP **is exposed neither to revenue risk nor to maintenance risk**: the management of the network will be entrusted to Île-de-France Mobilités (IDFM, the transport organising authority), which selects operators by competitive tender — RATP, ORA and Keolis have been designated for lines 14, 15 South, 16, 17 and 18 — and which bears the entirety of the rolling stock (rolling stock expenditure, 405 million in 2024, is entirely borne by IDFM and does not enter the balance sheet of the Société des Grands Projets (2025)).

For the institutional investor, the GPE illustrates the mechanism of **land value capture**: the accessibility value created by the infrastructure is reflected, with a lag, in the prices of neighbouring property assets. The study by the Grand Paris (2022) thus shows that, over the period 2008–2021 (the construction phase), neighbourhoods less than 800 metres from the future stations progressed by +57.7 % on average, against +60.8 % for neighbourhoods situated between 800 and 1500 metres. The future presence of stations therefore has no impact, for the moment at least, on the increase in property prices. However, when one considers not the neighbourhoods around the stations but the municipalities as a whole, it appears that municipalities entirely restructured by the GPE — Saint-Ouen, Bois-Colombes, Rosny-sous-Bois, Saint-Denis — have experienced unprecedented rises in both existing and new-build property, suggesting that the valuation operates at the municipal scale before concentrating around the stations as commissioning approaches, as the study by the Grand Paris (2022) supposes.

### Key Takeaways



Infrastructure assets designate the physical equipment essential to the functioning of the economy — energy networks, transport, utilities, social infrastructure — whose common feature is to generate stable income flows over 20 to 40 years, backed by concession contracts, regulated tariffs or *take-or-pay* clauses [Melchior (2021), Légifrance (2025), Ruester (2014)]. Their attraction for institutional investors rests on three differentiating properties: a natural monopoly position conferring resistance to cyclicity (inelasticity of demand for water, electricity or public transport), contractual indexation to inflation (transport assets displayed quarterly returns more than one percentage point higher in periods of high inflation), and low correlation with listed markets, improving the efficient frontier of a diversified portfolio [BlackRock (2023), Medda & Panayiotou (2014), Chen & Shen (2025)]. The risk spectrum is ordered into four segments: *core* (mature regulated assets), *core+* (the dominant sector), *value-add* (operational transformation) and *greenfield* (construction risk) [Infrastructure Investor (2025), InfraRed Capital Partners (2025), Cambridge Associates (2025)]. In general, infrastructure outperforms *buyout* in a stress phase by its near-zero dispersion and its inflationary resilience [McKinsey & Company (2026), Cambridge Associates (2025)], subject to the smoothing bias of quarterly valuations brought to light by the EDHEC (2025).



#### À propos de ce tableau

Ce tableau périodique organise les principales classes d'actifs financiers selon leur catégorie et leur niveau de risque. Chaque cellule contient un numéro dans le coin supérieur gauche qui indique le niveau de risque : plus ce numéro est élevé, plus l'actif est considéré comme risqué. Les actifs sont organisés de gauche à droite par liquidité décroissante. Les bordures deviennent progressivement plus épaisses vers la droite pour symboliser l'illiquidité croissante. De haut en bas, le risque augmente, visualisé par un dégradé de couleur qui devient plus foncé à mesure que l'on descend dans le tableau.

Figure 3.6: Periodic table of financial assets after Kador & Kowal (2020)

# Derivatives and the actors of finance

As Jégourel (2017) indicates, a **derivative** is a financial instrument whose value depends on that of another asset, called the **underlying asset**, which may be a share, a bond, a commodity, or again a stock market index. Derivatives are used to hedge against the risks of price fluctuation or to speculate on those variations; they thus play an important role in modern financial markets, allowing investors to manage risk and to exploit the opportunities offered by market movements. Otherwise put, with derivatives, one no longer buys an underlying asset but an **exposure to a market**.

## 1 Options

With optional derivatives, the contracting parties (buyer and seller) have the possibility during the life of the contract of not executing certain clauses of the corresponding contract (whence the name option) if its underlying asset or assets do not fulfil certain conditions.

### 1.1 A great variety of optional derivatives

More precisely, as Hull (2012) indicates, an **option** is a security conferring the right (and not the obligation) to buy or to sell an underlying asset, at a price agreed in advance (the **strike**  $K$ , fixed at date  $t = 0$ ), during or at the expiry of a period defined beforehand  $t = T$ : the buyer pays the seller of the option a **premium**  $\pi$  at the initial date, and receives in return, at a future date, a positive or nil flow (**payoff**) whose amount depends on the evolution of an underlying asset. The option is said to be **standard** if the underlying asset delivers no cash flow before expiry. For any date  $t$ , we denote by  $S_t$  the value at instant  $t$  of the underlying asset.

#### 1.1.1 European call and put

Among options, the most classic are **vanilla options**. A vanilla option is said to be *European* when it can be exercised only at the maturity date  $t = T$ . Two categories of option may then be distinguished, as Hull (2012) specifies:

- **European purchase options, or calls**, confer the right (and not the obligation) to buy an underlying asset at the exercise price  $K$  fixed at issue  $t = 0$ . The purchase option is therefore of interest only if  $S_T > K$  (the *call* is then said to be **in the money**): in that case, the position of the holder of the option will be  $S_T - K$ ; otherwise, the holder of the *call* will abandon the option and his position will be nil (the *call* is then said to be **out of the money** if  $S_T < K$  and **at the money** if  $S_T = K$ ). Thus, the **intrinsic value** of a European *call* is  $\Psi_T^C = \max(S_T - K, 0)$  (one must indeed not forget to subtract the premium paid in order to know how much the investor gains *in fine*). From then on, for a *call*, the *payoff* is nil as long as  $S_T < K$ , then grows with the value of the underlying asset (see figure 4.1).
- **European sale options, or puts**, confer the right (and not the obligation) to sell an underlying asset at the exercise price  $K$  fixed at issue  $t = 0$ . The sale option is therefore of interest only if  $S_T < K$  (the *put* is then said to be **in the money**): in that case, the position of the holder of the option will be  $K - S_T$ ; otherwise, the holder of the *put* will abandon the option and his position will be nil (the *put* is then said to be **out of the money** if  $S_T > K$  and **at the money** if  $S_T = K$ ). Thus, the **intrinsic value** of a European *put* is  $\Psi_T^P = \max(K - S_T, 0)$  (one must not forget to subtract the premium paid in order to know how much the investor gains *in fine*). From then on, for a *put*, the *payoff* decreases as long as  $S_T < K$ , until it reaches  $K$  where it then becomes nil (see figure 4.3).

It should be noted that, unlike the buyer of the option, the seller finds himself in a position where he is subject to the decisions of the buyer, who has the freedom to exercise the option or not. This situation therefore differs from bilateral contracts such as futures or *forwards*, where both parties are bound to respect the terms of the contract

symmetrically. Within the framework of an option, the seller must comply with the buyer's decision whether or not to exercise his option, which may entail potentially unlimited losses for the seller if market conditions evolve unfavourably. Indeed, for a European *call*, the seller's loss is given by  $\pi - \Psi_T^C$ , that is to say  $\pi - (S_T - K)$  when  $S_T > K$ : the more the price of the underlying  $S_T$  increases, the greater the potential loss for the seller; by contrast, if  $S_T \leq K$ , the seller realises a net gain  $P$ , corresponding to the premium received (see figure 4.2). Likewise, for a European *put*, the seller's loss is given by  $\pi - \Psi_T^P$ , that is to say  $\pi - (K - S_T)$  when  $S_T < K$ : the more the price of the underlying  $S_T$  falls, the greater the potential loss for the seller; by contrast, if  $S_T \geq K$ , the seller realises a net gain  $P$ , corresponding to the premium received (see figure 4.4). Consequently, it is essential for the seller to put in place **hedging strategies** in order to limit his financial risks. Conversely, the buyer of the option benefits from a loss limited to the amount of the premium paid, which in itself constitutes a form of protection against excessive risks.

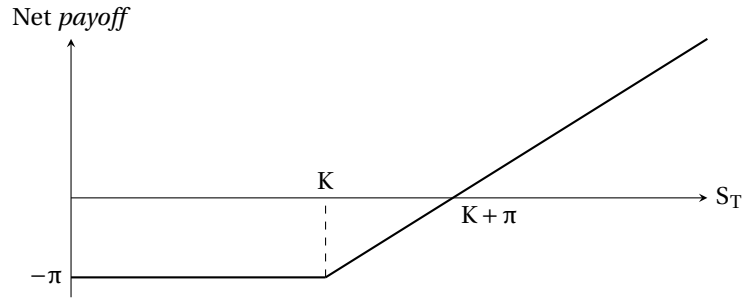


Figure 4.1: Net *payoff*  $\max(S_T - K, 0) - \pi$  at expiry  $t = T$  of a European *call* of price  $\pi$  and *strike*  $K$  as a function of the value of the underlying  $S_T$  at date  $t = T$  (buyer's side: *long call*)

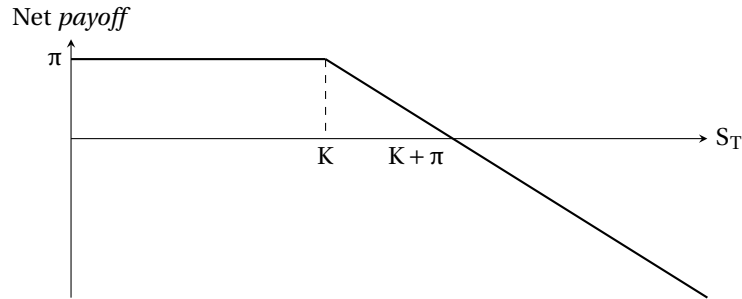


Figure 4.2: Net *payoff*  $\pi - \max(S_T - K, 0)$  at expiry  $t = T$  of a European *call* of price  $\pi$  and *strike*  $K$  as a function of the value of the underlying  $S_T$  at date  $t = T$  (seller's side: *short call*)

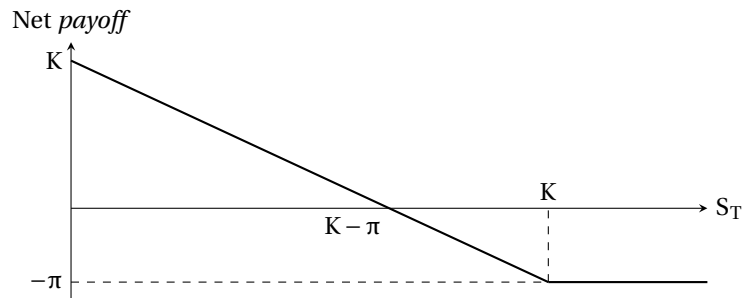


Figure 4.3: Net *payoff*  $\max(K - S_T, 0) - \pi$  at expiry  $t = T$  of a European *put* of price  $\pi$  and *strike*  $K$  as a function of the value of the underlying  $S_T$  at date  $t = T$  (buyer's side: *long put*)

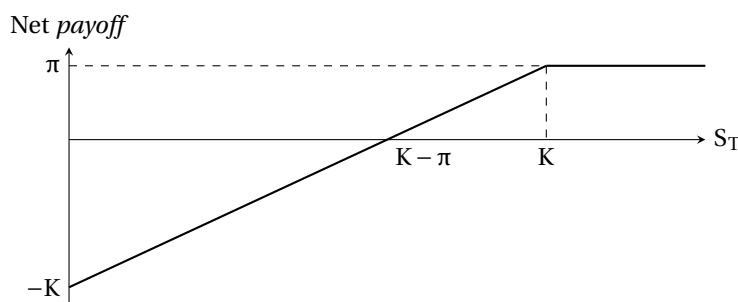


Figure 4.4: Net *payoff*  $\pi - \max(K - S_T, 0)$  at expiry  $t = T$  of a European *put* of price  $\pi$  and *strike*  $K$  as a function of the value of the underlying  $S_T$  at date  $t = T$  (seller's side: *short put*)

### Key Takeaways



A European option is a security conferring the right (and not the obligation) to buy or to sell an asset at a price agreed in advance (the *strike*  $K$ , fixed at date  $t = 0$ ) at the expiry of a period defined beforehand  $t = T$ : the buyer pays the seller of the option a *premium* at the initial date, and receives in return, at a future date  $t = T$ , a positive or nil flow (*payoff*) whose amount depends on the evolution of an underlying asset. Two categories of option are distinguished:

- **Purchase options, or calls**, confer the right (and not the obligation) to buy an underlying asset at the exercise price  $K$  fixed at issue  $t = 0$ . The purchase option is therefore of interest only if  $S_T > K$  (*call* in the money): in that case, the position of the holder of the option will be  $S_T - K$ ; otherwise, the holder of the *call* will abandon the option and his position will be nil (*call* out of the money if  $S_T < K$  and at the money if  $S_T = K$ ). Thus, the intrinsic value of a European *call* is  $\Psi_T^C = \max(S_T - K, 0)$ . From then on, for a *call*, the *payoff* is nil as long as  $S_T < K$ , then grows with the value of the underlying asset.
- **Sale options, or puts**, confer the right (and not the obligation) to sell an underlying asset at the exercise price  $K$  fixed at issue  $t = 0$ . The sale option is therefore of interest only if  $S_T < K$  (*put* in the money): in that case, the position of the holder of the option will be  $K - S_T$ ; otherwise, the holder of the *put* will abandon the option and his position will be nil (*put* out of the money if  $S_T > K$  and at the money if  $S_T = K$ ). Thus, the intrinsic value of a European *put* is  $\Psi_T^P = \max(K - S_T, 0)$ . From then on, for a *put*, the *payoff* decreases as long as  $S_T < K$ , until it reaches  $K$  where it then becomes nil.

### 1.1.2 American options and barrier options

An **American option**, for its part, may be exercised at any instant  $t$  of the interval  $]0, T]$ . Adopting the same reasoning as previously, the intrinsic value of an American *call*, which confers the right (and not the obligation) to buy an underlying asset at the exercise price  $K$  fixed at issue  $t = 0$  at any date  $t$  of the interval  $]0, T]$ , may then be defined, for any date  $t$  of the interval  $]0, T]$ , by  $\Psi_t^{CUSA} = \max(S_t - K, 0)$  (one must indeed not forget to subtract the premium paid in order to know how much the investor gains *in fine*). Likewise, the intrinsic value of an American *put*, which confers the right (and not the obligation) to sell an underlying asset at the exercise price  $K$  fixed at issue  $t = 0$  at any date  $t$  of the interval  $]0, T]$ , may then be defined, for any date  $t$  of the interval  $]0, T]$ , by  $\Psi_t^{PUSA} = \max(K - S_t, 0)$ . The terms ‘at the money’, ‘out of the money’ and ‘in the money’ likewise apply analogously. The price  $\pi$  (the premium) of an American option, as Hull (2012) explains, is generally higher than for a European option on the same underlying, with the same *strike* and the same maturity, since it offers its buyer greater freedom of action. While we shall determine the price of European options in this course, the valuation of the other types of option will remain outside our scope.

### Exercise



**Binary options.** Consider a European digital option (or binary option) on the S&P 500 (an index assumed to be quoted at 6600 today) with a *strike* of 6600 and a maturity of one month. Otherwise put, the acquirer of the option will gain 1 if the S&P is quoted above 6600 in one month, 0 otherwise. The trend of the S&P 500 is assumed to be nil, as is the interest rate.

- What would in your view be the fair price of this European option?
- Same question for a *strike* of 6800. It may be assumed that the monthly volatility of the S&P is 100 points.

Consider now an American digital option on the S&P 500 (an index assumed to be quoted at 6600 today) with a *strike* of 6800 and a maturity of one month. Otherwise put, the acquirer of the option will gain 1 if the S&P is quoted above 6800 at least once in the coming month, 0 otherwise.

- (i) Compare the price of the American option and that of the European option.
- (ii) Give an approximation of the ratio of the price of the American digital option with a *strike* of 6800 to that of the European digital option with the same *strike*.

**Solution.** Consider a European digital option on the S&P 500 (an index assumed to be quoted at 6600 today) with a *strike* of 6600 and a maturity of one month.

- (i) We may model the situation by a Bernoulli trial whose success (which pays precisely 1) is that the index is quoted at at least 6600 in one month (failure being that the S&P is quoted below 6600 in one month, which pays 0). From then on, we are prepared to pay the discounted expected gain of this financial product. But since the interest rate is assumed to be nil, the discounted expected gain is quite simply the expected gain of the product, namely  $\mathbb{P}([S_1 > 6600] | [S_0 = 6600])$ , where  $S_1$  is the price of the S&P in one month and  $S_0$  its current price. Since the price of the index today is precisely 6600 and since the *drift* is assumed to be nil, there is *a priori* as much chance that the S&P will close below 6600 as above 6600 in one month. So  $\mathbb{P}([S_1 > 6600] | [S_0 = 6600]) = \frac{1}{2}$ . We are therefore prepared to pay 50 cents for this option (this is the expected gain of the option, which therefore pays 1 euro in 50 % of cases and 0 otherwise).
- (ii) We may still model the situation by a Bernoulli trial whose success (which pays precisely 1) is that the index is quoted at at least 6800 in one month (failure being that the S&P is quoted below 6800 in one month, which pays 0). From then on, we are prepared to pay the discounted expected gain of this financial product. But since the interest rate is assumed to be nil, the discounted expected gain is quite simply the expected gain of the product, namely  $\mathbb{P}([S_1 > 6800] | [S_0 = 6600])$ , where  $S_1$  is the price of the S&P in one month and  $S_0$  its current price. For simplicity we assume that the price of the S&P follows a normal distribution, with expectation 6600 (the current value given that the *drift* is nil) and volatility 100. We then have:

$$\begin{aligned}\mathbb{P}([S_1 > 6800] | [S_0 = 6600]) &= \mathbb{P}\left(\left[\frac{S_1 - 6600}{100} > \frac{6800 - 6600}{100}\right] \middle| [S_0 = 6600]\right) \\ &= \mathbb{P}\left(\left[\frac{S_1 - 6600}{100} > 2\right] \middle| [S_0 = 6600]\right) \\ &= \mathbb{P}(\mathcal{N}(0, 1) > 2) \approx 2\%\end{aligned}$$

We are therefore prepared to pay 2 cents for this option (this is the expected gain of the option, which therefore pays 1 euro in 2 % of cases and 0 otherwise).

Consider now an American digital option on the S&P 500 (an index assumed to be quoted at 6600 today) with a *strike* of 6800 and a maturity of one month.

- (i) There is much more chance that the S&P will be quoted at least once above 6800 during the month than there is that the S&P will be quoted above 6800 at the end of the month exactly. The American option is therefore structurally more expensive than the European option, since it has more chance of delivering its *payoff*.
- (ii) Let us conduct the thought experiment that during the month the S&P reaches 6800. Then the American option will deliver 1 euro for certain: at that instant, the American option is therefore worth 1 euro. What of the European option with a strike of 6800? At that precise instant, the European option is at the money: by reasoning analogous to question 1 of the first part, we find that the European option is worth 50 cents (there is as much chance that the price will close above 6800 as below, given that the absence of *drift* is assumed). Otherwise put, the American option is worth roughly twice as much as the European option: roughly 4 cents.  $\square$

Other types of option, said to be **exotic**, exist and are studied notably by Hull (2012). For example, the **barrier option** is exercised at expiry only, and the terminal payment is that of a European option (*call* or *put*) conditional on passing above or below a barrier  $B$ . In particular, the intrinsic values of barrier *calls*, from which one will not forget to subtract the premium paid in order to know how much the investor gains *in fine*, are then determined as follows:



- **Up-and-out call:**  $\Psi_T^{\text{UOC}} = \max(S_T - K, 0) \times \mathbb{1}_{\max_{t \leq T} S_t \leq B}$ : the *call* pays only if the price of the underlying has never exceeded  $B$  between its purchase date and the maturity date of the *call*. Buying a UOC is appropriate when one anticipates a moderate rise in the underlying, without a surge.
- **Down-and-out call:**  $\Psi_T^{\text{DOC}} = \max(S_T - K, 0) \times \mathbb{1}_{\min_{t \leq T} S_t \geq B}$ : the *call* pays only if the price of the underlying has always exceeded  $B$  between its purchase date and the maturity date of the *call*. Buying a DOC is appropriate when one is bullish and rules out any pronounced retreat.
- **Up-and-in call:**  $\Psi_T^{\text{UIC}} = \max(S_T - K, 0) \times \mathbb{1}_{\max_{t \leq T} S_t \geq B}$ : the *call* pays only if the price of the underlying has at least once exceeded  $B$  between its purchase date and the maturity date of the *call*. Buying a UIC is appropriate when one anticipates a strong and durable upward movement, capable of crossing the threshold  $B > K$ .
- **Down-and-in call:**  $\Psi_T^{\text{DIC}} = \max(S_T - K, 0) \times \mathbb{1}_{\min_{t \leq T} S_t \leq B}$ : the *call* pays only if the price of the underlying has at least once fallen below  $B$  between its purchase date and the maturity date of the *call*. The profile is paradoxical: the investor hopes for a trough followed by a vigorous rebound beyond  $K$  (a ‘*buy the dip*’ strategy).
- **Up-and-out put:**  $\Psi_T^{\text{UOP}} = \max(K - S_T, 0) \times \mathbb{1}_{\max_{t \leq T} S_t \leq B}$ : the *put* pays only if the price of the underlying has never exceeded  $B$  between its purchase date and the maturity date of the *put*. Buying a UOP is appropriate when one seeks cover against a moderate fall, without fearing a prior strong rise.
- **Down-and-out put:**  $\Psi_T^{\text{DOP}} = \max(K - S_T, 0) \times \mathbb{1}_{\min_{t \leq T} S_t \geq B}$ : the *put* pays only if the price of the underlying has always exceeded  $B$  between its purchase date and the maturity date of the *put*. Buying a DOP is appropriate in order to hedge against moderate falls while accepting not to be protected in the event of a crash.
- **Up-and-in put:**  $\Psi_T^{\text{UIP}} = \max(K - S_T, 0) \times \mathbb{1}_{\max_{t \leq T} S_t \geq B}$ : the *put* pays only if the price of the underlying has at least once exceeded  $B$  between its purchase date and the maturity date of the *put*. Buying a UIP is appropriate in a ‘peak then crash’ scenario: the operator anticipates an overheating of the market (beyond  $B$ ) followed by a reversal below  $K$ .
- **Down-and-in put:**  $\Psi_T^{\text{DIP}} = \max(K - S_T, 0) \times \mathbb{1}_{\min_{t \leq T} S_t \leq B}$ : the *put* pays only if the price of the underlying has at least once fallen below  $B$  between its purchase date and the maturity date of the *put*. The purchase of a DIP is appropriate when one anticipates a severe fall in the market.

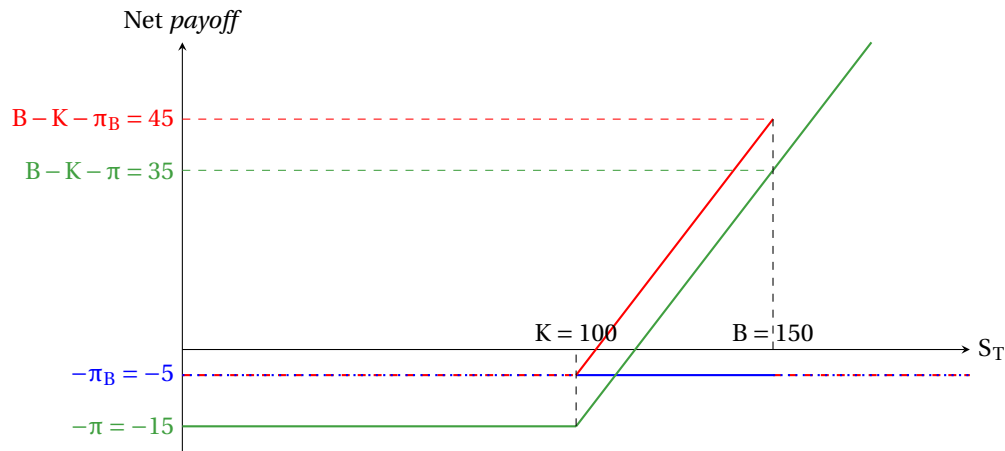
Offering weaker *payoffs* than a European option on account of the barrier, barrier options are therefore cheaper than their European counterparts on the same underlying, with the same *strike* and the same maturity.

### Exercise



**Barrier option.** Consider an *up-and-out call* of maturity  $T = 1$  year, *strike*  $K = 100$ , barrier  $B = 150$ , on the Amazonia share, whose current spot price is  $S_0 = 105$ . This option is sold for  $\pi_B = 5$  dollars, against  $\pi = 15$  dollars for a European *call* of the same *strike* and the same maturity (but without a barrier, therefore). You wish to bet on a reasonable rise in the Amazonia share; you do not, however, believe in a rise of more than 50 % over one year. You therefore decide to buy this *up-and-out call*. Recall on what conditions the *up-and-out call* will generate a payment for the buyer. Then justify the intrinsic value and the *payoff* of the *up-and-out call* and of the European *call* as a function of these conditions. Finally, plot on a single graph the *payoff* at maturity of the *up-and-out call* if the barrier is crossed, the *payoff* at maturity of the *up-and-out call* if it is not, and the *payoff* at maturity of the European *call*.

**Solution.** Consider an *up-and-out call* of maturity  $T = 1$  year, *strike*  $K = 100$ , barrier  $B = 150$ , on the Amazonia share, whose current spot price is  $S_0 = 105$ : this means that this product will pay only if the price of the Amazonia share remains below  $B = 150$  during the year following the acquisition of the *up-and-out call*. From then on, the intrinsic value of the *up-and-out call* will be  $\Psi_T^{\text{UOC}} = 0$  if the price of Amazonia exceeds  $B = 150$  at least once during the year; the *payoff* of the *up-and-out call* will stand at  $\Psi_T^{\text{UOC}} - \pi_B = -5$  (an outright loss). By contrast, if the price of the Amazonia share remains below  $B = 150$  during the year following the acquisition of the *up-and-out call*, then the intrinsic value of the latter will be  $\Psi_T^{\text{UOC}} = \max(S_T - K, 0)$ , that is  $\Psi_T^{\text{UOC}} = S_1 - 100$  (whence a *payoff* of  $\Psi_T^{\text{UOC}} - \pi_B = S_1 - 105$ ) if the spot price at expiry exceeds  $K = 100$ , and  $\Psi_T^{\text{UOC}} = 0$  (whence a *payoff* of  $\Psi_T^{\text{UOC}} - \pi_B = -5$ ). Subsequently, the European *call* will have an intrinsic value at maturity of  $\Psi_T^{\text{C}} = \max(S_T - K, 0)$ , that is  $\Psi_T^{\text{C}} = S_1 - 100$  if the spot price at expiry exceeds  $K = 100$  (whence a *payoff* of  $\Psi_T^{\text{C}} - \pi = S_1 - 115$ ) and  $\Psi_T^{\text{C}} = 0$  if the spot price at expiry does not exceed  $K = 100$  (whence a *payoff* of  $\Psi_T^{\text{C}} - \pi = -15$ ). We then deduce the following graph with, in green, the *payoff* of the European *call*, in blue, that of the *up-and-out call* when the barrier is crossed and, in red, that of the *up-and-out call* when it is not:



□

### 1.1.3 A few words on the options market in France

The market in options on French shares is marked by a strong **concentration of trading**, distributed mainly between two large platforms: the **MONEP** (*marché des options négociables de Paris*, the Paris traded options market, created in 1987), which forms part of the Euronext group, and **Eurex** (created in 1998), which belongs to the Deutsche Börse group. In 2019, according to Le Moign & Ezra (2021), 49 % of volumes were traded on the MONEP, while 51 % were traded on Eurex, a distribution stable relative to 2018. In terms of transaction mode, a significant share of trading takes place outside the order book<sup>1</sup>: on the MONEP, 67 % of volumes were traded in the form of blocks, with an average size of 1794 contracts, while on Eurex, 55 % of volumes took the form of blocks of 1454 contracts on average. These **exchanges outside the order book** reflect significant activity by institutional investors. El Karoui (2003) specifies that the options listed on the Paris market are American, but that over-the-counter options are often European. Despite the wide range of instruments available (several tens of thousands), only 2 % of the instruments offered each day on Eurex and 2.4 % on Euronext are actually traded, revealing a concentration of volumes on a small number of options, often those with short maturity and whose exercise price is close to the price of the underlying. In addition, on Eurex, 29 % of the volumes traded in 2019 concerned options with a maturity of less than one month, while 17 % of the volumes of options traded had an exercise price between 99 % and 101 % of the closing price of the underlying.

#### Focus...



**What is an option of French jurisdiction?** As Le Moign & Ezra (2021) emphasise, the notion of French nationality relates to jurisdiction as defined by Regulation (EU) 600/2014 on **markets in financial instruments** (MiFIR) as regards transaction reporting: an issuer is regarded as having '**French jurisdiction**' when the most liquid market for its shares is situated in France. This means that the Autorité des marchés financiers (AMF) is the competent authority for its supervision and receives all the transaction reports required by MiFIR. Consequently, options on shares linked to these issuers likewise fall under French supervision.

The market in options of French jurisdiction is likewise characterised by a notable concentration of trading on a limited number of underlyings. Le Moign & Ezra (2021) specify that in 2019, on Eurex, **the ten leading underlyings represented 73 % of the volumes traded**, against 63 % on Euronext. The most traded options relate to major issuers such as Total, LVMH, Axa, Société Générale and BNP Paribas. This concentration is explained by the high liquidity of these securities as well as by their importance on the market. On average, 1689 option contracts were traded per day on Eurex, against 102.4 million shares for their underlyings on Euronext. Moreover, the amount traded in notional equivalent on options of French jurisdiction in 2019 represented roughly 20.6 % of the amounts traded on their underlying shares on Euronext. Le Moign & Ezra (2021) add that this distribution shows that, although the volume of

<sup>1</sup>Otherwise put, many exchanges of option contracts do not take place directly via the order book of the market platforms, which is the place where buyers and sellers publish their intentions to buy or sell transparently. Instead, these exchanges are carried out in the form of bilateral or 'over-the-counter' transactions between market participants, often in large quantities, called 'blocks'. These transactions are generally negotiated directly between the parties, often institutional investors, before being registered on the market platform. This type of exchange makes it possible to execute large orders without this influencing too strongly the prices visible on the public market, while meeting the specific needs of participants in terms of the size or the conditions of the transaction.

options is low relative to shares, they play a crucial role in investors' strategy, notably for issuers with eventful news, such as Casino and Carrefour, where options represented a significant share of trading.

### Focus...



**The notional equivalent of an option.** As Hull (2012) recalls, for each option, the **notional equivalent** is computed by multiplying the exercise price of the option by the number of units of the underlying covered by the option (often a hundred shares for an option on a share). This amount is called 'notional' because it is a theoretical value, which does not necessarily involve an actual exchange of that amount of shares, unless the option is exercised. For example, if an option contract covers a hundred shares of a company with an exercise price of fifty euros, the notional equivalent of this contract would be  $100 \times 50 = 5000$  euros. The amount traded in notional equivalent on a market thus groups together the total theoretical value of the transactions carried out on that market, offering a measure of the potential exposure and of the *trading* activity in options, even if not all these options will be exercised.

The analysis of the characteristics of the options traded on French markets finally shows a strong **preference among investors for options with short maturity** and with a minimal gap between the exercise price and the price of the underlying. In 2019, on Eurex, 5 % of the volumes traded concerned options with a maturity of less than five days, while 29 % of the volumes involved options of less than one month's maturity. On Euronext, these proportions were even higher, with 7 % of the volumes traded for options of less than five days' maturity and 38 % for those of less than one month. These figures, transcribed by Le Moign & Ezra (2021), reflect a marked preference among investors for short-term options, where visibility on the price and volatility of the underlying is better. Moreover, 67 % of the instruments traded on Eurex in 2019 were dealt in fewer than ten times during the year, and 80 % of them were traded on fewer than ten days, confirming the **rapid turnover** of instruments on this market. This dynamic is likewise observable on Euronext, where 66 % of the instruments were traded fewer than ten times and 80 % on fewer than ten days. These trends show that investors favour options with a high sensitivity to movements of the underlying, notably those whose exercise price is close to the closing price of the share, which is illustrated by the fact that 17 % of the volumes traded on Eurex in 2019 concerned options whose exercise price lay between 99 % and 101 % of the closing price of the underlying. These strategic choices show that the options market in France is strongly influenced by the volatility of the underlying, the maturity of the options and investors' appetite for instruments allowing them to capitalise on rapid market movements.

### Key Takeaways



The market in options of French jurisdiction — in the sense of the MiFIR regulation, which attributes supervision to the AMF as soon as an issuer's most liquid market is situated in France — is concentrated on two platforms, Euronext (MONEP) and Eurex, which shared volumes almost equally in 2019; in both cases, a majority of trading takes place outside the order book in the form of blocks, reflecting the predominance of institutional investors dealing in large sizes without exposing their intention to the public market [Le Moign & Ezra (2021)]. The structural preferences of investors reveal a strong attraction to options with short maturity and at the money, reflecting a strategy of capitalising on rapid movements of the underlying rather than long-term risk management [Le Moign & Ezra (2021)].

## 1.2 Determining the price of a European option

The price (the premium) of an option is equal to the sum of the intrinsic value of the option and a **time value**: indeed, the more distant the expiry of the option, the more this time value increases, since the higher is the probability that prices will vary in the direction favourable to the buyer. This probability determines the risk taken by the seller of the option, which the latter passes on in the premium he demands. Other parameters influence the price of an option, such as the market interest rate and volatility. The objective of the lines which follow will be to determine the price of European *calls* and *puts*.

### 1.2.1 Absence of arbitrage opportunity (AOA)

The assumption of the absence of arbitrage opportunity (**AOA**) is a fundamental assumption in finance, which underpins numerous models for valuing financial assets. As El Karoui (2003) recalls, this assumption stipulates that in an **efficient financial market**, there exists no possibility of realising a riskless profit from an initially nil outlay and without any probability of loss. An arbitrage is thus described as a '*free lunch*', inasmuch as an investor can obtain a gain without risk and without any initial investment. The absence of arbitrage opportunity then

guarantees that the prices of financial assets are coherent and do not permit such opportunities. Formally, we say that there is an arbitrage opportunity at time horizon  $T$  on a financial market if there exists an asset (or a portfolio of assets) with an initial price  $X_0 = 0$  at instant  $t = 0$  and a future value  $X_T$  at instant  $T$  satisfying  $X_T \geq 0$  almost surely, and  $\mathbb{P}([X_T > 0]) > 0$ : this means that the investor disburses nothing initially ( $X_0 = 0$ ), that the future value of the portfolio cannot be negative ( $X_T \geq 0$ ), and that there is a non-zero probability of gain ( $\mathbb{P}([X_T > 0]) > 0$ ). The assumption of AOA is then crucial in order to ensure the coherence of prices on financial markets: it guarantees that the prices of financial assets are established in such a way as not to permit the realisation of riskless profits without initial investment. Thus, AOA requires that for any portfolio generating a non-negative future gain with a non-zero probability of positive gain, there must exist a positive initial cost for that portfolio. If this is not the case, the market would permit arbitrage, which would lead to incoherent prices and to the inefficiency of the market. Thus, consider two financial assets with the same maturity  $T$  and future payments  $X_T$  and  $Y_T$ . Let  $X_0$  and  $Y_0$  be their respective prices at date  $t = 0$ . If  $X_T \geq Y_T$  in all states of the world, then the inequality  $X_0 \geq Y_0$  must necessarily follow. Indeed, suppose for the sake of contradiction that  $X_0 < Y_0$ . In that case, an investor could buy the product  $X_T$  and simultaneously sell the product  $Y_T$ , thus realising an immediate riskless profit of  $Y_0 - X_0$  at  $t = 0$ . Moreover, since  $X_T \geq Y_T$  holds in all states of the world, the investor would likewise receive  $X_T - Y_T \geq 0$  at expiry  $T$ : this would constitute an arbitrage opportunity, which is contrary to our assumption of AOA; consequently, the inequality  $X_0 \geq Y_0$  must hold.

The assumption of the absence of arbitrage opportunity makes it possible to determine the relation which exists between the price of a European *call* and that of a European *put* on the same underlying, with the same maturity and the same *strike*. Let  $B_t$  be the price at date  $t$  of the zero-coupon bond of maturity  $T$ , therefore an asset paying 1 at expiry  $T$ . Let us then show that the respective prices  $C_t$  and  $P_t$  of a European *call* and *put* of maturity  $T$  and *strike*  $K$  satisfy at every date  $t$  the equality  $P_t = C_t - S_t + KB_t$ . In order to demonstrate this relation, let us begin by recalling the definition of the values of these options at expiry  $T$ : the *payoff* of a *call* at expiry  $T$  is given by  $\max(S_T - K, 0)$ , while that of a *put* at expiry satisfies  $\max(K - S_T, 0)$ . Now  $\max(K - S_T, 0) = \max(S_T - K, 0) - S_T + K$ : indeed, if  $S_T > K$ , then  $\max(S_T - K, 0) = S_T - K$  and  $\max(K - S_T, 0) = 0$ ; and if  $S_T \leq K$ , then  $\max(S_T - K, 0) = 0$  and  $\max(K - S_T, 0) = K - S_T$ . Now, let us consider a portfolio constructed as follows:

- A long position in a European *put* with *strike*  $K$  and maturity  $T$ : acquisition cost  $P_t$ , *payoff* at expiry  $\max(K - S_T, 0)$ .
- A short position in a European *call* with the same *strike*  $K$  and maturity  $T$ : receipt obtained  $C_t$ , *payoff* at expiry  $-\max(S_T - K, 0)$ .
- A long position in the underlying share: acquisition cost  $S_t$ , *payoff* at expiry  $S_T$ .

At the acquisition date  $t$ , this portfolio therefore yields  $-P_t - S_t + C_t$ , while at date  $T$  it will offer a payment of  $\max(K - S_T, 0) + S_T - \max(S_T - K, 0)$ , that is to say  $\max(S_T - K, 0) - S_T + K - \max(S_T - K, 0) + S_T = K$  by virtue of the relation demonstrated *supra*. Now the acquisition of  $K$  zero-coupon bonds of maturity  $T$  yields  $-KB_t$  on acquisition (purchase cost) and generates  $K$  at date  $T$ . Otherwise put, the bonds yield the same as the portfolio thus constructed. The assumption of AOA then requires equality between the acquisition cost of the portfolio,  $P_t + S_t - C_t$  and that of the bonds,  $KB_t$ . Indeed:

- If  $P_t + S_t - C_t > KB_t$  (the bonds are cheaper to acquire than the portfolio), then the acquisition of  $K$  bonds and the short sale of the portfolio costs  $KB_t - (P_t + S_t - C_t) < 0$  (therefore yields money) at purchase and is neutral at maturity (a payment of  $-K + K = 0$ : one buys the portfolio back at maturity, which costs  $K$ , and one recovers  $K$  at maturity on settling the bonds).
- If  $P_t + S_t - C_t < KB_t$  (the bonds are more expensive to acquire than the portfolio), then the short sale of  $K$  bonds and the acquisition of the portfolio costs  $(P_t + S_t - C_t) - KB_t < 0$  (therefore yields money) at purchase and is neutral at maturity (a payment of  $K - K = 0$ : one sells the portfolio back at maturity, which yields  $K$ , and one buys the bonds back at maturity, which costs  $K$ ).

In both of the aforementioned cases, there is arbitrage: the only way of satisfying the assumption of AOA is therefore indeed to satisfy **put-call parity**  $P_t + S_t - C_t = KB_t$ , that is to say  $P_t = C_t - S_t + KB_t$  at every date  $t$ . **Careful!** It is important to note that this relation of put-call parity does not hold for American options, since these options may be exercised at any moment before expiry, which introduces differences in the value of the options which are not captured by this simple relation.

## Key Takeaways



The assumption of the absence of arbitrage opportunity (AOA) stipulates that, in an efficient financial market, there exists no possibility of realising a riskless profit from an initially nil outlay and without any probability of loss [El Karoui (2003)]. This is how one can determine a put-call parity for European options: denoting by  $B_t$  the price at date  $t$  of the zero-coupon bond of maturity  $T$  (an asset paying 1 at expiry  $T$ ), one shows that the respective prices  $C_t$  and  $P_t$  of a European *call* and *put* of maturity  $T$  and *strike*  $K$  satisfy at every date  $t$  the relation  $P_t = C_t - S_t + KB_t$ .

## 1.2.2 The Cox-Ross-Rubinstein binomial model

Put-call parity therefore makes it possible, knowing the price of a European *call*, to determine that of a European *put* on the same underlying, with the same *strike* and the same maturity. How, then, is the price of a *call* to be determined? A first method is to take up the **binomial model** theorised by Cox et al. (1979). This model supposes that over a given period, the price of an underlying asset can evolve according to two possible scenarios: either it rises, or it falls. More precisely, consider an underlying asset whose current price is  $S_0$ . At the end of the period, the price of the underlying may either rise to  $S_u = S_0 \times u$ , or fall to  $S_d = S_0 \times d$ , where  $u > 1$  and  $d < 1$  are the up and down factors respectively. We likewise assume the existence of a riskless asset (for example, a zero-coupon bond) offering a certain return,  $r$  denoting the riskless interest rate at horizon  $T$ ; we set  $R = 1 + r$  (the case of a discrete rate) or  $R = e^r$  (the case of a continuous rate). Consider a European purchase option (*call*) on this asset with an exercise price  $K$ . The *payoff* of the option at the end of the period is therefore given by  $C_u = \max(S_u - K, 0)$  if the price of the underlying asset rises and by  $C_d = \max(S_d - K, 0)$  if the price of the underlying asset plunges.

In order to determine the price of the option today  $C_0$ , let us construct a **replicating portfolio** composed of  $\Delta$  units of the underlying asset and  $B$  units of the riskless asset, that is to say a portfolio whose *payoffs* must be equal to those of the option in each state of the world. In particular, if the price of the asset rises, this portfolio will be worth  $\Delta S_u + BR$  while, if it falls, the replicating portfolio will be worth  $\Delta S_d + BR$ . The replicating character of the portfolio then implies satisfying the system of equations, in the unknowns  $\Delta$  and  $B$ , formed by  $\Delta S_u + BR = C_u$  on the one hand and  $\Delta S_d + BR = C_d$  on the other (the same values at the final date):

- Let us subtract the second equation from the first in order to eliminate  $B$ : we find  $\Delta S_u + BR - (\Delta S_d + BR) = C_u - C_d$ , that is to say  $\Delta(S_u - S_d) = C_u - C_d$ , which makes it possible to obtain as value  $\Delta = \frac{C_u - C_d}{S_u - S_d}$ . The real number  $\Delta$  is then known as the **hedge ratio**.
- Now, let us substitute  $\Delta$  into one of the original equations in order to find  $B$ . Let us use for example the first equation  $\Delta S_u + BR = C_u$ . It therefore follows that  $\frac{C_u - C_d}{S_u - S_d} S_u + BR = C_u$  that is to say, isolating  $B$ ,  $BR = C_u - \frac{C_u - C_d}{S_u - S_d} S_u$ , whence  $B = \frac{1}{R} \left( C_u - \frac{S_u(C_u - C_d)}{S_u - S_d} \right) = \frac{1}{R} \left( \frac{C_u(S_u - S_d) - S_u(C_u - C_d)}{S_u - S_d} \right) = \frac{1}{R} \left( \frac{C_d S_u - C_u S_d}{S_u - S_d} \right)$ .

These solutions represent respectively the proportion of underlying assets ( $\Delta$ ) and the amount invested in the riskless asset ( $B$ ) in order to construct the replicating portfolio. By virtue of the assumption of AOA, the initial price of the option  $C_0$  must then be equal to the value of the replicating portfolio, whence  $C_0 = \Delta S_0 + B$  (the same values at the initial date). It then follows that:

$$\begin{aligned}
 C_0 = \Delta S_0 + B &\iff C_0 = \frac{C_u - C_d}{S_u - S_d} S_0 + \frac{C_d S_u - C_u S_d}{R(S_u - S_d)} \\
 &\iff C_0 = \frac{1}{R} \left( \frac{R(C_u - C_d)S_0 + (C_d S_u - C_u S_d)}{S_u - S_d} \right) \\
 &\iff C_0 = \frac{1}{R} \left( \frac{(RS_0 - S_d)C_u + (S_u - RS_0)C_d}{S_u - S_d} \right) \\
 &\iff C_0 = \frac{1}{R} \left( \frac{RS_0 - S_d}{S_u - S_d} C_u + \frac{S_u - RS_0}{S_u - S_d} C_d \right) \\
 &\iff C_0 = \frac{1}{R} \left( \frac{R - S_d/S_0}{S_u/S_0 - S_d/S_0} C_u + \frac{S_u/S_0 - R}{S_u/S_0 - S_d/S_0} C_d \right) \\
 &\iff C_0 = \frac{1}{R} \left( \frac{R - d}{u - d} C_u + \frac{u - R}{u - d} C_d \right)
 \end{aligned}$$

Let us then set  $q_u = \frac{R - d}{u - d}$  and  $q_d = \frac{u - R}{u - d} = 1 - q_u$ : we then verify  $C_0 = \frac{1}{R}(q_u C_u + q_d C_d)$ . The values  $q_u$  and  $q_d$  are then known as the **risk-neutral probabilities**  $q_u$  and  $q_d$ , since they are computed in such a way as to render the process of evolution of the price of the underlying asset free of arbitrage opportunity. As Hull (2012) recalls, risk-neutral probabilities may therefore differ from the historical frequencies of rise and fall of the underlying. Let us note finally, as El Karoui (2003) signals, that the *pricing* of the *call* is valid only if the risk-neutral probabilities are effectively probabilities, therefore if and only if  $0 < q_u < 1$ , that is to say if and only if  $d < R < u$  (which is in general the case in practice, since otherwise one could prove the existence of an arbitrage opportunity).

## Exercise



**One-period trinomial model.** Consider a one-period financial market, composed of a riskless asset  $S^0$  and a risky asset  $S$ . The riskless asset is worth  $S_0^0 = 1$  in period  $t = 0$  and  $S_1^0 = 1 + r$  in period  $t = 1$ , where  $r$  is a positive real number representing the (discrete) riskless rate. The risky asset is worth  $S_0 > 0$  at date  $t = 0$  and its price at date  $t = 1$  is a random variable  $S_1$  defined on a sample space  $\{\omega_d, \omega_m, \omega_u\}$  equipped with a probability  $\mathbb{P}$  defined by  $\mathbb{P}(\{\omega_u\}) = p_u$ ,  $\mathbb{P}(\{\omega_d\}) = p_d$  and  $\mathbb{P}(\{\omega_m\}) = p_m$ , where  $p_u, p_m$  and  $p_d$  are real numbers between 0 and 1 satisfying  $p_u + p_m + p_d = 1$ . The random variable  $S_1$  then takes the following values:  $S_1(\omega_u) = uS_0$ ,  $S_1(\omega_m) = mS_0$  and  $S_1(\omega_d) = dS_0$ , where  $d, m$  and  $u$  are constants satisfying  $u > m > d \geq 0$ . A portfolio strategy consists of a pair  $(x, \Delta)$  of real numbers, where  $x$  is the initial capital and  $\Delta$  the number of units of the risky asset  $S$  bought at  $t = 0$  (if  $\Delta < 0$ , then  $\Delta$  represents the number of units of  $S$  sold short at  $t = 0$ ). The value at  $t$  of the portfolio of strategy  $(x, \Delta)$  is then denoted  $X_t^{x, \Delta}$ , for every  $t$  in  $\{0, 1\}$ .

- (i) Recall the definition of an arbitrage strategy in this model and write the assumption of the absence of arbitrage opportunity (AOA).
- (ii) For a strategy  $(x, \Delta)$ , give the expression of  $X_1^{x, \Delta}$ .
- (iii) The aim of this question is to give a necessary condition for AOA as a function of the coefficients  $r, d, m$  and  $u$  of the model.
  - (a) In the case  $1 + r \leq d$ , show that there is an arbitrage opportunity.
  - (b) In the case  $1 + r \geq u$ , show that there is an arbitrage opportunity.
  - (c) Deduce from this a necessary condition for AOA.
- (iv) The aim is now to show that the condition of the previous question is sufficient. To that end, we study the existence of a risk-neutral probability. We assume in this question that the condition obtained in question (iii)-(c) is satisfied.
  - (a) Recall the definition of a risk-neutral probability.
  - (b) Let  $\mathbb{Q}$  be such a probability measure, with  $q_u := \mathbb{Q}(\{\omega_u\}) > 0$ ,  $q_m := \mathbb{Q}(\{\omega_m\}) > 0$ ,  $q_d := \mathbb{Q}(\{\omega_d\}) > 0$ . Give a system of two equations satisfied by  $(q_u, q_m, q_d)$ .
  - (c) Give the expressions of  $q_u$  and  $q_d$  as functions of  $q_m, r, u, d$  and  $m$ .
  - (d) Deduce from this that there exist infinitely many risk-neutral probabilities.
- (v)
  - (a) Show that the existence of a risk-neutral probability implies the assumption of AOA.
  - (b) Deduce from this that the necessary condition obtained in question (iii)-(c) is sufficient.
  - (c) Recall the condition obtained in the one-period binomial model and compare it with the condition obtained *supra*.

**Solution.** (i) We call an arbitrage opportunity any strategy which makes it possible to realise a gain for certain without taking any risk, that is to say without mobilising any money initially. In this model, we say that there is no arbitrage if there exists no real number  $\Delta$  satisfying  $\mathbb{P}([X_1^{0, \Delta} \geq 0]) = 1$  (mobilising  $x = 0$ , if I acquire  $\Delta$  units of risky assets initially, I never lose money at date  $t = 1$ ) and  $\mathbb{P}([X_1^{0, \Delta} > 0]) > 0$  (there even exists a non-zero probability that I realise a net gain at date  $t = 1$ ).

- (ii) Suppose that we invest  $x$  euros initially distributed between the risky asset and the riskless asset. We suppose that we acquire  $\Delta$  units of the risky asset. These  $\Delta$  units therefore cost  $\Delta S_0$ . There therefore remain in our portfolio  $(x - \Delta S_0)$  euros which we shall be able to invest in the riskless asset, placed at the riskless rate  $r$ : at date  $t = 1$ , this amount is therefore worth  $(x - \Delta S_0)(1 + r)$ . As for the  $\Delta$  units of risky assets, they are worth  $\Delta S_1$  at date  $t = 1$ . Thus, we have  $X_1^{x, \Delta} = \Delta S_1 + (x - \Delta S_0)(1 + r)$ .
- (iii)
  - (a) In the case  $1 + r \leq d$ , let us show that there is an arbitrage opportunity. The intuition behind this result is the following: in the worst case (event  $\omega_d$ ), the risky asset yields  $d$ , which remains higher than the return  $1 + r$  of the riskless asset: one may as well put all one's wealth in the risky asset and nothing in the riskless asset. Let us show that such a strategy constitutes an arbitrage. Let  $x$  be the initial wealth. Suppose that we invest it entirely in the risky asset. We want to acquire  $\Delta$  units of risky assets (where  $\Delta$  is a strictly positive real number), which costs us  $\Delta S_0$ . In order to finance this purchase, we sell short  $\Delta S_0$  units of



the riskless asset: we therefore recover  $-\Delta S_0$  euros from this sale. We therefore mobilise no capital at the initial date. At date  $t = 1$ , we must therefore repay the riskless asset, which costs us  $(1 + r)\Delta S_0$ . But we also recover  $\Delta S_1$  by liquidating our risky asset. In the worst case (event  $\omega_d$ , which occurs with probability  $p_d > 0$ ),  $S_1$  is worth  $dS_0$ : we therefore have  $X_1^{0,\Delta} = \Delta dS_0 - (1 + r)\Delta S_0 = \Delta S_0(d - (1 + r)) \geq 0$  by assumption. This justifies the relation  $\mathbb{P}([X_1^{0,\Delta} \geq 0]) = 1$ : mobilising  $x = 0$ , if one acquires  $\Delta$  units of risky assets initially, one never loses money at date  $t = 1$ . In the best case (event  $\omega_u$ , which occurs with probability  $p_u > 0$ ),  $S_1$  is worth  $uS_0$ : we therefore have  $X_1^{0,\Delta} = \Delta uS_0 - (1 + r)\Delta S_0 = \Delta S_0(u - (1 + r)) > \Delta S_0(d - (1 + r)) \geq 0$ , since  $u > d$ , whence  $X_1^{0,\Delta} = \Delta uS_0 - (1 + r)\Delta S_0 > 0$ . This justifies the relation  $\mathbb{P}([X_1^{0,\Delta} > 0]) > 0$ : there even exists a non-zero probability of gaining money at date  $t = 1$ .

- (b) In the case  $1 + r \geq u$ , let us show that there is an arbitrage opportunity. The intuition behind this result is the following: in the best case (event  $\omega_u$ ), the risky asset yields  $u$ , which remains lower than the return  $1 + r$  of the riskless asset: one may as well put all one's wealth in the riskless asset and nothing in the risky asset. Let us show that such a strategy constitutes an arbitrage. Let  $x$  be the initial wealth. Suppose that we invest it entirely in the riskless asset. We sell short  $-\Delta$  units of risky assets (where  $\Delta$  is a strictly positive real number), which yields us  $\Delta S_0$ . This contribution makes it possible to finance the acquisition of  $\Delta S_0$  units of the riskless asset. We therefore mobilise no capital at the initial date. At date  $t = 1$ , we must therefore repay the risky asset, which costs us  $\Delta S_1$ . But we also recover  $(1 + r)\Delta S_0$  by liquidating the riskless asset. In the worst case (event  $\omega_u$ , which occurs with probability  $p_u > 0$ ),  $S_1$  is worth  $uS_0$ : we therefore have  $X_1^{0,\Delta} = (1 + r)\Delta S_0 - \Delta uS_0 = \Delta S_0((1 + r) - u) \geq 0$  by assumption. This justifies the relation  $\mathbb{P}([X_1^{0,\Delta} \geq 0]) = 1$ : mobilising  $x = 0$ , if one acquires  $\Delta S_0$  units of riskless assets initially, one never loses money at date  $t = 1$ . In the best case (event  $\omega_d$ , which occurs with probability  $p_d > 0$ ),  $S_1$  is worth  $dS_0$ : we therefore have  $X_1^{0,\Delta} = (1 + r)\Delta S_0 - \Delta dS_0 = \Delta S_0((1 + r) - d) > \Delta S_0((1 + r) - u) \geq 0$ , since  $u > d$ , whence  $X_1^{0,\Delta} = (1 + r)\Delta S_0 - \Delta dS_0 > 0$ . This justifies the relation  $\mathbb{P}([X_1^{0,\Delta} > 0]) > 0$ : there even exists a non-zero probability of gaining money at date  $t = 1$ .
- (c) By contraposition of the two previous sub-questions, for there to be absence of arbitrage opportunity, the inequality  $d < 1 + r < u$  must therefore necessarily hold.
- (iv) (a) A risk-neutral probability is a probability measure which makes it possible to equalise the expected gain of a portfolio strategy with the gain which would result from placing the same amount in the riskless asset. Mathematically, a risk-neutral probability  $\mathbb{Q}$  is a probability measure on  $\Omega$  equivalent to  $\mathbb{P}$  satisfying  $\mathbb{Q}(\{\omega_u\}) > 0$ ,  $\mathbb{Q}(\{\omega_m\}) > 0$ ,  $\mathbb{Q}(\{\omega_d\}) > 0$  and  $\mathbb{E}_{\mathbb{Q}}[X_1^{x,\Delta}] = (1 + r)x$ , for every portfolio strategy  $(x, \Delta)$ .
- (b) We know that  $\mathbb{Q}$  is a probability measure on  $\Omega$ : we therefore have  $q_u + q_m + q_d = 1$ . We know moreover that  $\mathbb{E}_{\mathbb{Q}}[X_1^{x,\Delta}] = (1 + r)x$ , for every portfolio strategy  $(x, \Delta)$  (this is what is called the 'martingale property'). Now  $X_1^{x,\Delta} = \Delta S_1 + (x - \Delta S_0)(1 + r)$ . We therefore have  $\mathbb{E}_{\mathbb{Q}}[X_1^{x,\Delta}] = (1 + r)x$  if, and only if,  $\Delta \mathbb{E}_{\mathbb{Q}}[S_1] + (x - \Delta S_0)(1 + r) = (1 + r)x$  if, and only if,  $\Delta \mathbb{E}_{\mathbb{Q}}[S_1] - \Delta S_0(1 + r) = 0$  if, and only if,  $\Delta (q_u u S_0 + q_m m S_0 + q_d d S_0) - \Delta S_0(1 + r) = 0$  if, and only if (since  $\Delta S_0$  is non-zero),  $(q_u u + q_m m + q_d d) - (1 + r) = 0$  if, and only if,  $q_u u + q_m m + q_d d = 1 + r$ . Thus, the risk-neutral probability  $\mathbb{Q}$  satisfies the following system of equations:

$$(\Sigma): \begin{cases} q_u + q_m + q_d = 1 \\ q_u u + q_m m + q_d d = 1 + r \end{cases}$$

(c) We then have:

$$\begin{aligned} (\Sigma) &\Leftrightarrow \begin{cases} q_u + q_d = 1 - q_m \\ q_u u + q_d d = 1 + r - q_m m \end{cases} \\ &\Leftrightarrow \begin{cases} q_d = 1 - q_m - q_u \\ q_u u + (1 - q_m - q_u)d = 1 + r - q_m m \end{cases} \\ &\Leftrightarrow \begin{cases} q_d = 1 - q_m - q_u \\ (u - d)q_u = 1 + r - d - (m - d)q_m \end{cases} \\ &\Leftrightarrow \begin{cases} q_d = \frac{u - 1 - r}{u - d} - q_m \frac{u - m}{u - d} \\ q_u = \frac{u - d}{u - d} - q_m \frac{m - d}{u - d} \end{cases} \end{aligned}$$

- (d) Thus, we find ourselves with a system of two equations in three unknowns  $q_d$ ,  $q_m$  and  $q_u$ , therefore presenting *a priori* infinitely many solutions. We further impose that  $q_d$ ,  $q_m$  and  $q_u$  be strictly positive. Now on the one hand:

$$\begin{aligned}
 q_d > 0 &\iff \frac{u-1-r}{u-d} - q_m \frac{u-m}{u-d} > 0 \\
 &\iff \frac{u-1-r}{u-d} > q_m \frac{u-m}{u-d} \\
 &\iff q_m < \frac{u-1-r}{u-m} \\
 &\iff q_m < \frac{u-(1+r)}{u-m}
 \end{aligned}$$

Note that  $u > 1+r$  and  $u > m$ , by assumption, whence  $\frac{u-(1+r)}{u-m} > 0$ . We must likewise satisfy  $q_d < 1$ . Now we have:

$$\begin{aligned}
 q_d < 1 &\iff \frac{u-1-r}{u-d} - q_m \frac{u-m}{u-d} < 1 \\
 &\iff \frac{u-1-r}{u-d} < q_m \frac{u-m}{u-d} + 1 \\
 &\iff u-1-r < q_m(u-m) + u-d \\
 &\iff q_m > \frac{d-1-r}{u-m}
 \end{aligned}$$

Since  $d < 1+r$ ,  $d-1-r < 0$ , whence  $\frac{d-1-r}{u-m} < 0$  and this constraint is void: it suffices to assume  $q_m > 0$ . On the other hand, we have:

$$\begin{aligned}
 q_u > 0 &\iff \frac{1+r-d}{u-d} - q_m \frac{m-d}{u-d} > 0 \\
 &\iff \frac{1+r-d}{u-d} > q_m \frac{m-d}{u-d} \\
 &\iff q_m < \frac{(1+r)-d}{m-d}
 \end{aligned}$$

Note that  $1+r < u$  and  $m > d$ , by assumption, whence  $\frac{1+r-d}{m-d} > 0$ . We must likewise satisfy  $q_u < 1$ . Now we have:

$$\begin{aligned}
 q_u < 1 &\iff \frac{1+r-d}{u-d} - q_m \frac{m-d}{u-d} < 1 \\
 &\iff \frac{1+r-d}{u-d} < q_m \frac{m-d}{u-d} + 1 \\
 &\iff 1+r-d < q_m(m-d) + u-d \\
 &\iff q_m > \frac{1+r-u}{u-d}
 \end{aligned}$$

Note that  $u > 1+r$  and  $u > d$ , by assumption, whence  $\frac{1+r-u}{u-d} < 0$ . In reality, this is therefore not even a constraint: it suffices to assume  $q_m > 0$ . *In fine*, the solutions  $(q_d, q_m, q_u)$  of the system  $(\Sigma)$  which further satisfy the conditions that  $q_u, q_m$  and  $q_d$  lie strictly between 0 and 1 belong to the following set, of infinite cardinality:

$$\mathcal{S} := \left\{ \left( \frac{u-(1+r)}{u-d} - q_m \frac{u-m}{u-d}, q_m, \frac{1+r-d}{u-d} - q_m \frac{m-d}{u-d} \right); q_m \in \left] 0, \min \left( \frac{(1+r)-d}{m-d}, \frac{u-(1+r)}{u-m} \right) \right[ \right\}$$

- (v) (a) Let us show that the existence of a risk-neutral probability implies the assumption of the absence of arbitrage opportunity. Suppose for the sake of contradiction that there exists a risk-neutral probability  $\mathbb{Q}$  satisfying, for a certain portfolio strategy  $(0, \Delta)$ ,  $\mathbb{Q}([X_1^{0,\Delta} \geq 0]) = 1$  and  $\mathbb{Q}([X_1^{0,\Delta} > 0]) > 0$  (the strategy never loses money and gains some with a non-zero probability). On the one hand, the assumption  $\mathbb{Q}([X_1^{0,\Delta} \geq 0]) = 1$  implies that the random variable  $X_1^{0,\Delta}$  is almost surely positive. But, since  $\mathbb{Q}$  is risk-neutral, we have  $\mathbb{E}_{\mathbb{Q}}[X_1^{0,\Delta}] = (1+r) \times 0 = 0$ . Now a finite random variable which takes only positive values and which has zero expectation is necessarily almost surely zero. We therefore have  $\mathbb{Q}([X_1^{0,\Delta} = 0]) = 1$ : it therefore follows that  $\mathbb{Q}([X_1^{0,\Delta} > 0]) = 0$ , which contradicts our initial assumption. By contradiction, we have therefore just demonstrated that there existed no arbitrage opportunity in the presence of a risk-neutral probability  $\mathbb{Q}$ .
- (b) In question (iii), we proved that, for there to be absence of arbitrage opportunity, it was necessary that the inequality  $d < 1+r < u$  hold. Conversely, we showed in question (iv) that, if the inequality  $d < 1+r < u$  held, then there existed a risk-neutral probability. Now the previous sub-question shows that the existence of a risk-neutral probability implies the absence of arbitrage opportunity. Thus, we have demonstrated



that, in the one-period trinomial model, there is absence of arbitrage opportunity if, and only if, the inequality  $d < 1 + r < u$  holds.

- (c) We observe that the AOA condition in the trinomial model is exactly the same as in the binomial model:  $d < 1 + r < u$ . There is therefore no impact of the existence of a 'median state'  $\omega_m$  on the definition of arbitrage opportunities.  $\square$

### Exercise



**Determining the price of a call in a two-period model.** Consider two assets: one riskless which gives a continuous return of 8 % per year, and the other risky, offered at the unit price  $S_0 = 40$  euros at date  $t = 0$  and giving the value  $S_1 = 42$  euros (that is a monthly rise of 5 %: we then set  $u = 1.05$  so as to satisfy  $uS_0 = 42$ ) and  $S_1 = 38$  euros (that is a monthly fall of 5 %: we then set  $d = 0.95$  so as to satisfy  $dS_0 = 38$ ) one month later (date  $t = 1$ ). Determine the value of a European *call* with an expiry of one month whose exercise price is  $K = 39$ . Comment.

**Solution.** Let us note first of all that the monthly return of the riskless asset satisfies  $e^{0.08/12}$  (since one month represents one twelfth of a year), that is roughly  $r = 0.67$  %, whence  $R = 1 + r \approx 1.0067$ . Since  $d < R < u$ , there is no arbitrage opportunity, and it is therefore possible to determine the price of the aforementioned *call*.

- If the state of the world at  $t = 1$  is bullish (that is to say if  $S_u = uS_0$ ), then the intrinsic value of the *call* will be  $C_u = \max(uS_0 - K, 0) = \max(42 - 39, 0) = 3$ . By contrast, if the state of the world at  $t = 0$  is bearish (that is to say if  $S_d = dS_0$ ), then the intrinsic value of the *call* will be  $C_d = \max(dS_0 - K, 0) = \max(38 - 39, 0) = 0$ .
- Subsequently, let us compute the risk-neutral probability  $q$  of such an option. Knowing that the monthly riskless rate is  $e^{0.08/12}$ , we have  $q = \frac{e^{0.08/12} - d}{u - d} \approx \frac{1.0067 - 0.95}{1.05 - 0.95} \approx 0.5669$ . Under the risk-neutral probability, we therefore have  $\mathbb{E}[C_1] = qC_u + (1 - q)C_d \approx 0.5669 \times 3 + (1 - 0.5669) \times 0 = 1.70$  euros.
- The value of the option is then given by the expectation, discounted at the riskless rate  $r$ , of its *payoff*  $C_1$  at date  $t = 1$ , computed under the risk-neutral probability  $q$ . Otherwise put, the value  $C_0$  of the *call* at date  $t = 0$  is  $C_0 = \frac{\mathbb{E}[C_1]}{e^{0.08/12}} \approx 1.69$  euros. In conclusion, in order to acquire this *call*, one will have to disburse 1.69 euros. In return, one will have the right to buy a share one month later at the price of 39 euros.
- If one anticipates that the share will be worth *in fine* more than  $39 + 1.69 = 40.69$  euros, then one will have gained money, even though one will have had to pay this premium of 1.69 euros. Otherwise one will be at a loss: the *call* is therefore of interest only if one bets that the share will be worth more than 40.69 euros per unit one month later.  $\square$

The binomial model offers a simple and intuitive method for valuing options while taking account of the possible variations in the price of the underlying asset over one period. The notion of risk-neutral probabilities, justified by the absence of arbitrage opportunity, makes it possible to ensure that the price of the option correctly reflects the risks associated with the evolution of the underlying asset. The binomial model as described previously may be extended to value the price of an option over several periods. Cox et al. (1979) then propose a generalisation which makes it possible better to capture the evolution of the price of an option over several periods. Let us thus consider an underlying asset whose current price is  $S_0$ . The binomial model with  $n \geq 2$  periods supposes that, at each period, the price of the underlying may either rise by a factor  $u > 1$ , or fall by a factor  $d < 1$ , and that the fact that the price of the asset falls or rises in a given period does not influence the movement of the price in the following period (independence of variations). At the end of  $n$  periods, there are therefore  $2^n$  possible paths for the price of the underlying, each corresponding to a sequence of  $n$  rises or falls. For a European purchase option (*call*), the *payoff* at expiry  $T$  (after  $n$  periods) is given by  $C_T = \max(S_T - K, 0)$ , where  $S_T$  is the price of the underlying at expiry, which may take  $n + 1$  distinct values corresponding to the different possible scenarios. In order to value the price of the option  $C_0$  today (at  $t = 0$ ), Cox et al. (1979) use a process of **backward induction**:

- At expiry  $t = T$ , compute the *payoff* of the option for each possible value of the underlying. These values are given by  $S_T = S_0 u^i d^{T-i}$  for every integer  $i$  between 0 and  $T$ , where  $i$  represents the number of rises in the price.
- At the previous period  $t = T - 1$ , compute the price of the option by taking the expectation of future prices, discounted at the riskless rate  $R = 1 + r$ , using the risk-neutral probabilities  $q_u$  and  $q_d$  defined identically to the simple case: we therefore have  $C_{T-1} = \frac{1}{R} (q_u C_u + q_d C_d)$ , where  $C_u$  and  $C_d$  are the prices of the option in the two possible scenarios at period  $t = T - 1$ .

— Repeat the process for each earlier period, until returning to the initial period  $t = 0$ .

Cox et al. (1979) then show that the initial price of the option  $C_0$ , obtained by applying the process of *backward induction* under the risk-neutral measure, satisfies:

$$C_0 = \frac{1}{R^T} \sum_{t=0}^T \binom{T}{t} q_u^t q_d^{T-t} \max(S_0 u^t d^{T-t} - K, 0)$$

This formula thus integrates all the possible trajectories of the underlying asset and makes it possible to obtain an approximation of the price of the option using a discrete multi-period model. A natural generalisation of the one-period model, the Cox-Ross-Rubinstein model makes it possible to capture more finely the evolution of the price of the underlying and to value options within a multi-period framework. As Hull (2012) emphasises, this model is particularly useful for understanding the dynamics of option prices and is often used as the basis for more complex models, such as the Black-Scholes model, in the limit where the number of periods tends to infinity.

### Exercise



**Determining the price of a call in a three-period model.** Consider three periods  $t = 0$ ,  $t = 1$  (six months after period 0) and  $t = 2$  (six months after period 1). Consider two assets: one riskless which gives a continuous annual return of 8 %, and the other risky, offered at the unit price  $S_0 = 100$  at date  $t = 0$  and able to rise or fall in each period by 10 % (we set  $d = 0.9$  and  $u = 1.1$ ). We suppose that the rise or fall between period 0 and period 1 is independent of the rise or fall between period 1 and period 2. Determine the value of a European *call* with an expiry of one year whose exercise price is  $K = 100$ .

**Solution.** Let us note first of all that the half-yearly return of the riskless asset satisfies  $e^{0.08/2}$  (since six months represent half a year), that is roughly  $r = 4.08$  %, whence  $R = 1 + r \approx 1.0408$ . Since  $d < R < u$ , there is no arbitrage opportunity. The risk-neutral probability  $q$  is then well defined and satisfies  $q = \frac{e^{0.08/2} - d}{u - d} \approx \frac{e^{0.08/2} - 0.9}{1.1 - 0.9}$ , that is  $q \approx 0.7041$ . For  $i$  in  $\{0, 1\}$ , we denote by  $U_i$  (resp.  $D_i$ ) the event ‘the price rises (resp. falls) between periods  $i$  and  $i + 1$ ’. By definition of the risk-neutral probability, we know that  $\mathbb{P}(U_1) = q$ ,  $\mathbb{P}(D_1) = 1 - q$ ,  $\mathbb{P}(U_2) = q$  and  $\mathbb{P}(D_2) = 1 - q$ .

— Suppose  $D_1$  satisfied (an event of probability  $1 - q$ ). Let us then denote by  $C_1^d$  the value of the *call*. We have, under the risk-neutral probability,  $C_1^d = \max(dS_0 - K, 0) = \max(0.9 \times 100 - 100, 0) = 0$ . We then set  $S_1^d = 90$ .

- Given the event  $D_1$  satisfied, we suppose that  $D_2$  occurs (an event of probability  $1 - q$ ). Since the events concerning the evolution between period 0 and period 1 are independent of those concerning the evolution between period 1 and period 2, the event  $D_1 \cap D_2$  has probability  $\mathbb{P}(D_1 \cap D_2) = (1 - q)^2 \approx 0.0875$ . Let us then denote by  $C_2^{d,d}$  the value of the *call* in this case. We have, under the risk-neutral probability,  $C_2^{d,d} = \max(dS_1^d - K, 0) = \max(0.9 \times 90 - 100, 0) = 0$ .
- Given the event  $D_1$  satisfied, we suppose that  $U_2$  occurs (an event of probability  $q$ ). Since the events concerning the evolution between period 0 and period 1 are independent of those concerning the evolution between period 1 and period 2, the event  $D_1 \cap U_2$  has probability  $\mathbb{P}(D_1 \cap U_2) = q(1 - q) \approx 0.2084$ . Let us then denote by  $C_2^{d,u}$  the value of the *call* in this case. We have, under the risk-neutral probability,  $C_2^{d,u} = \max(uS_1^d - K, 0) = \max(1.1 \times 90 - 100, 0) = 0$ .

— Suppose  $U_1$  satisfied (an event of probability  $q$ ). Let us then denote by  $C_1^u$  the value of the *call*. We have, under the risk-neutral probability,  $C_1^u = \max(uS_0 - K, 0) = \max(1.1 \times 100 - 100, 0) = 10$ . We then set  $S_1^u = 110$ .

- Given the event  $U_1$  satisfied, we suppose that  $U_2$  occurs (an event of probability  $q$ ). Since the events concerning the evolution between period 0 and period 1 are independent of those concerning the evolution between period 1 and period 2, the event  $U_1 \cap U_2$  has probability  $\mathbb{P}(U_1 \cap U_2) = q^2 \approx 0.4957$ . Let us then denote by  $C_2^{u,u}$  the value of the *call* in this case. We have, under the risk-neutral probability,  $C_2^{u,u} = \max(uS_1^u - K, 0) = \max(1.1 \times 110 - 100, 0) = 21$ .
- Given the event  $U_1$  satisfied, we suppose that  $D_2$  occurs (an event of probability  $1 - q$ ). Since the events concerning the evolution between period 0 and period 1 are independent of those concerning the evolution between period 1 and period 2, the event  $U_1 \cap D_2$  has probability  $\mathbb{P}(U_1 \cap D_2) = q(1 - q) \approx 0.2084$ . Let us then denote by  $C_2^{u,d}$  the value of the *call* in this case. We have, under the risk-neutral probability,  $C_2^{u,d} = \max(dS_1^u - K, 0) = \max(0.9 \times 110 - 100, 0) = 0$ .

Thus, the *call* takes the value 0 with probability  $2 \times 0.2084 + 0.0875 = 0.5043$  and the value 21 with probability 0.4957. Consequently, the expectation of the value of the *call* at maturity is  $\mathbb{E}[C_2] \approx 21 \times 0.4957 + 0 \times 0.5043$  (under the risk-neutral probability), that is 10.41 euros. From then on, the value of this purchase option at  $t = 0$  is  $C_0 = \frac{\mathbb{E}[C_2]}{(e^{0.08/2})^2} \approx 9.61$  euros.  $\square$

### 1.2.3 The Black-Scholes model

The Cox-Ross-Rubinstein binomial model, although effective for valuing options over a discrete number of periods, presents certain limitations when it comes to modelling real markets where price variations are continuous. In order to remedy these limitations, Black & Scholes (1973) developed a more sophisticated analytical model which makes it possible to determine the price of an option in a continuous framework. The model of Black & Scholes (1973) rests on several fundamental assumptions. First of all, the price of the underlying asset follows a continuous stochastic process of **geometric Brownian motion** type, that is to say that price variations are normally distributed, with a mean and a variance proportional to the square of the volatility of the underlying. In addition, the market is assumed to be **frictionless**, which means that there are no transaction costs, no restrictions on short selling, and no dividends on the underlying asset during the life of the option. Finally, it is assumed that it is possible perfectly to replicate the *payoff* of an option by a dynamic portfolio composed of the underlying asset and a riskless bond (the existence of a **replicating portfolio**). Under these assumptions, Black & Scholes (1973) derived an analytical formula for the price  $C_0$  of a European purchase option (*call*) with *strike*  $K$  and maturity  $T$ :

$$C_0 = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2) \quad \text{with} \quad \begin{cases} d_1 = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T \right) \\ d_2 = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T \right) \end{cases}$$

where  $\Phi$  denotes the cumulative distribution function of the standard normal distribution,  $S_0$  the current price of the underlying asset,  $r$  is the riskless interest rate and  $\sigma$  is the volatility of the underlying. Moreover, denoting by  $\delta_T$  the time interval between two periods, if we set, with Cox et al. (1979),  $u = \exp(\sigma\sqrt{\delta_T})$  and  $d = \exp(-\sigma\sqrt{\delta_T})$  in the formula of the Cox-Ross-Rubinstein model studied *supra*, it can be shown that the price of the *call* according to the said formula converges towards the price of the *call* according to the formula of Black & Scholes (1973) when the number  $T$  of periods tends to  $+\infty$ . Finally, the price  $P_0$  of the European *put* with *strike*  $K$ , maturity  $T$  and the same underlying as the aforementioned *call* is determined thanks to the relation of parity between *call* and *put* (noting that  $\Phi(x) + \Phi(-x) = 1$  for every real number  $x$  by symmetry of the normal distribution), and we find  $P_0 = -S_0 \Phi(-d_1) + Ke^{-rT} \Phi(-d_2)$ .

#### Debate



**Are market returns Gaussian?** The assumption of the normality of markets rests on the idea that the returns of financial assets follow a normal (or Gaussian) distribution. This assumption is widely used in financial models, notably in the model of Black & Scholes (1973) and other frameworks for *option pricing*, where it greatly simplifies computations and analyses. However, as Walter (2001) emphasises, this assumption presents significant limitations. Indeed, financial markets are often characterised by **extreme events** (the ‘fat tails’ of the distribution of returns) which are not captured by the normal distribution. These events include stock market crashes, speculative bubbles, and other sudden and severe price fluctuations.

Consequently, the assumption of normality tends to **underestimate the risk** of speculative bubbles and panic movements, thus leading to an inadequate assessment of risk in portfolio management and in the *pricing* of financial assets. Walter (2001) brings out the fact that this underestimation of risk is particularly problematic for the investment management industry, which often relies on models based on normality in order to take crucial decisions. Thus, although the assumption of normality is useful for certain theoretical aspects, it must be used with caution and complemented by other approaches which take account of the empirical characteristics of financial returns, such as fat-tailed distributions (like the log-normal distribution rather than the normal distribution) and stochastic volatility (which considers volatility not as constant but as a random variable).

Although the model of Black & Scholes (1973) is a powerful tool for the *pricing* of options, it presents practical difficulties, notably as regards the estimation of the volatility parameter. Indeed, the volatility used in the formula of Black & Scholes (1973) is an **implied volatility**, that is to say that it is deduced from the market price of the option, making this parameter directly unobservable. A commonly used proxy for estimating this implied volatility is **historical volatility**, computed from past fluctuations in the price of the underlying asset. However, this proxy has its limitations, especially in periods of high volatility. As Hull (2012) recalls, historical volatility rests on the assumption that the conditions of the past will be maintained in the future, which may be misleading in periods of financial stress or crisis. During such periods, market volatility may increase suddenly and significantly, making estimates based on historical data obsolete or insufficient. Consequently, relying solely on historical volatility may lead to an under-valuation or an over-valuation of options, especially when the market anticipates extreme movements.

## Focus...



**How is a historical volatility to be computed?** Historical volatility is a measure of the variation in the prices of a financial asset over a given period. It is computed using the logarithmic returns of the past prices of the asset. The process for computing historical volatility unfolds in several stages, recalled by Hull (2012):

- **Collection of data:** one collects the closing prices of the asset underlying the European option over a period of  $T$  days ( $T$  being fixed by the modeller). These prices will be denoted  $S_1, S_2, \dots, S_T$ , where  $S_t$  is the closing price at date  $t$ .
- **Computation of logarithmic returns:** for every date  $t$  between 2 and  $T$ , one computes the daily logarithmic returns between date  $t-1$  and date  $t$ , denoted  $r_t$ , which are given by  $r_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$ .
- **Computation of the variance of returns:** one computes the mean of the returns, denoted  $\bar{r}$ , by  $\bar{r} = \frac{1}{T-1}(r_2 + \dots + r_T)$ , and one deduces from it their variance, denoted  $s^2$ , by  $s^2 = \frac{1}{T-1}[(r_2 - \bar{r})^2 + \dots + (r_T - \bar{r})^2]$ .
- **Computation of historical volatility:** the annualised historical volatility, denoted  $\sigma$ , is obtained by taking the square root of the variance and adjusting it for the number of *trading* days in the year (generally 252 days), that is  $\sigma = \sqrt{s^2 \times 252}$ .

The historical volatility thus computed represents the annualised standard deviation of past returns, providing a measure of the risk of the asset based on its history. However, as El Karoui (2003) signals, it rests on the assumption that past conditions will continue into the future, which may not be the case in periods of high volatility: in such situations, historical volatility may underestimate or overestimate real future volatility.

## Exercise



**The Black-Scholes formula.** Let  $\Phi$  be the cumulative distribution function of a standard normal distribution.

- (i) Consider a European *call* on a share whose current price is  $S_0 = 26$  euros with an exercise price  $K = 25$  euros, for which the riskless interest rate is  $r = 10\%$  per year and the volatility is  $\sigma = 30\%$  per year. Suppose that this *call* has a maturity of 3 months, that is  $T = 0.25$  year. Determine the price of this *call* using the Black-Scholes formula. Take  $\Phi(d_1) \approx 0.693$  and  $\Phi(d_2) \approx 0.638$  after having defined and determined the values of  $d_1$  and  $d_2$ .
- (ii) Consider a European *put* on a share whose current price is  $S_0 = 69$  euros with an exercise price  $K = 70$  euros, for which the riskless interest rate is  $r = 5\%$  per year and the volatility is  $\sigma = 35\%$  per year. Suppose that this *call* has a maturity of 6 months, that is  $T = 0.5$  year. Determine the price of this *put* using the Black-Scholes formula. Take  $\Phi(d_1) \approx 0.565$  and  $\Phi(d_2) \approx 0.468$  after having defined and determined the values of  $d_1$  and  $d_2$ .

**Solution.** Let  $\Phi$  be the cumulative distribution function of a standard normal distribution.

- (i) **European call.** We set:

$$\begin{cases} d_1 = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T \right) \approx 0.503 \\ d_2 = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T \right) \approx 0.353 \end{cases}$$

Then  $\Phi(d_1) \approx 0.693$  and  $\Phi(d_2) \approx 0.638$ . From then on, the price  $C_0$  of the *call* at date  $t = 0$  is, by virtue of the Black-Scholes formula,  $C_0 = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2) \approx 2.45$  euros (to the nearest cent on account of roundings and approximations).

- (ii) **European put.** We set:

$$\begin{cases} d_1 = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T \right) \approx 0.167 \\ d_2 = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T \right) \approx -0.081 \end{cases}$$

Then  $\Phi(-d_1) = 1 - \Phi(d_1) \approx 0.434$  and  $\Phi(-d_2) = 1 - \Phi(d_2) \approx 0.532$ . From then on, the price  $P_0$  of the *put* at date  $t = 0$  is, by virtue of the Black-Scholes formula,  $P_0 = -S_0\Phi(-d_1) + Ke^{-rT}\Phi(-d_2) \approx 6.40$  euros (to the nearest cent on account of roundings and approximations).  $\square$

## Key Takeaways



Under the assumptions of an absence of friction, geometric Brownian motion of the underlying and the existence of a replicating portfolio, Black & Scholes (1973) establish the analytical formula for the price of a European *call*  $C_0 = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)$  with  $d_1 = \frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T\right)$  and  $d_2 = \frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T\right)$ , where  $S_0$  is the current price of the underlying,  $K$  the exercise price (*strike*),  $T$  the maturity in years,  $r$  the continuous riskless rate,  $\sigma$  the annualised volatility of the underlying, and  $\Phi$  the cumulative distribution function of the standard normal distribution. The price of the European *put* with the same characteristics is deduced from it by put-call parity:  $P_0 = -S_0\Phi(-d_1) + Ke^{-rT}\Phi(-d_2)$ . The volatility  $\sigma$  is the central and unobservable parameter of the model: in practice, it is estimated either by historical volatility — the standard deviation of past logarithmic returns annualised by  $\sqrt{252}$  — or by implied volatility, deduced by inverting the formula from the observed market price [Hull (2012)]. The underlying Gaussian assumption constitutes the main limitation of the model: real returns present fat tails and a negative asymmetry which the normal distribution does not capture, leading to a systematic underestimation of the risk of extreme movements [Walter (2001)].

## 1.3 Optional hedging strategies

Optional hedging strategies make it possible to construct asymmetric exposures to risk by exploiting the non-linear properties of options, whose sensitivity to market parameters is measured by the Greeks — delta, gamma, vega, theta — which ground the techniques of **delta hedging** permitting the dynamic neutralisation of the directional exposure of a portfolio. Other, less technical hedging techniques exist, among them combinations of shares and options and combinations of options with one another, whose profit-loss profiles make it possible to bet on volatility or to bound exposure at lesser cost.

1.3.1 From the Greeks to *delta hedging*

The two *pricing* formulas (European *call* and European *put*) demonstrated by Black & Scholes (1973) make it possible to study the sensitivity of the prices of an option (purchase or sale) to the various market parameters. Let  $\pi$  be the price of an option with *strike*  $K$  and maturity  $T$  on an underlying whose initial price is  $S_0$  and whose implied volatility is  $\sigma$ . The study of the comparative statics of the price of options then defines the **Greeks**, analysed for example by Hull (2012):

- **Delta** ( $\Delta$ ). The delta measures the sensitivity of the price of the option to variations in the price of the underlying asset and is therefore defined by  $\Delta = \frac{\partial \pi}{\partial S_0}$ . Concretely, it indicates by how much the price of the option will change for a variation of one unit in the price of the underlying. For a European *call*, the delta is given by  $\Delta_{call} = \Phi(d_1)$ , where  $\Phi$  is the cumulative distribution function of the standard normal distribution, and  $d_1$  is defined by  $d_1 = \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S_0}{K} + \left(r + \frac{\sigma^2}{2}\right)T\right)\right]$ . The delta of a *call* always lies between 0 and 1, since the price of the *call* increases when the price of the underlying increases (the rise in the price of the underlying increasing the chances that the option will end in the money), but never by more than one unit (this is understood by taking the extreme case of a purchase option with a zero exercise price, on an underlying which cannot have a negative price). For a European *put*, the delta is given by  $\Delta_{put} = \Phi(d_1) - 1$ , which makes it always negative and between  $-1$  and 0. This is explained intuitively by the fact that a rise in the price of the underlying reduces the probability that the option will be exercised, thereby reducing the value of the *put*.
- **Gamma** ( $\Gamma$ ). The gamma measures the sensitivity of the delta to variations in the price of the underlying and is therefore defined by  $\Gamma = \frac{\partial \Delta}{\partial S_0} = \frac{\partial^2 \pi}{\partial S_0^2}$ . It is therefore the second derivative of the price of the option with respect to the price of the underlying. The gamma is identical for European *calls* and *puts*, and is given by  $\Gamma = \frac{\phi(d_1)}{S_0\sigma\sqrt{T}}$ , where  $\phi = \Phi'$  is the density of the standard normal distribution. The gamma is always positive, which means that the delta of an option increases when the price of the underlying increases. This intuitive behaviour is linked to the fact that an option becomes more sensitive to the evolution of the underlying as its price approaches the exercise price, making the gamma particularly important for options which are at the money, that is to say when the delta is worth 0.5 for the *call* or  $-0.5$  for the *put* (that is to say where the price of the underlying is close to the exercise price).

## Focus...



**The delta of an at-the-money option.** When an option is at the money, this means that the price of the underlying asset  $S_0$  is at the level of the exercise price  $K$ . In this context, the probability that the option will be exercised at expiry

is approximately 50 %: there is an equal chance that the price of the underlying will end above (for a *call*) or below (for a *put*) the exercise price.

- For a European *call*, the delta ( $\Delta_{call} = \Phi(d_1)$ ) measures the sensitivity of the price of the option to a variation in the price of the underlying. Now, when the option is at the money ( $S_0 \approx K$ ), the term  $\ln(S_0/K)$  in the definition of  $d_1$  is close to zero: it is therefore the factor  $\sigma\sqrt{T}$  (which represents the uncertainty due to volatility over the time until expiry) which dominates. This means that  $d_1$  approaches 0, and therefore that  $\Phi(d_1) \approx \Phi(0) = 0.5$ . Intuitively, this means that when the *call* is at the money, a small increase in the price of the underlying has roughly a 50 % chance of taking the option into the money, thereby increasing its value significantly. This is why the delta of the *call* is worth roughly 0.5.
- For a European *put*, the delta ( $\Delta_{put} = \Phi(d_1) - 1$ ) measures the sensitivity of the price of the option to a variation in the price of the underlying. Now, when the option is at the money, as for the *call*,  $d_1$  is close to 0 (at least if the level of volatility  $\sigma$  is not explosive), so  $\Phi(d_1) \approx 0.5$ . Thus, for the *put*, it follows that  $\Delta_{put} \approx 0.5 - 1 = -0.5$ . The delta of the *put* is negative because a rise in the price of the underlying reduces the probability that the *put* will be exercised (since for a *put*, the holder gains if the price of the underlying is lower than the exercise price). When the option is at the money, there is roughly a 50 % chance that the *put* will remain in the money or move out of the money, whence a delta of  $-0.5$ .
- **Theta** ( $\Theta$ ). The theta measures the sensitivity of the price of the option to the passage of time, that is to say the speed at which the price of the option decreases as expiry approaches, all other things being equal: we therefore have  $\Theta = \frac{\partial \pi}{\partial T}$ . For a European *call*, the theta is given by  $\Theta_{call} = -\frac{S_0\Phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}\Phi(d_2)$ , while for a European *put* it is worth  $\Theta_{put} = -\frac{S_0\Phi(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}\Phi(-d_2)$ . The theta is therefore negative for *calls* and generally negative for *puts*, since the time value of the option decreases as expiry approaches; this decrease is more marked for options which are at the money, since these options have a greater time value.
- **Vega** ( $v$ : it is therefore the Greek letter ‘nu’ that is used). The vega measures the sensitivity of the price of the option to the implied volatility of the underlying: we therefore have  $v = \frac{\partial \pi}{\partial \sigma}$ . It indicates by how much the price of the option will change for a variation of one percentage point in volatility. The vega is identical for European *calls* and *puts*, and is given by  $v = S_0\Phi(d_1)\sqrt{T}$ . The vega is always positive, since an increase in volatility increases the value of the option, both for a *call* and for a *put*. Indeed, the higher the volatility, the more the probabilities of large movements of the underlying increase, which makes the option more likely to end in the money (therefore in profit for the holder). Unlike the gamma and the theta, the vega is an increasing function of maturity. Thus a parallel increase in volatility will have more impact on options whose expiry date is distant than on those whose expiry date is close.
- **Rho** ( $\rho$ ). The rho measures the sensitivity of the price of the option to variations in the riskless interest rate: we therefore have  $\rho = \frac{\partial \pi}{\partial r}$ . It indicates by how much the price of the option will change for a variation of one percentage point in the interest rate. For a European *call*, the rho is given by  $\rho_{call} = KTe^{-rT}\Phi(d_2)$  while, for a European *put*, the rho is equal to  $\rho_{put} = -KTe^{-rT}\Phi(-d_2)$ . The rho is therefore positive for *calls* and negative for *puts*, which is explained by the fact that a rise in the riskless interest rate increases the value of *calls* (since it reduces the discounted value of the exercise price to be paid at expiry), while it decreases the value of *puts* (for the same reason).

**Delta hedging** is an essential hedging strategy within the framework of volatility trading, where the objective is to profit from the fluctuations of the underlying while minimising the risks associated with those movements. Indeed, when a *trader* is long an option (that is to say that he has bought the option), he often finds himself with a position which is gamma positive and theta negative. The positive gamma implies that the sensitivity of the delta of the option to variations in the underlying is high, which means that the delta will increase with rises in the underlying and decrease with its falls. By contrast, the negative theta reflects the loss of value of the option with the passage of time, a major disadvantage for long positions. Holding an option over time can therefore cause it to lose value, on account of the decrease in the value of the option due to the passing of time. From then on, as El Karoui (2003) emphasises, the *trader* will have to see to it every day that he ‘pays his daily theta’ by profiting from his long gamma position: if the underlying option is sufficiently volatile, the frequent and large fluctuations in the price of the underlying will allow the *trader* to profit from the positive gamma: in other words, these fluctuations allow the *trader* to realise small regular profits by adjusting the position, which can offset (or even surpass) the daily loss due to the negative theta. Otherwise put, *delta hedging* consists in adjusting the position in underlying shares in order



to offset variations in the delta, and thus to neutralise the effects of small market fluctuations: if the price of the underlying increases, the delta of the *call* will increase, obliging the *trader* to sell more shares; conversely, if the price decreases, the delta likewise decreases, and the *trader* will have to buy back shares in order to maintain the balance. For example, suppose that a *trader* buys a purchase option (*call*) with an exercise price of 100 dollars, and that he holds a single option.

- At the outset, suppose that the price of the underlying share is likewise 100 dollars, which places the option at the money. The delta of the option is then 0.5, indicating that for each variation of one dollar in the price of the share, the price of the option will change by 0.5 dollar. The gamma of the option is assumed to be equal to 0.1, which means that the delta will increase by 0.1 for each dollar of variation in the underlying. The theta, for its part, is assumed to be equal to  $-0.05$ , thus reflecting a daily loss of value of the option due to the passage of time. In order to neutralise the effect of the delta and to render the position insensitive to small variations in the price of the underlying, the *trader* sells 0.5 share short for each option held: he thus initially creates a delta-neutral position, with one *call* option and a short sale of 0.5 share.
- On day 2, the price of the share rises to 102 dollars, which takes the delta of the option from 0.5 to 0.7. The option then becomes more sensitive to variations in the underlying. In order to maintain delta neutrality, the *trader* must adjust his position by selling 0.2 additional share, bringing the short sale to a total of 0.7 share. This sale of additional shares at the price of 102 dollars allows the *trader* to realise a small profit, but the short position in shares now entails an unrealised loss, since the price of the shares has risen from 100 to 102 dollars.
- On day 3, the price of the share falls back to 100 dollars, and the delta of the option returns to its initial level of 0.5. In order to remain delta-neutral, the *trader* must then buy back 0.2 share, which he does at the price of 100 dollars. This transaction makes it possible to close the position taken on day 2. But since the *trader* had sold these shares short at 102 dollars, he realises a profit of 2 dollars per share on this buyback.

The principle of *delta hedging*, which therefore consists in neutralising the delta of the portfolio in order to protect oneself against small variations in the price of the underlying, can therefore, as Bennett (2014) emphasises, be associated with an adjustment of one's position in shares in order to take advantage of fluctuations in the price of the underlying. Thus, even if the position in shares is short, frequent readjustments make it possible to realise small profits which can offset the potential losses due to the short position. Although the theta of the option is negative (meaning, we recall, that the value of the option decreases each day on account of the passage of time), the positive gamma therefore makes it possible to profit from market fluctuations in order to adjust the position in shares and generate profits, which can offset the loss due to the theta: this is the principle of ***gamma scalping***. *In fine*, although short selling exposes the *trader* to risks, the regular readjustments of his position make it possible to realise profits which can offset the potential losses, making the strategy profitable in a volatile market, where frequent and significant movements of the underlying offer numerous opportunities for readjusting the portfolio.

### Key Takeaways



The Greeks measure the sensitivity of the price  $\pi$  of an option to market parameters [Hull (2012)] :

- $\Delta = \partial\pi/\partial S_0$ : this is the sensitivity to the price of the underlying. We have  $\Delta_{\text{call}} = \Phi(d_1) \in [0, 1]$  and  $\Delta_{\text{put}} = \Phi(d_1) - 1 \in [-1, 0]$ . In the neighbourhood of the money ( $S_0 \approx K$ ), we have  $\Delta \approx 0.5$  for a *call* and  $\Delta \approx -0.5$  for a *put*.
- $\Gamma = \partial^2\pi/\partial S_0^2$ : this is the sensitivity of the delta to the underlying, identical for *call* and *put*; the gamma is always positive and maximal at the money; it measures the convexity of the option.
- $\Theta = \partial\pi/\partial T$ : this is the parameter which measures the erosion of the time value, negative for *calls* and also for *puts* in general; it is more marked at the money.
- $\nu = \partial\pi/\partial\sigma$  ('vega'): this is the sensitivity to implied volatility: identical for *call* and *put*, the vega is always positive and increasing with maturity.
- $\rho = \partial\pi/\partial r$ : this is the sensitivity to the riskless rate; it is positive for *calls* (a rise in  $r$  reduces the discounted value of the *strike*) and negative for *puts*.

*Delta hedging* consists in permanently maintaining a delta-neutral position by adjusting the quantity of shares held in proportion to the delta of the option: if the delta increases (underlying rising), the trader sells shares; if it decreases, he buys some back [El Karoui (2003)]. A trader long an option is structurally gamma positive (he benefits from large fluctuations of the underlying) and theta negative (he pays the time value each day): ***gamma scalping***



designates precisely the strategy consisting in financing this daily theta by the readjustment profits generated by the oscillations of the underlying [Bennett (2014)].

### 1.3.2 Combinations of shares and options

In addition to *delta hedging*, which neutralises the sensitivity of a portfolio to variations in the underlying, investors may use strategies combining shares and options in order to protect themselves against adverse market movements while profiting from potential variations. In environments where uncertainty is high, it is indeed fundamental to seek protection (a hedge) against adverse movements. Thus, the strategy of the **protective put** consists in buying a *put* on a share that one already holds. This makes it possible to protect oneself against a large fall in the price of the share while retaining the potential for gain in the event of a rise. Indeed, suppose that one buys a share at a price  $S_0$  and that one wishes to resell it at horizon  $T$ . Potentially, one may lose up to  $S_0$  (if the price of the share falls to 0 at date  $T$ ). More generally, the *payoff* at date  $T$  of the share will be  $S_T - S_0$ , where  $S_T$  denotes the price of the share at date  $T$  (which corresponds to a loss of  $S_0 - S_T$ ). Suppose that one wishes to limit one's potential loss. Let us then buy a *put* with *strike*  $K$  on the share at maturity  $T$ . Suppose that this option costs  $\pi$ . At maturity, the latter will pay us  $\max(K - S_T, 0)$ , whence a final *payoff* of  $\max(K - S_T, 0) - \pi$ . Two cases are then possible:

- Either  $S_T > K$ : one will not exercise the sale option and one will obtain 0 from it at maturity, that is a final *payoff* of  $-\pi$ . Taking account of the purchase of the share which yields  $S_T - S_0$ , this gives us a final financial flow of  $(S_T - S_0) - \pi > K - S_0 - \pi$ .
- Or  $S_T \leq K$ : one will then exercise the sale option and one will obtain  $K - S_T$  from it at maturity, that is a final *payoff* of  $K - S_T - \pi$ . Taking account of the purchase of the share which yields  $S_T - S_0$ , this gives us a constant final financial flow of  $(S_T - S_0) + (K - S_T - \pi) = K - \pi - S_0$ .

We then observe that the final financial flow is limited to a potential loss of  $K - S_0 - \pi$ . Relative to a situation without hedging which would yield  $S_T - S_0$ , we see that the purchase of a *put* costs  $\pi$  but limits the potential loss. In reality, as figure 4.5 shows, buying a *protective put* amounts exactly to buying a *call* of the same *strike* and the same maturity as the *put* constituting the protective arrangement. This result is not a graphical coincidence: it is precisely the statement of put-call parity, a fundamental arbitrage relation in option theory. Denoting by  $C$  the price of the *call* and by  $P$  that of the *put* of the same *strike*  $K$  and the same maturity  $T$ , and designating by  $B(0, T) = (1 + r)^{-T}$  the riskless discount factor, it is written  $S_0 + P = C + KB(0, T)$ . The left-hand side represents exactly the cost of the *protective put* (purchase of the share at the price  $S_0$  and purchase of the *put* at the price  $P$ ), while the right-hand side represents an alternative strategy of identical total cost (purchase of the *call* at the price  $C$  and placement of the discounted value of the *strike*  $KB(0, T)$  in the riskless asset, so as to have exactly  $K$  euros available at maturity  $T$ ). Why then, if the *protective put* costs exactly as much as the purchase of the *call* alone, would an investor ever choose to tie up  $S_0$  euros in the share in order to hold from it only a *payoff* identical to that of a *call* which costs only  $C < S_0$ ? Several frictions make the *protective put* rational.

- **The investor already holds the share.** The most common case is that of a shareholder who does not wish to dispose of his holding — for tax reasons (deferral of taxation of an unrealised capital gain), regulatory reasons (a director subject to *lock-up* clauses), or strategic reasons (maintenance of a controlling block) — but wishes to protect himself against a temporary fall: he then buys a *put* on the shares he already owns, without having to sell them, nor to reconstruct a synthetic position via a *call* and a zero-coupon.
- **The *call* is not available or is illiquid.** For small-capitalisation shares, or on certain emerging markets, the listed *call* quite simply does not exist or presents prohibitive *bid-ask spreads*; the *put* may be more liquid (especially in phases of stress, where demand for downside protection is strong).

Analogously, the strategy of the **protective call** consists in buying a *call* on a share that one sells short. This makes it possible to protect oneself against a large rise in the price of the share while retaining the potential for gain in the event of a fall. Indeed, suppose that one sells a share short at a price  $S_0$  and that one wishes to buy it back at horizon  $T$ . Potentially, one may have an unlimited loss (if the price of the share shoots off into space). More generally, the *payoff* at date  $T$  of the share will be  $S_0 - S_T$ , where  $S_T$  denotes the price of the share at date  $T$  (which corresponds to a loss of  $S_T - S_0$ ). Suppose that one wishes to limit one's potential loss. Let us then buy a *call* with *strike*  $K$  on the share at maturity  $T$ . Suppose that this option costs  $\pi$ . At maturity, the latter will pay us  $\max(S_T - K, 0)$ , whence a final *payoff* of  $\max(S_T - K, 0) - \pi$ . Two cases are then possible:

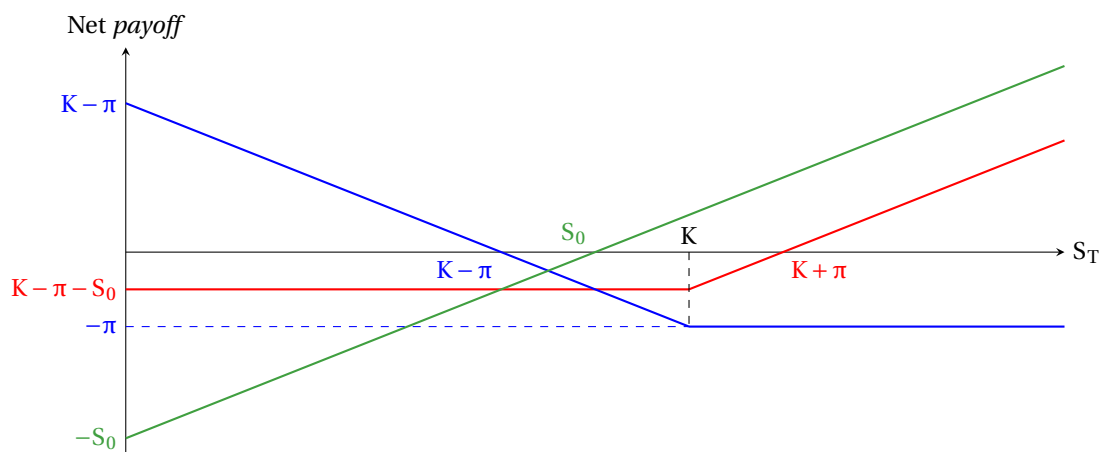


Figure 4.5: *Payoff* of a *protective put* as a function of the final price  $S_T$  of a share at date  $T$ : purchase of a *share* at the initial price  $S_0$  and of a *put* of price  $\pi$ , *strike*  $K$  and maturity  $T$

- Either  $S_T < K$ : one will not exercise the purchase option and one will obtain 0 from it at maturity, that is a final *payoff* of  $-\pi$ . Taking account of the purchase of the share which yields  $S_0 - S_T$ , this gives us a final financial flow of  $(S_0 - S_T) - \pi > S_0 - K - \pi$ .
- Or  $S_T \geq K$ : one will then exercise the purchase option and one will obtain  $S_T - K$  from it at maturity, that is a final *payoff* of  $S_T - K - \pi$ . Taking account of the purchase of the share which yields  $S_0 - S_T$ , this gives us a constant final financial flow of  $(S_0 - S_T) + (S_T - K - \pi) = S_0 - K - \pi$ .

We then observe that the final financial flow is limited to a potential loss of  $S_0 - K - \pi$ . Relative to a situation without hedging which would yield  $S_0 - S_T$ , we see that the purchase of a *call* costs  $\pi$  but limits the potential loss. In reality, as figure 4.6 shows, buying a *protective call* amounts exactly to buying a *put* of the same *strike* and the same maturity as the *call* constituting the protective arrangement. Acquiring a *protective call* at the net cost  $\pi - S_0$  — that is to say selling the share short at the price  $S_0$  and simultaneously buying a *call* of *strike*  $K$  and maturity  $T$ , for a net cost of  $C - S_0$  — amounts exactly, by the rearranged put-call parity  $C - S_0 = P - KB(0, T)$ , to buying a *put* of the same *strike* and the same maturity at the price  $P$  and simultaneously borrowing  $KB(0, T)$  at the riskless rate. The left-hand side represents the net cost of the *protective call* (receipt of  $S_0$  on the short sale, payment of the premium  $C$ ), while the right-hand side represents an alternative strategy of identical cost (purchase of the *put* at the price  $P$ , receipt of the borrowing  $KB(0, T)$  repayable by  $K$  at maturity). Why then would a bearish investor choose to sell the share short and buy a *call* rather than simply buying a *put* backed by a borrowing, which is rigorously equivalent for him? Several frictions make the *protective call* rational:

- **The investor is already short and wants to cap his risk.** This is the dominant use case: an operator having initiated a short sale in anticipation of a fall in the price is exposed to a theoretically unbounded loss if the price soars. Buying a *call* of *strike*  $K$  on the existing short position caps this maximum loss at  $K - S_0 + C$ : beyond  $K$ , the gain on the *call* exactly offsets the loss on the short sale. The operator has no reason to unwind his short position in order to reconstruct a *put* synthetically via a borrowing.
- **The *put* is not available or is illiquid.** Symmetrically to the *protective put* for the seller, the listed *put* may be illiquid or non-existent on certain underlyings — small capitalisations, shares of emerging markets — notably outside the phases of stress where demand for downside protection is strong; the *call* is in that case often more liquid since bullish speculative demand is structurally deeper than bearish hedging demand on most equity markets.

As a complement to strategies such as the *protective put*, investors may likewise use optional hedging strategies such as the *covered call* and the *covered put*. These strategies make it possible to generate additional income or to protect oneself against adverse market movements while retaining a certain exposure to variations in the underlying. The strategy of the *covered call* consists in selling a *call* on a share that one already holds. This approach makes it possible to generate additional income via the premium received on the sale of the option, while nevertheless limiting the potential gain if the price of the share exceeds the *strike* of the *call*. Concretely, suppose that one owns a

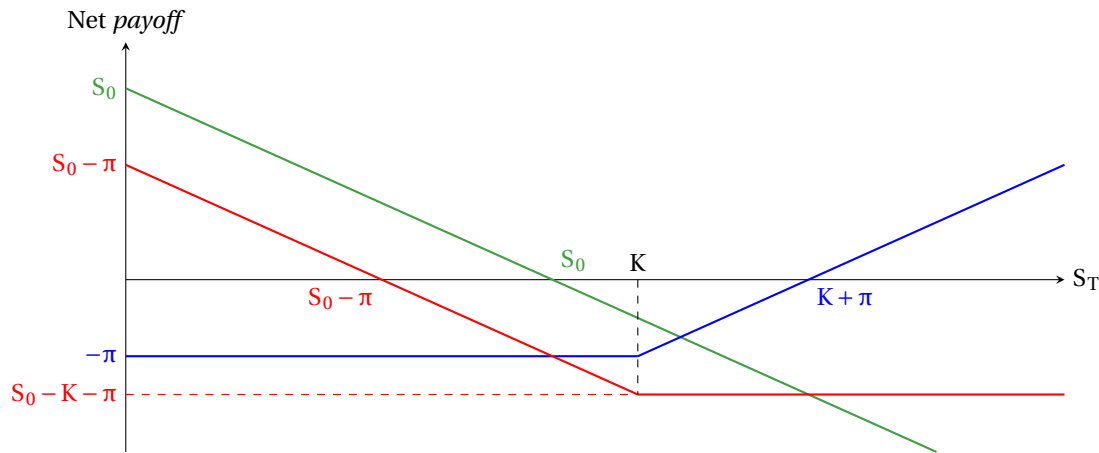


Figure 4.6: *Payoff* of a **protective call** as a function of the final price  $S_T$  of a share at date  $T$ : short sale of a **share** at the initial price  $S_0$  and purchase of a **call** of price  $\pi$ , *strike*  $K$  and maturity  $T$ . The maximum loss is capped at  $S_0 - K - \pi$  if the price soars beyond  $K$ ; the maximum gain is attained for  $S_T = 0$  and equals  $S_0 - \pi$ .

share bought at a price  $S_0$  and that one sells a *call* of *strike*  $K$  for a premium of  $\pi$ . At the maturity of the option, two scenarios present themselves:

- If  $S_T \leq K$ , the *call* option will not be exercised, and the investor will keep the premium of  $\pi$  in addition to the potential gain on the share ( $S_T - S_0$ ). The total *payoff* is therefore  $(S_T - S_0) + \pi$ .
- If  $S_T > K$ , the option will be exercised, and the investor will have to sell the share at the price of  $K$ . The gain on the share is then limited to  $K - S_0$ , but the premium of the option is still retained. The total *payoff* is therefore  $(K - S_0) + \pi$ , which constitutes the maximum gain in this case.

As figure 4.7 shows, being long a *covered call* amounts to being *short* the *put* of the same *strike* and the same maturity as the *call* constituting the hedging arrangement. More precisely, acquiring a *covered call* at the net cost  $S_0 - C$  — that is to say buying the share at the price  $S_0$  and simultaneously selling a *call* of *strike*  $K$  and maturity  $T$  — amounts exactly, by the rearranged put-call parity  $S_0 - C = KB(0, T) - P$ , to acquiring a zero-coupon paying  $K$  at maturity at the price  $KB(0, T)$  and simultaneously selling a *put* of the same *strike* and the same maturity at the price  $P$ . The left-hand side represents exactly the net cost of the *covered call* (purchase of the share at the price  $S_0$ , receipt of the premium  $C$  on the *call* sold), while the right-hand side represents an alternative strategy of identical total cost (purchase of a zero-coupon of final value  $K$  at the price  $KB(0, T)$ , receipt of the premium  $P$  on the *put* sold). Why then would an investor choose to tie up  $S_0$  euros in the share and sell a *call* rather than simply selling a *put* backed by a zero-coupon, which is rigorously equivalent for him? Several frictions here again make the *covered call* rational:

- **The investor already holds the share and wants to improve its return.** This is the dominant use case (*buy-write* or *yield enhancement*): a long holder who anticipates a stagnation of the price in the medium term agrees to cap his potential gain beyond  $K$  in exchange for an immediate receipt of the premium  $C$ , thereby reducing his effective cost price from  $S_0$  to  $S_0 - C$ . He has no reason to dispose of his shares in order to construct an equivalent synthetic position.
- **Selling a naked *put* is constrained by regulation.** The equivalent strategy ‘zero-coupon + *short put*’ requires selling a *put* without cover in shares: under Solvency II, this position generates a high SCR capital requirement on account of its theoretically unbounded exposure; under the UCITS regulation, it consumes a significant fraction of the derivatives exposure budget, potentially blocking other positions in the portfolio. By contrast, the sale of a *call* covered by the physical holding of the share creates no additional net exposure in the sense of the commitment approach, since the delivery risk is entirely neutralised by the underlying asset held in the portfolio.

Conversely, the strategy of the **covered put** consists in selling a *put* on a share that one has sold short. This strategy is used in order to generate additional income while protecting oneself against a fall in the price of the share. Suppose that one sells a share short at a price  $S_0$  and that one likewise sells a *put* of *strike*  $K$  for a premium of  $\pi$ . At the maturity of the option, two scenarios present themselves:

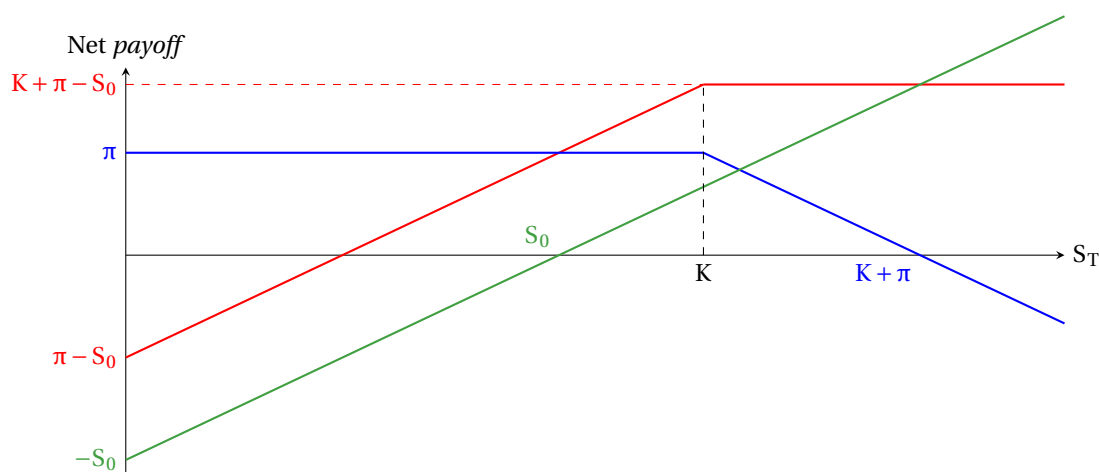


Figure 4.7: *Payoff* of a *covered call* as a function of the final price  $S_T$  of a share at date  $T$ : purchase of a *share* at the initial price  $S_0$  and short sale of a *call* of price  $\pi$ , *strike*  $K$  and maturity  $T$

- If  $S_T \geq K$ , the *put* option will not be exercised, and the investor will keep the premium of  $\pi$  in addition to the gain realised on the short sale ( $S_0 - S_T$ ). The total *payoff* is therefore  $(S_0 - S_T) + \pi$ .
- If  $S_T < K$ , the option will be exercised, and the investor will have to buy the share at the price of  $K$ , thereby limiting the gain of the short sale to  $(S_0 - K)$ . The total *payoff* is therefore  $(S_0 - K) + \pi$ , which constitutes the maximum gain in this case.

As figure 4.8 shows, being long a *covered put* amounts to being *short* the *call* of the same *strike* and the same maturity as the *put* constituting the hedging arrangement. More precisely, acquiring a *covered put* at the net cost  $KB(0, T) - P$  — that is to say placing  $KB(0, T)$  in the riskless asset and simultaneously selling a *put* of *strike*  $K < S_0$  and maturity  $T$ , receiving the premium  $P$  — amounts exactly, by the rearranged put-call parity  $KB(0, T) - P = S_0 - C$ , to buying the share at the price  $S_0$  and simultaneously selling a *call* of the same *strike* and the same maturity. The left-hand side represents the net cost of the *covered put* (placement of the zero-coupon, receipt of the premium  $P$ ), while the right-hand side is its synthetic equivalent (purchase of the share, receipt of the premium  $C$  on the *call* sold). These strategies are particularly useful in market environments where volatility is moderate and where investors wish to improve the return of their portfolio while retaining an exposure to the underlying shares. Otherwise put, as Hull (2012) emphasises, the limitation of gains is often perceived as acceptable in markets where strong rises are judged unlikely: the investor chooses to secure a more moderate but more certain gain, rather than to speculate on a significant rise.

### 1.3.3 Combinations of options

As a complement to strategies combining shares and options such as *protective puts* and *covered calls*, investors may likewise use pure option strategies in order to position themselves according to their market expectations. Among these strategies, *straddles*, *strangles*, *bull spreads*, and *bear spreads* are particularly popular for taking advantage of expected (or unexpected) variations in the underlying. Thus, the strategy of the *straddle* consists in simultaneously buying a *call* and a *put* with the same *strike*  $K$  and the same expiry  $T$ . As Hull (2012) indicates, this strategy is used when the investor anticipates strong volatility of the underlying, without having any particular conviction about the direction of the movement. Indeed, the final *payoff* will be the sum of the *payoff* of the *call* and the *payoff* of the *put*, that is  $\max(S_T - K, 0) + \max(K - S_T, 0)$ . Two cases are then possible: if  $S_T = K$ , both options expire *out of the money*, and the *payoff* is negative, equal to  $C - P$ , where  $C$  and  $P$  represent respectively the price of the *call* and of the *put*; by contrast, if  $S_T \neq K$ , one of the options will expire *in the money*, and the *payoff* will increase as a function of the gap between  $S_T$  and  $K$ , potentially to the point of offsetting the initial loss and generating a profit. The *payoff* of a *straddle* is illustrated in figure 4.9. The strategy of the *strangle* is similar to the *straddle*, but with different *strikes* for the *call* and the *put*. The idea is to buy a *call* with a *strike*  $K_C$  higher than the *strike*  $K_P$  of the *put*. Hull (2012) specifies that the *strangle* is less costly than a *straddle* since the options are bought *out of the money*, but it requires a larger movement of the underlying in order to be profitable. The final *payoff* is therefore  $\max(S_T - K_C, 0) + \max(K_P - S_T, 0)$ . The two scenarios are the following: if  $S_T$  remains between  $K_P$  and  $K_C$ , both options expire *out of the money*, and the

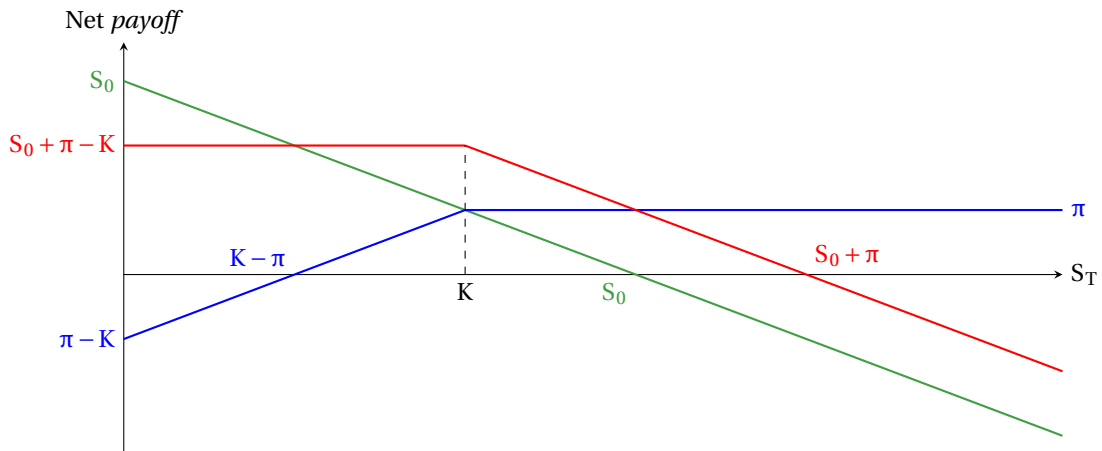


Figure 4.8: *Payoff* of a **covered put** as a function of the final price  $S_T$  of a share at date  $T$  : short sale of a **share** at the initial price  $S_0$  and short sale of a **put** of price  $\pi$ , *strike*  $K$  and maturity  $T$ . The gain is capped at  $S_0 + \pi - K$  if the price falls below  $K$  ; the strategy is loss-making beyond the *break-even* threshold  $S_0 + \pi$  and the loss is unbounded if the price soars.

*payoff* is negative, equal to  $-C - P$ ; by contrast, if  $S_T$  leaves this range, one of the options will expire *in the money*, and the *payoff* will increase, potentially to the point of generating a net profit. The *payoff* of a *strangle* is illustrated in figure 4.10.

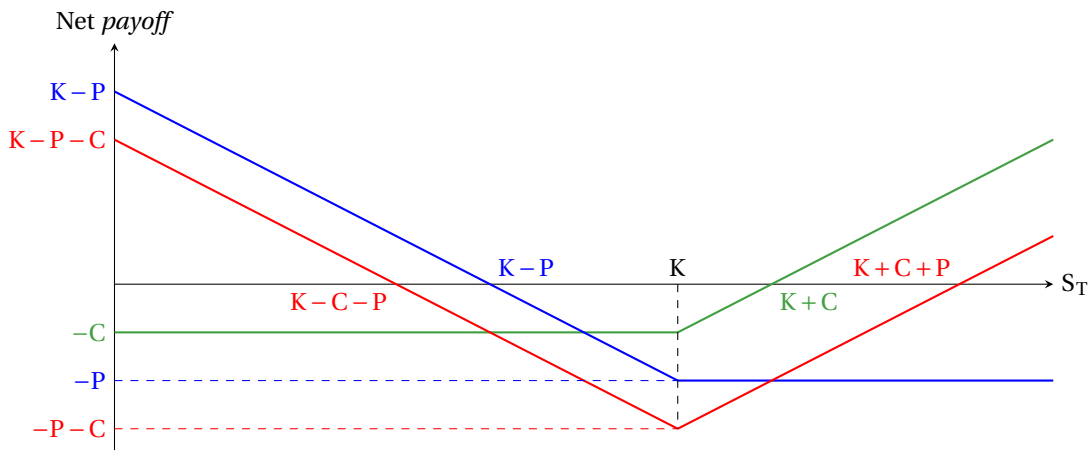


Figure 4.9: *Payoff* of a **straddle** as a function of the final price  $S_T$  of the underlying at date  $T$ : simultaneous purchase of a **call** and a **put** with the same *strike*  $K$  and the same maturity  $T$ , at the respective prices  $C$  and  $P$

The *bull spread* and *bear spread* strategies use *calls* or *puts* in order to profit from a moderate directional expectation. A **bull spread** is created by buying a *call* of *strike*  $K_1$  and selling a *call* of *strike*  $K_2$  with  $K_2 > K_1$ . This strategy is profitable if the underlying rises, but the gains are limited by the sale of the *call* at  $K_2$ . The *payoff* of such a combination of options is illustrated in figure 4.11. Conversely, a **bear spread** (figure 4.12) is constructed by buying a *put* of *strike*  $K_2$  and selling a *put* of *strike*  $K_1$  ( $K_2 > K_1$ ). This strategy is profitable if the underlying falls, but the gains are limited by the sale of the *put* at  $K_1$ . As Hull (2012) explains, these strategies are often used in order to limit risks while retaining a reasonable potential for gain. They are particularly suited to markets where the investor has a clear, but moderate, expectation about the direction of the underlying. From a practical point of view, these strategies present the advantage of a reduced initial cost relative to the simple purchase of an option : the premium received on the option sold partially finances the option bought, so that the investor sacrifices part of the potential for gain in exchange for a less capital-intensive entry into the position.

Finally, a notable combination of options is the **synthetic long**, which consists in buying a *call* and selling a *put* of the same *strike*  $K$  and the same maturity  $T$ . By put-call parity, the *payoff* at maturity of this combination is worth exactly  $S_T - K$ , which is identical to that of a long position on the underlying bought at the price  $K$ ; symmetrically, the

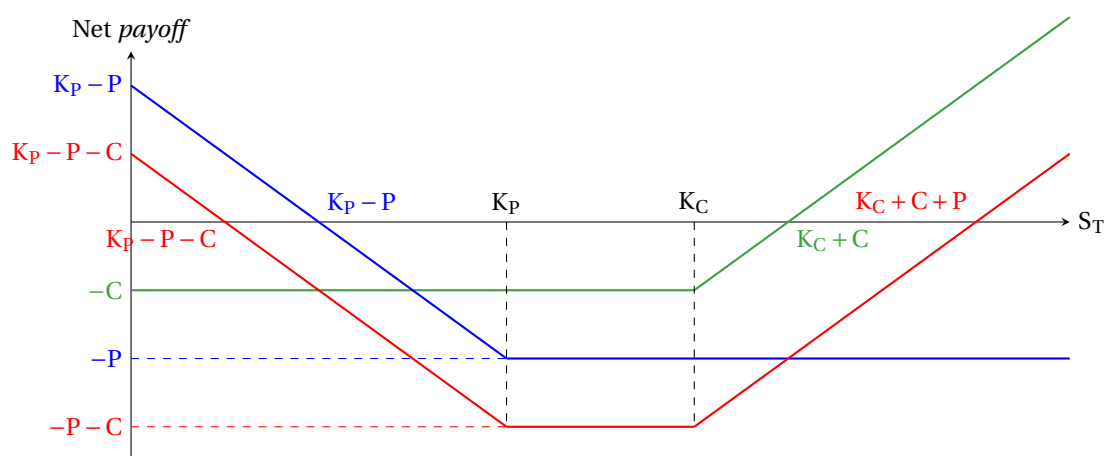


Figure 4.10: Payoff of a **strangle** as a function of the final price  $S_T$  of the underlying at date T: simultaneous purchase of a **call** with strike  $K_C$  and of a **put** with strike  $K_P$  ( $K_P < K_C$ ), both of maturity T and at the respective prices C and P

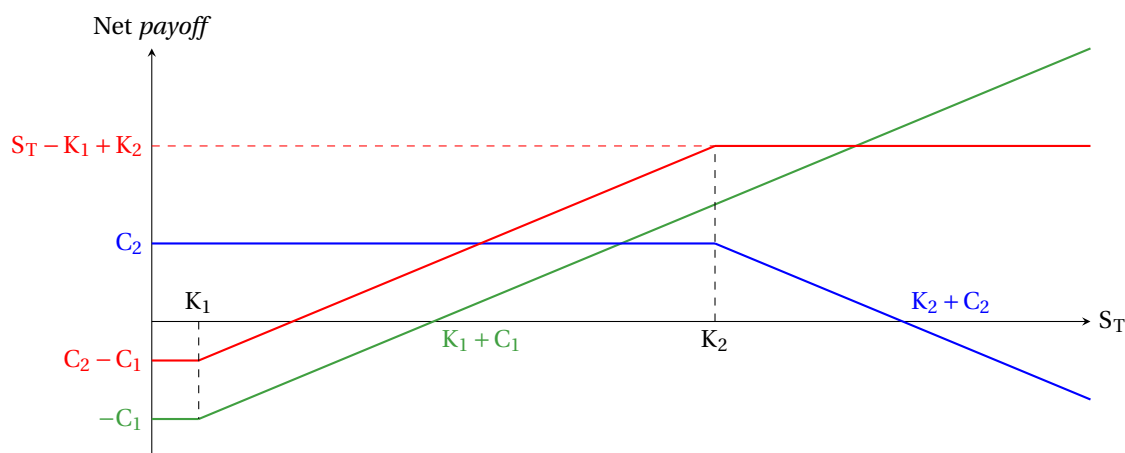


Figure 4.11: Payoff of a **bull spread** as a function of the final price  $S_T$  of the underlying at date T: purchase of a **call** with strike  $K_1$  of maturity T at the price  $C_1$  and sale of a **call** with strike  $K_2$  of maturity T at the price  $C_2$ , with  $K_2 > K_1$

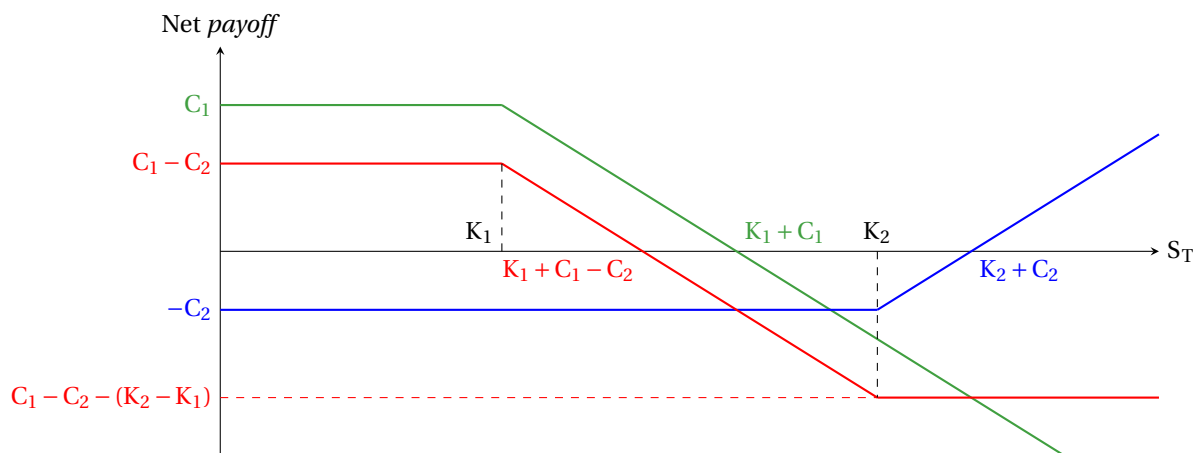


Figure 4.12: Payoff of a **bear spread** as a function of the final price  $S_T$  of the underlying at date T: sale of a **call** with strike  $K_1$  of maturity T at the price  $C_1$  and purchase of a **call** with strike  $K_2$  of maturity T at the price  $C_2$ , with  $K_2 > K_1$

**synthetic short** — sale of a **call** and purchase of a **put** of the same **strike** — replicates a short position. The practical

interest of these constructions lies in their capital efficiency: whereas a direct long position on the underlying ties up the entirety of the price  $S_0$ , the *synthetic long* requires only an initial flow equal to the difference of premiums  $C - P$ , which is generally small or even close to nil when the *strike*  $K$  is chosen close to the money (*at the money*). The investor thus benefits from an economic exposure identical to a direct position on the underlying for a far smaller outlay, which amounts to introducing an **implicit leverage effect**: any variation of  $S_T$  is passed on in full to the profit of the strategy, but on a markedly smaller base of capital committed, mechanically amplifying the return on capital invested — and the risk of loss in proportion.

### Exercise



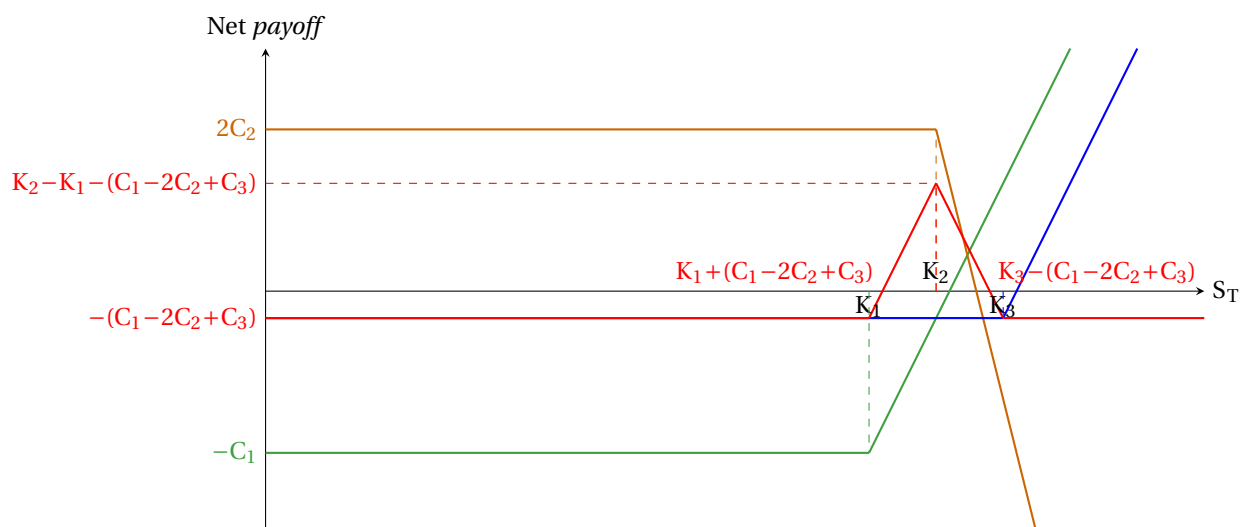
**Butterfly strategy on purchase options.** An investor anticipates low volatility of the XYZ security, whose current price is  $S_0 = 100$  euros. He puts in place a *long butterfly* using European purchase options of the same expiry  $T = 3$  months, at the exercise prices  $K_1 = 90$  euros,  $K_2 = 100$  euros and  $K_3 = 110$  euros, with the following premiums:  $C_1 = 12$  euros,  $C_2 = 6$  euros and  $C_3 = 2$  euros. The strategy consists in (i) buying a *call* of *strike*  $K_1 = 90$  €; (ii) selling two *calls* of *strike*  $K_2 = 100$  euros; (iii) buying a *call* of *strike*  $K_3 = 110$  euros.

- (i) Compute the net cost of setting up the strategy. Interpret its sign.
- (ii) Express the profit at maturity  $\Pi(S_T)$  as a function of the price  $S_T$ , distinguishing the cases  $S_T \leq K_1$ ,  $K_1 < S_T \leq K_2$ ,  $K_2 < S_T \leq K_3$  and  $S_T > K_3$ .
- (iii) Determine the two break-even points of the strategy.
- (iv) Compute the maximum gain and the maximum loss, and specify at which points they are attained. Comment.
- (v) Represent graphically the profit profile  $\Pi(S_T)$  as a function of  $S_T$ .
- (vi) Compare this profile with that of a *short straddle* of the same central *strike*  $K_2 = 100$  euros, assuming that the *put* of *strike*  $K_2$  costs 4 euros. What are the respective advantages and disadvantages of the two strategies?

**Solution.** (i) The net cost of setting up is worth  $C_1 - 2C_2 + C_3 = 12 - 2 \times 6 + 2 = 2$  euros. The strategy is a net debit: the investor pays 2 euros in order to put the *butterfly* in place.

- (ii) The profit at maturity is written  $\Pi(S_T) = \max(S_T - K_1, 0) - 2\max(S_T - K_2, 0) + \max(S_T - K_3, 0) - 2$ . Distinguishing the zones: if  $S_T \leq 90$ : all the *calls* expire worthless and  $\Pi = -2 < 0$ ; if  $90 < S_T \leq 100$ : only the *call*  $K_1$  is exercised and  $\Pi = (S_T - 90) - 2 = S_T - 92 \in ]-2, 8]$ ; if  $100 < S_T \leq 110$ : the *calls*  $K_1$  and  $K_2$  are exercised and  $\Pi = (S_T - 90) - 2(S_T - 100) - 2 = -S_T + 108 \in ]-2, 8]$ ; if  $S_T > 110$ : all three *calls* are exercised and  $\Pi = (S_T - 90) - 2(S_T - 100) + (S_T - 110) - 2 = -2$ .
- (iii) The break-even points are obtained by solving  $\Pi(S_T) = 0$ . For the left branch ( $S_T$  between 90 and 100),  $S_T - 92 = 0$  if, and only if,  $S_T^* = 92$  euros while, for the right branch ( $S_T$  between 100 and 110),  $-S_T + 108 = 0$  if, and only if,  $S_T^{**} = 108$  euros. The two break-even points are therefore symmetric about the central *strike*.
- (iv) The maximum gain is attained at  $S_T = K_2 = 100$  euros: we then have  $\Pi_{\max} = 100 - 92 = 8$  euros. Conversely, the maximum loss is attained for  $S_T \leq K_1$  or  $S_T \geq K_3$ : we then have  $\Pi_{\min} = -2$  euros. The gain-to-loss ratio is therefore  $8/2 = 4$ , which illustrates the attraction of the strategy for an expectation of low volatility.
- (v) The graph of the profit is the following:





- (vi) The *short straddle* consists in simultaneously selling a *call* and a *put* of the same *strike*  $K_2 = 100$  euros and the same maturity. The total premium received is worth  $C_2 + P_2 = 6 + 4 = 10$  euros, and the profit at maturity is written:

$$\Pi_{straddle}(S_T) = \begin{cases} S_T - 90 & \text{if } S_T \leq 100 \\ 110 - S_T & \text{if } S_T > 100 \end{cases}$$

The break-even points are therefore  $S_T^* = 90$  euros and  $S_T^{**} = 110$  euros, symmetric about  $K_2$ ; the maximum gain of 10 euros is attained at  $S_T = K_2 = 100$  euros, while the loss is unlimited on both sides. The two strategies therefore share the same directional intuition — betting on a low realisation of volatility around  $K_2$  — but differ radically in their risk profile. On the side of the *long butterfly*, the maximum loss is strictly bounded by the net premium paid (2 euros), whatever the variations of the underlying, which distinguishes it fundamentally from the *short straddle* whose sold legs expose it to theoretically unlimited losses in the event of a marked departure of  $S_T$  from  $K_2$ . The *short straddle* nevertheless presents its own advantages which justify its use by operators with sufficient own funds. Its maximum gain of 10 euros is higher than that of the *butterfly* (8 euros), and its zone of profit — the corridor  $]90, 110[$  of width 20 — is more extensive than that of the *butterfly*  $]92, 108[$ . Moreover, the immediate receipt of a premium of 10 euros at initiation constitutes a significant cash flow advantage: this sum may be placed over the duration of the strategy and generate an intermediate return, which the *butterfly*, at a net debit, does not permit.  $\square$

### Exercise



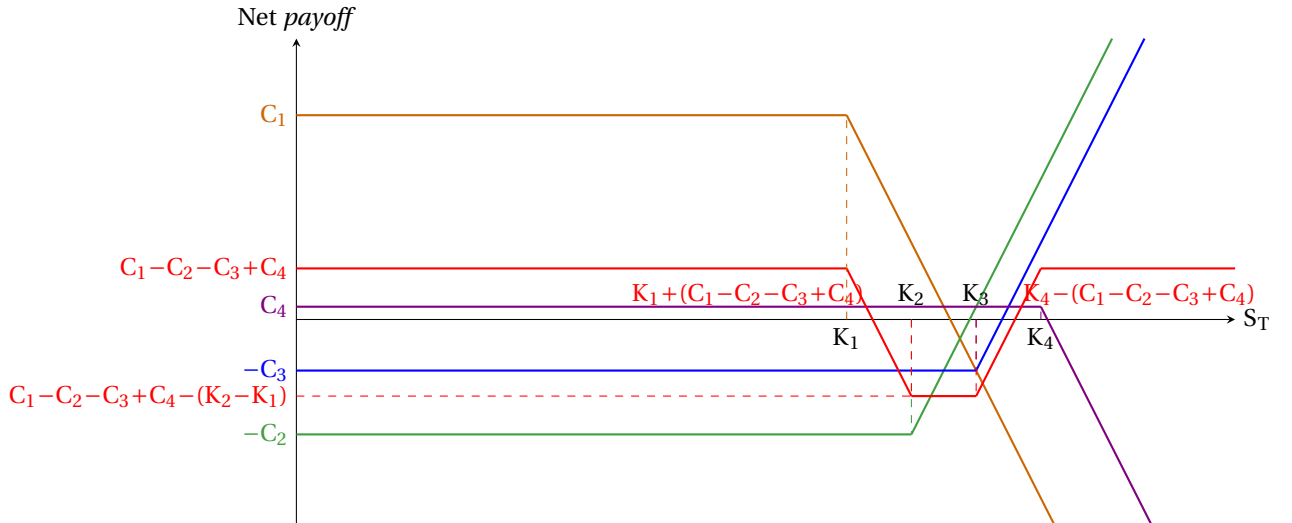
**Short condor strategy on purchase options.** An investor anticipates strong volatility of the XYZ security, whose current price is  $S_0 = 100$  euros. He puts in place a *short condor* using European purchase options of the same expiry  $T = 3$  months, at the exercise prices  $K_1 = 85$  euros,  $K_2 = 95$  euros,  $K_3 = 105$  euros and  $K_4 = 115$  euros, with the following premiums:  $C_1 = 16$  euros,  $C_2 = 9$  euros,  $C_3 = 4$  euros and  $C_4 = 1$  euro. The strategy consists in (i) selling a *call* of *strike*  $K_1 = 85$  euros; (ii) buying a *call* of *strike*  $K_2 = 95$  euros; (iii) buying a *call* of *strike*  $K_3 = 105$  euros; (iv) selling a *call* of *strike*  $K_4 = 115$  euros.

- Compute the net credit received on setting up the strategy. Interpret its sign and compare with the *long butterfly* of the previous exercise.
- Express the profit at maturity  $\Pi(S_T)$  as a function of the price  $S_T$ , distinguishing the five cases  $S_T \leq K_1$ ,  $K_1 < S_T \leq K_2$ ,  $K_2 < S_T \leq K_3$ ,  $K_3 < S_T \leq K_4$  and  $S_T > K_4$ .
- Determine the two break-even points of the strategy.
- Compute the maximum gain and the maximum loss, and specify at which points they are attained. Comment on the economic intuition.
- Represent graphically the profit profile  $\Pi(S_T)$  as a function of  $S_T$ .

- (vi) Compare this profile with that of a *strangle* bought with strikes  $K_A = 90$  euros (*put* bought, premium  $P_A = 5$  euros) and  $K_B = 110$  euros (*call* bought, premium  $C_B = 3$  euros). What are the respective advantages and disadvantages of the two strategies?

**Solution.** (i) The net credit received at initiation is worth  $C_1 - C_2 - C_3 + C_4 = 16 - 9 - 4 + 1 = +4$  euros. The strategy is a net credit: unlike the *long butterfly* of the previous exercise which requires a debit of 2 euros, the *short condor* generates an immediate receipt of 4 euros. This credit constitutes the maximum gain of the strategy, attained when the underlying moves very markedly, confirming that the *short condor* is a long volatility strategy.

- (ii) The profit at maturity is written  $\Pi(S_T) = -\max(S_T - K_1, 0) + \max(S_T - K_2, 0) + \max(S_T - K_3, 0) - \max(S_T - K_4, 0) + 4$ . Distinguishing the five zones: if  $S_T \leq 85$ , no *call* is exercised and  $\Pi = +4$  euros; if  $85 < S_T \leq 95$ , only the sold *call*  $K_1$  is exercised against us and  $\Pi = -(S_T - 85) + 4 = -S_T + 89 \in [-6, 4]$ ; if  $95 < S_T \leq 105$ , the *calls*  $K_1$  (sold) and  $K_2$  (bought) are exercised and  $\Pi = -(S_T - 85) + (S_T - 95) + 4 = -6$  euros; if  $105 < S_T \leq 115$ : the *calls*  $K_1$  (sold),  $K_2$  and  $K_3$  (bought) are exercised and  $\Pi = -(S_T - 85) + (S_T - 95) + (S_T - 105) + 4 = S_T - 111 \in [-6, 4]$ ; if  $S_T > 115$ , all four *calls* are exercised and  $\Pi = -(S_T - 85) + (S_T - 95) + (S_T - 105) - (S_T - 115) + 4 = +4$  euros.
- (iii) The break-even points are obtained by solving  $\Pi(S_T) = 0$ . For the left branch ( $S_T$  between 85 and 95),  $-S_T + 89 = 0$  if, and only if,  $S_T^* = 89$  euros; for the right branch ( $S_T$  between 105 and 115),  $S_T - 111 = 0$  if, and only if,  $S_T^{**} = 111$  euros. The two break-even points are symmetric about the centre  $\frac{K_2 + K_3}{2} = 100$  euros, at a distance of  $(K_2 - K_1) - (C_1 - C_2 - C_3 + C_4) = 10 - 4 = 6$  euros on either side.
- (iv) The maximum gain is worth  $\Pi_{\max} = +4$  euros; it is attained for  $S_T \leq K_1 = 85$  euros or  $S_T \geq K_4 = 115$  euros, that is to say when the underlying moves sufficiently far from the central corridor. Conversely, the maximum loss is worth  $\Pi_{\min} = -6$  euros; it is attained for every  $S_T \in [K_2, K_3] = [95, 105]$  euros, that is to say when the underlying remains perfectly stable around  $S_0 = 100$  euros. The loss-to-gain ratio is worth  $6/4 = 1.5$ : the investor risks 1.5 euros in order to gain 1, which is the counterpart of a net credit strategy; unlike the *long butterfly*, the gains are here capped but the maximum loss is likewise bounded.
- (v) The graph of the profit is the following:



- (vi) The *long strangle* consists in simultaneously buying a *put* of strike  $K_A = 90$  euros and a *call* of strike  $K_B = 110$  euros, both being out of the money (*out-of-the-money*). The total premium paid is worth  $P_A + C_B = 5 + 3 = 8$  euros, and the profit at maturity is written:

$$\Pi_{\text{strangle}}(S_T) = \begin{cases} 82 - S_T & \text{if } S_T < 90 \\ -8 & \text{if } 90 \leq S_T \leq 110 \\ S_T - 118 & \text{if } S_T > 110 \end{cases}$$

The break-even points are  $S_T^* = 82$  euros and  $S_T^{**} = 118$  euros, symmetric about 100 euros; the **maximum loss** of 8 euros is attained over the whole range [90, 110] euros, while the gain is unlimited beyond the break-even points. Thus, the two strategies share the same directional intuition – betting on a strong realisation of volatility, that is to say on the underlying leaving a corridor – but differ fundamentally in their risk profile. The *long strangle* generates a net debit of 8 euros, whereas the *short condor* generates a net credit of 4 euros, and its zone of maximum loss is wider since any price between  $K_A = 90$  euros and  $K_B = 110$  euros leads to losing the entire premium; moreover, its break-even points (82 euros and 118 euros) are further from the current price, requiring a larger movement in order to be profitable. These disadvantages nevertheless have a major counterpart: the gain of the *long strangle* is theoretically unlimited in both directions beyond the break-even points, and the bought positions generate no margin requirement. Ultimately, the *long strangle* suits an investor anticipating a violent movement of the underlying and seeking a potentially unlimited gain, at the price of a higher premium and a wider zone of loss. Moreover, the immediate receipt of a premium of 4 euros at the initiation of the *short condor* constitutes a significant cash flow advantage: this sum may be placed over the duration of the strategy and generate an intermediate return, which the *long strangle*, at a net debit, does not permit.  $\square$

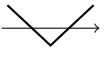
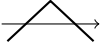
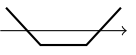
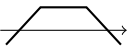
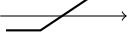
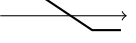
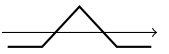
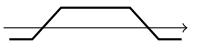
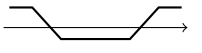
| Strategy                | P&L profile                                                                         | Composition                                                                                                 | Market expectation                                                 |
|-------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Straddle bought</b>  |    | Purchase of a <i>call</i> and purchase of a <i>put</i> (same strike)                                        | Large fluctuations expected, direction unknown                     |
| <b>Straddle sold</b>    |    | Sale of a <i>call</i> and sale of a <i>put</i> (same strike)                                                | Small fluctuations expected, direction unknown                     |
| <b>Strangle bought</b>  |    | Purchase of a <i>call</i> and purchase of a <i>put</i> with different strikes                               | Large fluctuation, direction unknown; capping of the maximum loss  |
| <b>Strangle sold</b>    |    | Sale of a <i>call</i> and sale of a <i>put</i> with different strikes                                       | Small fluctuation, direction unknown; capping of the zone of gains |
| <b>Bull spread</b>      |  | Purchase of a <i>call</i> and sale of a <i>call</i> with a higher strike                                    | Moderate rise anticipated; limited losses                          |
| <b>Bear spread</b>      |  | Sale of a <i>call</i> and purchase of a <i>call</i> with a higher strike                                    | Moderate fall anticipated; limited losses                          |
| <b>Butterfly bought</b> |  | Purchase of one <i>call</i> , sale of 2 <i>calls</i> and purchase of one <i>call</i> with different strikes | Small fluctuations, direction unknown; limited gains and losses    |
| <b>Butterfly sold</b>   |  | Sale of one <i>call</i> , purchase of 2 <i>calls</i> and sale of one <i>call</i> with different strikes     | Large fluctuations, direction unknown; limited gains and losses    |
| <b>Condor bought</b>    |  | Sale of 2 <i>calls</i> and purchase of 2 <i>calls</i> at four different strikes                             | Small fluctuations, direction unknown; small gains and losses      |
| <b>Condor sold</b>      |  | Purchase of 2 <i>calls</i> and sale of 2 <i>calls</i> at four different strikes                             | Large fluctuations, direction unknown; small gains and losses      |

Table 4.1: Option combination strategies

## 2 Other common derivatives

Derivatives markets today represent considerable volumes: at the end of 2024, the notional amount of over-the-counter derivatives alone reached roughly 728 trillion dollars according to the BIS, of which 548 trillion dollars for interest rate products alone according to ISDA (2025). On organised markets, FIA (2025) for its part records 28.22 billion *futures* contracts traded in 2024.

### 2.1 Firm directional commitment: *futures* and *forwards*

Unlike the options studied *supra*, which confer on their buyer a **right** — and not an obligation — to buy or to sell an asset at an agreed price, forward and futures contracts are **firm and symmetric commitments**: both parties are

irrevocably bound to execute the transaction at the date and price fixed at the conclusion of the contract, whatever the market conditions at expiry. This symmetry of rights and obligations — which removes the optional value and therefore the premium to be paid — is the structuring characteristic of these instruments: it implies a linear *payoff*, without asymmetry of gain or loss, which makes them both simpler to value than options and more dangerous to handle for an uncovered operator, since the potential loss is theoretically unbounded.

### 2.1.1 Classic forward contracts

A **forward contract** is an agreement by which two parties commit themselves irrevocably, at the date of conclusion of the contract (date 0), to exchange a specified underlying asset — a share, an index, an interest rate, a currency, a commodity — at a future date  $T$  (the maturity) and at a price  $F_0$  fixed from the moment of conclusion, called the **forward price**. The buyer of the contract takes a **long position**: he commits himself to buying the asset at the price  $F_0$  at maturity, whatever the market value  $S_T$  at that date; the seller for his part takes a **short position**: he commits himself to delivering the asset at the price  $F_0$ . The *payoff* of the long position at maturity is therefore  $S_T - F_0$  (and  $F_0 - S_T$  for the short position), linear and with no premium to be paid at the conclusion — unlike options. This symmetry is the fundamental characteristic of forward contracts: the buyer's gain is exactly the seller's loss and conversely, making these instruments zero-sum games in market value. Two large families of forward contracts are distinguished, whose difference lies essentially in their place and mode of execution:

- **Forwards**, over-the-counter (OTC) forward contracts, negotiated directly between two counterparties, with characteristics (notional, maturity, reference asset, settlement arrangements) entirely bespoke. The bulk of forward contracts on currencies, on interest rates (*forward rate agreements*, FRA), on commodities and on non-standardised shares belongs to this category.
- **Futures**, standardised forward contracts, traded on an organised market: the characteristics of the contract are entirely fixed by the exchange (size, expiries, reference asset, mode of delivery). In return for this standardisation, *futures* benefit from high liquidity and from a near-absence of counterparty risk thanks to the mechanism of the **clearing house**, detailed *infra*.

#### Exercise



**Forward contract.** An investor concludes a *forward* contract bearing on the sale of 100 000 pounds sterling (GBP) against American dollars (USD), at an exchange rate fixed at  $F_0 = 1.4000$  USD/GBP. What is the investor's gain or loss if the exchange rate at the expiry of the contract is  $S_T = 1.3900$  USD/GBP? And if the exchange rate at expiry is  $S_T = 1.4200$  USD/GBP?

**Solution.** Consider an investor who, on 01/01/ $N$ , buys a *forward* contract which *obliges* him on 01/01/ $N + 1$  to sell 100 000 pounds sterling against American dollars at an exchange rate of 1.4000 dollars per pound. So, on 01/01/ $N + 1$ , the seller will therefore recover  $100\,000 \times 1.4000 = 140\,000$  dollars.

- Suppose that on 01/01/ $N + 1$ , the *spot* exchange rate is 1.3900 dollars per pound. Now, the seller must on that day compulsorily sell 100 000 pounds. He therefore acquires them for  $100\,000 \times 1.3900 = 139\,000$  dollars. The seller then resells the 100 000 pounds thus acquired and he then recovers, in accordance with the *forward* contract, 140 000 dollars. Assuming that there have been no additional fees, the net gain is therefore  $140\,000 - 139\,000 = 1\,000$  dollars for this operation.
- Suppose that on 01/01/ $N + 1$ , the *spot* exchange rate is 1.4200 dollars per pound. Now, the seller must on that day compulsorily sell 100 000 pounds. He therefore acquires them for  $100\,000 \times 1.4200 = 142\,000$  dollars. The seller then resells the 100 000 pounds thus acquired and he then recovers, in accordance with the *forward* contract, 140 000 dollars. Assuming that there have been no additional fees, the net loss is therefore  $142\,000 - 140\,000 = 2\,000$  dollars for this operation. □

The most important organisational difference between *forwards* and *futures* is the intervention of a **clearing house** (*central counterparty clearing house*, CCP) on *futures* markets. Through the mechanism of **novation**, the CCP interposes itself between the buyer and the seller from the moment the contract is concluded: it becomes simultaneously the seller facing the buyer, and the buyer facing the seller, assuming the counterparty risk of each party. This mechanism eliminates the bilateral risk of default which is, by contrast, entirely present in an OTC *forward*: if the counterparty of a *forward* defaults before maturity, the *in-the-money* party may find itself uncovered, as Bank for International Settlements (2018) explains. In order to cover this risk, the CCP requires two elements from each participant:

- An **initial margin**: an amount deposited before the opening of any position, sized so as to cover the potential losses over a liquidation period (typically one to two days of adverse variation). At CME Group (2023), the SPAN 2 methodology calibrates the initial margin call so as to cover losses at a confidence level of 99 %, with a historical coverage rate of 99.97 % for standardised products;
- A daily **variation margin** call: each day, the position is revalued at the market price (*mark-to-market*). If the position has lost value, the holder must pay the difference; if he has gained, he receives it. This mechanism prevents the accumulation of unrealised losses and makes the risk of default structurally marginal.

This arrangement was strengthened after the 2008 crisis : the Pittsburgh G20 (September 2009) required that all standardised OTC derivatives be centrally cleared via CCPs by the end of 2012 at the latest, a reform translated in Europe by the **EMIR** regulation (*European market infrastructure regulation*) and in the United States by the *Dodd-Frank Wall Street Reform Act*, as Financial Stability Board (FSB) (2010) relates.

### Exercise



**Computing a margin call.** An investor takes a short position on a *futures* contract on silver metal for July delivery, at the price of  $F_0 = 17.20$  dollars per ounce. The size of the contract is 5000 ounces. The initial margin is 4000 dollars and the maintenance margin is 3000 dollars. What variation in the *futures* price will trigger a margin call?

**Solution.** Suppose that one sells short a *futures* contract on silver which will reach expiry six months after the short sale. The *futures* price is 17.20 dollars per ounce and the size of the contract is 5 000 ounces: the volume of the contract at expiry is therefore  $17.20 \times 5\,000 = 86\,000$  dollars. Otherwise put, six months later, one *will have to* (it is an obligation, unlike the option which leaves the choice) sell 5 000 ounces of silver at 17.20 dollars per ounce.

- The initial margin required by the broker is 4 000 dollars: in order to have the right to buy such a *futures* contract, one must hold *a minima* 4 000 dollars in the account which will be dedicated to this position. The margin required by the broker in order to maintain the position is 3 000 dollars: otherwise put, as soon as the investor has lost more than 1000 dollars, say an amount  $x > 1000$  dollars, he will be in a situation of margin call: he will have to add  $x$  dollars to his account as soon as possible in order to maintain his position, failing which the broker will liquidate the latter (and will therefore take the money directly from the investor's account).
- In practice, at what *spot* price for the ounce of silver does the margin call occur? Since this is a short sale contract, the investor loses money when the price of the ounce of silver rises. If he is constrained, at the expiry of the contract, to buy on the *spot* market 5 000 ounces of silver for a total volume of 87 000 dollars, which he will only be able to resell, in accordance with the *futures* contract, at 86 000 dollars, then he will lose 1000 dollars. Now, knowing that one contract represents 5000 ounces, the price  $p$  triggering the margin call will satisfy  $p \times 5000 = 87\,000$ , that is to say  $p = 17.40$  dollars: as soon as the ounce of silver exceeds 17.40 dollars on the *spot* market, the investor will therefore be in a margin call. The investor will then have to fund his account so as to bring it back to the level of the initial deposit of 4000 dollars, failing which his position will be liquidated.  $\square$

### Focus...



**The spot-forward relation and the cost of carry model.** The determination of the forward price  $F_0$  rests on a pure arbitrage argument, independent of any expectation about the future price of the asset. Suppose that an asset is quoted at  $S_0$  today, that the continuous riskless rate is  $r$ , and that the asset distributes a continuous yield  $q$  (a dividend for a share, the foreign interest rate for a currency, a convenience yield for a commodity). Then the unique forward price free of arbitrage opportunity is  $F_0 = S_0 e^{(r-q)T}$ . Indeed, if  $F_0 > S_0 e^{(r-q)T}$ , there exists an arbitrage: borrow  $S_0$  at the riskless rate  $r$  for the duration  $T$ , buy the asset *spot* at the price  $S_0$  (receiving along the way the income  $e^{qT}$  attached to the asset, of a discounted amount  $S_0 e^{qT}$ ), and simultaneously sell the *forward* at the price  $F_0$ ; at maturity, one delivers the asset against the payment of  $F_0$ , and one repays the borrowing (which has become  $S_0 e^{rT}$ ); the riskless net gain is then  $F_0 + S_0 e^{qT} - S_0 e^{rT} > 0$ . Conversely, if  $F_0 < S_0 e^{(r-q)T}$ , there again exists an arbitrage: sell the asset short at the price  $S_0$ , place the proceeds of the sale at the riskless rate  $r$ , and buy the *forward* at the price  $F_0$ ; at maturity, use the funds placed (which have become  $S_0 e^{(r-q)T}$ ) in order to pay the forward price, receive the asset, return it to the securities lender, and pocket the difference  $S_0 e^{(r-q)T} - F_0 > 0$  without risk.

Forward contracts fulfil two distinct but complementary economic functions. The first is **hedging**: an operator exposed to the risk of variation in a price (an airline on the price of kerosene, an exporter on the exchange rate, a bond manager on interest rates) takes a forward position in the opposite direction to his *spot* exposure, in such a

way that the gains on the one offset the losses on the other. The second is **directional speculation**: an operator who anticipates a rise (respectively a fall) in the underlying takes a long (respectively short) position in *futures*, benefiting from a substantial leverage effect since only the initial margin is tied up — a limited fraction of the notional, calibrated by the clearing house as a function of the historical volatility of the contract, as CME Group (2023) explains. Hedging by forward contracts is, however, rarely perfect: it generates a **basis risk**, defined as the residual uncertainty due to the fact that the variation in the *spot* price of the asset to be hedged and the variation in the forward price of the contract used as a hedge are not perfectly correlated, as Figlewski (1984) analyses. The basis is defined at every date  $t$  by  $b_t = S_t - F_t$ . If the hedge is imperfect (asset hedged different from the underlying of the *future*, or maturity not aligned), the basis may vary unpredictably, reducing the effectiveness of the hedge. For example, an airline which hedges the price of kerosene with *futures* on crude oil (WTI or Brent) exposes itself to the basis risk linked to the refining margin (*crack spread*), which may widen in periods of tension on refining capacity, as Corporate Finance Institute (2024) details. The shape of the forward price curve (*futures curve*), that is to say the evolution of the current prices of *futures*  $F_0(T)$  as a function of their maturity  $T$ , then informs us about the balance between supply and demand for the physical asset and about market expectations. Two configurations are distinguished:

- **Contango** if  $F_0(T) > S_0$ , which corresponds to the ‘normal’ situation for an asset without a convenience yield and whose cost of carry is positive (storage, financing); the *futures* curve is increasing with maturity;
- **Backwardation** if  $F_0(T) < S_0$ , a situation observed for commodities under supply tension, where the *convenience yield*  $q$  (the value of holding the physical asset immediately) exceeds the riskless rate  $r$ .

Keynes (1930) theorised ‘*normal backwardation*’ as the natural equilibrium situation of forward markets, in which producer-*hedgers* agree to sell forward below the expected *spot* price in order to transfer their price risk to speculators. These two configurations have direct implications for the *roll yield* realised by commodity index funds, which must renew their positions at each expiry: a curve in *contango* mechanically generates a negative *roll yield* (one buys dear in order to replace the expiring contract), whereas a curve in *backwardation* generates a positive *roll yield*.

### Key Takeaways



A forward contract is an irrevocable commitment, concluded at date 0, to exchange an underlying asset (of *spot* price  $S_t$  at every date  $t$ ) at maturity  $T$  at the price  $F_0 = S_0 e^{(r-q)T}$  fixed from the moment of conclusion; the *payoff* is linear ( $S_T - F_0$  for the long,  $F_0 - S_T$  for the short) and symmetric, with no premium to be paid at initiation, which distinguishes it fundamentally from the option. A distinction is drawn between *forwards* (OTC, bespoke, exposed to bilateral counterparty risk) and *futures* (standardised, listed, whose counterparty risk is almost eliminated by the clearing house which interposes itself by novation between the two parties) [Bank for International Settlements (2018)]. Forward contracts fulfil two complementary functions: hedging, by taking a position opposite to the *spot* exposure, and directional speculation, amplified by the leverage effect of the initial margin [CME Group (2023)]; hedging nevertheless generates a residual basis risk  $b_t = S_t - F_t$ , linked to the imperfect correlation between the variation in the *spot* of the asset hedged and that of the *future* used [Figlewski (1984)]. Finally, the shape of the curve of *futures* prices  $F_0(T)$  as a function of their maturity  $T$  informs us about the supply-demand balance: contango ( $F_0(T) > S_0$ , an increasing curve) corresponds to the normal situation of an asset with a positive cost of carry; backwardation ( $F_0(T) < S_0$ ) is observed for commodities under tension where the *convenience yield* exceeds the riskless rate [Keynes (1930)].

#### 2.1.2 Money management measures

Every operator engaged in *futures* contracts is exposed to a potentially large daily risk of loss on account of the leverage inherent in these instruments. The central risk measure for quantifying and controlling this risk is **value-at-risk** (VaR). For a portfolio of positions and a time horizon  $h$  (typically one trading day), the VaR at the confidence level  $\alpha \in ]0, 1[$  is defined as the maximum loss which will not be exceeded with a probability  $\alpha$ : formally, if  $L$  denotes the loss (a random variable) of the portfolio over the period  $h$ , then  $\text{VaR}(\alpha) = \inf\{\ell \in \mathbb{R}; \mathbb{P}([L > \ell]) \leq 1 - \alpha\} = q_\alpha(L)$ , where  $q_\alpha(L)$  denotes the quantile of order  $\alpha$  of the distribution of losses. Concretely, a one-day 99 % VaR means that, in 99 % of market scenarios, the daily loss of the portfolio will not exceed this threshold; *a contrario*, in 1 % of cases, the loss exceeds the VaR, without anything being said about the magnitude of that excess. The link between VaR and the margin mechanisms seen previously is direct: the SPAN (*standard portfolio analysis of risk*) initial margin used by CME Group (2024) is precisely a 99 % VaR calibrated over a one-day liquidation horizon. Concretely, the CCP computes the potential losses of the portfolio under sixteen standardised market scenarios (combinations of



variations in price and implied volatility) and retains the maximum loss as the margin required, as Artzner et al. (1999) explain. The initial margin is therefore the oldest and most widespread practical application of VaR: well before this term was formalised by the RiskMetrics group of J.P. Morgan in 1994, clearing houses were already calibrating their margins according to a logic of loss quantiles, as CFTC (2001) explains.

VaR nevertheless suffers from a fundamental limitation, brought to the fore by Artzner et al. (1999): it is not a coherent risk measure in the axiomatic sense, since it violates the property of **subadditivity** — the VaR of a portfolio may be greater than the sum of the VaRs of its components, which penalises diversification. Above all, VaR is blind to the magnitude of losses beyond the quantile: two portfolios whose losses exceed the 99 % VaR respectively by 1 % and by 50 % of capital present the same VaR, whereas their catastrophic risk profile is radically different. **Expected shortfall** (ES), also called *conditional value-at-risk* (CVaR) or *expected tail loss* (ETL), fills this gap. For a confidence level  $\alpha$ , the ES is defined as the expected loss conditional on exceeding the VaR:

$$ES(\alpha) = \mathbb{E}[L \mid L > VaR(\alpha)] = \frac{1}{1-\alpha} \int_{\alpha}^1 VaR(u) du$$

ES is a coherent measure in the sense of Artzner et al. (1999): as Acerbi & Tasche (2002) explain, it satisfies the four axioms of monotonicity, subadditivity, positive homogeneity and translation invariance. In the context of *futures*, ES captures precisely the risk of a market *gap* — an opening of the session far away from the previous close, for example following an overnight macroeconomic or geopolitical announcement — which may project the loss beyond the 99 % VaR and lead to a negative margin balance despite the previous day's margin call. It is the institutional recognition of this insufficiency which led the Basel Committee to replace the 99 % VaR by the 97.5 % ES as the benchmark measure for banks' market capital requirements within the framework of the *Fundamental review of the trading book* (FRTB, the revised Basel III framework), fully applicable from 2025, as Bank Policy Institute (2023) details: the move to ES entailed an estimated rise of 63.2 % in market capital requirements for group 1 banks according to the latest Basel III monitoring report according to Bank Policy Institute (2023).

#### Focus...



**Stop-loss and stop-limit orders.** In the operational management of a position in forward contracts, the control of the risk of loss does not rest solely on *ex ante* sizing rules: it likewise requires mechanisms of automatic execution permitting the closing of a position as soon as a predefined threshold of loss is reached, independently of any human intervention.

- **The stop-loss order** (or *stop-market*) is a closing order triggered automatically when the price of the underlying crosses a predefined threshold — the 'trigger price' or *stop price* — in the direction unfavourable to the position. As soon as this threshold is touched, the order becomes a market order and is executed immediately at the best price available. The advantage is the certainty of closing; the disadvantage is the **risk of slippage**: in the event of an illiquid market or an overnight *gap*, the effective execution price may be significantly lower (for a long position) than the trigger price, amplifying the loss beyond the anticipated threshold. For example, an operator long a contract on the E-mini S&P 500 bought at 5200 points who places a stop-loss at 5150 points will be executed at market as soon as the price falls to 5150 points; if an overnight macroeconomic announcement provokes an opening of the session at 5100 points (a *gap*), the stop will be triggered at the opening but executed at 5100 points, that is 50 points below the intended threshold, representing an additional loss of 2500 dollars per contract (the value of an E-mini point being 50 dollars).
- **The stop-limit order** combines a *stop price* and a *limit price*: when the trigger price is reached, the order does not become a market order but a limit order, which will be executed only at a price equal to or better than the limit price fixed. This structure eliminates the risk of excessive *slippage* — the operator is assured of never being executed at a price below his limit — but introduces in return a **risk of non-execution**: if the market collapses brutally beyond the limit price without ever returning to that level, the order remains pending and the position remains open, exposing the operator to losses well above those anticipated. The *stop-limit* order is therefore suited to liquid markets with low *intraday* volatility; it is inadvisable on illiquid commodity forward contracts or at the time of high-impact announcements (*non-farm payrolls*, Fed decisions), where *gaps* are frequent and order books can empty rapidly.
- **The trailing stop** is a dynamic variant of the *stop-loss*: instead of being fixed at an absolute level, the trigger price moves automatically in the direction favourable to the position by a fixed distance (the *trailing distance*), while remaining stationary if the market evolves unfavourably. An operator long with a *trailing stop* of 50 points sees his stop rise from 5150 to 5200 when the underlying rises to 5250 points, then to 5220 if the price reaches 5270 points:



he thus mechanically protects the gains accumulated without having to monitor his position permanently, which makes it a favoured tool of *trend following* strategies on forward contracts.

Beyond statistical measures of risk, the management of positions in *futures* relies on simple but disciplining operational rules, grouped under the term **money management**. The fixed fractional rule (*fixed fractional position sizing*) consists in risking on each position only a fixed fraction  $f$  of total capital  $W$ , typically between 0.5 % and 2 %: if the maximum acceptable loss on a position is  $W \times f$ , and the *futures* contract presents a daily VaR of  $v$  dollars per contract, the number of contracts to hold is  $\lfloor W \times f / v \rfloor$ . This rule guarantees that no individual position can, even in the event of an extreme loss scenario (an overnight *gap*), wipe out a significant share of capital. By way of illustration, an operator with 100 000 dollars of capital applying the 2 % rule will never risk more than 2000 dollars per position; faced with an E-mini S&P 500 contract whose 99 % daily VaR is roughly 7000 dollars in normal conditions, this rule quite simply forbids him to take a position on this contract with such limited capital, which is precisely its preventive role, as Hull (2012) explains. Subsequently, **drawdown** (or maximum cumulative loss) is the second usual operational constraint: defined as the maximum fall in capital between a peak and the subsequent trough, namely  $DD = \max_{t \leq s} (W_t - W_s)$ , it measures the psychological and financial resilience required in order to survive a sequence of consecutive losses before the return to equilibrium. *Drawdown* is all the more dangerous the higher the leverage: with a leverage of 10, a fall of 10 % in the underlying wipes out the entirety of the capital. A disciplined operator sets himself a maximum acceptable *drawdown* threshold — varying between 5 % and 7.5 % of capital for derivatives *traders* at Millennium to more than 20 % for funds specialising in cryptocurrencies — beyond which he mechanically reduces his exposure, independently of his directional convictions. This rule is complementary to VaR: where VaR controls daily risk position by position, the *drawdown* threshold controls the cumulative risk over a series of successive positions, which is the true measure of the soundness of a *trading* strategy on forward contracts in the long run, as Hull (2012) details.

### Key Takeaways



*Value-at-risk* (VaR) at the confidence level  $\alpha$  is the quantile of order  $\alpha$  of the distribution of losses  $L$  over a horizon  $h$ :  $\text{VaR}(\alpha) = q_\alpha(L)$ . A one-day 99 % VaR therefore means that the daily loss exceeds this threshold only in 1 % of scenarios — without saying anything about the magnitude of the excess. The initial margin of clearing houses is its direct practical application [CME Group (2024)]. VaR nevertheless suffers from two major limitations: it violates subadditivity (diversification may be penalised) and it is blind to the magnitude of losses beyond the quantile [Artzner et al. (1999)]. *Expected shortfall* (ES) then corrects these limitations by measuring the expected loss conditional on exceeding the VaR:

$$\text{ES}(\alpha) = \mathbb{E}[L \mid [L > \text{VaR}(\alpha)]] = \frac{1}{1 - \alpha} \int_{\alpha}^1 \text{VaR}(u) \, du$$

## 2.2 The exchange of continuous flows: *swaps*

**Swaps** constitute the most used derivative instrument in the world, with a gross notional of 573 000 billion dollars for OTC interest rate derivatives alone according to Bank for International Settlements (2023): they allow two counterparties periodically to exchange financial flows of a different nature — fixed rate against floating rate for **interest rate swaps** (IRS), flows in different currencies for **cross-currency basis swaps** (CCBS), and so on — without necessarily exchanging the notional.

### 2.2.1 Interest rate swaps (IRS)

An **interest rate swap** (IRS) is an over-the-counter (OTC) contract by which two counterparties undertake to exchange, at periodic intervals and over a determined duration, interest flows computed on the same notional  $N$  but according to two distinct rate benchmarks: one pays a fixed rate agreed at initiation, the other pays a variable (or floating) rate indexed to a short-term market rate. The notional  $N$  is never exchanged: only the **flow differential** is settled at each payment date, which makes the nominal size of the market (the gross notional) far greater than the real economic exposure. The IRS market is the largest segment of the world OTC derivatives market: according to Bank for International Settlements (2023), the gross notional of OTC interest rate derivatives reached 573 000 billion dollars, representing 80 % of total OTC derivatives (715 000 billion dollars). The *plain vanilla* IRS is the most

elementary form: at each date  $T_i$  (where  $i$  is an integer between 1 and  $n$ , the maturity of the *swap*), the fixed payer pays  $N \times c \times \delta_i$  and receives  $N \times L_{i-1} \times \delta_i$ , where  $c$  is the fixed rate of the swap (the *swap rate*),  $L_{i-1}$  is the floating rate observed at the previous date  $T_{i-1}$  (typically the *secured overnight financing data* or the *euro short-term rate* for contracts in euros or in dollars), and  $\delta_i = (T_i - T_{i-1})/360$  is the year fraction according to the applicable day count convention, as Hull (2012) explains. Since the end of the LIBOR transition for the American dollar on 30 June 2023, USD IRS are almost exclusively indexed to **SOFR** (*secured overnight financing rate*), a benchmark rate computed daily by the Fed and explained by Federal Reserve Bank of New York (2023) on the basis of overnight repurchase operations on American Treasury bills. In the euro area, it is **€STR** (*euro short-term rate*), published by BCE (2026a) since October 2019, which fulfils this function.

In order to understand how to value an IRS at any date  $t$ , one must start from a simple observation: a *swap* is a contract which generates two series of future financial flows in opposite directions — the fixed flows on one side, the variable flows on the other. Valuing the *swap* therefore amounts to computing, separately, what each of these two series is worth today, then taking the difference. Let us denote by  $P(t, T)$  the discount factor corresponding to the price at date  $t$  of a zero-coupon bond of maturity  $T$  (that is to say  $P(t, T) = e^{-r(t, T)(T-t)}$  under continuous compounding, where  $r(t, T)$  is the annualised interest rate at date  $t$  of the zero-coupon bond of maturity  $T > t$ ). In practice, this rate is not directly observable for all maturities: it is deduced (*bootstrapped*) from the prices of liquid market instruments — overnight deposits, futures on short rates, quoted swaps — and interpolated for intermediate maturities, which constitutes the zero-coupon yield curve. In the context of post-2008 swaps, this curve is systematically the **OIS curve** (*overnight indexed swap*), constructed from €STR or SOFR rates: it is regarded as the ‘counterparty-risk-free’ benchmark for discounting the flows of a collateralised *swap*, since the *overnight* operations on which it rests expose one to practically no risk of default, as Henrard (2014) explains.

- The **fixed leg** is the simplest to value. The firm which pays fixed has undertaken to pay, at each payment date  $T_i$ , a flow equal to  $N \times c \times \delta_i$ , where  $N$  is the notional,  $c$  is the contractual fixed rate, and  $\delta_i = (T_i - T_{i-1})/360$  is the year fraction between two consecutive payment dates (according to the applicable day count convention). Each of these future flows is certain from the initiation of the contract: it therefore suffices to discount each flow to date  $t$  by multiplying it by  $P(t, T_i)$ , then to sum the whole:

$$V_{\text{fix}}(t) = N \times c \times \sum_{i=1}^n \delta_i P(t, T_i)$$

This formula is merely the sum of the present values of  $n$  fixed interest flows, each discounted at its own zero-coupon rate corresponding to its expiry date: it is exactly the same logic as for valuing a fixed-rate bond (except that the notional is not exchanged at maturity).

- The **floating leg** is more delicate, since the variable flows are not known in advance: only the first flow (indexed to the rate observed at initiation) is certain, the following ones will depend on future market rates. In order to value them, we use *forward* rates: the **forward rate**  $F(T_{i-1}, T_i)$  is the interest rate that the market ‘anticipates’ (more precisely: that it guarantees without arbitrage) for the period running from  $T_{i-1}$  to  $T_i$ , computed from today on the basis of the discount factors, that is  $F(T_{i-1}, T_i) = \frac{1}{\delta_i} \left( \frac{P(0, T_{i-1})}{P(0, T_i)} - 1 \right)$ . The intuition is the following: placing one euro from 0 to  $T_i$  must give the same result as placing one euro from 0 to  $T_{i-1}$ , then replacing the proceeds from  $T_{i-1}$  to  $T_i$  at the *forward* rate  $F(T_{i-1}, T_i)$ ; otherwise, an arbitrage opportunity would exist. We may therefore replace each unknown future floating rate by its *forward* rate, then discount each flow thus estimated as if it were a fixed flow:

$$V_{\text{flt}}(0) = N \sum_{i=1}^n \delta_i F(T_{i-1}, T_i) P(0, T_i)$$

We may then simplify the whole: substituting the expression for the *forward* rates into the relation above, the terms telescope and the sum reduces to:

$$\begin{aligned} V_{\text{flt}}(0) &= N \sum_{i=1}^n \left[ \delta_i \times \frac{1}{\delta_i} \left( \frac{P(0, T_{i-1})}{P(0, T_i)} - 1 \right) \times P(0, T_i) \right] \\ &= N \sum_{i=1}^n \left[ \left( \frac{P(0, T_{i-1})}{P(0, T_i)} - 1 \right) \times P(0, T_i) \right] \\ &= N \sum_{i=1}^n (P(0, T_{i-1}) - P(0, T_i)) = N(P(0, T_0) - P(0, T_n)) \end{aligned}$$

This result has an immediate economic interpretation: a floating-rate bond always trades **at par** (that is to say at its nominal value) at each coupon reset date, since its future flows adjust exactly so as to offset any variation in

rates. The floating leg of the *swap* therefore amounts to receiving the notional  $N \times P(0, T_0) = N$  today and repaying it in discounted  $P(0, T_n)$  — that is exactly the difference of discount factors in the previous relation.

- The **par swap rate**  $S$  is the fixed rate  $c$  which renders the initial value of the *swap* nil, that is to say which equalises the value of the two legs at initiation. This condition  $V_{\text{flt}}(0) = V_{\text{fix}}(0)$  is the universal market convention: a *swap* contract is always initiated *at-the-money*, without any exchange of premium, each counterparty accepting a fixed rate such that nobody is advantaged at signature. Equalising the value of the fixed leg and that of the floating leg and isolating  $c$ , we obtain:

$$\begin{aligned} V_{\text{fix}}(0) = V_{\text{flt}}(0) &\iff N \times c \sum_{i=1}^n \delta_i P(t, T_i) = N(P(0, T_0) - P(0, T_n)) \\ &\iff c \sum_{i=1}^n \delta_i P(t, T_i) = N(P(0, T_0) - P(0, T_n)) \\ &\iff c = \frac{P(0, T_0) - P(0, T_n)}{\sum_{i=1}^n \delta_i P(0, T_i)} \end{aligned}$$

The denominator is called the **annuity factor**  $A$ : it measures the present value of a series of payments of one euro made at each date  $T_i$ . The larger  $A$  (long maturity, low rates), the lower the par rate, since the fixed flows are spread over many heavily discounted periods. The par rate is therefore entirely determined by the OIS curve, which operators bootstrap daily from liquid instruments (interbank deposits, Euribor/SOFR *futures*, quoted *swaps*) and interpolate via parametric or spline methods, as Brigo & Mercurio (2006) explain.

Moreover, the **DV01** (*dollar value of a basis point*), sometimes denoted PV01 (*price value of a basis point*) in euros, measures the **sensitivity** of the value of the *swap* to an infinitesimal variation in rates. Concretely, it answers the question: ‘if all the rates of the curve rise by one basis point (+0.01 percentage point), by how much does the value of the *swap* vary?’ For the *fixed payer*, a rise in rates raises the future floating flows that he receives, so his *swap* gains value: the DV01 is positive for the fixed payer. Differentiating the value of the swap with respect to a parallel shift  $\Delta r$  of the curve, it therefore follows that  $DV01 = N \times A \times 0.0001$ , where  $A$  is the *annuity factor* of the par rate formula. To fix ideas: an IRS of notional 10 million, of maturity 10 years, with an *annuity factor*  $A \approx 8.5$  (a typical value for a flat curve at 3 %), has a DV01 of  $10\,000\,000 \times 8.5 \times 0.0001 = 8500$  euros per basis point: a rise of 100 basis points would therefore entail a gain of roughly 850 000 euros for the fixed payer, symmetrically an identical loss for the fixed receiver. It is this DV01 that rates *desks* seek to render nil (‘hedging the DV01’) in order to immunise their portfolio against parallel movements of the curve. The primary use of the IRS is therefore the **hedging of interest rate risk**. A firm which has borrowed at a variable rate is exposed to a rise in rates: it enters into a *swap* in which it **pays fixed and receives variable**, thus transforming its liability into synthetic fixed-rate debt. Symmetrically, a bank whose balance sheet is structurally *long duration* (long assets such as property loans financed by short deposits) uses the IRS in order to shorten the duration of its assets without disposing of the loans, as Bank for International Settlements (2023) analyses. Bond fund managers, finally, have recourse to IRS in order to adjust the sensitivity of their portfolio rapidly and at low transaction cost, without having to trade the underlying securities.

### Exercise



**Valuation of a plain vanilla IRS.** On 1 January  $N$  a firm contracted a loan of 5 000 000 euros at a variable rate (annual €STR plus 80 basis points), repayable *in fine* in 3 years, with annual payment of interest. On 1 January  $N$ , €STR is at 2.40 %. In order to hedge against a rise in rates, the firm simultaneously concludes an IRS of maturity 3 years on a notional of 5 million euros in a fixed payer / variable receiver position (annual €STR). The zero-coupon OIS yield curve observed at the date of setting up is the following:

| Maturity | Zero-coupon rate | Discount factor     |
|----------|------------------|---------------------|
| 1 year   | 2.60 %           | $P(0, 1) = 0.97467$ |
| 2 years  | 2.85 %           | $P(0, 2) = 0.94528$ |
| 3 years  | 3.05 %           | $P(0, 3) = 0.91333$ |

- Compute the par rate of the three-year IRS (the *swap rate*  $S$ ). Verify that the value of the IRS is nil at initiation.
- At the end of year  $N$  (one year after setting up), the €STR observed on 1 January  $N$  is 2.40 %. Compute the net flow exchanged between the firm and the bank on 1 January  $N+1$ .

- (iii) If the yield curve has shifted in parallel by +75 basis points one year after setting up, compute the market value of the residual *swap* (two years) for the firm at that date.

**Solution.** (i) By definition of the par rate  $S$ , with  $T_0 = 0$ ,  $P(0,0) = 1$ ,  $\delta_i = 1$ , we obtain:

$$S = \frac{P(0,0) - P(0,3)}{\delta_1 P(0,1) + \delta_2 P(0,2) + \delta_3 P(0,3)} = \frac{1 - 0.91333}{0.97467 + 0.94528 + 0.91333} = \frac{0.08667}{2.83328} \approx 3.059\%$$

We then verify that the value of the fixed leg is  $V_{\text{fix}}(0) = 5\,000\,000 \times 3.059\% \times 2.83328 \approx 433\,200$  euros. Likewise, the value of the variable leg is  $V_{\text{flt}}(0) = 5\,000\,000 \times (1 - 0.91333) = 5\,000\,000 \times 0.08667 \approx 433\,350$  euros. The residual gap (below 200 euros) is due solely to the roundings in the discount factors: the value is indeed nil.

- (ii) At the end of year  $N$  (one year after setting up), suppose that the €STR observed is 2.40 %. The firm pays fixed: it pays  $5\,000\,000 \times 3.059\% = 152\,950$  euros. It receives variable: it receives  $5\,000\,000 \times 2.40\% = 120\,000$  euros. The net flow exchanged is therefore the difference between the two, namely  $152\,950 - 120\,000 = 32\,950$  euros. This negative flow represents the cost of the hedge: the firm pays more than the floating rate since rates have not (yet) risen.
- (iii) Suppose now that the yield curve has shifted in parallel by +75 basis points one year after setting up. The new curve (at  $N+1$ ) is  $r'(1) = 2.60\% + 0.75\% = 3.35\%$  and  $r'(2) = 2.85\% + 0.75\% = 3.60\%$ . The new discount factors are then  $P'(1) = e^{-3.35\% \times 1} \approx 0.96702$  and  $P'(2) = e^{-3.60\% \times 2} \approx 0.93239$ . The new two-year par rate is then  $S'_2 = \frac{1 - 0.93239}{0.96702 + 0.93239} = \frac{0.06761}{1.89941} \approx 3.559\%$ . The value of the *swap* for the fixed payer is the present value of the difference between the *forward* rate (the new par rate) and the contractual fixed rate, discounted over the residual *annuity factor*:

$$\begin{aligned} V_{\text{swap}} &= N \times (S'_2 - S) \times A'_2 \\ &= N \times (S'_2 - S) \times (\delta_2 P'(1) + \delta_3 P'(2)) \\ &= 5\,000\,000 \times (3.559\% - 3.059\%) \times 1.89941 \\ &= 5\,000\,000 \times 0.500\% \times 1.89941 \approx +47\,485 \text{ euros} \end{aligned}$$

The rise in rates has therefore created a positive value for the fixed-paying firm: the *swap*, initially of nil value, is now worth roughly +47 500 euros in *mark-to-market* — exactly the counterpart of the rise in the financial cost of its variable-rate loan.  $\square$

### Key Takeaways



An interest rate *swap* (IRS) is an over-the-counter contract by which two counterparties exchange, over a given duration, interest flows computed on the same notional but according to two different benchmarks: a fixed rate agreed at initiation against a variable rate indexed to a short-term market rate (SOFR for the dollar, €STR for the euro), the notional never being exchanged and only the flow differential being settled at each maturity, which explains why the gross notional (roughly 573 000 billion dollars for OTC interest rate derivatives in 2023) is far greater than the real economic exposure, representing 80 % of total OTC derivatives according to Bank for International Settlements (2023). Valuing an IRS amounts to computing separately the present value of the fixed leg — the discounted sum of the coupons  $N \times c \times \delta_i$  on the zero-coupon OIS curve — and of the floating leg — which may be assimilated, after telescoping the *forward* rates, to the difference  $N(P(0, T_0) - P(0, T_n))$ , a reflection of the fact that a floating-rate bond always trades at par at each reset date — then to take the difference [Hull (2012)]. Subsequently, the par rate  $S$  is the fixed rate which renders this initial value nil:

$$S = \frac{P(0, T_0) - P(0, T_n)}{\sum_{i=1}^n \delta_i P(0, T_i)}$$

Thus, every *plain vanilla* IRS is initiated without any entry premium, the OIS curve (€STR or SOFR) serving as the counterparty-risk-free discounting benchmark [Henrard (2014)]. Thanks to IRS, a firm indebted at a variable rate can thus transform its liability into synthetic fixed-rate debt by taking the fixed payer / variable receiver leg, while a bank or a bond manager adjusts the duration of its portfolio rapidly and at low transaction cost thanks to standardised IRS.

### 2.2.2 Currency swaps (CCS)

A **currency swap** (*cross-currency swap*, **CCS**) is an OTC contract by which two counterparties undertake to exchange interest flows — and above all notionals — denominated in two different currencies, over a determined duration and at an exchange rate fixed from the moment of initiation. It is precisely this last characteristic which structurally distinguishes the CCS from the IRS: whereas the notional of an IRS is never exchanged (only the differentials of interest flows are), a CCS involves a **triple exchange of notionals** — at initiation, during its life (for the so-called *resettable* variants), and at expiry — which creates a concrete balance sheet exposure to exchange rate risk. The first formalised currency swap is historically the one concluded in August 1981 between IBM and the World Bank, arranged by Salomon Brothers: the World Bank, which had reached its borrowing ceilings in Swiss francs (CHF) and in Deutsche Marks (DM), issued a bond in dollars and exchanged the flows against the bonds in CHF (200 million at 6.1875 %) and in DM (300 million at 10 %) of which IBM for its part wished to divest itself, having benefited from a strong appreciation of the dollar, as Susmel (2015) relates. Since then, the OTC foreign exchange derivatives market — which includes currency swaps, *FX swaps* and *currency basis swaps* — has experienced strong growth: according to Bank for International Settlements (2023), the gross notional of OTC foreign exchange derivatives reached 120 000 billion dollars, corresponding to a rise in volumes of 12 % in the first half of 2023. Three main variants are distinguished according to the nature of the rates exchanged:

- The **fixed-fixed swap**: each counterparty pays a fixed rate in its own currency; this is the historic form illustrated by the swap between IBM and the World Bank;
- The **fixed-floating swap** (or *cross-currency coupon swap*): one pays a fixed rate, the other a variable rate in its own currency;
- The **floating-floating swap** (or *cross-currency basis swap*, CCBS): both legs are indexed to short-term variable rates in their respective currencies (typically €STR for the euro and SOFR for the dollar), to which is added a **basis spread**  $b$  on one of the legs; this is the most liquid and most traded form on contemporary institutional markets.

In every case, the mechanism is identical: at date  $T_0$ , counterparty A hands counterparty B a notional  $N_{\text{USD}}$  in dollars and receives a notional  $N_{\text{EUR}} = N_{\text{USD}}/S_0$  in euros, where  $S_0$  is the spot USD/EUR exchange rate observed at initiation. At each coupon date  $T_i$ , the interest flows are exchanged (in their respective currencies). At maturity  $T_n$ , the notionals are returned at the initial amounts, at the same exchange rate  $S_0$  — and not at the spot rate prevailing at expiry — which guarantees that the exchange rate risk on the notional is entirely hedged for both parties, as Hull (2012) explains.

The valuation of a CCS rests on the principle of **covered interest parity (CIP)**: in the absence of arbitrage opportunity, the forward exchange rate  $F(t, T)$  between two currencies must satisfy  $F(t, T) = S_t \times \frac{P_{\text{EUR}}(t, T)}{P_{\text{USD}}(t, T)} = S_t \times e^{(r_{\text{EUR}}(t, T) - r_{\text{USD}}(t, T))(T - t)}$ , where  $S_t$  is the *spot* exchange rate (in euros per dollar),  $r_{\text{EUR}}(t, T)$  and  $r_{\text{USD}}(t, T)$  are the respective zero-coupon OIS rates in each currency. The interpretation is immediate: if the interest rate in euros is lower than the rate in dollars, the euro ought to appreciate forward by exactly the difference, which cancels out any advantage in placing funds in one currency rather than the other. The value of the *fixed-fixed* CCS may then be decomposed into two zero-coupon bonds — one in each currency — whose present value is computed separately, and which are brought back to a common currency via the current spot exchange rate  $S_t$ :

$$V_{\text{CCS}}(t) = \underbrace{N_{\text{EUR}} \left( c_{\text{EUR}} \sum_{i=1}^n \delta_i P_{\text{EUR}}(t, T_i) + P_{\text{EUR}}(t, T_n) \right)}_{\text{euro leg (received)}} - \underbrace{\frac{S_t}{S_0} N_{\text{USD}} \left( c_{\text{USD}} \sum_{i=1}^n \delta_i P_{\text{USD}}(t, T_i) + P_{\text{USD}}(t, T_n) \right)}_{\text{dollar leg (paid)}}$$

where the factor  $S_t/S_0$  translates the effect of the variation in the exchange rate between the date of initiation and the date of valuation on the notional in dollars. We note that this formula naturally generalises the equation of the fixed leg of the IRS: each leg is valued as a fixed-rate bond in its own currency (with here an exchange of the notional at maturity, whence the term  $P(t, T_n)$ ), and the net value is expressed in a reference currency via the spot rate, as Brigo & Mercurio (2006) explain.

However, since the financial crisis of 2008, markets have observed a persistent and structural deviation from CIP: it is no longer equivalent to borrow dollars directly on the American money market and to borrow them synthetically via a currency swap. This deviation is measured by the **cross-currency basis spread**  $b$  (expressed in basis points), which is added to the non-dollar floating leg of a CCBS: a counterparty wishing to borrow dollars via an €STR/SOFR swap will have to pay SOFR *flat* and receive €STR less a spread  $b < 0$ , that is  $(e\text{STR} - |b|)$ . On the five-year EUR/USD

pair, this *spread*, structurally negative since 2008 (an average of  $-21$  basis points over the last four years), had almost vanished in November 2024 — reaching  $+3$  basis points, a sign of a temporary reversal linked to the strong movement of appreciation of the dollar after the American elections, as ING (2024) emphasises. Significant negative *basis spreads* nevertheless persisted on other major pairs:  $-51$  basis points for the USD/JPY pair at five years and  $-33$  basis points for the USD/DKK pair (Danish krone) at five years according to ING (2024). This deviation is attributed to the structural excess of world demand for dollar funding, to the Basel III regulatory constraints weighing on bank balance sheets, and to the flows of non-American issuers (notably Japanese and European) seeking to hedge after issuing bonds in dollars, as David-Pur et al. (2023) analyse. The primary use of the CCS is therefore the **hedging of exchange rate risk on debt**. A French firm which issues a bond in dollars (in order to access the American investor base, deeper at certain maturities) but whose revenues are in euros enters into a *fixed-fixed* CCS in order to exchange all its flows in dollars (coupons and repayment of principal) against flows in euros at a fixed rate: it thus transforms a dollar debt into synthetic euro debt, with no residual exposure to the EUR/USD rate. Central banks for their part use bilateral *swap* lines: on 15 March 2020, the Fed strengthened its permanent lines with five central banks (the ECB, the Bank of Japan, the Bank of England, the Bank of Canada and the Swiss National Bank), then opened on 19 March 2020 nine new temporary lines, bringing the total to 14 central banks; drawings peaked at 449 billion dollars in the week of 27 May 2020, as Hoffner (2023) relates.

### Exercise



**Valuation of a *fixed-fixed* EUR/USD currency swap.** On 1 January N, a French firm issues a bond in dollars for an amount of 11 000 000 USD at a fixed rate of 4.50 %, maturity three years, annual coupons, repayment *in fine*. The spot exchange rate is  $S_0 = 1.10$  USD/EUR, that is an equivalent notional of  $N_{\text{EUR}} = 10\,000\,000$  euros. In order to hedge its exchange rate risk in full, the firm simultaneously concludes a three-year *fixed-fixed* CCS in which it pays fixed EUR at the rate  $c_{\text{EUR}}$  and receives fixed USD at 4.50 %, with exchange of the notionals at initiation and at expiry at the exchange rate  $S_0$ . The zero-coupon OIS curves observed at that date are :

| Maturity | EUR ZC rate | $P_{\text{EUR}}(0, T_i)$ | $P_{\text{USD}}(0, T_i)$ |
|----------|-------------|--------------------------|--------------------------|
| 1 year   | 2.60 %      | 0.97467                  | 0.95694                  |
| 2 years  | 2.85 %      | 0.94528                  | 0.91574                  |
| 3 years  | 3.05 %      | 0.91333                  | 0.87810                  |

The USD factors correspond to SOFR zero-coupon rates of 4.50 %, 4.70 % and 4.85 % at maturities 1, 2 and 3 years respectively.

- Compute the EUR par rate  $c_{\text{EUR}}^*$  of the CCS, that is to say the fixed rate in euros which cancels the value of the *swap* at initiation.
- Compute the annual net flow in euros borne by the firm under the *swap* (excluding the exchange of notional).
- One year after setting up ( $t = 1$  year), the EUR curve has shifted by  $+50$  basis points and the exchange rate has moved to  $S_1 = 1.05$  USD/EUR (the dollar has strengthened). Compute the market value of the residual *swap* (two years) for the firm, assuming that the USD curves are unchanged.

**Solution.** (i) At initiation, the value of the CCS is nil: the USD leg received (valued at the spot rate  $S_0$ ) equalises the EUR leg paid. The USD leg corresponds to receiving a fixed bond at 4.50 % of notional 11 M USD ; its value in dollars is worth:

$$\begin{aligned}
 V_{\text{USD}}(0) &= 11\,000\,000 \times (4.50 \% \times (0.95694 + 0.91574 + 0.87810) + 0.87810) \\
 &= 11\,000\,000 \times (4.50 \% \times 2.75078 + 0.87810) \\
 &= 11\,000\,000 \times 1.00189 \approx 11\,020\,790 \text{ USD}
 \end{aligned}$$

Converted into euros at the spot rate  $S_0 = 1.10$ , we obtain  $V_{\text{USD}}^{\text{in EUR}}(0) = \frac{11\,020\,790}{1.10} \approx 10\,019\,000$  EUR. The EUR leg paid must then be worth exactly this amount; now for the EUR fixed payer (who pays  $c_{\text{EUR}}$  and receives the notional), its value is  $V_{\text{EUR}}(0) = N_{\text{EUR}} \times (c_{\text{EUR}} \times A_{\text{EUR}} + P_{\text{EUR}}(0, 3))$ , where  $A_{\text{EUR}} = 0.97467 + 0.94528 + 0.91333 = 2.83328$ . Setting  $V_{\text{EUR}}(0) = V_{\text{USD}}(0)/S_0$ , we must therefore solve the equation  $10\,000\,000 \times (c_{\text{EUR}} \times 2.83328 + 0.91333) = 10\,019\,000$ , that is to say  $c_{\text{EUR}} = \frac{10\,019\,000/10\,000\,000 - 0.91333}{2.83328} \approx 3.126 \%$ . The firm therefore pays 3.126 % in euros against 4.50 % in dollars: the rate differential (roughly 137 basis points) is precisely the reflection of the gap in levels between the EUR and USD curves (the SOFR curve being appreciably higher than the €STR curve), in accordance with CIP.



- (ii) The firm receives the coupons of the USD leg (which serve it to repay the holders of the bond):  $11\,000\,000 \times 4.50\% = 495\,000$  USD, that is  $495\,000/1.10 = 450\,000$  EUR at the initial exchange rate. It pays the coupons of the EUR leg:  $10\,000\,000 \times 3.126\% = 312\,600$  EUR. The annual net flow borne by the firm is therefore  $312\,600 - 450\,000 = -137\,400$  EUR (this is a favourable net flow: the firm receives 137 400 EUR net per year). This annual gain is the direct counterpart of the rate advantage obtained on the American bond market (4.50 % in USD whereas issuing directly in EUR would have cost it the EUR par rate, that is roughly 3.06 % according to the previous exercise, plus an issue premium).
- (iii) Let us determine the new EUR curve after a *shift* of +50 basis points): we therefore have  $r'_{\text{EUR}}(1) = 3.35\%$  and  $r'_{\text{EUR}}(2) = 3.60\%$ , which gives as new discount factors  $P'_{\text{EUR}}(1, 2) = e^{-3.35\% \times 1} \approx 0.96702$  and  $P'_{\text{EUR}}(1, 3) = e^{-3.60\% \times 2} \approx 0.93070$ . The residual value of the EUR leg paid (two years remaining, notional included) therefore satisfies  $V'_{\text{EUR}} = 10\,000\,000 \times (3.126\% \times (0.96702 + 0.93070) + 0.93070) \approx 990\,600$  EUR, while the residual value of the USD leg received (USD curves unchanged), converted at the new spot rate  $S_1 = 1.05$  USD/EUR, is now worth  $V_{\text{USD}}^{\text{in EUR}} = \frac{11\,000\,000 \times (4.50\% \times (0.91574 + 0.87810) + 0.87810)}{1.05} \approx 1\,035\,914$  EUR. The market value of the *swap* for the firm (which is a receiver of American dollars and a payer of euros) therefore becomes  $V_{\text{CCS}}(1) = V_{\text{USD}}^{\text{in EUR}} - V'_{\text{EUR}} \approx 1\,035\,914 - 990\,600$ . The positive value is explained by two combined effects: on the one hand, the rise in EUR rates has reduced the value of the EUR leg paid (advantageous for the payer); on the other hand, the appreciation of the dollar ( $S$  moves from 1.10 to 1.05 USD/EUR, that is a rise in the dollar) increases, once converted into euros, the value of the USD leg received — which illustrates the residual exchange rate risk borne by the *mark-to-market* of the *swap*, even when the notionals are fixed at the initial rate  $S_0$ .  $\square$

### Key Takeaways



A *cross-currency swap* (CCS) is fundamentally distinguished from an IRS by the triple exchange of notionals — at initiation, periodically for the *resettable* variants, and at expiry at the initial exchange rate  $S_0$  — which creates a concrete balance sheet exposure and entirely hedges the exchange rate risk on the notional for both counterparties [Hull (2012)]. Three variants are distinguished: *fixed-fixed* (the historic form), *fixed-floating* and *floating-floating* (CCBS, the most liquid form), in which both legs are indexed to variable rates (€STR against SOFR) to which is added a *basis spread*  $b$  on the non-dollar leg. The valuation rests on covered interest parity (CIP):

$$V_{\text{CCS}}(t) = N_{\text{EUR}} \underbrace{\left( c_{\text{EUR}} \sum_i \delta_i P_{\text{EUR}}(t, T_i) + P_{\text{EUR}}(t, T_n) \right)}_{\text{EUR leg (received)}} - \frac{S_t}{S_0} N_{\text{USD}} \underbrace{\left( c_{\text{USD}} \sum_i \delta_i P_{\text{USD}}(t, T_i) + P_{\text{USD}}(t, T_n) \right)}_{\text{USD leg (paid)}}$$

where the factor  $S_t/S_0$  translates the impact of the variation in the exchange rate on the *mark-to-market* value of the foreign leg, which explains why the CCS bears a residual exchange rate risk on its market value — distinct from the risk on the notional, hedged by the exchange at  $S_0$  [Brigo & Mercurio (2006)]. Since the 2008 crisis, CIP has been structurally violated: the *cross-currency basis spread*  $b$  measures the extra cost of borrowing dollars synthetically via a CCBS relative to the American money market [David-Pur et al. (2023)]. In practice, the primary use of the CCS is the hedging of exchange rate risk on debt: a French firm issuing in dollars in order to access the American investor base concludes a *fixed-fixed* CCS in order to transform its dollar debt entirely into synthetic euro debt.

### 2.2.3 Credit risk swaps (CDS)

A *credit default swap* (CDS) is an OTC contract by which the protection seller undertakes to indemnify the protection buyer in the event of the occurrence of a credit event affecting a reference entity, in exchange for the periodic payment of a premium (the *CDS spread*) expressed in annual basis points applied to a notional  $N$ . Unlike the IRS and CCS studied previously, the CDS is not an exchange of interest flows between two different rates: it is an **insurance contract against default risk**, whose underlying is the credit quality of an issuer (a firm, a sovereign or a structured entity), and not an interest rate or an exchange rate. The credit events recognised by ISDA (2003) include bankruptcy, failure to pay and restructuring of the debt of the reference entity. In the absence of a credit event, the protection seller receives the premium and pays nothing; in the event of a credit event, he pays the buyer the loss suffered: either the difference between the notional and the residual value of the reference bond (cash settlement), or the notional against delivery of the depreciated bond (physical settlement). The CDS market experienced sustained growth in the 2000s before being drastically contracted by the financial crisis of 2007–2008, whose systemic effects were amplified by CDS on *mortgage-backed securities*. The world gross notional of CDS, which reached 58 000 billion dollars at the end of 2007 according to Bank for International Settlements (2008), stood at 8 600 billion dollars at the



end of 2023, that is a contraction of 45.6 % relative to 2014 (16 000 billion), as IOSCO (2024) specifies. This reduction reflects both the compression of bilateral positions by novation and the effects of compulsory central clearing: the share of CDS cleared by a central counterparty (CCP) reached 69.9 % of gross notional (that is 6 300 billion) according to ISDA (2024b). Liquidity is today highly concentrated in **index CDS** — the CDX in North America and the iTraxx in Europe — which represent nearly 90 % of market activity by volume traded according to ISDA (2024a). Moreover, in April 2009, ISDA introduced a reform standardising CDS contracts, dubbed the *CDS big bang*, as Beurre et al. (2009) explain. Before this reform, each CDS had its own *running spread*, fixed at initiation so that the value of the contract was nil. Since 2009, contracts have been standardised around **two fixed coupons** only: 100 basis points (*investment grade*) or 500 basis points (*high yield*). In return, an initial payment (*upfront*) is exchanged between the counterparties in order to compensate for the gap between the standardised coupon and the *fair spread* of the reference entity. This standardisation made the central clearing of CDS possible and facilitated their uniform valuation across the market. The *recovery rate* is likewise standardised: 40 % for *investment grade* entities and 20 % for *high yield* entities and emerging market sovereign CDS, as Beurre et al. (2009) detail.

The valuation of a CDS rests on the modelling of the **probability of default** of the reference entity. The standard approach, said to be reduced-form, postulates that default occurs as the first jump of a Poisson process of intensity  $h(t)$ , called the **hazard rate** or default intensity, as Jarrow & Turnbull (1995) explain. The probability of survival of the entity between 0 and  $T$  is then:

$$Q(0, T) = \mathbb{P}(\tau > T) = \exp\left(-\int_0^T h(u) du\right)$$

where  $\tau$  is the random variable of the time of default. Under the simplifying assumption of a constant hazard rate  $h$ , we obtain  $Q(0, T) = e^{-hT}$ , which formally recalls the structure of the discount factor  $P(0, T) = e^{-rT}$ : the hazard rate  $h$  plays for default risk the same role that the interest rate  $r$  plays for the time value of money. The probability of default between  $T_{i-1}$  and  $T_i$  conditional on survival until  $T_{i-1}$  is then worth  $\mathbb{P}([T_{i-1} < \tau] \leq [T_i | \tau > T_{i-1}]) = 1 - e^{-h\delta_i} \approx h\delta_i$  for  $h\delta_i$  close to 0. Subsequently, a CDS generates two legs whose present values must be equal at the market *fair spread*  $s^*$ :

- The **premium leg**: the protection buyer pays  $N \times s \times \delta_i$  at each date  $T_i$ , but only if the reference entity has not defaulted. The present value of the premium leg is therefore:

$$PL(s) = N \times s \sum_{i=1}^n \delta_i P(0, T_i) Q(0, T_i)$$

- The **protection leg**: the protection seller pays  $N(1 - R)$  at the moment of default  $\tau$  (where  $R$  is the recovery rate), if the latter occurs before maturity  $T_n$ . Its present value is:

$$\begin{aligned} DL &= N(1 - R) \int_0^{T_n} P(0, u) h e^{-hu} du \\ &= N(1 - R) \int_0^{T_n} P(0, u) \left(-\frac{d}{du} Q(0, u)\right) du \quad \text{since } Q(0, u) = e^{-hu}, \text{ so } Q'(0, u) = -h e^{-hu} \\ &= N(1 - R) \int_0^{T_n} P(0, u) (-dQ(0, u)) \quad (\text{Stieltjes integral}) \\ &= N(1 - R) \sum_{i=1}^n \int_{T_{i-1}}^{T_i} P(0, u) (-dQ(0, u)) \quad \text{by additivity of the integral} \\ &\approx N(1 - R) \sum_{i=1}^n P(0, T_i) \int_{T_{i-1}}^{T_i} (-dQ(0, u)) \quad \text{since } P(0, u) \approx P(0, T_i) \text{ on } [T_{i-1}, T_i] \\ &= N(1 - R) \sum_{i=1}^n P(0, T_i) \left[ -(Q(0, T_i) - Q(0, T_{i-1})) \right] \quad \text{since } -\int_{T_{i-1}}^{T_i} dQ = -(Q(0, T_i) - Q(0, T_{i-1})) \\ &= N(1 - R) \sum_{i=1}^n P(0, T_i) (Q(0, T_{i-1}) - Q(0, T_i)) \end{aligned}$$

The **fair spread** (or *par CDS spread*)  $s^*$  is then the *spread* which cancels the value of the CDS at initiation ( $PL(s^*) = DL$ ), that is to say:

$$s^* = \frac{(1 - R) \sum_{i=1}^n P(0, T_i) (Q(0, T_{i-1}) - Q(0, T_i))}{\sum_{i=1}^n \delta_i P(0, T_i) Q(0, T_i)}$$

In the limit of a constant hazard rate  $h$ , a flat yield curve  $r$  and a continuous maturity, the expression of the *fair spread* simplifies into an approximation due to Duffie (1999), namely  $s^* \approx h(1 - R)$ , which is one of the most used formulas in practice for bootstrapping the hazard rate  $h$  from quoted *spreads*: if the five-year CDS of a firm is quoted at  $s = 120$  basis points and the conventional recovery rate is  $R = 40\%$ , the implied hazard rate is  $h = 120/(1 - 40\%) = 200$  basis points per year, that is an annual probability of default of roughly  $1 - e^{-0.02} \approx 1.98\%$ , as Hull (2012) explains.

The protection buyer therefore uses a CDS in order to **hedge the credit exposure** of a bond position without having to dispose of the security: he keeps the carry of the bond (the coupon) and transfers the default risk to the protection seller in exchange for payment of the *spread*. The protection seller, for his part, receives the *spread* in exchange for a synthetic exposure to credit risk, without tying up the notional in capital (until the Basel III reforms): this is the equivalent of a leveraged bond loan, as Duffie (1999) explains. CDS are likewise used in **directional speculation** on the credit quality of an issuer: CDS on Greek sovereign debt, for example, were massively bought in 2010–2012 by operators anticipating a restructuring according to Hull (2012). Finally, **index CDS** (CDX IG, CDX HY, iTraxx Europe) make it possible to gain exposure to, or to hedge against, the credit risk of a basket of 125 standardised reference entities, offering far greater liquidity than *single-name* CDS.

### Exercise



**Valuation of a CDS and extraction of the hazard rate.** On 1 January N, a bond manager holds an *investment grade* bond of notional 10 000 000 euros, of maturity three years, and wishes to hedge his default risk via a standardised CDS (fixed coupon 100 basis points, *big bang*). The three-year CDS on the reference entity is quoted at a *fair spread* of  $s^* = 180$  basis points on the market. The conventional recovery rate is  $R = 40\%$ . The OIS yield curve is flat at  $r = 3\%$  (continuous compounding).

- (i) Using the approximation of Duffie (1999), compute the implied hazard rate  $h$  and the probability of survival at 1, 2 and 3 years.
- (ii) Compute the value of the premium leg  $PL(100bp)$  at the standardised coupon of 100 basis points (bp), discretising annually.
- (iii) Compute the value of the protection leg  $DL$ .
- (iv) Deduce from this the initial payment (*upfront*) that the protection buyer must pay the seller in order to enter into the standardised CDS. Interpret the sign.

**Solution.** (i) According to Duffie's approximation, the implied hazard rate satisfies  $h = \frac{s^*}{1-R} = \frac{0.180}{1-0.40} \approx 3.00\%$  per year. The probabilities of survival for their part satisfy  $Q(0, 1) = e^{-0.03 \times 1} \approx 0.97045$ ,  $Q(0, 2) = e^{-0.03 \times 2} \approx 0.94176$  and  $Q(0, 3) = e^{-0.03 \times 3} \approx 0.91393$ .

- (ii) The discount factors for an interest rate of  $r = 3\%$  (continuous) satisfy  $P(0, 1) = e^{-0.03} \approx 0.97045$ ,  $P(0, 2) \approx 0.94176$ ,  $P(0, 3) \approx 0.91393$  (here, the values  $P(0, T_i)$  and  $Q(0, T_i)$  are equal but this is a coincidence since  $r = h$  in this exercise). We may then evaluate the premium leg for a standardised coupon of  $s = 100$  basis points and  $\delta_i = 1$ :

$$\begin{aligned} PL(100bp) &= 10\,000\,000 \times 0.0100 \times \sum_{i=1}^3 P(0, T_i) Q(0, T_i) \\ &= 10\,000\,000 \times 0.0100 \times 0.97045^2 + 0.94176^2 + 0.91393^2 \\ &= 10\,000\,000 \times 0.01 \times 2.66395 \approx 266\,395 \text{ EUR} \end{aligned}$$

- (iii) We have first of all  $Q(0, 0) - Q(0, 1) = 1 - 0.97045 = 0.02955$ , then  $Q(0, 1) - Q(0, 2) = 0.97045 - 0.94176 = 0.02869$  and finally  $Q(0, 2) - Q(0, 3) = 0.94176 - 0.91393 = 0.02783$ . We deduce from this the value of the protection leg:

$$\begin{aligned} DL &= 10\,000\,000 \times (1 - 0.40) \times (P(0, 1) \times 0.02955 + P(0, 2) \times 0.02869 + P(0, 3) \times 0.02783) \\ &= 6\,000\,000 \times (0.97045 \times 0.02955 + 0.94176 \times 0.02869 + 0.91393 \times 0.02783) \\ &= 6\,000\,000 \times (0.02867 + 0.02701 + 0.02544) = 6\,000\,000 \times 0.08112 \approx 486\,720 \text{ EUR} \end{aligned}$$

- (iv) The *fair spread* is 180 basis points but the standardised coupon is 100 basis points. The protection seller receives a coupon too low relative to the risk taken: he therefore requires an initial payment from the buyer in compensation. The *upfront* is the present value of the gap between the protection leg and the premium

leg at the contractual coupon: it is worth  $DL - PL(100 \text{ bp}) = 486\,720 - 266\,395 \approx +220\,325$  EUR. The *upfront* is positive: the protection buyer therefore pays 220 325 EUR to the seller at the setting up of the contract, then 100 basis points annually for three years. This amount compensates exactly for the fact that the seller accepts a coupon of 100 basis points whereas the risk of the entity would justify 180 basis points: without this *upfront*, the seller would suffer a certain loss in present value of 220 325 EUR. We verify the coherence: if the *fair spread* had been lower than 100 basis points, the *upfront* would have been negative — the seller would then have paid the buyer in order to enter into a contract whose coupon is favourable to him.  $\square$

### Key Takeaways



A *credit default swap* (CDS) is an insurance contract against default risk: the protection buyer periodically pays a *spread*  $s$  (in basis points on a notional  $N$ ) to the protection seller, who undertakes in return to indemnify the loss  $N(1 - R)$  in the event of a credit event (bankruptcy, failure to pay, restructuring) recognised by ISDA (2003); since the *CDS big bang* of 2009, contracts have been standardised around two fixed coupons (100 basis points for *investment grade*, 500 basis points for *high yield*), an initial *upfront* payment compensating for the gap with the market *fair spread*, and conventional recovery rates of 40 % (*investment grade*) and 20 % (*high yield* and emerging sovereigns) [Beurre et al. (2009)]. The valuation rests on the reduced-form model of Jarrow & Turnbull (1995): default occurs as the first jump of a Poisson process of intensity  $h$  (the hazard rate), which gives a probability of survival  $Q(0, T) = e^{-hT}$  formally analogous to the discount factor  $P(0, T) = e^{-rT}$ ; the *fair spread*  $s^*$  equalises the present value of the premium leg (periodic flows conditional on survival) and of the protection leg (indemnification in the event of default):

$$s^* = \frac{(1 - R) \sum_{i=1}^n P(0, T_i) (Q(0, T_{i-1}) - Q(0, T_i))}{\sum_{i=1}^n \delta_i P(0, T_i) Q(0, T_i)} \approx h(1 - R)$$

The uses are three in number: credit hedging (the buyer keeps the carry of the bond and transfers the default risk without disposing of the security), the synthesis of leveraged credit exposure for the protection seller (the equivalent of a bond loan without tying up the notional), and directional speculation on the credit quality of an issuer [Hull (2012)].

#### 2.2.4 Swaps on commodities

A *commodity swap* (**commodity swap**) is an OTC contract by which two counterparties undertake to exchange, at periodic intervals, financial flows computed on a reference quantity  $Q$  of a commodity: one pays a fixed price  $K$  agreed at initiation, the other pays a variable price indexed to a market price observed at each maturity (or, in the so-called *average swap* variant, to the arithmetic mean of the *spot* price over the elapsed period). As for the IRS, no physical commodity is ever delivered: only the **monetary differential** between the fixed price and the variable price is settled at each payment date  $T_i$ , which makes it a purely financial instrument of hedging or speculation, as Hull (2012) specifies. The *commodity swap* is structurally distinguished from the *swaps* studied previously since its underlying is a non-financial asset, whose price is governed by physical supply and demand balances (stocks, seasonality, transport and storage constraints), and not by the term structure of interest rates alone. This specificity leads one to mobilise, for its valuation, the theory of the cost of carry and the notion of convenience yield, specific to commodity markets, as Schwartz (1997) analyses. The most traded underlying commodities are **energy** (WTI and Brent crude oil, natural gas, electricity), **metals** (aluminium, copper, gold) and **agricultural products** (wheat, maize, soya). At each payment date  $T_i$ , the net flow paid by the fixed payer is  $\text{Net flow}(T_i) = Q \times (K - S_i)$ , where  $S_i$  is the *spot* price (or the mean of the spot prices over the period  $[T_{i-1}, T_i]$  in the average variant) observed at date  $T_i$ . If  $S_i > K$  (a rise in the price of the commodity), the flow is negative for the fixed payer — he receives the difference from the protection seller; if  $S_i < K$  (a fall in the price), he pays the difference. In both cases, the effective price paid by the consumer is  $S_i + (K - S_i) = K$ , which illustrates the primary function of the *swap*: to transform an exposure to a variable price into a synthetic fixed price, independently of market fluctuations.

The fair price of a *commodity swap* is determined by the *forward* curve of the commodity in question. For a storable asset (oil, metals), the *forward* price at date  $T$  is related to the *spot* price  $S_0$  by the *cost of carry* relation, namely  $F(0, T) = S_0 e^{(r+u-y)T}$ , where  $r$  is the riskless rate (OIS),  $u$  is the annualised rate of storage costs (as a percentage of the value of the commodity), and  $y$  is the convenience yield: the latter measures the economic advantage conferred by physical holding of the commodity (the capacity to honour immediate deliveries, to keep a production facility running), as opposed to holding a financial contract. When stocks are abundant,  $y$  is low and the *forward* curve

is in contango ( $F(0, T)$  increasing in  $T$ ); when stocks are tight,  $y$  is high and the curve is in backwardation ( $F(0, T)$  decreasing in  $T$ ), a frequent configuration on oil markets at the time of supply shocks, as Schwartz (1997) explains. Subsequently, the value of a *commodity swap* at date  $t$  is obtained by discounting the difference between the fixed price  $K$  and the *forward* prices corresponding to each residual maturity:

$$V_{\text{swap}}(t) = Q \sum_{i=1}^n (F(t, T_i) - K) P(t, T_i)$$

where  $F(t, T_i)$  is the *forward* price observed at  $t$  for delivery at  $T_i$ , and  $P(t, T_i)$  is the OIS discount factor. At initiation ( $t = 0$ ), the market convention is to set the value of the *swap* to zero, which determines the **par fixed price** (*par commodity swap price*):

$$K^* = \frac{\sum_{i=1}^n F(0, T_i) P(0, T_i)}{\sum_{i=1}^n P(0, T_i)}$$

that is to say the weighted average of the *forward* prices, the weights being the discount factors. This formula is the exact analogue of the formula for the *swap* rate  $S^*$  for IRS: it is the 'commodities' version of it, where the *forward* rates  $F(0, T_i)$  replace the *forward* interest rates  $F(T_{i-1}, T_i)$ .

What use for *commodity swaps*? An **industrial consumer** (a refinery, an airline, a chemical company) may enter into a *swap* in a fixed payer / variable receiver position: it fixes its supply cost and transfers the risk of a price rise to the financial counterparty. Likewise, a **producer** (an independent oil company, a farmer) enters in a fixed receiver / variable payer position: it fixes its selling price and protects itself against a fall. **Financial intermediaries** (banks, funds) for their part act as counterparty on both sides, hedging their residual risk via futures listed on the CME or the ICE, as Hull (2012) explains. Thus, the company Southwest Airlines (2008) had fixed more than 70 % of its fuel consumption for 2008 at roughly 51 USD per barrel, whereas the price of crude peaked at 147 USD per barrel in July 2008: the cumulative saving over the period 1998–2008 is estimated at 3.5 billion dollars relative to the average cost of the industry according to Southwest Airlines (2021).

### Exercise



**Valuation of a commodity swap on crude oil.** On 1 January N, a European refiner plans to buy 120 000 barrels of Brent crude oil spread over three equal quarterly deliveries of 40 000 barrels at the dates  $T_1 = 3$  months,  $T_2 = 6$  months and  $T_3 = 9$  months. In order to fix its supply cost, it concludes a 9-month *commodity swap* in a fixed payer / variable receiver position (Brent ICE spot price at each maturity). The Brent *forward* price curve observed at the date of setting up is the following:

| Maturity         | Forward price            | Discount factor        | $F(0, T_i) \times P(0, T_i)$ |
|------------------|--------------------------|------------------------|------------------------------|
| $T_1 = 3$ months | 82.40 dollars per barrel | $P(0, 0.25) = 0.99256$ | 81.788                       |
| $T_2 = 6$ months | 81.10 dollars per barrel | $P(0, 0.50) = 0.98518$ | 79.898                       |
| $T_3 = 9$ months | 79.60 dollars per barrel | $P(0, 0.75) = 0.97783$ | 77.835                       |

- Compute the par fixed price  $K^*$  of the *swap*.
- At maturity  $T_1$ , the Brent spot price is 88.50 dollars per barrel. Compute the net flow exchanged between the refiner and the bank, and determine the effective price of the oil for the refiner.
- Six months after setting up, the residual *forward* curve has shifted upwards:  $F(0.5, T_2) = 91.20$  dollars per barrel and  $F(0.5, T_3) = 90.30$  dollars per barrel. Compute the market value of the residual *swap* (two remaining maturities) for the refiner at that date, with  $P(0.5, T_2) \approx 0.99254$  and  $P(0.5, T_3) \approx 0.98514$ .

**Solution.** (i) The par fixed price of the *swap* is:

$$\begin{aligned} K^* &= \frac{F(0, T_1)P(0, T_1) + F(0, T_2)P(0, T_2) + F(0, T_3)P(0, T_3)}{P(0, T_1) + P(0, T_2) + P(0, T_3)} \\ &= \frac{81.788 + 79.898 + 77.835}{0.99256 + 0.98518 + 0.97783} \approx 81.04 \end{aligned}$$

The refiner will pay a fixed price of 81.04 dollars per barrel, slightly below the current *spot* price (assumed equal to 82 dollars per barrel), which reflects the backwardation structure of the curve: the market anticipates a fall in *forward* prices over the next nine months.

- (ii) At maturity  $T_1$ , the net flow paid by the refiner is  $Q_1 \times (K^* - S_{T_1}) = 40\,000 \times (81.04 - 88.50) = -298\,400$  dollars. The flow is negative for the fixed-paying refiner: he receives 298 400 dollars from the bank, since the *spot* price has exceeded the fixed price. His effective supply price is then  $S_{T_1} + (K^* - S_{T_1}) = 88.50 - 7.46 = 81.04$  dollars per barrel. The hedge has perfectly fulfilled its role: despite a surge in crude to 88.50 dollars per barrel, the refiner will effectively have paid only 81.04 dollars per barrel.
- (iii) Six months after setting up, the residual *forward* curve has shifted upwards:  $F(0.5, T_2) = 91.20$  dollars per barrel and  $F(0.5, T_3) = 90.30$  dollars per barrel. The value of the residual *swap* for the fixed payer is then:

$$\begin{aligned} V(0.5) &= Q \left( (F(0.5, T_2) - K^*) P(0.5, T_2) + (F(0.5, T_3) - K^*) P(0.5, T_3) \right) \\ &= 40\,000 \times ((91.20 - 81.04) \times 0.99254 + (90.30 - 81.04) \times 0.98514) \\ &= 40\,000 \times 19.206 \approx +768\,240 \text{ dollars} \end{aligned}$$

The strong rise in the *forward* curve (from roughly 81 dollars to 91 dollars per barrel) has created a substantial positive value for the fixed-paying refiner: his *swap*, initiated at nil value, is now worth more than 768 000 dollars in *mark-to-market*, which exactly offsets the increased cost of his future physical supply.  $\square$

### Key Takeaways



A *commodity swap* is an OTC contract for the exchange of financial flows computed on a quantity  $Q$  of a commodity: the fixed payer pays  $Q \times K$  and receives  $Q \times S_i$  at each maturity  $T_i$  (or the mean of the *spot* prices over the period in the average variant), with no physical delivery, which guarantees him a synthetically fixed supply price whatever the market level — the effective value paid always being  $S_i + (K - S_i) = K$  [Hull (2012)]. Its valuation mobilises the **cost of carry** relation, namely  $F(0, T) = S_0 e^{(r+u-y)T}$ , where  $u$  denotes the storage costs and  $y$  the convenience yield, which measures the economic advantage of physical holding of the commodity [Schwartz (1997)]. The value of the *swap* at date  $t$  and the par fixed price  $K^*$  which cancels it at initiation are written:

$$V_{\text{swap}}(t) = Q \sum_{i=1}^n (F(t, T_i) - K) P(t, T_i) \quad \text{and} \quad K^* = \frac{\sum_{i=1}^n F(0, T_i) P(0, T_i)}{\sum_{i=1}^n P(0, T_i)}$$

$K^*$  is therefore the weighted average of the *forward* prices — the exact analogue of the *swap rate*  $S^*$  for IRS, where the *forward* interest rates are replaced by the *forward* prices of the commodity. In practice, the industrial consumer enters in a fixed payer / variable receiver position in order to fix its supply cost, the producer in a fixed receiver / variable payer position in order to secure its selling price, and financial intermediaries act as counterparty on both sides while hedging their residual risk on organised *futures* markets [Hull (2012)].

## 2.3 Packaged and securitised derivatives

Beyond the standardised derivatives traded on organised markets, a range of packaged or securitised instruments makes it possible to access sophisticated risk profiles in wrappers suited to institutional constraints or to specific client bases. **Swaptions** — options on interest rate *swaps* — offer corporate treasurers and bond managers a conditional flexibility for hedging against a future interest rate risk without committing themselves to it irrevocably; **contracts for difference** (CFD) for their part have democratised access to leverage by allowing individual investors to gain exposure to the variation of an underlying for a fraction of its value, at the price of an amplified risk of loss; **warrants** and **turbos**, finally, offer on regulated listed markets an access to the asymmetric profile of options, with reduced initial outlays and liquidity assured by a market maker.

### 2.3.1 Institutional flexibility: swaptions

A **swaption** (a contraction of *swap option*) is an option which confers on its holder the right, but not the obligation, to enter into a *swap* contract on conditions fixed in advance (fixed rate  $K$ , maturity, notional), at or before an expiry date  $T_{\text{exp}}$ . The *swaption* thus combines the structure of an option (asymmetry of profile, payment of a premium) with the underlying of an IRS (the *forward swap rate*). Two basic types are distinguished:

- A **payer swaption** (or *swaption call* on the fixed rate) gives the right to enter into a fixed payer position in the underlying *swap*: its holder benefits if rates rise (the market fixed rate  $S_T$  exceeds the strike  $K$ , which makes it advantageous to pay the fixed rate  $K$  lower than the market rate);

— A **receiver swaption** (or *swaption put* on the fixed rate) gives the right to enter into a fixed receiver position: its holder benefits if rates fall.

A distinction is likewise drawn between European *swaptions* (exercise possible only at expiry  $T_{\text{exp}}$ ), American *swaptions* (exercise possible at any instant) and Bermudan *swaptions* (exercise possible at predefined discrete dates, typically the coupon dates of the underlying *swap*): the latter are the most used in *callable* debt structures (a bond redeemable in advance), since they allow the issuer to redeem his bond at each coupon date from a certain horizon onwards, as Brigo & Mercurio (2006) explain.

The value of a European *swaption* is obtained by the model of Black (1976), which is a direct adaptation of the model of Black & Scholes (1973) to options whose underlying is a *forward* interest rate rather than a share price. The idea is the following: the *payer swaption* gives the right to enter into an IRS by paying the fixed rate  $K$  at date  $T_{\text{exp}}$ : it will be exercised if and only if the market *swap rate*  $S(T_{\text{exp}})$  exceeds the strike  $K$  at that date. Its *payoff* at expiry is:

$$\Pi_{\text{pay}}(T_{\text{exp}}) = N \times \underbrace{\sum_{i=1}^n \delta_i P(T_{\text{exp}}, T_i)}_{=A(T_{\text{exp}})} \times \max(S(T_{\text{exp}}) - K, 0)$$

We note that  $A(T_{\text{exp}})$  is the *annuity factor* of the underlying *swap* at the expiry date, that is to say the present value of a series of payments of one euro at each coupon date  $T_i$ , so that the term  $N(S - K)A(T_{\text{exp}})$  of the *payoff* of the *swaption* is indeed the value of a fixed payer IRS at the strike  $K$  when the market rate is worth  $S$ . This *payoff* is structurally identical to that of a European *call* of exercise price  $K$  and underlying  $S$ , except that the ‘discount factor’ is not  $e^{-rT}$  (continuous rate) but the *annuity factor*  $A$ . The formula of Black (1976) is obtained by applying exactly the same logic as that of Black & Scholes (1973):

$$V_{\text{pay}}(t) = N \times A(t) \times (S(t)\Phi(d_1) - K\Phi(d_2)) \quad \text{with} \quad \begin{cases} d_1 = \frac{\ln(S(t)/K) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{T_{\text{exp}}-t}} \\ d_2 = \frac{\ln(S(t)/K) - \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{T_{\text{exp}}-t}} \end{cases}$$

where  $\Phi$  is the cumulative distribution function of the standard normal distribution. The value of a *receiver swaption* is obtained by put-call parity applied to *swaps*: holding simultaneously a *payer* and a *receiver swaption* of the same strike  $K$  guarantees entry into a *swap* (one will exercise one or the other according to the direction of the market), which is worth today  $NA(t)(S(t) - K)$ . We deduce from this  $V_{\text{rec}}(t) = N \times A(t) \times (K\Phi(-d_2) - S(t)\Phi(-d_1))$ , with  $d_1$  and  $d_2$  defined exactly as previously, given that  $\Phi(-d_i) = 1 - \Phi(d_i)$ . In practice, *swaptions* are quoted on an **expiry**  $\times$  **maturity grid**: for example, a ‘5Y  $\times$  10Y’ *swaption* designates an option of five-year expiry on an IRS of ten-year maturity. For each point of this grid, the market quotes an **implied volatility**  $\sigma(K)$ , which varies with the strike  $K$ : this dependence constitutes the **volatility smile** of *swaptions*. The **SABR model** (*stochastic alpha beta rho*), introduced by Hagan et al. (2002), has become the market standard for parameterising and interpolating this *smile*.

What of institutional uses? The primary use of *swaptions* is the management of **callable bonds**: a firm which issues a fixed-rate bond with an early redemption clause (a *call*) implicitly incorporates a Bermudan *swaption* into its debt. From the issuer’s point of view, the *call* confers on him the right to refinance his debt if rates fall (he will exercise early redemption in order to re-borrow more cheaply): this is economically equivalent to holding a Bermudan *receiver swaption* of *strike* equal to the coupon of the bond, as Hull (2012) explains. On the buyer’s side, insurance companies and pension funds use *receiver swaptions* in order to hedge the reinvestment risk of their long liabilities: if rates fall, the discounted value of their commitments increases; the *receiver swaption* allows them to receive a high fixed rate (strike  $K$ ) in a *swap*, offsetting the fall in the yield of their bond assets. This institutional asymmetry — insurers buyers of *receiver swaptions*, issuers sellers of *payer swaptions* — durably structures the *swaptions* market and explains the persistent asymmetry observed in implied volatility surfaces, as Brigo & Mercurio (2006) analyse.

### Exercise



**Valuation of a payer swaption by the Black model.** On 1 January N, a firm plans to issue in two years a fixed-rate bond of notional 10 000 000 euros and of maturity five years. Fearing a rise in rates in the meantime, it buys a European *payer swaption* of two-year expiry on a five-year IRS (a ‘2Y  $\times$  5Y’ *swaption*) of strike  $K = 3.80\%$ . The five-year *forward swap rate* observed today for a start in two years is  $S_0 = 3.50\%$ . The five-year *forward annuity factor* is worth  $A_0 = 4.40$ . The Black volatility quoted on the market for this *swaption* is  $\sigma = 28\%$ .

- (i) Compute  $d_1$ ,  $d_2$  and the premium of the payer swaption. Interpret the relation economically. We shall assume  $\Phi(d_1) \approx 0.4960$  and  $\Phi(d_2) \approx 0.3424$ .



- (ii) Is the swaption *in-the-money*, *at-the-money* or *out-of-the-money*? What would its value be if  $S_0 = K = 3.80\%$ ? Compute the new values of  $d_1$  and of  $d_2$ . We shall take  $\Phi(d_1) \approx 0.5785$ .

**Solution.** (i) For  $\tau := T_{\text{exp}} - 0 = 2$  years (where  $T_{\text{exp}}$  denotes the expiry date of the *swap*),  $S_0 = 3.50\%$  (the current interest rate),  $K = 3.80\%$  (the *strike* of the *swap*) and  $\sigma = 28\%$  (the volatility measured by the market), we find  $d_1 = \frac{\ln(3.50/3.80) + 0.5 \times 0.28^2 \times 2}{0.28 \times \sqrt{2}} \approx -0.0096$ , whence we deduce  $d_2 = d_1 - \sigma\sqrt{\tau} = -0.0096 - 0.28\sqrt{2} \approx -0.4056$ . We then deduce:

$$\begin{aligned} V_{\text{pay}} &= N \times A_0 \times (S_0 \Phi(d_1) - K \Phi(d_2)) \\ &= 10\,000\,000 \times 4.40 \times (3.50\% \times 0.4960 - 3.80\% \times 0.3424) \approx 191\,356 \text{ euros} \end{aligned}$$

In this relation, the part  $N \cdot A_0 \cdot S_0 \cdot \mathcal{N}(d_1) \approx 762\,384$  euros measures the discounted expected value of the variable flows received (the floating leg of the underlying *swap*), weighted by the adjusted risk-neutral probability of exercise  $\Phi(d_1)$ , while the term  $N \cdot A_0 \cdot K \cdot \mathcal{N}(d_2) \approx 573\,028$  euros measures the discounted expected value of the fixed flows paid (the fixed leg at the strike  $K$ ), weighted by the risk-neutral probability of exercise  $\Phi(d_2)$ . The premium is then the difference between the two parts.

- (ii) Since we verify  $S_0 = 3.50\% < K = 3.80\%$ , the *swaption* is *out-of-the-money*: paying the fixed rate  $K = 3.80\%$  is not currently advantageous since the market rate is only  $3.50\%$ . If  $S_0 = K$  (*at-the-money*), the formula for  $d_1$  simplifies since  $\ln(S_0/K) = \ln(1) = 0$ . We then find  $d_1 = \frac{\sigma\sqrt{\tau}}{2} = \frac{0.28 \times \sqrt{2}}{2} \approx 0.1980$ , then  $d_2 = d_1 - \sigma\sqrt{\tau} = -\frac{\sigma\sqrt{\tau}}{2} = -d_1$ . It therefore follows that:

$$\begin{aligned} V_{\text{pay}}^{\text{ATM}} &= N \times A_0 \times (S_0 \Phi(d_1) - K \Phi(d_2)) \\ &= N \times A_0 \times (S_0 \Phi(d_1) - K \Phi(-d_1)) \\ &= N \times A_0 \times (S_0 \Phi(d_1) - K(1 - \Phi(d_1))) \\ &= 10\,000\,000 \times 4.40 \times 3.80\% \times (0.5785 - 0.4215) \approx 262\,504 \text{ euros} \end{aligned}$$

The ATM premium is naturally higher (262 504 euros against 191 356 euros), since the *swaption* is more deeply in the money in expectation.  $\square$

### Key Takeaways



A *swaption* is an option on an IRS: the *payer swaption* gives the right to enter as fixed payer (benefits from a rise in rates), the *receiver swaption* gives the right to enter as fixed receiver (benefits from a fall). The value of a European *payer swaption* is given by the model of Black (1976), a direct analogue of Black & Scholes (1973) where the underlying is the *forward swap rate*  $S(t)$  and the discount factor is the *annuity factor*  $A(t)$ :

$$V_{\text{pay}}(t) = N \times A(t) \times (S(t) \Phi(d_1) - K \Phi(d_2)) \quad \text{with} \quad \begin{cases} d_1 = \frac{\ln(S(t)/K) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{T_{\text{exp}}-t}} \\ d_2 = \frac{\ln(S(t)/K) - \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{T_{\text{exp}}-t}} \end{cases}$$

The *receiver swaption* is deduced from it by put-call parity:  $V_{\text{rec}}(t) = N \times A(t) \times (K \Phi(-d_2) - S(t) \Phi(-d_1))$ .

### 2.3.2 CFDs, or the democratisation of leverage

A **contract for difference (CFD)** is a bilateral contract by which two counterparties undertake to exchange the **difference between the price of an underlying asset at the opening of the position and its price at the close**, multiplied by the number of contracts held, as AMF (2018c) defines it. The buyer of a CFD pays if the price falls and receives if the price rises; the seller takes the opposite position. No ownership of the underlying asset is ever transferred: the CFD is a pure instrument of price differential, which allows it to replicate faithfully the economic exposure to any listed underlying (a share, an index, a currency pair, a commodity, a cryptocurrency) without the operational constraints linked to its physical holding or its delivery. If the underlying is worth  $S_t$  at the opening and  $S_T$  at the close, and the operator holds  $n$  CFDs in a long position, his result is  $\Pi = n \times (S_T - S_t) - F$  where the fees  $F$  include the **bid-ask spread** charged by the issuer (the difference between the purchase price and the sale price offered), any commissions, and the **overnight financing fees** (*swap points*) due for any CFD held beyond the



close of the session (the CFD being a leveraged instrument financed by the issuer, the operator pays a daily interest rate on the notional value of his position), as AMF (2018c) specifies. The cost of *overnight* holding is therefore an important element of valuation: for a long position held for  $d$  days, the financing cost is worth approximately  $n \times S_t \times \frac{r_{\text{ref}} + m}{360} \times d$ , where  $r_{\text{ref}}$  is the reference rate (the interbank *overnight* rate, typically SOFR or €STR) and  $m$  is the broker's commercial margin, generally between 200 and 300 basis points.

The distinctive characteristic of the CFD relative to the other derivatives studied *supra* is its **accessibility to high leverage for retail investors**. Whereas an E-mini S&P 500 *future* requires an initial margin of several thousand dollars and implies a fixed contract size, a CFD on an index may be opened with a margin as small as a few hundred euros, for a far greater notional exposure. Leverage  $L$  is defined as the ratio between the notional value of the position and the margin  $M$  deposited, that is  $L = \frac{n \times S_t}{M}$ , so that a variation of 1 % in the underlying translates into a variation of  $1 \% \times L$  on the margin. With a leverage of 20 (a margin rate of 5 %), an unfavourable variation of 5 % in the underlying entirely wipes out the outlay (liquidation of the position). It is precisely for this reason that CFDs are distinguished from institutional derivatives: their high leverage, combined with their accessibility (no minimum contract size, no clearing house, mass-market electronic interface), has made them the principal vector of the 'democratisation' of speculative access to financial markets for private individuals. More precisely, unlike *futures* traded on an organised market with a central clearing house, CFDs are therefore bilateral contracts between the investor and his **broker-issuer** (*CFD provider*), which plays the role of *market maker*: it permanently displays a buying price (*bid*) and a selling price (*ask*) and acts as counterparty to all orders. The broker may hedge its residual risk in two ways: either by taking an opposite position on the underlying market (*straight-through processing*), or by internalising the risk, betting that its clients' gains and losses offset one another statistically. This second model, more widespread among mass-market brokers, creates a **structural conflict of interest**: the issuer benefits from its clients' losses, which may give it an incentive to widen spreads or to execute orders at unfavourable prices, as AMF (2018c) warns. Confronted with a high rate of losses among retail investors — 74 to 89 % of client accounts were losing money on CFDs — European Securities and Markets Authority (2018) adopted in 2018, and the national authorities including the AMF subsequently made permanent, permanent restrictions on the marketing of CFDs to retail investors:

- **Leverage caps** differentiated by asset class: 30 for major currency pairs, 20 for major indices and gold, 10 for commodities other than gold and for minor indices, 5 for individual shares, 2 for cryptocurrencies and 5 for other products.
- An **automatic close-out** of positions at 50 % of margin (the *margin close-out rule*): the issuer is obliged to cut positions as soon as the residual margin falls below 50% of the initial margin required, thus limiting the risk of a negative balance.
- A **negative balance protection**: the retail investor cannot lose more than the balance of his account, even in the event of a market *gap* (a sharp jump in the quotation of an asset, notably between Friday evening and Monday morning);
- A **prohibition on bonuses** as an incentive to open an account.

### Exercise



**Analysis of a CFD position: leverage, margin and financing cost.** An investor opens on 1 January N a long position on 200 CFDs on the CAC 40 index, whose level at the opening is 7 200 points. Each CFD is worth 1 euro per point. The broker requires an initial margin of 5 % of the notional value. The *overnight* financing rate is €STR plus 250 basis points (we shall assume the €STR rate to be equal to 3 %). The *margin close-out* rule is triggered at 50 % of the initial margin.

- (i) Compute the notional of the position, the initial margin required, the effective leverage and the *margin close-out* threshold.
- (ii) The investor holds the position for 15 days. Compute the total *overnight* financing cost.
- (iii) At the end of 15 days, the CAC 40 is quoted at 7 450 points. Compute the net result of the position (excluding the *bid-ask spread*).

**Solution.** (i) The notional value of the position of 200 CFDs amounts to  $200 \times 7\,200 \times 1,00 = 1\,440\,000$  euros. The initial margin required in order to open such a position is therefore  $5 \% \times 1\,440\,000 = 72\,000$  euros, which

corresponds to a leverage of  $1\,440\,000/72\,000 = 20$ . Moreover, the *margin close-out* rule is triggered when unrealised losses reach 50 % of the initial margin, that is  $50\% \times 72\,000 = 36\,000$  euros. This loss corresponds to a fall in the index of  $36\,000/(200 \times 1) = 180$  points. The close-out threshold is therefore at  $7\,200 - 180 = 7\,020$  points, that is a fall of 2.5 % in the CAC 40.

- (ii) For a long position held for  $d$  days, the financing cost is worth approximately  $n \times S_t \times \frac{r_{\text{ref}} + m}{360} \times d$ , where  $r_{\text{ref}}$  is the reference rate and  $m$  is the broker's commercial margin. With  $r_{\text{ref}} + m = 3\% + 2.5\% = 5.5\%$  and  $d = 15$  days, this corresponds to a financing cost of  $1\,440\,000 \times \frac{5.5\%}{360} \times 15 = 1\,440\,000 \times 0.002292 \approx 3\,300$  euros. This is an amount which may seem negligible over a short position, but which accumulates significantly over long holdings and erodes the return of strategies which would consist in using CFDs as substitutes for long-term investments.
- (iii) The gross capital gain realised amounts to  $200 \times (7\,450 - 7\,200) \times 1 = 50\,000$  euros, from which the *overnight* financing cost must be deducted, which gives a net capital gain excluding the *bid-ask spread* of  $50\,000 - 3\,300 = +46\,700$  euros, that is a return of  $46\,700/72\,000 \approx +64.9\%$  on the margin committed, for a rise of only  $250/7\,200 \approx 3.5\%$  in the underlying: this is the leverage effect at work. Symmetrically, an identical fall of 250 points would have generated a loss of  $50\,000 + 3\,300 = 53\,300$  euros, that is  $-74.0\%$  of the margin if the *close-out* rule had not been triggered (the threshold was at  $-180$  points, that is  $-36\,000$  euros of loss, which would have closed the position before reaching  $-250$  points).  $\square$

### Focus...



**Access to the asymmetric profile: warrants and turbos.** Warrants and turbos are **exchange-listed derivatives**, issued by financial institutions (investment banks such as Société Générale, BNP Paribas or Commerzbank) and tradable on regulated markets such as Euronext Paris. Unlike standardised options traded on organised markets (CBOE, Eurex), they cannot be sold short by an investor: only the issuer acts as counterparty, ensuring liquidity via a *market making* mechanism. Each product is identified by an ISIN code and quoted continuously. These two families of instruments share the vocation of offering retail or semi-institutional investors an **access to the asymmetric gain profile** (potentially unlimited gains, loss limited to the premium), as Broyon (2026) explains.

- A **warrant** is economically identical to a European option (or American, according to the terms of the prospectus) issued directly by a bank: it confers on its holder the right, but not the obligation, to buy (*call warrant*) or to sell (*put warrant*) an underlying asset at a fixed exercise price (*strike*)  $K$ , at or before an expiry date  $T$ , in exchange for the payment of a premium  $C$  on purchase, determined by the formula of Black & Scholes (1973). What distinguishes the *warrant* from a standard exchange-listed option is not its legal nature but its **parity**  $p$ : a *warrant* does not give the right over one unit of the underlying, but over a fraction  $1/p$  of it (typically  $p = 10$  or  $p = 100$ ). Its quoted price is therefore  $C/p$ , which makes the entry ticket accessible at low unit amounts (often a few cents to a few euros), while preserving a significant leverage effect.
- A **turbo** for its part resembles a *down-and-out* barrier option, generally with a knock-out barrier equal to its *strike*, issued by an institution. In the event of knock-out, the holder therefore loses virtually all of his outlay, independently of any subsequent rebound of the underlying. Here again, the *turbo* is a leveraged product whose parity  $p$  is defined analogously to that of the *warrant*. There moreover exist **open-end turbos** with no fixed expiry date: the issuer then incorporates daily financing fees identical to those of CFDs: the barrier  $K$  is then adjusted each day by an amount equal to the financing cost multiplied by the notional value, as Swissquote (2025) explains.

### Key Takeaways



A CFD (*contract for difference*) is a bilateral contract replicating economic exposure to an underlying via the exchange of the price differential alone, without transfer of ownership; its distinctive characteristic is the high leverage accessible to private individuals, an unfavourable variation of  $1/L$  ( $L$  being the leverage) in the underlying being enough to wipe out the outlay [AMF (2018c)]. The broker-issuer plays the role of *market maker* and may internalise the risk (a structural conflict of interest: it benefits from client losses), while the cost of *overnight* holding makes any long-duration holding prohibitive. Faced with a loss rate of 74 to 89 % of retail accounts, European Securities and Markets Authority (2018) imposed in 2018 permanent restrictions: leverage caps differentiated by asset class (from 2 for cryptocurrencies to 30 for major currencies), automatic close-out at 50 % of residual margin, negative balance protection and a prohibition on incentive bonuses.

### 2.3.3 A few words on structured products

A **structured product** is a financial instrument whose redemption profile — coupon, capital protection, redemption date — is determined in advance by a contractual formula which combines a bond asset (generally a debt security) and one or more embedded derivatives (vanilla options, barriers, basket options). The issuer (an investment bank) finances itself via the bond component and finances the optionalities thanks to the coupon given up by the investor: a structured product is therefore always analysable as the sum of a zero-coupon bond and a portfolio of options, as Hull (2012) explains. This market is experiencing strong growth in France: annual inflows from a non-professional client base went from roughly 23 billion euros in 2021 to nearly 42 billion in 2023, distributed more than 80 % via life assurance contracts; the median annual return observed on the products redeemed over this period stood between 6 % and 7 % in a context of rising markets according to AMF (2025a).

The most widespread family in Europe is that of **autocalls** (*automatic callable*): these products with conditional early redemption represented the majority of the volumes issued in France in 2023 according to AMF (2025a), at the expense of capital-guaranteed products whose share collapsed with the rise in key rates. From a regulatory point of view, structured products marketed to non-professional investors are subject to the obligation to produce a **key information document** (KID) in conformity with the European PRIIP regulation (2014/1286/EU), which notably imposes the presentation of three performance scenarios (unfavourable, intermediate, favourable) and of the total cost of the product (the summary cost indicator).

- An **athena autocall** is a structured product which may be automatically redeemed in advance if, at a periodic observation date  $T_i$ , the level of the underlying (an index, a share or a basket) exceeds an **autocall barrier**, generally fixed at 100 % of the initial level. In the event of a call at date  $T_i$ , the investor receives his initial capital increased by a cumulative annual coupon  $c$  for each period elapsed since issue, that is  $N \times (1 + i \times c)$ , where  $i$  is the number of years elapsed and  $c$  the annual coupon. If no call occurs before the final maturity  $T_n$ , two scenarios present themselves: if the underlying is above a **protection barrier** (typically 60 to 70 % of the initial level), the investor recovers his capital in full; below it, he suffers a capital loss proportional to the fall in the underlying, as Société Générale (2022) explains.
- The **phoenix autocall** is a variant in which **conditional coupons** are paid periodically, at each observation date  $T_i$ , as soon as the underlying is above a coupon barrier  $b_c$  (often 70 to 80 % of the initial level, below the autocall barrier). The fundamental distinction from the athena is that the investor receives regular flows during the life of the product, as Nalo (2025) explains. In the **memory phoenix** variant, the coupons not paid during an unfavourable period (underlying below  $b_c$ ) are placed in ‘memory’ and made good in full at the first subsequent observation date at which the underlying moves back above  $b_c$ . This **memory effect** makes the phoenix particularly attractive in markets that fall slightly and then recover, and differentiates it from the athena whose coupons are paid only once, at the call.

All the preceding products may come in a **worst-of** (resp. *best-of*) version: instead of being indexed to a single underlying, the product is indexed to a **basket of underlyings**, and the performance retained at each observation date is that of the least performing (resp. the most performing) underlying of the basket. For example, a *worst-of* athena autocall on a basket of three shares (TotalEnergies, LVMH, BNP Paribas) will be called at  $T_i$  if and only if each of the three shares is above its autocall barrier; at maturity, the capital loss is computed on the performance of the worst of the three shares. The *worst-of* mechanism mechanically increases the risk (the product is more difficult to call, the potential loss higher), but in return makes it possible to offer a significantly higher annual coupon: the investor implicitly sells an **option on the correlation** between the underlyings. The investor is thus implicitly a seller of correlation: the lower the correlation between the assets of the basket, the higher the risk that a single asset will break down and drag the product with it, the more expensive the *worst-of* option is for the issuer to hedge, and the more generous the coupon offered must be — in return for an increased risk for the investor, as Finance de marché (2013) explains. Conversely, a basket of strongly correlated assets (for example three indices from the same sector) is almost equivalent to a single-underlying product: the coupon paid will then be close to that of a classic autocall, as Sherif (2019) analyses.

#### Key Takeaways



A structured product is analysable as the sum of a zero-coupon bond and a portfolio of options: the issuer finances the optionalities thanks to the coupon given up by the investor [Hull (2012)]. The dominant family is that of *autocalls*, of which two main variants are distinguished by their logic of coupon payment:

- The athena: automatic early call if the underlying exceeds the autocall barrier (100 % of the initial level) at an observation date; the cumulative coupon  $N \times (1 + i \times c)$  is paid only at the moment of the call; at maturity without

a call, the capital is protected above the protection barrier (60 % to 70 %), otherwise there is a loss proportional to the fall [Société Générale (2022)].

- The phoenix: the same call mechanics, but periodic conditional coupons are paid at each observation date as soon as the underlying exceeds the coupon barrier  $b_c$  (70 % to 80 % of the initial level); in the memory variant, the coupons not paid are made good at the first subsequent date at which  $b_c$  is crossed [Nalo (2025)].

These products exist in a *worst-of* version: the call condition and the computation of losses bear on the performance of the least performing underlying of the basket; the mechanism increases the risk (a call is more difficult, the potential loss amplified) but justifies an appreciably higher coupon, since the investor is implicitly a seller of correlation: the lower the correlation between the assets of the basket, the higher the risk that a single asset will break down and the more generous the coupon paid must be in return [Finance de marché (2013)].

### 3 The actors of alternative finance

The actors of alternative finance bring together a heterogeneous constellation of participants — *hedge funds*, private equity funds, private debt funds, funds of funds, *family offices*, crowdfunding, incubators... — who are situated outside the traditional circuits of collective management and classic banking intermediation. Their common feature is to mobilise **alternative management strategies** (leverage, arbitrage, derivatives, unlisted assets...) in the service of objectives of absolute performance and diversification for investors.

#### 3.1 Hedge funds

A *hedge fund* is an alternative investment fund, reserved mainly for professional or qualified investors, which collects capital in order to manage it in a highly flexible manner via so-called alternative management strategies (short selling, leverage, derivatives, arbitrage) with the objective of generating a so-called absolute performance, that is to say one little correlated with the general trend of financial markets. It is distinguished from traditional funds by a lighter regulation, a great freedom in the choice of assets and of the techniques used, as well as a higher risk profile in return for a superior potential return.

##### 3.1.1 Definition, origin and legal structure

In 1949, Alfred Winslow Jones, a sociologist by training who had converted into a financial analyst for *Fortune* magazine, constituted the first *hedge fund*, establishing himself as managing partner (*general partner*, GP), as CFA Institute (2020) recalls. His model rests on three founding principles: permanently maintaining **short positions** in order to offset long positions, having systematic recourse to **leverage**, and receiving a **variable remuneration** that is exclusively incentive-based, fixed at 20 % of profits, with no fixed management fee, as CFA Institute (2020) details. Short positions were supposed to immunise the portfolio against market reversals, conferring on the fund an ambition of absolute performance, that is to say one independent of the general direction of markets. While these three original precepts retain a historical relevance, the contemporary reality of *hedge funds* has largely freed itself from them according to CFA Institute (2020): not all maintain short positions, not all are massively indebted, and virtually all now receive fixed management fees in addition to performance fees. The rise of the industry has been particularly sustained: according to the HFRI cited by CFA Institute (2020), assets under management (**AUM**) went from roughly 39 billion dollars in 1990 to 491 billion in 2000, then to 2 900 billion dollars at the end of 2015, testifying to a multiplication by 74 of the assets managed in twenty-five years. Despite this growth, *hedge funds* and mutual funds represent only roughly 3 % of total assets managed worldwide in 2014, but captured more than 30 % of the total revenues from management fees in 2013, illustrating the profitability of their pricing model, as CFA Institute (2020) analyses.

From a legal and organisational point of view, hedge funds are structured almost universally in the form of **limited partnerships**, in which the manager occupies the position of GP — theoretically exposed to unlimited liability — while investors intervene as *limited partners* (LP), whose liability is confined to their contributions, as CFA Institute (2020) explains. These vehicles, reserved for a restricted number of qualified investors — institutions and *high-net-worth individuals* — benefit from a lighter regulation relative to funds open to the general public: in the United States, the *Securities Act* of 1933 authorises their marketing in the form of private placements (*Regulation D*), while the European *Alternative investment fund managers* directive (**AIFMD**), implemented in 2013, requires managers operating or marketing in the European Union to obtain administrative authorisation, as CFA Institute (2020) recalls.

In the United States, since 2012, *hedge funds* managing more than 100 million dollars are likewise required to register with the *Securities and Exchange Commission* (SEC) and to submit regularly the *Form PF* risk disclosure document to the *Financial Stability Oversight Council*, as CFA Institute (2020) explains. From the point of view of domiciliation, the Cayman Islands constitute the jurisdiction of choice for American, European and Asian managers, on account of their favourable taxation and the flexibility of their company law; some managers nevertheless choose a domestic domiciliation — in the United Kingdom or the United States — in order to benefit from the credibility conferred by registration with the *Financial Services Authority* (FSA) or the SEC, even if it means setting up parallel *offshore* funds in order to attract certain categories of foreign investor. This architecture confers on the manager a great operational latitude: investment mandates are deliberately formulated in a broad manner, and managers generally have a discretionary allowance — typically below 20 % of AUM — with no sectoral or geographical allocation constraint according to CFA Institute (2020). A single management group may moreover manage several *hedge funds* simultaneously with distinct strategies — a credit fund, a commodities fund, a multi-strategy fund — while possibly managing direct management mandates as well, which complicates the identification of conflicts of interest and reinforces the importance of a thorough operational **due diligence**, as CFA Institute (2020) explains.

The fee structure constitutes one of the most distinctive characteristics of *hedge funds* and the main source of their profitability for managers. The canonical **2–20** model combines a fixed *management fee* of 2 % of annual AUM — received independently of performance — and an *incentive fee* of 20 % of the profits realised, as CFA Institute (2020) explains: a fund starting with 100 million dollars which achieves a gross return of 30 % in the first year generates year-end AUM of 130 million dollars, that is a management fee of  $0.02 \times 130 = 2.6$  million and a performance fee of  $0.2 \times 30 = 6$  million, for total fees of  $2.6 + 6 = 8.6$  million dollars and a net return for the LP of only  $(30 - 2.6 - 6) / 100 = 21.4\%$  after deduction of fees against 30 % gross (which brings the value of the fund *in fine* to 121.4 million); if the performance fee is computed net of the management fee, it falls to  $0.2 \times (30 - 2.6) = 5.48$  million, bringing the net return of the LP to  $(30 - 2.6 - 5.48) / 100 = 21.92\%$ . Competitive pressure has since led to a compression of these fees: according to CFA Institute (2020), the average management fees of new funds stood at 1.6 % and performance fees at 17.75 % in 2015, against an average performance fee of 18.5 % in 2007. Moreover, two protective mechanisms frame the computation of performance fees. On the one hand, the **hurdle rate** requires that the fee be received only once a minimum return has been attained — often indexed to a riskless rate such as Libor or a Treasury bill rate increased by a premium, or fixed as an absolute nominal return target — according to two variants explained by CFA Institute (2020): the *hard hurdle* formula (fee computed on the excess return above the threshold alone) and the *soft hurdle* formula (fee computed on the whole of the return as soon as the threshold is exceeded). On the other hand, the **high-watermark** stipulates that the manager may receive a performance fee only on condition of having first wiped out past losses and restored the net asset value net of fees to its highest historical level: if the fund goes from 121.4 million to 110 million in the second year, only the management fee of  $0.02 \times 110 = 2.2$  million is received, with no *incentive fee*, bringing the net value of the fund to  $110 - 2.2 = 107.8$  million; in the third year, if the fund rises again to 128 million, the *management fees* stand at  $0.02 \times 128 = 2.56$  million, while the performance fee applies only to the excess of  $128 - 121.4 = 6.6$  million above the *high-watermark* of 121.4 million, that is  $0.2 \times 6.6 = 1.32$  million, for a net annual return of  $(128 - 2.56 - 1.32) / 107.8 = 15.14\%$  that year and a fund value of  $128 - 2.56 - 1.32 = 124.12$  million euros. Over the three years, the average net annual return for the LP comes out at only  $(124.12 / 100)^{1/3} - 1 \approx 7.47\%$ , for a capital gain of 24.12 million dollars, while the manager will have collected a total of  $(2.6 + 6) + 2.2 + (2.56 + 1.32) = 14.68$  million dollars of fees, that is  $14.68 / (24.12 + 14.68) = 37.4\%$  of the profit generated. Moreover, the conditions of redemption are likewise framed contractually: a **lockup period** obliges the investor to keep his units for a minimum duration, while a *notice period* of 30 to 90 days is generally required before any redemption, as CFA Institute (2020) explains.

#### Focus...



**Funds of funds.** Funds of funds (FoF) constitute a variant of indirect access to the *hedge fund* asset class. In its most common form, a FoF collects capital from investors — notably wealthy private individuals whose outlay would be insufficient to satisfy the minimum entry ticket of a first-rank *hedge fund* — and allocates this capital to a diversified portfolio of underlying funds, covering several strategies (*equity hedge*, *event-driven*, *relative value*, *macro*), several geographical regions and several management styles, as CFA Institute (2020) explains. The first structural advantage of FoF is **access to closed funds**: many of the best-performing managers no longer accept new direct capital and the FoF, whose relationship history is long-standing and whose assets are significant, may have negotiated preferential access to these vehicles according to CFA Institute (2020). The second advantage is the **mutualisation of due diligence**: correctly assessing a *hedge fund* implies analysing the strategy and the investment process, the performance history (gross and net of fees), the fee structure, the size and longevity of the fund, the management style and *key person risk*, the reputation, the quality of investor relations, the risk management systems,



the *prime brokerage* arrangements and the quality of the independent audit, as CFA Institute (2020) explains. The third advantage concerns **redemption conditions**: FoF may have negotiated shorter notice periods and reduced lockup periods with the underlying managers, and may have a liquidity reserve or access to temporary financing in order to honour the redemption requests of their own investors without having to liquidate their positions in the underlying funds prematurely, as CFA Institute (2020) analyses.

These operational advantages nevertheless have a direct and measurable cost: the **double layer of fees**. Each underlying fund levies its own management fee and its performance fee, typically according to the 2–20 model; the FoF superimposes on this its own fees, conventionally fixed at 1 % of *management fee* and 10 % of *incentive fee* according to CFA Institute (2020). Thus, supposing that an underlying fund generates on an AUM of 100 million a gross return of 20 %, the computation of fees at the level of the underlying fund is established as follows:  $0.02 \times 100 = 2$  million of *management fees* and  $0.2 \times 20 = 4$  million of *incentive fees*, that is a net amount of  $120 - 6 = 114$  million euros for an *a priori* net return of  $114/100 = 14$  %. But if the investor goes through a FoF in order to access this underlying fund, then a further  $0.01 \times 100 = 1$  million of *management fees* and  $0.1 \times 14 = 1.4$  million of *incentive fees* must be deducted, that is a net amount of  $114 - 2.4 = 111.6$  million for a net return on the investment via the FoF of  $111.6/100 = 11.6$  %. The gap of 2.4 percentage points (14 % against 11.6 %) represents the **pure cost of intermediation**. This gap is not negligible: compounded over ten years, it leads *a priori* to a final wealth of  $100 \times 1.14^{10} \approx 370.7$  million for the direct investor against  $100 \times 1.116^{10} \approx 299.7$  million for the one going through the FoF, that is a shortfall of 70.6 million for an initial capital of 100 million. Let us note, however, that this example is constructed under the favourable assumption that all the underlying funds deliver exactly the same gross return of 20 % per year: in practice, if the FoF selects managers whose gross performance offsets the additional cost of 2.4 points, intermediation becomes rational; otherwise, the added value of the FoF's *due diligence* is nil or even negative. One last behavioural aspect deserves to be signalled. As CFA Institute (2020) explains, hedge fund managers often describe FoF capital as *fast money*: in practice, FoF — notably those serving wealthy private individuals rather than institutions — tend to be the first to request their redemptions when the performance of the underlying funds deteriorates, increasing liquidity pressure on managers precisely at the moment when it is most destabilising. This procyclical behaviour may harm the investors remaining in the underlying fund and leads some first-rank managers to grant FoF less favourable redemption conditions than to direct institutional investors, paradoxically reducing one of the comparative advantages that FoF are supposed to bring.

### Key Takeaways



Born in 1949 under the impetus of Alfred Winslow Jones, the *hedge fund* designates a private investment vehicle structured as a *limited partnership* — the manager being the *general partner* (GP), the qualified investors being *limited partners* (LP) — which benefits from a lighter regulation, a great latitude in the strategies employed and deliberately broad mandates. Its canonical remuneration, the ‘2–20’ model (a *management fee* of 2 % of AUM and an *incentive fee* of 20 % of profits), is framed by two protections in favour of the LP: the *hurdle rate*, a minimum return threshold below which no performance fee is due, and the *high-watermark*, which obliges the manager to wipe out past losses before receiving an *incentive fee* again. Investors are moreover subject to a *lockup period* and to a *notice period* of 30 to 90 days, which limit liquidity but allow the manager to manage his positions in an orderly fashion. Moreover, *funds of funds* (FoF) offer indirect and diversified access to the asset class, at the price of a double layer of fees (1–10 in addition to the 2–20 of the underlying funds) which appreciably erodes the net return, unless the added value of their *due diligence* and their access to closed funds offsets this additional cost.

#### 3.1.2 A taxonomy of strategies: from *equity hedge* to *macro*

The richness and complexity of the *hedge fund* universe stem in large part from the almost unlimited diversity of the strategies they deploy. This heterogeneity makes any classification approximate and evolving by nature: the large data houses such as *Hedge Fund Research Inc.* (HFRI) or the *Chartered Alternative Investment Analyst* (CAIA) association propose competing taxonomies, and the boundaries between categories remain porous, so keen are managers to preserve the flexibility of their mandates, as CFA Institute (2020) explains. The HFRI classification, which prevails in the academic and practical literature, identifies four large families of strategies: *equity hedge*, *event-driven*, *relative value* and *macro*. The CAIA classification, which prefers the categories *corporate restructuring*, *convergence trading*, *opportunistic* and *market directional*, approximately overlaps the four HFRI families, underlining that while the labels vary, the underlying economic logics remain stable according to CFA Institute (2020).

The relative weights of these families in assets under management have undergone an important structural mutation over the last twenty-five years. Related to absolute assets, these percentages reveal a greater magnitude: in 1990, the

industry managed 39 billion dollars, rising to 2 900 billion in 2015 according to CFA Institute (2020). The 39 % share of the *macro* strategy in 1990 therefore represented roughly 15.2 billion dollars; its 27 % share in 2015 represents 783 billion, a multiplication by 51 in absolute value but a fall of 12 points in relative weight. Conversely, *relative value* went from 5.5 billion dollars (14 % of 39 billion dollars) to 812 billion (28 % of 2 900 billion dollars), that is a multiplication by 148 in absolute value, reflecting a growing sophistication of the industry and its shift towards strategies with lower market directionality CFA Institute (2020).

| Strategy              | 1990 | 2010 | 2015 | Change 1990–2015 |
|-----------------------|------|------|------|------------------|
| <i>Macro</i>          | 39 % | 20 % | 27 % | –12 points       |
| <i>Equity hedge</i>   | 37 % | 30 % | 19 % | –18 points       |
| <i>Event-driven</i>   | 10 % | 26 % | 26 % | +16 points       |
| <i>Relative value</i> | 14 % | 24 % | 28 % | +14 points       |

Table 4.2: Evolution of the distribution of AUM by *hedge fund* strategy according to CFA Institute (2020)

- (i) The first family, **equity hedge strategies**, constitutes the historic category of the sector, directly derived from the founding principles of Alfred Winslow Jones. These funds operate mainly on listed equity markets and their derivatives, adopting a *bottom-up* approach: the analysis starts from the individual firm and works up towards sectors and the market as a whole, unlike the *top-down* approach of *macro* strategies, as CFA Institute (2020) explains. Their share of the industry's total assets, which reached 37 % in 1990 (that is roughly 14.4 billion dollars), receded to 19 % in 2015 (551 billion dollars), a contraction in relative weight explained notably by the rise of more sophisticated *market neutral* strategies and of quantitative management according to CFA Institute (2020).
  - Within this family, **market neutral funds** represent the most accomplished expression of neutrality to market risk: they construct portfolios whose beta is kept as close as possible to zero, by combining long positions on undervalued securities and short positions on overvalued securities, according to signals arising from quantitative, technical or fundamental analysis, as CFA Institute (2020) explains. Neutrality is sometimes extended to the size, sector and style factors, so that the performance of the fund is theoretically immunised against any evolution of the leading market: the objective is a nil beta, and not simply a beta below 1.
  - **Fundamental growth and fundamental value strategies**, by contrast, adopt an avowedly directional approach: the former rely on fundamental analysis in order to identify firms with a strong potential for growth and capital revaluation, accumulating concentrated long positions; the latter seek assets discounted relative to their intrinsic value, in a tradition close to *deep value investing*, possibly extending to positions in distressed securities (*deep value*), as CFA Institute (2020) analyses.
  - **Quantitative directional strategies**, which have recourse to technical analysis in order to identify aberrant price-value pairs, are distinguished from the preceding ones by their systematic and algorithmic character: they vary their net long or short exposure as a function of the anticipated direction of the market and of the phase of the cycle, without discretionary judgement according to CFA Institute (2020).
  - **Sector specific funds** for their part exploit the manager's sectoral expertise in niches such as technology, life sciences (*biotech & life sciences*) or financial services, combining quantitative, technical and fundamental analysis according to CFA Institute (2020).
  - Finally, **short bias funds**, structurally net sellers, constitute one of the rare sources of pure hedging in periods of crash, at the price of a durably negative performance during prolonged bull markets: between 2010 and 2014, the MSCI ACWI (*all world country index*) delivered an annualised return of 10.7 % according to CFA Institute (2020), making the maintenance of net short positions structurally costly.
- (ii) The second family, **event-driven strategies**, seeks to exploit the valuation discontinuities generated by corporate restructuring events, according to an equally *bottom-up* approach according to CFA Institute (2020). Their progression — from 10 % of the industry's AUM in 1990 (3.9 billion dollars) to 26 % in 2015 (754 billion dollars), that is a multiplication by 193 in absolute value — is explained by the strong intensification of merger and acquisition and corporate restructuring activity from the 2000s onwards, which mechanically multiplied the opportunities accessible to these strategies.
  - The **merger arbitrage** strategy is the most classic illustration of this: when a takeover offer is announced, the target typically trades at a price slightly below the price offered, the *deal spread* reflecting the uncertainty as to the completion of the operation according to CFA Institute (2020). If a company A proposes to buy company B at 50 dollars per share while B is quoted at 47 dollars, the arbitrageur buys B at 47 dollars and sells A short, hoping



to pocket the spread of three dollars at the closing of the operation; his main risk — known as *deal-break risk* — is that of a failure of the transaction, which would entail a brutal return of B's price towards its pre-announcement levels, typically below 40 dollars, generating a loss asymmetrically greater than the hoped-for gain, as CFA Institute (2020) warns.

- **Distressed & restructuring funds** intervene for their part on the securities of issuers in a situation of financial distress, or even in insolvency proceedings: they buy *senior* bonds at a discount sometimes greater than 40 or even 50 basis points relative to par, anticipating that a favourable restructuring will make it possible to recover a significant fraction of the nominal value, or construct complex relative positions — long senior debt, *short* junior debt, or *long* preference shares, *short* ordinary shares — in order to profit from the expected widening of the *spreads* between instruments of different ranks, as CFA Institute (2020) explains.
  - **Activist funds** adopt a more interventionist posture: by acquiring a significant stake — typically between 5 % and 15 % of the capital — in a listed company, they seek to influence its strategy (disposals of assets, distribution of capital, renewal of management, optimisation of the financial structure), being distinguished from *private equity* in that they operate exclusively on public markets and therefore do not have to bear the illiquidity of an unlisted equity investment, as CFA Institute (2020) explains.
  - **Special situations strategies** finally target opportunities in issuers engaged in operations outside mergers and acquisitions or insolvency proceedings: issues or buybacks of securities, exceptional distributions of capital, *spin-offs*, all situations liable to generate a temporary desynchronisation between market price and fundamental value, as CFA Institute (2020) analyses.
- (iii) The third family, **relative value strategies**, rests on a fundamentally different principle: it is not a matter of anticipating the direction of the market nor a specific event, but of identifying and exploiting a price anomaly between closely related financial instruments, with the expectation that this divergence will be resolved in due course, as CFA Institute (2020) explains. Their share of the industry's AUM went from 14 % in 1990 (5.5 billion dollars) to 28 % in 2015 (812 billion dollars), the strongest relative progression among the four families, reflecting both the development of derivatives markets and the influx of capital attracted by risk-return profiles perceived as stable and decorrelated from the leading market according to CFA Institute (2020).
- **Convertible arbitrage** (*fixed-income convertible arbitrage*) constitutes the most widespread bond variant of this family: the fund buys a convertible bond — whose theoretical value breaks down into a bond component and a call option on the shares of the issuer — and simultaneously sells short the ordinary shares of the issuer, seeking to isolate the option component of the security, presumed to be mispriced relative to its theoretical value computed by option *pricing* models, as CFA Institute (2020) explains. This portfolio is theoretically of nil beta (*market neutral*), but requires a daily dynamic *delta hedging* in order to neutralise the variations in the price of the underlying share.
  - **Fixed-income asset-backed** strategies identify mispricings between different securitisation tranches (ABS, MBS, CDO<sup>2</sup>), while *general fixed-income* strategies construct relative positions between corporate issuers, between sovereign and private debt, or along the yield curve of a single issuer, possibly incorporating exchange rate and curve dynamics considerations, as CFA Institute (2020) explains.
  - **Volatility strategies** exploit the anomalies between implied volatility (reflected in option prices) and realised (historical) volatility: a long position on options will be profitable if realised volatility exceeds the implied volatility at the moment of purchase, while a short position will benefit from an overvalued implied volatility, in accordance with the model of Black & Scholes (1973).

It should be noted that *relative value* strategies, despite their asymptotically low-volatility profile in a normal regime, are exposed to severe tail risks: the 1998 crisis and the failure of Long-Term Capital Management (LTCM) — a *relative value* fund on sovereign bonds having accumulated a leverage estimated at more than twenty-five times its own funds — illustrate that the theoretical convergence of *spreads* may be durably deferred in periods of stress, trapping a heavily indebted portfolio, as CFA Institute (2020) recalls.

<sup>2</sup> Asset-backed securities (ABS), mortgage-backed securities (MBS) and collateralized debt obligations (CDO) are three securitisation instruments: a bank transfers a portfolio of claims to a dedicated legal entity (a *special purpose vehicle*, SPV), which issues securities cut into tranches (*senior*, *mezzanine*, *equity*) in decreasing order of repayment priority. ABS constitute the generic term and may be backed by any type of claim (consumer credit, car loans, credit card balances); MBS are a sub-case of these whose underlying consists exclusively of mortgage claims; CDO, of a superior complexity, have as their underlying not direct bank claims but other debt securities (corporate bonds, ABS, or even other CDO — so-called 'CDO squared'). It is this mechanism of cascading *repackaging* which amplified the 2008 crisis: the *equity* tranches of CDO backed by *subprime* MBS lost all value almost simultaneously, propagating the shock to the whole of the world financial system.

- (iv) The fourth family, **macro funds**, adopts a philosophy opposed to that of *bottom-up* strategies: they start from a global macroeconomic reading — analysis of economic cycles, of balance of payments imbalances, of inflationary dynamics, of monetary policies — in order to construct directional positions on rate, foreign exchange, equity and commodity markets, as CFA Institute (2020) explains. Their share of assets receded from 39 % in 1990 (15.2 billion dollars) to 20 % in 2010 (580 billion dollars) before rising again to 27 % in 2015 (783 billion dollars), a rebound which coincides with the European sovereign debt crises and the divergences of monetary policy between the major central banks, which revived macroeconomic directional opportunities according to CFA Institute (2020). The flexibility of these funds is maximal: they may use forward contracts, options, interest rate and currency swaps, and direct positions on commodities, often with considerable leverage reflecting the manager's conviction in his macroeconomic scenario, as CFA Institute (2020) analyses. Historically, it was the **Quantum Fund**, managed by George Soros and Stan Druckenmiller, which forced the exit of sterling from the European Monetary System on 16 September 1992 (*Black Wednesday*), by selling short more than 10 billion pounds sterling and generating a profit estimated at one billion dollars in a single day according to CFA Institute (2020). Moreover, *commodity trading advisors (CTA)*, sometimes classified as a *macro* sub-category, are distinguished from it by their entirely systematic character: their entry and exit signals are generated algorithmically, mainly from *trend following* models applied to a broad universe of forward markets — interest rates, currencies, equity indices, commodities. This systematic character confers on them asymmetric return profiles that are particularly precious during prolonged crises: the Prequin database cited by CFA Institute (2020) counted 1067 institutional investors active in CTA in 2015, against only 331 in 2008, illustrating the renewed interest in these strategies in the post-crisis period. In 2008, a year in which the MSCI ACWI lost –10.8 % in annualised return over the period Q4 2007–Q4 2009 with a volatility of 24.2 %, and in which the worst monthly *drawdown* of *hedge funds* in general reached –7.5 %, trend-following CTA for the most part displayed positive returns, profiting from prolonged directional trends on bond, foreign exchange and commodity markets, as CFA Institute (2020) recalls.

### Key Takeaways



*Hedge funds* are distributed into four large families according to the HFRI classification: *equity hedge* strategies (long and short positions on equities, a *bottom-up* approach, a share receding from 37 % to 19 % of AUM between 1990 and 2015), *event-driven* strategies (merger arbitrage, *distressed*, activism, *special situations*, a share rising from 10 % to 26 % of AUM), relative value strategies (arbitrage of convertibles, of the yield curve and of volatility, the strongest relative progression with 14 % to 28 % of AUM), and *macro* strategies (directional positions founded on a global macroeconomic reading of cycles and monetary policies, a share going from 39 % to 27 % of AUM). These four families are distinguished above all by their economic logic: net directional exposure for *equity hedge* and *macro*, market neutrality sought for relative value, and exploitation of a specific corporate catalyst for *event-driven*. The rise of relative value and of *event-driven* reflects the growing sophistication of the industry and its shift towards strategies of low directionality, while CTA — the systematic and algorithmic sub-category of *macro* — are distinguished by an asymmetric return profile that is precious in periods of prolonged crisis, as their positive performance in 2008 illustrated. Despite the usefulness of this taxonomy, the boundaries between categories remain porous and evolving: managers seek to preserve the flexibility of their mandates and the large multi-strategy funds combine several logics within a single vehicle.

### 3.1.3 Performance, measurement bias and portfolio integration

The rigorous assessment of the performance of *hedge funds* is made structurally difficult by a set of biases which contaminate the published historical data and lead to a systematic overestimation of the average returns of the asset class. The first of these biases is **survivorship bias**: the databases indexing *hedge funds* generally include only the funds still active at the moment of publication, retrospectively excluding all liquidated funds, whose performance was precisely insufficient to ensure their survival, as CFA Institute (2020) explains. Now the data of Brooks & Kat (2002) suggest that at least 25 % of the funds created are liquidated within the first three years of existence, for want of sufficient performance to cover operational costs or to attract new capital. The second bias, known as **backfill bias** or **instant history bias**, occurs at the entry of a new fund into a database: its past performance is then added retroactively, creating an upward bias since managers typically enter databases only after having established a favourable history, as CFA Institute (2020) explains. These two effects combined mean that the HFRI indices and their equivalents tend to reflect only the performance of funds in good health, mechanically inflating the average returns published. The *HFRI FoF Composite Index*, measuring the real performance of equally weighted portfolios of *hedge funds*, then provides a more conservative and more representative estimate of the effective experience of the average investor according to CFA Institute (2020).

Now, over the period 1990–2014, the data of the *HFRI FoF Composite Index* cited by CFA Institute (2020) show an annualised return of 7.2 % for an annualised volatility of 6.0 %, leading to a Sharpe ratio of 0.63 (a riskless rate considered to be roughly 3.4 %) — higher than that of world equities (0.21) and world bonds (0.49) over the same period; the proportion of positive months reaches 69.3 %, against 61.3 % for world equities, and the worst monthly performance is limited to –7.5 % against –19.8 % for equities. These figures must, however, be interpreted with caution. The Sharpe ratio is a measurement instrument ill-suited to the asymmetric and leptokurtic distributions (that is to say with fat tails in the distribution of returns) which characterise the returns of *hedge funds*: by penalising upside and downside volatility symmetrically, it ignores the particular structure of the distribution tails and may lead to a flattering representation of a strategy which generates regular returns but exposes itself to rare and severe extreme losses (*tail risk*), as CFA Institute (2020) explains. The **Sortino ratio**, which substitutes for the total standard deviation the downside deviation computed relative to a target return, then provides a more relevant measure of the risk-return profile: over the same period, the Sortino ratio of hedge funds (0.74) is lower than that of world bonds (1.09) and of *private equity* (1.19), reversing the ranking established by the Sharpe and tempering the image of excellence that the latter projects, as CFA Institute (2020) qualifies. Other complementary measures further reinforce this observation of caution: **value at risk** (VaR), if it is computed assuming a normal distribution of returns, will systematically underestimate the risk of loss for a negatively skewed distribution; *shortfall risk* measures and *stress testing* analyses therefore constitute indispensable complements to any rigorous assessment of the risk profile of a fund, as CFA Institute (2020) recalls.

The financial crisis of 2008 then constitutes an empirical test of the diversification properties attributed to *hedge funds*. Over the period Q4 2007–Q4 2009, the *maximum drawdown* of the *HFRI FoF Composite Index* stood at –22.2 %, that is a level appreciably higher than that of world bonds (–10.1 %) but markedly lower than the –54.6 % recorded by world equities according to CFA Institute (2020). This relative resilience does not, however, mean that *hedge funds* fulfilled their promise of decorrelation: the **correlation between the returns of hedge funds and those of equity markets**, historically moderate over the period 2000–2001, increased sharply in 2008, a well-documented phenomenon according to which correlations between risky assets tend to converge towards 1 in periods of crisis, precisely at the moment when diversification is most precious for the investor, as CFA Institute (2020) recalls. The claim of some managers to offer a performance totally uncorrelated with equity markets is therefore empirically unsupported at the scale of the industry, even if certain strategies — trend-following CTA, *volatility long* — did effectively deliver positive performances during the crisis according to CFA Institute (2020). A second factor amplifying losses in periods of stress is the procyclical mechanism of **margin calls**: a heavily indebted fund confronted with a *drawdown* may find itself constrained to unwind its positions in haste, exerting selling pressure on already illiquid markets, aggravating the fall in prices and triggering a spiral of *redemptions* which may lead to the liquidation of the fund, as CFA Institute (2020) explains. *Lockup* and *notice periods* aim precisely to attenuate this risk by allowing the manager to unwind his positions in an orderly fashion, but may prove insufficient when redemptions accumulate and margin calls require an immediate response, as CFA Institute (2020) qualifies.

From the point of view of portfolio construction, the integration of *hedge funds* into a traditional asset allocation rests on the modern portfolio theory of Markowitz (1952): an imperfect correlation with existing assets reduces the standard deviation of the portfolio below the weighted average of the individual standard deviations, making it possible to improve the overall Sharpe ratio for a given level of expected return. In practice, institutional managers complement this analysis with **tail risk** indicators — historical frequencies of monthly losses below –1 %, –5 % and –10 % — as well as with stress scenarios including assumptions of high correlation, so as not to rely exclusively on long-run averages that are little representative of crisis conditions, as CFA Institute (2020) explains. **Manager selection** constitutes the final critical dimension, and doubtless the most decisive for the *hedge fund* investor, so great is the dispersion of individual performances within each strategy. The **due diligence** process recommended by CFA Institute (2020) covers several complementary dimensions: the strategy and the investment process (markets addressed, *long & short* philosophy, alleged competitive advantage, benchmark index retained); the performance history (method of computing returns — gross or net of fees —, comparability with a relevant *benchmark*, persistence of outperformance); the fee structure and its effects on the investor's net return, incorporating the *hurdle rate* and the *high-water mark*; the size and longevity of the fund (risks linked to investor concentration, minimum asset threshold for covering fixed costs); risk and systems management (leverage, brokerage, *prime brokerage*, diversification of counterparties); and finally operational quality and independent audit, whose importance was dramatically underlined by the Madoff affair, where the absence of third-party verification of the assets declared permitted a pyramid fraud of 65 billion dollars before the mechanism collapsed in 2008, as CFA Institute (2020) recalls. The dispersion of performances is moreover made all the more difficult to anticipate in that persistence is empirically weak: Stewart et al. (2009) thus suggest that institutional investors display a mediocre record in selecting managers

on the basis of past history, which argues for a deliberate diversification between several managers and several strategies rather than for a concentration on a few prestigious names.

### Key Takeaways



The assessment of the performance of *hedge funds* is structurally biased upwards by two cumulative effects: survivorship bias, which retrospectively excludes liquidated funds from databases, and *backfill bias*, which introduces into those same databases a history selected *ex post* by managers having already established a favourable *track record* [CFA Institute (2020)]. These two biases mean that the published HFRI indices systematically overestimate the average return effectively received by the investor, whose experience is better approached by the more conservative *HFRI FoF Composite Index*. Over the period 1990–2014, the latter displays a Sharpe ratio of 0.63, higher than that of world equities (0.21) and world bonds (0.49). This result nevertheless calls for two major qualifications: on the one hand, the Sharpe ratio penalises upside and downside volatility symmetrically and therefore ignores the risk of rare extreme losses (*tail risk*); on the other hand, the Sortino ratio — which measures downside deviation — establishes the reverse ranking, placing *hedge funds* (0.74) behind bonds (1.09) and *private equity* (1.19), which considerably tempers the image of excellence that the Sharpe ratio alone projects [CFA Institute (2020)].

## 3.2 New actors and renewed practices

The strategies described so far — *hedge funds*, *private equity* funds, *master-feeder* structures — share a common trait: they presuppose a considerable initial capital, subscription thresholds often fixed at several million dollars and a so-called qualified or sophisticated investor profile, as defined by American, European or Caymanian regulations, as CFA Institute (2020) recalls. This pricing and regulatory architecture long confined alternative finance to a wealth elite: according to CFA Institute (2020), 69 % of American investors with more than 125 million dollars hold *hedge funds*, against only 42 % of those whose wealth exceeds 25 million. The last decade has, however, seen the emergence of a set of radically different structures and practices, which seek precisely to fill the *equity gap*, that is to say the financing *no man's land* between the classic bank loan and the entry of an institutional fund into the capital, in which thousands of entrepreneurial projects fail each year for want of interlocutors suited to their size and their stage of development. **Crowdfunding**, incubators, **corporate venture capital** and public seed arrangements thus form a complementary ecosystem, oriented towards the project owners that *hedge funds* and LBOs ignore by construction.

### 3.2.1 The rise of crowdfunding

**Crowdfunding** designates the whole set of mechanisms allowing a project owner to solicit, via a digital platform (Ulule, KissKissBank, Leetchi...), financial contributions from a large number of individuals, without going through the intermediation of a bank or an institutional fund. Born in the United States at the beginning of the 2000s with platforms such as Kickstarter (2009) or Indiegogo (2008), it rests on a fundamental principle of **disintermediation**: putting the supply of and demand for financing directly in contact, by exploiting the aggregated liquidity of a diffuse crowd of investors or donors. On a world scale, the crowdfunding market was estimated at 23.2 billion dollars in 2026 and should reach 60.3 billion dollars by 2035, that is a compound annual growth rate of 11.2 % according to Market Growth Reports (2026). In France, the benchmark annual barometer, published jointly by Forvis Mazars & France FinTech (2025), records 10.8 billion euros collected cumulatively since the establishment of a regulatory framework in 2015. If one keeps to current terminology, *crowdfunding* in reality covers three legally and economically distinct families. The first, the **donation** (*reward-based crowdfunding* in its variant with a counterpart, or *donation-based* without a counterpart), is the oldest and best-known form among the general public: the contributor finances a cultural, associative, humanitarian or entrepreneurial project in exchange for a symbolic reward — a copy of the product, a mention, an experience — or purely gratuitously. In France, in 2024, donation mobilised 178.9 million euros (125.7 million euros in donation without reward and 53.2 million euros in donation with reward) for 160 137 projects financed and more than 3.7 million funders, that is an average contribution of 41 euros per funder for donation without reward, according to Forvis Mazars & France FinTech (2025). The second family, **crowdlending** (financing by loan), groups together two sub-modalities: the interest-bearing loan (substituted bank debt, granted directly to an SME or a private individual) and bond financing (issue of mini-bonds or bonds by the borrowing company). This is by far the bulk of the French market, with 1349.6 million euros collected in 2024 (of which 294.3 million euros in interest-bearing loans and 1055.3 million euros in bond financing), bond financing being itself dominated 61 % by the property sector, renewable energy not this time offsetting the retreat of that underlying, as Forvis Mazars & France FinTech (2025) details. The third family, **crowdequity** (financing in equity or in *royalties*), allows retail investors to take a direct stake in the capital of an unlisted firm, generally a *start-up* or an

innovative SME, via a digital interface. In 2024, crowdfunding in equity investment thus mobilised 201.6 million euros for 420 projects financed by 38 541 investors, with an average amount collected per project of 480 000 euros and an average contribution per investor of 5231 euros — that is an investor profile more wealth-oriented than in donation, as Forvis Mazars & France FinTech (2025) explains. The selection rate in this segment is particularly severe: only 7 % of the files presented to platforms are actually put online for collection, against roughly 10 % for bond lending, reflecting the difficulty of assessing the potential of young ventures at often pre-revenue stages, as Forvis Mazars & France FinTech (2025) shows. As for the positive exit rate — that is to say the proportion of companies whose securities were able to be sold without capital loss — it reached 71.4 % on the exits recorded in 2024, a figure which should be read with caution inasmuch as numerous holdings remain awaiting liquidity, as Forvis Mazars & France FinTech (2025) qualifies.

#### Focus...



**The European regulatory framework: PSFP status (EU regulation 2020/1503).** European regulation no. 2020/1503, known as the 'PSFP regulation', has been applicable since 10 November 2021 and has imposed on all crowdfunding platforms operating in lending and in equity a prior authorisation as crowdfunding service provider (**PSFP**), issued in France by the Autorité des marchés financiers (AMF), as Parlement européen & Conseil de l'Union européenne (2020) explains. Since 10 November 2023, any platform not holding this authorisation is forbidden to operate. The regulation caps at five million euros over twelve rolling months the collection per project owner, and imposes a regulatory distinction between 'sophisticated' and 'non-sophisticated' investors: in France, in 2024, 76.1 % of the investors active on platforms were classified as non-sophisticated according to Forvis Mazars & France FinTech (2025). This framework likewise instituted a '**European passport**' allowing a platform authorised in one member state to operate throughout the Union: in 2024, 37 European platforms had applied for this mechanism in order to operate in France, for a still modest volume of 127 projects and 68 million euros, most of the passports issued remaining dormant according to Forvis Mazars & France FinTech (2025). At the end of 2024, 42 platforms had obtained PSFP authorisation in 2023, and 14 more in the course of 2024.

The trajectory of the French market illustrates both the maturity acquired by this mode of financing in a decade and the structural constraints which now weigh on its development. After a phase of almost uninterrupted growth — collection was multiplied by twelve between 2015 (167 million euros) and 2022 (2355 million euros) — the market entered a phase of correction: 2089 million euros in 2023 (–11.3 %) then 1731 million euros in 2024 (–17.1 %), marking the second consecutive year of retreat according to Forvis Mazars & France FinTech (2025). This reversal is mainly attributable to the crisis in the property sector : property crowdfunding, which constituted the backbone of bond *crowdlending*, was hit head-on by the **rise in interest rates** begun from 2022 onwards and by the collapse of housing starts. In the first half of 2025, property *crowdlending* still showed a retreat of –15 % to 388.2 million euros according to France FinTech & Forvis Mazars (2026). Paradoxically, it is crowdfunding by donation — the most 'popular' of the three families — which displays the most resilient dynamic: the number of projects financed grew by +22 % in the first half of 2025 (56 359 projects) according to France FinTech & Forvis Mazars (2026). This quantitative dynamism contrasts with the weakness of the average amounts (41 euros per contributor in donation without reward according to Forvis Mazars & France FinTech (2025)), underlining that the democratisation of crowdfunding is above all a democratisation of the act of financing, not necessarily of the volume financed. The analysis of the structure of the market moreover reveals a **notable concentration**, which tempers the discourse on the horizontality of this mode of financing. Three platforms, which exceed 100 million euros of annual collection (an average of 156.8 million euros), by themselves represent 27.1 % of total collection in 2024, and the top ten concentrate 56.4 % of it according to Forvis Mazars & France FinTech (2025); the intermediate-sized platforms (collection between 10 and 50 million euros) number 24, with an average collection of 25.9 million euros. The apparent atomicity of the ecosystem — more than sixty platforms responding to the barometer are counted — therefore masks a *de facto* oligopolistic structure, comparable to that observed in other *fintech* sectors. This concentration is explained in part by the fixed costs of acquiring investors and project owners, by the compliance requirements linked to PSFP authorisation (a PSFP employing on average 16 staff according to Forvis Mazars & France FinTech (2025)), and by the **network effect**: a well-established platform mechanically attracts more quality projects, which improves its performance statistics, which attracts more investors. Forvis Mazars & France FinTech (2025) finally notes that only 32 % of platforms are financially at break-even in 2024, a sign that the economic model of *crowdfunding* remains fragile for the great majority of actors, and that the sector is set to consolidate in the years to come.

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**Default risk in crowdlending.** Crowdfunding by loan is not free of default risk. In the **property** sector, repayment delays of more than six months concern 15 % to 20 %, insolvency proceedings affect 6 % to 8 % of ongoing projects and definitive losses concern 4 % to 6 % of projects according to Forvis Mazars & France FinTech (2025) — a significant deterioration relative to pre-crisis levels. For **micro-enterprises and SMEs**, delays have increased and 2 % to 4 % of the firms previously in insolvency proceedings in 2023 now appear as definitive losses in 2024. By contrast, the **renewable energy** sector displays a certain resilience: very few delays and no insolvency proceedings or definitive losses recorded according to Forvis Mazars & France FinTech (2025). These gaps underline the importance of sectoral diversification within *crowdlending* itself, and bring out the sensitivity of the property segment to interest rate cycles — a structural dependence which platforms seek to offset by a tightening of selection (roughly one project in ten presented to investors) and by strengthened security (first-rank mortgages, fiduciary arrangements), as Forvis Mazars & France FinTech (2025) explains.

The comparison between *crowdfunding* and the instruments of institutional alternative finance described previously brings out a fundamental **complementarity** rather than a substitution. *Crowdlending* and *crowdequity* typically intervene at stages — the seed stage, the first fundraising for a *start-up*, the refinancing of an SME of one to three million euros — which are structurally below the minimum ticket of a classic *private equity* fund or a *hedge fund*, but above the threshold of a conventional bank loan application for a firm without a history or without sufficient guarantees. It is precisely the **equity gap** that this mechanism seeks to plug. The France FinTech & Forvis Mazars (2026) barometer confirms this logic: the growing segments are those backed by the real economy of proximity — **crowdfactoring** for micro-enterprises and SMEs (invoice financing, up year on year by +69 % in the first half of 2025) and renewable energy (+7 %). *Crowdfunding* thus appears less as a competitor of institutional finance than as a complement to the architecture of the financing of the real economy, capable of mobilising an atomised citizen savings where institutional funds, by construction, cannot intervene.

## Key Takeaways



*Crowdfunding* designates a mechanism of financial disintermediation allowing project owners to raise funds from a diffuse crowd of investors via a digital platform, without having recourse to bank or institutional intermediation. It covers three economically distinct families: donation (with or without a counterpart, with a civic and cultural vocation), *crowdlending* (interest-bearing or bond lending, dominated by property and renewable energy), and *crowdequity* (a direct stake in the capital of unlisted companies, essentially *start-ups* and innovative SMEs). These three forms fulfil an **equity gap** function: they intervene at financing stages structurally below the minimum ticket of a *private equity* fund, but beyond the accessibility threshold of conventional bank credit for firms without a history or without sufficient guarantees [Forvis Mazars & France FinTech (2025)]. The European regulatory framework arising from the PSFP regulation (EU 2020/1503), applicable since November 2021, has professionalised the sector by imposing a prior authorisation and by instituting a European passport, at the price of a *de facto* oligopolistic concentration: the top ten French platforms concentrate more than half of total collection, and fewer than a third of platforms are financially at break-even [Forvis Mazars & France FinTech (2025)]. The French market, after having been multiplied by twelve between 2015 and 2022, has begun a structural retreat linked to the property crisis and to the rise in rates, underlining its sensitivity to interest rate cycles and the fragility of an economic model still in the course of consolidation [Forvis Mazars & France FinTech (2025)].

## 3.2.2 Incubators, accelerators and corporate venture capital

As Leitão et al. (2022) explain, an **incubator** is a long-term support structure (twelve to thirty-six months) which takes charge of a project owner from the stage of the idea or of the pre-constituted project: it generally offers him a shared workspace, access to mentors and experts, and sometimes a symbolic seed capital, in exchange for a minority stake or with no capital counterpart in public structures. According to Cohen (2013), an **accelerator**, by contrast, targets firms already constituted and endowed with a *minimum viable product* (MVP): it is an intensive programme, of short duration (three to six months in general), centred on rapid growth, access to financing and networking with investors. The selectivity criterion is markedly higher, and the programme concludes almost systematically with a *demo day* or *pitch day* before an audience of venture capital funds, as Leitão et al. (2022) mention. A **startup studio** (or *venture builder*, or even *company builder*) constitutes finally a third distinct category: it does not select existing firms but itself creates *start-ups ex nihilo* from ideas or market problems identified internally, contributing seed capital, a multidisciplinary operational team and shared resources, in exchange for a generally more substantial stake in the capital (15 % to 40 %) than in classic incubator or accelerator models, as Gomes Teixeira (2022) explains.



These three modalities coexist and complement one another in the French ecosystem, the boundary between them often being porous in practice.

The mapping of this ecosystem in France testifies to a substantial density, the fruit of voluntarist public policies since the beginning of the 2000s. The consultancy Xerfi (2024) counts, in 2024, more than 450 incubators, accelerators, *startup studios* and campuses active on national territory, of which 230 are concentrated in the Île-de-France region in order to support roughly 12 000 *start-ups* in that region according to Institut Paris Région (2025). At national level, of 133 accelerators identified, 66 are established in Paris or in Île-de-France — that is half of the acceleration ecosystem concentrated in a single territory, reflecting the polarisation of capital, talent and specialised media around the capital, as Mag Startup (2025) emphasises. This density must not, however, obscure the rise of regional ecosystems: the Auvergne-Rhône-Alpes, Occitanie and Hauts-de-France regions have developed significant sectoral clusters, notably in *deeptech*, *medtech* and the energy transition, relying on the competitiveness clusters and the technological research institutes (IRT) arising from the future investments programme, as Xerfi (2024) explains. On the level of measured impact, Butz & Mrozewski (2021) show that *start-ups* benefiting from support in an accelerator present a survival rate roughly 23 % higher than that of unsupported firms at a comparable stage, which confers on these structures a systemic role of **risk reduction in the innovation financing chain**. France was moreover ranked first in the ESNA *Startup nations standard* report for the second consecutive year in 2024, a sign according to Direction générale des entreprises (2025a) of the international recognition of the maturity of its ecosystem of support for innovative entrepreneurship.

#### Focus...



**Station F: the largest *start-up* campus in the world.** Inaugurated on 29 June 2017 in the renovated Halle Freyssinet (thirteenth arrondissement of Paris), **Station F** constitutes the world emblem of the densification of support ecosystems. Financed 90 % by Xavier Niel (founder of Iliad and Free) for an investment of 250 million euros, it extends over 34 000 square metres and simultaneously accommodates roughly 1000 *start-ups*, 3000 workstations, 26 international acceleration and incubation programmes (including those of Facebook, Microsoft, HEC Paris and numerous large industrial groups), an amphitheatre of 370 seats and a *fab lab*, as EGF (2021) details. The economic model rests on a monthly subscription of at least 259 euros excluding tax per workstation (the '*Founders program*'), with no automatic stake in the capital — which constitutes a break from the model of traditional accelerators which require in return between 5 % and 10 % of the capital. Station F (2026) aggregates in a single place the whole set of actors of the innovation value chain: investment funds, large groups, public services (Bpifrance is present there), which makes it less an incubator than a 'condensed ecosystem', whose value resides as much in the density of interactions as in the services provided individually.

The ecosystem of incubators and accelerators cannot be analysed without taking into account its **articulation with venture capital funds**, of which it constitutes the local *deal flow*. In 2024, French *venture capital* and *growth capital* actors invested five billion euros in 1354 operations, a progression of 38 % relative to 2023, driven by the recovery of investments above 15 million euros (65 operations in 2024 against 43 in 2022), as France Invest (2025) reports. In parallel, French management companies raised 6.8 billion euros of new capital (a progression of 34 % relative to 2023), of which 38 % in *venture capital* funds proper and 27 % in *growth capital*, bringing to 40.4 billion euros the cumulative total raised since 2016 by French *venture capital* and *growth capital* funds according to France Invest (2025). The digital sector remains preponderant with 532 operations in 2024, ahead of the medical, pharmaceutical and biotechnology sector (277 operations, in constant progression since 2021), as France Invest (2025) explains. The employment dynamic in the supported firms is particularly remarkable: the 2975 French *start-ups* in portfolio at the end of 2023 employed more than 140 000 people, with employment growth of +6.3 % in 2023 — that is more than fifteen times the pace of employment in the French market sectors (+0.4 %) over the same period according to France Invest (2025). The average net internal rate of return over ten years of *venture capital* and *growth capital* funds stood at 8.6 % at the end of 2024, a level which — for most funds — 'has nothing to envy in that of traditional VCs', according to Lefebvre (2025).

#### Focus...



**Corporate venture capital: a third of French funding rounds.** *Corporate venture capital (CVC)* designates the venture capital funds constituted and managed by large firms in order to invest, on a minority basis, in *start-ups* whose activities present a strategic or financial interest for the founding group, as Pinkow & Iversen (2020) explain. Distinct from classic M&A — it is not intended to lead to an acquisition — and distinct from independent *venture capitalism* (VC), it occupies a hybrid position which confers on it a growing legitimacy among entrepreneurs. According to the Asterfab barometer relayed by Lefebvre (2025), French CVCs now represent nearly 30 % by value



of the funding rounds carried out by French *start-ups*, testifying to an accelerated professionalisation since the pioneering creation of the Mobivia fund in 2010. Among the main active French CVC funds figure Opera Tech Ventures (BNP Paribas), Orange Digital Ventures, L'Oréal BOLD Ventures, 574 Invest (SNCF), Total Energy Ventures, Schneider Electric Ventures, Engie New Ventures and Société Générale Ventures, as Forinov (2025) enumerates. CVCs intervene mainly at the seed stage (30 % of cases) and in series A (the first significant institutional funding round of a *start-up*, occurring after the seed phase; 41 % of cases), for an average ticket of three million euros, a stake generally between 5 % and 15 % of the capital, and a contribution distinct from financial capital: sector experts, brand image, access to clients — all competitive advantages that independent VC funds cannot offer according to Lefebvre (2025). The growing pressure on financial returns constrains these structures to a twofold objective: financial performance comparable to independent VC and documented operational synergies with the industrial group, on pain of seeing their internal legitimacy called into question, as Lefebvre (2025) emphasises. Indeed, a CVC does not contribute capital alone; it contributes a commercial address book and sometimes a first client contract — which transforms the investment into a relationship of industrial alliance rather than of pure financial intermediation.

### Key Takeaways



Incubators, accelerators and *startup studios* constitute three complementary modalities of support for innovation, distinguished by the stage of intervention and the intensity of the support: the incubator supports a pre-constituted project over twelve to thirty-six months, the accelerator targets a firm possessing an MVP and propels it in three to six months towards a first institutional round, while the *start-up studio* creates companies *ex nihilo* from market opportunities identified internally, in exchange for a more substantial stake in the capital [Leitão et al. (2022)]. These structures fulfil a systemic function of risk reduction: *start-ups* supported in an accelerator display a survival rate roughly 23 % higher than that of unsupported firms at a comparable stage [Butz & Mrozewski (2021)]. In doing so, they feed the *deal flow* of venture capital funds; in 2024, French *venture capital* and *growth capital* invested five billion euros in 1354 operations, and the firms in portfolio created employment at a pace fifteen times higher than that of the market sectors [France Invest (2025)].

### 3.2.3 The role of the state and the public authorities

The intervention of the state in the financing of alternative finance and of the innovation economy rests, in France, on a pivotal operator: **Bpifrance**, the public investment bank created in 2012 by the merger of the Caisse des dépôts et consignations and OSEO, which today constitutes the principal institutional vector of public action in matters of venture capital, bank guarantee and support for internationalisation. In 2024, Bpifrance (2025b) thus injected in total 60 billion euros into the French economy, a level stabilised since 2022, for a net profit of 896 million euros, testifying to the soundness of its hybrid economic model between public service logic and financial performance. Its action is deployed according to three complementary modes. The first mode is the **direct financing of innovation**: Bpifrance mobilises subsidies, repayable advances and unsecured loans intended for innovative SMEs and mid-sized companies — notably honour loans (102 million euros distributed to 19 700 founders in 2024, up +9.6 % year on year), the innovation loan (up to five million euros, seven years including two of grace period, with no security over assets) and R&D aid; across the whole innovation strand, 5.2 billion euros were deployed to 5503 firms in 2024, as Bpifrance (2025b) details. The second mode is **equity investment**: in innovation capital (a direct stake in *start-ups* at seed stage or in series A and B), Bpifrance invested 572 million euros in 147 operations in 2024, of which 70 % directed towards *deeptech*, 235 million euros towards *greentech* and 130 million euros towards artificial intelligence *start-ups*, as Bpifrance (2025b) explains; in development capital, the amounts reach 1.8 billion euros in 169 firms. The third mode is the **fund of funds activity**: with 77 subscriptions for 1.7 billion euros in 2024 (up +7.5 % year on year), Bpifrance is the leading institutional subscriber to French *venture capital* and *growth capital* funds; since 2020, it has invested more than 7.5 billion euros in 650 partner funds, generating a leverage effect of 31 billion euros invested in French firms by co-investing funds in four years, that is a multiplier of more than four, as Bpifrance (2025a) explains. Finally, its **bank guarantee** function — nine billion euros of loans or investments guaranteed in favour of 65 750 firms in 2024 (up +4 % relative to 2023) — constitutes the most massive instrument in volume and the most directly accessible to micro-enterprises and SMEs which do not have sufficient collateral security to access ordinary bank credit, as Bpifrance (2025b) emphasises. The economic logic is that of a **market complement**: Bpifrance intervenes where market failure is established, that is to say at the stages where the risk is too high or too asymmetric for the private investor to agree to commit alone, without for all that substituting itself for him, as Banque de France (2024b) explains.

## Focus...



**The Bpifrance guarantee arrangement: architecture and leverage effects.** The range of Bpifrance guarantees covers the entire life cycle of the firm: the **creation guarantee** (up to 60 % of the loan, a ceiling of 200 000 euros, for firms less than three years old), the **micro-enterprise development guarantee** (up to 70 % of the loan, a ceiling of 1.5 million euros), the **innovation guarantee** (up to 60 % of the loan) and the **transmission guarantee** for disposals and takeovers of firms, as L'expert comptable (2026) details: the bank, if it judges the project viable but wishes to limit its risk, applies for the Bpifrance guarantee (it is the bank which makes the application to Bpifrance and not the entrepreneur); if the firm can no longer repay, Bpifrance then takes charge of its share of the loss. The cost for the entrepreneur is a guarantee commission of roughly 0.5 % to 1 % of the amount guaranteed, incorporated into the total cost of the loan: in the event of a claim, Bpifrance indemnifies the bank up to its share of the guarantee, the bank retaining by construction a fraction of the risk so that its incentive to select files remains entire. The mechanism thus produces a significant **budgetary leverage effect**: a public endowment at risk of one euro makes it possible to guarantee several euros of bank financing, the expectation of claims being mutualised over a portfolio of firms. In 2024, the nine billion euros of loans or investments guaranteed involved a public budgetary mobilisation below this notional amount, which makes the guarantee the most efficient instrument in relation to the fiscal cost borne by the community, as Bpifrance (2025b) explains.

Moreover, the **France 2030** plan, announced in October 2021 by Emmanuel Macron and endowed with an envelope of 54 billion euros over five years, constitutes the most recent expression of a French tradition of strategic public investment in innovation, the heir of the future investments programmes. Its underlying logic is that of the **catalyst**: the state does not finance the entirety of projects but provides a signal of selectivity and a co-financing which reduces the residual risk borne by private investors, generating a leverage effect of roughly 1.3 private euros for each public euro committed according to Ayming (2025). At the end of 2024, 40 billion euros had been committed of the 54 billion of the formal envelope (excluding the interest on non-consumable endowments and on the Innovation and Industry Fund), for the benefit of 7457 projects and 7002 unique beneficiaries, according to Secrétariat général pour l'investissement (2025). More precisely, the France 2030 plan is deployed according to ten strategic objectives among which figure, in the first rank, **digital and industrial sovereignty** (3.4 billion euros invested in artificial intelligence and 6103 patent filings generated according to Ayming (2025)) and massive support for industrial *start-ups* and *deeptech*. Thus, the **Deeptech Plan**, launched as early as 2019 with an initial endowment of 1.5 billion euros raised to 4.1 billion euros in 2023, according to Ministère de l'Économie et des Finances (2025a), set the ambition of creating 500 *deeptech start-ups* per year by 2030. In 2024, 385 were created (up +13 % relative to 2023), and more than 1300 have been created since 2019, according to Zabala Innovation Consulting (2025). The **French Tech 2030** sub-programme, complementary to and distinct from the *deeptech* plan, targets for its part a hundred or so strategic *deeptech start-ups* with a strong potential for disruption: launched in 2023 and active in 2025, it combines direct financing, personalised support and privileged access to public procurement, in a logic of *champion picking* avowed by the state, as Direction générale des entreprises (2025b) explains. Moreover, the governance of France 2030 rests on the **General Secretariat for Investment** (SGPI), placed under the Prime Minister, which coordinates the whole set of operators — Bpifrance for economic innovation, the Agence nationale de la recherche (ANR) for fundamental research, ADEME for the ecological transition — according to an architecture of delegated management whose establishment of a centralised information system Cour des comptes (2025b) has commended, while noting delays in the consolidation of performance indicators between operators. One of the critical issues raised by Cour des comptes (2025b) resides in the measurement of **additionality**: demonstrating that the public euro financed a project which would not have seen the light of day without it, and did not simply substitute the taxpayer for the private investor, remains methodologically delicate, in particular for intermediate-sized projects where public financing and private financing are in direct competition.

## Key Takeaways



The intervention of the state in the financing of innovation rests on two complementary levers: a pivotal operator, Bpifrance, and a direct industrial policy embodied by the France 2030 plan. Bpifrance articulates four instruments according to a logic of market complement: the direct financing of innovation, equity investment, the fund of funds activity and the bank guarantee [Bpifrance (2025b)]. France 2030 (40 billion euros committed at the end of 2024) prolongs this logic by a catalyst policy: the state co-finances in order to reduce the residual risk of private investors, generating a leverage effect of roughly 1.3 private euros per public euro [Ayming (2025)]. In particular, the *Deeptech* Plan (4.1 billion euros) and the French Tech 2030 programme illustrate the move from horizontal support to an avowed logic of sectoral *champion picking*.



# The blockchain turn

The forms of alternative finance described so far — *hedge funds*, crowdfunding, venture capital, incubation — have in common that they rest on financial and legal infrastructures inherited from the twentieth century : identified intermediaries (platforms, managers, public banks), regulated vehicles (PSFP, AIF, *limited partnerships*) and a regulatory state whose intervention — budgetary or normative — constitutes, as we have seen, the foundation of trust which conditions the functioning of the whole. These alternative forms are certainly disruptive relative to the classic deposit bank and bond market, but they remain inscribed within a framework of intermediation: a third party — whether a fund, a platform or a public operator — interposes itself between the supplier and the seeker of financing, authenticates the transaction, assumes a legal responsibility and bears, partially, the counterparty risk. **Distributed ledger technology (DLT)**, of which the *blockchain* constitutes the best-known variant, calls this presupposition of intermediation into question more radically than all previous innovations: it claims to make possible the conclusion and automatic execution of financial transactions between any agents whatsoever, without a central trusted third party, by means of a distributed, cryptographically secured and public register. **Cryptocurrencies** — Bitcoin since 2009, Ethereum since 2015 and several thousand digital assets since — constitute its most direct and most controversial financial application. They are not merely a new type of speculative asset: they carry an architectural ambition — to reconfigure the very functions of money, credit, account-keeping and settlement-delivery — whose implications for regulation, monetary policy and financial stability constitute today one of the major issues for central banks, regulators and governments.

## 1 What is a *blockchain*?

Adapting a definition from the fact-finding mission of the National Assembly conducted by De La Raudière & Mis (2006), a ***blockchain*** designates ‘*a register, a large [computer] database which has the particularity of being shared simultaneously with all its users, [among whom] the holders of this register [also have] the capacity to enter data in it according to specific rules fixed by a computer protocol [...] secured thanks to cryptography*’.

### 1.1 Public *blockchain* versus private *blockchain*

A *blockchain* is therefore akin to a large database aggregating a set of exchanges carried out between its users: it is therefore not *a priori* linked to a monetary dimension. Moreover, every *blockchain* rests on an algorithm which secures exchanges (exchange of information, carrying out of transactions or of services...). Finally, each exchange results, at the end of a **validation process**, in an entry — called precisely a ‘block’ — on the register, which then forms a **chain of timestamped blocks**, whence the name ‘*blockchain*’.

#### 1.1.1 The case of private *blockchains*: permissioned access

It is then appropriate to clear up a misunderstanding that the previous definition might suggest. Indeed, **being shared with all its users** does not necessarily mean being shared publicly. Thus, the content of certain *blockchains* is accessible only to certain persons. One speaks of a **private *blockchain*** or permissioned *blockchain*. For example, IBM Food Trust, a *blockchain* ensuring the traceability of food, is accessible only to a **consortium of firms** including notably the American company Walmart, the Dutch company Unilever, the Swiss company Nestlé and the French company Carrefour. As IBM (2022) recalls, ‘*Food Trust is the only network of its kind to connect the actors of the food supply chain by means of a permissioned, immutable and shared record of food system data. The result is a customisable suite of solutions which make it possible to increase the safety and freshness of foods, to improve the efficiency of the supply chain, to reduce waste [and] to improve the reputation of [the] brands [which use IBM Food Trust]*’.

As Hossen & Gauthier (2024) explain, unlike a piece of software which sometimes copes badly with system updates — let us think for example of Intel applications which, at the beginning of 2025, are only partially supported by the new M1 processors of Mac computers — or which is subject to risks of hacking — cracks of the Office suite proliferate on the internet without Microsoft, whose capitalisation nevertheless exceeds 2200 billion dollars, being able to fight them significantly —, a *blockchain* thus offers a durable and secure solution. Adapted to the world of business, a private *blockchain* moreover makes it possible to remove intermediaries while remaining closed and secure, as Hossen & Gauthier (2024) emphasise. From then on, each member of the consortium finding an interest in participating in the private *blockchain* — that of a reduction in the costs of recording operations in the first rank —, no private *blockchain* is **necessarily backed by a cryptocurrency**. The example of IBM Food Trust thus shows that the presence of a *blockchain* does not necessarily imply any monetary aspect: the *blockchain* is above all a **technological tool**.

### Focus...



**The use of the *blockchain* by the Estonian government: digital identity.** As Kalvet (2012) recalls, from the end of the 1990s onwards, Estonia made the choice of building an entirely dematerialised state infrastructure, resting on three technological pillars whose scope and limits Semenzin et al. (2022) detail — the **electronic identity** system (*e-ID*), the data exchange platform *X-Road* and the keyless signature infrastructure (**KSI *blockchain***). According to the data published by the institutional portal e-Estonia (2024), nearly 98 % of the 1.3 million Estonian citizens today hold a digital identity card, which allows them to access the whole set of dematerialised public services — tax return, consultation of medical records, electronic voting, signature of contracts —; this digital signature mechanism alone allows each user to save on average five working days per year, for a cumulative total of more than 800 million electronic signatures issued to date. The KSI *blockchain*, developed from 2008 onwards in partnership with the private company Guardtime, rests on a mechanism of **cryptographic hashing** making it possible to timestamp and seal the integrity of public registers without any private key being required in order to verify their authenticity: as Semenzin et al. (2022) emphasise, this system secures the health, property, commercial and succession registers, as well as the digital judicial system and the Estonian official journal. Commission européenne (2024) establishes the transnational scope of this arrangement: since 2014, more than 125 000 digital identities have been issued to foreign nationals from more than 180 countries within the framework of the *e-Residency* programme, which represents roughly 10 % of the Estonian population; the KSI system has moreover been deployed beyond Estonian borders, notably with the Dutch judicial information service and American federal agencies. It is nevertheless appropriate to qualify this model: Semenzin et al. (2022) signal that the KSI architecture, which is a permissioned *blockchain* controlled by the state, departs significantly from the decentralising principles of public *blockchains*; Kalvet (2012) had moreover already shown that the success of Estonian e-government rests on singular historical conditions — notably the absence of an inherited administrative infrastructure and the culture of ‘*ethical hacking*’ —, which makes any mechanical transposition of this model to other states particularly delicate.

### 1.1.2 From the openness of public *blockchains* to the necessity of cryptocurrencies

The use of the *blockchain* for the world of business is a use subsequent to the date of creation of the *blockchain*. When Bitcoin was created in 2009, the dichotomy between private *blockchain* and public *blockchain* did not exist, since — notably until Ripple appeared in 2012 — all *blockchains* were public, as Hossen & Gauthier (2024) mention. But what is a **public *blockchain***? It is a *blockchain* technology for which every person can validate an operation carried out on the register or access the register of exchanges effected, unlike IBM Food Trust where only a consortium of firms is authorised to validate operations on the *blockchain*. A public *blockchain* is therefore totally **decentralised, open and transparent**: any computer can potentially become a **validation node** of an exchange of information. One then understands why a public *blockchain* guarantees a high level of security: since validation nodes are multiple, it is difficult for a hacker to take control of the network, since it would be necessary to infect computers located all over the world with no connection between them — since there exists no ‘central server’ from which to infect all the validation nodes.

But then a practical problem appears, which Hossen & Gauthier (2024) emphasise: participants having *a priori* no intrinsic interest in taking part in the validation of exchanges, an incentive is needed. To be sure, the *blockchain* is initially the concretisation of an anarcho-capitalist desire, formalised for the first time by May (1988) in his ‘Crypto Anarchist Manifesto’: ‘*Crypto anarchy will allow national secrets to be traded freely and will allow illicit and stolen materials to be traded, [...] and cryptologic methods will fundamentally alter the nature of the interference of government and large corporations in economic transactions*’. But this flame of the **so-called ‘cypherpunk’ movement** — ‘*to cypher*’ meaning ‘to encipher’ or ‘to encrypt’ — is insufficient to create a real movement of adoption: as Hossen

& Gauthier (2024) indicate, not all individuals have a protest or even anarchist streak, and few are prepared to invest themselves free of charge in a project of such magnitude. Accordingly, in order to encourage participation in the life of the *blockchain*, that is to say concretely to validate information in order to add it to the register, a **monetary incentive** must be created: this is the reason why all the algorithms underlying a public *blockchain* then find themselves valued through a **cryptocurrency** whose tokens, a kind of monetary unit, are known as **tokens**.

Careful, this does not mean that all existing cryptocurrencies are backed by public *blockchains*. There do indeed exist private *blockchains* which have a cryptocurrency, as Hossen & Gauthier (2024) analyse: this is for example the case of Ripple, an American firm which created a simplified payment protocol, useful notably for the **foreign exchange market** (Forex) for which the protocol searches for the fastest and least costly path for exchanging two currencies with one another. Ripple chose to make its algorithm operate through a *blockchain* and to value it through a cryptocurrency. Thus, the meaning of a cryptocurrency is not univocal according to Hossen & Gauthier (2024): whereas the cryptocurrency is almost akin to the monetary valuation of a firm in the case of a private *blockchain*, like a share, it has essentially a utilitarian role in the case of public *blockchains*, namely to reward monetarily the involvement of validation nodes. But how, moreover, does this process of validating exchanges operate in practice?

### Key Takeaways



A *blockchain* is a distributed computer register, shared simultaneously between all its users, in which each exchange is validated according to a cryptographic protocol then entered in the form of a timestamped block, thus forming an immutable chain [De La Raudière & Mis (2006)]. The fundamental distinction opposes private (permissioned) *blockchains*, whose access is reserved to a consortium of identified actors, to public *blockchains*, entirely decentralised and open: any node may validate transactions there, which confers on the network a structural robustness in the face of attacks, since there exists no central server to compromise. This total openness nevertheless raises a problem of incentives: why would any given actor devote his computing resources to validating transactions for others? The answer is monetary: every public *blockchain* is backed by a cryptocurrency whose tokens remunerate the validation nodes, thus transforming participation in the network into an economically rational activity [Hossen & Gauthier (2024)]. This causal link — a public *blockchain* calls for a cryptocurrency — must not be reversed: not all cryptocurrencies rest on a public *blockchain* (Ripple is the canonical example), and the *blockchain* remains above all a technological tool whose monetary dimension is only one modality among others, not an intrinsic end.

## 1.2 Validation processes: proofs of work, of stake and of authority

In order to be added to the register, an exchange must compulsorily be validated, that is to say certified by a third party who authenticates it. It is indeed validation which guarantees the security and integrity of exchanges. Accordingly, the process of validating exchanges concerns all *blockchains* without exception, whether these be public or private. Who then can become a validation node? In simplified fashion, it is possible according to Hossen & Gauthier (2024) to answer in three different ways:

- For private *blockchains*, the computers which can validate transactions are in general pre-selected, which rests on reputation: one speaks of **proof of authority** (PoA). This is the example of Ripple, where only the company's computers can validate a transaction, or of IBM Food Trust, where only the members of the consortium can certify an exchange in order to add it to the register.
- Historically, this was not the answer given: the first cryptocurrencies were public *blockchains*, for which anyone could validate a transaction. Thus, with Bitcoin notably, it is necessary to use the computing power of one's computer in order to be the first to validate the transaction; the more powerful the machine, the faster validation can operate: one speaks of **proof of work** (PoW).
- However, the structural limits of proof of work were rapidly perceived. Today, most recent *blockchains* rest on **proof of stake** (PoS): the computer which will validate the transaction may be selected randomly, in proportion to its involvement in the blockchain, an involvement which may for example be materialised by the number of *tokens* placed in escrow.

### 1.2.1 The ECDSA algorithm

In every case, the validation of exchanges rests on a verification of '**digital signature**', known as the **ECDSA algorithm** (*elliptic curve digital signature algorithm*). Without entering into the detail of the complex mathematical formulas which govern this algorithm but which Johnson et al. (2001) for example analyse, it must be understood that every



‘account’ on a *blockchain*, called an ‘**address**’, is associated with two keys: one public and the other private. Everyone has the public key, like the bank account details associated with a classic bank account; by contrast, the private key is rigorously personal: it allows one, like the login details (customer number and password) of a banking website, to access all one’s accounts and to carry out sensitive operations. The **private key** therefore allows total access to one’s cryptocurrency wallet; one is therefore truly the owner of one’s *tokens* if, and only if, one holds the private key of one’s wallet: one speaks of asymmetric encryption. How does this work? Suppose for example that a person A wishes to send a message to a person B.

- On the one hand, person A uses person B’s public key in order to encode (that is to say to encrypt) the data. Person B will then use his private key in order to decode (that is to say to decrypt) the message. Person B will therefore indeed be the only one able to retrieve the message: this **asymmetry of the encryption process** guarantees the **confidentiality** of the data in the cryptographic sense of the term.
- On the other hand, person A uses his private key in order to sign his own message, which generates a fingerprint. Person B will then apply person A’s public key to this fingerprint in order to make sure that the sender of the message is indeed person A: this **digital signature** guarantees the **authenticity** of the data.

### Focus...



**On the importance of holding private keys for the protection of *tokens*.** There are two ways of keeping one’s *tokens*: on the one hand, crypto-assets may be kept by an intermediary such as a broker or more generally by a clearly identifiable legal person (a centralised exchange platform); on the other hand, crypto-assets may be held directly by private individuals on decentralised exchange platforms or on cold wallets of the Ledger USB key type, that is to say, as a first approximation, when the crypto-assets are directly accessible on the *blockchain* without needing to have recourse to a centralised platform. Now, as Wagstaff (2014) recall, on centralised exchange platforms such as Binance, Kraken, Crypto.com and so on, it is the exchange platform which keeps control of the private keys. This is the reason why, in the event of the hacking of platforms — the siphoning off of nearly 800 000 bitcoins from Mt. Gox between 2011 and 2014, which provoked the bankruptcy of this then number-one platform, is the canonical example — users find themselves blocked and their *tokens* are frozen by the platforms or stolen by hackers. Legal remedies are moreover slim, all the more so as in general the platforms hacked have their registered office located outside the United States or the European Union. This is why Andreas Antonopoulos popularised the expression ‘*Not your keys, not your coins*’. By contrast, the principle of decentralised exchange platforms such as PancakeSwap or UniSwap is never to take control of users’ private keys: more secure, decentralised exchange platforms (**decentralised exchanges** or DEX) are, however, slower, less ergonomic and less user-friendly than their centralised competitors (**centralised exchanges** or CEX).

Concretely, how does the validation of data proceed? The mathematical problem is extremely complex to solve, as Wang (2014) details. Let us give a metaphorical version of it, taken from the work of Hossen & Gauthier (2024). Let us imagine a transparent safe and suppose that a person 1 wishes to send  $x$  *tokens* to person 2.

- Person 1 places the  $x$  *tokens* in the transparent safe then locks it with the help of person 2’s public key, in order to make sure that only person 2 will be able to retrieve the money. He then sends the locked safe to the address of person 1, like a tracked letter at the post office whose content it would be impossible to read.
- Moreover, person 1 affixes a ‘seal’, like a signature on a bank cheque, to the safe with the help of his private key, in order to certify that it is indeed he who is the sender of the transaction. The miners then verify that the seal — that is to say the digital signature — is valid by comparing it with person 1’s public key.
- While all the information of the transaction is accessible (the safe is transparent), nobody can open the safe (in particular the computer validating the transaction cannot do so), since person 2’s private key is required. Once the latter has received the safe, she then opens it with her private key and retrieves the money of the transaction.

### Debate



**Is the *blockchain* anonymous?** Some will think that the *blockchain* is anonymous (incidentally, speaking of ‘the’ *blockchain* makes very little sense: it would have been better to specify ‘*blockchains*’, so diverse is this sector). Thus, Andrianto & Diputra (2017) consider that, inasmuch as Bitcoin rests on a cryptographic technology which guarantees the anonymity and confidentiality of transactions, the *blockchain* facilitates transactions and reduces



costs relative to foreign exchange markets. Banque de France (2013) even considers that, by their anonymous character, cryptocurrencies, in particular Bitcoin, favour the circumvention of the rules relating to the fight against money laundering and the financing of terrorism. The reality is more complex, as Hossen & Gauthier (2024) explain. To be sure, there exist *blockchains* built specifically to guarantee anonymity (Monero, Zcash, Dash...); by contrast, most cryptocurrencies, Bitcoin in particular, are **pseudonymous**: certainly, only the address is publicly accessible, on sites such as <https://www.blockchain.com/explorer/>, but if one knows which person is hiding behind this address, then one attains an extreme level of transparency: content of the wallet, log of operations, and so on. It is as if, by giving your bank account details to someone, the latter had access to the entirety of your bank statements since the opening of the account: such is the level of transparency of the *blockchain*. As Hossen & Gauthier (2024) emphasise, one must likewise be aware that, when one buys cryptocurrencies on a centralised exchange platform, the latter verifies our identity — this is what is called **KYC** (*know your client*) procedures. Since this platform holds our private keys, it knows who is behind each movement and it is then possible to trace all movements to and from the account; so much for anonymity...

### 1.2.2 Proof of work (PoW)

With PoW, the validation nodes constitute '**miners**'. Naturally, miners do not generally validate a single transaction: several transactions of a single block are generally validated. It is the first miner to validate all the transactions of the block — which he has constructed from the transactions awaiting validation — who receives a reward, the others receiving nothing. This reward is twofold: on the one hand, miners recover minimal **transaction fees** — this is what explains why, if one sends 1 bitcoin to another wallet, the recipient receives for example only 0.99 — and, on the other hand, miners earn *tokens* specially created by the mining process. For example, if a Bitcoin miner is the first to validate his block, he will receive 3.125 new bitcoins as a reward (which is worth more than 150 000 euros in August 2024). Otherwise put, as Hossen & Gauthier (2024) illustrate, it is as if the Bitcoin algorithm, like all the other proof-of-work algorithms (the current version of Ethereum, Cardano, Dogecoin, Litecoin...), were 'running the printing press' in order to print new liquidity.

Proof-of-work algorithms therefore naturally suffer from a **major ecological problem**, since energy is wasted — some miners validate for nothing, given that only the first to have validated all the transactions of his block recovers a reward —: thus, as Servet (2021) recalls, the annual consumption of the bitcoin network is said to be reaching 128 TWh per year, that is 0.6 % of world electricity consumption, the equivalent of the consumption of countries such as Norway, New Zealand or even Argentina. In order to increase the probability of being the first to solve the digital signature problem, miners therefore federate their computing power within cooperatives, called **mining farms** (*mining pools*), in order to mutualise costs and gains. This is the reason why, as the evolution of the *Cambridge Bitcoin electricity consumption index* (CBECI) published by CCAF (2022) shows, before the **prohibition of cryptocurrency mining** by China in July 2021, the country abounded in mining farms which resembled veritable factories; today, it is moreover in the United States that such factories are multiplying.

What of the security of *blockchains* founded on proof of work? Naturally, several validations are necessary actually to add the exchange to the register. More precisely, in absolute terms, half of the total computing power of the *blockchain* should certify the authenticity of the transaction. However, as Hossen & Gauthier (2024) specify, it is considered in practice that, when six different miners validate a transaction, the level of security is already extremely robust. This observation nevertheless leaves the door open to a possible hacking, which Aponte-Novoa et al. (2021) bring to light: if a miner holds at least 50 % of the computing capacity of the whole *blockchain*, he can hack it by creating for example fictitious transactions: 'I have received at my address B 1000 bitcoins from address A', whereas address A is false or is not funded; naturally, all the other miners will reject this transaction<sup>1</sup>, but since the hacker has half of the computing capacity, he will accept it and the transaction will be added 'by force' to the register; it will then no longer be possible to delete the block once it has been added to the register: this is a '**51 % attack**'. Naturally, this is a thought experiment and, for example, the Bitcoin *blockchain* has never been violated in its integrity in more than fifteen years of existence; as Hossen & Gauthier (2024) emphasise, tens of billions of dollars would have to be mobilised in order to create a mega-factory more powerful than all the miners combined. The security of *blockchains* founded on proof of work is therefore maximal.

<sup>1</sup>Indeed, since address A will be false, the 'digital signature' of the transaction determined by the ECDSA algorithm will be invalid, as Hossen & Gauthier (2024) explain. The protocol for verifying the digital signature will result in a rejection: honest miners will therefore signal that this transaction is invalid. But the *hacker*, for his part, although the ECDSA algorithm will indicate that the transaction is fictitious, will manually accept the transaction and, if he holds more than half of the computing power of the *blockchain*, this fictitious transaction will *de facto* be validated by a majority of the network and will then be wrongly added to the inviolable register of the *blockchain*.

## Focus...



**How is the computing power of a computer to be measured?** The computing power of a computer dedicated to mining is measured by its ‘**hash rate**’, that is to say the number of operations which it is possible for it to validate each second. The sum of all the computers dedicated to mining makes it possible to assess the maximum number of operations that the *blockchain* can validate each second; as Aponte-Novoa et al. (2021) indicate, holding half of the computing power of the *blockchain* therefore means concretely being able to validate operations faster than all the other miners combined. With mining *rigs* for example, one increases one’s computing power by multiplying the number of graphics cards dedicated to the process of verifying digital signatures.

### 1.2.3 Proof of stake (PoS)

With proof of stake, the principle is to show one’s **involvement in the *blockchain*** in order to maximise the probability of receiving transactions to validate. As Hossen & Gauthier (2024) explain, the probability  $p$  of being chosen is therefore on the one hand an increasing function of the quantity of tokens that one places in escrow: someone who ‘blocks’ 10 000 Avalanche (AVAX) tokens will thus have more chance of receiving a transaction to validate than another **forger** — this is the name attributed to validation nodes for PoS, like the miners of PoW — who has blocked only 1000 AVAX. This blocking takes the name of ‘**staking**’: to stake tokens is to block them in order to show one’s involvement and to maximise one’s chances of validating transactions. Naturally, Hossen & Gauthier (2024) add that, generally, the probability  $p$  of being chosen to validate is likewise a function of ‘loyalty’: someone who has staked 5000 AVAX for a month will for example have more chance of receiving a transaction to validate than another forger who has staked the same amount but for only two days.

Unlike proof-of-work algorithms, validation rewards consist solely of transaction fees. There are therefore no new tokens created which thereby increase the money supply, which generates a loss of value of each token taken individually<sup>2</sup>: cryptocurrencies based on a proof-of-stake *blockchain* are **not structurally inflationary**, as Hossen & Gauthier (2024) emphasise. Another major advantage of proof of stake: since there is no competition between validation nodes — since it is an algorithm which decides to whom to attribute the transaction as a function of the amount staked and of the duration of the *staking* — the environmental problem of mining is solved. Moreover, the speed of validation is far greater: since a validation occupies a single computer, unlike the case of PoW where many machines fight over the same transactions, multiplying tenfold the validation nodes in PoS means multiplying the computing capacity. In economic terms, it is said that **the returns to scale of PoW are decreasing** — even if there are more computers, processing capacity does not necessarily increase — while **the returns to scale of PoS are a minima constant** — increasing the number of validating computers means increasing the processing capacity of the *blockchain* —: it is said that proof-of-stake algorithms are ‘**scalable**’.

What of the security of *blockchains* founded on proof of stake? Here, as Aponte-Novoa et al. (2021) indicate, the possibility of a 51 % attack resides in the possibility of holding more than 50 % of the tokens: it will then suffice to stake all of these tokens in order to be with certainty the person most involved in the *blockchain*. All the transactions to be validated will then necessarily pass through the hegemonic forger, which leaves the door open to hacking if the latter is malicious. Naturally, this is here again a thought experiment. Let us take the example of TRON, the tenth cryptocurrency in terms of **market capitalisation** at 28 August 2024 according to CoinMarketCap (2026): this comes close to 14 billion dollars on that day. Recall that the market capitalisation of a cryptocurrency (as of a firm moreover) is defined by the product of the number of tokens (or of shares in the case of a firm) and the unit value of a token (or of a share in the case of a firm). The *market cap* therefore represents the total value of a cryptocurrency; in the field of cryptocurrencies, the unit value of a token is generally expressed in dollars; accordingly, the *market cap* of a cryptocurrency corresponds to its ‘total money supply’, expressed in dollars, which may be found live on CoinMarketCap (2026) for example. Thus, wanting to orchestrate a 51 % attack on this cryptocurrency requires putting nearly seven billion dollars on the table; an amount far too costly, which explains the very good resistance of the large cryptocurrencies founded on proof of stake. But let us not forget that there exist according to CoinMarketCap (2026) more than 2.4 million cryptocurrencies in August 2024; most have capitalisations of only a few thousand dollars. Holding half of them in order to carry out a 51 % attack is, this time, entirely conceivable.

<sup>2</sup>Let us take up the example given by Hossen & Gauthier (2024): consider a proof-of-work *blockchain* associated with a cryptocurrency for which it is supposed that there initially exist ten tokens available. Let us imagine that these tokens are associated with a reality such as the production of pizzas in a given economy. If a transaction is validated and generates an additional token through the intermediary of mining, then there are now ten tokens in the economy, but still as many pizzas: in order to acquire a pizza, more tokens will therefore have to be spent. This is the principle of inflation: one runs the printing press and one creates value. But, if the economy does not follow, prices increase, since this new value is not associated with any new economic production: money therefore loses its value.

### 1.2.4 Proof of authority (PoA)

With proof of authority, the validation nodes are predefined and determined by a central authority; this is the case of Ripple or of IBM Food Trust for example, where only the computers of the company or of partners designated by it have the right to validate a block of transactions. As Hossen & Gauthier (2024) emphasise, this has the effect of removing all competition between validation nodes. Thus, only one computer therefore has to validate a Ripple transaction for the latter to appear on the register, which takes barely five seconds, considerably faster than with proof of work: PoA algorithms are therefore **particularly scalable**. Hossen & Gauthier (2024) add that, like proof of stake, PoA algorithms are **little energy-consuming**, which greatly reduces the carbon footprint of the *blockchain*.

#### Focus...



**The Ripple protocol.** Launched in 2012 by Ripple Labs, the **XRP Ledger** protocol is structurally distinguished from proof-of-work or proof-of-stake *blockchains*: as Schwartz et al. (2014) formalise in the founding white paper of the protocol, the **Ripple protocol consensus algorithm** (RPCA) rests on a mechanism of iterative voting between validating nodes belonging to mutually recognised sub-networks of trust — the ‘*unique node lists*’ —, which makes it possible to reach consensus in three to five seconds without ever having recourse to mining, and therefore without significant energy consumption. On the performance side, FPA (2025) establishes that the XRP Ledger processes up to 1500 transactions per second for a unit cost below one hundredth of a cent, against seven transactions per second for the Bitcoin network and settlement times of one to five working days for the **SWIFT** network, whose international transfer fees reach on average between 26 and 50 dollars per operation depending on the currency corridors concerned. The declared ambition of Ripple Labs is precisely to compete with this interbank network by proposing a ‘bridge currency’: in the arrangement known as **on-demand liquidity** (ODL), the XRP token serves as an intermediary of instant liquidity, allowing a financial institution to convert, transmit and reconvert currencies without having to tie up foreign exchange reserves in the accounts of foreign correspondents; more than 300 financial institutions were partners of the RippleNet network at the end of 2024.

It is precisely the legal nature of this token which was at the heart of the litigation commenced on 20 December 2020, when the **Securities and Exchange Commission** (SEC) filed a civil complaint against Ripple Labs, on the ground that the company had proceeded to the offer and sale of unregistered financial securities — the XRP token being qualified as an ‘*investment contract*’ within the meaning of the **Howey test** (a legal criterion established by the Supreme Court of the United States in the judgment *SEC versus W.J. Howey Co.* of 1946, according to which a transaction constitutes an *investment contract*, and therefore falls within federal securities regulation, as soon as it brings together four cumulative conditions: an investment of money, in a common enterprise, accompanied by a reasonable expectation of profits, generated by the efforts of a third party and not by the investor himself) —, initially claiming more than two billion dollars in damages, as the summary decision published by SDNY (2023) records. The decision of 13 July 2023 handed down by SDNY (2023) then introduced an unprecedented legal distinction: while the institutional sales of XRP carried out by Ripple Labs did indeed constitute a violation of section 5 of the *Securities Act* of 1933, the transactions effected on public secondary markets did not satisfy the criteria of the Howey test and could not therefore be requalified as financial securities; this reasoning therefore operates a dissociation between the issuer and the secondary market of a single digital asset, opening a jurisprudential breach for the whole of the crypto-asset sector. In August 2024, the court therefore ordered Ripple Labs to pay a civil fine of 125 million dollars — a fraction of the initial claim of two billion made by the regulator — before a final agreement came, in March 2025, to reduce the amount effectively paid to 50 million dollars, the SEC abandoning its appeal and Ripple Labs withdrawing its own, as Stempel (2025) reports.

Proof of authority therefore rests entirely on the trust one places in the validation nodes, that is to say on the reputation enjoyed by the validators. Validators must therefore reveal their identity, which may be useful in the event of an attempt to corrupt the *blockchain*, with an interest for private firms in attributing responsibilities. This is why, according to Hossen & Gauthier (2024), PoA algorithms seem particularly suited notably to small private *blockchains* federating a consortium of firms exchanging sensitive information (IBM Food Trust) or to large private *blockchains* whose solution implies a trust of a legal nature for users — initially, Ripple addressed itself to banking establishments and the latter naturally require **legal responsibilities in the event of failure**, whence the need for a central authority to sue in the event of a service default.

By contrast, the strength of PoA is likewise its weakness. Indeed, as Aponte-Novoa et al. (2021) bring out, since there are fewer validation nodes than with PoW or PoS algorithms for which there is no barrier to entry — certainly, for PoW, without good equipment, it seems utopian to validate, but nothing prevents one from trying; likewise, for PoS, even by staking little, there remains (depending on the *blockchains*) a minute probability of receiving a transaction to validate —, it is easier for a hacker to corrupt a private *blockchain* since there are far fewer computers to hack.

## Focus...



**The ‘blockchain trilemma’.** For certain proof-of-stake *blockchains*, however, Hossen & Gauthier (2024) emphasise that there is sometimes a minimum of tokens to stake, that is to say to place in escrow, in order to begin validating transactions with a non-zero probability. For example, Ethereum, which rested on proof of work until 15 September 2022, put online at that latter date a version 2.0 which rests this time on proof of stake. Now, in order to have the right to become a forger on this new version, at least 32 ethers (the name of the tokens associated with Ethereum) must be placed in escrow, which represents roughly 80 000 dollars at 28 August 2024 according to CoinMarketCap (2026). In a certain way, such PoS algorithms likewise suffer from a certain centralisation. Otherwise put, Ethereum 2.0 therefore sacrifices to a certain extent decentralisation on the altar of scalability (the great black mark of PoW). It is this issue which characterises what Buterin (2023), the founder of Ethereum, called the ‘blockchain trilemma’: a *blockchain* cannot be simultaneously scalable, decentralised and secure; there is always one pole which falls short in this triangle of incompatibilities.

## Key Takeaways



In order to be added to the register, an exchange must compulsorily be validated, that is to say certified by a third party who authenticates it (a validating node). Each transaction on a blockchain rests on an asymmetric encryption (the ECDSA algorithm) combining a public key — accessible to all, analogous to bank account details — and a private key — rigorously personal — which guarantees simultaneously confidentiality (only the recipient can decipher the message) and authenticity (only the sender can sign the transaction), the possession of the private key therefore constituting the necessary and sufficient condition of effective ownership of the *tokens* [Wang (2014)]. According to Hossen & Gauthier (2024), there exist three large families of validation of exchanges on a *blockchain*:

|                    | Proof of work (PoW): miners                                                                                                                                  | Proof of stake (PoS): forgers or minters                                                                                                                            | Proof of authority (PoA): validators                                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Protocol</b>    | The greater the computing capacity of one's computer, the greater the probability of validating a block.                                                     | The quantity of tokens placed in escrow ( <i>staking</i> ) and the duration of the <i>staking</i> determine the probability of receiving a transaction to validate. | Validation is carried out by computers pre-selected according to their reputation : PoA is centralised.                                                     |
| <b>Competition</b> | Miners seek to be the first to validate their block of transactions (constructed from <i>pending</i> transactions).                                          | An algorithm chooses the forger who will validate a transaction as a function of his involvement : there is no competition.                                         | The validation nodes are pre-selected by a central authority : there is no competition between nodes.                                                       |
| <b>Reward</b>      | The first miner to validate his block receives <i>tokens</i> created <i>ex nihilo</i> (a <i>coin-base transaction</i> ) as well as minimal transaction fees. | The forger chosen to validate the transaction recovers transaction fees only.                                                                                       | The validating node may or may not charge transaction fees : this depends on each protocol.                                                                 |
| <b>Security</b>    | In order to hack the <i>blockchain</i> , a <i>hacker</i> must hold more than half of the total computing capacity ( <i>hashrate</i> ).                       | In order to hack the <i>blockchain</i> , a <i>hacker</i> must hold more than half of the tokens available on the market.                                            | The security protocol is implemented by the central organisation : the security of the <i>blockchain</i> therefore depends on the security of the protocol. |
| <b>Efficiency</b>  | Decentralised to be sure, PoW is also little <i>scalable</i> and very energy-consuming.                                                                      | Decentralised, <i>scalable</i> and little energy-consuming, PoS is dangerous for cryptocurrencies with a low <i>market cap</i> .                                    | <i>Scalable</i> and little energy-consuming, PoA is however centralised, which increases the risks of hacking.                                              |

### 1.3 A few words on the main *blockchains*

Bitcoin and Ethereum constitute respectively the first and second cryptocurrencies in terms of **market capitalisation** — computed by taking the product of the number of tokens available and the unit value, generally expressed in American dollars, of a token. Structurally and economically, these two cryptocurrencies are nevertheless very different and the uses made of them are not the same.

#### 1.3.1 Bitcoin, a libertarian asset and a new store of value

**Bitcoin** was created by an unknown developer — or several, given the complexity of the code<sup>3</sup> — under the pseudonym of Satoshi Nakamoto in 2009, in the wake of the *subprime* crisis: it is therefore a project which has its source in a **context of mistrust, or even of defiance, with regard to traditional finance and institutions**. Indeed,

<sup>3</sup>The millions of lines of code which define the Bitcoin *blockchain* are freely accessible (as for any public blockchain) on the site <https://github.com/bitcoin-core>. As Hossen & Gauthier (2024) specify, this code is constantly updated through improvement proposals (BIP: *Bitcoin improvement proposals*) formulated by volunteer developers, anonymous or not, whose list is for example accessible on <https://bitcoin.org/en/development> (around 350 Bitcoin contributors in August 2024).

Nakamoto (2008) regarded any centralised institutional system as a source of instability for the world monetary and financial equilibrium. More precisely, as Hossen & Gauthier (2024) specify, governments were accused of having sown the seeds of the crisis by allowing greater freedoms to banks and financial establishments, which were able to invest in extremely risky assets, notably with savers' deposits. The latter then discovered that their savings, which they thought were safe in bank accounts, were in reality at risk. The bankruptcy of Lehman Brothers on 15 September 2008, a bank as important as Goldman Sachs, Morgan Stanley or JP Morgan in their time, then constituted a veritable shock which lit the spark that set the whole financial system ablaze. But this crisis was not only a financial crisis; it was above all a crisis of confidence: as Hossen & Gauthier (2024) indicate, it was at that time that the **Occupy Wall Street movements** developed, as if to encourage citizens to take back control of the economy, or at least to have a right of scrutiny over the world of finance. Now the objective of Bitcoin, as a public *blockchain*, is precisely to offer everyone this right of scrutiny, by creating a secure, decentralised and inclusive payment system, in which anyone, provided he has an internet connection, can access the register of the entirety of transactions and even take part in their validation: thus can confidence be restored in a renewed financial system according to Hossen & Gauthier (2024).

But Bitcoin must not be envisaged as a simple payment system, otherwise it would be in direct competition with the more than two million other cryptocurrencies which can be used for payments while being far more efficient. Indeed, the efficiency of Bitcoin, and its scalability in particular, leave much to be desired: as Kim et al. (2018) recall, to date, Bitcoin can validate only seven transactions per second; a ridiculous figure if one compares it with the Visa protocol which today validates 1700 transactions per second, still well below the capacities of Visa which can support up to 24 000 transactions per second. It is therefore utopian according to Hossen & Gauthier (2024) to imagine that Bitcoin at the present time could one day be used as a standard means of payment. How then are we to explain the hegemony still today of Bitcoin — whose market capitalisation at 28 August 2024 represents more than 57 % of the aggregate capitalisation of the first one hundred and twenty-five cryptocurrencies according to Trading View (2024) — even though other cryptocurrencies such as Solana, an algorithm resting on proof of stake — and even on a refinement of proof of stake, *proof of history* — can validate up to 65 000 transactions per second? According to Hossen & Gauthier (2024), it is because Bitcoin must be envisaged today not as a simple means of payment, but above all as a **store of value**, a kind of 'digital gold'. How is such a point of view to be justified? In order to answer this question, we shall have to study the mechanism of the validation rewards of the Bitcoin blockchain.

### Focus...



**The question of Bitcoin's scalability.** Why only seven transactions per second? The initial idea was to limit the risks of centralisation. Since it is the miner who has the greatest computing capacity who succeeds in solving the validation problem first, one might imagine that a gigantic 'mining factory' might spring from the ground, with almost unlimited means, and validate at the same time all the transactions awaiting validation (*pending transactions*), leaving nothing for the other miners. In order to avoid too strong a centralisation, the size of blocks has been limited to one megabyte (MB) since 12 October 2010, as Hossen & Gauthier (2024) recall. Considering that a transaction — which comes down to a few pieces of information such as the sending address, the receiving address, the date, the amount exchanged and the amount of fees paid — weighs on average around 570 bytes, a block can therefore contain at most roughly 1750 transactions to validate. Now, nowadays, miners take between eight and eleven minutes to validate all the transactions of a block, which means that the maximum validation capacity of the Bitcoin *blockchain* is around three transactions per second. Improvements were subsequently proposed for Bitcoin, as Pérez-Solà et al. (2019) specify: the **Segwit protocol**, launched on 1 August 2017, thus made it possible to optimise the weight of transactions in a validation block so as to be able to create blocks containing up to 3500 or even 4000 transactions. This update of the Bitcoin protocol, described as a **soft fork**, made it possible to reach this figure of seven transactions per second at most today, a figure still well short of the Visa standards used for card payments.

The **scalability** constraint of the Bitcoin *blockchain* — which processes only three to four transactions per second, against roughly 1700 for the Visa network, on account of a block size limited to one megabyte and an average mining time of ten minutes — led Poon & Dryja (2016) to propose, in their white paper of January 2016, a **secondary layer** (*layer 2*) of Bitcoin resting on a network of **bidirectional payment channels** *off-chain*: two parties place in escrow an amount of bitcoins via a multi-signature contract entered on the main *blockchain*, then proceed to as many over-the-counter transactions as they wish — instantaneous and free of charge — before closing the channel by a single final transaction, the only one engraved on the chain. The security of the arrangement rests on **hashed timelock contracts (HTLC)**: if one of the parties fraudulently broadcasts an earlier state of the channel, its funds are automatically redistributed to the other party by way of penalty, making cheating economically irrational. On the macroeconomic level of the network, the institutional report of Fidelity Digital Assets (2024) establishes that the average capacity per payment channel has progressed by 214 % over the last four years; in December 2025,

according to the data published by Bitcoin Magazine (2025a), the Lightning network reached a historic record of 5637 bitcoins of total capacity, distributed across 48 678 active channels and 14 940 nodes. The commercial adoption of the protocol is moreover experiencing a sustained dynamic: as Wright (2025) establishes, the volume of payments routed via Lightning progressed by 1212 % between August 2021 and August 2023 ; in 2024, the payments platform CoinGate reported that Lightning represented more than 16 % of the bitcoin orders processed, against 6.5 % two years earlier. These performances nevertheless contrast with a structural limit which Poon & Dryja (2016) emphasise from the outset: the amount of an individual transaction within a channel is capped by the amount placed in escrow at the opening, making the protocol ill-suited to transactions of high amount; moreover, the number of nodes of the network has receded since its peak of 20 700 at the beginning of 2022, illustrating the persistent challenge of **decentralisation** as the network capitalises without the user base broadening in the same proportion.

While the Lightning network addresses itself to private individuals and to micro-payments, the **sidechain** Liquid answers properly institutional requirements of speed, confidentiality and issuance of digital assets. The very concept of a *sidechain* — an independent *blockchain* linked to a main chain by a mechanism of **bidirectional anchoring** (*two-way peg*) — was formalised as early as 2014 in the white paper of Back et al. (2014): like the chips of a casino, the *sidechain* rests on a mechanism of **collateralisation** (there are as many chips distributed by the casino as there are real notes and coins in its vaults). In September 2018, Blockstream deployed Liquid in production: as Blockstream (2018) details, the architecture rests on **proof of authority**, ensured by the *Liquid Federation*, a federation of recognised institutional actors — exchange platforms such as Bitfinex, financial service providers and *blockchain* companies — each of whose members holds a validation key. As soon as at least two thirds of these *functionaries* have co-signed a block, the latter is definitively validated, which makes it possible to reach a finality of transactions in less than two minutes, against ten minutes for a single block on the main chain. This gain in speed is accompanied by **confidential transactions**: unlike the Bitcoin *blockchain*, which is entirely public, Liquid cryptographically masks the amounts and the types of assets exchanged, a property which Blockstream (2018) presents as a *sine qua non* condition for attracting institutional investors anxious not to expose their market positions to their competitors. Besides Liquid Bitcoin (L-BTC), the network permits the issuance of native assets (*issued assets*), opening the way to the **tokenisation** of financial instruments and real assets; in July 2025, according to Bitcoin Magazine (2025b), the total value locked in the Liquid ecosystem amounted to 3.27 billion dollars. In March 2026, Binance (2026) announced the deployment of the first **post-quantum signature** transactions on the Liquid network in production, making the *sidechain* a technological laboratory for the future evolutions of the Bitcoin ecosystem. The main limit of this arrangement resides precisely in the centralisation that the federation implies: while the *functionaries* are institutional actors having every interest in maintaining confidence in the ecosystem, their role as trusted third parties constitutes a single point of failure which breaks with the founding principle of decentralisation of *blockchain* technology.

Every time a miner is the first to validate a block of transactions, he receives newly created bitcoins: the number of bitcoins in circulation therefore increases mechanically over time. And yet, the number of bitcoins in circulation will never be able to exceed 21 million. How is this possible? As Meynkhard (2019) explains, initially, being the first to mine a block of transactions awaiting validation (*pending transactions*) yielded fifty bitcoins. The number of bitcoins in circulation therefore increased more and more rapidly. This rapid money creation entailed a sustained inflation — the **inflation rate** is defined here as the ratio of the number of bitcoins created in one year to the total number of bitcoins in circulation. In order progressively to limit Bitcoin's inflation, each time 210 000 blocks have been mined — which occurs on average every four years, knowing that a block takes on average ten minutes to be validated — the mining reward is halved: this is the **halving** process. The first *halving* took place on 28 November 2012, and the mining reward was thus reduced to twenty-five bitcoins; the second took place on 9 July 2016 (the reward moving to 12.5 bitcoins after 420 000 blocks mined), the third on 11 May 2020 (moving to 6.25 bitcoins after 630 000 blocks mined), and the fourth on 19 April 2024 (moving to 3.125 bitcoins after 740 000 blocks mined). But while the mining reward is halved, this is not the case for the difficulty of the signature verification problem to be solved: after each *halving*, mining Bitcoin therefore becomes twice as expensive. Consequently, according to Meynkhard (2019), holding bitcoins becomes rarer and rarer, and the intrinsic value of Bitcoin increases. Indeed, supply begins to dry up while demand continues its growth: the law of supply and demand then predicts a rise in the price. Otherwise put, the price of Bitcoin therefore undergoes an **intrinsic upward pressure**; this is a behaviour typical of stores of value such as property or physical gold, according to Hossen & Gauthier (2024). A behaviour reinforced by the fact that Bitcoin, an asset already more than fifteen years old, can rely today on a powerful network effect: it is the most liquid cryptocurrency of all — that is to say the one which is the simplest to exchange on exchange platforms —; Bitcoin concentrates the most users and the most computer developers.

## Key Takeaways



Born in 2009 in the wake of the *subprime* crisis, Bitcoin embodies a libertarian response to the failure of centralised financial institutions: its proof-of-work protocol offers any user having an internet connection transparent access to the register of transactions and the possibility of validating its content [Nakamoto (2008)]. However, its low scalability — seven transactions per second at most, against 1700 for Visa — disqualifies it as a standard means of payment; its true value proposition is rather that of a digital store of value, analogous to gold [Hossen & Gauthier (2024)]. This status rests on the *halving* mechanism: every 210 000 blocks mined (that is roughly four years), the miners' reward is halved, compressing new supply while demand continues its growth — which generates an intrinsic upward pressure on the price [Meynkhart (2019)]. Thus, total supply is by construction capped at 21 million bitcoins, conferring on the asset an algorithmically guaranteed scarcity that no central bank can reproduce. These characteristics, combined with a dominant network effect — Bitcoin represented more than 57 % of the aggregate capitalisation of the first one hundred and twenty-five cryptocurrencies in August 2024 [Trading View (2024)] — explain its durable hegemony despite the emergence of technically superior protocols.

### 1.3.2 Ethereum and *smart contracts*: the world of decentralised finance

Created in 2015 notably by the Russian engineer Vitalik Buterin, the cryptocurrency **Ethereum**, today second in terms of market capitalisation, was not initially intended to be used as a decentralised means of payment. As Metcalfe (2020) emphasises, Ethereum was indeed conceived as a platform allowing its users to create decentralised applications (**dApps**) by encoding them thanks to Ethereum's computer language: **Solidity**. dApps are programs requiring no central servers or central network, whose parameters of code execution and data storage are managed in a decentralised manner, generally via **smart contracts**, that is to say computer protocols which facilitate, verify and execute the negotiation or the execution of a contract, or which render a contractual clause unnecessary. What more precisely are these dApps which can be encoded on the Ethereum blockchain? They are pieces of software which may include security programs, voting systems, payment methods... In every case, these dApps, in order to be perfectly decentralised, must respect a precise set of specifications which Metcalfe (2020) enumerates:

- dApps must be **transparent**: both the code and the *smart contracts* must be public and verifiable. The fact that the data are recorded on a blockchain, and therefore present on numerous nodes, then makes it possible to establish trust without requiring intermediaries. The multiplication of nodes across the world likewise guarantees a good **resistance to censorship**, that is to say a capacity to escape any risk of prohibition or governmental control.
- dApps must be **open source**: since there exists no court for each cryptocurrency and since there exists a risk of a computer flaw in the programming of *smart contracts* — whose exploitation could allow a *hacker* to seize the *tokens* — operating in *open source* permits the constant improvement of the code.
- Finally, dApps must be **open to any computer** having an internet connection, in order to allow users to retain control of their funds at all times, while limiting costs and delays.

## Focus...



**The beginnings of decentralisation on the internet.** As Hossen & Gauthier (2024) qualify, the idea of decentralisation on the internet did not, however, emerge with the *blockchain*: **torrent networks** already made it possible, as early as the summer of 2002, the date of the establishment of the BitTorrent protocol, to cut up information — a video, an installation file, a document... — into several segments and to distribute these to different interlocutors who can then in turn play the role of server (*seeder*) for a person wanting in turn to download the information (*leecher*). On the economic level, this peer-to-peer architecture rests on a logic of **network externalities**: the higher the number of *seeders* for a given file, the more the download speed of each *leecher* increases, so that the value of the service grows with the number of users, contrary to the traditional client-server model where bandwidth constitutes a limiting factor paid for by a single sender. This architecture nevertheless engenders a **free rider problem**: a significant fraction of users switch off the *torrent* client as soon as their download is finished, reducing by as much the seeding capacity of the network; the BitTorrent protocol incorporates an incentive mechanism known as '*tit-for-tat*' which modulates the download speed received as a function of the throughput shared by the user, precisely in order to discipline this opportunistic behaviour.

The most documented economic dimension of these networks nevertheless remains **intellectual property piracy**: as Danaher et al. (2013) establish, virtually all the econometric studies published before 2014 conclude in favour of a statistically significant negative effect of online piracy on the receipts of the cultural industries; their analysis of the French HADOPI law moreover shows that the establishment of the **graduated response** — a system of progressive



warning of infringing internet users: the internet user caught in the act of unauthorised sharing first receives a warning email, then a recorded-delivery letter, before a referral to the public prosecutor is contemplated — entailed a rise of roughly 25 % in digital music sales for the large record companies, making this arrangement one of the rare documented cases of proven effectiveness of an anti-piracy public policy. On the European scale, the report of Office de l'Union européenne pour la propriété intellectuelle (2024) reveals that European internet users access illicit online content on average ten times a month, and that visits to pirate IPTV sites progressed by 10 % in 2023 relative to the previous year. In France, ARCOM (2024) — the regulatory authority for audiovisual and digital communication, heir to the missions of HADOPI since 2022 — recorded no fewer than 2.6 million complaints filed by rights holders in the course of the single year 2023; of this total, roughly 137 000 first warnings were addressed to infringing internet users, followed by some 33 000 second notifications and 4000 third warnings resulting in more than 230 convictions with a fine, which testifies to a structural persistence of the phenomenon despite two decades of regulation. These figures illustrate the fundamental tension which Krishnan et al. (2004) brings to light: the BitTorrent protocol is in itself a technologically neutral tool, whose massive use for the purposes of circumventing copyright reflects less a technical failure than a persistent mismatch between the economic models of the cultural industries and the accessibility and price preferences of digital consumers.

Smart contracts, more precisely, are algorithms which verify conditions of the type ‘if such and such a condition of the contract is validated, then carry out such and such actions’ or ‘as long as such and such a condition of the contract is verified, then carry out such and such actions’; one speaks of conditional functions and loops in computing. To take up the example mentioned by Hossen & Gauthier (2024), one might imagine a smart contract which one buys at the same time as one books a flight; if this flight is more than three hours late at departure or arrival (condition of validity), then the passenger is indemnified in an amount of 300 euros paid in *tokens* (clause of the contract); a potential revolution in the field of insurance. This whole contract could thus be encoded on the secure and inviolable register of the Ethereum *blockchain* thanks to the Solidity computer language. Otherwise put, Ethereum serves as a *template* for encoding a precise contract, by facilitating the creation of these smart contracts while relying on the security of the Ethereum protocol, a *blockchain* which rests today on proof of work. Obviously, the contracts encoded can be far more complicated, but their great advantage is that they are univocal: the condition is verified or it is not; it is not subject to interpretation like a paper contract in which a badly placed comma could change the meaning. The more ‘complicated’ the *smart contract*, that is to say the more operations it requires for its conditions of validity to be verified or not, the more the validating nodes — here the miners — must supply a sustained computing effort. This last effort must be specifically remunerated; this is where **gas** comes in, the fuel which makes it possible to use the ‘motorway’ of the Ethereum network, the remuneration of the validating nodes. The price of gas is generally expressed in weis, and more specifically in billions of weis — one speaks of a gwei (for giga-wei). A billion billion ( $10^{18}$ ) weis, that is a billion gweis, corresponds to one **ether**, the name of the native *token* associated with Ethereum. Like Bitcoin, for which the miners are directly remunerated in bitcoins, with Ethereum the ‘fuel’ for making a smart contract work is therefore the ether, whose gas fees go directly into the pocket of the validating node. Thus, as Hossen & Gauthier (2024) recall, the more gweis a user agrees to pay in order to validate his transaction, the more he will attract the miners who will verify his *smart contract* and the more rapidly the transaction associated with the latter will be added to the *blockchain*.

### Focus...



**Inviolable *blockchains*, but vulnerable platforms.** While the Bitcoin *blockchain* has, since its creation in 2009, never been compromised in its fundamental cryptographic protocol, the **centralised exchange platforms** which constitute its peripheral ecosystem have by contrast constituted favoured targets for malicious actors, revealing a structural vulnerability between the robustness of the decentralised protocol and the fragility of the third-party infrastructures which ensure its liquidity. The canonical case remains that of Mt. Gox: founded in 2010 in Tokyo, this exchange platform processed at its apogee more than 70 % of world transactions in bitcoins before going bankrupt in 2014 following a large-scale hacking. Now, as Decker & Wattenhofer (2014) demonstrate by exploiting the transaction traces of the whole Bitcoin network over more than a year, the official account of Mt. Gox attributing its bankruptcy to **transaction malleability attacks** does not stand up to analysis: each Bitcoin transaction is uniquely identified by a **transaction identifier** (TXID), computed by hashing the entirety of the data of the transaction, digital signature included. Certain fields of the digital signature — notably the *scriptSig* field — are not entirely covered by the cryptographic protection, so that an ill-intentioned third party can slightly modify their binary representation without altering either their legal validity or their economic content — the sender, the recipient and the amount remain identical —, but by changing the bits sufficiently for the resulting *hash*, and therefore the TXID, to be entirely different (it is the same principle as when the recipient of a payment enters information on the back of a banker's

cheque: this changes nothing as to the validity of the said cheque, determined by the information and signatures of the issuer on the front). The attacker then broadcasts this mutated version of the transaction to the nodes of the network before the original transaction is confirmed: if it is the mutated version which is included first in a block, the original TXID is invalidated, and any computer system which uses this identifier to track the state of a payment — as Mt. Gox did — wrongly concludes that the transaction has failed, which may lead it to reimburse the sender and therefore to transfer the same funds twice: this is a disguised form of **double spending**. And yet, the data analysed by Decker & Wattenhofer (2014) testify to no generalised use of this attack vector prior to the closure of the platform in February 2014, suggesting that the misappropriation operated through more gradual internal mechanisms; the malleability vulnerability was definitively corrected only in August 2017 by the **Segwit** (*segregated witness*) *soft fork*, which separates the signature data from the rest of the transaction so that they no longer enter into the computation of the TXID. This is precisely what Gandal et al. (2018) confirm: by analysing the transaction data of Mt. Gox over the period from February to November 2013, they identify two distinct episodes in the course of which roughly 600 000 bitcoins of a cumulative value of 188 million dollars were fraudulently acquired by agents who in all likelihood did not pay their real price, and they establish that the price of bitcoin progressed on average by 4 % on the days when these suspicious transactions took place; in total, of the 850 000 bitcoins declared stolen at the time of the bankruptcy of February 2014 — of which 200 000 have since been recovered —, a substantial part had therefore already been withdrawn from the accounts since at least 2011, without the managers of the platform having sufficient internal audit systems to detect it.

According to Metcalfe (2020), Ethereum is thus akin to the **standard-bearer of decentralised finance** (DeFi). Even more than Bitcoin, which is only a currency without a central regulatory authority, the ambition of DeFi is thus to go further by proposing a whole ecosystem of financial services without any organ of control and free of use. DeFi then makes it possible to bring down barriers to entry: costs are reduced by virtue of the absence of intermediaries, which puts a wide range of financial services within everyone's reach, provided the user has an internet connection. Ethereum is, however, a victim of its success: with all the smart contracts which are encoded in Solidity, the Ethereum network finds itself receiving a growing number of transactions to validate. And while the Ethereum network is to date more efficient than the Bitcoin network, making it possible to validate up to forty-four transactions per second against seven for Bitcoin, it too rapidly reaches saturation. A logical economic consequence which Hossein & Gauthier (2024) bring out: since demand for validation exceeds the capacity of the miners, the latter choose to validate the most remunerative contracts, that is to say those which offer a generous payment of gwei, which entails in its wake a runaway increase in transaction fees, which may sometimes reach the equivalent of some thirty euros for a transaction worth some fifty for example. It is this pernicious effect which justified the desire to update the Ethereum *blockchain* in order to make it move from proof of work to proof of stake, marking the advent of Ethereum 2.0; this is a deep update of the *blockchain*, known as a **hard fork**, absolutely without precedent in the history of the *blockchain*.

### Focus...



**Hard fork versus soft fork.** A *fork* designates any modification of the consensus rules of a *blockchain* protocol, and the distinction between its two forms — the *hard fork* and the *soft fork* — rests on a single criterion: **backward compatibility**. A *hard fork* introduces a non-backward-compatible modification of the consensus rules, that is to say that the nodes which have not adopted the update no longer recognise as valid the blocks produced by the updated nodes: the two groups diverge irremediably, each continuing the extension of its own chain from a common pivot block, so that the network splits into two independent *blockchains* sharing the same history prior to the point of divergence. The canonical example is that of **Bitcoin Cash**: on 1 August 2017, at the height of block 478 559, a fraction of the Bitcoin community refused the Segwit approach to solving the problems of malleability and scalability and proceeded to a *hard fork* taking the maximum block size from one megabyte to eight megabytes, thus creating a new cryptocurrency, Bitcoin Cash (BCH), alongside the original Bitcoin (BTC), whose holders automatically received an equivalent amount in BCH.

A *soft fork*, by contrast, introduces a backward-compatible modification: the nodes not updated continue to recognise the blocks produced by the updated nodes as valid, since the new rules are strictly more restrictive than the old ones — they reduce the space of accepted blocks without widening its boundaries —, so that the network preserves its unity on condition that the majority of computing power adopts the update. The most emblematic illustration of this is precisely **Segwit** (*segregated witness*), activated on the Bitcoin network on 24 August 2017: by separating the signature data (*witness*) from the rest of the content of the transaction, this *soft fork* made it possible both to correct the malleability vulnerability mentioned previously and to increase the effective capacity of blocks

without formally modifying their size limit, the nodes not updated continuing to validate the new transactions as if they were ordinary transactions.

### Key Takeaways



Launched in 2015, Ethereum is conceived not as a simple means of payment but as a platform of decentralised applications (dApps), whose smart contracts encoded in the Solidity language make it possible to execute contractual clauses automatically as soon as a defined condition is verified — without any intermediary, without ambiguity of interpretation [Metcalf (2020)]. In so doing, Ethereum constitutes the technical foundation of decentralised finance (DeFi): by removing intermediaries, it reduces the costs of access to an extended range of financial services for any user having an internet connection. The remuneration of validating nodes rests on gas, a unit of fuel expressed in gwei (billions of wei), whose price adjusts to demand: the more complex a contract, the more gas it consumes, and the more generously the user must remunerate the validators in order to be processed as a priority [Hossen & Gauthier (2024)]. It is precisely this runaway increase in transaction fees — the direct consequence of the saturation of a network capped at forty-four transactions per second — which led to the migration of Ethereum towards proof of stake (the Ethereum 2.0 *hard fork*), a protocol rupture without precedent in the history of the *blockchain*.

### 1.3.3 Binance and the universe of centralised finance in the *blockchain*

Founded in July 2017 in Hong Kong by Changpeng Zhao, the centralised exchange platform **Binance** (for *binary finance*) conceived, from its first hours of existence, a **native utility token** — the *Binance Coin* (BNB) — whose economic logic exceeds the simple function of fee reduction. At the *initial coin offering* (ICO) of July 2017, Binance put into circulation an initial supply of 200 million BNB tokens whose distribution was established at 50 % for the public sale, 40 % for the foundation and the founding team, and 10 % for *angel* investors, as Binance (2017) records; holders of BNB benefited from a degressive reduction of transaction fees on the platform — 50 % in the first year, 25 % in the second, 12.5 % in the third —, thus creating a mechanism of **loyalty through the asset** which structurally linked the value of the token to the volume of activity of the platform. In September 2020, Binance launched the **BNB Smart Chain** (BSC), an independent *blockchain* whose consensus mechanism known as ‘*proof of staked authority*’ (PoSA) constitutes a hybrid between proof of stake and proof of authority: only twenty-one validators, selected from among the holders of BNB having placed the most tokens in escrow, are authorised to validate blocks, which makes it possible to reach an average block time of three seconds, as Monga & Singh (2022) measure in their comparative analysis of the performances of the BSC and of Ethereum — against several tens of seconds on the latter —, at the price, however, of a markedly reduced decentralisation, restricted to twenty-one permanent nodes. On the level of the monetary policy of the token, Binance put in place a **destruction mechanism** (*coin burn*) aiming to reduce the total supply from 200 million to 100 million tokens: each quarter, the platform allocates roughly 20 % of its profits to the buyback and permanent destruction of BNB by sending them to an unrecoverable address (*burn address*); the BEP-95 protocol, introduced subsequently, adds a real-time destruction of 10 % of the gas fees of the BSC, thus mechanically linking the deflation of the token to the activity of the decentralised ecosystem. According to the data published by Ainvest (2025), since 2023, this cumulative process has reduced the total *supply* by 31 %, while the cumulative value destroyed exceeded 58.5 billion dollars in the second quarter of 2025; BNB thus displays a **deflationary structure** that is rare among large-capitalisation tokens, in which the rise in the volume of Binance’s activity mechanically engenders a destruction of tokens, then a rarefaction of supply, and therefore — on the assumption of stable or growing demand — an appreciation of the residual value of each unit: this is what may be described as an **algorithmic scarcity rent**, which fundamentally distinguishes this model from the inflationary issues characteristic of proof-of-work *blockchains*.

More generally, the rise of **centralised exchange platforms** has generated, over the years, a concentration of the market which Banque du Canada (2024) describes as exceptional by the standards of traditional financial markets: over the period from July 2022 to July 2023, Binance alone controlled roughly 65 % of the *spot* volume of the main centralised exchange platforms (*centralised exchanges*, CEX), while the leading regulated platform in the United States, Coinbase, represented only 10 % of the same perimeter; the ecosystem counted more than 400 active CEX, but fewer than ten concentrated the bulk of volumes. This dominance has slightly eroded under the effect of the judicial affairs described *infra*: according to the data compiled by CoinGecko (2025), Binance held no more than 38 % of the world *spot* market in April 2025, its direct competitors — OKX, Bybit, Bitget — having absorbed the bulk of the share lost. What Banque du Canada (2024) emphasises is that this phenomenon of concentration is not the simple product of network externalities, but results from a structural **vertical integration** of financial functions within CEX: where traditional financial markets rigorously distinguish the exchange, the broker-dealer, the central

depository and the lender, the large CEX simultaneously assume all of these roles, creating synergies of revenue but also conflicts of interest and systemic risks without equivalent in regulated finance. On the microeconomic level, CEX moreover exercise market power over the issuers of new crypto-assets: as Hossen & Gauthier (2024) explain, creators of blockchain protocols have paid hundreds of thousands of dollars to Binance in order to list their tokens — Blockstack is said to have paid 250 000 dollars for the listing of its Stacks (STX) token in 2020 —, a practice which is economically akin to an entry fee to the most liquid market, and which creates considerable **barriers to entry** for the protocols whose founders do not have sufficient resources to access the dominant platforms. The tension between the commercial power of these platforms and the growing regulatory requirements constitutes the central knot of the governance of the sector: subject to anti-money-laundering obligations (*know your client*, KYC; *anti-money laundering*, AML) as soon as they accept flows in fiat currency, CEX in reality operate in a heterogeneous legal space — Binance has successively been domiciled in Hong Kong, in Malta, in the Cayman Islands and in the British Virgin Islands — which allows them to escape partially from national regulatory frameworks, which Dell’Erba (2023) describes as ‘**cross-border regulatory arbitrage**’ for the benefit of a business model resting on the opacity of the segregation of client assets.

### Focus...



**The FTX platform scandal.** Founded in May 2019 in Hong Kong by Sam Bankman-Fried (SBF) and Gary Wang, two MIT graduates, FTX (for *futures exchange*) became within the space of three years the second largest crypto-asset exchange platform in the world, valued at its apogee at 32 billion dollars and with a daily volume exceeding 10 billion dollars: as Investopedia (2022) indicate, FTX had carved out a reputation as a serious institutional platform, notably thanks to its offering of derivative and leveraged products, and benefited from a public image carefully maintained by SBF — who generously distributed funds to philanthropic causes and political campaigns — which the reality of its internal governance was rapidly to belie. The native token of the platform, the **FTX Token** (FTT), launched on 28 July 2019, was an ERC-20 asset deployed on the Ethereum *blockchain*: it allowed its holders to benefit from reductions on *trading* fees, to use it as margin collateral on forward contracts, and to receive *staking* rewards; Binance moreover held significant volumes of FTT, obtained at the time of its participation in the capital of FTX in 2021, before divesting itself of them. The crisis was triggered on 2 November 2022 when CoinDesk published an analysis of the balance sheet of **Alameda Research** — a quantitative *trading* fund co-founded by SBF — revealing that its assets consisted to the tune of roughly 30 % of FTT tokens, of a value then estimated at five billion dollars, exposing a worrying financial circularity: Alameda Research guaranteed a substantial part of its commitments with the token issued by SBF’s own platform, creating a circular correlation between the solvency of the fund and the market value of an asset whose liquidity itself depended on confidence in FTX. The reaction of Changpeng Zhao, who publicly announced the liquidation of Binance’s holdings in FTT, precipitated a massive wave of withdrawals: in two days, the price of FTT collapsed by more than 80 %, going from more than thirty dollars to less than two dollars. After the failure of the takeover attempt by Binance — which withdrew the day after the announcement of the non-binding offer, having consulted the accounts of the platform —, FTX, Alameda Research and more than one hundred affiliated entities filed for bankruptcy on 11 November 2022 under **chapter 11 of the American bankruptcy law**, as U.S. Department of Justice (2024) establishes. The judicial investigation revealed that FTX is said to have misappropriated roughly eight billion dollars of client deposits in order to finance the speculative bets of Alameda Research and the personal expenses of its managers; before the American Congress, the new CEO John J. Ray III — the insolvency administrator who had managed the Enron bankruptcy — declared: ‘*never in my career have I encountered such a complete failure of corporate controls and such a total absence of trustworthy financial information*’, thus summarising what Bryne (2023) describes as a ‘multidimensional governance failure’, simultaneously affecting the integrity of information systems, the segregation of client assets, internal control and regulatory supervision. Sentenced in March 2024 to twenty-five years of federal imprisonment and to repay eleven billion dollars, SBF embodies the textbook case of the systemic risks engendered by the **absence of asset segregation** and the unregulated integration of the functions of exchange, depository and investment fund within a single centralised actor.

In particular, on 21 November 2023, U.S. Department of Justice (2023) made public its indictment against Binance and its founder Changpeng Zhao, simultaneously with the announcement of a plea agreement negotiated by the platform: Binance notably pleaded guilty to violations of the *Bank Secrecy Act* (BSA) — a federal law imposing internal anti-money-laundering programmes —, of the *International Emergency Economic Powers Act* (IEEPA) — by permitting transactions involving nationals of countries under American embargo, notably Iran and Cuba — and of failure to register as a money transmitting business. The terms of the settlement stipulate, according to Roush (2024), that Binance must pay more than 4.3 billion dollars in total: 1.8 billion in fines and 2.5 billion in

forfeiture of assets, which constitutes the highest pecuniary settlement ever obtained by the American Department of Justice against a crypto-asset platform; Changpeng Zhao personally pleaded guilty to violation of the BSA for failure to put in place an effective anti-money-laundering programme and resigned from his post as CEO, giving way to Richard Teng, the platform's former director of regional markets. This affair illustrates the theoretical observation that Dell'Erba (2023) formulates: the concentration of multiple functions within a single unregulated actor — market maker, depository, lender and exchange platform — generates **systemic risks** which exceed the absorption capacity of market mechanisms alone. In this respect, in Europe, the **MiCA** regulation (*markets in crypto-assets*), whose provisions relating to crypto-asset service providers (**CASP**) became fully applicable in December 2024, constitutes the most structured institutional response to this risk: it imposes on CEX operating in the European Union the obtaining of an authorisation, the segregation of client assets from the platform's own assets — a measure which would precisely have made it possible to avoid the FTX scenario — and compliance with own funds requirements proportional to the volumes processed. The convergence of these evolutions — a massive American settlement, the entry into force of MiCA, growing requirements of transparency as regards reserves (*proof of reserves*) — thus sketches a movement of **regulated re-intermediation** of the sector, which constrains CEX to bring their governance model structurally closer to that of the traditional market infrastructures which they originally intended to supplant.

### Key Takeaways



Centralised exchange platforms (CEX) embody a constitutive tension of the cryptocurrency universe: they offer liquidity, ergonomics and access to derivatives markets, but do so by reconstituting exactly the structures of intermediation that the *blockchain* claimed to eliminate. Binance, which controlled up to 65 % of world *spot* volume in 2023, is the canonical example: its native BNB token rests on a deflationary *coin burn* mechanism which algorithmically links the scarcity of supply to the activity of the platform, while its plea agreement of November 2023 (4.3 billion dollars, violations of the *Bank Secrecy Act* and of the American embargo) illustrates the risk of cross-border regulatory arbitrage inherent in an opportunistic corporate domiciliation [U.S. Department of Justice (2023)]. Moreover, the bankruptcy of FTX in November 2022 exposed the structural systemic risk of CEX: the absence of segregation between client assets and the platform's own funds, combined with the vertical integration of the functions of exchange, depository and hedge fund, permitted the misappropriation of roughly eight billion dollars of client deposits [U.S. Department of Justice (2024)]. The most structured institutional response is the European MiCA regulation, fully applicable since December 2024, which imposes authorisation, segregation of assets and own funds requirements on CASP operating in the European Union: this movement of regulated re-intermediation constrains CEX to converge structurally towards the model of the traditional market infrastructures which they intended to surpass.

## 2 Nature, diversity and applications of cryptocurrencies

According to Hossen & Gauthier (2024), cryptocurrencies must present a certain number of characteristics in order to ensure the twofold imperative of the security and efficiency of transactions:

- **Transparency:** all transactions are public and the register must be accessible to all, and this even in the case of private *blockchains* backed by a cryptocurrency, such as Ripple. It is a question of trust: how otherwise is one to make sure that the transaction did indeed take place, if there exists no register to verify it?
- **Inviolability and unfalsifiability:** the algorithm which underlies the *blockchain* is resistant to the main computer attacks, notably 51 % attacks. For proof-of-authority *blockchains* in particular, unless the responsibility of the central authority is engaged, it is therefore for the latter to implement a resistant encryption algorithm.
- **Irreversibility:** the person who receives *tokens* cannot suffer a cancellation — a *recall*, as is the case after the paying in of a bounced cheque for example —, while the payer cannot retract his payment; an absolute necessity knowing that, for all public *blockchains*, no authority legally guarantees the transaction.
- **Disintermediation and deregulation:** the transaction is carried directly to the receiving address, which limits the transaction fees due to a clearing house; moreover, no state authority regulates the creation, the transactions or the price. What then of future state cryptocurrencies?

### 2.1 Can crypto-assets be likened to money?

According to article L54-10-1 of the French Monetary and Financial Code, a **cryptocurrency** designates ‘any digital representation of a value which is not issued or guaranteed by a central bank or by a public authority, which is not

*necessarily attached to a currency having legal tender status and which does not possess the legal status of a currency, but which is accepted by natural or legal persons as a means of exchange and which may be transferred, stored or exchanged electronically'.*

### 2.1.1 Crypto-assets tested against the Aristotelian definition of money

Thus, according to Hossen & Gauthier (2024), a cryptocurrency is structurally conceived as a **medium of exchange**, despite the inherent volatility of its price; it likewise functions as a **store of value** (notably for Bitcoin), the growing interest of institutional investors strengthening confidence in the *blockchain* and creating a virtuous circle. More precisely, the price of Bitcoin has been listed on the Chicago exchange, a benchmark notably for commodities, since 2015; in February 2021, Tesla had announced that it had invested 1.5 billion dollars in Bitcoin; since August 2021, PayPal has authorised Britons to pay for their online purchases with bitcoin (for merchants who accept it); the American regulator authorised in October 2021 the *listing* on the New York stock exchange of an ETF (a derivative product) on Bitcoin. . . All these signals of the '**normalisation** of Bitcoin' strengthen the idea that 'if the big players do it, why not me?', thus creating a virtuous circle which strengthens confidence in Bitcoin and supports those who invest in it as one would for a standard store of value such as property or gold. However, the former deputy director general for financial stability of the Banque de France, Odonnat (2021), qualifies this analysis by specifying that '*for cryptocurrencies, the means of exchange remains marginal: these crypto-assets represent 1200 billion euros, and Bitcoin 40 % of this mass; important to be sure, but still marginal relative to the world money supply of 45 000 billion euros. And whoever says store of value says stability; now one sees it with Bitcoin: these crypto-assets are very volatile*'.

#### Focus...



**Cryptocurrencies in everyday life.** The adoption of bitcoin as a means of payment in the real economy remains, despite a decade of development, a phenomenon whose scope remains marginal: according to the report published by ESRB (2025), transactions in *stablecoins* represented only 0.2 % of the total value of world e-commerce transactions in 2024, which illustrates the persistent gulf between the discourse on the payments revolution and the reality of everyday uses. The Parisian initiative of the '**Bitcoin boulevard**' constitutes in this respect an emblematic case of the limits of this adoption at the microeconomic level: in November 2016, at the initiative of the association Le Cercle du Coin, of the Maison du Bitcoin — today renamed Coinhouse — and of the payments processor BitPay, the passage du Grand-Cerf, situated in the second arrondissement of Paris, was inaugurated as the first 'Bitcoin boulevard' in France, bringing together some twenty merchants — jewellers, art galleries, fashion and decoration boutiques — accepting settlement in Bitcoin; as Le Monde (2020) relates, the surge and then the collapse of the price at the end of 2017, followed by years of bear market, progressively discouraged merchants from using this means of payment, weighed down by a volatility incompatible with the daily formation of prices, illustrating the theoretical tension between the function of **store of value** and the function of **medium of exchange** which Nakamoto (2008) nevertheless jointly assigned to bitcoin in his white paper. On the world scale, the attempts at adoption by large firms have followed a similar trajectory: Microsoft began to accept bitcoin in December 2014, via a partnership with the BitPay processor, for the purchase of content on the Windows Store and the Xbox platform, as BBC News (2014) confirmed as early as the date of the establishment of the arrangement; PayPal opened its crypto-asset purchase and sale services to its 300 million users from October 2021; the telecommunications operator AT&T has offered, since May 2019, the settlement of its bills in cryptocurrencies via BitPay. However, each of these initiatives remains confined to very specific uses: Microsoft limited the use of Bitcoin to the crediting of online accounts, Tesla suspended Bitcoin payments citing environmental concerns in May 2021, and the Visa and Mastercard networks in reality accept only *stablecoins* in their payment pilots, and not native Bitcoin. The ECB survey of November 2024 cited by Ledger (2025) reveals for its part that, while 9.7 % of European households declared that they held crypto-assets, their total holdings represented only 0.23 % of the financial assets of euro area households; in France specifically, only 6 % of households declared a holding, which, combined with the ESRB data on payment volumes, confirms that Bitcoin remains today more a **speculative investment asset** than an instrument of everyday payment: the properties of **divisibility** and **portability**, necessary conditions for any medium of exchange since the work of Jevons (1875), are certainly ensured technically by Bitcoin, but the volatility of its price neutralises in practice the **predictability of purchasing power** without which no rational economic agent could adopt it as a unit of account in his current transactions.

Moreover, can one say that cryptocurrencies likewise serve as a **unit of account**, the third Aristotelian function of money? At the present time, one can only answer in the negative, according to Odonnat (2021): indeed, cryptocurrencies do not have legal tender status in Europe, which means that a firm cannot for example keep its accounts in Bitcoin. And even if a firm accepts Bitcoin as a means of payment — which this time is perfectly authorised — its frame of reference for value always remains the euro: as Hossen & Gauthier (2024) recall, if on day  $j_0$ , Bitcoin is worth



30 000 euros a unit, a jeweller will sell a ring for 0.5 bitcoin; if on day  $j_1$ , Bitcoin is worth no more than 15 000 euros a unit, the jeweller will in all likelihood sell this same ring for one bitcoin. Thus, cryptocurrencies do not satisfy the three essential functions of money for Aristotle — medium of exchange, store of value, unit of account — the third characteristic falling short. This is why cryptocurrencies finally bear their name rather badly; the Banque de France therefore recommends rather the more rigorous use of the term ‘**crypto-assets**’, while the French tax authorities describe them as ‘digital assets’ in a more neutral manner.

To be sure, Alvarez et al. (2023) recall that, since June 2021, Bitcoin has had legal tender status in El Salvador: it is therefore henceforth a genuine official currency. Discussions, albeit abortive, were likewise entered into to make Bitcoin an official currency in Brazil in September 2021. Venezuela, under the impetus of N. Maduro, had also created in 2018 its own cryptocurrency indexed to the price of oil, the Petro — which was a stinging failure. But while it appears that Latin America presents itself *a priori* as fertile ground for the adoption of one or more cryptocurrencies as an official currency, it must be borne in mind that the official currencies in Latin America are of an ineffable fragility: inflation of 2700 % in Venezuela in 2021 or of more than 4 % in Brazil. This provokes a **dollarisation of the economy**, the dollar being judged more stable. Now, in Latin America, accepting the dollar as official currency, according to Hossen & Gauthier (2024), means bartering away one’s monetary sovereignty to Washington, which is politically unacceptable, the history between the United States and Latin America being difficult: numerous interferences during the Cold War, the overthrow of Allende by Pinochet with the blessing of the CIA in 1973 within the framework of Operation Condor... According to the analysis of Hossen & Gauthier (2024), between the American ‘devil’ embodied by the dollar and Bitcoin, neutral because decentralised, Latin American politicians save face by choosing Bitcoin. But, according to Hossen & Gauthier (2024), it seems improbable that the United States or the European Union, given the political weight of the Fed and the ECB, will accept sharing their monetary sovereignty with a libertarian competitor such as Bitcoin.

### Short-Answer Question



**Cryptocurrencies: institutionalisation or repression?** On 20 March 2023, the European Commission definitively adopted the MiCA regulation (*markets in crypto-assets*), the first harmonised framework at European Union scale for regulating crypto-assets, while the American SEC took the giants Coinbase and Binance to court for ‘sale of unregistered securities’, illustrating the tension between two divergent approaches to a market reaching 2500 billion dollars of capitalisation according to CoinMarketCap (2026). Cryptocurrencies designate digital assets issued on a *blockchain*, functioning according to a decentralised consensus (proof of work for Bitcoin, proof of stake for Ethereum 2.0), and whose value rests on programmed scarcity and the confidence of users rather than on a state guarantee. Institutionalisation refers to the progressive integration of cryptocurrencies into traditional financial systems via regulations, approved Bitcoin ETFs (January 2024 in the United States) and central bank digital currencies (CBDC), while repression designates the restrictive measures aiming to limit their use (prohibition of exchanges in China since 2021, OFAC sanctions against Tornado Cash in 2022). As Allnews (2023) analyses, this duality reflects a potential shift of control of the market from decentralised actors towards the giants of traditional finance, regulators oscillating between a desire to protect investors and a fear of loss of monetary sovereignty. Does the institutionalisation of cryptocurrencies, via frameworks such as MiCA or Bitcoin ETFs, prefigure their stable integration into the world economy, or does the persistent repression by states reflect a structural resistance to their disruptive potential? This question invites us first to examine the dynamics of institutionalisation which integrate cryptocurrencies into regulated financial markets, before analysing the logics of repression which aim to contain their systemic and geopolitical risks.

The institutionalisation of cryptocurrencies is observed in the adoption of structured regulatory frameworks and the massive entry of institutional actors, transforming a marginal speculative asset into a recognised component of diversified portfolios. The MiCA regulation, applicable since June 2024, frames crypto-asset service providers (formerly digital asset service providers, DASP) by requirements of capital (125 000 euros minimum), of cybersecurity and of anti-money laundering (AML), classifying *stablecoins* (USDT, USDC) into three categories according to their systemic risk and prohibiting unbacked algorithms such as TerraUSD, whose collapse in May 2022 had wiped out 40 billion dollars. As Cour des comptes (2023) emphasises in its report on crypto-assets, this regulation aims to reconcile innovation and investor protection, with 90 DASP/CASP registered with the French AMF between 2020 and September 2023. In parallel, institutionalisation is accelerating via derivative products: the approval by the SEC of eleven *spot* Bitcoin ETFs on 10 January 2024 (BlackRock’s *iShares Bitcoin Trust*, Fidelity’s *Wise Origin Bitcoin Fund*, ARK Invest’s *ARKB*) led to a total of 58.4 billion dollars of cumulative net inflows according to Journal du Coin (2025b). Central banks are likewise taking part in this normalisation: 130 countries are studying central bank digital currencies in 2026 (the Chinese e-CNY operational since 2022, the digital euro in the ECB pilot phase since November 2023), aiming to counter the competition of decentralised cryptocurrencies while retaining monetary



control. This trajectory, which Soana (2021) describes as ‘pragmatic regulation’, reflects a growing recognition of the technological maturity of *blockchains*, with Ethereum 2.0 (September 2022) reducing its energy consumption by 99.95 % and processing 2.4 million daily transactions according to Journal du Coin (2026).

This institutionalisation nevertheless remains conditional and confronted with logics of state repression aiming to contain the financial, criminal and geopolitical risks of cryptocurrencies. China, a pioneer in the total prohibition of exchanges and of mining since September 2021, justifies this repression by the loss of monetary control and by criminal use: according to Chainalysis (2025), illicit transactions in cryptocurrencies reached 40.9 billion dollars in 2024 (0.14 % of total volume). In the United States, Gary Gensler’s SEC has pursued a judicial offensive against 80 % of major cryptocurrencies described as ‘unregistered *securities*’, resulting in record fines (Binance: 4.3 billion dollars in November 2023; Coinbase: 100 million in 2024), while the American *Office of foreign asset control* (OFAC) sanctions transaction *mixers* (a source of anonymity on the *blockchain*) such as Tornado Cash (August 2022), the first *open-source* software targeted by sanctions. In Europe, MiCA is akin to a soft repression: significant *stablecoins* indexed to non-EU currencies — including Tether’s USDT (180 billion dollars of capitalisation in April 2026) — are subject to a ceiling of 200 million euros of daily transactions in the European Union, beyond which the issuer must suspend all new issuance, a deliberately low threshold in order to preserve liquidity in euros according to KuCoin (2026); in parallel, the ECB has supported since October 2025 a possible tightening aimed at banning non-compliant multi-issuance *stablecoins*, according to Journal du Coin (2025a). And yet, geopolitically, cryptocurrencies escape sanctions: according to Journal du Coin (2024), 20 billion dollars in USDT are said to have transited via the Russian platform Garantex since 2022, before OFAC (2022) and then the EU (sixteenth package of sanctions, February 2025) sanctioned it and Tether blocked its wallets; moreover, Cryptoast (2025) confirms that Russian oil companies use Bitcoin, Ethereum and USDT for their exchanges with China and India, the Russian oil market weighing 192 billion dollars in 2024. As Zola et al. (2024) establish in an analysis of all 43 entities entered on OFAC’s SDN list with Bitcoin addresses (February 2024), sanctions remain ineffective for roughly half of the entities targeted, which continue to carry out transactions via their sanctioned addresses, for a cumulative post-sanction volume of roughly 11.2 million dollars — violations linked to drug trafficking (eight million) and to cybercrime (three million) being the most resistant to sanctions.

Cryptocurrencies thus illustrate a dialectical tension between institutionalisation and repression which is resolved not into a binary alternative but into a progressive hybridisation. MiCA and Bitcoin ETFs mark an institutional recognition of their technological maturity and of their usefulness for the diversification of portfolios. However, this integration remains conditioned by a targeted repression of illicit uses and a monetary reassertion of control via CBDC, as Zola et al. (2024) document. The decisive shift towards institutionalisation will depend on the capacity of regulators to balance innovation and systemic stability: MiCA reduces the risks of fraud without stifling innovation, while Chinese repression displaced 70 % of the Bitcoin *hashrate* towards the United States without eradicating the phenomenon. The fundamental issue resides in monetary sovereignty: decentralised cryptocurrencies challenge the monopoly of central banks, forcing the latter to hybridise their systems (a piloted e-euro, an e-CNY with 300 billion yuan in circulation). Ultimately, it seems that institutionalisation will prevail in a regulated form, but at the price of an increased centralisation of *blockchains*, transforming the original decentralised promise into a hybrid financial infrastructure.

### Key Takeaways



Within the meaning of article L54-10-1 of the Monetary and Financial Code, a cryptocurrency is a digital representation of value neither issued nor guaranteed by a public authority, without legal tender status, but accepted as a means of exchange and electronically transferable. Confronted with the Aristotelian triptych, it fails the third test: while it partially fulfils the functions of medium of exchange and, for Bitcoin, of store of value, its structural volatility prevents it from serving as a unit of account — no firm can keep its accounts in Bitcoin, and any merchant who accepts it continues to set his prices in euros. This is why the Banque de France prefers the more rigorous term crypto-asset, while the French tax authorities opt for that of digital asset. On the geopolitical level, the adoption of Bitcoin as legal tender in El Salvador (2021) illustrates less a faith in the technology than a sovereignty trade-off: between dollarisation — which amounts to bartering away one’s monetary sovereignty to Washington — and a politically neutral decentralised asset, certain Latin American states choose Bitcoin in order to save face. It nevertheless remains improbable that the United States or the European Union, given the institutional weight of the Fed and the ECB, will one day accept sharing their monetary sovereignty with such a libertarian competitor.

### 2.1.2 The question of central bank digital currencies

A **central bank digital currency** (CBDC) is a digital form of fiat money (euro, dollar, yuan...) issued, controlled and regulated by the central bank of a country or of a monetary area. There exist two types of CBDC according to Hossen & Gauthier (2024): **retail CBDC**, intended for the general public, as a complement to current fiat money, and **wholesale CBDC**, accessible only to banks for the settlement and delivery of financial assets. On paper, as Ward & Rochemont (2019) analyse, CBDC make it possible to reduce intermediation costs, to facilitate and lower the costs of cross-border payments, and to fight money laundering and corruption through a perfect traceability of each transaction, down to the token. This is the whole problem of CBDC: at a time when one can pay instantaneously and free or almost free of charge (instant SEPA transfers, the Lydia or Revolut applications, PayPal between friends...), what would be the advantage of having recourse to CBDC?

CBDC can doubtless be of interest to banking establishments, with a view to reducing fees. By contrast, as Hossen & Gauthier (2024) emphasise, for private individuals, accepting CBDC means accepting an absolute 'policing' of every transaction, down to the token. On the central banks' side, the advantages are enormous: they will be able to deposit funds directly in consumers' digital wallets ('stimulus cheques' in a period of pandemic for example), or conversely to offer negative interest rates (concretely, to deduct interest from our digital wallets each month in order to encourage spending in the hope of stimulating the economy). Moreover, governments will be able to tax each transaction immediately, since they will directly have control of accounts via the holding of private keys (could one then move towards a deduction of VAT at source, like the deduction of income tax at source?). Hossen & Gauthier (2024) thus conclude that doing business outside government control will no longer be possible with CBDC: utopia or dystopia of a cashless society? In this context, Hossen & Gauthier (2024) doubt that Americans and Europeans will accept such control of their purchasing power.

#### Short-Answer Question



**Central bank digital currencies.** In November 2023, the European Central Bank (ECB) launched the preparation phase of the digital euro, while the People's Bank of China reached 300 billion yuan of e-CNY in circulation in 2025, attesting that 130 countries are actively exploring central bank digital currencies (CBDC), according to Bank for International Settlements (2025a). A central bank digital currency designates the digital form of central bank money, equivalent to banknotes and coins, issued by the central bank on a dematerialised medium (an individual account or a programmable *token*), founded on digital or *blockchain* technologies, and appearing on the liabilities side of its balance sheet. Unlike decentralised cryptocurrencies (Bitcoin, *stablecoins*), CBDC preserve state monetary sovereignty while responding to the digitisation of payments, where cashless transactions represent 92 % of volumes in Sweden in 2025. As De Sèze (2023) establishes in the ECB's prospective analysis, CBDC aim to preserve the role of central bank money as ultimate liquidity in the face of competition from *stablecoins* (Tether aggregating more than 180 billion dollars in circulation in April 2026) and of the bank disintermediation induced by *fintechs*. And yet, this project raises tensions between technological innovation, financial stability and monetary sovereignty, commercial banks fearing a 'digital *bank run*' while regulators anticipate gains in financial inclusion and in the efficiency of cross-border payments. To what extent do central bank digital currencies represent a natural evolution of central bank money adapted to the digital era, or an instrument of increased centralisation of monetary control? This question presupposes examining first the economic and technological motivations which justify their development, before analysing the systemic risks and institutional challenges that they pose.

The economic and technological motivations of CBDC belong to the resilience of the monetary system in the face of the accelerated digitisation of payments and the emergence of decentralised cryptocurrencies. Since the Covid-19 crisis, digital transactions have exploded: in France, contactless payments already represented 55 % of proximity card transactions in the first half of 2024, with a number of contactless mobile payments progressing by 61 % year on year, and cashless payments in the euro area reached 77.6 billion transactions in the second half of 2024, up 8.6 % year on year according to Banque de France (2025e). In parallel, the share of cash in retail transactions in the euro area went from 79 % in 2016 to 59 % in 2022, that is a fall of 20 points in six years according to the ECB's SPACE study cited by Format Research (2022), reflecting an accelerated dematerialisation of uses but still far from a cashless economy. Now, as De Sèze (2023) emphasises in his preparatory report on the digital euro, CBDC guarantee the physical presence of central bank money among households and firms, avoiding a scenario in which decentralised *stablecoins* (USDT, USDC) would replace the euro or the dollar as dominant unit of account. Technologically, CBDC exploit the programmability of *smart contracts* (like the Ethereum environment) for innovative uses: conditional payments (a wage paid only after performance), direct targeting of social benefits, or integration into regulated decentralised finance (DeFi) systems. China illustrates this maturity: the e-CNY, whose pilot began in 2019 in seventeen Chinese provinces, recorded 3.48 billion cumulative transactions for a total value of 16 700 billion yuan

(roughly 2300 billion dollars) up to November 2025, with 230 million wallets opened, according to MEXC (2025); on the cross-border side, the mBridge platform — co-developed with the BIS, Hong Kong, Thailand and the United Arab Emirates — has reached 55.5 billion dollars of cumulative payments since 2022, the e-CNY representing 95 % of this volume, in an explicit strategy of reducing dependence on SWIFT, as Reuters (2026a) relates. The Nordic central banks (Sweden and Norway) are exploring hybrid models combining distributed ledger technology (via a permissioned *blockchain*) and centralised infrastructures for a scalability of 100 000 transactions per second.

These technological and economic advances nevertheless come up against major systemic and institutional risks, justifying a prudent and differentiated implementation. First, the risk of a ‘digital *bank run*’: an attractive CBDC (zero fees, possible yield) could empty bank deposits. In the euro area, the proposed regulation on the digital euro provides for an individual holding ceiling of roughly 3000 euros — maintained by the ECB in the face of banking demands for a ceiling of 500 euros — in order to limit the risks of bank disintermediation, a maximum outstanding amount which would represent 1000 billion euros if all 340 million residents saturated their wallets, that is 25 % of retail deposits in the euro area, as BNP Paribas (2023) analyses. Access will be ensured by commercial banks and payment service providers, on whom the legislation imposes a legal obligation of distribution, as well as by post offices for unbanked persons, the ECB maintaining no direct relationship with private individuals, as Vie publique (2025b) explains. A second risk is cybersecurity: DDoS attacks against permissioned *blockchains* and the risks of *quantum computing* threaten the robustness of the systems. Third is the risk to privacy: an anonymous CBDC, like cash, would facilitate money laundering; ONUDC (2023) assesses world money laundering across all channels at between 800 billion and 2000 billion dollars annually, that is 2 % to 5 % of world GDP, crypto-assets representing only a fraction of this — 40.9 billion dollars in 2024 according to Chainalysis (2025). Conversely, traceable versions such as the e-CNY pose risks of mass surveillance: according to Le Rubicon (2023), the People’s Bank of China has put in place three dedicated centres (certification of identities, recording of transactions, big data analysis) making it possible to cross-reference all flows in real time, an architecture which had led the head of MI6 to describe the e-CNY as a ‘digital Trojan horse’ in December 2021. Finally, on the geopolitical level, CBDC directly challenge the dominance of the dollar via SWIFT — which handles 90 % of world cross-border payments through its 11 000 members in 200 countries and 10.6 billion messages in 2021 according to Le Monde (2022) — Russia and China having developed alternative systems (SPFS and CIPS) whose current marginality the Banque de France nevertheless notes: SPFS has only a limited number of members and CIPS represents barely 0.3 % of SWIFT’s activity, as Économie et politique (2022) explains, even though their announced merger and the mBridge project could in due course reshape world financial flows according to Fondation Robert Schuman (2025).

Central bank digital currencies represent an inescapable evolution of central bank money, justified by the necessity of preserving its sovereignty in the face of the digitisation of payments and of competition from private cryptocurrencies. CBDC guarantee universal access to central bank money (financial inclusion for 1.4 billion unbanked persons worldwide) while innovating via programmability. The Chinese e-CNY and the Nordic pilots demonstrate their technical feasibility. However, this transition remains conditioned by the resolution of systemic risks: a digital *bank run* limited by ceilings, quantum cybersecurity and a balance between privacy and surveillance. In the euro area, the digital euro (whose establishment is planned for 2029) will hybridise DLT and centralised systems for a scalability of 65 000 transactions per second, while preserving bank intermediation. *In fine*, as De Sèze (2023) theorises, success depends on a balance between *retail* accessibility (the majority of anticipated uses) and interbank *wholesale* use (the majority of the value generated), avoiding both disintermediation and financial exclusion.

### Key Takeaways



A central bank digital currency (CBDC) is the digital form of fiat money, issued and controlled by the central bank, which appears on the liabilities side of its balance sheet: it is fundamentally distinguished from decentralised cryptocurrencies by preserving state monetary sovereignty [De Sèze (2023)]. Its economic justification is defensive: in the face of disintermediation by *fintechs*, the progression of *stablecoins* and the retreat of cash (59 % of retail transactions in the euro area in 2022 against 79 % in 2016), central banks seek to maintain central bank money as ultimate liquidity for households. The programmability of *smart contracts* moreover opens up unprecedented uses — conditional payments, direct targeting of benefits — which the Chinese e-CNY (230 million wallets, 3.48 billion cumulative transactions) illustrates on a large scale [MEXC (2025)]. These advantages come up against two structural risks: on the one hand, the risk of a digital *bank run*, which the digital euro (planned for 2029) intends to contain by an individual holding ceiling of 3000 euros; on the other hand, the risk of mass surveillance, particularly documented in the architecture of the e-CNY, which illustrates the irreducible tension between anti-money-laundering traceability and the protection of privacy [Le Rubicon (2023)].

## 2.2 A few everyday applications of crypto-assets

Henry & Irrera (2017) recall that Jamie Dimon, the CEO of JP Morgan, had a famous word to discredit Bitcoin in September 2017: *'Bitcoin is a fraud. [...] It is impossible to create an industry where people invent a currency out of thin air and think that those who buy it are brilliant people'*. While this sentence caused a great stir, it testifies in any case to a certain ignorance — feigned on Jamie Dimon's part? — of the universe of cryptocurrencies. To be sure, as Hossen & Gauthier (2024) recall, 'real' firms such as Amazon rest on **tangible assets** (warehouses, delivery vehicles, servers...). But the valuation of a share likewise rests on **intangible** assets and elements (patents, expertise, reputation...). It is the combination of these two parameters which forms the value of an asset. Cryptocurrencies rest on an algorithm; it is an intangible asset to be sure, but a very real one. Now, without any tangible economic underlying, the price of cryptocurrencies then essentially models the **confidence** of investors in the algorithm; a versatile element, which explains the extreme volatility (that is to say the variability) of cryptocurrency prices. But this volatility and this immateriality do not for all that exclude the possibility of real applications in everyday life.

### 2.2.1 Towards a *blockchain* revolution?

According to Hossen & Gauthier (2024), in the **banking field**, the *blockchain* would offer the possibility of validating transactions without an intermediary, which limits transaction costs (Ripple); in the **field of insurance**, reimbursement procedures could be automated and formalities lightened, which would represent a gain in efficiency (why not imagine a smart contract which automatically compensates those who have suffered a flight delay?); the *blockchain* could also facilitate the monitoring of the **traceability** of a product within a production or distribution chain (VeChain); in the **field of property**, one could record notarial deeds on the inviolable register of the *blockchain*, which would be of interest in corrupt countries where private property is often flouted (Factom); one could likewise make available the computing power of one's computer when one is not using it in exchange for *tokens* (Golem) as one lets one's flat on Airbnb when one goes on holiday... While cryptocurrencies can therefore potentially revolutionise every aspect of our daily life, the fact remains that, according to Nakamoto (2008), they call into question the monopoly of established institutions as trusted third parties (banks, insurance companies, notaries...). And sometimes, these institutions put up resistance (*lobbying*) and prevent certain promising projects from emerging.

#### Focus...



**The question of smart grids.** What is one to think of a project which aims to implement **smart grids** in the energy field, that is to say concretely to be able to resell one's electricity to one's neighbours in exchange for *tokens*? Until February 2017 in France, as Hossen & Gauthier (2024) recall, one had imperatively to resell one's electricity to ERDF: even if it is my neighbour who needs the electricity from my solar panels, the electricity is sent back to an ERDF redistribution node before being redistributed to the network, which causes energy losses despite high voltage. This ecological issue was coupled with an economic issue: ERDF buys back electricity from private individuals for three or four cents per kilowatt-hour, while it resells it for between fifteen and twenty-five cents per kilowatt-hour. One could therefore imagine resales of electricity between private individuals at rates far more advantageous for all (eight or nine cents per kilowatt-hour for example), which is indeed the heart of the new '**industrial revolution**' for which Rifkin (2012) calls.

It is precisely for this reason that order no. 2016-1019 of 27 July 2016 relating to the self-consumption of electricity was adopted on 15 February 2017 within the framework of the law relating to the energy transition for green growth. The legislative provisions laid down by these texts have as their objective to facilitate the development of **self-consumption** and notably define the notion of collective self-consumption (CSC). As Ministère de la Transition écologique (2016) indicates, self-consumption designates the consumption of one's own production of electricity; associated with the notion of **self-production** (that is to say the production of one's own consumption), this practice is developing in a context in which the production costs of renewable electricity installations are decreasing and in which electricity prices are increasing. Self-consumption is **collective** when the producers or the final consumers are multiple: in that case they must group together within a legal entity (an association, a cooperative...) created for that purpose and, in order to ensure the character of proximity on the electricity network, the withdrawal and injection points must be situated downstream of the same medium-to-low voltage electricity transformation substation.

Order no. 2016-1019 moreover provides for the obligation on network operators to facilitate self-consumption operations (individual and collective), the establishment by the Commission de régulation de l'énergie of a **network usage tariff** adapted to self-consumption installations in order to take account of the reductions in network usage costs that these operations can bring, and the exemption, for small installations in self-consumption with injection of the surplus not benefiting from a purchase tariff on the electricity injected, from the obligation to be attached to a **balancing perimeter**, the surplus production being able to be allocated by default to the balancing perimeter of the

distribution network operators in respect of their network losses free of charge in order to facilitate the realisation of projects. Moreover, as EDF (2024) specifies, within the framework of the order of 6 October 2021 known as S21, the **surplus production** not allocated to the CSC operation may benefit from a purchase obligation contract: this arrangement allows eligible installations, holding an S21 purchase obligation contract for sale of surplus or sale in full, to sell to EDF Obligation d'Achat the electricity which has not been consumed by the participants in the CSC operation, at a price fixed by law and which may be guaranteed for twenty years.

Moreover, when in October 2021 Facebook changed its name to become Meta, a direct reference to the **metaverse**, this turn was extensively commented on in the press. Economically, it may be analysed in the light of the Schumpeterian theory of innovation: the heart of the dynamic of capitalism, innovation results for the economist Schumpeter (1934) from a dynamic of **creative destruction**; when clusters of innovations appear, the sectors which use them then have a comparative advantage over the historic firms which do not implement them; the latter are then downgraded and progressively disappear in favour of the most innovative firms. And while Facebook was still worth in 2021 nearly a thousand billion dollars, with 120 billion dollars of revenue in the year and gross margins of 81 % making it possible to generate a free cash flow for 2021 of 36.5 billion dollars according to Hossen & Gauthier (2024), this turn towards the metaverse, a universe totally new for Facebook — the leader of web 2.0 — illustrates well the fact that the historic giants are conscious of the potential of this new market and want to be a party to it, even when they are among the most prosperous in their sector of activity. The problem is that web 3.0 and even more so web 4.0 function more on decentralised logics which are the antipodes of the methods of Facebook, which made its fortune by collecting users' data — Facebook is thus a giant of *big data*. There is therefore a change of paradigm to be adopted if the firms of web 1.0 and web 2.0 are to succeed in the metaverse turn: whereas Facebook was contemplating until 2021 creating its own cryptocurrency — the Libra, subsequently renamed Diem and limited to simple *stablecoins*; this project having ended up being abandoned by Meta — the road is still long before the current giants of the web — the GAFAM (Google, Amazon, Facebook, Apple, Microsoft), to which X/Twitter should be added — avoid being overtaken by web 4.0 according to Hossen & Gauthier (2024).

More precisely, as Hossen & Gauthier (2024) explain, the **Libra** project, created in May 2019, renamed Diem in December 2020, then dropped by Facebook, which had become Meta, in January 2022, consisted in proposing several *stablecoins* — a diem euro aligned on the euro, a diem dollar aligned on the American dollar, and so on — collateralised with the help of a basket of currencies defined by Facebook so as to avoid Diem being too dependent on the price of a single fiat currency. Now, how is one to make sure that the collateralisation is complete, that is to say how is one to make sure that, behind a diem dollar, there is indeed a real American dollar in the coffers? Moreover, how is one to make sure that Facebook, whose financial model was built around the trade in users' data, would indeed respect the confidentiality of financial data when the firm would be subject to rigorous identity verification procedures (KYC)? *In fine*, Libra or Diem finally constituted a currency in its own right whose monetary authority would be ensured by the consortium of firms forming part of the Diem network (Facebook, PayU, Uber, Spotify, Shopify, Iliad, Coinbase...). Now, as the minister of the economy Bruno Le Maire had recalled in an interview in June 2019 cited by Hossen & Gauthier (2024), *'the capacity to issue securities, to constitute a reserve and to be a lender of last resort: all this is not conceivable [for private actors. Money should] remain in the hands of states and not of private firms which answer to private interests'*.

### Focus...



**From web 1.0 to web 4.0.** As Hossen & Gauthier (2024) explain, **web 1.0** is the name given to the most basic version of the internet, the one which appeared during the 1990s, made up of a network of websites formed of static web pages linked to one another by hyperlinks; web 1.0 therefore allowed users to read content online — this is the era of Google or Netscape —: web 1.0 therefore connects information; the verbs of web 1.0 are to **access** and to **find**. It is in the 2000s that the designation **web 2.0** emerges, which allows users themselves to publish content and internet users to connect with one another via social networks such as Facebook or Twitter, firms which likewise deploy mobile applications: as Hossen & Gauthier (2024) analyse, web 2.0 thus connects people; the verbs of web 2.0 are to **share** and to **collaborate**; one moves from the *world wide web* to the *social web*. The decade of the 2010s then marks the era of the *semantic web* or **web 3.0**; it is henceforth a matter of connecting knowledge with the help of artificial intelligence, of which voice recognition personal assistants such as Siri or Alexa are obvious examples; the verbs of web 3.0 are to **connect** and to **co-create** for Hossen & Gauthier (2024). The decade of the 2020s would finally mark a new frontier: the arrival of the metaverse, one of the keystones of the *intelligent web* or **web 4.0**, in which intelligences will be connected by notably creating decentralised communities or intelligent purchasing platforms — the idea would then be to attain an artificial intelligence which is truly at ease with natural language and which, when one asks a voice assistant for example to book a week's holiday in Naples, directly proposes a booking which

conforms to our tastes —; with web 4.0, one would therefore arrive at an ‘active’ web where it would be the network itself which would propose actions — the verb to be attained, still utopian in 2026 although the development of artificial intelligence is approaching it: to **reason**.

### 2.2.2 A few words on NFTs

A *non fungible token* (NFT) is a token issued on a *blockchain*. *Stricto sensu*, an NFT therefore corresponds precisely to a cryptocurrency. Nothing, technologically speaking at least, can therefore differentiate an NFT from a standard *token* such as a bitcoin for example, unless perhaps different validation procedures (*proof of work*, *proof of stake*, *proof of authority*) — which ultimately distinguishes each cryptocurrency from the others, independently of whether one is dealing with NFTs or not. The key to understanding the difference between a standard cryptocurrency such as Bitcoin or Ethereum and an NFT resides in the notion of uniqueness and of **fungibility**: as Hossen & Gauthier (2024) explain, whereas a standard *token* corresponds either to the reward of validators — for public *blockchains* — or to a pseudo-valuation of the firm which issues it — for a private *blockchain* — an NFT is a kind of unique and non-fungible collectible token. What does this last term mean? If one looks at a transaction in bitcoins, in order to settle for an electric car against one bitcoin, the token one uses is a matter of indifference, in the same way that if one buys a croissant for one euro, it makes no difference whether one pays with a French one-euro coin (whose face represents a tree) or with a German one-euro coin (whose face represents the Bundestag). The token in itself is therefore not linked to a unique use or product. With bitcoin, it is the same: the tokens are exchangeable and perfectly equivalent. One sees it too with banknotes in everyday life: a crumpled five-euro note makes it possible to buy exactly the same thing as a brand-new five-euro note; the idea of the fungibility of money thus suggests this notion of possible replacement, more precisely of interchangeability. By contrast, with non-fungible products, the interchangeable aspect disappears. **Non-fungibility** is a criterion which already exists in everyday life: when one buys a gift card in a specific shop, or when one receives at the supermarket a voucher for a specific product, one cannot exchange this card in order to buy a product in another shop and one cannot acquire another good — even of equivalent or lower value than the voucher — in the supermarket which issued the voucher. Let us then comment more specifically on the non-fungible aspect of NFTs. The idea of an NFT is to be a unique token: one may liken it, as a first approximation, to a cryptocurrency having only a single token — unlike Bitcoin for which up to 21 million *tokens* will be able to be mined — this token being moreover indivisible: while one can exchange down to a hundred-millionth of a bitcoin — what is called a ‘satoshi’ in homage to the anonymous inventor of the Bitcoin blockchain — it is impossible to exchange half an NFT for example. The fact that the total money supply of an NFT is equal to one unit, the latter being indivisible, is obviously specified in the code which defines the NFT.

#### Focus...



**NFTs and digital art.** The fact that the token is unique then makes it possible to understand why an NFT may be apprehended as a **collectible token**, and this makes it possible to understand the affinities between art and NFTs which Hossen & Gauthier (2024) bring out. To hold an NFT is to hold a unique product whose ownership is inviolable since it is entered in the immutable register of a *blockchain*. An NFT is thus akin to a kind of trading card unique of its kind and whose particularity is to be displayed in an open showcase that anyone can admire — provided that the *blockchain* which issues the NFT is public. To hold an NFT is therefore as if one held a work in a museum, which everyone can admire and even photograph, but of which only the holder referenced on the *blockchain* will be able to boast the ownership. One understands here why the NFT is naturally linked to digital art: *de facto*, a digital product is neither more nor less than a sequence of 0s and 1s, and it is easily copyable if it is an image or a video — one need only observe the proliferation of *streaming* sites which broadcast digital content without authorisation. But if one backs a work of digital art with an NFT, that is to say if one encodes the NFT to say ‘this *token* attributes ownership of the digital work XXX’, then one can let anyone admire the work, and one will be the only one to be its true owner. Obviously, one also understands the two structural limits of an NFT: on the one hand, it cannot solve the problems that rights holders face since it cannot prevent others from admiring the work on which the NFT is backed — which is what film producers in particular seek — but it can in any case resolve **conflicts of ownership rights**, by proposing, through the register of the *blockchain*, an inviolable history which, after public exploration, makes it possible to determine univocally the person who can boast of being the inventor or the owner of the work concerned. On the other hand, the NFT does not make it possible to **fight copying**: a work issued as an NFT can be copied and pasted or downloaded onto one’s computer, as one can hold at home a copy of the Mona Lisa or as one can take a photograph of the Rosetta Stone at the British Museum in London. But nobody would defend the idea that a photograph of the Rosetta Stone is equivalent to the original in the museum. The problem is quite different with NFTs: if an image or



a film is a sequence of zeros and ones, it is impossible to differentiate between the original and the copy on one's computer; it is rigorously the same file. But the NFT does in any case fill a gap by identifying both the creator of the work and its current owner. In a certain way, an NFT is the analogue of a **certificate of authenticity**: it guarantees, in an inviolable and univocal manner — since it benefits from the same technological advantages as any *blockchain* — that such and such a digital work was created by such and such an individual and is currently held by such and such another. In practical terms, the *blockchain* then makes it possible to identify univocally the current owner of the work, the only person legitimately entitled to transfer his ownership right, that is to say concretely to sell his NFT. The NFT market, while it does not make it possible to limit access to a work — it would then be rather on the side of private *blockchains* with their permissions that a solution would have to be contemplated — then makes it possible to renew the art market by avoiding both the risks of counterfeiting — since only the real work is backed by an NFT — and the risks of usurpation — since, on a *blockchain*, only the holder of its private key, therefore concretely only the legitimate owner of the NFT, has the power to transfer his tokens, as Hossen & Gauthier (2024) detail.

But the problem which arises today in art is not solely in the domain of ownership. What is one to think of **serial art**, whether limited series such as Andy Warhol's Marilyn or unlimited series as for certain works by Jeff Koons? How is one to restore the unity of the multiple in a contemporary art which today dilutes the unique flash of genius in a proliferation of identical or almost identical works? Here again, NFTs can provide an answer according to Hossen & Gauthier (2024). If the NFT is *a priori* a unique and indivisible token, how is one to conceive it as being backed by multiple works whose owners will in all likelihood be different? The idea is then to create **numbered tokens**: for a series of works, one will create a cryptocurrency of which several indivisible units will be issued, and these tokens will still be non-fungible, since the work numbered 001 will not be exactly the work numbered 003. Thus, the NFT, more than a simple certificate of authenticity, becomes the certificate of authenticity of a potentially numbered work. An aspect which already exists in everyday life: the first edition of Baudelaire's *Les Fleurs du Mal* was issued in very few numbered copies, which today fetch a fortune. Where NFTs even have a comparative advantage over numbered series is that there is no risk of falsification thanks to NFTs, since each numbered work is referenced with the same rigour on the *blockchain*. Naturally, a basic economic argument tells us that, the more editions a series comprises, the less rare it becomes and the less important the price of each work individually will be, all other things being equal. NFTs do not escape this logic, which is moreover splendidly apparent with '**sports NFTs**'. Since 2021, the latter have indeed proliferated, like the Panini trading cards of yesteryear, which arouses an enthusiasm among fans who fight over them: the French platform **Sorare** has thus renewed the field of trading cards, by proposing NFTs each of which represents a football player, and even raised 680 million dollars of funds at the end of 2021, as Hossen & Gauthier (2024) recall. In a certain way, trading cards likewise form a kind of art, whose marketing dimension is, in this field of sport, obviously omnipresent: taking advantage of the digitisation of sport — with for example Messi who in 2024 received part of his remuneration in cryptocurrencies according to Hossen & Gauthier (2024) — NFT projects around sport seek new sources of revenue around the globe. Obviously, the idea of selling collectible tokens makes it possible to avoid any problem of stock as with shirts for example — it suffices to issue more tokens, as certain Panini cards were made less rare by issuing more copies of them; the product will then become less rare and therefore potentially less expensive, but the irrational enthusiasm of fans will surely compensate. ... For example, the most fashionable players such as Messi, Ronaldo or Lewandowski see certain versions of their card issued in fewer than ten copies, or even in a single copy, on Sorare, and display extravagant prices — 45 000 euros for the unique-rarity Messi card (a single copy of the card) in 2024 for example — while other second-division players see their card issued in more than ten thousand copies, displaying a unit price of a few euros. But if an originally unknown player begins to excel, each of the cards representing him will arouse the interest of fans, and it is probable that the unit price of the card will soar. Be that as it may, NFTs bring out a genuine change of paradigm in the field of art — art in the broad sense if one includes the trading cards of sports NFTs. Indeed, NFTs are neither more nor less than kinds of certificates of authenticity which attest to the ownership of a digital work. Previously, the certificate of authenticity without the work on which it was backed was worth nothing: holding a certificate proving that a diamond is real without holding the corresponding diamond is useless. But with digital art, possession loses its material dimension according to Hossen & Gauthier (2024). What makes the unity and the uniqueness of the work is no longer the pixels which form it, but its certificate of authenticity, in other words the NFT itself. This logic of abstraction finds its culmination in the fact that it is the NFT itself which becomes the work of art, independently of the very reality of that work. An obvious example of this change of paradigm analysed by Hossen & Gauthier (2024) is the work — physical, tangible — *Morons* by Banksy (drawn by the *street artist* in 2006), of a value of 95 000 dollars, which was burnt by Injective Protocol in March 2021 in order to transform this work into a digital work thanks to NFTs. The tokenised work was then resold a few days later on the specialised platform OpenSea for 228.69 ETH, that is 380 000 dollars at the price of the time, as Hossen & Gauthier (2024) recall. A genuine movement of art towards a dematerialisation as we mentioned in the introduction, or a passing speculative movement? *Morons* shows in any



case the new porosity between physical art and digital art. Naturally, Banksy's choice is not innocent: an emblematic figure of anti-capitalist criticism, the *street artist* inscribes *Morons* in his line of protest works which castigate the irrational enthusiasm for art. Cynically, the work *Morons* is moreover the *mise en abyme* of an auction sale of a white canvas bearing only the following inscription: '*I can't believe you morons actually buy this shit*'.

Non-fungible *tokens* have experienced a genuine boom: between 23 June 2017 and 27 April 2021, 4.7 million NFTs were sold and resold on the Ethereum and WAX *blockchains*, totalling 6.1 million exchanges over the period according to Nadini et al. (2021). The enthusiasm for these digital assets at the crossroads of computing, art, sport, finance and the metaverse is, however, particularly recent: while the capitalisation of the NFT market in 2018 — the capitalisation of the NFT market being defined by the sum of the values of each NFT, these values being themselves estimated by CoinMarketCap on the basis of average sale prices — was close to 41 million dollars, this figure leapt to 142 million dollars at the end of 2019 then to 338 million dollars at the end of 2020 to reach 44.2 billion dollars at the end of 2021; as early as July 2021, the value of all the NFT sales carried out had moreover already exceeded 2.5 billion dollars over the year 2021, as Hossen & Gauthier (2024) explain. Hossen & Gauthier (2024) first of all distinguish four large domains of NFTs: those linked to art, those relating to video games, those relating to the metaverse and those linked to sport. For the latter, one could even be more general inasmuch as they are trading cards: this is why one could describe them as genuine collectible tokens (*collectible*). Taking up the segmentation carried out by Nadini et al. (2021), one could add to this list a more marginal category, that of *utility tokens*, that is to say utility tokens whose main function is to permit future access to a product or a service offered by a firm. Thus, Unstoppable Domains makes it possible to change the name of an internet domain to .crypto, thus transforming the website itself into an NFT, secured by the *blockchain*. Naturally, the segmentation of the NFT market into five categories — art, *collectible*, video games, metaverse and *utility* — to take up Nadini et al. (2021) may be discussed, all the more so as certain examples straddle two categories — Sorare is for example at the crossroads of the *collectible* and video game categories; the case of CryptoKitties may be considered both as art and as collectible. ... —; one may thus find other proposals for the segmentation of the market. More precisely, as Hossen & Gauthier (2024) detail, CryptoKitties are a series of male or female 'digital cats' issued by the company Dapper Labs, sold from 2017 onwards. Although having a childish design which casts doubt on their artistic dimension, CryptoKitties have seen their value explode: the CryptoKitty Genesis was thus sold for 117 000 dollars in December 2017; a work of art like Warhol's series (there may be up to 50 000 'first generation' CryptoKitties), a simple collectible token or even a video game (inasmuch as one takes care of these virtual cats and can even make them reproduce with one another)? Since then, other collections have emerged, between *collectible* and works of art, the best known being surely Bored Apes Yacht Club (BAYC) — ten thousand anthropomorphic apes, whose face almost recalls that of a drugged individual, declined in thousands of different outfits, issued by the company Yuga Labs — and CryptoPunks — ten thousand pixellated images of a punk man, generated automatically by an algorithm and issued by the studio Larva Labs. As Hossen & Gauthier (2024) explain, sold from April 2021 onwards, BAYC NFTs generated in barely ten months two billion dollars of sales, the most expensive of these apes even having been sold for 3.4 million dollars in October 2021. With an entry price of more than 70 ETH, that is more than 220 000 dollars at the Ethereum price of 27 March 2022, and a sale record in February 2022 of 8000 ETH, that is 23.7 million dollars at the time, CryptoPunks have likewise generated two billion dollars of sales since their creation in June 2017 according to Hossen & Gauthier (2024).

As Nadini et al. (2021) show, it is truly the arrival of CryptoKitties in 2017 which placed the world of NFTs in the limelight. The NFT market was then totally swamped by these CryptoKitties, whence the hegemony of the art category — in which CryptoKitties are classified for Nadini et al. (2021) — in the volume of sales linked to the NFT market. After this dazzling boom of 2017, the NFT market remained rather stable until mid-2020, concentrating 60 000 dollars of daily exchanges on average, even if other categories began to gain in popularity: between January 2019 and July 2020, roughly 90 % of the total volume traded on the NFT market was thus shared between the art, video game and metaverse categories, these three categories contributing respectively 18 %, 33 % and 39 % of the total. It is truly only in 2021 that the boom of the NFT market took place, even exceeding ten million dollars of daily exchanges from March 2021 onwards according to Nadini et al. (2021). Art then recovers a dominant position in the NFT market: from mid-July 2020 onwards, NFTs categorised as art thus contributed roughly 71 % of the total volume of transactions, followed by collectible assets — such as Sorare — representing 12 % of the total. The composition of the NFT market nevertheless turns out to be different if one focuses not on the volume — that is to say the sum of the amounts in dollars — of the transactions carried out but on their number: since July 2020, the most traded NFTs belong to the game and *collectible* categories, which represent respectively 44 % and 38 % of transactions according to Nadini et al. (2021); by contrast, only 10 % of transactions concern NFTs classified in the art category. On the whole, the share of the volume of NFT transactions linked to art has increased since 2020, while the share of the amount of transactions linked to art has decreased. This observation is fundamental for the investor: whereas

certain types of NFT are particularly liquid, and therefore traded very often, others, traded a restricted number of times, must rather be perceived as a store of value. NFTs linked to art, and to a lesser extent those linked to the metaverse and to *collectibles*, must in general be perceived rather as stores of value, which the investor keeps for a long time, while NFTs linked to video games are more ‘utilitarian’ — that is to say here linked to use in the game — and therefore associated rather with small short-term profits; the tokens of the *utility* category, for their part, are to be read rather as a niche market: if one takes up the example of Unstoppable Domains, the world of the sale of websites is a complex and illiquid market, displaying generally high prices — as is customary in niche markets. The study of NFT prices moreover seems to reinforce this intuition. Thus, in the database of Nadini et al. (2021), while the average sale price of NFTs is below 15 dollars for 75 % of assets (third quartile) and above 1594 dollars for 1 % of assets (ninety-ninth percentile), the picture is heterogeneous according to category: the 1 % most expensive assets of the art, metaverse and *utility* categories thus traded at an average sale price above 6290, 9485 and 12 756 dollars respectively. Another way of confirming the dichotomy between store of value — a domain rather captured by NFTs linked to art, to collectible tokens, to the metaverse — and the more utilitarian tokens perceived possibly as a small source of short-term profits — a category rather organised around NFTs linked to video games — is to measure the frequency with which assets are traded on the market: whereas a **store of value** is generally kept for a long time in an individual’s portfolio, more utilitarian assets are generally resold rapidly as soon as it is possible to derive a profit from them. In the database of Nadini et al. (2021), only 20 % of NFTs have been the object of a resale: this means that, for four NFTs out of five, the asset is still in the hands of the first buyer, who has not yet resold it. Now it seems legitimate to think that, if an investor has still not resold his asset, it is because he thinks its value is going to grow, thus likening the product to a store of value. And while only 0.07 % of the 4.7 million NFTs of the database — that is to say fewer than four thousand — have been resold more than ten times, no token of the *utility* category has been resold more than twelve times, while one counts several NFTs linked to video games which have been resold more than a thousand times. There therefore exists a **trade-off between liquidity and store of value** which the investor must always bear in mind when acquiring an NFT: the less expensive an NFT, the more easily it trades on the market and the more it offers the possibility of small short-term profits; conversely, the higher its price, the more it becomes linked to a niche market — more difficult to capture, by the simple fact of the financial barrier to entry alone — and the more its value will tend to remain stable or to climb slowly. Obviously, this is not an absolute observation: although CryptoKitties traded at astronomical prices at their creation, most of these digital cats are today sold at a loss.

### Focus...



**The collapse of the NFT bubble.** According to Furet (2024), a fall of 37 % in the total number of NFT transactions was observed between January 2022 and January 2023, while 30 % of NFT collections already displayed in 2024 a unit value below one dollar. The triggering factors of the bursting of this **speculative bubble** are multiple and interdependent according to Hossen & Gauthier (2024): the world monetary tightening of 2022, which made risky assets structurally less attractive for fixed-income investors, the mechanical correlation with the collapse of the prices of the underlying cryptocurrencies — Ethereum, the main hosting chain of NFTs, having lost more than 63 % of its value between November 2021 and June 2022 —, and finally the bankruptcy of FTX in November 2022 which brutally drained the liquidity of the secondary market. In November 2025, monthly NFT sales amounted to no more than 320 million dollars — that is a fall of 50 % relative to October 2025 — and sectoral capitalisation had fallen by 66 % over the year as a whole, going from 9.2 to 3.1 billion dollars according to Phemex (2025). This return to reality illustrates the fundamental economic principle according to which an asset whose price rests exclusively on the anticipation of resale at a higher price — without any flow of revenue or stable use value — satisfies the classic definition of a **greater fool theory** according to Hossen & Gauthier (2024): its sustainability depends on the continuous influx of new buyers, and its collapse is inescapable as soon as this influx dries up.

### Key Takeaways



An NFT (*non fungible token*) is a unique and indivisible token: unlike Bitcoin, all of whose units are interchangeable (fungible), each NFT constitutes an inviolable digital certificate of authenticity which identifies univocally the creator and the owner of a work, without for all that preventing the copying of the underlying file [Hossen & Gauthier (2024)]. The NFT market, which reached 44.2 billion dollars of capitalisation at the end of 2021, then underwent a brutal collapse — 3.1 billion at the end of 2025 — illustrating the **greater fool theory**: an asset whose value rests exclusively on the anticipation of resale at an increasing price collapses inescapably as soon as the influx of new buyers dries up [Phemex (2025)].

### 2.2.3 Cryptocurrency fundraising: ICO and STO

It happens that an individual or a company wishes to launch itself on the *blockchain*: it is then a matter of creating a new cryptocurrency. As at the moment of a company's stock market flotation, this creation of new tokens is generally associated with a fundraising which permits the financing of the project: *initial coin offerings (ICO)* — by analogy with *initial public offerings (IPO)* on the stock exchange. As Hossen & Gauthier (2024) explain, at the time of an ICO, the developers of a new *blockchain* solution progressively put tokens up for sale — which will be tradable on a secondary market (exchange platforms) some weeks, months or years later, as soon as the *blockchain* solution has been entirely put in place — at a promotional price — that is to say at a price below that which will be offered at the very moment when the solution is launched (even if nothing guarantees that the price of the *tokens* on sale will not immediately collapse thereafter) — which allows them to raise funds without going through other alternatives such as **crowdfunding**, that is to say the participatory financing which consists in collecting financial contributions from a large number of private investors, individuals or firms, by means of a dedicated internet platform. Toledano (2018), the chair of the working group on the *blockchain* of France Stratégie, specifies that ICO serve to finance projects concerning the improvement of infrastructures and *start-ups*, notably in the media, gaming and internet of things sectors. The ICO mode of financing has experienced a boom since 2017: as AMF (2018b) recalls, it was 1.6 % of equity financing in the world which was carried out through ICO in 2017, and eighteen ICO took place in France between 2017 and 2019; more than thirteen billion dollars were raised through ICO in the world in 2018. By way of comparison, in 2017, Kharif (2017) thus assessed the amount of ICO at between four and eight billion dollars, whereas the sum of traditional fundraisings from venture capital was of the order of a billion dollars.

#### Focus...



**Major projects financed via ICO.** As Hossen & Gauthier (2024) emphasise, Brave, a web browser created by the co-founder of Mozilla Firefox, raised through an ICO in 2017 in barely thirty seconds thirty-five million dollars; the firm Telegram, for its part, had raised 1.7 billion dollars at its ICO in 2018 — which moreover ended in failure and the abandonment of the Gram cryptocurrency project in 2020; investors therefore had to be reimbursed (losing 28 % of the amount invested for those who wanted to be reimbursed in May 2020; those who agreed to wait until the end of April 2021 by contrast received a compensation of 10 % of the amount invested in addition).

ICO are indeed an innovative mode of financing that is particularly effective for firms. Whereas the means available to a firm for raising funds are fairly limited — a direct or bond loan without an equity stake, a very illiquid venture capital investment in the initial phase, or a liquid investment in a second phase, generally after a stock market flotation — ICO permit a flexible transition between the three options mentioned above: the system of fundraising through the **progressive issuance of tokens** indeed permits the progressive establishment of a market for trading securities whose degree of liquidity grows as a function of the success of the firm. As Fisch et al. (2021) indicate, such a mode of financing then presents numerous advantages: besides the contribution of capital, the firm can play both on the liquidity and on the price of the *tokens* in order to reward its first investors and attract new ones; moreover, the firm retains a certain **control of its capital** with the possibility of buying back the tokens and reimbursing investors; finally, as a market operator, the firm directly acquires useful information about its financing conditions — which then allow it to assess its market value and to optimise its pace of growth as well as possible. The ICO mechanism likewise presents advantages for investors, notably in venture capital: as Hossen & Gauthier (2024) mention, the limitation of the number of tokens put up for sale acts as a signal and a guarantee of **non-dilution of capital**; ICO are indeed progressive — one begins by putting few tokens into circulation at a very preferential price, then the firm puts more and more of them up for sale (even if the quantity remains limited) at higher and higher prices.

#### Focus...



**From the AMF's optional visa to the regulation of ICO by the MiCA regulation.** Outside any pre-existing regulatory constraint, **initial coin offerings (ICO)** developed between 2016 and 2018 in an almost total legal vacuum: not being assimilable to financial instruments, **utility tokens** escaped the corpus of classic financial law, leaving investors without minimum protection in the face of fundraisings whose promoters could disappear overnight. France was the first country in the world to remedy this by an explicit legislative arrangement: article 85 of the **loi PACTE** (law no. 2019-486 of 22 May 2019) introduced into the Monetary and Financial Code articles L.552-1 to L.552-7, instituting an **optional visa** from AMF (2019b) for issuers of tokens wishing to proceed to a public offering. The optional character of the arrangement is deliberate: as the parliamentary report of Woerth (2021) notes, the legislature wished to avoid the systematic obligation of a visa giving project owners an incentive to relocate their operations to less demanding jurisdictions, notably Liechtenstein, Malta or Singapore, preferring to attract quality

ICO onto French soil rather than to drive away the whole market. The conditions for the award of the visa, specified by article L.552-5 of the Monetary and Financial Code and AMF instruction DOC-2019-06, are four in number: the issuer must be constituted in the form of a legal person established or registered in France; it must produce an information document (*white paper*) drawn up in conformity with the general regulation of the AMF and accompanied by a summary in French if it is drafted in a foreign language; it must put in place an **arrangement for monitoring and safeguarding the assets collected**; finally, it must demonstrate the existence of mechanisms making it possible to identify and to fight money laundering and the financing of terrorism (**LCB-FT**). The AMF has a period of twenty working days in which to decide, from the filing of the complete dossier. As Hossen & Gauthier (2024) explain, the legal scope of the visa must be precisely delimited: it does not constitute an approval of the economic merits of the project, nor an authentication of the financial or technical elements of the *white paper*, and does not amount to a verification of the underlying computer code; it signifies only that the issuer satisfies the minimum legal guarantees and that the information document is complete and comprehensible. Its practical value lies in its **canvassing effect**: only ICO having received the visa of the AMF may legally be the object of active solicitation of the public in France, which confers on the visa a significant commercial value for any project targeting a retail investor base. The first visa was issued on 18 December 2019 to the company French-ICO, whose fundraising target was set at one million euros, according to the official communiqué of AMF (2019a).

This national framework has since been superseded: order no. 2024-936 of 15 October 2024 transposing the European **MiCA** regulation repealed articles L.552-4 to L.552-7 of the Monetary and Financial Code, substituting for the national optional visa a harmonised European regime which is this time compulsory for offers of crypto-assets exceeding certain thresholds. More precisely, as regards ICO *stricto sensu* — that is to say public offers of *utility tokens*, to the exclusion of *asset-referenced tokens* and *e-money tokens* which come under distinct titles of the regulation — title II of the MiCA regulation retains, as ORWL (2025) analyses, three categories of exemption which delimit the field of application: private offers, addressed to fewer than 150 investors per member state; offers whose total consideration does not exceed one million euros over twelve months at European Union scale; and tokens offered free of charge (within the framework of operations known as *airdrops*). For ICO falling within the scope of MiCA, the issuer — which must imperatively be a legal person established in the European Union, in conformity with article 4 of the regulation — is required to notify in advance to the competent national authority a standardised white paper whose content is defined by ESMA's regulatory technical standards, to respect a **fourteen-day right of withdrawal** for the benefit of purchasers, and not to exceed certain collection thresholds without additional authorisation. The main procedural innovation relative to the AMF visa lies in the **European passport mechanism**: an ICO having notified its *white paper* to a national authority — the AMF for France — may legally be offered throughout all the member states without additional formality, which removes the previous regulatory fragmentation.

ICO thus make it possible to collect funds in a fairly deregulated environment. They must not be confused with *security token offerings* (**STO**) which are launched with a view to regulatory governance; registered with governmental bodies almost with the same rigour as fundraisings by capital increase in listed companies, STO certainly put *tokens* up for sale, like ICO, but the comparison stops here, according to Hossen & Gauthier (2024): with STO, investors acquire through the *tokens* bought a genuine **financial security**, which is akin to the holding of shares in the firm which issues them. Such offers are totally regulated and far more secure than ICO, but are very often reserved for experienced investors alone who can mobilise substantial sums of money — given the financial and time efforts mobilised by firms in order to carry out STO successfully and to obtain the agreement of all the regulatory bodies, one easily understands that the firms which implement them expect a generous contribution from their investors. And while one might think that the **Autorité des normes comptables** (ANC, the French accounting standards authority) does not explicitly distinguish between ICO and STO *a priori*, regulation no. 2018-07 of 10 December 2018 of that body is nevertheless clear: article 619-2 indeed distinguishes between '*on the one hand, offers of tokens presenting the characteristics of financial securities or financial contracts [...] or those of cash bonds*', that is to say precisely STO relating to **security tokens** and accounted for in the same way as '*financial securities, forward financial instruments or cash bonds, according to the nature and the specific characteristics of the corresponding instruments*' (article 619-3), and '*on the other hand, offers of tokens not presenting the characteristics of financial securities, financial contracts or cash bonds*', that is to say ICO in the proper sense, generally associated with **utility tokens** — that is to say tokens associated with a future service, which will therefore be rendered only when the firm has finished developing the solution for which it is precisely raising funds.



### Key Takeaways

ICO (*initial coin offerings*) are fundraisings carried out by the progressive issuance of tokens tradable subsequently, which finance *blockchain* projects without going through the traditional circuits of venture capital or *crowdfunding*, at the price of a long minimal framework [AMF (2018b)]. The MiCA regulation (applicable since 2024, transposed into French law by the order of 15 October 2024) has superseded the AMF's optional visa arising from the loi PACTE: for ICO bearing on *utility tokens* exceeding one million euros of collection or addressed to more than 150 investors per member state, the issuer — necessarily a legal person established in the EU — is henceforth required to notify a standardised *white paper* to the competent national authority and to respect a fourteen-day right of withdrawal for the benefit of purchasers [ORWL (2025)]. Moreover, ICO are distinguished from STO (*security token offerings*), where the tokens issued have the nature of fully regulated *financial securities* (within the meaning of ANC regulation 2018-07 and of MiCA), giving their holders genuine proprietary rights assimilable to shares in the issuer.

## 2.3 Towards the advent of decentralised finance (DeFi)

**Decentralised finance** (DeFi) aims to offer a 3.0 analogue of traditional finance thanks to *blockchain* technology: payment and lending systems, verification of contracts without having recourse to a traditional institution, cryptocurrency exchange platforms...

### 2.3.1 Ethereum and ERC-20 tokens

As we have seen, the Ethereum blockchain permits the implementation of decentralised finance with the help of **smart contracts**. These contracts are therefore specific and bear on a precise object: verifying a flight delay in order possibly to pay compensation, certifying a vote at an election, ensuring the traceability of a dairy product... Obviously, the transactions associated with these smart contracts could be carried out directly in ethers. But there is here an economic risk which Hossen & Gauthier (2024) bring out: let us imagine that the smart contract associated with compensation following a flight delay is badly encoded and ends up being hacked; if it were remunerated in ethers, this would affect by ricochet all the other smart contracts which are also remunerated in ethers by undermining the confidence placed in them. More precisely, the reasoning, detailed by Hossen & Gauthier (2024), is the following: let us imagine that an insurance company specialising in travel indemnifies its policyholders by 0.5 ETH when the associated flight (outward or return) is more than three hours late at destination. This indemnification obviously rests on a smart contract (if the plane is late, then one indemnifies). Unfortunately, this *smart contract* is badly coded and results in errors, and one never receives the 0.5 ETH promised. Psychologically, it is probable according to Hossen & Gauthier (2024) that the injured parties will conclude that Ethereum is not a reliable solution, whereas the fault must nevertheless be sought on the side of the encoder of the *smart contract*; the price of Ethereum will then in all likelihood suffer a negative externality due to the distrust with regard to the said smart contract. It is the same psychological mechanism which intervenes when it is announced that a centralised exchange platform has been hacked. Thus, while the Bitcoin *blockchain* has never been violated in its integrity, certain bitcoin exchange platforms have already been hacked, as Hossen & Gauthier (2024) recall: the theft of nearly 20 000 bitcoins (that is five million dollars at the time) in 2015 on Bitstamp (a Slovenian platform based in Luxembourg created in 2011), of nearly 120 000 bitcoins (72 million dollars at the time) in 2016 on Bitfinex (a Hong Kong platform created in 2012), not to mention the canonical example of the Mt. Gox hacking detailed *supra*. Whether one likes it or not, the announcement of a platform hacking undermines confidence in the products whose exchange it permits: this is a psychological bias which Hossen & Gauthier (2024) note.

This is why smart contracts generally have their own token based on the *template* of the Ethereum code: these are **ERC-20** tokens (twentieth *Ethereum request for comment*). Ethereum thus offers a whole ecosystem to developers of smart contracts, the latter benefiting from the security of the Ethereum *blockchain* and from the robustness of the Solidity language in order to execute their *smart contracts*. ERC-20 tokens thus make it possible to have a kind of 'parallel accounting' on the Ethereum network, which offers better legibility and visibility for each smart contract project. Indeed, the history of transactions for each ERC-20 token can be followed in complete transparency on the site <https://etherscan.io>. It is this legibility which makes the ERC-20 standard a genuine norm: uniformity of code, reduction of programming complexity, great security and above all **interoperability** (that is to say structural compatibility between the different ERC-20 tokens). This last point is fundamental since all ERC-20 tokens, in that they are encoded like Ethereum, can therefore be validated by Ethereum miners, as Hossen & Gauthier (2024) emphasise: someone who launches a smart contract through an ERC-20 token can therefore immediately benefit from the strength of the Ethereum network without having to build it *ex nihilo*. This is why speaking of 'cryptocurrencies' for ERC-20 tokens is somewhat an abuse of language inasmuch as these finally depend entirely on Ethereum, of which they are in a certain way a variant of the code applied to specific *smart contracts*. More precisely, the gas fees

for transferring ERC-20 tokens must in any case be paid in *gwei*s, that is to say in *ethers*: Ethereum therefore does indeed constitute the indispensable ‘fuel’ of all *smart contracts*, and no ERC-20 token is therefore truly independent of Ethereum.

### Focus...



**The ERC-721 standard for NFTs.** While the Ethereum *blockchain* permits the issuance of one’s own smart contracts through the ERC-20 norm, it is likewise possible to encode one’s own NFTs on it through the **ERC-721** norm. As Hossen & Gauthier (2024) explain, this new standard was proposed to the Ethereum community at the end of 2017 by the developer Dieter Shirley. Thus, while the NFT market surged above all in 2021, it was already possible to create them easily four years earlier through ERC-721 tokens. The particularity of these tokens is, as we have mentioned, to be unique — and this even if there exist several copies of a single work, each NFT then being backed by a unique number of the serial work — and indivisible: it can therefore neither be damaged nor destroyed in the course of its existence. ERC-721 tokens therefore allowed the Ethereum *blockchain* to develop on it the concept of **digital scarcity**. More precisely, ERC-721 tokens are defined on the *blockchain* notably by a name and a symbol, which permits the traceability of the NFT — for example on the EtherScan site which lists the more than 50 000 ERC-721 tokens created since December 2017 (<https://etherscan.io/tokens-nft>) — a maximum number of units or *market cap* — that is to say the number of numbered copies of the work (equal to 1 in the case of a unique work) — an ‘owner’ field which indicates the creator of the token, which makes it possible to distinguish the original work from its copies, an ‘approval’ field which defines the extent to which the token may be exchanged and a ‘holder’ field which designates the current owner of the NFT — who is therefore not necessarily the creator. All the structural properties of ERC-721 tokens — non-fungibility and indivisibility for example — are for their part defined in the metadata. The fact that NFTs were initially created on the Ethereum blockchain is interesting inasmuch as the transfer of a token requires forging fees. Now, since Ethereum is a proof-of-stake blockchain, forgers receive transaction fees — which explains why, if one sends a bitcoin to another address, the latter will receive only 0.9999 of it for example. Now, knowing that NFTs are indivisible, sending an ERC-721 token implies receiving the totality of it at another address. The fact that ERC-721 tokens rest on the Ethereum *blockchain* makes it possible to circumvent the problem: in order to transfer an NFT on the Ethereum *blockchain*, it suffices to pay the miner with the help of *ethers*; the gas fees as well as the mining reward will then be levied on the amount of *ethers* provisioned, without the integrity of the NFT being touched; if the amount of *ethers* provisioned turns out to be insufficient to complete the transaction, the latter is then rejected and the NFT remains in the hands of its owner, as Hossen & Gauthier (2024) explain.

The problem of ERC-721 tokens is, however, the same as that of all the tokens defined on the Ethereum blockchain, whether Ethereum itself or ERC-20 tokens: the rapid saturation of the *blockchain* which brings in its wake a **runaway increase in transaction fees**. Now if one wishes to transfer several ERC-721 NFTs, one must transfer them one after the other — since each token is unique — which means multiplying transactions and therefore multiplying tenfold the fees to be paid. Indeed, it is important to note, as Hossen & Gauthier (2024) emphasise, that transaction fees in *blockchains* are only slightly correlated with the amount or the equivalent value transferred. For a validating node, validating a transaction means verifying — thanks to the ECDSA algorithm — a digital signature of a transaction whose information consists of the sending address, the receiving address, the date, the number of tokens transferred, the fees paid to the validating node... Now, among all the information signed, the amount is only one parameter; transferring 0.01 or 10 bitcoins therefore changes nothing as to the complexity of the digital signature to be deciphered. Otherwise put, the **difficulty of validation** is independent of the amount transferred. This is why the fees for a transaction of 10 bitcoins are not *a priori* a thousand times higher than those paid for a transaction of 0.01 bitcoin — even if, as Hossen & Gauthier (2024) recall, inasmuch as it is the users themselves who decide the transaction fees to be paid (on centralised exchange platforms such as Binance or Kraken, one does not choose the fees; the platform does it for us), there is a tendency for users to agree to pay more fees for transactions of a significant value. In particular, within the framework of ERC-721 tokens, the individual transfer of 10 NFTs therefore costs far more than a possible — but impossible with ERC-721 tokens — single transfer of the 10 NFTs at the same time. It is with a view to economic optimisation that **ERC-1155** tokens were created which, as their name indicates, are still based on the Ethereum *blockchain*: as Hossen & Gauthier (2024) explain, these offer an interface of *smart contracts* which can represent and control a potentially unlimited number of fungible and non-fungible tokens; one speaks of a *multi-token*. Beyond the fact that ERC-1155 tokens make it possible to create a unified interface for all the *tokens* based on Ethereum, these make it possible, in the universe of NFTs, to contemplate mass transfers — including of several NFTs — in a single operation, thus considerably reducing transaction fees. Indeed, as Hossen & Gauthier (2024) detail, if one produces ten NFTs via the ERC-721 standard, each of these numbered tokens must be transferred one by one; with the ERC-1155 standard, one can manage in a single transaction a transfer of several NFTs at the same time, even if each of them is fundamentally unique and should therefore *a priori* require an



individual validation; this makes it possible considerably to optimise the resources devoted to mining. While the technical superiority of ERC-1155 *multi-tokens* over the ERC-721 standard seems obvious, especially for the creation of a ‘family’ (collection) of NFTs, the ERC-1155 standard, created in 2018, is still less used than the ERC-721 standard for the creation of NFTs according to Hossen & Gauthier (2024). The best-known ERC-1155 token in the field of NFTs to date is the Enjin Coin, the 442nd cryptocurrency in terms of capitalisation (*market cap*) at 7 April 2026 according to CoinMarketCap (2026).

### Key Takeaways



In order to avoid a flaw in a smart contract undermining confidence in the ether itself, each *smart contract* on Ethereum has its own ERC-20 token, encoded according to the Ethereum standard, that is to say in the Solidity language [Hossen & Gauthier (2024)]. These tokens offer a parallel accounting, a total interoperability between projects, and benefit from the outset from the security of the Ethereum network — on which they nevertheless remain dependent, transaction fees (gas) always having to be settled in ethers.

#### 2.3.2 How DEX work

A concrete example of an ERC-20 token is CAKE, the *token* of the decentralised exchange platform (**DEX**) PancakeSwap, on which thousands of different cryptocurrencies are traded for an amount equivalent to 1200 million euros each day on average according to Hossen & Gauthier (2024). On DEX, transactions are carried out over the counter directly between users, thanks to *smart contracts*, without having recourse to an exchange platform as a trusted intermediary: these are over-the-counter (OTC) transactions. Unlike centralised exchange platforms (**CEX**) — such as Binance, Coinbase or Kraken — which require a transfer of the *tokens* to the platform, DEX therefore only put a buyer and a seller in contact. Let us examine a concrete example of a transaction on a DEX such as PancakeSwap:

- A **maker** (here, a seller) records on a DEX an offer to sell, for example ‘I am selling 1 BTC for 10 ETH’ (the symbol of Bitcoin being BTC and that of Ethereum being ETH). This offer will then be signed by the *maker* thanks to his private key, and the maker’s bitcoin will then be locked in his wallet with the help of a *smart contract* acting directly on the *blockchain*, like a card payment authorisation for online purchases in order to verify the solvency of the account. This *smart contract* moreover leaves the possibility of cancelling this order at any moment and therefore of recovering the enjoyment of one’s funds.
- A **taker** (here, a buyer) likewise records on the DEX an offer to buy, for example ‘I am looking for 0.5 BTC for 5 ETH’. This offer will likewise be signed by the *taker* thanks to his private key, and the maker’s 5 ETH will then be locked in his wallet with the help of a *smart contract* acting directly on the *blockchain* — here again, the *smart contract* moreover leaves the possibility of cancelling this order at any moment and therefore of recovering the enjoyment of one’s funds. Moreover, any order (of the *maker* as of the *taker*) which does not find a taker will be automatically cancelled after a certain delay.
- A partial execution of the *maker’s* order is then possible: the *smart contract* verifies the signatures and makes sure that the offer of the *maker* and that of the *taker* have not expired. If these three points are valid, then the order is directly executed on the *blockchain* — the execution time may therefore be rather long depending on the congestion of the chain — and the *smart contract* automatically levies the fees — both the transaction fees of the validating node and the fees of the platform, the latter being paid in CAKE — which are recorded in the orders.

Hossen & Gauthier (2024) summarise the structure of the smart contract governing PancakeSwap as follows: (i) if a first person (*maker*) records an order, then confirmation of it is requested from him through a digital signature; (ii) if the signature is made, then the maker’s associated *tokens* are locked; (ii-bis) if the user requests the cancellation of the transaction, then the associated *tokens* are unlocked and the initial order (to sell, here) is deleted; (iii) if a second person (*taker*) records an order in relation to the *maker*, then confirmation of his order is requested from the *taker* through a digital signature; (iv) if the signature is made, then the taker’s associated *tokens* are locked; (v) if the *maker* and the *taker* have signed their orders, and their order is still valid (no request for cancellation, nor delay exceeded for example), then the order is automatically executed between the two parties and PancakeSwap levies a commission for this exchange of the order of 0.2 % paid in CAKE to the *liquidity providers*.



## Focus...



**The liquidity pools of DEX.** Decentralised exchange platforms (DEX), while they offer great security — provided that the *smart contract* which governs them is secure; otherwise how is one to make sure that the temporary locking of funds cannot result in a spoliation of them? — nevertheless suffer from a **lack of liquidity** which Hossen & Gauthier (2024) bring out. On centralised exchange platforms such as Binance, one transfers one's funds and one sees directly — thanks to the order book — the amount which is put up for sale or which is demanded for purchase for each pair of cryptocurrencies. But on decentralised platforms, since funds are never locked, how is one to know whether a *maker* order that one places will rapidly find a taker? A possible absence of liquidity would undermine the confidence placed in the decentralised platform according to Hossen & Gauthier (2024). In order to remedy this risk, PancakeSwap offers the mechanism of **liquidity pools** (LP). Users, called 'liquidity providers', contribute for each pair of cryptocurrencies X/Y both X *tokens* and Y *tokens* to a liquidity pool. A user who then wants to exchange his X *tokens* for Y *tokens* — this exchange takes the name of a **swap** — will therefore add X *tokens* to the pool and help himself in return to Y, by paying fees (which is fair since they unbalance the liquidity pool, unlike the *liquidity providers* who, for their part, feed it) which are precisely paid to the liquidity providers. There is therefore for liquidity providers a monetary interest in 'putting themselves on *stand-by*' while waiting for swappers to come and help themselves in their pool: this passive investment bears the name of **yield farming**. *Yield farming* must therefore not be confused with *staking*: while the effects are *in fine* the same (blocking *tokens* in order to receive interest), the mechanism and the meaning are for their part totally different: with *staking*, one takes advantage of proof of stake, one takes part in the validation of exchanges in a specific cryptocurrency; by contrast, with *yield farming*, one facilitates exchange transactions (*swaps*) between two cryptocurrencies on a decentralised platform and one does not take part in the entering of the resulting transactions on the *blockchain*. Whereas *staking* therefore necessarily bears on cryptocurrencies resting on proof of stake, *yield farming* can therefore concern *a priori* any pair of cryptocurrencies, as Hossen & Gauthier (2024) emphasise.

The use of a **wallet** is from then on indispensable in the universe of decentralised finance, as Hossen & Gauthier (2024) explain. Indeed, while in the case of a centralised exchange platform (CEX), one transfers money or *tokens* to an address belonging to the CEX (which then takes charge of associating it internally with its legitimate 'owner' — in inverted commas since it will be recalled that CEX keep the private keys of their users; the latter are not *stricto sensu* fully the owners of their funds) — by contrast, DEX never hold users' *tokens*. At most, when an order is issued on a DEX, the *smart contract* which governs the protocol of the DEX locks the funds, but this escrow is only temporary (possibility of cancellation by the issuer of the order or expiry of the order after a certain delay). How then is one to place orders on a DEX? It is therefore necessary, in addition to defining a username and a password, to associate with one's account a decentralised wallet, over which one retains control (holding of the private keys by the owner of the *tokens*): a wallet is therefore more exactly a 'key ring'; it plays the role of an interface between the holder of a private key and cryptocurrency protocols, thus allowing its user to access his *tokens*, to store them, to receive them and to send them. As Hossen & Gauthier (2024) detail, a decentralised wallet may for example take the form of an **internet browser extension** (generally free), the latter making it possible here again to access funds directly and to manage them, the user retaining control of his private keys. In order to use such an extension, it suffices to install it on one's browser. Among the best-known internet browser extensions figures MetaMask, which works with the Ethereum blockchain and therefore supports all ERC-20 tokens, but also ERC-721 and ERC-1155 tokens (Ethereum norms associated with NFTs).

## Focus...



**Decentralised lending.** Let us imagine that a person A wishes to borrow 30 000 euros: this is today possible without having recourse to banks with the help of decentralised applications and smart contracts, as Hossen & Gauthier (2024) explain — one may for example cite the ERC-20 token **Aave**. If person A has one bitcoin, initially worth 40 000 euros, she can then propose a smart contract in which she requests 30 000 euros; in exchange for which she places her bitcoin in escrow by way of guarantee — let us note that this is an over-guarantee given that the bitcoin is worth more than the amount desired. A person B may then decide to contract with person A by sending her 30 000 euros (in *stablecoins*). If person A defaults — that is to say if she does not repay person B one of the monthly instalments (defined with the help of a *smart contract*) — or if the price of bitcoin plunges to the point where the amount placed in escrow no longer offsets the debt remaining to be repaid, person B would recover the bitcoin placed in escrow. Such a system would possibly permit loans at very low rates, or again micro-loans which banks rarely grant.

One could, however, argue that this system admits of a great limitation: one must hold *tokens* in order to obtain euros. Two remarks then arise naturally, as Hossen & Gauthier (2024) emphasise. On the one hand, why not quite simply resell one's *tokens*? If one thinks that the price of Bitcoin is going to explode, it would be a shame to resell

it even if one needs money immediately; a loan makes it possible to place one's bitcoin in escrow while making it possible to obtain 'real' money, and once the loan is repaid, one recovers the enjoyment of the bitcoin. On the other hand, why not have recourse to a bank or a credit institution which *a priori* demands no collateralisation? The idea of an absence of collateralisation in centralised finance is often a decoy: in property loans, very often, the house is mortgaged, which means that the loan is in reality over-guaranteed; and for consumer credit, since there is no collateral, it is hardly surprising that rates reach astronomical values such as 20 % a year.

*In fine*, these decentralised loans constitute the decentralised analogue of **Lombard credit**. Lombard credit (also called an advance against pledge) is a short- or medium-term bank loan — generally one to four years — granted to a private individual or a firm in return for the pledging of liquid financial assets (shares, bonds, investment funds, life assurance contracts), of which the borrower remains the owner throughout the duration of the credit; structured most often as an *in fine* loan, it requires the repayment of the capital only at maturity — the borrower until then paying only the interest — which makes it possible to mobilise cash without having to dispose of one's portfolio of assets, as Boluze (2025) explains.

### Key Takeaways



A DEX (*decentralised exchange*) puts buyers and sellers of cryptocurrencies in contact over the counter via *smart contracts*, without ever holding users' funds — unlike centralised platforms (CEX) such as Binance [Hossen & Gauthier (2024)]. In order to remedy the structural lack of liquidity (owing to the fact that transactions take place *on-chain* without a visible order book), DEX rely on liquidity pools fed by *liquidity providers*, remunerated in native tokens for this passive deposit called *yield farming* — to be distinguished from *staking*, which takes part in the validation of transactions on the *blockchain* itself.

### 2.3.3 The rise of DAO

As we have seen, the world of DeFi rests on decentralised applications (dApps). Both the code and the *smart contracts* must be public and verifiable: a kind of **self-governance** and of steering by the community then takes shape; certain DeFi projects whose tokens are associated with governance rights (*governance tokens*) bear witness to this. The notion of **governance tokens** is then inseparable from that of a decentralised autonomous organisation (**DAO**). As Vitalik Buterin, the creator of Ethereum, wrote in 2014, a DAO is '*an entity that lives on the internet and exists autonomously, but also heavily relies on hiring individuals to perform certain tasks that the automaton itself cannot do*'. The idea of DAO is therefore, as Hossen & Gauthier (2024) explain, to substitute for traditional hierarchical organisations collaborative and horizontal human organisations. More precisely, a DAO represents the whole set of holders of its governance tokens, known or not, and all the other actors who interact directly or indirectly with these holders of these *tokens*: they work together to create value via the project and to derive profit from it. In a DAO, proposals are put forward by the members of the community; these proposals are then voted on by the holders of the DAO's token, the voting right being proportional to the number of tokens held. DAO thus create genuine **virtual communities** which escape any regulation coming from a central authority: certainly, the rules for allocating voting rights are managed by *smart contracts*, but in practice, the coordination of proposals and votes is organised *off chain* by means of alternative social networks such as Discord or Telegram, as Hossen & Gauthier (2024) set out. Beyond the protocol, the tokens and the alternative social networks, DAO likewise generally use a **multi-signature vault**, which allows participants spread across several countries to collect and manage collectively a treasury composed of cryptocurrencies. DAO thus permit the management of a **decentralised treasury**. Most DAO are said to be *permissionless*: one is free to join them or to leave them, which means that one can join them without asking anyone's opinion. The site <https://deepdao.io/organizations> thus lists all the DAO at the present time, as well as their treasury under management: it is the analogue of firms in the *blockchain*.

Now the rise of DAO poses a genuine regulatory issue, notably since nobody can claim to manage the treasury, since the latter is decentralised: the question 'in which country is the treasury located?' or 'under which legislation must the organisation be regulated?' does not even make sense, since the DAO is nationless according to Hossen & Gauthier (2024). And this absence of nationality poses a problem in the event of a default of the DAO. For example, **TheDAO**, the first DAO launched in May 2016 by the company Slock.It, aimed to finance projects on the Ethereum *blockchain* by acting as a decentralised investment fund, decisions being indeed taken collectively by the participants, holders of TheDAO tokens (the voting right being proportional to the number of tokens held). More precisely, the funds (collected in ethers, the cryptocurrency of Ethereum) were held in a *smart contract* (a multi-signature vault), and the governance mechanisms made it possible to vote on the allocation of funds to DeFi projects. However, in June 2016, after having collected 150 million dollars in ethers, TheDAO suffered a hack which

led to the extortion of roughly one third of its funds under management (which moreover provoked a fall in the price of Ethereum of more than 30 %), as Hossen & Gauthier (2024) recall. The failure of TheDAO may be explained by the lack of maturity of certain investors, by the absence of technological tools to help the community in decision-making and the management of funds, and by the absence of an audit process or of bug bounty programmes. To what point, then, is decentralisation desirable? In principle, governance tokens transcribe the idea of the Athenian agora according to Hossen & Gauthier (2024). However, the failure of TheDAO showed that this ideal was to a certain extent utopian, and that it was appropriate, while waiting for the universe of DAO to be more mature, to maintain a kind of expert control over DeFi projects. This is why many DeFi protocols today are not governed by the community, and not all protocol changes occur following votes whose voices are proportional to the number of governance tokens held. Indeed, certain entities retain the right of veto over the vote or the control of discussion forums dealing with modifications of the protocol.

### Focus...



**The hacking of TheDAO.** The hacking of TheDAO did not have as its source the exploitation of a flaw in a third-party centralised infrastructure, but of a vulnerability inherent in the code of a **smart contract** deployed directly on the Ethereum *blockchain*, which engages the responsibility of the *blockchain* itself in the eyes of its community of users. Launched in April 2016 in the form of a **decentralised autonomous organisation** (DAO) intended to function as an algorithmically managed venture capital fund, TheDAO had raised, during its *crowdfunding* campaign, some 11 million ethers — that is approximately 150 million dollars, which made it at the time the largest participatory fundraising in history. On 17 June 2016, only twenty days after its entry into production, an attacker exploited a **reentrancy vulnerability** (*reentrancy bug*): as Atzei et al. (2017) formalise in their taxonomy of attacks on Ethereum smart contracts, this type of flaw allows a malicious contract to call recursively the withdrawal function of a target contract before the state of the balance is updated, thus making it possible to siphon off the funds in an iterative and continuous manner; in this instance, 3.6 million ethers — that is roughly one third of TheDAO's funds — were thus stolen. Faced with the scale of the damage, the Ethereum community was confronted with a political dilemma which Atzei et al. (2017) describe as an 'irreducible tension between immutability and correction': on 20 July 2016, at the height of block 1 920 000, a controversial **hard fork** was executed in order retroactively to cancel the fraudulent transactions and to restore the funds to the injured investors; this decision, which violated the founding principle of the immutability of the *blockchain*, entailed the splitting of the network into two distinct chains — Ethereum (ETH), which adopted the forked version, and **Ethereum Classic** (ETC), which kept the original chain intact —, illustrating **decentralised governance** not as a guarantee of neutral arbitration, but as the site of a conflict of values between economic pragmatism and protocol integrity.

In practice, as Hossen & Gauthier (2024) set out, most DeFi projects are therefore **hybrid projects**, where the management and safekeeping of private keys, of funds or of users' information are managed by third-party nodes, which departs from the ideal of a genuine decentralisation. This was for example the case of the decentralised exchange platform (not so much so in reality) Celsius, which moreover went bankrupt in July 2022. But then, might this hybrid character in practice of DeFi not be a means of avoiding heavy regulatory compliance? As we have seen, under the MiCA regulation, crypto-asset service providers (DASP/CASP) will have to comply with security requirements in order to protect their clients' funds and to publish information about their activity; a considerable cost for the *start-ups* of web 3.0 according to Hossen & Gauthier (2024). Conversely, since DeFi projects are by definition not managed by a legal entity (a firm, and so on), the MiCA regulation does not apply to decentralised projects. Certain firms thus make the choice of combining a centralised architecture in the foreground with a DeFi architecture in the background in order to escape regulation: this is the case of numerous firms which offer, for a subscription, 'buy and sell signals' to be executed on a decentralised exchange platform (*social trading*). Hossen & Gauthier (2024) then wonder: in wanting to regulate too much, does the European Union, in any case powerless in the face of projects which cannot be ontologically associated with any territory, not risk shooting itself in the foot by imposing staggering costs on *blockchain* projects which would like to comply with the legislation, while others would exploit the nationless character of DeFi and the complaisance of tax havens in order to maintain unfair competition with regard to Europe?

### Key Takeaways



A DAO (*decentralised autonomous organisation*) is a horizontal and collaborative organisation governed by governance tokens: proposals are voted on there in proportion to the number of tokens held, the treasury is managed collectively via a multi-signature vault, and coordination is organised *off chain* on networks such as Discord or Telegram [Hossen & Gauthier (2024)]. The failure of TheDAO in 2016 — of which one third of the 150 million dollars

raised was stolen via a reentrancy vulnerability, forcing Ethereum into a controversial *hard fork* which provoked the split between ETH and Ethereum Classic — nevertheless illustrates that the ideal of the decentralised agora remains a partial utopia, justifying the maintenance of expert control in current DeFi projects. On the regulatory level, the nationless nature of DAO removes them *de facto* from the MiCA regulation, which creates unfair competition between compliant *blockchain* projects and hybrid structures which exploit this nationlessness in order to escape European prudential requirements according to Hossen & Gauthier (2024).

### 3 The economic, tax and legal issues linked to the *blockchain*

The *blockchain* and the crypto-assets associated with it do not constitute merely a technological innovation: they simultaneously shake three pillars of the contemporary economic and legal order — **taxation**, **monetary stability** and **state sovereignty** — without positive law having yet found a fully satisfactory answer to any of these three challenges, as OCDE (2024) recognised in admitting that the tax evasion issues specific to crypto-assets were largely unexplored during the first years of development of the ecosystem.

#### 3.1 A still hazy tax regime

While the MiCA regulation has brought a harmonised framework for the issuance and marketing of crypto-assets, the taxation applicable to them remains, for its part, piecemeal and often behind the real practices of investors. Since 1 January 2023, the **finance act for 2022** has certainly laid down essential markers, but whole swathes of the universe of DeFi still escape any clear tax qualification: *staking*, *yield farming* and *liquidity providing* remain in an almost total legal vacuum in France. According to Hossen & Gauthier (2024), this still hazy taxation is structurally due to the rapidity of technological innovation, which the legislature struggles to follow, and to the ontological nationlessness of assets whose value, transactions and issuers do not readily allow themselves to be enclosed within the categories of national tax law.

##### 3.1.1 New tax rules in France

According to article 79 of the finance act for 2022, all operations linked to cryptocurrencies since 1 January 2023 carried out by private individuals on a non-professional basis, that is to say within the framework of private wealth management, come under the **single flat-rate levy** (PFU), also known as the **flat tax**, that is to say a tax on the capital gain of 30 % at most. More precisely, any non-professional private individual carrying out operations of purchase and resale of cryptocurrencies since 1 January 2023 may choose either to be taxed at 31.4 % on the capital gain realised, or to renounce the tax of 12.8 % and opt for the application of the progressive scale of the tax on capital gains. This last option is of interest for non-taxable taxpayers who will then pay only 18.6 % of social levies; even for taxpayers whose marginal rate of taxation is 11 %, the option is of interest in that it permits a taxation of 29.6 % (that is 18.6 % of social levies and 11 % of income tax) against 31.4 % for the PFU. Moreover, as Hossen & Gauthier (2024) specify, the CSG paid, in an amount of 10.6 % of the capital gain, is deductible from taxable income (within the limit of 6.8 % of overall taxable income the following year) — which may result *in fine* in a taxation of roughly 29 % at most.

This provision thus clarifies article 41 of the finance act for 2019, which provided only for an overall tax rate of 30 % on the capital gains realised (therefore offering neither the choice of opting for the progressive income tax scale nor the possibility of deducting the CSG the following year), combining a rate of 12.8 % in respect of income tax and a rate of 17.2 % in respect of social levies for operations on digital assets carried out ‘on an occasional basis’ only (see article 150 VH bis of the General Tax Code). The legislation was then extremely imprecise: how is one to distinguish between occasional and habitual purchases and resales? While it is obviously possible to sketch out a **body of evidence** (frequency of disposals, purchases of crypto-assets with a view to their resale, duration of holding of crypto-assets between purchase and resale, use of derivative products or of leverage, use of trading robots to automate purchase and resale operations...), the legislation offered no clear threshold and the assessment then had to be made on a case-by-case basis.

#### Debate



**When is one considered a cryptocurrency professional?** As Hossen & Gauthier (2024) explain, a private individual might think that, whatever the way in which he buys, exchanges and resells his cryptocurrencies, he will be considered a non-professional by the tax administration, and all the more so as the finance act of 2022 removes the

dichotomy between occasional operations and habitual operations for distinguishing between non-professional and professional investor. This is a mistake: indeed, article 92 of the General Tax Code has been amended since 1 January 2023 so that operations of disposal of digital assets carried out ‘under conditions analogous to those which characterise an activity exercised by a person engaging in this type of operation on a professional basis’ are subject to the regime of **non-commercial profits** (BNC), that is to say a professional regime. Otherwise put, as Hossen & Gauthier (2024) analyse, if one uses the same techniques as professionals, for example automatic *trading* robots, there is a strong chance that the tax administration will requalify the private individual as a professional, especially if he makes thousands of operations in the year which cannot be assimilated to a normal use — that is to say a use not devoid of a certain expertise — of the *blockchain* and of cryptocurrencies. Obviously, as the fact-finding mission of the National Assembly relating to crypto-assets directed by Woerth (2021) recalls, ‘*in the BNC regime, the distinction between occasional activities and professional activities rests more on qualitative criteria — the superiority of the overall annual sum of the capital gains arising from the activity relative to the other income of the tax household, the use of professional tools, the complexity of the operations carried out — which counterbalance quantitative criteria alone*’. Thus, even if one has carried out 10 000 cryptocurrency-for-cryptocurrency exchanges for an average amount of a hundred or so euros and one realises an annual capital gain of the order of 500 euros, unless one is otherwise on income support, there is little chance of attracting the gaze of the tax authorities according to Hossen & Gauthier (2024) — moreover, for non-professionals, the capital gain on cryptocurrencies is even exempt from tax as long as it does not exceed 305 euros in the year (a very low figure if one compares it for example with the United Kingdom and its exemption threshold of 11 700 pounds sterling). By contrast, if the income one derives from cryptocurrencies exceeds roughly one third of one’s other income, one must expect according to Hossen & Gauthier (2024) to attract the eye of the tax administration, which will ask itself about the qualification of the investment: if one has realised one or two large capital gains in the year and no other operation besides, or if one makes multiple purchases at one go in order to resell them a few months later all at once — following a recommendation in a specialised publication for example — it is likely that one will succeed in convincing the tax inspector according to Hossen & Gauthier (2024); by contrast, if one buys and resells cryptocurrencies on the same day — one then speaks of *intraday trading*, or even of *scalping* if positions last less than an hour — on many days in the year, or if one uses *trading* robots — one need only consult the times of the transactions: if the history presents positions taken at 4 in the morning, 1 in the afternoon, 4 in the afternoon, then 11 at night, it is difficult to convince an inspector that it is indeed a single individual who took them all (when then does he sleep?) — one will in all likelihood fall into the boxes ‘use of professional tools’ and ‘complexity of the operations carried out’ according to Hossen & Gauthier (2024).

### Key Takeaways



Since 1 January 2023, capital gains on crypto-assets realised by non-professional private individuals come under the single flat-rate levy (*flat tax*) of 31.4 % (12.8 % of income tax and 18.6 % of social levies), with the possibility of opting for the progressive scale — an advantageous option for taxpayers whose marginal rate of taxation is less than or equal to 11 %; below 305 euros of annual capital gain, the exemption applies [Hossen & Gauthier (2024)]. A private individual may, however, be requalified as a professional (BNC regime) if his operations present the characteristics of a professional activity: recourse to *trading* robots, repeated *intraday* positions, or cryptocurrency capital gains significant relative to the other income of the tax household [Woerth (2021)].

#### 3.1.2 The question of *stablecoins*

**Stablecoins** are cryptocurrencies whose price does not vary, whatever the economic conjuncture; a major issue in the universe of decentralised finance, when one sees that certain cryptocurrencies sometimes fall by more than 50 % in a single day. As Baughman et al. (2022) specify, the best-known *stablecoins* are Tether (USDT) and USDC, which both rest on a mechanism of **collateralisation** with the dollar: if there exists a tether or a USDC token on the market, this means that the entity which issues it (the Hong Kong company Bitfinex notably for USDT; the American companies Circle, of which Visa is a partner, and Coinbase for USDC) is the holder (in theory) in its bank account of a ‘real’ dollar. Thus, even if demand for USDT or USDC falls (and a basic economic argument then predicts a fall in the price in order to rebalance supply and demand), thanks to collateralisation, the price remains stable since one can at any moment recover in exchange for one’s token a real dollar in the coffers of the entity which issues the *stablecoin*. It is a similar principle to that of the chips one buys in a casino: when one enters the casino, one buys chips with *fiat* money (euro, dollar, and so on), then one plays with one’s chips (one wins or loses some); when one leaves the place, one exchanges them for real money.



## Focus...



**Terra and the fall of algorithmic *stablecoins*.** As Hossen & Gauthier (2024) recall, certain cryptocurrencies provide for the possibility of destroying certain tokens (the *burn* phenomenon) or conversely of creating some (the *mint* phenomenon) in order to maintain a constant value of each token, like the ‘monetary snake’ phenomenon under Giscard d’Estaing which made it possible to maintain the price of the franc around a value fixed relative to other currencies (the American dollar and the German mark notably). These processes of creation and destruction of *tokens* may be directly implemented in the code which defines the cryptocurrency or initiated following a majority vote of the holders of the *tokens*. It is the first situation which occurred with the cryptocurrencies Luna and Terra (which have today disappeared from circulation): as Hossen & Gauthier (2024) explain, in June 2019, the Terra project created a *stablecoin* based on the dollar (that is to say a currency whose unit price was to be constantly equal to one dollar) of a new kind, which did not rest on a collateralisation with more or less transparent centralised bodies. Indeed, as a general rule, in order to guarantee parity between the *token* and the dollar, such centralised bodies had to hold reserves in dollars, each *token* having to be guaranteed by a dollar; but the audits on the transparency and the quality of the collateralisation remained hazy. The Terra project for its part proposed an **algorithmic *stablecoin*** resting on two cryptocurrencies: Luna (LUNA) and Terra (UST) (the *stablecoin*). But if there is no collateralisation with a ‘real’ dollar, how is the parity of UST to be ensured? Indeed, if the volumes of sale of UST were to increase, the law of supply and demand would mean that the price of UST would fall and move away from its parity with the dollar. It is precisely in order to avoid these fluctuations that Luna was created: as Hossen & Gauthier (2024) detail, thanks to an algorithm that the company had developed, Terra could constantly adjust the number of Luna in circulation, by creating some (*mint*) or destroying some (*burn*), in order to stabilise the price of UST. More precisely, if the price of UST were to fall below the dollar, then the **Luna Foundation Guard** (LFG), a non-profit association whose object is precisely to defend the parity between UST and the dollar, could — according to rules defined by the Anchor algorithmic protocol — buy back 1 UST (which therefore costs less than one dollar), destroy it (*burn*) and then create the equivalent of one dollar in Luna tokens (*mint*), before reselling the Luna tokens thus created for a real dollar in order to realise a profit (since the purchase of one UST cost less than one dollar, let us recall). Conversely, if the price of UST were to exceed the dollar, on account of strong demand for example, then the LFG bought one dollar’s worth of Luna, destroyed it in order to mint one UST, then resold this UST *token* in order to realise a profit (since it is worth here more than one dollar). This mechanism of **arbitrage** occurred until no further profits could be generated: the market therefore self-regulates until the arbitrage opportunity disappears, that is to say until UST recovers its anchor (*peg*) of ‘one UST for one dollar’. The problem brought out by Hossen & Gauthier (2024) is that, if the price of UST plunges too far, then one creates an enormous number of Luna tokens; now, if the supply of Luna increases enormously while demand does not follow, then the price of Luna falls. And this is the beginning of a vicious circle: if the price of Luna falls, more Luna tokens must be created in order to obtain the equivalent of a real dollar, so supply increases even faster while demand does not follow. From then on, the price of Luna continues its fall, so much so that the monetary creation of Luna tokens struggles more and more to ensure the stability of UST. The system collapses, and this is precisely what happened in May 2022: UST lost more than 99 % of its value, as did Luna. This is why the European MiCA regulation explicitly prohibits algorithmic *stablecoins*: **article 68 of EU regulation no. 2023/1114** provides that an *asset-referenced token* or an *e-money token* may not be issued if its stability rests on an algorithmic mechanism: every *stablecoin* circulating in the European Union must therefore be entirely backed by reserves of real assets in a one-to-one ratio, held by a third party and isolated from the issuer’s own holdings, which excludes by construction any model of endogenous stabilisation by automatic issuance or destruction of tokens.

The tax regime of private investors is particularly advantageous in that it makes *stablecoins* a genuine tax loophole: indeed, operations of purchase and resale between cryptocurrencies are quite simply not taxable, since the proceeds of their disposal are considered an unrealised capital gain or loss, as Hossen & Gauthier (2024) specify. Thus, let us imagine that one buys a *token* X on 1 January 2023 for 10 000 dollars. On 31 December 2023, one resells this *token* for 11 000 dollars: one has therefore realised a capital gain of 1000 dollars, which will be taxed at 30 % or less (depending on the marginal rate of taxation). But if one had resold on 31 December 2023 this *token* for 11 000 USDT or 11 000 USDC then, since one would have remained in the universe of cryptocurrency, the capital gain would be considered unrealised. While it is certainly impossible to escape taxation — sooner or later, one will have to move back into *fiat* currencies or buy goods and services with one’s *tokens* (which, as Hossen & Gauthier (2024) warn, is a taxable event triggering taxation: the use of a Binance payment card, for example, in order to spend one’s *tokens* directly is therefore not a means of escaping taxation) — the fact of using *stablecoins* as a tax loophole therefore makes it possible to decide when one will ‘settle up’. An issue that is all the more important in that **capital losses are not carried forward** from one tax year to the next, as Hossen & Gauthier (2024) emphasise. For example, let us imagine that on 1 January 2023, you buy a *token* X for 10 000 dollars; on 31 December 2023, you resell it for 8000 dollars; on 1 January 2024, you buy back a *token* X for 8500 dollars; on 31 December 2024, you resell this *token* X

for 9000 dollars: you will then have to pay tax on the capital gain of 500 dollars realised in 2023 and will not be able to carry forward the capital loss of 2000 dollars realised in 2024. By contrast, let us imagine that on 1 January 2023, you buy a *token X* for 10 000 dollars; on 31 December 2023, you resell it for 8000 USDT; on 1 January 2024, you buy back a *token X* for 8500 USDT; on 31 December 2024, you resell this *token X* for 9000 dollars: you then pay no tax, since you have not realised a capital gain on your *tokens*<sup>4</sup>.

By contrast, unlike non-professional private individuals for whom operations of exchanging cryptocurrencies for cryptocurrencies are not taxable, each exchange must be taxed for professionals. In particular, there is no 'tax loophole' for *stablecoins* for professionals; every operation must be accounted for, without exception, and gives rise to the recognition of a capital gain or loss realised. This drastic choice aims, as the fact-finding mission relating to crypto-assets conducted by Woerth (2021) mentioned above recalls, to '*avoid the risk of a de facto exemption of the gains realised by firms which would not convert, or would convert very little of, their digital assets into legal tender*'. Two problems then present themselves to firms, raised by the rapporteur Woerth (2021). The first point is '*a liquidity problem: the tax on the capital gain of which the firm is liable must be settled in euros, but the exchange between crypto-assets generates no receipts in euros for the firm, which is therefore constrained to convert part of its digital assets into legal tender*'. At the present time, no solution has yet been proposed, such as accounting for operations of exchange between digital assets only at the scale of a day, a week or a month in order to avoid the cumbersomeness of accounting entries. The second point is '*a problem linked to the volatility of digital assets: if the firm does not immediately convert the digital assets it has just acquired and the price of these assets decreases, it must pay the tax on a capital gain which has in the meantime decreased or disappeared*'.

### Key Takeaways



*Stablecoins* (USDT, USDC) are fixed-price cryptocurrencies backed by a real asset according to a mechanism of one-to-one collateralisation with the dollar; conversely, algorithmic *stablecoins* — such as TerraUSD/Luna, whose collapse in May 2022 wiped out 99 % of the value — rest on *mint* and *burn* mechanisms without any real reserve, which explains their explicit prohibition by article 68 of the MiCA regulation [Hossen & Gauthier (2024)]. On the tax side, exchanges between cryptocurrencies (including into *stablecoins*) trigger no taxation for non-professional private individuals, since the capital gain is considered unrealised: *stablecoins* thus constitute a tax loophole making it possible to secure one's gains while deferring taxation, provided one does not move back into *fiat* currency or use one's *tokens* to pay directly for goods and services [Hossen & Gauthier (2024)].

#### 3.1.3 Mining versus staking

As regards **mining**, no doubt is permitted according to Hossen & Gauthier (2024): every miner is necessarily a professional who will be taxed under the regime of non-commercial profits (BNC). As the *Bulletin officiel des finances publiques et des impôts* (BOFIP) recalls, '*the taxable result derived from this activity is determined in conformity with the ordinary law rules applicable to non-commercial profits, it being specified that the acquisition value retained for the computation of the taxable result is nil when the bitcoins have been allocated free of charge*' (BOI-BNC-CHAMP-10-10-20-40, paragraph 1080); the tax authorities therefore consider that the *tokens* are received by the miner at a nil price. For mining firms generating less than 83 600 euros of turnover — turnover then being determined by the sum of the values of the *tokens* received in the year at the price corresponding to the moment when these were received by the miner — it is then possible to be taxed directly on the amount of turnover, after an allowance of 34 %, according to the income tax scale of the manager — if it is a micro-enterprise or an EURL; otherwise, it is the firm itself which pays corporation tax (IS) and which must subsequently pay the money back in the form of dividends to the manager (but there is then a risk of double taxation: the IS on the one hand on the company, then the *flat tax* on the dividends for the director), as Hossen & Gauthier (2024) detail. This 'micro-BNC' regime certainly makes it possible to avoid an overly tedious accounting but, in return, no charges — acquisition of computers, cost of electricity... — are deductible. In order to deduct charges, one must imperatively request the *déclaration contrôlée* regime — which is in any case compulsory for firms generating more than 83 600 euros of annual turnover, as Direction de l'information légale et administrative (2026a) explains. Here we find the same **problems of liquidity**

<sup>4</sup>In reality, the example given is simplified since the capital gain must, in France, be expressed in euros: account must therefore additionally be taken of the variation in the euro/dollar exchange rate between 1 January 2023 and 31 December 2024 in order to obtain the exact value in euros of the capital gain or loss realised. And while it might be of interest to use a *stablecoin* based on the euro and not on the dollar in order to avoid these tedious computations, *stablecoins* based on the euro are according to Hossen & Gauthier (2024) far less reliable than those based on the dollar (Stasis Euro, a *stablecoin* based on the euro, is only the 357th cryptocurrency in terms of market capitalisation at 3 April 2022, and is not even available on Binance or Coinbase, the two leading centralised exchange platforms); it should be recalled that the exchange benchmark in the universe of cryptocurrency is rather the dollar than the euro: the choice of USDT or USDC as a *stablecoin* in order to escape the volatility of the market and secure the value of one's holdings while deferring taxation therefore seems a much better choice.



**and volatility** as for professional *traders*: miners pay in *fiat* currency a taxation as soon as the *tokens* are received, whereas the value of the *tokens* may vary over time — they therefore risk paying for a result which no longer exists on account of the depreciation of the tokens —; and while any unrealised capital loss is accounted for in the form of a provision for risk, here again the debate remains as to the tax deductibility of this provision in the absence of clear legislation or case law to date (a legal vacuum), as Hossen & Gauthier (2024) emphasise. One point, however, on which the legislation is now clear is the question of VAT: both for miners under the micro-BNC regime and for those under the *déclaration contrôlée* regime, the fact-finding mission relating to crypto-assets mentioned above recalls that the activity of mining is exempt from VAT. Indeed, miners *‘are remunerated, not by the parties to the transaction, but by the system which manages itself and only when they win the validation of a block. Remuneration by the allocation of cryptocurrencies, understood as a reward, therefore presents a random character. From then on, it does not appear possible to identify the existence of an individualised service performed by the miner for the benefit of a determined beneficiary. Accordingly, in the absence of a direct link between the consideration and the activity of mining, the latter does not fall within the scope of VAT and the miner will not have to collect tax in respect of the digital assets received as a reward’* (BOI-TVA-CHAMP-10-10-10, paragraph 69). However, tax coherence then implies that it is impossible to deduct VAT for all the expenditure relating to the activity of mining: *‘the miner, who carries out no taxable operations, is not entitled to exercise a right of deduction’* in respect of the expenditure incurred in order to carry out his mining activity and for which he receives a reward (BOI-TVA-CHAMP-10-10-10, paragraph 69). It therefore appears impossible to deduct notably the VAT on the mining computers acquired — whose value almost always exceeds 5000 euros according to Hossen & Gauthier (2024) — nor on electricity bills when the main object of the firm is the activity of mining.

Where the legislation presents a total haze at the present time is on the question of **staking**. While it is clear that a miner clearly has a professional ambition according to Hossen & Gauthier (2024): he must equip himself with a dedicated computer whose acquisition value often exceeds ten thousand euros, he must optimise his validation ‘work’ so as to cover his electricity charges... A miner is *in fine* a genuine manager. By contrast, many are those who proceed to the *staking* of their tokens because they simply prefer that these should earn them interest — the validation fees when one receives a transaction to validate, in this instance — rather than leaving them to sleep in a decentralised wallet or in one’s Binance account and earning nothing. Moreover, as Hossen & Gauthier (2024) emphasise, on exchange platforms, one does not even do *staking* properly speaking but rather ‘pseudo-*staking*’: one blocks a certain number of *tokens* on Binance (for example), but in reality, one entrusts tokens to Binance, and it is the whole set of tokens entrusted that Binance finally places in escrow so that the platform itself may then proceed to the *staking* — that is to say to the validation of the transactions distributed by the algorithm — in its own name; it is therefore Binance which earns the transaction fees, and the platform then pays back a certain percentage to the initial lender. This example shows well that *staking* represents more a logic of passive investment than of professional investment according to Hossen & Gauthier (2024). It is the same thing for *yield farming* or *liquidity providing* on decentralised exchange platforms: one contributes tokens in order to create liquidity on markets, in exchange for commissions on the *swap* operations carried out. Now neither *yield farming* nor *liquidity providing* is dealt with in the legislation at the present time; the legal vacuum is total according to Hossen & Gauthier (2024). As for *staking*, it seems that there is a great confusion on the part of the tax administration, which assimilates mining and *staking*: as the information report on the implementation of the conclusions of the fact-finding mission relating to crypto-assets by Woerth (2021) mentions, *‘validation by proof of interest (sic) [or] proof of stake requires only that the miner (sic) stake a certain amount of crypto-assets, which he recovers when he has correctly executed his mining (sic) work, with a much lower level of energy consumption’*. A dramatic confusion since it would assimilate from the outset every staker to a mining professional, and therefore to the BNC regime — with all the tax and accounting complexity which flows from it.

### Key Takeaways



Mining comes unambiguously under the regime of non-commercial profits (BNC): the *tokens* received are taxed at their value on receipt (a nil acquisition value according to the BOFIP), the activity is exempt from VAT — but without any right of deduction — and the problems of liquidity and volatility make the regime particularly penalising [Hossen & Gauthier (2024)]. By contrast, the tax treatment of *staking* remains a legal vacuum: the tax administration, confusing mining and *staking*, tends to assimilate every staker to a professional subject to the BNC [Woerth (2021)], whereas *staking* is more akin to a passive investment; *yield farming* and *liquidity providing* are for their part totally absent from the legislation [Hossen & Gauthier (2024)].

### 3.2 The rise of cryptocurrencies, a vector of tax optimisation?

While the decentralised and pseudonymous nature of cryptocurrencies does not *a priori* destine them to fraudulent use, their structural characteristics, notably their **nationlessness**, make them objectively conducive to optimisation, or even to tax evasion, as the considerable gap between the capital gains on crypto-assets estimated by Chainalysis (3.5 billion euros) and those actually declared to the Direction générale des finances publiques (0.4 billion euros), noted by Observatoire de la justice fiscale (2024), attests. OCDE (2024) confirms that the risks linked to the use of crypto-assets as a means of evading tax obligations are particularly high when taxpayers hold crypto-assets abroad, for example through **extraterritorial intermediaries**, and that these tax evasion issues were largely unexplored during the first years of development of the ecosystem, concentrating the bulk of regulatory attention on money laundering alone.

#### 3.2.1 The notion of a non-cooperative territory: what analogy for cryptocurrencies?

In order to establish an indicator making it possible to understand, and where appropriate to assess, the intrinsic predispositions of certain crypto-assets, let us start again from the definition of a **non-cooperative state and territory** (ETNC) present in the economic, legal and administrative literature. While, as Boulenguer (2022) explains, there exists no objective and clear test which permits the identification of a country as a tax haven, criteria have been put in place by institutions in order to assess the good tax governance of a state or a territory, and thus to draw up a blacklist of the zones which depart from this definition. In France, tax havens are described in the General Tax Code (CGI) as non-cooperative states and territories (**ETNC**), that is to say that they have '*not concluded with France a convention on administrative assistance permitting the exchange of any information necessary for the application of the tax legislation of the parties, nor signed such a convention with at least twelve states or territories*' (**article 238-0A of the CGI**). These tax jurisdictions are said to be non-cooperative when they possess legislation which harms exchanges of information with other states by keeping, for example, opacity about the origin of their clientele in order to escape the radars of administrations and regulators. A criterion of **respect for the principles of tax transparency** is thus written into annex V of the conclusions of Conseil de l'Union européenne (2017) relating to the European Union list of non-cooperative jurisdictions for tax purposes, to which is added a need to respect the principles of tax fairness. In order not to be entered on the European Union blacklist of tax havens, jurisdictions therefore undertake not to put in place preferential tax measures which could be considered harmful by facilitating the creation of offshore structures or arrangements intended to attract profits which do not reflect a real economic activity in that country or territory. Chavagneux (2018) thus concludes that the legal spaces having non-existent or insignificant taxes or implementing measures running counter to international tax agreements (of the BEPS type) by favouring for example the anonymisation of bank accounts and the lack of transparency of financial operations, are categorised as ETNC. Now, inasmuch as most cryptocurrencies are totally decentralised tools, there exists *a priori* no means for states to obtain information about the holders of cryptocurrencies, except for individuals and firms buying their *tokens* on platforms such as Coinbase or Binance which have automated procedures for the fight against money laundering and the financing of terrorism (*know your client* or KYC procedures, involving notably a verification of identity and of domicile): in this last case, the role of watchdog therefore falls not to the cryptocurrency protocols themselves, but indeed to the firms which make it possible to buy or exchange them. Now, there exist numerous other means of acquiring cryptocurrencies (the mining of *tokens* for example) without going through regulated platforms: in that case, the **surveillance of capital flows** becomes far more difficult for the public authorities to implement.

Moreover, in tax fraud practices, there exists according to Chavagneux (2018) a dissociation between the place where an economic transaction occurs (receiving a salary, an inheritance, profits...) and the place where it is recorded, therefore controlled and taxed, thereby blurring the geography of multinationals' investments abroad. Now the decentralisation which characterises public *blockchains* makes the associated cryptocurrencies ontologically nationless tools, without any link to any territory whatsoever: the dichotomy between the place of realisation of a transaction and its recording on a deterritorialised *blockchain* is therefore established. And while geographical (an island territory, a city district) or legislative (a regulator) separation is not a characteristic specific to tax havens according to Hampton (1996), the literature observes that the local economy in ETNC is mainly oriented towards non-residents. Otherwise put, zones of non-cooperation are places of temporary storage of funds, giving rise to a disproportion between the size of the financial sector and domestic financing needs, as Dufey & Giddy (1978) explain. In order to verify whether there exists an analogy with the *blockchain*, it is then appropriate to try to analyse as far as possible the movements of cryptocurrency flows. Moreover, Saez & Zucman (2019) insist on the numerous inequalities of wealth which underlie the existence of **tax havens**: the circumvention of tax indeed distorts the distribution of tax resources between countries and appreciably reduces the state's receipts, which

alters the redistribution mechanisms implemented by states in order to reduce inequalities on their territory. This phenomenon may be explained notably by the efforts made by non-cooperative territories to attract *offshore* firms, as McCarthy (1979) details. Indeed, these zones depend on their capacity to attract firms and to distinguish themselves from the competition with the help of aggressive legislative incentives in tax matters. Could there be an attraction of firms towards the relative legal vacuum that cryptocurrencies occupy today?

### Key Takeaways



Cryptocurrencies present a structural analogy with non-cooperative states and territories (ETNC): their decentralisation makes them ontologically nationless, dissociating by construction the place of realisation of a transaction from its recording on a deterritorialised *blockchain* — exactly as tax havens dissociate the place of economic activity from the place of taxation [Chavagneux (2018)]. Outside regulated platforms imposing KYC procedures, states have no direct means of monitoring cryptocurrency flows or of identifying their holders; the question then arises whether the current legal vacuum surrounding crypto-assets does not constitute, like the ETNC, a space of attractiveness for capital seeking to escape the mechanisms of tax redistribution [Saez & Zucman (2019)].

### 3.2.2 The *dark web* and cryptocurrencies

According to Clément & Lelieur (2021), Bitcoin rapidly became a ‘new playground for money laundering’. More precisely, the agency Tracfin (2016), the French intelligence service in charge of the fight against clandestine financial circuits, money laundering and the financing of terrorism, thus estimated in its report *Trends and analysis of the risks of money laundering and financing of terrorism in 2016* that ‘*blockchains do not really create new methods of fraud but offer a new field of application for proven methods*’. For example, as Pisani & Piotet (2008) indicate, from 70 to 75 % of web pages are not indexed by search engines, constituting the ‘deep web’ (*deep web*); among this deep web, some of the pages deal with illegal content (weapons, drugs, counterfeit money...) and constitute the *dark web*, accessible only with the help of the Tor network, with .onion addresses.

### Focus...



**How does the Tor network work?** As Hossen & Gauthier (2024) explain, numerous sites are blocked by internet service providers (ISPs) because they are identified as illegal: sites selling counterfeit notes, weapons, false papers, drugs... These websites, in order to be publicly accessible and not to be blocked by ISPs, must therefore use **overlay networks**, that is to say a computer network built on others, the nodes of the overlay network being interconnected by logical links to the underlying network. Now the complexity of the underlying network is not visible at the level of the overlay network; the latter therefore constitutes a kind of ‘shop window’ which is not identified by ISPs as illegal, but permits access to the underlying network, which for its part contains the illegal information. Access to the underlying network operates by deciphering the logical links binding it to the overlay network. This deciphering is accessible only with specific software, configurations or protocols; the Tor network is an example of this.

Thus, as Ghimiray (2024) explains on the blog associated with the Avast antivirus, **Silk Road**, the equivalent of Amazon on the *dark market* for counterfeit notes, weapons, drugs and other illegal products, used from 2011 to 2014 Bitcoin as the exclusive means of settling transactions; the payment system rested on an escrow deposit, and the amount of the transaction was blocked by the site pending receipt of the good or the service, the money being paid into the seller’s account only on receipt of the ‘product’. More precisely, *escrow* is an arrangement concluded between the parties to a transaction by which an independent trusted third party receives and disburses the money on their behalf. The term originally applied in the field of property purchases in the Anglo-Saxon world. On *dark web* sites, the term therefore describes an arrangement whereby, after agreement on a transaction, the buyer sends cryptocurrencies to an independent trusted third party; the latter transfers the funds to the seller only after having received confirmation from the buyer that the goods or services received correspond to the conditions agreed at the conclusion of the transaction. In exchange for this guarantee of the proper conclusion of the contract of sale, the *escrow* keeps a percentage of the funds paid. This system therefore aims to limit the risks of fraud. Ghimiray (2024) adds that the FBI was able to arrest at the end of 2013 one of the heads of the Silk Road network, Ross Ulbricht, in San Francisco, and to seize a large part of the bitcoins in his account, Silk Road indeed levying a commission of 8 to 15 % on each transaction.

In such a context of the necessity of fighting illicit trafficking by having recourse to the *blockchain*, the first texts and debates were issued by the Financial Action Task Force (**FATF**) in 2014. However, as Boulenguer (2022) specifies, it was not until December 2018 and the Buenos Aires G20 that the necessity of regulating crypto-assets was formulated for the first time, notably in order to fight money laundering and the financing of terrorism, in conformity with FATF

standards, and other measures were contemplated if necessary. It is likewise striking to observe that, despite the numerous works conducted to fight money laundering and the financing of criminal activities, the implications in terms of tax policy and tax evasion were only belatedly addressed: five years after the first FATF recommendations, the first works began. The reasons for this delay are difficult to identify, but OCDE (2020) confirms that the issues linked to tax optimisation, avoidance and evasion '*remained largely unexplored whereas they constitute an important aspect of the overall regulatory framework*', which aims notably to harmonise the **transmission of information** between the member states of the European Union and the OECD.

#### Focus...



**The limits of the transmission of information linked to movements of crypto-assets.** Although the OECD put in place a legal tool in 2014 allowing the jurisdictions of residence of taxpayers to obtain automatically tax information on an annual basis concerning *offshore* assets held with local institutions, this does not include crypto-assets. Indeed, in most cases, cryptocurrencies are not covered by the field of application of the *Common reporting standard (CRS)*, which applies only to traditional financial assets and to fiat currencies, as OCDE (2020) specifies: the 'standard for automatic exchange of financial account information in tax matters', the full name of the CRS, was drawn up by the OECD and was incorporated into European Council directive 2014/107/EU of 9 December 2014; the provisions are transposed by the law of 18 December 2015 relating to the common reporting standard (**NCD**), published in *Mémorial A* no. 244 of 24 December 2015. But the NCD does not apply to all cryptocurrency platforms. Indeed, as Boulenguer (2022) specifies, decentralised platforms have no reporting obligation under the CRS and the NCD, and are therefore unlikely to be reported to the tax authorities: this legal vacuum at the present time thus facilitates the use of cryptocurrencies for purposes of tax fraud and illicit activities.

#### Key Takeaways



Bitcoin was rapidly instrumentalised as a favoured means of payment on the *dark web* — illustrated by the Silk Road platform (2011–2014) — on account of the pseudonymity of transactions and the absence of an identifiable banking third party [Clément & Lelieur (2021)]. On the regulatory level, while the FATF initiated a reflection as early as 2014 on money laundering and the financing of terrorism, the tax evasion issues specific to crypto-assets were only belatedly addressed: decentralised platforms remain excluded from the scope of the *Common reporting standard (CRS)*, thus preserving a legal vacuum which facilitates the use of cryptocurrencies for illicit purposes without any obligation of reporting to the tax authorities [OCDE (2020)].

### 3.2.3 The difficult estimation of the fraudulent use of cryptocurrencies

Estimates of the volumes in transit on *blockchains* of a fraudulent character are essentially produced by institutions. Thus, each year Tracfin (2024) delivers its **LCB-FT** report (the fight against money laundering and the financing of terrorism) and estimates the volumes of money laundered by investigating the financing of terrorism and organised crime networks. As regards cryptocurrencies, its assessments have rested since 2019 on the data of specialised firms, such as Chainalysis, CipherTrace, Coinfirm, Elliptic, Merkle Science, Scorechain and TRM Labs. These latter companies, all American moreover, use tools for analysing *on chain* data<sup>5</sup> in order (i) to assess the risk of crypto-asset exchange platforms (such as Binance or Coinbase) and of other firms in the field of cryptography (such as Ledger), (ii) to ensure the compliance of crypto-asset wallets with LCB-FT standards relating to the execution of transactions and (iii) to investigate the source and destination of cryptocurrency funds.

In the latest report of Chainalysis (2024), an American firm specialising in the supply of *blockchain* investigation tools and services for police forces in the United States and Europe, in order to track the criminal use of cryptocurrencies and recover funds, the share of all cryptocurrency activity associated with illicit activities since 2017 is assessed each year at less than 2 %. However, since 2021, a **notable increase in the share of illicit activities in the volume of *on chain* transactions** is to be noted, going from 0.14 % in 2021 to 0.34 % in 2023. Just like the figures proposed by the FATF or Tracfin, Chainalysis nevertheless takes no account of the suspicious origin of the official (*fiat*) currency used at the time of acquisition. GAFI (2021) thus estimated the volume of doubtful, or even criminal, assets at between 0.62 % and 9.9 % of the total number of crypto-asset transactions; the amplitude of this range raises all the more questions in that some of the data on which it rests are based on the figures of Chainalysis... Hossen & Gauthier

<sup>5</sup>As Hossen & Gauthier (2024) indicate, an ***on chain transaction*** is carried out and recorded directly on the register of the *blockchain*: this is the case when one moves crypto-assets from one's Binance account to one's Coinbase account, for example. Such transactions are therefore publicly accessible on sites such as <https://www.blockchain.com/explorer>. Conversely, the ***off chain world*** designates the whole set of contracts, transactions and operations carried out outside the register of a *blockchain*; this is the case for all SEPA transfers, for example.

(2024) recall that this work of identifying provenance on the one hand, and on the other of classifying the illicit use of cryptocurrency funds, requires, besides very specific competences (extraction of the available data, or *scraping*), considerable technical means (a powerful computer with substantial storage space). Moreover, the methodology employed generally remains at the discretion of firms whose economic model rests entirely on the detection of these financial crimes. In addition, these firms are cautious so that their techniques may not be known to their targets.

### Key Takeaways



Estimates of the fraudulent use of cryptocurrencies, produced mainly by private American companies such as Chainalysis (2024), are methodologically fragile and hardly comparable: while the share of *on chain* transactions of an illicit character remains below 2 % according to Chainalysis (2024), GAFI (2021) estimates it in a range going from 0.62 to 9.9 %, an amplitude which reveals the absence of consensus — all the more so as these figures rest on opaque methodologies, take no account of the suspicious origin of the *fiat* currency used at acquisition, and are produced by actors whose economic model depends directly on the detection of these same crimes [Hossen & Gauthier (2024)].

## 3.3 The economic risks posed by cryptocurrencies

While the success of cryptocurrencies reflects a real demand for innovation in payment systems — in particular in emerging economies where they make up for the insufficiency of traditional banking infrastructures — their growing integration into the world financial system is accompanied by **multidimensional economic risks** which De Warren (2022) identifies as structural rather than cyclical: slowness of transactions, energy cost, deficient security, strong concentration of the market among a few actors, and high liquidity risks which bring **threats to financial stability**, although the current size of the crypto ecosystem still remains insufficient to trigger a systemic crisis on a world scale. On the level of monetary balances, the development of private cryptocurrencies calls into question the capacity of central banks to exercise their **monetary sovereignty**: the BCE (2026b) warns that the growth of *stablecoins* could, in the event of massive adoption, reduce bank deposits in the euro area and increase the dependence of credit institutions on more volatile resources — a risk today accentuated by the fact that more than 95 % of *stablecoins* are denominated in dollars, liable to import into the euro area monetary conditions defined outside its control.

### 3.3.1 Opportunities and challenges of payments in cryptocurrencies

The recommendations of the Landau (2018) report to the minister of the economy and finance excluded a direct regulation of activities linked to the *blockchain*, with the exception of illegal activities such as money laundering or the sale of illicit products, for example. But since 2021, one observes a **rise in agreements between payment cards and cryptocurrency platforms**. This evolution could in all likelihood make transactions in cryptocurrencies commonplace and consequently increase the number of transactions carried out with this digital means of payment. It is in this context of the normalisation of cryptocurrencies that governments are seeking to legislate in order to frame the *blockchain* sector. A world overview makes it possible to realise that emerging and developing countries do not hesitate to launch themselves into cryptocurrencies, as Bank for International Settlements (2022a) reports.

### Focus...



**Visa, Binance and European regulation.** As Boulenguer (2022) indicates, Visa joined forces with Binance in 2021 in order to propose a payment card which makes it possible to spend directly one's cryptocurrencies available on Binance, the 'Binance Card'. In reality, as Hossen & Gauthier (2024) specify, one does not pay directly in cryptocurrencies: Visa only sends an order to Binance so that the latter converts the funds available on the platform into euros; Visa then permits settlement in euros. The innovation resides in the fluidity of the operation: the user has no formality to carry out, and the payment is *in fine* as rapid as with a standard payment card. As Journal du Coin (2023) indicates, the objective of this partnership was to offer '*a tangible means of transacting, converting and spending cryptos for everyday use*'. This possibility was then wrongly perceived as a means of avoiding taxation by buying goods and services with one's capital gains realised directly in cryptocurrencies. The regulation is nevertheless clear, as Hossen & Gauthier (2024) recall, and requires private individuals to declare the capital gain realised on the resale of their *tokens* for *fiat* money (euro, dollar...) or for real goods. Regulatory pressure in the face of money laundering risks nevertheless pushed Visa to abandon on 20 December 2023 the Binance Card for the platform's European clients. More generally, as Binance (2024) indicates, Mastercard and Visa broke off their partnership with Binance amid the exchange's legal battles with the SEC and the *Commodity Futures Trading Commission* (CFTC), while in August 2023, Mastercard announced its decision to end its alliance with Binance in Argentina, Brazil, Colombia and Bahrain. Binance (2024) nevertheless announced in June 2024 the restoration of encrypted payment



services via Mastercard and Visa following ‘*a thorough review of the rigorous controls and processes put in place by Binance*’. The reactivation of the ‘Buy crypto’ option via Mastercard thus permits a maximum single purchase of up to 5000 euros for transactions in euros and up to 20 000 dollars for purchases in American dollars, as Binance (2024) specifies. The decision of Mastercard and Visa to resume their relationship with the crypto exchange thus signals a potential easing of Binance’s legal challenges, as it complies more fully with regulation.

While Europe and America therefore remain wary with regard to cryptocurrencies, in Africa a digital revolution has taken place and Bitcoin has accelerated this trend. For example, as Karayan (2022) indicates, since 27 April 2022, the Central African Republic has legalised bitcoin as an official currency alongside the CFA franc, which means that all economic agents may use Bitcoin for their daily transactions. The reasons emphasised for using digital innovations have been the fight against financial exclusion and, according to Adrian et al. (2021), ‘**cryptoisation**’ would permit the replacement of failing national currencies and the strengthening of an inclusive payment system. The most eloquent example is the digital revolution in Kenya, which Russon (2019), an economics journalist at BBC News, moreover describes as the ‘African Silicon Valley’. Indeed, in Kenya, mobile-phone-based technology (such as M-Pesa, a system of microfinancing and money transfer by mobile launched in 2007 by Vodacom and Vodafone, the main mobile operators in Africa) has reduced transaction costs and has been regarded as a concrete advance for the payment system, as Brunnermeier et al. (2021) explain: **M-Pesa** has thus allowed users to store value on their mobile phone or their mobile account in the form of electronic money. This currency may be used for several purposes, including transfers to other users, payments for goods and services and conversion to and from cash. More precisely, the empirical analysis of Ndung’u et al. (2016) has demonstrated that these currencies have permitted a reduction in transaction costs of 65 % relative to the period 2006-2013 and an annual contribution to GDP growth of roughly 0.45 percentage point. This result therefore strengthens the idea of the **role of financial innovations in growth**, through a fall in transaction costs and the emergence of alternative means of financing for firms.

### 3.3.2 Cryptocurrencies and stability risks for *fiat* currencies

However, this rise of cryptocurrencies in the world economy brings to the fore a stability risk for fiat currencies. The Chinese example is the most telling: whereas the large Chinese firms (WeChat with WeChatPay and Alibaba with AliPay) initially offered payments in cryptocurrencies, the Chinese government understood the issues raised by such financial innovations and toughened anti-trust measures in order to avoid being dispossessed of its monetary sovereignty, and notably had these practices banned, as Adrian et al. (2021) explain. Moreover, since 2014, the Chinese authorities have drawn up a project for a digital currency produced by the People’s Bank of China (the country’s central bank), the **e-yuan**, which made its official appearance during the 2022 winter Olympic Games in Beijing. This movement belongs more globally to the ever more marked desire of central banks to introduce their own currency into the digital landscape. Now the **coexistence of decentralised cryptocurrencies and CBDC** poses a new challenge to government, notably from the point of view of **competition between currencies**. While Hayek (1976) suggested that governments should not manage monetary questions since the market is capable of equilibrating itself without the intervention of a regulatory body, such an approach is today called into question on account of self-fulfilling expectations and of the state’s obligation to intervene in order to curb crises and improve the economic situation, among other things. A struggle between the old monetary and financial world and the new is then taking shape before our eyes with successive phases of groping and of hesitations, or even of inertia or procrastination on the part of public and private bodies, as Auer et al. (2022) show.

In order to shed light on this struggle, Fernández-Villaverde & Sanches (2018) used a theoretical model based on three agents (buyers, sellers and entrepreneurs) and two types of monetary asset (central bank and private money). According to their study, the equilibrium is not efficient in an environment of private currencies. Logically, private currencies negatively affect monetary policy based on the famous **monetary growth rule** (that is to say a monetary inflation) since the social equilibrium is not optimal. By contrast, the introduction of a ‘**legitimate currency**’ introduced by a monetary authority has a positive effect on economic growth and the effectiveness of monetary policy is restored. In the same vein, Kang & Lee (2022) a model in which Bitcoin and central bank currencies are a medium of exchange. Their model belongs to the framework of the approach of Hayek (1976): during the 1970s, inflation was high and the solution brought by Hayek was to introduce ‘non-central bank’ currencies in order to create competition. According to Hayek (1976), the latter is healthy since it would push banks to offer a more stable currency.



## Debate

**Is competition between an official currency and a cryptocurrency healthy?** The conclusion of Kang & Lee (2022) deduced from their modelling is the following: the welfare of an economy where only central bank money is available is higher than the bi-monetary model (Bitcoin and central bank money) when inflation is low; Bitcoins are in competition with the national currency only if inflation is relatively high. Moreover, when the inflation rate is below 2 %, there is a welfare loss of 0.0048 % measured by consumption in the case of a bi-monetary economy — the figure nevertheless depending on the calibration of the model. The welfare gap between a bi-currency economy and a monetary economy with central banks nevertheless narrows when inflation increases, the effect of monetary substitution explaining this result. In order to avoid this situation, must central banks then maintain a very low inflation rate? Is competition between currencies healthy for economies? The answer is probably negative today, since private sector money is not secured on account of the absence of a lender of last resort, and this is why Europe today aims to frame the blockchain sector more closely.

### 3.3.3 Sovereignty risks

The multiplication of ICO induces risks of anarchic, deregulated or unregulated development. These risks are exacerbated when a firm creates its own cryptocurrency, since such an act can potentially call into question the role of the state in its sovereign privilege of minting money. Now, as the French minister of the economy and finance Bruno Le Maire recalled in an interview of June 2019 relayed by Capital (2019), *'the capacity to issue securities, to constitute a reserve and to be a lender of last resort: all this is not conceivable [for private actors. Money should] remain in the hands of states and not of private firms which answer to private interests'*. This **conflict of sovereignty between states and private firms** was exacerbated by the Libra project, created in May 2019 then renamed **Diem** in December 2020, carried and then dropped by Facebook (Meta). Diem was to permit monetary exchanges from WhatsApp and Messenger (subsidiaries of Meta) through several *stablecoins*: a 'diem euro' aligned on the price of the euro, a 'diem dollar' aligned on that of the American dollar... The involvement of twenty-one firms in the **Calibra association** (Facebook, Vodafone, Uber, Spotify, Visa, PayPal, Mastercard...), the promoter of the Libra / Diem project, was naturally synonymous with a potential of several billion users becoming clients. But how could one make sure that, behind each diem euro, there was indeed a real euro? The necessity of making sure of the collateralisation of the cryptocurrency is indeed a crucial issue for regulators, as Hossen & Gauthier (2024) mention: if, behind a diem euro, there are only sixty cents in Facebook's coffers (therefore a collateralisation of the diem euro to the tune of 60 %), it is as if Facebook had created without any agreement the missing forty cents. And while the directors of Calibra had asserted, in order to reassure banking institutions, that the Facebook currency was not conceived to replace the American dollar or any other currency, political leaders became anxious about the influence of the holders of one cryptocurrency on the price of all cryptocurrencies or even of fiat currencies (that is to say currencies having legal tender status such as the euro or the dollar): *'it would suffice for private actors to decide to modify the euro share of the reserve fund for our common currency to depreciate or appreciate brutally'* Le Maire (2019) had thus asserted in an opinion piece.

Beyond the question of regulation, governments were likewise anxious about the **transparency of the project**. Indeed, the scope of the project and the capacity of the digital giants to create trust in their payment and credit services alerted the competent authorities, anxious to preserve their power in terms of money creation, and made it possible to initiate a reflection on the potential architectures of digital currencies, as Bédu et al. (2022) recall. More precisely, while the first generation of cryptocurrencies (represented notably by Bitcoin) had as its objective to eliminate financial intermediaries, Mark Zuckerberg's project for its part announces a new generation of cryptocurrencies which associates firms developing payment applications (Mastercard, PayPal, Visa) with a *blockchain*. What then of the twofold issue of respecting the transparency obligations applicable to the processing of personal data under the general data protection regulation (GDPR) and of the cyber-protection of users? While, according to an article by ORWL (2019), the stakeholders of Calibra had asserted that *'the account information or financial data of Calibra's clients will not be used to improve advertising targeting across Facebook Inc.'s range of products'*, Libra raises questions in terms of transparency between the entities of the group on the one hand, and between the different members of the consortium on the other. Indeed, the virtual currency project belongs to an overall strategy in which the user could simultaneously interact with his community on WhatsApp or Messenger, buy products online offered via sponsored posts on Facebook or Instagram and finalise his purchase in 'Facebook money', and this without ever leaving the Meta ecosystem. The fear of seeing the consortium use and resell to third parties the banking information of these clients is very much present — a fear to which is added that of cybersecurity: an intrusion could affect all the members of the consortium. Beyond this, the opacity of virtual currencies and the flaws in tax regulations that we have detailed *supra* suggest that Diem would have contributed to the circumvention of tax, a strategy already



present at Meta, a firm regularly accused by civil society of paying a derisory tax relative to its profits: these are the points that Le Maire (2019) notably raised in the aforementioned opinion piece.

#### Key Takeaways



The rise of cryptocurrencies in everyday payments — illustrated by the Binance Card and, in emerging countries, by initiatives such as M-Pesa — and the growth of ICO raise three major risks for states [Hossen & Gauthier (2024)]: a risk to monetary stability — private currencies weaken the effectiveness of monetary policy, except when inflation is high, according to Kang & Lee (2022) —; a risk to sovereignty — cryptoisation may short-circuit the sovereign privilege of minting money, as Facebook's Libra/Diem project illustrated, finally abandoned under pressure from regulators [Le Maire (2019)] —; and a risk to transparency and cybersecurity — the opacity of cryptocurrency flows and the legal vacuum surrounding digital assets would facilitate tax evasion and the circumvention of the GDPR [ORWL (2019)].



# Bibliography

- Acemoglu, D. (2024). The Simple Macroeconomics of AI. <https://economics.mit.edu/sites/default/files/2024-04/The%20Simple%20Macroeconomics%20of%20AI.pdf>. Consulté le 26/01/2026.
- Acerbi, C. & Tasche, D. (2002). On the coherence of expected shortfall. *Journal of Banking & Finance*, 26(7), 1487–1503. <https://www.financerisks.com/filedati/WP/EVT/CHOERENT%20EXPECTED%20SHORTFALL.pdf>. Consulté le 20/01/2026.
- ACPR (1991). Études. *Bulletin de la Commission bancaire*. [https://acpr.banque-france.fr/sites/default/files/media/2017/10/30/cb\\_bul\\_04\\_etu\\_01.pdf](https://acpr.banque-france.fr/sites/default/files/media/2017/10/30/cb_bul_04_etu_01.pdf). Consulté le 06/06/2024.
- ACT (2021). Cash management and forecasting: Best Practices for Corporate Treasury. <https://hub.treasurers.org/bedrock-of-treasury-cash-flow-forecasting/>. Consulté le 25/01/2026.
- Actu juridique (2024). Le régime fiscal des SIIC. <https://www.actu-juridique.fr/fiscalite/droit-fiscal/le-regime-fiscal-des-siic/>. Consulté le 01/03/2026.
- Adrian, T., He, D., & Narain, A. (2021). Cryptoactifs: pour une réglementation mondiale complète, cohérente et coordonnée. <https://www.imf.org/fr/Blogs/Articles/2021/12/09/blog120921-global-crypto-regulation-should-be-comprehensive-consistent-coordinated>. Consulté le 20/01/2026.
- Agarwal, S. (2025). Répartition du PIB mondial en 2025: Qui alimente l'économie mondiale? <https://indiamacroindicators.co.in/resources/blogs/world-gdp-breakdown-2025-who-powers-the-global-economy>. Consulté le 03/03/2026.
- Agence internationale de l'énergie (2024). World energy outlook 2024. <https://iea.blob.core.windows.net/assets/140a0470-5b90-4922-a0e9-838b3ac6918c/WorldEnergyOutlook2024.pdf>. Consulté le 26/02/2026.
- Aglietta, M. & Valla, N. (2016). Taux d'intérêt négatifs: décryptage d'une anomalie. *L'Économie politique*, 70(2), 8–23. <https://www.cairn.info/revue-l-economie-politique-2016-2-page-8.htm>. Consulté le 22/01/2026.
- Aglietta, M. & Valla, N. (2017). *Macroéconomie financière*. La Découverte, Paris. <http://livre21.com/LIVREF/F4/F004061.pdf>. Consulté le 20/01/2026.
- Ahmed, K. & Calice, G. (2024). The effects of the EBA's stress testing framework on banks' lending. *Economic Modelling*, 130, 106583. <https://pure.bangor.ac.uk/ws/portalfiles/portal/68735069/1-s2.0-S0264999323004364-main.pdf>. Consulté le 24/01/2026.
- Ainvest (2025). BNB tokenomics: How deflationary dynamics and ecosystem growth reshape regulated crypto era. <https://www.ainvest.com/news/bnb-tokenomics-deflationary-dynamics-ecosystem-growth-reshape-regulated-crypto-era-2508/>. Consulté le 05/04/2026.
- Akram, Q. F., Rime, D., & Sarno, L. (2008). Arbitrage in the foreign exchange market: Turning on the microscope. *Journal of International Economics*, 76(2), 237–253. <https://www.econstor.eu/bitstream/10419/209859/1/nb-wp2005-12.pdf>. Consulté le 20/01/2026.
- Albert, M. (1991). *Capitalisme contre capitalisme*. Paris: Seuil. <https://excerpts.numilog.com/books/9782021444896.pdf>. Consulté le 07/12/2025.

- All News (2023). Les points forts de la dette d'infrastructure. <https://www.allnews.ch/content/points-de-vue/les-points-forts-de-la-dette-dinfrastructure>. Consulté le 02/03/2026.
- Allard-Prigent, C., Audenis, C., Berger, K., Carnot, N., Duchêne, S., & Pesin, F. (2002). Présentation du modèle Mésange, modèle économétrique de simulation et d'analyse générale de l'économie. <https://www.semanticscholar.org/paper/PR%C3%89SENTATION-DU-MOD%C3%88LE-M%C3%89SANGE-Mod%C3%A8le-%C3%89conom%C3%A9trique-Allard-Prigent-Audenis/29c8231efef95a0212cab5eb9b5cb9e5f43833b5>. Consulté le 20/01/2026.
- Allayannis, G. & Weston, J. P. (2001). The use of foreign currency derivatives and firm market value. *The Review of Financial Studies*, 14(1), 243–276. [https://www.ruf.rice.edu/~westonj/pub/2001\\_RFS.pdf](https://www.ruf.rice.edu/~westonj/pub/2001_RFS.pdf). Consulté le 03/03/2026.
- Allianz Global Investors (2024). Dette d'infrastructure — stratégie *Core Infrastructure Debt*. <https://fr.allianzgi.com/fr-fr/investisseursprofessionnels/nos-expertises-cles/marches-privées/dette-d-infrastructure>. Consulté le 02/03/2026.
- Allnews (2023). Les États-unis et la crypto: entre institutionnalisation et répression. <https://www.allnews.ch/content/points-de-vue/les-etats-unis-et-la-crypto-entre-institutionnalisation-et-repression>. Consulté le 06/04/2026.
- Alstadsæter, A., Godar, S., Nicolaides, P., & Zucman, G. (2024). *Global Tax Evasion Report 2024*. EU Tax Observatory. [https://www.taxobservatory.eu/www-site/uploads/2023/10/global\\_tax\\_evasion\\_report\\_24.pdf](https://www.taxobservatory.eu/www-site/uploads/2023/10/global_tax_evasion_report_24.pdf). Consulté le 26/01/2026.
- Alvarez, F., Argente, D., & Van Patten, D. (2023). Are cryptocurrencies currencies? Bitcoin as legal tender in El Salvador. *Science*, 382(6677). <https://www.science.org/doi/abs/10.1126/science.add2844>. Consulté le 06/06/2024.
- Alvarez & Marsal (2024). A&m valuation insights: December 2024. <https://www.alvarezandmarsal.com/sites/default/files/2024-12/AM%20Spain%20-%20Valuation%20Insights%20-%20December%2024.pdf>. Consulté le 26/01/2026.
- Alvarez-Plata, P. & Garcia-Herrero, A. (2007). Consequences for monetary policy in dollarized economies. <https://www.bis.org/repofficepubl/arpresearch200709.1.pdf>. Consulté le 22/01/2026.
- AMF (2018a). Ce qu'il faut savoir sur les placements collectifs (fonds et SICAV). <https://www.amf-france.org/fr/espace-epargnants/comprendre-les-produits-financiers/placements-collectifs/ce-quil-faut-savoir-sur-les-placements-collectifs-fonds-et-sicav>. Consulté le 20/01/2026.
- AMF (2018b). L'AMF publie une analyse sur les tendances des initial coin offerings (ICO). <https://www.amf-france.org/fr/actualites-publications/communiqués/communiqués-de-lamf/lamf-publie-une-analyse-sur-les-tendances-des-initial-coin-offerings-ico>. Consulté le 11/08/2024.
- AMF (2018c). Les CFD (*contracts for difference*). <https://www.amf-france.org/fr/espace-epargnants/comprendre-les-produits-financiers/produits-complexes/cfd>. Consulté le 13/03/2026.
- AMF (2019a). L'amf délivre son premier visa sur une offre au public de jetons (ICO). <https://www.amf-france.org/fr/actualites-publications/communiqués/communiqués-de-lamf/lamf-delivre-son-premier-visa-sur-une-offre-au-public-de-jetons-ico>. Consulté le 07/04/2026.
- AMF (2019b). Vers un nouveau régime pour les crypto-actifs en france. <https://www.amf-france.org/fr/actualites-publications/actualites/vers-un-nouveau-regime-pour-les-crypto-actifs-en-france>. Consulté le 07/04/2026.
- AMF (2024). Cartographie 2024 des marchés et des risques. <https://www.amf-france.org/sites/institutionnel/files/private/2024-07/cartographie-2024-des-marchés-et-des-risques.pdf>. Consulté le 26/02/2026.
- AMF (2025a). Cartographie des produits structurés commercialisés auprès des investisseurs particuliers en france. <https://www.amf-france.org/sites/institutionnel/files/private/2025-03/cartographie-pole-commun-2024.pdf>. Consulté le 17/03/2026.

- AMF (2025b). ETF: l'activité des investisseurs particuliers en forte progression en 2024. <https://www.amf-france.org/fr/actualites-publications/actualites/etf-lactivite-des-investisseurs-particuliers-en-forte-progression-en-2024>. Consulté le 26/02/2026.
- AMIS (2011). Agricultural market information system (AMIS). <https://www.amis-outlook.org/home>. Consulté le 20/01/2026.
- Amundi (2025). Résultats annuels et du 4<sup>e</sup> trimestre 2024. <https://legroupe.amundi.com/article/resultats-annuels-et-du-4eme-trimestre-2024>. Consulté le 03/03/2026.
- Amundi Asset Management (2024). Lyxor PEA Monde (MSCI World) UCITS ETF – Capi. (FR0011869353). [https://www.amundiassetmanagement.fr/pdfDocuments/download/0e8aa8b4-6d9e-4508-9651-2ffd24befe15/PSC\\_1\\_UM94386\\_FRA\\_FRA\\_20240126\\_20240130.PDF](https://www.amundiassetmanagement.fr/pdfDocuments/download/0e8aa8b4-6d9e-4508-9651-2ffd24befe15/PSC_1_UM94386_FRA_FRA_20240126_20240130.PDF). Consulté le 26/02/2026.
- Amundi Asset Management (2025). Amundi PEA S&P 500 Screened UCITS ETF Acc (FR0013412285). [https://www.amundi.fr/fr\\_part/product/view/FR0013412285](https://www.amundi.fr/fr_part/product/view/FR0013412285). Consulté le 20/01/2026.
- Amzallag, C. (2022). *Mathématiques pour filières commerciales: cours et exercices corrigés pour étudiants en BUT et en BTS*, chapitre 2. Mathématiques financières, (pp. 31–69). EMS Editions. <https://stm.cairn.info/mathematiques-pour-filieres-commerciales--9782376876137-page-31>. Consulté le 20/01/2026.
- Andrianto, Y. & Diputra, Y. (2017). The effect of cryptocurrency on investment portfolio effectiveness. *Journal of finance and accounting*, 5(6), 229–238. <https://www.sciencepublishinggroup.com/article/10.11648/j.jfa.20170506.14>. Consulté le 06/06/2024.
- ANIL (2025). Modalités d'application du dispositif Loc'Avantages. <https://www.anil.org/aj-dispositif-loc-avantages/>. Consulté le 01/03/2026.
- Antonin, C. (2009). La dette souveraine française et les agences de notation. *OFCE Briefing Paper*. <https://www.ofce.sciences-po.fr/pdf/clair&net/clair&net-72.pdf>. Consulté le 20/01/2026.
- Aponte-Novoa, F. A., Orozco, A. L. S., Villanueva-Polanco, R., & Wightman, P. (2021). The 51 % attack on blockchains: a mining behavior study. *IEEE Access*, 9, 140 549–140 564. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9567686>. Consulté le 06/06/2024.
- AquaVest (2025). Top 4 water ETFs for 2025. <https://www.youtube.com/watch?v=BPPvKH4nVvg>. Consulté le 01/03/2026.
- Arcand, J. L., Berkes, E., & Panizza, U. (2012). Too much finance? *IMF Working Papers*, 12(161). <https://www.imf.org/external/pubs/ft/wp/2012/wp12161.pdf>. Consulté le 20/01/2026.
- ARCOM (2024). Rapport annuel 2023 de l'Arcom. <https://www.arcom.fr/se-documenter/etudes-et-donnees/etudes-bilans-et-rapports-de-larcom/rapport-annuel-2023-de-larcom>. Consulté le 05/04/2026.
- ART (2023). Les contrats de concession autoroutière en France. <https://www.autorite-transport.fr/les-autoroutes/les-contrats-de-concession/>. Consulté le 02/03/2026.
- Arthur Loyd (2023). Qu'est-ce que l'indice INSEE ILAT ? Arthur Loyd. <https://www.arthur-loyd.com/faq/a-quoi-servent-les-indices-insee-ilat>. Consulté le 01/03/2026.
- Artus, P. (2019). *Discipliner la finance*. Odile Jacob, Paris. [https://www.melchior.fr/sites/melchior/files/EEE/2019/intro\\_discipliner\\_la\\_finance\\_0.pdf](https://www.melchior.fr/sites/melchior/files/EEE/2019/intro_discipliner_la_finance_0.pdf). Consulté le 20/01/2026.
- Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203–228. [https://www.ressources-actuarielles.net/EXT/ISFA/1226.nsf/0/0bd2345fdff664eec1256e16004ff688/\\$FILE/ARTZNER%201998.pdf](https://www.ressources-actuarielles.net/EXT/ISFA/1226.nsf/0/0bd2345fdff664eec1256e16004ff688/$FILE/ARTZNER%201998.pdf). Consulté le 20/01/2026.
- Atradius (2025). B2B payment practices trends in Western Europe 2025. <https://group.atradius.com/knowledge-and-research/reports/b2b-payment-practices-trends-western-europe-2025>. Consulté le 25/01/2026.
- Atzei, N., Bartoletti, M., & Cimoli, T. (2017). A survey of attacks on Ethereum smart contracts (SoK). <https://eprint.iacr.org/2016/1007.pdf>. Consulté le 05/04/2026.

- Auer, R., Farag, M., Lewrick, U., Orazem, L., & Zoss, M. (2022). Banking in the shadow of bitcoin? The institutional adoption of cryptocurrencies. *BIS Working Papers*, 1012. <https://www.bis.org/publ/work1013.pdf>. Consulté le 06/06/2024.
- Aurissergues, E. (2022). Quels effets de la hausse des taux d'intérêt sur la croissance économique française? Un tour d'horizon des modèles macroéconométriques. *OFCE, le blog*. <https://urls.fr/NJN1ot>. Consulté le 06/06/2024.
- Autorité bancaire européenne (2023). EBA publishes the results of its 2023 EU-wide stress test. <https://www.eba.europa.eu/publications-and-media/press-releases/eba-publishes-results-its-2023-eu-wide-stress-test>. Consulté le 24/01/2026.
- Ayming (2025). France 2030: bilan 2024 des investissements et de l'innovation en France. <https://www.ayming.fr/newsroom/actualites-avis-dexpert/france-2030-bilan-2024/>. Consulté le 23/03/2026.
- Azzouzi Idrissi, Y. & Madiès, P. (2012). Les risques de liquidité bancaire: définitions, interactions et régulation. *Revue d'économie financière*, 3(107), 315–332. <https://shs.cairn.info/revue-d-economie-financiere-2012-3-page-315>. Consulté le 20/01/2026.
- Back, A., Corallo, M., Dashjr, L., Friedenbach, M., Maxwell, G., Miller, A., Poelstra, A., Timón, J., & Wuille, P. (2014). Enabling blockchain innovations with pegged sidechains. <https://blockstream.com/sidechains.pdf>. Consulté le 05/04/2026.
- Bain & Company (2025). Looking back at M&A in 2025: Behind the Great Rebound. <https://www.bain.com/insights/looking-back-m-and-a-report-2026/>. Consulté le 25/01/2026.
- Balaresque, N. & Oster, D. (2021). *Cartographie*. Nathan, Paris. <https://site.nathan.fr/livres/la-cartographie-9782091629834.html>. Consulté le 19/12/2025.
- Bank for International Settlements (2008). OTC derivatives market activity in the second half of 2007. [https://www.bis.org/publ/otc\\_hy0805.pdf](https://www.bis.org/publ/otc_hy0805.pdf). Consulté le 13/03/2026.
- Bank for International Settlements (2018). Incentives to centrally clear over-the-counter (OTC) derivatives. <https://www.bis.org/publ/othp29.pdf>. Consulté le 07/03/2026.
- Bank for International Settlements (2022a). CBDCs in emerging market economies. <https://www.bis.org/publ/bppdf/bisap123.pdf>. Consulté le 07/04/2026.
- Bank for International Settlements (2022b). OTC foreign exchange turnover in april 2022. [https://www.bis.org/statistics/rpfx22\\_fx.pdf](https://www.bis.org/statistics/rpfx22_fx.pdf). Consulté le 02/03/2026.
- Bank for International Settlements (2023). *OTC Derivatives Statistics at End-June 2023*. Technical report, BIS. [https://www.bis.org/publ/otc\\_hy2311.htm](https://www.bis.org/publ/otc_hy2311.htm). Consulté le 07/03/2026.
- Bank for International Settlements (2025a). Lessons learnt on CBDCs. <https://www.bis.org/publ/othp73.pdf>. Consulté le 06/04/2026.
- Bank for International Settlements (2025b). OTC derivatives statistics at end-June 2025. [https://www.bis.org/publ/otc\\_hy2512.htm](https://www.bis.org/publ/otc_hy2512.htm). Consulté le 03/03/2026.
- Bank for International Settlements (2025c). OTC foreign exchange turnover in april 2025. [https://www.bis.org/statistics/rpfx25\\_fx.pdf](https://www.bis.org/statistics/rpfx25_fx.pdf). Consulté le 02/03/2026.
- Bank Policy Institute (2023). *Why is the FRTB Expected Shortfall Calculation Designed as It Is?* Technical report, Bank Policy Institute. <https://bpi.com/why-is-the-frtb-expected-shortfall-calculation-designed-as-it-is/>. Consulté le 07/03/2026.
- Banque de France (2013). Les dangers liés au développement des monnaies virtuelles: l'exemple du bitcoin. *Focus*, 10. [https://publications.banque-france.fr/sites/default/files/medias/documents/focus-10\\_2013-12-05\\_fr.pdf](https://publications.banque-france.fr/sites/default/files/medias/documents/focus-10_2013-12-05_fr.pdf). Consulté le 06/06/2024.
- Banque de France (2021a). *Le référentiel de financement des entreprises*, chapitre 311. Les actions. Banque de France. <https://www.banque-france.fr/system/files/2024-04/823415%20Fiche-311.pdf>. Consulté le 06/06/2024.

- Banque de France (2021b). Le seignuriage. *ABC de l'économie*, 10. <https://www.banque-france.fr/fr/publications-et-statistiques/publications/le-seignuriage>. Consulté le 06/06/2024.
- Banque de France (2023a). Le lien entre monnaie et inflation depuis 2008. [https://www.banque-france.fr/system/files/2023-01/bdf\\_232-8\\_monnaie-inflation.pdf](https://www.banque-france.fr/system/files/2023-01/bdf_232-8_monnaie-inflation.pdf). Consulté le 22/01/2026.
- Banque de France (2023b). Masse monétaire, agrégats monétaires et base monétaire. [https://abc-economie.banque-france.fr/sites/default/files/medias/documents/eeb\\_masse-monetaire.pdf](https://abc-economie.banque-france.fr/sites/default/files/medias/documents/eeb_masse-monetaire.pdf). Consulté le 31/08/2024.
- Banque de France (2023c). Moyens de paiement. *Bulletin de la Banque de France*, 249(4). [https://www.banque-france.fr/system/files/2023-12/BDF249-4\\_Cartographie\\_web.pdf](https://www.banque-france.fr/system/files/2023-12/BDF249-4_Cartographie_web.pdf). Consulté le 31/08/2024.
- Banque de France (2024a). The impact of oil prices on inflation in France and the euro area. <https://www.banque-france.fr/en/publications-and-statistics/publications/impact-oil-prices-inflation-france-and-euro-area>. Consulté le 27/02/2026.
- Banque de France (2024b). Le financement des investissements par Bpifrance. <https://www.banque-france.fr/system/files/2024-04/823415%20Fiche-414.pdf>. Consulté le 23/03/2026.
- Banque de France (2024c). Les montages à effet de levier LBO. <https://www.banque-france.fr/system/files/2024-04/823415-Fiche-431.pdf>. Consulté le 20/01/2026.
- Banque de France (2024d). Les politiques monétaires non conventionnelles (2008-2022). <https://abc-economie.banque-france.fr/sites/default/files/medias/documents/240304-les-politiques-monetaires-non-conventionnelles.pdf>. Consulté le 20/01/2026.
- Banque de France (2024e). Les programmes d'achats d'actifs. <https://www.banque-france.fr/fr/strategie-monetaire/cadre-operationnel/programmes-achat-actifs>. Consulté le 20/01/2026.
- Banque de France (2024f). Les émissions de billets augmentent et leur vitesse de circulation se replie. *Bloc-notes Éco*, 316. <https://www.banque-france.fr/fr/publications-et-statistiques/publications/les-emissions-de-billets-augmentent-et-leur-vitesse-de-circulation-se-replie>. Consulté le 22/01/2026.
- Banque de France (2024g). Rapport annuel 2023 sur l'épargne réglementée. <https://www.banque-france.fr/fr/publications-et-statistiques/publications/rapport-sur-lepargne-reglementee-2023>. Consulté le 20/01/2026.
- Banque de France (2025a). Indices obligataires au 31 décembre 2025. <https://www.banque-france.fr/fr/statistiques/taux-et-cours/indices-obligataires-2025-12-31>. Consulté le 28/01/2026.
- Banque de France (2025b). La crise de 1929. <https://www.banque-france.fr/fr/publications-et-statistiques/publications/la-crise-de-1929>. Consulté le 22/01/2026.
- Banque de France (2025c). La politique monétaire conventionnelle. [https://www.banque-france.fr/system/files/2025-02/Pol-mon-conv\\_2025-02-12.pdf](https://www.banque-france.fr/system/files/2025-02/Pol-mon-conv_2025-02-12.pdf). Consulté le 20/01/2026.
- Banque de France (2025d). Les taux d'intérêt négatifs. <https://www.banque-france.fr/fr/publications-et-statistiques/publications/les-taux-dinteret-negatifs>. Consulté le 22/01/2026.
- Banque de France (2025e). Statistiques relatives aux paiements: deuxième semestre 2024. <https://www.banque-france.fr/fr/communiques-de-presse/statistiques-relatives-aux-paiements-deuxieme-semester-2024>. Consulté le 06/04/2026.
- Banque de France (2026). Crédits aux particuliers – Décembre 2025. <https://www.banque-france.fr/fr/statistiques/credit/credits-aux-particuliers-2025-12>. Consulté le 24/02/2026.
- Banque des Territoires (2021). Bilan des APL en temps réel : 1,1 milliard d'économies, plus de "perdants"... et plus de "gagnants"! <https://www.banquedesterritoires.fr/bilan-des-apl-en-temps-reel-11-milliard-deconomies-plus-de-perdants-et-plus-de-gagnants>. Consulté le 01/03/2026.
- Banque des Territoires (2025). Eau, assainissement: un prix global toujours en hausse. <https://www.banquedesterritoires.fr/eau-assainissement-un-prix-global-toujours-en-hausse>. Consulté le 23/03/2026.



- Banque du Canada (2024). Market structure of cryptoasset exchanges. [https://publications.gc.ca/collections/collection\\_2024/banque-bank-canada/FB3-7-2024-2-eng.pdf](https://publications.gc.ca/collections/collection_2024/banque-bank-canada/FB3-7-2024-2-eng.pdf). Consulté le 05/04/2026.
- Banque mondiale (1994). Croissance du PIB (% annuel) – Japon. <https://donnees.banquemondiale.org/indicateur/NY.GDP.MKTP.KD.ZG?locations=JP>. Consulté le 26/01/2026.
- Banque mondiale (2013). The social impact of financial crises: Evidence from the global financial crisis. <https://openknowledge.worldbank.org/bitstream/10986/16912/1/WPS6703.pdf>. Consulté le 22/01/2026.
- Banque mondiale (2023). Market capitalization of listed domestic companies (current us\$). <https://data.worldbank.org/indicator/CM.MKT.LCAP.CD>. Consulté le 26/02/2026.
- Bardaji, J., Campagne, B., Khder, M., Lafféter, Q., & Simon, O. (2017). Le modèle macroéconométrique mésange: réestimation et nouveautés. *Documents de travail de l'INSEE*, G2017(04). <https://www.insee.fr/fr/statistiques/2848300>. Consulté le 06/06/2024.
- Barro, R. J. & Gordon, D. B. (1983). Rules, discretion and reputation in a model of monetary policy. *Journal of monetary economics*, 12(1), 101–121. <https://www.sfu.ca/~kkasa/barro83.pdf>. Consulté le 20/01/2026.
- Baudot, J.-Y. (2024). Le taux d'intérêt continu. <http://www.jybaudot.fr/Gestionfi/tauxcontinu.html>. Consulté le 20/01/2026.
- Baughman, G., Carapella, F., Gerszten, J., & Mills, D. (2022). The stable in stablecoins. <https://www.federalreserve.gov/econres/notes/feds-notes/the-stable-in-stablecoins-20221216.html>. Consulté le 11/08/2024.
- Baumol, W. J. (1952). The transactions demand for cash: an inventory theoretic approach. *The Quarterly journal of economics*, 66(4), 545–556. <https://pages.stern.nyu.edu/~wbaumol/TheTransactionsDemandForCash.pdf>. Consulté le 06/06/2024.
- BBC News (2014). Microsoft to accept payments made in bitcoins. <https://www.bbc.com/news/technology-30377654>. Consulté le 05/04/2026.
- BCE (2015). Qu'est-ce qu'une banque centrale? <https://www.ecb.europa.eu/ecb-and-you/explainers/tell-me/html/what-is-a-central-bank.fr.html>. Consulté le 20/01/2026.
- BCE (2023). Annual report 2022. <https://www.ecb.europa.eu/press/annual-reports-financial-statements/annual/html/ecb.ar2022~8ae51d163b.en.html>. Consulté le 24/01/2026.
- BCE (2024a). Key ECB interest rates. <https://data.ecb.europa.eu/main-figures/ecb-interest-rates-and-exchange-rates/key-ecb-interest-rates>. Consulté le 20/01/2026.
- BCE (2024b). Souscription au capital. <https://www.ecb.europa.eu/ecb/orga/capital/html/index.fr.html>. Consulté le 12/12/2025.
- BCE (2024c). Study on the payment attitudes of consumers in the euro area (SPACE) 2024. [https://www.ecb.europa.eu/stats/ecb\\_surveys/space/html/ecb.space2024~19d46f0f17.en.html](https://www.ecb.europa.eu/stats/ecb_surveys/space/html/ecb.space2024~19d46f0f17.en.html). Consulté le 22/01/2026.
- BCE (2026a). Euro short-term rate (€Str). [https://www.ecb.europa.eu/stats/financial\\_markets\\_and\\_interest\\_rates/euro\\_short-term\\_rate/html/index.en.html](https://www.ecb.europa.eu/stats/financial_markets_and_interest_rates/euro_short-term_rate/html/index.en.html). Consulté le 07/03/2026.
- BCE (2026b). La BCE alerte sur l'impact potentiel des stablecoins sur la politique monétaire. <https://www.clubpatrimoine.com/contenus/stablecoins-bce-risque>. Consulté le 07/04/2026.
- Beaudry, P., Hou, C., & Portier, F. (2022). The dominant role of expectations and broad-based supply shocks in driving inflation. *NBER Working Paper*, w32322. [https://www.nber.org/system/files/working\\_papers/w32322/w32322.pdf](https://www.nber.org/system/files/working_papers/w32322/w32322.pdf). Consulté le 06/06/2024.
- Beck, T., Demirgüç-Kunt, A., & Levine, R. (2003). Law and finance: why does legal origin matter? *Journal of comparative economics*, 31(4), 653–675. [https://www.nber.org/system/files/working\\_papers/w9379/w9379.pdf](https://www.nber.org/system/files/working_papers/w9379/w9379.pdf). Consulté le 06/06/2024.
- Bédu, N., Granier, C., Malherbe, L., & Montalban, M. (2022). Les banques françaises face à l'émergence des FinTechs de paiement et de crédit: dynamique réglementaire et changement technique. *Revue d'économie industrielle*, (pp. 169–206). <https://journals.openedition.org/rei/11006>. Consulté le 20/01/2026.

- Bennett, C. (2014). Trading volatility. <https://www.trading-volatility.com/Trading-Volatility.pdf>. Consulté le 11/08/2024.
- Berend, L. & Prüser, J. (2024). The transmission of monetary policy via common cycles in the euro area. <https://arxiv.org/html/2410.05741v1>. Consulté le 22/01/2026.
- Berleemann, M., Freese, J., & Knoth, S. (2011). ECB presidency and inflation aversion among the citizens of European countries. *CESifo Forum*, 12(2), 88–94. <https://www.ifo.de/DocDL/forum2-11-special3.pdf>. Consulté le 22/01/2026.
- Bernanke, B. S. (1995). *Essays on the Great Depression*. Princeton University Press. <http://ndl.ethernet.edu.et/bitstream/123456789/62523/1/144.pdf.pdf>. Consulté le 26/01/2026.
- Bernanke, B. S. (2010). Central bank independence, transparency, and accountability. <https://www.federalreserve.gov/newsevents/speech/bernanke20100525a.htm>. Consulté le 24/01/2026.
- BestBrokers (2025). Gold's 2025 rally: Mapping central banks' demand. <https://www.bestbrokers.com/gold-brokers/golds-2024-rally-mapping-central-banks-demand-for-the-precious-metal-in-2024/>. Consulté le 27/02/2026.
- Beurre, J., Brigo, D., Schiemert, D., & Stoye, G. (2009). *The CDS Big Bang: Modelling, Pricing and Counterparty Valuation Implications of New CDS Market Standards*. Technical report, Imperial College London. <https://www.ma.imperial.ac.uk/~dbrigo/cdsbigbang.pdf>. Consulté le 12/03/2026.
- Binance (2017). Binance — ICO whitepaper. <https://www.exodus.com/assets/docs/binance-coin-whitepaper.pdf>. Consulté le 05/04/2026.
- Binance (2024). Les paiements Mastercard et Visa reviennent après une interruption sur Binance. <https://www.binance.com/fr/square/post/9140829848418>. Consulté le 20/01/2026.
- Binance (2026). Blockstream tests post-quantum Bitcoin security with live Liquid transactions. <https://www.binance.com/en/square/post/298611573466753>. Consulté le 05/04/2026.
- Bird & Bird (2025). 5<sup>e</sup> édition du baromètre Bird & Bird / France Invest: activité des fonds infrastructure en 2024. <https://www.twobirds.com/fr/insights/2025/france/5e-ed-barometre-bird-and-bird-france-invest-activite-fonds-infrastructure-rebond-confirme-fonds-fr>. Consulté le 02/03/2026.
- Bitcoin Magazine (2025a). Bitcoin's Lightning Network capacity hits new all-time high. <https://bitcoinmagazine.com/markets/bitcoins-lightning-network-capacity-hits-new-all-time-high>. Consulté le 05/04/2026.
- Bitcoin Magazine (2025b). Blockstream launches Simplicity smart contracts on Bitcoin's Liquid Network sidechain. <https://bitcoinmagazine.com/news/blockstream-launches-simplicity-smart-contracts-on-bitcoins-liquid-network-sidechain>. Consulté le 05/04/2026.
- Black, F. (1976). The pricing of commodity contracts. *Journal of Financial Economics*, 3(1–2), 167–179. <https://www.sciencedirect.com/science/article/abs/pii/0304405X76900246>. Consulté le 20/01/2026.
- Black, F. & Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654. <https://www.cs.princeton.edu/courses/archive/fall02/cs323/links/blackscholes.pdf>. Consulté le 06/06/2024.
- BlackRock (2023). Unlocking the power of infrastructure assets. <https://www.blackrock.com/institutions/en-global/investment-capabilities/strategies/infrastructure>. Consulté le 01/03/2026.
- BlackRock (2026). iShares Core S&P 500 UCITS ETF. <https://www.blackrock.com/fr/intermediaries/products/253743/ishares-sp-500-b-ucits-etf-acc-fund>. Consulté le 26/02/2026.
- Blaschke, W., Jones, M. T., Majnoni, G., & Peria, S. M. (2001). *Stress Testing of Financial Systems: An Overview of Issues, Methodologies, and FSAP Experiences*. Technical Report WP/01/88, International Monetary Fund. <https://www.imf.org/external/pubs/ft/wp/2001/wp0188.pdf>. Consulté le 24/01/2026.
- Blinder, A. S. (1979). *Economic policy and the great stagflation*. Elsevier. [https://api.pageplace.de/preview/DT0400.9781483264561\\_A23869070/preview-9781483264561\\_A23869070.pdf](https://api.pageplace.de/preview/DT0400.9781483264561_A23869070/preview-9781483264561_A23869070.pdf). Consulté le 20/01/2026.

- Blockstream (2018). Liquid: A Bitcoin sidechain. <https://blockstream.com/assets/downloads/pdf/liquid-whitepaper.pdf>. Consulté le 05/04/2026.
- Bloomberg (2015). BHP sounds warning as casualties mount in iron ore price war. <https://www.bloomberg.com/news/articles/2015-02-25/bhp-ceo-sounds-warning-as-casualties-mount-in-iron-ore-price-war>. Consulté le 27/02/2026.
- Bloomberg (2025). Gold revaluation risks "messy" outcome for Fed, Wrightson says. Bloomberg. <https://www.bloomberg.com/news/articles/2025-02-18/wrightson-says-gold-revaluation-a-messy-outcome-for-fed-policy>. Consulté le 28/02/2026.
- Blot, C. (2021). La politique monétaire de la BCE et la crise du Covid-19. *Revue de l'OFCE*, 172(2). <https://www.ofce.sciences-po.fr/pdf/revue/10-172OFCE.pdf>. Consulté le 06/06/2024.
- BMF Bourse (2025). Unibail-Rodamco-Westfield: après plusieurs mois encourageants, la valorisation boursières des groupes immobiliers peut-elle poursuivre sa remontada? <https://urls.fr/SU7PL->. Consulté le 01/03/2026.
- BNP Paribas (2022). Inflation et soutenabilité de la dette publique. <https://economic-research.bnpparibas.com/html/fr-FR/Inflation-soutenabilite-dette-publique-15/05/2022,46361>. Consulté le 22/01/2026.
- BNP Paribas (2023). La BCE s'accorde deux années supplémentaires avant de lancer l'euro numérique. <https://economic-research.bnpparibas.com/Media-Library/fr-FR/BCE-accorde-annees-supplementaires-lancer-euro-numerique-30/11/2023,c40162>. Consulté le 06/04/2026.
- BNP Paribas Real Estate (2022). At a glance T4 2021. [https://www.bnppre.fr/sites/default/files/2022-01/AAG%20INVESTISSEMENT%20T4%202021%20\(1\).pdf](https://www.bnppre.fr/sites/default/files/2022-01/AAG%20INVESTISSEMENT%20T4%202021%20(1).pdf). Consulté le 01/03/2026.
- BNP Paribas Real Estate (2024). At a Glance S1 2024. <https://www.bnppre.fr/sites/default/files/2024-07/AAG%20BUREAUX%20IDF%20-%20S1%202024.pdf>. Consulté le 01/03/2026.
- BNP Paribas Real Estate (2026). REview logistique – T4 2025. <https://www.realestate.bnpparibas.fr/fr/etudes-tendances/etudes-de-marche-france/review-logistique-t4-2025>. Consulté le 01/03/2026.
- Bodnar, G. M., Hayt, G. S., & Marston, R. C. (1998). 1998 Wharton survey of financial risk management by US non-financial firms. *Financial Management*, 27(4), 70–91. <https://www.jstor.org/stable/3666414>. Consulté le 03/03/2026.
- Boluze, L. (2025). Crédit lombard: définition et fonctionnement. <https://www.capital.fr/votre-argent/credit-lombard-1351756>. Consulté le 28/03/2026.
- Borders, K., Balladares, S., Barake, M., & Baselgia, E. (2023). Digital service taxes. [https://www.taxobservatory.eu/www-site/uploads/2023/06/EUTO\\_Digital-Service-Taxes\\_June2023.pdf](https://www.taxobservatory.eu/www-site/uploads/2023/06/EUTO_Digital-Service-Taxes_June2023.pdf). Consulté le 27/01/2026.
- Bordes, C. (2008). *La politique monétaire*, chapitre 1. Le cadre institutionnel de la politique monétaire, (pp. 10–23). La Découverte, Paris. <https://shs.cairn.info/la-politique-monetaire--9782707149527-page-10>. Consulté le 20/01/2026.
- Bordes, C., Clerc, L., & Marimoutou, V. (2007). Is there a structural break in equilibrium velocity in the euro area? *Notes d'études et de recherche de la Banque de France*, 165. <https://hal.science/hal-00308654/file/BordesClercMarimoutouner165.pdf>. Consulté le 06/06/2024.
- Borio, C. & Shim, I. (2003). What can (macro-)prudential policy do to support monetary policy? *BIS Working Papers*, 123. <https://www.bis.org/publ/work242.pdf>. Consulté le 20/01/2026.
- Bossu, S. & Henrotte, P. (2009a). *Finance des marchés: techniques quantitatives et applications pratiques*, chapitre 1. Taux d'intérêt, (pp. 1–17). Dunod, Paris.
- Bossu, S. & Henrotte, P. (2009b). *Finance des marchés: techniques quantitatives et applications pratiques*, chapitre 5. Théorie du portefeuille, (pp. 93–116). Dunod, Paris. <https://shs.cairn.info/finance-des-marches--9782100518128-page-93>. Consulté le 20/01/2026.
- Boulenguer, C. (2022). Portrait-type des entreprises qui fraudent en situation de monopole [conférence]. In *Congrès de l'AFEP*: Amiens, France.

- Bourguinat, H. (1986). *Les vertiges de la finance internationale*. Economica, Paris. <https://shs.cairn.info/les-vertiges-de-la-finance-internationale--9782717812145>. Consulté le 20/01/2026.
- Bpifrance (2023). Le PGE et le PGE saison. <https://bpifrance-creation.fr/encyclopedie/pge-pge-saison>. Consulté le 20/01/2026.
- Bpifrance (2025a). Bilan 2024 de l'activité fonds de fonds de Bpifrance (édition Capital Invest 2025). <https://urls.fr/kFJZMm>. Consulté le 23/03/2026.
- Bpifrance (2025b). Bilan d'activité 2024: 60 milliards d'euros injectés dans l'économie française et 896 millions d'euros de résultat net. <https://presse.bpifrance.fr/bilan-dactivite-2024-60-milliards-deuros-injectes-dans-leconomie-francaise-et-896-millions-deuros-de-resultat-net/>. Consulté le 23/03/2026.
- Brana, S. & Cazals, M. (2014a). *La monnaie*, chapitre 3. La création monétaire, (pp. 63–88). Dunod, Paris. <https://shs.cairn.info/la-monnaie--9782100708598-page-63>. Consulté le 20/01/2026.
- Brana, S. & Cazals, M. (2014b). *La monnaie*, chapitre 4. La politique monétaire, (pp. 89–124). Dunod, Paris. <https://shs.cairn.info/la-monnaie--9782100708598-page-89>. Consulté le 20/01/2026.
- Brandt, A. R. (2017). A critique of Hubbert's model for peak oil. <https://www.facetsjournal.com/doi/10.1139/facets-2017-0097>. Consulté le 26/02/2026.
- BRED (2024). Qu'est-ce que la règle des quinzaines pour les intérêts des livrets? <https://www.bred.fr/faq/epargner/livret/qu-est-ce-que-la-regle-des-quinzaines-pour-les-interets-des-livrets>. Consulté le 20/01/2026.
- Bricogne, J.-C., Lapègue, V., & Monso, O. (2009). La crise des supprimes: de la crise financière à la crise économique. [https://www.insee.fr/fr/statistiques/fichier/1407817/mars2009\\_d1.pdf](https://www.insee.fr/fr/statistiques/fichier/1407817/mars2009_d1.pdf). Consulté le 20/01/2026.
- BRICS Grain (2024). BRICS Clear: The future of cross-border settlements in digital trade. <https://www.bricsgrain.com/brics-clear-the-future-of-cross-border-settlements-in-digital-trade/>. Consulté le 27/01/2026.
- BRICS Grain (2025). BRICS nations accelerate de-dollarization through local currencies and digital finance. <https://www.bricsgrain.com/brics-nations-accelerate-de-dollarization-through-local-currencies-and-digital-finance/>. Consulté le 27/01/2026.
- Brigo, D. & Mercurio, F. (2006). *Interest Rate Models: Theory and Practice*. Springer Finance, 2 edition. <https://github.com/yc-liu/readings/blob/master/Brigo%20D.%2C%20Mercurio%20F.%20Interest%20Rate%20Models%20-%20Theory%20and%20Practice.pdf>. Consulté le 20/01/2026.
- Brooks, C. & Kat, H. M. (2002). The statistical properties of hedge fund index returns and their implications for investors. *Journal of Alternative Investments*, 5(2), 26–44. <https://www.globalriskguard.com/resources/hedge/Hedge%20fund%20strategies.pdf>. Consulté le 20/01/2026.
- Broyon, A. (2026). Comprendre et maîtriser les turbos et warrants. <https://neofa.com/fr/placement-financier/bourse/comprendre-et-maitriser-les-turbos-et-les-warrants/>. Consulté le 13/03/2026.
- Bruegel (2025). The oil price cap and embargo on russia work imperfectly, and defects must be fixed. <https://www.bruegel.org/analysis/oil-price-cap-and-embargo-russia-work-imperfectly-and-defects-must-be-fixed>. Consulté le 26/02/2026.
- Brunnermeier, M. K., James, H., & Landau, J.-P. (2021). The digitalization of money. *BIS Working Papers*, 941. <https://www.bis.org/publ/work941.pdf>. Consulté le 06/06/2024.
- Brunnermeier, M. K., Nagel, S., & Pedersen, L. H. (2009). Carry trades and currency crashes. *NBER Macroeconomics Annual*, 23, 313–347. <https://doi.org/10.1086/593088>. Consulté le 03/03/2026.
- Bryne, D. (2023). FTX collapse is a case study in bad governance. <https://www.thecorporategovernanceinstitute.com/insights/news-analysis/governance-causes-ftx-collapse/>. Consulté le 05/04/2026.
- Buchner, M. (2020). Fiscal policy in an age of secular stagnation. [https://uece2.rc.iseg.ulisboa.pt/events/2019/INFER%20REM%20sessions/Session%20A3\\_Chair%20Ana%20Ven%C3%A2ncio/2\\_18\\_Fiscal%20Policy\\_secular%20stagnation.pdf](https://uece2.rc.iseg.ulisboa.pt/events/2019/INFER%20REM%20sessions/Session%20A3_Chair%20Ana%20Ven%C3%A2ncio/2_18_Fiscal%20Policy_secular%20stagnation.pdf). Consulté le 22/01/2026.

- BusinessAM (2025). Examen de l'impact d'une réévaluation de l'or sur la dette souveraine américaine. <https://fr.businessam.be/examen-de-limpact-dune-reevaluation-de-lor-sur-la-dette-souveraine-americaine/>. Consulté le 28/02/2026.
- Buterin, V. (2023). The three transitions. [https://vitalik.eth.limo/general/2023/06/09/three\\_transitions.html](https://vitalik.eth.limo/general/2023/06/09/three_transitions.html). Consulté le 11/08/2024.
- Butz, H. & Mrozewski, M. J. (2021). The selection process and criteria of impact accelerators: An exploratory study. *Sustainability*, 13(12), 6617. <https://www.mdpi.com/2071-1050/13/12/6617>. Consulté le 20/01/2026.
- Bénassy-Quéré, A. (2015). *Économie monétaire mondiale*. Economica, Paris. <https://fr.scribd.com/document/959748347/E-conomie-Mone-taire-Internationale>. Consulté le 20/01/2026.
- Bénassy-Quéré, A., Brunnermeier, M., Enderlein, H., et al. (2018). Réconcilier solidarité et discipline de marché dans la zone euro. <https://www.telos-eu.com/fr/economie/reconcilier-solidarite-et-discipline-de-marche-dan.html>. Consulté le 11/08/2024.
- CaixaBank Research (2020). The ECB and the Fed: two mandates, one target. <https://www.caixabankresearch.com/en/economics-markets/financial-markets/ecb-and-fed-two-mandates-one-target>. Consulté le 24/01/2026.
- Calliess, C. & Tuncel, E. (2023). The role of Article 11 TFEU in the greening of the ECB's monetary policy. <https://www.cambridge.org/core/journals/german-law-journal/article/role-of-article-11-tfeu-in-the-greening-of-the-ecbs-monetary-policy/03A7D072E1F92542187FB92DC7079198>. Consulté le 24/01/2026.
- Cambridge Associates (2025). Powering the future : Infrastructure trends, performance, and portfolio impact. <https://www.cambridgeassociates.com/insight/powering-the-future-infrastructure-trends-performance-and-portfolio-impact/>. Consulté le 01/03/2026.
- Campbell, J. Y., Serfaty de Medeiros, K., & Viceira, L. M. (2010). Global currency hedging. *The Journal of Finance*, 65(1), 87–121. [https://www.nber.org/system/files/working\\_papers/w13088/w13088.pdf](https://www.nber.org/system/files/working_papers/w13088/w13088.pdf). Consulté le 03/03/2026.
- Capitaine, C. (2021). Transactions immobilières: vers une année 2021 record. <https://www.immomat.in.com/evaluation/services-evaluer/transactions-vers-une-annee-2021-record.html>. Consulté le 01/03/2026.
- Capital (2019). Bruno Le Maire découvre la Libra de Facebook (et il n'aime pas trop ça). <https://www.capital.fr/crypto/bruno-le-maire-decouvre-la-libra-de-facebook-et-il-naime-pas-trop-ca-1342323>. Consulté le 20/01/2026.
- Cassidy, J. (2002). Dot.com: The Greatest Story Ever Sold. <https://archive.org/details/dotcongreteatestst00cass/page/n391/mode/2up>. Consulté le 26/01/2026.
- Castel, M. (2020). La conversion de la France au libéralisme financier. *Revue d'économie financière*, 1(137), 23–34. <https://shs.cairn.info/revue-d-economie-financiere-2020-1-page-23>. Consulté le 20/01/2026.
- CATL (2025). Naxtra battery breakthrough & dual-power architecture. <https://www.catl.com/en/news/6401.html>. Consulté le 28/02/2026.
- CBRE Investment Management (2024). Infrastructure outlook H2 2024. <https://www.cbreim.com/insights/-/media/a74624ebf6a34ca487b569109607ca18.ashx>. Consulté le 02/03/2026.
- CCAF (2022). Bitcoin mining map. [https://ccaf.io/cbnsi/cbeci/mining\\_map](https://ccaf.io/cbnsi/cbeci/mining_map). Consulté le 11/08/2024.
- CCSF (2025). Rapport annuel de l'Observatoire des tarifs bancaires. <https://www.banque-france.fr/system/files/2025-10/OTB2025.pdf>. Consulté le 25/01/2026.
- CDC Habitat (2024). Réforme du mode de calcul des APL: ce qui a changé. <https://www.cdc-habitat.fr/infos-et-logements/reforme-apl-2021>. Consulté le 01/03/2026.
- Cecchetti, S. G. & Karras, G. (1992). Sources of Output Fluctuations During the Interwar Period: Further Evidence on the Causes of the Great Depression. *NBER Working Paper Series*, 4049. <http://www.nber.org/papers/w4049.pdf>. Consulté le 26/01/2026.
- Center for Strategic and International Studies (2026). Greenland, rare earths, and arctic security. <https://www.csis.org/analysis/greenland-rare-earths-and-arctic-security>. Consulté le 28/02/2026.

- CFA Institute (2020). Alternative investments and portfolio management. <https://cfatraininghub.com/staging/wp-content/uploads/2020/10/2020-L1V6.pdf>. Consulté le 17/03/2026.
- CFTC (2001). Review of standard portfolio analysis of risk (SPAN) margining system. [https://www.cftc.gov/sites/default/files/files/tm/tmspan\\_margining043001.pdf](https://www.cftc.gov/sites/default/files/files/tm/tmspan_margining043001.pdf). Consulté le 07/03/2026.
- Chaboud, A., Rime, D., & Sushko, V. (2023). The foreign exchange market. *BIS Working Paper*, 1094. <https://www.bis.org/publ/work1094.pdf>. Consulté le 03/03/2026.
- Chainalysis (2024). The 2024 crypto crime report. <https://www.chainalysis.com/wp-content/uploads/2024/06/the-2024-crypto-crime-report-release.pdf>. Consulté le 11/08/2024.
- Chainalysis (2025). Crypto crime trends for 2023: Illicit transaction activity reaches all-time high. <https://www.chainalysis.com/blog/2025-crypto-crime-report-introduction/>. Consulté le 06/04/2026.
- Chan-Lau, J. A. (2014). Identifying spillover risk in the international banking system. <https://www.elibrary.imf.org/display/book/9781484368589/ch021.xml>. Consulté le 22/01/2026.
- Chancellor, E. (2021). *Finance de marchés*, chapitre 7. Les mesures de performance de portefeuille, (pp. 129–147). Ellipses, Paris.
- Charreaux, G. (2014a). *Finance d'entreprise*, chapitre 4. La décision d'investissement, (pp. 93–110). EMS Editions.
- Charreaux, G. (2014b). *Finance d'entreprise*, chapitre 7. Évaluation de l'entreprise et création de valeur, (pp. 173–208). EMS Éditions. <https://shs.cairn.info/finance-d-entreprise--9782847696691-page-173>. Consulté le 25/01/2026.
- Charreaux, G. (2014c). *Finance d'entreprise*, chapitre 4. La gestion financière à court terme, (pp. 259–282). EMS Éditions. <https://shs.cairn.info/finance-d-entreprise--9782847696691-page-259>. Consulté le 25/01/2026.
- Charupat, N. & Miu, P. (2011). The pricing and performance of leverage exchange-traded funds. *Journal of Banking & Finance*, 35(4), 966–977. <https://www.sciencedirect.com/science/article/abs/pii/S0378426610003444>. Consulté le 26/02/2026.
- Chavagneux, C. (2018). Pour en finir avec les paradis fiscaux et l'opacité fiscale. *Constructif*, 3(51), 11–15. <https://shs.cairn.info/revue-constructif-2018-3-page-11?lang=fr>. Consulté le 20/01/2026.
- Chen, A. & Shen, J. (2025). Direct infrastructure equity: Performance, return attribution & inflation resilience. <https://www.pgim.com/fr/en/institutional/insights/asset-class/multi-asset/multi-asset-solutions/direct-infrastructure-equity-performance>. Consulté le 01/03/2026.
- China Briefing (2025). China's rare earth elements: What businesses need to know. <https://www.china-briefing.com/news/chinas-rare-earth-elements-dominance-in-global-supply-chains/>. Consulté le 27/02/2026.
- CISSM (2025). North Korean genocide, nuclear weapons, and food assistance. <https://cissm.umd.edu/research-impact/publications/north-korean-genocide-nuclear-weapons-and-food-assistance>. Consulté le 01/03/2026.
- Clavert, F. (2006). Hjalmar Schacht, financier et diplomate (1930-1950). <https://shs.hal.science/tel-01100444/file/these.pdf>. Consulté le 20/01/2026.
- CLFI (2025). Who's paying what? M&A deal multiples by buyer type and sector. <https://clfi.co.uk/insights/ma-ev-ebitda-multiples-2025-pe-vs-corporate/>. Consulté le 25/01/2026.
- Clower, R. W. (1964). Monetary history and positive economics. *The Journal of Economic History*, 24(3), 364–380. <https://www.cambridge.org/core/journals/journal-of-economic-history/article/abs/monetary-history-and-positive-economics/6CB3A40C31935687B073D444AF8E3B9A>. Consulté le 20/01/2026.
- Club Patrimoine (2025). La France est passée de AAA en 2010 à A+ en 2025.
- Clément, S. & Lelieur, J. (2021). La confirmité anti-blanchiment face aux crypto-actifs. *Revue de science criminelle et de droit pénal comparé*, 1(1), 15–27. <https://shs.cairn.info/revue-de-science-criminelle-et-de-droit-penal-compare-2021-1-page-15?lang=fr>. Consulté le 06/08/2024.

- CME Group (2020). Nasdaq Veles California Water Index Futures. <https://www.cmegroup.com/markets/equities/nasdaq/nasdaq-veles-california-water-index.html>. Consulté le 01/03/2026.
- CME Group (2023). CME clearing margining practices. [https://www.cftc.gov/media/10226/AAC121423presentation\\_SuzanneSprague/download](https://www.cftc.gov/media/10226/AAC121423presentation_SuzanneSprague/download). Consulté le 07/03/2026.
- CME Group (2024). Futures and options margin model. CME Group Risk Management. <https://www.cmegroup.com/solutions/risk-management/performance-bonds-margins/futures-and-options-margin-model.html>. Consulté le 07/03/2026.
- CNN (2026). Trump says rare earths are part of his Greenland deal: It's "completely bonkers". <https://www.cnn.com/2026/01/22/business/greenland-rare-earth-minerals-trump>. Consulté le 27/02/2026.
- Cohen, S. L. (2013). What do accelerators do? Insights from incubators and angels. *Innovations: Technology, Governance, Globalization*, 8(3–4), 19–25. <https://scholarship.richmond.edu/cgi/viewcontent.cgi?article=1055&context=management-faculty-publications>. Consulté le 20/01/2026.
- CoinGecko (2025). Market share of centralized crypto exchanges, by trading volume. <https://www.coingecko.com/research/publications/centralized-crypto-exchanges-market-share>. Consulté le 05/04/2026.
- CoinMarketCap (2026). Today's cryptocurrency prices by market cap. <https://coinmarketcap.com>. Consulté le 20/01/2026.
- Collain, B., Déjean, F., & Le Theule, M.-A. (2021). *Mini manuel de comptabilité générale: l'essentiel du cours, exercices corrigés*, chapitre 9. L'enregistrement des immobilisations financières et des placements financiers, (pp. 138–160). Dunod, Paris. <https://shs.cairn.info/mini-manuel-de-comptabilite-generale--9782100758715-page-138>. Consulté le 20/01/2026.
- Comité de Bâle sur le contrôle bancaire (2009). *Principles for Sound Stress Testing Practices and Supervision*. Technical report, Bank for International Settlements. <https://www.bis.org/publ/bcbs155.pdf>. Consulté le 24/01/2026.
- Commission européenne (2013). Entrée en vigueur du “two-pack”: le paquet législatif complète le cycle de surveillance budgétaire pour la zone euro et améliore encore la gouvernance économique. [https://ec.europa.eu/commission/presscorner/detail/fr/MEMO\\_13\\_457](https://ec.europa.eu/commission/presscorner/detail/fr/MEMO_13_457). Consulté le 20/01/2026.
- Commission européenne (2024). E-residency in Estonia: How the governmental initiative has transformed digital identity. <https://interoperable-europe.ec.europa.eu/collection/public-sector-tech-watch/use-national-blockchain-infrastructure-support-e-residency-initiative-estonia>. Consulté le 04/04/2026.
- Confluence Investment Management (2024). Asset allocation bi-weekly: The yield curve un-inverts. <https://www.confluenceinvestment.com/asset-allocation-bi-weekly-oct-14-2024/>. Consulté le 28/01/2026.
- Congressional Research Service (2014). Foreign assistance to North Korea. [https://www.congress.gov/crs\\_external\\_products/R/PDF/R40095/R40095.39.pdf](https://www.congress.gov/crs_external_products/R/PDF/R40095/R40095.39.pdf). Consulté le 01/03/2026.
- Connaissance des énergies (2017). Dunkerque : une nouvelle porte d'entrée pour le GNL en France. <https://www.connaissancedesenergies.org/dunkerque-une-nouvelle-porte-dentree-pour-le-gnl-en-france-241104>. Consulté le 02/03/2026.
- Connaissance des énergies (2025). L'union européenne, une puissance de l'uranium qui s'ignore. <https://www.connaissancedesenergies.org/tribune-actualite-energies/union-europeenne-une-puissance-de-luranium-qui-signore>. Consulté le 28/02/2026.
- Conseil de l'Union européenne (2017). Conclusions du Conseil relatives à la liste de l'Union européenne des pays et territoires non coopératifs à des fins fiscales (2017/C 438/04). [https://eur-lex.europa.eu/legal-content/FR/TXT/PDF/?uri=CELEX:52017XG1219\(01\)](https://eur-lex.europa.eu/legal-content/FR/TXT/PDF/?uri=CELEX:52017XG1219(01)). Consulté le 12/12/2025.
- Conseil supérieur du notariat (2025). Bilan immobilier des Notaires de France 2025 – Conférence de presse nationale du 8 décembre 2025. <https://www.calameo.com/books/005125198e0441dfb549f>. Consulté le 01/03/2026.
- Consumer Federation of America (2025). Curious How Much Banks Make from Overdraft Fees? We Have the Data. <https://consumerfed.org/curious-how-much-banks-make-from-overdraft-fees-we-have-the-data>. Consulté le 25/01/2026.



- Corporate Finance Institute (2024). Basis risk. <https://corporatefinanceinstitute.com/resources/career-map/sell-side/capital-markets/what-is-basis-risk/>. Consulté le 07/03/2026.
- Corsetti, G., Pesenti, P., & Roubini, N. (1998a). What Caused the Asian Currency and Financial Crisis? Part I: A Macroeconomic Overview. *NBER Working Paper Series*, 6833. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=145249](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=145249). Consulté le 26/01/2026.
- Corsetti, G., Pesenti, P., & Roubini, N. (1998b). What Caused the Asian Currency and Financial Crisis? Part II: The Policy Debate. *NBER Working Paper Series*, 6834. <https://yfpsresourcelibrary.blob.core.windows.net/fcic/YPFS/w6834.pdf>. Consulté le 26/01/2026.
- Coulter, J. M. & Vogel, T. J. (2004). Pets.com, Inc.: Assessing financial performance and risks in the e-commerce industry. *Issues in Accounting Education*, 19(4), 567–582. <https://publications.aaahq.org/iae/article-abstract/19/4/567/7470/Pets-com-Inc-Assessing-Financial-Performance-and>. Consulté le 20/01/2026.
- Coupey-Soubeyran, J. (2012a). *Monnaie, banques, finance*, chapitre 3. La monnaie, (pp. 113–158). PUF, Paris. <https://shs.cairn.info/monnaie-banques-finance--9782130593065-page-113>. Consulté le 20/01/2026.
- Coupey-Soubeyran, J. (2012b). *Monnaie, banques, finance*, chapitre 2. Les taux d'intérêt, (pp. 67–112). PUF, Paris. <https://shs.cairn.info/monnaie-banques-finance--9782130593065-page-67>. Consulté le 20/01/2026.
- Coupey-Soubeyran, J. (2012c). *Monnaie, banques, finance*, chapitre 5. Financement de l'économie et systèmes financiers, (pp. 201–226). PUF, Paris. <https://shs.cairn.info/monnaie-banques-finance--9782130593065-page-201>. Consulté le 20/01/2026.
- Coupey-Soubeyran, J. & Renault, T. (2018). 10 ans après la faillite de Lehman Brothers, le risque systémique a-t-il baissé? *La lettre du CEPII*, 391. [https://www.cepii.fr/PDF\\_PUB/lettre/2018/let391.pdf](https://www.cepii.fr/PDF_PUB/lettre/2018/let391.pdf). Consulté le 06/06/2024.
- Cour des comptes (2023). Les crypto-actifs: une régulation à renforcer. [https://www.ccomptes.fr/sites/default/files/2023-12/S2023\\_1247\\_Crypto\\_actifs.pdf](https://www.ccomptes.fr/sites/default/files/2023-12/S2023_1247_Crypto_actifs.pdf). Consulté le 06/04/2026.
- Cour des comptes (2025a). Analyse de l'exécution budgétaire 2024 – mission “cohésion des territoires”. [https://www.ccomptes.fr/sites/default/files/2025-04/NEB-2024-Cohesion-territoires\\_0.pdf](https://www.ccomptes.fr/sites/default/files/2025-04/NEB-2024-Cohesion-territoires_0.pdf). Consulté le 01/03/2026.
- Cour des comptes (2025b). *Analyse de l'exécution budgétaire 2024: Mission Investir pour la France de 2030*. Technical report, Cour des comptes. <https://www.ccomptes.fr/sites/default/files/2025-04/NEB-2024-Investir-pour-France-2030.pdf>. Consulté le 23/03/2026.
- Cour des comptes européenne (2026). Rapport spécial 04/2026: Matières premières critiques pour la transition énergétique. [https://www.eca.europa.eu/ECAPublications/SR-2026-04/SR-2026-04\\_FR.pdf](https://www.eca.europa.eu/ECAPublications/SR-2026-04/SR-2026-04_FR.pdf). Consulté le 28/02/2026.
- Cox, J. C., Ross, S. A., & Rubinstein, M. (1979). Option pricing: a simplified approach. *Journal of Financial Economics*, 7(3), 229–263.
- Critical Metals (2025). Technical report summary on the Tanbreez project, South Greenland, 12 march 2025. [https://www.sec.gov/Archives/edgar/data/1951089/000121390025023337/ea023391301ex96-1\\_critical.htm](https://www.sec.gov/Archives/edgar/data/1951089/000121390025023337/ea023391301ex96-1_critical.htm). Consulté le 28/02/2026.
- Cryptoast (2025). La Russie utiliserait du Bitcoin, de l'Ether et de l'USDT pour échanger du pétrole. <https://cryptoast.fr/russie-utilise-bitcoin-ether-usdt-pour-echanger-petrole/>. Consulté le 06/04/2026.
- Cushman & Wakefield (2026). Indicateurs investissement immobilier résidentiel: 4,3 milliards d'euros au S2 2025. <https://www.cushmanwakefield.com/fr-fr/france/news/2026/01/investissement-residentiel-t4-2025>. Consulté le 01/03/2026.
- CVCE (2014). Les mesures non-conventionnelles. <https://www.cvce.eu/education/unit-content/-/unit/7124614a-42f3-4ced-add8-a5fb3428f21c/db6ed04-b474-4987-b157-d09a613da094>. Consulté le 20/01/2026.
- Dai, M. (2009). La grande crise systémique de 2008: causes, conséquences et mesures de politique. <https://opee.unistra.fr/IMG/pdf/19-article1.pdf>. Consulté le 20/01/2026.
- Dall'Orto Mas, R., Vonessen, B., Fehlker, C., & Arnold, K. (2020). The case for central bank independence. <https://www.ecb.europa.eu/pub/pdf/scpops/ecb.op248~28bebb193a.en.pdf>. Consulté le 24/01/2026.

- Danaher, B., Smith, M. D., & Telang, R. (2013). Piracy and copyright enforcement mechanisms. [https://www.nber.org/system/files/working\\_papers/w19150/w19150.pdf](https://www.nber.org/system/files/working_papers/w19150/w19150.pdf). Consulté le 05/04/2026.
- David-Pur, L., Galil, K., Rosenboim, M., & Shapir, O. M. (2023). Cross-currency basis swap spreads and corporate dollar funding. *Journal of International Financial Markets, Institutions and Money*, 85. <https://www.sciencedirect.com/science/article/abs/pii/S1042443123000483>. Consulté le 20/01/2026.
- Davidson, P. (1983). *Financial Markets, Money and the Real World*. Macmillan, New York. <http://joseluisoreiro.com.br/site/link/1bb720e3376005f65afbdb4bcbf8b37e39ec01953.pdf>. Consulté le 20/01/2026.
- Davies, G. (2002). *A History of Money: From Ancient Times to the Present Day*. University of Wales Press, Cardiff. [https://library.uniteddiversity.coop/Money\\_and\\_Economics/A\\_History\\_of\\_Money-From\\_Ancient\\_Times\\_to\\_the\\_Present\\_Day.pdf](https://library.uniteddiversity.coop/Money_and_Economics/A_History_of_Money-From_Ancient_Times_to_the_Present_Day.pdf). Consulté le 20/01/2026.
- Davies, R. B., Martin, J., Parenti, M., & Toubal, F. (2018). Multinational firms and transfer pricing. *Journal of Political Economy*, 126(3), 1121–1157. [https://www.cepii.fr/PDF\\_PUB/wp/2014/wp2014-21.pdf](https://www.cepii.fr/PDF_PUB/wp/2014/wp2014-21.pdf). Consulté le 26/01/2026.
- De Grauwe, P. (2016). *Economics of monetary union*. Oxford University Press. <https://www.europeansources.info/record/economics-of-monetary-union-12th-edition/>. Consulté le 20/01/2026.
- De La Raudière, L. & Mis, J.-M. (2006). Rapport d'information déposé en application de l'article 145 du règlement par la mission d'information commune sur les chaînes de blocs (blockchains). *Assemblée Nationale*, 1501. [https://www.assemblee-nationale.fr/dyn/15/rapports/micblocs/l15b1501\\_rapport-information](https://www.assemblee-nationale.fr/dyn/15/rapports/micblocs/l15b1501_rapport-information). Consulté le 06/08/2024.
- De Long, B. & Shleifer, A. (1991). The stock market bubble of 1929: Evidence from closed-end mutual funds. *Journal of Economic History*, 51(3), 675–700. [https://dash.harvard.edu/bitstream/1/30703980/1/bubble\\_1929.pdf](https://dash.harvard.edu/bitstream/1/30703980/1/bubble_1929.pdf). Consulté le 26/01/2026.
- De Santis, R. A. (2012). The euro area sovereign debt crisis. *ECB Working Paper*, 1419. <https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp1419.pdf>. Consulté le 28/01/2026.
- De Sèze, N. (2023). Monnaies numériques de banque centrale: une mise en perspective des travaux 'à travers le monde. *Revue d'économie financière*, 149, 91–108. <https://www.cairn.info/revue-d-economie-financiere-2023-1-page-91.htm>. Consulté le 06/04/2026.
- De Warren, G. (2022). Enjeux et risques des crypto-actifs. <https://www.tresor.economie.gouv.fr/Articles/fd93c09f-4610-4dd4-bec2-50333afe4bde/files/82327107-08b9-4347-b16b-46f251fe3a1f>. Consulté le 07/04/2026.
- Decker, C. & Wattenhofer, R. (2014). Bitcoin transaction malleability and MtGox. <https://arxiv.org/pdf/1403.6676>. Consulté le 05/04/2026.
- Delaplace, M. (2021a). *Monnaie et financement de l'économie*, chapitre 2. Les formes anciennes de la monnaie, (pp. 20–32). Dunod, Paris. <https://shs.cairn.info/monnaie-et-financement-de-l-economie--9782100758722-page-20>. Consulté le 20/01/2026.
- Delaplace, M. (2021b). *Monnaie et financement de l'économie*, chapitre 3. Les fonctions et les formes actuelles de la monnaie, (pp. 33–50). Dunod, Paris. <https://shs.cairn.info/monnaie-et-financement-de-l-economie--9782100758722-page-33>. Consulté le 20/01/2026.
- Delaplace, M. (2021c). *Monnaie et financement de l'économie*, chapitre 4. Les motifs de détention de la monnaie: la demande de monnaie, (pp. 51–65). Dunod, Paris. <https://shs.cairn.info/monnaie-et-financement-de-l-economie--9782100758722-page-51>. Consulté le 20/01/2026.
- Delaplace, M. (2021d). *Monnaie et financement de l'économie*, chapitre 5. La création monétaire, (pp. 75–94). Dunod, Paris.
- Delaplace, M. (2021e). *Monnaie et financement de l'économie*, chapitre 6. La monnaie centrale et le besoin de refinancement du système bancaire, (pp. 69–88). Dunod, Paris. <https://shs.cairn.info/monnaie-et-financement-de-l-economie--9782100758722-page-69>. Consulté le 20/01/2026.
- Delaplace, M. (2021f). *Monnaie et financement de l'économie*, chapitre 8. La définition et la mise en œuvre de la politique monétaire, (pp. 141–168). Dunod, Paris. <https://shs.cairn.info/monnaie-et-financement-de-l-economie--9782100758722-page-141>. Consulté le 20/01/2026.

- Delaplace, M. (2021g). *Monnaie et financement de l'économie*, chapitre 9. La politique monétaire en France et au sein de la zone euro, (pp. 169–205). Dunod, Paris. <https://shs.cairn.info/monnaie-et-financement-de-l-economie--9782100758722-page-169>. Consulté le 20/01/2026.
- Dell'Erba, M. (2023). Crypto-trading platforms as exchanges. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=4405361](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4405361). Consulté le 05/04/2026.
- DeLong, J. B. (2006). A Short Note on the Size of the Dot-Com Bubble. *NBER Working Paper Series*, 12011. [https://www.nber.org/system/files/working\\_papers/w12011/w12011.pdf](https://www.nber.org/system/files/working_papers/w12011/w12011.pdf). Consulté le 26/01/2026.
- Deloof, M. (2003). Does Working Capital Management Affect Profitability of Belgian Firms? *Journal of Business Finance & Accounting*, 30(3-4), 573–587. <https://repository.uantwerpen.be/docman/irua/d22bc3/36276.pdf>. Consulté le 25/01/2026.
- D'Haese, W. (2024). Inversion de la courbe des taux: quelles implications pour les investisseurs? <https://www.deutschebank.be/fr/actu-et-conseils/articles/inversion-courbe-taux-implications-investisseurs.html>. Consulté le 28/01/2026.
- Diamond, D. W. (1984). Financial intermediation and delegated monitoring. *The Review of Economic Studies*, 51(3), 393–414. [http://www.rncarpio.com/teaching/MicroBanking/Diamond\\_1984\\_DelegatedMonitoring.pdf](http://www.rncarpio.com/teaching/MicroBanking/Diamond_1984_DelegatedMonitoring.pdf). Consulté le 20/01/2026.
- Diamond, D. W. & Dybvig, P. H. (1983). Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3), 401–419. <https://www.bu.edu/econ/files/2012/01/DD83jpe.pdf>. Consulté le 20/01/2026.
- Diamond, P. A. (1965). National debt in a neoclassical growth model. *American Economic Review*, 55(5), 1126–1150. <http://piketty.pse.ens.fr/files/Diamond1965.pdf>. Consulté le 24/01/2026.
- Direction de l'information légale et administrative (2024a). Livrets, plans et comptes d'épargne. <https://www.service-public.fr/particuliers/vosdroits/N20376>. Consulté le 20/01/2026.
- Direction de l'information légale et administrative (2024b). Plan d'épargne en actions (PEA). <https://www.service-public.fr/particuliers/vosdroits/F2385>. Consulté le 11/06/2024.
- Direction de l'information légale et administrative (2025). Revenus locatifs d'un logement conventionné Anah (Loc'Avantages). <https://www.service-public.gouv.fr/particuliers/vosdroits/F34115>. Consulté le 01/03/2026.
- Direction de l'information légale et administrative (2026a). Bénéfices non commerciaux (BNC): régime réel d'imposition. <https://entreprendre.service-public.gouv.fr/vosdroits/F32105>. Consulté le 12/12/2025.
- Direction de l'information légale et administrative (2026b). Quel est le barème de l'impôt sur le revenu? <https://www.service-public.gouv.fr/particuliers/vosdroits/F1419>. Consulté le 20/02/2026.
- Direction générale des entreprises (2025a). Développement des start-ups : la France première du classement 2024 de l'esna. <https://www.entreprises.gouv.fr/la-dge/actualites/developpement-des-start-ups-la-france-premiere-du-classement-2024-de-lesna>. Consulté le 20/01/2026.
- Direction générale des entreprises (2025b). French Tech 2030. <https://lafrenchtech.gouv.fr/fr/programme/french-tech-2030/>. Consulté le 23/03/2026.
- Direction générale du Trésor (2020). Signature d'un contrat pour la réservation de capacités de regazéification au terminal en France entre Qatar Petroleum et Elengy. <https://www.tresor.economie.gouv.fr/Articles/2020/02/24/signature-d-un-contrat-pour-la-reservation-de-capacites-de-regazeification>. Consulté le 02/03/2026.
- Draghi, M. (2018). Central bank independence. <https://www.bis.org/review/r181029d.htm>. Consulté le 24/01/2026.
- Du, W., Tepper, A., & Verdelhan, A. (2018). Deviations from covered interest rate parity. *The Journal of Finance*, 73(3), 915–957. [https://www.nber.org/system/files/working\\_papers/w23170/w23170.pdf](https://www.nber.org/system/files/working_papers/w23170/w23170.pdf). Consulté le 03/03/2026.
- Dufey, G. & Giddy, I. H. (1978). International financial planning: The use of market-based forecasts. *California Management Review*, 21(1), 69–81. [https://www.researchgate.net/publication/271956369\\_International\\_Financial\\_Planning\\_The\\_Use\\_of\\_Market-Based\\_Forecasts](https://www.researchgate.net/publication/271956369_International_Financial_Planning_The_Use_of_Market-Based_Forecasts). Consulté le 20/01/2026.

- Duffie, D. (1999). Credit swap valuation. *Financial Analysts Journal*, 55(1), 73–87. <https://www.darrellduffie.com/uploads/1/4/8/0/148007615/duffiecreditswapvaluation1999.pdf>. Consulté le 20/01/2026.
- Dyson, K. & Marcussen, M. (2009). *Central Banks in the Age of the Euro: Europeanization, Convergence, and Power*. Oxford University Press. <https://global.oup.com/academic/product/central-banks-in-the-age-of-the-euro-9780199218233>. Consulté le 22/01/2026.
- Décideurs (2010). Lion Capital fond pour Picard. [https://www.scottopartners.com/wp-content/uploads/2011/02/Picard\\_Decideurs-\\_oct-2010.pdf](https://www.scottopartners.com/wp-content/uploads/2011/02/Picard_Decideurs-_oct-2010.pdf). Consulté le 20/01/2026.
- e-Estonia (2024). ID-card. <https://e-estonia.com/solutions/estonian-e-identity/id-card/>. Consulté le 04/04/2026.
- Ecalfe, F. (2023). Les obligations indexées du Trésor. <https://fipeco.fr/commentaire/Les%20obligations%20index%20es%20du%20tr%20sor>. Consulté le 20/01/2026.
- EDF (2024). L'autoconsommation collective. <https://www.edf-oa.fr/content/autoconsommation-collective>. Consulté le 11/08/2024.
- EDF (2025). Le TURPE 7 entre en vigueur au 1<sup>er</sup> août 2025. <https://www.edf.fr/entreprises/decryptages/normes-et-reglementations/les-baremes-du-turpe-evoluent-au-1er-aout-2025>. Consulté le 02/03/2026.
- EDHEC (2019). Les investisseurs institutionnels doivent-ils revenir sur l'immobilier? [https://www.edhec.edu/sites/default/files/ckeditor-inlines/pp\\_investisseurs\\_institutionnels\\_immobilier\\_final.pdf](https://www.edhec.edu/sites/default/files/ckeditor-inlines/pp_investisseurs_institutionnels_immobilier_final.pdf). Consulté le 01/03/2026.
- EDHEC (2025). Fair value or fair guess ? inside the engines of infrastructure valuation. [https://edhecinfraprivateassets.com/wp-content/uploads/2025/06/2025\\_fair\\_value\\_or\\_fair\\_guess.pdf](https://edhecinfraprivateassets.com/wp-content/uploads/2025/06/2025_fair_value_or_fair_guess.pdf). Consulté le 01/03/2026.
- Efficio Consulting (2024). Working Capital Trends in Europe: 2023/2024 Report. <https://www.efficioconsulting.com/en-gb/resources/reports/working-capital-trends-in-europe-2023-report/>. Consulté le 25/01/2026.
- EGF (2021). La station f (paris). <https://www.egfbtp.com/la-station-f-paris/>. Consulté le 20/01/2026.
- EIOPA (2016). Solvency II legal provisions. [https://www.eiopa.europa.eu/browse/regulation-and-policy/solvency-ii\\_en](https://www.eiopa.europa.eu/browse/regulation-and-policy/solvency-ii_en). Consulté le 02/03/2026.
- El Karoui, N. (2003). Couverture des risques dans les marchés financiers. [https://math.uni.lu/thalmaier/finance/el\\_karoui.pdf](https://math.uni.lu/thalmaier/finance/el_karoui.pdf). Consulté le 11/08/2024.
- Elster, J. (1986). *An Introduction to Karl Marx*. Cambridge University Press. [https://proletarios.org/books/Elster-Introduccion\\_a\\_Karl\\_Marx.pdf](https://proletarios.org/books/Elster-Introduccion_a_Karl_Marx.pdf). Consulté le 20/01/2026.
- Energy Institute (2025). Statistical review of world energy 2025. <https://assets.kpmg.com/content/dam/kpmgsites/sg/pdf/2025/08/2025-statistical-review-of-world-energy.pdf>. Consulté le 27/02/2026.
- Engie (2006). Document de référence 2006 – Gaz de France. <https://www.engie.com/sites/default/files/assets/documents/2019-09/doc-de-ref-gdf-2006.pdf>. Consulté le 02/03/2026.
- Engle, R. F. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350. <https://www.jstor.org/stable/1392121>. Consulté le 26/02/2026.
- ESRB (2025). Crypto-assets and decentralised finance. [https://www.esrb.europa.eu/pub/pdf/reports/esrb.report202510\\_cryptoassets.en.pdf](https://www.esrb.europa.eu/pub/pdf/reports/esrb.report202510_cryptoassets.en.pdf). Consulté le 05/04/2026.
- Euronext (2016). Règles des indices de la famille CAC, version 16-02. [https://static.arnaudsylvain.fr/2023/03/cac\\_family\\_rules\\_version\\_17-01fr\\_mar\\_2017.pdf](https://static.arnaudsylvain.fr/2023/03/cac_family_rules_version_17-01fr_mar_2017.pdf). Consulté le 26/02/2026.
- European Securities and Markets Authority (2012). Guidelines on ETFs and other UCITS issues. [https://www.esma.europa.eu/sites/default/files/library/2015/11/2012-44\\_0.pdf](https://www.esma.europa.eu/sites/default/files/library/2015/11/2012-44_0.pdf). Consulté le 26/02/2026.
- European Securities and Markets Authority (2018). Decision renewing the temporary restriction on the marketing, distribution or sale of contracts for differences to retail clients. [https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32019X0430\(01\)](https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32019X0430(01)). Consulté le 13/03/2026.

- Eurostat (2023). Le taux d'inflation annuel de la zone euro en baisse à 2.4 %. <https://ec.europa.eu/eurostat/documents/2995521/18005473/2-30112023-AP-FR.pdf/b01fbb1d-bb08-1f83-67c8-1a5162d0d84a>. Consulté le 20/01/2026.
- Eurostat (2024). Financial corporations - Statistics on financial assets and liabilities. [https://ec.europa.eu/eurostat/statistics-explained/index.php/Financial\\_corporations\\_-\\_statistics\\_on\\_financial\\_assets\\_and\\_liabilities](https://ec.europa.eu/eurostat/statistics-explained/index.php/Financial_corporations_-_statistics_on_financial_assets_and_liabilities). Consulté le 22/01/2026.
- Eurostat (2025a). Government debt at 87.4% of GDP in euro area (Q4 2024). <https://ec.europa.eu/eurostat/web/products-euro-indicators/w/2-22042025-bp>. Consulté le 28/01/2026.
- Eurostat (2025b). Over 12 % of materials in the EU come from recycling. <https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20251119-1>. Consulté le 28/02/2026.
- Evans, M. D. D. & Lyons, R. K. (2002). Order flow and exchange rate dynamics. *Journal of Political Economy*, 110(1). <https://www.bis.org/publ/bppdf/bispap02j.pdf>. Consulté le 02/03/2026.
- Eyrolle, N. (2019). Cascades informationnelles, comportements moutonniers, aversion au risque et efficience des marchés financiers. <https://www.ceremade.dauphine.fr/~viger/memoire2019Eyrolle.pdf>. Consulté le 26/01/2026.
- Fama, E. F. (1965). The behavior of stock-market prices. *The Journal of Business*, 38(1), 34–105.
- Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. <https://people.hec.edu/rosu/wp-content/uploads/sites/43/2023/09/Fama-Efficient-capital-markets-1970.pdf>. Consulté le 26/02/2026.
- Fama, E. F. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, 14(3), 319–338. [https://eml.berkeley.edu/~craigne/EconH195/Fall\\_09/webpage/Fama\\_Foward%20Discount.pdf](https://eml.berkeley.edu/~craigne/EconH195/Fall_09/webpage/Fama_Foward%20Discount.pdf). Consulté le 03/03/2026.
- Fama, E. F. & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of financial economics*, 33(1), 3–56. [https://www.jufinance.com/mag/fin534\\_16/Common\\_risk\\_factors\\_Fama\\_French\\_JFE1993.pdf](https://www.jufinance.com/mag/fin534_16/Common_risk_factors_Fama_French_JFE1993.pdf). Consulté le 06/06/2024.
- Fama, E. F. & French, K. R. (2004). The capital asset pricing model: Theory and evidence. *Journal of Economic Perspectives*, 18(3), 25–46. <https://www.aeaweb.org/articles?id=10.1257%2F0895330042162430>. Consulté le 26/02/2026.
- FAO (2017). Water for sustainable food and agriculture. <https://openknowledge.fao.org/server/api/core/bitstreams/b48cb758-48bc-4dc5-a508-e5a0d61fb365/content>. Consulté le 01/03/2026.
- FAO (2024). Food outlook: Market summaries. <https://openknowledge.fao.org/server/api/core/bitstreams/3b08b8c9-b009-4ec5-b26e-aa0c3733c97b/content>. Consulté le 01/03/2026.
- Fastmarkets (2024). The lithium triangle: Three countries control more than 50 % of global resources. Fastmarkets. <https://www.fastmarkets.com/insights/the-lithium-triangle-three-countries-control-more-than-50-of-global-resources/>. Consulté le 28/02/2026.
- Federal Reserve (2024). *2024 Federal Reserve Stress Test Results*. Technical report, Federal Reserve System. <https://www.federalreserve.gov/publications/files/2024-dfast-results-20240626.pdf>. Consulté le 24/01/2026.
- Federal Reserve (2025a). Asian Financial Crisis. <https://www.federalreservehistory.org/essays/asian-financial-crisis>. Consulté le 26/01/2026.
- Federal Reserve (2025b). Oil Shock of 1973-1974. <https://www.federalreservehistory.org/essays/oil-shock-of-1973-74>. Consulté le 26/01/2026.
- Federal Reserve Bank of Dallas (2022). Energy indicators. <https://www.dallasfed.org/research/energy/indicators/2022/en2208>. Consulté le 27/02/2026.
- Federal Reserve Bank of New York (2023). Transition from LIBOR. Federal Reserve Bank of New York. <https://www.newyorkfed.org/arrc/sofr-transition>. Consulté le 07/03/2026.
- Federal Reserve Bank of St. Louis (2026). Crude oil prices: West Texas Intermediate (WTI). <https://fred.stlouisfed.org/series/DCOILWTICO>. Consulté le 27/02/2026.

- Feldstein, M. & Horioka, C. (1980). Domestic Saving and International Capital Flows. *The Economic Journal*, 90(358), 314–329. <https://academic.oup.com/ej/article-abstract/90/358/314/5156095>. Consulté le 26/01/2026.
- Ferguson, B. (2024). Trade frictions in surface water markets. [https://economics.yale.edu/sites/default/files/2024-12/Ferguson%20JMP\\_1111.pdf](https://economics.yale.edu/sites/default/files/2024-12/Ferguson%20JMP_1111.pdf). Consulté le 01/03/2026.
- Fernández-Villaverde, J. & Sanches, D. (2018). On the economics of digital currencies. *Federal Reserve Bank of Philadelphia Working Papers*, 18(07). <https://www.philadelphiafed.org/-/media/frbp/assets/working-papers/2018/wp18-07.pdf>. Consulté le 06/06/2024.
- FIA (2025). ETD volume – December 2024 (annual review). FIA Statistics. <https://www.fia.org/fia/articles/etd-volume-december-2024>. Consulté le 07/03/2026.
- Fidelity Digital Assets (2024). The Lightning Network: Expanding Bitcoin use cases. [https://www.fidelitydigitalassets.com/sites/g/files/djuvja3256/files/acquiadam/FDA\\_TheLightningNetwork\\_ExpandingBitcoinUseCases\\_1187503.1.0\\_V5.pdf](https://www.fidelitydigitalassets.com/sites/g/files/djuvja3256/files/acquiadam/FDA_TheLightningNetwork_ExpandingBitcoinUseCases_1187503.1.0_V5.pdf). Consulté le 05/04/2026.
- Figlewski, S. (1984). Hedging performance and basis risk in stock index futures. *Journal of Finance*, 39(3), 657–669. <https://acad.xlri.ac.in/ais/cms/ResDb/FTF175/figlewski%20basis%20risk.pdf>. Consulté le 20/01/2026.
- Finance de marché (2013). Qu'est-ce qu'un *Worst-of-Call*? Définition et payoff d'une option exotique sur panier. <http://financedemarche.fr/finance/quest-ce-quun-worst-of-call-definition-payoff-dune-option-exotique-sur-panier>. Consulté le 17/03/2026.
- Financial Stability Board (FSB) (2010). *Implementing OTC Derivatives Market Reforms*. Technical report, FSB. [https://www.fsb.org/uploads/r\\_101025.pdf](https://www.fsb.org/uploads/r_101025.pdf). Consulté le 07/03/2026.
- Fisch, C., Masiak, C., Vismara, S., & Block, J. (2021). Motives and profiles of ICO investors. *Journal of Business Research*, 125, 564–576. [https://pure.eur.nl/ws/portalfiles/portal/58014662/1\\_s2.0\\_S0148296319304539\\_main.pdf](https://pure.eur.nl/ws/portalfiles/portal/58014662/1_s2.0_S0148296319304539_main.pdf). Consulté le 06/06/2024.
- Fisher, I. (1911). *The Purchasing Power of Money*. Macmillan. <https://public.econ.duke.edu/~kdh9/Courses/Graduate%20Macro%20History/Readings-1/Fisher%20Purchasing%20Power%20of%20Money.pdf>. Consulté le 20/01/2026.
- Fisher, I. (1930). *The Theory of Interest*. Macmillan, New York. [https://files.libertyfund.org/files/1416/0219\\_Bk.pdf](https://files.libertyfund.org/files/1416/0219_Bk.pdf). Consulté le 20/01/2026.
- Flanders, A. (2016). Cointegration analysis of stocks-to-use ratios with price and acreage response for corn, soybeans, and wheat. *AgEcon Search*. <https://ageconsearch.umn.edu/record/302506/>. Consulté le 01/03/2026.
- FMI (2009). La riposte à la crise mondiale - Rapport annuel 2009 du FMI. [https://www.imf.org/-/media/websites/imf/imported-flagship-issues/external/french/pubs/ft/ar/2009/pdf/ar09\\_fra.pdf](https://www.imf.org/-/media/websites/imf/imported-flagship-issues/external/french/pubs/ft/ar/2009/pdf/ar09_fra.pdf). Consulté le 22/01/2026.
- FMI (2025). Currency composition of official foreign exchange reserves (COFER): US dollar's share of foreign reserves is little changed in exchange-rate-adjusted terms. *IMF Data*. <https://data.imf.org/en/news/october%201%202025%20cofer>. Consulté le 22/01/2026.
- Fondation de la Mer (2023). Les grands fonds marins: Ressources et enjeux. <https://www.fondationdelamer.org/wp-content/uploads/2023/12/Les-Grands-Fonds-Marins.pdf>. Consulté le 27/02/2026.
- Fondation Robert Schuman (2025). L'euro numérique à l'aube d'un nouveau monde monétaire: enjeux géopolitiques. <https://urls.fr/m1uLeX>. Consulté le 06/04/2026.
- Fonds monétaire international (2000). The impact of higher oil prices on the global economy. <https://www.imf.org/external/pubs/ft/oil/2000/oilrep.pdf>. Consulté le 27/02/2026.
- Fontana, G. (2006). The Federal Reserve and the European Central Bank: a theoretical comparison of their legislative mandates. *Journal of Post Keynesian Economics*, 28(3), 433–450. [https://www.researchgate.net/publication/5173001\\_The\\_Federal\\_Reserve\\_and\\_the\\_European\\_Central\\_Bank\\_A\\_theoretical\\_comparison\\_of\\_their\\_legislative\\_mandates](https://www.researchgate.net/publication/5173001_The_Federal_Reserve_and_the_European_Central_Bank_A_theoretical_comparison_of_their_legislative_mandates). Consulté le 24/01/2026.

- Forinov (2025). La liste des fonds de corporate venture capital français. <https://blog.forinov.com/outils-startups/la-liste-des-fonds-de-corporate-ventures-capital-francais/972/>. Consulté le 20/01/2026.
- Format Research (2022). Study on the payment attitudes of consumers in the euro area (SPACE) 2022. <https://www.formatresearch.com/fr/2022/12/21/Habitudes-de-paiement-des-consommateurs-de-la-BCE-dans-la-zone-euro/>. Consulté le 06/04/2026.
- Forvis Mazars & France FinTech (2025). Baromètre du crowdfunding en france 2024. [https://francefintech.org/wp-content/uploads/2025/02/BAROMETRE-CROWDFUNDING-2024\\_FRANCE-FINTECH-FORVIS-MAZARS\\_120225.pdf](https://francefintech.org/wp-content/uploads/2025/02/BAROMETRE-CROWDFUNDING-2024_FRANCE-FINTECH-FORVIS-MAZARS_120225.pdf). Consulté le 20/01/2026.
- FPA (2025). How Ripple (XRP) is building a bridge to the future of cross-border payments. <https://www.financialplanningassociation.org/learning/publications/journal/SEP25-how-ripple-xrp-building-bridge-future-cross-border-transactions-open>. Consulté le 04/04/2026.
- France FinTech & Forvis Mazars (2026). Baromètre du crowdfunding en france – 1<sup>er</sup> semestre 2025. <https://francefintech.org/barometre-du-crowdfunding-en-france-s1-2025/>. Consulté le 20/01/2026.
- France insoumise (2023). Proposition de loi n° 1774 visant à indexer les salaires sur l'inflation. *Assemblée Nationale*, 4753. [https://www.assemblee-nationale.fr/dyn/16/textes/l16b1774\\_proposition-loi.pdf](https://www.assemblee-nationale.fr/dyn/16/textes/l16b1774_proposition-loi.pdf). Consulté le 06/08/2024.
- France Invest (2025). Activité 2024 des acteurs français du venture & growth – 3<sup>e</sup> édition. [https://www.franceinvest.eu/wp-content/uploads/2025/06/France-Invest-2024\\_Etude-Venture-Growth\\_VDEF.pdf](https://www.franceinvest.eu/wp-content/uploads/2025/06/France-Invest-2024_Etude-Venture-Growth_VDEF.pdf). Consulté le 20/01/2026.
- France Rénov' (2023). Louer solidaire avec Loc'Avantages. <https://france-renov.gouv.fr/aides/loc-avantages>. Consulté le 01/03/2026.
- Freihat, A. R. F. (2019). Factors Affecting Price to Earnings Ratio (P/E). *Review of Integrative Business and Economics Research*, 9(2), 35–46. <https://virtusinterpress.org/IMG/pdf/rgcv9i2p.4.pdf>. Consulté le 26/01/2026.
- Friedman, M. (1956). *Studies in the Quantity Theory of Money*. University of Chicago Press, Chicago. <https://fr.scribd.com/document/559015786/Milton-Friedman-Studies-in-the-Quantity-Theory-of-Money-University-of-Chicago-Press-1956>. Consulté le 20/01/2026.
- Friedman, M. (1957). *A Theory of the Consumption Function*. Princeton University Press. <https://www.nber.org/system/files/chapters/c4402/c4402.pdf>. Consulté le 26/01/2026.
- Friedman, M. (1968). The role of monetary policy. *American Economic Review*, 58(1). <https://www.aeaweb.org/aer/top20/58.1.1-17.pdf>. Consulté le 09/12/2025.
- Friedman, M. (1970). The counter-revolution in monetary theory. *American Economic Review*. <https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119205814.app2>. Consulté le 09/12/2025.
- Frézal, S. (2021). *Financiariser l'assurance: enquête sur Solvabilité II*, chapitre 2. Solvabilité I, Solvabilité II, (pp. 67–88). Presses de Sciences Po, Paris. <https://shs.cairn.info/financiariser-l-assurance--9782724627459-page-67>. Consulté le 20/01/2026.
- Funds Magazine (2025). Les infras non cotées à l'heure de la démocratisation. <https://fundsmagazine.optionfinance.fr/strategie/les-infras-non-cotees-a-lheure-de-la-democratisation.html>. Consulté le 01/03/2026.
- Furet, A. (2024). Après l'engouement des NFT, l'heure de vérité: quel avenir pour ces actifs numériques? <https://urls.fr/Fjjh4t>. Consulté le 07/04/2026.
- Fédération internationale pour les droits humains (2023). Ivory Coast: Victory at the African court for victims of the Trafigura toxic waste dump. <https://www.fidh.org/en/issues/business-human-rights-environment/business-and-human-rights/ivory-coast-victory-at-the-african-co>. Consulté le 27/02/2026.
- GAFI (2021). Second 12-month review of revised fatf standards: virtual assets and VASPs. <https://www.fatf-gafi.org/content/dam/fatf-gafi/guidance/Second-12-Month-Review-Revised-FATF-Standards-Virtual-Assets-VASPs.pdf.coredownload.pdf>. Consulté le 11/08/2024.



- Gandal, N., Hamrick, J., Moore, T., & Oberman, T. (2018). Price manipulation in the Bitcoin ecosystem. *Journal of Monetary Economics*, 95, 86–96. <https://tylermoore.utulsa.edu/jme17.pdf>. Consulté le 05/04/2026.
- Garcia, J. A. & Van Rixtel, A. (2008). Inflation-linked bonds from a central bank perspective. <https://www.ecb.europa.eu/pub/pdf/scpops/ecbocp62.pdf>. Consulté le 22/01/2026.
- Gartner (2025). Hype cycle de Gartner. <https://testrigor.com/blog/gartner-hype-cycle-for-ai-2025>. Consulté le 26/01/2026.
- Ghimiray, D. (2024). Un guide du site web Silk Road. <https://www.avast.com/fr-fr/c-silk-road-dark-web-market>. Consulté le 11/08/2024.
- Giannone, D. & Lenza, M. (2008). The Feldstein-Horioka Fact. *ECB Working Paper Series*, 873. <https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp873.pdf>. Consulté le 26/01/2026.
- Glachant, J., Lorenzi, J.-H., & Trainar, P. (2008). Private equity et capitalisme français. *Conseil d'Analyse Économique*, 75. <http://cae-eco.fr/static/pdf/075.pdf>. Consulté le 20/01/2026.
- Glauben, T., Svanidze, M., Götz, L., Prehn, S., Jaghdani, T. J., Duric, I., & Kuhn, L. (2022). The war in Ukraine, agricultural trade and risks to global food security. *Intereconomics*, 57(3). <https://www.intereconomics.eu/contents/year/2022/number/3/article/the-war-in-ukraine-agricultural-trade-and-risks-to-global-food-security.html>. Consulté le 01/03/2026.
- Glauber, J., McNamara, B., & Olivetti, E. (2023). La Russie met fin à l'Initiative céréalière de la mer Noire. <https://ssa.foodsecurityportal.org/fr/node/2486>. Consulté le 01/03/2026.
- Glencore (2025). Preliminary results 2024. <https://www.glencore.com/media-and-insights/news/preliminary-results-2024>. Consulté le 27/02/2026.
- Goldgar, A. (2007). *Tulipmania: Money, Honor, and Knowledge in the Dutch Golden Age*. University of Chicago Press. [https://www.researchgate.net/publication/37693065\\_Tulipmania\\_Money\\_Honor\\_and\\_Knowledge\\_in\\_the\\_Dutch\\_Golden\\_Age](https://www.researchgate.net/publication/37693065_Tulipmania_Money_Honor_and_Knowledge_in_the_Dutch_Golden_Age). Consulté le 26/01/2026.
- Goldman Sachs (2026). AI: In a bubble? <https://www.gartner.fr/fr/methodologies/hype-cycle>. Consulté le 26/01/2026.
- Goldsmith, R. W. (1959). *Financial structure and development*. Yale University, New Haven. <https://www.nber.org/system/files/chapters/c4417/c4417.pdf>. Consulté le 20/01/2026.
- Gomes Teixeira, S. (2022). Start-up studio, accélérateurs et incubateurs: quelles différences?
- Google Finance (2026). NASDAQ. <https://urls.fr/1fLpm9>. Consulté le 26/01/2026.
- Gordon, M. J. (1959). Dividends, Earnings, and Stock Prices. *The Review of Economics and Statistics*, 41(2), 99–105. <http://www.piketty.pse.ens.fr/files/Gordon1959.pdf>. Consulté le 26/01/2026.
- Gourinchas, P.-O., Martin, P., & Messer, T. (2023). The economics of sovereign debt, bailouts, and the euro-zone crisis. *IMF Working Paper*, 177. <https://www.imf.org/-/media/files/publications/wp/2023/english/wpia2023177-print-pdf.pdf>. Consulté le 28/01/2026.
- Grand Paris (2022). Évolution des prix immobiliers depuis le début des travaux du Grand Paris Express. <https://www.grandparis.com/marche-de-limmobilier/etude-de-marche/evolution-des-prix-de-limmobilier-depuis-le-debut-des-travaux-d>. Consulté le 02/03/2026.
- Gregory, F. & Milas, P. J. (2024). China in the Democratic Republic of the Congo: A new dynamic in critical mineral competition. <https://ssi.armywarcollege.edu/SSI-Media/Recent-Publications/Article/3938204/>. Consulté le 28/02/2026.
- Grischenko, O. V. & Huang, J.-z. (2012). Inflation risk premium: Evidence from the tips market. *Federal Reserve Board Finance and Economics Discussion Series*, 6. <https://www.federalreserve.gov/pubs/feds/2012/201206/201206pap.pdf>. Consulté le 22/01/2026.
- Grjebine, T. & Tripier, F. (2016). Finance et croissance: le court terme aux dépens du long terme? *La lettre du CEPPI*, 364. [https://www.cepii.fr/PDF\\_PUB/lettre/2016/let364.pdf](https://www.cepii.fr/PDF_PUB/lettre/2016/let364.pdf). Consulté le 20/01/2026.

- Groupe d'experts sur le SMIC (2023). Rapport sur le salaire minimum interprofessionnel de croissance. <https://www.tresor.economie.gouv.fr/Articles/284b121f-b187-4280-b327-05f18064c3fa/files/dd527e16-c2ab-441b-8fd0-e02eca058761>. Consulté le 20/01/2026.
- Guillaumin, G. (2018). *Macroéconomie*, chapitre 4. La monnaie et le système bancaire, (pp. 140–161). Dunod, Paris. <https://shs.cairn.info/macroeconomie--9782100793259-page-140>. Consulté le 20/01/2026.
- Géoconfluences (2024). Soudure. <https://geoconfluences.ens-lyon.fr/glossaire/soudure>. Consulté le 01/03/2026.
- Hache, E. & Normand, M. (2024). Critical materials: Assessing the EU strategy. [https://www.veblen-institute.org/IMG/pdf/hache-normand\\_march\\_2024\\_eng.pdf](https://www.veblen-institute.org/IMG/pdf/hache-normand_march_2024_eng.pdf). Consulté le 28/02/2026.
- Hackett Group (2025). EU Firms with Idle €1.3 Trillion Working Capital. <https://www.thehackettgroup.com/insights/eu-companies-have-1-3-trillion-euros/>. Consulté le 25/01/2026.
- Hadjjemmanuil, C. (2022). Bail-in in the European banking union: A close reading of Article 27 SRMR. *European Banking Institute*, 116. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=4047044](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4047044). Consulté le 20/01/2026.
- Hagan, P. S., Kumar, D., Lesniewski, A. S., & Woodward, D. E. (2002). Managing smile risk. *Wilmott Magazine*, (pp. 84–108). [https://www.next-finance.net/IMG/pdf/pdf\\_SABR.pdf](https://www.next-finance.net/IMG/pdf/pdf_SABR.pdf). Consulté le 20/01/2026.
- Hamilton, J. D. (2009). Causes and consequences of the oil shock of 2007–08. *Brookings Papers on Economic Activity*, 40(1), 215–283. [https://www.brookings.edu/wp-content/uploads/2009/03/2009a\\_bpea\\_hamilton.pdf](https://www.brookings.edu/wp-content/uploads/2009/03/2009a_bpea_hamilton.pdf). Consulté le 26/02/2026.
- Hampton, M. P. (1996). Creating spaces: The political economy of island offshore finance centres: the case of Jersey. *Geographische Zeitschrift*, 84(2), 103–113. <http://geoprodig.cnrs.fr/items/show/86431>. Consulté le 20/01/2026.
- Harbert, T. (2019). Here's how much the 2008 bailouts really cost. <https://mitsloan.mit.edu/ideas-made-to-matter/heres-how-much-2008-bailouts-really-cost>. Consulté le 20/01/2026.
- Harris, L. & Gurel, E. (1986). Price and volume effects associated with changes in the S&P 500 list: New evidence for the existence of price pressures. *The Journal of Finance*, 41(4), 815–829. <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-6261.1986.tb04550.x>. Consulté le 26/02/2026.
- Hasbrouck, J. (2007). *Empirical Market Microstructure*. New York: Oxford University Press. <https://www.acsu.buffalo.edu/~keechung/MGF743/Readings/Hasbrouck's%20book.pdf>. Consulté le 02/03/2026.
- Hayek, F. (1976). *The denationalization of money*. The Institute of Economics Affairs, Londres. [https://cdn.mises.org/Denationalisation%20of%20Money%20The%20Argument%20Refined\\_5.pdf](https://cdn.mises.org/Denationalisation%20of%20Money%20The%20Argument%20Refined_5.pdf). Consulté le 20/01/2026.
- Henrard, M. (2014). *Interest Rate Modelling in the Multi-Curve Framework: Foundations, Evolution and Implementation*. Palgrave Macmillan. <https://link.springer.com/book/10.1057/9781137374660>. Consulté le 20/01/2026.
- Henrat-Lemée, C. (2021). *Géopolitique et géoéconomie du monde contemporain: puissance et conflits*, chapitre 10. La mondialisation financière: de l'efficience à l'instabilité, (pp. 244–264). La Découverte, Paris. <https://shs.cairn.info/geopolitique-et-geoeconomie-du-monde-contemporain--9782348086670-page-244>. Consulté le 20/01/2026.
- Henry, D. & Irrera, A. (2017). JP Morgan's Dimon says bitcoin "is a fraud". <https://www.reuters.com/article/legal/jpmorgans-dimon-says-bitcoin-is-a-fraud-idUSKCN1BN2PM/>. Consulté le 11/08/2024.
- Herman, J. D. (2024). Seasonal forecast of the California water price index. *Journal of Water Resources Planning and Management*, 150(1). <https://ascelibrary.org/doi/abs/10.1061/JWRMD5.WRENG-6239>. Consulté le 01/03/2026.
- Hicks, J. R. (1974). *The Crisis in Keynesian Economics*. Basic Books. <https://archive.org/details/crisisinkeynesia0000john/page/n3/mode/2up>. Consulté le 20/01/2026.
- Hilscher, J., Raviv, A., & Reis, R. (2021). Inflating away the public debt? An empirical assessment. *Review of Financial Studies*, (pp. 1–43). <https://eprints.lse.ac.uk/107543/2/hhab018.pdf>. Consulté le 22/01/2026.
- Hines, J. R. & Rice, E. M. (1994). Fiscal paradise: Foreign tax havens and American business. *Quarterly Journal of Economics*, 109(1), 149–182. <https://scispace.com/pdf/fiscal-paradise-foreign-tax-havens-and-american-business-6du899u5li.pdf>. Consulté le 26/01/2026.

- Hodes Weill / Cornell University (2024). 2024 institutional infrastructure allocations monitor. <https://www.hodesweill.com/single-post/2024-institutional-infrastructure-allocations-monitor>. Consulté le 01/03/2026.
- Hoffner, B. (2023). United States: Central bank swaps to 14 countries, 2020. *Journal of Financial Crises*, 5(1). <https://elischolar.library.yale.edu/journal-of-financial-crises/vol5/iss1/21/>. Consulté le 12/03/2026.
- Hofstede, G. (1984). *Culture's consequences: International differences in work-related values*, volume 5. Sage, Beverly Hills. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=1496209](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1496209). Consulté le 12/12/2025.
- Hossen, T. P. (2020). Le triangle d'incompatibilité des stratégies boursières. <https://www.thomashossen.fr/trading>. Consulté le 28/08/2024.
- Hossen, T. P. & Gauthier, T. (2024). *Blockchain, art numérique et métavers: la révolution des "crypto-arts" ou non-fungible tokens (NFT)*. Thomas Hossen, Paris. <https://www.amazon.fr/Blockchain-art-numérique-métavers-non-fungible-ebook/dp/B0CST3KT1J>. Consulté le 09/09/2024.
- Hull, J. C. (2012). *Options, futures and other derivatives*. Prentice Hall, Boston. [https://fac.ksu.edu.sa/sites/default/files/options\\_futures\\_and\\_other\\_derivatives\\_8th\\_ed\\_part1.pdf](https://fac.ksu.edu.sa/sites/default/files/options_futures_and_other_derivatives_8th_ed_part1.pdf). Consulté le 09/09/2024.
- Hwang, M. & Rogers, D. (2017). The shift away from *Take-or-Pay* contracts in LNG. <https://www.jdsupra.com/legalnews/the-shift-away-from-take-or-pay-42077/>. Consulté le 02/03/2026.
- Högenauer, A.-L. & Howarth, D. (2016). Unconventional monetary policies and the European Central Bank's problematic democratic legitimacy. *Zeitschrift für öffentliche und gemeinwirtschaftliche Unternehmen*, 39(2-3), 248-265. <https://orbilu.uni.lu/bitstream/10993/28059/1/>. Consulté le 22/01/2026.
- IATP (2008). Commodities market speculation: The risk to food security and agriculture. <https://www.iatp.org/documents/commodities-market-speculation-the-risk-to-food-security-and-agriculture>. Consulté le 01/03/2026.
- IBM (2022). IBM Food Trust: une nouvelle ère dans l'approvisionnement alimentaire mondial. <https://www.ibm.com/fr-fr/blockchain/solutions/food-trust>. Consulté le 11/08/2024.
- ICMA, Banque islamique de développement, & LSEG (2024). Guidance on green, social and sustainability Sukuk. <https://www.icmagroup.org/assets/documents/Sustainable-finance/ICMA-IsDB-LSEG-Guidance-on-Green-Social-and-Sustainability-Sukuk-April-2024.pdf>. Consulté le 26/01/2026.
- iFRAP (2023). Le spread de taux avec l'Allemagne: un thermomètre de la tension sur la dette française. <https://www.ifrap.org/budget-et-fiscalite/le-spread-de-taux-avec-lallemagne-un-thermometre-de-la-tension-sur-la-dette-francaise>. Consulté le 20/01/2026.
- IG (2024a). Qu'est-ce que l'effet de levier? <https://www.ig.com/fr/formation-trading/qu-est-ce-que-l-effet-de-levier>. Consulté le 11/08/2024.
- IG (2024b). Vente à découvert (short selling): définition. <https://www.ig.com/fr/glossaire-trading/vente-a-decouvert-definition>. Consulté le 11/08/2024.
- IGEDD (2026). Prix immobilier – Évolution à long terme. <https://www.igedd.developpement-durable.gouv.fr/prix-immobilier-evolution-a-long-terme-a1048.html>. Consulté le 01/03/2026.
- Ikonen, P. & Lehmus, M. (2024). Fiscal policy and debt sustainability in the euro area since the COVID-19 pandemic and energy crisis. *Bank of Finland Bulletin*. <https://www.bofbulletin.fi/en/2024/1/fiscal-policy-and-debt-sustainability-in-the-euro-area-since-the-covid-19-pandemic-and-energy-crisis/>. Consulté le 28/01/2026.
- Immobilier danger (2026). Évolution du taux immobilier: historique et prévisions 2026. <https://www.immobilier-danger.com/evolution-taux-immobilier.html>. Consulté le 01/03/2026.
- InfraRed Capital Partners (2025). Infrastructure strategic outlook 2025. <https://www.ircp.com/wp-content/uploads/2025/01/InfraRed-Outlook-2025.pdf>. Consulté le 02/03/2026.
- Infrastructure Investor (2025). Key trends in infrastructure strategies. <https://www.infrastructureinvestor.com/key-trends-in-infrastructure-strategies/>. Consulté le 02/03/2026.

- ING (2024). *EUR/USD Cross-Currency Swap*. Technical report, ING Bank. <https://research.ing.com/docs/9B434B85-7765-401C-9EA5-1B9D1E7C3A4A.pdf>. Consulté le 12/03/2026.
- INSEE (2021). Classification des fonctions de consommation des ménages – COICOP 1999. <https://www.insee.fr/fr/information/2408172>. Consulté le 07/06/2024.
- INSEE (2023). L'indice de référence des loyers du premier trimestre 2023. <https://www.insee.fr/fr/statistiques/7344857>. Consulté le 01/03/2026.
- INSEE (2024a). Agriculture et prélèvements d'eau douce. <https://www.insee.fr/fr/statistiques/7728881?sommaire=7728903>. Consulté le 23/03/2026.
- INSEE (2024b). Indice de référence des loyers (IRL). <https://www.insee.fr/fr/statistiques/serie/001515333>. Consulté le 01/03/2026.
- INSEE (2024c). L'indice de référence des loyers du quatrième trimestre 2023. <https://www.insee.fr/fr/statistiques/7749702>. Consulté le 01/03/2026.
- INSEE (2024d). Taux d'intérêt par pays: données annuelles de 2008 à 2023. *Chiffres-clés*. <https://www.insee.fr/fr/statistiques/2130712>. Consulté le 06/06/2024.
- INSEE (2025a). Emploi, chômage, revenus du travail. <https://www.insee.fr/fr/statistiques/8376888?sommaire=8376908>. Consulté le 28/01/2026.
- INSEE (2025b). Indice des loyers commerciaux (ILC): documentation méthodologique. INSEE, Métadonnées. <https://www.insee.fr/fr/metadonnees/source/indicateur/p1642/documentation-methodologique>. Consulté le 01/03/2026.
- INSEE (2025c). Parc de logements au 1<sup>er</sup> janvier 2025. *INSEE Focus*, 359. <https://www.insee.fr/fr/statistiques/8640662>. Consulté le 01/03/2026.
- Insights Global / Financial Times (2024). World's top oil trader Vitol earns \$13 billion in another strong year. <https://www.insights-global.com/vitol-reports-13-billion-profits-in-2023/>. Consulté le 27/02/2026.
- Institut Paris Région (2025). Incubateurs et accélérateurs de start-up en Île-de-france. <https://www.institutparisregion.fr/cartographies-interactives/incubateurs-et-accelerateurs-de-start-up-en-ile-de-france/>. Consulté le 20/01/2026.
- Institute of Energy Economics Japan (2024). The OPEC+ agreed on production cuts and Saudi Arabia. <https://eneken.ieej.or.jp/data/12231.pdf>. Consulté le 26/02/2026.
- Interactive Brokers (2025). Gold and the U.S. dollar: An evolving relationship? <https://www.interactivebrokers.com/campus/traders-insight/securities/commodities/gold-and-the-u-s-dollar-an-evolving-relationship/>. Consulté le 27/02/2026.
- Invesco (2025). Invesco water resources ETF (PHO) : Fact sheet q4 2025. <https://www.invesco.com/us-rest/contentdetail?contentId=d15407c649400410VgnVCM10000046f1bf0aRCRD&dnsName=us>. Consulté le 01/03/2026.
- Investing News Network (2025). Top 5 palladium and platinum countries by production. <https://investingnews.com/daily/resource-investing/precious-metals-investing/platinum-investing/top-platinum-palladium-producing-countries/>. Consulté le 28/02/2026.
- Investopedia (2022). FTX crypto exchange collapse: Causes, consequences, and lessons. <https://www.investopedia.com/what-went-wrong-with-ftx-6828447>. Consulté le 05/04/2026.
- IOSCO (2024). Single-name credit default swaps market. <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD806.pdf>. Consulté le 12/03/2026.
- Iqbal, M. & Llewellyn, D. T., Eds. (2002). *Islamic Banking and Finance: New Perspectives on Profit-Sharing and Risk*. Edward Elgar Publishing. <https://urls.fr/o9LeDv>. Consulté le 26/01/2026.

- Ireland, P. N. (2024). Money growth and inflation in the Euro Area, UK, and USA: measurement issues and recent results. *Macroeconomic Dynamics*, (pp. 1–27). <https://www.cambridge.org/core/journals/macroeconomic-dynamics/article/money-growth-and-inflation-in-the-euro-area-uk-and-usa-measurement-issues-and-recent-results/962C8408B6CF1D69CE6DF8C61C698F55>. Consulté le 22/01/2026.
- IRIS (2024). Les BRICS+ : vers une domination sans limites sur les marchés des matières premières ? <https://www.iris-france.org/186697-les-brics-vers-une-domination-sans-limites-sur-les-marches-des-matieres-premieres-2/>. Consulté le 27/01/2026.
- ISDA (2003). ISDA 2003 credit derivatives definitions. ISDA. <https://danskeci.com/-/media/pdf/danskeci-com/dodd-frank/isda-definitions/isda-credit-2003.pdf>. Consulté le 12/03/2026.
- ISDA (2021). Legal guidelines for smart derivatives contracts: Foreign exchange. <https://www.isda.org/a/bPYTE/ISDA-Legal-Guidelines-for-Smart-Derivatives-Contracts-FX.pdf>. Consulté le 03/03/2026.
- ISDA (2024a). Analyzing trends in single-name CDS and index CDS market activity. <https://www.isda.org/a/U3bgE/CDS-Market-Dynamics-Analyzing-Trends-in-Single-name-CDS-and-Index-CDS-Market-Activity.pdf>. Consulté le 12/03/2026.
- ISDA (2024b). Key trends in the size and composition of OTC derivatives markets in the second half of 2024. <https://urls.fr/ltENe3>. Consulté le 12/03/2026.
- ISDA (2025). Key trends in the size and composition of OTC derivatives markets in the second half of 2024. <https://www.isda.org/a/1rjgE/Key-Trends-in-the-Size-and-Composition-of-OTC-Derivatives-Markets-in-the-Second-Half-of-2024.pdf>. Consulté le 07/03/2026.
- Jackson, M. O. & Pernoud, A. (2020). Systemic risk in financial networks: A survey. *Annual Review of Economics*, 13, 171–202. <https://arxiv.org/pdf/2012.12702.pdf>. Consulté le 22/01/2026.
- Jarrow, R. A. & Turnbull, S. M. (1995). Pricing derivatives on financial securities subject to credit risk. *Journal of Finance*, 50(1), 53–85. <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-6261.1995.tb05167.x>. Consulté le 20/01/2026.
- Jevons, W. S. (1875). *Money and the Mechanism of Exchange*. New York: D. Appleton and Company. [https://cdn.mises.org/Money%20and%20the%20Mechanism%20of%20Exchange\\_2.pdf](https://cdn.mises.org/Money%20and%20the%20Mechanism%20of%20Exchange_2.pdf). Consulté le 05/04/2026.
- Johnson, D., Menezes, A., & Vanstone, S. (2001). The elliptic curve digital signature algorithm (ECDSA). *International Journal of Information Security*, 1, 36–63. <https://www.cs.miami.edu/home/burt/learning/Csc609.142/ecdsa-cert.pdf>. Consulté le 20/01/2026.
- Jokung Nguéna, O. (2004). *Mathématiques et gestion financière*, chapitre 4. L'évaluation des obligations à taux fixes; le risque de taux, (pp. 173–230). De Boeck Supérieur, Louvain-la-Neuve. <https://stm.cairn.info/mathematiques-et-gestion-financiere--9782804144029-page-173>. Consulté le 20/01/2026.
- Journal du Coin (2023). Fin de la Binance Card en Europe: l'échange en plein tournant réglementaire. <https://journalducoin.com/exchanges/binance-card-payé-facture-reglementaire-europe/>. Consulté le 20/01/2026.
- Journal du Coin (2024). 20 milliards de dollars en USDT vers la Russie: Londres et Washington lancent une enquête. <https://journalducoin.com/actualites/20-milliards-dollars-usdt-vers-la-russie-londres-et-washington-lancent-enquete/>. Consulté le 06/04/2026.
- Journal du Coin (2025a). Crypto en Europe: Christine Lagarde appelle à une réglementation stricte des stablecoins étrangers. <https://journalducoin.com/actualites/crypto-europe-christine-lagarde-appelle-a-reglementation-strict-stablecoins-etangers/>. Consulté le 06/04/2026.
- Journal du Coin (2025b). ETF Bitcoin: les institutionnels sont de retour à l'achat avec +675 millions de dollars pour les fonds BTC. <https://www.binance.com/fr/square/post/10-02-2025-bitcoin-spot-etfs-record-676m-net-inflow-in-a-day-as-institutional-demand-acce>. Consulté le 06/04/2026.
- Journal du Coin (2026). Ethereum atteint un nouveau record: 2,4 millions de transactions quotidiennes. <https://journalducoin.com/ethereum/ethereum-eth-nouveau-record-2-millions-transactions-quotidiennes-frais-bas/>. Consulté le 06/04/2026.

- J.P. Morgan (2026). Understanding gold and its role in a portfolio. <https://am.jpmorgan.com/fr/en/asset-management/institutional/insights/market-insights/market-updates/on-the-minds-of-investors/understanding-gold-in-portfolio/>. Consulté le 27/02/2026.
- Jung, A. (2024). The quantity theory of money, 1870-2020. <https://www.ecb.europa.eu/pub/pdf/scpwps/ecb.wp.2940.en.pdf>. Consulté le 22/01/2026.
- Jégourel, Y. (2017). *Les produits financiers dérivés*. La Découverte, Paris, troisième édition.
- Kador, J. & Kowal, K. (2020). Periodic table of asset classes. <https://kathleenkowal.com/portfolio/table-of-asset-classes/>. Consulté le 26/02/2026.
- Kaika, M. (2003). The Water Framework Directive: A new directive for a changing social, political and economic European framework. *European Planning Studies*, 11(3), 299–316. <https://www.tandfonline.com/doi/abs/10.1080/09654310303640>. Consulté le 23/03/2026.
- Kaldor, N. (1961). *The theory of capital*, chapitre 10. Capital accumulation and economic growth, (pp. 177–222). Springer. <http://gesd.free.fr/kaldor61.pdf>. Consulté le 06/06/2024.
- Kalvet, T. (2012). Innovation: a factor explaining e-government success in Estonia. *Electronic Government, an International Journal*, 9(2), 142–157. [https://www.researchgate.net/publication/235347972\\_Innovation\\_A\\_factor\\_explaining\\_e-government\\_success\\_in\\_Estonia](https://www.researchgate.net/publication/235347972_Innovation_A_factor_explaining_e-government_success_in_Estonia). Consulté le 04/04/2026.
- Kang, J.-K. & Stulz, R. M. (1997). In bank-centered corporate governance worth it? A cross-sectional analysis of the performance of Japanese firms during the asset price deflation. [https://www.nber.org/system/files/working\\_papers/w6238/w6238.pdf](https://www.nber.org/system/files/working_papers/w6238/w6238.pdf). Consulté le 26/01/2026.
- Kang, K.-Y. & Lee, S. (2022). Money, bitcoin, and monetary policy. *Journal of Money, Credit and Banking*, 56(1), 225–256. [https://swb.skku.edu/\\_res/sier/etc/WP2019-03.pdf](https://swb.skku.edu/_res/sier/etc/WP2019-03.pdf). Consulté le 06/06/2024.
- Karayan, R. (2022). La République Centrafricaine fait du bitcoin une monnaie légale. <https://www.usine-digitale.fr/article/la-republique-centrafricaine-fait-du-bitcoin-une-monnaie-legale.N1999987>. Consulté le 20/01/2026.
- Karnik, A. (2025). Mastering Cash Forecasting in Corporate Treasury. <https://www.linkedin.com/pulse/mastering-cash-forecasting-corporate-treasury-anuj-karnik-jy5mc>. Consulté le 25/01/2026.
- Kehoe, T. J., Arellano, C., & Conesa, J. C. (2012). Chronic sovereign debt crises in the eurozone, 2010–2012. <https://www.minneapolisfed.org/article/2012/chronic-sovereign-debt-crises-in-the-eurozone-20102012-epp>. Consulté le 28/01/2026.
- Kekre, R., Lenel, M., & Mainardi, F. (2024). Monetary policy, segmentation, and the term structure. *Becker Friedman Institute for Economics*, 45. [https://bfi.uchicago.edu/wp-content/uploads/2024/04/BFI\\_WP\\_2024-45.pdf](https://bfi.uchicago.edu/wp-content/uploads/2024/04/BFI_WP_2024-45.pdf). Consulté le 28/01/2026.
- Keynes, J. M. (1930). *A Treatise on Money*, volume 2. London: Macmillan. <https://ia601502.us.archive.org/4/items/in.ernet.dli.2015.28575/2015.28575.A-Treatise-On-Money.pdf>. Consulté le 20/01/2026.
- Keynes, J. M. (1933). *The Means to Prosperity*. Macmillan. <https://gutenberg.ca/ebooks/keynes-means/keynes-means-00-h.html>. Consulté le 26/01/2026.
- Keynes, J. M. (1936). Théorie générale de l'emploi, de l'intérêt et de la monnaie. *The Quarterly Journal of Economics*, 51(2). <http://gesd.free.fr/keynes36f.pdf>. Consulté le 30/11/2025.
- Kharif, O. (2017). Only one in 10 tokens is in use following initial coin offerings. <https://www.bloomberg.com/news/articles/2017-10-23/only-one-in-10-tokens-is-in-use-following-initial-coin-offerings>. Consulté le 20/01/2026.
- Khoury, S. J. & Pal, P. C. (2020). Negative interest rates. *Journal of Risk and Financial Management*, 13(5), 90. <https://www.mdpi.com/1911-8074/13/5/90>. Consulté le 22/01/2026.
- Kilian, L. (2009). Oil Price Shocks, Monetary Policy and Stagflation. *Journal of Money, Credit and Banking*, 40(6), 1079–1122. <https://www.rba.gov.au/publications/confs/2009/pdf/kilian.pdf>. Consulté le 26/01/2026.
- Kilian, L. & Park, C. (2009). The impact of oil price shocks on the U.S. stock market. *International Economic Review*, 50(4), 1267–1287. <https://onlinelibrary.wiley.com/doi/10.1111/j.1468-2354.2009.00568.x>. Consulté le 26/02/2026.

- Kim, S., Kwon, Y., & Cho, S. (2018). A survey of scalability solutions on blockchain. *International Conference on ICTC*. [https://www.researchgate.net/publication/329491655\\_A\\_Survey\\_of\\_Scalability\\_Solutions\\_on\\_Blockchain](https://www.researchgate.net/publication/329491655_A_Survey_of_Scalability_Solutions_on_Blockchain). Consulté le 20/01/2026.
- King, R. G. & Levine, R. (1993). Finance and growth: Schumpeter might be right. *The Quarterly Journal of Economics*, 108(3), 717–737. [https://www.researchgate.net/publication/24091420\\_Finance\\_and\\_Growth\\_Schumpeter\\_Might\\_Be\\_Right](https://www.researchgate.net/publication/24091420_Finance_and_Growth_Schumpeter_Might_Be_Right). Consulté le 22/01/2026.
- Kishimoto, S. (2016). Remunicipalization: A practical guide for communities and policy makers. <https://blueplanetproject.net/wp-content/uploads/2023/02/aa974-publicwaterforall-mar14-remun.pdf>. Consulté le 23/03/2026.
- Knight Frank (2021). Investment market France, Q1 2021. <https://content.knightfrank.com/research/2216/documents/en/investment-market-france-may-2021-8056.pdf>. Consulté le 01/03/2026.
- Knight Frank (2025). The greater Paris region office market, Q1 2025. <https://www.knightfrank.fr/fichiers/publications2020/file/kf-greater-paris-office-market-q1-2025-682ef7d8af306462868521.pdf>. Consulté le 01/03/2026.
- Knight Frank (2026). Les marchés immobiliers français : bilan 2025 et perspectives 2026. <https://www.knightfrank.fr/actualites/les-marches-immobiliers-francais-bilan-2025-et-perspectives-2026>. Consulté le 01/03/2026.
- Kose, M. A., Prasad, E. S., Rogoff, K., & Wei, S.-J. (2009). Financial Globalization and Economic Policies. *Brookings Global Economy & Development*, 34. [https://www.brookings.edu/wp-content/uploads/2016/06/04\\_financial\\_globalization\\_prasad.pdf](https://www.brookings.edu/wp-content/uploads/2016/06/04_financial_globalization_prasad.pdf). Consulté le 26/01/2026.
- Krishnan, R., Smith, M. D., & Telang, R. (2004). BitTorrent: The economics of peer-to-peer networks. *Journal of Information, Law and Technology*, 1. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=504062](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=504062). Consulté le 05/04/2026.
- Krohn, I., Schrimpf, A., & Sushkov, V. (2025). The FX trade execution landscape through the prism of the 2025 BIS triennial survey. *BIS Quarterly Review*, 4. [https://www.bis.org/publ/qtrpdf/r\\_qt2512v.htm](https://www.bis.org/publ/qtrpdf/r_qt2512v.htm). Consulté le 03/03/2026.
- Krugman, P. R. & Wells, R. (2019). *Macroéconomie*. De Boeck Supérieur, Louvain-la-Neuve. <https://studyroombd.wordpress.com/wp-content/uploads/2014/08/macroeconomics-krugman-wells.pdf>. Consulté le 20/01/2026.
- KuCoin (2026). Qu'est-ce que la loi MiCA pour les stablecoins et quel est son impact sur les cryptomonnaies ? <https://www.kucoin.com/fr/blog/fr-what-is-mica-act-and-its-impact-on-cryptocurrency>. Consulté le 06/04/2026.
- Kwok, C. & Tadesse, S. (2006). National culture and financial systems. *Journal of International business studies*, 37, 227–247. <https://deepblue.lib.umich.edu/bitstream/handle/2027.42/57264/wp884%20.pdf>. Consulté le 06/06/2024.
- Kydland, F. E. & Prescott, E. C. (1977). Rules rather than discretion: The inconsistency of optimal plans. *Journal of Political Economy*, 85(3). [https://www.sfu.ca/~kkasa/prescott\\_77.pdf](https://www.sfu.ca/~kkasa/prescott_77.pdf). Consulté le 09/12/2025.
- La Finance pour tous (2023). Comment expliquer les taux d'intérêt négatifs ? <https://www.lafinancepourtous.com/decryptages/finance-perso/banque-et-credit/taux-d-interet/comment-expliquer-les-taux-dinteret-negatifs/>. Consulté le 22/01/2026.
- La Finance pour tous (2024). Federal Reserve system (FED). <https://www.lafinancepourtous.com/decryptages/marches-financiers/acteurs-de-la-finance/federal-reserve-system-fed/>. Consulté le 24/01/2026.
- Laeven, L. & Valencia, F. (2020). Systemic banking crises database II. *IMF Economic Review*, 68, 307–361. <https://link.springer.com/article/10.1057/s41308-020-00107-3>. Consulté le 22/01/2026.
- Lagarde, C. (2022). La politique monétaire dans la zone euro (Conférence Karl Otto Pöhl). <https://www.ecb.europa.eu/press/key/date/2022/html/ecb.sp220920~c3afc1a441.fr.html>. Consulté le 20/01/2026.
- Landau, J.-P. (2018). Les crypto-monnaies: rapport au Ministre de l'économie et des finances. [https://minefi.hosting.augure.com/Augure\\_Minefi/r/ContenuEnLigne/Download?id=6F566715-5AA6-4550-9384-D97D9783D437&filename=Synth%C3%A8se%20du%20rapport%20Landau.pdf](https://minefi.hosting.augure.com/Augure_Minefi/r/ContenuEnLigne/Download?id=6F566715-5AA6-4550-9384-D97D9783D437&filename=Synth%C3%A8se%20du%20rapport%20Landau.pdf). Consulté le 20/01/2026.



- Le Bourva, J. (1962). Création de monnaie et multiplicateur du crédit. *Revue Économique*, 13(1), 29–56. [https://www.persee.fr/doc/reco\\_0035-2764\\_1962\\_num\\_13\\_1\\_407496](https://www.persee.fr/doc/reco_0035-2764_1962_num_13_1_407496). Consulté le 20/01/2026.
- Le Figaro (2024). Nouvelle-Calédonie: l'usine de nickel Prony Resources a redémarré après six mois de blocage. <https://www.lefigaro.fr/societes/nouvelle-caledonie-l-usine-de-nickel-prony-resources-redemarre-apres-six-mois-de-blocage-20241118>. Consulté le 27/02/2026.
- Le Maire, B. (2019). Libra est une menace pour la souveraineté des États. <https://fr.linkedin.com/pulse/libra-est-une-menace-pour-la-souverainete-des-etats-bruno-le-maire>. Consulté le 20/01/2026.
- Le Moign, C. & Ezra, A. (2021). Caractéristiques du marché des options sur actions françaises. <https://www.amf-france.org/sites/institutionnel/files/private/2021-07/caracteristiques-du-marche-des-options-francaises-a-mono-sous-jacents.pdf>. Consulté le 11/04/2024.
- Le Monde (2001). Carrefour cède Picard Surgelés. *Le Monde*. [https://www.lemonde.fr/archives/article/2001/02/21/carrefour-cede-picard-surgeles\\_4177509\\_1819218.html](https://www.lemonde.fr/archives/article/2001/02/21/carrefour-cede-picard-surgeles_4177509_1819218.html). Consulté le 20/01/2026.
- Le Monde (2020). À Paris, le « boulevard du bitcoin » boude la cryptomonnaie. *Le Monde*. [https://www.lemonde.fr/m-le-mag/article/2020/12/25/a-paris-le-boulevard-du-bitcoin-boude-la-cryptomonnaie\\_6064512\\_4500055.html](https://www.lemonde.fr/m-le-mag/article/2020/12/25/a-paris-le-boulevard-du-bitcoin-boude-la-cryptomonnaie_6064512_4500055.html). Consulté le 05/04/2026.
- Le Monde (2022). Guerre en Ukraine: peut-on écarter la Russie du réseau SWIFT? <https://urls.fr/1sDpT0>. Consulté le 06/04/2026.
- Le Monde (2023). Faut-il indexer les salaires sur l'inflation, comme le proposent les députés "insoumis"? [https://www.lemonde.fr/les-decodeurs/article/2023/11/30/faut-il-indexer-les-salaires-sur-l-inflation-comme-le-proposent-les-deputes-insoumis\\_6146875\\_4355771.html](https://www.lemonde.fr/les-decodeurs/article/2023/11/30/faut-il-indexer-les-salaires-sur-l-inflation-comme-le-proposent-les-deputes-insoumis_6146875_4355771.html). Consulté le 20/01/2026.
- Le Point (2017). UE: 1600 milliards d'euros d'aides d'État pour les banques. [https://www.lepoint.fr/economie/ue-1-600-milliards-d-euros-d-aides-d-etat-pour-les-banques-21-12-2012-1605016\\_28.php](https://www.lepoint.fr/economie/ue-1-600-milliards-d-euros-d-aides-d-etat-pour-les-banques-21-12-2012-1605016_28.php). Consulté le 20/01/2026.
- Le Rubicon (2023). Le yuan numérique: un péril pour l'hégémonie du dollar. <https://lerubicon.org/le-yuan-numerique-un-peril-pour-lhegemonie-du-dollar/>. Consulté le 06/04/2026.
- Ledger (2025). EU central bank sees crypto adoption growth, raises stability concerns. <https://www.ledgerinsights.com/eu-central-bank-sees-crypto-adoption-growth-raises-stability-concerns/>. Consulté le 05/04/2026.
- Lefevbre, A. (2025). Le corporate venture monte en puissance en France. <https://www.574invest.com/paru-dans-option-finance-le-corporate-venture-monte-en-puissance-en-france/>. Consulté le 20/01/2026.
- Leitão, J., Pereira, D., & Gonçalves, A. (2022). Business incubators, accelerators, and performance of technology-based ventures: A systematic literature review. *Journal of Open Innovation: Technology, Market, and Complexity*, 8(1), 46. <https://www.mdpi.com/2199-8531/8/1/46>. Consulté le 20/01/2026.
- Leland, H. E. & Pyle, D. H. (1977). Informational asymmetries, financial structure, and financial intermediation. *The Journal of Finance*, 32(2), 371–387. [https://www.researchgate.net/publication/24139319\\_Informational\\_Asymmetries\\_Financial\\_Structure\\_and\\_Financial\\_Intermediation](https://www.researchgate.net/publication/24139319_Informational_Asymmetries_Financial_Structure_and_Financial_Intermediation). Consulté le 20/01/2026.
- Levine, R. (2004). Finance and growth: Theory and evidence. [https://www.nber.org/system/files/working\\_papers/w10766/w10766.pdf](https://www.nber.org/system/files/working_papers/w10766/w10766.pdf). Consulté le 22/01/2026.
- Levy Yeyati, E. (2006). Financial dollarization: evaluating the consequences. *Economic Policy*, 21(45), 61–118. [https://www.researchgate.net/publication/4758439\\_Financial\\_Dollarization\\_Evaluating\\_the\\_Consequences](https://www.researchgate.net/publication/4758439_Financial_Dollarization_Evaluating_the_Consequences). Consulté le 22/01/2026.
- L'expert comptable (2026). Financements de la BPI : guide complet des aides et prêts (2026). <https://www.l-expert-comptable.com/a/532145-quel-role-de-la-bpi-pour-l-aide-au-financement-des-petites-entreprises.html>. Consulté le 23/03/2026.

- Lintner, J. (1965). Security prices, risk, and maximal gains from diversification. *The Journal of Finance*, 20(4), 587–615. <https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.1965.tb02930.x>. Consulté le 26/02/2026.
- London Mining Network (2013). Glencore Xstrata: the birth of a mining monster. <https://londonminingnetwork.org/wp-content/uploads/2013/10/Glencore-Xstrata-merger-final-131004.doc>. Consulté le 27/02/2026.
- LSEG (2025). ICD – LSEG Islamic finance development report 2025. [https://www.lseg.com/content/dam/data-analytics/en\\_us/documents/reports/lseg-islamic-finance-development-indicator-2025.pdf](https://www.lseg.com/content/dam/data-analytics/en_us/documents/reports/lseg-islamic-finance-development-indicator-2025.pdf). Consulté le 26/01/2026.
- Lucas, R. E. (1990). Why Doesn't Capital Flow from Rich to Poor Countries? *The American Economic Review*, 80(2), 92–96. [http://qed.econ.queensu.ca/pub/faculty/sumon/lucas\\_richpoor.pdf](http://qed.econ.queensu.ca/pub/faculty/sumon/lucas_richpoor.pdf). Consulté le 26/01/2026.
- Lucotte, Y. (2018). Obligations et taux d'intérêt. [https://celene.univ-orleans.fr/pluginfile.php/513483/mod\\_resource/content/1/Chapitre%201\\_Obligations%20et%20taux%20dint%C3%A9r%C3%AAt\\_partie%201.pdf](https://celene.univ-orleans.fr/pluginfile.php/513483/mod_resource/content/1/Chapitre%201_Obligations%20et%20taux%20dint%C3%A9r%C3%AAt_partie%201.pdf). Consulté le 20/01/2026.
- Lustig, H., Roussanov, N., & Verdelhan, A. (2011). Common risk factors in currency markets. *The Review of Financial Studies*, 24(11), 3731–3777. [https://www.nber.org/system/files/working\\_papers/w14082/w14082.pdf](https://www.nber.org/system/files/working_papers/w14082/w14082.pdf). Consulté le 03/03/2026.
- Lustig, H. & Verdelhan, A. (2007). The cross section of foreign currency risk premia and consumption growth risk. *American Economic Review*, 97(1), 89–117. [https://web.mit.edu/adrienv/www/FX\\_Xsection.pdf](https://web.mit.edu/adrienv/www/FX_Xsection.pdf). Consulté le 03/03/2026.
- Lyons, R. K. (2001). *The Microstructure Approach to Exchange Rates*. Cambridge (Massachusetts): MIT Press. <https://direct.mit.edu/books/monograph/2004/The-Microstructure-Approach-to-Exchange-Rates>. Consulté le 02/03/2026.
- Légifrance (2019). Décret n° 2019-771 du 23 juillet 2019 relatif aux obligations d'actions de réduction de la consommation d'énergie finale dans des bâtiments à usage tertiaire. <https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000038812251>. Consulté le 01/03/2026.
- Légifrance (2022). Loi n° 2022-1158 du 16 août 2022 portant mesures d'urgence pour la protection du pouvoir d'achat. [https://www.legifrance.gouv.fr/jorf/article\\_jo/JORFARTI000046186741](https://www.legifrance.gouv.fr/jorf/article_jo/JORFARTI000046186741). Consulté le 01/03/2026.
- Légifrance (2023). Loi n° 2023-568 du 7 juillet 2023 visant à protéger les logements contre l'occupation illicite. <https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000047800299>. Consulté le 01/03/2026.
- Légifrance (2025). Délibération n° 2025-78 du 13 mars 2025 — TURPE 7 HTA-BT. <https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000051587195>. Consulté le 02/03/2026.
- Létard, V. (2025). Bilan 2024 du logement social: Valérie Létard souhaite poursuivre une dynamique de relance de la production de logements sociaux en 2025. <https://www.ecologie.gouv.fr/presse/bilan-2024-du-logement-social-valerie-letard-souhaite-poursuivre-dynamique-relance>. Consulté le 01/03/2026.
- Mackay, C. (1856). *Extraordinary Popular Delusions and the Madness of Crowds*. Richard Bentley. <https://peachv.org/images/Human-1/MSPopularDelusionsMackayII.pdf>. Consulté le 26/01/2026.
- Macron, E. (2023). Présentation du Plan eau. <https://www.elysee.fr/emmanuel-macron/2023/03/30/presentation-du-plan-eau>. Consulté le 23/03/2026.
- Mag Startup (2025). Paris : 66 accélérateurs, saturation ou opportunité ? <https://www.magstartup.com/accelerateurs-paris/>. Consulté le 20/01/2026.
- Mahanand, P. K. & Naik, A. (2025). Towards de-dollarization and a new global financial order. *International Journal of Political Science and Governance*. <https://www.journalofpoliticalscience.com/uploads/archives/7-9-18-680.pdf>. Consulté le 27/01/2026.
- Major Prépa (2026). Les suites arithmético-géométriques. <https://major-prepa.com/mathematiques/suites-arithmetico-geometriques/>. Consulté le 20/01/2026.

- Maloney, M. T. & Mulherin, J. H. (2003). The complexity of price discovery in an efficient market: the stock market reaction to the Challenger crash. *Journal of Corporate Finance*, 9(4), 453–479.
- Malthus, T. R. (1846). *Principes d'économie politique considérés sous le rapport de leur application pratique: suivis des définitions en économie politique*, volume 8. Guillaumin. <https://fr.scribd.com/document/949828115/Principes-d-Economie-Politique-Thomas-Malthus>. Consulté le 20/01/2026.
- Market Growth Reports (2026). Crowdfunding: Global market size, share, growth and industry analysis, forecasts to 2033. <https://www.marketgrowthreports.com/fr/market-reports/crowdfunding-global-market-103509>. Consulté le 20/01/2026.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91. <http://finance.martinsewell.com/capm/Markowitz1952.pdf>. Consulté le 26/02/2026.
- Marshall, A. (1922). *Money, Credit and Commerce*. Macmillan. <https://archive.org/details/moneycreditcomme00mars/page/n17/mode/2up>. Consulté le 20/01/2026.
- Martin, P., Monnet, E., & Ragot, X. (2021). Que peut encore faire la Banque centrale européenne? *Les notes du CAE*, 65. <https://www.cae-eco.fr/staticfiles/pdf/cae-note065.pdf>. Consulté le 06/06/2024.
- Massa, O. I., Good, D. L., & Irwin, S. H. (2009). Empirical confidence intervals for WASDE forecasts of corn, soybean, and wheat prices. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=1392388](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1392388). Consulté le 01/03/2026.
- May, T. (1988). The crypto-anarchist manifesto. <https://groups.csail.mit.edu/mac/classes/6.805/articles/crypto/cypherpunks/may-crypto-manifesto.html>. Consulté le 11/08/2024.
- McCarthy, I. S. (1979). Hosting offshore banks: Benefits and costs. *Finance and Development*, 16(4), 45–48. <https://www.elibrary.imf.org/view/journals/022/0016/004/article-A012-en.xml>. Consulté le 20/01/2026.
- McKinsey & Company (2023). The Economic Potential of Generative AI: The Next Productivity Frontier. <https://urls.fr/IEzYcN>. Consulté le 26/01/2026.
- McKinsey & Company (2026). Global private markets report 2026. <https://www.mckinsey.com/industries/private-capital/our-insights/global-private-markets-report>. Consulté le 01/03/2026.
- McPhail, L. L. & Babcock, B. A. (2012). Impact of US biofuel policy on US corn and gasoline price variability. *Energy*, 37(1). <https://www.sciencedirect.com/science/article/abs/pii/S0360544211007225>. Consulté le 01/03/2026.
- Mecardo (2025). Record crops, tight stocks. <https://mecardo.com.au/record-crops-tight-stocks/>. Consulté le 01/03/2026.
- Medda, F. & Panayiotou, A. (2014). Portfolio of infrastructure investments : An analysis of European pension funds. *Journal of Infrastructure Systems*. [https://discovery.ucl.ac.uk/1508416/3/Medda\\_A.P\\_F.M\\_Portfolio%20of%20Infrastructure%20Investments\\_JIS.pdf](https://discovery.ucl.ac.uk/1508416/3/Medda_A.P_F.M_Portfolio%20of%20Infrastructure%20Investments_JIS.pdf). Consulté le 02/03/2026.
- Meissner, C. M. (2015). The limits of bimetallism. *NBER Working Paper Series*, 20852. <https://www.nber.org/papers/w20852>. Consulté le 22/01/2026.
- Meketa Investment (2025). The spectrum of infrastructure assets. [https://meketa.com/wp-content/uploads/2025/07/MEKETA\\_Infrastructure-core-vs-non-core.pdf](https://meketa.com/wp-content/uploads/2025/07/MEKETA_Infrastructure-core-vs-non-core.pdf). Consulté le 02/03/2026.
- Meketa Investment Group (2023). Treasury inflation protected securities (TIPS). [https://meketa.com/wp-content/uploads/2023/08/MEKETA\\_TIPS.pdf](https://meketa.com/wp-content/uploads/2023/08/MEKETA_TIPS.pdf). Consulté le 22/01/2026.
- Melchior (2021). Comment évaluer le modèle de la concession d'autoroutes? le cas Vinci. <https://www.melchior.fr/etude-de-cas/comment-evaluer-le-modele-de-la-concession-d-autoroutes-le-cas-de-vinci>. Consulté le 02/03/2026.
- Mendret, J. (2022). Réutilisation des eaux usées: que va changer le nouveau règlement européen ? <https://www.umontpellier.fr/articles/reutilisation-des-eaux-usees-que-va-changer-le-nouveau-reglement-europeen>. Consulté le 23/03/2026.
- Menkhoff, L., Sarno, L., Schmeling, M., & Schrimpf, A. (2012). Currency momentum strategies. *Journal of Financial Economics*, 106(3), 660–684. <https://www.bis.org/publ/work366.pdf>. Consulté le 03/03/2026.

- Merton, R. C. (1972). An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis*, 7(4), 1851–1872. [http://www.stat.ucla.edu/~nchristo/statistics\\_c183\\_c283/analytic\\_derivation\\_frontier.pdf](http://www.stat.ucla.edu/~nchristo/statistics_c183_c283/analytic_derivation_frontier.pdf). Consulté le 26/02/2026.
- Metcalfe, W. (2020). *Blockchain and crypt-currency*, chapitre 5. Ethereum, smart contracts, dApps, (pp. 77–93). Springer. [https://library.oapen.org/bitstream/handle/20.500.12657/37713/2020\\_Book\\_BlockchainAndCryptCurrency.pdf](https://library.oapen.org/bitstream/handle/20.500.12657/37713/2020_Book_BlockchainAndCryptCurrency.pdf). Consulté le 06/06/2024.
- MEXC (2025). Le cadre du yuan numérique chinois permet les paiements d'intérêts à partir de 2026. <https://www.mexc.co/fr/news/369806>. Consulté le 06/04/2026.
- Meynkhard, A. (2019). Fair market value of bitcoin: halving effect. *Investment Management & Financial Innovations*, 16(4), 72. <https://pdfs.semanticscholar.org/3686/3ee36a6e246b23c5baf2ba1e623ffd1fa832.pdf>. Consulté le 06/06/2024.
- Mining Beyond Hot Air (2024). Le secteur du nickel en crise: exemple de la Nouvelle-Calédonie. <https://miningbeyondhotair.org/2024/09/11/le-secteur-du-nickel-en-crise-exemple-de-la-nouvelle-caledonie-kanaky/>. Consulté le 27/02/2026.
- Ministère de la Transition écologique (2016). Systèmes d'autoconsommation. <https://www.ecologie.gouv.fr/politiques-publiques/systemes-dautoconsommation>. Consulté le 11/08/2024.
- Ministère de la Transition écologique (2023a). Financement des services publics d'eau potable et d'assainissements. [https://www.ecologie.gouv.fr/gestion-leau-en-france#scroll-nav\\_\\_7](https://www.ecologie.gouv.fr/gestion-leau-en-france#scroll-nav__7). Consulté le 23/03/2026.
- Ministère de la Transition écologique (2023b). Le plan d'action pour une gestion résiliente et concertée de l'eau. <https://www.ecologie.gouv.fr/politiques-publiques/plan-daction-gestion-resiliente-concertee-leau>. Consulté le 23/03/2026.
- Ministère de la Transition écologique (2024). Ressources minérales des grands fonds marins. <https://www.statistiques.developpement-durable.gouv.fr/edition-numerique/chiffres-cles-mer-littoral/64-ressources-minerales-des-grands-fonds>. Consulté le 27/02/2026.
- Ministère de l'Économie (2014). Facileco: existe-t-il aujourd'hui un risque de déflation en zone euro? <https://www.economie.gouv.fr/facileco/deflation-risque-zone-euro>. Consulté le 20/01/2026.
- Ministère de l'Économie (2024). Facileco: comment est fixé le taux du livret A? <https://www.economie.gouv.fr/facileco/livret-a-taux-calcul>. Consulté le 20/01/2026.
- Ministère de l'Économie et des Finances (2025a). La Direction générale des entreprises (DGE) publie son rapport sur l'écosystème deeptech en France. <https://www.economie.gouv.fr/actualites/la-direction-generale-des-entreprises-dge-publie-son-rapport-sur-lecosysteme-deeptech-en>. Consulté le 01/03/2026.
- Ministère de l'Économie et des Finances (2025b). Épargne réglementée: le taux du livret A fixé à 1,7 %, le taux du LEP fixé à 2,7 %. <https://presse.economie.gouv.fr/epargne-reglementee-le-taux-du-livret-a-fixe-a-17-le-taux-du-livret-depargne-populaire-fixe-a-27>. Consulté le 01/03/2026.
- Minsky, H. P. (1975). *John Maynard Keynes*. Columbia University Press. <http://digamo.free.fr/minsky75.pdf>. Consulté le 26/01/2026.
- Miranda-Agrippino, S. & Ricco, G. (2021). The transmission of monetary policy shocks. *American Economic Journal: Macroeconomics*, 13(3), 74–107. <https://www.ofce.sciences-po.fr/pdf/dtravail/WP2017-15.pdf>. Consulté le 20/01/2026.
- Mishkin, F. S. (2000). Lessons from the Asian Crisis. *NBER Working Paper Series*, 7102. [https://www.nber.org/system/files/working\\_papers/w7102/w7102.pdf](https://www.nber.org/system/files/working_papers/w7102/w7102.pdf). Consulté le 26/01/2026.
- Mishkin, F. S. (2007). *The economics of money, banking, and financial markets*. Pearson Education. <https://unitimesofficial.wordpress.com/wp-content/uploads/2020/06/the-economics-of-money-banking-and-financial-markets-11th-edition-global-edition.pdf>. Consulté le 20/01/2026.

- Mister Prépa (2024). Les crises de change asiatiques des années 1990: un séisme économique. <https://misterprepa.net/crises-de-change-asiatiques-annees-1990/>. Consulté le 26/01/2026.
- Modigliani, F. & Miller, M. H. (1958). The cost of capital, corporation finance and the theory of investment. *The American economic review*, 48(3), 261–297. <https://www.jstor.org/stable/1809766>. Consulté le 19/01/2026.
- Mody, A., Woong Kang, D., & Ligthart, N. (2020). The ECB and the Fed: a comparative narrative. <https://www.bruegel.org/blog-post/ecb-and-fed-comparative-narrative>. Consulté le 24/01/2026.
- Monga, S. & Singh, D. (2022). MRBSChain: a novel scalable medical records Binance smart chain blockchain framework. *Scientific Reports*, 12(17660). <https://www.nature.com/articles/s41598-022-22569-3>. Consulté le 05/04/2026.
- Mordor Intelligence (2026). Middle East Islamic finance market growth report 2031. <https://www.mordorintelligence.com/industry-reports/middle-east-islamic-finance-market>. Consulté le 26/01/2026.
- Morlais, M.-A. (2010). Introduction aux mathématiques financières. [http://marieamelie.morlais.free.fr/FICHIERS/mathfinance\\_cours201011.pdf](http://marieamelie.morlais.free.fr/FICHIERS/mathfinance_cours201011.pdf). Consulté le 24/01/2026.
- Moschetto, B. & Deguen, N. (2022). *L'organisation monétaire européenne*, chapitre 4. La réalité de l'Union économique et monétaire, (pp. 53–74). PUF, Paris. <https://shs.cairn.info/l-organisation-monetaire-europeenne--9782130813118-page-53>. Consulté le 20/01/2026.
- Mundell, R. A. (1963). Capital mobility and stabilization policy under fixed and flexible exchange rates. *Canadian Journal of Economics and Political Science*, 29(4), 475–485. [https://www.depfe.unam.mx/actividades/11/macro-abierta/11-2\\_majr\\_05\\_mundell\\_1963.pdf](https://www.depfe.unam.mx/actividades/11/macro-abierta/11-2_majr_05_mundell_1963.pdf). Consulté le 20/01/2026.
- Nadini, M., Alessandretti, L., Di Giacinto, F., Martino, M., Aiello, L. M., & Baronchelli, A. (2021). Mapping the NFT revolution: Market trends, trade networks, and visual features. *Scientific Reports*, 11, 20902. <https://doi.org/10.1038/s41598-021-00053-8>. Consulté le 07/04/2026.
- Nagel, J. (2022). The ECB's mandate: maintaining price stability in the euro area. <https://www.bundesbank.de/en/press/speeches/the-ecb-s-mandate-maintaining-price-stability-in-the-euro-area--898824>. Consulté le 24/01/2026.
- Nakamoto, S. (2008). Bitcoin: un système de paiement électronique pair-à-pair. <https://bitcoin.org/bitcoin.pdf>. Consulté le 28/08/2024.
- Nalo (2025). Différence entre produits structurés athena et phoenix : le guide. Blog Nalo. <https://blog.nalo.fr/produits-structures-athena-phoenix/>. Consulté le 17/03/2026.
- NAREIT (2025). Nareit media fact sheet, December 2024. [https://www.reit.com/sites/default/files/2025-01/MediaFactSheet\\_Dec-2024.pdf](https://www.reit.com/sites/default/files/2025-01/MediaFactSheet_Dec-2024.pdf). Consulté le 01/03/2026.
- Nasdaq Global Index Watch (2026). The Nasdaq Veles California Water Index (NQH2O): Historical data. <https://indexes.nasdaqomx.com/Index/Overview/NQH2O>. Consulté le 01/03/2026.
- Navarro, M. (2008). Les accords de Bâle. *Regards croisés sur l'économie*, 1(3). <https://urls.fr/ofwhH->. Consulté le 20/01/2026.
- Ndung'u, N., Morales, A., & Ndirangu, L. (2016). Cashing in on the digital revolution. <https://www.elibrary.imf.org/view/journals/022/0053/002/article-A005-en.xml>. Consulté le 20/01/2026.
- Network, T. J. (2024). *State of Tax Justice 2024*. Tax Justice Network. <https://taxjustice.net/wp-content/uploads/2024/11/State-of-Tax-Justice-2024-French-Tax-Justice-Network.pdf>. Consulté le 26/01/2026.
- Nexialog Consulting (2022). Notre offre Bâle 4. <https://www.nexialog.com/wp-content/uploads/2022/09/Plquette-Offre-Bale4-Nexialog-Consulting.pdf>. Consulté le 20/01/2026.
- Nexity (2024). L'évolution des prix de l'immobilier sur 10 ans. <https://www.nexity.fr/guide-immobilier/conseils-achat/tendance-du-marche/evolution-des-prix-de-l-immobilier-sur-10-ans>. Consulté le 01/03/2026.
- Noble, G. (2014). Terminal méthanier de Dunkerque: dans les coulisses du chantier. <https://www.batiactu.com/edito/terminal-methanier-de-dunkerque---dans-les-couliss-37317.php>. Consulté le 02/03/2026.

- Norchaeva, S. (2024). Islamic finance: Principles, governance, sustainability and investment insights. *European Journal of Management, Economics and Business*, 1(3). <https://ejmeb.com/index.php/journal/article/view/38/26>. Consulté le 26/01/2026.
- Norges Bank (2014). The currency carry trade. *Discussion note*, 3. [https://www.nbim.no/globalassets/documents/discussion-paper/2014/nbim\\_discussionnotes\\_3-14.pdf](https://www.nbim.no/globalassets/documents/discussion-paper/2014/nbim_discussionnotes_3-14.pdf). Consulté le 03/03/2026.
- Notaires de France (2024). Bilan immobilier de l'année 2024. <https://www.notaires.fr/fr/actualites/bilan-immobilier-de-lannee-2024>. Consulté le 01/03/2026.
- Nugrahaeni, R. (2025). BRICS and the shift away from dollar dependence. *Chicago Policy Review*. <https://chicagopolicyreview.org/2025/10/08/brics-and-the-shift-away-from-dollar-dependence/>. Consulté le 27/01/2026.
- Observatoire Crédit logement (2026). Tableau de bord trimestriel – 4<sup>e</sup> trimestre 2025. [https://www.observatoirecreditlogement.fr/uploads/obs\\_publications/63284409-Tableau\\_Bord\\_L\\_Obs\\_Credit\\_Logement\\_Csa\\_T4\\_2025.pdf](https://www.observatoirecreditlogement.fr/uploads/obs_publications/63284409-Tableau_Bord_L_Obs_Credit_Logement_Csa_T4_2025.pdf). Consulté le 01/03/2026.
- Observatoire de la justice fiscale (2024). Crypto actifs/monnaies et délinquance financière. <https://obs-justice-fiscale.attac.org/comprendre-la-fiscalite/article/crypto-actifs-monnaies-et-delinquance-financiere>. Consulté le 07/04/2026.
- Observatoire des services publics d'eau et d'assainissement (2025). Prix moyen global de l'eau au 1<sup>er</sup> janvier 2024. <https://www.eaufrance.fr/chiffres-cles/prix-moyen-global-de-leau-au-1er-janvier-2024>. Consulté le 23/03/2026.
- Observatoire français des conjonctures économiques (1986). Le contre-choc pétrolier et la baisse du dollar. *Revue de l'OFCE*, 15, 117–148. <https://www.ofce.sciences-po.fr/pdf/revue/2-015.pdf>. Consulté le 27/02/2026.
- Obstfeld, M. & Rogoff, K. (1996). *Foundations of International Macroeconomics*. Cambridge (Massachusetts): MIT Press. <https://fr.scribd.com/doc/56605459/Obstfeld-Rogoff-Foundations-of-International-Macroeconomics>. Consulté le 03/03/2026.
- OCDE (2012). Oecd environmental outlook to 2050: The consequences of inaction. [https://www.oecd.org/content/dam/oecd/en/publications/reports/2012/03/oecd-environmental-outlook-to-2050\\_g1g14e69/9789264122246-en.pdf](https://www.oecd.org/content/dam/oecd/en/publications/reports/2012/03/oecd-environmental-outlook-to-2050_g1g14e69/9789264122246-en.pdf). Consulté le 01/03/2026.
- OCDE (2020). Les défis fiscaux soulevés par la numérisation de l'économie – Évaluation d'impact économique: cadre inclusif sur le BEPS. [https://www.oecd.org/fr/publications/les-defis-fiscaux-soulevés-par-la-numerisation-de-l-economie-rapport-sur-l-evaluation-d-impact-economique\\_044011c1-fr.html](https://www.oecd.org/fr/publications/les-defis-fiscaux-soulevés-par-la-numerisation-de-l-economie-rapport-sur-l-evaluation-d-impact-economique_044011c1-fr.html). Consulté le 11/08/2024.
- OCDE (2024). Apporter la transparence fiscale dans le monde des crypto-actifs: une mise à jour. [https://www.oecd.org/content/dam/oecd/fr/publications/reports/2024/07/bringing-tax-transparency-to-crypto-assets-an-update\\_fa222834/1247fef1-fr.pdf](https://www.oecd.org/content/dam/oecd/fr/publications/reports/2024/07/bringing-tax-transparency-to-crypto-assets-an-update_fa222834/1247fef1-fr.pdf). Consulté le 07/04/2026.
- Odonnat, I. (2021). Entretien: “les cryptomonnaies ne sont pas des monnaies”. *Ouest France*. <https://www.ouest-france.fr/economie/banques-finance/entretien-les-cryptomonnaies-ne-sont-pas-des-monnaies-801f4320-31b0-11ec-897d-2a6a1867e3db>. Consulté le 06/06/2024.
- Office de l'Union européenne pour la propriété intellectuelle (2024). Les Européens accèdent à des contenus illicites en ligne 10 fois par mois: rapport sur la violation du droit d'auteur en ligne 2024. [https://euipo.europa.eu/tunnel-web/secure/webdav/guest/document\\_library/observatory/documents/reports/2024\\_online\\_copyright\\_infringement/2024\\_online\\_copyright\\_infringement\\_in\\_the\\_EU\\_PressR\\_fr.pdf](https://euipo.europa.eu/tunnel-web/secure/webdav/guest/document_library/observatory/documents/reports/2024_online_copyright_infringement/2024_online_copyright_infringement_in_the_EU_PressR_fr.pdf). Consulté le 05/04/2026.
- Office international de l'eau (2023). La réutilisation des eaux usées traitées en France métropolitaine. <https://www.oieau.org/actualites/le-chiffre-cle-de-la-semaine-la-reutilisation-des-eaux-usees-traitees-en-france-metropolitaine>. Consulté le 23/03/2026.
- Office international de l'eau (2024). Part du volume d'eau mis en distribution perdu par fuite dans les canalisations en 2022. <https://chiffrecle.oieau.fr/2833>. Consulté le 23/03/2026.

- Oil & Gas Journal (2024). OJG survey shows global oil reserves increase while natural gas reserves decline. <https://www.ogj.com/exploration-development/reserves/article/55250010/ogj-survey-shows-global-oil-reserves-increase-while-natural-gas-reserves-decline>. Consulté le 26/02/2026.
- ONUDC (2023). Money laundering. <https://www.unodc.org/unodc/en/money-laundering/overview.html>. Consulté le 06/04/2026.
- Oppers, S. (2000). A model of the bimetallic system. *Journal of Economic Dynamics and Control*, 24(9–10), 1595–1619. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=883290](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=883290). Consulté le 22/01/2026.
- Option finance (2025). Immobilier institutionnel: vers un nouveau paradigme en Europe. <https://www.optionfinance.fr/rubrique-asset-management/immobilier-institutionnel-vers-un-nouveau-paradigme-en-europe.html>. Consulté le 01/03/2026.
- Organisation météorologique mondiale (2022). L'Europe connaît un réchauffement climatique plus de deux fois plus élevé que la moyenne mondiale. <https://wmo.int/fr/news/media-centre/leurope-connaît-un-rechauffement-climatique-plus-de-deux-fois-plus-eleve-que-la-moyenne-mondiale>. Consulté le 23/03/2026.
- ORWL (2019). Libra: les enjeux juridiques de la cryptomonnaies de Facebook. <https://www.orwl.fr/libra-les-enjeux-juridiques-de-la-cryptomonnaie-de-facebook/>. Consulté le 20/01/2026.
- ORWL (2025). MiCA: l'encadrement européen des ICO. <https://www.orwl.fr/mica-lencadrement-europeen-des-ico/>. Consulté le 07/04/2026.
- Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press. [https://www.actu-environnement.com/media/pdf/ostrom\\_1990.pdf](https://www.actu-environnement.com/media/pdf/ostrom_1990.pdf). Consulté le 08/12/2025.
- Our World in Data (2024). The world's lithium is mined in just a handful of countries. <https://ourworldindata.org/data-insights/the-worlds-lithium-is-mined-in-just-a-handful-of-countries>. Consulté le 28/02/2026.
- Ozkan, E. U. (2010). The Transformation of the Ownership Structure of Japanese Financial and Commercial Institutions. [https://en.apu.ac.jp/rcaps/uploads/fckeditor/publications/journal/RJAPS\\_V30\\_Ozkan2.pdf](https://en.apu.ac.jp/rcaps/uploads/fckeditor/publications/journal/RJAPS_V30_Ozkan2.pdf). Consulté le 26/01/2026.
- Papaioannou, M. G. (2006). *Exchange Rate Risk Measurement and Management: Issues and Approaches for Firms*. IMF Working Paper 6, Fonds monétaire international. <https://www.elibrary.imf.org/view/journals/001/2006/255/article-A001-en.xml>. Consulté le 03/03/2026.
- Papin, C. (2025). Comprendre le lissage du PTZ en 5 minutes: simulation et calcul. <https://www.pretto.fr/pret-immobilier/pret-taux-zero-ptz/calcul-lissage-ptz/>. Consulté le 20/01/2026.
- Park, Y. C. (1988). The variability of velocity: An international comparison. *IMF Staff Papers*, 17(3), 620–637. <https://www.elibrary.imf.org/view/journals/024/1970/003/article-A006-en.xml>. Consulté le 24/01/2026.
- Parlement européen & Conseil de l'Union européenne (2020). Règlement (UE) n° 2020/1503 du Parlement européen et du Conseil du 7 octobre 2020 relatif aux prestataires européens de services de financement participatif pour les entreprises. <https://eur-lex.europa.eu/legal-content/FR/ALL/?uri=CELEX%3A32020R1503>. Consulté le 20/01/2026.
- Passenger Project (2025). Standardisation and rare-earth permanent magnets. <https://passenger-project.eu/project-news/standardisation-and-rare-earth-permanent-magnets/>. Consulté le 28/02/2026.
- Paulson, N., Janzen, J., Swanson, K., & Schnitkey, G. (2022). Revisiting Ukraine, Russia, and agricultural commodity markets. <https://farmdocdaily.illinois.edu/2022/02/revisiting-ukraine-russia-and-agricultural-commodity-markets.html>. Consulté le 01/03/2026.
- Penn State College of Earth and Mineral Sciences (2009). The process of crude oil refining. EME 801 Energy Markets, Policy, and Regulation, Pennsylvania State University. <https://courses.ems.psu.edu/eme801/node/636>. Consulté le 26/02/2026.
- Pérez-Solà, C., Delgado-Segura, S., Herrera-Joancomartí, J., & Navarro-Arribas, G. (2019). Analysis of the SegWit adoption in Bitcoin. <https://deic.uab.cat/~guille/files/papers/2018.recsi.segwit.pdf>. Consulté le 11/08/2024.



- Pernaud-Orliac, P. (2012). Titre subordonné. <https://www.pernaud.fr/titre-subordonne/>. Consulté le 11/08/2024.
- Petrusheva, N. & Jordanoski, I. (2016). Comparative analysis between the fundamental and technical analysis of stocks. *Journal of Process Management. New Technologies*, 4(2), 26–31. <https://pdfs.semanticscholar.org/d429/04808773e006b5a1328d5b7b1ee133a8136a.pdf>. Consulté le 06/06/2024.
- Phemex (2025). Marché NFT: ventes au plus bas en 2025. <https://phemex.com/fr/news/article/nft-market-plummets-to-lowest-sales-volume-in-2025-43272>. Consulté le 07/04/2026.
- Phillips, A. W. (1958). The relation between unemployment and the rate of change of money wage rates in the United Kingdom, 1861–1957. *Economica*, 25(100), 283–299. <http://pombo.free.fr/phillips1958.pdf>. Consulté le 20/01/2026.
- Pigou, A. (1917). The value of money. *The Quarterly Journal of Economics*, 32(1), 38–65. <https://www.jstor.org/stable/1885078>. Consulté le 20/01/2026.
- Pike, R. & Neale, B. (2012). *Corporate Finance and Investment: Decisions and Strategies*. Pearson Education. [https://api.pageplace.de/preview/DT0400.9781292208589\\_A35840178/preview-9781292208589\\_A35840178.pdf](https://api.pageplace.de/preview/DT0400.9781292208589_A35840178/preview-9781292208589_A35840178.pdf). Consulté le 25/01/2026.
- Pinkow, F. & Iversen, J. (2020). Strategic objectives of corporate venture capital as a tool for open innovation. *Journal of Open Innovation: Technology, Market, and Complexity*, 6(4), 157. <https://www.sciencedirect.com/science/article/pii/S2199853122011325>. Consulté le 20/01/2026.
- Pisani, F. & Piotet, D. (2008). *Comment le web change le monde: l'alchimie des multitudes*. Pearson France, Paris. [https://ekladata.com/t3vvPPP5XdNzCCT\\_2sdb\\_tvg9ys/commentlewebchangelemonde-introetchap1.1222126314.pdf](https://ekladata.com/t3vvPPP5XdNzCCT_2sdb_tvg9ys/commentlewebchangelemonde-introetchap1.1222126314.pdf). Consulté le 20/01/2026.
- Poon, J. & Dryja, T. (2016). The Bitcoin Lightning Network: Scalable off-chain instant payments. <https://lightning.network/lightning-network-paper.pdf>. Consulté le 05/04/2026.
- Porta, R. L., Lopez-de Silanes, F., Shleifer, A., & Vishny, R. W. (1998). Law and finance. *Journal of political economy*, 106(6), 1113–1155. [https://www.nber.org/system/files/working\\_papers/w5661/w5661.pdf](https://www.nber.org/system/files/working_papers/w5661/w5661.pdf). Consulté le 06/06/2024.
- PPIC (2025). California's water market. <https://www.ppic.org/publication/californias-water-market/>. Consulté le 01/03/2026.
- Prasad, E. S., Rogoff, K. S., Wei, S.-J., & Kose, M. A. (2004). Financial Globalization, Growth and Volatility in Developing Countries. *Journal of Economic Literature*, 42(4), 887–935. [https://www.nber.org/system/files/working\\_papers/w10942/w10942.pdf](https://www.nber.org/system/files/working_papers/w10942/w10942.pdf). Consulté le 26/01/2026.
- Preqin (2025). Preqin benchmarks: Private markets performance data, Q1 2025. [https://assets.ctfassets.net/zf87m07ner47/3nTjBzw9f22WiUrYCMqYZo/f9fce304c3de67578b7714805175cd10/PM\\_Performance\\_Data\\_Report\\_Q1\\_2025\\_final.pdf](https://assets.ctfassets.net/zf87m07ner47/3nTjBzw9f22WiUrYCMqYZo/f9fce304c3de67578b7714805175cd10/PM_Performance_Data_Report_Q1_2025_final.pdf). Consulté le 01/03/2026.
- Private Corner (2023). Investir en infrastructures *Brownfield & Greenfield*. <https://private-corner.eu/newsroom/guides/investir-dans-les-infrastructures-brownfield-greenfield>. Consulté le 02/03/2026.
- Programme alimentaire mondial (2025). Bilan annuel 2024. <https://publications.wfp.org/2024/fr/bilan-annuel/>. Consulté le 01/03/2026.
- Public Citizen (2002). Cochabamba, Bolivia: Water privatization case study. [https://www.citizen.org/wp-content/uploads/Bolivia\\_PDF.pdf](https://www.citizen.org/wp-content/uploads/Bolivia_PDF.pdf). Consulté le 01/03/2026.
- Public Eye (2023). Switzerland: The commodities hub. <https://www.publiceye.ch/en/topics/commodities/switzerland/commodities-hub>. Consulté le 27/02/2026.
- Raabe, K. & Burgstaller, J. (2004). The impact of oil price changes on growth and inflation. [https://www.oenb.at/dam/jcr:db783f84-1942-48d3-bfeb-02bff0d3a525/mop\\_20042\\_the\\_impact\\_tcm16-19678.pdf](https://www.oenb.at/dam/jcr:db783f84-1942-48d3-bfeb-02bff0d3a525/mop_20042_the_impact_tcm16-19678.pdf). Consulté le 27/02/2026.

- Reinhart, C. M., Rogoff, K. S., & Savastano, M. A. (2003). Addicted to dollars. *NBER Working Paper Series*, 10015. <https://www.nber.org/papers/w10015>. Consulté le 22/01/2026.
- Renversez, F. (2008). De l'économie d'endettement à l'économie de marchés financiers. *Regards croisés sur l'économie*, 1(3), 54–64. <https://cemi.ehess.fr/docannexe/file/2772/renversez.pdf>. Consulté le 20/01/2026.
- Reuters (2024). Taux: le spread France-Allemagne inchangé après le second tour des législatives. [https://urls.fr/sZ\\_rLa](https://urls.fr/sZ_rLa). Consulté le 20/01/2026.
- Reuters (2025a). Exclusive: Trump administration eyes stake in company developing greenland rare earths mine. <https://www.reuters.com/business/trump-administration-eyes-stake-company-developing-greenland-rare-earth-mine-2025-10-03/>. Consulté le 27/02/2026.
- Reuters (2025b). Gold's rise in central bank reserves appears unstoppable. <https://www.reuters.com/markets/commodities/golds-rise-central-bank-reserves-appears-unstoppable-2025-09-04/>. Consulté le 27/02/2026.
- Reuters (2026a). La plateforme de monnaie numérique transfrontalière dirigée par la Chine connaît une croissance record. <https://urls.fr/47mfV4>. Consulté le 06/04/2026.
- Reuters (2026b). Venezuela to export \$2 billion worth of oil to US in deal with Trump. <https://www.reuters.com/business/energy/venezuela-us-talks-export-venezuelan-oil-us-sources-say-2026-01-06/>. Consulté le 26/02/2026.
- Rifkin, J. (2012). *La troisième révolution industrielle: comment le pouvoir latéral va transformer l'énergie, l'économie et le monde*. Les liens qui libèrent, Paris. <https://excerpts.numilog.com/books/9782918597476.pdf>. Consulté le 20/01/2026.
- Robinson, J. (1952). *The rate of interest and other essays*. MacMillan, Londres. [https://link.springer.com/chapter/10.1007/978-1-349-16188-1\\_4](https://link.springer.com/chapter/10.1007/978-1-349-16188-1_4). Consulté le 20/01/2026.
- Rogoff, K. (1985). The optimal degree of commitment to an intermediate monetary target. *The quarterly journal of economics*, 100(4), 1169–1189. <https://www.jstor.org/stable/1885679>. Consulté le 20/01/2026.
- Rolnick, A. J. & Weber, W. E. (1986). Gresham's Law or Gresham's Fallacy? *Journal of Political Economy*, 94(1), 185–199. <https://researchdatabase.minneapolisfed.org/concern/publications/5q47rn90w>. Consulté le 22/01/2026.
- Roush, T. (2024). Judge approves Binance's \$4.3 billion settlement for anti-money laundering, sanctions violations. <https://www.forbes.com/sites/tyleroush/2024/02/23/judge-approves-binance-43-billion-settlement-for-anti-money-laundering-sanctions-violations/>. Consulté le 05/04/2026.
- Rousseau, P. L. & Wachtel, P. (2011). What is happening to the impact of financial deepening on economic growth? *Economic inquiry*, 49(1), 276–288. [https://pages.stern.nyu.edu/~pwachtel/images/RW\\_EI\\_Jan2008.pdf](https://pages.stern.nyu.edu/~pwachtel/images/RW_EI_Jan2008.pdf). Consulté le 20/01/2026.
- Ruester, S. (2014). *Financing LNG Projects and the Role of Long-Term Sales-and-Purchase Agreements*. Technical report, DIW Berlin. <https://www.econstor.eu/bitstream/10419/106474/1/816088659.pdf>. Consulté le 02/03/2026.
- Russon, M.-A. (2019). Crypto-currencies gaining popularity in Kenya. <https://www.bbc.com/news/business-47307575>. Consulté le 20/01/2026.
- Rystad Energy (2024). Upstream breakeven: Shale oil inflation — onshore middle east cheapest source of new production at \$27/bbl. <https://www.rystadenergy.com/news/upstream-breakeven-shale-oil-inflation>. Consulté le 27/02/2026.
- Saez, E. & Zucman, G. (2019). Progressive wealth taxation. *Brookings Papers on Economic Activity*, 2019(2), 437–533. <https://www.brookings.edu/wp-content/uploads/2020/10/Saez-Zuchman-final-draft.pdf>. Consulté le 20/01/2026.
- Samuelson, P. A. (1958). An exact consumption-loan model of interest with or without the social contrivance of money. *Journal of Political Economy*, 66(6), 467–482. <http://www.econ2.jhu.edu/courses/336/06-Samuelson-Paper.pdf>. Consulté le 24/01/2026.

- Samuelson, P. A. & Solow, R. M. (1960). Analytical aspects of anti-inflation policy. *The American Economic Review*, 50(2), 177–194. <http://joseluisoreiro.com.br/site/link/9bcb1774222d83fdd31ef1b7acd68832e6ac0ba4.pdf>. Consulté le 20/01/2026.
- Santander (2026). Overdraft Charges Calculator — Help & Support. <https://www.santander.co.uk/personal/support/current-accounts/overdraft-cost-calculator>. Consulté le 25/01/2026.
- Sargent, T. J. & Wallace, N. (1981). Some unpleasant monetarist arithmetic. *Federal Reserve Bank of Minneapolis Quarterly Review*, 5(3), 1–17. <http://reparti.free.fr/sargent81.pdf>. Consulté le 20/01/2026.
- Sassen, S. (2018). The Global City. <https://erikafontanez.com/wp-content/uploads/2017/09/sassen-the-global-city-introducing-a-concept.pdf>. Consulté le 26/01/2026.
- Scanlon, K., Lunde, J., & Whitehead, C. (2008). Mortgage product innovation in advanced economies: More choice, more risk? *European Journal of Housing Policy*, 8(2), 109–131. <https://researchonline.lse.ac.uk/id/eprint/29889/>. Consulté le 24/02/2026.
- Scher, M. J. (2001). Bank-firm Cross-shareholding in Japan. *DESA discussion papers*, 15. <https://www.un.org/esa/desa/papers/2001/esa01dp15.pdf>. Consulté le 26/01/2026.
- Schlesinger, A. M. (1957). *The Age of Roosevelt: The Crisis of the Old Order*. Houghton Mifflin. <https://ia802908.us.archive.org/35/items/in.ernet.dli.2015.183858/2015.183858.The-Crisis-Of-The-Old-Order1919-1933.pdf>. Consulté le 26/01/2026.
- Schmidt, V. A. (2015). The eurozone's crisis of democratic legitimacy: Can the EU rebuild public trust and support for European economic integration? *European Economy Discussion Papers*, 15. [https://economy-finance.ec.europa.eu/document/download/95e529ef-923d-4d14-b225-09e6dbe9be57\\_en?filename=dp015\\_en.pdf](https://economy-finance.ec.europa.eu/document/download/95e529ef-923d-4d14-b225-09e6dbe9be57_en?filename=dp015_en.pdf). Consulté le 22/01/2026.
- Schnabel, I. (2023). Quantitative tightening: rationale and market impact. <https://www.ecb.europa.eu/press/key/date/2023/html/ecb.sp230302~41273ad467.en.html>. Consulté le 20/01/2026.
- Schroders (2025). Les investisseurs privés peuvent désormais accéder aux opportunités de la dette infrastructure. <https://urls.fr/rLRSQz>. Consulté le 02/03/2026.
- Schumpeter, J. A. (1911). *The theory of economic development*. Routledge.
- Schumpeter, J. A. (1934). *The Theory of Economic Development: An Inquiry into Profits, Capital, Credit, Interest, and the Business Cycle*. Cambridge: Harvard University Press. <https://cruel.org/books/hy/shortschumpeter/SchumpeterTheoryofEconDev.pdf>. Consulté le 06/12/2025.
- Schwager, J. D. (1995). *Technical analysis*, volume 43. John Wiley & Sons.
- Schwartz, D., Youngs, N., & Britto, A. (2014). *The Ripple Protocol Consensus Algorithm*. Technical report, Ripple Labs Inc. [https://ripple.com/files/ripple\\_consensus\\_whitepaper.pdf](https://ripple.com/files/ripple_consensus_whitepaper.pdf). Consulté le 04/04/2026.
- Schwartz, E. S. (1997). The stochastic behavior of commodity prices: Implications for valuation and hedging. *Journal of Finance*, 52(3), 923–973. <https://static.twentyoverten.com/593e8a9e7299b471eaecf644/H1tGPLaXM/The-Stochastic-Behavior-of-Commodity-Prices-Implications-for-Valuation-and-Hedging.pdf>. Consulté le 20/01/2026.
- SDNY (2023). SEC v. Ripple Labs, inc.: Order on cross-motions for summary judgment. <https://www.nysd.uscourts.gov/sites/default/files/2023-07/SEC%20vs%20Ripple%207-13-23.pdf>. Consulté le 04/04/2026.
- Secrétariat général pour l'investissement (2025). France 2030: rapport d'activité 2024. <https://www.info.gouv.fr/actualite/france-2030-publication-du-rapport-dactivite-2024>. Consulté le 23/03/2026.
- Selgin, G. (1996). Salvaging Gresham's Law: The good, the bad, and the illegal. *Journal of Money, Credit and Banking*, 28(4), 637–649. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=3731](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3731). Consulté le 22/01/2026.
- Semenzin, S., Rozas, D., & Hassan, S. (2022). Blockchain-based application at a governmental level: disruption or illusion? The case of Estonia. *Policy and Society*, 41(3), 386–401. <https://academic.oup.com/policyandsociety/article/41/3/386/6566828>. Consulté le 04/04/2026.

- Servet, J.-M. (2021). Le bitcoin, mirage monétaire et désastre écologique. <https://institut-rousseau.fr/le-bitcoin-mirage-monetaire-et-desastre-ecologique/>. Consulté le 11/08/2024.
- Service de la donnée et des études statistiques (2025a). La dépense de gestion des eaux usées en 2022. <https://www.statistiques.developpement-durable.gouv.fr/la-depense-de-gestion-des-eaux-usees-en-2022>. Consulté le 23/03/2026.
- Service de la donnée et des études statistiques (2025b). L'eau en france: ressource et utilisation – Extrait du bilan environnemental 2024. <https://www.statistiques.developpement-durable.gouv.fr/leau-en-france-ressource-et-utilisation-extrait-du-bilan-environnemental-2024>. Consulté le 23/03/2026.
- Service de la donnée et des études statistiques (2025c). Les prélèvements d'eau douce par usage en France en 2022. <https://www.statistiques.developpement-durable.gouv.fr/les-prelevements-deau-douce-par-usage-en-france-en-2022>. Consulté le 23/03/2026.
- Shaarawy, I. (2024). BRICS Summit 2024 in Russia: Advancing de-dollarization and cooperative initiatives. *Egypt Oil and Gas*. <https://egyptoil-gas.com/features/brics-summit-2024-in-russia-advancing-de-dollarization-and-cooperative-initiatives/>. Consulté le 27/01/2026.
- Sharpe, W. F. (1964). Capital asset prices: a theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3), 425–442. <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-6261.1964.tb02865.x>. Consulté le 06/06/2024.
- Sherif, N. (2019). Calling out autocallable pricing. <https://www.risk.net/our-take/6348336/calling-out-autocallable-pricing>. Consulté le 17/03/2026.
- Shiller, R. J. (2000). *Irrational exuberance*. Princeton University Press. <http://www.library.fu.ru/files/Shiller2.pdf>. Consulté le 20/01/2026.
- Shleifer, A. (1986). Do demand curves for stocks slope down? *The Journal of Finance*, 41(3), 579–590. [https://static1.squarespace.com/static/5e6033a4ea02d801f37e15bb/t/5f6246569d7a1247b45cb93d/1600276058968/shleifer\\_demand\\_curves\\_jf.pdf](https://static1.squarespace.com/static/5e6033a4ea02d801f37e15bb/t/5f6246569d7a1247b45cb93d/1600276058968/shleifer_demand_curves_jf.pdf). Consulté le 26/02/2026.
- Shleifer, A. & Vishny, R. W. (1997). The limits of arbitrage. *The Journal of Finance*, 52(1), 35–55. <https://web.stanford.edu/~piazzesi/Reading/ShleiferVishny1997.pdf>. Consulté le 03/03/2026.
- Sia Partners (2017). De Bâle I à Bâle IV: chronique d'une saga réglementaire. <https://www.sia-partners.com/fr/publications/publications-de-nos-experts/de-bale-1-a-bale-4-chronique-dune-saga-reglementaire>. Consulté le 20/01/2026.
- Sierra Daily News (2025). California farmland values plummet as water concerns and market pressures intensify. <https://www.sierradailynews.com/state/california-farmland-values-plummet-as-water-concerns-and-market-pressure-intensify/>. Consulté le 01/03/2026.
- Simmel, G. (1987). *Philosophie de l'argent*. PUF, Paris. <https://fr.scribd.com/document/382010362/Simmel-Philosophie-de-l-Argent>. Consulté le 20/01/2026.
- Smith, C. W. & Stulz, R. M. (1985). The determinants of firms' hedging policies. *Journal of Financial and Quantitative Analysis*, 20(4), 391–405. <https://bpb-us-w2.wpmucdn.com/u.osu.edu/dist/0/30211/files/2016/05/determinantsofirms-29oslo5.pdf>. Consulté le 03/03/2026.
- Soana, G. (2021). Regulating cryptocurrencies checkpoints: Fighting a trench war with cavalry? *Economic Notes*. <https://onlinelibrary.wiley.com/doi/10.1111/ecno.12195>. Consulté le 06/04/2026.
- Société des Grands Projets (2023). L'essentiel du Grand Paris Express. <https://www.grandparisexpress.fr/gpe-essentiel>. Consulté le 02/03/2026.
- Société des Grands Projets (2025). Société des Grands Projets – Investir présentation. [https://media-mediatheque.sgp.fr/pm\\_12814\\_219\\_219569-si0ycigflv.pdf](https://media-mediatheque.sgp.fr/pm_12814_219_219569-si0ycigflv.pdf). Consulté le 02/03/2026.
- Société Générale (2022). Structured products: Athena autocall. <https://www.sgforge.com/product-structured-products-athena/>. Consulté le 17/03/2026.

- Sortino, F. A. & Price, L. N. (1994). Performance measurement in a downside risk framework. *The Journal of Investing*, 3(3), 59–64. <https://www.pm-research.com/content/iiinvest/3/3/59>. Consulté le 26/02/2026.
- Southwest Airlines (2008). Current report filing (Form 8-K), april 17, 2008. U.S. Securities and Exchange Commission. [https://www.southwestairlinesinvestorrelations.com/sec-filings/all-sec-filings/content/0000092380-08-000007/ex99\\_1.htm](https://www.southwestairlinesinvestorrelations.com/sec-filings/all-sec-filings/content/0000092380-08-000007/ex99_1.htm). Consulté le 13/03/2026.
- Southwest Airlines (2021). The southwest jet fuel hedge strategy. Southwest50.com. <https://southwest50.com/our-stories/the-southwest-jet-fuel-hedge-strategy/>. Consulté le 13/03/2026.
- S&P Dow Jones Indices (2023). Float adjustment methodology. Document méthodologique, S&P Dow Jones Indices. <https://www.spglobal.com/spdji/en/documents/index-policies/methodology-sp-float-adjustment.pdf>. Consulté le 26/02/2026.
- S&P Global (2023a). Previewing the 2024 S&P GSCI rebalance: Big oil's mojo is back. <https://www.indexologyblog.com/2023/11/02/previewing-the-2024-sp-gsci-rebalance-big-oils-mojo-is-back/>. Consulté le 27/02/2026.
- S&P Global (2023b). S&p U.S. indices methodology. <https://www.spglobal.com/spdji/en/methodology/article/sp-us-indices-methodology/>. Consulté le 26/02/2026.
- S&P Global (2025a). Islamic finance 2025–2026: Resilient growth amid upcoming headwinds. <https://www.spglobal.com/ratings/en/regulatory/article/250417-islamic-finance-2025-2026-resilient-growth-amid-upcoming-headwinds-s13453768>. Consulté le 26/01/2026.
- S&P Global (2025b). SPIVA. <https://www.spglobal.com/spdji/en/research-insights/spiva/>. Consulté le 26/02/2026.
- Stall, K. (2023). Household savings and negative interest rates. *Journal of Macroeconomics*, 75, 103483. [https://www.researchgate.net/publication/369983787\\_Household\\_Savings\\_and\\_Negative\\_Interest\\_Rates](https://www.researchgate.net/publication/369983787_Household_Savings_and_Negative_Interest_Rates). Consulté le 22/01/2026.
- State Street Investment Management (2025). SPDR® Gold Shares (GLD). <https://www.ssga.com/library-content/products/factsheets/etfs/us/factsheet-us-en-gld.pdf>. Consulté le 28/02/2026.
- Station F (2026). Station F: Programs. <https://stationf.co/programs/in-house>. Consulté le 20/01/2026.
- Statista (2024a). Value of money supply M1 in euro area 2001-2024. <https://www.statista.com/statistics/254220/money-supply-m1-eurozone/>. Consulté le 24/01/2026.
- Statista (2024b). Value of money supply M2 in euro area 2001-2024. <https://www.statista.com/statistics/254226/money-supply-m2-eurozone/>. Consulté le 24/01/2026.
- Stempel, J. (2025). Ripple labs says it settles with US SEC, will pay reduced \$50 million. <https://finance.yahoo.com/news/ripple-labs-says-settles-us-195706883.html>. Consulté le 04/04/2026.
- Sterdyniak, H. & Debonneuil, M. (1984). La boucle prix-salaires dans l'inflation. *Revue économique*, 35(2). [https://www.persee.fr/doc/reco\\_0035-2764\\_1984\\_num\\_35\\_2\\_408779](https://www.persee.fr/doc/reco_0035-2764_1984_num_35_2_408779). Consulté le 20/01/2026.
- Stewart, S. D., Neumann, J. J., Knittel, C. R., & Heisler, J. (2009). Absence of value: An analysis of investment allocation decisions by institutional plan sponsors. *Financial Analysts Journal*, 65(6), 34–51. [https://www.fi360.com/uploads/media/2013\\_handouts\\_watanabe.pdf](https://www.fi360.com/uploads/media/2013_handouts_watanabe.pdf). Consulté le 20/01/2026.
- STSA (2023). Commodities: The Swiss commodity sector in figures. <https://suissenegoce.ch/app/uploads/2024/12/Magazine-2023-ENG.pdf>. Consulté le 27/02/2026.
- Susmel, R. (2015). Case study: Currency swaps — IBM and the World Bank. <https://www.bauer.uh.edu/rsusmel/7386/Case%20-%20IBM%20WB%20Swap.pdf>. Consulté le 12/03/2026.
- Swissquote (2025). Turbo warrants: comprendre les transactions à fort effet de levier. <https://www.swissquote.com/fr-ch/private/inspire/blog/markets-instruments/turbo-warrants-high-leverage-trading-without-options>. Consulté le 13/03/2026.
- SWP (2025). The strategic raw material partnership between the EU and Zambia. <https://www.swp-berlin.org/10.18449/2025C19/>. Consulté le 28/02/2026.

- Sénat (2016). La gestion de l'eau: Étude de législation comparée n°268. <https://www.senat.fr/notice-rapport/2015/lc268-notice.html>. Consulté le 23/03/2026.
- Sénat (2024). Projet de loi de finances pour 2025: Cohésion des territoires. [https://www.senat.fr/rap/a24-145-6/a24-145-6\\_mono.html](https://www.senat.fr/rap/a24-145-6/a24-145-6_mono.html). Consulté le 01/03/2026.
- Taylor, J. B. (1993). Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39. [https://web.stanford.edu/~johntayl/Onlinepaperscombinedbyyear/1993/Discretion\\_vs\\_Policy\\_Rules\\_in\\_Practice.pdf](https://web.stanford.edu/~johntayl/Onlinepaperscombinedbyyear/1993/Discretion_vs_Policy_Rules_in_Practice.pdf). Consulté le 09/12/2025.
- Tearline (2025). DRC: Expansion of Chinese-owned cobalt and copper mines in the Lualaba province. <https://urls.fr/otuSJH>. Consulté le 28/02/2026.
- Temin, P. (1975). *Lessons for the present from the Great Depression*. MIT Press. <https://dspace.mit.edu/bitstream/handle/1721.1/63865/lessonsforpresen00temi.pdf?sequence=1>. Consulté le 26/01/2026.
- The Curious Economist (2024). Corporate tax haven in ireland. <https://thecuriouseconomist.com/corporate-tax-haven-in-ireland/>. Consulté le 27/01/2026.
- The Daily Economy (2025). The return of quantitative easing. <https://thedailyeconomy.org/article/the-return-of-quantitative-easing/>. Consulté le 24/01/2026.
- The Democracy Center (2006). Bechtel vs. bolivia : Details of the case and the campaign. <https://www.democracyctr.org/bechtel-vs-bolivia>. Consulté le 01/03/2026.
- The Diplomat (2016). A great leap to nowhere: Remembering the US-North Korea "leap day" deal. The Diplomat. <https://thediplomat.com/2016/02/a-great-leap-to-nowhere-remembering-the-us-north-korea-leap-day-deal/>. Consulté le 01/03/2026.
- The Motley Fool (2025). Are REITs a good investment ? <https://www.fool.com/investing/stock-market/market-sectors/real-estate-investing/reit/are-reits-a-good-investment/>. Consulté le 01/03/2026.
- The Week (2011). The federal reserve's 'breathhtaking' \$7.7 trillion bank bailout. <https://theweek.com/articles/479867/federal-reserves-breathhtaking-77-trillion-bank-bailout>. Consulté le 20/01/2026.
- Thiria, P. (2017). Prix de transfert et BEPS. *Revue française de finances publiques*, 138(2), 113–122. <https://droit.cairn.info/revue-revue-francaise-de-finances-publiques-2017-2-page-113>. Consulté le 26/01/2026.
- Tinbergen, J. (1959). Economic policy: principles and design. *The Economic Journal*, 69(274), 353–356. <https://repub.eur.nl/pub/16740>. Consulté le 20/01/2026.
- Tobin, J. (1956). The interest elasticity of the transactions demand for cash. *The Review of Economics and Statistics*, 38(3), 241–247. [https://fraser.stlouisfed.org/files/docs/meltzer/res\\_1956\\_tobin\\_interest\\_elasticity.pdf](https://fraser.stlouisfed.org/files/docs/meltzer/res_1956_tobin_interest_elasticity.pdf). Consulté le 20/01/2026.
- Tobin, J. (1958). Liquidity preference as behavior towards risk. *Review of Economic Studies*, 25, 65–86. <https://www.efalken.com/LowVolClassics/tobin1958.pdf>. Consulté le 20/01/2026.
- Tokarski, P. (2019). Divergence and diversity in the euro area. [https://www.swp-berlin.org/publications/products/research\\_papers/2019RP06\\_tks.pdf](https://www.swp-berlin.org/publications/products/research_papers/2019RP06_tks.pdf). Consulté le 22/01/2026.
- Toledano, J. (2018). Les enjeux des blockchains. <https://www.vie-publique.fr/files/rapport/pdf/184000413.pdf>. Consulté le 11/08/2024.
- Tool Advisor (2025). 12 statistiques sur l'intelligence artificielle (IA) à connaître en 2026. <https://tool-advisor.fr/blog/statistiques-intelligence-artificielle-ia/>. Consulté le 19/01/2026.
- Toute l'Europe (2018). Crise de la dette grecque: quelle situation après 9 ans de bras de fer? <https://www.touteleurope.eu/economie-et-social/crise-de-la-dette-grecque-quelle-situation-apres-9-ans-de-bras-de-fer/>. Consulté le 28/01/2026.
- Toute l'Europe (2025). Paradis fiscaux: la liste de l'Union européenne. <https://www.touteleurope.eu/economie-et-social/paradis-fiscaux-la-liste-de-l-union-europeenne/>. Consulté le 19/12/2025.

- Tracfin (2016). Tendances et analyse des risques de blanchiments de capitaux et de financement du terrorisme en 2016. [https://www.economie.gouv.fr/files/rapport-analyse-tracfin-2016\\_0.pdf?v=1675079461](https://www.economie.gouv.fr/files/rapport-analyse-tracfin-2016_0.pdf?v=1675079461). Consulté le 11/08/2024.
- Tracfin (2024). LCB-FT: état de la menace 2023-2024. [https://www.economie.gouv.fr/files/directions\\_services/tracfin/TRACFIN\\_2023\\_Tome3\\_FR\\_WEB.pdf?v=1722352627](https://www.economie.gouv.fr/files/directions_services/tracfin/TRACFIN_2023_Tome3_FR_WEB.pdf?v=1722352627). Consulté le 11/08/2024.
- Trading Economics (2025a). Zone euro agrégat monétaire M1. <https://fr.tradingeconomics.com/euro-area/money-supply-m1>. Consulté le 22/01/2026.
- Trading Economics (2025b). Zone euro agrégat monétaire M3. <https://fr.tradingeconomics.com/euro-area/money-supply-m3>. Consulté le 22/01/2026.
- Trading View (2024). Market cap BTC dominance. <https://www.tradingview.com/symbols/BTC.D/>. Consulté le 28/08/2024.
- Trafigura (2024). Annual report 2024. [https://www.trafigura.com/media/svxjrm1/2024\\_trafigura\\_annual\\_report.pdf](https://www.trafigura.com/media/svxjrm1/2024_trafigura_annual_report.pdf). Consulté le 27/02/2026.
- Transport & Environment (2023). EU strategic partnerships: How to shape secure, diverse and sustainable trade in critical minerals. [https://www.transportenvironment.org/uploads/files/2023\\_11\\_Briefing\\_EU\\_strategic\\_partnerships.pdf](https://www.transportenvironment.org/uploads/files/2023_11_Briefing_EU_strategic_partnerships.pdf). Consulté le 28/02/2026.
- Transtainer (2015). Iron ore drops below US\$39/t. <https://www.transtainer.com.au/news/iron-ore-drops-below-us39t-2/>. Consulté le 27/02/2026.
- Treynor, J. L. (1965). How to rate management of investment funds. *Harvard Business Review*, 43(1), 63–75. <https://onlinelibrary.wiley.com/doi/epdf/10.1002/9781119196679.ch10>. Consulté le 26/02/2026.
- Uddin, M. A. (2015). Principles of Islamic finance: Prohibition of Riba, Gharar and Maysir. *MPRA Paper*, 67711. [https://mpra.ub.uni-muenchen.de/67711/1/MPRA\\_paper\\_67711.pdf](https://mpra.ub.uni-muenchen.de/67711/1/MPRA_paper_67711.pdf). Consulté le 26/01/2026.
- Union sociale pour l'habitat (2023). Hausse de 3 % du taux du livret A: le Mouvement HLM prend acte d'une décision équilibrée. <https://www.union-habitat.org/hausse-de-3-du-taux-du-livret-le-mouvement-hlm-prend-acte-d-une-decision-equilibree>. Consulté le 01/03/2026.
- Union sociale pour l'habitat (2025a). La demande et les attributions: situation à mi-2025 et perspectives. *USH Statistiques*, 3. [https://www.union-habitat.org/sites/default/files/articles/pdf/2025-12/ush\\_stats\\_3\\_la\\_demande\\_et\\_les\\_attributions\\_situation\\_a\\_mi-2025\\_et\\_perspectives\\_2025-12-19.pdf](https://www.union-habitat.org/sites/default/files/articles/pdf/2025-12/ush_stats_3_la_demande_et_les_attributions_situation_a_mi-2025_et_perspectives_2025-12-19.pdf). Consulté le 01/03/2026.
- Union sociale pour l'habitat (2025b). Les HLM en chiffres – Édition 2025. [https://www.union-habitat.org/sites/default/files/articles/pdf/2025-09/ush\\_reperes\\_158\\_les\\_hlm\\_en\\_chiffres\\_2025\\_2025-09.pdf](https://www.union-habitat.org/sites/default/files/articles/pdf/2025-09/ush_reperes_158_les_hlm_en_chiffres_2025_2025-09.pdf). Consulté le 01/03/2026.
- Université numérique (2009). Les types d'obligations. [https://moodle.luniversitenumerique.fr/pluginfile.php/3855/mod\\_resource/content/0/Leçon%20-%20Les%20types%20d'obligations.pdf](https://moodle.luniversitenumerique.fr/pluginfile.php/3855/mod_resource/content/0/Leçon%20-%20Les%20types%20d'obligations.pdf). Consulté le 20/01/2026.
- Upflow (2024). 13 Accounts Receivable Stats You Should Know in 2024. <https://upflow.io/blog/ar-collections/13-accounts-receivable-cash-collection-statistics-2024>. Consulté le 25/01/2026.
- U.S. Department of Agriculture (2008). High US exports drive record prices and low 2007/08 ending stocks. [https://esmis.nal.usda.gov/sites/default/files/xs55mc046/0p0967691/rb68xc63c/WHS-yearbook-11-13-2008\\_Special\\_Report.pdf](https://esmis.nal.usda.gov/sites/default/files/xs55mc046/0p0967691/rb68xc63c/WHS-yearbook-11-13-2008_Special_Report.pdf). Consulté le 01/03/2026.
- U.S. Department of Agriculture (2024). Wheat outlook: September 2024 (WHS-24i). [https://ers.usda.gov/sites/default/files/\\_laserfiche/outlooks/110018/WHS-24i.pdf](https://ers.usda.gov/sites/default/files/_laserfiche/outlooks/110018/WHS-24i.pdf). Consulté le 01/03/2026.
- U.S. Department of Justice (2024). Samuel Bankman-Fried sentenced to 25 years for his orchestration of multiple fraudulent schemes. <https://www.justice.gov/opa/pr/samuel-bankman-fried-sentenced-25-years-prison>. Consulté le 05/04/2026.
- U.S. Department of Justice (2023). Binance and CEO plead guilty to federal charges. <https://www.justice.gov/archives/opa/pr/binance-and-ceo-plead-guilty-federal-charges-4b-resolution>. Consulté le 05/04/2026.



- U.S. Energy Information Administration (2022a). Europe is a key destination for Russia's energy exports. <https://www.eia.gov/todayinenergy/detail.php?id=55021>. Consulté le 27/02/2026.
- U.S. Energy Information Administration (2022b). Low inventories and high demand boost global distillate crack spreads. <https://www.eia.gov/todayinenergy/detail.php?id=52718>. Consulté le 27/02/2026.
- U.S. Energy Information Administration (2023). Crude oil prices increased in first-half 2022 and declined in second half. <https://www.eia.gov/todayinenergy/detail.php?id=55079>. Consulté le 27/02/2026.
- U.S. Energy Information Administration (2025a). Country analysis brief : Russia. [https://www.eia.gov/international/content/analysis/countries\\_long/Russia/pdf/Russia%20CAB\\_2025.pdf](https://www.eia.gov/international/content/analysis/countries_long/Russia/pdf/Russia%20CAB_2025.pdf). Consulté le 27/02/2026.
- U.S. Energy Information Administration (2025b). Venezuela – Analysis. <https://www.eia.gov/international/analysis/country/VEN>. Consulté le 26/02/2026.
- U.S. Geological Survey (2025). Mineral commodity summaries 2025: Rare earths. <https://pubs.usgs.gov/periodicals/mcs2025/mcs2025-rare-earth.pdf>. Consulté le 27/02/2026.
- U.S. Trade Representative (2023). United States and Japan sign critical minerals agreement. <https://ustr.gov/about-us/policy-offices/press-office/press-releases/2023/march/united-states-and-japan-sign-critical-minerals-agreement>. Consulté le 28/02/2026.
- US Treasury Borrowing Advisory Committee (2023). TIPS issuance, demand, and level of supply. <https://home.treasury.gov/system/files/221/TBACCharge2Q22023.pdf>. Consulté le 22/01/2026.
- Valla, N. (2014). Banques centrales: les défis de la sortie de crise. *Revue d'économie financière*, 2014/1(113), 227–242. <https://shs.cairn.info/revue-d-economie-financiere-2014-1-page-227>. Consulté le 20/01/2026.
- Vie publique (2020). Grand Paris Express: une estimation des coûts en hausse. <https://www.vie-publique.fr/en-bref/277104-grand-paris-express-augmentation-des-couts>. Consulté le 02/03/2026.
- Vie publique (2025a). Dette publique: le rôle des agences de notation en six questions. <https://www.vie-publique.fr/questions-reponses/289661-dette-publique-le-role-des-agences-de-notation-en-six-questions>. Consulté le 20/01/2026.
- Vie publique (2025b). Euro numérique: de quoi s'agit-il ? <https://www.vie-publique.fr/questions-reponses/297781-euro-numerique-euro-mnbc-de-quoi-sagit-il>. Consulté le 06/04/2026.
- Vie publique (2025c). Sept questions sur les concessions d'autoroutes en France. <https://www.vie-publique.fr/questions-reponses/291620-concessions-dautoroutes-tout-comprendre-en-7-questions>. Consulté le 02/03/2026.
- Village de la justice (2021). Fonctionnement et qualification juridique de la clause *take-or-pay* en droit des contrats. <https://www.village-justice.com/articles/fonctionnement-qualification-juridique-clause-take-pay-droit-des-contrats,37637.html>. Consulté le 02/03/2026.
- Villeroy de Galhau, F. (2022). Quel narratif de politique monétaire après la forward guidance? <https://www.banque-france.fr/fr/governors-interventions/quel-narratif-de-politique-monetaire-apres-la-forward-guidance>. Consulté le 20/01/2026.
- Voulvoulis, N., Arpon, K. D., & Giakoumis, T. (2016). The EU Water Framework Directive: from great expectations to problems with implementation. *Science of The Total Environment*, 575, 358–366. <https://bura.brunel.ac.uk/bitstream/2438/27732/1/FullText.pdf>. Consulté le 23/03/2026.
- Waemustafa, W. & Sukri, S. (2017). Theory of gharar and its interpretation of risk and uncertainty from the perspectives of Authentic Hadith and the Holy Quran: A qualitative analysis. *International Journal of Economic Perspectives*, 10(2), 1–27. <https://mpira.ub.uni-muenchen.de/78316/>. Consulté le 26/01/2026.
- Wagstaff, J. (2014). Mt. gox bitcoin debacle: huge heist or sloppy glitch. [https://people.duke.edu/~charvey/Media/2014/Reuters\\_February\\_28\\_2014.pdf](https://people.duke.edu/~charvey/Media/2014/Reuters_February_28_2014.pdf). Consulté le 11/08/2024.
- Walsh, C. E. (1995). Optimal contracts for central bankers. *American Economic Review*, 85(1), 150–167. <https://people.ucsc.edu/~walshc/MyPapers/OptimalContracts.pdf>. Consulté le 24/01/2026.

- Walter, C. P. (2001). The efficient market hypothesis, the gaussian assumption, and the investment management industry. In *EFMA Lugano Meetings*. <https://hal.science/hal-04574614/document>. Consulté le 06/06/2024.
- Wang, D. (2014). Secure implementation of ECDSA signatures in bitcoin. [http://www.nicolascourtois.com/bitcoin/thesis\\_Di\\_Wang.pdf](http://www.nicolascourtois.com/bitcoin/thesis_Di_Wang.pdf). Consulté le 11/08/2024.
- Ward, O. & Rochemont, S. (2019). Understanding central bank digital currencies (CBDC). *Institute and Faculty of Actuaries*, 13(2), 263–268. <https://cointhinktank.com/upload/CBDC%20-%20Understanding%20CBDCs.pdf>. Consulté le 06/06/2024.
- We Are Water (2024). Water waste : Ending water losses, a global priority. We Are Water Foundation. <https://www.wearewater.org/en/insights/water-waste-ending-water-losses-a-global-priority/>. Consulté le 01/03/2026.
- WEED (2008). Food speculation: The main factor of the price bubble in 2008. [https://steuer-gegen-armut.org/fileadmin/Dateien/Kampagnen-Seite/Unterstuetzung\\_Wissenschaft/WEED\\_Food\\_Speculation.pdf](https://steuer-gegen-armut.org/fileadmin/Dateien/Kampagnen-Seite/Unterstuetzung_Wissenschaft/WEED_Food_Speculation.pdf). Consulté le 01/03/2026.
- Weelim (2025). Bilan du marché des SCPI en 2024: collectes, performances et perspectives. <https://www.weelim.fr/scpi/bilan-du-marche-des-scpi-en-2024-collectes-performances-et-perspectives/>. Consulté le 01/03/2026.
- Weil, P. (2008). Overlapping generations: The first jubilee. *Journal of Economic Perspectives*, 22(4), 115–134. <https://pubs.aeaweb.org/doi/10.1257/jep.22.4.115>. Consulté le 24/01/2026.
- Wheelock, D. C. (1992). Monetary Policy in the Great Depression: What the Fed Did, and Why. *Federal Reserve Bank of St. Louis Review*, 74(2), 3–28. <https://fraser.stlouisfed.org/files/docs/meltzer/whemon92.pdf>. Consulté le 26/01/2026.
- White & Case (2023). Will the United States' new Critical Minerals Agreements shape electric vehicle trade? <https://www.whitecase.com/insight-alert/will-united-states-new-critical-minerals-agreements-shape-electric-vehicle>. Consulté le 28/02/2026.
- Williams, J. B. (1938). *The Theory of Investment Value*. Cambridge, MA: Harvard University Press. <https://ia601506.us.archive.org/18/items/in.ernet.dli.2015.225177/2015.225177.The-Theory.pdf>. Consulté le 26/02/2026.
- Wilmarth, A. E. (2011). The Dodd-Frank Act: A flawed and inadequate response to the too-big-to-fail problem. *Oregon Law Review*, 89(3), 951–1057. <https://scispace.com/pdf/the-dodd-frank-act-a-flawed-and-inadequate-response-to-the-2py9yf9faw.pdf>. Consulté le 20/01/2026.
- Wise, H. (2025). What islamic finance offers the modern investor. <https://www.juliusbaer.com/en/insights/wealth-insights/how-to-invest/what-islamic-finance-offers-the-modern-investor/>. Consulté le 26/01/2026.
- Woerth, E. (2021). Rapport d'information déposé en application de l'article 145-8 du règlement par la commission des finances, de l'économie générale et du contrôle budgétaire sur la mise en œuvre des conclusions de la mission d'information relative aux crypto-actifs. *Assemblée Nationale*, 4753. [https://www.assemblee-nationale.fr/dyn/15/rapports/cion\\_fin/l15b4753\\_rapport-information](https://www.assemblee-nationale.fr/dyn/15/rapports/cion_fin/l15b4753_rapport-information). Consulté le 06/08/2024.
- World Gold Council (2025a). Central banks — gold demand trends : Full year 2024. <https://www.gold.org/goldhub/research/gold-demand-trends/gold-demand-trends-full-year-2024/central-banks>. Consulté le 28/02/2026.
- World Gold Council (2025b). Gold demand trends : Full year 2024. <https://www.gold.org/goldhub/research/gold-demand-trends/gold-demand-trends-full-year-2024>. Consulté le 28/02/2026.
- Wright, L. (2025). Bitcoin's Lightning Network capacity declined 20 % in 2025 but it's not as bad as it sounds. <https://cryptoslate.com/why-lightning-network-capacity-declining-20-in-2025-is-not-as-bad-as-it-sounds/>. Consulté le 05/04/2026.
- Xerfi (2024). Les incubateurs et accélérateurs. [https://www.xerfi.com/presentationetude/les-incubateurs-et-accelerateurs\\_SAE57](https://www.xerfi.com/presentationetude/les-incubateurs-et-accelerateurs_SAE57). Consulté le 20/01/2026.
- Yahoo Finance (2024). Cours journaliers du S&P 500. <https://fr.finance.yahoo.com/quote/%5EGSPC/history/?period1=-1325635200&period2=1726099200&interval=1d&filter=history&frequency=1d&includeAdjustedClose=true>. Consulté le 12/09/2024.

Young, T. W. (1991). Calmar ratio: A smoother tool. *Futures*, 20(1), 40.

Yun, S.-J., Choi, S.-Y., & Kim, Y. S. (2023). Examining the hedge performance of US dollar, VIX, and gold against S&P 500 volatility. *PLOS One*, 18(10). <https://pmc.ncbi.nlm.nih.gov/articles/PMC10553206/>. Consulté le 27/02/2026.

Z2Data (2026). Why critical minerals are the crown jewel of Trump's Bid for Greenland. <https://www.z2data.com/insights/why-critical-minerals-are-crown-jewel-of-trumps-bid-for-greenland>. Consulté le 27/02/2026.

Zabala Innovation Consulting (2025). Bilan France 2030: projets financés et opportunités. <https://www.zabala.fr/actualites/bilan-france-2030-projets-opportunites/>. Consulté le 23/03/2026.

Zola, F., Medina, J. A., & Orduna, R. (2024). Assessing the impact of sanctions in the crypto ecosystem: Effective measures or ineffective deterrents? <https://arxiv.org/pdf/2409.10031.pdf>. Consulté le 06/04/2026.

Économie de l'Eau (2024). Recettes globales des agences de l'eau en 2023. <https://economie.eaufrance.fr/chiffres-cles/recettes-globales-des-agences-de-leau-en-2023>. Consulté le 23/03/2026.

Économie et politique (2022). Swift, la russie, l'europe et le dollar. <https://www.economie-et-politique.org/2022/03/01/swift-la-russie-leurope-et-le-dollar/>. Consulté le 06/04/2026.



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